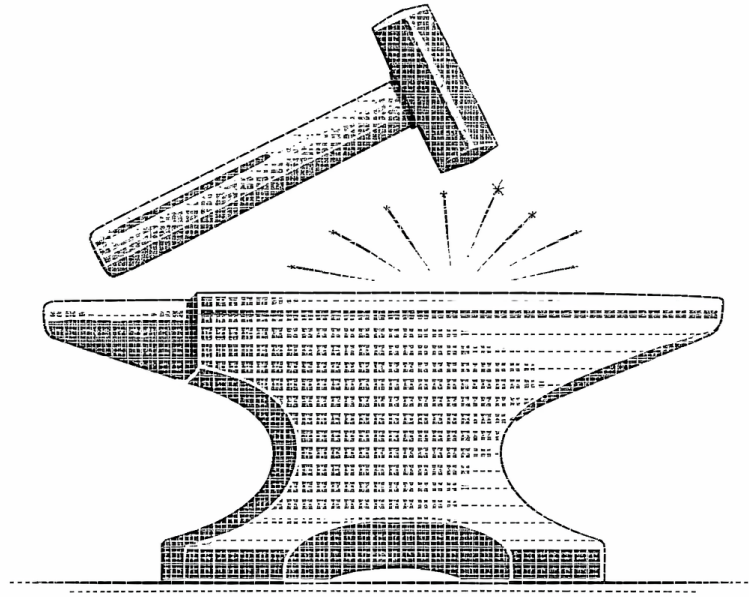


CRYSTALS, GREEN-FUNCTIONS, THERMAL QUANTUM FIELD THEORY AND APPLICATIONS IN QUASI-EQUILIBRIUM MATERIALS SCIENCE



by

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James Gleick about Isaac Newton in [18]:

“Newton formed his argument in classic greek geometrical style. Axioms, Lemmas, Corollaries, QED. As the best model available for perfection in knowledge, it gave his physical program the stamp of certainty. He proved facts about triangles and tangents, chords and parallelograms and from there, by a long chain of arguments, proved facts about the moon and the tides.”

David Hilbert in a lecture¹ given at the International Congress of Mathematicians in Paris 1900 [22]:

"Zudem ist es ein Irrtum zu glauben, dass die Strenge in der Beweisführung die Feindin der Einfachheit wäre."

Galileo Galilei (not verbatim translated) in [11]:

The book of nature is written in the language of mathematics.

Lucius Annaeus Seneca in his Hercules Furens, see also [45]:

"Non est ad astra mollis e terris via."

If you find any mistakes, please contact² me so that I can correct them.

¹The explicit sentence can be found on page 257.

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This still needs to be done or digitalized

Legend:

- colorless = Not important for the doctoral thesis.
-  = Important for the doctoral thesis but very (too?) difficult / tedious to do cleanly.

Chapter 5:

- section 5.2: the derivation of the electron-phonon-coupling, as in Felicianos article.

Chapter 6: The derivation of the Bloch-Theorem and the reference to Slater-determinants solving the non-interacting problem.

Chapter 7:

- Proof of Corollary 7.33.
- Proof of Corollary 7.37.
- Looking up what a derivation is in mathematics and adding a remark on that, as in Remark 7.37.1.
- Proof of Corollary 7.41.
- Proof of Lemma 7.42 (only needs to be digitalized).
- Looking up what a derivation is in mathematics and adding a remark on that in Corollary 7.45.
- Elaborating on Definition 7.47.
- Proof of Corollary 7.49.

Chapter 8:

- Updating Definition 8.14 (is important for Eliashberg-theory from the Operator-formalism)
- Proof of Lemma 8.31 (only needs to be digitalized).
- Proving the identity $[c(\tau)]^\dagger = c^\dagger(-\tau)$, etc. (is important for Eliashberg-theory from the Operator-formalism)

Chapter 9:

- Proof of Lemma 9.6.
- Proof of Theorem 9.7.
- Proof of Theorem 9.10.
- Elaborating on Remark 9.11.1.
- Proof of Corollary 9.17.
- Definition 9.19 (Thermal equilibrium of two systems).
- Proof of Theorem 9.20.
- Elaborating on Hypothesis 9.25.
- Proof of Theorem 9.27.

-
- Proof of Theorem 9.29.
 - Introduction to the grand canonical ensemble in section 9.5.
 - Proof of the equivalence of ensembles in section 9.6.
 - Elaborating on Legendre-transformations in thermodynamics in section 9.7.
 - Elaborating on the transition from microcanonical to canonical ensemble in section 9.8.2.
 - Proof of Lemma 9.36.
 - Elaborating on Theorem 9.42.
 - Elaborating on the transition from canonical to grand canonical ensemble in section 9.8.3.

Chapter 10:

- Thermodynamic relations for mixed-systems as in Remark 10.7.3.

Chapter 16:

- Going through the proof of Lemma 16.33.
- Proof of the trace-log-identity in Corollary 16.19;
- Elaborating on Remark 16.33.2.
- Updating section 16.2.
- Implementing and Updating Section 16.3 from my Master-thesis
- Updating the proofs in subsections 16.1.7 and 16.1.8.

Chapter 19:

- Calculating the polarization in Theorem 19.4.

Chapter 17:

- Just fixing the things from my master-thesis and then referencing the Thomale-paper for the rest of the details.
- Implementing the implications of the symmetries in section 17.1.2 (only needs to be copied from the Thomale-paper).

Chapter 18:

- Elaborating on Theorem 18.26.
- Elaborating on subsection 18.2.3.
- Adding the value in Remark 18.14.1.
- Figuring out which identity I was referring to in Remark 18.14.3.

Chapter 20:

- Checking if the details are really compatible with the definition of the Green-functions introduced earlier.

- Adding references to earlier identities (?).

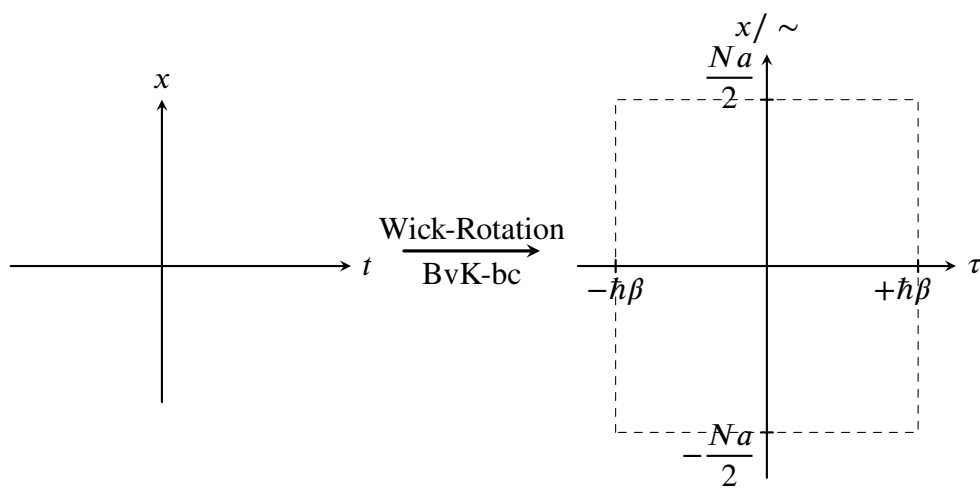
Chapter 22:

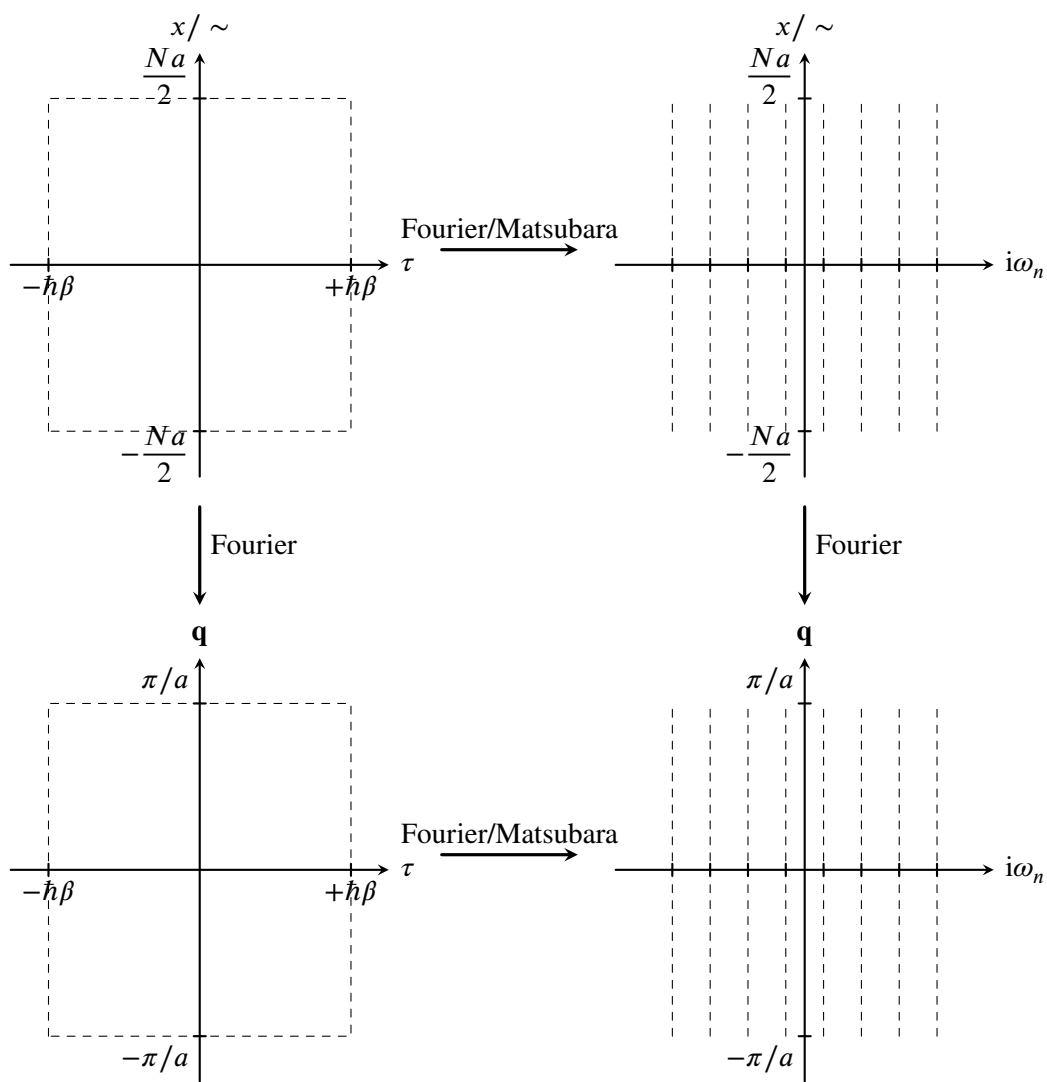
- Adding the equations also in the interacting case.

Chapter 24:

- Cleaning the whole thing up.
- Doing the calculation for the equation of motion for the last Green-function.

Chapter 37: Getting a clean setup of Feynman-diagrams in condensed matter theory and drawing their pictures.





Part I.

**Fundamentals on Quantum Mechanics and
Crystals**

Motivation and Overview

After this part of the notes you will understand ...

- what donuts in Figure 1 have to do with materials theory,
- how and why one determines the band-structure of a crystal, and how and why it is usually represented by pictures like in Figure 2,

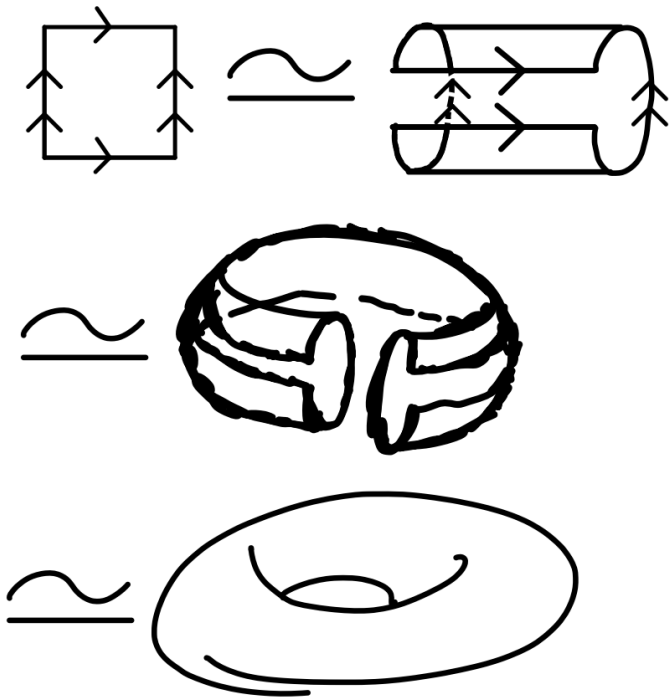


Figure 1.: Identifying the boundaries of a square yields a torus („donut“). One also says that the two objects are „homeomorphic“ to each other.

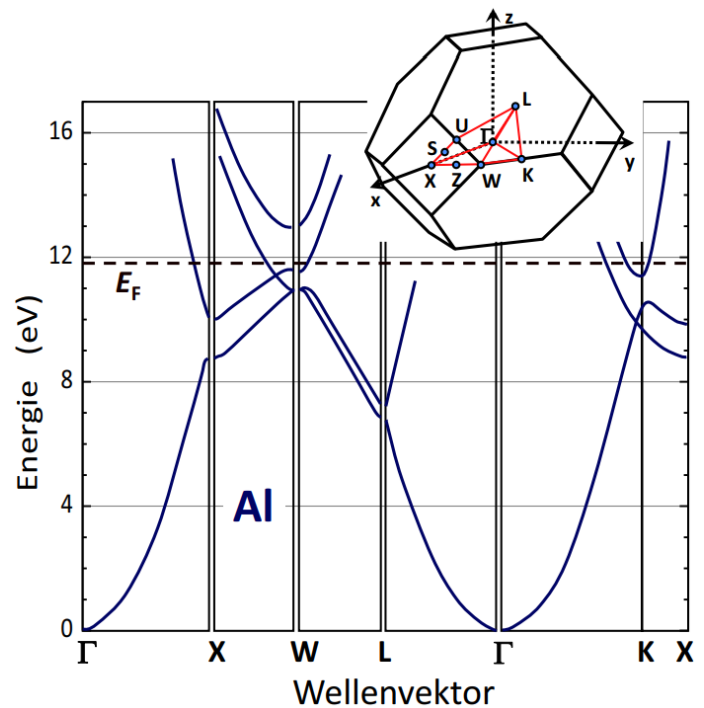


Figure 2.: The Band-Structure of Aluminium along the high-symmetry $\mathbf{k}$ -points. Taken from [GM].

1. Fourier-Expansions

Fact 1.1 The Fourier-transformation exists in all kinds of shapes and colours. One can expand

- finite sequences of numbers in the complex plane ¹
- periodic functions on a finite interval $(0, L)^2$,
- antiperiodic functions on a finite interval $(0, \hbar\beta)$ leading to the so-called Matsubara-frequencies³.
- sufficiently fast decaying functions⁴ on $\mathbb{R}$,

Here we will get to know the different versions that are most important for materials theory.

Remark 1.1.1 We wish to warn the reader, that in various different situations various *different* standard-conventions have established throughout the scientific literature. This makes it particular difficult for newcomers. For example the Fourier-transformed dynamical matrices have a different prefactor conventions than the Fourier-transformed ionic elongations although both of them are just discrete Fourier-transformations on the underlying lattice [9].

1.1. Fourier for a Discrete Function on a Finite Set of Points

Definition 1.2 For a sequence of N (complex or real-valued) numbers $\{\xi_1, \dots, \xi_N\} \subset \mathbb{C}^{d_x}$, one introduces their *discrete Fourier-transformed sequence* as another sequence of numbers $\{\hat{\xi}_1, \dots, \hat{\xi}_N\} \subset \mathbb{C}^{d_x}$ as

$$\hat{\xi}_j = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} \xi_n e^{+i2\pi \frac{n}{N}j} \quad (1.1)$$

Remark 1.2.1 There are many prefactor-conventions in this definition, which can be quite annoying when one looks at literature, and the corresponding authors don't state which one they used. The sign in the exponent could have been chosen the other way around in Equations (1.1) and (1.2). The sign-convention chosen above refers usually to j representing a „space-like“-coordinate, whereas the other sign-convention refers to „time-like“-coordinates. The prefactor $\frac{1}{\sqrt{N}}$ in Equation (1.1) can also be exchanged for 1, or $\frac{1}{N}$, or sometimes $\frac{1}{\sqrt{Na}} \equiv 1/\sqrt{L}$ or $\frac{1}{Na} \equiv 1/L$ with corresponding changes in the Equation (1.2) accordingly, whereas a represents a so-called "lattice constant" (will be introduced later). If one chooses the last prefactor convention, one also sometimes encounters

¹Cf. Definition 1.2

²Cf. Theorem 1.7

³Cf. Lemma 1.11

⁴Notice that these functions coincide at $x = \pm\infty$, such that they are "periodic to zero at infinity", which is the main idea behind the derivation of the underlying formulae. Cf. also Notation 1.17.

integral notations, although one is strictly speaking working with discrete variables. This trend is particularly present in thermal quantum field theory (cf. Equation (18.19) in Chapter 18), when one introduces Grassmann-variables in integral notation for their temporal argument, although they actually refer to discrete variables. Notice further that one of the prefactor-conventions here corresponds to the usual Fourier-transformation for periodic functions (with slightly changed notation).

Remark 1.2.2 As we will see below, one introduces the same notation for position-operators of ions that are centered around a crystal-site. Consequently, one finds Fourier-transforms of operators and the inverse transformations.

Lemma 1.3 The inverse transformation reads

$$\xi_j = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} \hat{\xi}_n e^{-i2\pi \frac{n}{N} j} \quad (1.2)$$

Proof 1.3 See Appendix, Proofs, page 393.

1.2. Fourier for a Continuous, Periodic Function on a Finite Interval

This section briefly lists basic terms and concepts that are mostly assumed to be known to clarify the notation.

Notation 1.4 We consider the set of periodic, square-integrable functions on (a, b) as

$$L_{\text{per}}^2(a, b) = \left\{ f : (a, b) \rightarrow \mathbb{C} \mid f(a) = f(b) \right\} \quad (1.3)$$

Lemma 1.5 The set $L_{\text{per}}^2(a, b)$ is, equipped with point-wise addition and scalar-multiplication a complex vector space.

Proof 1.5 One easily verifies the different properties.

Lemma 1.6 The mapping

$$\begin{aligned} \langle \cdot, \cdot \rangle : L_{\text{per}}^2(a, b) \times L_{\text{per}}^2(a, b) &\rightarrow \mathbb{C} \\ (f, g) &\mapsto \langle f, g \rangle \equiv \langle f | g \rangle \end{aligned} \quad (1.4)$$

with

$$\langle f, g \rangle := \int_a^b \overline{f(x)} g(x) dx.$$

defines a hermitian scalar-product on $L_{\text{per}}^2(a, b)$.

Proof 1.6 See Kaul, Fischer, Band 2.

Theorem 1.7 For $f \in L_{\text{per}}^2(a, b)$ one finds the *Fourier-expansion* in terms of complex exponentials as

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{\pm i \frac{2\pi n x}{b-a}}, \quad (1.5)$$

where the Fourier coefficients c_n are given by

$$c_n = \frac{1}{b-a} \int_a^b f(x) e^{\mp i \frac{2\pi n x}{b-a}} dx. \quad (1.6)$$

which converges in the L^2 -sense⁵.

Proof 1.7 A proof, that periodic functions can be expanded in this way, is given in the literature⁶.

Remark 1.7.1 The sign is convention; In analogy to electromagnetic waves one usually agrees to use the one sign-convention " $\pm \equiv +$ " if ξ represents a space-like-argument, and the other sign-convention " $\pm \equiv -$ " if it is a time-like-argument. Notice that just as in the discrete case, one oftentimes shifts around prefactors into / out of the Fourier-coefficients.

Remark 1.7.2 One immediately finds for the functions defined on a three-torus (i.e. periodic in three arguments⁷) similar expansions.

Notation 1.8 For $f \in L_{\text{per}}^2([a, b])$ we denote its periodic standard-continuation on $\mathbb{R}$ with the same symbol for notational simplicity.

Fact 1.9 The plane-waves provide a basis for $L_{\text{per}}^2([a, b])$. Together with the scalar-product, this turns the space into a Hilbert-space.

⁵This means that

$$\lim_{N \rightarrow \infty} \int_a^b \left| f(x) - \sum_{n=-N}^N c_n e^{\pm i \frac{2\pi n x}{b-a}} \right|^2 dx = 0. \quad (1.7)$$

⁶In the book [14], one finds the expansion and the identities for the coefficients for periodic functions that are defined on the interval $[0, 2\pi]$. So one will not circumvent a little bit of effort with the substitution rule, in order to reproduce the results above.

⁷Notice that wave-functions in a crystal (after applying of Born-von-Karman-boundary-conditions fall into this category.

1.3. Fourier for a Continuous, Statistical Function on a Finite Interval

Notation 1.10 For reasons, that are difficult to motivate several chapters in advance, we consider functions

$$\mathcal{G}_\zeta \in L^2_{\text{per}}(-\hbar\beta, +\hbar\beta) \quad (1.8)$$

with time-like argument

$$\begin{aligned} \mathcal{G}_\zeta : (-\hbar\beta, +\hbar\beta) &\rightarrow \mathbb{C} \\ \tau &\mapsto \mathcal{G}_\zeta(\tau) \end{aligned}$$

and which carry a *statistical index* $\zeta \in \{-1, +1\}$ such that the *statistics-condition* is fulfilled

$$\forall \tau \in (-\hbar\beta, 0) : \quad \mathcal{G}_\zeta(\tau + \hbar\beta) = \zeta \mathcal{G}_\zeta(\tau). \quad (1.9)$$

Remark 1.10.1 We call a function that obeys this condition to be ζ -periodic.

Remark 1.10.2 The statistical index ζ represents *bosonic* ($\zeta = +1$) and *fermionic* ($\zeta = -1$) functions.

Remark 1.10.3 Denoting it's analytic standard-continuation on $\mathbb{R}$ with the same symbol, we obtain the property

$$\forall \tau \in \mathbb{R} : \quad \mathcal{G}_\zeta(\tau + \hbar\beta) = \zeta \mathcal{G}_\zeta(\tau). \quad (1.10)$$

Lemma 1.11 By a Fourier-expansion one finds the expansion for the periodic standard-continuation as

$$\mathcal{G}_\zeta(\tau) = \frac{1}{\hbar\beta} \sum_{\omega_n \in \Omega_\zeta} \mathcal{G}_\zeta(i\omega_n) e^{-i\omega_n \tau} \quad (1.11)$$

with coefficients

$$\mathcal{G}_\zeta(i\omega_n) = \int_0^{\hbar\beta} d\tau \mathcal{G}_\zeta(\tau) e^{i\omega_n \tau}. \quad (1.12)$$

and *Matsubara-frequencies* that depend on the statistics

$$\Omega_\zeta = \frac{2\mathbb{Z} + 1 - \theta(\zeta)}{\hbar\beta} \pi. \quad (1.13)$$

Proof 1.11 See Appendix, Proofs, page 393.

Corollary 1.12 For future reference we note the identity

$$\forall \omega_m \in \Omega_{+1} : \int_0^{\hbar\beta} d\tau e^{i\tau\omega_m} = \delta_{\omega_m,0} \hbar\beta \quad (1.14)$$

Proof 1.12 See Appendix, Proofs, page 395.

Remark 1.12.1 Notice that the difference of two fermionic Matsubara-frequencies is again a bosonic Matsubara-frequency.

For reasons that are difficult to motivate several chapters in advance, we introduce the following Lemma.

Lemma 1.13 For a Matsubara-frequency $\omega_n \in \Omega_{S_{X,X'}}$ with $S_{X,X'} \in \{-1, +1\}$ it holds

$$\frac{1}{\hbar} \int_0^{\hbar\beta} d\tau e^{i\omega_n \tau} e^{\tau(\xi_n - \xi_m)/\hbar} = \frac{1}{\hbar} \frac{S_{X,X'} e^{\beta(\xi_n - \xi_m)} - 1}{i\omega_n + (\xi_n - \xi_m)/\hbar} \quad (1.15)$$

Proof 1.13 With the definition of the Matsubara-frequencies in Equation (1.13), we remark the trivial identity

$$e^{i\hbar\omega_n\beta} = S_{X,X'} \quad (1.16)$$

and then we compute straightforwardly

$$\begin{aligned} \frac{1}{\hbar} \int_0^{\hbar\beta} d\tau e^{i\omega_n \tau} e^{\tau(\xi_n - \xi_m)/\hbar} &= \frac{1}{\hbar} \int_0^{\hbar\beta} d\tau e^{\tau[i\omega_n + (\xi_n - \xi_m)/\hbar]} \\ &= \frac{1}{\hbar} \frac{1}{i\omega_n + (\xi_n - \xi_m)/\hbar} \left[e^{\tau[i\omega_n + (\xi_n - \xi_m)/\hbar]} \right]_0^{\hbar\beta} \\ &= \frac{1}{\hbar} \frac{1}{i\omega_n + (\xi_n - \xi_m)/\hbar} \left[S_{X,X'} e^{\beta(\xi_n - \xi_m)} - 1 \right] \\ &= \frac{1}{\hbar} \frac{S_{X,X'} e^{\beta(\xi_n - \xi_m)} - 1}{i\omega_n + (\xi_n - \xi_m)/\hbar} \end{aligned}$$

Definition 1.14 For each $\zeta \in \{-1, +1\}$ we denote the corresponding subspace of $L^2_{\text{per}}(-\hbar\beta, +\hbar\beta)$ for which Equation (1.9) is fulfilled with $L_B \equiv L_1$ and $L_F \equiv L_{-1}$ respectively.

Lemma 1.15 The space of periodic functions can be decomposed into the direct sum of bosonic and fermionic functions as

$$L_{\text{per}}^2(-\hbar\beta, +\hbar\beta) = L_B \oplus L_F \equiv \bigoplus_{\zeta \in \{-1, +1\}} L_{\zeta} \quad (1.17)$$

Proof 1.15 The plane-waves can be divided into bosonic and fermionic parts by considering the corresponding Matsubara-frequencies. The bosonic and fermionic plane-waves are linearly independent, and they span the whole space of periodic functions.

Lemma 1.16 On L_{ζ} we obtain the resolution of identity

$$\mathbb{1}_{L_{\zeta}} \equiv \sum_{\omega_n \in \Omega_{\zeta}} |\mathrm{i}\omega_n\rangle \langle \mathrm{i}\omega_n| \quad (1.18)$$

with plane waves

$$\langle \tau | \mathrm{i}\omega_n \rangle \equiv \mathrm{e}^{-\mathrm{i}\omega_n \tau}, \quad \langle \mathrm{i}\omega_n | \tau \rangle \equiv \mathrm{e}^{+\mathrm{i}\omega_n \tau} \quad (1.19)$$

and the delta-function

$$\delta(\tau) = \frac{1}{\hbar\beta} \sum_{\omega_n \in \Omega_{\zeta}} \mathrm{e}^{-\mathrm{i}\omega_n \tau} \quad (1.20)$$

Proof 1.16 See Appendix, Proofs, page 395.

1.4. Fourier for a Continuous, Decaying Function on an Infinite Interval

Notation 1.17 For „sufficiently smooth“ complex-valued functions and distributions that decay „sufficiently fast“ at space-time-infinity

$$f : \mathbb{R}^3 \times \mathbb{R} \rightarrow \mathbb{C}$$

$$(\vec{x}, t) \mapsto f(\vec{x}, t) \equiv \int \frac{\mathrm{d}^3 k}{(2\pi)^3} \int \frac{\mathrm{d}\omega}{2\pi} \mathrm{e}^{\mathrm{i}(\vec{k} \cdot \vec{x} - \omega t)} f(\vec{k}, \omega)$$

we write for the ("physical") *Fourier-transformation*

$$(\mathcal{F}f) : \mathbb{R}^3 \times \mathbb{R} \rightarrow \mathbb{C}$$

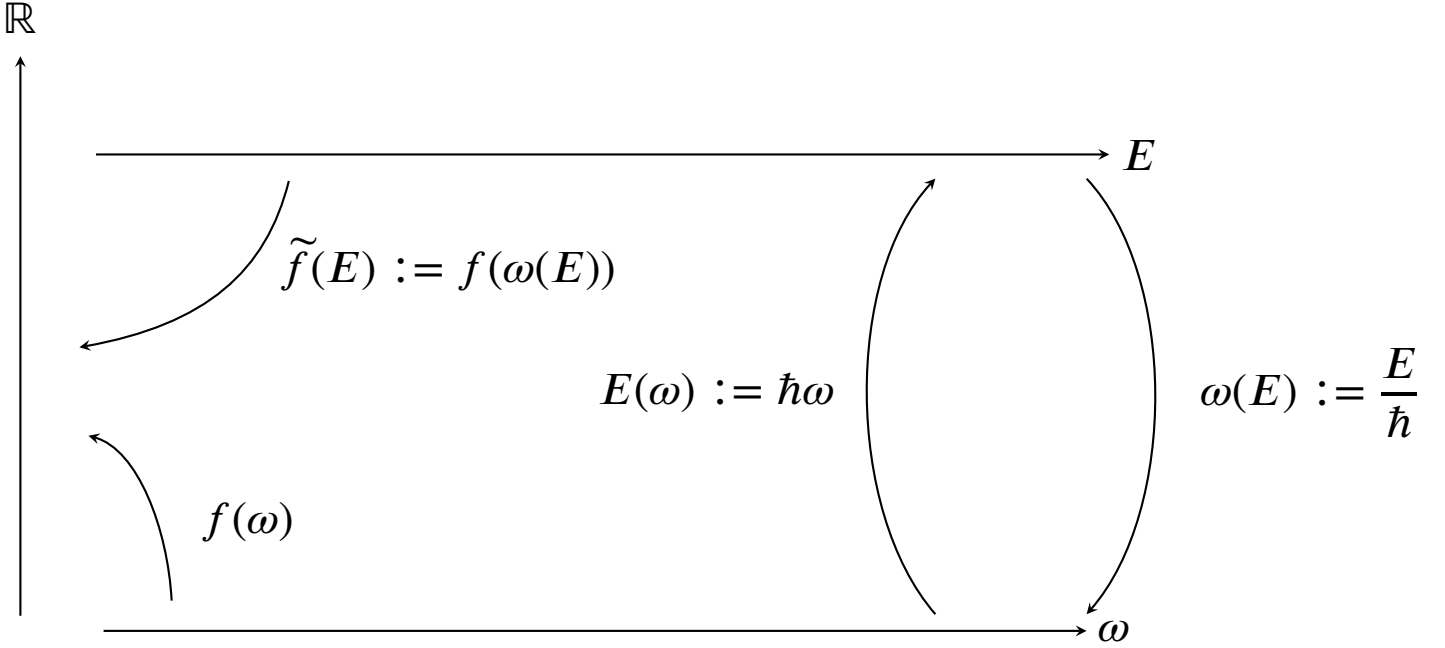
$$(\vec{k}, \omega) \mapsto (\mathcal{F}f)(\vec{k}, \omega) \equiv f(\vec{k}, \omega) \equiv \int d^3x \int dt e^{-i(\vec{k} \cdot \vec{x} - \omega t)} f(\vec{x}, t)$$

Proof 1.17 A proof for this expansion, can be found in [14].

Fact 1.18 Transitioning from frequency to energy integrals happens so often, that we consider it to be reasonable to add the transition-formula here

$$\int_{\mathbb{R}} d\omega f(\omega) \stackrel{\text{SR.}}{=} \int_{\mathbb{R}} dE f(\omega(E)) \omega'(E) = \int_{\mathbb{R}} dE f\left(\frac{E}{\hbar}\right) \frac{1}{\hbar} \equiv \frac{1}{\hbar} \int_{\mathbb{R}} dE \tilde{f}(E). \quad (1.21)$$

The figure illustrates the transition.



For reasons that are impossible to understand several chapters in advance, we compute the following integrals.

Lemma 1.19 Let $\xi_n \in \mathbb{R}$ and $\epsilon > 0$. Then we obtain

$$\frac{i}{\hbar} \int dt e^{i\omega t} e^{-\epsilon|t|} e^{it(\xi_n - \xi_m)/\hbar} \theta(t) = \frac{-1}{\hbar} \frac{1}{\omega + i\epsilon + (\xi_n - \xi_m)/\hbar} \quad (1.22)$$

Proof 1.19 We compute straightforwardly

$$\begin{aligned}
 \frac{i}{\hbar} \int dt e^{i\omega t} e^{-\epsilon|t|} e^{it(\xi_n - \xi_m)/\hbar} \theta(t) &= \frac{i}{\hbar} \int_0^\infty dt e^{t(-\epsilon + i[\omega + (\xi_n - \xi_m)/\hbar])} \\
 &= \frac{i}{\hbar} \frac{-1}{-\epsilon + i[\omega + (\xi_n - \xi_m)/\hbar]} \\
 &= \frac{-1}{\hbar} \frac{1}{\omega + i\epsilon + (\xi_n - \xi_m)/\hbar}
 \end{aligned}$$

Lemma 1.20 Let $\xi_\bullet \in \mathbb{R}$. Then we obtain

$$\frac{i}{\hbar} \int dt e^{i\omega t} e^{-\epsilon|t|} e^{it(\xi_n - \xi_m)/\hbar} \theta(-t) = \frac{1}{\hbar} \frac{1}{\omega - i\epsilon + (\xi_n - \xi_m)/\hbar} \quad (1.23)$$

Proof 1.20 We compute straightforwardly

$$\begin{aligned}
 \frac{i}{\hbar} \int dt e^{i\omega t} e^{-\epsilon|t|} e^{it(\xi_n - \xi_m)/\hbar} \theta(-t) &= \frac{i}{\hbar} \int_{-\infty}^0 dt e^{t(+\epsilon + i[\omega + (\xi_n - \xi_m)/\hbar])} \\
 &= \frac{i}{\hbar} \frac{1}{+\epsilon + i[\omega + (\xi_n - \xi_m)/\hbar]} \\
 &= \frac{1}{\hbar} \frac{1}{\omega - i\epsilon + (\xi_n - \xi_m)/\hbar}
 \end{aligned}$$

Fact 1.21 Notice that these results can be summarized by the identity

$$\frac{i}{\hbar} \int dt e^{i\omega t} e^{-\epsilon|t|} e^{it(\xi_n - \xi_m)/\hbar} \theta(\pm t) = \frac{\mp 1}{\hbar} \frac{1}{\omega \pm i\epsilon + (\xi_n - \xi_m)/\hbar} \quad (1.24)$$

Notice further the standard-identity

Theorem 1.22 We therefore note the standard-identities for test-functions that decay sufficiently fast at infinity

$$\delta(\omega) = \frac{1}{2\pi} \int dt e^{i\omega t} \quad (1.25)$$

Proof 1.22 See, for example, [14], page 324.

2. Quantum Mechanics

There is at least three formalisms with which one can describe quantum-mechanical systems:

1. The „wavefunction-formalism“,
2. The „density-matrix-formalism“,
3. The „path-integral-formalism“.

In this chapter we will focus on the wavefunction-formalism, the other formalisms build upon it.

2.1. Fundamentals about Hilbert-Spaces

Definition 2.1 A Hilbert-space $\mathcal{H}$ is a complex vector space, with a hermitian¹ scalar-product

$$\begin{aligned}\langle \cdot, \cdot \rangle : \mathcal{H} \times \mathcal{H} &\rightarrow \mathbb{C} \\ (\psi, \chi) &\mapsto \langle \psi, \chi \rangle \equiv \langle \psi | \chi \rangle\end{aligned}$$

which is complete and has a countable orthonormal basis².

Notation 2.2 The countable orthonormal basis from the definition above is usually denoted as $\{|1\rangle, |2\rangle, \dots, |d_{\mathcal{H}}\rangle\}$ such that

$$\mathcal{H} = \text{span}_{\mathbb{C}}\{|1\rangle, |2\rangle, \dots, |d_{\mathcal{H}}\rangle\} \quad (2.1)$$

and most oftentimes the basis is specified further, depending on the context of the problem.

Fact 2.3 Some common single-particle Hilbert-spaces³ to work with are in $d_{\mathbf{x}} \in \{1, 2, 3\}$ spatial-dimensions ...

- $\mathcal{H} = L^2(\mathbb{R}^{d_{\mathbf{x}}})$ with $\langle \psi | \chi \rangle = \int d^{d_{\mathbf{x}}} \mathbf{x} \bar{\psi}(\mathbf{x}) \chi(\mathbf{x})$ for a Schrödinger-particle in space,
- $\mathcal{H} = L^2(\mathbb{T}^{d_{\mathbf{x}}})$ with $\langle \psi | \chi \rangle = \int_{\text{vol}} d^{d_{\mathbf{x}}} \mathbf{x} \bar{\psi}(\mathbf{x}) \chi(\mathbf{x})$ for a Schrödinger-particle in a $d_{\mathbf{x}}$ -dimensional „crystal“⁴,

¹Recall that a hermitian scalar-product is linear w.r.t. the second argument, obeys $\langle \psi_1, \psi_2 \rangle = \langle \psi_2, \psi_1 \rangle^*$ and is positive semidefinite.

²Note that the latter condition is equivalent to the Hilbert-space being „separable“. The precise meaning of this can be looked up. Usually (in almost all situations) one is however doing quite well with the abovementioned definition.

³Proving that these spaces fall into the definition above is far from trivial. For example the separability of $L^2(\mathbb{R})$ takes (according to "folklore") about 6 pages in J. von Neumanns book.

⁴The precise meaning of the word „crystal“ will be introduced later. At this point we note, that one can identify coordinates $\mathbf{x}$ in a „crystal“ (with periodic boundary conditions) with angles on a torus.

- $\mathcal{H} = \mathbb{C}^2$ with complex standard-scalar-product for a two-level-system,
- $\mathcal{H} = L^2(\mathbb{R}^{d_x}) \otimes \mathbb{C}^2$ with scalar-product $\langle \psi | \chi \rangle = \sum_{\sigma \in \{\uparrow, \downarrow\}} \int d^{d_x} \mathbf{x} \bar{\psi}_\sigma(\mathbf{x}) \chi_\sigma(\mathbf{x}) \equiv \oint_{\sigma, \mathbf{x}} \bar{\psi}_\sigma(\mathbf{x}) \chi_\sigma(\mathbf{x})$ for a Pauli-particle in space,
- $\mathcal{H} = L^2(\mathbb{T}^{d_x}) \otimes \mathbb{C}^2$ with scalar-product $\langle \psi | \chi \rangle = \sum_{\sigma \in \{\uparrow, \downarrow\}} \int_{\text{vol}} d^{d_x} \mathbf{x} \bar{\psi}_\sigma(\mathbf{x}) \chi_\sigma(\mathbf{x}) \equiv \oint_{\sigma, \mathbf{x}} \bar{\psi}_\sigma(\mathbf{x}) \chi_\sigma(\mathbf{x})$ for a Pauli-particle in a „crystal“,

Remark 2.3.1 For the Hilbert space in physics, it is common practice to consider those elements as equivalent / equal, that differ only on sets of measure zero, thereby working with equivalence classes. However, since these distinctions have no effect on the observables considered in this work, we shall not concern ourselves with them here.

Remark 2.3.2 When there is more than one particle present, new structures and phenomena related to permutation symmetry appear. It turns out that the generalization to n -particle quantum mechanics depends on whether the particles under consideration are distinguishable or indistinguishable. In the former case the Hilbert space $\mathcal{H}$ of the many-particle system is simply the tensor product

$$\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \cdots \otimes \mathcal{H}_n. \quad (2.2)$$

Remark 2.3.3 The Hilbert-spaces for indistinguishable many-particle-systems are a little bit tedious to introduce (approximately 20 pages in the book from Negele and Orland). We will consider them in a later chapter. Apart from the single-particle-examples however, everything in this chapter also applies for many-particle-systems.

Notation 2.4 For vectors $\psi \in \mathcal{H}$ one agrees on the notation $|\psi\rangle \equiv \psi$. One calls these kind of vectors „ket“-vectors. Through the canonical isomorphism⁵ to the dual-space

$$\begin{aligned} J_{\mathcal{H}} : \mathcal{H} &\rightarrow \mathcal{H}^* \\ \psi &\mapsto J_{\mathcal{H}}(\psi) \equiv \langle \psi, \bullet \rangle \equiv \langle \psi | \bullet \rangle \end{aligned}$$

one introduces so-called „bra“-vectors. Taking a bra- and a ket-vector $\langle \chi |$ and $|\psi\rangle$ and „calculating their bra-c-ket“ $\langle \chi | \psi \rangle$ is therefore equivalent to calculating the scalar-product between two vectors.

Notation 2.5 An operator A is a linear mapping defined on the Hilbert-space

$$\begin{aligned} A : \mathcal{H} &\mapsto \mathcal{H} \\ |\psi\rangle &\mapsto A|\psi\rangle \end{aligned}$$

⁵Isomorph, as a *real* vector space, not as a complex vector-space, since the mapping $J_{\mathcal{H}}$ is anti-linear. (Cf. Zirnbauer-Lecture-Notes.)

Remark 2.5.1 In the case that $\mathcal{H}$ represents a single-particle Hilbert-space, one also speaks of a *single-particle-operator*. In the case that $\mathcal{H}$ represents a many-particle Hilbert-space, and the action of A is not trivial on the different sectors, one also speaks of a *many-particle-operator*.

Remark 2.5.2 Sometimes one also encounters notations as $|A\psi\rangle \equiv A|\psi\rangle$. For the corresponding bra-vector one writes $J_{\mathcal{H}}(A\psi) \equiv \langle A\psi|\bullet\rangle$.

Notation 2.6 Let $\mathcal{H} = \text{span}_{\mathbb{C}}\{|1\rangle, |2\rangle, \dots\}$ be spanned by an orthonormal basis⁶. Then we have the matrix of $\text{mat } A$ w.r.t. this basis defined as

$$(\text{mat } A)_{i,j} \equiv \langle i|A|j\rangle. \quad (2.3)$$

Remark 2.6.1 As a special case of an operator, let us introduce the *projector* $P_{\chi} \equiv |\chi\rangle\langle\chi|$ of a state $\chi \in \mathcal{H}$ as a linear mapping

$$\begin{aligned} P_{\chi} : \mathcal{H} &\rightarrow \mathcal{H} \\ \psi &\mapsto P_{\chi}(\psi) \equiv |\chi\rangle\langle\chi|\psi\rangle. \end{aligned}$$

Remark 2.6.2 In Section 9.1 we will understand how the matrix-representation of projectors are related to dyadic products in linear algebra.

Remark 2.6.3 Note that

$$\mathbb{1} \equiv \text{Id}_{\mathcal{H}} \equiv \sum_n |n\rangle\langle n|, \quad (2.4)$$

for an orthonormal system of the Hilbert-space, which can serve as a useful mnemonic in the following sense:

- A vector $\psi \equiv |\psi\rangle$ can be expanded w.r.t. the basis as

$$|\psi\rangle = \mathbb{1}|\psi\rangle = \sum_n |n\rangle\langle n|\psi\rangle \quad (2.5)$$

which is the same expression that one encounters in linear algebra for finite-dimensional vector spaces.

- The quantity $A|\psi\rangle$ represents again a vector that can be expanded w.r.t. the basis as

$$A|\psi\rangle = A\mathbb{1}|\psi\rangle = \sum_n A|n\rangle\langle n|\psi\rangle \quad (2.6)$$

this yields for the m -th component of the vector $A|\psi\rangle$

⁶Case-dependent: Could be a finite basis, but also infinite.

$$\langle m|A|\psi\rangle = \sum_n \langle m|A|n\rangle \langle n|\psi\rangle \quad (2.7)$$

with matrix-elements $\langle m|A|n\rangle \in \mathbb{C}$.

Notation 2.7 In the case of a single, spinless particle, i.e. for $V \in \{\mathbb{T}^{d_x}, \mathbb{R}^{d_x}\}$, with $d_x \in \{1, 2, 3\}$, as a mnemonic, one usually introduces for $\mathcal{H} = L^2(V)$ so-called *position-eigenbasis* $|\mathbf{x}\rangle$ with $\mathbf{x} \in V$, giving a special version of the unity-operator

$$\text{Id}_{\mathcal{H}} \equiv \int_V d^{d_x} \mathbf{x} |\mathbf{x}\rangle \langle \mathbf{x}|. \quad (2.8)$$

By direct comparison of the expressions

$$\int d^{d_x} \mathbf{x} \bar{\chi}(\mathbf{x}) \psi(\mathbf{x}) = \langle \chi | \psi \rangle = \langle \chi | \text{Id}_{\mathcal{H}} | \psi \rangle = \int d^{d_x} \mathbf{x} \langle \chi | \mathbf{x} \rangle \langle \mathbf{x} | \psi \rangle, \quad (2.9)$$

one sees that it must hold

$$\langle \chi | \mathbf{x} \rangle = \bar{\chi}(\mathbf{x}), \quad \langle \mathbf{x} | \psi \rangle = \psi(\mathbf{x}). \quad (2.10)$$

Remark 2.7.1 Although one finds in almost every physics-textbook that one must handle these "functions" with a lot of care, note that a regularization of a delta-function is square-integrable. Within the framework of solid-state-physics, one finds the well-defined, discrete „twins“ of position-eigenstates in so-called „Wannier-states“. Adopting a continuum-notation one finds well defined expressions and justifications for the „state-vectors“ $|\mathbf{x}\rangle$ above.

Notation 2.8 In the case of a single, spin- $\frac{1}{2}$ particle, i.e. for $\mathcal{H} = L^2(V) \otimes \mathbb{C}^2$, with $V \in \{\mathbb{T}^{d_x}, \mathbb{R}^{d_x}\}$, and $d_x \in \{1, 2, 3\}$ and some basis for $\mathbb{C}^2 \equiv \text{span}_{\mathbb{C}}\{|\uparrow\rangle, |\downarrow\rangle\}$ (let it be the eigenstates of σ_z here for concreteness), one also finds a resolution of identity w.r.t. to the product-basis

$$\text{Id}_{\mathcal{H}} \equiv \oint_{\substack{\mathbf{x} \in V \\ \sigma \in \{\uparrow, \downarrow\}}} |\mathbf{x}, \sigma\rangle \langle \mathbf{x}, \sigma| \equiv \oint |x\rangle \langle x| \quad (2.11)$$

with coordinate-representation of states $|\psi\rangle \in \mathcal{H}$ w.r.t. our chosen basis given by

$$\langle x | \psi \rangle \equiv \psi(x) \equiv \psi_{\sigma}(\mathbf{x}). \quad (2.12)$$

Remark 2.8.1 In the case of more than one particle (with, or without spin) present, one can also introduce position-eigenstates as a product-basis of single-particle-position-eigenstates. This is elaborated in section 27.1.1.

Notation 2.9 Let A be a linear operator. The hermitian adjoint of A is the unique linear mapping $A^{\dagger} : \mathcal{H} \rightarrow \mathcal{H}$ that fulfills the following condition:

$$\forall \psi, \chi \in \mathcal{H} \times \mathcal{H} : \quad \langle \chi | A^\dagger | \psi \rangle = \langle \psi | A | \chi \rangle^* \equiv \langle \psi | A \chi \rangle^* = \langle A \chi | \psi \rangle \quad (2.13)$$

where we introduce the notations

$$\langle \psi | A^\dagger \equiv \langle A \psi | \equiv J_{\mathcal{H}}(A | \psi) \rangle. \quad (2.14)$$

Remark 2.9.1 If $A = A^\dagger$, then⁷ we call the operator A *self-adjoint* or *hermitian*⁸. For reasons that will become clear in the next chapter, we will call self-adjoint operators *observables*⁹

Remark 2.9.2 Some properties of hermitian operators are:

- Their eigenvalues are real,
- eigenvectors associated to different eigenvalues are orthogonal,
- they can be diagonalized in an orthonormal basis of eigenvectors (spectral-theorem).

2.2. Canonical Quantization (from Classical Physics to Quantum Physics)

Hypothesis 2.10 Let

$$\begin{aligned} H &\equiv H(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N, \mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_N, t) \\ &\equiv H(x_{1,1}, \dots, x_{1,d_x}, x_{2,1}, \dots, x_{2,d_x}, \dots, p_{1,1}, \dots, p_{1,d_x}, p_{2,1}, \dots, p_{2,d_x}, \dots, t) \\ &\equiv H(x_1, x_2, \dots, x_{d_x N}, p_1, p_2, \dots, p_{d_x N}, t) \\ &\equiv H(x, p, t) \end{aligned} \quad (2.15)$$

be the Hamilton-function of a classical, non-relativistic, N -(Newton-)particle system in d_x dimensions. Let us further assume, that H is a linear-combination of expressions of the type

1. $p_i p_j$,
2. $p_i f_j(x_1, x_2, \dots, t)$
3. $F(x_1, x_2, \dots, t)$

with functions f_i and F . The *canonically-quantized Hamilton-operator*, which we also denote by H^{10} , is obtained by replacing the expressions in the following way:

1. $p_i p_j \mapsto \frac{\hbar}{i} \frac{\partial}{\partial x_i} \frac{\hbar}{i} \frac{\partial}{\partial x_j} = -\hbar^2 \frac{\partial}{\partial x_i} \frac{\partial}{\partial x_j}$,
2. $p_i f_j(x_1, x_2, \dots, t) = \frac{1}{2} (p_i f_j(x_1, x_2, \dots, t) + f_j(x_1, x_2, \dots, t) p_i) \mapsto \frac{1}{2} \left(\frac{\hbar}{i} \frac{\partial}{\partial x_i} f_j(x_1, x_2, \dots, t) + f_j(x_1, x_2, \dots, t) \frac{\hbar}{i} \frac{\partial}{\partial x_i} \right)$

⁷Recall that two operators are considered to be identical if their action on an arbitrary vector of the vector space is the same.

⁸As we are physicist, we throw both of these kinds of operators into the same pot.

⁹Mnemonic: „[observ]able-vari[able]“.

¹⁰One has to conclude from the context, whether one is speaking about a Hamilton-function or a Hamilton-operator.

$$3. F(x_1, x_2, \dots, t) \mapsto F(x_1, x_2, \dots, t)$$

Remark 2.10.1 There is no mathematical way of proving that this is the correct technique to quantize a classical system. The only possibility is to compare the results of the quantization scheme with the results of experiments.

Remark 2.10.2 Note that the canonical quantization is a mapping from functions to operators. Sometimes one encounters the notation $x \equiv \hat{x}$ or $p_x \equiv \hat{p}_x \equiv \hbar/i\partial_x$ to highlight this fact.

Remark 2.10.3 As an example, consider the Hamilton-function of a single, non-relativistic particle in a time-independent potential $V \equiv V(\mathbf{x})$

$$H(\mathbf{x}, \mathbf{p}) = \frac{\mathbf{p}^2}{2m} + V(\mathbf{x}). \quad (2.16)$$

Then, the canonically-quantized Hamilton-operator reads

$$H = -\hbar^2 \frac{\Delta}{2m} + V(\mathbf{x}). \quad (2.17)$$

Lemma 2.11 Notice further, that one has on the operator-level the so-called canonical commutation relations (CCR)

$$[x_i, p_j] = i\hbar\delta_{i,j}. \quad (2.18)$$

Proof 2.11 See Appendix, Proofs, page 395.

Definition 2.12 We consider an N -particle system in $d_{\mathbf{x}} \in \{1, 2, 3\}$ dimensions. The Schrödinger-equation associated to this system reads

$$i\hbar (\partial_t \psi)(\mathbf{x}, t) = (H\psi)(\mathbf{x}, t). \quad (2.19)$$

with canonically-quantized (and potentially time-dependent) Hamilton-operator $H(t)$.

Remark 2.12.1 Particles¹¹ (with mass m etc.), that obey the Schrödinger-equation, are called *Schrödinger-particles*.

Theorem 2.13 The formal solution to equation (2.19) is given by application of *time-evolution operator*

¹¹In the broadest sense of the word.

$$\psi(\mathbf{x}, t_f) = U(t_f, t_i)\psi(\mathbf{x}, t_i) \quad (2.20)$$

with

$$U(t_f, t_i) = T_t \exp \left(-\frac{i}{\hbar} \int_{t_i}^{t_f} dt' H(t') \right) = \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{i}{\hbar} \right)^n \int_{[t_i, t_f]^n} dt_1 \cdots dt_n T_t \prod_{j=1}^n H(t_j). \quad (2.21)$$

Remark 2.13.1 Notice, that the formal solution to this equation is for time-independent potentials given by

$$\psi(\mathbf{x}, t) = e^{-\frac{i}{\hbar} H(t-t_0)} \psi(\mathbf{x}, t_0) \equiv U(t, t_0) \psi(\mathbf{x}, t_0) \quad (2.22)$$

assuming we have $\psi(\mathbf{x}, t_0)$ given (which is most often not the case). The operator U is called *time-evolution-operator* and can be generalized for time-dependent potentials.

Definition 2.14 A time-dependent pure state $t \mapsto \psi(t) \equiv \psi_t \in \mathcal{H}$ is called *stationary*, if it satisfies

$$\frac{d}{dt} \langle \psi_t | A | \psi_t \rangle \equiv \frac{d}{dt} \langle A \rangle_{\psi_t} = 0 \quad \text{for all hermitian operators } A. \quad (2.23)$$

Remark 2.14.1 As examples:

- An electron moving through free space is not in stationary state (the expectation-value of the position operator changes with time).
- The ground state of the electronic system of a solid is (in the wavefunction-formalism (and as we shall see later, in the "density-matrix" formalism as well), and in absence of interactions) a stationary state (in the non-interacting case: a product of a so-called „Slater-determinant“ (to be introduced later) and plane wave for the spatial and temporal part respectively; the expectation-value of all observables are then constant in time, if there is no external perturbation).

Theorem 2.15 For a time-independent potential-function $V \equiv V(\mathbf{x}, t) \equiv V(\mathbf{x})$ a product ansatz

$$\psi(\mathbf{x}, t) \equiv \psi(\mathbf{x}) \phi(t) \quad (2.24)$$

leads to¹² the time-independent-Schrödinger-equation

$$E\psi(\mathbf{x}) = (H\psi)(\mathbf{x}), \quad (2.25)$$

with

$$\phi(t) = e^{-iEt/\hbar}. \quad (2.26)$$

¹²With "leads to" we mean: We can solve the time-dependent Schrödinger-equation iff we can solve the time-independent Schrödinger-equation.

Proof 2.15 See Appendix, Proofs, page 396.

Corollary 2.16 The product ansatz above yields stationary-states.

Proof 2.16 See Appendix, Proofs, page 397.

Hypothesis 2.17 The product-ansatz in Equation (2.24) gives the „correct“ solution of the Schrödinger-equation, iff the state of interest is a stationary-state that can be described in the wavefunction-formalism.

Remark 2.17.1 Notice that the canonically quantized Hamiltonian of a single particle in $d_{\mathbf{x}} \in \{1, 2, 3\}$ dimensions with Hamilton-function given as $H = \frac{p^2}{2m} + V(\mathbf{x})$ is thus given by

$$H = -\frac{\hbar^2}{2m} \Delta_{\mathbf{x}} + V(\mathbf{x}). \quad (2.27)$$

Definition 2.18 The time-dependent Pauli equation for an N -particle system reads

$$i \frac{\partial}{\partial t} \psi(\mathbf{x}_1, s_1; \dots; \mathbf{x}_N, s_N; t) = \hat{H} \psi(\mathbf{x}_1, s_1; \dots; \mathbf{x}_N, s_N; t), \quad (2.28)$$

where the Hamiltonian is

$$\hat{H} = \sum_{j=1}^N \left[\frac{1}{2m_j} (-i\nabla_j - q_j \mathbf{A}(\mathbf{x}_j, t))^2 + q_j \phi(\mathbf{x}_j, t) - \frac{q_j}{2m_j} \sigma_j \cdot \mathbf{B}(\mathbf{x}_j, t) \right] + \hat{V}_{\text{int}}. \quad (2.29)$$

Here, σ_j are the Pauli matrices acting on the spin degree of freedom of particle j , $\mathbf{A}$ and ϕ denote the vector and scalar potentials, and $\mathbf{B} = \nabla \times \mathbf{A}$ is the magnetic field.

Remark 2.18.1 For a system of N electrons (all of mass m and charge $-e$), this simplifies to

$$\hat{H} = \sum_{j=1}^N \left[\frac{1}{2m} (-i\nabla_j + e\mathbf{A}(\mathbf{x}_j, t))^2 - e\phi(\mathbf{x}_j, t) + \frac{e}{2m} \sigma_j \cdot \mathbf{B}(\mathbf{x}_j, t) \right] + \sum_{i < j} \frac{e^2}{|\mathbf{x}_i - \mathbf{x}_j|}. \quad (2.30)$$

Remark 2.18.2 The single-particle Pauli-equation reads

$$i\hbar \partial_t \psi = \left(\frac{1}{2m} [(\mathbf{p} - q\mathbf{A})^2 - q\hbar \boldsymbol{\sigma} \cdot \mathbf{B}] + q\phi \right) \psi \quad (2.31)$$

with $\psi(\mathbf{x}, t) \equiv (\psi_{\uparrow}, \psi_{\downarrow})^T(\mathbf{x}, t)$ etc.

Remark 2.18.3 Analogously one speaks about *Pauli*-particles.

Fact 2.19 A time-independent vector potential $\mathbf{A}(\mathbf{x})$ leads to a time-independent Hamiltonian $\hat{H}$ in the Pauli equation, and thus allows for a product ansatz of the form $\psi(\mathbf{x}_1, s_1; \dots; \mathbf{x}_N, s_N; t) = \psi(\mathbf{x}_1, s_1; \dots; \mathbf{x}_N, s_N)\phi(t)$, which yields the time-independent Pauli equation for a spatial wavefunction ψ .

2.3. The Postulates (for Wavefunction-States)

I A *wavefunction-state* of a physical system is a normalized solution $\psi(\bullet, t)$ of the underlying Schrödinger- / Pauli-equation, s.t. for each $t \in \mathbb{R}$, the coordinates¹³ of the vector are complex-valued

$$\begin{aligned} \mathbf{x} &\mapsto \psi(\mathbf{x}, t) \equiv \langle \mathbf{x} | \psi_t \rangle \in \mathbb{C} && \text{for Schrödinger-equations} \\ x &\mapsto \psi(x, t) \equiv \langle x | \psi_t \rangle \in \mathbb{C} && \text{for Pauli-equations} \end{aligned} \quad (2.32)$$

and the vector is normalized, i.e. $\langle \psi_t | \psi_t \rangle = 1$.

II a) For any measurable quantity¹⁴ there is an observable A .

b) The outcome of a measurement of A , i.e. the measured quantity, is one of the eigenvalues α_n of A .

c) Let $\{|\phi_{n,i}\rangle\}_{i=1,\dots,m}$ denote an orthonormal basis of the eigenspace of the eigenvalue (of A) α_n . A measurement of the observable A with a system in the state $|\psi_t\rangle$ gives the measured value α_n with probability

$$p_n = \sum_{i=1}^m |\langle \phi_{n,i} | \psi_t \rangle|^2. \quad (2.33)$$

Remark 2.19.1 For states that are not represented by wavefunctions, one obtains generalizations of the postulates. We will elaborate on this at a later point (when we acutally need it).

Remark 2.19.2 For wavefunctions obeying the Pauli-equation, the *Born-probability-interpretation* can be summarized as follows:

$$|\psi(x_1, x_2, \dots, x_N)|^2 \prod_{j=1}^N dx_j = \left\{ \begin{array}{l} \text{The probability for finding the } N \text{ particles in the } 3N\text{-dimensional volume } \prod_{j=1}^N dx_j, \\ \text{surrounding the point } (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N) \\ \text{in the } 3N\text{-dimensional configuration space.} \end{array} \right\}.$$

For wavefunctions obeying the Schrödinger-equation, the same interpretation holds, but with $x_i \mapsto \mathbf{x}_i$ on the left hand side of the equation.

Remark 2.19.3 One finds these postulates in different books in all kinds of colours. It is, in my opinion, very difficult, if not impossible, to give a pedagogical and simultaneously complete introduction / description of quantum mechanics.

Notation 2.20 Further one finds the *expectation-value* of the observable A in the state $|\psi_t\rangle$ as

$$\langle A \rangle = \langle \psi_t | A | \psi_t \rangle. \quad (2.34)$$

¹³Notice that, depending on the system (with or without spin) and the number of particles, these coordinates are vectors with entries as many as $\mathbf{x}$ has.

¹⁴For example position, momentum, spin, angular-momentum, etc. There are also more complicated observables, such as Fermi-surfaces (through ARPES-measurements). Proving that these observables are indeed observables in the mathematical sense is a non-trivial, but straightforward task.

2.4. Time-Dependencies in Many-Body Quantum Mechanics

Notation 2.21 Let $\mathcal{H}$ denote the Hilbert-space (of a single- or many-particle) quantum system with a (possibly spin-dependent) time-independent Hamiltonian H (in the Schrödinger-picture) and consider the time-evolution in the Schrödinger- / Pauli-equation

$$i\hbar\partial_t|\psi(t)\rangle = H|\psi(t)\rangle.$$

Definition 2.22 For arbitrary times $t_i, t_f \in \mathbb{R}$ we define the time-evolution operator $U(t_f, t_i)$ of the system as

$$U(t_f, t_i) = \exp - \frac{i}{\hbar} (t_f - t_i) H.$$

Theorem 2.23 The time-evolution operator defined above gives a formal solution to the Schrödinger-equation for a given initial-condition.

Proof 2.23 By taking the derivative, one sees that one indeeds obtains a solution for a given initial condition.

Definition 2.24 Let $X \equiv X_S \equiv X_S(t)$ be a possibly time-dependent operator, then one defines the corresponding Heisenberg-operator as

$$X_H(t) := U(t, 0)^\dagger X U(t, 0).$$

Theorem 2.25 One immediately concludes the Heisenberg equation of motions

$$\frac{d}{dt}X_H(t) = -\frac{i}{\hbar}[X, H]_H(t) + \left(\frac{\partial X_S}{\partial t}\right)_H(t) \quad (2.35)$$

Proof 2.25 See Appendix, Proofs, page 397.

Definition 2.26 In the so-called *Schrödinger-picture* one evaluates expectation-values of an observable X as

$$\langle X \rangle(t) \equiv \langle \psi_t | X | \psi_t \rangle, \quad (2.36)$$

In the *Heisenberg-picture* one evaluates expectation-values of an observable as

$$\langle X \rangle(t) \equiv \langle \psi_{t=0} | X_H(t) | \psi_{t=0} \rangle \quad (2.37)$$

whereas $|\psi_{t=0}\rangle$ refers to some (known) state at time equal zero.

Remark 2.26.1 For pure states, both pictures are equivalent.

Remark 2.26.2 Sometimes it turns out to be useful to switch to yet another picture, the so-called „interaction-picture“. As the number of occasions where this is useful is rare (e.g. „Fermi’s golden rule“, or „one-body-Green-functions for interacting systems“) we will not introduce it at this point.

3. Crystals

We focus on „crystalline solids“ (elaborated below) in this lecture.

3.1. Crystals, Lattices and Bases

Definition 3.1 For $d_x \in \{1, 2, 3\}$ and linearly independent vectors $\bigcup_{j=1}^{d_x} \{\mathbf{a}_j\}$ we introduce the corresponding *infinite point lattice*

$$L_\infty \equiv \left\{ \mathbf{R}_n \equiv \sum_{i=1}^{d_x} n_i \mathbf{a}_i \quad | \quad \mathbf{n} \in \mathbb{Z}^{d_x} \right\} \subset \mathbb{R}^{d_x}. \quad (3.1)$$

Remark 3.1.1 The vectors $\bigcup_{j=1}^{d_x} \{\mathbf{a}_j\}$ are usually called *lattice vectors* or *basis (vectors)*.

Definition 3.2 We define the *unit-cell* UC of a lattice w.r.t. the basis $\bigcup_{j=1}^{d_x} \{\mathbf{a}_j\}$ as the set

$$\text{UC} \equiv \text{span}_{[0,1]} \bigcup_{j=1}^{d_x} \{\mathbf{a}_j\} \subset \mathbb{R}^{d_x} \quad (3.2)$$

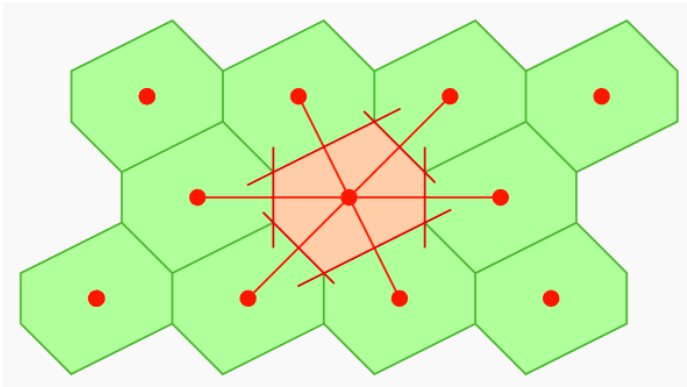
and we define its volume as

$$\text{vol}(\text{UC}) = \int d^{d_x} \mathbf{r} \, 1_{\text{UC}}(\mathbf{r}). \quad (3.3)$$

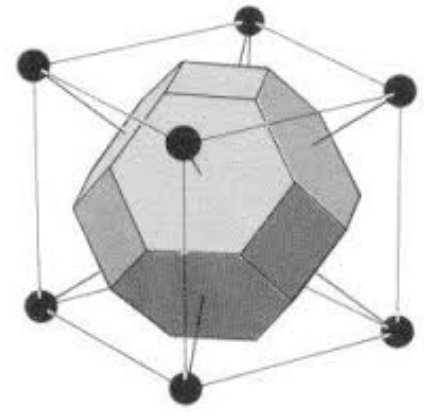
Definition 3.3 The *Wigner-Seitz-cell* around the lattice point $\mathbf{R}_m$ is defined as the set

$$A_{\mathbf{R}_m} \equiv A(\mathbf{R}_m) \equiv A_m = \{ \mathbf{r} \in \mathbb{R}^d \mid \forall \mathbf{n} \neq \mathbf{m} : |\mathbf{r} - \mathbf{R}_m| < |\mathbf{r} - \mathbf{R}_n| \}. \quad (3.4)$$

Remark 3.3.1 Therefore, all points in $A_{\mathbf{R}_m}$ are closer to the point $\mathbf{R}_m$ than to the neighboring grid-points.



(a) 2D construction for an „oblique-lattice“.



(b) 3D construction for a „bcc-lattice“.

Figure 3.1.: (a) 2D and (b) 3D constructions of a Wigner-Seitz-cell.

Definition 3.4 In one dimension, $d_x = 1$, there exists only one kind of (*Bravais*-)lattice: Fixing the lattice constant a determines the lattice completely.

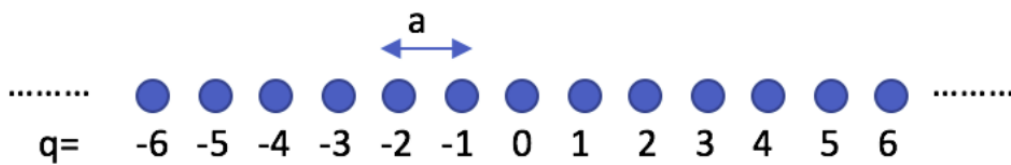


Figure 3.2.: One dimensional lattice (schematically).

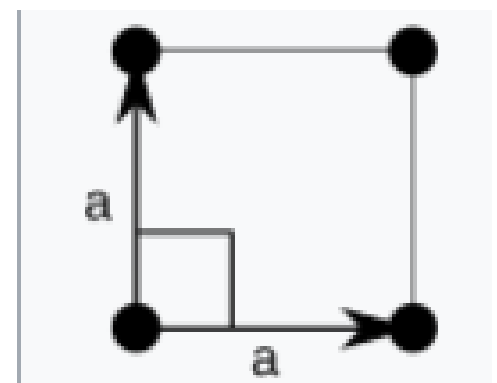
Definition 3.5 In two dimension, $d_x = 2$, one introduces 5 so-called (*infinite*) *Bravais-lattices*¹, by considering the following cases of unit-vector combinations: Let $\varphi = \angle(\mathbf{a}_1, \mathbf{a}_2)$. Then one introduces ...

1. Square lattice

One choice of basis vectors is:

$$\mathbf{a}_1 = (a, 0), \quad \mathbf{a}_2 = (0, a),$$

where $a = b$, and the angle $\varphi = 90^\circ$.



The square lattice.

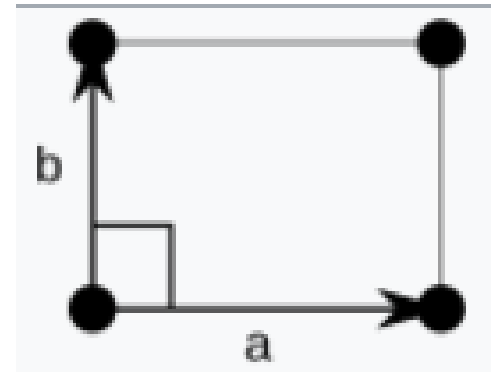
2. Rectangular lattice

¹International Tables A page 720: There are 5 Bravais-Lattices in two dimensions.

One choice of basis vectors is:

$$\mathbf{a}_1 = (a, 0), \quad \mathbf{a}_2 = (0, b),$$

where $a \neq b$, and the angle $\varphi = 90^\circ$.



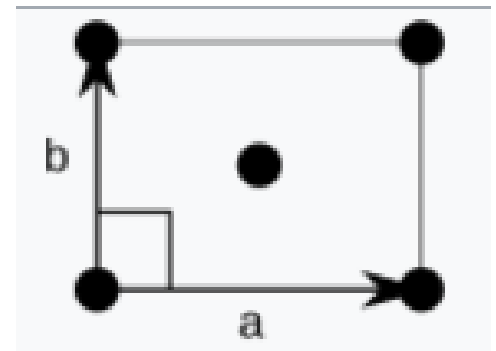
The rectangular lattice.

3. Centered rectangular lattice

One choice of basis vectors is:

$$\mathbf{a}_1 = \left(\frac{a}{2}, \frac{b}{2}\right), \quad \mathbf{a}_2 = (a, 0),$$

where $a \neq b$, and the angle $\varphi \neq 90^\circ$.



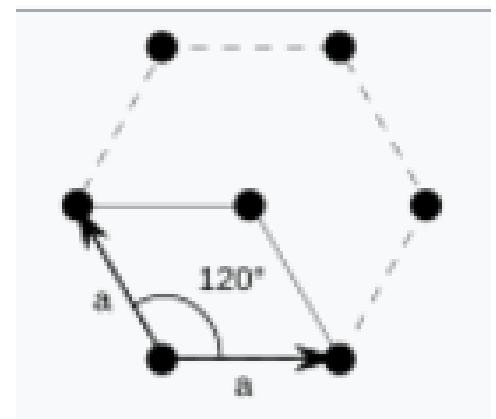
The centered rectangular lattice.

4. Triangular (hexagonal) lattice

One choice of basis vectors is:

$$\mathbf{a}_1 = (a, 0), \quad \mathbf{a}_2 = \left(-\frac{a}{2}, \frac{\sqrt{3}}{2}a\right),$$

and the angle $\varphi = 120^\circ$.



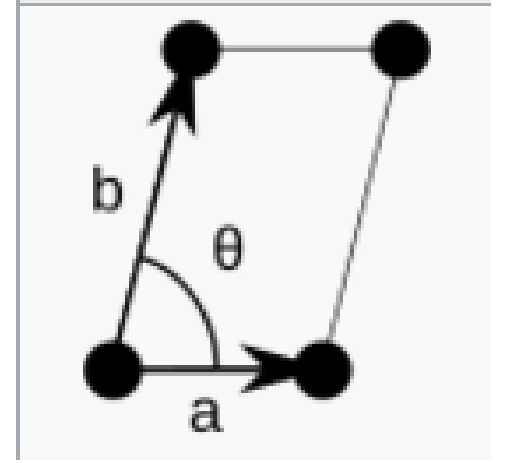
The triangular lattice.

5. Oblique lattice

One choice of basis vectors is:

$$\mathbf{a}_1 = (a, 0), \quad \mathbf{a}_2 = (b \cos \varphi, b \sin \varphi),$$

where $a \neq b$, and $0^\circ < \varphi < 180^\circ$.



The oblique lattice.

Definition 3.6 In three dimensions, $d_x = 3$, one introduces 14 so-called *Bravais-lattices*, by considering the following cases of unit-vector combinations: We denote the angles as

$$\alpha = \angle(\mathbf{a}_2, \mathbf{a}_3), \quad \beta = \angle(\mathbf{a}_1, \mathbf{a}_3), \quad \gamma = \angle(\mathbf{a}_1, \mathbf{a}_2) \quad (3.5)$$

and introduce the notation

$$\mathbf{a}_1 = \mathbf{a}, \quad \mathbf{a}_2 = \mathbf{b}, \quad \mathbf{a}_3 = \mathbf{c} \quad (3.6)$$

then one introduces the different lattices² ...

1: the *triclinic lattice*: $a \neq b \neq c$, $\alpha \neq \beta \neq \gamma$,

2-3: The *monoclinic* lattices

a) the *monoclinic primitive lattice*: $a \neq b \neq c$, $\alpha = \gamma = 90^\circ \neq \beta$

b) the *monoclinic base-centered lattice*: $a \neq b \neq c$, $\alpha = \gamma = 90^\circ \neq \beta$

4-7: The *orthorhombic* lattices

a) the *orthorhombic primitive lattice*: $a \neq b \neq c$, $\alpha = \beta = \gamma = 90^\circ$

b) the *orthorhombic base-centered lattice*: $a \neq b \neq c$, $\alpha = \beta = \gamma = 90^\circ$

c) the *orthorhombic body-centered lattice*: $a \neq b \neq c$, $\alpha = \beta = \gamma = 90^\circ$

d) the *orthorhombic face-centered lattice*: $a \neq b \neq c$, $\alpha = \beta = \gamma = 90^\circ$

8: the *hexagonal lattice*: $a = b \neq c$, $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$

9: the *trigonal lattice*: $a = b = c$, $\alpha = \beta = \gamma \neq 90^\circ$

10-11: The *tetragonal* lattices

a) the *tetragonal primitive lattice*: $a = b \neq c$, $\alpha = \beta = \gamma = 90^\circ$

b) the *tetragonal body-centered lattice*: $a = b \neq c$, $\alpha = \beta = \gamma = 90^\circ$

12-14: The *cubic* lattices

²Notice that e.g. the equations $a \neq b \neq c$ allow (strictly speaking) for the case $a = c \neq b$. But this is just poorly written in the books.

- a) the *cubic primitive lattice*: $a = b = c$, $\alpha = \beta = \gamma = 90^\circ$
 b) the *cubic body-centered lattice*: $a = b = c$, $\alpha = \beta = \gamma = 90^\circ$
 c) the *cubic face-centered lattice*: $a = b = c$, $\alpha = \beta = \gamma = 90^\circ$

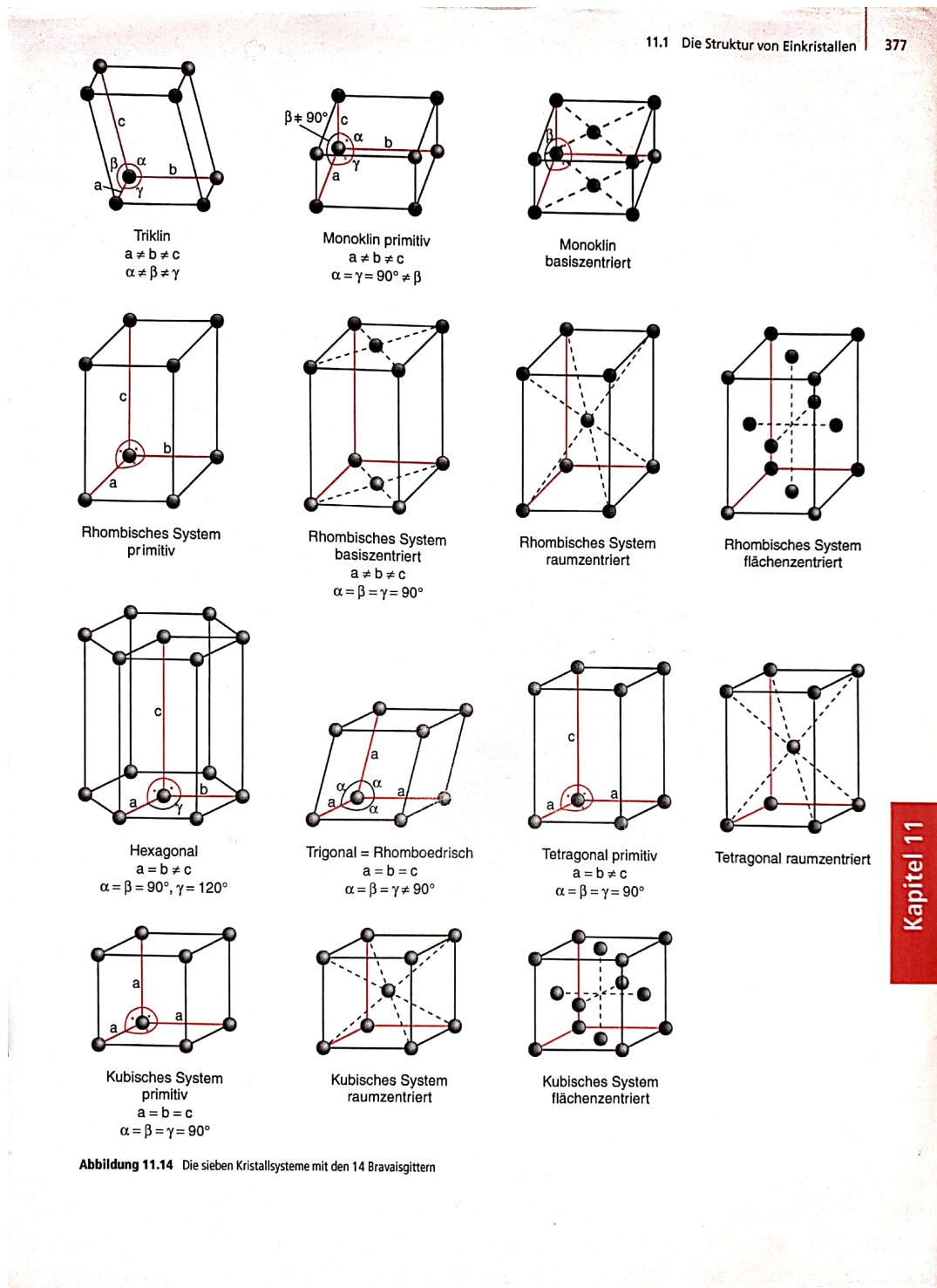


Figure 3.3.: Excerpt from Wolfgang Demtröder's Experimentalphysik 3.

Quoting from [Ashcroft]:

The seven crystal systems and fourteen Bravais lattices described above exhaust the possibilities. This

is far from obvious [...]. However, it is of no practical importance to understand why these are the only distinct cases. It is enough to know why the categories exist, and what they are.

Definition 3.7 We define an *infinite crystal* C_∞ by taking an infinite Bravais-lattice L_∞ , and enlarge the number of points in the following way

$$C_\infty \equiv \left\{ \mathbf{R}_{\mathbf{n}\kappa} \equiv \mathbf{R}_{\mathbf{n}} + \mathbf{R}_\kappa \equiv \mathbf{R}_{\mathbf{n},\kappa,\alpha} \mid \mathbf{R}_{\mathbf{n}} \in L_\infty \wedge \kappa \in \{1, 2, \dots, r\} \wedge \alpha \in \{1, \dots, d_{\mathbf{x}}\} \right\} \subset \mathbb{R}^{d_{\mathbf{x}}} \quad (3.7)$$

with *atomic basis-vectors* $\{\mathbf{R}_1, \dots, \mathbf{R}_r\}$ satisfying the equation

$$\mathbf{R}_\mu \in \{\mathbf{R}_1, \dots, \mathbf{R}_r\} \subset \text{UC}. \quad (3.8)$$

Remark 3.7.1 The *atomic basis-vectors* $\{\mathbf{R}_1, \dots, \mathbf{R}_r\}$ represent the individual positions of the different atoms within the unit cell, and must not be confused with the primitive-vectors of the underlying lattice.

Remark 3.7.2 Notice that therefore, every infinite Bravais-lattice can be interpreted as a crystal with a trivial mono-atomic basis, i.e. $r = 1$ and $\mathbf{R}_1 = 0$.

Remark 3.7.3 A simple mnemonic for the definitions reads

$$\text{Crystal} = \text{Lattice} + \text{Basis}. \quad (3.9)$$

Remark 3.7.4 Note further, that this decomposition is not unique. For example potassium in its crystalline-form at room-temperature can be considered as a fcc-lattice with a mono-atomic basis *or* as a sc-lattice with a diatomic basis.

Remark 3.7.5 There are little pitfalls in the phraseology of the community... Graphene's crystal-structure is an example. It's realized as a triangular lattice with two basis-atoms. Because of the point-structure that one ends up with, one usually calls this crystal the *honeycomb-lattice*, although it is neither a Bravais-lattice, nor an infinite point-lattice in the sense above.

Definition 3.8 We call the lattice vectors $\bigcup_{j=1}^{d_{\mathbf{x}}} \{\mathbf{a}_j\}$ *primitive*, if r in Equation (3.7) is minimal. The corresponding unit-cell is called a *primitive unit-cell*.

Remark 3.8.1 A *primitive unit-cell* is the smallest unit cell that one can obtain for a crystal.

Definition 3.9 For a Bravais-lattice L_∞ with basis $\bigcup_{j=1}^{d_x} \{\mathbf{a}_j\}$ we introduce the corresponding *reciprocal basis* in different dimensions

- $d = 1$: $\mathbf{b}_1 = 2\pi \frac{\mathbf{a}_1}{|\mathbf{a}_1|}$.
- $d = 2$: $\mathbf{b}_1 = 2\pi \frac{(-a_{2y}, a_{2x})}{|\mathbf{a}_1 \wedge \mathbf{a}_2|}$, $\mathbf{b}_2 = 2\pi \frac{(a_{1y}, -a_{1x})}{|\mathbf{a}_1 \wedge \mathbf{a}_2|}$.
- $d = 3$: $\mathbf{b}_1 = 2\pi \frac{\mathbf{a}_2 \times \mathbf{a}_3}{\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)}$, $\mathbf{b}_2 = 2\pi \frac{\mathbf{a}_3 \times \mathbf{a}_1}{\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)}$, $\mathbf{b}_3 = 2\pi \frac{\mathbf{a}_1 \times \mathbf{a}_2}{\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)}$.

Subsequently we define the corresponding *infinite reciprocal lattice* RL_∞ as

$$RL_\infty \equiv \left\{ \mathbf{G} \in \mathbb{R}^{d_x} \mid \mathbf{G} = \sum_{j=1}^{d_x} n_j \mathbf{b}_j \text{ with } \mathbf{n} \in \bigtimes_{j=1}^{d_x} \mathbb{Z} \right\} = \text{span}_{\mathbb{Z}^{d_x}} \bigcup_{j=1}^{d_x} \{\mathbf{b}_j\}. \quad (3.10)$$

Remark 3.9.1 Recall that the wedge-product and the triple-product measure the (signed) volume of the parallelogram and the parallelepiped spanned by the vectors they are being applied on.

Corollary 3.10 It holds that

$$RL_\infty \equiv \left\{ \mathbf{G} \in \mathbb{R}^{d_x} \mid \forall \mathbf{R} \in L_\infty : \mathbf{G} \cdot \mathbf{R} \in 2\pi\mathbb{Z} \right\}. \quad (3.11)$$

Proof 3.10 With the anticommutativity-rules for the triple-product, one immediately sees that these vectors fulfill the defining condition.

Definition 3.11 The *first Brillouin-zone* FBZ is defined as

$$\text{FBZ} = \left\{ \mathbf{r} \in \mathbb{R}^{d_x} \mid \forall \mathbf{G}_n \in RL_\infty : |\mathbf{r} - \mathbf{0}| < |\mathbf{r} - \mathbf{G}_n| \right\}. \quad (3.12)$$

Remark 3.11.1 Therefore, the first Brillouin-zone is the Wigner-Seitz-cell of the origin of the reciprocal lattice.

Definition 3.12 A *unit-cell-periodic-function* or also *lattice-periodic function* is a smooth function

$$\begin{aligned} f : \mathbb{R}^{d_x} &\rightarrow \mathbb{C} \\ \mathbf{x} &\mapsto f(\mathbf{x}) \end{aligned}$$

satisfying

$$f(\mathbf{x} + \mathbf{R}_n) = f(\mathbf{x}) \quad \text{for } \mathbf{R}_n \in L_\infty. \quad (3.13)$$

Remark 3.12.1 Examples include

- The potential $\mathbf{x} \mapsto V(\mathbf{x}) \in \mathbb{R}$ that a "Bloch-electron"³ experiences is a lattice-periodic function.
- A "Bloch-wavefunction"⁴ $\mathbf{x} \mapsto \psi_{n,\mathbf{k}}(\mathbf{x}) \in \mathbb{C}$ can be factorized into a plane-wave $e^{i\mathbf{k}\cdot\mathbf{x}}$ and a lattice-periodic function $u_{n\mathbf{k}}(\mathbf{x})$.
- The "charge-density"⁵ $\mathbf{x} \mapsto \rho(\mathbf{x}) \in \mathbb{R}$ is a lattice-periodic function.

Corollary 3.13 For a lattice-periodic function f it follows

$$f(\mathbf{x}) = \sum_{\mathbf{G} \in \text{RL}_\infty} f_{\mathbf{G}} e^{i\mathbf{G}\cdot\mathbf{x}} \quad (3.14)$$

with expansion coefficients defined by

$$f_{\mathbf{G}} = \frac{1}{\text{Vol}(\text{UC})} \int_{\text{UC}} d^d \mathbf{x} f(\mathbf{x}) e^{-i\mathbf{G}\cdot\mathbf{x}} \quad (3.15)$$

Proof 3.13 See Appendix, Proofs, page 399.

Symmetries in Crystals

Symmetry-considerations are not trivial matters. There are endless tables on which kind of lattices imply what kind of behaviour for different kinds of experiments. (Laue-backscattering, Raman-scattering etc.) These behaviours can be concluded from symmetry-considerations and group-theoretical methods corresponding to the underlying lattice. Those who are interested can search for "International Tables for Crystallography".

3.2. Boundaries, Equivalences and Fourier

Hypothesis 3.14 We consider a real material embedded into $\mathbb{R}^{d_x}$ with some finite volume $0 < V_{d_x} < \infty$ and some underlying crystal(-structure), and introduce the numbers $N_1, \dots, N_{d_x} \in \mathbb{N}$ such that the scalar-product ($d_x = 1$), wedge-product ($d_x = 2$) or triple-product ($d_x = 3$) coincides the volume

- $d_x = 1$: $V_{d_x=1} \equiv L = N_1 \sqrt{\mathbf{a}_1^2}$.
- $d_x = 2$: $V_{d_x=2} \equiv A = N_1 N_2 \mathbf{a}_1 \wedge \mathbf{a}_2 = N_1 N_2 \text{Det} \begin{pmatrix} a_{1,x} & a_{1,y} \\ a_{2,x} & a_{2,y} \end{pmatrix}$.
- $d_x = 3$: $V_{d_x=3} \equiv V = N_1 N_2 N_3 \mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3) = N_1 N_2 N_3 \text{Det} \begin{pmatrix} a_{1,x} & a_{1,y} & a_{1,z} \\ a_{2,x} & a_{2,y} & a_{2,z} \\ a_{3,x} & a_{3,y} & a_{3,z} \end{pmatrix}$.

³This term will be defined later.

⁴This term will be defined later.

⁵This term will be defined later.

and we denote the total number of cells as

$$\prod_{i=1}^{d_x} N_i = N_p \quad (3.16)$$

Remark 3.14.1 Recall that the wedge product captures the (signed) area spanned by the two vectors in the plane, whereas the triple-product is the (signed) volume of the parallelepiped defined by the three vectors given.

Remark 3.14.2 When people speak about infinite crystals, they are usually considering the so-called *thermodynamic limit* in which $V_{d_x} \rightarrow \infty$ and $\prod_{i=1}^{d_x} N_i \rightarrow \infty$ with the additional condition that the size of the unit-cell is a constant $\frac{1}{V_{d_x}} \prod_{j=1}^{d_x} N_j = \text{const}$. In almost every (DFT) computer simulation however, crystals have a finite size with "periodic boundaries" (see below.)⁶ One obtains „good approximations“ to densities of states and other observables by setting at least $N_i \approx 40$, unless there are some symmetries to consider.

We envision to catch the following two birds with one stone:

1. If we consider a piece of solid, and if we „look“ far into the bulk of the underlying material, we observe that in the Schrödinger-equation the potential arising from the ionic cores on the lattice site that a single-electron experiences „looks“ (at least within a finite region inside the bulk) like a periodic⁷ function.
2. If we consider a piece of solid, it has a finite size.

Fact 3.15 One can model both these attributes in harmony to each other, by introducing periodic boundary conditions for crystal as a whole and restricting attention to solutions of the Schrödinger-eq. (4.2) which are periodic with the finite Bravais-lattice. We therefore identify wavefunctions in the bulk of the crystal as wavefunctions on a three-torus.

Definition 3.16 On $\mathbb{R}^{d_x}$ we define the following relation

$$\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^{d_x} : \mathbf{x} \sim_x \mathbf{y} \quad :\Longleftrightarrow \quad \exists \mathbf{n} \in \mathbb{Z}^{d_x} : \mathbf{x} = \mathbf{y} + \sum_{i=1}^{d_x} n_i N_i \mathbf{a}_i. \quad (3.17)$$

Introducing these periodic boundary conditions is known as *Born- von Karman-boundary-conditions*.

Lemma 3.17 This defines an equivalence-relation.

Proof 3.17 Reflexivity is clear. For symmetry, we note that

$$\mathbf{x} = \mathbf{y} + \sum_{i=1}^{d_x} n_i N_i \mathbf{a}_i \Longleftrightarrow \mathbf{y} = \mathbf{x} + \sum_{i=1}^{d_x} (-n_i) N_i \mathbf{a}_i. \quad (3.18)$$

⁶Exceptions: quasicrystals, crystals with impurities, open boundary conditions.

⁷Notice that there are many useful mathematical formulae for periodic functions.

For transitivity one adds the coefficients.

Remark 3.17.1 Recall that a set on which an equivalence-relation is defined, can be partitioned into its equivalence-classes. This is an integral insight, in the construction of the Hilbert-space and the finite point lattice, and the Fourier-transform.

Lemma 3.18 It holds

$$\mathbb{R}^{d_x} / \sim_x \equiv \left\{ [\mathbf{x}] = \sum_{i=1}^{d_x} [c_i \mathbf{a}_i] \mid c_i \in [0, N_i) \right\} \subset \mathbb{R}^{d_x}. \quad (3.19)$$

Proof 3.18 We have to show, that for $\tilde{\mathbf{x}} \in \mathbb{R}^{d_x}$ there is a unique representative $[\mathbf{x}] \in \mathbb{R}^{d_x} / \sim_x$, such that $\tilde{\mathbf{x}} \in [\mathbf{x}]$. For this, we note that the lattice-vectors, being linearly independent, form a basis of $\mathbb{R}^{d_x}$. Therefore, we can decompose $\tilde{\mathbf{x}}$ as

$$\tilde{\mathbf{x}} = \sum_{i=1}^{d_x} \tilde{c}_i \mathbf{a}_i \quad \text{with } \tilde{c}_i \in \mathbb{R}. \quad (3.20)$$

Now we define

$$c_i = \tilde{c}_i \bmod N_i \in [0, N_i) \quad (3.21)$$

and

$$\mathbf{x} = \sum_{i=1}^{d_x} c_i \mathbf{a}_i \quad \text{with } [\mathbf{x}] \in \mathbb{R}^{d_x} / \sim_x. \quad (3.22)$$

It follows further that $\tilde{\mathbf{x}} \sim_x \mathbf{x}$ and therefore $[\tilde{\mathbf{x}}] = [\mathbf{x}] \in \mathbb{R}^{d_x} / \sim_x$.

Corollary 3.19 Since $L_\infty \subset \mathbb{R}^{d_x}$, this induces also a partitioning of the lattice into equivalence classes as

$$L_f \equiv L_\infty / \sim_x \equiv \left\{ [\mathbf{R}_n] = \sum_{i=1}^{d_x} [n_i \mathbf{a}_i] \mid 0 \leq n_i < N_i \wedge n_i \in \mathbb{N} \right\} \quad (3.23)$$

where we introduced the *finite point lattice* L_f .

Proof 3.19 Analogously to the proof of Lemma 3.18.

Remark 3.19.1 Analogously we introduce the *finite crystal* C_f as

$$C_f \equiv \left\{ \mathbf{R}_{\mathbf{n}\kappa} \equiv \mathbf{R}_{\mathbf{n}} + \mathbf{R}_{\kappa} \equiv \mathbf{R}_{\mathbf{n},\kappa,\alpha} \mid \mathbf{R}_{\mathbf{n}} \in L_f \wedge \kappa \in \{1, 2, \dots, r\} \wedge \alpha \in \{1, \dots, d_x\} \right\} \subset \mathbb{R}^{d_x}. \quad (3.24)$$

In order to write down neat and simple expressions for the „discrete Fourier-transform“ (will be introduced later), it is also convenient to introduce yet another lattice.

Definition 3.20 For a Bravais-lattice L_∞ with infinite reciprocal lattice and corresponding vectors $\mathbf{b}_i$ as in TBD, we introduce the *fine infinite reciprocal lattice* FRL_∞ as

$$\text{FRL}_\infty \equiv \left\{ \mathbf{q} \equiv \sum_{i=1}^{d_x} \frac{n_i}{N_i} \mathbf{b}_i \mid n_i \in \mathbb{Z} \right\} \subset \mathbb{R}^{d_x} \quad (3.25)$$

Definition 3.21 On FRL_∞ we define the following relation

$$\forall \mathbf{q}, \mathbf{q}' \in \text{FRL}_\infty : \mathbf{q} \sim_q \mathbf{q}' \iff \exists \mathbf{n} \in \mathbb{Z}^{d_x} : \mathbf{q} = \mathbf{q}' + \sum_{i=1}^{d_x} n_i N_i \mathbf{b}_i \quad (3.26)$$

Lemma 3.22 This is an equivalence relation on FRL_∞ .

Proof 3.22 Analogously to the proof of Lemma 3.18.

Corollary 3.23 It holds

$$Q \equiv \text{FRL}_\infty / \sim_q \equiv \left\{ [\mathbf{q}] = \sum_{i=1}^{d_x} \left[\frac{n_i}{N_i} \mathbf{b}_i \right] \mid 0 \leq n_i < N_i \right\} \quad (3.27)$$

Proof 3.23 Analogously to the proof of Lemma 3.18.

Notation 3.24 This set Q above, is what within the physics-community called the *allowed wave-vectors within the first Brillouin-zone*. Oftentimes one re-centers the representatives of the equivalence-classes around the origin, mostly for aesthetic reasons. There are however exceptions where it has computational reasons, for example when one introduces approximations. Then, considering for example acoustic phonons, for which one has $\hbar\omega_{\mathbf{q}} \approx \text{const. } q$ for $q \ll 1$ ("Goldstone-theorem"), it is favourable to consider this „symmetric“ choice of equivalence-classes.

Notation 3.25 Summing over these equivalence-classes is usually denoted with an unobtrusive

$$\sum_{\mathbf{q}} f(\mathbf{q}) \quad (3.28)$$

instead of

$$\sum_{[\mathbf{q}] \in Q} f(\mathbf{q}) \quad (3.29)$$

with functions, that are independent on the choice of representative.

In order to proceed we now switch to the physics-notation.

Notation 3.26 In physics, one introduces the notation for the actual position(-operators) of the ionic coordinates (their average oscillates around the corresponding points in the crystal)

$$\left\{ \tau_{\mathbf{n}\kappa} \equiv \mathbf{R}_{\mathbf{n}\kappa} + \Delta \tau_{\mathbf{n}\kappa} \mid [\mathbf{R}_{\mathbf{n}}] \in L_f \quad \wedge \quad \kappa \in \{1, 2, \dots, r\} \right\}$$

one introduces their Fourier-transforms as another sequence of numbers

$$\left\{ \tau_{\mathbf{q}\kappa} \mid [\mathbf{q}] \in Q \quad \wedge \quad \kappa \in \{1, 2, \dots, r\} \right\}$$

as finite sums

$$\tau_{\mathbf{q}\kappa} = \frac{1}{\sqrt{N}} \sum_{[\mathbf{R}_{\mathbf{n}}] \in L_f} \tau_{\mathbf{n}\kappa} e^{+i\mathbf{q} \cdot \mathbf{R}_{\mathbf{n}}} \quad (3.30)$$

whereas $N \equiv N_p \equiv \prod_{i=1}^{d_x} N_i$.

Remark 3.26.1 Notice that this definition is well-defined, as both sides are independent of the choice of the representative that one chooses.

Lemma 3.27 The inverse transformation reads

$$\tau_{\mathbf{n}\kappa} = \frac{1}{\sqrt{N}} \sum_{[\mathbf{q}] \in Q} \tau_{\mathbf{q}\kappa} e^{-i\mathbf{q} \cdot \mathbf{R}_{\mathbf{n}}} \quad (3.31)$$

Proof 3.27 With the geometric series (again).

High Symmetry Points

Fact 3.28 For each lattice, one introduces within the first Brillouin-zone so-called „*high-symmetry-points*“ in order to plot band-structures.

Remark 3.28.1 These high-symmetry-points for each individual crystal are not unique. In different literatures one finds different high-symmetry-points.

Remark 3.28.2 It will turn out later, that fundamental quantities as so-called “phononic dispersion-relations” and “electronic band-structures”, are (for each respective “branch” or “band”) actually functions which take arguments from the first Brillouin-zone of the underlying crystal. Let us use ξ as a generic symbol for such an object of interest with domain located within the first Brillouin-zone:

$$\begin{aligned}\xi : Q &\rightarrow \mathbb{R} \\ \mathbf{q} &\mapsto \xi(\mathbf{q})\end{aligned}$$

with $Q \subset \text{FBZ}$. As one can hardly draw the graph of a function that takes three-dimensional arguments, one usually depicts the values of these functions on $\mathbf{k}$ -path. What one then usually sees is a picture with the $\mathbf{k}$ -path on the abscissa with the corresponding values of ξ on the ordinate. Hence the picture advertised in the beginning.

Remark 3.28.3 One can also rewrite this using the usual tools (and notations) that one knows from math-lectures by introducing a concrete curve $[0, 1] \ni s \mapsto \gamma(s) \in \text{FBZ}$ that connects the different high-symmetry $\mathbf{k}$ -points and then interpreting the pictures as image of a composite mapping $[0, 1] \ni s \mapsto \xi(\gamma(s)) \in \mathbb{R}$. However...

Remark 3.28.4 These high-symmetry-points are actually a topic on their own... The mentality is, to get a glimpse of for example the band-structure of a certain material. (Keep in mind that the band-structure is a four-dimensional object, taking three arguments and giving one number for each argument.) By reducing the dimension to a two-dimensional picture in a neat-fashion, one can partly try to draw conclusions about this material. (For example, if there is a band-gap along the high-symmetry-points, it indicates that there is a band-gap in the density of states, leading to the classification of the material being metallic or insulating. There are however exceptions when this does not work...



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High-throughput electronic band structure calculations: Challenges and tools

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Figure 3.4.: A paper that lists high symmetry points for the different Bravais-lattices.

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The following protocol should be followed. Unit cells must first be reduced to standard primitives, then they should be appropriately relaxed (if needed). Before the static run, the cells should be reduced again to standard primitive (symmetry and orientation might have changed during the relaxation). The user should then perform the static run and then project the eigenvalues along the directions which are specified in the “kpath” option. If the user is running AFLOW and VASP, the web interface can also prepare a template input file “aflow.in” which performs all the mentioned tasks.

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Appendix A

The choice of lattice vectors implemented in AFLOW is given here. When the primitive lattice is the same as the conventional one, it is simply called “lattice”. Variables $a, b, c, \alpha, \beta, \gamma$ denote the axial lengths and interaxial angles of the conventional lattice vectors, while $k_x, k_y, k_z, k_x, k_y, k_z$ are those of the primitive reciprocal lattice vectors $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$. The coordinates of symmetry $\mathbf{k}$ -points are given in fractions of $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$.

A.1. Cubic (CUB, cP)

Lattice (see Fig. 1 and Table 2)

$$\begin{aligned}\mathbf{a}_1 &= (a, 0, 0) \\ \mathbf{a}_2 &= (0, a, 0) \\ \mathbf{a}_3 &= (0, 0, a)\end{aligned}$$

A.2. Face-centered cubic (FCC, cF)

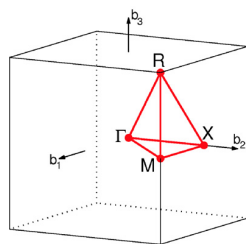
(See Fig. 2 and Table 3.)

Conventional lattice

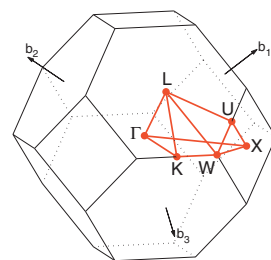
$$\begin{aligned}\mathbf{a}_1 &= (a, 0, 0) \\ \mathbf{a}_2 &= (0, a, 0) \\ \mathbf{a}_3 &= (0, 0, a)\end{aligned}$$

Primitive lattice

$$\begin{aligned}\mathbf{a}_1 &= (0, a/2, a/2) \\ \mathbf{a}_2 &= (a/2, 0, a/2) \\ \mathbf{a}_3 &= (a/2, a/2, 0)\end{aligned}$$

Fig. 1. Brillouin zone of CUB lattice. Path: Γ -X-M- Γ -R-X|M-R. An example of band structure using this path is given in Fig. 26.Table 2
Symmetry $\mathbf{k}$ -points of CUB lattice.

$\times \mathbf{b}_1$	$\times \mathbf{b}_2$	$\times \mathbf{b}_3$		$\times \mathbf{b}_1$	$\times \mathbf{b}_2$	$\times \mathbf{b}_3$	
0	0	0	Γ	1/2	1/2	1/2	R
1/2	1/2	0	M	0	1/2	0	X

Fig. 2. Brillouin zone of FCC lattice. Path: Γ -X-W-K- Γ -L-U-W-L-K|U-X. An example of band structure using this path is given in Fig. 27.Table 3
Symmetry $\mathbf{k}$ -points of FCC lattice.

$\times \mathbf{b}_1$	$\times \mathbf{b}_2$	$\times \mathbf{b}_3$		$\times \mathbf{b}_1$	$\times \mathbf{b}_2$	$\times \mathbf{b}_3$	
0	0	0	Γ	5/8	1/4	5/8	U
3/8	3/8	3/4	K	1/2	1/4	3/4	W
1/2	1/2	1/2	L	1/2	0	1/2	X

A.3. Body-centered cubic (BCC, cI)

(See Fig. 3 and Table 4.)

Conventional lattice

$$\begin{aligned}\mathbf{a}_1 &= (a, 0, 0) \\ \mathbf{a}_2 &= (0, a, 0) \\ \mathbf{a}_3 &= (0, 0, a)\end{aligned}$$

Primitive lattice

$$\begin{aligned}\mathbf{a}_1 &= (-a/2, a/2, a/2) \\ \mathbf{a}_2 &= (a/2, -a/2, a/2) \\ \mathbf{a}_3 &= (a/2, a/2, -a/2)\end{aligned}$$

A.4. Tetragonal (TET, tP)

Lattice (see Fig. 4 and Table 5)

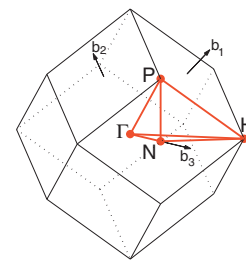
Fig. 3. Brillouin zone of BCC lattice. Path: Γ -H-N- Γ -P-H|P-N. An example of band structure using this path is given in Fig. 28.

Figure 3.5.: Excerpt of the paper above.

4. Quantum Mechanics of Crystals - Electrons and Cores

4.1. The General Problem

Notation 4.1 We consider a charge neutral *piece of solid material* and model it as a finite crystal as in Remark 3.19.1. Notice that this crystal has ...

- No defects, no impurities, no disorder, no surfaces, no interfaces, no grain-boundaries, no dislocations, no vacancies, ...
- a dimension $d_x \in \{1, 2, 3\}$,
- an underlying crystal-structure, with lattice-vectors $\bigcup_{j=1}^{d_x} \{\mathbf{a}_j\}$, and atomic basis-vectors $\{\mathbf{R}_1, \dots, \mathbf{R}_r\}$ according to Definition 3.7,
- a finite size, which will determine the numbers of unit-cells in each direction $N_1, \dots, N_{d_x}$ and the total number of unit cells $\prod_{i=1}^{d_x} N_i = N_p$ in Hypothesis 3.14,
- atoms or (inclusive) ionic cores with (individual) masses $\{M_1, \dots, M_r\}$ charges $\{Z_1e, \dots, Z_re\}$ centered located on the crystal-points.
- $N_e = (Z_1 + \dots + Z_r) \prod_{j=1}^{d_x} N_j$ (indistinguishable) electrons with (individual) mass m and charge $-e$.

Remark 4.1.1 Notice that within these notations, this gives us a total of $rN_p = N_a$ ions to take care of, which results in $rd_xN_p = d_xN_a$ ionic coordinates. To distinguish the ionic coordinates from the lattice-points, we denote them (the coordinates) with¹ τ (with proper indices that are clear in their meaning from the context).

Remark 4.1.2 Similarly, we have to take care of d_xN_e electronic coordinates². Neglecting the spin-coordinate, we collect the spatial coordinates of the electrons and ionic cores in the collective vectors

$$\mathbf{r} \equiv (\mathbf{r}_1, \dots, \mathbf{r}_{N_e}) = \text{„electronic coordinates“}$$

$$\boldsymbol{\tau} \equiv (\boldsymbol{\tau}_1, \dots, \boldsymbol{\tau}_{N_a}) = \text{„ionic coordinates“}$$

¹Notice that we denote the coordinates of the ions with another symbol than the lattice-vectors. This notation (" τ ") was (probably) introduced by Feliciano Giustino (one of the main developers of the EPW-software) in his 2017-article „Electron-Phonon interactions from first principles“.

²We neglect the spin-coordinates for the moment. The fact, that in usual textbooks, for example, [Czycholl], the spin is being neglected is actually not really trivial. It is usually later re-implemented in the formulation of Bloch-wavefunctions by just „attaching“ the spin-part of a wavefunction to its spatial component. There are though exceptions, for example when the lattice-periodic potential possesses a non-trivial spin-dependence. Usual codes, for example [ELK, QE] possess an exchange-correlation functional with this property. This is the so-called „LSDA“-approximation.

Remark 4.1.3 From X-ray-spectroscopy, one can really conclude, that we are (in general) working with partially but not fully ionized cores. The ionic cores can be very well described with the orbital-model. Innermost shells are very similar to the shells of the generalized hydrogen-problem. One can however also compute their band-structure and sees that it is very flat.

Remark 4.1.4 With powder-diffraction (also called „Debye-Scherrer-method“) or with the „Laue-Backscattering-Method“ one can also conclude that the ions are really centered around lattice points.

Notation 4.2 We write for the kinetic-energy-operators

$$\begin{aligned} T_e &= \sum_{i=1}^{N_e} -\frac{\hbar^2}{2m} \Delta_{\mathbf{r}_i}, \\ T_a &= \sum_{i=1}^{N_a} -\frac{\hbar^2}{2M_i} \Delta_{\boldsymbol{\tau}_i} = \sum_{\substack{\mathbf{R}_n \in L_f \\ \alpha \in \{1, \dots, d_x\} \\ \kappa \in \{1, \dots, r\}}} -\frac{\hbar^2}{2M_\kappa} \frac{\partial^2}{\partial \tau_{n\kappa\alpha}^2} \end{aligned}$$

and for the interaction-operators

$$\begin{aligned} V_{e-e} &\equiv \sum_{1 \leq i < j \leq N_e} v_{e-e}(\mathbf{r}_i - \mathbf{r}_j) \\ V_{a-a} &\equiv \sum_{1 \leq i < j \leq N_a} v_{a_i-a_j}(\boldsymbol{\tau}_i - \boldsymbol{\tau}_j) \\ V_{e-a} &\equiv \sum_{1 \leq i \leq N_e} \sum_{1 \leq j \leq N_a} v_{e-a_j}(\mathbf{r}_i - \boldsymbol{\tau}_j) \end{aligned}$$

Remark 4.2.1 In modern density functional theory codes, one oftentimes approximates the interaction-potentials above through so-called „pseudopotentials“, instead of the proper Coulomb-interactions³, which is mainly for computational convenience. (Leitmotiv: „Just solve it!“)

Hypothesis 4.3 Assuming the solid to be in a stationary state, the time-independent Schrödinger-equation now reads

³That would read in the current notation

$$\begin{aligned} V_{e-e} &\equiv \sum_{1 \leq i < j \leq N_e} v_{e-e}(\mathbf{r}_i - \mathbf{r}_j) && \approx \sum_{1 \leq i < j \leq N_e} \frac{e^2}{4\pi\epsilon_0 |\mathbf{r}_i - \mathbf{r}_j|} \\ V_{a-a} &\equiv \sum_{1 \leq i < j \leq N_a} v_{a_i-a_j}(\boldsymbol{\tau}_i - \boldsymbol{\tau}_j) && \approx \sum_{1 \leq i < j \leq N_a} \frac{Z_i Z_j e^2}{4\pi\epsilon_0 |\boldsymbol{\tau}_i - \boldsymbol{\tau}_j|} \\ V_{e-a} &\equiv \sum_{1 \leq i \leq N_e} \sum_{1 \leq j \leq N_a} v_{e-a_j}(\mathbf{r}_i - \boldsymbol{\tau}_j) && \approx \sum_{1 \leq i \leq N_e} \sum_{1 \leq j \leq N_a} \frac{-Z_j e^2}{4\pi\epsilon_0 |\mathbf{r}_i - \boldsymbol{\tau}_j|} \end{aligned} \tag{4.1}$$

$$H\psi(\mathbf{r}, \tau) = E\psi(\mathbf{r}, \tau) \quad (4.2)$$

with

$$H \equiv T_e + T_a + V_{e-e} + V_{a-a} + V_{e-a} \quad (4.3)$$

Remark 4.3.1 This Schrödinger-equation is impossible to solve exactly (neither analytically nor numerically) for crystals that are bigger than a couple of unit-cells in each direction. One needs to refer to approximation-schemes.

4.2. The Adiabatic / Born-Oppenheimer-Approximation

The general aim is to „decouple“ the dynamics (in the sense of a differential equation) of the electrons from the dynamics of the lattice.

Notation 4.4 We decompose the Hamiltonian above as

$$\begin{aligned} H &= T_e + V_{e-e} + V_{a-a} + V_{e-a} & + & T_a \\ &\equiv H_0 & + & V \end{aligned}$$

Definition 4.5 Applying now a perturbative treatment and neglecting the terms in equation (30.25) (see below) is called the *Born-Oppenheimer-Approximation* a.k.a *adiabatic approximation*

Notation 4.6 Let us assume, that for a fixed ion-configuration τ we are able to determine all the orthonormal solutions

$$\left\{ \mathbb{R}^{d_x N_e} \ni \mathbf{r} \mapsto \phi_{\alpha, \tau}(\mathbf{r}) \in \mathbb{C} \right\}_{\alpha \in \mathcal{A}} \quad (4.4)$$

of the equation

$$(H_0 \phi_{\alpha, \tau})(\mathbf{r}) = ([T_e + V_{e-e} + V_{a-a} + V_{e-a}] \phi_{\alpha, \tau})(\mathbf{r}) = \epsilon_{\alpha, \tau} \phi_{\alpha, \tau}(\mathbf{r}) \quad (4.5)$$

Remark 4.6.1 This assumption, of being able to solve the electronic problem, was actually quite optimistic when it was proposed in 1927. In fact, even nowadays, only perturbative solutions to this problem exist.

Remark 4.6.2 In order to solve the problem

$$(H_0 \phi_{\alpha, \tau})(\mathbf{r}) \equiv \epsilon_{\alpha, \tau} \phi_{\alpha, \tau}(\mathbf{r}) \quad (4.6)$$

perturbatively, it is nowadays usual to do a mean-field-approximation: Instead of considering the full electron-electron-interaction in Equation (4.1), one determines (with density functional theory) an effective single-particle-interaction, such that one has

$$\left(\left[T_e + V_{e-e} + V_{a-a} + V_{e-a} \right] \phi_{\alpha,\tau} \right) (\mathbf{r}) = \epsilon_{\alpha,\tau} \phi_{\alpha,\tau}(\mathbf{r}) \quad \xLeftrightarrow{\text{Approximation}} \quad \left(\left[T_e + V_{\text{eff}} + V_{a-a} + V_{e-a} \right] \phi_{\alpha,\tau} \right) (\mathbf{r}) = \epsilon_{\alpha,\tau} \phi_{\alpha,\tau}(\mathbf{r}) \quad (4.7)$$

whereas the effective one-particle-potentials

$$V_{\text{eff}}(\mathbf{r}) \equiv \sum_{i=1}^{N_e} V_{\text{eff}}(\mathbf{r}_i) \equiv \sum_{i=1}^{N_e} V_H(\mathbf{r}_i) + V_{\text{XC}}(\mathbf{r}_i) \quad (4.8)$$

is composed of the so-called „Hartree“ and „exchange-correlation“-contributions. These potentials are determined through density functional theory, on which we will not elaborate further in this course.

Remark 4.6.3 With the procedure elucidated above, it becomes clear, that $\alpha \in \mathcal{A}$ represents a multi-index and the wavefunction $\mathbf{r} \mapsto \phi_{\alpha,\tau}(\mathbf{r})$ represents a Slater-determinant. These Slater-determinants are being build with single-particle-wavefunctions $\psi_{n\mathbf{k}\sigma}(x)$ (*Bloch-functions* or *Kohn-Sham-wavefunctions*, or, even more precise, *Bloch-Kohn-Sham-wavefunctions*) which will turn out to obey the one-particle Schrödinger-equation

$$\left(-\frac{\hbar^2}{2m} \Delta + V_H(\mathbf{x}) + V_{\text{XC}}(\mathbf{x}) + V_{e-a}(\mathbf{x}) + \text{const.} \right) \psi_{n\mathbf{k}\sigma}(\mathbf{x}) = E_{n\mathbf{k}} \psi_{n\mathbf{k}\sigma}(\mathbf{x}) \quad (4.9)$$

with $n \in \{1, 2, \dots\}$, $\mathbf{k} \in \mathcal{Q}$ and $\sigma \in \{\downarrow, \uparrow\}$ in the absence of spin-orbit-coupling⁴.

Remark 4.6.4 One notices, that in order to tackle the Schrödinger-equations of crystals, one divides the problem into further and further smaller (more or less managable) problems:

- We will be able to solve the time-independent Schrödinger-equation for the full (electronic and ionic system) in Equation (2.19) ...
 ...if we are able to solve the time-independent Schrödinger-equation for the effective electronic problem for some fixed ion-configuration in Equation (4.2).
- We will be able to solve the time-independent Schrödinger-equation for the effective electronic problem for some fixed ion-configuration in Equation (4.2) ...
 ...if we are able to find a (lattice-periodic) single-particle-potential that represents a good approximation to the electron-electron-interactions in Equation (4.7),
 ...and (inclusive) if we are able solve the time-independent Schrödinger-equation for a single-electron moving in a lattice-periodic potential with an effective one-particle-interaction in Equation (4.9).

Remark 4.6.5 There are also problems with the procedure outlined above... For example band-gaps and energies might be calculated inaccurately by a poor approximation of the electron-electron-interactions. One cure / catchphrase is here the so-called „GW-approximation“, on which we will not elaborate further in this document.

⁴In the presence of spin-orbit coupling the energies acquire a spin-dependence $E_{n\mathbf{k}} \mapsto E_{n\mathbf{k}\sigma}$.

Remark 4.6.6 Although the laborious procedure outlined above results in „concrete“ (numerical) expressions for the wavefunctions, it will later on turn out to be appropriate to not work directly with the wavefunctions themselves, but with the matrix-elements (w.r.t. those wavefunctions above) of the Hamiltonian and a „density“ that represents „the state of the thermodynamic equilibrium in the grand canonical ensemble in absence of spontaneous symmetry breaking“. We will briefly touch this formalism, when we compute specific heats in the low-temperature-limit in the formalism of second-quantization.

Notation 4.7 Having determined the wave-functions $\{\mathbf{r} \mapsto \phi_{\alpha,\tau}(\mathbf{r})\}_{\alpha \in \mathcal{A}}$ with eigenenergies $\{\epsilon_{\alpha,\tau} \equiv \epsilon_{\alpha}(\tau)\}_{\alpha \in \mathcal{A}}$ for an (arbitrarily) fixed ion position τ , we now consider the full time-independent Schrödinger-equation

$$H\psi(\mathbf{r}, \tau) = E\psi(\mathbf{r}, \tau) \quad (4.10)$$

and expand the wavefunctions w.r.t. complete basis

$$\psi(\mathbf{r}, \tau) = \sum_{\alpha \in \mathcal{A}} \chi_{\alpha}(\tau) \phi_{\alpha,\tau}(\mathbf{r}) \quad (4.11)$$

Theorem 4.8 The expansion in Equation (4.11) is a solution to the Schrödinger-Equation (4.10) in frame of the adiabatic approximation, if the coefficients $\chi_{\alpha}(\tau)$ fulfill the equation

$$(T_a + \epsilon_{\alpha,\tau}) \chi_{\alpha}(\tau) = E \chi_{\alpha}(\tau) \quad (4.12)$$

Proof 4.8 See Appendix, Proofs, page 402.

Remark 4.8.1 Since the electronic energies represent for each index α a mapping $\tau \mapsto \epsilon_{\alpha,\tau} \equiv \epsilon_{\alpha}(\tau)$ one usually interpretes Equation (30.27) as the *ionic Schrödinger-equation*.

5. The Ionic Problem

5.1. The Schrödinger-Equation of the Ionic-System in the Harmonic-Approximation

We start with the effective Schrödinger-equation for the wavefunctions of the ions

$$H\chi(\tau) = E\chi(\tau) \quad \text{with} \quad H = \sum_{\substack{\mathbf{R}_n \in \mathbf{L}_f \\ \alpha \in \{1, \dots, d_x\} \\ \kappa \in \{1, \dots, r\}}} -\frac{\hbar^2}{2M_\kappa} \frac{\partial^2}{\partial \tau_{n\kappa\alpha}^2} + V_{\text{eff}}(\tau) \quad (5.1)$$

with an effective potential¹

$$\tau \mapsto V_{\text{eff}}(\tau) \in \mathbb{R} \quad (5.2)$$

that is the result of the solution to the electronic problem for different ion configurations τ .

Remark 5.0.1 Recall from Lemma ?? that we have the canonical commutation relations for the ionic coordinates

$$[\tau_{n\kappa\alpha}, p_{\tau_{n'\kappa'\alpha'}}] = i\hbar \delta_{n,n'} \delta_{\kappa,\kappa'} \delta_{\alpha,\alpha'} \quad (5.3)$$

Intermezzo: Coordinate-Change

Lemma 5.1 If one changes coordinates as

$$x_i = x_i(x'_1, \dots, x'_n) \quad (5.4)$$

then it follows for the derivatives of the corresponding coordinates that

$$\frac{\partial}{\partial x'_i} = \sum_{j=1}^n \frac{\partial x_j}{\partial x'_i} \frac{\partial}{\partial x_j} \quad (5.5)$$

Proof 5.1 See any textbook on calculus in more than one dimensions.

¹Notice that we changed the notation of the effective potential, relative to the last section.

Hypothesis 5.2 We assume that there is an minimal equilibrium-position for the ions τ s.t.

$$\min_{\tau \in \mathbb{R}^{d_x N_a}} V_{\text{eff}}(\tau) \equiv V_{\text{eff}}(\mathbf{R}_{\text{eq}}) = \text{const.} \stackrel{\text{w.l.o.g.}}{=} 0 \quad \text{with} \quad (\nabla_{\tau} V_{\text{eff}})(\mathbf{R}_{\text{eq}}) = 0. \quad (5.6)$$

Notation 5.3 We change coordinates to the elongation of the ions from the individual lattice-sites

$$\Delta \tau_{\mathbf{n}\kappa} \equiv \tau_{\mathbf{n}\kappa} - \mathbf{R}_{\mathbf{n}\kappa} \quad \Longleftrightarrow \quad \Delta \tau_{\mathbf{n}\kappa\alpha} \equiv \tau_{\mathbf{n}\kappa\alpha} - R_{\mathbf{n}\kappa\alpha}. \quad (5.7)$$

Lemma 5.4 The derivatives change according to the chain-rule

$$\frac{\partial}{\partial \tau_{\mathbf{n}\kappa\alpha}} \equiv \frac{\partial}{\partial \Delta \tau_{\mathbf{n}\kappa\alpha}}. \quad (5.8)$$

and that this transformation preserves the CCR; it holds

$$[\Delta \tau_{\mathbf{n}\kappa\alpha}, p_{\Delta \tau_{\mathbf{n}'\kappa'\alpha'}}] = i\hbar \delta_{\mathbf{n},\mathbf{n}'} \delta_{\kappa,\kappa'} \delta_{\alpha,\alpha'} \quad (5.9)$$

Proof 5.4 Follows immediately from the linearity of the commutator together with Lemma 5.1.

Corollary 5.5 Changing the coordinates and expanding the effective potential in a series up to the second-order simplifies the Hamiltonian to

$$\begin{aligned} T_a = & \sum_{\substack{\mathbf{R}_{\mathbf{n}} \in \mathbf{L}_{\text{f}} \\ \kappa \in \{1, \dots, r\} \\ \alpha \in \{1, \dots, d_x\}}} -\frac{\hbar^2}{2M_{\kappa}} \frac{\partial^2}{\partial \Delta \tau_{\mathbf{n}\kappa\alpha}^2} \\ V_{\text{eff}}(\Delta \tau) \equiv V_{\text{eff}}(0) + & \sum_{\substack{\mathbf{n} \in \mathbf{L}_{\text{f}} \\ \kappa \in \{1, \dots, r\} \\ \alpha \in \{1, \dots, d_x\}}} \left(\frac{\partial V_{\text{eff}}}{\partial \Delta \tau_{\mathbf{n}\kappa\alpha}} \right) (0) \Delta \tau_{\mathbf{n}\kappa\alpha} + \frac{1}{2} \sum_{\substack{\mathbf{n}, \mathbf{n}' \in \mathbf{L}_{\text{f}} \\ \kappa, \kappa' \in \{1, \dots, r\} \\ \alpha, \alpha' \in \{1, \dots, d_x\}}} \left(\frac{\partial^2 V_{\text{eff}}}{\partial \Delta \tau_{\mathbf{n}\kappa\alpha} \partial \Delta \tau_{\mathbf{n}'\kappa'\alpha'}} \right) (0) \Delta \tau_{\mathbf{n}\kappa\alpha} \Delta \tau_{\mathbf{n}'\kappa'\alpha'} + \dots \\ \approx 0 \quad + 0 & \quad + \frac{1}{2} \sum_{\substack{\mathbf{n}, \mathbf{n}' \in \mathbf{L}_{\text{f}} \\ \kappa, \kappa' \in \{1, \dots, r\} \\ \alpha, \alpha' \in \{1, \dots, d_x\}}} \left(\frac{\partial^2 V_{\text{eff}}}{\partial \Delta \tau_{\mathbf{n}\kappa\alpha} \partial \Delta \tau_{\mathbf{n}'\kappa'\alpha'}} \right) (0) \Delta \tau_{\mathbf{n}\kappa\alpha} \Delta \tau_{\mathbf{n}'\kappa'\alpha'} \end{aligned}$$

Proof 5.5 Follows immediately by the assumptions and the Taylor-expansion.

Notation 5.6 From the Hesse-matrix at the equilibrium-position, one defines the *real-space-dynamical-matrix* as

$$D_{\mathbf{n}\kappa\alpha, \mathbf{n}'\kappa'\alpha'} \equiv \frac{1}{\sqrt{M_\kappa M_{\kappa'}}} \left(\frac{\partial^2 V_{\text{eff}}}{\partial \Delta \tau_{\mathbf{n}\kappa\alpha} \partial \Delta \tau_{\mathbf{n}'\kappa'\alpha'}} \right) (0). \quad (5.10)$$

Remark 5.6.1 Notice that ...

- D is real-valued (the potential is real-valued),
- D is symmetric since the Schwarz-theorem applies

$$\frac{\partial^2 V_{\text{eff}}}{\partial \Delta \tau_{\mathbf{n}\kappa\alpha} \partial \Delta \tau_{\mathbf{n}'\kappa'\alpha'}} (\bullet) = \frac{\partial^2 V_{\text{eff}}}{\partial \Delta \tau_{\mathbf{n}'\kappa'\alpha'} \partial \Delta \tau_{\mathbf{n}\kappa\alpha}} (\bullet),$$

which implies for the dynamical matrix the relation

$$D_{\mathbf{n}\kappa\alpha, \mathbf{n}'\kappa'\alpha'} = D_{\mathbf{n}'\kappa'\alpha', \mathbf{n}\kappa\alpha} \quad (5.11)$$

meaning that the real-space-dynamical matrix is symmetric.

- D is positive definite, i.e. $\mathbf{v}^T \cdot D \cdot \mathbf{v} > 0$ for $\mathbf{v} \neq 0$, since we assume an equilibrium-position that minimizes the potential energy.

At this point we are facing a crossroad. We can now continue with the solution of the Schrödinger-equation in two ways. The first is valuable for pedagogical reasons, and the other is valuable for computational (i.e. numerical) reasons. We will present both. In either way, we have to go through a series of coordinate-transformations, which is quite exhausting to keep track of.

5.1.1. Road 1 through Real-Space: A Poor Man's Phonons.

Notation 5.7 We introduce new coordinates as

$$\tilde{u}_{\mathbf{n}\kappa\alpha} \equiv \sqrt{M_\kappa} \Delta \tau_{\mathbf{n}\kappa\alpha} \quad (5.12)$$

Lemma 5.8 We obtain for the new derivatives the relation

$$\frac{\partial}{\partial \tilde{u}_{\mathbf{n}\kappa\alpha}} = \frac{1}{\sqrt{M_\kappa}} \frac{\partial}{\partial \Delta \tau_{\mathbf{n}\kappa\alpha}} \quad (5.13)$$

and this transformation preserves the CCR, i.e. it holds

$$[\tilde{u}_{\mathbf{n}\kappa\alpha}, p_{\tilde{u}_{\mathbf{n}'\kappa'\alpha'}}] = i\hbar \delta_{\mathbf{n}, \mathbf{n}'} \delta_{\kappa, \kappa'} \delta_{\alpha, \alpha'} \quad (5.14)$$

Proof 5.8 Follows immediately from the linearity of the commutator, and the chain-rule for derivatives in Lemma 5.1.

Corollary 5.9 Switching to the new coordinates simplifies the Hamiltonian in the harmonic approximation to

$$\begin{aligned}
 H_{\text{harm}} &= T_{\text{ions}} + V_{\text{eff}}(\Delta \mathbf{r}) \\
 &= \frac{1}{2} \sum_{\substack{\mathbf{R}_n \in L_f \\ \kappa \in \{1, \dots, r\} \\ \alpha \in \{1, \dots, d\}}} -\hbar^2 \frac{\partial^2}{\partial \tilde{u}_{\mathbf{n}\kappa\alpha}^2} + \frac{1}{2} \sum_{\substack{\mathbf{n}, \mathbf{n}' \in L_f \\ \kappa, \kappa' \in \{1, \dots, r\} \\ \alpha, \alpha' \in \{1, \dots, d\}}} D_{\mathbf{n}\kappa\alpha, \mathbf{n}'\kappa'\alpha'} \tilde{u}_{\mathbf{n}\kappa\alpha} \tilde{u}_{\mathbf{n}'\kappa'\alpha'} \\
 &= \frac{1}{2} \mathbf{p}_{\tilde{\mathbf{u}}}^T \cdot \mathbb{1} \cdot \mathbf{p}_{\tilde{\mathbf{u}}} + \frac{1}{2} \tilde{\mathbf{u}}^T \cdot D \cdot \tilde{\mathbf{u}}
 \end{aligned}$$

Proof 5.9 Follows immediately by insertion and applying the correspondence principle backwards.

Lemma 5.10 Since the dynamical matrix is symmetric, one can find an orthogonal² matrix $C \in \text{Mat}(\mathbb{R}^{d_x N_a} \times \mathbb{R}^{d_x N_a})$ s.t.

$$C D C^T = \Omega = \text{diag}(\omega_1^2, \omega_2^2, \dots, \omega_{d_x N_a}^2) \quad (5.16)$$

with frequencies that are chosen to be positive $\omega_i \in \mathbb{R}_{>0}$.

Proof 5.10 See any textbook on linear algebra.

Notation 5.11 We introduce new coordinates as

$$\mathbf{u} \equiv C \tilde{\mathbf{u}} \quad (5.17)$$

Lemma 5.12 We obtain new momenta as

$$\mathbf{p}_{\mathbf{u}} = C^T \mathbf{p}_{\tilde{\mathbf{u}}} \iff \mathbf{p}_{\tilde{\mathbf{u}}} = C \mathbf{p}_{\mathbf{u}} \quad (5.18)$$

and this coordinate-transformation is compatible with the CCR; it holds

$$[u_{\mathbf{n}\kappa\alpha}, p_{u_{\mathbf{n}'\kappa'\alpha'}}] = i\hbar \delta_{\mathbf{n}, \mathbf{n}'} \delta_{\kappa, \kappa'} \delta_{\alpha, \alpha'} \quad (5.19)$$

Proof 5.12 Follows immediately from the linearity of the commutator, and the chain-rule for derivatives in Lemma 5.1.

²Recall that this means that the transpose of the matrix is its inverse

$$C \cdot C^T = C^T \cdot C = \mathbb{1}. \quad (5.15)$$

Corollary 5.13 Changing to the new coordinates brings the Hamiltonian into the diagonal form

$$H_{\text{harm.}} = \frac{1}{2} \mathbf{p}_{\mathbf{u}}^T \cdot \mathbb{1} \cdot \mathbf{p}_{\mathbf{u}} + \frac{1}{2} \mathbf{u}^T \cdot \Omega \cdot \mathbf{u}$$

Proof 5.13 By direct calculation, we obtain

$$\begin{aligned} H_{\text{harm.}} &= \frac{1}{2} \tilde{\mathbf{p}}^T \cdot C^T \cdot C \cdot \tilde{\mathbf{p}} + \frac{1}{2} \tilde{\mathbf{u}}^T C^T \cdot C \cdot D \cdot C^T \cdot C \tilde{\mathbf{u}} \\ &= \frac{1}{2} \tilde{\mathbf{p}}^T \cdot C^T \cdot C \cdot \tilde{\mathbf{p}} + \frac{1}{2} \tilde{\mathbf{u}}^T C^T \cdot \Omega \cdot C \tilde{\mathbf{u}} \end{aligned}$$

Remark 5.13.1 Notice therefore, that within the harmonic approximation, the ionic system is can be described as a sum of harmonic oscillators

$$H_{\text{harm.}} = \frac{1}{2} \mathbf{p}_{\mathbf{u}}^T \cdot \mathbb{1} \cdot \mathbf{p}_{\mathbf{u}} + \frac{1}{2} \mathbf{u}^T \cdot \Omega \cdot \mathbf{u} = \frac{1}{2} \sum_{i=1}^{d_{\mathbf{x}} N_a} p_i^2 + \omega_i^2 u_i^2 \equiv \sum_{i=1}^{d_{\mathbf{x}} N_a} H_i$$

Corollary 5.14 Using the standard-techniques from the single-particle-harmonic-oscillator, i.e. introducing ladder-operators

$$b_j = \sqrt{\frac{\omega_j}{2\hbar}} u_j + i \sqrt{\frac{1}{2\hbar\omega_j}} p_j, \quad b_j^\dagger = \sqrt{\frac{\omega_j}{2\hbar}} u_j - i \sqrt{\frac{1}{2\hbar\omega_j}} p_j \quad (5.20)$$

with commutation relations

$$[b_j, b_k^\dagger] = \delta_{j,k}, \quad [b_j, b_k] = [b_j^\dagger, b_k^\dagger] = 0, \quad (5.21)$$

one arrives at the semi-final form the Hamilton-operator in the harmonic approximation

$$H_{\text{harm.}} = \sum_{j=1}^{d_{\mathbf{x}} N_a} \hbar \omega_j \left(b_j^\dagger b_j + \frac{1}{2} \right). \quad (5.22)$$

Proof 5.14 By using standard-theorems that we proved earlier in the lecture and on the exercise sheets, we immediately obtain the result.

Remark 5.14.1 We therefore see, that we can solve the effective Schrödinger-equation of the ionic problem iff we can diagonalize the real-space-dynamical-matrix D (which is a task that we can't for realistic system sizes). One circumvents this problem by going to reciprocal space.

5.1.2. Road 2 through Reciprocal-Space: State of the Art Description and Identities.

Theorem 5.15 Through the Coulomb-potential and the periodic boundary-conditions the dynamical matrix inherits the translational invariance

$$D_{\mathbf{n}\kappa\alpha, \mathbf{n}'\kappa'\alpha'} \equiv D_{\kappa\alpha, \kappa'\alpha'}(\mathbf{R}_{\mathbf{n}} - \mathbf{R}_{\mathbf{n}'}) \quad (5.23)$$

Proof 5.15 Follows from the distance-dependence of the Coulomb-interaction, when one explicitly writes down the sums.

Remark 5.15.1 Notice that this implies that the dynamical matrix is a function of the difference of the lattice-vectors, which implies the relations

$$D_{\mathbf{n}\kappa\alpha, \mathbf{n}'\kappa'\alpha'} = D_{\mathbf{n}-\mathbf{n}'\kappa\alpha, 0\kappa'\alpha'} = D_{0\kappa\alpha, -(\mathbf{n}-\mathbf{n}')\kappa'\alpha'} \quad (5.24)$$

Notation 5.16 We change the coordinates and consider discrete Fourier-transforms with prefactor-conventions, to introduce the *Fourier-transformed Dynamical Matrix*

$$\begin{aligned} \Delta\tau_{\mathbf{n}\kappa\alpha} &\equiv \frac{1}{\sqrt{N_p}} \sum_{\mathbf{q} \in Q} \Delta\tau_{\mathbf{q}\kappa\alpha} e^{-i\mathbf{q} \cdot \mathbf{R}_{\mathbf{n}}} &\iff & \Delta\tau_{\mathbf{q}\kappa\alpha} \equiv \frac{1}{\sqrt{N_p}} \sum_{\mathbf{R}_{\mathbf{n}} \in L_f} \Delta\tau_{\mathbf{n}\kappa\alpha} e^{+i\mathbf{q} \cdot \mathbf{R}_{\mathbf{n}}} \\ D_{\kappa\alpha, \kappa'\alpha'}(\mathbf{R}_{\mathbf{n}} - \mathbf{R}_{\mathbf{n}'}) &\equiv \frac{1}{N_p} \sum_{\mathbf{q} \in Q} D_{\kappa\alpha, \kappa'\alpha'}(\mathbf{q}) e^{-i\mathbf{q} \cdot (\mathbf{R}_{\mathbf{n}} - \mathbf{R}_{\mathbf{n}'})} &\iff & D_{\kappa\alpha, \kappa'\alpha'}(\mathbf{q}) \equiv \sum_{\mathbf{R}_{\mathbf{n}} \in L_f} D_{\kappa\alpha, \kappa'\alpha'}(\mathbf{R}_{\mathbf{n}}) e^{+i\mathbf{q} \cdot \mathbf{R}_{\mathbf{n}}} \end{aligned} \quad (5.25)$$

Remark 5.16.1 One might wonder how it comes that one uses in two subsequent lines two different prefactor-conventions for the definition of the Fourier-transform (of a quantity). It is this choice in particular that makes the resulting final expression for the Hamiltonian look „particularly simple“. This charm has prevailed and these two prefactor-conventions are therefore in all the books and more importantly in all the codes for first principles and ab initio calculations and one has to get used to them.

Remark 5.16.2 Notice that since $\Delta\tau_{\mathbf{n}\kappa\alpha} = \Delta\tau_{\mathbf{n}\kappa\alpha}^\dagger$ (as an operator) it follows that

$$\Delta\tau_{\mathbf{q}\kappa\alpha}^\dagger = \frac{1}{\sqrt{N_p}} \sum_{\mathbf{R}_{\mathbf{n}} \in L_f} \Delta\tau_{\mathbf{n}\kappa\alpha}^\dagger e^{-i\mathbf{q} \cdot \mathbf{R}_{\mathbf{n}}} = \frac{1}{\sqrt{N_p}} \sum_{\mathbf{R}_{\mathbf{n}} \in L_f} \Delta\tau_{\mathbf{n}\kappa\alpha} e^{+i(-\mathbf{q}) \cdot \mathbf{R}_{\mathbf{n}}} = \Delta\tau_{-\mathbf{q}\kappa\alpha} \quad (5.26)$$

(as an operator again) and one obtains the further auxiliary-identity

$$\Delta\tau_{\mathbf{n}\kappa\alpha} = \Delta\tau_{\mathbf{n}\kappa\alpha}^\dagger = \frac{1}{\sqrt{N_p}} \sum_{\mathbf{q} \in Q} \Delta\tau_{\mathbf{q}\kappa\alpha}^\dagger e^{+i\mathbf{q} \cdot \mathbf{R}_{\mathbf{n}}} = \frac{1}{\sqrt{N_p}} \sum_{\mathbf{q} \in Q} \Delta\tau_{-\mathbf{q}\kappa\alpha} e^{+i\mathbf{q} \cdot \mathbf{R}_{\mathbf{n}}} \quad (5.27)$$

which will turn out to be good to use in the following.

Lemma 5.17 The derivatives change accordingly again:

$$\frac{\partial}{\partial \Delta \tau_{\mathbf{n}\kappa\alpha}} = \frac{1}{\sqrt{N_p}} \sum_{\mathbf{q} \in Q} e^{+i\mathbf{q}\mathbf{R}_n} \frac{\partial}{\partial \Delta \tau_{\mathbf{q}\kappa\alpha}} \iff \frac{\partial}{\partial \Delta \tau_{\mathbf{q}\kappa\alpha}} = \frac{1}{\sqrt{N_p}} \sum_{\mathbf{R}_n \in L_f} e^{-i\mathbf{q}\mathbf{R}_n} \frac{\partial}{\partial \Delta \tau_{\mathbf{n}\kappa\alpha}} \quad (5.28)$$

and this transformation preserves the CCR, i.e. one has

$$\begin{aligned} \left[\Delta \tau_{\mathbf{q}\kappa\alpha}, \frac{\partial}{\partial \Delta \tau_{\mathbf{q}'\kappa'\alpha'}} \right] &= \frac{1}{N_p} \sum_{\mathbf{R}_n, \mathbf{R}_{n'} \in L_f} e^{-i\mathbf{q}\mathbf{R}_n} e^{+i\mathbf{q}'\mathbf{R}_{n'}} \left[\Delta \tau_{\mathbf{n}\kappa\alpha}, \frac{\partial}{\partial \Delta \tau_{\mathbf{n}'\kappa'\alpha'}} \right] \\ &= \frac{\delta_{\kappa, \kappa'} \delta_{\alpha, \alpha'}}{N_p} \sum_{\mathbf{R}_n \in L_f} e^{i\mathbf{R}_n \cdot (\mathbf{q} - \mathbf{q}')} \\ &= \delta_{\mathbf{q}, \mathbf{q}'} \delta_{\kappa, \kappa'} \delta_{\alpha, \alpha'} \end{aligned} \quad (5.29)$$

Proof 5.17 Follows immediately from the linearity of the commutator, and the chain-rule for derivatives in Lemma 5.1.

Corollary 5.18 It follows for the Hamiltonian

$$T_{\text{ions}} + V_{\text{eff}}(\tau) \approx \sum_{\substack{\mathbf{q} \in Q \\ \kappa \in \{1, \dots, r\} \\ \alpha \in \{1, \dots, d_x\}}} -\frac{\hbar^2}{2M_\kappa} \frac{\partial}{\partial \Delta \tau_{-\mathbf{q}\kappa\alpha}} \frac{\partial}{\partial \Delta \tau_{\mathbf{q}\kappa\alpha}} + \frac{1}{2} \sum_{\substack{\mathbf{q} \in Q \\ \kappa, \kappa' \in \{1, \dots, r\} \\ \alpha, \alpha' \in \{1, \dots, d_x\}}} \Delta \tau_{-\mathbf{q}\kappa\alpha} \frac{D_{\kappa\alpha, \kappa'\alpha'}(\mathbf{q})}{\sqrt{M_\kappa M_{\kappa'}}^{-1}} \Delta \tau_{\mathbf{q}\kappa'\alpha'}.$$

Proof 5.18 See Appendix, p. 61.

Lemma 5.19 For each $\mathbf{q} \in Q$ the matrix $D(\mathbf{q})$ with matrix entries $D_{\kappa\alpha, \kappa'\alpha'}(\mathbf{q})$ is hermitian.

Proof 5.19 See Appendix, p. 403.

Corollary 5.20 Since $D(\mathbf{q})$ is hermitian, there is a unitary matrix $U(\mathbf{q})$ consisting of orthonormal eigenvectors (as columns) that diagonalizes the dynamical matrix

$$U^\dagger(\mathbf{q}) D(\mathbf{q}) U(\mathbf{q}) = \text{diag} \left(\omega_1^2(\mathbf{q}), \dots, \omega_{rd_x}^2(\mathbf{q}) \right). \quad (5.30)$$

Proof 5.20 This proof is carried out in any textbook on linear algebra.

Notation 5.21 Recall from any linear algebra course, that the matrix-entries of $U(\mathbf{q})$ are given by the orthonormal eigenvectors of the dynamical matrix $D(\mathbf{q})$. We introduce for each $\mathbf{q} \in Q$ its eigenvectors as

$$\{\mathbf{e}_{:,1}(\mathbf{q}), \mathbf{e}_{:,2}(\mathbf{q}), \dots, \mathbf{e}_{:,rd_x}(\mathbf{q})\}.$$

This gives the equation for their components

$$\sum_{\kappa'\alpha'} D_{\kappa\alpha,\kappa'\alpha'}(\mathbf{q}) e_{\kappa'\alpha',\nu}(\mathbf{q}) = \omega_\nu^2(\mathbf{q}) e_{\kappa\alpha,\nu}(\mathbf{q}) \iff D(\mathbf{q})\mathbf{e}_{:, \nu}(\mathbf{q}) = \omega_\nu^2(\mathbf{q})\mathbf{e}_{:, \nu}(\mathbf{q}) \quad (5.31)$$

Lemma 5.22 The unitarity of the transformation-matrix $U(\mathbf{q})$ implies the relations

$$\begin{aligned} \sum_{\kappa\alpha} e_{\kappa\alpha,\nu}^*(\mathbf{q}) e_{\kappa\alpha,\nu'}(\mathbf{q}) &= \delta_{\nu\nu'} \\ \sum_{\nu} e_{\kappa\alpha,\nu}^*(\mathbf{q}) e_{\kappa'\alpha',\nu}(\mathbf{q}) &= \delta_{\kappa',\kappa} \delta_{\alpha',\alpha} \end{aligned} \quad (5.32)$$

whereas the first relation merely resembles the orthonormality of the eigenvectors w.r.t. the standard scalar-product on the complex numbers, and the second equation is a version of the completeness-relation for the eigenvectors.

Proof 5.22 See any textbook on linear algebra.

Lemma 5.23 For the eigenvectors and eigenvalues of the dynamical matrix in reciprocal space one finds the relations

$$\omega_{-\mathbf{q}\nu}^2 = \omega_{\mathbf{q}\nu}^2, \quad e_{\kappa\alpha,\nu}(-\mathbf{q}) = e_{\kappa\alpha,\nu}^*(\mathbf{q}). \quad (5.33)$$

Proof 5.23 See Appendix, p. 403.

Notation 5.24 For reasons that will become clear in a moment, we introduce coordinates according to

$$\Delta\tau_{\mathbf{q}\kappa\alpha} = \sum_{\nu'} e_{\kappa\alpha,\nu'}(\mathbf{q}) \frac{u_{\mathbf{q}\nu'}}{\sqrt{M_\kappa}}, \quad \iff \quad u_{\mathbf{q}\nu} = \sum_{\kappa'\alpha'} e_{\kappa'\alpha',\nu}^*(\mathbf{q}) \sqrt{M_{\kappa'}} \Delta\tau_{\mathbf{q}\kappa'\alpha'} \quad (5.34)$$

Lemma 5.25 We will prove ...

- that the inverse-relation in Equation (5.34) is correct,
- the auxiliary-identity

$$\Delta\tau_{\mathbf{q}\kappa\alpha}^\dagger = \sum_{\nu'} e_{\kappa\alpha,\nu'}^*(\mathbf{q}) \frac{u_{\mathbf{q}\nu'}^\dagger}{\sqrt{M_\kappa}} = \Delta\tau_{-\mathbf{q}\kappa\alpha}, \quad (5.35)$$

- the derivatives change according to

$$\frac{\partial}{\partial \Delta \tau_{\mathbf{q}\kappa\alpha}} = \sum_{\nu'} e_{\kappa\alpha, \nu'}^*(\mathbf{q}) \sqrt{M_\kappa} \frac{\partial}{\partial u_{\mathbf{q}\nu'}} \iff \frac{\partial}{\partial u_{\mathbf{q}\nu}} = \sum_{\kappa'\alpha'} e_{\kappa'\alpha', \nu}(\mathbf{q}) \frac{1}{\sqrt{M_{\kappa'}}} \frac{\partial}{\partial \Delta \tau_{\mathbf{q}\kappa'\alpha'}}. \quad (5.36)$$

- and this transformation preserves the CCR, i.e. one has

$$[u_{\mathbf{q}\nu}, \frac{\partial}{\partial u_{\mathbf{q}'\nu'}}] = -\delta_{\nu, \nu'} \delta_{\mathbf{q}, \mathbf{q}'} \quad (5.37)$$

Proof 5.25 See Appendix, p. 404.

Lemma 5.26 Summarizing all the coordinate-changes, we obtain in total

$$\tau_{\mathbf{n}\kappa\alpha} = R_{\mathbf{n}\kappa\alpha} + \frac{1}{\sqrt{N_p}} \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ \nu' \in \{1, \dots, rd_x\}}} e_{\kappa\alpha, \nu'}(\mathbf{q}) \frac{e^{i\mathbf{q}\mathbf{R}_n}}{\sqrt{M_\kappa}} u_{\mathbf{q}\nu'} \iff u_{\mathbf{q}\nu} = \frac{1}{\sqrt{N_p}} \sum_{\substack{\mathbf{R}_n \in \mathbf{L}_f \\ \kappa'\alpha'}} e_{\kappa'\alpha', \nu}^*(\mathbf{q}) \sqrt{M_{\kappa'}} e^{-i\mathbf{q}\mathbf{R}_n} (\tau_{\mathbf{n}\kappa'\alpha'} - R_{\mathbf{n}\kappa'\alpha'}) \quad (5.38)$$

which implies the equations

$$\frac{1}{\partial \tau_{\mathbf{n}\kappa\alpha}} = \frac{1}{\sqrt{N_p}} \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ \nu' \in \{1, \dots, rd_x\}}} e_{\kappa'\alpha', \nu'}^*(\mathbf{q}) \sqrt{M_\kappa} e^{-i\mathbf{q}\mathbf{R}_n} \frac{\partial}{\partial u_{\mathbf{q}\nu'}} \iff \frac{\partial}{\partial u_{\mathbf{q}\nu}} = \frac{1}{\sqrt{N_p}} \sum_{\substack{\mathbf{R}_n \in \mathbf{L}_f \\ \kappa'\alpha'}} e_{\kappa'\alpha', \nu}(\mathbf{q}) \frac{e^{+i\mathbf{q}\mathbf{R}_n}}{\sqrt{M_{\kappa'}}} \frac{\partial}{\partial \tau_{\mathbf{n}\kappa'\alpha'}}$$

Proof 5.26 See Appendix, p. 405.

Corollary 5.27 It follows

$$T_{\text{ions}} + V_{\text{eff}}(\tau) \approx \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ \nu \in \{1, \dots, rd_x\}}} -\frac{\hbar^2}{2} \frac{\partial}{\partial u_{-\mathbf{q}\nu}} \frac{\partial}{\partial u_{\mathbf{q}\nu}} + \frac{\omega_\nu^2}{2}(\mathbf{q}) u_{-\mathbf{q}\nu} u_{\mathbf{q}\nu}.$$

Proof 5.27 See Appendix, p. 407.

Definition 5.28 We introduce the ladder-operators

$$a_{\mathbf{q}\nu} = \sqrt{\frac{\omega_\nu(\mathbf{q})}{2\hbar}} \left(u_{\mathbf{q}\nu} + \frac{\hbar}{\omega_\nu(\mathbf{q})} \frac{\partial}{\partial u_{-\mathbf{q}\nu}} \right), \quad a_{\mathbf{q}\nu}^\dagger = \sqrt{\frac{\omega_\nu(\mathbf{q})}{2\hbar}} \left(u_{-\mathbf{q}\nu} - \frac{\hbar}{\omega_\nu(\mathbf{q})} \frac{\partial}{\partial u_{+\mathbf{q}\nu}} \right) \quad (5.39)$$

Remark 5.28.1 Although this is simply a neat rewriting of operators, one oftentimes calls this step *canonical quantization* of position operators or *second quantization*.

Lemma 5.29 The ladder-operators fulfill the CCR

$$[a_{\mathbf{q}\nu}, a_{\mathbf{q}'\nu'}^\dagger] = \delta_{\nu,\nu'} \delta_{\mathbf{q},\mathbf{q}'}, \quad [a_{\mathbf{q}\nu}, a_{\mathbf{q}'\nu'}] = [a_{\mathbf{q}\nu}^\dagger, a_{\mathbf{q}'\nu'}^\dagger] = 0 \quad (5.40)$$

and one finds the inversion formulae

$$u_{\mathbf{q}\nu} = \sqrt{\frac{\hbar}{2\omega_\nu(\mathbf{q})}} (a_{\mathbf{q}\nu} + a_{-\mathbf{q}\nu}^\dagger), \quad \frac{\partial}{\partial u_{\mathbf{q}\nu}} = \sqrt{\frac{\omega_\nu(\mathbf{q})}{2\hbar}} (a_{-\mathbf{q}\nu} - a_{\mathbf{q}\nu}^\dagger) \quad (5.41)$$

Proof 5.29 See Appendix, p. 409.

Corollary 5.30 Inserting this in the Hamiltonian above yields the final expression

$$H = \sum_{\substack{\mathbf{q} \in Q \\ \nu \in \{1, \dots, rd_x\}}} \hbar \omega_\nu(\mathbf{q}) \left(a_{\mathbf{q}\nu}^\dagger a_{\mathbf{q}\nu} + \frac{1}{2} \right). \quad (5.42)$$

Proof 5.30 See Appendix, p. 410.

Remark 5.30.1 Notice that we used $\omega_\nu(-\mathbf{q}) = \omega_\nu(+\mathbf{q})$ within the proof.

Lemma 5.31 Notice the important formula that is an integral component on any derivation of electron-phonon-couplings:

$$\Delta \tau_{\mathbf{n}\kappa\alpha} = \frac{1}{\sqrt{N_p}} \sum_{\substack{\mathbf{q} \in Q \\ \nu' \in \{1, \dots, rd_x\}}} \sqrt{\frac{\hbar}{2M_\kappa \omega_{\nu'}(\mathbf{q})}} e_{\kappa\alpha, \nu'}(\mathbf{q}) e^{i\mathbf{q}\mathbf{R}_n} (a_{\mathbf{q}\nu'} + a_{-\mathbf{q}\nu'}^\dagger) \quad (5.43)$$

Proof 5.31 See Appendix, p. 411.

5.2. Electron-Phonon-Coupling

Cf. Waste-Folder 6, topic 10: For Felicianos derivation of electron-phonon-coupling.

5.3. Summary

The Main Tricks

By using a Taylor-expansion around the crystal-coordinates one has to simply diagonalize a matrix in order to solve the Schrödinger-equation (within the harmonic approximation). By going to reciprocal space, we have reduced the problem

$$\text{“Diagonalizing a symmetric } rd_{\mathbf{x}}N_p \times rd_{\mathbf{x}}N_p \text{-matrix”} \quad (5.44)$$

with $N_p \in [10^{20}, 10^{24}]$ to the *equivalent* problem

$$\text{“Diagonalizing } N_p \text{ hermitian } rd_{\mathbf{x}} \times rd_{\mathbf{x}} \text{-matrices”} \quad (5.45)$$

with $rd_{\mathbf{x}} \leq 100$ (in general...).

Thus the problem of

“Solving the effective Schrödinger-equation for the wavefunctions of the ions with an effective potential that is the result of the solution to the electronic problem for different ion configurations”

turned out to be equivalent³ (within the validity of the harmonic-approximation) to the problem

“Solving the Schrödinger-equation of harmonic-oscillators and simultaneously diagonalizing $rd_{\mathbf{x}} \times rd_{\mathbf{x}}$ -matrices for different values of $\mathbf{q} \in \mathbf{Q}$.”

Eigen-energies and eigenstates of the ionic Schrödinger-equation

Let us now introduce an enumeration of the cross-product

$$\left\{ \lambda_1, \lambda_2, \dots, \lambda_{rd_{\mathbf{x}}N_p} \right\} \equiv \{1, 2, \dots, rd_{\mathbf{x}}\} \times \mathbf{Q}$$

such that $\lambda \equiv (j, \mathbf{q})$. Now each harmonic oscillator H_{λ} appearing in the harmonic-approximation can be diagonalized individually with standard-techniques, such that the total Hamiltonian

$$H \approx H_{\text{ionic harmonic}} = \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ v \in \{1, \dots, rd_{\mathbf{x}}\}}} \hbar \omega_v(\mathbf{q}) \left(a_{\mathbf{q}v}^{\dagger} a_{\mathbf{q}v} + \frac{1}{2} \right) \equiv \sum_{\lambda \in \{1, 2, \dots, rd_{\mathbf{x}}N_p\}} \hbar \omega_{\lambda} \left(a_{\lambda}^{\dagger} a_{\lambda} + \frac{1}{2} \right). \quad (5.46)$$

can be diagonalized with a product-ansatz. This yields a basis for the underlying Hilbert-space

$$\bigotimes_{j=1}^{rd_{\mathbf{x}}N_p} L^2(\mathbb{R}) \simeq L^2(\mathbb{R}^{rd_{\mathbf{x}}N_p}) \equiv \mathcal{H}$$

a basis by products of wavefunctions

³In the sense: If we have found the solutions, i.e. wavefunctions, to the latter problem, then we can find solutions to the prior problem by simply changing the coordinates and expanding w.r.t. these wavefunctions.

$$\mathcal{B} = \left\{ |\Psi_{n_{\lambda_1}, n_{\lambda_2}, \dots, n_{\lambda_{rd_x N_p}}}\rangle \equiv |n_{\lambda_1}, n_{\lambda_2}, \dots, n_{\lambda_{rd_x N_p}}\rangle \mid n_{\lambda_i} \in \{0, 1, 2, \dots, \infty\} \right\} \quad (5.47)$$

We thus conclude that the total energies that the system can have (corresponding to the individual eigenstates that the system can be in) are

$$E_{n_{\lambda_1}, n_{\lambda_2}, \dots, n_{\lambda_{rd_x N_p}}} = \sum_{j=1}^{rd_x N_p} \hbar \omega_{\lambda_j} \left(n_{\lambda_j} + \frac{1}{2} \right) \equiv \sum_{\mathbf{q} \in Q} \sum_{v=1}^{rd_x} \hbar \omega_{v\mathbf{q}} \left(n_{v\mathbf{q}} + \frac{1}{2} \right). \quad (5.48)$$

Remark 5.31.1 Notice that each of these wavefunctions has $rd_x N_p$ -positional-arguments! However... Usually one does not work so much about the ionic wavefunctions themselves anymore in solid-state-theory. It will turn out that the energies are quite enough to get a good picture of what is happening inside a solid in a lot of situations.

Definition of Phonons

Definition 5.32 For each $v \in \{1, 2, \dots, rd_x\}$ one categorizes the *phonon-branches* as mappings

$$Q \ni \mathbf{q} \mapsto \omega_v(\mathbf{q}) \in \mathbb{R}_{>0}. \quad (5.49)$$

Remark 5.32.1 A phonon branch ω_v is called *acoustical* (or *optical*), if

$$\lim_{\mathbf{q} \rightarrow 0} \omega_v(\mathbf{q}) = 0 \quad \left(\text{or} \quad \lim_{\mathbf{q} \rightarrow 0} \omega_v(\mathbf{q}) \neq 0 \right). \quad (5.50)$$

Fact 5.33 For a crystal, there are d_x acoustical phonon-branches and $rd_x - d_x$ optical branches. (Without Proof.)

Notation 5.34 Without loss of generality we enumerate the phonon-branches

$$\{\omega_1, \dots, \omega_{d_x}, \omega_{d_x+1}, \dots, \omega_{rd_x}\} \quad (5.51)$$

in such a way, that

- the first d_x branches are acoustical,
- the remaining $rd_x - d_x$ branches are optical.

Interpretation of Phonons as Quasiparticles

For this we borrow the words from Alexander Altland in his book about condensed matter field theory:

Consider a standard harmonic oscillator (HO) Hamiltonian [...]. The HO has, of course, the status of a single-particle problem. However, the equidistance of its energy levels suggests an alternative interpretation. One can think of a given energy state ϵ_n as an accumulation of n elementary entities, or quasi-particles, each having energy ω . What can be said about the features of these new objects? First, they are structureless, i.e. the only “quantum number” identifying the quasi-particles is their energy ω (otherwise n -particle states formed of the quasi-particles would not be equidistant). This implies that the quasi-particles must be bosons. (The same state ω can be occupied by more than one particle [...]).

About the rewritten form of the HO-Hamiltonian

$$H_{\text{HO}} = \frac{p^2}{2m} + \frac{m\omega^2}{2}x^2 = \omega \left(a^\dagger a + \frac{1}{2} \right) \quad (5.52)$$

[...] its “real” advantage is that it naturally affords a many-particle interpretation. To this end, let us declare $|0\rangle$ to represent a “vacuum” state, i.e. a state with zero particles present. Next, imagine that $a^\dagger|0\rangle$ is a state with a single featureless particle [...] of energy ω . Similarly $(a^\dagger)^n|0\rangle$ is considered as a many-body state with n particles, i.e. within the new picture [...]. [...] While, at first sight, this may look unfamiliar, the new interpretation is internally consistent. Moreover, it achieves what we had asked for above, i.e. it allows an interpretation of the HO states as a superposition of independent structureless entities. Physically, the quasi-particles [...] are identified with the phonon modes of the solid.

6. The Electronic Problem

To Be Done... Bloch-Theorem and reference to Slater-determinants solving the non-interacting problem.

6.1. The Many-Electron Problem

6.2. The One-Electron Problem (Bloch-Theorem)

6.3. Density Functional Theory - Kohn-Sham-Equations

Part II.

Green-Function-Techniques

Motivation and Overview

After this part of the notes you will understand ...

- what Hans Bethe meant, when he said¹ that

"[The fact that] Particles can be created and annihilated, is the bread and butter of todays physics and has remained so."

and why this is usually visualized through pictures like the one in Figure 6.1.

- How Green-Functions (in this work) are defined by combining two big pillars of modern physics, namely *Quantum Many Body Physics* and *Equilibrium Statistical Physics*.

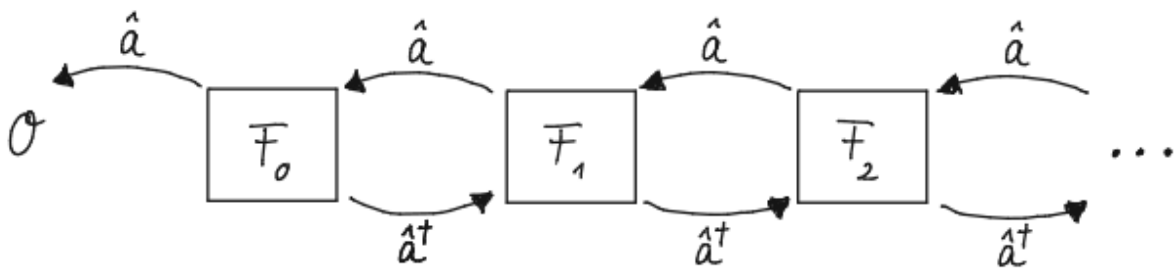
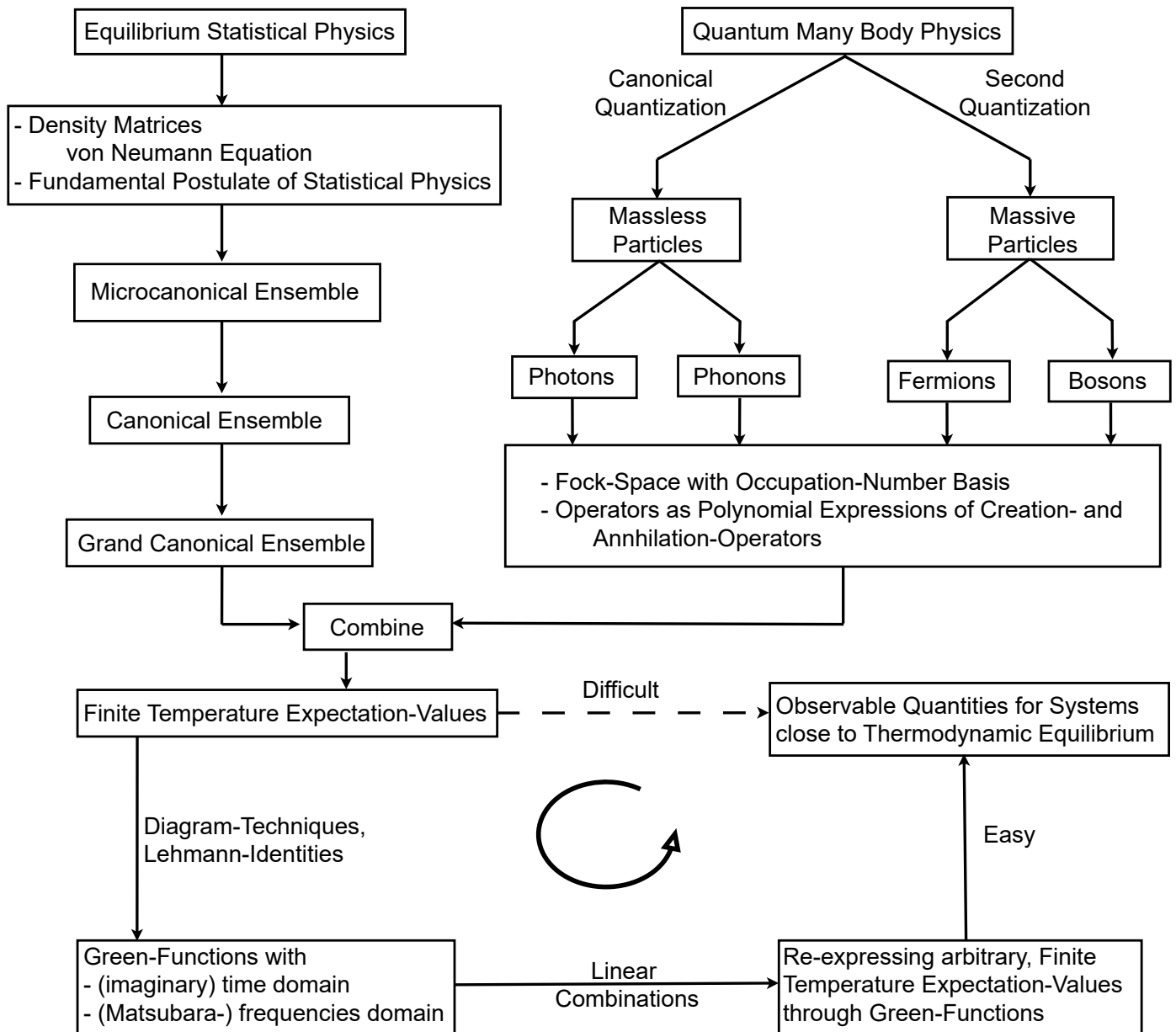


Figure 6.1.: Particles being created and annihilated, mapping from one sub-Hilbert-space to another.

¹Hans Bethe lecture held at MIT Nov. 21 1977 <https://www.youtube.com/watch?v=E61UR4Lbifo>, timestamp: 01h 11min

Outline of the Presentation



7. Fundamentals I - Hilbert-Spaces, States and Operators

7.1. The Many-Body-Formalism for Distinguishable Objects

Fact 7.1 Let $\mathcal{H}_1, \dots, \mathcal{H}_n$ denote the single-particle-Hilbert-spaces with respective scalar-products of n (pairwise) distinguishable *objects*¹. Then the Hilbert-space $\mathcal{H}$ of the total system is given as tensor product

$$\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \dots \otimes \mathcal{H}_n \equiv \bigotimes_{j=1}^n \mathcal{H}_j \quad (7.1)$$

with sesqui-linear scalar-product defined on product-states as

$$\langle \psi_1 \otimes \dots \otimes \psi_n, \psi'_1 \otimes \dots \otimes \psi'_n \rangle_{\mathcal{H}} := \langle \psi_1, \psi'_1 \rangle_{\mathcal{H}_1} \dots \langle \psi_n, \psi'_n \rangle_{\mathcal{H}_n} \equiv \prod_{j=1}^n \langle \psi_j, \psi'_j \rangle_{\mathcal{H}_j}. \quad (7.2)$$

which is extended by linearity on the entire tensor-product.

Remark 7.1.1 Notice also that if $e_1^i, e_2^i, \dots, e_{d_i}^i$ denote bases of $\mathcal{H}_i$, then the products

$$\left\{ e_{l_1}^1 \otimes \dots \otimes e_{l_n}^n \mid (l_1, \dots, l_n) \in \bigtimes_{j=1}^n \mathbb{N}_{d_j} \right\} \quad (7.3)$$

form a basis of $\mathcal{H}$.

Fact 7.2 Phonons, photons and electrons are pairwise distinguishable from another.

¹We are using the term "object", in order to represent a "wave" or a "particle" object.

7.2. The Many-Body-Formalism for Indistinguishable Particles (Second Quantization, Algebraic Formalism)

7.2.1. Algebras in Many Particle Physics

Definition 7.3 Let $\mathbb{K}$ be a field and $\mathcal{A}$ be a vector space over $\mathbb{K}$ equipped with an additional binary operation

$$\begin{aligned} \odot : \mathcal{A} \times \mathcal{A} &\rightarrow \mathcal{A} \\ (x, x') &\mapsto x \odot x'. \end{aligned} \tag{7.4}$$

Then $\mathcal{A}$ is an *associative algebra over $\mathbb{K}$* , if for all $x, y, z \in \mathcal{A}$ and $a, b \in \mathbb{K}$ the following identities hold

1. Right distributivity: $(x + y) \odot z = x \odot z + y \odot z$,
2. Left distributivity: $z \odot (x + y) = z \odot x + z \odot y$,
3. Compatibility with scalars: $(ax) \odot (by) = (ab)(x \odot y)$,
4. Associativity²: $(x \odot y) \odot z = x \odot (y \odot z)$.

Remark 7.3.1 For $x, y \in \mathcal{A}$ one calls $x \odot y$ the product of x and y and the binary operation is oftentimes referred to as *multiplication*. An associative algebra over a field $\mathbb{K}$ is sometimes called a *associative $\mathbb{K}$ -algebra*, and $\mathbb{K}$ is called the *base-field*.

Remark 7.3.2 The algebra $\mathcal{A}$ is called *unitary*, if there exists an element $1 \in \mathcal{A}$ such that

$$\forall x \in \mathcal{A} : 1 \odot x = x \odot 1 = x. \tag{7.5}$$

Remark 7.3.3 When we construct the Fock-space in Definition 7.6, the product will be denoted by the so-called "zeta-product" $\odot \equiv \zeta$. This will stress the interpretation of objects in the Fock-space as wavefunctions with coordinate-representations. In the constructions of the so-called " ζ -algebra" and the "polynomial algebra" in Definitions 7.15, 16.1 we will use the usual product symbol.

Definition 7.4 Let $\mathcal{A}$ be a vector-space over the field $\mathbb{K}$ with a basis. We define³ the *associative, unitary algebra generated by $\mathbb{K} \oplus \mathcal{A}$* , as all formal polynomial expressions that one can build with the elements (or, equivalently, the basis elements) of $\mathcal{A}$, given coefficients from the field $\mathbb{K}$ and obeying the relations in Definition 7.3.

The initial situation (which one encounters quite naturally for particle-like objects in physics) is as follows.

²This requirement is not unique in the literature. There are non-associative algebras. We will however not be concerned with them.

³We don't give this object a symbol, because we will make use of this object three times. Depending on the situation it will be referenced with a different symbol. In order to avoid confusion, we leave out a symbol in this definition.

Notation 7.5 Let $\mathcal{H} = \text{span}_{\mathbb{C}} \{|1\rangle, |2\rangle, \dots, |d_{\mathcal{H}}\rangle\}$ be a single-particle Hilbert-space with (not necessarily, but oftentimes orthonormal) basis, corresponding to wavefunctions

$$\langle x | i \rangle \equiv \psi_i(x). \quad (7.6)$$

Definition 7.6 We define the *Fock-space* $\mathcal{F}_{\zeta} \equiv \mathcal{F}_{\zeta}(\mathcal{H})$ of the complex vector space $\mathcal{H}$ as the associative, unitary algebra generated by $\mathbb{C} \oplus \mathcal{H}$ with binary operation $\odot_{\zeta}$ which we call ζ -product and relations

$$\forall |\psi\rangle, |\psi'\rangle \in \mathcal{H} : \quad |\psi\rangle \odot_{\zeta} |\psi'\rangle - \zeta |\psi'\rangle \odot_{\zeta} |\psi\rangle = 0. \quad (7.7)$$

Remark 7.6.1 One distinguishes two cases:

- For $\zeta = -1$, the Fock-space is called the *exterior algebra* and it is commonly denoted with $\wedge(\mathcal{H})$.
- For $\zeta = +1$, the Fock-space is called the *symmetric algebra* and commonly denoted with $S(\mathcal{H})$.

Remark 7.6.2 We will occasioanlly denote the vectors and the algebra-product in a different way (without the Bra-Ket-Notation, or with adapted Bra-Ket-Notation) as

$$|\psi\rangle \odot_{\zeta} |\psi'\rangle \equiv |\psi \odot_{\zeta} \psi'\rangle = \psi \odot_{\zeta} \psi'. \quad (7.8)$$

We do this out of notational convenience, for example if double-bra-kets appear otherwise, as in Equations (7.15), (7.30). The meaning of the notation will be clear from the respective context.

Theorem 7.7 The Fock space factorizes into subspaces s.t.

$$\mathcal{F}_{\zeta}(\mathcal{H}) = \mathcal{F}_{\zeta}^0(\mathcal{H}) \oplus \mathcal{F}_{\zeta}^1(\mathcal{H}) \oplus \mathcal{F}_{\zeta}^2(\mathcal{H}) \oplus \dots \equiv \begin{cases} \bigoplus_{n=0}^{d_{\mathcal{H}}} \wedge^n(\mathcal{H}) & \text{for } \zeta = -1 \\ \bigoplus_{n=0}^{\infty} S^n(\mathcal{H}) & \text{for } \zeta = +1 \end{cases} \quad (7.9)$$

with convention $\mathcal{F}_{\zeta}^0(\mathcal{H}) \equiv \mathbb{C}$ and dimensions

$$\dim_{\mathbb{C}} \mathcal{F}_{\zeta}^n(\mathcal{H}) = \begin{cases} \binom{d_{\mathcal{H}}}{n} & \text{for } \zeta = -1 \\ \binom{d_{\mathcal{H}} + n - 1}{n} & \text{for } \zeta = +1 \end{cases} \quad (7.10)$$

Proof 7.7 See Appendix, Proofs, page 412.

Remark 7.7.1 Note further the identity for fermions

$$\dim_{\mathbb{C}} \mathcal{F}_{-1}^n(\mathcal{H}) = \binom{d_{\mathcal{H}}}{n} = \binom{d_{\mathcal{H}}}{d_{\mathcal{H}} - n} = \dim_{\mathbb{C}} \mathcal{F}_{-1}^{d_{\mathcal{H}} - n}(\mathcal{H}) \quad (7.11)$$

Corollary 7.8 One finds the asymptotic behaviour

$$\dim_{\mathbb{C}} \mathcal{F}_{\zeta}^n(\mathcal{H}) \stackrel{1 \ll d_{\mathcal{H}}}{\approx} \frac{1}{n!} (\dim_{\mathbb{C}} \mathcal{H})^n. \quad (7.12)$$

Proof 7.8 Follows by expanding the binomial coefficients, cancelling terms and using the asymptotic behaviour of the factorials.⁴

Lemma 7.9 It follows for the dimension of the Fock-space

$$\dim_{\mathbb{C}} \mathcal{F}_{\zeta}(\mathcal{H}) = \begin{cases} 2^{d_{\mathcal{H}}} & \text{for } \zeta = -1, \\ \infty & \text{for } \zeta = +1. \end{cases} \quad (7.13)$$

Proof 7.9 The bosonic case is trivial. For the dimension of the fermionic Fock-space one finds

$$\dim_{\mathbb{C}} \Lambda(\mathcal{H}) = \sum_{n=0}^{d_{\mathcal{H}}} \binom{d_{\mathcal{H}}}{n} = 2^{d_{\mathcal{H}}}. \quad (7.14)$$

Fact 7.10 By specifying the product of the algebra, we can interpret it's elements as many-particle-wavefunctions. As the single-particle-wavefunctions are already defined by assumption, this procedure happens inductively.

Definition 7.11 Consider two elements $\psi \in \mathcal{F}_{\zeta}^p(\mathcal{H})$ and $\phi \in \mathcal{F}_{\zeta}^q(\mathcal{H})$ of the corresponding degree subspaces in the algebra. We define the *coordinate-representation* of the corresponding wave-function for $p + q$ particles as

$$\begin{aligned} \langle x_1, \dots, x_{p+q} | \psi \circledast \phi \rangle &\equiv (\psi \circledast \phi)(x_1, \dots, x_{p+q}) \\ &:= \frac{1}{p!q!} \sum_{\pi \in S_{p+q}} \text{sign}(\pi)^{1-\theta(\zeta)} \psi(x_{\pi(1)}, \dots, x_{\pi(p)}) \phi(x_{\pi(p+1)}, \dots, x_{\pi(p+q)}). \end{aligned} \quad (7.15)$$

Remark 7.11.1 Notice that, depending on the nature of the single-particle wavefunctions (scalar-wavefunctions that solve a Schrödinger-equation or spinor-wavefunctions that solve a Pauli-equation) the concrete meaning of the "coordinates" $x_{\bullet}$ is being specified from the context.

⁴Cf. [58].

Corollary 7.12 For $\psi_i \in \mathcal{F}_\zeta^k(\mathcal{H})$, $\phi \in \mathcal{F}_\zeta^l(\mathcal{H})$, $\chi \in \mathcal{F}_\zeta^m(\mathcal{H})$ one finds for the ζ -product the following rules for the wavefunctions w.r.t. their coordinate representation:

1. $(\psi_1 + \psi_2) \circledcirc \phi = \psi_1 \circledcirc \phi + \psi_2 \circledcirc \phi$ (linearity),
2. $(\psi \circledcirc \phi) \circledcirc \chi = \psi \circledcirc (\phi \circledcirc \chi)$ (associativity),
3. $\psi \circledcirc \phi = \zeta^{kl} \phi \circledcirc \psi$ (Koszul-sign-rule).

Proof 7.12 See Appendix, Proofs, page 414.

Remark 7.12.1 Notice that the third equation gives the compatibility of the ζ -product with eq. (7.7).

Definition 7.13 On the direct sum $\mathcal{H} \oplus \mathcal{H}^*$ we define the ζ -bilinearform B_ζ as mapping

$$\begin{aligned} B_\zeta : \mathcal{H} \oplus \mathcal{H}^* \times \mathcal{H} \oplus \mathcal{H}^* &\rightarrow \mathbb{C} \\ (|v\rangle + \langle\phi|, |v'\rangle + \langle\phi'|) &\mapsto B_\zeta(|v\rangle + \langle\phi|, |v'\rangle + \langle\phi'|) \equiv \langle\phi|v'\rangle - \zeta\langle\phi'|v\rangle \end{aligned}$$

Lemma 7.14 Notice that this bilinearform is (anti-)symmetric depending on the statistics of the system:

$$\forall u, u' \in \mathcal{H} \oplus \mathcal{H}^* : \quad B_\zeta(u, u') + \zeta B_\zeta(u', u) = 0 \quad (7.16)$$

Proof 7.14 See Appendix, Proofs, page 414.

Definition 7.15 We define the ζ -algebra $\mathcal{A}_\zeta(\mathcal{H} \oplus \mathcal{H}^*, B_\zeta)$ as the associative algebra generated by $\mathbb{C} \oplus \mathcal{H} \oplus \mathcal{H}^*$ with relations

$$\forall u, u' \in \mathcal{H} \oplus \mathcal{H}^* : \quad uu' - \zeta u'u = B_\zeta(u, u') \quad (7.17)$$

Remark 7.15.1 Again, one distinguishes two cases:

- For $\zeta = -1$, the underlying algebra is a so-called *Clifford-algebra*.
- For $\zeta = +1$, the underlying algebra is a so-called *Weyl-algebra*.

Definition 7.16 We define the *statistical commutator*, or ζ -commutator as

$$[A, B]_\zeta \equiv AB - \zeta BA. \quad (7.18)$$

Remark 7.16.1 This definition is a little bit tricky since it is a magnet for confusion and errors. We are following the prefactor-convention from [Altland p. 44, Negele, Orland p. 13], in contrast to the prefactor-convention from [Coleman p. 43, Negele, Orland p. 15]⁵.

Remark 7.16.2 Notice that this definition becomes generalized later on in Definition 8.12.

Lemma 7.17 A useful identity relates the ordinary commutator to statistical commutators

$$[A, BC] = [A, B]_{\zeta} C + \zeta B[A, C]_{\zeta}. \quad (7.19)$$

Proof 7.17 See Appendix, Proofs, page 414.

7.2.2. Representation and Scalar-Product on the Fock-Space

In order to define operators on the Fock-space, we will define a representation

$$\begin{aligned} \mathcal{A}_{\zeta}(\mathcal{H} \oplus \mathcal{H}^*, B_{\zeta}) \times \mathcal{F}_{\zeta} &\rightarrow \mathcal{F}_{\zeta} \\ (x, \xi) &\mapsto x \cdot \xi. \end{aligned} \quad (7.20)$$

We will do so inductively.

Fact 7.18 Since $\mathcal{A}_{\zeta}(\mathcal{H} \oplus \mathcal{H}^*, B_{\zeta})$ is generated by $\mathbb{C} \oplus \mathcal{H} \oplus \mathcal{H}^*$, it suffices to specify the *action / multiplication* in Equation (7.20) for

$$(1) \text{ scalars: } x \equiv k \in \mathbb{C} \subset \mathcal{A}_{\zeta}(\mathcal{H} \oplus \mathcal{H}^*, B_{\zeta}) \quad (7.21)$$

$$(2) \text{ vectors: } x = |\psi\rangle \equiv c_{|\psi\rangle}^+ \in \mathcal{H} \subset \mathcal{A}_{\zeta}(\mathcal{H} \oplus \mathcal{H}^*, B_{\zeta}) \quad (7.22)$$

$$(3) \text{ dual-vectors: } x = \langle\psi| \equiv c_{\langle\psi|} \in \mathcal{H}^* \subset \mathcal{A}_{\zeta}(\mathcal{H} \oplus \mathcal{H}^*, B_{\zeta}) \quad (7.23)$$

whereas *multiplication by scalars* is given by the structure of a complex vector space.

Definition 7.19 *Multiplication by vectors* is defined by the statistical product computed inside the algebra

$$c_{|\psi\rangle}^+ \cdot |\xi\rangle \equiv c_{|\psi\rangle}^+ |\xi\rangle := |\psi\rangle \odot_{\zeta} |\xi\rangle \equiv |\psi \odot_{\zeta} \xi\rangle. \quad (7.24)$$

Remark 7.19.1 The notation $c_{|\psi\rangle}^+$ is used in order to distinguish the algebra element $|\psi\rangle \equiv |\psi\rangle + 0 \in \mathcal{A}_{\zeta}(\mathcal{H} \oplus \mathcal{H}^*, B_{\zeta})$ from its concrete realization as a linear operator $c_{|\psi\rangle}^+$ on $\mathcal{F}_{\zeta}$.

Remark 7.19.2 Note also that this multiplication increases the degree of a state in a statistical sense (in physics language: the number of particles):

⁵Notice that there is an inconsistency within the book from Negele and Orland!

- For $\zeta = +1$ we obtain a mapping $c_{|\psi\rangle}^+ : S^l(\mathcal{H}) \rightarrow S^{l+1}(\mathcal{H})$.
- For $\zeta = -1$ we obtain a mapping $c_{|\psi\rangle}^+ : \Lambda^l(\mathcal{H}) \rightarrow \Lambda^{l+1}(\mathcal{H}) \cup \{0\}$, that either increases the degree of a state, or maps it to zero.

Remark 7.19.3 Notice further, that one immediately finds the linearity

$$c_{\lambda|\psi\rangle + \lambda'|\psi'\rangle}^+ = \lambda c_{|\psi\rangle}^+ + \lambda' c_{|\psi'\rangle}^+. \quad (7.25)$$

for $\lambda, \lambda' \in \mathbb{C}$.

Definition 7.20 *Multiplication by co-vectors* is defined recursively and denoted as

$$\begin{aligned} c_{\langle\phi|} \cdot 1 &:= 0 \\ c_{\langle\phi|} \cdot |\psi\rangle &:= B_\zeta(\langle\phi|, |\psi\rangle) = \langle\phi|\psi\rangle && \text{for } |\psi\rangle \in \mathcal{H} \subset \mathcal{F}_\zeta \\ c_{\langle\phi|} \cdot |\xi \mathbin{\text{\textcircled{\tiny \zeta}}} \eta\rangle &:= (c_{\langle\phi|} \cdot |\xi\rangle) \mathbin{\text{\textcircled{\tiny \zeta}}} |\eta\rangle + \zeta' |\xi\rangle \mathbin{\text{\textcircled{\tiny \zeta}}} (c_{\langle\phi|} \cdot |\eta\rangle) && \text{for } |\xi\rangle, |\eta\rangle \in \mathcal{F}_\zeta \end{aligned} \quad (7.26)$$

whereas $|\xi\rangle \in \mathcal{F}_\zeta^l(\mathcal{H})$ in the last equation.

Remark 7.20.1 Here, again, the notation $c_{\langle\phi|}$ is used in order to distinguish the algebra element $\langle\phi| \equiv 0 + \langle\phi| \in \mathcal{A}_\zeta(\mathcal{H} \oplus \mathcal{H}^*, B_\zeta)$ from its concrete realization as a linear operator $c_{\langle\phi|}$ on $\mathcal{F}_\zeta$.

Remark 7.20.2 Notice that due to linearity, we obtain the action

$$(k + c_{|\psi\rangle}^+ + c_{\langle\phi|}) \cdot |\xi\rangle \equiv k|\xi\rangle + c_{|\psi\rangle}^+ |\xi\rangle + c_{\langle\phi|} |\xi\rangle \quad (7.27)$$

Remark 7.20.3 Notice that the commutation relations in Equation (7.17) imply the relations

$$\begin{aligned} 1_{\mathcal{F}_\zeta} \langle\phi|\psi\rangle &= c_{\langle\phi|} c_{|\psi\rangle}^+ - \zeta c_{|\psi\rangle}^+ c_{\langle\phi|} \\ 0 &= c_{\langle\phi|} c_{\langle\phi'|} - \zeta c_{\langle\phi'|} c_{\langle\phi|} = c_{|\psi\rangle}^+ c_{|\psi'\rangle}^+ - \zeta c_{|\psi'\rangle}^+ c_{|\psi\rangle}^+ \end{aligned}$$

Notation 7.21 Let us fix an orthonormal basis $\mathcal{H} = \text{span}_{\mathbb{C}} \{|1\rangle, |2\rangle, \dots, |d_{\mathcal{H}}\rangle\}$ of the single-particle Hilbert-space with dual basis $\mathcal{H}^* = \text{span}_{\mathbb{C}} \{\langle 1|, \langle 2|, \dots, \langle d_{\mathcal{H}}|\}$, and let us write $\langle i| \equiv |c_{\langle i|} \equiv c_i$ and $|i\rangle \equiv c_{|i}^+ \equiv c_i^+$.

Lemma 7.22 Using the duality-relation $\langle i|j\rangle = \delta_{ij}$ we obtain the relations (7.17) in the form

$$[c_i, c_j^+]_{\zeta} = \delta_{ij}, \quad [c_i, c_j]_{\zeta} = [c_i^+, c_j^+]_{\zeta} = 0 \quad (7.28)$$

Proof 7.22 See Appendix, Proofs, page 414.

Remark 7.22.1 In the next subsection, we will elucidate how c_i and $c_i^\dagger$ will be acting as linear operators on the Fock-space with its canonical Hermitian scalar-product. In that setting, the operator $c_i^+ \equiv c_i^\dagger$ will turn out to be the „Hermitian adjoint“ of c_i .

Theorem 7.23 Equations (7.26), (7.24) define a representation of the algebra $\mathcal{A}_\zeta(\mathcal{H} \oplus \mathcal{H}^*, B_\zeta)$ on the complex vector-space $\mathcal{F}_\zeta$.

Proof 7.23 See Appendix, Proofs, page 414.

Theorem 7.24 One obtains the identities

$$\begin{aligned}
 c_j^\dagger \cdot (c_{i_1}^\dagger c_{i_2}^\dagger \cdots c_{i_l}^\dagger |0\rangle) &= c_j^\dagger c_{i_1}^\dagger c_{i_2}^\dagger \cdots c_{i_l}^\dagger |0\rangle \\
 c_j \cdot (c_{i_1}^\dagger c_{i_2}^\dagger \cdots c_{i_l}^\dagger |0\rangle) &= \delta_{ji_1} c_{i_2}^\dagger c_{i_3}^\dagger \cdots c_{i_l}^\dagger |0\rangle + \zeta \delta_{ji_2} c_{i_1}^\dagger c_{i_3}^\dagger \cdots c_{i_l}^\dagger |0\rangle + \cdots + \zeta^{l-1} \delta_{ji_l} c_{i_1}^\dagger c_{i_2}^\dagger \cdots c_{i_{l-1}}^\dagger |0\rangle \\
 &= \sum_{k=1}^l \zeta^{k+1} \delta_{j,i_k} c_{i_1}^\dagger \cdots c_{i_{k-1}}^\dagger c_{i_{k+1}}^\dagger \cdots c_{i_l}^\dagger |0\rangle \\
 &= \sum_{k=1}^l \zeta^{k+1} \delta_{j,i_k} c_{i_1}^\dagger \cdots \widehat{c_{i_k}^\dagger} \cdots c_{i_l}^\dagger |0\rangle.
 \end{aligned}$$

Proof 7.24 See Appendix, Proofs, page 415.

Definition 7.25 We introduce a mapping on the subspace

$$\langle \cdot, \cdot \rangle_{\mathcal{F}_\zeta^n} : \mathcal{F}_\zeta^n \times \mathcal{F}_\zeta^n \rightarrow \mathbb{C} \quad (7.29)$$

by defining it through permanents ($\zeta = +1$) and determinants ($\zeta = -1$) as

$$\begin{aligned}
 \langle \psi_1 \oslash \cdots \oslash \psi_n, \psi'_1 \oslash \cdots \oslash \psi'_n \rangle_{\mathcal{F}_\zeta(\mathcal{H})} &:= \sum_{\pi \in S_n} \text{sign}(\pi)^{1-\theta(\zeta)} \prod_{j=1}^n \langle \psi_n, \psi'_{\pi(n)} \rangle_{\mathcal{H}} \\
 &= \begin{vmatrix} \langle \psi_1, \psi'_1 \rangle_{\mathcal{H}} & \cdots & \langle \psi_1, \psi'_n \rangle_{\mathcal{H}} \\ \vdots & \ddots & \vdots \\ \langle \psi_n, \psi'_1 \rangle_{\mathcal{H}} & \cdots & \langle \psi_n, \psi'_n \rangle_{\mathcal{H}} \end{vmatrix}_\zeta
 \end{aligned} \quad (7.30)$$

and sesqui-linear continuation.

Corollary 7.26 This mapping is a well-defined hermitian scalar-product on $\mathcal{F}_\zeta^n$, meaning that it is sesqui-linear, positive definite.

Proof 7.26 See Appendix, Proofs, page 540.

Remark 7.26.1 We continue this hermitian scalar-product on $\mathcal{F}_\zeta^n$ in the canonical way on the entire Fock-space $\mathcal{F}_\zeta$, by defining the scalar-product of different particle-number subspaces to vanish. Equipped with a scalar-product, the Fock-space is a Hilbert-space.

7.2.3. Occupation-Number States and Bra-Ket-Isomorphism

Definition 7.27 We introduce the *statistical occupation-numbers* as set

$$\mathbb{N}_\zeta = \begin{cases} \{0, 1\} & \text{for } \zeta = -1, \\ \{0, 1, 2, \dots, \infty\} & \text{for } \zeta = +1. \end{cases} \quad (7.31)$$

Definition 7.28 We define the *occupation-number-states* as elements of the Fock-space

$$\mathcal{B}_\zeta = \left\{ \frac{(c_1^\dagger)^{n_1}}{\sqrt{n_1!}} \dots \frac{(c_{d_H}^\dagger)^{n_{d_H}}}{\sqrt{n_{d_H}!}} |0\rangle =: |n_1, \dots, n_{d_H}\rangle \mid n_1, \dots, n_{d_H} \in \mathbb{N}_\zeta \right\} \subset \mathcal{F}_\zeta. \quad (7.32)$$

with the so-called *vacuum-state* a.k.a *the empty state*

$$|0\rangle \equiv |\text{vac.}\rangle \equiv 1 \in \mathbb{C}. \quad (7.33)$$

Remark 7.28.1 In the *physics-jargon*, one usually says that such a state possesses n_1 particles in the first state, n_2 particles in the second state, etc.

Remark 7.28.2 Notice that this definition comes with different prefactor-conventions in the literature. We are following the convention of [56], in contrast to, for example, the convention of [46].

Recall now that in a complex vector space with sesqui-linear scalar-product there is a canonically defined operation of taking the hermitian adjoint⁶ of an operator, denoted by the symbol $\dagger$.

⁶The hermitian adjoint of an operator A w.r.t. a scalar-product is defined such that the following equation

$$TBD \quad (7.34)$$

is fulfilled.

Corollary 7.29 One has

$$(c_i^+)^{\dagger} = c_i \iff c_i^+ = c_i^{\dagger}. \quad (7.35)$$

Proof 7.29 See Appendix, Proofs, page 544.

Fact 7.30 The Hermitian scalar-product on the single-particle-Hilbert-space $\langle \cdot, \cdot \rangle : \mathcal{H} \times \mathcal{H} \rightarrow \mathbb{C}$ determines a one-to-one-correspondence between the Hilbert-space and the dual-space known as *ket-bra-bijection* a.k.a. *Riesz-isomorphism*

$$\begin{aligned} J_{\mathcal{H}} : \mathcal{H} &\rightarrow \mathcal{H}^* \\ \lambda |\psi\rangle &\mapsto J_{\mathcal{H}}(\lambda\psi) \equiv \lambda^* \langle \psi | \equiv \lambda^* \langle \psi | \bullet \rangle \end{aligned}$$

for $\lambda \in \mathbb{C}$.

We will introduce this one-to-one-correspondence as well for the Fock-space.

Lemma 7.31 It holds $\mathcal{F}_{\zeta}(\mathcal{H})^* \simeq \mathcal{F}_{\zeta}(\mathcal{H}^*)$ as a vector-space.

Proof 7.31 See Appendix, Proofs, page 544.

Definition 7.32 We introduce the *ket-bra-correspondence of the Fock-space* as

$$\begin{aligned} J_{\mathcal{F}} : \mathcal{F}_{\zeta}(\mathcal{H}) &\rightarrow \mathcal{F}_{\zeta}(\mathcal{H}^*) \\ \lambda |\psi_1\rangle \otimes \dots \otimes |\psi_n\rangle &\mapsto \lambda^* J_{\mathcal{H}}(\psi_1) \otimes \dots \otimes J_{\mathcal{H}}(\psi_n) = \lambda^* \langle \psi_1 | \otimes \dots \otimes \langle \psi_n | \end{aligned}$$

with $\lambda \in \mathbb{C}$.

Corollary 7.33 In *physics-notation*, this implies the ket-bra-correspondence

$$\mathcal{F}(\mathcal{H}) \ni (c_1^{\dagger})^{n_1} \dots (c_{d_{\mathcal{H}}}^{\dagger})^{n_{d_{\mathcal{H}}}} |0\rangle \mapsto J_{\mathcal{F}} \left((c_1^{\dagger})^{n_1} \dots (c_{d_{\mathcal{H}}}^{\dagger})^{n_{d_{\mathcal{H}}}} |0\rangle \right) = \langle 0 | c_{d_{\mathcal{H}}}^{n_{d_{\mathcal{H}}}} \dots c_1^{n_1} \in \mathcal{F}(\mathcal{H}^*) \quad (7.36)$$

with $\lambda \in \mathbb{C}$.

Proof 7.33 Recall that, by the Riesz-isomorphism $J_{\mathcal{H}}$ and its multiplicative extension $J_{\mathcal{F}}$, the bra $J_{\mathcal{F}}(\psi) \equiv \langle \psi |$ associated to a state $\psi \in \mathcal{F}_{\zeta}(\mathcal{H})$ acts on kets by pairing them with ψ via the Hermitian scalar-product of Equation (7.30), i.e.

$$(J_{\mathcal{F}}(\psi))(\xi) = \langle \psi, \xi \rangle_{\mathcal{F}_{\zeta}(\mathcal{H})} \quad \text{for all } \xi \in \mathcal{F}_{\zeta}(\mathcal{H}). \quad (7.37)$$

In *physics-notation*, the right-hand-side of the claimed identity, $\langle 0 | c_{d_H}^{n_{d_H}} \cdots c_1^{n_1}$, is by convention precisely the functional

$$\xi \mapsto \langle |0\rangle, c_{d_H}^{n_{d_H}} \cdots c_1^{n_1} \xi \rangle_{\mathcal{F}_{\zeta}(\mathcal{H})}. \quad (7.38)$$

Hence, by Equation (7.37), it suffices to establish, for every $\xi \in \mathcal{F}_{\zeta}(\mathcal{H})$, the identity

$$\langle (c_1^{\dagger})^{n_1} \cdots (c_{d_H}^{\dagger})^{n_{d_H}} |0\rangle, \xi \rangle_{\mathcal{F}_{\zeta}(\mathcal{H})} = \langle |0\rangle, c_{d_H}^{n_{d_H}} \cdots c_1^{n_1} \xi \rangle_{\mathcal{F}_{\zeta}(\mathcal{H})}. \quad (7.39)$$

By Corollary 7.29 one has $(c_i^{\dagger})^{\dagger} = c_i$ for every $i = 1, \dots, d_H$, and hence also, since the dagger is applied repeatedly to one and the same operator,

$$((c_i^+)^{n_i})^{\dagger} = (c_i^{+\dagger})^{n_i} = c_i^{n_i}, \quad n_i \in \mathbb{N}_{\zeta}. \quad (7.40)$$

By the defining property of the Hermitian adjoint (see footnote above), this translates into

$$\langle (c_i^+)^{n_i} x, y \rangle_{\mathcal{F}_{\zeta}(\mathcal{H})} = \langle x, c_i^{n_i} y \rangle_{\mathcal{F}_{\zeta}(\mathcal{H})} \quad \text{for all } x, y \in \mathcal{F}_{\zeta}(\mathcal{H}). \quad (7.41)$$

Now define, for $k = 0, 1, \dots, d_H$, the auxiliary states

$$\Psi_k := (c_{k+1}^+)^{n_{k+1}} \cdots (c_{d_H}^+)^{n_{d_H}} |0\rangle, \quad (7.42)$$

so that $\Psi_0 = (c_1^{\dagger})^{n_1} \cdots (c_{d_H}^{\dagger})^{n_{d_H}} |0\rangle$ is precisely the state occurring on the left-hand-side of Equation (7.39), while $\Psi_{d_H} = |0\rangle$.

We show by induction on k that

$$\langle \Psi_0, \xi \rangle_{\mathcal{F}_{\zeta}(\mathcal{H})} = \langle \Psi_k, c_k^{n_k} c_{k-1}^{n_{k-1}} \cdots c_1^{n_1} \xi \rangle_{\mathcal{F}_{\zeta}(\mathcal{H})} \quad \text{for all } \xi \in \mathcal{F}_{\zeta}(\mathcal{H}), \quad (7.43)$$

where, for $k = 0$, the operator-string $c_k^{n_k} \cdots c_1^{n_1}$ denotes the empty product, i.e. the identity on $\mathcal{F}_{\zeta}(\mathcal{H})$.

For $k = 0$, Equation (7.43) is the trivial identity $\langle \Psi_0, \xi \rangle = \langle \Psi_0, \xi \rangle$. For the induction step $k \rightarrow k+1$, write $\Psi_k = (c_{k+1}^+)^{n_{k+1}} \Psi_{k+1}$ and apply Equation (7.41) with $i = k+1$, $x = \Psi_{k+1}$ and $y = c_k^{n_k} \cdots c_1^{n_1} \xi$:

$$\langle \Psi_k, c_k^{n_k} \cdots c_1^{n_1} \xi \rangle_{\mathcal{F}_{\zeta}(\mathcal{H})} = \langle (c_{k+1}^+)^{n_{k+1}} \Psi_{k+1}, c_k^{n_k} \cdots c_1^{n_1} \xi \rangle_{\mathcal{F}_{\zeta}(\mathcal{H})} = \langle \Psi_{k+1}, c_{k+1}^{n_{k+1}} c_k^{n_k} \cdots c_1^{n_1} \xi \rangle_{\mathcal{F}_{\zeta}(\mathcal{H})}, \quad (7.44)$$

which, combined with the induction hypothesis for k applied to the left-hand-side, is precisely Equation (7.43) for $k+1$. This proves the claim for all $k = 0, \dots, d_H$.

Taking $k = d_H$ in Equation (7.43), and using $\Psi_{d_H} = |0\rangle$, yields exactly Equation (7.39) for every $\xi \in \mathcal{F}_{\zeta}(\mathcal{H})$. By Equation (7.37), this means that $J_{\mathcal{F}}((c_1^{\dagger})^{n_1} \cdots (c_{d_H}^{\dagger})^{n_{d_H}} |0\rangle)$ and $\langle 0 | c_{d_H}^{n_{d_H}} \cdots c_1^{n_1}$ act as the same linear functional on $\mathcal{F}_{\zeta}(\mathcal{H})$, hence are equal.

Theorem 7.34 The states defined in Equation (7.32) are an orthonormal basis $\mathcal{F}_\zeta(\mathcal{H})$

Proof 7.34 See Appendix, Proofs, page 543.

7.2.4. Operators on the Fock-Space

One-Body-Operators

Notation 7.35 Let $L \in \text{End}(\mathcal{H})$ be a linear operator on the single-particle-Hilbert-space $\mathcal{H}$.

Definition 7.36 The second quantized version of L is given as element of the ζ -algebra (which we denote by the same symbol for notational simplicity):

$$\text{End}(\mathcal{H}) \ni L \mapsto L^{2Q} \equiv \sum_{j=1}^{d_{\mathcal{H}}} c_{L|j}^+ c_j \in \mathcal{A}_\zeta(\mathcal{H} \oplus \mathcal{H}^*, B_\zeta) \quad (7.45)$$

Remark 7.36.1 For notational simplicity, one oftentimes simply drops the upper index and writes $L^{2Q} \equiv L$ as it is clear from the context what is meant. The same holds true for other operators in second quantization.

Corollary 7.37 On the one-particle-subspace of the Fock-space, this mapping acts precisely as its first quantized version, i.e. **TBMathematised.**

Proof 7.37 See Appendix, Proofs, page 416.

Remark 7.37.1 This implies, that the second-quantized-version of L is mathematically considered to be a so-called *derivation*⁷.

Lemma 7.38 This is an hermitian operator on the Fock-space.

Proof 7.38 See Appendix, Proofs, page 416.

Lemma 7.39 Switching to standard-physics-notation, we obtain the second-quantized version of the one-body-operator L as

$$\text{End}(\mathcal{H}) \ni L \mapsto L^{2Q} \equiv \sum_{i,j=1}^{d_{\mathcal{H}}} \langle i|L|j \rangle c_i^\dagger c_j \in \mathcal{A}_\zeta(\mathcal{H} \oplus \mathcal{H}^*, B_\zeta). \quad (7.46)$$

⁷ **TBElaborated?**

Proof 7.39 See Appendix, Proofs, page 546.

Definition 7.40 We define the *number-operator* as the second-quantized version of the identity-operator on the single-particle-Hilbert-space, i.e.

$$\text{End}(\mathcal{H}) \ni \mathbb{1} \mapsto \mathbb{1}^{2Q} \equiv \sum_{j=1}^{d_H} c_j^\dagger c_j \equiv N \in \mathcal{A}_\zeta(\mathcal{H} \oplus \mathcal{H}^*, B_\zeta). \quad (7.47)$$

Corollary 7.41 For basis-states as in equation (7.32) we have

$$\hat{N}|n_1, \dots, n_{d_H}\rangle = \sum_{j=1}^{d_H} n_j |n_1, \dots, n_{d_H}\rangle. \quad (7.48)$$

Proof 7.41 TBD

Lemma 7.42 For fermionic or bosonic operators, one finds in both cases the identity

$$[a_j^\dagger a_k, a_l^\dagger a_l] = a_j^\dagger a_l \delta_{k,l} - a_l^\dagger a_k \delta_{l,j} \quad (7.49)$$

Proof 7.42 Cf. Gross, AQM2013, Sheet 5, Exercise 1 a), c). (TBDigitalized)

Two-Body-Operators

Notation 7.43 Let $V \in \text{End}(\mathcal{H} \otimes \mathcal{H})$ be a linear operator on the tensor product of two single-particle Hilbert-spaces.

Definition 7.44 The *second quantized version* of V is given as an element of the ζ -algebra

$$\text{End}(\mathcal{H} \otimes \mathcal{H}) \ni V \mapsto V^{2Q} \equiv \frac{1}{2} \sum_{ijkl} \langle i, j | L | k, l \rangle c_i^\dagger c_j^\dagger c_l c_k \in \mathcal{A}_\zeta(\mathcal{H} \oplus \mathcal{H}^*, B_\zeta). \quad (7.50)$$

Remark 7.44.1 Notice also the peculiarity that the two latest indices in the expression are contra-symmetrically exchanged in order to be a derivation of its first-quantized version.

Corollary 7.45 On the two-particle-subspace the second-quantized-version of V acts precisely as its first-quantized-version. TBMathematised.

Proof 7.45 See Appendix, Proofs, page 547.

Remark 7.45.1 This again implies, that the second-quantized-version of V is mathematically considered to be a so-called derivation.

Lemma 7.46 This is an hermitian operator on the Fock-space.

Proof 7.46 See Appendix, Proofs, page 416.

n -Body-Operators

Definition 7.47 TBD Extension to higher-order-cases, even if they are not so relevant, to give an interpretation for SC.

Theorem 7.48 An Hermitian n -body-operator on $\bigotimes_{j=1}^n \mathcal{H}$ will be also Hermitian in its second quantized form as an operator on $\mathcal{F}_\zeta(\mathcal{H})$.

Proof 7.48 See Appendix, Proofs, page 416.

7.2.5. Useful Formulae

===== | Begin: To Be Updated / Corrected for n -body operators | =====

The template was the clean derivation of the Landauer-formula; Cf. also Zirnbauer p. 67, 68.

Probably this holds for Fermions *and* Bosons... One step needs to be filled.

Corollary 7.49 Let A and B be linear operators on $\mathcal{H}$. Denote by $[A, B] = AB - BA$ their commutator. Then

$$\begin{aligned} [A, B]^{2Q} &= [A^{2Q}, B^{2Q}] \\ e^{A^{2Q}} B^{2Q} e^{-A^{2Q}} &= (e^A B e^{-A})^{2Q}, \end{aligned} \quad (7.51)$$

Proof 7.49 The first identity is easily proven using the anticommutation relations

$$\begin{aligned} [A^{2Q}, B^{2Q}] &= \sum_{k,j,m,n} \langle e_k, A e_j \rangle \langle e_m, B e_n \rangle [a_k^\dagger a_j, a_m^\dagger a_n] \\ &= \sum_{k,j,m,n} \langle e_k, A e_j \rangle \langle e_m, B e_n \rangle (a_k^\dagger a_m \delta_{j,n} - a_m^\dagger a_j \delta_{k,n}) \\ &= \text{TBD} \\ &= \sum_{k,j} \langle e_k, (AB - BA) e_j \rangle a_k^\dagger a_j = [A, B]^{2Q}, \end{aligned} \quad (7.52)$$

while the second one is implied by the first equality, the Baker–Campbell–Hausdorff formula

$$e^{A^{2Q}} B^{2Q} e^{-A^{2Q}} = B^{2Q} + [A^{2Q}, B^{2Q}] + \frac{1}{2} [B^{2Q}, [B^{2Q}, A^{2Q}]] + \dots,$$

and the linearity of the scalar-product.

===== | End: To Be Updated / Corrected for n -body operators | =====

7.3. Examples of Operators on the Fock-Space

7.3.1. One-Body-Operators

Notation 7.50 We consider the Hilbert-space of a crystal $\mathcal{H} = L^2(\mathbb{R}^3 / \sim)$. The most important operator on the single-particle Hilbert-space is a Schrödinger-Kohn-Sham-Bloch-Hamiltonian H , containing kinetic, potential, Hartree- and exchange-potential as

$$H \equiv -\hbar^2 \frac{\Delta}{2m} + V_{\text{ionic}} + V_{\text{Hartree}} + V_{\text{XC}} \quad (7.53)$$

with action on the single-particle Hilbert-space as

$$\langle \mathbf{r} | H | \psi \rangle \equiv -\frac{\hbar^2}{2m} (\Delta \psi)(\mathbf{r}) + (V_{\text{ionic}} + V_{\text{Hartree}} + V_{\text{XC}})(\mathbf{r}) \psi(\mathbf{r}) \quad (7.54)$$

Corollary 7.51 Using the basis in which the Schrödinger-Kohn-Sham-Bloch-Hamiltonian is diagonal, we obtain

$$H^{2Q} \equiv \sum_{\substack{\mathbf{k}, \mathbf{k}' \in Q \\ n, n' \in \{1, 2, 3, \dots\}}} \langle n, \mathbf{k} | H | n', \mathbf{k}' \rangle c_{n, \mathbf{k}}^\dagger c_{n', \mathbf{k}'} \equiv \sum_{\substack{\mathbf{k} \in Q \\ n \in \{1, 2, 3, \dots\}}} \epsilon_n(\mathbf{k}) c_{n, \mathbf{k}}^\dagger c_{n, \mathbf{k}} \quad (7.55)$$

Remark 7.51.1 Later on, we will consider the sub-Hilbert-space that is spanned by the so-called Wannier-states of a selected band-subspace. Then we will also consider individual term-contributions of the Hamiltonian in the Wannier-basis.

7.3.2. Two-Body-Operators

Notation 7.52 We consider the Hilbert-space of a crystal $\mathcal{H} = L^2(\mathbb{R}^3 / \sim)$. The most important operator on the single-particle Hilbert-space is the Coulomb-interaction energy between two particles with charge e

$$V_{\text{Coulomb}}(\psi \otimes \psi')(\mathbf{x}, \mathbf{x}') = \frac{e^2}{4\pi\epsilon_0 |\mathbf{x} - \mathbf{x}'|} \psi(\mathbf{x}) \psi'(\mathbf{x}') \quad (7.56)$$

Fact 7.53 After choosing a basis of $\mathcal{H}$ (and therefore, by extension, of its tensor-product with itself) we obtain the Coulomb-interaction in second quantization as

$$V_{\text{Coulomb}}^{2Q} \equiv \frac{1}{2} \sum_{ijkl \in \mathcal{A}} \langle i, j | V_{\text{Coulomb}} | k, l \rangle c_i^\dagger c_j^\dagger c_l c_k \quad (7.57)$$

with matrix elements

$$\langle i, j | V_{\text{Coulomb}} | k, l \rangle = \iint_{\mathbb{R}^3 / \sim} \overline{\psi_i(\mathbf{x}) \psi_j(\mathbf{x}')} V_{\text{Coulomb}}(\psi_k \otimes \psi_l)(\mathbf{x}, \mathbf{x}') d^3 \mathbf{x} d^3 \mathbf{x}' \quad (7.58)$$

7.3.3. n -Body-Operators

Fact 7.54 There are not so many n -body operators that appear "naturally" in physical systems. Most n -body operators (like the gravitational- or electrostatic- interaction between n -particles) can be traced back to sums of two-body operators. One of the only few "true" n -body operators that the author is aware of, is found after one integrates out the electronic system in a BCS-like theory and finds higher-order terms in the purely bosonic action. For more details on this, see Section 28.2 and more specifically Lemma 28.29.

8. Fundamentals II - Many-Particle Hilbert-Spaces, Time-Dependencies and Examples

8.1. Summary of the Many-Particle Hilbert-Spaces - Pure and Mixed Systems

Notation 8.1 When we speak about "Green-functions" (to be introduced later), we consider one of the following Hilbert-spaces:

- (1) $\mathcal{H} \equiv \mathcal{H}_{m=0} = \text{span}_{\mathbb{C}} \left\{ \prod_{j=1}^d \frac{1}{\sqrt{n_j!}} (c_j^\dagger)^{n_j} | \text{vac.} \rangle \equiv |n_1, n_2, \dots, n_d\rangle \mid n_1, n_2, \dots, n_d \in \mathbb{N}_\zeta \right\}$ (massless particles),
- (2) $\mathcal{H} \equiv \mathcal{H}_{m>0} = \text{span}_{\mathbb{C}} \left\{ \prod_{j=1}^{d_H} \frac{1}{\sqrt{n_j!}} (c_j^\dagger)^{n_j} | \text{vac.} \rangle \equiv |n_1, n_2, \dots, n_{d_H}\rangle \mid n_1, n_2, \dots, n_{d_H} \in \mathbb{N}_\zeta \right\}$ (massive particles),
- (3) $\mathcal{H} = \mathcal{H}_{m>0} \otimes \mathcal{H}_{m=0}$ (massive-massless-mixtures),
- (4) $\mathcal{H} = \mathcal{H}_{m_1>0} \otimes \mathcal{H}_{m_2=0} \otimes \mathcal{H}_{m_3=0}$ (massive-massless-massless-mixtures),
- (5) $\mathcal{H} = \mathcal{H}_{m_1>0} \otimes \mathcal{H}_{m_2>0} \otimes \mathcal{H}_{m_3=0}$ (massive-massive-massless-mixtures),
- $\vdots$

Remark 8.1.1 Notice that in every case except the first two cases, every Hilbert-space is composed of tensor-products of the particle-like or wave-like systems. In those cases, a basis of the total Hilbert-space is given by taking tensor-products of the respective occupation-number basis-elements. In turn, the basis of the total Hilbert-space is again given as an occupation-number basis (with a more tedious enumeration).

Remark 8.1.2 In the first two cases (i.e. in the case of exclusively massless or massive particles in the system), we will be speaking of a *pure system*.

Definition 8.2 Depending on the mass of the underlying particles in a system, we introduce a number η called *nature* as

$$\eta = \begin{cases} 0 & \text{for massless particles} \\ 1 & \text{for massive particles} \end{cases} \quad (8.1)$$

Remark 8.2.1 The table summarizes some examples of pure systems with their associated statistics ζ , nature η .

Particle / System (Example)	Nature η	Statistics ζ
Electron-Gas	1	-1
Phonon- & Photon-Gas	0	+1
Rubidium atoms / Bose-Einstein-Condensate	1	+1
XXZ-Model + Jordan-Wigner-Transformation (???)	0	-1

Definition 8.3 We introduce the so-called *ensemble Hamiltonian* Ξ as

$$\Xi = \begin{cases} H & \text{for } \eta = 0 \text{ (massless particles),} \\ H - \mu N & \text{for } \eta = 1 \text{ (massive particles),} \\ H_{m>0} - \mu N_{m>0} + H_{m=0} + V_{\text{int}} & \text{for massive-massless-mixtures,} \\ \vdots & \end{cases} \quad (8.2)$$

whereas the role of the so-called "chemical potential" μ will become clear when we introduce the different ensembles that one uses.

Remark 8.3.1 Notice that one can rewrite the equation above in the concise form

$$\Xi = \begin{cases} H_\eta - \eta \mu N & \text{for a pure system with nature } \eta, \\ H_{m>0} - \mu N_{m>0} + H_{m=0} + V_{\text{int}} & \text{for massive-massless-mixtures.} \\ \vdots & \end{cases} \quad (8.3)$$

Remark 8.3.2 For other cases, such as massive-massive-mixtures, one has to account for several chemical potentials.

Fact 8.4 The Hamiltonian of the system is usually expressed as a combination of a well-understood reference term, H_0 , and a perturbation term, V , whereas the perturbation may represent a superposition of one-body interactions (such as external potentials or impurities in a crystal) and two-body interactions (such as Coulomb interactions)... Hence we concretely want to cover the cases ...

1. systems with massless particles

$$\begin{aligned} H &= \sum_{i,j=1}^d H_{i,j} c_i^\dagger c_j + \sum_{i,j,k=1}^d \left(H_{i,j,k} c_i^\dagger c_j^\dagger c_k + H_{i,j,k} c_i^\dagger c_j c_k \right) + \dots, \\ &= H_0 + V \end{aligned} \quad (8.4)$$

2. systems with massive particles

$$\begin{aligned} H &= \sum_{i,j=1}^d H_{i,j} c_i^\dagger c_j + \frac{1}{2} \sum_{i,j,k,l=1}^d H_{i,j,kl} c_i^\dagger c_j^\dagger c_l c_k + \dots, \\ &= H_0 + V \end{aligned} \quad (8.5)$$

whereas we emphasize the reversed order in the last two indices,

3. and for the interactions of, for example, massive-massless-systems

$$V_{\text{int}} = \sum_{i,j=1}^{d_{m>0}} \sum_{k=1}^{d_{m=0}} \left(V_{i,j;k}^{\text{int}} c_i^\dagger c_j c_k^\dagger + \tilde{V}_{i,j;k}^{\text{int}} c_i^\dagger c_j c_k \right) + \dots, \quad (8.6)$$

with

$$H_0 \equiv H_{m=0} + H_{m>0}, \quad V \equiv V_{m=0} + V_{m>0} + V_{\text{int}} \quad (8.7)$$

4. here and for the remaining cases, we consider similar Hamiltonians ...

Remark 8.4.1 Notice that for systems with massive particles the Hamiltonians are always particle-number conserving, as they emerge from a second quantization scheme.

8.2. Time-Dependencies in Many-Body Quantum Mechanics

8.2.1. Real Time

Notation 8.5 Let $\mathcal{H}$ a many-particle Hilbert-space of a quantum system with a (possibly spin-dependent) time-independent ensemble Hamiltonian Ξ (in the Schrödinger-picture) and consider the time-evolution in the Schrödinger- / Pauli-equation

$$i\hbar \partial_t |\psi(t)\rangle = \Xi |\psi(t)\rangle.$$

Definition 8.6 For arbitrary times $t_i, t_f \in \mathbb{R}$ we define the *ensemble-time-evolution operator* $U_\Xi(t_f, t_i)$ of the system as

$$U_\Xi(t_f, t_i) = \exp - \frac{i}{\hbar} (t_f - t_i) \Xi.$$

Remark 8.6.1 Notice that one has the identity

$$U_\Xi(t_f, t_i)^\dagger = U_\Xi(t_i, t_f). \quad (8.8)$$

Remark 8.6.2 For notational convenience we will oftentimes drop the index in the time-evolution-operator as $U_\Xi \equiv U$.

Theorem 8.7 The time-evolution operator defined above gives a formal solution to a Schrödinger- / Pauli-equation (with ensemble-Hamiltonian) for a given initial-condition.

Proof 8.7 By taking the derivative, one sees that one indeeds obtains a solution for a given initial condition.

Definition 8.8 Let $X \equiv X_S \equiv X_S(t)$ be a possibly time-dependent operator in the [S]chrödinger-picture, then one defines the corresponding *ensemble-[H]eisenberg-representation* as

$$X_H(t) := U(t, 0)^\dagger X U(t, 0).$$

Remark 8.8.1 If we want to emphasize from which ensemble-Hamiltonian the time-evolution originates from (this is, for example, important in Theorem 15.8) then we might also switch to the notation

$$X_H(t) \equiv X_\Xi(t) \equiv X_{H_\Xi} \quad (8.9)$$

etc.

Theorem 8.9 One immediately concludes the *ensemble-Heisenberg-"equation of motion"*

$$\frac{d}{dt} X_H(t) = -\frac{i}{\hbar} [X, \Xi]_H(t) + \left(\frac{\partial X}{\partial t} \right)_H(t) \quad (8.10)$$

Proof 8.9 See Appendix, Proofs, page 416.

Remark 8.9.1 Most of the operators we consider are creation- and annihilation-operators $c, c^\dagger$, which are time-independent in the Schrödinger-picture. But one oftentimes simply drops the index, even if they are written in the Heisenberg-picture. This doesn't lead to any confusion since they acquire a time-argument then $c_H(t) \equiv c(t)$, etc.

Definition 8.10 We consider a monomial in fermionic ($\zeta_\bullet \equiv -1$) and / or bosonic ($\zeta_\bullet \equiv +1$) creation and annihilation-operators

$$X \equiv c_{\alpha_1}^\dagger \cdots c_{\alpha_n}^\dagger c_{\alpha'_1} \cdots c_{\alpha'_m}, \quad (8.11)$$

with statistics $\zeta_{\alpha_1}, \dots, \zeta_{\alpha_n}, \zeta_{\alpha'_1}, \dots, \zeta_{\alpha'_m} \in \{-1, +1\}$ then we define the *statistics of the operator X* as a sign

$$S(X) := \prod_{i=1}^n \zeta_{\alpha_i} \cdot \prod_{j=1}^m \zeta_{\alpha'_j} \in \{-1, +1\}. \quad (8.12)$$

Remark 8.10.1 Analogous one defines the *statistics of a polynomial expression in creation- and annihilation-operators*, if every monomial has the same statistics. For a polynomial expression in creation- and annihilation-operators with monomials that differ (individually) in their statistics, there is no reasonable definition of the statistics.

Remark 8.10.2 Notice that one finds for creation- and annihilation-operators $S(c_\bullet) = S(c_\bullet^\dagger) = \zeta$.

Definition 8.11 For a pair of operators X, X' , we define the *pair-statistics* $S \equiv S(X, X') \in \{-1, +1\}$ as

$$S(X, X') \equiv S_{X, X'} := \begin{cases} -1 & \text{if } S(X) = S(X') = -1, \\ +1 & \text{otherwise.} \end{cases} \quad (8.13)$$

Remark 8.11.1 Notice that the only case, in which the pair-statistics is equal to minus one, is if X and X' are (sums of) odd monomial expressions in fermionic operators.

Remark 8.11.2 Note further that $S(c_\bullet, c_\bullet^\dagger) = \zeta$.

Definition 8.12 For a pair of operators X, X' , we define the *pair-commutator* as

$$[X, X']_S \equiv XX' - S_{X, X'} X'X. \quad (8.14)$$

Definition 8.13 We consider a pair of operators X, X' . Then we define the *time-ordered-pair-product* of their ensemble-Heisenberg-representation as

$$T_t [X_H(t) X'_H(t')] \equiv \theta(t - t') X_H(t) X'_H(t') + S_{X, X'} \theta(t' - t) X'_H(t') X_H(t) \quad (8.15)$$

Remark 8.13.1 The definition of the pair-commutator and time-ordered-pair-product seem suspicious at first, but they will be integral in the formulation of Green-functions.

Remark 8.13.2 For X, X' being creation and annihilation-operators of the same statistics one has $S_{X, X'} \equiv \zeta$ and it follows

$$T_t c_\alpha(t) c_{\alpha'}^\dagger(t') \equiv \theta(t - t') c_\alpha(t) c_{\alpha'}^\dagger(t') + \zeta \theta(t' - t) c_{\alpha'}^\dagger(t') c_\alpha(t). \quad (8.16)$$

Remark 8.13.3 Analogously one also introduces an *anti-time-ordering-operator* $\bar{T}_t$; For creation and annihilation-operators it follows

$$\bar{T}_t [X_H(t) X'_H(t')] \equiv \theta(t' - t) X_H(t) X'_H(t') + S_{X, X'} \theta(t - t') X'_H(t') X_H(t). \quad (8.17)$$

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Definition 8.14 We consider a family of statistical operators $\{X_\alpha\}_{\alpha \in \mathcal{A}}$ with statistics $\{S_\alpha \equiv S(X_\alpha)\}_{\alpha \in \mathcal{A}}$. Then we define the *time-ordered-product* of their ensemble-Heisenberg-representation as

$$T_t \left[X_{\alpha_1}(t_1) X_{\alpha_2}(t_2) \cdots X_{\alpha_n}(t_n) \right] \equiv \sum_{\pi \in S_n} \chi_\pi \theta_\pi(t_1, t_2, \dots, t_n) X_{\alpha_{\pi(1)}}(t_{\pi(1)}) X_{\alpha_{\pi(2)}}(t_{\pi(2)}) \cdots X_{\alpha_{\pi(n)}}(t_{\pi(n)}) \quad (8.18)$$

with a product of Heaviside-functions

$$\theta_\pi(t_1, t_2, \dots, t_n) = \prod_{j=1}^{n-1} \theta(t_{\pi(j)} - t_{\pi(j+1)}) \quad (8.19)$$

enforcing the time-ordering $t_{\pi(1)} \geq t_{\pi(2)} \geq \dots \geq t_{\pi(n)}$, and *Koszul-sign*

$$\chi_\pi = \sum_{\substack{1 \leq i < j \leq n \\ \pi(i) > \pi(j)}} S_{\alpha_{\pi(i)}} S_{\alpha_{\pi(j)}} \quad (8.20)$$

counting the number of statistical transpositions required to bring the product into the order specified by the permutation π .

Lemma 8.15 If all operators have the same statistics $S = S_\alpha$ for all $\alpha \in \mathcal{A}$, then one obtains the simple formula

$$T_t \left[X_{\alpha_1}(t_1) X_{\alpha_2}(t_2) \cdots X_{\alpha_n}(t_n) \right] \equiv \text{sign}(\pi)^{1-\theta(S)} X_{\alpha_{\pi(1)}}(t_{\pi(1)}) X_{\alpha_{\pi(2)}}(t_{\pi(2)}) \cdots X_{\alpha_{\pi(n)}}(t_{\pi(n)}) \quad (8.21)$$

where

$$t_{\pi(1)} > t_{\pi(2)} > \dots > t_{\pi(n)}. \quad (8.22)$$

Remark 8.15.1 When it cannot lead to any confusion, it is convenient to omit the brackets, with the notation that the time-ordering-operator acts (as a derivative) to everything that is „positioned right to it“

|=====| End: To Be Updated / Corrected |=====|

8.2.2. „Imaginary“ Time

For reasons that are difficult to motivate several chapters in advance, we introduce another kind of time-evolution-operator.

Definition 8.16 For arbitrary values $\tau_i, \tau_f \in (-\hbar\beta, +\hbar\beta)$ we define the *ensemble-imaginary-time-evolution operator* $U_{\Xi}(\tau_f, \tau_i)$ of the system as

$$U_{\Xi}(\tau_f, \tau_i) = \exp - \frac{1}{\hbar} (\tau_f - \tau_i) \Xi.$$

Remark 8.16.1 Notice that the phraseology „imaginary“ time is a little bit misleading: τ defines a real variable. The thing that changed w.r.t. the previous subsection is the evolution-operator, not the nature of the variables.

Remark 8.16.2 Notice further that we have, in contrast to the real-time case

$$U_{\Xi}(\tau_f, \tau_i) \neq U_{\Xi}(\tau_i, \tau_f)^{\dagger}. \quad (8.23)$$

Remark 8.16.3 Just as in the real-time case, we usually drop the index $U_{\Xi} \equiv U$.

Definition 8.17 Let $X \equiv X_S \equiv X_S(\tau)$ be the Wick-rotated version of a possibly time-dependent operator. Then one defines the corresponding *ensemble-Heisenberg-representation in imaginary time* as

$$X_H(\tau) := U(0, \tau) X U(\tau, 0).$$

Theorem 8.18 One immediately concludes the *ensemble-Heisenberg-"equation of motion" in imaginary time*

$$\frac{d}{d\tau} X_H(\tau) = -\frac{1}{\hbar} [X, \Xi]_H(\tau) + \left(\frac{\partial X}{\partial \tau} \right)_H(\tau) \quad (8.24)$$

Proof 8.18 See Appendix, Proofs, page 417.

Remark 8.18.1 Accordingly one introduces also the *imaginary-time-ordering operator* T_{τ} , analog to Definition 8.13.

8.3. Examples - Pure Systems

To remain as general as possible (and simultaneously reasonable) we specify neither the statistics nor the nature of the systems in this section.

8.3.1. Non-Interacting Cases

Example: A Quantum-Dot with Nature η and Statistics ζ

Definition 8.19 A *quantum-dot* is a system described by the Hamiltonian

$$H_0 = \epsilon c^\dagger c$$

and Fock-space

$$\mathcal{F}_\zeta \equiv \text{span}_{\mathbb{C}} \left\{ |n\rangle \mid n \in \mathbb{N}_\zeta \right\}.$$

Remark 8.19.1 One finds for the Heisenberg-time-evolution in the grand canonical ensemble as a special case of Lemma 8.22.

Example: A Two-Level-System with Nature η and Statistics ζ

Definition 8.20 A *two-level-system* is a system described by the Hamiltonian

$$H_0 \equiv \begin{pmatrix} c_\uparrow \\ c_\downarrow \end{pmatrix}^\dagger \begin{pmatrix} \epsilon_0 + \epsilon_z & -i\epsilon_y + \epsilon_x \\ +i\epsilon_y + \epsilon_x & \epsilon_0 - \epsilon_z \end{pmatrix} \begin{pmatrix} c_\uparrow \\ c_\downarrow \end{pmatrix}$$

Remark 8.20.1 Notice that a *spin- $\frac{1}{2}$ -particle*, described through the Zeeman-contribution to the Pauli-equation leads (after going over to the second quantized formalism) to a two-level-system.

Remark 8.20.2 Computing the Heisenberg-time-evolution for the creation- and annihilation-operators analogously to Lemma 8.22 requires in this case a basis-change. This is done in Lemma TBReferenced.

Example: A Diagonalized, Non-Interacting Gas with Nature η and Statistics ζ

Definition 8.21 A *diagonalized, non-interacting "gas"*¹ is a system described by the Hamiltonian

$$H_0 = \sum_{\alpha} \epsilon_{\alpha} c_{\alpha}^{\dagger} c_{\alpha}$$

and Fock-space $\mathcal{F}_{\zeta}$ spanned by occupation-number-states as usual

$$\mathcal{F}_{\zeta} = \text{span}_{\mathbb{C}} \left\{ |n_{\alpha_1}, n_{\alpha_2}, \dots, n_{\alpha_{d_H}}\rangle \mid n_{\alpha_1}, n_{\alpha_2}, \dots, n_{\alpha_{d_H}} \in \mathbb{N}_{\zeta} \right\} \quad (8.25)$$

whereas the state with $n_{\alpha_1} = n_{\alpha_2} = \dots = n_{\alpha_{d_H}} = 0$ represents the *vacuum-state* a.k.a *the empty state*.

Lemma 8.22 One finds for the ensemble-Heisenberg-time-evolutions the identities

$$\begin{aligned} c_{\bullet}(t) &= e^{-it(\epsilon_{\bullet} - \eta\mu)/\hbar} c_{\bullet}, & c_{\bullet}^{\dagger}(t) &= e^{it(\epsilon_{\bullet} - \eta\mu)/\hbar} c_{\bullet}^{\dagger} \\ c_{\bullet}(\tau) &= e^{-\tau(\epsilon_{\bullet} - \eta\mu)/\hbar} c_{\bullet}, & c_{\bullet}^{\dagger}(\tau) &= e^{\tau(\epsilon_{\bullet} - \eta\mu)/\hbar} c_{\bullet}^{\dagger} \end{aligned} \quad (8.26)$$

Proof 8.22 See Appendix, Proofs, page 417.

Remark 8.22.1 A fermionic, particle-like example of this Hamiltonian is the *free electron gas in the LDA-approximation* in a crystal

$$H_0 = \sum_{\substack{\mathbf{k} \in \mathcal{Q} \\ \sigma \in \{\downarrow, \uparrow\} \\ n \in \{1, \dots, \infty\}}} \epsilon_{n\sigma}(\mathbf{k}) c_{n\sigma}^{\dagger}(\mathbf{k}) c_{n\sigma}(\mathbf{k}).$$

The states in the Fock-space $\mathcal{F}$ are then written as

$$\mathcal{F} = \text{span}_{\mathbb{C}} \left\{ |n_{\alpha_1}, n_{\alpha_2}, \dots, n_{\alpha_{d_H}}\rangle \mid n_{\alpha_i} \in \{0, 1\}, \alpha_i \equiv (m_i, \sigma_i, \mathbf{k}_i) \in \{1, \dots, \infty\} \times \{\downarrow, \uparrow\} \times \mathcal{Q} \right\} \quad (8.27)$$

Notice that the dimension d_H of the single-particle Hilbert-space is given as

$$d_H = \dim \mathcal{H} = \left| \{1, \dots, \infty\} \times \{\downarrow, \uparrow\} \times \mathcal{Q} \right| = \infty \quad (8.28)$$

Usually one introduces a cutoff for the bands, and one considers crystals that are „large“, but finite size, such that one can work with a finite-dimensional basis.

Remark 8.22.2 Without proof, we remark that ...

¹Note that this unfortunately a poor convention regarding the phraseology. Systems that one describes with this formalism are not necessarily gases in the traditional sense of having a gas-phase (like steam). One treats electrons and phonons in solid materials with this formalism.

- for a crystal with time-reversal-symmetry one obtains $\epsilon_{m,\uparrow}(+\mathbf{k}) = \epsilon_{m,\downarrow}(-\mathbf{k})$, which is also called a *two-fold Kramers-degeneracy of the bands*.
- for a crystal with inversion-symmetry one obtains $\epsilon_{m,\sigma}(+\mathbf{k}) = \epsilon_{m,\sigma}(-\mathbf{k})$.

Remark 8.22.3 In the derivation of the Eliashberg-equations, it is explicitly used that one has a *two-fold Kramers-degeneracy of the bands*, i.e. $\epsilon_m(\mathbf{k}) \equiv \epsilon_{m,\uparrow}(\mathbf{k}) = \epsilon_{m,\downarrow}(\mathbf{k})$, which holds for systems that are time-reversal symmetric and inversion-symmetric [2].

Remark 8.22.4 A bosonic example of this Hamiltonian is the *free phonon gas* in a d_x -dimensional crystal (within the harmonic approximation)

$$H_0 = \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ \nu \in \{1, \dots, rd_x\}}} \hbar \omega_\nu(\mathbf{q}) c_\nu^\dagger(\mathbf{q}) c_\nu(\mathbf{q}) \quad (8.29)$$

The Hilbert-space $\mathcal{H}$ of the system reads

$$\mathcal{H} = \text{span}_{\mathbb{C}} \left\{ |n_{\alpha_1}, n_{\alpha_2}, \dots, n_{\alpha_{d_H}}\rangle \mid n_{\alpha_i} \in \{0, 1, 2, \dots\}, \alpha_i \equiv (\nu_i, \mathbf{q}_i) \in \{1, \dots, rd_x\} \times \mathbf{Q} \right\} \quad (8.30)$$

Remark 8.22.5 Usually one denotes the phononic creation- and annihilation operators with the symbol b instead of c .

Remark 8.22.6 Notice that we have proven in the earlier chapters that the time-reversal-symmetry of the crystal implies for the dispersions $\omega_\nu(+\mathbf{q}) = \omega_\nu(-\mathbf{q})$.

Example: A Non-Interacting Gas with Nature η and Statistics ζ

Definition 8.23 A *quadratic, non-interacting gas* is a system described by the Hamiltonian

$$H_0 = \sum_{\alpha, \beta} H_{\alpha, \beta} c_\alpha^\dagger c_\beta \equiv \mathbf{c}^\dagger \cdot \mathbf{H}_0 \cdot \mathbf{c} \quad (8.31)$$

and Fock-space $\mathcal{F}_\zeta$ spanned by occupation-number-states as usual

$$\mathcal{F}_\zeta = \text{span}_{\mathbb{C}} \left\{ |n_{\alpha_1}, n_{\alpha_2}, \dots, n_{\alpha_{d_H}}\rangle \mid n_{\alpha_1}, n_{\alpha_2}, \dots, n_{\alpha_{d_H}} \in \mathbb{N}_\zeta \right\} \quad (8.32)$$

whereas the state with $n_{\alpha_1} = n_{\alpha_2} = \dots = n_{\alpha_{d_H}} = 0$ represents the vacuum-state.

Remark 8.23.1 Computing the Heisenberg-time-evolution for the creation- and annihilation-operators analogously to Lemma 8.22 requires in this case a basis-change. We will not elaborate on this.

²Cf. Waste-Folder 1 book reference; Cf. also Lemmas 24.19, 24.22 where we used $\omega(+\mathbf{q}) = \omega(-\mathbf{q})$

8.3.2. Interacting Cases

Definition 8.24 The Hamiltonian of a *diagonal gas with quadratic interactions* reads

$$\begin{aligned} H &\equiv H_0 + V \\ &\equiv \sum_{\alpha \in \mathcal{A}} \epsilon_{\alpha} c_{\alpha}^{\dagger} c_{\alpha} + \sum_{\alpha_1, \alpha_2 \in \mathcal{A}} V_{\alpha_1 \alpha_2} c_{\alpha_1}^{\dagger} c_{\alpha_2}. \end{aligned}$$

Remark 8.24.1 A typical example of this model is a free electron gas in the LDA-approximation with non-local potentials (e.g. defects in the crystal).

Definition 8.25 The Hamiltonian of a diagonal gas with cubic interactions reads

$$H \equiv H_0 + V$$

$$\equiv \sum_{\alpha \in \mathcal{A}} \epsilon_{\alpha} c_{\alpha}^{\dagger} c_{\alpha} + \sum_{\substack{n_1, n_2, n_3 \in \mathbb{N}_{\mathcal{C}} \\ \overline{n_1}, \overline{n_2}, \overline{n_3} \in \mathbb{N}_{\mathcal{C}}}} \delta_{n_1+n_2+n_3+\overline{n_1}+\overline{n_2}+\overline{n_3}=3} \sum_{\alpha_1, \alpha_2, \alpha_3 \in \mathcal{A}} V_{\alpha_1 \alpha_2 \alpha_3} \frac{\left(c_{\alpha_1}^{\dagger}\right)^{\overline{n_1}} \left(c_{\alpha_2}^{\dagger}\right)^{\overline{n_2}} \left(c_{\alpha_3}^{\dagger}\right)^{\overline{n_3}}}{\frac{n_1 n_2 n_3}{\overline{n_1} \overline{n_2} \overline{n_3}}} c_{\alpha_3}^{n_3} c_{\alpha_2}^{n_2} c_{\alpha_1}^{n_1}.$$

Remark 8.25.1 Recall from the scheme of second quantization that cubic interactions are relevant only for systems with nature $\eta = 1$, since the Hamiltonian of a system with nature $\eta = 0$ must be even in the creation- and annihilation-operators (i.e. particle number conserving).

Remark 8.25.2 A typical example of this model is a free phonon gas in a crystal with anharmonic interactions, i.e. the *phono3py-model*³

$$= \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ v \in \{1, \dots, r d_{\mathbf{x}}\}}} H_0 + \sum_{\substack{\mathbf{q}, \mathbf{q}', \mathbf{q}'' \in \mathbf{Q} \\ v, v', v'' \in \{1, \dots, r d_{\mathbf{x}}\}}} V \Phi_{vv'v''}(\mathbf{q}, \mathbf{q}', \mathbf{q}'') (b_v(\mathbf{q}) + b_v^{\dagger}(-\mathbf{q})) (b_{v'}(\mathbf{q}') + b_{v'}^{\dagger}(-\mathbf{q}')) (b_{v''}(\mathbf{q}'') + b_{v''}^{\dagger}(-\mathbf{q}'')). \quad (8.33)$$

whereas we relabeled the creation- and annihilation-operators with the "usual" symbol b instead of c .

Remark 8.25.3 For future reference we abbreviate the matrix-elements as

$$\Phi_{vv'v''}(\mathbf{q}, \mathbf{q}', \mathbf{q}'') \equiv \Phi_{\lambda_1 \lambda_2 \lambda_3} \quad (8.34)$$

and emphasize that they obey the symmetries

$$\forall \pi \in S_3 : \Phi_{\lambda_1 \lambda_2 \lambda_3} = \Phi_{\lambda_{\pi(1)} \lambda_{\pi(2)} \lambda_{\pi(3)}} \quad (8.35)$$

and

$$\Phi_{vv'v''}(\mathbf{q}, \mathbf{q}', \mathbf{q}'') = \Phi_{vv'v''}^*(-\mathbf{q}, -\mathbf{q}', -\mathbf{q}'') \quad (8.36)$$

Definition 8.26 The Hamiltonian of a diagonal gas in a crystal with quartic interactions reads

$$H \equiv H_0 + V$$

$$\equiv \sum_{\alpha \in \mathcal{A}} \epsilon_{\alpha} c_{\alpha}^{\dagger} c_{\alpha} + \frac{1}{2} \sum_{\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathcal{A}} V_{\alpha_1 \alpha_2 \alpha_3 \alpha_4} c_{\alpha_1}^{\dagger} c_{\alpha_2}^{\dagger} c_{\alpha_4} c_{\alpha_3}$$

³Cf. phono3py-paper.

with the additional property

$$V_{\alpha_1\alpha_2\alpha_3\alpha_4} = V_{\alpha_2\alpha_1\alpha_4\alpha_3} \quad \text{for all } \alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathcal{A}. \quad (8.37)$$

Remark 8.26.1 A typical example of this model is the "GW-system"⁴ for electrons in a crystal with screened Coulomb-interactions.

Definition 8.27 The Hamiltonian of a non-diagonal gas in a crystal with quartic interactions reads

$$\begin{aligned} H &\equiv H_0 + V \\ &\equiv \sum_{\alpha, \beta \in \mathcal{A}} H_{\alpha, \beta} c_{\alpha}^{\dagger} c_{\beta} + \frac{1}{2} \sum_{\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathcal{A}} V_{\alpha_1\alpha_2\alpha_3\alpha_4} c_{\alpha_1}^{\dagger} c_{\alpha_2}^{\dagger} c_{\alpha_4} c_{\alpha_3} \end{aligned}$$

with the additional property

$$V_{\alpha_1\alpha_2\alpha_3\alpha_4} = V_{\alpha_2\alpha_1\alpha_4\alpha_3} \quad \text{for all } \alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathcal{A}. \quad (8.38)$$

Remark 8.27.1 A typical example of this model is a tight-binding model for electrons in a crystal with Hubbard-interactions.

8.4. Examples - Mixed Systems

8.4.1. Non-Interacting Cases

Definition 8.28 We consider a spin unpolarized, diagonal electron-gas and a diagonal phonon-gas that do not interact with each other

$$\begin{aligned} H &= H_0 \\ &= H_{0,\text{el}} + H_{0,\text{ph}} \\ &= \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} c_{m\mathbf{k}\sigma}^{\dagger} c_{m\mathbf{k}\sigma} + \sum_{\substack{\mathbf{q} \in Q \\ \nu \in \{1, \dots, r d_x\}}} \Omega_{\mathbf{q}, \nu} b_{\mathbf{q}\nu}^{\dagger} b_{\mathbf{q}\nu} \end{aligned} \quad (8.39)$$

with $\epsilon_{m\mathbf{k}\uparrow} = \epsilon_{m\mathbf{k}\downarrow}$.

⁴Cf. TReferenced

8.4.2. Interacting Cases

Definition 8.29 We consider the *EPW-system*, i.e. a spin unpolarized, diagonal electron-gas and a diagonal phonon-gas interacting with each other over a spin-independent-interaction as

$$\begin{aligned}
 H &= H_0 + V \\
 &= \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} c_{m\mathbf{k}\sigma}^\dagger c_{m\mathbf{k}\sigma} + \sum_{\substack{\mathbf{q} \in Q \\ v \in \{1, \dots, rd_x\}}} \hbar \omega_{\mathbf{q},v} b_{\mathbf{q},v}^\dagger b_{\mathbf{q},v} + \sum_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ v \in \{1, \dots, rd_x\} \\ m, n \in \{1, \dots, \infty\}}} \frac{g_{mnv}(\mathbf{k}, \mathbf{q})}{\sqrt{N_p}} c_{\mathbf{k}+\mathbf{q},m\sigma}^\dagger c_{\mathbf{k},n\sigma} (b_{\mathbf{q},v} + b_{-\mathbf{q},v}^\dagger) \quad (8.40)
 \end{aligned}$$

with $\epsilon_{m\mathbf{k}\uparrow} = \epsilon_{m\mathbf{k}\downarrow}$.

Lemma 8.30 The hermiticity of the Hamiltonian implies for the coupling

$$g_{mnv}(\mathbf{k}, \mathbf{q}) = g_{nmv}^*(\mathbf{k} + \mathbf{q}, -\mathbf{q}). \quad (8.41)$$

Proof 8.30 See Appendix, Proofs, page 418.

Remark 8.30.1 Notice that this property implies that $g_{mnv}(\mathbf{k}, 0) = g_{nmv}^*(\mathbf{k}, 0) \in \mathbb{R}$, i.e. the diagonal coupling at zero phonon-momentum is real-valued.

Lemma 8.31 Notice that for inversion-symmetric crystals one has the additional symmetry

$$g_{mnv}(\mathbf{k}, \mathbf{q}) = g_{mnv}^*(-\mathbf{k}, -\mathbf{q}). \quad (8.42)$$

Proof 8.31 ToBeDigitalized from Waste-Folder 1.

Definition 8.32 We consider the *EPy3W-system*, i.e. a spin unpolarized, diagonal electron-gas and an interacting phonon-gas interacting with each other over a spin-independent-interaction as

$$\begin{aligned}
 H &= H_0 + V \\
 &= \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} c_{m\mathbf{k}\sigma}^\dagger c_{m\mathbf{k}\sigma} + \sum_{\substack{\mathbf{q} \in Q \\ v \in \{1, \dots, rd_x\}}} \hbar \omega_{\mathbf{q},v} b_{\mathbf{q},v}^\dagger b_{\mathbf{q},v} + V_{\text{el-ph}} + V_{\text{ph-ph}}
 \end{aligned}$$

with $\epsilon_{m\mathbf{k}\uparrow} = \epsilon_{m\mathbf{k}\downarrow}$ and

$$\begin{aligned}
 V_{\text{el-ph}} &= \sum_{\substack{\mathbf{k}, \mathbf{q} \in \mathbb{Q} \\ \sigma \in \{\uparrow, \downarrow\} \\ \nu \in \{1, \dots, rd_x\} \\ m, n \in \{1, \dots, \infty\}}} \frac{g_{mn\nu}(\mathbf{k}, \mathbf{q})}{\sqrt{N_p}} c_{\mathbf{k}+\mathbf{q}m\sigma}^\dagger c_{\mathbf{k}n\sigma} \left(b_{\mathbf{q}\nu} + b_{-\mathbf{q}\nu}^\dagger \right) \\
 V_{\text{ph-ph}} &= \sum_{\substack{\mathbf{q}, \mathbf{q}', \mathbf{q}'' \in \mathbb{Q} \\ \nu, \nu', \nu'' \in \{1, \dots, rd_x\}}} \Phi_{\nu\nu'\nu''}(\mathbf{q}, \mathbf{q}', \mathbf{q}'') \left(b_{\nu}(\mathbf{q}) + b_{\nu}^\dagger(-\mathbf{q}) \right) \left(b_{\nu'}(\mathbf{q}') + b_{\nu'}^\dagger(-\mathbf{q}') \right) \left(b_{\nu''}(\mathbf{q}'') + b_{\nu''}^\dagger(-\mathbf{q}'') \right).
 \end{aligned} \tag{8.43}$$

Definition 8.33 We consider the *CEPy3W-system*, i.e. a spin unpolarized, diagonal *[C]oulomb-interacting* electron-gas and an interacting phonon-gas interacting with each other over a spin-independent-interaction as

$$\begin{aligned}
 H &= H_0 + V \\
 &= \sum_{\substack{\mathbf{k} \in \mathbb{Q} \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} c_{m\mathbf{k}\sigma}^\dagger c_{m\mathbf{k}\sigma} + \sum_{\substack{\mathbf{q} \in \mathbb{Q} \\ \nu \in \{1, \dots, rd_x\}}} \hbar\omega_{\mathbf{q},\nu} b_{\mathbf{q}\nu}^\dagger b_{\mathbf{q}\nu} + V_{\text{el-el}} + V_{\text{el-ph}} + V_{\text{ph-ph}}
 \end{aligned}$$

with $\epsilon_{m\mathbf{k}\uparrow} = \epsilon_{m\mathbf{k}\downarrow}$ and the precise form of the Coulomb-interaction depends on the level of accuracy that we seek to acquire.

9. Fundamentals III - Quantum Statistical Mechanics - Pure Systems

Quoting from [Negele, Orland, p. 47]:

In quantum statistical mechanics, the thermodynamic properties of a system in equilibrium are specified by the assumption of the equal occupation of accessible states. The physical problem is idealized by considering an appropriate ensemble of identical systems, and the probability of observing a particular state is given by the ratio of the number of systems in the ensemble in that state to the total number of systems in the ensemble. Three ensembles are commonly used. The microcanonical ensemble describes an isolated system, which has fixed energy E and fixed particle number. The probability of observing a state of energy $E' \neq E$ is zero and that of observing a state of energy $E' = E$ is constant, equal to the inverse of the total number of states of energy E . The canonical ensemble describes a system of fixed particle number in equilibrium with a thermal reservoir with which it may exchange energy at a specified temperature. By enumerating the number of states in which the system and the reservoir combined have a fixed total energy, it follows that the probability of observing this system alone with energy E is proportional to $e^{-E/k_B T} \equiv e^{-\beta E}$ where k_B is Boltzmann's constant. The average energy of the system is fixed and is, controlled by the temperature T . The grand canonical ensemble describes a system in equilibrium with a particle reservoir, with which it may change particles, as well as a thermal reservoir. Enumeration of the states in which the system and reservoir have fixed total energy and particle number leads to the result that the probability of observing the system alone with energy E and particle number N is proportional to $e^{-\beta(E-\mu N)}$. In addition to the average energy of the system, the average particle number is now fixed and is controlled by the chemical potential. The thermodynamic limit for a physical system of N particles contained in a volume V is defined by taking the limit as N and V go to infinity such that the ratio $\rho = N/V$ remains constant. An essential result of thermodynamics is that the three ensembles are equivalent for describing the macroscopic properties of most systems in the thermodynamic limit. One is then free to select the ensemble on the basis of formal convenience, and as in the case of classical statistical mechanics, the grand canonical ensemble offers the greatest formal simplicity.

9.1. Linear Mappings

In this section, we will carefully explain how expressions of the type "ket-bra" are to be understood.

Definition 9.1 Let V, W denote vector-spaces over a field $\mathbb{K}$. Let us write further $\text{Hom}(V, W)$ for the $\mathbb{K}$ -vector space of linear mappings from V to W , and V^* for the dual vector space of V .

Theorem 9.2 The mapping

$$\begin{aligned} I : W \otimes V^* &\rightarrow \text{Hom}(V, W) \\ w \otimes f &\mapsto I(w \otimes f) \end{aligned}$$

with

$$\forall v \in V : I(w \otimes f)(v) := f(v)w \quad (9.1)$$

and linear extension over the remaining tensor-product, is an isomorphism.

Proof 9.2 The linearity is given by construction:

$$I(w_1 \otimes f_1 + w_2 \otimes f_2) = I(w_1 \otimes f_1) + I(w_2 \otimes f_2) \quad (9.2)$$

For the linearity w.r.t. $\mathbb{K}$ -multiplication one finds for $\lambda \in \mathbb{K}$

$$I((\lambda w) \otimes f)(v) = f(v) \cdot \lambda \cdot w = I(w \otimes (\lambda f))(v) = (\lambda \cdot f)(v) \cdot w = \lambda \cdot f(v) \cdot w \quad (9.3)$$

For the injectivity, let us assume that $I(w \otimes f) = 0$, meaning that

$$\forall v \in V : I(w \otimes f)(v) = f(v)w = 0. \quad (9.4)$$

There is two solutions for this: Either $\forall v \in V : f(v) = 0$ or $w = 0$. Since f is a linear map, both cases imply that $w \otimes f = 0$, that only the zero gets mapped to zero.

For the surjectivity, we take $A \in \text{Hom}(V, W)$ and prove that there is a decomposition, such that

$$\forall v \in V : Av = I\left(\sum_i w_i \otimes f_i\right)(v) \quad (9.5)$$

For this, we choose a basis $e_1, \dots, e_n$ of V with dual basis $e^1, \dots, e^n$ of V^* , such that $e^i(e_j) = \delta_{i,j}$. Next, we decompose v w.r.t. this basis, i.e.

$$v = \sum_{i=1}^n e^i(v)e_i \quad \left(\equiv \sum_{i=1}^n v_i e_i \right) \quad (9.6)$$

This yields, using the linearity of A ,

$$Av = \sum_{i=1}^n e^i(v) Ae_i = \dots \quad (9.7)$$

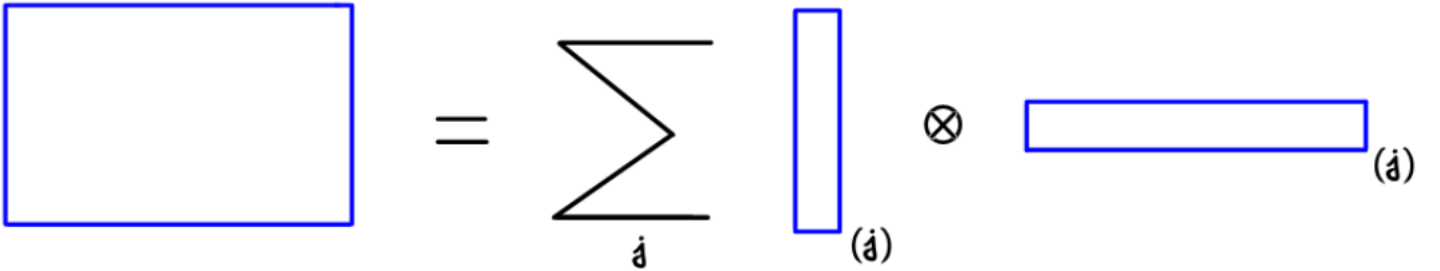
Since $Ae_i \in W$, and $e^i \in V^*$ we can rewrite this expression as

$$\dots = \sum_{i=1}^n I(Ae_i \otimes e^i)(v) = I\left(\sum_{i=1}^n Ae_i \otimes e^i\right)(v) \quad (9.8)$$

proving the surjectivity.

Remark 9.2.1 Since, after referring to a basis, vectors are usually interpreted as standing vectors, and dual-vectors are usually interpreted as lying-vectors, one sees that each matrix of size $m \times n$ can be decomposed uniquely (up to an ambiguity due to scalar factors moving in the tensor product) as a sum of terms each of which is the tensor product (sometimes called a 'dyadic' product in physics textbooks) of an m -component column vector with an n -component row vector.

Remark 9.2.2 The figure elucidates the isomorphism.



Notation 9.3 Switching to the physics-conventions, we realize that in the case of $V = W \equiv \mathcal{H}$ for some (single- or many-particle) Hilbert-space, $V^* \equiv \mathcal{H}^*$ its dual space, and $\text{End}(\mathcal{H})$ the set of all endomorphisms on the Hilbert-space, one can, for arbitrary $\psi, \chi \in \mathcal{H}$, interpret expressions of the type

$$|\psi\rangle\langle\chi| \quad (9.9)$$

as linear mappings, and by referring to a basis, their underlying matrix can be determined from a dyadic-product.

9.2. Density Matrices

Motivation: The need for a new Technique

Consider a Hilbert-space $\mathcal{H}$ with scalar-product $\langle \cdot, \cdot \rangle$. So far, we have only considered state-vectors in this Hilbert-space, so called *pure states*. But with this idea, one can not tackle all physically relevant situations:

Let us assume that $\{|\psi_i\rangle\}_{i=1,\dots,N} \subset \mathcal{H}$ is an arbitrary series of pure states, and let us assume a situation, in which a pure state $|\psi_i\rangle$ is created with probability $q_i \in [0, 1]$, s.t. $\sum_i q_i = 1$. Then, common-sense tells us, that we have for the expectation-value of an observable A the following equation

$$\langle A \rangle = \sum_i q_i \langle \psi_i | A | \psi_i \rangle. \quad (9.10)$$

Now let's rewrite the r.h.s.:

$$\langle A \rangle = \sum_i q_i \langle \psi_i | A | \psi_i \rangle = \sum_i q_i \text{Tr} (|\psi_i\rangle \langle \psi_i| A) = \text{Tr} \left(\sum_i q_i |\psi_i\rangle \langle \psi_i| A \right) =: \text{Tr} (\rho A) \quad (9.11)$$

Remark 9.3.1 In this example it immediately follows that $\text{Tr} \rho = 1$.

Remark 9.3.2 One calls such an operator ρ , that describes the systems state, a *density(-operator)*. We will introduce them now more carefully.

The Formalism of Density Matrices

Quantum mechanics distinguishes between two types of states that physical systems can be in: "pure" and "mixed states". In this section we will introduce both.

Definition 9.4 A **pure state** is a state that can be described by a ket vector $|\psi\rangle$ in a Hilbert-space $\mathcal{H}$.

Remark 9.4.1 Notice that this Hilbert-space can be either a single-particle Hilbert-space or a many-particle Hilbert-space.

Definition 9.5 We assume that the system is with probabilities

$$p_i \in [0, 1) \quad \text{and} \quad \sum_{i=1}^n p_i = 1 \quad (9.12)$$

in orthonormalized pure states

$$|\psi_i\rangle \quad \text{with} \quad \langle \psi_i | \psi_j \rangle = \delta_{ij}. \quad (9.13)$$

The system is then described by the *density matrix*, or *density* for short,

$$\begin{aligned}\rho : \mathcal{H} &\rightarrow \mathcal{H} \\ |\chi\rangle &\mapsto \rho|\chi\rangle\end{aligned}$$

which is defined as a sum of *projectors*

$$\hat{\rho} = \sum_{i=1}^n p_i |\psi_i\rangle\langle\psi_i|. \quad (9.14)$$

Remark 9.5.1 For $n = 1$, the density matrix still represents a pure state, for $n > 1$, we shall refer to it as describing a *mixed state*.

Remark 9.5.2 One can also introduce densities for states that are not orthonormalized. We will however not do so, since we are not interested in these cases.

Lemma 9.6 We have the following properties for the density matrix $\hat{\rho}$:

1. $\hat{\rho}$ is hermitian, i.e. $\hat{\rho} = \hat{\rho}^\dagger$.
2. $\hat{\rho}$ has the eigenvalues p_i with eigenstates $|\psi_i\rangle$.
3. $\hat{\rho}$ is positive, i.e. $\langle\chi|\hat{\rho}|\chi\rangle \geq 0$ for all $|\chi\rangle$.
4. $\hat{\rho}$ is *normalized*, i.e. $\text{Tr}(\hat{\rho}) = 1$.

Proof 9.6 TBDigitalized from Nolting p. 106 f.

Theorem 9.7 For time-dependent states $|\psi_i\rangle \equiv |\psi_i(t)\rangle$ that obey the Schrödinger-equation

$$i\hbar \frac{d}{dt} |\psi_i(t)\rangle = \hat{H} |\psi_i(t)\rangle \quad (9.15)$$

we find that $\hat{\rho} \equiv |\psi_i\rangle\langle\psi_i|$ obeys the von Neumann-equation

$$\frac{d}{dt} \hat{\rho} = -\frac{i}{\hbar} [\hat{H}, \hat{\rho}]. \quad (9.16)$$

Proof 9.7 TBDigitalized from Nolting p. 108

Remark 9.7.1 We call a solution a *stationary state* if $\hat{\rho}$ is time-independent.

Definition 9.8 We define the expectation value of an operator $\hat{A}$ in the state $\hat{\rho}$ as

$$\langle \hat{A} \rangle := \text{Tr}(\hat{\rho} \hat{A}). \quad (9.17)$$

Lemma 9.9 It follows that

$$\langle \hat{A} \rangle^* = \langle \hat{A}^\dagger \rangle \quad (9.18)$$

Proof 9.9 Follows from the hermiticity of the density matrix.

Theorem 9.10 The expectation value of an operator $\hat{A}$ in the state $\hat{\rho}$ is given by

$$\langle \hat{A} \rangle = \sum_{i=1}^n p_i \langle \psi_i | \hat{A} | \psi_i \rangle. \quad (9.19)$$

Proof 9.10 TBDigitalized from Nolting p. 109

Remark 9.10.1 Notice that this expression is completely intuitive, given the situation from Definition 9.5.

Definition 9.11 Given a state ρ , we define the *entropy* as

$$S(\hat{\rho}) = -k_B \text{Tr}(\hat{\rho} \ln \hat{\rho}). \quad (9.20)$$

Remark 9.11.1 Notice that in the template¹ it is also proven that this identity holds for all ensembles.

The subsequent sections will become a bit messy with notation, so we believe that it is appropriate to end this section with the following quote (again).

"Zudem ist es ein Irrtum zu glauben, dass die Strenge in der Beweisführung die Feindin der Einfachheit wäre."

David Hilbert in a lecture given at the International Congress of Mathematicians in Paris 1900

¹Cf. TBDigitalized from Nolting p. 151.

9.3. The Microcanonical Ensemble

Fact 9.12 For macroscopically large systems, it is impossible to determine the exact state of the system by solving the time-dependent Schrödinger-equation for some given initial value. In statistical physics, it is then common to "guess" / postulate the state of the system, by considering a more or less reasonable density matrix $\hat{\rho}$.

Definition 9.13 We define the *natural variables of the microcanonical ensemble* as the triplet $(E, V, N) \in (-\infty, +\infty) \times (0, +\infty) \times \mathbb{N}_0$, whereas ...

- $E \in (-\infty, +\infty)$ is the energy (relative to some reference-energy) of the system,
- $V \in (0, +\infty)$ is the volume of the system, and
- $N \in \mathbb{N}_0 = \{0, 1, 2, \dots, \infty\}$ is the number of particles.

Notation 9.14 For a fixed particle number N and a fixed volume V we enumerate the set of eigen-energies $\{E_1, E_2, \dots\}$ and orthonormal eigenstates $\{|E_1\rangle, |E_2\rangle, \dots\}$ of the Hamilton-operator of the system

$$\hat{H}|E_i\rangle = E_i|E_i\rangle \quad \text{with} \quad \langle E_i|E_j\rangle = \delta_{ij} \quad \text{and} \quad \mathcal{H}_N = \text{span}_{\mathbb{C}}\{|E_1\rangle, |E_2\rangle, \dots\}, \quad (9.21)$$

whereas $\mathcal{H}_N$ represents the underlying N -particle Hilbert-space.

Remark 9.14.1 Notice that the volume influences the eigen-energies, and that the eigen-states above are such that they correspond to N -particle states

$$\langle E_i|\hat{N}|E_i\rangle = N. \quad (9.22)$$

with the particle-number operator $\hat{N}$ from Definition 7.40. Thus, the eigenenergies actually depend on the volume V and the particle number N as

$$E_i \equiv E_i(V, N). \quad (9.23)$$

However, we will suppress this dependence for notational convenience.

Definition 9.15 For an uncertainty Δ , that corresponds to the energy-uncertainty in the measuring-process, we define the subset

$$\mathcal{E} = \{E. \in \{E_1, E_2, \dots\} \mid E. \in [E - \Delta, E]\}. \quad (9.24)$$

The corresponding states $\{|E.\rangle \mid E. \in \mathcal{E}\}$ is called *the set of microstates corresponding to the triplet (E, V, N)* . Further we introduce the *total number of accessible microstates in the microcanonical ensemble* $\Omega_{\Delta}(E, V, N)$ as the cardinal number of $\mathcal{E}$, i.e.

$$\Omega_{\Delta}(E, V, N) = \sum_{E. \in \mathcal{E}} 1 = |\mathcal{E}|. \quad (9.25)$$

Remark 9.15.1 When one explicitly computes the total number of microstates for some simple toy-models, one finds that Ω depends only lightly on Δ . One usually then *postulates* that this is also the case for more complicated systems and drops the index $\Omega_\Delta(E, V, N) \equiv \Omega(E, V, N)$.

Hypothesis 9.16 We *postulate* that the system is in a state that is described by the density matrix

$$\hat{\rho}_{\text{MCE}} = \sum_{m=1,2,\dots} p_m^{\text{MCE}} |E_m\rangle\langle E_m| \quad (9.26)$$

with the probabilities

$$p_m^{\text{MCE}} = \begin{cases} \Omega(E, V, N)^{-1} & \text{for } E_m \in \mathcal{E}. \\ 0 & \text{otherwise.} \end{cases} \quad (9.27)$$

Remark 9.16.1 This assumption is sometimes referred to as the *principle of equal a priori probabilities* or the *fundamental postulate of statistical mechanics*. It is oftentimes formulated in prose as

A closed system in equilibrium is equally probable in each of its accessible microstates.

We further say, that the system is in the *thermal state* if the system is properly described by the density above.

Remark 9.16.2 The fact that this is the correct state to describe the system is proven a posteriori by the fact that it leads to the correct experimental results. Thinking about it longer, it is a rather unreasonable assumption.

Corollary 9.17 The following statements are immediate

- $\hat{\rho}$ is a density matrix in the sense of Definition TBReferenced.
- $\hat{\rho}$ solves the stationary von Neumann equation (TBReferenced).
- $\hat{\rho}$ describes a system with energy E , i.e. $\langle \hat{H} \rangle \approx E$

Proof 9.17 TBDigitalized from Nolting p. 113 The only non-trivial thing is the normalization, i.e. $\text{Tr}(\hat{\rho}) = 1$. For this we ...

Definition 9.18 We define the *Boltzmann-entropy* in the microcanonical ensemble as

$$S_{\text{MCE}}(E, V, N) = k_B \ln \Omega(E, V, N). \quad (9.28)$$

and the *temperature* $T \equiv T(E, V, N)$ as

$$\frac{1}{T(E, V, N)} = \left(\frac{\partial S_{\text{MCE}}}{\partial E} \right)_{V, N} (E, V, N), \quad (9.29)$$

and the *pressure* $P \equiv P(E, V, N)$ as

$$P(E, V, N) = T(E, V, N) \cdot \left(\frac{\partial S_{\text{MCE}}}{\partial V} \right)_{E, N} (E, V, N), \quad (9.30)$$

and the *chemical potential* $\mu \equiv \mu(E, V, N)$ as

$$\mu(E, V, N) = -T(E, V, N) \cdot \left(\frac{\partial S_{\text{MCE}}}{\partial N} \right)_{E, V} (E, V, N). \quad (9.31)$$

Remark 9.18.1 The entropy is one of several so-called *state functions* or *thermodynamic potentials*. The temperature, pressure and chemical potential are then the corresponding *conjugate variables* to the energy, volume and particle number.

Remark 9.18.2 The appearance of the temperature in the definition of the pressure and chemical potential might seem to complicate matters at first glance. This will however turn out to be a very neat definition, as one can attribute physical interpretations to the pressure and the chemical potential, as we will see in the examples and in the following sections.

Remark 9.18.3 The chemical potential has, in physical terms, the meaning of the energy increase when one particle is added (at constant entropy and constant volume). This will become clear in Corollary 9.34.

Remark 9.18.4 Notice that one can rewrite the partial derivatives as

$$\left(\frac{\partial S_{\text{MCE}}}{\partial V} \right)_{E, N} (E, V, N) = \frac{P(E, V, N)}{T(E, V, N)}, \quad \left(\frac{\partial S_{\text{MCE}}}{\partial N} \right)_{E, V} (E, V, N) = -\frac{\mu(E, V, N)}{T(E, V, N)}, \quad (9.32)$$

such that we can write for the differential of the entropy

$$dS_{\text{MCE}} = \frac{1}{T}dE + \frac{P}{T}dV - \frac{\mu}{T}dN \iff dE = TdS_{\text{MCE}} - PdV + \mu dN \quad (9.33)$$

Definition 9.19 Definition of two systems being in thermal equilibrium.

Theorem 9.20 Zustandssumme des idealen Gases: Fließbach, Gleichung (6.20).

Proof 9.20 TBDigitalized.

9.4. The Canonical Ensemble

To introduce the canonical ensemble, we start from a description that uses the microcanonical ensemble.

Notation 9.21 We consider a isolated system $\Sigma_{I \cup II} \equiv \Sigma_I \cup \Sigma_{II}$ that consists of two parts Σ_I and Σ_{II} .

Here are some further assumptions and notations:

1. The Hilbert-space of the total system is given as a tensor-product $\mathcal{H}_{N_{I \cup II}} = \mathcal{H}_{N_I} \otimes \mathcal{H}_{N_{II}}$.
2. The Hamilton-operator of the total system is given by

$$H_{I \cup II} = H_I + H_{II} + H_{I \cap II}, \quad (9.34)$$

where H_I and H_{II} are the Hamilton-operators of Σ_I and Σ_{II} and $H_{I \cap II}$ is the interaction² Hamilton-operator.

3. The total system has an energy, volume and particle-number $E_{I \cup II}$, $V_{I \cup II}$ and $N_{I \cup II}$, respectively.
4. The subsystems Σ_I and Σ_{II} have energies E_1 and E_2 and volumes V_1 and V_2 and particle-numbers N_1 and N_2 , such that

$$\begin{aligned} E_I + E_{II} &= E_{I \cup II}, \\ V_I + V_{II} &= V_{I \cup II}, \\ N_I + N_{II} &= N_{I \cup II}. \end{aligned}$$

5. This total system $\Sigma_{I \cup II}$ is assumed to be properly described in the microcanonical ensemble, such that entropy $S_{I \cup II}$ and a temperature $T_{I \cup II}$ can be defined.
6. We further assume that Σ_I and Σ_{II} are in thermal equilibrium with each other *and* have the same temperature T as the total system $T \equiv T_{I \cup II} \equiv T_I \equiv T_{II}$.
7. For reasons that will become clear later, we call the second subsystem Σ_{II} the *reservoir* or *heat bath*.

The first subsystem $\Sigma_I \equiv \Sigma$ with Hilbert-space $\mathcal{H}_{N_I} \equiv \mathcal{H}_N$ and Hamilton-operator $H_I \equiv H$ is the system of interest, which we ultimately seek to describe. Thus we write for its volume $V_I \equiv V$ and particle number $N_I \equiv N$. The system Σ is considered to be a "very small", yet "macroscopic", part of the "very large" system $\Sigma_{I \cup II}$.

Hypothesis 9.22 We assume that the interaction between Σ_I and Σ_{II} is so weak that the energy exchange between the two systems is negligible $H_{I \cap II} \approx 0$ in the following discussions.

Definition 9.23 We define the *natural variables of the canonical ensemble* as the triplet (T, V, N) , where T is the temperature (of a so-called "reservoir", see below), V is the volume and N is the number of particles.

Notation 9.24 Just as in Notation 9.14: For a fixed particle number N and a fixed volume V we enumerate the set of eigen-energies $\{E_1, E_2, \dots\}$ and orthonormal eigenstates $\{|E_1\rangle, |E_2\rangle, \dots\}$ of the Hamilton-operator H of the system Σ as

$$\hat{H}|E_i\rangle = E_i|E_i\rangle \quad \text{with} \quad \langle E_i|E_j\rangle = \delta_{ij} \quad \text{and} \quad \mathcal{H}_N = \text{span}_{\mathbb{C}}\{|E_1\rangle, |E_2\rangle, \dots\}, \quad (9.35)$$

²Notice that this interaction in the Hamiltonian is necessary for the two systems to reach thermal equilibrium.

whereas $\mathcal{H}_N$ represents the underlying N -particle Hilbert-space.

Hypothesis 9.25 We assume, that the energies E_* of the system are so small, such that the following approximation is valid

$$\text{TBDone.} \quad (9.36)$$

Fact 9.26 The number of states of the total system at energy $E \approx E_{II} \gg E_*$, surpressing the volumes and particle-numbers for notational convenience, is given by

$$\Omega_{I \cup II}(E) = \sum_{E_* \in \{E_1, E_2, \dots\}} \Omega_I(E_*) \Omega_{II}(E - E_*) \quad (9.37)$$

whereas we identify

$\Omega_I(E_*) \Omega_{II}(E - E_*) =$ "number of ways to configure the system + reservoir with the system in microstate $|E_*\rangle$ ".

Thus we obtain in the expansion of the density matrix in the canonical ensemble

$$\hat{\rho}_{\text{CE}} = \sum_{*=1,2,\dots} p_*^{\text{CE}} |E_*\rangle \langle E_*|. \quad (9.38)$$

the ansatz

$$p_*^{\text{CE}} = \frac{\Omega_I(E_*) \Omega_{II}(E - E_*)}{\Omega_{I \cup II}(E)} \propto \Omega_{II}(E - E_*). \quad (9.39)$$

Theorem 9.27 One finds for the density $\rho \equiv \rho(T, V, N)$ in the canonical ensemble

$$\hat{\rho}_{\text{CE}} = \frac{1}{Z} \sum_{*=1,2,\dots} e^{-\beta E_*} |E_*\rangle \langle E_*| \quad (9.40)$$

with

$$Z = \sum_{*=1,2,\dots} e^{-\beta E_*}. \quad (9.41)$$

where $\beta = 1/k_B T$ is the inverse temperature.

Proof 9.27 TBDigitalized.

Remark 9.27.1 Notice that we can rewrite the density as

$$\hat{\rho}_{\text{CE}} = \frac{1}{Z} \sum_{\bullet=1,2,\dots} e^{-\beta E_{\bullet}} |E_{\bullet}\rangle \langle E_{\bullet}| = \frac{1}{Z} e^{-\beta \hat{H}} \quad (9.42)$$

Corollary 9.28 The following statements are immediate

- $\hat{\rho}$ is a density matrix in the sense of Definition TBReferenced.
- $\hat{\rho}$ solves the stationary von Neumann equation (TBReferenced).

Proof 9.28 The normalization is trivial. The fact that it solves the von Neumann equation as well.

Theorem 9.29 The (Helmholtz) free energy is minimal in the canonical ensemble.

Proof 9.29 03 page 28 for minimality of (Helmholtz) free energy in the canonical ensemble.

9.5. The Grand Canonical Ensemble

TBD

See also 03 page 35 for a derivation.

Minimality of the grand canonical potential a.k.a. Landau-potential : Nolting page 157.

Ehrenfest Categorization of Phase Transitions: Some book page 314.

Theorem 9.30 For fixed (T, V) , the function

$$\mu \mapsto N(\mu) \equiv - \left(\frac{\partial \Omega}{\partial \mu} \right) (\mu, T, V) \quad (9.43)$$

is monotonically increasing.

Proof 9.30 See [Salmhofer, page 118].

9.6. Equivalence of Ensembles

9.6.1. Equivalence of the Microcanonical and Canonical Ensemble

TBDigitalized from Nolting p. 74.

9.6.2. Equivalence of the Microcanonical and Canonical Ensemble

TBDigitalized from Nolting p. 88 f. (partly).

9.7. Legendre Transformations

TBCopied from... Was it Schwabl that did a great job on this?

9.8. Thermodynamics

9.8.1. Deductions in the Microcanonical Ensemble

Notation 9.31 We consider a system that is properly described in the microcanonical ensemble by the natural variables (E, V, N) , with entropy-function S_{MCE} as in Definition 9.18.

Remark 9.31.1 For notational convenience, we will in the following suppress the index "MCE" in $S_{\text{MCE}} \equiv S$.

Corollary 9.32 Inverting the entropy S w.r.t. the energy E we can conclude that there exists an energy-function E such that

$$(S, V, N) \mapsto E(S, V, N) \in \mathbb{R} \quad (9.44)$$

with differential as in Equation (9.33) as

$$dE = TdS - PdV + \mu dN \quad (9.45)$$

Proof 9.32 Follows from the theorem of local invertibility.

Hypothesis 9.33 We keep in mind all possible ways of supplying energy to a system. This can occur through ...

1. through contact with other bodies from which heat is transferred,
2. the performance of work,
3. through the addition of material (i.e. increasing the particle number).

The total change in energy therefore takes the form

$$dE = \delta Q + \delta W + \delta E_N \quad (9.46)$$

with ...

1. *heat supply* δQ ,
2. *mechanical work* δW ,
3. *energy increase due to material supply* δE_N .

Remark 9.33.1 It is Equation (9.46), that is usually referred to as the *first law of thermodynamics*. Notice however that Equation (9.45) is equivalent to it.

Remark 9.33.2 One usually says that the energy of a system changes according to Equation (9.46) when passing from one equilibrium state to another, infinitesimally close one.

Corollary 9.34 By comparing these two expressions for dE we find the identifications ...

1. $\delta Q = T dS$ (the change of energy due to heat supply),
2. $\delta W = -P dV$ (the work performed on the system),
3. $\delta E_N = \mu dN$ (the change of the energy when the particle number increases).

Proof 9.34 We interpret the individual terms in Equation (9.45) as in Equation (9.46).

Remark 9.34.1 The relationship

$$\delta Q = T dS \quad (9.47)$$

is called *the second law of thermodynamics for reversible changes*. Later, we shall formulate the second law in its general form.

9.8.2. Transitioning from Microcanonical to Canonical Ensemble

Legendre transformation to the free energy etc...

Definition 9.35 We define the *heat capacity* of a system in the canonical ensemble as derivative

$$C_{V,N}(T) = T \left(\frac{\partial S}{\partial T} \right)_{V,N} (T) \quad (9.48)$$

Lemma 9.36 The following identities are immediate

$$\begin{aligned} \bullet \quad C_{V,N}(T) &= \partial_T \langle H \rangle_{V,N}(T) \\ \bullet \quad C_{V,N}(T) &= -T \left(\partial_T^2 F \right) (N, T, V) \\ \bullet \quad C_{V,N}(T) &= k_B \beta^2 \left(\langle H^2 \rangle_{N,V}(T) - \langle H \rangle_{N,V}^2(T) \right) = k_B \beta^2 \langle (H - \langle H \rangle)^2 \rangle_{N,V,T} \geq 0 \end{aligned} \quad (9.49)$$

Proof 9.36 TBD; Schwabl for the first two, Nolting page 73 for the third one.

Exception for positivity of the heat capacity: Black holes. Cf. Tong Lecture notes, under Equation (1.11).

Notation 9.37 For a crystal³ at hand we introduce ...

- the *total mass of the crystal* m (with unit kg)
- the *molar mass of the crystal* M (with unit kg/mol)
- the *number of moles of the crystal* n (with unit mol)

³Notice that one can also introduce these definitions for materials that are not crystalline. However, in this document, we will not be concerned with those materials.

Fact 9.38 One has for the number of moles the identity

$$n = \frac{m}{M} \quad (9.50)$$

Notation 9.39 We enumerate the masses of the r atoms in the unit cell as $m_1, m_2, \dots, m_r$.

Fact 9.40 The total mass m of the crystal is given as

$$m = N_x N_y N_z \cdot \sum_{j=1}^r m_j. \quad (9.51)$$

Example: A monoatomic crystal, e.g. lead (Pb) in an fcc-lattice with a single atom per unit cell.

- The molar mass M of lead is

$$M_{\text{Pb}} = 207.2 \text{ g/mol} = 207.2 \cdot 10^{-3} \text{ kg/mol} \quad (9.52)$$

- The total mass m of a lead-crystal (with monoatomic basis in an fcc-lattice) is given as the total number of cells multiplied with the number of lead-atoms per cell (i.e. one) multiplied with the mass of a lead atom
i.e.

$$m_{\text{Pb}} = N_x N_y N_z \cdot m_{\text{Pb-atom}} \quad (9.53)$$

with

$$m_{\text{Pb-atom}} = 207.2 \text{ u} = 207.2 \cdot 1.66 \cdot 10^{-27} \text{ kg} \quad (9.54)$$

Combining these two equations yields for the number of moles n in a crystal with mass m the equation

$$n = \frac{m_{\text{Pb}}}{M_{\text{Pb}}} = N_x N_y N_z \cdot \frac{1.66 \cdot 10^{-27}}{10^{-3}} \text{ mol} = N_x N_y N_z \cdot 1.66 \cdot 10^{-24} \text{ mol} \quad (9.55)$$

Definition 9.41 We define the *molar heat capacity* $C_{V,N,\text{molar}}(T)$ as the amount of heat required to raise the temperature of one mole of a substance by one Kelvin, i.e.

$$C_{V,N,\text{molar}}(T) = \frac{C_{V,N}(T)}{n} \quad (9.56)$$

whereas n is the amount of substance in moles.

Remark 9.41.1 Identities analogous to Lemma 9.36 follow immediately.

Theorem 9.42 TBD

See 03 page 41, 42 for a nice summary of potentials and partition functions.

TBD: 02: page 129 (useful formulae).

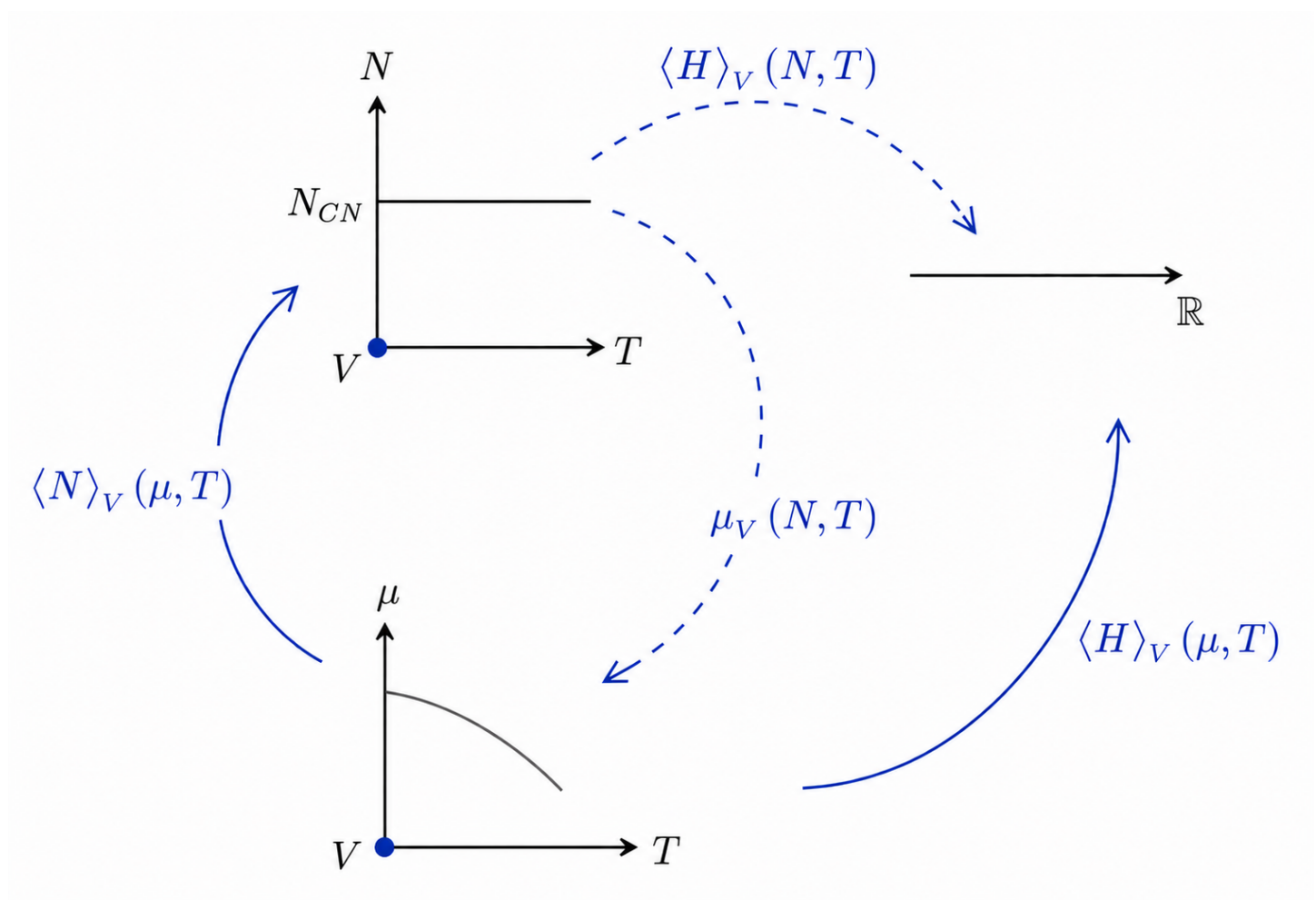
TBD: 03: page 22 (equilibrium is found by maximizing the entropy).

9.8.3. Transitioning from Canonical Ensemble to the Grand Canonical Ensemble

TBElaborated.

Corollary 9.43 Computing the specific heat capacity in the grand canonical ensemble one finds

$$\begin{aligned} C_{N,V}(T) &= \partial_T \langle H \rangle(N, T, V) \\ &= (\partial_T \langle H \rangle)(\mu(T, V), T, V) + (\partial_\mu \langle H \rangle)(\mu(T, V), T, V) \cdot (\partial_T \mu)(N, V, T) \end{aligned} \quad (9.57)$$

 whereas the particle number N is kept constant.

Proof 9.43 For a fixed N_{CN} we introduce the map $(T, V) \mapsto \mu(T, V)$ such that

$$\langle N \rangle(\mu(T, V), T, V) = N_{\text{CN}} \quad (9.58)$$

and we omit the N_{CN} in μ for notational simplicity. Now we recall, that by equivalence of the ensembles, one finds

$$\langle H \rangle(N, T, V) \equiv \langle H \rangle(\mu(T, V), T, V) \quad (9.59)$$

Now the promised equality follows from the chain-rule, i.e. by introducing the map $T \mapsto g(T) \equiv (\mu(T), T)$ etc.

Might be Interesting later:

- Callen - Thermodynamics and an Introduction to Thermostatistics, Wiley Sons
- Texier Roux - Physique Statistique, Dunod
- Diu, Guthmann, Lederer, Roulet - Physique Statistique, Hermann
- Jancovici - Statistical Physics and Thermodynamics, McGraw-Hill
- Landau Lifschitz - Statistical Physics, Pergamon Press
- K. Huang - Statistical Mechanics, Wiley
- Barrat Hansen - Basic Concepts for Simple and Complex Fluids, Cambridge University Press
- Somewhere page 13, 14: Stirlings approximation.
- Somewhere page 19, 20, 22: Entropy and Temperature for two combined systems.
- Somewhere page 27, 28: Chapter 4 Canonical Ensemble; Same book page 35, 36 Chapter 5 Grand canonical ensemble.
- Somewhere pdfpage44: "Ueberlässt man ein abgeschlossenes Vielteilchensystem sich selbst, so streben die makroskopisch messbaren Groeszen gegen zeitlich konstante Werte. Dies ist ein Erfahrungssatz."
- Somewhere page 39: "Die Energie E kann nur mit einer endlichen Genauigkeit δE bestimmt werden (...)"
- Page 157 in the template: minimization of the grand canonical potential.

9.9. Auxiliary Identities

For later use we introduce the following notation

Notation 9.44 We introduce analoga for the [F]ermi- / [B]ose- / [ζ]-functions that take dimensionless arguments as

$$F(x) = \frac{1}{e^x + 1} \quad / \quad B(x) = \frac{1}{e^x - 1} \quad / \quad n^\zeta(x) = \frac{1}{e^x - \zeta} \quad (9.60)$$

with $\zeta \in \{-1, +1\}$ as usual.

Lemma 9.45 One obtains the identities

$$\frac{d}{dx} n^\zeta(x) = -n^\zeta(x) (1 + \zeta n^\zeta(x)) \equiv -\frac{1}{4} \zeta \operatorname{ch}^2(x/2) \equiv \begin{cases} -\frac{1}{4} \operatorname{sech}^2(x/2) & \text{for } \zeta = -1 \\ -\frac{1}{4} \operatorname{csch}^2(x/2) & \text{for } \zeta = +1 \end{cases} \quad (9.61)$$

Proof 9.45 See Appendix, Proofs, page 418.

Notation 9.46 For a temperature $\beta = \frac{1}{k_B T}$ we write for the [F]ermi- / [B]ose- / [ζ]-functions

$$F_{\hbar\beta}(\omega) = \frac{1}{e^{\hbar\beta\omega} + 1} \quad / \quad B_{\hbar\beta}(\omega) = \frac{1}{e^{\hbar\beta\omega} - 1} \quad / \quad n_{\hbar\beta}^\zeta(\omega) = \frac{1}{e^{\hbar\beta\omega} - \zeta} \quad (9.62)$$

with $\zeta \in \{-1, +1\}$ as usual.

Remark 9.46.1 Notice that one has the relation $n_{\hbar\beta}^\zeta(\omega) = n^\zeta(\beta\hbar\omega)$, etc.

For reasons that are difficult to motivate in advance⁴, we will need the following identities in later chapters.

Lemma 9.47 For $i\omega_n \in i\Omega_{-1}$ one has the identities

$$F_{\hbar\beta}(\bullet + i\omega_n) = -B_{\hbar\beta}(\bullet), \quad F_{\hbar\beta}(i\omega_n - \bullet) = B_{\hbar\beta}(\bullet) + 1 \quad (9.63)$$

further one has

$$F_{\hbar\beta}(-\bullet) = 1 - F_{\hbar\beta}(\bullet). \quad (9.64)$$

Proof 9.47 See Appendix, Proofs, page 419.

⁴These identities will be needed in different situations, for example to compute the self-energy or the heat capacity of an interacting electron-phonon-system.

Remark 9.47.1 Notice further that one also has the trivial identities for $\omega_n \in \Omega_{+1}$

$$F_{\hbar\beta}(\bullet \pm i\omega_n) = F_{\hbar\beta}(\bullet) \quad (9.65)$$

Lemma 9.48 One has the derivatives

$$\frac{d}{dT} n_{\beta}^{\zeta}(\hbar\omega) = \frac{\hbar\omega}{k_B T^2} n_{\beta}^{\zeta}(\hbar\omega) \left(1 + \zeta n_{\beta}^{\zeta}(\hbar\omega) \right) = \begin{cases} \frac{\hbar\omega}{4k_B T^2} \operatorname{sech}^2 \left(\frac{\hbar\omega}{2k_B T} \right) & \text{for } \zeta = -1 \\ \frac{\hbar\omega}{4k_B T^2} \operatorname{csch}^2 \left(\frac{\hbar\omega}{2k_B T} \right) & \text{for } \zeta = +1 \end{cases} \quad (9.66)$$

Proof 9.48 See Appendix, Proofs, page 419.

Corollary 9.49 For $z \in \mathbb{C}$ we find the identity

$$\frac{d}{dz} n_{\hbar\beta}^{\zeta}(z) = -\hbar\beta n_{\hbar\beta}^{\zeta}(z) \left(1 + \zeta n_{\hbar\beta}^{\zeta}(z) \right) \equiv -\frac{\hbar\beta}{4} \zeta \operatorname{ch}^2(\hbar\beta z/2) \equiv \begin{cases} -\frac{\hbar\beta}{4} \operatorname{sech}^2(\hbar\beta z/2) & \text{for } \zeta = -1 \\ -\frac{\hbar\beta}{4} \operatorname{csch}^2(\hbar\beta z/2) & \text{for } \zeta = +1 \end{cases} \quad (9.67)$$

Proof 9.49 Follows directly from Lemma 9.45 with the chain-rule. Alternatively, by direct computation, see Appendix, Proofs, page 419.

10. Teaser for Equilibrium Green-Functions

“The Green’s function method is one of the most powerful and versatile formalisms in physics”

The very first sentence in the book from Stefanucci and van Leuween [46].

The ultimate goal of equilibrium Green-function-techniques is to *describe* (in the widest sense of the word) systems as in Notation 8.1 in the canonical ensemble, or in the grand canonical ensemble, or in hybrid ensembles.

10.1. General Setup - Pure Systems

Ingredient I: Many-Particle-Hilbert-Space, States and Operators

Notation 10.1 We consider a system with statistics ζ and nature η , and an underlying many-particle Hilbert-space $\mathcal{H} \equiv \mathcal{H}_\eta$ that is generated by creation- and annihilation-operators

$$\mathcal{H} = \text{span}_{\mathbb{C}} \left\{ \prod_{j=1}^{d_\eta} \frac{1}{\sqrt{n_j!}} \left(c_j^\dagger \right)^{n_j} | \text{vac.} \rangle \equiv | n_1, n_2, \dots, n_{d_\eta} \rangle \mid n_1, n_2, \dots, n_{d_\eta} \in \mathbb{N}_\zeta \right\} \quad (10.1)$$

with a number of states to occupy

$$d_\eta = \begin{cases} d_{m=0} & \text{for } \eta = 0, \\ d_{\mathcal{H}} & \text{for } \eta = 1, \end{cases} \quad (10.2)$$

generated through creation- and annihilation-operators acting on a vacuum. The Hamiltonian of the system is then given by a contribution H_0 that we believe to understand and perturbations V as

$$\begin{aligned} H &= H_0 & + & V \\ &= \sum_{i,j} H_{ij} c_i^\dagger c_j & + & V(\mathbf{c}^\dagger, \mathbf{c}) \end{aligned} \quad (10.3)$$

with creation- and annihilation-operators $\{c_1, c_1^\dagger, c_2, c_2^\dagger, \dots, c_{d_\eta}, c_{d_\eta}^\dagger\}$ obey the statistical commutation-relations

$$[c_i, c_j]_\zeta = [c_i^\dagger, c_j^\dagger]_\zeta = 0, \quad [c_i, c_j^\dagger]_\zeta = \delta_{i,j} \quad (10.4)$$

Remark 10.1.1 Usually one switches the notation to a more *generic case*, and just writes generalized (system-specific) indices $\{c_1, c_1^\dagger, c_2, c_2^\dagger, \dots, c_{d_\eta}, c_{d_\eta}^\dagger\} \equiv \bigcup_{\alpha \in \mathcal{A}} \{c_\alpha, c_\alpha^\dagger\}$ with some index-set $\mathcal{A}$.

Ingredient II: The Natural Ensemble for Pure Systems

For systems with nature $\eta = 0$, we want to use the canonical ensemble and ignore the particle-number dependence, which is (on an enumerational level) equivalent to the grand canonical ensemble with $\mu = 0$. For systems with nature $\eta = 1$, we want to use the grand canonical ensemble with μ dictating the particle number. We combine these two cases into a single notation, by introducing the "natural ensemble".

Notation 10.2 We consider a system with nature $\eta \in \{0, 1\}$ in the grand canonical ensemble, with a volume V , temperature T and chemical potential

$$\begin{cases} \mu = 0 & \text{for } \eta = 0 \\ \mu \stackrel{\text{in general}}{\neq} 0 & \text{for } \eta = 1. \end{cases} \quad (10.5)$$

This gives us $2 + \eta$ natural variables.

Fact 10.3 In the absence of „spontaneous symmetry breaking“¹ expectation-values in the grand canonical ensemble are evaluated w.r.t. the density

$$\rho = \frac{e^{-\beta(H-\eta\mu N)}}{\text{Tr}_{\mathcal{H}} e^{-\beta(H-\eta\mu N)}} \equiv \frac{1}{Z_{\eta}} e^{-\beta(H-\eta\mu N)} \equiv \begin{cases} \frac{1}{Z} e^{-\beta H} & \text{for } \eta = 0 \\ \frac{1}{Z} e^{-\beta(H-\mu N)} & \text{for } \eta = 1. \end{cases}, \quad (10.6)$$

whereas we introduced the *natural partition-function*

$$Z_{\eta} \equiv \begin{cases} Z(T, V) & \text{for } \eta = 0 \\ Z(\mu, T, V) & \text{for } \eta = 1. \end{cases} \quad (10.7)$$

Remark 10.3.1 Recall that the state above represents a solution to the stationary von Neumann equation.

Remark 10.3.2 Although in the case of „spontaneous symmetry breaking“, the evaluation of observables w.r.t. the state above yields incorrect results, one can oftentimes however use the state above, and break the underlying symmetry „by hand“, in order to recover the „correct“ physical (ground) state (or, in the case that we will encounter later, the correct physical „field theory“) again.

Fact 10.4 The thermodynamic potential² in the natural ensemble is given by the so-called *natural energy*

$$\Omega_{\eta} \equiv \begin{cases} \Omega(T, V) = -\frac{\text{Log } Z(T, V)}{\beta} & \text{for } \eta = 0 \\ \Omega(\mu, T, V) = -\frac{\text{Log } Z(\mu, T, V)}{\beta} & \text{for } \eta = 1, \end{cases} \quad (10.8)$$

¹We will introduce this concept in the chapter about superconductivity.

²Recall that the thermodynamic potential is a scalar-quantity which is used to represent the thermodynamic state of a system.

from which one can conclude the usual relations for the entropy, pressure and, in case of $\eta = 1$, the conjugated variable of N as

$$\frac{\partial \Omega_\eta}{\partial T} = -S, \quad \frac{\partial \Omega_\eta}{\partial V} = -P, \quad \text{and if } \eta = 1 : \quad \frac{\partial \Omega_\eta}{\partial \mu} = -N \quad (10.9)$$

whereas both sides of these respective equations depend on the $2 + \eta$ variables, which has been omitted for notational clarity.

Remark 10.4.1 Notice that the thermodynamic potential reduces to the Helmholtz free energy F for $\eta = 0$, and to the Landau-potential Ω for $\eta = 1$.

10.2. General Setup - Mixed Systems

We discuss the case of mixed systems at the example of mixture with massive and massless particles. The generalizations to other mixtures is straight-forward.

Ingredient I: Hilbert-Space, States and Operators

Notation 10.5 We consider a system, which is made of two subsystems, one with Fock-space $\mathcal{H}_{m>0} \equiv \mathcal{F}_\zeta$ with generators $\{c_{\alpha_1}, c_{\alpha_1}^\dagger, c_{\alpha_2}, c_{\alpha_2}^\dagger, \dots, c_{\alpha_{d_{m>0}}}, c_{\alpha_{d_{m>0}}}^\dagger\}$ and statistical operators, and another one with Hilbert-space $\mathcal{H}_{m=0}$ with bosonic generators $\{c_{\beta_1}, c_{\beta_1}^\dagger, c_{\beta_2}, c_{\beta_2}^\dagger, \dots, c_{\beta_{d_{m=0}}}, c_{\beta_{d_{m=0}}}^\dagger\}$. The total Hilbert-space $\mathcal{H}$ of the system is given as tensor-product

$$\mathcal{H} = \mathcal{H}_{m>0} \otimes \mathcal{H}_{m=0}$$

with basis³ and scalar-product as usual. The generic Hamiltonian reads

$$\begin{aligned} H_{\text{tot}} &\equiv H_{m>0} + H_{m=0} + V_{\text{int}} \\ &\equiv H_{m>0} + V_{m>0} + H_{m=0} + V_{m=0} + V_{\text{int}} \\ &\equiv \sum_{i,j} H_{0,ij} c_{\alpha_i}^\dagger c_{\alpha_j} + V_{m>0}(\mathbf{c}_\alpha^\dagger, \mathbf{c}_\alpha) + \sum_{i,j} H_{0,ij} c_{\beta_i}^\dagger b_{\beta_j} + V_{m=0}(\mathbf{c}_\beta^\dagger, \mathbf{c}_\beta) + V_{\text{int}}(\mathbf{c}^\dagger, \mathbf{c}) \end{aligned} \quad (10.10)$$

with interactions as in Fact 8.4.

Ingredient II: The Hybrid Ensemble

Notation 10.6 We further restrict our attention to the case in which the above mentioned system is made up of two sub-systems which have a mutual volume V and temperature T and a chemical potential μ for the massive particles. This gives us the three natural variables (μ, T, V) .

³Cf. Flensburg page 276.

Fact 10.7 In the absence of „spontaneous symmetry breaking“ (to be introduced in a later chapter) expectation-values in the hybrid ensemble are evaluated w.r.t. the *hybrid density*

$$\rho = \frac{e^{-\beta(H_{m>0} - \mu N_{m>0} + H_w + V_{\text{int}})}}{\text{Tr}_{\mathcal{H}} e^{-\beta(H_p - \mu N + H_w + V_{\text{int}})}} \equiv \frac{1}{Z} e^{-\beta(H_p - \mu N + H_w + V_{\text{int}})}, \quad (10.11)$$

whereas we introduced the *mixed partition-function* $Z \equiv Z(N, \mu, T, V)$.

Remark 10.7.1 Recall that the state above represents a solution to the stationary von Neumann equation.

Remark 10.7.2 Although in the case of „spontaneous symmetry breaking“, the evaluation of observables w.r.t. the state above yields incorrect results, one can oftentimes however use the state above, and break the underlying symmetry „by hand“, in order to recover the „correct“ physical state (or, in the case that we will encounter later, the correct physical „field theory“) again.

Remark 10.7.3 Thermodynamic Relations?

10.3. Applications (Teaser)

In [35]:

"The ultimate objective of the quantum theory of many-particle systems is to understand experimentally observable properties of a diverse range of physical systems."

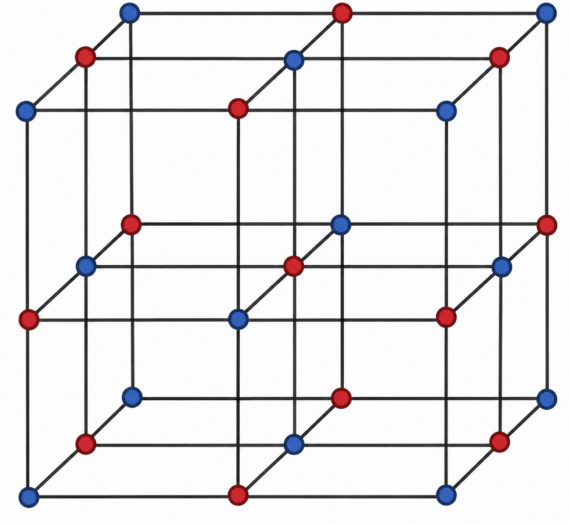
This quote from Negele and Orland is hitting very hard, because most of the theory and formalism presented in this work can be applied to *any crystalline system* (there are more than 200.000 known crystals in the Inorganic Crystal Structure Database⁴). Lying out the tools to study all of these systems is apparently a very powerful formalism.

It is a standard-fact, that in many experimental setups that are in some sense "close" to thermal equilibrium, observable quantities can be expressed as "equilibrium Green-functions" (will be introduced later). We list some examples whereas the meaning of the individual symbols will become clear throughout this document.

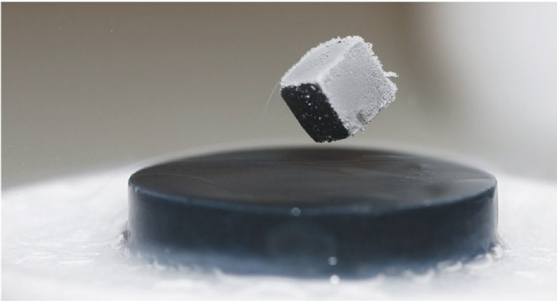
⁴<https://icsd.fiz-karlsruhe.de/>

System ("Cartoon"):

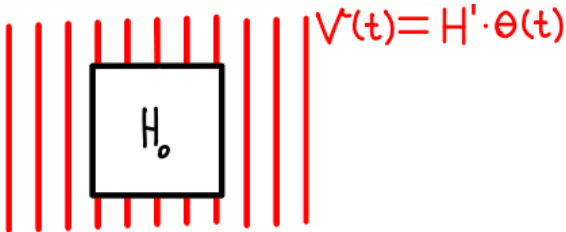
- Any arbitrary Crystal:



- Superconductor⁷:



- An *arbitrary* observable A of a System subjected to a „weak“ Perturbation H' :



Observable(s):

- Electronic Density of States / Band-Structure:

$$\rho_{nk}(\omega) = -\frac{\hbar}{\pi} \text{Im} \lim_{\epsilon \rightarrow 0} G_{n,n}^{\text{ret}}(\omega + i\epsilon)$$

- Magnetization⁵:

$$\langle M \rangle \propto \sum_{\substack{\mathbf{k} \in \text{BZ} \\ n \in \{1,2,\dots\} \\ \omega \in (-\infty, +\infty)}} G_{n,n}^<(\omega) - G_{n,n}^<(\omega)$$

$\uparrow, \uparrow$ $\downarrow, \downarrow$

- Electronic / Phononic / Total Heat Capacity:

$$C_{N,V}(T) = \partial_T \langle H \rangle_{N,T} = \text{"sum of Green-functions"}^6$$

- Electronic Conductivity Tensor:

$$\sigma(\omega) = \text{"Green-function with current operators"}$$

- Thermal Conductivity Tensor:

$$\kappa = \text{"Green-function with energy-current operators"}$$

- Order Parameter⁸ (in Eliashberg-Theory):

$$\Delta_n(k) \propto \hat{\mathcal{G}}_{\text{Gorkov}}^n(k)$$

- Kubo-Formula:

$$\langle A \rangle(t) = \langle A \rangle(t = -\infty) + \int dt' G_{AH'}^{\text{ret}}(t, t') + \mathcal{O}(H')^2$$

⁵Note that for this to be unequal to zero one has to have spin orbit coupling or spontaneous symmetry breaking or another mechanism.

⁶Depending on the model, there can be a different number and different types of Green-functions.

⁷The picture was taken from [51].

⁸Here again, one has to ("spontaneously") break a symmetry in order to get non-zero values.

11. Green-Functions at Zero Temperature

Can be obtained from Green-Functions at Finite-Temperature (see next chapter) and taking the limit $T \rightarrow 0$, as elucidated in the Appendix, Section 27.3. A derivation of some standard-identities that one oftentimes encounters in literature on many-body-Green-function-techniques is also provided in the Appendix.

12. Real-Time-Two-Operator-Green-Functions at Finite Temperature

12.1. Two-Operator Green-Functions without Periodicity

For reasons that are impossible to motivate in advance, we introduce the following notations and functions.

Definition 12.1 For a pair of two operators (X, X') and times $t, t' \in \mathbb{R}$ we define the *real-time-two-operator-Green-function* in its various notations as *expectation-value*

$$\begin{aligned}
 G_{XX'}^{++}(t, t') &\equiv G_{XX'}^{\mathbb{T}}(t, t') \equiv G_{XX'}(t, t') \\
 &\equiv -\frac{i}{\hbar} \langle T_t X(t) X'(t') \rangle \\
 &\equiv -\frac{i}{\hbar} \frac{\text{Tr}_{\mathcal{H}} e^{-\beta \Xi} T_t X(t) X'(t')}{\text{Tr}_{\mathcal{H}} e^{-\beta \Xi}} \\
 &\equiv -\frac{i}{\hbar} \frac{\text{Tr}_{\mathcal{H}} e^{-\beta \Xi} \left[\theta(t - t') X(t) X'(t') + S_{X, X'} \theta(t' - t) X'(t') X(t) \right]}{\text{Tr}_{\mathcal{H}} e^{-\beta \Xi}}
 \end{aligned} \tag{12.1}$$

with time-ordering-operator and ensemble-Heisenberg-time-evolution as usual.

Definition 12.2 We introduce various *Green-functions* corresponding to the operator-pair (X, X') as

- $G_{X, X'}(t, t') \equiv G_{X, X'}^{++}(t, t') \equiv G_{X, X'}^{\mathbb{T}}(t, t') \equiv -\frac{i}{\hbar} \langle T_t X(t) X'(t') \rangle$
- $G_{X, X'}^{+-}(t, t') \equiv G_{X, X'}^{<}(t, t') \equiv -\frac{i}{\hbar} S_{X, X'} \langle X'(t') X(t) \rangle$
- $G_{X, X'}^{-+}(t, t') \equiv G_{X, X'}^{>}(t, t') \equiv -\frac{i}{\hbar} \langle X(t) X'(t') \rangle$
- $G_{X, X'}^{--}(t, t') \equiv G_{X, X'}^{\bar{\mathbb{T}}}(t, t') \equiv -\frac{i}{\hbar} \langle \bar{T}_t X(t) X'(t') \rangle$
- $G_{X, X'}^{\text{ret}}(t, t') = -\frac{i}{\hbar} \theta(t - t') \langle [X(t), X'(t')]_S \rangle$
- $G_{X, X'}^{\text{adv}}(t, t') = +\frac{i}{\hbar} \theta(t' - t) \langle [X(t), X'(t')]_S \rangle$

Lemma 12.3 The Green-functions defined above only depend on the time-difference, hence one can write

$$G_{X,X'}^{\square}(t, t') \equiv G_{X,X'}^{\square}(t - t'). \quad (12.2)$$

Proof 12.3 See Appendix, Proofs, page 420.

Remark 12.3.1 For the discussion of Feynman-diagrams, it is fruitful to introduce this redundant, extra time-argument.

Lemma 12.4 The following identities are immediate, if one neglects boundary-conditions¹

- $G_{X,X'}^{\top}(t, t') = +\theta(t - t') G_{X,X'}^{>}(t, t') + \theta(t' - t) G_{X,X'}^{<}(t, t'),$
- $G_{X,X'}^{\bar{\top}}(t, t') = +\theta(t' - t) G_{X,X'}^{>}(t, t') + \theta(t - t') G_{X,X'}^{<}(t, t'),$
- $G_{X,X'}^{\text{ret}}(t, t') = \left(G_{X,X'}^{++} - G_{X,X'}^{+-} \right)(t, t') = \left(G_{X,X'}^{-+} - G_{X,X'}^{--} \right)(t, t') = \theta(t - t') \left(G_{X,X'}^{-+} - G_{X,X'}^{+-} \right)(t, t'),$
- $G_{X,X'}^{\text{adv}}(t, t') = \left(G_{X,X'}^{++} - G_{X,X'}^{-+} \right)(t, t') = \left(G_{X,X'}^{+-} - G_{X,X'}^{--} \right)(t, t') = \theta(t' - t) \left(G_{X,X'}^{+-} - G_{X,X'}^{-+} \right)(t, t'),$
- $G_{X,X'}^{\top}(t, t') + G_{X,X'}^{\bar{\top}}(t, t') = G_{X,X'}^{>}(A, A') + G_{X,X'}^{<}(t, t'),$
- $G_{X,X'}^{\text{ret}}(t, t') - G_{X,X'}^{\text{adv}}(t, t') = G_{X,X'}^{>}(t, t') - G_{X,X'}^{<}(t, t'),$

and if one has $X^{\dagger} = X'$ one also finds

- $G_{X,X'}^{\text{ret}}(t, t') = G_{X,X'}^{\text{adv}}(t', t)^{\dagger}.$

whereas the last equation is to be understood as a generalized matrix.

Proof 12.4 See Appendix, Proofs, page 420.

Remark 12.4.1 We emphasize a comment from Kamenev [26]:

"I can see how one can be overwhelmed by a number of different Green-functions, rules to follow and by the length of the calculations."

¹This is what one usually does in thermal quantum field theory. There are some comments about this in [Mahan].

12.2. Pure One-Body Green-Functions without Periodicity

Definition 12.5 We consider $X = c_\alpha$ and $X' = c_{\alpha'}^\dagger$ for multi-indices $A \equiv (\alpha, t) \in \mathcal{A} \times \mathbb{R}$, with

$$[c_\alpha, c_{\alpha'}^\dagger]_\zeta = \delta_{\alpha, \alpha'}, \quad S(c_\alpha) = S(c_{\alpha'}^\dagger) = \zeta.$$

Then we write for the *real-time-one-body-Green-function*², the lesser, greater, anti-time-ordered, retarded and advanced Green's-functions in their various shapes as expectation-value

$$\begin{aligned} \bullet \quad G(A, A') &\equiv G^{++}(A, A') \equiv G^\mathbb{T}(A, A') \equiv -\frac{i}{\hbar} \left\langle T_t c_\alpha(t) c_{\alpha'}^\dagger(t') \right\rangle, \\ \bullet \quad G^{+-}(A, A') &\equiv G^<(A, A') \equiv -\frac{i}{\hbar} \zeta \left\langle c_{\alpha'}^\dagger(t') c_\alpha(t) \right\rangle, \\ \bullet \quad G^{-+}(A, A') &\equiv G^>(A, A') \equiv -\frac{i}{\hbar} \left\langle c_\alpha(t) c_{\alpha'}^\dagger(t') \right\rangle, \\ \bullet \quad G^{--}(A, A') &\equiv G^{\bar{\mathbb{T}}}(A, A') \equiv -\frac{i}{\hbar} \left\langle \bar{T}_t c_\alpha(t) c_{\alpha'}^\dagger(t') \right\rangle, \\ \bullet \quad G^{\text{ret}}(A, A') &= -\frac{i}{\hbar} \theta(t - t') \left\langle [c_\alpha(t), c_{\alpha'}^\dagger(t')]_\zeta \right\rangle, \\ \bullet \quad G^{\text{adv}}(A, A') &= +\frac{i}{\hbar} \theta(t' - t) \left\langle [c_\alpha(t), c_{\alpha'}^\dagger(t')]_\zeta \right\rangle. \end{aligned}$$

with time-ordering-operator and ensemble-Heisenberg-time-evolution as usual.

Remark 12.5.1 Among physicists, the usual *interpretation* of the causal Green-function is as follows³

The one-particle Green's function describes a Gedankenexperiment which can never happen in practice. One creates a particle at time t' in the state α' and then (if $t > t'$) annihilates another particle at time t in the state α .

Lemma 12.6 We obtain for the time-ordered, the anti-time-ordered, the retarded and the advanced Green-functions the equations of motion, and write them in wise foresight in sets as

$$\partial_t G_{\alpha, \alpha'}^\mathbb{T}(t) = -\frac{i}{\hbar} \delta(t) \delta_{\alpha, \alpha'} - \frac{1}{\hbar} \left\langle T_t [c_\alpha, \frac{\Xi}{\hbar}]_H(t) c_{\alpha'}^\dagger(0) \right\rangle, \quad (12.3)$$

$$\partial_t G_{\alpha, \alpha'}^{\text{ret}}(t) = -\frac{i}{\hbar} \delta(t) \delta_{\alpha, \alpha'} - \frac{1}{\hbar} \theta(t) \left\langle [[c_\alpha, \frac{\Xi}{\hbar}]_H(t), c_{\alpha'}^\dagger(0)]_\zeta \right\rangle, \quad (12.4)$$

and as

$$\partial_t G_{\alpha, \alpha'}^{\bar{\mathbb{T}}}(t) = \frac{i}{\hbar} \delta(t) \delta_{\alpha, \alpha'} - \frac{1}{\hbar} \left\langle \bar{T}_t [c_\alpha, \frac{\Xi}{\hbar}]_H(t) c_{\alpha'}^\dagger(0) \right\rangle, \quad (12.5)$$

²Other names of this function are *time-ordered Green-function*, or *causal Green-function*. One also calls this Green-function the *propagator* of the underlying physical system.

³From [30], p. 167, with adapted notation to fit our purposes.

$$\partial_t G_{\alpha,\alpha'}^{\text{adv}}(t) = -\frac{i}{\hbar} \delta(-t) \delta_{\alpha,\alpha'} + \frac{1}{\hbar} \theta(-t) \left\langle \left[\left[c_\alpha, \frac{\Xi}{\hbar} \right]_{\text{H}}(t), c_{\alpha'}^\dagger(0) \right]_\zeta \right\rangle. \quad (12.6)$$

Proof 12.6 See Appendix, Proofs, page 422.

Remark 12.6.1 Notice the symmetric behaviour of total minus-signs in these expressions.

Change of Basis

Notation 12.7 We consider a change of basis⁴ with a unitary transformation

$$\tilde{c}_\alpha^\dagger = \sum_i U_{i,\alpha} c_i^\dagger, \quad \tilde{c}_\alpha = \sum_i U_{i,\alpha}^* c_i \quad (12.7)$$

Remark 12.7.1 Notice that this can equivalently be rewritten by regarding the total sets of operators as column-vectors

$$\tilde{\mathbf{c}} = U^\dagger \mathbf{c}, \quad \tilde{\mathbf{c}}^\dagger = \mathbf{c}^\dagger U \quad (12.8)$$

Corollary 12.8 We obtain for the two Green-functions w.r.t. to the two basis sets the relationship

$$\tilde{G}^{\blacksquare} = U^\dagger G^{\blacksquare} U \iff U \tilde{G}^{\blacksquare} U^\dagger = G^{\blacksquare} \quad (12.9)$$

whereas $\blacksquare$ represents *any* of the double upper indices in Definition 12.5.

Proof 12.8 See Appendix, Proofs, page 424.

⁴Notice that in the case of $\eta = 1$ this might originate from a basis-change of the underlying single-particle Hilbert-space.

Examples: Pure, Non-Interacting Systems

Example: A Quantum Dot with Nature η and Statistics ζ

We consider the system from Definition 8.19. It's Green-functions can be inferred as a special case of the diagonal gas in Theorem 12.9 below.

Example: A Diagonal Gas with Nature η and Statistics ζ

We consider the system from Definition 8.21.

Theorem 12.9 Introducing the variables $\xi_\alpha \equiv \epsilon_\alpha - \eta\mu$ it follows

$$\begin{aligned}
 G_{\alpha,\alpha'}^{\mathbb{T}}(t) &= -\frac{i}{\hbar} e^{-it\xi_\alpha/\hbar} \delta_{\alpha,\alpha'} \left[\theta(t) \left(1 + \zeta n_\beta^\zeta(\xi_\alpha) \right) + \theta(-t) \zeta n_\beta^\zeta(\xi_\alpha) \right], & G_{\alpha,\alpha'}^{\mathbb{T}}(\omega) &= +\frac{1}{\hbar} \delta_{\alpha,\alpha'} \lim_{\epsilon \rightarrow 0} \frac{1 + \zeta n_{h\beta}^\zeta(\omega)}{(\omega + i\epsilon - \xi_\alpha/\hbar)} - \frac{\zeta n_{h\beta}^\zeta(\omega)}{(\omega - i\epsilon - \xi_\alpha/\hbar)}, \\
 G_{\alpha,\alpha'}^<(t) &= -\frac{i}{\hbar} e^{-it\xi_\alpha/\hbar} \delta_{\alpha,\alpha'} \zeta n_\beta^\zeta(\xi_\alpha), & G_{\alpha,\alpha'}^<(\omega) &= -\frac{i}{\hbar} \delta_{\alpha,\alpha'} \zeta n_\beta^\zeta(\xi_\alpha) 2\pi \delta(\omega - \xi_\alpha/\hbar), \\
 G_{\alpha,\alpha'}^>(t) &= -\frac{i}{\hbar} e^{-it\xi_\alpha/\hbar} \delta_{\alpha,\alpha'} \left(1 + \zeta n_\beta^\zeta(\xi_\alpha) \right), & G_{\alpha,\alpha'}^>(\omega) &= -\frac{i}{\hbar} \delta_{\alpha,\alpha'} \left(1 + \zeta n_\beta^\zeta(\xi_\alpha) \right) 2\pi \delta(\omega - \xi_\alpha/\hbar), \\
 G_{\alpha,\alpha'}^{\overline{\mathbb{T}}}(t) &= -\frac{i}{\hbar} e^{-it\xi_\alpha/\hbar} \delta_{\alpha,\alpha'} \left[\theta(-t) \left(1 + \zeta n_\beta^\zeta(\xi_\alpha) \right) + \theta(+t) \zeta n_\beta^\zeta(\xi_\alpha) \right], & G_{\alpha,\alpha'}^{\overline{\mathbb{T}}}(\omega) &= +\frac{1}{\hbar} \delta_{\alpha,\alpha'} \lim_{\epsilon \rightarrow 0} -\frac{1 + \zeta n_{h\beta}^\zeta(\omega)}{(\omega - i\epsilon - \xi_\alpha/\hbar)} + \frac{\zeta n_{h\beta}^\zeta(\omega)}{(\omega + i\epsilon - \xi_\alpha/\hbar)}, \\
 G_{\alpha,\alpha'}^{\text{ret}}(t) &= -\frac{i}{\hbar} e^{-it\xi_\alpha/\hbar} \delta_{\alpha,\alpha'} \theta(t), & G_{\alpha,\alpha'}^{\text{ret}}(\omega + i\epsilon) &= +\frac{1}{\hbar} \delta_{\alpha,\alpha'} \frac{1}{\omega + i\epsilon - \xi_\alpha/\hbar}, \\
 G_{\alpha,\alpha'}^{\text{adv}}(t) &= -\frac{i}{\hbar} e^{-it\xi_\alpha/\hbar} \delta_{\alpha,\alpha'} \theta(-t), & G_{\alpha,\alpha'}^{\text{adv}}(\omega - i\epsilon) &= +\frac{1}{\hbar} \delta_{\alpha,\alpha'} \frac{1}{\omega - i\epsilon - \xi_\alpha/\hbar}.
 \end{aligned} \tag{12.10}$$

Proof 12.9 See Appendix, Proofs, page 424.

Remark 12.9.1 Notice that we can conclude the expectation-value for the number-operator from the lesser Green-function as

$$\langle c_\alpha^\dagger c_\alpha \rangle = -\frac{\hbar}{i} \zeta G_{\alpha,\alpha}^<(t=0) = n_\beta^\zeta(\xi_\alpha). \tag{12.11}$$

which will come handy in many situations.

Remark 12.9.2 Notice that this can be rewritten as a matrix-equation:

- We introduce the matrix $\Xi_{\alpha,\alpha'} \equiv \delta_{\alpha,\alpha'} \xi_\alpha$ and correspondingly we define the matrix-valued distributions

$$\mathbf{n}_\beta^\zeta(\Xi) \equiv \left(\mathbf{n}_\beta^\zeta(\Xi) \right)_{\alpha,\alpha'} = \delta_{\alpha,\alpha'} n_\beta^\zeta(\xi_\alpha), \quad \delta(\omega \mathbf{1} - \Xi/\hbar) \equiv (\delta(\omega \mathbf{1} - \Xi/\hbar))_{\alpha,\alpha'} = \delta_{\alpha,\alpha'} \delta(\omega - \xi_\alpha/\hbar) \tag{12.12}$$

- and matrix-valued notation

$$\left[\left((\omega \pm i\epsilon) \mathbf{1} - \frac{\Xi}{\hbar} \right)^{-1} \right]_{\alpha,\alpha'} = \frac{\delta_{\alpha,\alpha'}}{\omega \pm i\epsilon - \xi_\alpha/\hbar} \tag{12.13}$$

Corollary 12.10 Introducing the matrix-notation as above we find

$$\begin{aligned}
 \mathbf{G}_0^{\top}(t) &= -\frac{i}{\hbar} e^{-it\Xi/\hbar} \left[\theta(t) \left(\mathbf{1} + \zeta \mathbf{n}_{\beta}^{\zeta}(\Xi) \right) + \theta(-t) \zeta \mathbf{n}_{\beta}^{\zeta}(\Xi) \right], & \mathbf{G}_0^{\top}(\omega) &= +\frac{1}{\hbar} \lim_{\epsilon \rightarrow 0} \frac{1 + \zeta \mathbf{n}_{\beta}^{\zeta}(\omega)}{(\omega + i\epsilon) \mathbf{1} - \Xi/\hbar} - \frac{\zeta \mathbf{n}_{\beta}^{\zeta}(\omega)}{(\omega - i\epsilon) \mathbf{1} - \Xi/\hbar}, \\
 \mathbf{G}_0^{<}(t) &= -\frac{i}{\hbar} e^{-it\Xi/\hbar} \zeta \mathbf{n}_{\beta}^{\zeta}(\Xi), & \mathbf{G}_0^{<}(\omega) &= -\frac{i}{\hbar} \zeta \mathbf{n}_{\beta}^{\zeta}(\Xi) 2\pi \delta(\omega \mathbf{1} - \Xi/\hbar), \\
 \mathbf{G}_0^{>}(t) &= -\frac{i}{\hbar} e^{-it\Xi/\hbar} \left(\mathbf{1} + \zeta \mathbf{n}_{\beta}^{\zeta}(\Xi) \right), & \mathbf{G}_0^{>}(\omega) &= -\frac{i}{\hbar} \left(\mathbf{1} + \zeta \mathbf{n}_{\beta}^{\zeta}(\Xi) \right) 2\pi \delta(\omega \mathbf{1} - \Xi/\hbar), \\
 \mathbf{G}_0^{\bar{\top}}(t) &= -\frac{i}{\hbar} e^{-it\Xi/\hbar} \left[\theta(-t) \left(\mathbf{1} + \zeta \mathbf{n}_{\beta}^{\zeta}(\Xi) \right) + \theta(+t) \zeta \mathbf{n}_{\beta}^{\zeta}(\Xi) \right], & \mathbf{G}_0^{\bar{\top}}(\omega) &= +\frac{1}{\hbar} \lim_{\epsilon \rightarrow 0} -\frac{1 + \zeta \mathbf{n}_{\beta}^{\zeta}(\omega)}{(\omega - i\epsilon) \mathbf{1} - \Xi/\hbar} + \frac{\zeta \mathbf{n}_{\beta}^{\zeta}(\omega)}{(\omega + i\epsilon) \mathbf{1} - \Xi/\hbar}, \\
 \mathbf{G}_0^{\text{ret}}(t) &= -\frac{i}{\hbar} e^{-it\Xi/\hbar} \theta(t), & \mathbf{G}_0^{\text{ret}}(\omega + i\epsilon) &= +\frac{1}{\hbar} \frac{1}{(\omega + i\epsilon) \mathbf{1} - \Xi/\hbar}, \\
 \mathbf{G}_0^{\text{adv}}(t) &= -\frac{i}{\hbar} e^{-it\Xi/\hbar} \theta(-t), & \mathbf{G}_0^{\text{adv}}(\omega - i\epsilon) &= +\frac{1}{\hbar} \frac{1}{(\omega - i\epsilon) \mathbf{1} - \Xi/\hbar}.
 \end{aligned} \tag{12.14}$$

Proof 12.10 Follows from applying the matrix-identities.

Example: A Non-Diagonal Gas with Nature η and Statistics ζ

We consider the system from Definition 8.23.

Hypothesis 12.11 A direct (analytical) computation of all the individual Green-functions from the definition is (to the authors knowledge) impossible (unless one is referring to the matrices that diagonalize the Hamiltonian). One can however determine the individual Green-functions by first determining the Matsubara- or the imaginary time Green-function and subsequently using Green's chart (Figure 14.2) to determine the individual Green-functions.

Fermionic Examples

Example: A Spin- $\frac{1}{2}$ -Particle

We consider the system from Definition 8.20. Notice that this is an obvious special case of the system considered in [24].

Fact 12.12 We can compute the Green-functions of this system very efficiently, by combining Corollaries 12.8 and 12.10. To this end, we only need to find the transformation matrices that diagonalize the Hamiltonian.

Lemma 12.13 In order to find the eigenvalues of

$$H \equiv \begin{pmatrix} c_{\uparrow} \\ c_{\downarrow} \end{pmatrix}^{\dagger} \begin{pmatrix} \epsilon_0 + \epsilon_1^z & -i\epsilon_1^y + \epsilon_1^x \\ +i\epsilon_1^y + \epsilon_1^x & \epsilon_0 - \epsilon_1^z \end{pmatrix} \begin{pmatrix} c_{\uparrow} \\ c_{\downarrow} \end{pmatrix} \tag{12.15}$$

it suffices to find the eigenvalues of the matrix

$$h = \begin{pmatrix} n_z & -in_y + n_x \\ +in_y + n_x & -n_z \end{pmatrix}, \quad \text{with } \vec{n} = \epsilon/\epsilon \in S^2. \tag{12.16}$$

Proof 12.13 See Appendix, Proofs, page 427.

Fact 12.14 The unit-vector $\mathbf{n}$ can be expressed in spherical coordinates $(r, \phi, \theta) \in 1 \times (0, 2\pi) \times (0, \pi)$ as:

$$\mathbf{n} = \begin{pmatrix} \cos(\phi) \sin(\theta) \\ \sin(\phi) \sin(\theta) \\ \cos(\theta) \end{pmatrix}.$$

Lemma 12.15 The matrix

$$h = \begin{pmatrix} n_z & -in_y + n_x \\ +in_y + n_x & -n_z \end{pmatrix}, \quad \text{with } |\vec{n}| = 1. \quad (12.17)$$

possesses the eigenvalues ± 1 with eigenvectors

$$|\uparrow_n\rangle = \cos \frac{\theta}{2} |\uparrow\rangle + \sin \frac{\theta}{2} e^{i\phi} |\downarrow\rangle, \quad |\downarrow_n\rangle = \sin \frac{\theta}{2} |\uparrow\rangle - \cos \frac{\theta}{2} e^{i\phi} |\downarrow\rangle \quad (12.18)$$

whereas

$$\left\{ |\uparrow\rangle \equiv \mathbf{e}_x = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, |\downarrow\rangle \equiv \mathbf{e}_y = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\} \subset \mathbb{R}^2$$

represent the eigenstates of the third Pauli-matrix.

Proof 12.15 See Appendix, Proofs, page 427.

Corollary 12.16 It follows

$$|\uparrow\rangle = \cos \frac{\theta}{2} |\uparrow_n\rangle + \sin \frac{\theta}{2} |\downarrow_n\rangle, \quad |\downarrow\rangle = \sin \frac{\theta}{2} e^{-i\phi} |\uparrow_n\rangle - \cos \frac{\theta}{2} e^{-i\phi} |\downarrow_n\rangle$$

Proof 12.16 See Appendix, Proofs, page 429.

Remark 12.16.1 Notice that we obtain for the corresponding creation- (and annihilation-) operators

$$c_\uparrow = \cos \frac{\theta}{2} c_{\uparrow_n} + \sin \frac{\theta}{2} c_{\downarrow_n}, \quad c_\downarrow = \sin \frac{\theta}{2} e^{-i\phi} c_{\uparrow_n} - \cos \frac{\theta}{2} e^{-i\phi} c_{\downarrow_n}$$

whereas we denote $c_{\uparrow_n} \equiv c_{\uparrow_n}$ etc for notational convenience.

Remark 12.16.2 Since this is a unitary transformation, these new operators $\left\{ c_{\sigma_n}, c_{\sigma_n}^\dagger \right\}_{\sigma \in \{\downarrow, \uparrow\}}$ also obey the CAR.

Remark 12.16.3 Notice that we have now found the spectral decomposition of the Hamiltonian

$$H = \sum_{\sigma \in \{\uparrow, \downarrow\}} E_\sigma c_{\sigma_n}^\dagger c_{\sigma_n} \quad (12.19)$$

with $E_\sigma \equiv \epsilon_0 \pm \epsilon$ and we write as usual $\xi_\sigma \equiv E_\sigma - \mu$.

Remark 12.16.4 It follows from Lemma 8.22 for the grand-canonical-time-evolution

$$c_{\sigma_n}(t) = e^{-i\xi_\sigma t/\hbar} c_{\sigma_n}, \quad c_{\sigma_n}^\dagger(t) = e^{+i\xi_\sigma t/\hbar} c_{\sigma_n}^\dagger$$

Fact 12.17 We can now use Corollary 12.8 by remarking that the expansion in Equation (12.7) is represented exactly through the one in Equation (12.18). This gives us the unitary transformation-matrix

$$U = \begin{pmatrix} \cos \frac{\theta}{2} & \sin \frac{\theta}{2} e^{i\phi} \\ \sin \frac{\theta}{2} & -\cos \frac{\theta}{2} e^{i\phi} \end{pmatrix} \iff U^\dagger = \begin{pmatrix} \cos \frac{\theta}{2} & \sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} e^{-i\phi} & -\cos \frac{\theta}{2} e^{-i\phi} \end{pmatrix} \quad (12.20)$$

it is therefore sufficient to determine the individual Green-functions in the diagonal basis. For this, we can refer to Theorem 12.9.

Remark 12.17.1 Notice that we could have also computed the individual Green-functions in the "pedestrian-way". This is elaborated in the Appendix, subsection 27.10.4.

12.3. Phononic Displacement-Green-Functions without Periodicity

We consider ...

- either the system from Remark 8.22.4 (which represents a special case from the system in Definition 8.21),
- or the system from Remark 8.25.2 (which represents a special case from the system in Definition 8.25),
- or the phononic part of the system from Definitions 8.28, 8.29, 8.32, 8.33
- or reasonable generalizations or special cases of these systems.

Notice that we have $\zeta = +1$ and $\eta = 0$ for all of these systems.

Definition 12.18 For $\nu \in \{1, \dots, rd_x\}$ and $\mathbf{q} \in Q$, we introduce the *phononic-displacement-operator* $A_\nu(\mathbf{q})$ as

$$A_\nu(\mathbf{q}) = c_\nu(\mathbf{q}) + c_\nu^\dagger(-\mathbf{q}). \quad (12.21)$$

Remark 12.18.1 Notice that it is a little bit *misleading* to call these objects displacement / position-operators, since in Lemma 5.29, in particular in Equation (5.41) we see that there are prefactors missing. However... This notation is all over the literature, which is why we will also stick to it, leaving it with this word of warning.

Remark 12.18.2 Notice that $A_\nu(\mathbf{q}) = A_\nu^\dagger(-\mathbf{q}) \iff A_\nu(-\mathbf{q}) = A_\nu^\dagger(\mathbf{q})$

Definition 12.19 We consider $X = A_\nu(\mathbf{q})$ and $X' = A_{\nu'}^\dagger(\mathbf{q}')$ for multi-indices

$$(\alpha, t) \equiv (\mathbf{q}, \nu, t) \in Q \times \{1, \dots, rd_x\} \times \mathbb{R} \quad (12.22)$$

we introduce the different *real-time-[D]isplacement-Green-functions*⁵

$$\begin{aligned} \bullet \quad D(A, A') &\equiv D^{++}(A, A') \equiv D^{\mathbb{T}}(A, A') \equiv -\frac{i}{\hbar} \left\langle T_t A_\alpha(t) A_{\alpha'}^\dagger(t') \right\rangle, \\ \bullet \quad D^{+-}(A, A') &\equiv D^<(A, A') \equiv -\frac{i}{\hbar} \left\langle A_{\alpha'}^\dagger(t') A_\alpha(t) \right\rangle, \\ \bullet \quad D^{-+}(A, A') &\equiv D^>(A, A') \equiv -\frac{i}{\hbar} \left\langle A_\alpha(t) A_{\alpha'}^\dagger(t') \right\rangle, \\ \bullet \quad D^{--}(A, A') &\equiv D^{\bar{\mathbb{T}}}(A, A') \equiv -\frac{i}{\hbar} \left\langle \bar{T}_t A_\alpha(t) A_{\alpha'}^\dagger(t') \right\rangle, \\ \bullet \quad D^{\text{ret}}(A, A') &= -\frac{i}{\hbar} \theta(t - t') \left\langle [A_\alpha(t), A_{\alpha'}^\dagger(t')] \right\rangle, \\ \bullet \quad D^{\text{adv}}(A, A') &= +\frac{i}{\hbar} \theta(t' - t) \left\langle [A_\alpha(t), A_{\alpha'}^\dagger(t')] \right\rangle, \end{aligned}$$

with canonical time-evolution-operator as usual.

⁵Cf. Cuevas p. 233 vs Mahan p. 99 and Negele p. 245, whereas there is a typo (missing minus-sign) in the first literature.

Remark 12.19.1 The fact that this is a reasonable definition is far from obvious and will be discussed further later on. See also Remark 13.9.1.

Remark 12.19.2 Since these Green-functions represent a special usecase of Lemma 12.3 one also obtains (in the stationary states) the simplification for the temporal argument

$$D_{\nu,\nu'}^{\overline{\square}}(t, \mathbf{q} | t', \mathbf{q}') \equiv D_{\mathbf{q},\mathbf{q}'}^{\overline{\square}}(t | t') \equiv D_{\mathbf{q},\mathbf{q}'}^{\overline{\square}}(t - t', 0) \equiv D_{\mathbf{q},\mathbf{q}'}^{\overline{\square}}(t - t') \quad (12.23)$$

Remark 12.19.3 In Chapter 17 we will see (i.e. *prove*), that due to the translation-invariance of crystals, typical Hamiltonians, as the ones in Equations (8.29), (8.33), (8.39), (8.40), one obtains phononic Green-functions which are *diagonal in the crystal-momentum*, implying that one can write

$$D^{\overline{\square}}(A, A') \equiv D_{\nu,\nu'}^{\overline{\square}}(t, \mathbf{q} | t', \mathbf{q}') = \delta_{\mathbf{q},\mathbf{q}'} D_{\nu,\nu'}^{\overline{\square}}(t, \mathbf{q} | t', \mathbf{q}') \equiv \delta_{\mathbf{q},\mathbf{q}'} D_{\nu,\nu'}^{\overline{\square}}(t | t') \quad (12.24)$$

Since these Green-functions represent a special usecase of Lemma 12.3 one also obtains (in the stationary states) the further simplification

$$D^{\overline{\square}}(A, A') \equiv \delta_{\mathbf{q},\mathbf{q}'} D_{\nu,\nu'}^{\overline{\square}}(t | t') \equiv \delta_{\mathbf{q},\mathbf{q}'} D_{\nu,\nu'}^{\overline{\square}}(t - t', 0) \equiv \delta_{\mathbf{q},\mathbf{q}'} D_{\nu,\nu'}^{\overline{\square}}(t - t') \quad (12.25)$$

Remark 12.19.4 In [39], page 66, it is elaborated, that the free phonon Green-functions obey in the so-called "position basis" (to be introduced later) common differential equations.

Example: A Diagonal Phonon Gas

We consider the system Remark 8.22.4 (which represents a special case from the system in Definition 8.21).

Fact 12.20 Notice first that due to the structure of the non-interacting Hamiltonian the phononic Green-functions are diagonal in the branches, i.e.

$$D_{\nu,\nu'}^{\overline{\square}}(t) \equiv \delta_{\nu,\nu'} D_{\nu}^{\overline{\square}}(t) \quad (12.26)$$

Theorem 12.21 It follows⁶

$$\begin{aligned}
 D_{\alpha,\alpha'}^{\top}(t) &= -\frac{i}{\hbar} \delta_{\nu,\nu'} \delta_{\mathbf{q},\mathbf{q}'} \sum_{\sigma \in \{+,-\}} \theta(\sigma t) \left[(1 + B_{\hbar\beta}(\omega_{\mathbf{q}_\nu})) e^{-\sigma\omega_{\mathbf{q}_\nu} i t} + B_{\hbar\beta}(\omega_{\mathbf{q}_\nu}) e^{+\sigma\omega_{\mathbf{q}_\nu} i t} \right], & D_{\alpha,\alpha'}^{\top}(\omega) &= +\frac{1}{\hbar} \delta_{\nu,\nu'} \delta_{\mathbf{q},\mathbf{q}'} \lim_{\eta \rightarrow 0^+} \sum_{\sigma \in \{+,-\}} \left[\sigma \frac{B_{\hbar\beta}(\Omega_{\mathbf{q}_\nu}) + 1}{\omega - \sigma\omega_{\mathbf{q}_\nu} + \sigma i \eta} + \sigma \frac{B_{\hbar\beta}(\Omega_{\mathbf{q}_\nu})}{\omega + \sigma\omega_{\mathbf{q}_\nu} + \sigma i \eta} \right], \\
 D_{\alpha,\alpha'}^<(t) &= -\frac{i}{\hbar} \delta_{\nu,\nu'} \delta_{\mathbf{q},\mathbf{q}'} \left[(B_{\beta}(\Omega_{\mathbf{q}_\nu}) + 1) e^{+i\omega_{\mathbf{q}_\nu} t} + B_{\beta}(\Omega_{\mathbf{q}_\nu}) e^{-i\omega_{\mathbf{q}_\nu} t} \right], & D_{\alpha,\alpha'}^<(\omega) &= -\frac{i}{\hbar} \delta_{\nu,\nu'} \delta_{\mathbf{q},\mathbf{q}'} 2\pi \left[(B_{\hbar\beta}(\omega_{\mathbf{q}_\nu}) + 1) \delta(\omega + \omega_{\mathbf{q}_\nu}) + B_{\hbar\beta}(\omega_{\mathbf{q}_\nu}) \delta(\omega - \omega_{\mathbf{q}_\nu}) \right], \\
 D_{\alpha,\alpha'}^>(t) &= -\frac{i}{\hbar} \delta_{\nu,\nu'} \delta_{\mathbf{q},\mathbf{q}'} \left[(B_{\beta}(\omega_{\mathbf{q}_\nu}) + 1) e^{-i\omega_{\mathbf{q}_\nu} t} + B_{\beta}(\omega_{\mathbf{q}_\nu}) e^{+i\omega_{\mathbf{q}_\nu} t} \right], & D_{\alpha,\alpha'}^>(\omega) &= -\frac{i}{\hbar} \delta_{\nu,\nu'} \delta_{\mathbf{q},\mathbf{q}'} 2\pi \left[(B_{\hbar\beta}(\omega_{\mathbf{q}_\nu}) + 1) \delta(\omega - \omega_{\mathbf{q}_\nu}) + B_{\hbar\beta}(\omega_{\mathbf{q}_\nu}) \delta(\omega + \omega_{\mathbf{q}_\nu}) \right], \\
 D_{\alpha,\alpha'}^{\bar{\top}}(t) &= -\frac{i}{\hbar} \delta_{\nu,\nu'} \delta_{\mathbf{q},\mathbf{q}'} \sum_{\sigma \in \{+,-\}} \theta(-\sigma t) \left[(1 + B_{\hbar\beta}(\omega_{\mathbf{q}_\nu})) e^{-\sigma\omega_{\mathbf{q}_\nu} i t} + B_{\hbar\beta}(\omega_{\mathbf{q}_\nu}) e^{+\sigma\omega_{\mathbf{q}_\nu} i t} \right], & D_{\alpha,\alpha'}^{\bar{\top}}(\omega) &= +\frac{1}{\hbar} \delta_{\nu,\nu'} \delta_{\mathbf{q},\mathbf{q}'} \lim_{\eta \rightarrow 0^+} \sum_{\sigma \in \{+,-\}} \left[\sigma \frac{B_{\hbar\beta}(\Omega_{\mathbf{q}_\nu})}{\omega - \sigma\omega_{\mathbf{q}_\nu} + \sigma i \eta} + \sigma \frac{B_{\hbar\beta}(\Omega_{\mathbf{q}_\nu}) + 1}{\omega + \sigma\omega_{\mathbf{q}_\nu} + \sigma i \eta} \right], \\
 D_{\alpha,\alpha'}^{\text{ret}}(t) &= -\frac{2}{\hbar} \delta_{\nu,\nu'} \delta_{\mathbf{q},\mathbf{q}'} \theta(+t) \sin(\omega_{\mathbf{q}_\nu} t), & D_{\alpha,\alpha'}^{\text{ret}}(\omega + i\epsilon) &= +\frac{1}{\hbar} \delta_{\nu,\nu'} \delta_{\mathbf{q},\mathbf{q}'} \frac{2\omega_{\nu}(\mathbf{q})}{(\omega + i\epsilon)^2 - \omega_{\nu}^2(\mathbf{q})}, \\
 D_{\alpha,\alpha'}^{\text{adv}}(t) &= +\frac{2}{\hbar} \delta_{\nu,\nu'} \delta_{\mathbf{q},\mathbf{q}'} \theta(-t) \sin(\omega_{\mathbf{q}_\nu} t), & D_{\alpha,\alpha'}^{\text{adv}}(\omega - i\epsilon) &= +\frac{1}{\hbar} \delta_{\nu,\nu'} \delta_{\mathbf{q},\mathbf{q}'} \frac{2\omega_{\nu}(\mathbf{q})}{(\omega - i\epsilon)^2 - \omega_{\nu}^2(\mathbf{q})}.
 \end{aligned} \tag{12.27}$$

Proof 12.21 See Appendix, Proofs, page 430.

⁶Notice that we have a different sign than in the literature [30] page 99 for the advanced Green-function $D_{\mathbf{q}_\nu}^{\text{adv}}(t)$. Notice apart from this typo from Mahan, our results are in harmony with the literature: For D^{ret} and $D^<$: See [21] page 73 and also [30] page 116 with Wick-transformation.

13. Imaginary-Time-Two-Operator-Green-Functions at Finite Temperature

13.1. Two-Operator Green-Functions with Periodicity

Definition 13.1 For a pair of two operators (X, X') and some imaginary time $\tau \in \mathbb{R}$ we define the *imaginary-time two-operator Green-function* in its various shapes

$$\begin{aligned}\mathcal{G}_{X,X'}(\tau, \tau') &\equiv -\frac{1}{\hbar} \langle T_\tau X(\tau) X'(\tau') \rangle \\ &\equiv -\frac{1}{\hbar} \frac{\text{Tr}_{\mathcal{H}} e^{-\beta\Xi} T_\tau X(\tau) X'(\tau')}{\text{Tr}_{\mathcal{H}} e^{-\beta\Xi}} \\ &\equiv -\frac{1}{\hbar} \frac{\text{Tr}_{\mathcal{H}} e^{-\beta\Xi} \left[\theta(\tau - \tau') X(\tau) X'(\tau') + S_{X,X'} \theta(\tau' - \tau) X'(\tau') X(\tau) \right]}{\text{Tr}_{\mathcal{H}} e^{-\beta\Xi}}\end{aligned}$$

with imaginary-time-ordering-operator and ensemble-Heisenberg-imaginary-time-evolution as usual.

Theorem 13.2 It follows

$$\mathcal{G}_{X,X'}(\tau, \tau') \equiv \mathcal{G}_{X,X'}(\tau - \tau', 0) \equiv \mathcal{G}_{X,X'}(\tau - \tau') \equiv S_{X,X'} \mathcal{G}_{X,X'}(\tau - \tau' + \hbar\beta), \quad (13.1)$$

which implies that the function

$$(-\hbar\beta, \hbar\beta) \ni \tau \mapsto \mathcal{G}_{X,X'}(\tau) \in \mathbb{C} \quad (13.2)$$

is periodic.

Proof 13.2 See Appendix, Proofs, page 440

Remark 13.2.1 This periodicity-property is known as the *Kubo-Martin-Schwinger-relation* [34].

Remark 13.2.2 Notice that this implies, with Lemma 1.11, that there exists a Matsubara-Fourier-transformation of the Green-function

$$i\Omega_{S_{X,X'}} \ni i\omega_n \mapsto \mathcal{G}_{X,X'}(i\omega_n) \equiv \int_0^{\hbar\beta} d\tau \mathcal{G}_{X,X'}(\tau) e^{i\omega_n \tau} \in \mathbb{C}. \quad (13.3)$$

13.2. Pure One-Body Green-Functions with Periodicity

Definition 13.3 We consider $X = c_\alpha$ and $X' = c_{\alpha'}^\dagger$ for multi-indices $A \equiv (\alpha, t) \in \mathcal{A} \times \mathbb{R}$, with

$$[c_\alpha, c_{\alpha'}^\dagger]_\zeta = \delta_{\alpha, \alpha'}, \quad S(c_\alpha) = S(c_{\alpha'}^\dagger) = \zeta.$$

Then we write for the *imaginary-time one-body Green-function* in its various shapes

$$\mathcal{G}(A, A') \equiv \mathcal{G}_{\alpha, \alpha'}(\tau, \tau') \equiv -\frac{1}{\hbar} \frac{\text{Tr}_{\mathcal{H}} e^{-\beta \Xi} \left[\theta(\tau - \tau') c_\alpha(\tau) c_{\alpha'}^\dagger(\tau') + \zeta \theta(\tau' - \tau) c_{\alpha'}^\dagger(\tau') c_\alpha(\tau) \right]}{\text{Tr}_{\mathcal{H}} e^{-\beta \Xi}} \quad (13.4)$$

with *imaginary-time-ordering-operator* and ensemble-Heisenberg-imaginary-time-evolution as usual¹

Theorem 13.4 It follows

$$\mathcal{G}_{\alpha, \alpha'}(\tau, \tau') \equiv \mathcal{G}_{\alpha, \alpha'}(\tau - \tau', 0) \equiv \mathcal{G}_{\alpha, \alpha'}(\tau - \tau') \equiv \zeta \mathcal{G}_{\alpha, \alpha'}(\tau - \tau' + \hbar\beta), \quad (13.5)$$

which implies that the function

$$(-\hbar\beta, \hbar\beta) \ni \tau \mapsto \mathcal{G}_{\alpha, \alpha'}(\tau) \in \mathbb{C} \quad (13.6)$$

is periodic.

Proof 13.4 See Appendix, Proofs, page 440

Lemma 13.5 An useful identity is the so-called *equation of motion*

$$\partial_\tau \mathcal{G}_{\alpha, \alpha'}(\tau) = -\frac{1}{\hbar} \delta(\tau) \delta_{\alpha, \alpha'} + \frac{1}{\hbar} \left\langle T_\tau \left[c_\alpha, \frac{\Xi}{\hbar} \right]_{\text{H}}(\tau) c_{\alpha'}^\dagger(0) \right\rangle. \quad (13.7)$$

Proof 13.5 See Appendix, Proofs, page 441

Remark 13.5.1 A common strategy to solve this differential equation is the so-called "integrating-factor method".

¹Notice that in the imaginary-time case, the creation- and annihilation-operators are no longer complex-conjugates to each other:

$$c_\bullet(\tau) = e^{\tau(H - \eta\mu N)/\hbar} c_\bullet e^{-\tau(H - \eta\mu N)/\hbar}, \quad c_\bullet^\dagger(\tau) = e^{\tau(H - \eta\mu N)/\hbar} c_\bullet^\dagger e^{-\tau(H - \eta\mu N)/\hbar}.$$

One of the (in the authors opinion) very few instances where this becomes actually an important subtlety regarding a calculation is, for example, in the derivation of the Eliashberg-equations [32].

Example: Pure, Non-Interacting Systems

Example: A Quantum Dot with Nature η and Statistics ζ

We consider the system from Definition 8.19. The Green-functions of this system can be inferred as a special case of the diagonal gas below.

Example: A Diagonal Gas with Nature η and Statistics ζ

We consider the system from Definition 8.21.

Theorem 13.6 Omitting convergence-generating factors, and introducing the variable $\xi_\alpha \equiv \epsilon_\alpha - \eta\mu$, it follows

$$\mathcal{G}_{\alpha,\alpha'}^0(i\omega_n) = \frac{1}{\hbar} \frac{\delta_{\alpha,\alpha'}}{i\omega_n - \xi_\alpha/\hbar}, \quad \mathcal{G}_{\alpha,\alpha'}^0(\tau) = -\frac{1}{\hbar} e^{-\tau\xi_\alpha/\hbar} \delta_{\alpha,\alpha'} \left[\theta(\tau) \left(1 + \zeta n_\beta^\zeta(\xi_\alpha) \right) + \theta(-\tau) \zeta n_\beta^\zeta(\xi_\alpha) \right]. \quad (13.8)$$

Proof 13.6 See Appendix, Proofs, page 441

Example: A Non-Interacting Gas with Nature η and Statistics ζ

We consider the system from Definition 8.23.

Theorem 13.7 Omitting convergence-generating factors it follows

$$\mathcal{G}_{\alpha,\alpha'}^0(i\omega_n) = \frac{1}{\hbar} \left(\left[i\omega_n - \frac{H_0 - \eta\mu}{\hbar} \right]^{\text{inv}} \right)_{\alpha,\alpha'}, \quad \mathcal{G}_{\alpha,\alpha'}^0(\tau) = \frac{1}{\hbar\beta} \sum_{\omega_n \in \Omega_\zeta} \frac{1}{\hbar} \left(\frac{e^{-i\omega_n\tau}}{i\omega_n - \frac{H_0 - \eta\mu}{\hbar}} \right)_{\alpha,\alpha'}. \quad (13.9)$$

Proof 13.7 See Appendix, Proofs, page 442

Remark 13.7.1 Notice that we can write this in matrix-notation as

$$\mathcal{G}_0(i\omega_n) = \frac{1}{\hbar} \frac{1}{i\omega_n - (H_0 - \eta\mu)/\hbar} \quad (13.10)$$

Remark 13.7.2 Note that within the proof of this we found the solution of the differential equation²

$$\left(\partial_\tau + \frac{H - \eta\mu}{\hbar} \right) \mathcal{G}(\tau) = -\delta(\tau) \delta_{\alpha,\alpha'} / \hbar \quad (13.11)$$

with $\mathcal{G}$ defined on the interval $(-\hbar\beta, +\hbar\beta)$, subject to the boundary-condition

$$\mathcal{G}(\tau + \hbar\beta) = -\mathcal{G}(\tau). \quad (13.12)$$

²Cf. [35], p. 73 and [7] p. Exercises for Chapter 9

Fermionic Systems

Example: A Spin- $\frac{1}{2}$ -Particle

We consider the system from Definition 8.20.

Theorem 13.8 It follows for $\omega_- \neq \omega_+$

$$\mathcal{G}(\tau) = \frac{1}{\hbar} \sum_{k=\pm} \frac{(z - (\epsilon_0 - \mu)/\hbar) \sigma_0 + \epsilon \cdot \sigma / \hbar}{(z - \omega_+) (z - \omega_-)} e^{-z\tau} (z - \omega_k) F_{h\beta}(z) \Big|_{z=\omega_k}, \quad \mathcal{G}(i\omega_n) = \frac{1}{\hbar} \frac{\left(i\omega_n - \frac{\epsilon_0 - \mu}{\hbar}\right) \sigma_0 + (\epsilon/\hbar) \cdot \sigma}{(i\omega_n - \omega_+) (i\omega_n - \omega_-)}, \quad (13.13)$$

with

$$\omega_\xi \equiv \frac{\xi\epsilon + \epsilon_0 - \mu}{\hbar} \quad \text{for } \xi \in \{-, +\}.$$

Proof 13.8 See Appendix, Proofs, page 444

13.3. Phononic Displacement-Green-Functions with Periodicity

We consider ...

- either the system from Remark 8.22.4 (which represents a special case from the system in Definition 8.21),
- or the system from Remark 8.25.2 (which represents a special case from the system in Definition 8.25),
- or the phononic part of the system from Definitions 8.28, 8.29, 8.32, 8.33
- or reasonable generalizations or special cases of these systems.

Notice that we have $\zeta = +1$ and $\eta = 0$ for all of these systems.

Definition 13.9 We consider $X = A_v(\mathbf{q})$ and $X' = A_{v'}^\dagger(\mathbf{q}')$ for multi-indices

$$(\alpha, \tau) \equiv (\mathbf{q}, v, \tau) \in Q \times \{1, \dots, rd_x\} \times \mathbb{R} \quad (13.14)$$

we introduce the *imaginary-time-[D]isplacement-Green-function* as

$$\mathcal{D}_{\alpha, \alpha'}(\tau, \tau') = -\frac{1}{\hbar} \langle T_\tau A_\alpha(\tau) A_{\alpha'}^\dagger(\tau') \rangle \iff \mathcal{D}_{v, v'}(\tau, \mathbf{q} | \tau', \mathbf{q}') = -\frac{1}{\hbar} \langle T_\tau A_v(\tau, \mathbf{q}) A_{v'}^\dagger(\tau', \mathbf{q}') \rangle. \quad (13.15)$$

Remark 13.9.1 It is far from obvious that this definition is a reasonable choice for a Green-function. It will turn out however, that this Green-function appears quite naturally in the functional integral formalism of the theory, after one has performed a suitable change of variables resulting in Theorem 27.76. It will also become useful in the discussion of Feynman-diagrams, cf. the proof of Theorem 19.3.

Remark 13.9.2 Analogously to Remark 12.19.2 we obtain with Theorem 13.2 that this Green-function is periodic and only depends on the time-difference, such that we can write

$$\mathcal{D}_{v, v'}(\tau, \mathbf{q} | \tau', \mathbf{q}') \equiv \mathcal{D}_{\mathbf{q}, \mathbf{q}'}(\tau, \tau') \equiv \mathcal{D}_{\mathbf{q}, \mathbf{q}'}(\tau - \tau') \quad (13.16)$$

and

$$\mathcal{D}_{\alpha, \alpha'}(\tau, \tau') \equiv \mathcal{D}_{\alpha, \alpha'}(\tau - \tau', 0) \equiv \mathcal{D}_{\alpha, \alpha'}(\tau - \tau') = +\mathcal{D}_{\alpha, \alpha'}(\tau - \tau' + \hbar\beta) \quad (13.17)$$

which implies that the function

$$(0, \hbar\beta) \ni \tau \mapsto \mathcal{D}_{\alpha, \alpha'}(\tau) \in \mathbb{C} \quad (13.18)$$

is periodic.

Remark 13.9.3 Analogously to Remark 12.19.3 we promise to prove in the later chapters that one obtains in typical crystalline systems simplifications as

$$\mathcal{D}_{\mathbf{q}, \mathbf{q}'}(\tau, \tau') \equiv \delta_{\mathbf{q}, \mathbf{q}'} \mathcal{D}_{\mathbf{q}}(\tau, \tau') \quad (13.19)$$

Example: A d -dimensional Phononic Gas

We consider the system from Remark 8.22.4.

Theorem 13.10 It follows

$$\begin{aligned}\mathcal{D}_{\alpha,\alpha'}^0(\tau) &= -\frac{1}{\hbar}\delta_{\alpha,\alpha'} \sum_{\nu \in \{+,-\}} \theta(\nu\tau) \left[(1 + B_{\hbar\beta}(\omega_\alpha)) e^{-\nu\omega_\alpha\tau} + B_{\hbar\beta}(\omega_\alpha) e^{\nu\omega_\alpha\tau} \right] \\ \mathcal{D}_{\alpha\alpha'}^0(i\omega_n) &= -\frac{1}{\hbar} \frac{2\omega_\alpha\delta_{\alpha,\alpha'}}{\omega_n^2 + \omega_\alpha^2}\end{aligned}\tag{13.20}$$

Proof 13.10 See Appendix, Proofs, page 445

Remark 13.10.1 Notice that this can be rewritten using hyperbolic functions³.

Remark 13.10.2 Notice also, that in the non-interacting case, the periodic standard-continuation of the function is symmetric, i.e. satisfies the equation

$$\mathcal{D}_{\alpha\alpha'}^0(\tau) = \mathcal{D}_{\alpha\alpha'}^0(-\tau).\tag{13.21}$$

This relation is integral, when one writes down and computes Feynman-diagrams related to electron-phonon theories.

Remark 13.10.3 We now recall that we can rewrite the bare phonon-propagator

$$\mathcal{D}_{0,\nu}(i\omega_n, \mathbf{q}) = \frac{1}{\hbar} \frac{2\omega_\nu(\mathbf{q})}{(i\omega_n)^2 - \omega_\nu^2(\mathbf{q})} = \frac{1}{\hbar} \int_0^\infty d\omega \frac{2\omega}{(i\omega_n)^2 - \omega^2} \delta(\omega - \omega_\nu(\mathbf{q}))\tag{13.22}$$

and therefore conclude, that the electron-phonon coupling-strength contains the phonon-propagator as a factor of the integrand.

Remark 13.10.4 Notice further the decomposition

$$\begin{aligned}\mathcal{D}_{0,\nu}(i\omega_n, \mathbf{q}) &= -\frac{1}{\hbar} \left(\frac{1}{i\omega_n - \omega_\nu(\mathbf{q})} - \frac{1}{i\omega_n + \omega_\nu(\mathbf{q})} \right) \\ &= -\frac{1}{\hbar} \left(\frac{-i\omega_n - \omega_\nu(\mathbf{q})}{(i\omega_n)^2 - \omega_\nu^2(\mathbf{q})} - \frac{-i\omega_n + \omega_\nu(\mathbf{q})}{(i\omega_n)^2 - \omega_\nu^2(\mathbf{q})} \right) \\ &= -\frac{1}{\hbar} \frac{2\omega_\nu(\mathbf{q})}{\omega_n^2 + \omega_\nu^2(\mathbf{q})}\end{aligned}$$

³<https://physics.stackexchange.com/questions/202672/free-phonon-propagator-in-imaginary-time?rq=1>

which is useful when one has the intention of continuing this expression analytically to the real axis.

Theorem 13.11 The imaginary-time-displacement-Green-function of a phononic gas solves the differential equation

$$(\partial_\tau^2 - \omega_\alpha^2) \mathcal{D}_{\alpha, \alpha'}(\tau) = 2 \frac{\omega_\alpha}{\hbar} \delta(\tau) \delta_{\alpha, \alpha'} \quad (13.23)$$

with $\mathcal{D}$ defined on the interval $(0, +\hbar\beta)$, subject to the boundary-condition

$$\mathcal{D}(\tau + \hbar\beta) = +\mathcal{D}(\tau). \quad (13.24)$$

Proof 13.11 See Appendix, Proofs, page 446

Lemma 13.12 It follows

$$2\langle a_{\nu, \mathbf{q}}^\dagger a_{\nu, \mathbf{q}} \rangle + 1 = \langle A_{\nu, \mathbf{q}}^\dagger A_{\nu, \mathbf{q}} \rangle = \lim_{\epsilon \rightarrow 0} \int_{-\infty}^{+\infty} d\omega B_{\hbar\beta}(\omega) \left(-\frac{\hbar}{\pi} \text{Im} D_{\nu, \nu}^{\text{ret}}(\omega + i\epsilon) \right) \quad (13.25)$$

Remark 13.12.1 Notice that this is the diagonal element ($\nu' = \nu$, $\mathbf{q}' = \mathbf{q}$) of the phononic spectral matrix from Section 14.3, and that the prefactor is $d\omega$, not $d\omega/(2\pi)$, since the latter would be off by a factor of 2π , as one can check explicitly using Theorem 13.10.

Proof 13.12 See Appendix, Proofs, page 448

14. Lehmann-Representation and more Green-Functions

Notation 14.1 The *Lehmann-representation*, refers to the following algebraic trick: We introduce the exact eigenbasis of the Hamiltonian

$$\mathcal{B} \equiv \{|0\rangle, |1\rangle, |2\rangle, \dots\} \quad \text{s.t.} \quad \mathcal{H} = \text{span}_{\mathbb{C}} \mathcal{B} \quad (14.1)$$

with eigenvalues

$$\xi_n |n\rangle \equiv \Xi |n\rangle \equiv \begin{cases} (H - \eta\mu N) |n\rangle & \text{for systems with nature } \eta \\ (H_p - \mu N + H_w + V_{\text{int}}) |n\rangle & \text{for wave-particle-mixtures} \\ \vdots & \end{cases} \quad (14.2)$$

Remark 14.1.1 The existence of this basis is guaranteed, if we introduce some cutoff for our basis, since then we have an hermitian operator on a finite dimensional complex vector space.

Remark 14.1.2 Notice that we capture all three cases from Notation 8.1.

Remark 14.1.3 Notice further, that ξ denotes here *not* the non-interacting energies relative to the chemical potential, but the relative energies of the full many body Hamiltonian!

With this eigenbasis, we can now formally derive integral identities that connect the various Green-functions.

Fact 14.2 Recall also the *Sokhotski-Plemelj-theorem* a.k.a. the *Dirac-identity* for integrals over the frequency ω

$$\frac{1}{\omega \pm i\epsilon + \text{const.}} = \text{P} \frac{1}{\omega + \text{const.}} \mp i\pi\delta(\omega + \text{const.}) \quad (14.3)$$

in the distributional sense for $\epsilon \rightarrow 0$.

Proof 14.2 For a fully fleshed proof, see, for example, [16] in the Appendix.

Remark 14.2.1 Notice that in the literature it seems to be quite "popular" to state this identity without any proof. See for example [5] Equation (7.19) or [2] above Equation (7.6) or [28] Equation (31.24), or [36] Equation (3.152), or [12] Equation (1.20), or [30] below Equation (3.115).

14.1. Identities for Two-Operator-Green-Functions

Definition 14.3 For reasons that are difficult to motivate in advance, we introduce the *spectral function corresponding to the operator pair* (X, X') as

$$A_{X,X'}(\omega) \equiv -\frac{\hbar}{\pi} \operatorname{Im} \lim_{\epsilon \rightarrow 0} G_{X,X'}^{\text{ret}}(\omega + i\epsilon). \quad (14.4)$$

Whereas this limit is to be understood in the distributional sense¹.

Remark 14.3.1 Notice that we defined these functions by their frequency-dependencies, and not by their time-dependencies (in contrast to the Green-functions).

Remark 14.3.2 Special cases of this spectral function will correspond to so-called "densities of states" for various systems. We will elaborate on this in the next sections.

Lemma 14.4 With the Lehmann-representation one finds the expansion

$$\mathcal{G}_{X,X'}(i\omega_l) = \frac{1}{Z} \sum_{(n,m) \in \mathbb{N}^2} \langle n|X|m\rangle \langle m|X'|n\rangle \frac{e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}}{\hbar} \frac{1}{i\omega_l + (\xi_n - \xi_m)/\hbar} \quad (14.5)$$

and similar identities for all the other Green-functions (and the spectral function).

Proof 14.4 See Appendix, Proofs, page 450

Theorem 14.5 The Matsubara-Green-function can be analytically continued down to the real axis to the retarded / advanced Green-functions as

$$\begin{aligned} i\omega_n \mapsto \omega + i\epsilon &\Rightarrow \mathcal{G}_{X,X'}(i\omega_n) \mapsto G_{X,X'}^{\text{ret}}(\omega + i\epsilon), \\ i\omega_n \mapsto \omega - i\epsilon &\Rightarrow \mathcal{G}_{X,X'}(i\omega_n) \mapsto G_{X,X'}^{\text{adv}}(\omega - i\epsilon). \end{aligned}$$

Proof 14.5 Follows immediately from the Lehmann-representations in Eqs. (31.150)-(31.157) and the Carlson-theorem².

Theorem 14.6 One also obtains an adapted version of the *Fluctuation-Dissipation-Theorem*:

¹This means, that this object needs to be integrated over.

²Cf. [35].

$$\begin{aligned}
 G_{X,X'}^<(\omega) &= 2i S_{X,X'} n_{\hbar\beta}^{S_{X,X'}}(\omega) \operatorname{Im} \lim_{\epsilon \rightarrow 0} G_{X,X'}^{\text{ret}}(\omega + i\epsilon), \\
 G_{X,X'}^>(\omega) &= 2i \left(1 + S_{X,X'} n_{\hbar\beta}^{S_{X,X'}}(\omega) \right) \operatorname{Im} \lim_{\epsilon \rightarrow 0} G_{X,X'}^{\text{ret}}(\omega + i\epsilon).
 \end{aligned} \tag{14.6}$$

Proof 14.6 See Appendix, Proofs, page 459

Theorem 14.7 One finds the *Lehmann-identities*

$$\begin{aligned}
 \operatorname{Re} G_{X,X'}(\omega) &= + \operatorname{Re} G_{X,X'}^{\text{ret}}(\omega) &= + \operatorname{Re} G_{X,X'}^{\text{adv}}(\omega), \\
 \operatorname{Im} G_{X,X'}(\omega) &= + \operatorname{Im} G_{X,X'}^{\text{ret}}(\omega) \cdot \left(\tanh \frac{\beta \hbar \omega}{2} \right)^{S_{X,X'}} &= - \operatorname{Im} G_{X,X'}^{\text{adv}}(\omega) \cdot \left(\tanh \frac{\beta \hbar \omega}{2} \right)^{S_{X,X'}},
 \end{aligned} \tag{14.7}$$

Proof 14.7 See Appendix, Proofs, page 460

Theorem 14.8 We conclude the so-called *Kramers-Kronig-relations* a.k.a. *dispersion-relations* as

$$\begin{aligned}
 \operatorname{Re} G_{X,X'}^{\text{ret}}(\omega) &= \frac{1}{\pi} \int d\omega' \operatorname{Im} G_{X,X'}^{\text{ret}}(\omega') \mathcal{P} \frac{1}{\omega' - \omega} \\
 \operatorname{Im} G_{X,X'}^{\text{ret}}(\omega) &= \frac{1}{\pi} \int d\omega' \operatorname{Re} G_{X,X'}^{\text{ret}}(\omega') \mathcal{P} \frac{1}{\omega' - \omega}
 \end{aligned}$$

Proof 14.8 See, for example, [5], Equation (7.34).

Remark 14.8.1 There are more identities that one can conclude, for example, by combining the results of this section with Theorem 12.21. See, for example, [30] Equations (3.327) and (3.328) or [43] Equations (4.2.14a) and (4.2.14b). They are however not needed for any further discussions, which is the reason why we omit them.

Theorem 14.9 In the case that $X' = X^\dagger$ we obtain ...

- for $S_{X,X^\dagger} = -1$ the implication

$$\omega \in \mathbb{R} \quad \Rightarrow \quad 0 < A_{X,X^\dagger}(\omega) \tag{14.8}$$

- for $S_{X,X^\dagger} = +1$ the implication

$$\omega > 0 \quad \Rightarrow \quad 0 < A_{X,X^\dagger}(\omega) \tag{14.9}$$

Proof 14.9 Follows immediately by combining Remark 8.11.2 with Equation (31.149) since $\exp - \hbar\beta\omega < 1$ for $\omega > 0$.

Remark 14.9.1 We emphasize at this point, something, that is usually swiped under the carpet: In the case of *bosonic* statistics, it is completely fine for a bosonic "density of states" to be *negative* at negative frequencies. There is nothing wrong about this! The *physically reasonable region* however is usually (for example in the case of phonons) at positive frequencies.

We summarize the different identities in *Green's Chart* which graphically depicts how one can determine every two-operator Green-function from the imaginary-time or (inclusive) Matsubara-Green-function.

Green's Chart

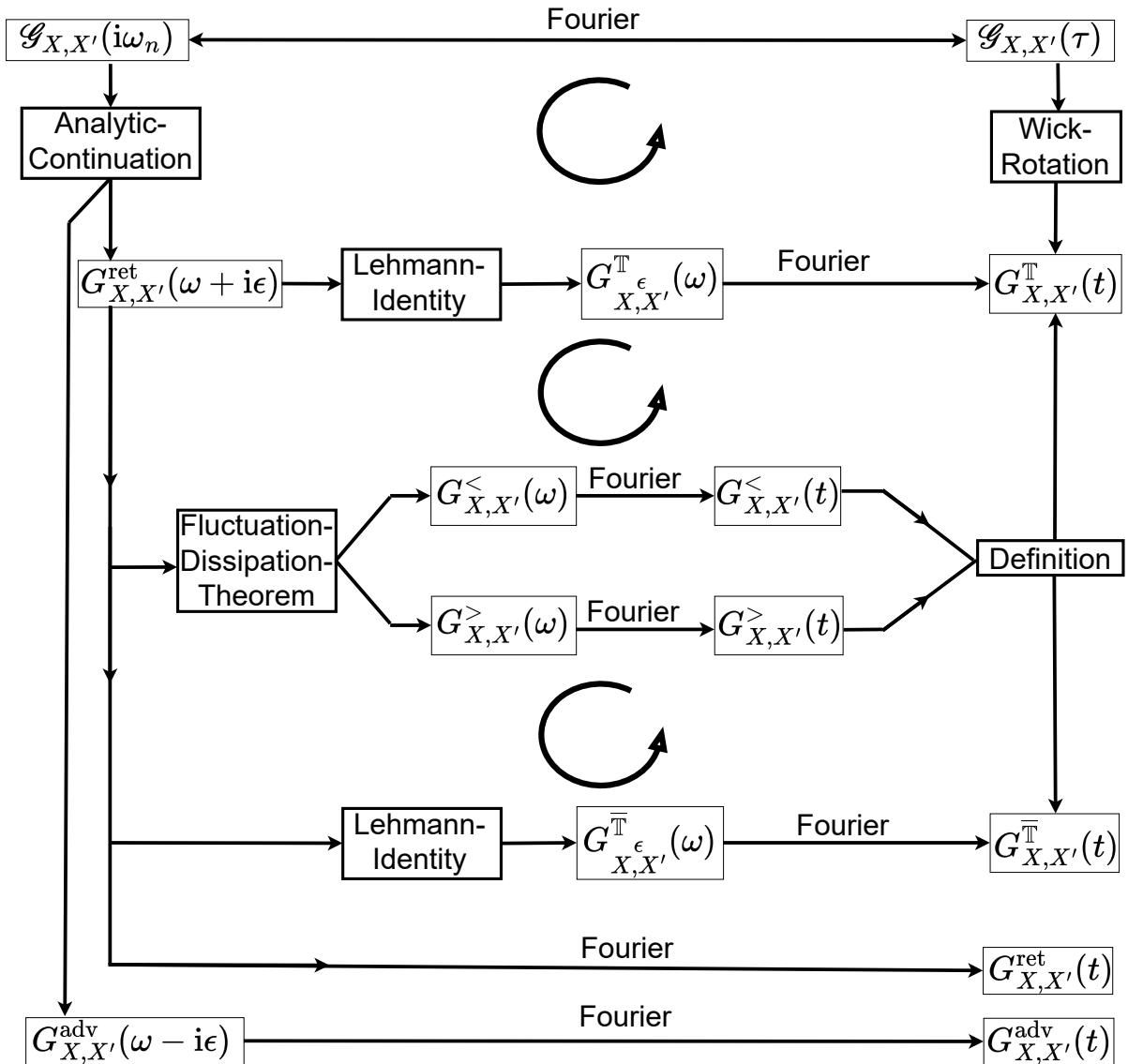


Figure 14.1.: Summarizing the different relationships of the various two-operator Green-functions.

14.2. Identities for Pure One-Body-Green-Functions

In this section, we consider $X = c_\alpha$ and $X' = c_{\alpha'}^\dagger$ for multi-indices $A \equiv (\alpha, t) \in \mathcal{A} \times \mathbb{R}$, with

$$[c_\alpha, c_{\alpha'}^\dagger]_\zeta = \delta_{\alpha, \alpha'}, \quad S(c_\alpha) = S(c_{\alpha'}^\dagger) = \zeta.$$

Definition 14.10 For reasons that are difficult to motivate in advance, we introduce the *spectral matrix* or the *generalized spectral functions* $A_{\alpha\alpha'}(\omega)$ defined by

$$A_{\alpha\alpha'}(\omega) := -\frac{\hbar}{\pi} \operatorname{Im} \lim_{\epsilon \rightarrow 0} G_{\alpha\alpha'}^{\text{ret}}(\omega + i\epsilon).$$

Whereas this limit is to be understood in the distributional sense³.

Remark 14.10.1 Notice that we defined these functions by their frequency-dependencies, and not by their time-dependencies (in contrast to the Green-functions).

Lemma 14.11 With the Lehmann-representation one finds the expansion

$$\mathcal{G}_{\alpha, \alpha'}(i\omega_l) = \frac{1}{Z} \sum_{(n, m) \in \mathbb{N}^2} \langle n | c_\alpha | m \rangle \langle m | c_{\alpha'}^\dagger | n \rangle \frac{e^{-\beta\xi_n} - \zeta e^{-\beta\xi_m}}{\hbar} \frac{1}{i\omega_l + (\xi_n - \xi_m)/\hbar} \quad (14.10)$$

and similar identities for all the other Green-functions (and the generalized spectral functions).

Proof 14.11 See Appendix, Proofs, page 450

Remark 14.11.1 In the derivation of these identities, it also becomes clear why there will always be the need to introduce convergence-generating factors for some of the Green-functions.

Theorem 14.12 The Matsubara-Green-function can be analytically continued down to the real axis to the retarded / advanced Green-functions as

$$\begin{aligned} i\omega_n \mapsto \omega + i\epsilon &\Rightarrow \mathcal{G}_{\alpha, \alpha'}(i\omega_n) \mapsto G_{\alpha, \alpha'}^{\text{ret}}(\omega + i\epsilon) \\ i\omega_n \mapsto \omega - i\epsilon &\Rightarrow \mathcal{G}_{\alpha, \alpha'}(i\omega_n) \mapsto G_{\alpha, \alpha'}^{\text{adv}}(\omega - i\epsilon) \end{aligned}$$

Proof 14.12 Follows immediately from Theorem 14.5.

³As explained earlier.

Remark 14.12.1 Notice however, that one has to be a little bit careful when one wants to perform the analytic continuation. The function has to be rewritten w.r.t. simple poles, or functions that decay „fast enough“ at infinity.

Theorem 14.13 The *fluctuation-dissipation-theorem*⁴ states the two *Lehmann-identities*

$$\begin{aligned} G_{\alpha\alpha'}^<(\omega) &= 2i \zeta n_{h\beta}^{\zeta}(\omega) \operatorname{Im} \lim_{\epsilon \rightarrow 0} G_{\alpha\alpha'}^{\operatorname{ret}}(\omega + i\epsilon), \\ G_{\alpha\alpha'}^>(\omega) &= 2i \left(1 + \zeta n_{h\beta}^{\zeta}(\omega) \right) \operatorname{Im} \lim_{\epsilon \rightarrow 0} G_{\alpha\alpha'}^{\operatorname{ret}}(\omega + i\epsilon). \end{aligned} \quad (14.11)$$

Proof 14.13 Follows from Theorem 14.6.

Lemma 14.14 For a given chemical potential μ , we can compute the average number of particles as

$$\langle N \rangle = \frac{-1}{\pi} \lim_{\epsilon \rightarrow 0} \sum_{\substack{\alpha \in \mathcal{A} \\ E \in (-\infty, +\infty)}} n_{\beta}^{\zeta}(E) \operatorname{Im} G_{\alpha,\alpha}^{\operatorname{ret}}(E + i\epsilon)$$

Proof 14.14 See Appendix, Proofs, page 462

Remark 14.14.1 Analogously one can compute the expectation-value for the particle numbers associated to a certain type of state, for example for electrons in a crystal, one can compute the number of spin-up or spin-down electrons. One can also compute the expectation-value of the holes.

Lemma 14.15 The spectral function fulfills the following identities

- $\delta_{\alpha,\alpha'} = \int d\omega A_{\alpha,\alpha'}(\omega)$
- $d = \int d\omega \operatorname{Tr} A(\omega) \equiv \int d\omega \rho(\omega)$
- For fermions ($\zeta = -1$) one finds the density

$$0 < \frac{dN_{\text{states}}}{d\omega} \equiv \rho(\omega) \equiv \operatorname{Tr} A(\omega) = -\frac{\hbar}{\pi} \operatorname{Im} \lim_{\epsilon \rightarrow 0} \operatorname{Tr} G^{\operatorname{ret}}(\omega + i\epsilon) \quad \text{for all } \omega \in \mathbb{R} \quad (14.12)$$

Proof 14.15 See Appendix, Proofs, page 462

Remark 14.15.1 Noting that the spectral function is in the fermionic case positive definite, one usually⁵ proposes the interpretation of the spectral function as a *density (of states)* or (after renormalization) *probability-density* by generalizing from the non-interacting case. Although these are, strictly speaking, just single-particle-states, and not many-particle-states, that are being counted when one integrates this object.

⁴In [21], p. 46 the origin of this name is motivated. Another proof is provided (for the fermionic case) in [7] page 136.

⁵Cf. [30], p. 123

Remark 14.15.2 For systems, where the index can be decomposed one can also introduce specific densities. One has additional indices $i \in I$, for example ...

- electronic spin-up and spin-down-component $\mathcal{A} \ni \alpha \equiv (i, \sigma) \in I \times \{\downarrow, \uparrow\}$,
- electronic Wannier-states $\mathcal{A} \ni \alpha \equiv (\mathbf{R}_j, n, i) \in L_f \times \{1, 2, 3, \dots\} \times I$,
- electronic Bloch-states $\mathcal{A} \ni \alpha \equiv (\mathbf{k}, n, i) \in Q \times \{1, 2, 3, \dots\} \times I$.

Corollary 14.16 Notice that this interpretation allows us, in the fermionic case, to write the expectation-value of the particle-number in the intuitive form

$$\langle N \rangle = \int_{-\infty}^{+\infty} dE F_\beta(E) \rho(E) \quad (14.13)$$

Proof 14.16 Follows immediately from combining the lemmata.

Theorem 14.17 Another useful set of *Lehmann-identities* is

$$\begin{aligned} \operatorname{Re} G_{\alpha, \alpha'}^\epsilon(\omega) &= + \operatorname{Re} G_{\alpha, \alpha'}^{\operatorname{ret}}(\omega + i\epsilon) &= + \operatorname{Re} G_{\alpha, \alpha'}^{\operatorname{adv}}(\omega - i\epsilon), \\ \operatorname{Im} G_{\alpha, \alpha'}^\epsilon(\omega) &= + \operatorname{Im} G_{\alpha, \alpha'}^{\operatorname{ret}}(\omega + i\epsilon) \cdot \left(\coth \frac{\beta \hbar \omega}{2} \right)^\zeta &= - \operatorname{Im} G_{\alpha, \alpha'}^{\operatorname{adv}}(\omega - i\epsilon) \cdot \left(\coth \frac{\beta \hbar \omega}{2} \right)^\zeta, \end{aligned}$$

in the limit $\epsilon \rightarrow 0$.

Proof 14.17 Follows from Theorem 14.7.

We summarize the different identities in *Green's Chart* which graphically depicts how one can determine every one-body Green-function from the imaginary-time or (inclusive) Matsubara-Green-function.

Green's Chart

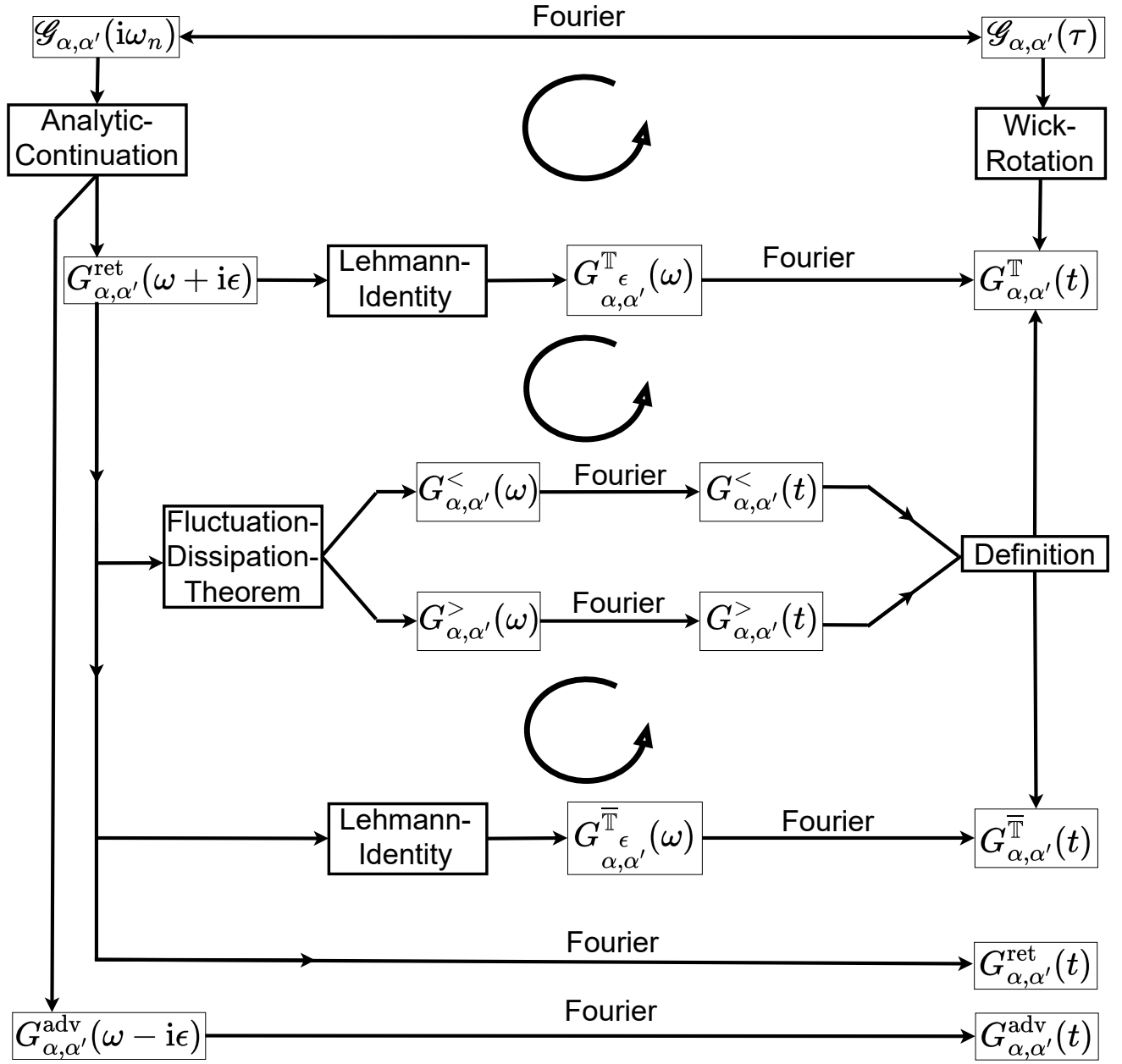


Figure 14.2.: Summarizing the different relationships of the various pure one-body Green-functions.

14.3. Identities for Phononic Displacement-Green-Functions

Definition 14.18 For reasons that are difficult to motivate in advance, we introduce the *phononic spectral matrix* or the *phononic generalized spectral functions* $\rho_{\text{ph}}(\omega)$ defined by

$$\rho_{\text{ph}}(\omega) \equiv -\frac{\hbar}{\pi} \text{Im} \lim_{\epsilon \rightarrow 0} D_{\alpha, \alpha'}^{\text{ret}}(\omega + i\epsilon).$$

Whereas this limit is to be understood in the distributional sense⁶.

Remark 14.18.1 Notice that we defined these functions by their frequency-dependencies, and not by their time-dependencies (in contrast to the Green-functions).

Remark 14.18.2 Notice that emphasizing all the individual indices that are inside this function, it actually reads as

$$\rho_{\text{ph}}(\omega) \equiv A_{A_v(\mathbf{q}), A_{v'}^\dagger(\mathbf{q}')}(\omega) \equiv -\frac{\hbar}{\pi} \text{Im} \lim_{\epsilon \rightarrow 0} D_{v, v'}^{\text{ret}}(\omega + i\epsilon).$$

Theorem 14.19 One obtains that

$$0 < \omega \quad \Rightarrow \quad 0 < \rho_{\text{ph}}(\omega) \quad (14.14)$$

Proof 14.19 This is simply a special case of Theorem 14.9.

Remark 14.19.1 Notice that the phononic spectral function is only necessarily positive definite for $\omega > 0$. For frequencies that are smaller than zero it can (from a mathematical perspective) surely happen that it becomes negative, and *there is nothing unphysical about this*, if one restricts the interpretation of it being a probability distribution to positive frequencies. It is actually itself *unphysical to even consider negative frequencies* here, since the argument ω corresponds to the oscillation-frequency with which the ions in a crystal are moving.

Remark 14.19.2 Following the literature [30] (page 125), the factor $B_{\hbar\beta}(\omega) \rho_{\text{ph}}(\omega)$ is interpreted as *the temperature-dependent probability of having phonons with crystal-momentum $\mathbf{q}$ and frequency ω in the branch v* .

⁶As explained earlier.

Theorem 14.20 The Matsubara-Green-function can be analytically continued down to the real axis to the retarded / advanced Green-functions as

$$\begin{aligned} i\omega_n \mapsto \omega + i\epsilon &\Rightarrow \mathcal{D}_{\alpha,\alpha'}(i\omega_n) \mapsto D_{\alpha,\alpha'}^{\text{ret}}(\omega + i\epsilon) \\ i\omega_n \mapsto \omega - i\epsilon &\Rightarrow \mathcal{D}_{\alpha,\alpha'}(i\omega_n) \mapsto D_{\alpha,\alpha'}^{\text{adv}}(\omega - i\epsilon) \end{aligned}$$

Proof 14.20 Follows immediately from Theorem 14.5.

Remark 14.20.1 Notice however, that one has to be a little bit careful when one wants to perform the analytic continuation. The function has to be rewritten w.r.t. simple poles, or functions that decay „fast enough“ at infinity.

Theorem 14.21 The *fluctuation-dissipation-theorem*⁷ states the two *Lehmann-identities*

$$\begin{aligned} D_{\alpha\alpha'}^<(\omega) &= 2iB_{\hbar\beta}(\omega) \text{Im} \lim_{\epsilon \rightarrow 0} D_{\alpha\alpha'}^{\text{ret}}(\omega + i\epsilon), \\ D_{\alpha\alpha'}^>(\omega) &= 2i(1 + B_{\hbar\beta}(\omega)) \text{Im} \lim_{\epsilon \rightarrow 0} D_{\alpha\alpha'}^{\text{ret}}(\omega + i\epsilon). \end{aligned} \tag{14.15}$$

Proof 14.21 Follows from Theorem 14.6.

Theorem 14.22 Another useful set of *Lehmann-identities* is

$$\begin{aligned} \text{Re } D_{\alpha,\alpha'}^<(\omega) &= +\text{Re } D_{\alpha,\alpha'}^{\text{ret}}(\omega + i\epsilon) = +\text{Re } D_{\alpha,\alpha'}^{\text{adv}}(\omega - i\epsilon), \\ \text{Im } D_{\alpha,\alpha'}^<(\omega) &= +\text{Im } D_{\alpha,\alpha'}^{\text{ret}}(\omega + i\epsilon) \cdot \coth \frac{\beta\hbar\omega}{2} = -\text{Im } D_{\alpha,\alpha'}^{\text{adv}}(\omega - i\epsilon) \cdot \coth \frac{\beta\hbar\omega}{2}, \end{aligned}$$

in the limit $\epsilon \rightarrow 0$.

Proof 14.22 Follows from Theorem 14.7.

We summarize the different identities in *Green's Chart* which graphically depicts how one can determine every one-body Green-function from the imaginary-time or (inclusive) Matsubara-Green-function.

⁷In [21], p. 46 the origin of this name is motivated.

Green's Chart

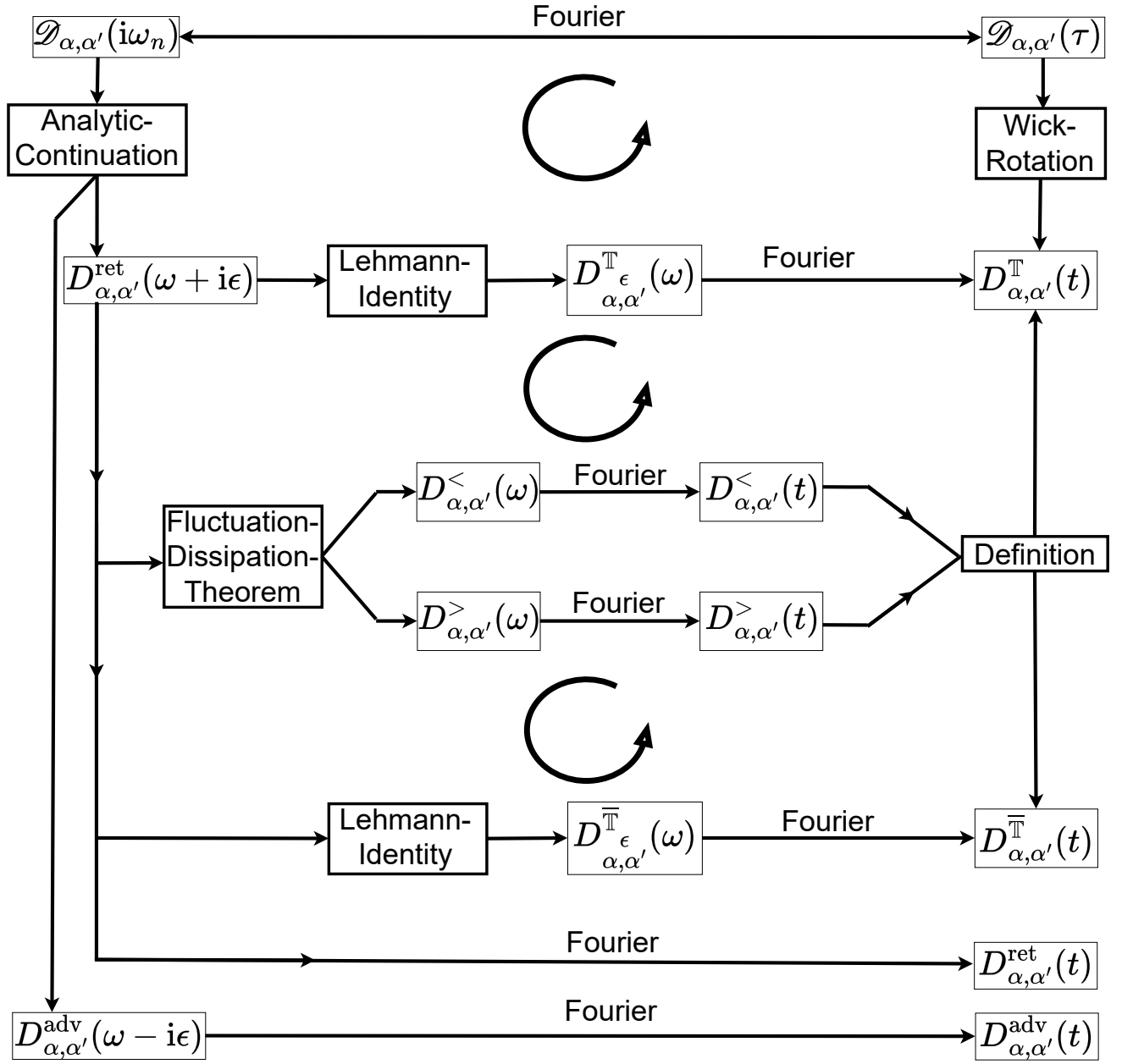


Figure 14.3.: Summarizing the different relationships of the various phononic Green-functions.

14.4. Definitions of Pure n -Body Green-Functions

For reasons that are difficult to motivate in advance, we introduce generalize the one-body Green-functions from Equation (13.4) to n -body Green-functions as follows.

Definition 14.23 We define *real- and imaginary-time n -body Green-functions* as

$$G^n(A_1, \dots, A_n | A'_n, \dots, A'_1) \equiv \left(-\frac{i}{\hbar}\right)^n \langle T_t c_{\alpha_1}(t_1) \dots c_{\alpha_n}(t_n) c_{\alpha'_1}^\dagger(t'_1) \dots c_{\alpha'_n}^\dagger(t'_n) \rangle, \quad (14.16)$$

$$\mathcal{G}^n(A_1, \dots, A_n | A'_n, \dots, A'_1) \equiv \left(-\frac{1}{\hbar}\right)^n \langle T_\tau c_{\alpha_1}(\tau_1) \dots c_{\alpha_n}(\tau_n) c_{\alpha'_1}^\dagger(\tau'_1) \dots c_{\alpha'_n}^\dagger(\tau'_n) \rangle. \quad (14.17)$$

Remark 14.23.1 Notice that we use another sign-convention here than Negele and Orland on page 53.

15. Systematic Calculation of Imaginary-Time and Matsubara-Green-Functions for Interacting Systems

15.1. S-Matrix-Expansion for the Two-Operator Green-Functions

In this whole section, we consider a situation as in Equation (10.3) or its mixed system generalization, where the total system is written as

$$H = H_0 + V \iff \Xi = \Xi_0 + V \quad (15.1)$$

whereas H_0 is a Hamiltonian of which the Green-functions are accessible, and V is a perturbation of some kind.

Definition 15.1 For $\tau \in \mathbb{R}$, we define the *time-evolution operator in the interaction representation* as

$$U_{\text{int}}(\tau) = e^{\tau\Xi_0/\hbar} e^{-\tau\Xi/\hbar} \iff U_{\text{int}}^{-1}(\tau) = e^{\tau\Xi/\hbar} e^{-\tau\Xi_0/\hbar} \quad (15.2)$$

Lemma 15.2 We obtain for the derivative of the time-evolution operator in the interaction representation the equation

$$\partial_\tau U_{\text{int}}(\tau) = -\frac{1}{\hbar} V_{H_{\Xi_0}}(\tau) U_{\text{int}}(\tau) \quad (15.3)$$

Proof 15.2 See Appendix, Proofs, page 463

Corollary 15.3 One can (formally) solve the differential-equation for $U_{\text{int}}(\tau)$ for the boundary condition $U(0) = 1$ with the *time-ordered exponential* as

$$U_{\text{int}} = \sum_{n=0}^{\infty} \left(-\frac{1}{\hbar}\right)^n \frac{1}{n!} \int_0^\tau d\tau_1 \cdots \int_0^\tau d\tau_n T_\tau \left[V_{H_{\Xi_0}}(\tau_1) \cdots V_{H_{\Xi_0}}(\tau_n) \right] \equiv T_\tau \exp \left[-\frac{1}{\hbar} \int_0^\tau d\tilde{\tau} V_{H_{\Xi_0}}(\tilde{\tau}) \right] \quad (15.4)$$

Proof 15.3 See Appendix, Proofs, page 463

Lemma 15.4 We can rewrite a two-operator Green-function from Definition 13.1 for $\tau \in (0, \hbar\beta)$ as

$$\begin{aligned}\mathcal{G}_{X,X'}(\tau) &\equiv -\frac{1}{\hbar} \langle T_\tau X(\tau) X'(0) \rangle \\ &= -\frac{1}{\hbar} \frac{\text{Tr}_{\mathcal{H}} e^{-\beta\Xi_0} U_{\text{int}}(\beta) U_{\text{int}}^{-1}(\tau) X_{H_{\Xi_0}}(\tau) U_{\text{int}}(\tau) X'(0)}{\text{Tr}_{\mathcal{H}} e^{-\beta\Xi_0} U(\beta)}\end{aligned}\quad (15.5)$$

Proof 15.4 See Appendix, Proofs, page 463

Definition 15.5 For $\tau_1, \tau_2 \in (0, \hbar\beta)$, we now introduce the *S-matrix* as

$$S(\tau_1, \tau_2) = T_\tau \exp \left[-\frac{1}{\hbar} \int_{\tau_1}^{\tau_2} d\tilde{\tau} V_{H_{\Xi_0}}(\tilde{\tau}) \right] \quad (15.6)$$

Remark 15.5.1 Notice that we obtain immediately the equality

$$U_{\text{int}}(\tau) = S(\tau, 0) \equiv S(\tau). \quad (15.7)$$

Lemma 15.6 For $\tau_{1,2,3} \in \mathbb{R}$ we obtain the identities

$$S(\tau_2, \tau_1) = U_{\text{int}}(\tau_2) U_{\text{int}}^{-1}(\tau_1), \quad S(\tau_3, \tau_2) S(\tau_2, \tau_1) = S(\tau_3, \tau_1) \quad (15.8)$$

Proof 15.6 See, for example, [30], page 130.

Theorem 15.7 For $\tau \in (0, \hbar\beta)$ the two-operator Green-function can be rewritten as

$$\mathcal{G}_{X,X'}(\tau) \equiv -\frac{1}{\hbar} \langle T_\tau X(\tau) X'(0) \rangle = -\frac{1}{\hbar} \frac{\langle T_\tau S(\beta) X_{H_{\Xi_0}}(\tau) X'_{H_{\Xi_0}}(0) \rangle_0}{\langle S(\beta) \rangle_0} \quad (15.9)$$

Proof 15.7 See Appendix, Proofs, page 464

Corollary 15.8 One finds for $\tau \in (0, \hbar\beta)$ the *S*-matrix-expansion of the various Green-functions

$$\begin{aligned}\mathcal{G}_{X,X'}(\tau) &= -\frac{1}{\hbar} \frac{\left\langle T_\tau S(\beta) X_{\Xi_0}(\tau) X'_{\Xi_0}(0) \right\rangle_0}{\langle S(\beta) \rangle_0} \\ \mathcal{G}_{\alpha\alpha'}(\tau) &= -\frac{1}{\hbar} \frac{\left\langle T_\tau S(\beta) c_{\alpha\Xi_0}(\tau) c_{\alpha'\Xi_0}^\dagger(0) \right\rangle_0}{\langle S(\beta) \rangle_0} \\ \mathcal{D}_{\alpha\alpha'}(\tau) &= -\frac{1}{\hbar} \frac{\left\langle T_\tau S(\beta) \hat{A}_{\alpha\Xi_0}(\tau) \hat{A}_{\alpha'\Xi_0}^\dagger(0) \right\rangle_0}{\langle S(\beta) \rangle_0}\end{aligned}\tag{15.10}$$

Proof 15.8 Follows by considering special cases of the Theorem above.

Remark 15.8.1 The *S*-matrix-expansion of the Green-functions can be derived in a much more straightforward way, by referring to the quantum field theory of the underlying system. We will present this in Equation (16.36).

Remark 15.8.2 Notice a peculiarity here that is oftentimes swiped under the carpet: This expansion holds *only* for $\tau \in (0, \hbar\beta)$, i.e. $\mathcal{G}_{X,X'}(\tau) \equiv \mathcal{G}_{X,X'}(\tau, 0)$, etc. This same peculiarity will re-appear (in the same way!), when we rederive the imaginary-time correlation-function in the field-integral formalism.

Theorem 15.9 *Wick-Theorem from an operator-formalism.* states that

$$\mathcal{G}_0^n(A_1, \dots, A_n | A'_1, \dots, A'_n) = \begin{vmatrix} \mathcal{G}_0(A_1 | A'_1) & \cdots & \mathcal{G}_0(A_1 | A'_n) \\ \vdots & \ddots & \vdots \\ \mathcal{G}_0(A_n | A'_1) & \cdots & \mathcal{G}_0(A_n | A'_n) \end{vmatrix}_\zeta.\tag{15.11}$$

Proof 15.9 We refer to [7]: Notice first that Equation (14.17) in our document coincides (up to a factor of $\hbar$) with Equation (10.65) in the literature, and Equation (13.4) in our document coincides (again, up to the factor of $\hbar$) with Equation (10.33b) in the literature. We refer then to section 10.6 in the literature for a proof with the equation of motion technique.

Theorem 15.10 The *linked cluster theorem* states that the formulae in eq. (15.10) can be simplified to

$$\begin{aligned}\mathcal{G}_{X,X'}(\tau) &= -\frac{1}{\hbar} \sum_{n=0}^{\infty} \left(\frac{-1}{\hbar}\right)^n \int_0^{\hbar\beta} d\tau_1 \cdots \int_0^{\hbar\beta} d\tau_n \left\langle T_\tau X_{\Xi_0}(\tau) X'_{\Xi_0}(0) V_{\Xi_0}(\tau_1) \cdots V_{\Xi_0}(\tau_n) \right\rangle_0^{\text{DCD}} \\ \mathcal{G}_{\alpha\alpha'}(\tau) &= -\frac{1}{\hbar} \sum_{n=0}^{\infty} \left(\frac{-1}{\hbar}\right)^n \int_0^{\hbar\beta} d\tau_1 \cdots \int_0^{\hbar\beta} d\tau_n \left\langle T_\tau c_{\alpha\Xi_0}(\tau) c_{\alpha'\Xi_0}^\dagger(0) V_{\Xi_0}(\tau_1) \cdots V_{\Xi_0}(\tau_n) \right\rangle_0^{\text{DCD}} \\ \mathcal{D}_{\alpha,\alpha'}(\tau) &= -\frac{1}{\hbar} \sum_{n=0}^{\infty} \left(\frac{-1}{\hbar}\right)^n \int_0^{\hbar\beta} d\tau_1 \cdots \int_0^{\hbar\beta} d\tau_n \left\langle T_\tau A_{\alpha\Xi_0}(\tau) A_{\alpha'\Xi_0}^\dagger(\tau') V_{\Xi_0}(\tau_1) \cdots V_{\Xi_0}(\tau_n) \right\rangle_0^{\text{DCD}}\end{aligned}$$

where only $\text{DCD} \equiv \text{"[D]ifferent [C]onnected [D]iagrams"}$ are evaluated

Proof 15.10 This is (unfortunately) a folklore-theorem that is always stated, but never proven in the generality in which one claims its radius of applicability: In the book from Mahan (cf. [30], page 132) this theorem is basically stated as a fact without a proof, and also only for the case of an electron-gas. In the book from Bruus and Flensberg (cf. [7], chapter 12) this theorem is only proven in the case of an electron-gas with electron-electron-interactions (and even there are, in my opinion, some gaps in the argument). In the book from Altland (cf. [5], chapter 5) this theorem is only proven with several examples, but one time without a proof, and another time (in the case of an electron gas) only until second order. In the book from Salmhofer (cf. [41], chapter 2) this statement *is* proven for a general case, but only for scalar field theories, not fermionic or mixed field theories. In the book from Negele and Orland (cf. [35], chapter 2 section 3) it is also only proven for pure systems (not for mixed systems). The list goes on... We will use this theorem therefore without a rigorous proof, and when we come to the computation of self-energies (where this theorem does matter) we will notice that a straight-forward expansion of the numerators and denominators indeed gives such a factorization that allows for a cancellation.

Remark 15.10.1 The straight-forward evaluation of these infinite series is cumbersome and basically never done in practice. There is a much more elegant way to compute Green-functions by identifying proper kernels in the so-called "Dyson-equation" (to be introduced below). This process involves the identification of "good" Green-functions, which can be linked to many observable quantities one the one hand, and possess a kernel that is easy to compute on the other hand.

15.2. The Dyson-Equation for Pure One-Body Green-Functions

Notation 15.11 For reasons that will become clear in a later chapter¹ we assume the existence of a function²

$$\begin{aligned}\Sigma : i\Omega_\xi &\rightarrow \text{Mat}_{d \times d}(\mathbb{C}) \\ i\omega_n &\mapsto \Sigma(i\omega_n)\end{aligned}$$

which we call the (*statistical*) *self-energy* s.t. the *Dyson-equation* for the one-body Green-function is fulfilled

$$\mathcal{G}(i\omega_n) = \mathcal{G}_0(i\omega_n) + (\mathcal{G}_0 \cdot \Sigma \cdot \mathcal{G})(i\omega_n), \quad (15.12)$$

where the dot inside the parenthesis corresponds to matrix-multiplication.

Remark 15.11.1 It should be noted, that there are different sign-conventions for the self-energy in the literature. We are following the one from [5, 38]³ in contrast to [35, 27]⁴. We will however motivate the here chosen convention below.

¹When we have introduced the effective action formalism, it will become clear that one can conclude the existence and shape of this function from symmetry-considerations for *most* fermionic and bosonic systems. For phonons however one can not conclude the existence of a self-energy. Phonons will be discussed in the next section.

²Note that the self-energy depends on the statistics of the system $\Sigma_\zeta \equiv \Sigma$.

³See explicitly in [5] Equation (5.38) and in [38] under Equation (25).

⁴See explicitly in [35] under Equation (3.37) and in [27] Equation (4.35).

Remark 15.11.2 Recall from Theorem 27.99 that this implies the existence of a corresponding imaginary-time-version of this equation⁵ and that the Matsubara-Fourier-transformed self-energy does not have the unit of an energy.

Remark 15.11.3 The justification for the name „self-energy“ can be seen in the following way: A formal inversion of the equation above yields

$$\begin{aligned} \mathcal{G}(i\omega_n) &= (1 - \mathcal{G}_0 \cdot \Sigma)^{-1} (i\omega_n) \cdot \mathcal{G}_0(i\omega_n) = \frac{1}{(\mathcal{G}_0^{-1} - \Sigma)(i\omega_n)} = \frac{1}{\hbar i\omega_n - (H_0 + \Sigma(i\omega_n) - \eta\mu) / \hbar}, \\ &= \frac{1}{i\hbar\omega_n - (H_0 + \Sigma(i\omega_n) - \eta\mu)}. \end{aligned} \quad (15.13)$$

The self-energy is therefore a term, that is formally being added on top of the Hamiltonian-matrix.

Remark 15.11.4 Notice that in the equation-system of the last remark we found the identity

$$\mathcal{G}^{-1}(i\omega_n) = \mathcal{G}_0^{-1}(i\omega_n) - \Sigma(i\omega_n) \quad (15.14)$$

If we now generalize the one-body Green-function to possess two Matsubara-frequencies, i.e.

$$\mathcal{G}(A|A') \equiv \mathcal{G}(i\omega_n, \alpha | i\omega_{n'}, \alpha') \equiv \mathcal{G}_{\substack{\alpha, \alpha' \\ i\omega_n, i\omega_{n'}}} \equiv \delta_{\omega_n, \omega_{n'}} \mathcal{G}_{\alpha, \alpha'}(i\omega_n), \quad \text{etc.} \quad (15.15)$$

then we see, that we can write the *generalized Dyson-equation*

$$\mathcal{G}^{-1} = \mathcal{G}_0^{-1} - \Sigma. \quad (15.16)$$

Corollary 15.12 With the von Neumann-series, one concludes

$$\mathcal{G}(i\omega_n) = (1 - \mathcal{G}_0 \cdot \Sigma)^{-1} (i\omega_n) \cdot \mathcal{G}_0(i\omega_n) = \sum_{n=0}^{\infty} ([\mathcal{G}_0 \cdot \Sigma]^n \cdot \mathcal{G}_0)(i\omega_n),$$

Proof 15.12 Follows immediately from using the von Neumann-series.

⁵More concretely

$$\begin{aligned} \Sigma &: (0, \hbar\beta)^2 \rightarrow \text{Mat}_{d_H \times d_H}(\mathbb{C}) \\ (\tau, \tau') &\mapsto \Sigma(\tau, \tau') \end{aligned}$$

s.t. the Dyson-equation in (imaginary-)time-space the (connected or unconnected) one-body Green-function is fulfilled

$$\mathcal{G}(\tau, \tau') = \mathcal{G}_0(\tau, \tau') + (\mathcal{G}_0 \cdot \Sigma \cdot \mathcal{G})(\tau, \tau'),$$

where the dot inside the parenthesis is to be understood as a generalized product as in Definition 27.95.

Remark 15.12.1 It will turn out, that depending on the situation, it is advantageous to make use of the first or second equation-sign.

Remark 15.12.2 It will also turn out, that the self-energy of the system, which seems like a very abstract concept here, can be calculated quite straight-forward. The self-energy Σ of a particle will turn out to be the sum of „all the connected irreducible diagrams“, whereas we will carefully explain what is meant by this statement in the following subsections. We will need all the identities presented here for the computation of the self-energy.

15.3. The Dyson-Equation for Phononic Displacement-Green-Functions

Notation 15.13 For reasons that will become clear in a later chapter we assume the existence of a function

$$\begin{aligned}\Pi : i\Omega_{+1} &\rightarrow \text{Mat}_{d_{\max} \times d_{\max}}(\mathbb{C}) \\ i\omega_n &\mapsto \Pi(i\omega_n)\end{aligned}$$

with $d_{\max} = rd_x N_p$, which we call the *polarization* s.t. the *Dyson-equation* of the phononic displacement-Green-function is fulfilled

$$\mathcal{D}(i\omega_n) = \mathcal{D}_0(i\omega_n) + (\mathcal{D}_0 \cdot \Pi \cdot \mathcal{D})(i\omega_n),$$

where the dot inside the parenthesis corresponds to matrix-multiplication.

Remark 15.13.1 In contrast to its electronic counterpart, this equation is unique within the literature.

Remark 15.13.2 Recall from Theorem 27.99 that this implies the existence of a corresponding imaginary-time-version version of this equation⁶ which again has not the unit of an energy.

Remark 15.13.3 Notice that a formal inversion of the equation above yields⁷

⁶More concretely

$$\begin{aligned}\Pi : (0, \hbar\beta)^2 &\rightarrow \text{Mat}_{d_{\max} \times d_{\max}}(\mathbb{C}) \\ (\tau, \tau') &\mapsto \Pi(\tau, \tau')\end{aligned}$$

s.t. the Dyson-equation in (imaginary-)time for phononic displacement-Green-function is fulfilled

$$\mathcal{D}(\tau, \tau') = \mathcal{D}_0(\tau, \tau') + (\mathcal{D}_0 \cdot \Pi \cdot \mathcal{D})(\tau, \tau'),$$

where the dot inside the parenthesis is to be understood as a generalized product as in Definition 27.95.

⁷Cf. [4].

$$\begin{aligned} \mathcal{D}(i\omega_n) &= (1 - \mathcal{D}_0 \cdot \Pi)^{-1}(i\omega_n) \cdot \mathcal{D}_0(i\omega_n) = \frac{1}{(\mathcal{D}_0^{-1} - \Pi)(i\omega_n)} \\ \Leftrightarrow \mathcal{D}_{v,v'}^{-1}(\mathbf{q}, i\omega_n) &= \mathcal{D}_{0,v,v'}^{-1}(\mathbf{q}, i\omega_n) - \Pi_{v,v'}(\mathbf{q}, i\omega_n) \end{aligned} \quad (15.17)$$

Lemma 15.14 For typical phononic systems in crystals, this equation can be rewritten to become

$$\mathcal{D}(\mathbf{q}, i\omega_n) = -\frac{1}{\hbar} \frac{2\omega(\mathbf{q})}{\left(\omega_n^2 \mathbb{1}_{rd_x \times rd_x} + \omega^2(\mathbf{q})\right) + \frac{2}{\hbar} \omega(\mathbf{q}) \Pi(\mathbf{q}, i\omega_n)} \quad (15.18)$$

Proof 15.14 See Appendix, Proofs, page 464

Corollary 15.15 With the von Neumann-series, one concludes

$$\mathcal{D}(i\omega_n) = (1 - \mathcal{D}_0 \cdot \Pi)^{-1}(i\omega_n) \cdot \mathcal{D}_0(i\omega_n) = \sum_{n=0}^{\infty} ([\mathcal{D}_0 \cdot \Pi]^n \cdot \mathcal{D}_0)(i\omega_n),$$

Proof 15.15 Follows from a formal inversion with the von Neumann-series.

Remark 15.15.1 It will turn out, that depending on the situation, it is advantageous to make use of the first or second equation-sign.

Remark 15.15.2 Just as for the self-energy of the system, the polarization can be calculated quite straight-forward.

15.4. The Bethe-Salpeter-Equation for Two-Body Green-Functions of Particle-Nature

For pedagogical reasons (we just presented a formula with which one can compute the one-body Green-function of an interacting system; Now we should present the formula for the two-body Green-function of an interacting system) we will now present the Bethe-Salpeter-Equation for the two-body Green-function and refer to the proof that is being carried out several chapters later.

Theorem 15.16 Under the proper circumstances, we can conclude the existence of a so-called *effective interaction* Vert which solves the equation

$$\begin{aligned} \mathcal{G}(i_1, i_2 | i'_1, i'_2) &= \mathcal{G}(i_1 | i'_1) \mathcal{G}(i_2 | i'_2) + \zeta \mathcal{G}(i_1 | i'_2) \mathcal{G}(i_2 | i'_1) \\ &\quad - \oint_{j_1, j_2, j'_1, j'_2} \mathcal{G}(i_1 | j_1) \mathcal{G}(i_2 | j_2) \text{Vert}(j_1, j_2 | j'_1, j'_2) \mathcal{G}(j'_1 | i'_1) \mathcal{G}(j'_2 | i'_2) \end{aligned} \quad (15.19)$$

Proof 15.16 Can be concluded from Theorem 16.48.

Remark 15.16.1 Just like the self-energy and the polarization are not quantities that are "easy to identify", the effective interaction Vert is another quantity that needs to be computed. We will learn later how to do it.

15.5. Matsubara-Summations

Without any motivation, we will elucidate how to compute summations of the type

$$S_\zeta = \sum_{\omega_n \in \Omega_\zeta} h(\omega_n). \quad (15.20)$$

since we will encounter summations of this type quite often in the following chapters.

Notation 15.17 For reasons, that will become clear soon, we assume that the function

$$\mathbb{C} - \mathbb{Z} \ni z \mapsto h(-iz) \in \mathbb{C} \quad (15.21)$$

is complex differentiable in the entire complex plane, with the exception of a finite number of poles of finite order which are no Matsubara-frequencies, which we denote with

$$\mathbb{Z} \equiv \{z_1, \dots, z_N\}. \quad (15.22)$$

Lemma 15.18 The complex function

$$\mathbb{C} - i\Omega_\zeta \ni z \mapsto g_\zeta(z) = \frac{\hbar\beta}{e^{\hbar\beta z} - \zeta} \in \mathbb{C} \quad (15.23)$$

has simple poles as

$$\forall i\omega_n \in i\Omega_\zeta : \lim_{z \rightarrow i\omega_n} (z - i\omega_n) g_\zeta(z) = \zeta. \quad (15.24)$$

Proof 15.18 See Appendix, Proofs, page 465

Remark 15.18.1 Notice that now the function

$$\mathbb{C} - P \ni z \mapsto g_\zeta(z) h(-iz) \equiv f_\zeta(z) \in \mathbb{C} \quad (15.25)$$

is complex differentiable as a product of complex differentiable functions, with the pole-set $P = i\Omega_\zeta \cup Z$.

Remark 15.18.2 Notice further the identity

$$g_\zeta(z) = \frac{\hbar\beta}{e^{\hbar\beta z} - \zeta} = \hbar\beta n_{\hbar\beta}^\zeta(z) \quad (15.26)$$

Theorem 15.19 Consider a complex function $f : \mathbb{C} \rightarrow \mathbb{C}$, which has poles at the points $\{z_1, \dots, z_N\} \subset \mathbb{C}$, and is holomorphic on $\mathbb{C} \setminus \{z_1, \dots, z_N\}$. Let γ be a closed curve in the complex plane. Then the *residue theorem* states that

$$\oint_\gamma dz f(z) = 2\pi i \sum_{z_n} I(\gamma, z_n) \text{Res}(f, z_n),$$

where

$$I(\gamma, z) = \begin{cases} 0, & \text{if } z \text{ is not enclosed by } \gamma, \\ +n, & \text{if } z \text{ is enclosed } n \text{ times counter clockwise by } \gamma, \\ -n, & \text{if } z \text{ is enclosed } n \text{ times clockwise by } \gamma, \end{cases}$$

and the *residue* $\text{Res}(f, z_0)$ of the function f at point z_0 is defined as follows. If f has pole of order n at z_0 , meaning that

$$f(z) = \frac{g(z)}{(z - z_0)^n},$$

where $g(z)$ is holomorphic in the vicinity of z_0 , then:

$$\text{Res}(f, z_0) = \frac{1}{(n-1)!} \frac{d^{n-1}}{dz^{n-1}} \left((z - z_0)^n f(z) \right) \Big|_{z=z_0}.$$

Proof 15.19 See in the literature, for example [3], Theorem 17 on page 150, or for example [40].

Theorem 15.20 If f decays sufficiently fast⁸ at complex infinity, then it follows

$$S_\zeta = \sum_{\omega_n \in \Omega_\zeta} h(\omega_n) \stackrel{\text{Claim}}{=} -\zeta \sum_{k=1}^N \text{Res}(f_\zeta(z), z = z_k) \quad (15.27)$$

⁸This "roughly" that it decays "faster" than $z \rightarrow 1/z$, whereas there are exceptions, for example for the free propagator one obtains a regularization through infinitesimal phases, cf. [5].

Proof 15.20 See Appendix, Proofs, page 465

Lemma 15.21 In the case that the complex function can be written as a product of simple poles

$$\mathbb{C} - Z \ni z \mapsto h(-iz) \equiv \text{const.} \prod_{k=1}^N \frac{1}{(z - z_k)} \in \mathbb{C} \quad (15.28)$$

one has

$$S_\zeta = \sum_{\omega_n \in \Omega_\zeta} h(\omega_n) \stackrel{\text{Claim}}{=} -\zeta \hbar \beta \text{const} \sum_{k=1}^N n_{\hbar\beta}^\zeta(z_k) \prod_{\substack{l=1 \\ l \neq k}}^N \frac{1}{z_k - z_l} \quad (15.29)$$

whereas an empty product is as usual defined to be equal to one.

Proof 15.21 Follows immediately by insertion.

Corollary 15.22 For

$$h(\omega_n) = \frac{1}{\hbar\beta} \frac{1}{i\omega_n - C} \quad (15.30)$$

one finds the identity

$$S_\zeta = \sum_{\omega_n \in \Omega_\zeta} h(\omega_n) \stackrel{\text{Claim}}{=} -\zeta n_{\hbar\beta}^\zeta(C). \quad (15.31)$$

Proof 15.22 See Appendix, Proofs, page 466

Remark 15.22.1 Notice that this is exactly the type of summation that appears when evaluating Matsubara-sums of the type

$$S_\zeta = \frac{1}{\beta} \sum_{\omega_n \in \Omega_\zeta} \mathcal{G}_0(i\omega_n). \quad (15.32)$$

Corollary 15.23 For $\omega_m \in \Omega_{-1}$ and summation-function

$$h(\omega_n) = \frac{1}{\hbar^2\beta} \frac{1}{i\omega_n - C_1} \frac{-2C_2}{(\omega_m - \omega_n)^2 + C_2^2} \quad (15.33)$$

one finds the identity

$$S_{-1} = \sum_{\omega_n \in \Omega_{-1}} h(\omega_n) \stackrel{\text{Claim}}{=} \frac{-1}{\hbar} \left[\frac{B_{\hbar\beta}(C_2) + F_{\hbar\beta}(C_1)}{i\omega_m - C_1 + C_2} + \frac{B_{\hbar\beta}(C_2) + 1 - F_{\hbar\beta}(C_1)}{i\omega_m - C_1 - C_2} \right] \quad (15.34)$$

Proof 15.23 See Appendix, Proofs, page 466

Remark 15.23.1 Notice that this is exactly the type of summation that appears when evaluating Matsubara-sums of the type

$$\frac{1}{\beta} \sum_{\omega_{n_2} \in \Omega_{-1}} \mathcal{G}_0(\mathbf{i}\omega_{n_2}) \mathcal{D}_0(\mathbf{i}(\omega_n - \omega_{n_2})). \quad (15.35)$$

Part III.

(Thermal) Quantum Field Theory

16. Algebraic Concepts in Quantum Field Theory

“Quantum field theory (QFT) provides an extremely powerful set of computational methods that have yet to find any fundamental limitations. It has led to the most fantastic agreement between theoretical predictions and experimental data in the history of science. It provides deep and profound insights into the nature of our universe, and into the nature of other possible self-consistent universes. On the other hand, the subject is a mess.”

M. D. Schwartz in [44], p. xv

“Go for the messes - that’s where the action is.”

Steven Weinberg in [1], p. 389.

16.1. Quantum Field Theory - Pure Systems

*“The formal calculus in Grassmann algebra, which was developed by Berezin in his first book *Method of the Second Quantization*, made him sure that a nontrivial analogue of analysis exists in which anticommuting variables appear on the same footing as the usual ones.”*

Felix Berezin in [6].

Supermathematics is a branch of mathematical physics, which (among many other applications) applies algebraic concepts to the study of bosonic and in particular fermionic systems. The mathematical rigorous introduction of these concepts is (at least for the purposes that we have in mind) not very complicated, but it is a lot of simple definitions and theorems.

Since there exists already so much great introductory literature to these concepts we will not prove all the standard-theorems but just list those that we need, referring to the literature¹ for the proofs.

16.1.1. Fundamentals

Definition 16.1 We associate generators $U \cup \overline{U} = \{\psi_1, \overline{\psi}_1, \dots, \psi_D, \overline{\psi}_D\}$ with annihilation-operators (ψ) and creation-operators ($\overline{\psi}$). Then we define the *algebra of statistical polynomials* $\mathcal{P}_\zeta$ as the associative algebra generated by $\mathbb{C} \oplus U \cup \overline{U}$ with relations

$$\psi_i \psi_j = \zeta \psi_j \psi_i, \quad \overline{\psi}_i \psi_j = \zeta \psi_j \overline{\psi}_i, \quad \overline{\psi}_i \overline{\psi}_j = \zeta \overline{\psi}_j \overline{\psi}_i. \quad (16.1)$$

for $i, j \in \{1, \dots, D\}$.

¹Cf. [5, 54, 55] etc.

Remark 16.1.1 One distinguishes between two cases:

- For $\zeta = -1$ this algebra is called *Graßmann-algebra* and oftentimes denotes as $\mathcal{P}_{-1} \equiv \mathcal{G}$. The generators $\{\psi_i, \bar{\psi}_i\}_{i \in \{1, \dots, D\}}$ are referred to as *Graßmann-numbers*.
- For $\zeta = +1$ this algebra is called *symmetric algebra*. The generators $\{\psi_i, \bar{\psi}_i\}_{i \in \{1, \dots, D\}}$ are referred to as *complex-numbers*.

Remark 16.1.2 Notice that these commutation-relations are the same as in Equation (7.7).

Remark 16.1.3 For the moment, we focus on the case $D = d_\eta$, such that for each creation- and annihilation-operator there is a Graßmann-variable associated. In the later application, it will be the case that $D \equiv d_\eta N$ with $N \rightarrow \infty$, such that for each creation- and annihilation-operator, there will be infinitely many Graßmann-variables associated. Typically one has a notation as $\psi_{n\alpha}$ etc. with $\alpha \in \mathcal{A}$ and $n \in \{1, 2, \dots, N\}$. The index n is either associated with an imaginary time $\tau_n \in (0, \hbar\beta)$ (such that $\psi_{n\alpha} \equiv \psi_\alpha(\tau_n)$) or a Matsubara-frequency $i\omega_n \in i\Omega_\zeta$ (such that $\psi_{n\alpha} \equiv \psi_\alpha(i\omega_n)$). When we consider different Graßmann-variables $\{\psi_1, \dots, \psi_{d_\eta}\}$ and $\{\phi_1, \dots, \phi_{d_\eta}\}$ that anticommute with each other (and correspond to different „timestamps“ or „Matsubara-frequencies“), it is this situation just elucidated that we have in mind.

Lemma 16.2 A basis of the algebra $\mathcal{P}_\zeta$ is given by the monomials

$$\left\{ \psi_1^{n_1} \cdots \psi_D^{n_D} \bar{\psi}_1^{\bar{n}_1} \cdots \bar{\psi}_D^{\bar{n}_D} \mid n_1, \bar{n}_1, \dots, n_D, \bar{n}_D \in \mathbb{N}_\zeta \right\} \subset \mathcal{P}_\zeta \quad (16.2)$$

Proof 16.2 Follows from introductory courses on linear algebra.

Corollary 16.3 Every element $f(\psi, \bar{\psi}) \equiv f \in \mathcal{P}_\zeta$ with $\psi \equiv (\psi_1, \dots, \psi_D)$ can be written as uniquely as

$$f(\psi, \bar{\psi}) = \sum_{\substack{n_1, \dots, n_D \in \mathbb{N}_\zeta \\ \bar{n}_1, \dots, \bar{n}_D \in \mathbb{N}_\zeta}} c_{n_1, \dots, n_D, \bar{n}_1, \dots, \bar{n}_D} \psi_1^{n_1} \cdots \psi_D^{n_D} \bar{\psi}_1^{\bar{n}_1} \cdots \bar{\psi}_D^{\bar{n}_D} \quad (16.3)$$

with complex numbers $c_{n_1, \dots, n_D, \bar{n}_1, \dots, \bar{n}_D} \in \mathbb{C}$.

Proof 16.3 Follows from introductory courses on linear algebra.

Remark 16.3.1 We decompose f as

$$\begin{aligned} f(\psi, \bar{\psi}) &= c_{0, \dots, 0, 0, \dots, 0} + \left(f(\psi, \bar{\psi}) - c_{0, \dots, 0, 0, \dots, 0} \right) \\ &\equiv \text{body}_f + \text{soul}_f(\psi, \bar{\psi}) \\ &\equiv f_0 + f_{\text{nil}}(\psi, \bar{\psi}) \end{aligned} \quad (16.4)$$

whereas $c_{0,\dots,0} \equiv f_0 \equiv \text{body}_f \in \mathbb{C}$ is called the *body* and $\text{soul}_f \equiv f_{\text{nil}}$ the *soul* of f .

Lemma 16.4 In the case that $\zeta = -1$, one finds that the soul of f is *nilpotent*, i.e.

$$\exists N_f \in \mathbb{N} : f_{\text{nil}}^{N_f}(\psi, \bar{\psi}) = 0. \quad (16.5)$$

whereas we define N_f as the *nilpotency index* of f .

Proof 16.4 The fact that f_{nil} is nilpotent is easily seen by writing down the polynomial expansion and taking the square... The exact value of N_f depends on the structure of the soul.

Remark 16.4.1 Notice that $N_f \leq 2D$.

Definition 16.5 We call a polynomial $f \in \mathcal{P}_\zeta$ *invertible*, if there is another element $f^{-1} \equiv \frac{1}{f} \in \mathcal{P}_\zeta$ such that

$$f f^{-1} = f^{-1} f = 1. \quad (16.6)$$

Remark 16.5.1 We call f^{-1} the *inverse element* of f .

Lemma 16.6 If $\zeta = +1$ a polynomial is always invertible. If $\zeta = -1$, a polynomial f is invertible, iff $f_0 \neq 0$ and then the inverse element is given by

$$f^{-1}(\psi, \bar{\psi}) = \frac{1}{f_0} \sum_{n=0}^{N_f-1} \left(-\frac{f_{\text{nil}}(\psi, \bar{\psi})}{f_0} \right)^n. \quad (16.7)$$

Proof 16.6 See Appendix, Proofs, page 469.

Remark 16.6.1 Notice further, that this is "simply" a Graßmann-version of the geometric series, formally

$$f^{-1}(\psi, \bar{\psi}) \equiv \frac{1}{f_0 + f_{\text{nil}}(\psi, \bar{\psi})} \equiv \frac{1}{f_0} \sum_{n=0}^{\infty} \left(-\frac{f_{\text{nil}}(\psi, \bar{\psi})}{f_0} \right)^n. \quad (16.8)$$

Remark 16.6.2 Notice that the inverse element is also again a polynomial.

Definition 16.7 We define derivatives for $\zeta = +1$ in the usual way. For $\zeta = -1$ we define the *left-derivative* for each $j \in \{1, \dots, D\}$ as

$$f \equiv f_0 + \psi_j f_1 \quad \Rightarrow \quad \frac{\partial}{\partial \psi_j} f := f_1 \quad (16.9)$$

whereas f_1 is independent of ψ_j .

Remark 16.7.1 Something peculiar about Graßmann-derivatives is that one can also define a *right-derivative* as

$$f \equiv f_0 + f_1 \psi_j \quad \Rightarrow \quad \left(\partial f / \partial \psi_j \right) \equiv f \frac{\bar{\partial}}{\partial \psi_j} := f_1 \quad (16.10)$$

Remark 16.7.2 The easiest way to see why one can do this, is to realize that derivatives are basically just linear maps on $\mathcal{P}_\zeta$. For $\zeta = 1$ this map is the same, no matter if one defines it as a left- or right-derivative. For $\zeta = -1$ one can define two different maps, which are related to each other.

Definition 16.8 For a monomial expression

$$m \equiv \psi_1^{n_1} \cdots \psi_D^{n_D} \bar{\psi}_1^{\bar{n}_1} \cdots \bar{\psi}_D^{\bar{n}_D} \quad (16.11)$$

we define the *degree* of m as

$$\deg(m) := \sum_{j=1}^D (n_j + \bar{n}_j) \quad (16.12)$$

Theorem 16.9 Let m_1 and m_2 be two monomial expressions in $\mathcal{P}_\zeta$. Then the *graded chain-rule* reads

$$\frac{\partial}{\partial \psi_j} (m_1 m_2) (\psi, \bar{\psi}) = \left[\left(\frac{\partial m_1}{\partial \psi_j} \right) m_2 + \zeta^{\deg(m_1)} m_1 \left(\frac{\partial m_2}{\partial \psi_j} \right) \right] (\psi, \bar{\psi}) \quad (16.13)$$

Proof 16.9 See, for example, [52] Equation (3.5).

Definition 16.10 For a monomial expression

$$m \equiv \psi_1^{n_1} \cdots \psi_D^{n_D} \bar{\psi}_1^{\bar{n}_1} \cdots \bar{\psi}_D^{\bar{n}_D} \quad (16.14)$$

we define² *conjugation in the polynomial algebra* as a mapping

$$\begin{aligned} * \equiv \dagger : \mathcal{P}_\zeta &\rightarrow \mathcal{P}_\zeta \\ \psi_1^{n_1} \cdots \psi_D^{n_D} \bar{\psi}_1^{\bar{n}_1} \cdots \bar{\psi}_D^{\bar{n}_D} &\mapsto \left(\psi_1^{n_1} \cdots \psi_D^{n_D} \bar{\psi}_1^{\bar{n}_1} \cdots \bar{\psi}_D^{\bar{n}_D} \right)^\dagger \equiv \psi_D^{\bar{n}_D} \cdots \psi_1^{\bar{n}_1} \bar{\psi}_D^{n_D} \cdots \bar{\psi}_1^{n_1} \end{aligned} \quad (16.15)$$

²Cf. [35], p. 26.

Lemma 16.11 The *statistics neutral commutation relations* read

$$\left[\frac{\partial}{\partial \psi_i}, \psi_j \right]_{\zeta} = \delta_{i,j}, \quad \left[\frac{\partial}{\partial \psi_i}, \bar{\psi}_j \right]_{\zeta} = 0 = \left[\frac{\partial}{\partial \bar{\psi}_i}, \psi_j \right]_{\zeta} \quad (16.16)$$

Proof 16.11 See, for example, [52] Equation (3.8)

Lemma 16.12 The *statistics neutral Schwartz-rule* reads

$$0 = \left[\frac{\partial}{\partial \psi_i}, \frac{\partial}{\partial \psi_j} \right]_{\zeta} = \left[\frac{\partial}{\partial \psi_i}, \frac{\partial}{\partial \bar{\psi}_j} \right]_{\zeta} = \left[\frac{\partial}{\partial \bar{\psi}_i}, \frac{\partial}{\partial \bar{\psi}_j} \right]_{\zeta} \quad (16.17)$$

Proof 16.12 See, for example, [52] Equation (3.8)

Remark 16.12.1 Notice that these commutation-relations are the same as in Notation 7.21. In fact, one can introduce the statistical algebra of polynomials and their derivatives as ζ -algebras quite straightforwardly by following the same steps that we did in the previous chapters. It becomes though a little bit difficult to follow on the notation (one needs, for the purposes that we have in mind, eight³ „copies“ of generators that correspond to creation- and annihilation-operators), and therefore we will not do so here.

Notation 16.13 By inclusion of another set of generators $\{j_1, \bar{j}_1, \dots, j_D, \bar{j}_D\}$, which are commonly referred to as *source-terms* we can consider combined polynomial expressions in even another polynomial algebra, which we don't give a name for notational simplicity.

Remark 16.13.1 The introduction of these so-called source-terms is the main trick when it comes to the computation of Green-functions within the framework of "Thermal Quantum Field Theory" (which will be introduced in the next chapter).

Remark 16.13.2 We introduce the notation

$$\bar{\psi}^T \equiv \psi^\dagger \quad \text{and} \quad j^\dagger \psi \equiv \sum_{k=1}^D \bar{j}_k \psi_k \quad (16.18)$$

Lemma 16.14 The derivative of a source-term reads

³Two copies for polynomials in variables and their conjugates. Two more for their derivatives. And again four more for „source-terms“ and their derivatives (to be introduced soon).

$$\begin{aligned}\frac{\partial}{\partial \bar{j}_i} \exp \sum_k \bar{j}_k \psi_k &= \psi_i \exp \sum_k \bar{j}_k \psi_k \\ \frac{\partial}{\partial j_{i'}} \exp \sum_k \bar{\psi}_k j_k &= \zeta \bar{\psi}_{i'} \exp \sum_k \bar{\psi}_k j_k\end{aligned}$$

Proof 16.14 See Appendix, Proofs, page 469.

Corollary 16.15 It holds that

$$\begin{aligned}\frac{\partial^n}{\partial \bar{j}_{i_1} \cdots \partial \bar{j}_{i_n}} \exp \sum_k \bar{j}_k \psi_k &= \psi_{i_1} \cdots \psi_{i_n} \exp \sum_k \bar{j}_k \psi_k \\ \frac{\partial^{n'}}{\partial j_{i'_1} \cdots \partial j_{i'_{n'}}} \exp \sum_k \bar{\psi}_k j_k &= \zeta^n \bar{\psi}_{i'_1} \cdots \bar{\psi}_{i'_{n'}} \exp \sum_k \bar{\psi}_k j_k\end{aligned}$$

Proof 16.15 Using the Lemma above proves the corollary.

Remark 16.15.1 Notice that the commutation-relations imply

$$\frac{\partial^{n+n'}}{\partial \bar{j}_{i_1} \cdots \partial \bar{j}_{i_n} \partial j_{i'_1} \cdots \partial j_{i'_{n'}}} e^{\sum_k \bar{j}_k \psi_k + \bar{\psi}_k j_k} = \zeta^n \psi_{i_1} \cdots \psi_{i_n} \bar{\psi}_{i'_1} \cdots \bar{\psi}_{i'_{n'}} e^{\sum_k \bar{j}_k \psi_k + \bar{\psi}_k j_k} \quad (16.19)$$

Definition 16.16 For $\zeta = +1$ we define integration as the usual Riemann- / Lebesgue-integral. For $\zeta = -1$ we define for a function as in Equation (16.3) the *Berezin-integral* as

$$\int d_\zeta(\psi, \bar{\psi}) f(\psi, \bar{\psi}) := \frac{\partial^{2D}}{\partial \psi_1 \cdots \partial \psi_D \partial \bar{\psi}_1 \cdots \partial \bar{\psi}_D} f \quad (16.20)$$

with integral-measure⁴

$$d_\zeta(\bar{\psi}, \psi) \equiv \prod_{j=1}^D \frac{d\bar{\psi}_j d\psi_j}{(2\pi i)^{\theta(\zeta)}} \quad (16.22)$$

⁴Notice that in the case $\zeta = +1$, one has

$$\frac{d\bar{\psi}_j d\psi_j}{2\pi i} \equiv \frac{d \operatorname{Re} \psi_j d \operatorname{Im} \psi_j}{\pi} \quad (16.21)$$

16.1.2. Partition-Functions and Expectation-Values for Non-Interacting Models

Notation 16.17 Let $A \in \text{GL}_n(\mathbb{C})$ be an invertible, quadratic matrix. Then we consider *the partition-function corresponding to A with statistics ζ* as

$$Z_0 := \int d_\zeta(\bar{\psi}, \psi) \exp - \psi^\dagger A \psi \equiv \int d_\zeta(\bar{\psi}, \psi) \exp - S_0[\psi, \bar{\psi}] \quad (16.23)$$

for complex- or Graßmann-variables.

Remark 16.17.1 Equivalently, we also speak about the *partition-function w.r.t. the action S* .

Remark 16.17.2 Note that we are, following the convention for notational simplicity, surpressing the dependence on ζ in the symbol Z . We will oftentimes surpress this dependence also on other quantities throughout this section (and work).

Lemma 16.18 The partition function can be calculated to

$$Z_0 = \frac{1}{\text{Det } A^\zeta}. \quad (16.24)$$

Proof 16.18 For bosons, see [5] Chapter 3 Section 2, for fermions see Chapter 4 Section 1 or, for example for fermions also [57].

Corollary 16.19 It follows also

$$Z_0 = \exp(-\zeta \text{Tr Log } A) \quad (16.25)$$

Proof 16.19 Follows from the identity $\text{Tr Log } \bullet = \text{Log Det } \bullet$.

Definition 16.20 We consider a monomial expression in the integration-variables, and define its *expectation value w.r.t. A*

$$\left\langle \psi_{i_1} \cdots \psi_{i_n} \bar{\psi}_{i'_1} \cdots \bar{\psi}_{i'_m} \right\rangle_0 := \frac{1}{Z} \int d_\zeta(\bar{\psi}, \psi) \psi_{i_1} \cdots \psi_{i_n} \bar{\psi}_{i'_1} \cdots \bar{\psi}_{i'_m} \exp - \psi^\dagger A \psi. \quad (16.26)$$

Remark 16.20.1 By $\mathbb{C}$ -linearity, one defines *expectation-values w.r.t. A* for polynomial expressions in the integration-variables $P(\bar{\psi}, \psi)$.

Definition 16.21 We define the n -body Green-function $\mathcal{G}_0$ w.r.t. A as expectation-value

$$\mathcal{G}_0(i_1, \dots, i_n | i'_1, \dots, i'_n) = \left(-\frac{1}{c}\right)^n \left\langle \psi_{i_1} \cdots \psi_{i_n} \bar{\psi}_{i'_1} \cdots \bar{\psi}_{i'_n} \right\rangle_0 \quad (16.27)$$

Remark 16.21.1 The constant c is arbitrary. For the setting that we have in mind later, it is *convention* to set it to $c \equiv \hbar$, such that the dimension of the Green-function becomes 1/Energy.

Remark 16.21.2 It is a priori completely unclear, that this definition of the Green-function will turn out to have anything to do with the Green-functions that we defined in earlier chapters. We will however prove in the next chapter, that the Green-functions defined in Equation (14.17) can be re-expressed as correlation-functions of integrals of the type in Equation (16.27).

Theorem 16.22 For monomial expressions, which contain the same number of ψ -variables as $\bar{\psi}$ -variables, one finds the *Wick-theorem*

$$\left\langle \psi_{i_1} \cdots \psi_{i_n} \bar{\psi}_{i'_1} \cdots \bar{\psi}_{i'_n} \right\rangle_0 = \sum_{\pi \in S_n} \text{sign}(\pi)^{1-\theta(\zeta)} \prod_{k=1}^n \left\langle \psi_{i_{\pi(k)}} \bar{\psi}_{i'_k} \right\rangle_0 = \sum_{\pi \in S_n} \text{sign}(\pi)^{1-\theta(\zeta)} \prod_{k=1}^n A_{i_k i'_{\pi(k)}}^{-1} \quad (16.28)$$

Proof 16.22 See, for example, [55] p. 46.

Remark 16.22.1 Notice the special case for $m = n = 1$

$$-c\mathcal{G}_0(i | i') = \left\langle \psi_i \bar{\psi}_{i'} \right\rangle_0 = A_{i,i'}^{-1} \iff \mathcal{G}_0(i | i') = -\frac{1}{c} A_{i,i'}^{-1} \iff A_{i,i'} = -\frac{1}{c} \mathcal{G}_0^{-1}(i | i') \quad (16.29)$$

which allows us to write formally in the non-interacting case

$$Z_0 = \int d_\zeta(\bar{\psi}, \psi) \exp - \psi^\dagger \left(-\frac{1}{c} \mathcal{G}_0^{-1}\right) \psi \equiv \int d_\zeta(\bar{\psi}, \psi) \exp - S_0[\psi, \bar{\psi}] \quad (16.30)$$

Corollary 16.23 The above theorem implies an identity for the calculation of Green-functions as

$$\mathcal{G}_0(i_1, \dots, i_n | i'_1, \dots, i'_n) = \left(-\frac{1}{c}\right)^n \left\langle \psi_{i_1} \cdots \psi_{i_n} \bar{\psi}_{i'_1} \cdots \bar{\psi}_{i'_n} \right\rangle_0 = \sum_{\pi \in S_n} \text{sign}(\pi)^{1-\theta(\zeta)} \prod_{k=1}^n \mathcal{G}_0(i_k | i'_{\pi(k)}) \quad (16.31)$$

Proof 16.23 Using the definition, apparently gives the desired result.

Remark 16.23.1 Notice the reversed order in the second argument-series, in contrast to Eq. (16.27).

16.1.3. Partition-Functions and Expectation-Values for Interacting Models

Notation 16.24 Let $A \in \text{GL}_n(\mathbb{C})$ be an invertible, quadratic matrix, and $P[\bar{\psi}, \psi]$ an arbitrary⁵ polynomial expression in the integration-variables. For reasons that are impossible to clarify in advance, we consider the quantity

$$\begin{aligned} Z &\equiv \int d_\zeta(\bar{\psi}, \psi) \exp - (\psi^\dagger A \psi + P[\psi, \bar{\psi}]) \\ &\equiv \int d_\zeta(\bar{\psi}, \psi) \exp - (S_0[\psi, \bar{\psi}] + S_{\text{int}}[\psi, \bar{\psi}]) \\ &\equiv \int d_\zeta(\bar{\psi}, \psi) \exp - S[\psi, \bar{\psi}] \end{aligned}$$

for complex- or Graßmann-variables.

Notation 16.25 We consider a monomial expression in the integration-variables, and define its *expectation value w.r.t. S* as

$$\left\langle \psi_{i_1} \cdots \psi_{i_n} \bar{\psi}_{i'_1} \cdots \bar{\psi}_{i'_m} \right\rangle := \frac{1}{Z} \int d_\zeta(\bar{\psi}, \psi) \psi_{i_1} \cdots \psi_{i_n} \bar{\psi}_{i'_1} \cdots \bar{\psi}_{i'_m} \exp - S[\psi, \bar{\psi}] \quad (16.32)$$

Remark 16.25.1 Accordingly, one also introduces the *n-body-Green-function w.r.t. S* as expectation-value

$$\mathcal{G}(i_1, \dots, i_n | i'_n, \dots, i'_1) = \left(-\frac{1}{c}\right)^n \left\langle \psi_{i_1} \cdots \psi_{i_n} \bar{\psi}_{i'_1} \cdots \bar{\psi}_{i'_n} \right\rangle. \quad (16.33)$$

Remark 16.25.2 Notice that this allows us to write down a *formal* equation that is *exclusive* for one-body Green-functions

$$\mathcal{G}(i | i') \equiv -\frac{1}{c} \frac{1}{Z} \int d_\zeta(\bar{\psi}, \psi) \psi_i \bar{\psi}_{i'} \exp - \psi^\dagger \left(-\frac{1}{c} \mathcal{G}^{-1}\right) \psi. \quad (16.34)$$

Remark 16.25.3 Combining Equations (16.63) and Equation (16.34) yields that we can rewrite correlation functions as functional integrals over the so-called "quadratic part of the effective action $\text{Vert}(i, i') \equiv \text{Vert}$ " (which will be introduced later) as

$$\mathcal{G}(i, i') = -\frac{1}{c} \frac{\int \mathcal{D}(\eta, \bar{\eta}) \eta(i) \bar{\eta}(i') \exp - \eta^\dagger \left(-\frac{1}{c} \text{Vert}(:, :)\right) \eta}{\int \mathcal{D}(\eta, \bar{\eta}) \exp - \eta^\dagger \left(-\frac{1}{c} \text{Vert}(:, :)\right) \eta} \quad (16.35)$$

⁵Exception: For systems with fermionic statistics (i.e. $\zeta = -1$) we demand that P is even in the fields $\bar{\psi}$ and ψ , meaning that it only contains polynomials with equal number of Graßmann-variables and conjugated Graßmann-variables.

Fact 16.26 In general, there is no possibility to calculate this partition-function or these expectation-values w.r.t. S in closed-form. But it will turn out to be exactly these expectation-values that we need to compute in order to make predictions about the behaviour of non-interacting systems. There are several standard-techniques to evaluate integrals of this kind, for example

- Expanding in the interaction and using the so-called "Wick-theorem" (see below),
- performing a so-called "Hubbard-Stratanovich-transformation" (will be introduced later),
- applying a so-called "saddle-point approximation" (will be introduced later, works only for $\zeta = 1$).

Assuming strong-convergence⁶ one obtains for the expectation-value

$$\left\langle \psi_{i_1} \cdots \psi_{i_n} \bar{\psi}_{i'_1} \cdots \bar{\psi}_{i'_m} \right\rangle = \frac{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \left\langle \psi_{i_1} \cdots \psi_{i_n} \bar{\psi}_{i'_1} \cdots \bar{\psi}_{i'_m} \prod_{l=0}^k S_{\text{int}}[\psi, \bar{\psi}] \right\rangle_0}{\sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \left\langle \prod_{l=0}^k S_{\text{int}}[\psi, \bar{\psi}] \right\rangle_0} \quad (16.36)$$

For the first option (expanding terms) we will see that Feynman-diagrams built a system to efficiently organize the appearing terms.

Remark 16.26.1 For monomial expressions with $n = m = 1$, we introduced the self-energy-concept in order to simplify the computation of one-body Green-functions. For monomial expressions with $n = m = 2$, one can introduce the so-called "Bethe-Salpeter-equation" in order to simplify the computation of two-body Green-functions. For general cases, one can simplify this expression with the so-called linked-cluster-theorem, which leads to the expression in Theorem (15.10).

We conclude with a remark from the literature⁷

"One often needs to sum the perturbation expansion up to infinite order. This is done by identifying the most important subset of diagrams and including them in a self-energy. A warning is in place: the correct identification of the most important diagrams can be very tricky, even in the presence of a small parameter. The reason is that some high-order diagrams, which are not included in a particular model, even though formally smaller, may occur in such large numbers that they yield a contribution that must be included. Any perturbation theory may be just an asymptotic series and not converging in the strict mathematical sense. Finally, it is by no means clear that a "good" self-energy always even exists. Strongly interacting systems of this kind need a nonperturbative approach."

16.1.4. External Sources and Connected Green-Functions

Notation 16.27 We equip the partition-function Z in Notation 16.24 with external sources $j, \bar{j}$, by making the replacement

$$Z \mapsto Z[j, \bar{j}] \equiv \int d_{\zeta}(\bar{\psi}, \psi) \exp \left[- (S_0[\psi, \bar{\psi}] + S_{\text{int}}[\psi, \bar{\psi}]) + j^{\dagger} \psi + \bar{\psi}^{\dagger} j \right] \quad (16.37)$$

⁶In the sense, that integral and infinite series can be exchanged without altering the result.

⁷Cf. [21], p. 52.

Remark 16.27.1 We remark, that the sign with which the source-terms enter is convention, which is very error-prone. We emphasize therefore also that even *within* [35] one finds *different* prefactor-conventions, see for example pages 70, 75, 76 with the one convention, and then pages 105 and 107 for the other one.

Theorem 16.28 One finds in the non-interacting case

$$Z_0[j, \bar{j}] = \int d_\zeta(\bar{\psi}, \psi) \exp(-\psi^\dagger A \psi + j^\dagger \psi + \psi^\dagger j) = \frac{\exp(-j^\dagger A^{-1} j)}{\text{Det } A^\zeta} \quad (16.38)$$

Proof 16.28 For the bosonic case ($\zeta = +1$) see, for example, [35] page 22 or [5] page 104. For the fermionic case ($\zeta = -1$) see, for example, [35] page 34 or [5] page 165.

Remark 16.28.1 Notice, that this identity can be combined with Equation (16.29) such that one obtains

$$\begin{aligned} \int d_\zeta(\bar{\psi}, \psi) \exp\left(-\psi^\dagger \left(-\frac{1}{c} \mathcal{G}_0^{-1}\right) \psi + j^\dagger \psi + \psi^\dagger j\right) &= \text{Det} \left(-\frac{\mathcal{G}_0^\zeta}{c^{-1}} \right) \exp\left(-\frac{j^\dagger \mathcal{G}_0 j}{c^{-1}}\right) \\ &= \exp\left(-\zeta \text{Tr Log} \left(-\frac{1}{c} \mathcal{G}_0^{-1}\right) - \frac{j^\dagger \mathcal{G}_0 j}{c^{-1}}\right) \end{aligned}$$

whereas the individual Equation-signs can be advantageous in different situations.

Definition 16.29 We introduce the generating functional

$$\frac{Z[\bar{j}, j]}{Z} = \frac{1}{Z} \int \mathcal{D}_\zeta(\psi, \bar{\psi}) e^{-S[\psi, \bar{\psi}] + j^\dagger \psi + \psi^\dagger j} = \left\langle \exp\left[j^\dagger \psi + \psi^\dagger j\right] \right\rangle \quad (16.39)$$

Definition 16.30 We also introduce the *analogue of the Helmholtz free energy / Landau-potential*⁸

$$W[\bar{j}, j] = + \log Z[\bar{j}, j] \quad (16.40)$$

Remark 16.30.1 Notice that this is at zero external sources proportional (with proportionality constant $k_B T$) to the natural energy (a.k.a. grand canonical potential for $\eta = 1$ and free energy for $\eta = 0$) of a system; Cf. Fact 10.4.

Remark 16.30.2 In the literature [37] this quantity is known as "generating functional for connected correlation functions". In the following we will elaborate on what this exactly means. The is also the reason why one omits the factor $k_B T$: This quantity is mostly considered to be an algebraic object, rather than a thermodynamic.

⁸Notice that we are using a different sign-convention relative to [25].

Notation 16.31 We introduce the *field expectation values*

$$\begin{aligned}\phi[\bullet, j, \bar{j}] &= \langle \psi(\bullet) \rangle_{j, \bar{j}} = 1 \frac{\partial}{\partial \bar{j}(\bullet)} W[j, \bar{j}], \\ \bar{\phi}[\bullet, j, \bar{j}] &= \langle \bar{\psi}(\bullet) \rangle_{j, \bar{j}} = \zeta \frac{\partial}{\partial j(\bullet)} W[j, \bar{j}].\end{aligned}\tag{16.41}$$

Remark 16.31.1 For $j, \bar{j} \neq 0$ it follows $\phi, \bar{\phi} \stackrel{\text{in general}}{\neq} 0$.

Lemma 16.32 In the absence of spontaneous symmetry breaking, the field-expectation values $\phi, \bar{\phi}$ are functionals of the sources $j, \bar{j}$, and they are made up of monomials that are *odd* in the sources $j, \bar{j}$.

Proof 16.32 See Appendix, Proofs, page 469.

Lemma 16.33 The mapping $(j, \bar{j}) \mapsto (\phi, \bar{\phi})$ is *invertible*, in the sense that we can find expansions

$$j[\bullet, \phi', \bar{\phi}'], \quad \bar{j}[\bullet, \phi', \bar{\phi}']\tag{16.42}$$

such that

$$\phi\left[\bullet, j[\bullet, \phi', \bar{\phi}'], \bar{j}[\bullet, \phi', \bar{\phi}']\right] = \phi', \quad \bar{\phi}\left[\bullet, j[\bullet, \phi', \bar{\phi}'], \bar{j}[\bullet, \phi', \bar{\phi}']\right] = \bar{\phi}.\tag{16.43}$$

Proof 16.33 See Appendix, Proofs, page 470.

Remark 16.33.1 Notice that in the construction of the inverse, we realized that the expansions for the sources is *odd* in the fields ϕ and $\bar{\phi}$, which implies that the derivatives (w.r.t. to the ϕ -fields) are even in the fields and commute with the fields. This is a crucial observation for the effective action.

Intermezzo: Vectors with Graßmann-Entries and Chain-Rule

Remark 16.33.2

- **Definitions ...** of a module over a Graßmann-algebra $V_{\mathcal{P}_\zeta} = V \otimes \mathcal{P}_\zeta$
- **Derivatives ...** at a point $X \in V_{\mathcal{P}_\zeta}$ are defined as linear maps $dF_X : V_{\mathcal{P}_\zeta} \rightarrow V_{\mathcal{P}_\zeta}$, so derivatives live in $\text{Hom}_{\mathcal{P}_\zeta}(V_{\mathcal{P}_\zeta}, V_{\mathcal{P}_\zeta})$

Notation 16.34 With the assumption that the mapping $(j, \bar{j}) \mapsto (\phi, \bar{\phi})$ is invertible, and with suppression of the dependencies $j \equiv j[\bullet, \phi, \bar{\phi}]$, we write the fermionic effective action (being the analog of the Gibbs-potential) as a Legendre-transformation of W

$$\Gamma[\phi, \bar{\phi}] = -W[j, \bar{j}] + \phi^\dagger j + j^\dagger \phi. \quad (16.44)$$

Remark 16.34.1 For the Legendre-transformation, there is a diversity in signs throughout the literature... Basically every author seems to be using a different convention.

Lemma 16.35 One immediately concludes that

$$\begin{aligned} \frac{\partial}{\partial \bar{\phi}(\bullet)} \Gamma[\phi, \bar{\phi}] &= 1j[\bullet, \phi, \bar{\phi}], \\ \frac{\partial}{\partial \phi(\bullet)} \Gamma[\phi, \bar{\phi}] &= \zeta \bar{j}[\bullet, \phi, \bar{\phi}]. \end{aligned}$$

Proof 16.35.0 See Appendix, Proofs, page 474.

16.1.5. Series-Expansions

Notation 16.36 After one has established an order relation⁹ for the generalized indices, one introduces the series-expansions for the functionals

$$\begin{aligned} F[\pi, \bar{\pi}] &\equiv \sum_{n, m=0}^{\infty} \sum_{\substack{i_1 < \dots < i_n \\ i'_1 < i'_2 < \dots < i'_m}} \bar{\pi}(i_1) \cdots \bar{\pi}(i_n) f(i_1, \dots, i_n, i'_1, \dots, i'_m) \pi(i'_m) \cdots \pi(i'_1) \\ &= \sum_{n, m=0}^{\infty} \frac{1}{n! m!} \sum_{\substack{i_1, \dots, i_n \\ i'_1, i'_2, \dots, i'_m}} \bar{\pi}(i_1) \cdots \bar{\pi}(i_n) f(i_1, \dots, i_n, i'_1, \dots, i'_m) \pi(i'_m) \cdots \pi(i'_1) \end{aligned} \quad (16.45)$$

with complex-valued coefficients $f(i_1, \dots, i_n, i'_1, \dots, i'_m) \in \mathbb{C}$, such that

$$f(i_{\pi(1)}, \dots, i_{\pi(n)}, i'_{\sigma(1)}, \dots, i'_{\sigma(m)}) = \text{sgn}(\pi)^{1-\theta(\zeta)} \text{sgn}(\sigma)^{1-\theta(\zeta)} f(i_1, \dots, i_n, i'_1, \dots, i'_m)$$

for $(\pi, \sigma) \in S_n \times S_m$.

Remark 16.36.1 Later we will also encounter the situation that the coefficients f take values in an algebra. In this case one has to pay special attention to differentiation-rules **TBElaborated**.

⁹To avoid double counting.

Remark 16.36.2 Sometimes¹⁰ one might find oneself in the situation that one has access to a series-expansion of the type

$$F[\pi, \bar{\pi}] \equiv \sum_{n,m=0}^{\infty} \sum_{\substack{i_1, \dots, i_n \\ i'_1, i'_2, \dots, i'_m}} \bar{\pi}(i_1) \cdots \bar{\pi}(i_n) \tilde{f}(i_1, \dots, i_n, i'_1, \dots, i'_m) \pi(i'_m) \cdots \pi(i'_1) \quad (16.46)$$

with complex-valued coefficients $\tilde{f}(i_1, \dots, i_n, i'_1, \dots, i'_m) \in \mathbb{C}$ that do not obey any symmetry properties. In this case, one has to identify the coefficients f in the expansion in Equation (16.45) as the properly antisymmetrized contributions.

Remark 16.36.3 Sometimes (but not in this work ;-)) the prefactors $(n!m!)^{-1}$ are included in the vertex-functions, such that the definition changes slightly.

Remark 16.36.4 One notates the individual coefficients depending on the functional of consideration, whereas the exact meaning of the word "generate" will be elucidated below.

Functional F	Fields $\pi, \bar{\pi}$	Coefficients f	Purpose
$S[\psi, \bar{\psi}]$	$\psi, \bar{\psi}$	s	Defines the System
$Z[j, \bar{j}]/Z$	$j, \bar{j}$	z	"Generates" Green-functions
$W[j, \bar{j}]$	$j, \bar{j}$	w	"Generates" connected Green-functions
$\Gamma[\phi, \bar{\phi}]$	$\phi, \bar{\phi}$	γ	"Generates" vertex-functions

Table 16.1.: Summary of the functional expansions and their roles.

Lemma 16.37 For $n = m = 2$ one obtains f from $\tilde{f}$ as

$$f_{1,2,3,4} = \frac{1}{4} \left(\tilde{f}_{1,2,3,4} + \zeta \tilde{f}_{2,1,3,4} + \zeta \tilde{f}_{1,2,4,3} + \tilde{f}_{2,1,4,3} \right) \quad (16.47)$$

Proof 16.37 See Appendix, Proofs, page 472.

Remark 16.37.1 Notice that we thus obtain only one surviving contribution is for $\zeta = +1$ given by $f^{\text{sym, sym}}$ and for $\zeta = -1$ it is $f^{\text{as, as}}$.

Fact 16.38 Without proof, we inform the reader that one can apply the same symmetrization-scheme for $n = m = 3$ as

¹⁰For example, in the later chapter on electron-phonon-systems.

$$f_{1,2,3,4,5,6} = \left(\frac{1}{3!}\right)^2 \sum_{\sigma, \tau \in S_3} [\text{sgn}(\sigma) \text{sgn}(\tau)]^{\frac{1-\zeta}{2}} \tilde{f}_{\sigma(1), \sigma(2), \sigma(3), \tau(4), \tau(5), \tau(6)} \quad (16.48)$$

and generalizations of this formula for the general case.

Lemma 16.39 Notice that we obtain the identity

$$f(i_1, \dots, i_n | i'_1, \dots, i'_m) = \zeta^{nm+n(n-1)/2+m(m-1)/2} \frac{\partial^{n+m}}{\partial \bar{\pi}(i_1) \dots \partial \bar{\pi}(i_n) \partial \pi(i'_1) \dots \partial \pi(i'_m)} F[\bar{\pi}, \pi] \Big|_{\bar{\pi}=\pi=0} \quad (16.49)$$

Proof 16.39 See Appendix, Proofs, page 473.

Remark 16.39.1 Notice that in the special case that $m = n$ one obtains

$$\zeta^{nm+n(n-1)/2+m(m-1)/2} = \zeta^{n^2+n(n-1)} = \zeta^{n^2} = \zeta^n \quad (16.50)$$

whereas we used in the second last step that the number $n(n-1)$ is even and in the last step, we used that n is even / odd if n^2 is even / odd, which is something that one learns in the first semester of mathematics (usually when one proves that $\sqrt{2}$ is irrational). Thus one finds

$$f(i_1, \dots, i_n | i'_1, \dots, i'_n) = \zeta^n \frac{\partial^{2n}}{\partial \bar{\pi}(i_1) \dots \partial \bar{\pi}(i_n) \partial \pi(i'_1) \dots \partial \pi(i'_n)} F[\bar{\pi}, \pi] \Big|_{\bar{\pi}=\pi=0} \quad (16.51)$$

Lemma 16.40 The n -body Green-function as functional derivative

$$\mathcal{G}(i_1, \dots, i_n | i'_1, \dots, i'_n) = \left(-\frac{\zeta}{c}\right)^n \frac{\partial^{2n}}{\partial \bar{j}(i_1) \dots \partial \bar{j}(i_n) \partial j(i'_1) \dots \partial j(i'_n)} \frac{Z[\bar{j}, j]}{Z} \Big|_{\bar{j}=j=0} \quad (16.52)$$

Proof 16.40 See Appendix, Proofs, page 474.

Remark 16.40.1 One obtains thus

$$\mathcal{G}(i_1, \dots, i_n | i'_1, \dots, i'_n) = \left(-\frac{1}{c}\right)^n z(i_1, \dots, i_n | i'_1, \dots, i'_n) \quad (16.53)$$

and one says that the functional $Z[\bar{j}, j]/Z$ generates the Green-functions, since it's expansion coefficients are equal to the Green-functions.

Definition 16.41 We introduce the *[c]onnected Green-functions* as

$$\mathcal{G}_c(i_1, \dots, i_n | i'_1, \dots, i'_n) := \left(-\frac{\zeta}{\hbar} \right)^n \frac{\partial^{2n}}{\partial \bar{j}(i_1) \dots \partial \bar{j}(i_n) \partial j(i'_1) \dots \partial j(i'_n)} W[\bar{j}, j] \Big|_{\bar{j}=j=0} \quad (16.54)$$

Remark 16.41.1 The fact that this functional generates all connected diagrams is far from trivial. We will come back to this statement in a later chapter.¹¹

Remark 16.41.2 One obtains thus

$$\mathcal{G}_c(i_1, \dots, i_n | i'_1, \dots, i'_n) = \left(-\frac{1}{\hbar} \right)^n w(i_1, \dots, i_n | i'_1, \dots, i'_n) \quad (16.55)$$

and one says that the functional $W[\bar{j}, j]$ *generates* the *connect* Green-functions, since it's expansion coefficients are equal to the Green-functions.

Definition 16.42 We introduce the *vertex-functions* Vert by taking successive derivatives of Γ at the point $\phi = \bar{\phi} = 0$ as

$$\text{Vert}(i_1, \dots, i_n, i'_1, \dots, i'_n) := \left(-\frac{\hbar}{\zeta} \right)^n \frac{\partial^{2n}}{\partial \bar{\phi}(i_1) \dots \partial \bar{\phi}(i_n) \partial \phi(i'_1) \dots \partial \phi(i'_n)} \Gamma[\bar{\phi}, \phi] \Big|_{\bar{\phi}=\phi=0} \quad (16.56)$$

Remark 16.42.1 Notice that this implies for the coefficients

$$\text{Vert}(i_1, \dots, i_n, i'_1, \dots, i'_n) = (-\hbar)^n \gamma(i_1, \dots, i_n, i'_1, \dots, i'_n) \quad (16.57)$$

Remark 16.42.2 The particular vertex-function $\gamma(i_1, i_2, i'_1, i'_2)$ is oftentimes [27, 5] referred to as the *effective interaction*. It is said to ...

"describe the true interaction between two particles in the many-body system".

This is however irritating because it doesn't necessarily have the unit of an energy. Only after appropriate rescaling / change of basis can this vertex-function be interpreted as an interaction (with unit of energy). We will come back to this point in a later chapter.

Remark 16.42.3 One also says that the functional $\Gamma[\phi, \bar{\phi}]$ *generates* the vertex-functions.

¹¹There is a lot about this to read in the literature, with no book proving the identity in the generality in which its circle of applicability is being claimed.

- See, for example, [5] p. 215 for an instruction-manual on how to construct the diagrams.
- See, for example, [35] p. 79 for control-result in first order.
- See, for example [35] p. 97 for the statement itself.
- See, for example [35] p. 107 for more information.

Corollary 16.43 It holds

$$\begin{aligned}\mathcal{G}_c(i_1 | i'_1) &= \mathcal{G}(i_1 | i'_1) \\ \mathcal{G}_c(i_1, i_2 | i'_1, i'_2) &= \mathcal{G}(i_1, i_2 | i'_1, i'_2) - \left[\mathcal{G}(i_1 | i'_1) \mathcal{G}(i_2 | i'_2) + \zeta \mathcal{G}(i_1 | i'_2) \mathcal{G}(i_2 | i'_1) \right]\end{aligned}\quad (16.58)$$

Proof 16.43 See Appendix, Proofs, page 476.

Lemma 16.44 One finds for a functional $F \equiv F[\bar{j}, j]$ the chain-rule

$$\begin{aligned}\frac{\partial}{\partial \phi(\bullet)} F[\bar{j}, j] &= \sum_i \zeta \frac{\partial F}{\partial \bar{j}(i)} \frac{\partial^2 \Gamma}{\partial \phi(\bullet) \partial \phi(i)} + \frac{\partial F}{\partial j(i)} \frac{\partial^2 \Gamma}{\partial \phi(\bullet) \partial \bar{\phi}(i)} \\ \frac{\partial}{\partial \bar{\phi}(\bullet)} F[\bar{j}, j] &= \sum_i \zeta \frac{\partial F}{\partial \bar{j}(i)} \frac{\partial^2 \Gamma}{\partial \bar{\phi}(\bullet) \partial \phi(i)} + \frac{\partial F}{\partial j(i)} \frac{\partial^2 \Gamma}{\partial \bar{\phi}(\bullet) \partial \bar{\phi}(i)}\end{aligned}\quad (16.59)$$

Proof 16.44 See Appendix, Proofs, page 476.

Corollary 16.45 It follows that

$$\sum_{i_2} \begin{pmatrix} \frac{\partial^2 W}{\partial \bar{j}(i_3) \partial j(i_2)} & \zeta \frac{\partial^2 W}{\partial \bar{j}(i_3) \partial \bar{j}(i_2)} \\ \zeta \frac{\partial^2 W}{\partial j(i_3) \partial j(i_2)} & \frac{\partial^2 W}{\partial j(i_3) \partial \bar{j}(i_2)} \end{pmatrix} \begin{pmatrix} \frac{\partial^2 \Gamma}{\partial \bar{\phi}(i_2) \partial \phi(i_1)} & \frac{\partial^2 \Gamma}{\partial \bar{\phi}(i_2) \partial \bar{\phi}(i_1)} \\ \frac{\partial^2 \Gamma}{\partial \phi(i_2) \partial \phi(i_1)} & \frac{\partial^2 \Gamma}{\partial \phi(i_2) \partial \bar{\phi}(i_1)} \end{pmatrix} = \delta_{i_1, i_3} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (16.60)$$

Proof 16.45 See Appendix, Proofs, page 477.

Lemma 16.46 In the absense of "spontaneous symmetry breaking"¹², we can write down the quadratic part of the effective action $\text{Vert}(:, :)$ by simple inversion of a matrix:

$$\text{Vert}(:, :) = \mathcal{G}_c^{-1}(:, :) = \mathcal{G}^{-1}(:, :) \quad (16.61)$$

Proof 16.46 Using that the effective action has the same symmetries as the original action (cf. TBReferenced) we obtain from Equation (32.41) the simplified identity

$$\delta_{1,3} = \sum_2 \left(-\frac{c}{\zeta} \mathcal{G}(2|3) \left(-\frac{\zeta}{c} \frac{\partial^2 \Gamma}{\partial \bar{\phi}(1) \partial \phi(2)} \right) \right) = \sum_2 \mathcal{G}(2|3) \text{Vert}(1|2) = \sum_2 \text{Vert}(1|2) \mathcal{G}(2|3) \quad (16.62)$$

¹²This will be introduced later.

Thus we obtain

$$\left[\text{Vert}(:, :) \cdot \mathcal{G}(:, :) \right]_{n,m} = \delta_{n,m} \iff \text{Vert} = \mathcal{G}^{-1} \quad (16.63)$$

which yields in matrix-notation the desired result.

Remark 16.46.1 Notice that from here, we can easily obtain the generic version of the Dyson-equation (cf. Equation (15.16))

$$\mathcal{G}^{-1} = \mathcal{G}_0^{-1} - \Sigma \quad (16.64)$$

by introducing the self-energy Σ as the difference between the inverse of the free Green-function and the inverse of the full Green-function. This result alone is already quite nice, but we will see (in the next chapter) that one can conclude the explicit shape of the self-energy as well from symmetry-principles which will be quite a revealing moment for the interested reader.

Remark 16.46.2 One can¹³ also rewrite this equation in a more general way for system that have a broken symmetry, but we will not do this here for pedagogical purposes.

Theorem 16.47 In the absence of spontaneous symmetry breaking, another useful identity¹⁴. is

$$\mathcal{G}_c(i_1, i_2 | i'_1, i'_2) = - \sum_{l_1, l_2, l'_1, l'_2} \mathcal{G}_c(i_1 | l_1) \mathcal{G}_c(i_2 | l_2) \text{Vert}(l_1, l_2 | l'_1, l'_2) \mathcal{G}_c(l'_1 | i'_1) \mathcal{G}_c(l'_2 | i'_2) \quad (16.65)$$

Proof 16.47 See Appendix, Proofs, page 477.

Remark 16.47.1 In a later chapter we will see how this can be rewritten diagrammatically¹⁵.

Theorem 16.48 In the absence of spontaneous symmetry breaking, one concludes the *Bethe-Salpeter-equation* for the two-body Green-function as

$$\begin{aligned} \mathcal{G}(i_1, i_2 | i'_1, i'_2) &= \mathcal{G}(i_1 | i'_1) \mathcal{G}(i_2 | i'_2) + \zeta \mathcal{G}(i_1 | i'_2) \mathcal{G}(i_2 | i'_1) \\ &\quad - \sum_{l_1, l_2, l'_1, l'_2} \mathcal{G}(i_1 | l_1) \mathcal{G}(i_2 | l_2) \text{Vert}(l_1, l_2 | l'_1, l'_2) \mathcal{G}(l'_1 | i'_1) \mathcal{G}(l'_2 | i'_2) \end{aligned} \quad (16.66)$$

¹³Cf. [35] Equation (7.52).

¹⁴Other identities that are relevant in this context are given in the literature. Cf. for example

- [35]. Negele p. 108.
- [35] Equation (6.79 a,b) Identity for quartic green-function as correlation-functions
- For the relations between $\mathcal{G}^{(6)}$ and $\Gamma^{(6)}$: see, for example, [33] Equation (12).

¹⁵See also [35] p. 118.

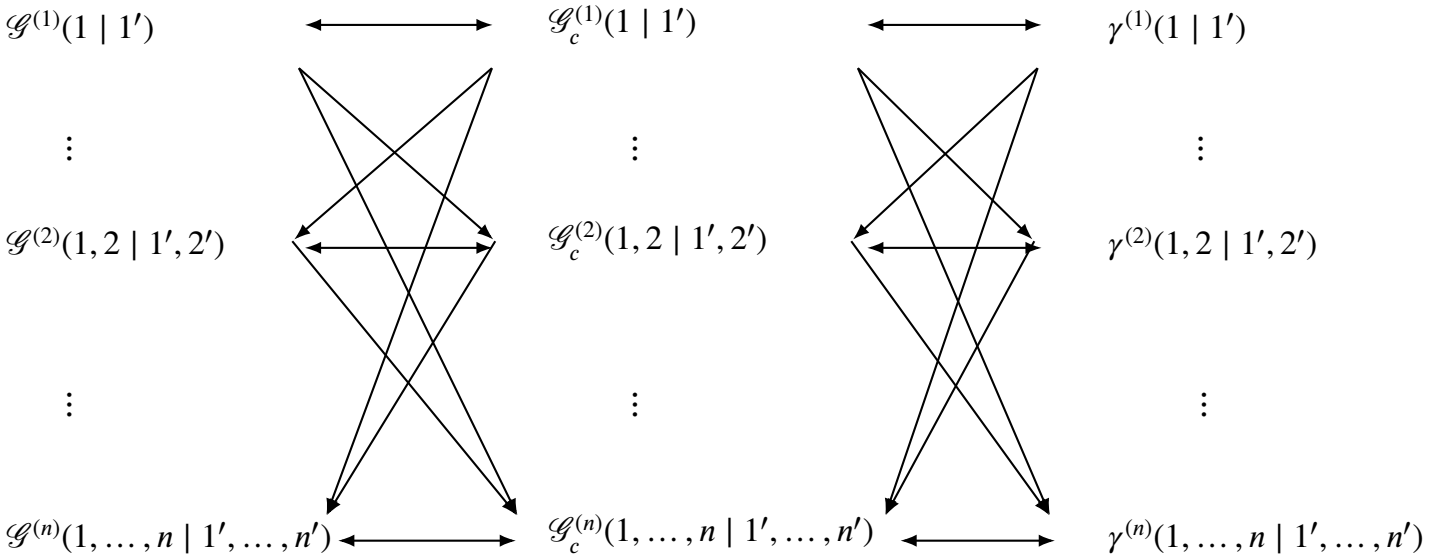
Proof 16.48 Combining Corollary 16.43 and Theorem 16.47 and rearranging terms gives the desired result.

Hypothesis 16.49 With induction, one sees that the following three problems are equivalent, in the sense that if one of them is solved, the other two are solved as well:

- *Finding all many-body Green-Functions* $\mathcal{G}^{(1)}(i_1, i'_1), \mathcal{G}^{(2)}(i_1, i_2, i'_1, i'_2), \dots, \mathcal{G}^{(n)}(i_1, \dots, i_n, i'_1, \dots, i'_n)$
- *Finding all connected many-body Green-Functions* $\mathcal{G}_c^{(1)}(i_1, i'_1), \mathcal{G}_c^{(2)}(i_1, i_2, i'_1, i'_2), \dots, \mathcal{G}_c^{(n)}(i_1, \dots, i_n, i'_1, \dots, i'_n)$
- *Finding all many-body vertex-Functions* $\gamma^{(1)}(i_1, i'_1), \gamma^{(2)}(i_1, i_2, i'_1, i'_2), \dots, \gamma^{(n)}(i_1, \dots, i_n, i'_1, \dots, i'_n)$

Remark 16.49.1 For the lowest orders, this follows from the Equations derived in this section. It is a standard-folklore-statement that this is a general theorem (see, for example, [35]).

Remark 16.49.2 The picture visualizes the equivalence of the three problems.



16.1.6. Change of Coordinates

Notation 16.50 We consider an invertible linear mapping M (with complex conjugate $\overline{M}$) between integration-variables („coordinate-change“)

$$\psi_i \mapsto (M \cdot \psi)_i \equiv \sum_{i'} M_{i,i'} \psi_{i'}, \quad \overline{\psi}_i \mapsto (\overline{M} \cdot \overline{\psi})_i \equiv \sum_{i'} \overline{M}_{i,i'} \overline{\psi}_{i'}$$

Remark 16.50.1 Notice that in the notation of physicists, this would be written as

$$\psi'_i = \psi'_i(\psi) = \sum_i M_{i',i} \psi_i, \quad \overline{\psi}'_i = \overline{\psi}'_i(\psi) = \sum_i \overline{M}_{i',i} \overline{\psi}_i \quad (16.67)$$

Remark 16.50.2 Notice that this mapping changes a monomial expression in the integration-variables according to

$$\psi_{i_1} \cdots \psi_{i_n} \bar{\psi}_{i'_1} \cdots \bar{\psi}_{i'_m} \mapsto \sum_{\substack{j_1, \dots, j_n \\ j'_1, j'_2, \dots, j'_{m-1}, j'_m}} \left(\prod_{l=1}^n M_{i_l, j_l} \right) \left(\prod_{l=1}^m \bar{M}_{i'_l, j'_l} \right) \psi_{j_1} \cdots \psi_{j_n} \bar{\psi}_{j'_1} \cdots \bar{\psi}_{j'_m} \quad (16.68)$$

Remark 16.50.3 The most important transformation will turn out to be the Fourier-series. It removes the derivatives (if one concatenates the variable-transformation with a standard-approximation) s.t. we can simply invert the matrix appearing in the action.

Lemma 16.51 Changing the coordinates according to Equation (16.67) yields for the derivatives

$$\frac{\partial}{\partial \psi'_i} = \sum_i \frac{\partial \psi_i}{\partial \psi'_i} \frac{\partial}{\partial \psi_i}, \quad \frac{\partial}{\partial \bar{\psi}'_i} = \sum_i \frac{\partial \bar{\psi}_i}{\partial \bar{\psi}'_i} \frac{\partial}{\partial \bar{\psi}_i} \quad (16.69)$$

Proof 16.51 See, for example, [54] p. 11.

Lemma 16.52 Changing the coordinates according to Equation (16.67) changes the integration-measure as

$$d_\zeta(\bar{\psi}', \psi') \equiv \text{Det} \left(\frac{\partial \psi'_i}{\partial \psi_j} \right)^\zeta \text{Det} \left(\frac{\partial \bar{\psi}'_i}{\partial \bar{\psi}_j} \right)^\zeta d_\zeta(\bar{\psi}, \psi) \quad (16.70)$$

Proof 16.52 TBDone.

Theorem 16.53 Assuming that one has statistical variables

$$\phi = \phi(\eta, \bar{\eta}), \quad \bar{\phi} = \bar{\phi}(\eta, \bar{\eta}) \quad (16.71)$$

we introduce the jacobian J as

$$J = \frac{\partial(\phi, \bar{\phi})}{\partial(\eta, \bar{\eta})} \quad (16.72)$$

then we obtain the *transformation-formula* as

$$\int d(\bar{\phi}, \phi) F(\bar{\phi}, \phi) = (\det J)^\zeta \int d(\bar{\eta}, \eta) F(\bar{\phi}(\bar{\eta}, \eta), \phi(\bar{\eta}, \eta)) \quad (16.73)$$

Proof 16.53 See in the literature [TBReferenced]

Corollary 16.54 Emphasizing the role of the matrices A and $M(A) \equiv M^\dagger A M$ into the notation, we obtain the important result

$$\langle \psi_{i_1} \cdots \psi_{i_n} \bar{\psi}_{i'_1} \cdots \bar{\psi}_{i'_m} \rangle_\zeta[A] = \sum_{\substack{j_1, \dots, j_n \\ j'_1, j'_2, \dots, j'_{m-1}, j'_m}} \left(\prod_{l=1}^n M_{i_l, j_l} \right) \left(\prod_{l=1}^m \bar{M}_{i'_l, j'_l} \right) \langle \psi_{j_1} \cdots \psi_{j_n} \bar{\psi}_{j'_1} \cdots \bar{\psi}_{j'_m} \rangle_\zeta[M(A)]$$

Proof 16.54 See Appendix, Proofs, page ??.

Using this mapping as transformation-matrix, we obtain for the partition-function the scaling-factor

$$Z_\zeta[A] \equiv \int d_\zeta(\bar{\psi}, \psi) e^{-\psi^\dagger A \psi} \equiv \text{Det } M^{-\zeta} \int d_\zeta(\bar{\psi}, \psi) e^{-\psi^\dagger M^\dagger A M \psi} \equiv \text{Det } M^{-\zeta} \cdot Z_\zeta[\tilde{A}]$$

and equally we obtain another scaling-factor for the numerator in the expression of the expectation-value.

Remark 16.54.1 Notice that in the case of a simple transformation as

$$\begin{pmatrix} \phi \\ \bar{\phi} \end{pmatrix} = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} \begin{pmatrix} \eta \\ \bar{\eta} \end{pmatrix} \quad (16.74)$$

such that one has

$$\det J = \det A \det B. \quad (16.75)$$

16.1.7. The Tree-Level-Approximation for the Effective Action

Approximation 16.55 In the so-called Tree-Level-Approximation, one obtains for the fermionic effective action

$$\Gamma[\phi, \bar{\phi}] \approx S[\phi, \bar{\phi}]. \quad (16.76)$$

Proof 16.55 See Appendix, Proofs, page 482.

Remark 16.55.1 This approximation is usually depicted with Feynman-diagrams as in the figure

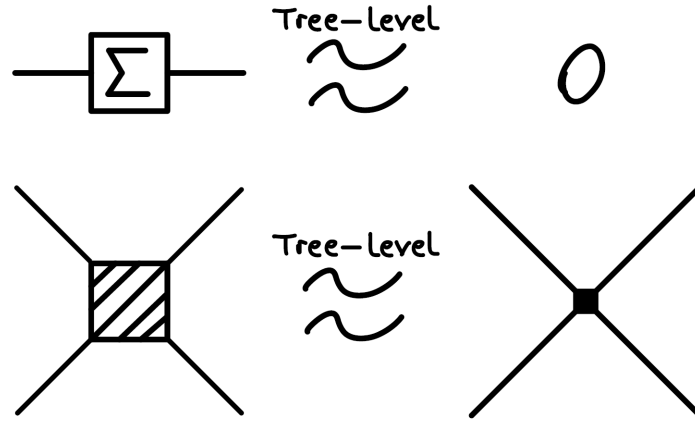


Figure 16.1.: Feynman-Diagrams (schematically) for the Tree-Level-Approximation.

Corollary 16.56 Within the Tree-Level-Approximation, equations (16.44) and (16.76) can be combined, which gives

$$\frac{\partial}{\partial \phi} S[\phi, \bar{\phi}] = \bar{j} \quad \text{and} \quad \frac{\partial}{\partial \bar{\phi}} S[\phi, \bar{\phi}] = -j. \quad (16.77)$$

Proof 16.56 Follows immediately from the approximation and Remark 16.35.

16.1.8. The One-Loop-Approximation for the Effective Action

Approximation 16.57 In the so-called One-Loop-Approximation, one obtains for the fermionic effective action

$$\Gamma[\phi, \bar{\phi}] \approx S[\phi, \bar{\phi}] - \frac{1}{2} \text{Tr} \log \begin{pmatrix} \frac{\partial^2}{\partial \phi \partial \phi} S[\phi, \bar{\phi}] & \frac{\partial^2}{\partial \phi \partial \bar{\phi}} S[\phi, \bar{\phi}] \\ \frac{\partial^2}{\partial \bar{\phi} \partial \phi} S[\phi, \bar{\phi}] & \frac{\partial^2}{\partial \bar{\phi} \partial \bar{\phi}} S[\phi, \bar{\phi}] \end{pmatrix}, \quad (16.78)$$

where

$$\left(\frac{\partial^2}{\partial \phi \partial \phi} S[\phi, \bar{\phi}] \right) \equiv \left(\frac{\partial^2}{\partial \phi \partial \phi} S[\phi, \bar{\phi}] \right) (i, j) \equiv \frac{\partial^2}{\partial \phi(i) \partial \phi(j)} S[\phi, \bar{\phi}], \quad \text{etc.}$$

Proof 16.57 See Appendix, Proofs, page 484.

Approximation 16.58 Expanding the r.h.s. of eq. (16.78) up to quartic order in the Grassmann-fields and neglecting higher order terms yields

$$\begin{aligned}
 \Gamma[\phi, \bar{\phi}] &\approx S[\phi, \bar{\phi}] - \text{Tr} \left[G_0 \cdot \frac{\partial^2}{\partial \bar{\phi} \partial \phi} S_{\text{int}}[\phi, \bar{\phi}] \right] \\
 &- \frac{1}{2} \text{Tr} \left[G_0 \cdot \frac{\partial^2}{\partial \bar{\phi} \partial \phi} S_{\text{int}}[\phi, \bar{\phi}] \cdot G_0^T \cdot \frac{\partial^2}{\partial \phi \partial \bar{\phi}} S_{\text{int}}[\phi, \bar{\phi}] \right] \\
 &+ \frac{1}{2} \text{Tr} \left[G_0 \cdot \frac{\partial^2}{\partial \bar{\phi} \partial \phi} S_{\text{int}}[\phi, \bar{\phi}] \cdot G_0 \cdot \frac{\partial^2}{\partial \phi \partial \bar{\phi}} S_{\text{int}}[\phi, \bar{\phi}] \right]
 \end{aligned} \tag{16.79}$$

Proof 16.58 See Appendix, Proofs, page 485.

Remark 16.58.1 This approximation is usually depicted with Feynman-diagrams as in the figure. The corrections to the Tree-Level contribution of the effective interaction $\gamma(x_1, x_2, x_3, x_4)$ are the particle-particle- and particle-hole-diagrams respectively.

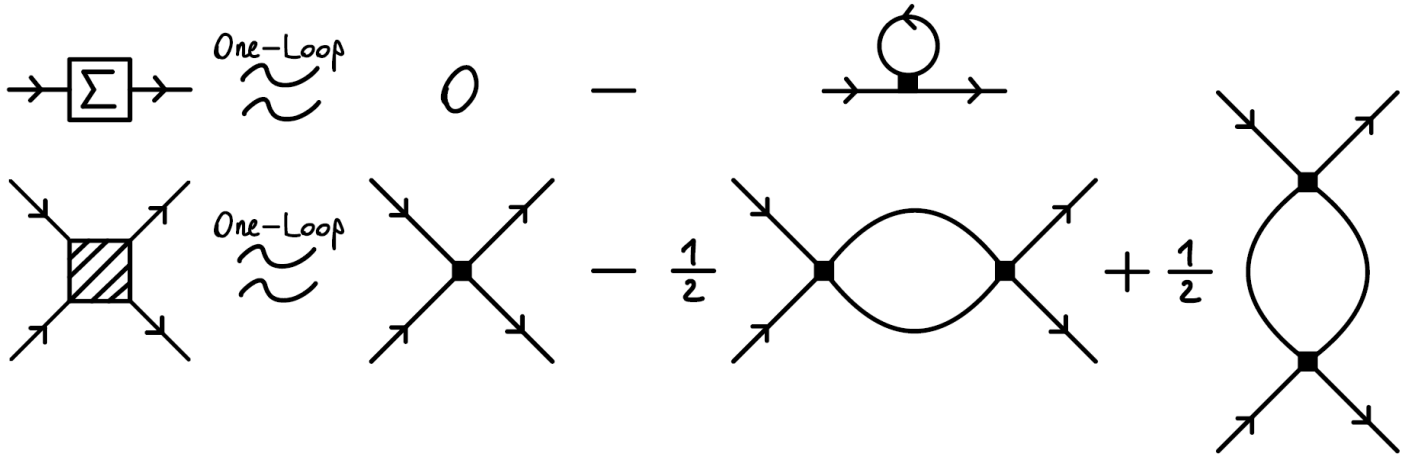


Figure 16.2.: Feynman-Diagrams (schematically) for the One-Loop-Approximation.

16.2. Quantum Field Theory - Mixed Systems

16.2.1. Fundamentals

TBCopied (a lot from Zinn-Justin).

16.2.2. Partition-Functions and Expectation-Values for Non-Interacting Models

TBD

16.2.3. Partition-Functions and Expectation-Values for Interacting Models

TBD

16.2.4. Gauss-Identities

TBD

16.2.5. External Sources, Connected Green-Functions and Vertex-Functions

TBD

16.3. Functional Renormalization

16.3.1. Functional Differential Equation for the Free Energy

16.3.2. Functional Differential Equation for the Effective Action

16.3.3. An Example for Flow-Equations

To Be Copied (and generalized for self-energy feedback) from [25]

17. Symmetry-Transformations

17.1. Pure Systems

17.1.1. Introduction

In this section we will see how symmetry considerations will constrain the shape of the coefficients γ .

Definition 17.1 We call an invertible linear mapping M (with complex conjugate $\overline{M}$) between Graßmann-fields

$$\psi(x) \mapsto (M \cdot \psi)(i) \equiv \sum_j M(i, j) \psi(j), \quad \overline{\psi}(i) \mapsto (\overline{M} \cdot \overline{\psi})(i) \equiv \sum_j \overline{M}(i, j) \overline{\psi}(j)$$

a *symmetry transformation* if both, the integration measure and the action remain unchanged under the action of the mapping M (and $\overline{M}$):

$$\mathcal{D}(\psi, \overline{\psi}) = \mathcal{D}(M \cdot \psi, \overline{M} \cdot \overline{\psi}) \quad \text{and} \quad S[\psi, \overline{\psi}] = S[M \cdot \psi, \overline{M} \cdot \overline{\psi}]. \quad (17.1)$$

Remark 17.1.1 The set of all symmetry transformations, equipped with composition as binary-operation, is a group. For each action and integration measure one refers to this group as the “symmetry group” of the problem [27].

Lemma 17.2 For a symmetry transformation M one has

$$W[j, \overline{j}] = W[\overline{M}^T \cdot j, M^T \cdot \overline{j}]. \quad (17.2)$$

implying the equations

$$\begin{aligned} \phi[\bullet, \overline{M}^T \cdot j, M^T \cdot \overline{j}] &= (M^{-1} \cdot \phi)[\bullet, j, \overline{j}], \\ \overline{\phi}[\bullet, \overline{M}^T \cdot j, M^T \cdot \overline{j}] &= (\overline{M}^{-1} \cdot \overline{\phi})[\bullet, j, \overline{j}]. \end{aligned}$$

Proof 17.2 See Appendix, Proofs, page 487.

Theorem 17.3 With the above preparations one obtains for a symmetry transformation M the equation

$$\Gamma[\phi, \bar{\phi}] = \Gamma[M^{-1} \cdot \phi, \bar{M}^{-1} \cdot \bar{\phi}]. \quad (17.3)$$

Proof 17.3 See Appendix, Proofs, page 488.

Remark 17.3.1 Thus, by equation (17.3) we obtain for each symmetry transformation M (respectively its inverse M^{-1}), by comparing coefficients, restricting conditions for the vertex-functions:

$$\gamma(i_1, \dots, i_n, j_1, \dots, j_m) = \sum_{\substack{\tilde{i}_1, \dots, \tilde{i}_n \\ \tilde{j}_1, \tilde{j}_2, \dots, \tilde{j}_{m-1}, \tilde{j}_m}} \left(\prod_{l=1}^n \bar{M}(i_l, \tilde{i}_l) \right) \left(\prod_{l=1}^m M(j_l, \tilde{j}_l) \right) \gamma(\tilde{i}_1, \dots, \tilde{i}_n, \tilde{j}_1, \dots, \tilde{j}_m)$$

The shape of the coefficients is therefore dictated by the symmetries of the underlying physical system. This realization will simplify concrete calculations later.

Remark 17.3.2 The above results can be memorized by the following sentence from [10]:

The effective action Γ shares all the symmetries of the action S .

Remark 17.3.3 Equation (17.2) gives analogous results for the free energy W .

Remark 17.3.4 The figure summarizes the results pictographically.

$$\left. \begin{aligned} S[\underline{y}, \bar{\underline{y}}] &= S[\underline{M} \cdot \underline{y}, \bar{\underline{M}} \cdot \bar{\underline{y}}] \\ \mathcal{D}(\underline{y}, \bar{\underline{y}}) &= \mathcal{D}(\underline{M} \cdot \underline{y}, \bar{\underline{M}} \cdot \bar{\underline{y}}) \end{aligned} \right\} \Leftrightarrow W[\underline{j}, \bar{\underline{j}}] = W[\bar{\underline{M}}^T \cdot \underline{j}, \underline{M}^T \cdot \bar{\underline{j}}] \Leftrightarrow \Gamma[\phi, \bar{\phi}] = \Gamma[\underline{M}^{-1} \cdot \phi, \bar{\underline{M}}^{-1} \cdot \bar{\phi}]$$

Figure 17.1.: Summary for Symmetry Transformations.

17.1.2. Implications

TBCopied from [25, 38].

17.2. Mixed Systems

TBD.

17.2.1. Introduction

TBD.

17.2.2. Implications

TBD.

18. Thermal Quantum Field Theory

18.1. Many-Body Quantum Field Theory - Pure Systems

18.1.1. Coherent States and Resolutions of Identity

In the previous chapter we learned many techniques on how to evaluate (functional) integrals. In this chapter, we will provide the bridge between the many-body-concepts developed earlier, and the (functional) integral formalism. For this we still require some tools.

Definition 18.1 We consider a polynomial algebra $\mathcal{P}_\zeta$ as in Definition 16.1 and a Fock-space $\mathcal{H}_\eta$ that is generated by creation and annihilation-operators, as in Equation (10.4). We then construct¹ a [G]eneralized [H]ilbert-space $\mathcal{GH}_{\eta,\zeta}$ as the set of linear combinations of states of the Hilbert-space $\mathcal{H}_\eta$ with coefficients in the polynomial algebra $\mathcal{P}_\zeta$.

Remark 18.1.1 Notice that this implies, that any vector $|\psi\rangle$ in the generalized Hilbert-space can be expanded as

$$|\psi\rangle = \sum_{\alpha} \chi_{\alpha} |\phi_{\alpha}\rangle \quad (18.1)$$

where the χ_{α} are elements of the polynomial algebra and the $|\phi_{\alpha}\rangle$ vectors of the Hilbert-space.

In order to treat expressions containing combinations of polynomial-algebra-generators and creation- and annihilation-operators, it is necessary to specify commutation-relations between the ψ 's and c 's.

Definition 18.2 In the generalized Hilbert-space, we define commutation-relations as

$$[\psi_i, c_j]_{\zeta} := 0 \quad (18.2)$$

etc. for the other combinations.

Definition 18.3 We define a *coherent state* $|\psi\rangle \in \mathcal{GH}_{\eta,\zeta}$ as an eigenstate of the statistical annihilation-operator, i.e. fulfilling the equation

$$c_j |\psi\rangle = \psi_j |\psi\rangle \quad (18.3)$$

with $\psi_j \in \mathcal{P}_{\zeta}$.

¹Cf. [35] p. 29 for this part, or [31] p. 64 for the enlarged Fock-space for fermionic coherent states.

Theorem 18.4 The state $|\psi\rangle$ defined through

$$|\psi\rangle = \exp\left(\sum_{j=1}^{d_\eta} c_j^\dagger \psi_j\right) |0\rangle \quad (18.4)$$

is a *coherent state with respect to generators* $\{c_1, c_1^\dagger, \dots, c_{d_\eta}, c_{d_\eta}^\dagger\}$ of $\mathcal{H}$ and $\{\psi_1, \bar{\psi}_1, \dots, \psi_{d_\eta}, \bar{\psi}_{d_\eta}\}$ of $\mathcal{P}_\zeta$, i.e. it is an eigenstate of the statistical annihilation-operators fulfilling

$$c_j |\psi\rangle = \psi_j |\psi\rangle \quad \text{for all } j \in \{1, \dots, d_\eta\} \quad (18.5)$$

Proof 18.4 See Appendix, Proofs, page 489.

Remark 18.4.1 Notice that we have $D = d_\eta$ in the sense of Definition 16.1.

Corollary 18.5 By taking the hermitian conjugate of Equation (18.4) we find that the „bra“ $\langle\psi|$ associated with the „ket“ $|\psi\rangle$ is a left-eigenstate of the set of creation-operators, i.e. for

$$\langle\psi| = \langle 0| \exp\left(\sum_{j=1}^{d_\eta} \bar{\psi}_j c_j\right) \quad (18.6)$$

it holds

$$\langle\psi| c_j^\dagger = \langle\psi| \bar{\psi}_j \quad (18.7)$$

Proof 18.5 See Appendix, Proofs, page 489.

Lemma 18.6 One finds the *derivative-identities* for the coherent states as

$$\langle\psi| c_j = \frac{\partial}{\partial \bar{\psi}_j} \langle\psi|, \quad c_j^\dagger |\psi\rangle = \zeta \frac{\partial}{\partial \psi_j} |\psi\rangle \quad (18.8)$$

Proof 18.6 See Appendix, Proofs, page 489.

Theorem 18.7 One finds for the overlap between two coherent states

$$\langle\psi'|\psi\rangle = \exp\left(\sum_{j=1}^{d_\eta} \bar{\psi}'_j \psi_j\right) \quad (18.9)$$

Proof 18.7 See, for example, [5], Chapter 4.

Theorem 18.8 Taking $D = d_\eta$ in Definition 16.1, we obtain the *completeness-relation*

$$\mathbb{1}_{\mathcal{H}} = \int d_\zeta(\bar{\psi}, \psi) e^{-\sum_{j=1}^{d_\eta} \bar{\psi}_j \psi_j} |\psi\rangle\langle\psi| \quad (18.10)$$

with integration-measure as in Definition 16.16

$$d_\zeta(\bar{\psi}, \psi) \equiv \prod_{j=1}^{d_\eta} \frac{d\bar{\psi}_j d\psi_j}{(2\pi i)^{\theta(\zeta)}} \quad (18.11)$$

Proof 18.8 See Appendix, Proofs, page 489.

Lemma 18.9 For $\eta = 1$, the Hamiltonian in Equation (10.3) is block-diagonal w.r.t. the number of particles.

Proof 18.9 See Appendix, Proofs, page 489.

Remark 18.9.1 Notice that this theorem implies, that $\langle N|H|N'\rangle = 0$ if $|N\rangle$ and $|N'\rangle$ represents occupation number states with different numbers of particles.

Lemma 18.10 Let $|n\rangle$ and $|n'\rangle$ denote two N -particle states in their occupation-number representation. Then it follows that

$$\langle n|\psi\rangle\langle\psi|n'\rangle = \langle\zeta\psi|n'\rangle\langle n|\psi\rangle. \quad (18.12)$$

Proof 18.10 See Appendix, Proofs, page 489.

Lemma 18.11 For an operator X that is given as a polynomial expression in ladder-operators $X \equiv X(c, c^\dagger)$, we write for the normalized overlap between coherent states

$$\frac{\langle\psi|X|\psi'\rangle}{\langle\psi|\psi'\rangle} = X(\bar{\psi}, \psi'), \quad (18.13)$$

i.e. the normalized overlap is *the* polynomial expression in Graßmann-variables, in which one replaced each ladder-operator by its Graßmann-counterpart.

Proof 18.11 See Appendix, Proofs, page 489.

Remark 18.11.1 Notice that for the Hamiltonian-operators in Fact 8.4, one obtains the identities

$$\frac{\langle \psi | H | \psi' \rangle}{\langle \psi | \psi' \rangle} \equiv H(\bar{\psi}, \psi') \equiv \begin{cases} \sum_{i,j=1}^d H_{i,j} \bar{\psi}_i \psi_j + \sum_{i,j,k=1}^d (H_{i,j;k} \bar{\psi}_i \bar{\psi}_j \psi_k + H_{i,j,k} \bar{\psi}_i \psi_j \psi_k) + \dots & \text{for } \eta = 0 \\ \sum_{i,j=1}^d H_{i,j} \bar{\psi}_i \psi_j + \frac{1}{2} \sum_{i,j,k,l=1}^d H_{i,j;kl} \bar{\psi}_i \bar{\psi}_j \psi_l \psi_k + \dots & \text{for } \eta = 1 \end{cases} \quad (18.14)$$

and the interacting part will be considered in the next subsection.

Lemma 18.12 One finds for the trace of an operator $X \equiv X(c, c^\dagger)$ that is block-diagonal in the number of fermionic particle-number (and arbitrary in the bosonic particle-number)

$$\text{Tr}_{\mathcal{F}_\zeta} X(c, c^\dagger) = \int d_\zeta(\bar{\psi}, \psi) e^{-\sum_{j=1}^{d_\eta} \bar{\psi}_j \psi_j} \langle \zeta \psi | X | \psi \rangle. \quad (18.15)$$

with

$$\langle \zeta \psi | \equiv \langle 0 | \exp \left(\zeta \sum_{j=1}^{d_\eta} \bar{\psi}_j c_j \right) \quad (18.16)$$

Proof 18.12 See Appendix, Proofs, page 489.

18.1.2. The Functional Field Integral of the Partition Function

Theorem 18.13 One can rewrite the grand canonical partition function Z as a functional field integral

$$Z = \text{Tr}_{\mathcal{H}_\eta} e^{-\beta(H - \eta \mu N)} = \int \mathcal{D}_\zeta(\bar{\psi}, \psi) e^{-S_\eta[\bar{\psi}, \psi]}$$

where the action is written in its *continuum-notation*

$$\begin{aligned} S_\eta[\bar{\psi}, \psi] &\equiv \int_0^{\hbar\beta} d\tau \left[\bar{\psi} \cdot \partial_\tau \psi + \frac{H(\bar{\psi}, \psi) - \eta \mu N(\bar{\psi}, \psi)}{\hbar} \right] \\ &\equiv \int_0^{\hbar\beta} d\tau \left[\bar{\psi} \cdot \partial_\tau \psi + \frac{H_0(\bar{\psi}, \psi) - \eta \mu N(\bar{\psi}, \psi)}{\hbar} \right] + \int_0^{\hbar\beta} d\tau \frac{V(\bar{\psi}(\tau), \psi(\tau))}{\hbar} \end{aligned} \quad (18.17)$$

and the other symbols will be introduced and explained in the course of the text.

Proof 18.13 See Appendix, Proofs, page 489.

Remark 18.13.1 Notice that the continuum-notation is a bit misleading, as we are strictly speaking dealing with finite sums. However, the charm of the continuum-notation has prevailed.

Remark 18.13.2 We notice, that the field integral above is *almost* in the form of the generic field integral that was introduced in Chapter 16. The only difference is that (in our notation) the action has (or seems to have) a continuous variable, whereas we based our formalism only on the discrete case. To this end, we need to translate the action into a discrete form by performing an appropriate variable transformation.

Remark 18.13.3 Notice that for the generic Hamilton-operators in Fact 8.4, one obtains the identities

$$S_\eta[\psi, \bar{\psi}] = \begin{cases} \int \bar{\psi}_\alpha(\tau) \left[\partial_\tau \delta_{\alpha\alpha'} + H_{\alpha\alpha'}^0 / \hbar \right] \psi_{\alpha'}(\tau) & + \int \frac{V_{\alpha;\beta;\gamma}}{\hbar} \bar{\psi}_\alpha(\tau) \bar{\psi}_\beta(\tau) \psi_\gamma(\tau) + \frac{V_{\alpha;\beta;\gamma}}{\hbar} \bar{\psi}_\alpha(\tau) \psi_\beta(\tau) \psi_\gamma(\tau) + \dots & \text{for } \eta = 0 \\ \int \bar{\psi}_\alpha(\tau) \left[\left(\partial_\tau - \frac{\mu}{\hbar} \right) \delta_{\alpha\alpha'} + H_{\alpha\alpha'}^0 / \hbar \right] \psi_{\alpha'}(\tau) + \frac{1}{2} \int \frac{V_{\alpha_1\alpha_2\alpha_3\alpha_4}}{\hbar} \bar{\psi}_{\alpha_1}(\tau) \bar{\psi}_{\alpha_2}(\tau) \psi_{\alpha_4}(\tau) \psi_{\alpha_3}(\tau) & + \dots & \text{for } \eta = 1 \end{cases} \quad (18.18)$$

Lemma 18.14 The following transformation is unitary up to a (diverging) scaling-factor

$$\begin{aligned} \psi(\tau) &= \frac{1}{\sqrt{\hbar\beta}} \sum_{\omega_n \in \Omega_\zeta} \psi(i\omega_n) e^{-i\omega_n\tau}, & \psi(i\omega_n) &= \frac{1}{\sqrt{\hbar\beta}} \int_0^{\hbar\beta} d\tau \psi(\tau) e^{i\omega_n\tau} \\ \bar{\psi}(\tau) &= \frac{1}{\sqrt{\hbar\beta}} \sum_{\omega_n \in \Omega_\zeta} \bar{\psi}(i\omega_n) e^{i\omega_n\tau}, & \bar{\psi}(i\omega_n) &= \frac{1}{\sqrt{\hbar\beta}} \int_0^{\hbar\beta} d\tau \bar{\psi}(\tau) e^{-i\omega_n\tau} \end{aligned} \quad (18.19)$$

where

$$\mathbb{Z} \ni n \mapsto \omega_n = \frac{2n+1-\theta(\zeta)}{\hbar\beta} \pi = \begin{cases} 2n\pi/\hbar\beta & \text{for bosons} \\ (2n+1)\pi/\hbar\beta & \text{for fermions} \end{cases} \quad (18.20)$$

Proof 18.14 See Appendix, Proofs, page 490.

Remark 18.14.1 Notice that this is a change of coordinates in the sense of Equation (16.67) with Jacobian-matrix

$$\text{Det} \left(\frac{\partial \psi(\tau_i)}{\partial \psi(i\omega_j)} \right) = \text{Det} \left(\frac{\partial \bar{\psi}(\tau_i)}{\partial \bar{\psi}(i\omega_j)} \right) = \text{TBLookedUp} \quad (18.21)$$

Remark 18.14.2 This diverging factor is mostly irrelevant. There are though situations, where it becomes important to keep the diverging factor inside the Equations. Examples are when one computes the natural energy for a non-interacting system in Lemma TBReferenced, or when one considers corrections to the partition-function in the superconducting state.

Remark 18.14.3 Notice that we found in the proof of this Lemma the identity

$$\text{Kronecker-Delta relations for zeta-transformations.} \quad (18.22)$$

Corollary 18.15 Transforming to Matsubara-frequencies gives in the non-interacting case ($V \equiv 0$)

$$S[\psi, \bar{\psi}] = \sum_{\substack{\omega_n \in \Omega_\zeta \\ \alpha, \alpha' \in \mathcal{A}}} \bar{\psi}_\alpha(i\omega_n) \left[\left(-i\omega_n - \frac{\eta\mu}{\hbar} \right) \delta_{\alpha\alpha'} + H_{\alpha\alpha'}/\hbar \right] \psi_{\alpha'}(i\omega_n) \quad (18.23)$$

and for the Hamiltonians in Fact 8.4 ...

- in the massless case ($\eta = 0$)

$$\begin{aligned} S[\psi, \bar{\psi}] &= \sum_{\substack{\omega_n \in \Omega_\zeta \\ \alpha, \alpha' \in \mathcal{A}}} \bar{\psi}_\alpha(i\omega_n) \left[-i\omega_n \delta_{\alpha\alpha'} + H_{\alpha\alpha'}/\hbar \right] \psi_{\alpha'}(i\omega_n) \\ &+ \frac{1}{\hbar} \sum_{\substack{\omega_n, \omega_{n'}, \omega_{n''} \in \Omega_\zeta \\ \alpha, \alpha', \alpha'' \in \mathcal{A}}} \left(H_{\alpha, \alpha'; \alpha''} \bar{\psi}_\alpha(\omega_n) \bar{\psi}_{\alpha'}(\omega_{n'}) \psi_{\alpha''}(\omega_{n''}) \delta_{\omega_n + \omega_{n'}, \omega_{n''}} + H_{\alpha; \alpha', \alpha''} \bar{\psi}_\alpha(\omega_n) \psi_{\alpha'}(\omega_{n'}) \psi_{\alpha''}(\omega_{n''}) \delta_{\omega_n, \omega_{n'} + \omega_{n''}} \right) \end{aligned} \quad (18.24)$$

- in the massless case ($\eta = 1$)

$$\begin{aligned} S[\psi, \bar{\psi}] &= \sum_{\substack{\omega_n \in \Omega_\zeta \\ \alpha, \alpha' \in \mathcal{A}}} \bar{\psi}_\alpha(i\omega_n) \left[\left(-i\omega_n - \frac{\mu}{\hbar} \right) \delta_{\alpha\alpha'} + H_{\alpha\alpha'}/\hbar \right] \psi_{\alpha'}(i\omega_n) \\ &+ \frac{1}{2} \frac{1}{\hbar\beta} \sum_{\substack{\alpha_1 \alpha_2 \alpha_3 \alpha_4 \in \mathcal{A} \\ \omega_{n_1} \omega_{n_2} \omega_{n_3} \omega_{n_4} \in \Omega_\zeta}} \frac{V_{\alpha_1 \alpha_2 \alpha_3 \alpha_4} \delta_{n_1 + n_2, n_3 + n_4}}{\hbar} \bar{\psi}_{\alpha_1}(i\omega_{n_1}) \bar{\psi}_{\alpha_2}(i\omega_{n_2}) \psi_{\alpha_4}(i\omega_{n_4}) \psi_{\alpha_3}(i\omega_{n_3}) \end{aligned} \quad (18.25)$$

with rescaled² partition-function.

Proof 18.15 See Appendix, Proofs, page 493.

Lemma 18.16 The natural energy from Equation (10.8) is ...

- in the non-interacting case $V \equiv 0$ given as

$$\Omega_0 = \frac{\zeta}{\beta} \sum_{\alpha \in \mathcal{A}} \log(1 - \zeta e^{-\beta \xi_\alpha}) \quad (18.26)$$

whereas $\xi_{\alpha_1}, \xi_{\alpha_2}, \dots, \xi_{\alpha_{d_H}}$ denote the eigenvalues of $H_0 - \eta\mu$.

- in the interacting case $V \neq 0$ given as

$$\Omega = \Omega_0 - \frac{1}{\beta} \sum \text{"all connected graphs"} \quad (18.27)$$

²Notice that this does not change the expectation-value, due to the quotient nature of the generating functional in Equation (16.39).

Proof 18.16 See Appendix, Proofs, page 494.

18.1.3. Imaginary-Time Generalized Green-Functions

From the proof of Theorem 18.17 we see that expectation-values of operators can be computed within the functional integral formalism only if they are normal ordered. Therefore one can compute any generalized Green-function but one has to re-express X and X' through their normal-ordered expressions.

18.1.4. Imaginary-Time Pure n -Body Green-Functions

Theorem 18.17 For $\tau_1, \dots, \tau_n, \tau'_1, \dots, \tau'_n \in (0, \hbar\beta)$ one finds for the imaginary-time n -body Green-Functions

$$\begin{aligned} \mathcal{G}(A_1, \dots, A_n | A'_1, \dots, A'_n) &= \left(-\frac{1}{\hbar}\right)^n \frac{1}{Z} \int \mathcal{D}_\zeta(\psi, \bar{\psi}) e^{-S[\psi, \bar{\psi}]} \psi_{\alpha_1}(\tau_1) \cdots \psi_{\alpha_n}(\tau_n) \bar{\psi}_{\alpha'_1}(\tau'_1) \cdots \bar{\psi}_{\alpha'_n}(\tau'_n) \\ &\equiv \left(-\frac{1}{\hbar}\right)^n \langle \psi(A_1) \cdots \psi(A_n) \bar{\psi}(A'_1) \cdots \bar{\psi}(A'_n) \rangle. \end{aligned}$$

Proof 18.17 See Appendix, Proofs, page 494.

Theorem 18.18 For the one-body-Green-function of a non-interacting (fermionic or bosonic) gas one finds³

$$\mathcal{G}_{\alpha, \alpha'}^0(\tau) = \frac{1}{\hbar\beta} \sum_{\omega_n \in \Omega_\zeta} \mathcal{G}_{\alpha, \alpha'}^0(i\omega_n) e^{-i\omega_n \tau} \quad (18.29)$$

with

$$\mathcal{G}_{\alpha, \alpha'}^0(i\omega_n) = \frac{1}{\hbar} \left(\left[i\omega_n - \frac{H_0 - \eta\mu}{\hbar} \right]_{\alpha, \alpha'}^{\text{inv}} \right) \iff \mathcal{G}_0(i\omega_n) \equiv \frac{1}{i\hbar\omega_n - (H_0 - \eta\mu)} \quad (18.30)$$

Proof 18.18 See Appendix, Proofs, page 494.

³Notice that we are dropping the infinitesimal phases for notational convenience. Keeping track of them yields

$$\mathcal{G}_{\alpha, \alpha'}^0(i\omega_n) = \frac{1}{\hbar} \left(\left[i\omega_n - \frac{H_0 - \eta\mu}{\hbar} e^{i\omega_n \delta} \right]_{\alpha, \alpha'}^{\text{inv}} \right) \quad (18.28)$$

18.1.5. Intermezzo: The Continuum-Notation with it's Perks and Problems

We introduced the mathematical framework of quantum field theory in Chapter 16 based on discrete indices. In the derivation of the field-integral however, we noticed that the resulting expressions are more conveniently written in a continuum-notation. This is a re-occurring phenomenon in physics. For example the action in Equation (18.17) or (equivalently) in Equation (18.18) can also be re-expressed with field operators (which are introduced in Appendix, section 27.8) in a continuum-notation over space, such that the entire action becomes an integral over four variables. The only thing one needs is some mild assumptions about the completeness of the basis-set.

Definition 18.19 We equip our partition-function with external sources and introduce the generating functional with continuum-notation

$$\frac{Z[\bar{j}, j]}{Z} = \frac{1}{Z} \int \mathcal{D}_\zeta(\psi, \bar{\psi}) e^{-S[\psi, \bar{\psi}] + \oint j^\dagger \psi + \psi^\dagger j} = \left\langle \exp \oint j^\dagger \psi + \psi^\dagger j \right\rangle \quad (18.31)$$

whereas the source-terms are given in their continuum-notation as

$$\oint j^\dagger \psi \equiv \int_0^{\hbar\beta} d\tau j^\dagger(\tau) \psi(\tau) \equiv \int_0^{\hbar\beta} d\tau \sum_{\alpha \in \mathcal{A}} \bar{j}_\alpha(\tau) \psi_\alpha(\tau) \equiv \Delta\tau \sum_{\substack{\alpha \in \mathcal{A} \\ n=1, \dots, N_\tau}} \bar{j}_\alpha(\tau_n) \psi_\alpha(\tau_n) \quad (18.32)$$

Lemma 18.20 One concludes, that one can determine the n -body Green-function as functional derivative

$$\mathcal{G}(A_1, \dots, A_n | A'_1, \dots, A'_n) = \left(-\frac{\zeta}{\hbar} \right)^n \frac{\delta^{2n}}{\delta \bar{j}(A_1) \dots \delta \bar{j}(A_n) \delta j(A'_1) \dots \delta j(A'_n)} \frac{Z[\bar{j}, j]}{Z} \Big|_{\bar{j}=j=0} \quad (18.33)$$

Proof 18.20 See Appendix, Proofs, page 474.

Notation 18.21 Just as in Definition 16.30, we introduce the analogue of the Gibbs free energy / grand canonical potential

$$W[\bar{j}, j] = + \log Z[\bar{j}, j] \quad (18.34)$$

Notation 18.22 Just as in Definition 16.41 we introduce the [c]onnected Green-functions

$$\mathcal{G}^c(A_1, \dots, A_n | A'_1, \dots, A'_n) = \left(-\frac{\zeta}{\hbar} \right)^n \frac{\delta^{2n}}{\delta \bar{j}(A_1) \dots \delta \bar{j}(A_n) \delta j(A'_1) \dots \delta j(A'_n)} W[\bar{j}, j] \Big|_{\bar{j}=j=0} \quad (18.35)$$

Theorem 18.23 One has the identity

$$\langle c_{\alpha'}^\dagger(0) c_\alpha(0) \rangle = \frac{\delta^2}{\delta \bar{j}(\tau = 0-, \alpha) \delta j(\tau' = 0, \alpha')} \frac{Z[\bar{j}, j]}{Z} \Big|_{\bar{j}=j=0} \quad (18.36)$$

Proof 18.23 Recall first that from the definition of the Green-function

$$\mathcal{G}(A, A') \equiv \mathcal{G}_{\alpha, \alpha'}(\tau, \tau') \equiv -\frac{1}{\hbar} \frac{\text{Tr}_{\mathcal{F}_\zeta} e^{-\beta(H-\mu N)} \left[\theta(\tau - \tau') c_\alpha(\tau) c_{\alpha'}^\dagger(\tau') + \zeta \theta(\tau' - \tau) c_{\alpha'}^\dagger(\tau') c_\alpha(\tau) \right]}{\text{Tr}_{\mathcal{F}_\zeta} e^{-\beta(H-\mu N)}}$$

one finds the identity

$$\langle c_{\alpha'}^\dagger(0) c_\alpha(0) \rangle = -\frac{\hbar}{\zeta} \mathcal{G}_{\alpha, \alpha'}(\tau = 0-, \tau' = 0) \quad (18.37)$$

This then implies with Lemma 16.40

$$\begin{aligned} \langle c_{\alpha'}^\dagger(0) c_\alpha(0) \rangle &= -\frac{\hbar}{\zeta} \mathcal{G}_{\alpha, \alpha'}(\tau = 0-, \tau' = 0) \\ &= -\frac{\hbar}{\zeta} \left(-\frac{\zeta}{\hbar} \right)^1 \frac{\delta^2}{\delta \bar{j}(\tau = 0-, \alpha) \delta j(\tau' = 0, \alpha')} \frac{Z[\bar{j}, j]}{Z} \Big|_{\bar{j}=j=0} \\ &= \frac{\delta^2}{\delta \bar{j}(\tau = 0-, \alpha) \delta j(\tau' = 0, \alpha')} \frac{Z[\bar{j}, j]}{Z} \Big|_{\bar{j}=j=0} \end{aligned} \quad (18.38)$$

Lemma 18.24 One can rewrite the source-term derivatives as

$$\frac{\delta}{\delta \bar{j}(\tau = 0-, \alpha)} = \frac{1}{\sqrt{\hbar \beta}} \sum_{\omega_n \in \Omega_\zeta} e^{+i\omega_n 0^+} \frac{\partial}{\partial \bar{j}(i\omega_n, \alpha)} \quad (18.39)$$

Proof 18.24 Follows by combining Lemma 16.51 with Lemma 18.14.

18.2. Many-Body Quantum Field Theory - Mixed Systems

We present the formalism at the example of **massive-massless-particles-mixtures**. The generalization to other mixtures is straight-forward.

18.2.1. Coherent States and Resolutions of Identity

In one to one correspondence to the respective creation- and annihilation-operators we again introduce Graßmann-numbers $\{\psi_1, \bar{\psi}_1, \dots, \psi_{d_{m>0}}, \bar{\psi}_{d_{m>0}}\}$ and complex-numbers $\{\phi_1, \bar{\phi}_1, \dots, \phi_{d_{m=0}}, \bar{\phi}_{d_{m=0}}\}$ to form polynomial expressions with them and creation- and annihilation-operators.

Theorem 18.25 The bra- and ket-states $\langle \chi |$ and $|\chi\rangle$ as

$$\langle \chi | \equiv \langle \text{vac.} | \exp \left(+ \sum_{j=1}^{d_{m=0}} b_j \bar{\phi}_j - \sum_{j=1}^{d_{m>0}} c_j \bar{\psi}_j \right), \quad |\chi\rangle \equiv \exp \left(+ \sum_{j=1}^{d_{m=0}} \phi_j b_j^\dagger - \sum_{j=1}^{d_{m>0}} \psi_j c_j^\dagger \right) | \text{vac.} \rangle,$$

and are left- and right-eigenstates of creation- and annihilation-operators, implying

$$\langle \chi | b_\beta^\dagger = \langle \chi | \bar{\phi}_\beta, \quad \langle \chi | c_\alpha^\dagger = \langle \chi | \bar{\psi}_\alpha, \quad b_\beta |\chi\rangle = \phi_\beta |\chi\rangle \quad \text{and} \quad c_\alpha |\chi\rangle = \psi_\alpha |\chi\rangle, \quad (18.40)$$

for all $\alpha \in \{1, \dots, d_{m>0}\}$, $\beta \in \{1, \dots, d_{m=0}\}$.

Proof 18.25 TBD.

Theorem 18.26 It follows

- Overlap
- Completeness Relation for the total Hilbert-space.

Proof 18.26 Follows from the generalization of the identities by straight forward computations.

18.2.2. The Functional Field Integral of the Partition Function

Theorem 18.27 One can rewrite the mixed partition function Z as a functional field integral

$$Z = \text{Tr}_{\mathcal{H}} \exp - \beta (H_{m>0} - \mu N_{m>0} + H_{m=0}) = \int \mathcal{D}(\chi, \bar{\chi}) e^{-S[\chi, \bar{\chi}]}$$

where

$$\begin{aligned}
 S[\chi, \bar{\chi}] &\equiv S_0[\chi, \bar{\chi}] + S_{\text{int}, m>0}[\psi, \bar{\psi}] + S_{\text{int}, m=0}[\phi, \bar{\phi}] + S_{\text{int}}[\chi, \bar{\chi}] \\
 &= \int_0^{\hbar\beta} d\tau \begin{pmatrix} \psi \\ \phi \end{pmatrix}^\dagger \cdot \begin{pmatrix} \partial_\tau + \frac{H_0 - \mu}{\hbar} & 0 \\ 0 & \partial_\tau + \frac{H_0}{\hbar} \end{pmatrix} \cdot \begin{pmatrix} \psi \\ \phi \end{pmatrix} + \int_0^{\hbar\beta} d\tau \frac{V_{m>0}(\bar{\psi}, \psi) + V_{m=0}(\bar{\phi}, \phi) + V_{\text{int}}(\bar{\chi}, \chi)}{\hbar}
 \end{aligned}$$

Proof 18.27 Follows from the generalization of the identities by straight forward computations.

Remark 18.27.1 Going over to Matsubara-frequencies turns this expression into a real mess, as one has to account for the different statistics within the respective summations. It is however common to use Matsubara-frequencies, as it is still the most tractable way to deal with these integrals.

18.2.3. Pure Imaginary-Time Generalized Green-Functions

TBDigitalized for interaction-expectation-values.

18.2.4. Pure Imaginary-Time n -Body Green-Functions

19. Examples of Self-Energies and Polarizations for Interacting Systems

In this whole section, we consider a situation as in Equation (10.3) or its mixed system generalization, where the total system is written as

$$H = H_0 + V \quad (19.1)$$

whereas H_0 is a Hamiltonian of which the Green-functions are accessible, and V is a perturbation of some kind. The electronic self-energy is the main object of interest. It will be always written in a series

$$\Sigma(i\omega_n) = \sum_{j=1}^{\infty} \Sigma^{(j)}(i\omega_n) = \Sigma^{(1)}(i\omega_n) + \Sigma^{(2)}(i\omega_n) + \Sigma^{(3)}(i\omega_n) + \mathcal{O}(V^4) \quad (19.2)$$

in the interaction-vertices.

19.1. Example: A Diagonalized Gas with Single-Particle-Interactions

Theorem 19.1 The self-energy reads

$$\Sigma_{\beta\beta'}(i\omega_n) = V_{\beta\beta'} = \text{„frequency-independent“} . \quad (19.3)$$

Proof 19.1 See Appendix, Proofs, page 305.

Remark 19.1.1 Notice that with the Dyson-equation (15.12) one can conclude that this is actually the only term that appears in the self-energy, i.e. $\Sigma_{\beta\beta'}^{(1)} \equiv \Sigma_{\beta\beta'}$ and $\Sigma_{\beta\beta'}^{(1)} = \Sigma_{\beta,\beta'}^{(2)} = \dots = 0$. When dealing with quadratic perturbations, self-energy concepts are basically useless. It's only when one has other, higher order perturbations on top of it, where it becomes reasonable to compute self-energies.

19.2. Example: A Diagonalized Gas with Quartic Interactions

Theorem 19.2 The electronic self-energy of this system reads can be approximated in the series expansion as¹

$$\Sigma_{\alpha,\alpha'}^{(1)}(i\omega_n) \equiv \zeta \sum_{\beta \in \mathcal{A}} (V_{\alpha\beta\beta\alpha'} - V_{\alpha\beta\alpha'\beta}) n_{\hbar\beta}^{\zeta} (\epsilon_{\beta} - \mu) = \text{„frequency-independent“} . \quad (19.4)$$

Proof 19.2 See Appendix, Proofs, page 308.

¹Cf. also [47] p. 167 - 169.

19.3. Example: A Spin-Independent-Interaction between an Electron-Gas and a Phonon-Gas

Theorem 19.3 The electronic self-energy of this system reads can be approximated in the series expansion as

$$\Sigma_{\mathbf{k},\mathbf{k}'}^{(1)}(i\omega_n) = 0, \quad (19.5)$$

$$\Sigma_{\mathbf{k},\mathbf{k}'}^{(2)}(i\omega_n) = \Sigma_{\mathbf{k},\mathbf{k}'}^{\text{Hartree}}(i\omega_n) + \Sigma_{\mathbf{k},\mathbf{k}'}^{\text{Fock}}(i\omega_n), \quad (19.6)$$

with Hartree-contribution

$$\begin{aligned} \Sigma_{\mathbf{k},\mathbf{k}'}^{\text{Hartree}}(i\omega_n) &= \delta_{\mathbf{k},\mathbf{k}'} \delta_{\mu,\mu'} \sum_{\substack{\mathbf{q} \in Q \\ \sigma \in \{\downarrow, \uparrow\} \\ n \in \{1, 2, \dots\} \\ v \in \{1, 2, \dots, rd_x\}}} \frac{g_{mm'v}(\mathbf{k}, 0) g_{nnv}(\mathbf{q}, 0)}{N_p} \mathcal{D}_{\mathbf{0}, \mathbf{0}}^0(0) F_{\beta}(\xi_{n\sigma\mathbf{q}}) \\ &= -\delta_{\mathbf{k},\mathbf{k}'} \delta_{\mu,\mu'} \sum_{\substack{\mathbf{q} \in Q \\ \sigma \in \{\downarrow, \uparrow\} \\ n \in \{1, 2, \dots\} \\ v \in \{1, 2, \dots, rd_x\}}} \frac{g_{mm'v}(\mathbf{k}, 0) g_{nnv}(\mathbf{q}, 0)}{N_p} \frac{F_{\beta}(\xi_{n\sigma\mathbf{q}})}{\Omega_v(\mathbf{q}' = 0)} \end{aligned} \quad (19.7)$$

and Fock-contribution

$$\Sigma_{\mathbf{k},\mathbf{k}'}^{\text{Fock}}(i\omega_n) = \delta_{\mathbf{k},\mathbf{k}'} \delta_{\mu,\mu'} \sum_{\substack{\mathbf{q} \in Q \\ n \in \{1, 2, \dots\} \\ v \in \{1, 2, \dots, rd_x\}}} \frac{g_{nmv}^*(\mathbf{k}, \mathbf{q}) g_{nm'v'}(\mathbf{k}, \mathbf{q})}{N_p} \left[\frac{B_{\beta}(\Omega_v(\mathbf{q})) + F_{\beta}(\xi_{\mathbf{q}+\mathbf{k}n\mu})}{i\hbar\omega_n - \xi_{\mathbf{q}+\mathbf{k}n\mu} + \Omega_v(\mathbf{q})} + \frac{B_{\beta}(\Omega_v(\mathbf{q})) + 1 - F_{\beta}(\xi_{\mathbf{q}+\mathbf{k}n\mu})}{i\hbar\omega_n - \xi_{\mathbf{q}+\mathbf{k}n\mu} - \Omega_v(\mathbf{q})} \right] \quad (19.8)$$

Proof 19.3 See Appendix, Proofs, page 312.

Remark 19.3.1 In the *diagonal band approximation*, one assumes that the self-energy is diagonal in the band indices, i.e.

$$\Sigma_{\mathbf{k},\mathbf{k}'}(i\omega_n) \approx \delta_{m,m'} \delta_{\mu,\mu'} \Sigma_{\mathbf{k}m\sigma}(i\omega_n) \quad \text{with} \quad \Sigma_{\mathbf{k}m\sigma}(i\omega_n) \approx \Sigma_{\mathbf{k}m\sigma}^{\text{Hartree}}(i\omega_n) + \Sigma_{\mathbf{k}m\sigma}^{\text{Fock}}(i\omega_n), \quad (19.9)$$

and with Hartree-contribution

$$\Sigma_{m\sigma\mathbf{k}}^{\text{Hartree}}(\bullet) = - \sum_{\substack{\mathbf{q} \in Q \\ \sigma' \in \{\downarrow, \uparrow\} \\ n \in \{1, 2, \dots\} \\ v \in \{1, 2, \dots, rd_x\}}} \frac{g_{nmv}(\mathbf{k}, 0) g_{nnv}(\mathbf{q}, 0)}{N_p} \frac{F_{\beta}(\epsilon_{n\sigma'\mathbf{q}} - \mu)}{\Omega_v(\mathbf{q}' = 0)} \in \mathbb{R} \quad (19.10)$$

and Fock-contribution

$$\Sigma_{m\sigma\mathbf{k}}^{\text{Fock}}(i\omega_n) = \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ n \in \{1,2,\dots\} \\ v \in \{1,2,\dots,rd_x\}}} \frac{|g_{nmv}(\mathbf{k}, \mathbf{q})|^2}{N_p} \left[\frac{B_\beta(\Omega_v(\mathbf{q})) + F_\beta(\xi_{\mathbf{q}+\mathbf{k}n\sigma})}{i\omega_n - \xi_{\mathbf{q}+\mathbf{k}n\sigma} + \Omega_v(\mathbf{q})} + \frac{B_\beta(\Omega_v(\mathbf{q})) + 1 - F_\beta(\xi_{\mathbf{q}+\mathbf{k}n\sigma})}{i\omega_n - \xi_{\mathbf{q}+\mathbf{k}n\sigma} - \Omega_v(\mathbf{q})} \right] \quad (19.11)$$

which is usually motivated by looking at the complex numbers appearing in Equations (19.7), (19.8) and using Remark 8.30.1.

Remark 19.3.2 Another common approximation is the *Migdal approximation*, which neglects the Hartree-term, which is good, when the change of the chemical potential as a function of temperature is small enough.

The Polarization

Theorem 19.4 The polarization of the system is given as

$$\Pi = 2i \int \frac{d^2}{(2\pi)^4} |g(1, 2)|^2 G_0(1) G_0(2) \quad (19.12)$$

or, in coordinates

$$\Pi_{\mathbf{q}, \mathbf{q}'}(i\omega_n) = 2\delta_{\mathbf{q}, \mathbf{q}'} \delta_{v, v'} \int \frac{d\mathbf{q}}{\Omega_{\text{BZ}}} |g_{mv}(\mathbf{k}, \mathbf{q})|^2 \frac{F_\beta(\epsilon_{m\mathbf{k}}) - F_\beta(\epsilon_{m\mathbf{k}+\mathbf{q}})}{\epsilon_{m\mathbf{k}+\mathbf{q}} - \epsilon_{m\mathbf{k}} - i\omega_n} \quad (19.13)$$

Proof 19.4 See Appendix, Proofs, page 321.

20. Quasi-Equilibrium-Formalism (Kubo-Formalism)

20.1. Kubo-Formula for arbitrary Perturbations

This argument is evidently quite general, and a large class of experiments in which a system is subjected to a weak, controllable external probe is characterized by simple correlation functions of operators at different times (...) - Negele Orland page 51.

Notation 20.1 We consider a time-dependent Hamiltonian $H(t)$ as

$$H(t) = H_0 + V(t) = H_0 + H'(t)\theta(t - t_0) \quad (20.1)$$

and some observable of interest A , given in second quantization as

$$A \equiv A(c, c^\dagger). \quad (20.2)$$

Remark 20.1.1 Recall that the expectation value of an operator A is (in the absence of spontaneous symmetry breaking) given by

$$\langle A(t) \rangle = \text{Tr}_{\mathcal{H}} (\rho(t)A). \quad (20.3)$$

Theorem 20.2 The *Kubo-formula* gives an expression for the change of the expectation value of an operator A due to a perturbation $V(t)$ as

$$\langle A(t) \rangle = \langle A \rangle_0(t) + \int_{t_0}^{\infty} dt' G_{A, H'(t')}^{\text{ret}}(t, t') + \mathcal{O}(V^2) \quad (20.4)$$

with

$$G_{A, H'(t')}^{\text{ret}}(t, t') = -\frac{i}{\hbar} \theta(t - t') \langle [A(t), H'(t')] \rangle_0 = -\frac{i}{\hbar} \theta(t - t') \text{Tr}_{\mathcal{H}} \left[\rho_0 [A(t), H'(t')] \right]. \quad (20.5)$$

and we use the notation $\rho_0 \equiv \rho(t_0)$.

Remark 20.2.1 Cf. also Schwabl 2 page 88. „dynamic susceptibility“.

Proof 20.2 First we note that with the Lehmann-representation we obtain

$$\rho \equiv \rho(t_0) = \frac{1}{Z} e^{-\beta(H-\mu N)} = \frac{1}{Z} \sum_n |n\rangle \langle n| e^{-\beta \xi_n} \quad (20.6)$$

whereas $|n\rangle \equiv |n(t_0)\rangle$. Now, applying the time-evolution of the von Neumann equation onto the density-matrix ρ we obtain

$$\rho(t) = \frac{1}{Z} \sum_n |n(t)\rangle \langle n(t)| e^{-\beta \xi_n} \quad (20.7)$$

Now we rewrite the ket-vectors $|n(t)\rangle$ in the interaction picture as

$$|n(t)\rangle = e^{-iH_0(t-t_0)/\hbar} \hat{U}(t, t_0) |n(t_0)\rangle \quad (20.8)$$

with

$$\hat{U}(t, 0) = e^{iH_0(t-t_0)/\hbar} U(t, 0) \quad (20.9)$$

Before we continue, note the general expression for the time-evolution-operator from Theorem 2.13. Therefore, using now the identity

$$\partial_t U(t, 0) = H(t) U(t, 0). \quad (20.10)$$

This yields therefore

$$\begin{aligned} \partial_t \hat{U}(t, 0) &= i e^{iH_0(t-t_0)/\hbar} (H_0 - H(t)) U(t, 0) \\ &= -\frac{i}{\hbar} e^{iH_0(t-t_0)/\hbar} V(t) U(t, 0) \\ &= -\frac{i}{\hbar} e^{iH_0(t-t_0)/\hbar} V(t) e^{-iH_0(t-t_0)/\hbar} \hat{U}(t, 0) \\ &= -\frac{i}{\hbar} V_{H_0}(t) \hat{U}(t, 0) \end{aligned} \quad (20.11)$$

and with this we obtain

$$\begin{aligned} \hat{U}(t, 0) - \hat{U}(0, 0) &= -\frac{i}{\hbar} \int_{t_0}^t dt' V_{H_0}(t') \hat{U}(t', 0) \\ \Leftrightarrow \hat{U}(t, 0) &= 1 - \frac{i}{\hbar} \int_{t_0}^t dt' V_{H_0}(t') \hat{U}(t', 0) \\ \Leftrightarrow \hat{U}(t, 0) &= 1 - \frac{i}{\hbar} \int_{t_0}^t dt' V_{H_0}(t') + \mathcal{O}(V^2) \end{aligned} \quad (20.12)$$

this gives us then

$$|n(t)\rangle = e^{-iH_0(t-t_0)/\hbar} \left(1 - \frac{i}{\hbar} \int_{t_0}^t dt' V_{H_0}(t') + \mathcal{O}(V^2) \right) |n(t_0)\rangle \quad (20.13)$$

This gives us, inserting it into the definition of the expectation value,

$$\begin{aligned}
 \langle A(t) \rangle &= \text{Tr}_{\mathcal{H}} (\rho(t) A) \\
 &= \frac{1}{Z} \sum_n e^{-\beta \xi_n} \text{Tr}_{\mathcal{H}} (|n(t)\rangle \langle n(t)| A) \\
 &= \dots
 \end{aligned} \tag{20.14}$$

now we compute the trace in the last line as

$$\begin{aligned}
 &\text{Tr}_{\mathcal{H}} (|n(t)\rangle \langle n(t)| A) \\
 &= \text{Tr}_{\mathcal{H}} \left(e^{-iH_0(t-t_0)/\hbar} \left(1 - \frac{i}{\hbar} \int_{t_0}^t dt' V_{H_0}(t') \right) |n(t_0)\rangle \langle n(t_0)| \left(1 + \frac{i}{\hbar} \int_0^t dt'' V_{H_0}(t'') \right) e^{+iH_0(t-t_0)/\hbar} \right) + \mathcal{O}(V^3) \\
 &= \text{Tr}_{\mathcal{H}} (e^{-iH_0(t-t_0)/\hbar} |n(t_0)\rangle \langle n(t_0)| e^{+iH_0(t-t_0)/\hbar} A) \\
 &\quad - \frac{i}{\hbar} \int_{t_0}^t dt' \text{Tr}_{\mathcal{H}} (e^{-iH_0(t-t_0)/\hbar} V_{H_0}(t') |n(t_0)\rangle \langle n(t_0)| e^{+iH_0(t-t_0)/\hbar} A - e^{-iH_0(t-t_0)/\hbar} |n(t_0)\rangle \langle n(t_0)| V_{H_0}(t') e^{+iH_0(t-t_0)/\hbar} A) \\
 &= \text{Tr}_{\mathcal{H}} (e^{-iH_0(t-t_0)/\hbar} |n(t_0)\rangle \langle n(t_0)| e^{+iH_0(t-t_0)/\hbar} A) \\
 &\quad - \frac{i}{\hbar} \int_{t_0}^t dt' \langle n(t_0) | A_{H_0}(t) V_{H_0}(t') - V_{H_0}(t') A_{H_0}(t) | n(t_0) \rangle
 \end{aligned} \tag{20.15}$$

inserting this above yields

$$\begin{aligned}
 \dots &= \langle A \rangle_0(t - t_0) \\
 &\quad - \frac{i}{\hbar} \frac{1}{Z} \sum_n e^{-\beta \xi_n} \int_{t_0}^t dt' \langle n(t_0) | A_{H_0}(t) V_{H_0}(t') - V_{H_0}(t') A_{H_0}(t) | n(t_0) \rangle + \mathcal{O}(V^2) \\
 &= \langle A \rangle_0(t - t_0) - \frac{i}{\hbar} \int_{t_0}^t dt' \langle [A_{H_0}(t), V_{H_0}(t')] \rangle_0 + \mathcal{O}(V^2) \\
 &= \langle A \rangle_0(t - t_0) - \frac{i}{\hbar} \int_{t_0}^t dt' \theta(t' - t_0) \langle [A_{H_0}(t), H'_{H_0}(t')] \rangle_0 + \mathcal{O}(V^2) \\
 &= \dots
 \end{aligned} \tag{20.16}$$

now we use

$$\theta(t' - t_0) = \begin{cases} 1 & t' > t_0 \\ 0 & t' < t_0 \end{cases} \tag{20.17}$$

which is, within the integral, always fulfilled, and this gives

$$\begin{aligned}
 \dots &= \langle A \rangle_0(t - t_0) - \frac{i}{\hbar} \int_{t_0}^t dt' \langle [A_{H_0}(t), H'_{H_0}(t')] \rangle_0 + \mathcal{O}(V^2) \\
 &= \dots
 \end{aligned} \tag{20.18}$$

now we will extend the integral further, by introducing the theta-function as

$$\theta(t - t') = \begin{cases} 1 & t > t' \\ 0 & t < t' \end{cases} \quad (20.19)$$

to rewrite the expression above as

$$\begin{aligned} \dots &= \langle A \rangle_0(t - t_0) - \frac{i}{\hbar} \int_{t_0}^{\infty} dt' \theta(t - t') \langle [A_{H_0}(t), H'_{H_0}(t')] \rangle_0 + \mathcal{O}(V^2) \\ &= \langle A \rangle_0(t - t_0) - \int_{t_0}^{\infty} dt' G_{A, H'(t')}^{\text{ret}}(t, t') + \mathcal{O}(V^2) \end{aligned} \quad (20.20)$$

Part IV.

Applications to Real Materials

Comment on the Applications

In this entire part, we will seek to model real, crystalline materials. To this end we remark, that that we will essentially be focusing on three (four) systems (depending on how one counts). We consider ...

Non-Interacting Electrons

We consider the system from Remark 8.22.1. (which represents a special case of the system in Definition 8.21). We therefore consider the Hamiltonian

$$H_{\text{el}} = \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} c_{m\mathbf{k}\sigma}^\dagger c_{m\mathbf{k}\sigma} \quad (20.21)$$

Non-Interacting Phonons

We consider the system from Remark 8.22.4 (which represents a special case of the system in Definition 8.21). We therefore consider the Hamiltonian

$$H_{\text{ph}} = \sum_{\substack{\mathbf{q} \in Q \\ v \in \{1, 2, \dots, rd_x\}}} \Omega_v(\mathbf{q}) b_{\mathbf{q}v}^\dagger b_{\mathbf{q}v} \quad (20.22)$$

Non-Interacting Electrons and Phonons (Total Crystal)

We consider the system from Definitions 8.28. We therefore consider the Hamiltonian

$$H = H_{\text{el}} + H_{\text{ph}} = \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} c_{m\mathbf{k}\sigma}^\dagger c_{m\mathbf{k}\sigma} + \sum_{\substack{\mathbf{q} \in Q \\ v \in \{1, \dots, rd_x\}}} \Omega_{\mathbf{q},v} b_{\mathbf{q}v}^\dagger b_{\mathbf{q}v} \quad (20.23)$$

Interacting Electrons and Phonons (Total Crystal)

We consider the system from Definitions 8.29. We therefore consider the Hamiltonian

$$H = \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} c_{m\mathbf{k}\sigma}^\dagger c_{m\mathbf{k}\sigma} + \sum_{\substack{\mathbf{q} \in Q \\ v \in \{1, \dots, rd_x\}}} \hbar \omega_{\mathbf{q},v} b_{\mathbf{q}v}^\dagger b_{\mathbf{q}v} + \sum_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ v \in \{1, \dots, rd_x\} \\ m, n \in \{1, \dots, \infty\}}} \frac{g_{mnv}(\mathbf{k}, \mathbf{q})}{\sqrt{N_p}} c_{\mathbf{k}+\mathbf{q}m\sigma}^\dagger c_{\mathbf{k}n\sigma} \left(b_{\mathbf{q}v} + b_{-\mathbf{q}v}^\dagger \right) \quad (20.24)$$

21. Thermal Quantum Field Theories of Real Materials

21.1. Non-Interacting Systems

21.1.1. Non-Interacting Electrons

Fact 21.1 Recall that the retarded Green-function for the electrons is given by analytical continuation of the Matsubara-Green-function according to Theorem 14.20 which is determined as

$$\mathcal{G}_{m,m'}^{\sigma,\sigma'}(i\omega_n, \mathbf{k}) = \frac{\delta_{\sigma,\sigma'}}{\hbar} \left(\frac{1}{i\omega_n - (H_{0,\sigma}(\mathbf{k}) - \mu) / \hbar} \right)_{m,m'} = \frac{\delta_{\sigma,\sigma'}}{i\hbar\omega_n - (\epsilon_{m\sigma}(\mathbf{k}) - \mu)} \quad (21.1)$$

with non-interacting band-dispersion-matrix $H_{0,\sigma}(\mathbf{k}) \equiv \text{diag}(\epsilon_{1,\sigma}(\mathbf{k}), \dots, \epsilon_{n_{\text{bnds}},\sigma}(\mathbf{k}))$, whereas n_{bnds} represents the number of bands that are under consideration¹.

Fact 21.2 Going over to the field-integral-description of the problem gives the action

$$S[\psi, \bar{\psi}] \equiv S_{\text{el}}[\psi, \bar{\psi}] \quad (21.2)$$

with

$$\bullet \quad S_{\text{el}}[\psi, \bar{\psi}] = \int \prod_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\downarrow, \uparrow\} \\ \tau \in (0, \hbar\beta) \\ m \in \{1, 2, \dots, \infty\}}} \bar{\psi}_{m\sigma}(\tau, \mathbf{k}) \left(\partial_\tau + \frac{\epsilon_{m\sigma}(\mathbf{k}) - \mu}{\hbar} \right) \psi_{m\sigma}(\tau, \mathbf{k})$$

Notation 21.3 We implement external sources into the partition-function as

$$Z \mapsto Z[j_{\text{el}}, \bar{j}_{\text{el}}] \equiv \int \mathcal{D}(\bar{\psi}, \psi) \exp \left[-S_{\text{el}}[\psi, \bar{\psi}] + \bar{j}_{\text{el}} \psi + \bar{\psi} j_{\text{el}} \right] \quad (21.3)$$

with abbreviations

$$\begin{aligned} \bullet \quad \bar{j}_{\text{el}} \psi &\equiv \int \prod_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\downarrow, \uparrow\} \\ \tau \in (0, \hbar\beta) \\ m \in \{1, 2, \dots, \infty\}}} \bar{j}_{m\sigma}(\tau, \mathbf{k}) \psi_{m\sigma}(\tau, \mathbf{k}) \\ \bullet \quad \bar{\psi} j_{\text{el}} &\equiv \int \prod_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\downarrow, \uparrow\} \\ \tau \in (0, \hbar\beta) \\ m \in \{1, 2, \dots, \infty\}}} \bar{\psi}_{m\sigma}(\tau, \mathbf{k}) j_{m\sigma}(\tau, \mathbf{k}) \end{aligned}$$

¹Usually this is the number of bands were are reproduced in the Wannierization-procedure.

Lemma 21.4 Going over to Matsubara-frequencies and introducing the four-momenta $k \equiv (i\omega_n, \mathbf{k}) \in i\Omega_{-1} \times Q$, we obtain

$$S_{[j, \bar{j}]}[\psi, \bar{\psi}] = S_{\text{el}}[\psi, \bar{\psi}] - \left(\bar{j}_{\text{el}} \psi + \bar{\psi} j_{\text{el}} \right)$$

with

$$\bullet S_{\text{el}}[\psi, \bar{\psi}] = \sum_{\substack{\sigma \in \{\downarrow, \uparrow\} \\ k \in i\Omega_{-1} \times Q \\ m \in \{1, 2, \dots, \infty\}}} \bar{\psi}_{m\sigma}(k) \left(-i\omega_k + \frac{\epsilon_{m\sigma}(\mathbf{k}) - \mu}{\hbar} \right) \psi_{m\sigma}(k) \equiv \sum_{\substack{\sigma \in \{\downarrow, \uparrow\} \\ k \in i\Omega_{-1} \times Q \\ m \in \{1, 2, \dots, \infty\}}} \bar{\psi}_{m\sigma}(k) \left(-\frac{1}{\hbar} \mathcal{G}_{m, \sigma}^{-\text{el}}(k) \right) \psi_{m\sigma}(k)$$

with negative four-momentum $-q \equiv (-i\omega_m, -\mathbf{q})$.

Proof 21.4 Follows from Corollary 18.15.

21.1.2. Non-Interacting Phonons

Fact 21.5 Going over to the field-integral-description of the problem gives the action

$$S[\phi, \bar{\phi}] = S_{\text{ph}}[\phi, \bar{\phi}] \quad (21.4)$$

with

$$\bullet S_{\text{ph}}[\phi, \bar{\phi}] = \oint_{\substack{\mathbf{q} \in Q \\ \tau \in (0, \hbar\beta) \\ v \in \{1, 2, \dots, rd_x\}}} \bar{\phi}_v(\tau, \mathbf{q}) (\partial_\tau + \omega_v(\mathbf{q})) \phi_v(\tau, \mathbf{q})$$

Notation 21.6 We implement external sources into the partition-function as

$$Z \mapsto Z[j_{\text{ph}}, \bar{j}_{\text{ph}}] \equiv \int \mathcal{D}(\bar{\phi}, \phi) \exp \left[-S_{\text{ph}}[\phi, \bar{\phi}] + \sum_{\bullet \in \{\text{el}, \text{ph}\}} \bar{j}_{\text{ph}} \phi_\bullet + \bar{\phi} j_{\text{ph}} \right] \quad (21.5)$$

with abbreviations

$$\bullet \bar{j}_{\text{ph}} \phi \equiv \oint_{\substack{\mathbf{q} \in Q \\ \tau \in (0, \hbar\beta) \\ v \in \{1, 2, \dots, rd_x\}}} \bar{j}_v(\tau, \mathbf{q}) \phi_v(\tau, \mathbf{q})$$

$$\bullet \bar{\phi} j_{\text{ph}} \equiv \oint_{\substack{\mathbf{q} \in Q \\ \tau \in (0, \hbar\beta) \\ v \in \{1, 2, \dots, rd_x\}}} \bar{\phi}_v(\tau, \mathbf{q}) j_v(\tau, \mathbf{q})$$

Lemma 21.7 Going over to Matsubara-frequencies and introducing the four-momenta $q \equiv (i\omega_m, \mathbf{q}) \in i\Omega_{+1} \times Q$ and $k \equiv (i\omega_n, \mathbf{k}) \in i\Omega_{-1} \times Q$, we obtain

$$S_{[j,\bar{j}]}[\phi, \bar{\phi}] = S_{\text{ph}}[\phi, \bar{\phi}] - \left(\bar{j}_{\text{ph}} \phi_{\text{ph}} + \bar{\phi}_{\text{ph}} j_{\text{ph}} \right)$$

with

$$\bullet S_{\text{ph}}[\phi, \bar{\phi}] = \sum_{\substack{q \in i\Omega_{+1} \times Q \\ v \in \{1,2,\dots,rd_x\}}} \bar{\phi}_v(q) (-i\omega_q + \omega_v(\mathbf{q})) \phi_v(q) \equiv \sum_{\substack{q \in i\Omega_{+1} \times Q \\ v \in \{1,2,\dots,rd_x\}}} \bar{\phi}_v(q) \left(-\frac{1}{\hbar} \mathcal{G}_{0,v}^{-\text{ph}}(q) \right) \phi_v(q)$$

with negative four-momentum $-q \equiv (-i\omega_m, -\mathbf{q})$.

Proof 21.7 Follows from Corollary 18.15.

Notice that in the case of phonons, one can also simplify the action by changing the field-variables. This is elaborated in Section 27.6.

21.1.3. Non-Interacting Electrons and Phonons (Total Crystal)

We consider the system from Definition 8.28.

Fact 21.8 Going over to the field-integral-description of the problem gives the action

$$S[\chi, \bar{\chi}] = S_{\text{el}}[\psi, \bar{\psi}] + S_{\text{ph}}[\phi, \bar{\phi}] \quad (21.6)$$

with

$$\bullet S_{\text{el}}[\psi, \bar{\psi}] = \int \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\downarrow, \uparrow\} \\ \tau \in (0, \hbar\beta) \\ m \in \{1,2,\dots,\infty\}}} \bar{\psi}_{m\sigma}(\tau, \mathbf{k}) \left(\partial_\tau + \frac{\epsilon_{m\sigma}(\mathbf{k}) - \mu}{\hbar} \right) \psi_{m\sigma}(\tau, \mathbf{k})$$

$$\bullet S_{\text{ph}}[\phi, \bar{\phi}] = \int \sum_{\substack{\mathbf{q} \in Q \\ \tau \in (0, \hbar\beta) \\ v \in \{1,2,\dots,rd_x\}}} \bar{\phi}_v(\tau, \mathbf{q}) (\partial_\tau + \omega_v(\mathbf{q})) \phi_v(\tau, \mathbf{q})$$

Notation 21.9 We implement external sources into the partition-function as

$$Z \mapsto Z[j_{\text{el}}, \bar{j}_{\text{el}}, j_{\text{ph}}, \bar{j}_{\text{ph}}] \equiv \int \mathcal{D}(\bar{\chi}, \chi) \exp \left[- \left(S_{\text{el}}[\psi, \bar{\psi}] + S_{\text{ph}}[\phi, \bar{\phi}] \right) + \sum_{\bullet \in \{\text{el}, \text{ph}\}} \bar{j} \cdot \chi + \bar{\chi} \cdot j \right] \quad (21.7)$$

with abbreviations

$$\bullet \bar{j}_{\text{el}} \chi_{\text{el}} \equiv \int \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\downarrow, \uparrow\} \\ \tau \in (0, \hbar\beta) \\ m \in \{1,2,\dots,\infty\}}} \bar{j}_{m\sigma}(\tau, \mathbf{k}) \psi_{m\sigma}(\tau, \mathbf{k})$$

$$\bullet \bar{j}_{\text{ph}} \chi_{\text{ph}} \equiv \int \sum_{\substack{\mathbf{q} \in Q \\ \tau \in (0, \hbar\beta) \\ v \in \{1,2,\dots,rd_x\}}} \bar{j}_v(\tau, \mathbf{q}) \phi_v(\tau, \mathbf{q})$$

- etc...

Lemma 21.10 Going over to Matsubara-frequencies and introducing the four-momenta $q \equiv (i\omega_m, \mathbf{q}) \in i\Omega_{+1} \times Q$ and $k \equiv (i\omega_n, \mathbf{k}) \in i\Omega_{-1} \times Q$, we obtain

$$S_{[j,\bar{j}]}[\chi, \bar{\chi}] = S_{\text{el}}[\psi, \bar{\psi}] + S_{\text{ph}}[\phi, \bar{\phi}] - \sum_{\bullet \in \{\text{el}, \text{ph}\}} \left(\bar{j} \cdot \chi + \bar{\chi} \cdot j \right)$$

with

$$\begin{aligned} \bullet S_{\text{el}}[\psi, \bar{\psi}] &= \sum_{\substack{\sigma \in \{\downarrow, \uparrow\} \\ k \in i\Omega_{-1} \times Q \\ m \in \{1, 2, \dots, \infty\}}} \bar{\psi}_{m\sigma}(k) \left(-i\omega_k + \frac{\epsilon_{m\sigma}(\mathbf{k}) - \mu}{\hbar} \right) \psi_{m\sigma}(k) \equiv \sum_{\substack{\sigma \in \{\downarrow, \uparrow\} \\ k \in i\Omega_{-1} \times Q \\ m \in \{1, 2, \dots, \infty\}}} \bar{\psi}_{m\sigma}(k) \left(-\frac{1}{\hbar} \mathcal{G}_{0, m\sigma}^{-\text{el}}(k) \right) \psi_{m\sigma}(k) \\ \bullet S_{\text{ph}}[\phi, \bar{\phi}] &= \sum_{\substack{q \in i\Omega_{+1} \times Q \\ v \in \{1, 2, \dots, rd_x\}}} \bar{\phi}_v(q) (-i\omega_q + \omega_v(\mathbf{q})) \phi_v(q) \equiv \sum_{\substack{q \in i\Omega_{+1} \times Q \\ v \in \{1, 2, \dots, rd_x\}}} \bar{\phi}_v(q) \left(-\frac{1}{\hbar} \mathcal{G}_{0, v}^{-\text{ph}}(q) \right) \phi_v(q) \end{aligned}$$

with negative four-momentum $-q \equiv (-i\omega_m, -\mathbf{q})$.

Proof 21.10 See Appendix, Proofs, page 393.

21.2. Interacting Electrons and Phonons (Total Crystal)

21.2.1. Introduction

Fact 21.11 Recall that the retarded Green-function for the electrons is given by analytical continuation of the Matsubara-Green-function according to Theorem 14.20 which is determined by solving the Dyson-equation (15.13) as

$$\mathcal{G}_{m, m'}^{\sigma, \sigma'}(i\omega_n, \mathbf{k}) = \frac{\delta_{\sigma, \sigma'}}{\hbar} \left(\frac{1}{i\omega_n - (H_{0, \sigma}(\mathbf{k}) + \Sigma_{\sigma}(\mathbf{k}, i\omega_n) - \mu) / \hbar} \right)_{m, m'} \quad (21.8)$$

with non-interacting band-dispersion-matrix $H_{0, \sigma}(\mathbf{k}) \equiv \text{diag}(\epsilon_{1, \sigma}(\mathbf{k}), \dots, \epsilon_{n_{\text{bnds}}, \sigma}(\mathbf{k}))$, whereas n_{bnds} represents the number of bands that are under consideration².

Remark 21.11.1 Notice further that we have determined the self-energy in this system up to second / third order perturbation theory in Theorem 19.3.

Remark 21.11.2 Usually, we enumerate the indices in such a way, that the matrix-representation of the Green-function becomes

²Usually this is the number of bands were are reproduced in the Wannierization-procedure.

$$\mathcal{G}(i\omega_n, \mathbf{k}) = \begin{pmatrix} \mathcal{G}_{\uparrow,\uparrow}(i\omega_n, \mathbf{k}) & \mathbf{0} \\ \mathbf{0} & \mathcal{G}_{\downarrow,\downarrow}(i\omega_n, \mathbf{k}) \end{pmatrix}.$$

with individual blocks for spin-up and spin-down electrons. More explicitly, this means

$$\mathcal{G}(i\omega_n, \mathbf{k}) = \begin{pmatrix} \mathcal{G}_{1,1}^{\uparrow,\uparrow} & \mathcal{G}_{1,2}^{\uparrow,\uparrow} & \cdots & \mathcal{G}_{1,n_{\text{bnds}}}^{\uparrow,\uparrow} & 0 & 0 & \cdots & 0 \\ \mathcal{G}_{2,1}^{\uparrow,\uparrow} & \mathcal{G}_{2,2}^{\uparrow,\uparrow} & \cdots & \mathcal{G}_{2,n_{\text{bnds}}}^{\uparrow,\uparrow} & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathcal{G}_{n_{\text{bnds}},1}^{\uparrow,\uparrow} & \mathcal{G}_{n_{\text{bnds}},2}^{\uparrow,\uparrow} & \cdots & \mathcal{G}_{n_{\text{bnds}},n_{\text{bnds}}}^{\uparrow,\uparrow} & 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & \mathcal{G}_{1,1}^{\downarrow,\downarrow} & \mathcal{G}_{1,2}^{\downarrow,\downarrow} & \cdots & \mathcal{G}_{1,n_{\text{bnds}}}^{\downarrow,\downarrow} \\ 0 & 0 & \cdots & 0 & \mathcal{G}_{2,1}^{\downarrow,\downarrow} & \mathcal{G}_{2,2}^{\downarrow,\downarrow} & \cdots & \mathcal{G}_{2,n_{\text{bnds}}}^{\downarrow,\downarrow} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & \mathcal{G}_{n_{\text{bnds}},1}^{\downarrow,\downarrow} & \mathcal{G}_{n_{\text{bnds}},2}^{\downarrow,\downarrow} & \cdots & \mathcal{G}_{n_{\text{bnds}},n_{\text{bnds}}}^{\downarrow,\downarrow} \end{pmatrix} (i\omega_n, \mathbf{k}).$$

Notation 21.12 We simplify the notation further, by recalling that the self-energy is, within the diagonal-band-approximation (cf. Remark 19.3.1), diagonal in spin and wave-vector indices, such that we can write

$$G_{\substack{m,m' \\ \sigma,\sigma' \\ \mathbf{k},\mathbf{k}'}}^{\text{ret}}(E + i\delta) \equiv \delta_{\mathbf{k},\mathbf{k}'} \delta_{\sigma,\sigma'} G_{m,m'}^{\text{ret}}(E + i\delta, \mathbf{k}) \quad (21.9)$$

whereas we keep the double spin-indices explicit for reasons that will become clear in the chapter about superconductivity.

Fact 21.13 Going over to the field-integral-description of the problem gives the action

$$S[\chi, \bar{\chi}] = S_{\text{el}}[\psi, \bar{\psi}] + S_{\text{ph}}[\phi, \bar{\phi}] + S_{\text{el-ph-int}}[\chi, \bar{\chi}] \quad (21.10)$$

with

$$\begin{aligned} \bullet S_{\text{el}}[\psi, \bar{\psi}] &= \int \prod_{\substack{\mathbf{k} \in \mathbf{Q} \\ \sigma \in \{\downarrow, \uparrow\} \\ \tau \in (0, \hbar\beta) \\ m \in \{1, 2, \dots, \infty\}}} \bar{\psi}_{m\sigma}(\tau, \mathbf{k}) \left(\partial_\tau + \frac{\epsilon_{m\sigma}(\mathbf{k}) - \mu}{\hbar} \right) \psi_{m\sigma}(\tau, \mathbf{k}) \\ \bullet S_{\text{ph}}[\phi, \bar{\phi}] &= \int \prod_{\substack{\mathbf{q} \in \mathbf{Q} \\ \tau \in (0, \hbar\beta) \\ v \in \{1, 2, \dots, rd_{\mathbf{x}}\}}} \bar{\phi}_v(\tau, \mathbf{q}) (\partial_\tau + \omega_v(\mathbf{q})) \phi_v(\tau, \mathbf{q}) \\ \bullet S_{\text{el-ph-int}}[\chi, \bar{\chi}] &= \int \prod_{\substack{\mathbf{k}, \mathbf{q} \in \mathbf{Q} \\ \tau \in (0, \hbar\beta) \\ v \in \{1, 2, \dots, rd_{\mathbf{x}}\} \\ m, n \in \{1, 2, \dots, \infty\}}} \frac{g_{mnv}(\mathbf{k}, \mathbf{q})}{\hbar \sqrt{N_p}} \left(\phi_v(\tau, \mathbf{q}) + \bar{\phi}_v(\tau, -\mathbf{q}) \right) \bar{\psi}_{m\sigma}(\tau, \mathbf{k} + \mathbf{q}) \psi_{n\sigma}(\tau, \mathbf{k}) \end{aligned}$$

Notation 21.14 We implement external sources into the partition-function as

$$Z \mapsto Z[j_{\text{el}}, \bar{j}_{\text{el}}, j_{\text{ph}}, \bar{j}_{\text{ph}}] \equiv \int \mathcal{D}(\bar{\chi}, \chi) \exp \left[- \left(S_{\text{el}}[\psi, \bar{\psi}] + S_{\text{ph}}[\phi, \bar{\phi}] + S_{\text{el-ph-int}}[\chi, \bar{\chi}] \right) + \sum_{\bullet \in \{\text{el, ph}\}} \bar{j}_{\bullet} \chi_{\bullet} + \bar{\chi}_{\bullet} j_{\bullet} \right] \quad (21.11)$$

with abbreviations

- $\bar{j}_{\text{el}} \chi_{\text{el}} \equiv \sum_{\substack{\mathbf{k} \in \mathbf{Q} \\ \sigma \in \{\downarrow, \uparrow\} \\ \tau \in (0, \hbar\beta) \\ m \in \{1, 2, \dots, \infty\}}} \bar{j}_{m\sigma}(\tau, \mathbf{k}) \psi_{m\sigma}(\tau, \mathbf{k})$
- $\bar{j}_{\text{ph}} \chi_{\text{ph}} \equiv \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ \tau \in (0, \hbar\beta) \\ v \in \{1, 2, \dots, rd_{\mathbf{x}}\}}} \bar{j}_v(\tau, \mathbf{q}) \phi_v(\tau, \mathbf{q})$
- etc...

Lemma 21.15 Going over to Matsubara-frequencies and introducing the four-momenta $q \equiv (i\omega_m, \mathbf{q}) \in i\Omega_{+1} \times \mathbf{Q}$ and $k \equiv (i\omega_n, \mathbf{k}) \in i\Omega_{-1} \times \mathbf{Q}$, we obtain

$$S_{[j, \bar{j}]}[\chi, \bar{\chi}] = S_{\text{el}}[\psi, \bar{\psi}] + S_{\text{ph}}[\phi, \bar{\phi}] + S_{\text{el-ph-int}}[\chi, \bar{\chi}] - \sum_{\bullet \in \{\text{el, ph}\}} (\bar{j}_{\bullet} \chi_{\bullet} + \bar{\chi}_{\bullet} j_{\bullet})$$

with

- $S_{\text{el}}[\psi, \bar{\psi}] = \sum_{\substack{\sigma \in \{\downarrow, \uparrow\} \\ k \in i\Omega_{-1} \times \mathbf{Q} \\ m \in \{1, 2, \dots, \infty\}}} \bar{\psi}_{m\sigma}(k) \left(-i\omega_k + \frac{\epsilon_{m\sigma}(\mathbf{k}) - \mu}{\hbar} \right) \psi_{m\sigma}(k) \equiv \sum_{\substack{\sigma \in \{\downarrow, \uparrow\} \\ k \in i\Omega_{-1} \times \mathbf{Q} \\ m \in \{1, 2, \dots, \infty\}}} \bar{\psi}_{m\sigma}(k) \left(-\frac{1}{\hbar} \mathcal{G}_{0, m\sigma}^{-\text{el}}(k) \right) \psi_{m\sigma}(k)$
- $S_{\text{ph}}[\phi, \bar{\phi}] = \sum_{\substack{q \in i\Omega_{+1} \times \mathbf{Q} \\ v \in \{1, 2, \dots, rd_{\mathbf{x}}\}}} \bar{\phi}_v(q) (-i\omega_q + \omega_v(\mathbf{q})) \phi_v(q) \equiv \sum_{\substack{q \in i\Omega_{+1} \times \mathbf{Q} \\ v \in \{1, 2, \dots, rd_{\mathbf{x}}\}}} \bar{\phi}_v(q) \left(-\frac{1}{\hbar} \mathcal{G}_{0, v}^{-\text{ph}}(q) \right) \phi_v(q)$
- $S_{\text{el-ph-int}}[\chi, \bar{\chi}] = \frac{1}{\sqrt{\hbar\beta}} \sum_{\substack{k \in i\Omega_{-1} \times \mathbf{Q} \\ q \in i\Omega_{+1} \times \mathbf{Q} \\ v \in \{1, 2, \dots, rd_{\mathbf{x}}\} \\ m, n \in \{1, 2, \dots, \infty\}}} \frac{g_{mnv}(\mathbf{k}, \mathbf{q})}{\hbar \sqrt{N_p}} (\phi_v(q) + \bar{\phi}_v(-q)) \bar{\psi}_{m\sigma}(q+k) \psi_{n\sigma}(k)$

with negative four-momentum $-q \equiv (-i\omega_m, -\mathbf{q})$.

Proof 21.15 See Appendix, Proofs, page 393.

Lemma 21.16 We can rewrite the interaction-term as

$$\begin{aligned}
 S_{\text{el-ph-int}}[\chi, \bar{\chi}] &= \left(S_{\text{el-ph-int}} + S_{\text{ph-sources}} \right)_{[j_{\text{ph}}, \bar{j}_{\text{ph}}]}[\chi, \bar{\chi}] \\
 &= \sum_{\substack{q \in i\Omega_{+1} \times Q \\ v \in \{1, \dots, rd_x\}}} \phi_v(q) \left(\rho_v(q) - \bar{j}_v(q) \right) + \bar{\phi}_v(q) \left(\rho_v(-q) - j_v(q) \right)
 \end{aligned}$$

with density-like-interaction

$$\rho_v(q) = \frac{1}{\sqrt{\hbar\beta N_p}} \sum_{\substack{\sigma \in \{\downarrow, \uparrow\} \\ k \in i\Omega_{-1} \times Q \\ n, m \in \{1, \dots, \infty\}}} \frac{g_{mnv}(\mathbf{k}, \mathbf{q})}{\hbar} \bar{\psi}_{m\sigma}(k+q) \psi_{n\sigma}(k) \quad (21.12)$$

Proof 21.16 See Appendix, Proofs, page 496.

Corollary 21.17 One finds

$$\overline{\rho_v(-q)} = \rho_v(q) \quad (21.13)$$

Proof 21.17 We compute straightforwardly

$$\begin{aligned}
 \overline{\rho_v(-q)} &= \frac{1}{\sqrt{\hbar\beta N_p}} \sum_{\substack{\sigma \in \{\downarrow, \uparrow\} \\ k \in i\Omega_{-1} \times Q \\ n, m \in \{1, 2, \dots, \infty\}}} g_{mnv}(\mathbf{k}, -\mathbf{q})^* \left(\bar{\psi}_{m\sigma}(k-q) \psi_{n\sigma}(k) \right)^\dagger \\
 &\stackrel{\text{conj. rule}}{=} \frac{1}{\sqrt{\hbar\beta N_p}} \sum_{\substack{\sigma \in \{\downarrow, \uparrow\} \\ k \in i\Omega_{-1} \times Q \\ n, m \in \{1, 2, \dots, \infty\}}} g_{mnv}(\mathbf{k}, -\mathbf{q})^* \bar{\psi}_{n\sigma}(k) \psi_{m\sigma}(k-q) \\
 &\stackrel{\substack{m \mapsto \tilde{n} \\ n \mapsto \tilde{m}}}{=} \frac{1}{\sqrt{\hbar\beta N_p}} \sum_{\substack{\sigma \in \{\downarrow, \uparrow\} \\ k \in i\Omega_{-1} \times Q \\ \tilde{n}, \tilde{m} \in \{1, 2, \dots, \infty\}}} g_{\tilde{n}\tilde{m}v}(\mathbf{k}, -\mathbf{q})^* \bar{\psi}_{\tilde{m}\sigma}(k) \psi_{\tilde{n}\sigma}(k-q) \\
 &\stackrel{\text{Lemma 8.30}}{=} \frac{1}{\sqrt{\hbar\beta N_p}} \sum_{\substack{\sigma \in \{\downarrow, \uparrow\} \\ k \in i\Omega_{-1} \times Q \\ \tilde{n}, \tilde{m} \in \{1, 2, \dots, \infty\}}} g_{\tilde{m}\tilde{n}v}(\mathbf{k} - \mathbf{q}, +\mathbf{q}) \bar{\psi}_{\tilde{m}\sigma}(k) \psi_{\tilde{n}\sigma}(k-q) \\
 &\stackrel{k \mapsto k+q}{=} \frac{1}{\sqrt{\hbar\beta N_p}} \sum_{\substack{\sigma \in \{\downarrow, \uparrow\} \\ k \in i\Omega_{-1} \times Q \\ \tilde{n}, \tilde{m} \in \{1, 2, \dots, \infty\}}} g_{\tilde{m}\tilde{n}v}(\mathbf{k}, \mathbf{q}) \bar{\psi}_{\tilde{m}\sigma}(k+q) \psi_{\tilde{n}\sigma}(k) \\
 &= \rho_v(q)
 \end{aligned} \quad (21.14)$$

21.2.2. Integrating out the Phonons

Lemma 21.18 Changing variables in a standard way allows us to integrate out the phonons, such that we obtain the effective fermionic action

$$\begin{aligned}
 S_{[j, \bar{j}]}[\psi, \bar{\psi}] &= S_{\text{el}}[\psi, \bar{\psi}] + S_{\text{el-el-int}}[\psi, \bar{\psi}]_{[j_{\text{ph}}, \bar{j}_{\text{ph}}]} - \left(\bar{j}_{\text{el}} \psi + \bar{\psi} j_{\text{el}} \right) \\
 &= \sum_{\substack{\sigma \in \{\downarrow, \uparrow\} \\ k \in i\Omega_{-1} \times Q \\ m \in \{1, 2, \dots, \infty\}}} \bar{\psi}_{m\sigma}(k) \left(-\frac{1}{\hbar} \mathcal{G}_{m, \sigma}^{-\text{el}}(k) \right) \psi_{m\sigma}(k) + S_{\text{el-el-int}}[\psi, \bar{\psi}]_{[j_{\text{ph}}, \bar{j}_{\text{ph}}]} - \left(\bar{j}_{\text{el}} \psi + \bar{\psi} j_{\text{el}} \right) \quad (21.15)
 \end{aligned}$$

giving four terms, whereas the term that is quartic in the fermionic fields can be simplified to

$$\begin{aligned}
 S_{\text{el-el-int}}[\psi, \bar{\psi}]_{[j_{\text{ph}}, \bar{j}_{\text{ph}}]} &= - \sum_{\substack{q \in i\Omega_{+1} \times Q \\ v \in \{1, \dots, rd_{\mathbf{x}}\}}} \frac{\omega_v(\mathbf{q})}{\omega_m^2 + \omega_v^2(\mathbf{q})} \rho_v(q) \rho_v(-q) \\
 &\quad - \hbar \sum_{\substack{q \in i\Omega_{+1} \times Q \\ v \in \{1, \dots, rd_{\mathbf{x}}\}}} \rho_v(q) \mathcal{G}_{0,v}(q) j_v(q) - \hbar \sum_{\substack{q \in i\Omega_{+1} \times Q \\ v \in \{1, \dots, rd_{\mathbf{x}}\}}} \bar{j}_v(q) \mathcal{G}_{0,v}(q) \rho_v(-q) \\
 &\quad + \hbar \sum_{\substack{q \in i\Omega_{+1} \times Q \\ v \in \{1, \dots, rd_{\mathbf{x}}\}}} \bar{j}_v(q) \mathcal{G}_{0,v}(q) j_v(q) \quad (21.16)
 \end{aligned}$$

Proof 21.18 See Appendix, Proofs, page 414.

Remark 21.18.1 Notice that this could be computed much simpler with Remark 16.28.1.

Lemma 21.19 Introducing the multi-index

$$\alpha \equiv (m_\alpha, \sigma_\alpha, i\omega_\alpha, \mathbf{k}_\alpha) \in \{1, 2, \dots\} \times \{\downarrow, \uparrow\} \times i\Omega_{-1} \times Q \equiv \Xi, \quad (21.17)$$

one can rewrite the interaction-term-contribution that is quartic in the Graßmann-variables as generalised bilinear-form

$$S_{\text{el-el-int}}[\psi, \bar{\psi}] = \sum_{\alpha_1, \dots, \alpha_4 \in \Xi} \bar{\psi}_{\alpha_1} \bar{\psi}_{\alpha_2} V_{\alpha_1, \alpha_2; \alpha_3, \alpha_4} \psi_{\alpha_3} \psi_{\alpha_4}$$

with

$$\begin{aligned}
 & V_{\alpha_1, \alpha_2; \alpha_3, \alpha_4} \\
 &= \frac{1}{\hbar \beta N_p} \sum_{\nu \in \{1, \dots, r d_x\}} \frac{\omega_\nu(\mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4})}{\omega_{\alpha_1 - \alpha_4}^2 + \omega_\nu^2(\mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4})} \frac{g_{m_{\alpha_1} m_{\alpha_4} \nu}(\mathbf{k}_{\alpha_4}, \mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4}) g_{m_{\alpha_3} m_{\alpha_2} \nu}^*(\mathbf{k}_{\alpha_2}, \mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4})}{\hbar^2} \delta_{k_{\alpha_3} + k_{\alpha_4}, k_{\alpha_2} + k_{\alpha_1}} \delta_{\sigma_{\alpha_1}, \sigma_{\alpha_4}} \delta_{\sigma_{\alpha_2}, \sigma_{\alpha_3}} \\
 &= \frac{1}{\hbar \beta N_p} \sum_{\nu \in \{1, \dots, r d_x\}} \frac{\omega_\nu(\mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4})}{\omega_{\alpha_1 - \alpha_4}^2 + \omega_\nu^2(\mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4})} \frac{g_{m_{\alpha_1} m_{\alpha_4} \nu}(\mathbf{k}_{\alpha_4}, \mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4}) g_{m_{\alpha_2} m_{\alpha_3} \nu}(\mathbf{k}_{\alpha_2} + \mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4}, -\mathbf{k}_{\alpha_1} + \mathbf{k}_{\alpha_4})}{\hbar^2} \delta_{k_{\alpha_3} + k_{\alpha_4}, k_{\alpha_2} + k_{\alpha_1}} \delta_{\sigma_{\alpha_1}, \sigma_{\alpha_4}} \delta_{\sigma_{\alpha_2}, \sigma_{\alpha_3}}
 \end{aligned} \tag{21.18}$$

Proof 21.19 See Appendix, Proofs, page 414.

Corollary 21.20 This³ action is invariant under conjugation.

Proof 21.20 See Appendix, Proofs, page 415.

21.2.3. Expectation-Values as Source-Derivatives

Notation 21.21 We assume that we have a partition-function with $(Z[j_{\text{el}}, \bar{j}_{\text{el}}, j_{\text{ph}}, \bar{j}_{\text{ph}}])$ and without (Z) electronic and phononic source-terms consider the ratio as generating functional

$$\frac{Z[j_{\text{el}}, \bar{j}_{\text{el}}, j_{\text{ph}}, \bar{j}_{\text{ph}}]}{Z} \equiv \frac{Z[j, \bar{j}]}{Z} \tag{21.19}$$

whereas the external sources enter the partition function as in Notation 16.27. After obtaining a final expression for the partition function, we compute various quantities by taking functional derivatives.

Remark 21.21.1 Notice that the partition function can be simplified, by integrating out the phonons. Then one obtains Equation (21.15), which we rewrite here symbolically as

$$\begin{aligned}
 Z[j, \bar{j}] &\equiv \int \mathcal{D}(\bar{\chi}, \chi) \exp \left[- \left(S_{\text{el}}[\psi, \bar{\psi}] + S_{\text{ph}}[\phi, \bar{\phi}] + S_{\text{el-ph-int}}[\chi, \bar{\chi}] \right) + \sum_{\bullet \in \{\text{el, ph}\}} \bar{j} \cdot \chi + \bar{\chi} \cdot j \right] \\
 &= \int \mathcal{D}(\bar{\psi}, \psi) \exp - S_{[j, \bar{j}]}[\psi, \bar{\psi}] \\
 &= \int \mathcal{D}(\bar{\psi}, \psi) \exp - \left(S_{\text{el}}[\psi, \bar{\psi}] + S_{\text{el-el-int}}[\psi, \bar{\psi}] - \left(\bar{j}_{\text{el}} \psi + \bar{\psi} j_{\text{el}} \right) \right) \\
 &= \int \mathcal{D}(\bar{\psi}, \psi) \exp \left(\bar{j}_{\text{el}} \psi + \bar{\psi} j_{\text{el}} \right) \exp - \left(S_{\text{el}} + S_{\text{el-el-int}} \right)_{[j_{\text{ph}}, \bar{j}_{\text{ph}}]}[\psi, \bar{\psi}]
 \end{aligned} \tag{21.20}$$

³ This is actually a much more general theorem that should hold IN GENERAL. This should be implemented somewhere in the text...

Corollary 21.22 Under the assumption, that the Matsubara-Green-function is diagonal w.r.t. the Matsubara-frequencies, we can re-express the following quantities as functional derivatives of the partition function:

Electronic Observables:

- The expectation-value of an occupation-number-operator w.r.t. the electronic state $(n, \sigma, \mathbf{k})$

$$\langle n_{n\sigma}(\mathbf{k}) \rangle = \frac{1}{\hbar\beta} \sum_{\omega_m \in \Omega_{-1}} e^{+i\omega_m 0^+} \frac{\partial^2}{\partial \bar{j}_{n,\sigma}(i\omega_m, \mathbf{k}) \partial j_{n,\sigma}(i\omega_m, \mathbf{k})} \frac{Z[j_{\text{el}}, \bar{j}_{\text{el}}, j_{\text{ph}}, \bar{j}_{\text{ph}}]}{Z} \Big|_{\bar{j}=j=0}, \quad (21.21)$$

- the total particle number

$$\langle N \rangle = \frac{1}{\hbar\beta} \sum_{\substack{\sigma \in \{\downarrow, \uparrow\} \\ k \in i\Omega_{-1} \times Q \\ n \in \{1, 2, \dots, \infty\}}} e^{+i\omega_k 0^+} \frac{\partial^2}{\partial \bar{j}_{n,\sigma}(k) \partial j_{n,\sigma}(k)} \frac{Z[j_{\text{el}}, \bar{j}_{\text{el}}, j_{\text{ph}}, \bar{j}_{\text{ph}}]}{Z} \Big|_{\bar{j}=j=0}, \quad (21.22)$$

- the electronic contribution to the total energy we have

$$\langle H_{\text{el}} \rangle = \frac{1}{\hbar\beta} \sum_{\substack{\sigma \in \{\downarrow, \uparrow\} \\ k \in i\Omega_{-1} \times Q \\ n \in \{1, 2, \dots, \infty\}}} e^{+i\omega_k 0^+} \epsilon_{n\sigma}(\mathbf{k}) \frac{\partial^2}{\partial \bar{j}_{n,\sigma}(k) \partial j_{n,\sigma}(k)} \frac{Z[j_{\text{el}}, \bar{j}_{\text{el}}, j_{\text{ph}}, \bar{j}_{\text{ph}}]}{Z} \Big|_{\bar{j}=j=0}, \quad (21.23)$$

Phononic Observables:

- the phononic occupation-number-operator

$$\langle n_v(\mathbf{q}) \rangle = \frac{1}{\hbar\beta} \sum_{\omega_m \in \Omega_{+1}} e^{+i\omega_m 0^+} \frac{\partial^2}{\partial \bar{j}_v(i\omega_m, \mathbf{q}) \partial j_v(i\omega_m, \mathbf{q})} \frac{Z[j_{\text{el}}, \bar{j}_{\text{el}}, j_{\text{ph}}, \bar{j}_{\text{ph}}]}{Z} \Big|_{\bar{j}=j=0},$$

- the phononic contribution to the total energy we have

$$\langle H_{\text{ph}} \rangle = \frac{1}{\hbar\beta} \sum_{\substack{q \in i\Omega_{+1} \times Q \\ v \in \{1, 2, \dots, r_{d_x}\}}} e^{+i\omega_q 0^+} \Omega_v(\mathbf{q}) \frac{\partial^2}{\partial \bar{j}_v(q) \partial j_v(q)} \frac{Z[j_{\text{el}}, \bar{j}_{\text{el}}, j_{\text{ph}}, \bar{j}_{\text{ph}}]}{Z} \Big|_{\bar{j}=j=0}. \quad (21.24)$$

Interaction Observables:

- the electron-phonon-interaction energy is given by

$$\left\langle c_{\mathbf{k}+\mathbf{q}m\sigma}^\dagger c_{\mathbf{k}n\sigma} (b_{\mathbf{q}v} + b_{-\mathbf{q}v}^\dagger) \right\rangle = \frac{1}{\sqrt{\hbar\beta}^3} \sum_{\substack{\omega_m \in \Omega_{+1} \\ \omega_n, \omega_{n'} \in \Omega_{-1}}} e^{i\omega_n 0^+} \frac{\partial^2}{\partial \bar{j}_{n\sigma\mathbf{k}}(i\omega_n) \partial j_{m\sigma\mathbf{k}+\mathbf{q}}(i\omega_{n'})} \left(\frac{\partial}{\partial \bar{j}_v(i\omega_m, \mathbf{q})} + \frac{\partial}{\partial j_v(-i\omega_m, -\mathbf{q})} \right) \frac{Z[j_{\text{el}}, \bar{j}_{\text{el}}, j_{\text{ph}}, \bar{j}_{\text{ph}}]}{Z} \Big|_{\bar{j}=j=0}. \quad (21.25)$$

Proof 21.22 See Appendix, Proofs, page 503.

Remark 21.22.1 Equations (21.21), (21.21) and (21.23) are merely listed for completeness. The Green-function-techniques developed in the earlier chapters allow us a much more elegant computation of the internal energy, as will become clear in Theorem 23.16.

22. Density of States, Thermodynamic Potential & Conjugated Variables

Motivation and Overview

After this part of the notes you will understand how to calculate for arbitrary materials and for arbitrary temperatures the Band-Structure and [D]ensity [O]f [S]tates (DOS) ...

- ... for a typical non-interacting model, resulting in a plot as in Figure 22.1.
- ... for a typical interacting model, resulting in a plot as in the literature [29].

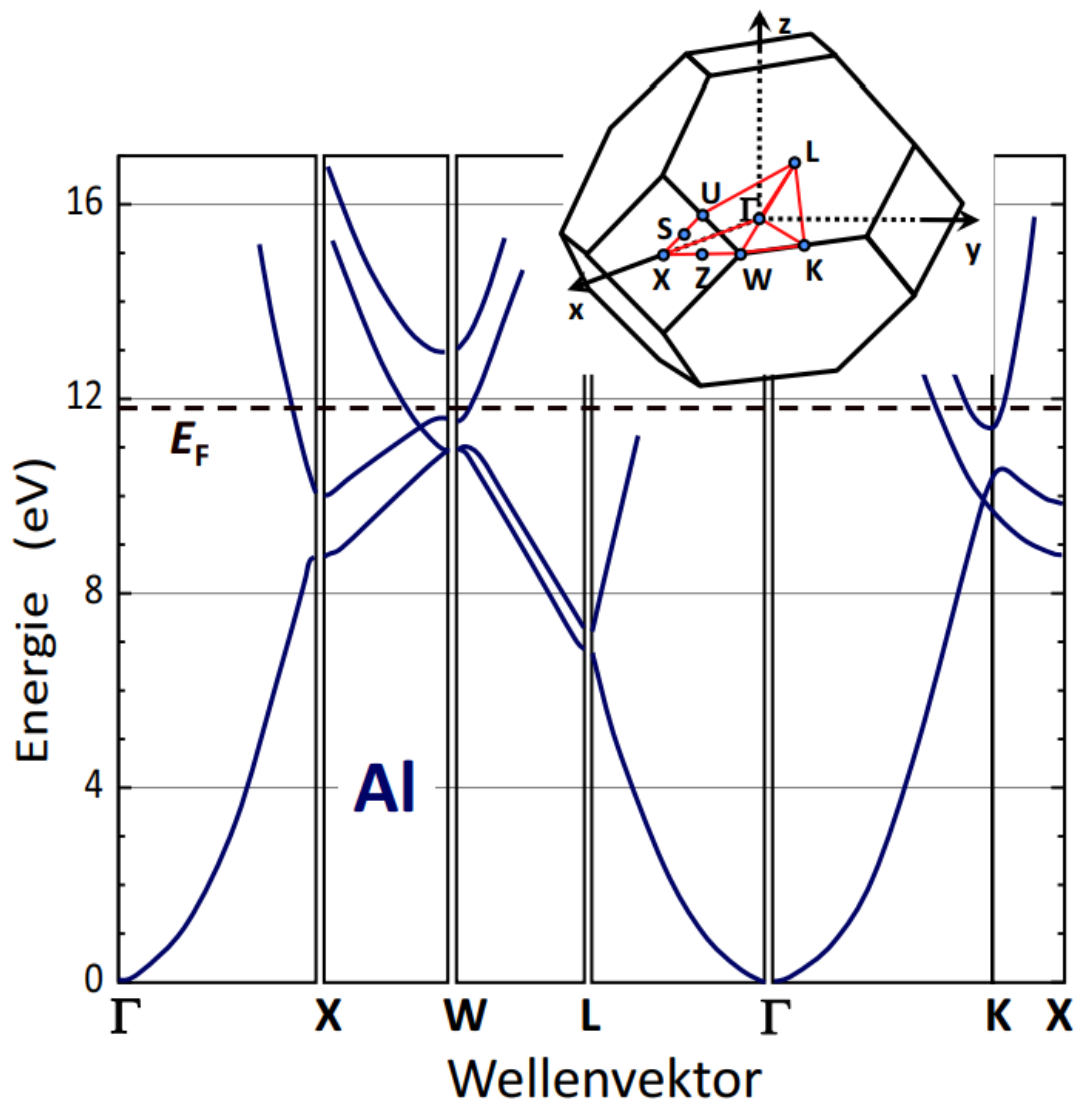


Figure 22.1.: The Band-Structure of Aluminium along the high-symmetry $\mathbf{k}$ -points. Taken from [20].

22.1. Non-Interacting Gas-Models

22.1.1. Non-Interacting Electrons

Electronic Density of States

Fact 22.1 Recall¹ that using Equation (14.12) we obtain the electronic density of states $\rho_0(E)$ for the system as

$$\rho_0(E) = \sum_{\substack{\mathbf{k} \in \mathbf{Q} \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \delta(E - (\epsilon_{m\mathbf{k}\sigma} - \mu)) \quad (22.1)$$

Proof 22.1 Follows immediately from the Sokhotski-Plemelj-Theorem in Equation (14.3).

Remark 22.1.1 Notice that this density of states is formally just a sum of Dirac-delta-peaks. However, when physicists *depict* (i.e. plot) the density of states, they recall that this model is just an approximation to a more realistic model which includes interactions. Since interactions usually imply the presence of smearing, one usually "artificially" introduces smearing in the non-interacting density of states by replacing the delta-peaks with Lorentzians or Gaussians.

Remark 22.1.2 We remark that regularizing the delta-functions with Gaussians converges "much faster" (in the number of k -points) than regularizing the delta-functions with Lorentzians.

Notation 22.2 Notice that this quantity is (after introducing cutoffs for the number of bands) normalized to

$$\int dE \rho_0(E) = N_x N_y N_z \cdot N_{\text{band}} \cdot 2 \quad (22.2)$$

Notice that the factor of 2 originates from spin-degeneracy. The density of states, as computed in [Q]uantum [E]spresso (QE) within the [L]ocal [D]ensity [A]pproximation (LDA), $\rho_{\text{QE}}(E)$ is normalized to the number of bands, i.e.

$$\frac{1}{N_x N_y N_z} \rho_0(E - \mu) = \frac{1}{N_x N_y N_z} \sum_{\substack{\mathbf{k} \in \mathbf{Q} \\ \sigma \in \{\downarrow, \uparrow\} \\ n \in \{1, 2, \dots\}}} \delta(E - \epsilon_{n\mathbf{k}}) \mapsto \frac{1}{N_x N_y N_z} \sum_{\substack{\mathbf{k} \in \mathbf{Q} \\ \sigma \in \{\downarrow, \uparrow\} \\ n \in \{1, 2, \dots, N_{\text{band}}\}}} \delta(E - \epsilon_{n\mathbf{k}}) \equiv \rho_{\text{QE, LDA}}(E) \quad (22.3)$$

such that it is normalized to

$$\int dE \rho_{\text{QE, LDA}}(E) = 2 N_{\text{band}} \quad (22.4)$$

¹Cf. Lemma 14.15.

Electronic Thermodynamic Potential - Landau Potential

Lemma 22.3 The grand canonical potential $\Omega_{0,\text{el}}(\mu, T, V)$ for the electronic system is given as

$$\Omega_{0,\text{el}}(\mu, T, V) = \frac{-1}{\beta} \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \text{Log} (1 + e^{-\beta(\epsilon_{m\mathbf{k}\sigma} - \mu)}) = \frac{-1}{\beta} \int dE \text{Log} (1 + e^{-\beta E}) \rho_0(E) \quad (22.5)$$

Proof 22.3 See Appendix, Proofs, page 504.

Electronic Thermodynamic Conjugated Variable - Chemical Potential

Fact 22.4 The chemical potential has to be computed by constraining the number of particles according to Lemma 14.14 as

$$\langle N \rangle_0 = \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} F_\beta(\epsilon_{m\mathbf{k}\sigma} - \mu) = \int_{-\infty}^{+\infty} dE F_\beta(E) \rho_0(E) \iff \frac{\langle N \rangle_0}{N_x N_y N_z} = \int_{-\infty}^{+\infty} dE F_\beta(E - \mu) \rho_{\text{QE, LDA}}(E) \quad (22.6)$$

Remark 22.4.1 Notice that we implicitly assumed that the number of bands is large enough to accommodate all electrons in the system.

Fact 22.5 We first notice that dictating the particle number (for example by requiring [C]harge [N]eutrality of the crystal) through Equation (22.6) can be rewritten as

$$\begin{aligned} F_0(\mu, T) &\equiv \langle N \rangle_0(\mu, T) - N_{\text{CN}} \\ &= \left[\int_{-\infty}^{+\infty} dE F_\beta(E) \rho_0(E) \right] - N_{\text{CN}} \\ &= \left[\int_{-\infty}^{+\infty} dE F_\beta(E - \mu) \rho_{\text{QE, LDA}}(E) \right] N_x N_y N_z - N_{\text{CN}} = 0. \end{aligned} \quad (22.7)$$

Remark 22.5.1 Notice that this is an implicit function, meaning that we can use the Theorem for implicit functions to compute, e.g. the derivative of the chemical potential as a function of temperature. Notice further that we dropped the volume-dependence for notational convenience.

Remark 22.5.2 Notice that the chemical potential that is defined through this Equation is unique due to Theorem 9.30.

Remark 22.5.3 Notice further that the partial derivatives of the function F_0 are given as the partial derivatives of the expected particle number $\langle N \rangle_0$, i.e.

$$(\partial_T F_0)(\mu, T) = \partial_T \langle N \rangle_0(\mu, T), \quad (\partial_\mu F_0)(\mu, T) = \partial_\mu \langle N \rangle_0(\mu, T). \quad (22.8)$$

Lemma 22.6 The temperature-derivative of the function F_0 can be computed with Lemma 9.45 and Equation (22.3) as

$$\begin{aligned} (\partial_T F_0)(\mu, T) &= \partial_T \langle N \rangle_0(\mu, T) = \frac{\beta}{4T} \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \text{sech}^2 \left(\frac{\beta \xi_{m\mathbf{k}\sigma}}{2} \right) \xi_{m\mathbf{k}\sigma} \\ &= N_x N_y N_z \frac{\beta}{4T} \int_{-\infty}^{+\infty} dE \text{sech}^2 \left(\beta \frac{E - \mu}{2} \right) (E - \mu) \rho_{\text{QE, LDA}}(E) \\ &= \frac{\beta}{4T} \int_{-\infty}^{+\infty} dE \text{sech}^2 \left(\frac{\beta E}{2} \right) E \rho_0(E) \\ &= \frac{\beta}{T} \int_{-\infty}^{+\infty} dE F_\beta(E) (1 - F_\beta(E)) E \rho_0(E) \end{aligned} \quad (22.9)$$

Proof 22.6 The individual identities are proven by referring to Lemma 9.48 and Notation 22.2.

Lemma 22.7 The chemical-potential-derivative of the function F can be computed as ...

$$\begin{aligned} (\partial_\mu F_0)(\mu, T) &= \partial_\mu \langle N \rangle_0(\mu, T) = \frac{\beta}{4} \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \text{sech}^2 \left(\frac{\xi_{m\mathbf{k}\sigma}}{2k_B T} \right) \\ &= N_x N_y N_z \frac{\beta}{4} \int_{-\infty}^{+\infty} dE \text{sech}^2 \left(\beta \frac{E - \mu}{2} \right) \rho_{\text{QE}}(E) \\ &= \frac{\beta}{4} \int_{-\infty}^{+\infty} dE \text{sech}^2 \left(\frac{\beta E}{2} \right) \rho_0(E) \end{aligned} \quad (22.10)$$

Proof 22.7 We compute the derivative of the Fermi-function w.r.t. the chemical potential with Equation (22.6) by noticing

$$F_\beta(\epsilon_{n\mathbf{k}\sigma} - \mu) \equiv n^{-1}(\beta(\epsilon_{n\mathbf{k}\sigma} - \mu)) \quad (22.11)$$

which implies with Lemma 9.45 and the chain-rule and $\xi_{n\mathbf{k}\sigma} \equiv \epsilon_{n\mathbf{k}\sigma} - \mu$ that

$$\partial_\mu n^{-1}(\beta(\epsilon_{n\mathbf{k}\sigma} - \mu)) = -\frac{1}{4} \text{sech}^2(\beta \xi_{n\mathbf{k}\sigma}/2) \cdot (-\beta) = \frac{\beta}{4} \text{sech}^2(\beta \xi_{n\mathbf{k}\sigma}/2) \quad (22.12)$$

proving that the first Equation-sign. The second and the third equation-sign follow from Equation (22.3).

Corollary 22.8 Assuming now we have access² to the values $\mu(T)$ that fulfill Equation (22.7), we can compute the temperature derivative of the chemical potential as

$$\left(\frac{\partial \mu}{\partial T}\right)(T) = -\left(\frac{\partial_T F_0}{\partial_\mu F_0}\right)(\mu(T), T) = -\left(\frac{\partial_T \langle N \rangle_0}{\partial_\mu \langle N \rangle_0}\right)(\mu(T), T). \quad (22.13)$$

Proof 22.8 Follows from the Theorem of implicit functions with Equation (22.7).

Remark 22.8.1 Notice that this equation can be rewritten in the physics-notation as

$$\frac{\partial \mu}{\partial T} = -\frac{\partial N}{\partial T} \frac{\partial \mu}{\partial N} \iff \frac{\partial \mu}{\partial T} \frac{\partial T}{\partial N} \frac{\partial N}{\partial \mu} = -1 \quad (22.14)$$

Corollary 22.9 Notice that this implies for Equation (22.13) in the non-interacting case the formula

$$\left(\frac{\partial \mu}{\partial T}\right)(T) = -\frac{1}{T} \frac{\int_{-\infty}^{+\infty} dE \operatorname{sech}^2\left(\beta \frac{E-\mu(T)}{2}\right) \rho_{\text{QE}}(E) [E - \mu(T)]}{\int_{-\infty}^{+\infty} dE \operatorname{sech}^2\left(\beta \frac{E-\mu(T)}{2}\right) \rho_{\text{QE}}(E)} \quad (22.15)$$

Proof 22.9 Follows immediately by combining Equation (22.13) with Lemmas 22.7 and 22.7.

Electronic Thermodynamic Conjugated Variable - Pressure

Fact 22.10 Notice that one can compute the pressure $P \equiv P(\mu, T, V)$ with Equations (10.9) and (22.5) as

$$-\frac{\Omega_0(\mu, T, V + \Delta V) - \Omega_0(\mu, T, V)}{\Delta V} \quad (22.16)$$

with $V \equiv N_p V_{\text{UC}}$, $N_p \equiv N_x N_y N_z$ and unit-cell volume

$$V_{\text{UC}} = \int_{\text{span}_{[0,1]\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}}} 1 d^3 x = \det(\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3) \quad (22.17)$$

²Notice that this is the case, as we compute the chemical potential numerically.

Electronic Thermodynamic Conjugated Variable - Entropy

We combine Equations (10.9) and (22.5) and compute the temperature-derivative

$$\begin{aligned} \partial_T \log(1 + e^{-\beta E}) &= \frac{1}{1 + e^{-\beta E}} \cdot e^{-\beta E} \cdot \left(\frac{E}{k_B T^2} \right) \\ &\stackrel{\text{Eq. (9.62)}}{=} F_\beta(E) \cdot \left(\frac{E}{k_B T^2} \right) \end{aligned} \quad (22.18)$$

which gives us

$$S = -\partial_T \Omega = \text{TBD} \quad (22.19)$$

22.1.2. Non-Interacting Phonons

Phononic Density of States

Definition 22.11 We introduce the *non-interacting phononic density of states* ρ_{ph} as

$$\rho_{\text{ph}}(E) = \frac{1}{N_x N_y N_z} \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ \nu \in \{1, 2, \dots, r d_x\}}} \delta(E - \Omega_\nu(\mathbf{q})) = \frac{1}{N_x N_y N_z} \frac{-1}{\pi} \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ \nu \in \{1, 2, \dots, r d_x\}}} \lim_{\eta \rightarrow 0^+} \text{Im} \frac{1}{E + i\eta - \Omega_\nu(\mathbf{q})} \quad (22.20)$$

Phononic Thermodynamic Potential - Helmholtz Free Energy

Lemma 22.12 The grand canonical potential $\Omega_{0,\text{el}}(\mu, T, V)$ for the electronic system is given as

$$\Omega_{0,\text{ph}}(T, V) = \frac{-1}{\beta} \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ \nu \in \{1, \dots, r d_x\}}} \text{Log}(1 - e^{-\beta \Omega_{\nu \mathbf{q}}}) = \frac{-1}{\beta} \int dE \text{Log}(1 - e^{-\beta E}) \rho_{0,\text{ph}}(E) \quad (22.21)$$

Proof 22.12 Analogous to the electronic case; Cf. Appendix, Proofs, page 504.

TBD

Phononic Thermodynamic Conjugated Variable - Chemical Potential

This is something that we will not elaborate in the PhD-thesis... However... It's not zero!

Phononic Thermodynamic Conjugated Variable - Pressure

This can be computed in complete analogy to Fact 22.10.

Phononic Thermodynamic Conjugated Variable - Entropy

We combine Equations (10.9) and (22.21) and compute the temperature-derivative

$$\begin{aligned} \partial_T \log(1 - e^{-\beta E}) &= \frac{1}{1 - e^{-\beta E}} \cdot (-e^{-\beta E}) \cdot \left(\frac{E}{k_B T^2} \right) \\ &\stackrel{\text{Eq. (9.62)}}{=} -B_\beta(E) \cdot \left(\frac{E}{k_B T^2} \right) \end{aligned} \quad (22.22)$$

which gives us

$$S = -\partial_T \Omega = \text{TBD} \quad (22.23)$$

22.1.3. Non-Interacting Electrons and Phonons (Total Crystal)

Total Crystal Thermodynamic Potential

Fact 22.13 The grand canonical potential $\Omega_0(\mu, T, V)$ is given as

$$\Omega_0(\mu, T, V) = \Omega_{0,\text{el}}(\mu, T, V) + \Omega_{0,\text{ph}}(T, V) \quad (22.24)$$

$$= \frac{-1}{\beta} \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \text{Log} \left(1 + e^{-\beta(\epsilon_{m\sigma}(\mathbf{k}) - \mu)} \right) + \frac{1}{\beta} \sum_{\substack{\mathbf{q} \in Q \\ v \in \{1, \dots, rd_{\mathbf{x}}\}}} \text{Log} \left(1 - e^{-\beta \Omega_v(\mathbf{q})} \right) \quad (22.25)$$

Proof 22.13 Follows from combining

22.2. Interacting Electrons and Phonons (Total Crystal)

22.2.1. Interacting Electrons

Fact 22.14 We compute the chemical potential, by constraining the number of particles according to Lemma 14.14 as

$$\langle N \rangle = \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \langle n_{m\mathbf{k}\sigma} \rangle = \frac{-1}{\pi} \lim_{\delta \rightarrow 0} \oint \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\} \\ E \in (-\infty, +\infty)}} F_\beta(E) \text{Im} G_{m,\sigma}^{\text{ret}}(E + i\delta) \quad (22.26)$$

Remark 22.14.1 The electronic self-energy depends on the specific system / interaction under consideration, but the formula is always "proper to use".

22.2.2. Interacting Phonons

We only remark that the computation of the occupation-number of phonons is not as straight-forward as in the electronic case. One can write on the one hand

$$\langle N_{\text{ph}} \rangle = \sum_{\substack{\mathbf{q} \in Q \\ v \in \{1, \dots, d_{\mathbf{x}}\}}} \langle n_{v\mathbf{q}} \rangle = \dots ? \quad (22.27)$$

22.2.3. Total Crystal

Theorem 22.15 For the grand canonical potential of the interacting system we have in the lowest order of perturbation theory

$$\text{TBDigitalized from Pinboard, Topic 17} \quad (22.28)$$

Proof 22.15 See Appendix, Proofs, page 504.

Remark 22.15.1 Notice further, that one can compute the pressure $P \equiv P(\mu, T, V)$ as

$$- \frac{\Omega(\mu, T, V + \Delta V) - \Omega(\mu, T, V)}{\Delta V} \quad (22.29)$$

with $V \equiv N_p V_{\text{UC}}$, $N_p \equiv N_x N_y N_z$ and unit-cell volume

$$V_{\text{UC}} = \int_{\text{span}_{[0,1]\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}}} 1 d^3x = \det(\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3) \quad (22.30)$$

23. Internal Energy & Specific Heat (Normal State)

In this part of the notes we will discuss how to calculate the specific heat for arbitrary materials in a typical non-interacting model at low temperatures as in Figures 23.1 and 23.2. We will then extend the discussion to interacting models as well.

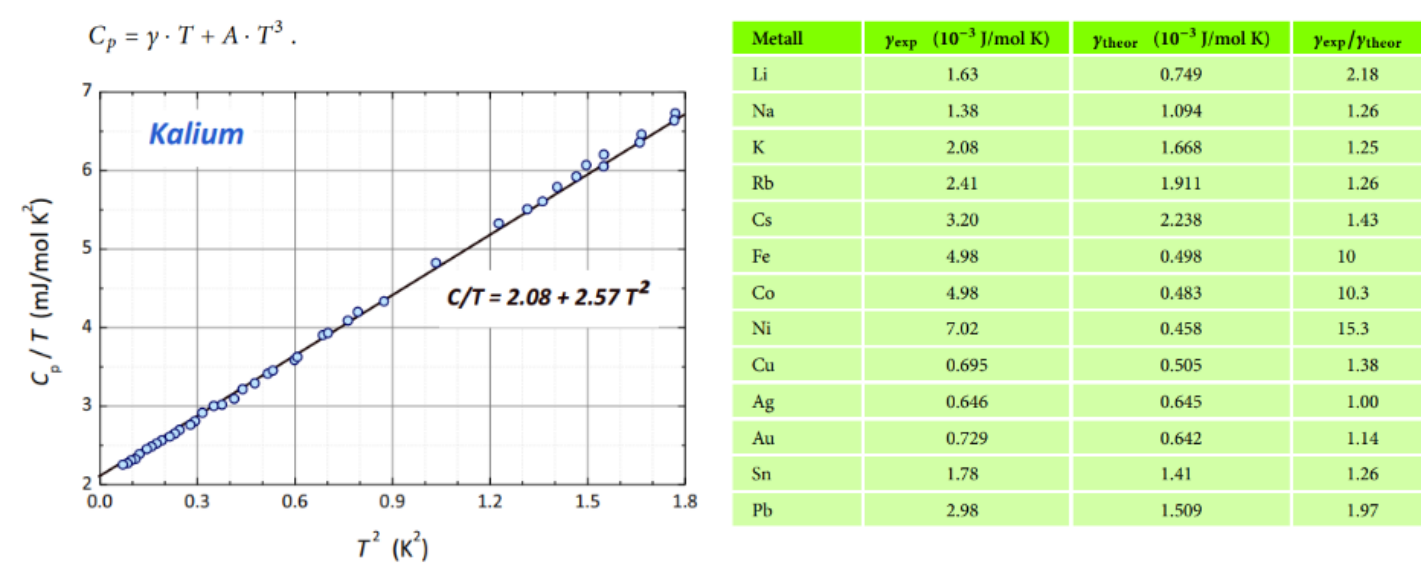


Figure 23.1.: Specific heat of various materials at low temperatures, taken from [20].

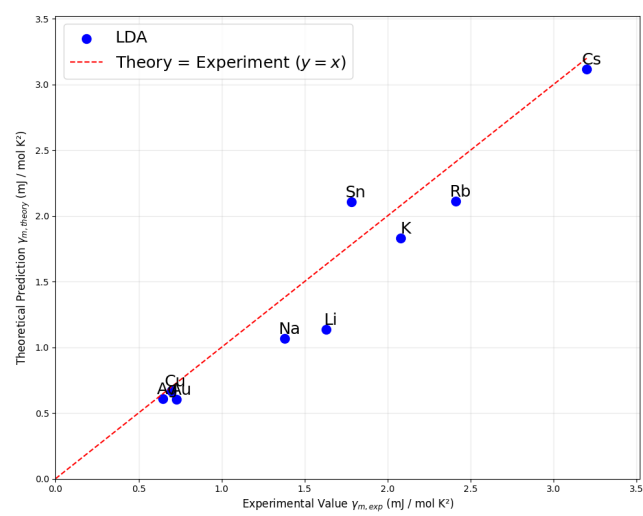


Figure 23.2.: Specific heat of various materials contrasting LDA and Experimental Results. Produced with quantum espresso v7.2 and norm conserving pseudopotentials from sssp.

23.1. Non-Interacting Gas-Models

23.1.1. Non-Interacting Electrons

Theorem 23.1 The expectation-value of the electronic internal energy can be computed as

$$\langle H_{\text{el}} \rangle_0 = \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} \langle c_{m\mathbf{k}\sigma}^\dagger c_{m\mathbf{k}\sigma} \rangle_0 = \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} F_\beta(\epsilon_{m\mathbf{k}\sigma} - \mu) \quad (23.1)$$

Corollary 23.2 Notice, that in the non-interacting case, this formula simplifies to

$$\langle H_{\text{el}} \rangle_0 = \int dE E F_\beta(E) \rho_0(E) \iff \frac{\langle H_{\text{el}} \rangle_0}{N_x N_y N_z} = \int dE (E - \mu) F_\beta(E - \mu) \rho_{\text{QE}}(E) \quad (23.2)$$

Proof 23.2 Follows with the Sokhotski-Plemelj theorem from Equation (23.10).

Lemma 23.3 In the absence of interactions one obtains the simple formula

$$\partial_T \frac{\langle H_{\text{el}} \rangle_0}{N_x N_y N_z} = \int dE \frac{E}{4} \text{sech}^2 \left(\frac{E - \mu(T)}{2k_B T} \right) \left[\frac{E - \mu(T)}{k_B T^2} \right] \rho_{\text{QE}}(E) \quad (23.3)$$

Proof 23.3 See Appendix, Proofs, page 507.

Lemma 23.4 In the absence of interactions one obtains the simple formula

$$\begin{aligned} \partial_\mu \frac{\langle H_{\text{el}} \rangle_0}{N_x N_y N_z} &= \frac{1}{N_x N_y N_z} \frac{\beta}{4} \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} \text{sech}^2 \left(\frac{\epsilon_{m\mathbf{k}\sigma} - \mu}{2k_B T} \right) \\ &= \frac{\beta}{4} \int dE E \text{sech}^2 \left(\frac{E - \mu}{2k_B T} \right) \rho_{\text{QE}}(E) \end{aligned} \quad (23.4)$$

Proof 23.4 See Appendix, Proofs, page 508.

23.1.2. Non-Interacting Phonons

Fact 23.5 The expectation-value of the phononic internal energy can be computed as

$$\langle H_{\text{ph}} \rangle_0 = \sum_{\substack{\mathbf{q} \in Q \\ v \in \{1,2,\dots,rd_x\}}} \Omega_v(\mathbf{q}) B_\beta(\Omega_v(\mathbf{q})) \quad (23.5)$$

Lemma 23.6 The temperature-derivatives of the non-interacting phononic contribution to the internal energy can be computed as

$$\begin{aligned} \partial_T \langle H_{\text{ph}} \rangle_0 &= \frac{\beta}{4T} \sum_{\substack{\mathbf{q} \in Q \\ v \in \{1,2,\dots,rd_x\}}} \left[\Omega_v(\mathbf{q}) \operatorname{csch}(\beta \Omega_v(\mathbf{q})/2) \right]^2 \\ &= \frac{\beta}{4T} \int dE \left[E \operatorname{csch}(\beta E/2) \right]^2 \rho_{\text{ph}}(E) \\ &= \frac{\beta}{4T} \lim_{\eta \rightarrow 0^+} \int dE \left[E \operatorname{csch}(\beta E/2) \right]^2 \left(-\frac{1}{\pi} \right) \sum_{\substack{\mathbf{q} \in Q \\ v \in \{1,2,\dots,rd_x\}}} \operatorname{Im} \frac{1}{E + i\eta - \Omega_v(\mathbf{q})} \end{aligned} \quad (23.6)$$

Proof 23.6 Follows by using Definition 22.11.

23.1.3. Non-Interacting Electrons and Phonons (Total Crystal)

Fact 23.7 The expectation-value of the total internal energy can be computed as

$$\begin{aligned} \langle H \rangle_0 &= \langle H_{\text{el}} \rangle_0 + \langle H_{\text{ph}} \rangle_0 \\ &= \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} F_\beta(\epsilon_{m\mathbf{k}\sigma} - \mu) + \sum_{\substack{\mathbf{q} \in Q \\ v \in \{1, \dots, rd_x\}}} \Omega_v(\mathbf{q}) B_\beta(\Omega_v(\mathbf{q})) \end{aligned} \quad (23.7)$$

Remark 23.7.1 Notice that these identities can be further simplified by using the density of states for the electronic and / or the phononic system.

Fact 23.8 The basic formula from is according to Equation (9.57)

$$\begin{aligned} c_{N,V}(T) &= \frac{C_{N,V}(T)}{n} = \frac{1}{n} \partial_T \langle H \rangle_{0,N,T,V} \\ &= \frac{1}{n} \left[(\partial_T \langle H \rangle_0)(\mu, T, V) + (\partial_\mu \langle H \rangle_0)(\mu, T, V) \cdot (\partial_T \mu)(N(\mu), V, T) \right] \end{aligned} \quad (23.8)$$

The Hamiltonian is made up of two contributions

$$H_{\text{tot}} = H_{\text{el}} + H_{\text{ph}} \quad (23.9)$$

Therefore, also for the specific heat, we have contributions from the electronic part and from the phononic part. We further need to account for the temperature derivative of the chemical potential. We consider all three quantities separately and then merge the results.

23.2. Interacting Electrons and Phonons (Total Crystal)

23.2.1. Interacting Electrons

Theorem 23.9 One finds the identity

$$\langle H_{\text{el}} \rangle = \sum_{\substack{\mathbf{k} \in \mathbf{Q} \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} \langle n_{m\mathbf{k}\sigma} \rangle = \frac{-1}{\pi} \lim_{\delta \rightarrow 0} \oint_{\substack{\mathbf{k} \in \mathbf{Q} \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\} \\ E \in (-\infty, +\infty)}} \epsilon_{m\mathbf{k}\sigma} F_{\beta}(E) \text{Im} G_{m,m}^{\text{ret}}(E + i\delta, \mathbf{k}) \quad (23.10)$$

Proof 23.9 See Appendix, Proofs, page 509.

The Electronic Contribution

Theorem 23.10 One finds for the interacting system two equivalent identities

$$\frac{\partial^2}{\partial \bar{j}_{n,\sigma}(k) \partial j_{n,\sigma}(k)} \frac{Z[j_{\text{el}}, \bar{j}_{\text{el}}, j_{\text{ph}}, \bar{j}_{\text{ph}}]}{Z} \Big|_{\bar{j}=j=0} = +\hbar \mathcal{G}_{\sigma,\sigma}^{n,n}(k, k) \quad (23.11)$$

$$\begin{aligned} \frac{\partial^2}{\partial \bar{j}_{n,\sigma}(k) \partial j_{n,\sigma}(k)} \frac{Z[j_{\text{el}}, \bar{j}_{\text{el}}, j_{\text{ph}}, \bar{j}_{\text{ph}}]}{Z} \Big|_{\bar{j}=j=0} &= +\hbar \mathcal{G}_{\sigma,\sigma}^{n,n}(k, k) \\ &- \frac{\hbar}{\beta N_p} \sum_{\substack{k \in i\Omega_{-1} \times \mathbf{Q} \\ n' \in \{1, \dots, \infty\} \\ v \in \{1, \dots, rd_x\}}} |g_{nn'v}(\mathbf{k}, \mathbf{k}' - \mathbf{k})|^2 \mathcal{G}_{0,n,\sigma}^2(k) \mathcal{G}_{0,n',\sigma}(k') \mathcal{D}_{0,v}(k' - k) \\ &+ \mathcal{O}(g^4) \end{aligned} \quad (23.12)$$

Proof 23.10 See Appendix, Proofs, page 508.

Remark 23.10.1 We only mention the second identity for completeness; In practice, it is difficult to converge this expression. The second identity can be simplified further by evaluating the occurring Matsura-sums (cf. B-sides). The first identity is the one that is used in practice, since it can be simplified with the fluctuation-dissipation-theorem.

23.2.2. Interacting Phonons

The Phononic Contribution

We first focus on the term in Equation (21.24).

Theorem 23.11 One finds for the interacting system the identity

$$\begin{aligned}
 \frac{\partial^2}{\partial \bar{j}_v(q) \partial j_v(q)} \frac{Z[j_{\text{el}}, \bar{j}_{\text{el}}, j_{\text{ph}}, \bar{j}_{\text{ph}}]}{Z} \Big|_{\bar{j}=j=0} &\approx -\hbar \mathcal{G}_{0,v}(q) \\
 &+ \frac{\hbar}{\beta N_p} \mathcal{G}_{0,v}^2(q) \delta_{q,0} \sum_{\substack{k,k' \in i\Omega_{-1} \times Q \\ n,n' \in \{1,\dots,\infty\} \\ \sigma,\sigma' \in \{\downarrow,\uparrow\}}} g_{nnv}(\mathbf{k}, 0) g_{n'n'v}(\mathbf{k}', 0) \mathcal{G}_{0,n,\sigma}(k) \mathcal{G}_{0,n',\sigma'}(k') \\
 &- \frac{\hbar}{\beta N_p} \mathcal{G}_{0,v}^2(q) \sum_{\substack{k \in i\Omega_{-1} \times Q \\ n,n' \in \{1,\dots,\infty\} \\ \sigma \in \{\downarrow,\uparrow\}}} |g_{n'nv}(\mathbf{k}, \mathbf{q})|^2 \mathcal{G}_{0,n,\sigma}(k) \mathcal{G}_{0,n',\sigma}(k+q) + \mathcal{O}(g^4)
 \end{aligned} \tag{23.13}$$

whereas the terms of order $\mathcal{O}(g^4)$ are those that are being neglected when one approximates the Bethe-Salpeter-equation.

Proof 23.11 See Appendix, Proofs, page 509.

Lemma 23.12 For the phononic occupation numbers of the interacting system we have the identity

$$\begin{aligned}
 \langle H_{\text{ph}} \rangle &\approx -\frac{1}{\beta} \sum_{\substack{q \in i\Omega_{+1} \times Q \\ v \in \{1,2,\dots,rd_{\mathbf{x}}\}}} e^{+i\omega_q 0^+} \Omega_v(\mathbf{q}) \mathcal{G}_{0,v}(q) \\
 &+ \frac{1}{\beta^2 N_p} \sum_{\substack{\sigma,\sigma' \in \{\downarrow,\uparrow\} \\ k,k' \in i\Omega_{-1} \times Q \\ n,n' \in \{1,\dots,\infty\} \\ v \in \{1,2,\dots,rd_{\mathbf{x}}\}}} \Omega_v(\mathbf{q} = \mathbf{0}) g_{nnv}(\mathbf{k}, 0) g_{n'n'v}(\mathbf{k}', 0) \mathcal{G}_{0,v}^2(q = 0) \mathcal{G}_{0,n,\sigma}(k) \mathcal{G}_{0,n',\sigma'}(k') \\
 &- \frac{1}{\beta^2 N_p} \sum_{\substack{k \in i\Omega_{-1} \times Q \\ q \in i\Omega_{+1} \times Q \\ v \in \{1,2,\dots,rd_{\mathbf{x}}\} \\ n,n' \in \{1,\dots,\infty\} \\ \sigma \in \{\downarrow,\uparrow\}}} \Omega_v(\mathbf{q}) |g_{n'nv}(\mathbf{k}, \mathbf{q})|^2 \mathcal{G}_{0,v}^2(q) \mathcal{G}_{0,n,\sigma}(k) \mathcal{G}_{0,n',\sigma}(k+q) + \mathcal{O}(g^4)
 \end{aligned} \tag{23.14}$$

whereas the terms of order $\mathcal{O}(g^4)$ are those that are being neglected when one approximates the Bethe-Salpeter-equation.

Proof 23.12 See Appendix, Proofs, page 511.

Remark 23.12.1 Notice that the individual terms have the unit of an energy and are hence consistent.

Corollary 23.13 Evaluating the Matsubara-sums simplifies the individual terms as

$$\begin{aligned} \langle H_{\text{ph}} \rangle \approx & \sum_{\substack{\mathbf{q} \in Q \\ v \in \{1,2,\dots,rd_x\}}} \Omega_v(\mathbf{q}) B_\beta(\Omega_v(\mathbf{q})) + \sum_{v \in \{1,2,\dots,rd_x\}} A_v B_v^2 \\ & - \sum_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ \sigma \in \{\downarrow, \uparrow\} \\ v \in \{1,2,\dots,rd_x\} \\ n, n' \in \{1, \dots, \infty\}}} \frac{\Omega_v(\mathbf{q})}{N_p} |g_{n'n v}(\mathbf{k}, \mathbf{q})|^2 (F_\beta(\xi_{n, \sigma, \mathbf{k}}) - F_\beta(\xi_{n', \sigma, \mathbf{k}+\mathbf{q}})) \left[\frac{B_\beta(\Omega_v(\mathbf{q})) - B_\beta(\xi_{n', \sigma, \mathbf{k}+\mathbf{q}} - \xi_{n, \sigma, \mathbf{k}})}{(\Omega_v(\mathbf{q}) + \xi_{n, \sigma, \mathbf{k}} - \xi_{n', \sigma, \mathbf{k}+\mathbf{q}})^2} + \frac{1}{4} \frac{\beta \text{csch}(\beta \Omega_v(\mathbf{q})/2)^2}{\Omega_v(\mathbf{q}) + \xi_{n, \sigma, \mathbf{k}} - \xi_{n', \sigma, \mathbf{k}+\mathbf{q}}} \right] \\ & + \mathcal{O}(g^4) \end{aligned}$$

with real coefficients A_v and B_v defined as

$$A_v = \Omega_v(\mathbf{q} = \mathbf{0}) \mathcal{G}_{0,v}^2(q = 0) = \Omega_v^{-1}(\mathbf{q} = \mathbf{0}), \quad B_v = \frac{1}{\sqrt{N_p}} \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\downarrow, \uparrow\} \\ n \in \{1, \dots, \infty\}}} g_{nnv}(\mathbf{k}, 0) F_\beta(\xi_{n\mathbf{k}\sigma}) \quad (23.15)$$

whereas the terms of order $\mathcal{O}(g^4)$ are those that are being neglected when one approximates the Bethe-Salpeter-equation.

Proof 23.13 See Appendix, Proofs, page 511.

23.2.3. Total Crystal

The Interaction Contribution

Lemma 23.14 One obtains for the electron-phonon-interaction energy the identity

$$\begin{aligned} \langle H_{\text{int}} \rangle &= \sum_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ \sigma \in \{\downarrow, \uparrow\} \\ v \in \{1, \dots, rd_x\} \\ m, n \in \{1, \dots, \infty\}}} \frac{g_{mnv}(\mathbf{k}, \mathbf{q})}{\sqrt{N_p}} \left\langle c_{\mathbf{k}+\mathbf{q}m\sigma}^\dagger c_{\mathbf{k}n\sigma} (b_{\mathbf{q}v} + b_{-\mathbf{q}v}^\dagger) \right\rangle \\ &= \sum_{\substack{k \in i\Omega_{-1} \times Q \\ q \in i\Omega_{+1} \times Q \\ \sigma \in \{\downarrow, \uparrow\} \\ v \in \{1, 2, \dots, rd_x\} \\ n, m \in \{1, \dots, \infty\} \\ \omega_{n'} \in \Omega_{-1}}} \frac{g_{mnv}(\mathbf{k}, \mathbf{q}) \hbar}{\sqrt{N_p} \sqrt{\hbar \beta}}^3 \left\langle \bar{\psi}_{m\sigma}(i\omega_{n'}, \mathbf{k} + \mathbf{q}) \psi_{n\sigma}(i\omega_n, \mathbf{k}) \left[\rho_v(q) \mathcal{G}_{0,v}(q) + \mathcal{G}_{0,v}(-q) \rho_v(q) \right] \right\rangle \quad (23.16) \end{aligned}$$

Proof 23.14 See Appendix, Proofs, page 514.

Corollary 23.15 Simplifying the above expression with an approximation of the Bethe-Salpeter-equation, and evaluating the Matsubara-sums gives the two contributions

$$\langle H_{\text{int}} \rangle = \text{I} + \text{II} \quad (23.17)$$

Proof 23.15 See Appendix, Proofs, page 515.

Collecting all Contributions

Fact 23.16 For the electronic and phononic occupation numbers of the interacting system we have the identity

$$\langle H_{\text{el}} \rangle = \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} \langle n_{m\mathbf{k}\sigma} \rangle = \frac{-1}{\pi} \lim_{\delta \rightarrow 0} \oint_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\} \\ E \in (-\infty, +\infty)}} \epsilon_{m\mathbf{k}\sigma} F_{\beta}(E) \text{Im} G_{m,m}^{\text{ret}}(E + i\delta, \mathbf{k}) \quad (23.18)$$

$$\begin{aligned} \langle H_{\text{ph}} \rangle &= \sum_{\substack{\mathbf{q} \in Q \\ v \in \{1, 2, \dots, rd_x\}}} \Omega_v(\mathbf{q}) \langle n_v(\mathbf{q}) \rangle \\ &\approx \sum_{\substack{\mathbf{q} \in Q \\ v \in \{1, 2, \dots, rd_x\}}} \Omega_v(\mathbf{q}) B_{\beta}(\Omega_v(\mathbf{q})) + \sum_{v \in \{1, 2, \dots, rd_x\}} A_v B_v^2 \\ &\quad - \sum_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ v \in \{1, 2, \dots, rd_x\} \\ n, n' \in \{1, \dots, \infty\} \\ \sigma \in \{\downarrow, \uparrow\}}} \frac{\Omega_v(\mathbf{q})}{N_p} |g_{n'n v}(\mathbf{k}, \mathbf{q})|^2 (F_{\beta}(\xi_{n,\sigma,\mathbf{k}}) - F_{\beta}(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}})) \left[\frac{B_{\beta}(\Omega_v(\mathbf{q})) - B_{\beta}(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}} - \xi_{n,\sigma,\mathbf{k}})}{(\Omega_v(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}} - \xi_{n',\sigma,\mathbf{k}+\mathbf{q}})^2} + \frac{1}{4} \frac{\beta \text{csch}(\beta \Omega_v(\mathbf{q})/2)^2}{\Omega_v(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}} - \xi_{n',\sigma,\mathbf{k}+\mathbf{q}}} \right] \\ &\quad + \mathcal{O}(g^4) \end{aligned} \quad (23.19)$$

$$\langle H_{\text{int}} \rangle = \text{TBDigitalized from PP3.} \quad (23.20)$$

Proof 23.16 See Appendix, Proofs, page 517.

The Molar Heat Capacity of the Interacting System

We combine Equations (9.56) and (9.49)

$$c_{N,T,V} = \frac{C_{N,V}(T)}{n} = \frac{1}{n} \partial_T \langle H \rangle_{N,T,V}. \quad (23.21)$$

Using now Equation (9.57) yields

$$\begin{aligned} C_{N,V}(T) &= \partial_T \langle H \rangle(N, T, V) \\ &= (\partial_T \langle H \rangle)(\mu(T, V), T, V) + (\partial_{\mu} \langle H \rangle)(\mu(T, V), T, V) \cdot (\partial_T \mu)(N, V, T) \end{aligned} \quad (23.22)$$

whereas the Hamiltonian is made up of three contributions

$$H = H_{\text{el}} + H_{\text{ph}} + H_{\text{int}}. \quad (23.23)$$

Therefore, also for the molar heat capacity, we have contributions from the electronic part, from the phononic part, and from the interaction part. We further need to account for the temperature derivative of the chemical potential. We consider all four quantities separately and then merge the results.

23.2.3.1. Temperature-Derivative of the Chemical Potential

Fact 23.17 We first notice that dictating the particle number (for example by requiring [C]harge [N]eutrality of the crystal) through Equation (22.26) can be rewritten as

$$F(\mu, T) \equiv \langle N \rangle(\mu, T) - N_{\text{CN}} = 0. \quad (23.24)$$

Remark 23.17.1 Notice that the chemical potential that is defined through this Equation is unique due to Theorem 9.30.

Corollary 23.18 Assuming now we have access¹ to the values $\mu(T)$ that fulfill Equation (23.24), we can compute the temperature derivative of the chemical potential as

$$\left(\frac{\partial \mu}{\partial T} \right) (T) = - \left(\frac{\partial_T F}{\partial_\mu F} \right) (\mu(T), T). \quad (23.25)$$

Proof 23.18 Follows from the Theorem of implicit functions.

Lemma 23.19 The temperature-derivative of the function F can be computed as

$$\begin{aligned} (\partial_T F)(\mu, T) = & \frac{-1}{\pi} \frac{\beta}{T} \lim_{\delta \rightarrow 0} \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\} \\ E \in (-\infty, +\infty)}} F_\beta(E) \operatorname{Im} \left[(1 - F_\beta(E)) E \frac{1}{E + i\delta - (\xi_{m\mathbf{k}\sigma} + \Sigma_{m\sigma\mathbf{k}}^{\text{ret}}(E + i\delta))} \right. \\ & + \left. \frac{k_B T^2 \partial_T \Sigma_{m\sigma\mathbf{k}}^{\text{ret}}(E + i\delta)}{(E + i\delta - (\xi_{m\mathbf{k}\sigma} + \Sigma_{m\sigma\mathbf{k}}^{\text{ret}}(E + i\delta)))^2} \right] \end{aligned} \quad (23.26)$$

with partial derivative of the retarded self-energy

$$k_B T^2 \partial_T \Sigma_{m\sigma\mathbf{k}}^{\text{ret}}(E + i\delta) = k_B T^2 \partial_T \Sigma_{m\sigma\mathbf{k}}^{\text{ret}, \text{H}}(E + i\delta) + k_B T^2 \partial_T \Sigma_{m\sigma\mathbf{k}}^{\text{ret}, \text{F}}(E + i\delta) \quad (23.27)$$

and individual contributions

¹Notice that this is the case, as we compute the chemical potential numerically.

$$k_B T^2 \partial_T \Sigma_{m\sigma\mathbf{k}}^{\text{ret, H}}(\bullet) = - \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ \sigma' \in \{\downarrow, \uparrow\} \\ n \in \{1, 2, \dots\} \\ v \in \{1, 2, \dots, rd_x\}}} \frac{g_{mmv}(\mathbf{k}, 0) g_{nnv}(\mathbf{q}, 0)}{4N_p} \frac{\xi_{n\sigma'\mathbf{q}}}{\Omega_v(\mathbf{q}' \equiv 0)} \text{sech}^2 \left(\frac{\xi_{n\sigma'\mathbf{q}}}{2k_B T} \right) \quad (23.28)$$

and

$$k_B T^2 \partial_T \Sigma_{m\sigma\mathbf{k}}^{\text{ret, F}}(E + i\delta) = \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ n \in \{1, 2, \dots, \infty\} \\ v \in \{1, 2, \dots, rd_x\}}} \frac{|g_{nmv}(\mathbf{k}, \mathbf{q})|^2}{4N_p} \left[\frac{\Omega_v(\mathbf{q}) \text{csch}^2(\beta\Omega_v(\mathbf{q})/2) + \xi_{n,\sigma,\mathbf{k}+\mathbf{q}} \text{sech}^2(\beta\xi_{n,\sigma,\mathbf{k}+\mathbf{q}}/2)}{E + i\delta - \xi_{n,\sigma,\mathbf{k}+\mathbf{q}} + \Omega_v(\mathbf{q})} + \frac{\Omega_v(\mathbf{q}) \text{csch}^2(\beta\Omega_v(\mathbf{q})/2) - \xi_{n,\sigma,\mathbf{k}+\mathbf{q}} \text{sech}^2(\beta\xi_{n,\sigma,\mathbf{k}+\mathbf{q}}/2)}{E + i\delta - \xi_{n,\sigma,\mathbf{k}+\mathbf{q}} - \Omega_v(\mathbf{q})} \right] \quad (23.29)$$

Proof 23.19 See Appendix, Proofs, page 518.

Lemma 23.20 The chemical-potential-derivative of the function F can be computed as

$$(\partial_\mu F)(\mu, T) = \frac{-1}{\pi} \frac{\beta}{4} \lim_{\delta \rightarrow 0} \oint \sum_{\substack{\mathbf{k} \in \mathbf{Q} \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\} \\ E \in (-\infty, +\infty)}} \text{Im} \left[\text{sech}^2 \left(\frac{\beta E}{2} \right) \frac{1}{E + i\delta - (\xi_{m\mathbf{k}\sigma} + \Sigma_{m\sigma\mathbf{k}}^{\text{ret}}(E + i\delta))} + F_\beta(E) \cdot \frac{\sigma_{m\sigma\mathbf{k}}(E + i\delta)}{(E + i\delta - (\xi_{m\mathbf{k}\sigma} + \Sigma_{m\sigma\mathbf{k}}^{\text{ret}}(E + i\delta)))^2} \right] \quad (23.30)$$

with auxiliary function

$$\sigma_{m\sigma\mathbf{k}}(E + i\delta) = \sigma_{m\sigma\mathbf{k}}^{\text{Hartree}}(E + i\delta) + \sigma_{m\sigma\mathbf{k}}^{\text{Fock}}(E + i\delta), \quad (23.31)$$

with the Hartree-contribution

$$\sigma_{m\sigma\mathbf{k}}^{\text{Hartree}}(E + i\delta) = - \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ \sigma' \in \{\downarrow, \uparrow\} \\ n \in \{1, 2, \dots\} \\ v \in \{1, 2, \dots, rd_x\}}} \frac{g_{mmv}(\mathbf{k}, 0) g_{nnv}(\mathbf{q}, 0)}{N_p} \frac{1}{\Omega_v(\mathbf{q}' = 0)} \text{sech}^2(\beta\xi_{n\sigma'\mathbf{q}}/2) \quad (23.32)$$

and the Fock-contribution

$$\sigma_{m\sigma\mathbf{k}}^{\text{Fock}}(E + i\delta) = \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ n \in \{1, 2, \dots\} \\ v \in \{1, 2, \dots, rd_x\}}} \frac{|g_{nmv}(\mathbf{k}, \mathbf{q})|^2}{N_p} \text{sech}^2(\beta\xi_{\mathbf{q}+\mathbf{k}n\sigma}/2) \left[\frac{1}{E + i\delta - \xi_{\mathbf{q}+\mathbf{k}n\sigma} + \Omega_v(\mathbf{q})} - \frac{1}{E + i\delta - \xi_{\mathbf{q}+\mathbf{k}n\sigma} - \Omega_v(\mathbf{q})} \right] \quad (23.33)$$

Proof 23.20 See Appendix, Proofs, page 519.

23.2.3.2. Temperature-Derivative of the Electronic Energy

Lemma 23.21 The temperature-derivatives of the electronic contribution to the internal energy can be computed as

$$\partial_T \langle H_{\text{el}} \rangle = \frac{-1}{\pi} \frac{1}{k_B T^2} \lim_{\delta \rightarrow 0} \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\} \\ E \in (-\infty, +\infty)}} \epsilon_{m\mathbf{k}\sigma} F_\beta(E) \text{Im} \left[(1 - F_\beta(E)) E \frac{1}{E + i\delta - (\xi_{m\mathbf{k}\sigma} + \Sigma_{m\sigma\mathbf{k}}^{\text{ret}}(E + i\delta))} \right. \\ \left. + \frac{k_B T^2 \partial_T \Sigma_{m\sigma\mathbf{k}}^{\text{ret}}(E + i\delta)}{(E + i\delta - (\xi_{m\mathbf{k}\sigma} + \Sigma_{m\sigma\mathbf{k}}^{\text{ret}}(E + i\delta, \mathbf{k})))^2} \right] \quad (23.34)$$

with partial derivative of the retarded self-energy as in Equation (23.29)

Proof 23.21 See Appendix, Proofs, page 521.

Remark 23.21.1 Notice that this is the very same Equation as (23.26), except for the factor of $\epsilon_{m\mathbf{k}\sigma}$ in the integrand-summand.

Lemma 23.22 The chemical-potential-derivative of the electronic contribution to the internal energy can be computed as

$$\partial_\mu \langle H_{\text{el}} \rangle = \frac{-1}{\pi} \frac{\beta}{4} \lim_{\delta \rightarrow 0} \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\} \\ E \in (-\infty, +\infty)}} \epsilon_{m\mathbf{k}\sigma} \text{Im} \left[\text{sech}^2 \left(\frac{\beta E}{2} \right) \frac{1}{E + i\delta - (\xi_{m\mathbf{k}\sigma} + \Sigma_{m\sigma\mathbf{k}}^{\text{ret}}(E + i\delta))} \right. \\ \left. + F_\beta(E) \cdot \frac{\sigma_{m\sigma\mathbf{k}}(E + i\delta)}{(E + i\delta - (\xi_{m\mathbf{k}\sigma} + \Sigma_{m\sigma\mathbf{k}}^{\text{ret}}(E + i\delta)))^2} \right] \quad (23.35)$$

with auxiliary function

$$\sigma_{m\sigma\mathbf{k}}(E + i\delta) = \sigma_{m\sigma\mathbf{k}}^{\text{Hartree}}(E + i\delta) + \sigma_{m\sigma\mathbf{k}}^{\text{Fock}}(E + i\delta), \quad (23.36)$$

as above.

Proof 23.22 Completely analogous to the proof of Lemma 23.20.

23.2.3.3. Temperature-Derivative of the Phononic Energy

Lemma 23.23 The temperature-derivatives of the non-interacting phononic contribution to the internal energy can be computed as

$$\partial_T \langle H_{\text{ph}} \rangle_0 = \frac{1}{4k_B T^2} \sum_{\substack{\mathbf{q} \in Q \\ v \in \{1,2,\dots,rd_x\}}} \left[\Omega_v(\mathbf{q}) \operatorname{csch}(\beta \Omega_v(\mathbf{q})/2) \right]^2 \quad (23.37)$$

and for the interacting phononic correction contribution

$$\begin{aligned} & \partial_T \delta \langle H_{\text{ph}} \rangle \\ & \approx 2 \sum_{v \in \{1,2,\dots,rd_x\}} A_v B_v \partial_T B_v \\ & + \frac{-1}{N_p} \sum_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ n, n' \in \{1, \dots, \infty\} \\ \sigma \in \{\downarrow, \uparrow\} \\ v \in \{1,2,\dots,rd_x\}}} \frac{\Omega_v(\mathbf{q}) |g_{n'nv}(\mathbf{k}, \mathbf{q})|^2}{4k_B T^2} (F_\beta(\xi_{n,\sigma,\mathbf{k}}) - F_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}})) \frac{\operatorname{csch}^2(\beta \Omega_v(\mathbf{q})/2)}{\Omega_v(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}} - \xi_{n',\sigma,\mathbf{k}+\mathbf{q}}} (\beta \Omega_v(\mathbf{q}) \coth(\beta \Omega_v(\mathbf{q})/2) - 1) \\ & + \frac{-1}{N_p} \sum_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ n, n' \in \{1, \dots, \infty\} \\ \sigma \in \{\downarrow, \uparrow\} \\ v \in \{1,2,\dots,rd_x\}}} \frac{\Omega_v(\mathbf{q}) |g_{n'nv}(\mathbf{k}, \mathbf{q})|^2}{4k_B T^2} (F_\beta(\xi_{n,\sigma,\mathbf{k}}) - F_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}})) \left[\frac{\Omega_v(\mathbf{q}) \operatorname{csch}^2(\beta \Omega_v(\mathbf{q})/2) - (\xi_{n',\sigma,\mathbf{k}+\mathbf{q}} - \xi_{n,\sigma,\mathbf{k}}) \operatorname{csch}^2\left(\frac{\xi_{n',\sigma,\mathbf{k}+\mathbf{q}} - \xi_{n,\sigma,\mathbf{k}}}{2k_B T}\right)}{(\Omega_v(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}} - \xi_{n',\sigma,\mathbf{k}+\mathbf{q}})^2} \right] \\ & + \frac{-1}{N_p} \sum_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ n, n' \in \{1, \dots, \infty\} \\ \sigma \in \{\downarrow, \uparrow\} \\ v \in \{1,2,\dots,rd_x\}}} \frac{\Omega_v(\mathbf{q}) |g_{n'nv}(\mathbf{k}, \mathbf{q})|^2}{4k_B T^2} (\xi_{n,\sigma,\mathbf{k}} \operatorname{sech}^2(\beta \xi_{n,\sigma,\mathbf{k}}/2) - \xi_{n',\sigma,\mathbf{k}+\mathbf{q}} \operatorname{sech}^2(\beta \xi_{n',\sigma,\mathbf{k}+\mathbf{q}}/2)) \left[\frac{B_\beta(\Omega_v(\mathbf{q})) - B_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}} - \xi_{n,\sigma,\mathbf{k}})}{(\Omega_v(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}} - \xi_{n',\sigma,\mathbf{k}+\mathbf{q}})^2} + \frac{1}{4} \frac{\beta \operatorname{csch}(\beta \Omega_v(\mathbf{q})/2)^2}{\Omega_v(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}} - \xi_{n',\sigma,\mathbf{k}+\mathbf{q}}} \right] \end{aligned} \quad (23.38)$$

with partial derivative

$$\partial_T B_v = \frac{1}{\sqrt{N_p}} \frac{\xi_{n,\sigma,\mathbf{k}}}{4k_B T^2} \sum_{\substack{\mathbf{k} \in Q \\ n \in \{1,2,\dots,\infty\} \\ \sigma \in \{\downarrow, \uparrow\}}} g_{mv}(\mathbf{k}, 0) \operatorname{sech}^2(\beta \xi_{n,\sigma,\mathbf{k}}/2) \quad (23.39)$$

Proof 23.23 See Appendix, Proofs, page 522.

Lemma 23.24 For the chemical potential-derivative of the phononic correction-term we obtain

$$\begin{aligned} & \partial_\mu \delta \langle H_{\text{ph}} \rangle \\ & \approx 2 \sum_{v \in \{1,2,\dots,rd_x\}} A_v B_v \partial_\mu B_v \\ & - \sum_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ v \in \{1,2,\dots,rd_x\} \\ n, n' \in \{1, \dots, \infty\} \\ \sigma \in \{\downarrow, \uparrow\}}} \frac{\beta \Omega_v(\mathbf{q})}{4} \frac{|g_{n'nv}(\mathbf{k}, \mathbf{q})|^2}{N_p} (\operatorname{sech}^2(\beta \xi_{n,\sigma,\mathbf{k}}/2) - \operatorname{sech}^2(\beta \xi_{n',\sigma,\mathbf{k}+\mathbf{q}}/2)) \left[\frac{B_\beta(\Omega_v(\mathbf{q})) - B_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}} - \xi_{n,\sigma,\mathbf{k}})}{(\Omega_v(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}} - \xi_{n',\sigma,\mathbf{k}+\mathbf{q}})^2} + \frac{1}{4} \frac{\beta \operatorname{csch}(\beta \Omega_v(\mathbf{q})/2)^2}{\Omega_v(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}} - \xi_{n',\sigma,\mathbf{k}+\mathbf{q}}} \right] \\ & + \mathcal{O}(g^4) \end{aligned}$$

and partial derivative

$$\partial_\mu B_\nu = \frac{1}{\sqrt{N_p}} \frac{\beta}{4} \sum_{\substack{\mathbf{k} \in Q \\ n \in \{1,2,\dots,\infty\} \\ \sigma \in \{\downarrow, \uparrow\}}} g_{nn\nu}(\mathbf{k}, 0) \operatorname{sech}^2(\beta \xi_{n,\sigma,\mathbf{k}}/2) \quad (23.40)$$

Proof 23.24 We use Equation (23.19) and note that we obtain with Lemma (9.45) and the chain-rule the identity

$$\partial_\mu F_\beta(\xi_{n,\sigma,\cdot}) = \frac{\beta}{4} \operatorname{sech}^2(\beta \xi_{n,\sigma,\cdot}/2) \quad (23.41)$$

23.2.3.4. Temperature-Derivative of the Interaction Energy

Since this contribution only becomes relevant for crystals with more than one atom in the unit-cell (which we are not considering here, for simplicity), we omit a final, perfectly simplified expression and simply point out that one obtains an expression by taking the temperature-derivative of Equation (23.20).

24. Superconductivity

General Remarks

Apart from the toy-model for BCS-theory in section 28.2, we will introduce two formalisms for superconductivity.

1. The Eliashberg-Theory (ET for short),
2. The Path-Integral-Formalism for ([G]eneralized) [E]liashberg-[T]heory (GET / ET for short).

The strength and beauty in the path-integral-approach lies in the fact that it is carried out analogously to BCS-theory and reproduces the already established behavior for the order-parameters as a function of temperature. Further it reproduces and even opens the possibility to generalize Eliashberg-Theory in the appropriate limits. The path-integral-formalism can be considered as a generalization / unification of these theories. Another folklore-statement that is being reproduced is that superconductivity originates from an attractive interaction¹ that is mediated through phonons.

Fact 24.1 For "Eliashberg-theory" (to be introduced below), we consider the Hamiltonian from Definition 8.29, i.e. a system of electrons and phonons with only electron-phonon-interactions

$$H = \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \epsilon_{m\sigma}(\mathbf{k}) c_{m\sigma}^\dagger(\mathbf{k}) c_{m\sigma}(\mathbf{k}) + \sum_{\substack{\mathbf{q} \in Q \\ v \in \{1, \dots, rd_x\}}} \Omega_v(\mathbf{q}) b_v^\dagger(\mathbf{q}) b_v(\mathbf{q}) + \sum_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ v \in \{1, \dots, rd_x\} \\ m, n \in \{1, \dots, \infty\}}} \frac{g_{mnv}(\mathbf{k}, \mathbf{q})}{\sqrt{N_p}} c_{m\sigma}^\dagger(\mathbf{k} + \mathbf{q}) c_{n\sigma}(\mathbf{k}) (b_v(\mathbf{q}) + b_v^\dagger(-\mathbf{q})) \quad (24.1)$$

Remark 24.1.1 The concept of "superconductivity" also extends to generalizations of this Hamiltonian. E.g. Hamiltonians that include more interactions, as in Definitions 8.32 and 8.33.

24.1. Phononic Eliashberg-Theory in the Operator-Formalism

24.1.1. An Operator-Expectation-Value like no Other

Fact 24.2 For normal-state systems, we introduced an expectation-value w.r.t. a density matrix in Definition 9.8. Then we introduced the natural density matrix in the natural ensemble in Equation (10.6) as *the* density matrix that we need to use in equilibrium systems in which no symmetries are being broken spontaneously, and we realized, that in hybrid ensembles one needs to have a generalization as in Equation (10.11). It was exactly *this* density

¹Note however that contrary to the folklore, this interaction is attractive *not only* close to the Fermi-surface, but for all energies. But it will turn later out, that only those electronic energies close to the Fermi-surface and hence only those interactions will contribute to the emergence of superconductivity, cf. Theorem 24.69.

matrix, that allowed us to conclude the translational invariance of the Green-functions and their (anti)-periodicity in Theorem 13.2. It was also *this* density matrix, that allowed us to draw *all* of the conclusions in Chapter 14.

Now we need to introduce the concept of spontaneous symmetry breaking, which is basically impossible to do properly (i.e. mathematically "clean") in the operator-formalism. (In contrast: In the path-integral description it will come quite natural from saddle-point-approximations.)

Definition 24.3 For an operator in second quantization $A \equiv A(c, c^\dagger, b, b^\dagger)$ and for *low enough temperatures*, i.e. $0 < T < T_{\text{SC}}$ we introduce a mapping

$$A \mapsto \langle A \rangle_{\text{SC}} \in \mathbb{C} \quad (24.2)$$

which we call *the expectation-value in the [S]uper-[C]onducting state*, and we assume that for this expectation-value „the U(1) symmetry is spontaneously broken“. We further *assume* ...

1. that this mapping is compatible with linearity and scalar-multiplication, i.e. for scalars $a, b \in \mathbb{C}$ and operators A, B we have

$$\langle aA + bB \rangle_{\text{SC}} = a \langle A \rangle_{\text{SC}} + b \langle B \rangle_{\text{SC}} \quad (24.3)$$

2. that this mapping obeys

$$\langle A \rangle_{\text{SC}}^* = \langle A^\dagger \rangle_{\text{SC}} \quad (24.4)$$

Remark 24.3.1 Notice again that the second requirement is completely easy to prove in the normal state (cf. Lemma 9.9), for which one has the density in Equation (10.11), but it is a priori not clear why it would also hold in the superconducting state, since we do not know a priori whether there exists a density for the superconducting state. It is the authors opinion, that the only way of describing superconductors "appropriately" is through the field integral formalism.

Remark 24.3.2 Now we come into the problematic situation that we need to explain the implications of the vague statement "broken U(1) symmetry" above, which is completely natural to see in the field integral formalism, but it seems like stating arbitrary facts in the operator-formalism. It is also in the field-integral formalism, where it is justified that we look at the so-called "Gorkov-Green-function" and why we obtain generalizations of the Bethe-Salpeter-equation (due to the effective action formalism!), but it seems almost random to do these things if one works in the operator-formalism.

24.1.2. Gorkovs-Green-Functions

Definition 24.4 We define² the *single-band Gorkov-Green-function* as

$$\hat{\mathcal{G}}_{\text{Gorkov}}^m(\tau, \mathbf{k}; \tau' \mathbf{k}') = -\frac{1}{\hbar} \begin{pmatrix} \left\langle T_{\tau} c_{m,\uparrow}(\tau, \mathbf{k}) c_{m,\uparrow}^{\dagger}(\tau', \mathbf{k}') \right\rangle_{\text{SC}} & \left\langle T_{\tau} c_{m,\uparrow}(\tau, \mathbf{k}) c_{m,\downarrow}(\tau', -\mathbf{k}') \right\rangle_{\text{SC}} \\ \left\langle T_{\tau} c_{m,\downarrow}^{\dagger}(\tau, -\mathbf{k}) c_{m,\uparrow}^{\dagger}(\tau', \mathbf{k}') \right\rangle_{\text{SC}} & \left\langle T_{\tau} c_{m,\downarrow}^{\dagger}(\tau, -\mathbf{k}) c_{m,\downarrow}(\tau', -\mathbf{k}') \right\rangle_{\text{SC}} \end{pmatrix} \quad (24.5)$$

Remark 24.4.1 Notice that inside the time-ordering-operator, the exchange of two fermionic operators yields a minus sign.

Remark 24.4.2 The off-diagonal components are usually called *Gorkov's anomalous Green-functions*.

Remark 24.4.3 Gorkov's anomalous Green-functions are only non-zero in the superconducting phase³. Evaluating these expressions w.r.t. the thermal state yields zero⁴.

Fact 24.5 What we mean, with a "broken U(1) symmetry", is that the off-diagonal components of this expectation-value are non-zero. All the other symmetries of the underlying action (and therefore, due to section 24.2, also of the underlying Green-function) are preserved. We have in particular ...

- that this Green-function is diagonal in the momentum-indices, i.e. it is spatial-translation symmetric

$$\hat{\mathcal{G}}_{\text{Gorkov}}^m(\tau, \mathbf{k}; \tau' \mathbf{k}') \equiv \delta_{\mathbf{k}, \mathbf{k}'} \hat{\mathcal{G}}_{\text{Gorkov}}^{m\mathbf{k}}(\tau, \tau') \quad (24.6)$$

- that this Green-function depends only on time-differences, i.e. the system is time-translation symmetric, i.e.

$$\hat{\mathcal{G}}_{\text{Gorkov}}^{m\mathbf{k}}(\tau, \tau') = \hat{\mathcal{G}}_{\text{Gorkov}}^{m\mathbf{k}}(\tau - \tau', 0) = \hat{\mathcal{G}}_{\text{Gorkov}}^{m\mathbf{k}}(0, \tau' - \tau) \equiv \hat{\mathcal{G}}_{\text{Gorkov}}^{m\mathbf{k}}(\tau - \tau') \quad (24.7)$$

- that this Green-function obeys the anti-periodicity relation that one has in the normal state: It holds for all wavevectors $\mathbf{k} \in Q$ and bands $m \in \{1, 2, 3, \dots, \infty\}$ the identity

$$\forall \tau \in (-\hbar\beta, 0) : \quad \hat{\mathcal{G}}_{\text{Gorkov}}^{m\mathbf{k}}(\tau + \hbar\beta) = -\hat{\mathcal{G}}_{\text{Gorkov}}^{m\mathbf{k}}(\tau) \quad (24.8)$$

²Notice that we are following the prefactor-convention of 03-Marsiglio, Eqs. (43-44), and Coleman page 519 Equations (14.158), or Polaron-Physics 07, p. 16 in contrast to Mahan3rd page 635, Eq. (10.31). Notice further that our Equations are in harmony with Allens long article on pages 8 and 33 and Equation (7.2) in particular.

³Quoting Mahan p. 635:

A basic feature of the BCS ground state (...) is that it is composed of a superposition of electronic states containing a different number of electrons. Such a formulation is possible in a grand canonical ensemble.

⁴Easy to prove, but also quoting Negele and Orland p. 77:

The expectation value of two creation or annihilation operators is zero in a state of definite particle number.

Remark 24.5.1 We again emphasize: A priori it is not clear why these relations would hold. When we re-derive the Eliashberg-equations later on in the path-integral formalism (when we introduce a so-called "Hubbard-Stratanovich-transformation" and apply a so-called "saddle-point-approximation") we will realise that the preservation of translational invariance and antiperiodicity is a natural consequence of breaking the $U(1)$ -symmetry but no other symmetry. Strictly speaking this assumption was a leap of faith when Eliashberg derived his equations (because the functional integral formalism was not invented back then).

24.1.3. Gorkovs Generalized Hermiticity Relation

Diagonal Components

Notation 24.6 We write

$$\hat{\mathcal{G}}_{11}^m(\tau, \mathbf{k}) \equiv \mathcal{G}_{\text{Gorkov}}^m(\tau, \mathbf{k}) \quad (24.9)$$

Lemma 24.7 It follows that

$$\hat{\mathcal{G}}_{22}^m(\tau, \mathbf{k}) \equiv -\mathcal{G}_{\text{Gorkov}}^m(-\tau, -\mathbf{k}) \quad (24.10)$$

Proof 24.7 TBDone.

Off Diagonal Components

Corollary 24.8 The translation invariance implies

$$\hat{\mathcal{G}}_{\text{Gorkov}}^m(\tau, 0, \mathbf{k}) \equiv \hat{\mathcal{G}}_{\text{Gorkov}}^m(\tau, \mathbf{k}) \equiv \hat{\mathcal{G}}_{\text{Gorkov}}^m(0, -\tau, \mathbf{k}) \quad (24.11)$$

Proof 24.8 Follows immediately from repeated application of Equation (24.7).

Remark 24.8.1 Notice that Equation (24.11) yields e.g. for the "2,1"-component of the Gorkov's matrix

$$\left\langle T_\tau c_{m,\downarrow}^\dagger(\tau, -\mathbf{k}) c_{m,\uparrow}^\dagger(0, \mathbf{k}) \right\rangle_{\text{SC}} = \left\langle T_\tau c_{m,\downarrow}^\dagger(0, -\mathbf{k}) c_{m,\uparrow}^\dagger(-\tau, \mathbf{k}) \right\rangle_{\text{SC}} \quad (24.12)$$

which yields for $\tau > 0$ the identity

$$\left\langle c_{m,\downarrow}^\dagger(\tau, -\mathbf{k}) c_{m,\uparrow}^\dagger(0, \mathbf{k}) \right\rangle_{\text{SC}} = \left\langle c_{m,\downarrow}^\dagger(0, -\mathbf{k}) c_{m,\uparrow}^\dagger(-\tau, \mathbf{k}) \right\rangle_{\text{SC}} \quad (24.13)$$

which is a highly non-trivial identity / assumption.

Notation 24.9 We now introduce the anomalous Gor'kov Green-function as

$$\hat{\mathcal{G}}_{12}^m(\tau, \mathbf{k}) \equiv \mathcal{F}_m(\tau, \mathbf{k}) \quad (24.14)$$

Gorkov

Before we prove the one relation that nobody in the literature seems to be able to write down properly, we need to recall two facts

- For an arbitrary operator one finds the identity $\langle A \rangle^* = \langle A^\dagger \rangle$,
- For creation and annihilation-operators, we find their hermitian adjoints as $[c(\tau)]^\dagger = c^\dagger(-\tau)$, etc.

Lemma 24.10 We will prove now that

$$\hat{\mathcal{G}}_{21}^m(\tau, \mathbf{k}) = \hat{\mathcal{G}}_{12}^{*m}(\tau, \mathbf{k}) = \mathcal{F}_m^*(\tau, \mathbf{k}) \equiv \mathcal{F}_m^\dagger(\tau, \mathbf{k}) \quad (24.15)$$

Gorkov Gorkov

Proof 24.10 See Appendix, Proofs, page 523.

24.1.4. Eliashbergs Preparations

Notation 24.11 We recall⁵ from the last subsection, the *single-band Gorkov-Green-function* as

$$\hat{\mathcal{G}}_{\text{Gorkov}}^m(\tau, \mathbf{k}) = -\frac{1}{\hbar} \begin{pmatrix} \left\langle T_\tau c_{m,\uparrow}(\tau, \mathbf{k}) c_{m,\uparrow}^\dagger(0, \mathbf{k}) \right\rangle_{\text{SC}} & \left\langle T_\tau c_{m,\uparrow}(\tau, \mathbf{k}) c_{m,\downarrow}(0, -\mathbf{k}) \right\rangle_{\text{SC}} \\ \left\langle T_\tau c_{m,\downarrow}^\dagger(\tau, -\mathbf{k}) c_{m,\uparrow}^\dagger(0, \mathbf{k}) \right\rangle_{\text{SC}} & \left\langle T_\tau c_{m,\downarrow}^\dagger(\tau, -\mathbf{k}) c_{m,\downarrow}(0, -\mathbf{k}) \right\rangle_{\text{SC}} \end{pmatrix} \equiv \begin{pmatrix} \mathcal{G}_{\text{Gorkov}}^m(\tau, \mathbf{k}) & \mathcal{F}_m(\tau, \mathbf{k}) \\ \mathcal{F}_m^\dagger(\tau, \mathbf{k}) & -\mathcal{G}_{\text{Gorkov}}^m(-\tau, -\mathbf{k}) \end{pmatrix}$$

Matsubara-Formalism

Lemma 24.12 In the Matsubara-representation one has the equation

$$\hat{\mathcal{G}}_{\text{Gorkov}}^m(i\omega_n, \mathbf{k}) = \begin{pmatrix} \mathcal{G}_{\text{Gorkov}}^m(i\omega_n, \mathbf{k}) & \mathcal{F}_m(i\omega_n, \mathbf{k}) \\ \mathcal{F}_m^\dagger(i\omega_n, \mathbf{k}) & -\mathcal{G}_{\text{Gorkov}}^m(-i\omega_n, -\mathbf{k}) \end{pmatrix} \quad (24.16)$$

⁵Notice that we are following the prefactor-convention of 03-Marsiglio, Eqs. (43-44), and Coleman page 519 Equations (14.158), or Polaron-Physics 07, p. 16 in contrast to Mahan3rd page 635, Eq. (10.31). Notice further that our Equations are in harmony with Allens long article on pages 8 and 33 and Equation (7.2) in particular.

Proof 24.12 See Appendix, Proofs, page 523.

Remark 24.12.1 We want to emphasize at this point, that $\mathcal{F}_m^\dagger(i\omega_n, \mathbf{k})$ refers to the Matsubara-representation of the mapping $\tau \mapsto \mathcal{F}_m^\dagger(\tau, \mathbf{k})$, and *not* the hermitian conjugate of $\mathcal{F}_m(i\omega_n, \mathbf{k})$. In order to emphasize it once more:

$$\mathcal{F}_m^\dagger(i\omega_n, \mathbf{k}) \neq [\mathcal{F}_m(i\omega_n, \mathbf{k})]^* \quad (24.17)$$

In fact, the correct identity can be concluded easily with Equation (1.12) to be

$$\mathcal{F}_m^\dagger(i\omega_n, \mathbf{k}) = [\mathcal{F}_m(-i\omega_n, \mathbf{k})]^*, \quad (24.18)$$

but we shall refrain from using this identity for the derivation of processing of the Gorkov-Green-function.

Lemma 24.13 Assuming a spin-degenerate and inversion-symmetric system, we obtain for the Green-function in the non-superconducting state

$$\begin{aligned} \hat{\mathcal{G}}_{0,m}^{-1} \underset{\text{Gorkov}}{(i\omega_n, \mathbf{k})} &= i\hbar\omega_n \sigma_0 - (\epsilon_m(\mathbf{k}) - \mu) \sigma_3 \\ \Leftrightarrow \hat{\mathcal{G}}_{0,m} \underset{\text{Gorkov}}{(i\omega_n, \mathbf{k})} &= \begin{pmatrix} 1 & 0 \\ i\hbar\omega_n - (\epsilon_m(\mathbf{k}) - \mu) & 1 \\ 0 & i\hbar\omega_n + (\epsilon_m(\mathbf{k}) - \mu) \end{pmatrix}, \end{aligned} \quad (24.19)$$

whereas we introduced $\epsilon_m(\mathbf{k}) = \epsilon_{m,\uparrow}(\pm\mathbf{k}) = \epsilon_{m,\downarrow}(\pm\mathbf{k})$.

Proof 24.13 See Appendix, Proofs, page 524.

Notation 24.14 We write for the Dyson-equation (cf. Remark 16.46.1) of the Gorkov-Green-function in the superconducting state

$$\hat{\mathcal{G}}_m^{-1} \underset{\text{Gorkov}}{(i\omega_n, \mathbf{k})} = \hat{\mathcal{G}}_{0m}^{-1} \underset{\text{Gorkov}}{(i\omega_n, \mathbf{k})} - \hat{\Sigma}_m \underset{\text{Gorkov}}{(i\omega_n, \mathbf{k})} \quad (24.20)$$

whereas $\hat{\mathcal{G}}_{0m} \underset{\text{Gorkov}}$ represents the Gorkov-Green-function in the non-superconducting state and absence of electron-phonon-interactions.

Remark 24.14.1 Notice further, that this equation can be rewritten in several ways, e.g. as

$$\hat{\mathcal{G}}^{-1} = \hat{\mathcal{G}}_0^{-1} - \hat{\Sigma} \quad \Leftrightarrow \quad \hat{\mathcal{G}}_0^{-1} = \hat{\mathcal{G}}^{-1} + \hat{\Sigma} \quad \Leftrightarrow \quad \hat{\mathcal{G}}_0^{-1} \hat{\mathcal{G}} = 1 + \hat{\Sigma} \hat{\mathcal{G}} \quad \Leftrightarrow \quad \hat{\mathcal{G}} = \hat{\mathcal{G}}_0 + \hat{\mathcal{G}}_0 \hat{\Sigma} \hat{\mathcal{G}} \quad (24.21)$$

whereas we lifted the multitude of arguments to lighten the notation.

Remark 24.14.2 Notice that just as the anti-periodicity of the Gorkov-Green-function, the existence (not to mention the computability) of the self-energy is a priori not clear.

Corollary 24.15 This gives for the Dyson-equation in the superconducting state the set of equations (for each component)

$$\begin{aligned}
 \mathcal{G}_{\text{Gorkov}}^m(k) &= +\mathcal{G}_{\text{Gorkov}}^{0m}(k) + \mathcal{G}_{\text{Gorkov}}^{0m}(k) \begin{bmatrix} \Sigma_{11}^m(k) \mathcal{G}_{\text{Gorkov}}^m(k) & + \Sigma_{12}^m(k) \mathcal{F}_m^\dagger(k) \\ \Sigma_{11}^m(k) \mathcal{F}_m(k) & - \Sigma_{12}^m(k) \mathcal{G}_{\text{Gorkov}}^m(-k) \end{bmatrix} \\
 \mathcal{F}_m(k) &= +\mathcal{G}_{\text{Gorkov}}^{0m}(k) \begin{bmatrix} \Sigma_{11}^m(k) \mathcal{F}_m(k) & - \Sigma_{12}^m(k) \mathcal{G}_{\text{Gorkov}}^m(-k) \\ \Sigma_{21}^m(k) \mathcal{G}_{\text{Gorkov}}^m(k) & - \Sigma_{22}^m(k) \mathcal{F}_m^\dagger(k) \end{bmatrix} \\
 \mathcal{F}_m^\dagger(k) &= -\mathcal{G}_{\text{Gorkov}}^{0m}(-k) \begin{bmatrix} \Sigma_{11}^m(k) \mathcal{G}_{\text{Gorkov}}^m(k) & - \Sigma_{12}^m(k) \mathcal{F}_m^\dagger(k) \\ \Sigma_{21}^m(k) \mathcal{F}_m(k) & - \Sigma_{22}^m(k) \mathcal{G}_{\text{Gorkov}}^m(-k) \end{bmatrix} \\
 -\mathcal{G}_{\text{Gorkov}}^m(-k) &= -\mathcal{G}_{\text{Gorkov}}^{0m}(-k) - \mathcal{G}_{\text{Gorkov}}^{0m}(-k) \begin{bmatrix} \Sigma_{11}^m(k) \mathcal{F}_m(k) & - \Sigma_{12}^m(k) \mathcal{G}_{\text{Gorkov}}^m(-k) \\ \Sigma_{21}^m(k) \mathcal{F}_m^\dagger(k) & - \Sigma_{22}^m(k) \mathcal{F}_m(k) \end{bmatrix}
 \end{aligned} \tag{24.22}$$

Proof 24.15 See Appendix, Proofs, page 524.

24.1.5. Derivation of the Self-Energy-Expressions in the Superconducting State

The main-result of this subsection is the following Theorem.

Theorem 24.16 With the equation of motion technique, one obtains the individual components of the Dyson-equation as

$$\mathcal{G}_{\text{Gorkov}}^n(i\omega_{n_1}, \mathbf{k}) = \mathcal{G}_{\text{Gorkov}}^{0,n}(i\omega_{n_1}, \mathbf{k}) \left[1 + \sum_{\substack{\omega_{n_2} \in \Omega_{-1} \\ \mathbf{k}' \in \mathbf{Q} \\ n' \in \{1, \dots, \infty\} \\ v' \in \{1, \dots, rd_{\mathbf{x}}\}}} \frac{-|g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{N_p} \left(+\frac{1}{\beta} \right) \mathcal{D}_{0,v'}(i(\omega_{n_1} - \omega_{n_2})) \left[\mathcal{G}_{\text{Gorkov}}^{n'}(i\omega_{n_2}, \mathbf{k}') \mathcal{G}_{\text{Gorkov}}^n(i\omega_{n_1}, \mathbf{k}) - \mathcal{F}_n^\dagger(i\omega_{n_1}, \mathbf{k}) \mathcal{F}_{n'}(i\omega_{n_2}, \mathbf{k}') \right] \right] \tag{24.23}$$

$$\mathcal{F}_n(i\omega_{n_2}, \mathbf{k}) = \mathcal{G}_{\text{Gorkov}}^{0,n}(i\omega_{n_2}, \mathbf{k}) \frac{-1}{\beta} \sum_{\substack{\omega_{n_1} \in \Omega_{-1} \\ \mathbf{k}' \in \mathbf{Q} \\ v' \in \{1, 2, \dots, rd_{\mathbf{x}}\} \\ n' \in \{1, 2, \dots\}}} \frac{|g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{N_p} \mathcal{D}_{0,v'}(i(\omega_{n_2} - \omega_{n_1})) \left[\mathcal{F}_{n'\mathbf{k}'}(i\omega_{n_1}) \mathcal{G}_{n-\mathbf{k}}(-i\omega_{n_2}) + \mathcal{F}_{n\mathbf{k}}(i\omega_{n_2}) \mathcal{G}_{n'\mathbf{k}'}(i\omega_{n_1}) \right] \tag{24.24}$$

TBD

(24.25)

TBD

(24.26)

This theorem is proven throughout the subsection.

Corollary 24.17 We obtain for the individual components of the self-energy the identities

$$\Sigma_{11}(k) = \sum_{\substack{n \\ i\omega_{n_2} \in \Omega_{-1} \\ \mathbf{k}' \in Q \\ n' \in \{1,2,3,\dots\} \\ v \in \{1,2,\dots,rd_x\}}} \frac{-|g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{\beta N_p} \mathcal{D}_{0,v'}(\mathbf{k} - \mathbf{k}') \mathcal{G}_{n'}(i\omega_{n_2}, \mathbf{k}') \quad (24.27)$$

and

$$\Sigma_{12}(k) = \sum_{\substack{n \\ i\omega_{n_2} \in \Omega_{-1} \\ \mathbf{k}' \in Q \\ n' \in \{1,2,3,\dots\} \\ v \in \{1,2,\dots,rd_x\}}} \frac{+|g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{\beta N_p} \mathcal{D}_{0,v'}(\mathbf{k} - \mathbf{k}') \mathcal{F}_{n'}(i\omega_{n_2}, \mathbf{k}') \quad (24.28)$$

$$\Sigma_{21}(i\omega_n, \mathbf{k}) = T B D \quad (24.29)$$

$$\Sigma_{22}(i\omega_n, \mathbf{k}) = T B D \quad (24.30)$$

Proof 24.17 We combine Equation (24.22) with Equations (24.23) - (24.26) which yields the desired result.

Preparation 0/4: Commutators and Equations of Motion for Creation- and Annihilation-Operators

Lemma 24.18 One finds for the imaginary-time-evolution of an operator X the Heisenberg-equation of motion

$$\partial_\tau X_H(\xi\tau) = -\frac{\xi}{\hbar} [H - \mu N, X]_H(\xi\tau) + \xi (\partial_\tau X)_H(\xi\tau) \quad (24.31)$$

Proof 24.18 See Appendix, Proofs, page 525.

Lemma 24.19 One finds the identities

$$\begin{aligned} \partial_\tau b_v(\xi\tau, \bullet) &= -\xi\omega_v(\bullet)b_v(\xi\tau, \bullet) - \xi \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m, n \in \{1, \dots, \infty\}}} \frac{g_{mnv}(\mathbf{k}, -\bullet)}{\hbar \sqrt{N_p}} c_{m\sigma}^\dagger(\xi\tau, \mathbf{k} - \bullet) c_{n\sigma}(\xi\tau, \mathbf{k}) \\ \partial_\tau b_v^\dagger(\xi\tau, -\bullet) &= +\xi\omega_v(-\bullet)b_v^\dagger(\xi\tau, -\bullet) + \xi \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m, n \in \{1, \dots, \infty\}}} \frac{g_{mnv}(\mathbf{k}, -\bullet)}{\hbar \sqrt{N_p}} c_{m\sigma}^\dagger(\xi\tau, \mathbf{k} - \bullet) c_{n\sigma}(\xi\tau, \mathbf{k}) \end{aligned}$$

for $\xi \in \{-1, +1\}$.

Proof 24.19 See Appendix, Proofs, page 525.

Remark 24.19.1 Notice that we used in the proof that $\omega_v(\mathbf{q}) = \omega_v(-\mathbf{q})$, and hence used the time-reversal-symmetry of the crystal.

Remark 24.19.2 Notice further, that the $\xi\tau$ dependence of the operators on the right-hand side of the equations can be simplified to refer to the operator-pair $c^\dagger c$.

Preparation 1/4: Equations of Motion for the normal, spin-up Green-Function

Lemma 24.20 One finds for the Gorkov-Green-function

$$\begin{aligned} & \left(\partial_\tau + \frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar} \right) \mathcal{G}_{\text{Gorkov}}^n(\tau, \mathbf{k}) \\ &= -\frac{1}{\hbar} \delta(\tau) + \sum_{\substack{\mathbf{k}' \in \mathcal{Q} \\ n' \in \{1, \dots, \infty\} \\ v' \in \{1, \dots, rd_x\}}} \frac{g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')}{\hbar^2 \sqrt{N_p}} \left\langle T_\tau c_{n', \uparrow}(\tau, \mathbf{k}') c_{n, \uparrow}^\dagger(\mathbf{k}, 0) (b_{v'}(\tau, \mathbf{k} - \mathbf{k}') + b_{v'}^\dagger(\tau, \mathbf{k}' - \mathbf{k})) \right\rangle_{\text{SC}} \end{aligned} \quad (24.32)$$

Proof 24.20 See Appendix, Proofs, page 527.

Definition 24.21 We define the *auxiliary-Green-function* for $\tau, \tau' \in (0, \hbar\beta)$

$$\mathcal{G}_{n, n', v'}^{\text{aux}}(\tau, \tau', \mathbf{k}, \mathbf{k}') = -\frac{1}{\hbar} \left\langle T_\tau c_{n', \uparrow}(\tau', \mathbf{k}') c_{n, \uparrow}^\dagger(0, \mathbf{k}) (b_{v'}(\tau, \mathbf{k} - \mathbf{k}') + b_{v'}^\dagger(\tau, \mathbf{k}' - \mathbf{k})) \right\rangle_{\text{SC}} \quad (24.33)$$

Remark 24.21.1 Notice that this Green-function has, just like a "regular" Green-function the unit of 1/Energy $\times$ Time.

Remark 24.21.2 Notice that Equation (24.32) can now be rewritten as

$$\left(\partial_\tau + \frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar} \right) \mathcal{G}_{\text{Gorkov}}^n(\tau, \mathbf{k}) = -\frac{1}{\hbar} \delta(\tau) - \sum_{\substack{\mathbf{k}' \in \mathcal{Q} \\ n' \in \{1, \dots, \infty\} \\ v \in \{1, \dots, rd_x\}}} \frac{g_{nn'v}(\mathbf{k}', \mathbf{k} - \mathbf{k}')}{\hbar \sqrt{N_p}} \mathcal{G}_{n, n', v}^{\text{aux}}(\tau, \tau, \mathbf{k}, \mathbf{k}') \quad (24.34)$$

Lemma 24.22 For $\mathbf{k} \neq \mathbf{k}'$ we obtain the equations for the first partial derivative

$$\partial_\tau \mathcal{G}_{n,n',v'}^{\text{aux}}(\tau, \tau', \mathbf{k}, \mathbf{k}') = + \frac{\omega_{v'}(\mathbf{k} - \mathbf{k}')}{\hbar} \left\langle T_\tau (b_{v'}(\tau, \mathbf{k} - \mathbf{k}') - b_{v'}^\dagger(\tau, \mathbf{k}' - \mathbf{k})) c_{n',\uparrow}(\tau', \mathbf{k}') c_{n,\uparrow}^\dagger(0, \mathbf{k}) \right\rangle_{\text{SC}} \quad (24.35)$$

and for the second partial derivative⁶

$$\begin{aligned} & (\partial_\tau^2 - \omega_{v'}^2(\mathbf{k} - \mathbf{k}')) \mathcal{G}_{n,n',v'}^{\text{aux}}(\tau, \tau', \mathbf{k}, \mathbf{k}') \\ &= \frac{-2\omega_{v'}(\mathbf{k} - \mathbf{k}')}{\hbar} \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ m,m' \in \{1, \dots, \infty\} \\ \sigma \in \{\uparrow, \downarrow\}}} \frac{g_{mm'v'}(\mathbf{q}, \mathbf{k}' - \mathbf{k})}{\hbar \sqrt{N_p}} \left\langle T_\tau c_{m,\sigma}^\dagger(\tau, \mathbf{q} + \mathbf{k}' - \mathbf{k}) c_{m',\sigma}(\tau, \mathbf{q}) c_{n',\uparrow}(\tau', \mathbf{k}') c_{n,\uparrow}^\dagger(0, \mathbf{k}) \right\rangle_{\text{SC}} \end{aligned} \quad (24.36)$$

Proof 24.22 See Appendix, Proofs, page 528.

Remark 24.22.1 Notice that we used $\omega_v(\mathbf{q}) = \omega_v(-\mathbf{q})$, and hence used the time-reversal-symmetry of the crystal.

Corollary 24.23 It follows for the equal-time Green-function⁷ the equation

$$\mathcal{G}_{n,n',v'}^{\text{aux}}(\tau, \tau, \mathbf{k}, \mathbf{k}') = - \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ \tau' \in [0, \hbar\beta] \\ m,m' \in \{1, \dots, \infty\} \\ \sigma \in \{\uparrow, \downarrow\}}} \mathcal{D}_{0,v'}(\tau - \tau') \frac{g_{mm'v'}(\mathbf{q}, \mathbf{k}' - \mathbf{k})}{\hbar \sqrt{N_p}} \left\langle T_\tau c_{m,\sigma}^\dagger(\tau', \mathbf{q} + \mathbf{k}' - \mathbf{k}) c_{m',\sigma}(\tau', \mathbf{q}) c_{n',\uparrow}(\tau, \mathbf{k}') c_{n,\uparrow}^\dagger(0, \mathbf{k}) \right\rangle_{\text{SC}}$$

Proof 24.23 See Appendix, Proofs, page 530.

Remark 24.23.1 Notice that now the equation of motion (24.34) can be rewritten as

$$\begin{aligned} & \left(\partial_\tau + \frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar} \right) \mathcal{G}_{\text{Gorkov}}^n(\tau, \mathbf{k}) \\ &= -\frac{1}{\hbar} \delta(\tau) \\ &+ \sum_{\substack{\mathbf{q}, \mathbf{k}' \in \mathbf{Q} \\ \tau' \in (0, \hbar\beta) \\ v' \in \{1, \dots, rd_{\mathbf{x}}\} \\ m,m',n' \in \{1, \dots, \infty\} \\ \sigma \in \{\uparrow, \downarrow\}}} \frac{g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}') g_{mm'v'}(\mathbf{q}, \mathbf{k}' - \mathbf{k})}{\hbar^2 N_p} \mathcal{D}_{0,v'}(\tau - \tau') \left\langle T_\tau c_{m,\sigma}^\dagger(\tau', \mathbf{q} + \mathbf{k}' - \mathbf{k}) c_{m',\sigma}(\tau', \mathbf{q}) c_{n',\uparrow}(\tau, \mathbf{k}') c_{n,\uparrow}^\dagger(0, \mathbf{k}) \right\rangle_{\text{SC}} \end{aligned} \quad (24.37)$$

⁶Note that this is in perfect harmony with Marsiglios article, except that he has typos in ω and g .

⁷Notice that Marsiglio has exchanged values for the couplings and he has no global minus-sign, which is because we defined our auxiliary-Green-function with a global minus-sign, which is more consistent with the usual definition of Green-functions.

Theorem 24.24 Neglecting vertex-corrections allows us to rewrite the equation of motion as

$$\begin{aligned} & \left(\partial_\tau + \frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar} \right) \mathcal{G}_{\text{Gorkov}}^n(\tau, \mathbf{k}) \\ &= -\frac{1}{\hbar} \delta(\tau) + \sum_{\substack{\tau' \in (0, \hbar\beta) \\ \mathbf{k}' \in \mathbf{Q} \\ n' \in \{1, \dots, \infty\} \\ v' \in \{1, \dots, rd_x\}}} \frac{|g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{N_p} \mathcal{D}_{0,v'}(\tau - \tau') \left[\mathcal{G}_{\text{Gorkov}}^n(\tau', \mathbf{k}) \mathcal{G}_{\text{Gorkov}}^{n'}(\tau - \tau', +\mathbf{k}') - \mathcal{F}_{n'}(\tau - \tau', \mathbf{k}') \mathcal{F}_n^\dagger(\tau', \mathbf{k}) \right] \end{aligned} \quad (24.38)$$

Remark 24.24.1 Notice that both sides of this Equation have the (correct) unit $\text{Energy}^{-1} \times \text{Time}^{-2}$.

Remark 24.24.2 Notice that the terms on the r.h.s. of this Equation are (II,3) and (I,2).

Proof 24.24 See Appendix, Proofs, page 531.

Theorem 24.25 Going to Matsubara-space gives the equation of motion for the Green-function as

$$\mathcal{G}_{\text{Gorkov}}^n(i\omega_{n_1}, \mathbf{k}) = \mathcal{G}_{\text{Gorkov}}^{0,n}(i\omega_{n_1}, \mathbf{k}) \left[1 + \sum_{\substack{\omega_{n_2} \in \Omega_{-1} \\ \mathbf{k}' \in \mathbf{Q} \\ n' \in \{1, \dots, \infty\} \\ v' \in \{1, \dots, rd_x\}}} \frac{-|g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{N_p} \left(+\frac{1}{\beta} \right) \mathcal{D}_{0,v'}(i(\omega_{n_1} - \omega_{n_2})) \left[\mathcal{G}_{\text{Gorkov}}^{n'}(i\omega_{n_2}, \mathbf{k}') \mathcal{G}_{\text{Gorkov}}^n(i\omega_{n_1}, \mathbf{k}) - \mathcal{F}_n^\dagger(i\omega_{n_1}, \mathbf{k}) \mathcal{F}_{n'}(i\omega_{n_2}, \mathbf{k}') \right] \right] \quad (24.39)$$

Proof 24.25 See Appendix, Proofs, page 534.

Preparation 2/4: Equations of Motion for the other anomalous Green-Function

Fact 24.26 The equation of motion for the anomalous Green-function in imaginary time reads

$$\left(\partial_\tau + \frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar} \right) \mathcal{F}_n(\tau, \mathbf{k}) = \sum_{\substack{\mathbf{k}' \in \mathbf{Q} \\ n' \in \{1, 2, 3, \dots\} \\ v' \in \{1, 2, \dots, rd_x\}}} \frac{g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')}{\hbar^2 N_p} \left\langle T_\tau c_{n'\uparrow}(\tau, \mathbf{k}') c_{n\downarrow}(0, -\mathbf{k}) (b_{v'}(\tau, \mathbf{k} - \mathbf{k}') + b_{v'}^\dagger(\tau, \mathbf{k}' - \mathbf{k})) \right\rangle \quad (24.40)$$

Proof 24.26 We compute the equation of motion for the anomalous Green-function in the 1,2-component of the Green-function. We obtain analogous to the Lemma in the 1,1-component

$$\begin{aligned}\partial_\tau \mathcal{F}_n(\tau, \mathbf{k}) &= \partial_\tau \left(-\frac{1}{\hbar} \right) \langle T_\tau c_{n\uparrow}(\tau, \mathbf{k}) c_{n\downarrow}(0, -\mathbf{k}) \rangle \\ &= \frac{1}{\hbar} \left\langle \left[c_{n,\uparrow}(\mathbf{k}), \frac{H - \mu N}{\hbar} \right] c_{n,\downarrow}(0, -\mathbf{k}) \right\rangle\end{aligned}\quad (24.41)$$

This gives, using the same calculation as before, an equation analogous to "filled square equation"

$$\left(\partial_\tau + \frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar} \right) \mathcal{F}_n(\tau, \mathbf{k}) = \sum_{\substack{\mathbf{k}' \in Q \\ n' \in \{1,2,3,\dots\} \\ v' \in \{1,2,\dots,rd_x\}}} \frac{g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')}{\hbar^2 N_p} \langle T_\tau c_{n'\uparrow}(\tau, \mathbf{k}') c_{n\downarrow}(0, -\mathbf{k}) (b_{v'}(\tau, \mathbf{k} - \mathbf{k}') + b_{v'}^\dagger(\tau, \mathbf{k}' - \mathbf{k})) \rangle \quad (24.42)$$

Notation 24.27 We introduce

$$\mathcal{G}_{\text{aux2}}(\tau, \tau', \mathbf{k}, \mathbf{k}') = -\frac{1}{\hbar} \langle T_\tau (b_{v'}(\tau, \mathbf{k} - \mathbf{k}') - b_{v'}^\dagger(\tau, \mathbf{k}' - \mathbf{k})) c_{n'\uparrow}(\tau', \mathbf{k}) c_{n\downarrow}(0, -\mathbf{k}) \rangle \quad (24.43)$$

Remark 24.27.1 Notice that this allows us to rewrite the equation of motion as

$$\left(\partial_\tau + \frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar} \right) \mathcal{F}_n(\tau, \mathbf{k}) = - \sum_{\substack{\mathbf{k}' \in Q \\ n' \in \{1,2,3,\dots\} \\ v' \in \{1,2,\dots,rd_x\}}} \frac{g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')}{\hbar N_p} \mathcal{G}_{\text{aux2}}(\tau, \tau, \mathbf{k}, \mathbf{k}') \quad (24.44)$$

Lemma 24.28 We obtain

$$\partial_\tau \mathcal{G}_{\text{aux2}}(\tau, \tau', \mathbf{k}, \mathbf{k}') = \frac{\omega_{v'}(\mathbf{k} - \mathbf{k}')}{\hbar} \langle T_\tau (b_{v'}(\tau, \mathbf{k} - \mathbf{k}') - b_{v'}^\dagger(\tau, \mathbf{k}' - \mathbf{k})) c_{n'\uparrow}(\tau', \mathbf{k}') c_{n\downarrow}(0, -\mathbf{k}) \rangle \quad (24.45)$$

and

$$\left(\partial_\tau^2 - \omega_{v'}^2(\mathbf{k} - \mathbf{k}') \partial_\tau \mathcal{G}_{\text{aux2}} \right) = \dots \quad (24.46)$$

Proof 24.28 Follows analogously as in the proof of Lemma TBReferenced.

Remark 24.28.1 This allows us to rewrite the equation of motion as

$$\left(\partial_\tau + \frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar} \right) \mathcal{F}_n(\tau, \mathbf{k}) = \sum_{\substack{\tau' \in (0, \hbar\beta) \\ \mathbf{q}, \mathbf{k}' \in Q \\ v' \in \{1,2,\dots,rd_x\} \\ m, m', n' \in \{1,2,3,\dots\} \\ \sigma \in \{\downarrow, \uparrow\}}} \frac{g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}') g_{mm'v'}(\mathbf{q}, \mathbf{k}' - \mathbf{k})}{\hbar^2 N_p} \mathcal{D}_{0,v'}(\tau - \tau') \langle T_\tau c_{m,\sigma}^\dagger(\tau', \mathbf{q} + \mathbf{k}' - \mathbf{k}) c_{m'\sigma}(\tau', \mathbf{q}) c_{n'\uparrow}(\tau, \mathbf{k}') c_{n\downarrow}(0, -\mathbf{k}) \rangle \quad (24.47)$$

Notation 24.29 Now we obtain for the individual creation- and annihilation-operators

- for the first one

$$c_{m,\sigma}^\dagger(\tau', \mathbf{q} + \mathbf{k}' - \mathbf{k}) = \begin{cases} \bar{\eta}_{m,1}(\tau', \mathbf{q} + \mathbf{k}' - \mathbf{k}), & \sigma = \uparrow \\ \eta_{m,2}(\tau', -\mathbf{q} - \mathbf{k}' + \mathbf{k}), & \sigma = \downarrow \end{cases} \quad (24.48)$$

- for the second one

$$c_{m',\sigma}(\tau', \mathbf{q}) = \begin{cases} \eta_{m',1}(\tau', \mathbf{q}), & \sigma = \uparrow \\ \bar{\eta}_{m',2}(\tau', -\mathbf{q}), & \sigma = \downarrow \end{cases} \quad (24.49)$$

- for the third one

$$c_{n'\uparrow}(\tau, \mathbf{k}') = \eta_{n'1}(\tau, \mathbf{k}') \quad (24.50)$$

- for the fourth one

$$c_{n\downarrow}(0, -\mathbf{k}) = \bar{\eta}_{n,2}(0, +\mathbf{k}) \quad (24.51)$$

Corollary 24.30 We obtain the equation of motion for $\mathcal{F}_n$ therefore reads now

$$\left(\partial_\tau + \frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar} \right) \mathcal{F}_n(\tau, \mathbf{k}) = \sum_{\substack{\mathbf{k}' \in \mathbb{Q} \\ \tau' \in (0, \hbar\beta) \\ v' \in \{1, 2, \dots, rd_x\} \\ n' \in \{1, 2, \dots\}}} \frac{|g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{N_p} \mathcal{D}_{0,v'}(\tau - \tau') \left[\mathcal{F}_{n'\mathbf{k}'}(\tau - \tau') \mathcal{G}_{n-\mathbf{k}}(-\tau') + \mathcal{F}_{n\mathbf{k}}(\tau') \mathcal{G}_{n'-\mathbf{k}'}(\tau - \tau') \right] \quad (24.52)$$

Proof 24.30 Now we approximate the resulting expression using the Bethe-Salpeter-equation again, whereas the first line corresponds to the "spin down" contribution and the second one to the "spin up" contribution, resulting in:

$$\begin{aligned} \text{CORR} &= \langle \eta_{m,2}(\tau', -\mathbf{q} - \mathbf{k}' + \mathbf{k}) \bar{\eta}_{m',2}(\tau', -\mathbf{q}) \eta_{n'1}(\tau, \mathbf{k}') \bar{\eta}_{n,2}(0, +\mathbf{k}) \rangle_{\text{SC}} \\ &+ \langle \bar{\eta}_{m,1}(\tau', +\mathbf{q} + \mathbf{k}' - \mathbf{k}) \eta_{m',1}(\tau', +\mathbf{q}) \eta_{n'1}(\tau, \mathbf{k}') \bar{\eta}_{n,2}(0, +\mathbf{k}) \rangle_{\text{SC}} \\ &= \langle \eta_{n'1}(\tau, \mathbf{k}') \eta_{m,2}(\tau', -\mathbf{q} - \mathbf{k}' + \mathbf{k}) \bar{\eta}_{m',2}(\tau', -\mathbf{q}) \bar{\eta}_{n,2}(0, +\mathbf{k}) \rangle_{\text{SC}} \\ &+ \langle \eta_{m',1}(\tau', +\mathbf{q}) \eta_{n'1}(\tau, \mathbf{k}') \bar{\eta}_{m,1}(\tau', +\mathbf{q} + \mathbf{k}' - \mathbf{k}) \bar{\eta}_{n,2}(0, +\mathbf{k}) \rangle_{\text{SC}} \\ &= \hbar^2 \mathcal{G}(i_1 = [n', 1, \tau, \mathbf{k}'], i_2 = [m, 2, \tau', -\mathbf{q} - \mathbf{k}' + \mathbf{k}], i'_1 = [n, 2, 0, \mathbf{k}], i'_2 = [m', 2, \tau', -\mathbf{q}]) \\ &+ \hbar^2 \mathcal{G}(i_1 = [m', 1, \tau', \mathbf{q}], i_2 = [n', 1, \tau, \mathbf{k}'], i'_1 = [n, 2, 0, \mathbf{k}], i'_2 = [m, 1, \tau', \mathbf{q} + \mathbf{k}' - \mathbf{k}]) \\ &= \text{I} + \text{II} \end{aligned} \quad (24.53)$$

Now we get two terms each

$$\begin{aligned} \text{I} &= \hbar^2 \hat{\mathcal{G}}(i_1, i'_1) \hat{\mathcal{G}}(i_2, i'_2) - \hbar^2 \hat{\mathcal{G}}(i_1, i'_2) \hat{\mathcal{G}}(i_2, i'_1) \\ &= \hbar^2 \hat{\mathcal{G}}_{\substack{n',n \\ 1,2 \\ \mathbf{k}',\mathbf{k}}}(\tau - 0) \hat{\mathcal{G}}_{\substack{m',m \\ 2,2 \\ -\mathbf{q}, -\mathbf{q} - \mathbf{k}' + \mathbf{k}}} (0) - \hbar^2 \hat{\mathcal{G}}_{\substack{n',m' \\ 1,2 \\ \mathbf{k}', -\mathbf{q}}}(\tau - \tau') \hat{\mathcal{G}}_{\substack{m,n \\ 2,2 \\ +\mathbf{k}, -\mathbf{q} - \mathbf{k}' + \mathbf{k}}}(\tau') = \dots \end{aligned} \quad (24.54)$$

and the other term gives

$$\begin{aligned} \Pi &= \hbar^2 \hat{\mathcal{G}}(i_1, i'_1) \hat{\mathcal{G}}(i_2, i'_2) - \hbar^2 \hat{\mathcal{G}}(i_1, i'_2) \hat{\mathcal{G}}(i_2, i'_1) \\ &= \hbar^2 \hat{\mathcal{G}}_{\substack{m',n \\ 1,2}}(\tau' - 0) \hat{\mathcal{G}}_{\substack{n',m \\ 1,1}}(\tau - \tau') - \hbar^2 \hat{\mathcal{G}}_{\substack{m',m \\ 1,1}}(0) \hat{\mathcal{G}}_{\substack{n',n \\ 1,2}}(\tau - 0) = \dots \end{aligned} \quad (24.55)$$

Now we use the definition of the Gorkov-Green-functions and the fact that translational invariance is preserved

$$\begin{aligned} \text{I} &= \hbar^2 \delta_{n',n} \delta_{m',m} \delta_{\mathbf{k}',\mathbf{k}} \hat{\mathcal{G}}_{n\mathbf{k}'}(\tau) \hat{\mathcal{G}}_{m\mathbf{q}}(0) - \hbar^2 \delta_{n',m'} \delta_{n,m} \delta_{\mathbf{k}',-\mathbf{q}} \hat{\mathcal{G}}_{n'\mathbf{k}'}(\tau - \tau') \hat{\mathcal{G}}_{n\mathbf{k}}(\tau') \\ &= \hbar^2 \delta_{n',n} \delta_{m',m} \delta_{\mathbf{k}',\mathbf{k}} \mathcal{F}_{n\mathbf{k}'}(\tau) (-1) \mathcal{G}_{m\mathbf{q}}(0) - \hbar^2 \delta_{n',m'} \delta_{n,m} \delta_{\mathbf{k}',-\mathbf{q}} \mathcal{F}_{n'\mathbf{k}'}(\tau - \tau') (-1) \mathcal{G}_{n\mathbf{k}}(-\tau') \end{aligned} \quad (24.56)$$

and the other term gives

$$\begin{aligned} \Pi &= \hbar^2 \delta_{m',n} \delta_{n',m} \delta_{\mathbf{q},\mathbf{k}} \hat{\mathcal{G}}_{n\mathbf{k}}(\tau') \hat{\mathcal{G}}_{n'\mathbf{q}}(\tau - \tau') - \hbar^2 \delta_{m,m'} \delta_{n,n'} \delta_{\mathbf{k},\mathbf{k}'} \hat{\mathcal{G}}_{m\mathbf{q}}(0) \hat{\mathcal{G}}_{n\mathbf{k}'}(\tau) \\ &= \hbar^2 \delta_{m',n} \delta_{n',m} \delta_{\mathbf{q},\mathbf{k}} \mathcal{F}_{n\mathbf{k}}(\tau') \mathcal{G}_{n'\mathbf{q}}(\tau - \tau') - \hbar^2 \delta_{m,m'} \delta_{n,n'} \delta_{\mathbf{k},\mathbf{k}'} \mathcal{G}_{m\mathbf{q}}(0) \mathcal{F}_{n\mathbf{k}'}(\tau') \end{aligned} \quad (24.57)$$

Therefore we obtain

$$\begin{aligned} \text{r.h.s.}(\star \star \star) &= \sum_{\substack{\mathbf{q}, \mathbf{k}' \in \mathbf{Q} \\ \tau' \in (0, \hbar\beta) \\ v' \in \{1, 2, \dots, r d_x\} \\ m, m', n' \in \{1, 2, \dots\}}} \frac{g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}') g_{mm'v'}(\mathbf{q}, \mathbf{k}' - \mathbf{k})}{\hbar^2 N_p} \mathcal{D}_{0,v'}(\tau - \tau') (1 + 2 + 3 + 4) \\ &\equiv \sum_{i=1}^4 \boxed{i} \end{aligned} \quad (24.58)$$

Then we get

$$\begin{aligned} \boxed{1} &= - \sum_{\substack{\mathbf{q}, \mathbf{k}' \in \mathbf{Q} \\ \tau' \in (0, \hbar\beta) \\ v' \in \{1, 2, \dots, r d_x\} \\ m, m', n' \in \{1, 2, \dots\}}} \frac{g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}') g_{mm'v'}(\mathbf{q}, \mathbf{k}' - \mathbf{k})}{N_p} \mathcal{D}_{0,v'}(\tau - \tau') \delta_{n',n} \delta_{m',m} \delta_{\mathbf{k}',\mathbf{k}} \mathcal{F}_{n\mathbf{k}'}(\tau) \mathcal{G}_{m\mathbf{q}}(0) \\ &\stackrel{\text{kill } n', m', \mathbf{k}'}{=} - \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ \tau' \in (0, \hbar\beta) \\ v' \in \{1, 2, \dots, r d_x\} \\ m \in \{1, 2, \dots\}}} \frac{g_{nnv'}(\mathbf{k}, 0) g_{mmv'}(\mathbf{q}, 0)}{N_p} \mathcal{D}_{0,v'}(\tau - \tau') \mathcal{F}_{n\mathbf{k}}(\tau) \mathcal{G}_{m\mathbf{q}}(0) \end{aligned} \quad (24.59)$$

and

$$\begin{aligned}
 \boxed{2} &= \sum_{\substack{\mathbf{q}, \mathbf{k}' \in \mathbf{Q} \\ \tau' \in (0, \hbar\beta) \\ v' \in \{1, 2, \dots, rd_x\} \\ m, m', n' \in \{1, 2, \dots\}}} \frac{g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}') g_{mm'v'}(\mathbf{q}, \mathbf{k}' - \mathbf{k})}{N_p} \mathcal{D}_{0, v'}(\tau - \tau') \delta_{n', m'} \delta_{n, m} \delta_{\mathbf{k}', -\mathbf{q}} \mathcal{F}_{n'\mathbf{k}'}(\tau - \tau') \mathcal{G}_{n-\mathbf{k}}(-\tau') \\
 &\stackrel{\text{kill } m, m', \mathbf{q}}{=} \sum_{\substack{\mathbf{k}' \in \mathbf{Q} \\ \tau' \in (0, \hbar\beta) \\ v' \in \{1, 2, \dots, rd_x\} \\ n' \in \{1, 2, \dots\}}} \frac{g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}') g_{nn'v'}(-\mathbf{k}', -\mathbf{k} + \mathbf{k}')}{N_p} \mathcal{D}_{0, v'}(\tau - \tau') \mathcal{F}_{n'\mathbf{k}'}(\tau - \tau') \mathcal{G}_{n-\mathbf{k}}(-\tau') \\
 &\stackrel{\text{symmetry}}{=} \sum_{\substack{\mathbf{k}' \in \mathbf{Q} \\ \tau' \in (0, \hbar\beta) \\ v' \in \{1, 2, \dots, rd_x\} \\ n' \in \{1, 2, \dots\}}} \frac{|g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{N_p} \mathcal{D}_{0, v'}(\tau - \tau') \mathcal{F}_{n'\mathbf{k}'}(\tau - \tau') \mathcal{G}_{n-\mathbf{k}}(-\tau')
 \end{aligned} \tag{24.60}$$

and

$$\begin{aligned}
 \boxed{3} &= \sum_{\substack{\mathbf{q}, \mathbf{k}' \in \mathbf{Q} \\ \tau' \in (0, \hbar\beta) \\ v' \in \{1, 2, \dots, rd_x\} \\ m, m', n' \in \{1, 2, \dots\}}} \frac{g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}') g_{mm'v'}(\mathbf{q}, \mathbf{k}' - \mathbf{k})}{N_p} \mathcal{D}_{0, v'}(\tau - \tau') \delta_{m', n} \delta_{n', m} \delta_{\mathbf{k}, \mathbf{q}} \mathcal{F}_{n\mathbf{k}}(\tau') \mathcal{G}_{n'\mathbf{k}'}(\tau - \tau') \\
 &\stackrel{\text{kill } m, m', \mathbf{q}}{=} \sum_{\substack{\mathbf{k}' \in \mathbf{Q} \\ \tau' \in (0, \hbar\beta) \\ v' \in \{1, 2, \dots, rd_x\} \\ n' \in \{1, 2, \dots\}}} \frac{g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}') g_{n'n v'}(\mathbf{k}, \mathbf{k}' - \mathbf{k})}{N_p} \mathcal{D}_{0, v'}(\tau - \tau') \mathcal{F}_{n\mathbf{k}}(\tau') \mathcal{G}_{n'\mathbf{k}'}(\tau - \tau') \\
 &\stackrel{\text{the other symmetry}}{=} \sum_{\substack{\mathbf{k}' \in \mathbf{Q} \\ \tau' \in (0, \hbar\beta) \\ v' \in \{1, 2, \dots, rd_x\} \\ n' \in \{1, 2, \dots\}}} \frac{|g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{N_p} \mathcal{D}_{0, v'}(\tau - \tau') \mathcal{F}_{n\mathbf{k}}(\tau') \mathcal{G}_{n'\mathbf{k}'}(\tau - \tau')
 \end{aligned} \tag{24.61}$$

and

$$\begin{aligned}
 \boxed{4} &= - \sum_{\substack{\mathbf{q}, \mathbf{k}' \in \mathbf{Q} \\ \tau' \in (0, \hbar\beta) \\ v' \in \{1, 2, \dots, rd_x\} \\ m, m', n' \in \{1, 2, \dots\}}} \frac{g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}') g_{mm'v'}(\mathbf{q}, \mathbf{k}' - \mathbf{k})}{N_p} \mathcal{D}_{0, v'}(\tau - \tau') \delta_{n', n} \delta_{m', m} \delta_{\mathbf{k}', \mathbf{k}} \mathcal{F}_{n\mathbf{k}'}(\tau) \mathcal{G}_{m\mathbf{q}}(0) \\
 &\stackrel{\text{kill } n', m', \mathbf{k}'}{=} - \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ \tau' \in (0, \hbar\beta) \\ v' \in \{1, 2, \dots, rd_x\} \\ m \in \{1, 2, \dots\}}} \frac{g_{nnv'}(\mathbf{k}, 0) g_{mmv'}(\mathbf{q}, 0)}{N_p} \mathcal{D}_{0, v'}(\tau - \tau') \mathcal{F}_{n\mathbf{k}}(\tau) \mathcal{G}_{m\mathbf{q}}(0)
 \end{aligned} \tag{24.62}$$

now we do the "usual thing" and neglect the terms $\boxed{1}$ and $\boxed{4}$, which means that we neglect contributions from optical phonons. The equation of motion for $\mathcal{F}_n$ therefore reads now

$$\left(\partial_\tau + \frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar} \right) \mathcal{F}_n(\tau, \mathbf{k}) = \sum_{\substack{\mathbf{k}' \in \mathbf{Q} \\ \tau' \in (0, \hbar\beta) \\ v' \in \{1, 2, \dots, rd_x\} \\ n' \in \{1, 2, \dots\}}} \frac{|g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{N_p} \mathcal{D}_{0, v'}(\tau - \tau') \left[\mathcal{F}_{n'\mathbf{k}'}(\tau - \tau') \mathcal{G}_{n-\mathbf{k}}(-\tau') + \mathcal{F}_{n\mathbf{k}}(\tau') \mathcal{G}_{n'-\mathbf{k}'}(\tau - \tau') \right] \tag{24.63}$$

Corollary 24.31 Going to Matsubara-space, we obtain

$$\mathcal{F}_n(i\omega_{n_2}, \mathbf{k}) = \mathcal{G}_{\text{Gorkov}}^{0,n}(i\omega_{n_2}, \mathbf{k}) \frac{-1}{\beta} \sum_{\substack{\omega_{n_1} \in \Omega_{-1} \\ \mathbf{k}' \in \mathbf{Q} \\ v' \in \{1,2,\dots,rd_x\} \\ n' \in \{1,2,\dots\}}} \frac{|g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{N_p} \mathcal{D}_{0,v'}(i(\omega_{n_2} - \omega_{n_1})) [\mathcal{F}_{n'\mathbf{k}'}(i\omega_{n_1}) \mathcal{G}_{n-\mathbf{k}}(-i\omega_{n_2}) + \mathcal{F}_{n\mathbf{k}}(i\omega_{n_2}) \mathcal{G}_{n'\mathbf{k}'}(i\omega_{n_1})] \quad (24.64)$$

Proof 24.31

Now we switch to Matsubara-frequencies yielding

$$\text{l.h.s.} = \frac{1}{\hbar\beta} \sum_{\omega_{n_1} \in \Omega_{+1}} \left(-\hbar \mathcal{G}_{\text{Gorkov}}^{0,n}(i\omega_{n_1}, \mathbf{k}) \right)^{-1} \mathcal{G}_{\text{Gorkov}}^n(i\omega_{n_1}, \mathbf{k}) e^{-i\omega_{n_1} \tau} \quad (24.65)$$

For the r.h.s. we consider only the temporal components at first...

First Term:

We consider

$$\begin{aligned} \int_0^{\hbar\beta} d\tau' \mathcal{D}_0(\tau - \tau') \mathcal{F}(\tau - \tau') \mathcal{G}(-\tau') &= \oint_{\substack{\tau' \in (0, \hbar\beta) \\ \omega_{n_{1,2}} \in \Omega_{-1} \\ \omega_{n_3} \in \Omega_{+1}}} e^{-i\omega_{n_3} + i\omega_{n_1}(\tau' - \tau) - i\omega_{n_2} \tau'} \mathcal{D}_0(i\omega_{n_3}) \mathcal{F}(i\omega_{n_1}) \mathcal{G}(-i\omega_{n_2}) \cdot \left(\frac{1}{\hbar\beta} \right)^3 \\ &= \dots \\ &= \left(\frac{1}{\hbar\beta} \right)^2 \sum_{\omega_{n_{1,2}} \in \Omega_{-1}} e^{-i\omega_{n_2} \tau} \mathcal{D}_0(i(\omega_{n_2} - \omega_{n_1})) \mathcal{F}(i\omega_{n_1}) \mathcal{G}(-i\omega_{n_2}) \end{aligned} \quad (24.66)$$

this gives therefore for both components on the r.h.s.

$$\left(\frac{1}{\hbar\beta} \right)^2 \sum_{\substack{\omega_{n_{1,2}} \in \Omega_{-1} \\ \mathbf{k}' \in \mathbf{Q} \\ v' \in \{1,2,\dots,rd_x\} \\ n' \in \{1,2,\dots\}}} \frac{|g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{N_p} \mathcal{D}_{0,v'}(i(\omega_{n_2} - \omega_{n_1})) e^{-\omega_{n_2} \tau} [\mathcal{F}_{n'\mathbf{k}'}(i\omega_{n_1}) \mathcal{G}_{n-\mathbf{k}}(-i\omega_{n_2}) + \mathcal{F}_{n\mathbf{k}}(i\omega_{n_2}) \mathcal{G}_{n'\mathbf{k}'}(i\omega_{n_1})] \quad (24.67)$$

Comparing coefficients yields thus

$$\left(\mathcal{G}_{\text{Gorkov}}^{0,n}(i\omega_{n_2}, \mathbf{k}) \right)^{-1} \mathcal{F}_n(i\omega_{n_2}, \mathbf{k}) = \frac{-1}{\beta} \sum_{\substack{\omega_{n_1} \in \Omega_{-1} \\ \mathbf{k}' \in \mathbf{Q} \\ v' \in \{1,2,\dots,rd_x\} \\ n' \in \{1,2,\dots\}}} \frac{|g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{N_p} \mathcal{D}_{0,v'}(i(\omega_{n_2} - \omega_{n_1})) [\mathcal{F}_{n'\mathbf{k}'}(i\omega_{n_1}) \mathcal{G}_{n-\mathbf{k}}(-i\omega_{n_2}) + \mathcal{F}_{n\mathbf{k}}(i\omega_{n_2}) \mathcal{G}_{n'\mathbf{k}'}(i\omega_{n_1})] \quad (24.68)$$

or (equivalently)

$$\mathcal{F}_n(i\omega_{n_2}, \mathbf{k}) = \mathcal{G}_{0,n}^{\text{Gorkov}}(i\omega_{n_2}, \mathbf{k}) \frac{-1}{\beta} \sum_{\substack{\omega_{n_1} \in \Omega_{-1} \\ \mathbf{k}' \in \mathbf{Q} \\ v' \in \{1,2,\dots,rd_x\} \\ n' \in \{1,2,\dots\}}} \frac{|g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{N_p} \mathcal{D}_{0,v'}(i(\omega_{n_2} - \omega_{n_1})) [\mathcal{F}_{n'\mathbf{k}'}(i\omega_{n_1}) \mathcal{G}_{n-\mathbf{k}}(-i\omega_{n_2}) + \mathcal{F}_{n\mathbf{k}}(i\omega_{n_2}) \mathcal{G}_{n'\mathbf{k}'}(i\omega_{n_1})] \quad (24.69)$$

From here, with Equation (Dyson-Equation in the SC-case) we can read off

$$\Sigma_{11}(i\omega_{n_2}, \mathbf{k}) = \sum_{\substack{i\omega_{n_1} \in \Omega_{-1} \\ \mathbf{k}' \in \mathbf{Q} \\ n' \in \{1,2,3,\dots\} \\ v \in \{1,2,\dots,rd_x\}}} \frac{-|g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{\beta N_p} \mathcal{D}_{0,v'}(i(\omega_{n_2} - \omega_{n_1})) \mathcal{G}_{n'}(i\omega_{n_1}, \mathbf{k}') \quad (24.70)$$

and

$$\Sigma_{12}(i\omega_{n_2}, \mathbf{k}) = \sum_{\substack{i\omega_{n_1} \in \Omega_{-1} \\ \mathbf{k}' \in \mathbf{Q} \\ n' \in \{1,2,3,\dots\} \\ v \in \{1,2,\dots,rd_x\}}} \frac{+|g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{\beta N_p} \mathcal{D}_{0,v'}(i(\omega_{n_2} - \omega_{n_1})) \mathcal{F}_{n'}(i\omega_{n_1}, \mathbf{k}') \quad (24.71)$$

notice the poorly exchanged roles of ω_{n_1} and ω_{n_2} w.r.t. the equations at the beginning of the section...

But it is ok, since this is the first time latexing this stuff.

Preparation 3/4 and 4/4: Equations of Motion for the normal, spin-down Green-Function and the first anomalous Green-Function

We look at the remaining equation of motions. First let's recall

$$-\mathcal{G}_{\text{Gorkov}}^m(-\tau, -\mathbf{k}) = -\frac{1}{\hbar} \left\langle T_\tau c_{n,\downarrow}^\dagger(\tau, -\mathbf{k}) c_{n,\downarrow}(0, -\mathbf{k}) \right\rangle \quad (24.72)$$

and analogously for $\mathcal{F}_n^\dagger$.

Definition 24.32 We introduce the function

$$\tilde{\mathcal{G}}_X^n(\tau, \mathbf{k}) = -\frac{1}{\hbar} \left\langle T_\tau c_{n,\downarrow}^\dagger(\tau, -\mathbf{k}) X(0) \right\rangle \quad (24.73)$$

Remark 24.32.1 Notice that this implies

$$-\mathcal{G}_{\text{Gorkov}}^m(-\tau, -\mathbf{k}) = \tilde{\mathcal{G}}_{X=c_{n,\downarrow}(-\mathbf{k})}^n(\tau, \mathbf{k}) \quad \text{and} \quad \mathcal{F}_n^\dagger = \tilde{\mathcal{G}}_{X=c_{n,\uparrow}^\dagger(+\mathbf{k})}^n(\tau, \mathbf{k}) \quad (24.74)$$

Lemma 24.33 We obtain the derivative

$$\begin{aligned}
 \partial_\tau \tilde{\mathcal{G}}_X^n(\tau, \mathbf{k}) &= -\frac{1}{\hbar^2} (\epsilon_n(\mathbf{k}) - \mu) \left\langle T_\tau c_{n,\downarrow}^\dagger(\tau, -\mathbf{k}) X(0) \right\rangle \\
 &\quad - \frac{1}{\hbar^2} \sum_{\substack{\mathbf{k}' \in Q \\ \nu' \in \{1,2,\dots,rd_x\} \\ n' \in \{1,2,\dots\}}} \frac{g_{nn'\nu'}^*(-\mathbf{k}', \mathbf{k}' - \mathbf{k})}{\hbar \sqrt{N_p}} \left\langle (b_{\nu'}(\tau, \mathbf{k} - \mathbf{k}') + b_{\nu'}^\dagger(\tau, \mathbf{k}' - \mathbf{k})) c_{n',\downarrow}^\dagger(\tau, -\mathbf{k}') X(0) \right\rangle + \begin{cases} \frac{1}{\hbar} \delta(\tau), & X = c_{n,\downarrow}(-\mathbf{k}) \\ 0, & X = c_{n,\uparrow}^\dagger(+\mathbf{k}) \end{cases} \\
 &\equiv \begin{cases} \partial_\tau \mathcal{G}_{\text{Gorkov}}^n(-\tau, -\mathbf{k}), & X = c_{n,\downarrow}(-\mathbf{k}) \\ \partial_\tau \mathcal{F}_n^\dagger(\tau, \mathbf{k}), & X = c_{n,\uparrow}^\dagger(+\mathbf{k}) \end{cases}
 \end{aligned} \tag{24.75}$$

Proof 24.33 Therefore we now consider the derivative

$$\begin{aligned}
 \partial_\tau \tilde{\mathcal{G}}_X^n(\tau, \mathbf{k}) &\stackrel{\text{for } X \in \{c, c^\dagger\}}{=} -\frac{\partial_\tau}{\hbar} \left(\theta(\tau) \left\langle c_{n,\downarrow}^\dagger(\tau, -\mathbf{k}) X(0) \right\rangle + \theta(-\tau) \zeta \left\langle X(0) c_{n,\downarrow}^\dagger(\tau, -\mathbf{k}) \right\rangle \right) \\
 &\stackrel{\text{cf. Def. 8.12 or so}}{=} -\frac{1}{\hbar} \left(\delta(\tau) [c_{n,\downarrow}^\dagger(0, -\mathbf{k}), X(0)]_\zeta \right) + \frac{1}{\hbar} \left\langle T_\tau \left[c_{n,\downarrow}^\dagger(-\mathbf{k}), \frac{H - \mu N}{\hbar} \right]_H(\tau) X(0) \right\rangle \\
 &= \frac{1}{\hbar} \left\langle T_\tau \left[c_{n,\downarrow}^\dagger(-\mathbf{k}), \frac{H - \mu N}{\hbar} \right]_H(\tau) X(0) \right\rangle + \begin{cases} \frac{1}{\hbar} \delta(\tau), & X = c_{n,\downarrow}(-\mathbf{k}) \\ 0, & X = c_{n,\uparrow}^\dagger(+\mathbf{k}) \end{cases}
 \end{aligned} \tag{24.76}$$

Therefore we need to compute the commutator⁸ whereas we use in the second equation sign the results from earlier calculations and in the third equation sign we make the replacement $\mathbf{k}^\dagger \mapsto -\mathbf{k}'$ and we also use the symmetry of ϵ w.r.t. inversion symmetry such that we obtain

$$\begin{aligned}
 [c_{n,\downarrow}^\dagger(-\mathbf{k}), \frac{H - \mu N}{\hbar}] &= [\frac{H - \mu N}{\hbar}, c_{n,\downarrow}(-\mathbf{k})]^\dagger \\
 &= \left(-\frac{\epsilon_n(-\mathbf{k}) - \mu}{\hbar} c_{n,\downarrow}(-\mathbf{k}) - \sum_{\substack{\mathbf{k}' \in Q \\ \nu' \in \{1,2,\dots,rd_x\} \\ n' \in \{1,2,\dots\}}} \frac{g_{nn'\nu'}(-\mathbf{k}', -\mathbf{k} - \mathbf{k}')}{\hbar \sqrt{N_p}} (b_{\nu'}(-\mathbf{k} - \mathbf{k}') + b_{\nu'}^\dagger(+\mathbf{k} + \mathbf{k}')) c_{n',\downarrow}(\mathbf{k}') \right)^\dagger \\
 &= -\frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar} c_{n,\downarrow}^\dagger(-\mathbf{k}) - \sum_{\substack{\mathbf{k}' \in Q \\ \nu' \in \{1,2,\dots,rd_x\} \\ n' \in \{1,2,\dots\}}} \frac{g_{nn'\nu'}^*(-\mathbf{k}', \mathbf{k}' - \mathbf{k})}{\hbar \sqrt{N_p}} (b_{\nu'}(\mathbf{k} - \mathbf{k}') + b_{\nu'}^\dagger(+\mathbf{k}' - \mathbf{k})) c_{n',\downarrow}^\dagger(-\mathbf{k}')
 \end{aligned} \tag{24.78}$$

This gives us (depending on the choice of X) two equations of motion

⁸Notice (as a Lemma) that one has on the one hand

$$[A, B] = AB - BA \Rightarrow [A, B]^\dagger = (AB - BA)^\dagger = B^\dagger A^\dagger - A^\dagger B^\dagger = -[A^\dagger, B^\dagger] = [B^\dagger, A^\dagger] \tag{24.77}$$

$$\begin{aligned}
 \partial_\tau \widetilde{\mathcal{G}}_X^n(\tau, \mathbf{k}) &= -\frac{1}{\hbar^2} (\epsilon_n(\mathbf{k}) - \mu) \left\langle T_\tau c_{n,\downarrow}^\dagger(\tau, -\mathbf{k}) X(0) \right\rangle \\
 &\quad - \frac{1}{\hbar^2} \sum_{\substack{\mathbf{k}' \in Q \\ v' \in \{1,2,\dots,rd_X\} \\ n' \in \{1,2,\dots\}}} \frac{g_{nn'v'}^*(-\mathbf{k}', \mathbf{k}' - \mathbf{k})}{\hbar \sqrt{N_p}} \left\langle (b_{v'}(\tau, \mathbf{k} - \mathbf{k}') + b_{v'}^\dagger(\tau, \mathbf{k}' - \mathbf{k})) c_{n',\downarrow}^\dagger(\tau, -\mathbf{k}') X(0) \right\rangle + \begin{cases} \frac{1}{\hbar} \delta(\tau), & X = c_{n,\downarrow}(-\mathbf{k}) \\ 0, & X = c_{n,\uparrow}^\dagger(+\mathbf{k}) \end{cases} \\
 &\equiv \begin{cases} \partial_\tau \mathcal{G}_{\text{Gorkov}}^n(-\tau, -\mathbf{k}), & X = c_{n,\downarrow}(-\mathbf{k}) \\ \partial_\tau \mathcal{F}_n^\dagger(\tau, \mathbf{k}), & X = c_{n,\uparrow}^\dagger(+\mathbf{k}) \end{cases}
 \end{aligned} \tag{24.79}$$

Definition 24.34 Now we introduce the auxiliary-Green-function: For $\tau, \tau' \in (0, \hbar\beta)$ we have

$$\widetilde{\mathcal{G}}_{\substack{v'v'n' \\ X}}^{\text{aux}}(\tau, \tau', \mathbf{k}, \mathbf{k}') = -\frac{1}{\hbar} \left\langle (b_{v'}(\mathbf{k} - \mathbf{k}') + b_{v'}^\dagger(\mathbf{k}' - \mathbf{k})) c_{n',\downarrow}^\dagger(\tau', -\mathbf{k}') X(0) \right\rangle \tag{24.80}$$

Remark 24.34.1 Notice that this allows us to rewrite the equations of motion as

$$\partial_\tau \widetilde{\mathcal{G}}_X^n(\tau, \mathbf{k}) = \frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar} \widetilde{\mathcal{G}}_X^n(\tau, \mathbf{k}) + \sum_{\substack{\mathbf{k}' \in Q \\ v' \in \{1,2,3,\dots,rd_X\} \\ n' \in \{1,2,\dots\}}} \frac{g_{nn'v'}(-\mathbf{k}', \mathbf{k}' - \mathbf{k})}{\hbar \sqrt{N_p}} \widetilde{\mathcal{G}}_{\substack{v'v'n' \\ X}}^{\text{aux}}(\tau, \tau', \mathbf{k}, \mathbf{k}') + \begin{cases} \frac{1}{\hbar} \delta(\tau), & X = c_{n,\downarrow}(-\mathbf{k}) \\ 0, & X = c_{n,\uparrow}^\dagger(+\mathbf{k}) \end{cases} \tag{24.81}$$

Lemma 24.35 We obtain the τ -derivative of $\widetilde{\mathcal{G}}_{\substack{v'v'n' \\ X}}^{\text{aux}}$ as

$$\begin{aligned}
 &(\partial_\tau^2 - \omega_{v'}^2(\mathbf{k} - \mathbf{k}')) \widetilde{\mathcal{G}}_{\substack{v'v'n' \\ X}}^{\text{aux}}(\tau, \tau', \mathbf{k}, \mathbf{k}') \\
 &= -\frac{2\omega_{v'}(\mathbf{k} - \mathbf{k}')}{\hbar} \sum_{\substack{\mathbf{q} \in Q \\ m, m' \in \{1,2,\dots\} \\ \sigma \in \{\downarrow, \uparrow\}}} \frac{g_{mm'v'}(\mathbf{q}, \mathbf{k}' - \mathbf{k})}{\hbar \sqrt{N_p}} \left\langle T_\tau c_{m,\sigma}^\dagger(\tau, \mathbf{q} + \mathbf{k}' - \mathbf{k}) c_{m',\sigma}(\tau, \mathbf{q}) c_{n',\downarrow}^\dagger(-\mathbf{k}', \tau') X(0) \right\rangle
 \end{aligned} \tag{24.82}$$

Proof 24.35 Completely analogous to the previous Lemma [TBReferenced](#) giving

Corollary 24.36 This gives the equation

$$\widetilde{\mathcal{G}}_{\substack{v'v'n' \\ X}}^{\text{aux}}(\tau, \tau', \mathbf{k}, \mathbf{k}') = \sum_{\substack{\mathbf{q} \in Q \\ \tau' \in (0, \hbar\beta) \\ m, m' \in \{1,2,\dots\} \\ \sigma \in \{\downarrow, \uparrow\}}} (-1) \mathcal{D}_{0,v'}(\tau - \tau') \frac{g_{mm'v'}(\mathbf{q}, \mathbf{k}' - \mathbf{k})}{\hbar \sqrt{N_p}} \left\langle T_\tau c_{m,\sigma}^\dagger(\tau, \mathbf{q} + \mathbf{k}' - \mathbf{k}) c_{m',\sigma}(\tau, \mathbf{q}) c_{n',\downarrow}^\dagger(-\mathbf{k}', \tau') X(0) \right\rangle \tag{24.83}$$

Proof 24.36 Analogous to Corollary [TBReferenced](#).

Notation 24.37 Now we simplify this again, using the Nambu-formalism. Therefore we rewrite (analogous to page twohundred-fifty-something)

$$c_{n'\downarrow}(\tau, -\mathbf{k}') = \eta_{n',2}(\tau, \mathbf{k}') \quad (24.84)$$

and obtain, with

$$\widetilde{\text{CORR}}(\sigma, X) = \left\langle T_\tau c_{m,\sigma}^\dagger(\tau, \mathbf{q} + \mathbf{k}' - \mathbf{k}) c_{m'\sigma}(\tau, \mathbf{q}) c_{n',\downarrow}^\dagger(-\mathbf{k}', \tau') X(0) \right\rangle \quad (24.85)$$

the different cases

$$\{\widetilde{\text{CORR}}(\sigma, X)\}_{\substack{\sigma \in \{\downarrow, \uparrow\} \\ X \in \{c_{n,\downarrow}(-\mathbf{k}), c_{n,\uparrow}(\mathbf{k})\}}} \quad (24.86)$$

i.e. four terms in total.

Lemma 24.38 For $X = c_{n,\downarrow}(-\mathbf{k})$ we obtain

$$\begin{aligned} \sum_{\sigma \in \{\downarrow, \uparrow\}} \widetilde{\text{CORR}}(\sigma, X) &= \delta_{n,n'} \delta_{m,m'} \delta_{\mathbf{k},\mathbf{k}'} \mathcal{G}_{\text{Gorkov}}^n(-\tau, -\mathbf{k}) \mathcal{G}_{\text{Gorkov}}^m(0, \mathbf{q}) - \delta_{n',m'} \delta_{n,m} \delta_{-\mathbf{q},\mathbf{k}'} \mathcal{G}_{\text{Gorkov}}^{n'}(\tau' - \tau, -\mathbf{k}') \mathcal{G}_{\text{Gorkov}}^n(-\tau', -\mathbf{k}') \\ &+ \delta_{n,m'} \delta_{n',m} \delta_{\mathbf{q},\mathbf{k}} \mathcal{F}_n(\tau', \mathbf{k}) \mathcal{F}_{n'}^\dagger(\tau - \tau', \mathbf{k}') - \delta_{m,m'} \delta_{n,n'} \delta_{\mathbf{k},\mathbf{k}'} \mathcal{G}_{\text{Gorkov}}^m(0, \mathbf{q}) (-1) \mathcal{G}_{\text{Gorkov}}^n(-\tau, -\mathbf{k}) \end{aligned} \quad (24.87)$$

Proof 24.38 We find with $c_{n,\downarrow}(-\mathbf{k}) = \bar{\eta}_{n,2}(\mathbf{k})$ for the correlators

$$\begin{aligned} &\sum_{\sigma \in \{\downarrow, \uparrow\}} \left\langle T_\tau c_{m,\sigma}^\dagger(\tau', \mathbf{q} + \mathbf{k}' - \mathbf{k}) c_{m'\sigma}(\tau', \mathbf{q}) c_{n'\downarrow}(\tau, -\mathbf{k}') c_{n,\downarrow}(0, -\mathbf{k}) \right\rangle \\ &= \left\langle T_\tau c_{m,\downarrow}^\dagger(\tau', \mathbf{q} + \mathbf{k}' - \mathbf{k}) c_{m'\downarrow}(\tau', \mathbf{q}) c_{n'\downarrow}(\tau, -\mathbf{k}') c_{n,\downarrow}(0, -\mathbf{k}) \right\rangle + \left\langle T_\tau c_{m,\uparrow}^\dagger(\tau', \mathbf{q} + \mathbf{k}' - \mathbf{k}) c_{m'\uparrow}(\tau', \mathbf{q}) c_{n'\downarrow}(\tau, -\mathbf{k}') c_{n,\downarrow}(0, -\mathbf{k}) \right\rangle \\ &= \left\langle T_\tau \eta_{m,2}(\tau', -\mathbf{q} - \mathbf{k}' + \mathbf{k}) \bar{\eta}_{m',2}(\tau', -\mathbf{q}) \eta_{n',2}(\tau, +\mathbf{k}') \bar{\eta}_{n,2}(0, \mathbf{k}) \right\rangle + \left\langle T_\tau \bar{\eta}_{m,1}(\tau', \mathbf{q} + \mathbf{k}' - \mathbf{k}) \eta_{m',1}(\tau', \mathbf{q}) \eta_{n',2}(\tau, \mathbf{k}') \bar{\eta}_{n,2}(0, \mathbf{k}) \right\rangle \\ &= \left\langle T_\tau \eta_{n',2}(\tau, +\mathbf{k}') \eta_{m,2}(\tau', -\mathbf{q} - \mathbf{k}' + \mathbf{k}) \bar{\eta}_{m',2}(\tau', -\mathbf{q}) \bar{\eta}_{n,2}(0, \mathbf{k}) \right\rangle + \left\langle T_\tau \eta_{m',1}(\tau', \mathbf{q}) \eta_{n',2}(\tau, \mathbf{k}') \bar{\eta}_{m,1}(\tau', \mathbf{q} + \mathbf{k}' - \mathbf{k}) \bar{\eta}_{n,2}(0, \mathbf{k}) \right\rangle \\ &= \hat{\mathcal{G}}(i_1 \equiv [n', 2, \tau, \mathbf{k}'], i_2 \equiv [m, 2, \tau', -\mathbf{q} - \mathbf{k}' + \mathbf{k}] | i'_1 \equiv [n, 2, 0, \mathbf{k}], i'_2 \equiv [m', 2, \tau', -\mathbf{q}]) \\ &+ \hat{\mathcal{G}}(i_1 \equiv [m', 1, \tau', \mathbf{q}], i_2 \equiv [n', 2, \tau, \mathbf{k}'] | i'_1 \equiv [n, 2, 0, \mathbf{k}], i'_2 \equiv [m, 1, \tau', \mathbf{q} + \mathbf{k}' - \mathbf{k}]) \\ &= \hat{\mathcal{G}}_{\substack{n',n \\ 2,2}}(\tau - 0) \hat{\mathcal{G}}_{\substack{m,m' \\ 2,2}}^m(0) - \hat{\mathcal{G}}_{\substack{n',m' \\ 2,2}}^m(\tau - \tau') \hat{\mathcal{G}}_{\substack{m,n \\ 2,2}}^m(\tau' - 0) + \hat{\mathcal{G}}_{\substack{m',n \\ 1,2}}^m(\tau' - 0) \hat{\mathcal{G}}_{\substack{n',m \\ 2,1}}^m(\tau - \tau') - \hat{\mathcal{G}}_{\substack{m',m \\ 1,1}}^m(0) \hat{\mathcal{G}}_{\substack{n',n \\ 2,2}}^m(\tau - 0) \\ &= \delta_{n,n'} \delta_{m,m'} \delta_{\mathbf{k},\mathbf{k}'} \hat{\mathcal{G}}_{\substack{n,k \\ 2,2}}^n(\tau - 0) \hat{\mathcal{G}}_{\substack{m,-q \\ 2,2}}^m(0) - \delta_{n',m'} \delta_{n,m} \delta_{-\mathbf{q},\mathbf{k}'} \hat{\mathcal{G}}_{\substack{n,k' \\ 2,2}}^n(\tau - \tau') \hat{\mathcal{G}}_{\substack{n,k \\ 2,2}}^n(\tau') + \delta_{n,m} \delta_{n',m} \delta_{\mathbf{q},\mathbf{k}} \hat{\mathcal{G}}_{\substack{n,k \\ 1,2}}^n(\tau') \hat{\mathcal{G}}_{\substack{n',k' \\ 2,1}}^n(\tau - \tau') - \delta_{m,m'} \delta_{n,n'} \delta_{\mathbf{k},\mathbf{k}'} \hat{\mathcal{G}}_{\substack{m,q \\ 1,1}}^m(0) \hat{\mathcal{G}}_{\substack{n,k \\ 2,2}}^n(\tau) \\ &= \delta_{n,n'} \delta_{m,m'} \delta_{\mathbf{k},\mathbf{k}'} \mathcal{G}_{\text{Gorkov}}^n(-\tau, -\mathbf{k}) \mathcal{G}_{\text{Gorkov}}^m(0, \mathbf{q}) - \delta_{n',m'} \delta_{n,m} \delta_{-\mathbf{q},\mathbf{k}'} \mathcal{G}_{\text{Gorkov}}^{n'}(\tau' - \tau, -\mathbf{k}') \mathcal{G}_{\text{Gorkov}}^n(-\tau', -\mathbf{k}') \\ &+ \delta_{n,m'} \delta_{n',m} \delta_{\mathbf{q},\mathbf{k}} \mathcal{F}_n(\tau', \mathbf{k}) \mathcal{F}_{n'}^\dagger(\tau - \tau', \mathbf{k}') - \delta_{m,m'} \delta_{n,n'} \delta_{\mathbf{k},\mathbf{k}'} \mathcal{G}_{\text{Gorkov}}^m(0, \mathbf{q}) (-1) \mathcal{G}_{\text{Gorkov}}^n(-\tau, -\mathbf{k}) \end{aligned} \quad (24.88)$$

implying that we managed to simplify the complicated correlator.

Lemma 24.39 For $X = c_{n,\uparrow}(\mathbf{k})$ we obtain

$$\begin{aligned}
 \sum_{\sigma \in \{\downarrow, \uparrow\}} \widetilde{\text{CORR}}(\sigma, X) &= \delta_{n,n'} \delta_{m,m'} \delta_{\mathbf{k},\mathbf{k}'} \mathcal{F}_n^\dagger(\tau, \mathbf{k}) (-1) \mathcal{G}_{\text{Gorkov}}^m(0, \mathbf{q}) - \delta_{n',m'} \delta_{n,m} \delta_{-\mathbf{q},\mathbf{k}'} (-1) \mathcal{G}_{\text{Gorkov}}^{n'}(\tau' - \tau, -\mathbf{k}') \mathcal{F}_n(\tau', \mathbf{k}') \\
 &+ \delta_{n,m'} \delta_{n',m} \delta_{\mathbf{q},\mathbf{k}} \mathcal{G}_{\text{Gorkov}}^n(\tau', \mathbf{k}) \mathcal{F}_{n'}^\dagger(\tau - \tau', \mathbf{k}') - \delta_{m,m'} \delta_{n,n'} \delta_{\mathbf{k},\mathbf{k}'} \mathcal{G}_{\text{Gorkov}}^m(0, \mathbf{q}) \mathcal{F}_n^\dagger(\tau, \mathbf{k})
 \end{aligned} \quad (24.89)$$

Proof 24.39 Then we find with $c_{n,\uparrow}^\dagger(+\mathbf{k}) = \bar{\eta}_{n,1}(\mathbf{k})$ for the correlators

$$\begin{aligned}
 &\sum_{\sigma \in \{\downarrow, \uparrow\}} \left\langle T_\tau c_{m,\sigma}^\dagger(\tau', \mathbf{q} + \mathbf{k}' - \mathbf{k}) c_{m',\sigma}(\tau', \mathbf{q}) c_{n',\downarrow}(\tau, -\mathbf{k}') c_{n,\uparrow}^\dagger(0, -\mathbf{k}) \right\rangle \\
 &= \dots \text{(analogous to above)} \\
 &= \langle T_\tau \eta_{n',2}(\tau, \mathbf{k}') \eta_{m,2}(\tau', -\mathbf{q} - \mathbf{k}' + \mathbf{k}) \bar{\eta}_{m',2}(\tau', -\mathbf{q}) \bar{\eta}_{n,1}(0, \mathbf{k}) \rangle + \langle T_\tau \eta_{m',1}(\tau', \mathbf{q}) \eta_{n',2}(\tau, \mathbf{k}') \bar{\eta}_{m,1}(\tau', \mathbf{q} + \mathbf{k}' - \mathbf{k}) \bar{\eta}_{n,1}(0, \mathbf{k}) \rangle \\
 &= \hat{\mathcal{G}}(i_1 \equiv [n', 2, \tau, \mathbf{k}'], i_2 \equiv [m, 2, \tau', -\mathbf{q} - \mathbf{k}' + \mathbf{k}] | i'_1 \equiv [n, 1, 0, \mathbf{k}], i'_2 \equiv [m', 2, \tau', -\mathbf{q}]) \\
 &+ \hat{\mathcal{G}}(i_1 \equiv [m', 1, \tau', \mathbf{q}], i_2 \equiv [n', 2, \tau, \mathbf{k}'] | i'_1 \equiv [n, 1, 0, \mathbf{k}], i'_2 \equiv [m, 1, \tau', \mathbf{q} + \mathbf{k}' - \mathbf{k}]) \\
 &= \hat{\mathcal{G}}_{n',n}^{2,1}(\tau) \hat{\mathcal{G}}_{m,m'}^{2,2}(0) - \hat{\mathcal{G}}_{n',m'}^{2,2}(\tau - \tau') \hat{\mathcal{G}}_{m,n}^{2,1}(\tau') + \hat{\mathcal{G}}_{m',n}^{1,1}(\tau') \hat{\mathcal{G}}_{n',m}^{2,1}(\tau - \tau') - \hat{\mathcal{G}}_{m',m}^{1,1}(0) \hat{\mathcal{G}}_{n',n}^{2,1}(\tau) \\
 &= \delta_{n,n'} \delta_{m,m'} \delta_{\mathbf{k},\mathbf{k}'} \hat{\mathcal{G}}_{n,k}^{2,1}(\tau) \hat{\mathcal{G}}_{m,-q}^{2,2}(0) - \delta_{n',m'} \delta_{n,m} \delta_{-\mathbf{q},\mathbf{k}'} \hat{\mathcal{G}}_{n',k'}^{2,1}(\tau - \tau') \hat{\mathcal{G}}_{n,k}^{2,1}(\tau') + \delta_{n,m'} \delta_{n',m} \delta_{\mathbf{q},\mathbf{k}} \hat{\mathcal{G}}_{n,k}^{1,1}(\tau') \hat{\mathcal{G}}_{n',k'}^{2,1}(\tau - \tau') - \delta_{m,m'} \delta_{n,n'} \delta_{\mathbf{k},\mathbf{k}'} \hat{\mathcal{G}}_{m,q}^{1,1}(0) \hat{\mathcal{G}}_{n,k}^{2,1}(\tau) \\
 &= \delta_{n,n'} \delta_{m,m'} \delta_{\mathbf{k},\mathbf{k}'} \mathcal{F}_n^\dagger(\tau, \mathbf{k}) (-1) \mathcal{G}_{\text{Gorkov}}^m(0, \mathbf{q}) - \delta_{n',m'} \delta_{n,m} \delta_{-\mathbf{q},\mathbf{k}'} (-1) \mathcal{G}_{\text{Gorkov}}^{n'}(\tau' - \tau, -\mathbf{k}') \mathcal{F}_n(\tau', \mathbf{k}') \\
 &+ \delta_{n,m'} \delta_{n',m} \delta_{\mathbf{q},\mathbf{k}} \mathcal{G}_{\text{Gorkov}}^n(\tau', \mathbf{k}) \mathcal{F}_{n'}^\dagger(\tau - \tau', \mathbf{k}') - \delta_{m,m'} \delta_{n,n'} \delta_{\mathbf{k},\mathbf{k}'} \mathcal{G}_{\text{Gorkov}}^m(0, \mathbf{q}) \mathcal{F}_n^\dagger(\tau, \mathbf{k})
 \end{aligned} \quad (24.90)$$

implying that we managed to simplify the complicated correlator. Now we can write down the equation of motion for the Gorkov-Green-function in imaginary time.

Remark 24.39.1 Notice that we obtain for $X = c_{n,\downarrow}(-\mathbf{k})$ the equation we obtain

$$\begin{aligned}
 &-\partial_\tau \mathcal{G}_{\text{Gorkov}}^n(-\tau, -\mathbf{k}) \\
 &= +\frac{1}{\hbar} \delta(\tau) + \frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar} \mathcal{G}_{\text{Gorkov}}^n(-\tau, -\mathbf{k}) \\
 &+ \sum_{\substack{\mathbf{q}, \mathbf{k}' \in \mathcal{Q} \\ v' \in \{1, 2, \dots, rd_x\} \\ m, m', n' \in \{1, 2, 3, \dots\} \\ \tau' \in (0, \hbar\beta)}} \frac{g_{nm'v'}^*(-\mathbf{k}', \mathbf{k}' - \mathbf{k})}{\hbar \sqrt{N_p}} (-1) \mathcal{D}_{0,v'}(\tau - \tau') \frac{g_{mm'v'}(\mathbf{q}, \mathbf{k}' - \mathbf{k})}{\hbar \sqrt{N_p}} \left[\delta_{n,n'} \delta_{m,m'} \delta_{\mathbf{k},\mathbf{k}'} \mathcal{G}_{\text{Gorkov}}^n(-\tau, -\mathbf{k}) \mathcal{G}_{\text{Gorkov}}^m(0, \mathbf{q}) \right. \\
 &\quad - \delta_{n',m'} \delta_{n,m} \delta_{-\mathbf{q},\mathbf{k}'} \mathcal{G}_{\text{Gorkov}}^{n'}(\tau' - \tau, -\mathbf{k}') \mathcal{G}_{\text{Gorkov}}^n(-\tau', -\mathbf{k}) \\
 &\quad + \delta_{n,m'} \delta_{n',m} \delta_{\mathbf{q},\mathbf{k}} \mathcal{F}_n(\tau', \mathbf{k}) \mathcal{F}_{n'}^\dagger(\tau - \tau', \mathbf{k}') \\
 &\quad \left. - \delta_{m,m'} \delta_{n,n'} \delta_{\mathbf{k},\mathbf{k}'} \mathcal{G}_{\text{Gorkov}}^m(0, \mathbf{q}) (-1) \mathcal{G}_{\text{Gorkov}}^n(-\tau, -\mathbf{k}) \right]
 \end{aligned} \quad (24.91)$$

Theorem 24.40 For $X = c_{n,\downarrow}(-\mathbf{k})$ this can be used to conclude (...)

Proof 24.40 We obtain

$$\begin{aligned}
 & -\partial_\tau \mathcal{G}_{\text{Gorkov}}^n(-\tau, -\mathbf{k}) \\
 & = +\frac{1}{\hbar} \delta(\tau) + \frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar} \mathcal{G}_{\text{Gorkov}}^n(-\tau, -\mathbf{k}) \\
 & + \sum_{\substack{\mathbf{q}, \mathbf{k}' \in \mathbf{Q} \\ v' \in \{1, 2, \dots, r_{d_x}\} \\ m, m', n' \in \{1, 2, 3, \dots\} \\ \tau' \in (0, \hbar\beta)}} \frac{g_{nn'v'}^*(-\mathbf{k}', \mathbf{k}' - \mathbf{k})}{\hbar \sqrt{N_p}} (-1) \mathcal{D}_{0, v'}(\tau - \tau') \frac{g_{mm'v'}(\mathbf{q}, \mathbf{k}' - \mathbf{k})}{\hbar \sqrt{N_p}} \left[\delta_{n, n'} \delta_{m, m'} \delta_{\mathbf{k}, \mathbf{k}'} \mathcal{G}_{\text{Gorkov}}^n(-\tau, -\mathbf{k}) \mathcal{G}_{\text{Gorkov}}^m(0, \mathbf{q}) \right. \\
 & \quad - \delta_{n', m'} \delta_{n, m} \delta_{-\mathbf{q}, \mathbf{k}'} \mathcal{G}_{\text{Gorkov}}^{n'}(\tau' - \tau, -\mathbf{k}') \mathcal{G}_{\text{Gorkov}}^n(-\tau', -\mathbf{k}) \\
 & \quad + \delta_{n, m'} \delta_{n', m} \delta_{\mathbf{q}, \mathbf{k}} \mathcal{F}_n(\tau', \mathbf{k}) \mathcal{F}_{n'}^\dagger(\tau - \tau', \mathbf{k}') \\
 & \quad \left. - \delta_{m, m'} \delta_{n, n'} \delta_{\mathbf{k}, \mathbf{k}'} \mathcal{G}_{\text{Gorkov}}^m(0, \mathbf{q}) (-1) \mathcal{G}_{\text{Gorkov}}^n(-\tau, -\mathbf{k}) \right] \quad (24.92)
 \end{aligned}$$

which can (equivalently) be rewritten as

$$\left[-\partial_\tau - \frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar} \right] \mathcal{G}_{\text{Gorkov}}^n(-\tau, -\mathbf{k}) = +\frac{1}{\hbar} \delta(\tau) + 1 + 2 + 3 + 4 \quad (24.93)$$

Now we will simplify the individual terms as

$$1^{\text{kill } n', m', \mathbf{k}'} = \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ \tau' \in (0, \hbar\beta) \\ v' \in \{1, 2, 3, \dots, r_{d_x}\} \\ m \in \{1, 2, 3, \dots\}}} (-1) \frac{g_{nn'v'}^*(-\mathbf{k}, 0)}{\hbar \sqrt{N_p}} \frac{g_{mm'v'}(\mathbf{q}, 0)}{\hbar \sqrt{N_p}} \mathcal{D}_{0, v'}(\tau - \tau') \mathcal{G}_{\text{Gorkov}}^n(-\tau, -\mathbf{k}) \mathcal{G}_{\text{Gorkov}}^m(0, \mathbf{q}) \quad (24.94)$$

and

$$\begin{aligned}
 2^{\text{kill } m, m', \mathbf{q}} &= \sum_{\substack{\mathbf{k}' \in \mathbf{Q} \\ \tau' \in (0, \hbar\beta) \\ v' \in \{1, 2, 3, \dots, r_{d_x}\} \\ n' \in \{1, 2, 3, \dots\}}} \frac{g_{nn'v'}^*(-\mathbf{k}', \mathbf{k}' - \mathbf{k})}{\hbar \sqrt{N_p}} \frac{g_{nn'v'}(-\mathbf{k}', \mathbf{k}' - \mathbf{k})}{\hbar \sqrt{N_p}} \mathcal{D}_{0, v'}(\tau - \tau') \mathcal{G}_{\text{Gorkov}}^{n'}(\tau' - \tau, -\mathbf{k}') \mathcal{G}_{\text{Gorkov}}^n(-\tau', \mathbf{k}) \\
 &= \sum_{\substack{\mathbf{k}' \in \mathbf{Q} \\ \tau' \in (0, \hbar\beta) \\ v' \in \{1, 2, 3, \dots, r_{d_x}\} \\ n' \in \{1, 2, 3, \dots\}}} \frac{|g_{nn'v'}(-\mathbf{k}', \mathbf{k}' - \mathbf{k})|^2}{\hbar^2 N_p} \mathcal{D}_{0, v'}(\tau - \tau') \mathcal{G}_{\text{Gorkov}}^{n'}(\tau' - \tau, -\mathbf{k}') \mathcal{G}_{\text{Gorkov}}^n(-\tau', \mathbf{k}) \quad (24.95)
 \end{aligned}$$

and

$$\begin{aligned}
 3^{\text{kill } m, m', \mathbf{q}} &= \sum_{\substack{\mathbf{k}' \in \mathbf{Q} \\ \tau' \in (0, \hbar\beta) \\ v' \in \{1, 2, 3, \dots, r_{d_x}\} \\ n' \in \{1, 2, 3, \dots\}}} \frac{g_{nn'v'}^*(-\mathbf{k}', \mathbf{k}' - \mathbf{k})}{\hbar \sqrt{N_p}} \frac{g_{nn'v'}(-\mathbf{k}', \mathbf{k}' - \mathbf{k})}{\hbar \sqrt{N_p}} \mathcal{D}_{0, v'}(\tau - \tau') \mathcal{F}_n(\tau', \mathbf{k}) \mathcal{F}_{n'}(\tau - \tau', \mathbf{k}') \\
 &= \sum_{\substack{\mathbf{k}' \in \mathbf{Q} \\ \tau' \in (0, \hbar\beta) \\ v' \in \{1, 2, 3, \dots, r_{d_x}\} \\ n' \in \{1, 2, 3, \dots\}}} \frac{|g_{nn'v'}(-\mathbf{k}', \mathbf{k}' - \mathbf{k})|^2}{\hbar^2 N_p} \mathcal{D}_{0, v'}(\tau - \tau') \mathcal{F}_n(\tau', \mathbf{k}) \mathcal{F}_{n'}(\tau - \tau', \mathbf{k}') \quad (24.96)
 \end{aligned}$$

and

$$4 \stackrel{\text{kill } n', m', \mathbf{k}'}{=} \sum_{\substack{\mathbf{q} \in Q \\ \tau' \in (0, \hbar\beta) \\ v' \in \{1, 2, 3, \dots, rd_x\} \\ m \in \{1, 2, 3, \dots\}}} \frac{g_{nnv'}^*(-\mathbf{k}, 0)}{\hbar\sqrt{N_p}} \frac{g_{mmv'}(\mathbf{q}, 0)}{\hbar\sqrt{N_p}} \mathcal{D}_{0, v'}(\tau - \tau')(-1) \mathcal{G}_{\text{Gorkov}}^m(0, \mathbf{q}) \mathcal{G}_{\text{Gorkov}}^n(-\tau, -\mathbf{k}) \quad (24.97)$$

Theorem 24.41 For $X = c_{n, \uparrow}^\dagger(\mathbf{k})$ we are able to conclude now (...)

Proof 24.41 Then we find

$$\partial_\tau \mathcal{F}_n^\dagger(\tau, \mathbf{k}) = \frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar} \mathcal{F}_n^\dagger(\tau, \mathbf{k}) + \dots \quad (24.98)$$

with

$$\dots = \sum_{\substack{\mathbf{k}', \mathbf{q} \in Q \\ v' \in \{1, 2, \dots, rd_x\} \\ m, m', n' \in \{1, 2, 3, \dots\} \\ \tau' \in (0, \hbar\beta)}} \frac{g_{nn'v'}^*(-\mathbf{k}', \mathbf{k}' - \mathbf{k})}{\hbar\sqrt{N_p}} (-1) \frac{g_{mm'v'}(\mathbf{q}, \mathbf{k}' - \mathbf{k})}{\hbar\sqrt{N_p}} \mathcal{D}_{0, v'}(\tau - \tau') \sum_{i=1}^4 [i] \quad (24.99)$$

This can (equivalently) be rewritten as

$$\left(\partial_\tau - \frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar} \right) \mathcal{F}_n^\dagger(\tau, \mathbf{k}) = \sum_{i=1}^4 i \quad (24.100)$$

with the individual terms given as

$$\begin{aligned} \sum_{i=1}^4 i = \sum_{\substack{\mathbf{k}', \mathbf{q} \in Q \\ v' \in \{1, 2, \dots, rd_x\} \\ m, m', n' \in \{1, 2, 3, \dots\} \\ \tau' \in (0, \hbar\beta)}} \frac{g_{nn'v'}^*(-\mathbf{k}', \mathbf{k}' - \mathbf{k})}{\hbar\sqrt{N_p}} (-1) \frac{g_{mm'v'}(\mathbf{q}, \mathbf{k}' - \mathbf{k})}{\hbar\sqrt{N_p}} \mathcal{D}_{0, v'}(\tau - \tau') \cdot \left[\delta_{n, n'} \delta_{m, m'} \delta_{\mathbf{k}, \mathbf{k}'} \mathcal{F}_n^\dagger(\tau, \mathbf{k}) (-1) \mathcal{G}_{\text{Gorkov}}^n(0, \mathbf{q}) \right. \\ \left. + \delta_{n', m'} \delta_{n, m} \delta_{\mathbf{k}', -\mathbf{q}} \mathcal{G}_{\text{Gorkov}}^{n'}(\tau' - \tau, -\mathbf{k}^\dagger) \mathcal{F}_n^\dagger(\tau^\dagger, \mathbf{k}^\dagger) \right. \\ \left. + \delta_{n, m'} \delta_{m, n'} \delta_{\mathbf{q}, \mathbf{k}} \mathcal{G}_{\text{Gorkov}}^n(\tau', \mathbf{k}) \mathcal{F}_n^\dagger(\tau^\dagger - \tau, \mathbf{k}^\dagger) \right. \\ \left. + \delta_{m, m'} \delta_{n, n'} \delta_{\mathbf{k}, \mathbf{k}'} \mathcal{G}_{\text{Gorkov}}^m(0, \mathbf{q}) \mathcal{F}_n^\dagger(\tau, \mathbf{k}) \right] \quad (24.101) \end{aligned}$$

therefore we obtain the individual contributions

$$1 \stackrel{\text{kill } n', m', \mathbf{k}'}{=} \sum_{\substack{\mathbf{q} \in Q \\ \tau' \in (0, \hbar\beta) \\ v' \in \{1, 2, 3, \dots, rd_x\} \\ m \in \{1, 2, 3, \dots\}}} \frac{g_{nnv'}^*(-\mathbf{k}, 0) g_{mmv'}(\mathbf{q}, 0)}{\hbar^2 N_p} \mathcal{D}_{0, v'}(\tau - \tau') \mathcal{F}_n^\dagger(\tau, \mathbf{k}) \mathcal{G}_{\text{Gorkov}}^m(0, \mathbf{q}) \quad (24.102)$$

and

$$\begin{aligned}
 2^{\text{kill } m, m', \mathbf{q}} &= \sum_{\substack{\mathbf{k}' \in Q \\ \tau' \in (0, \hbar\beta) \\ v' \in \{1, 2, 3, \dots, rd_x\} \\ n' \in \{1, 2, 3, \dots\}}} \frac{-g_{nn'v'}^*(-\mathbf{k}', \mathbf{k}' - \mathbf{k})g_{nn'v'}(-\mathbf{k}', \mathbf{k}' - \mathbf{k})}{\hbar^2 N_p} \mathcal{D}_{0, v'}(\tau - \tau') \mathcal{G}_{\text{Gorkov}}^{n'}(\tau' - \tau, \mathbf{k}') \mathcal{F}_n^\dagger(\tau^\dagger, \mathbf{k}^\dagger) \\
 &= \sum_{\substack{\mathbf{k}' \in Q \\ \tau' \in (0, \hbar\beta) \\ v' \in \{1, 2, 3, \dots, rd_x\} \\ n' \in \{1, 2, 3, \dots\}}} \frac{-|g_{nn'v'}^*(-\mathbf{k}', \mathbf{k}' - \mathbf{k})|^2}{\hbar^2 N_p} \mathcal{D}_{0, v'}(\tau - \tau') \mathcal{G}_{\text{Gorkov}}^{n'}(\tau' - \tau, \mathbf{k}') \mathcal{F}_n^\dagger(\tau^\dagger, \mathbf{k}^\dagger)
 \end{aligned} \tag{24.103}$$

and

$$3^{\text{kill } m, m', \mathbf{q}} = \sum_{\substack{\mathbf{k}' \in Q \\ \tau' \in (0, \hbar\beta) \\ v' \in \{1, 2, 3, \dots, rd_x\} \\ n' \in \{1, 2, 3, \dots\}}} \frac{-g_{nn'v'}^*(-\mathbf{k}', \mathbf{k}' - \mathbf{k})g_{n'n v'}(\mathbf{k}, \mathbf{k}' - \mathbf{k})}{\hbar^2 N_p} \mathcal{D}_{0, v'}(\tau - \tau') \mathcal{G}_{\text{Gorkov}}^n(\tau', \mathbf{k}) \mathcal{F}_{n'}^\dagger(\tau - \tau^\dagger, \mathbf{k}^\dagger) \tag{24.104}$$

and

$$4^{\text{kill } n', m', \mathbf{k}'} = \sum_{\substack{\mathbf{q} \in Q \\ \tau' \in (0, \hbar\beta) \\ v' \in \{1, 2, 3, \dots, rd_x\} \\ m \in \{1, 2, 3, \dots\}}} \frac{g_{nnv'}^*(-\mathbf{k}, 0)g_{mmv'}(\mathbf{q}, 0)}{\hbar^2 N_p} \mathcal{D}_{0, v'}(\tau - \tau') \mathcal{G}_{\text{Gorkov}}^m(0, \mathbf{q}) \mathcal{F}_n^\dagger(\tau, \mathbf{k}) \tag{24.105}$$

Now we do "the usual thing" and neglect the terms in $g(\bullet, \Gamma)$ and take the Matsubara-transformation of these equations. (To-be-continued, but easy!).

Fact 24.42 The equation of motion for the anomalous Green-function in imaginary time reads

$$\partial_\tau \mathcal{F}_n^\dagger(\tau, \mathbf{k}) = \text{See above} \tag{24.106}$$

Everything from here onwards has to be double-checked. Especially Thm 3.13

24.1.6. Derivation of the Eliashberg-Equations

Notation 24.43 For the self-energy we introduce abbreviations as

$$\Sigma_m(i\omega_n, \mathbf{k}) = i\hbar\omega_n \left[1 - Z_m(i\omega_n, \mathbf{k}) \right] \sigma_0 + \phi_m(i\omega_n, \mathbf{k}) \sigma_1 + \chi_m(i\omega_n, \mathbf{k}) \sigma_3 \stackrel{***}{=} \begin{pmatrix} \Sigma_m(i\omega_n, \mathbf{k}) & \Xi_m(i\omega_n, \mathbf{k}) \\ \Xi_m^*(i\omega_n, \mathbf{k}) & -\Sigma_m(-i\omega_n, -\mathbf{k}) \end{pmatrix} \quad (24.107)$$

whereas $***$ only holds in the case that ϕ is real. (Is this the case???)

Corollary 24.44 Combining the Dyson-equation (24.20) with Eq. (24.107) for the self-energy yields for the full Green-function

$$\mathcal{G}_{\text{Gorkov}}^m(i\omega_n, \mathbf{k}) = \frac{-1}{\theta_m(i\omega_n, \mathbf{k})} \left[\hbar i\omega_n Z_m(i\omega_n, \mathbf{k}) \sigma_0 + \phi_m(i\omega_n, \mathbf{k}) \sigma_1 + (\chi_m(i\omega_n, \mathbf{k}) + (\epsilon_m(\mathbf{k}) - \mu)) \sigma_3 \right] \quad (24.108)$$

with

$$\theta_m(i\omega_n, \mathbf{k}) = \hbar^2 \omega_n^2 Z_m(\mathbf{k}, i\omega_n)^2 + \left[\chi_m(\mathbf{k}, i\omega_n) + (\epsilon_m(\mathbf{k}) - \mu) \right]^2 + \phi_m(i\omega_n, \mathbf{k})^2 \quad (24.109)$$

Proof 24.44 See Appendix, Proofs, page 538.

Theorem 24.45 The self-energies for electron-phonon- and electron-electron-interactions are in first (non-vanishing) order perturbation theory given by

$$\Sigma_{\text{Gorkov}}^n(i\omega_n, \mathbf{k}) = \Sigma_n^{\text{ep}}(i\omega_n, \mathbf{k}) + \Sigma_n^{\text{c}}(i\omega_n, \mathbf{k}) \quad (24.110)$$

with *electron-phonon* and *Coulomb self energies*

$$\Sigma_n^{\text{ep}}(i\omega_n, \mathbf{k}) = -T \sum_{\substack{\mathbf{q} \in Q \\ \omega_{n'} \in \Omega_{-1} \\ v \in \{1, \dots, r d_x\} \\ m \in \{1, \dots, \infty\}}} \sigma_3 \mathcal{G}_{\text{Gorkov}}^m(i\omega_{n'}, \mathbf{q} + \mathbf{k}) \sigma_3 |g_{mnv}(\mathbf{k}, \mathbf{q})|^2 \mathcal{D}_v(i\omega_n - i\omega_{n'}, \mathbf{q}) \quad (24.111)$$

$$\Sigma_n^{\text{c}}(i\omega_n, \mathbf{k}) = -T \sum_{\substack{\mathbf{q} \in Q \\ \omega_{n'} \in \Omega_{-1} \\ m \in \{1, \dots, \infty\}}} \sigma_3 \mathcal{G}_{\text{Gorkov}}^{\text{od}m}(i\omega_{n'}, \mathbf{q}) \sigma_3 V_{n,m}(\mathbf{k}, \mathbf{k} + \mathbf{q}, i\omega_n - i\omega_{n'}) \quad (24.112)$$

Proof 24.45 These equations will be proven in section 24.1.5.

Lemma 24.46 Writing out these equations gives for the electron-phonon self-energy

$$\Sigma_m^{\text{ep}}(i\omega_n, \mathbf{k}) = -T \sum_{\substack{\mathbf{q} \in \mathcal{Q} \\ \omega_{n'} \in \Omega_{-1} \\ v \in \{1, \dots, rd_{\mathbf{x}}\} \\ n \in \{1, \dots, \infty\}}} \begin{pmatrix} \mathcal{G}_{\text{Gorkov}}^n(\bullet) & -\mathcal{F}_n(\bullet) \\ -\mathcal{F}_n^*(\bullet) & -\mathcal{G}_{\text{Gorkov}}^n(-\bullet) \end{pmatrix}_{\bullet=(i\omega_{n'}, \mathbf{q}+\mathbf{k})} |g_{mnv}(\mathbf{k}, \mathbf{q})|^2 \mathcal{D}_v(i\omega_n - i\omega_{n'}, \mathbf{q}) \quad (24.113)$$

Proof 24.46 See Appendix, Proofs, page 538.

Definition 24.47 We introduce the *density of states per spin at the Fermi-level* as

$$N_F = \text{TBD.} \quad (24.114)$$

Definition 24.48 We introduce⁹ the *anisotropic electron-phonon coupling strength*

$$\begin{aligned} \lambda_{\mathbf{k}, \mathbf{k}+\mathbf{q}}^{n,m}(\omega_n) &= \frac{N_F}{\hbar} \oint_{\substack{\omega \in (0, \infty) \\ v \in \{1, \dots, rd_{\mathbf{x}}\}}} \frac{2\omega}{\omega_n^2 + \omega^2} |g_{mnv}(\mathbf{k}, \mathbf{q})|^2 \delta(\omega - \omega_v(\mathbf{q})) \\ &= \frac{N_F}{\hbar} \sum_{v \in \{1, \dots, rd_{\mathbf{x}}\}} \frac{2\omega_v(\mathbf{q})}{\omega_n^2 + \omega_v^2(\mathbf{q})} |g_{mnv}(\mathbf{k}, \mathbf{q})|^2 \\ &= -N_F \sum_{v \in \{1, \dots, rd_{\mathbf{x}}\}} \mathcal{D}_{0,v}(i\omega_n, \mathbf{q}) |g_{mnv}(\mathbf{k}, \mathbf{q})|^2 \end{aligned} \quad (24.115)$$

Remark 24.48.1 Notice that the electron-phonon coupling strength has the unit of an energy.

Corollary 24.49 This definition allows us to rewrite the electron-phonon self-energy as

$$\Sigma_n^{\text{ep}}(i\omega_n, \mathbf{k}) = \frac{T}{\hbar N_F} \sum_{\substack{\mathbf{q} \in \mathcal{Q} \\ \omega_{n'} \in \Omega_{-1} \\ m \in \{1, \dots, \infty\}}} \sigma_3 \mathcal{G}_{\text{Gorkov}}^m(i\omega_{n'}, \mathbf{q} + \mathbf{k}) \sigma_3 \lambda_{\mathbf{k}, \mathbf{k}+\mathbf{q}}^{n,m}(\omega_n - \omega_{n'}) \quad (24.116)$$

This expression has the incorrect unit.

Proof 24.49 Follows from TBD.

⁹Cf. [06 Superhydrides] in "Success of Eliashberg".

Theorem 24.50 Assuming all expressions are exact, let us conclude *the Eliashberg-equations*

$$Z_n(i\omega_n, \mathbf{k}) = 1 + \frac{T}{\omega_n} \sum_{\substack{\mathbf{q} \in Q \\ \omega_{n'} \in \Omega_{-1} \\ m \in \{1, \dots, \infty\}}} \frac{Z_m(i\omega_{n'}, \mathbf{k} + \mathbf{q})}{\theta_m(i\omega_{n'}, \mathbf{k} + \mathbf{q})} \frac{\lambda_{n,m}(i\omega_n - i\omega_{n'}; \mathbf{k}, \mathbf{k} + \mathbf{q})}{N_F} \quad (24.117)$$

$$\chi_n(i\omega_n, \mathbf{k}) = -T \sum_{\substack{\mathbf{q} \in Q \\ \omega_{n'} \in \Omega_{-1} \\ m \in \{1, \dots, \infty\}}} \frac{\epsilon_m(\mathbf{k} + \mathbf{q}) - \mu + \chi_m(i\omega_{n'}, \mathbf{k} + \mathbf{q})}{\theta_m(i\omega_{n'}, \mathbf{k} + \mathbf{q})} \frac{\lambda_{n,m}(i\omega_n - i\omega_{n'}; \mathbf{k}, \mathbf{k} + \mathbf{q})}{N_F} \quad (24.118)$$

$$\phi_n(i\omega_n, \mathbf{k}) = T \sum_{\substack{\mathbf{q} \in Q \\ \omega_{n'} \in \Omega_{-1} \\ m \in \{1, \dots, \infty\}}} \frac{\phi_m(i\omega_{n'}, \mathbf{k} + \mathbf{q})}{\theta_m(i\omega_{n'}, \mathbf{k} + \mathbf{q})} \left[\frac{\lambda_{n,m}(i\omega_n - i\omega_{n'}; \mathbf{k}, \mathbf{k} + \mathbf{q})}{N_F} + V_{n,m}(i\omega_n - i\omega_{n'}; \mathbf{k}, \mathbf{k} + \mathbf{q}) \right] \quad (24.119)$$

Proof 24.50 See Appendix, Proofs, page 538.

Lemma 24.51 Fixing the number of particles n by demanding charge-neutrality yields the additional equation

$$n = 1 - 2T \sum_{\substack{\mathbf{q} \in Q \\ \omega_{n'} \in \Omega_{-1} \\ m \in \{1, \dots, \infty\}}} \frac{\epsilon_m(\mathbf{k} + \mathbf{q}) - \mu + \chi_m(i\omega_{n'}, \mathbf{k} + \mathbf{q})}{\theta_m(i\omega_{n'}, \mathbf{k} + \mathbf{q})} \quad (24.120)$$

Proof 24.51 See Appendix, Proofs, page 539.

From here onwards everything should be fine again.

24.1.7. Simplified Equations

For reasons that will be clear in the subsequent chapters, we consider the following limiting case.

Corollary 24.52 In the limiting case $Z \equiv 1$, $\chi \equiv 0$ we obtain, changing the notation $\phi \equiv \delta$, the system of gap equations

$$\bar{\delta}_n(k') = \sum_{\substack{k \in i\Omega_{-1} \times Q \\ v \in \{1, \dots, rd_x\} \\ m \in \{1, 2, 3, \dots, \infty\}}} \frac{\bar{\delta}_m(k)}{\omega_k^2 + \frac{(\epsilon_m(\mathbf{k}) - \mu)^2}{\hbar^2} + |\delta(k)|^2} \frac{2\omega_v(\mathbf{k} - \mathbf{k}')}{\omega_{k-k'}^2 + \omega_v^2(\mathbf{k} - \mathbf{k}')} \frac{|g_{mnv}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{\hbar^3 \beta N_p} \quad (24.121)$$

Proof 24.52 See Appendix, Proofs, page 539.

Remark 24.52.1 Notice that this can be rewritten, using $\hbar\delta \equiv \Delta$ and $\xi_m(\mathbf{k}) \equiv \epsilon_m(\mathbf{k}) - \mu$ in energy-units as

$$\bar{\Delta}_n(k') = \frac{1}{N_x N_y N_z} \frac{1}{\hbar\beta} \sum_{\substack{k \in i\Omega_{-1} \times Q \\ v \in \{1, \dots, rd_x\} \\ m \in \{1, 2, 3, \dots, \infty\}}} \frac{\bar{\Delta}_m(k)}{\Omega_k^2 + \xi_m^2(\mathbf{k}) + |\Delta_m(k)|^2} \frac{2\Omega_v(\mathbf{k} - \mathbf{k}')}{\Omega_{k-k'}^2 + \Omega_v^2(\mathbf{k} - \mathbf{k}')} |g_{mnv}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2 \quad (24.122)$$

Remark 24.52.2 This is a convolution of discrete functions! Can one now use smart convolution-theorems? ¹⁰

Remark 24.52.3 Notice that we can get an upper bound for gap-function by introducing the maximal value

$$|\delta_{\max}| = \max \{ |\delta_m(k)| \mid k \in \Omega_{-1} \times Q \wedge m \in \{1, 2, 3, \dots, \infty\} \} \quad (24.123)$$

which allows us to estimate

$$\begin{aligned} 0 \leq |\delta_m(k)| &\leq \sum_{\substack{k \in i\Omega_{-1} \times Q \\ v \in \{1, \dots, rd_x\} \\ m \in \{1, 2, 3, \dots, \infty\}}} \frac{|\delta_m(k)|}{\omega_k^2 + \frac{(\epsilon_m(\mathbf{k}) - \mu)^2}{\hbar^2} + |\delta_m(k)|^2} \frac{2\omega_v(\mathbf{k} - \mathbf{k}')}{\omega_{k-k'}^2 + \omega_v^2(\mathbf{k} - \mathbf{k}')} \frac{|g_{mnv}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{\hbar^3 \beta N_p} \\ &\leq |\delta_{\max}| \sum_{\substack{k \in i\Omega_{-1} \times Q \\ v \in \{1, \dots, rd_x\} \\ m \in \{1, 2, 3, \dots, \infty\}}} \frac{1}{\omega_k^2 + \frac{(\epsilon_m(\mathbf{k}) - \mu)^2}{\hbar^2}} \frac{2\omega_v(\mathbf{k} - \mathbf{k}')}{\omega_{k-k'}^2 + \omega_v^2(\mathbf{k} - \mathbf{k}')} \frac{|g_{mnv}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{\hbar^3 \beta N_p} \end{aligned} \quad (24.124)$$

which can be evaluated further with the Residue-theorem **TBDone**.

¹⁰ For example the derivative of the temperature!!!

24.2. Eliashberg-Theory in the Field Integral Formalism

We divide the whole procedure into several steps. All this time we will have a mean-field-theory for the bosonic field in mind (in analogy to Altlands description of superconductivity). The second quantized Hamiltonian of the underlying system is as in section 24.1.4.

24.2.1. The Electron-Electron-Interactions originating from the Phonons

Theorem 24.53 The effective electron-electron interaction can be decomposed into so-called *Spin-Singlet*-contributions („ $\downarrow\uparrow$ “) and two *Spin-Triplet*-contributions („ $\downarrow\downarrow$ “, „ $\uparrow\uparrow$ “) as

$$\begin{aligned}
 S_{\text{el-el-int}}[\psi, \bar{\psi}] &= \sum_{\alpha_1, \dots, \alpha_4 \in \Xi} \bar{\psi}_{\alpha_1} \bar{\psi}_{\alpha_2} V_{\alpha_1, \alpha_2; \alpha_3, \alpha_4} \psi_{\alpha_3} \psi_{\alpha_4} \\
 &= \sum_{\substack{q, k, k' \in \mathbb{Q} \times \Omega_{-1} \\ m_{\alpha_1}, \dots, m_{\alpha_4} \in \{1, 2, 3, \dots\}}} \bar{\psi}_{m_{\alpha_1} \uparrow}(q+k) \bar{\psi}_{m_{\alpha_2} \downarrow}(-k) V_{m_{\alpha_1}, m_{\alpha_2}; m_{\alpha_3}, m_{\alpha_4}}^{\downarrow\uparrow}(q, k, k') \psi_{m_{\alpha_3} \downarrow}(q-k') \psi_{m_{\alpha_4} \uparrow}(k') \\
 &+ \sum_{\substack{q, k, k' \in \mathbb{Q} \times \Omega_{-1} \\ m_{\alpha_1}, \dots, m_{\alpha_4} \in \{1, 2, 3, \dots\}}} \bar{\psi}_{m_{\alpha_1} \uparrow}(q+k) \bar{\psi}_{m_{\alpha_2} \uparrow}(-k) V_{m_{\alpha_1}, m_{\alpha_2}; m_{\alpha_3}, m_{\alpha_4}}^{\uparrow\uparrow}(q, k, k') \psi_{m_{\alpha_3} \uparrow}(q-k') \psi_{m_{\alpha_4} \uparrow}(k') \\
 &+ \sum_{\substack{q, k, k' \in \mathbb{Q} \times \Omega_{-1} \\ m_{\alpha_1}, \dots, m_{\alpha_4} \in \{1, 2, 3, \dots\}}} \bar{\psi}_{m_{\alpha_1} \downarrow}(q+k) \bar{\psi}_{m_{\alpha_2} \downarrow}(-k) V_{m_{\alpha_1}, m_{\alpha_2}; m_{\alpha_3}, m_{\alpha_4}}^{\downarrow\downarrow}(q, k, k') \psi_{m_{\alpha_3} \downarrow}(q-k') \psi_{m_{\alpha_4} \downarrow}(k') \\
 &= S_{\downarrow\uparrow}[\psi, \bar{\psi}] + S_{\uparrow\uparrow}[\psi, \bar{\psi}] + S_{\downarrow\downarrow}[\psi, \bar{\psi}]
 \end{aligned}$$

with spin-triplet vertices as above and spin-singlet matrix-elements (after adding up and collecting the individual contributions) given as

$$\begin{aligned}
 &2V_{m_{\alpha_1}, m_{\alpha_2}; m_{\alpha_3}, m_{\alpha_4}}^{\sigma\sigma}(q, k, k') \\
 &= V_{m_{\alpha_1}, m_{\alpha_2}; m_{\alpha_3}, m_{\alpha_4}}^{\downarrow\uparrow}(q, k, k') \\
 &= \frac{-1}{\hbar\beta N_p} \sum_{v \in \{1, \dots, rd_x\}} \frac{2\omega_v(\mathbf{k} + \mathbf{q} - \mathbf{k}')}{\omega_{k+q-k'}^2 + \omega_v^2(\mathbf{k} + \mathbf{q} - \mathbf{k}')} \frac{g_{m_{\alpha_1} m_{\alpha_4} v}(\mathbf{k}', \mathbf{k} + \mathbf{q} - \mathbf{k}') g_{m_{\alpha_3} m_{\alpha_2} v}^*(-\mathbf{k}, \mathbf{k} + \mathbf{q} - \mathbf{k}')}{\hbar^2} \quad (24.125) \\
 &= \frac{-1}{\hbar\beta N_p} \sum_{v \in \{1, \dots, rd_x\}} \frac{2\omega_v(\mathbf{k} + \mathbf{q} - \mathbf{k}')}{\omega_{k+q-k'}^2 + \omega_v^2(\mathbf{k} + \mathbf{q} - \mathbf{k}')} \frac{g_{m_{\alpha_1} m_{\alpha_4} v}(\mathbf{k}', \mathbf{k} + \mathbf{q} - \mathbf{k}') g_{m_{\alpha_2} m_{\alpha_3} v}(\mathbf{q} - \mathbf{k}', -(\mathbf{k} + \mathbf{q} - \mathbf{k}'))}{\hbar^2}
 \end{aligned}$$

Proof 24.53 See Appendix, Proofs, page 540.

Lemma 24.54 The vertices in Equation (24.125) are lesser than zero iff

- either $\mathbf{k} = -\mathbf{k}'$ AND $m_{\alpha_1} = m_{\alpha_3}$ AND $m_{\alpha_4} = m_{\alpha_2}$ (upper equation - „[N]on-[E]liashberg-[T]heory“ cases),
- or $\mathbf{q} = 0$ AND $m_{\alpha_1} = m_{\alpha_2}$ AND $m_{\alpha_4} = m_{\alpha_3}$ (lower equation - „[E]liashberg-[T]heory“ cases).

Proof 24.54 Referring to Equations (8.41), (8.42) proves the theorem.

Remark 24.54.1 Notice that the BCS-action in Equation (28.4) turns out to be a special case of the spin-singlet contributions in the limit of constant vertices.

Theorem 24.55 We decompose the spin-singlet contributions as

$$S[\psi, \bar{\psi}] \equiv S^{\text{ET}}[\psi, \bar{\psi}] + S^{\text{NET}}[\psi, \bar{\psi}] + S^{\text{RPC}}[\psi, \bar{\psi}] \quad (24.126)$$

whereas for the ([N]on-)[E]liashberg-[T]heory contribution

$$\begin{aligned} S^{\text{ET}}[\psi, \bar{\psi}] = & \sum_{\substack{k, k' \in \Omega_{-1} \times Q \\ m, n \in \{1, 2, 3, \dots\}}} \bar{\psi}_{m\uparrow}(k) \bar{\psi}_{m\downarrow}(-k) V_{\downarrow\uparrow}^{\text{ET}}(k, k') \psi_{n\downarrow}(-k') \psi_{n\uparrow}(k') \\ & + \sum_{\substack{\sigma \in \{\downarrow, \uparrow\} \\ k, k' \in \Omega_{-1} \times Q \\ m, n \in \{1, 2, 3, \dots\}}} \bar{\psi}_{m\sigma}(k) \bar{\psi}_{m\sigma}(-k) V_{\sigma\sigma}^{\text{ET}}(k, k') \psi_{n\sigma}(-k') \psi_{n\sigma}(k') \end{aligned}$$

whereas the first term produces the off-diagonal contributions to the Gorkov-Green-functions.

$$\begin{aligned} S_{\downarrow\uparrow}^{\text{NET}}[\psi, \bar{\psi}] = & \sum_{\substack{q, k \in i\Omega_{-1} \times Q \\ m, n \in \{1, 2, 3, \dots\}}} \bar{\psi}_{m\uparrow}(q) \psi_{m\downarrow}(q) V_{\downarrow\uparrow}^{\text{NET}}(q + k, -k) \psi_{n\uparrow}(k) \bar{\psi}_{n\downarrow}(k) \\ & + \sum_{\substack{\sigma \in \{\downarrow, \uparrow\} \\ q, k \in i\Omega_{-1} \times Q \\ m, n \in \{1, 2, 3, \dots\}}} \bar{\psi}_{m\sigma}(q) \psi_{m\sigma}(q) V_{\sigma\sigma}^{\text{NET}}(q - k, k) \psi_{n\sigma}(-k) \bar{\psi}_{n\sigma}(-k) \end{aligned}$$

whereas the terms in the second line **should be able to** produce the diagonal contributions to the Gorkov-Green-functions. The remaining terms are the [R]andom [P]hase [C]ancellations. (We assume them to be random phases that don't contribute to the physical observables.) Then we have, when we take the double contributions into account, the vertices

$$2V_{\sigma\sigma}^{\text{ET}}(k, k') = V_{\downarrow\uparrow}^{\text{ET}}(k, k') = \frac{-2}{\hbar\beta N_p} \sum_{v \in \{1, \dots, rd_x\}} \frac{\omega_v(\mathbf{k} - \mathbf{k}')}{\omega_{k-k'}^2 + \omega_v^2(\mathbf{k} - \mathbf{k}')} \frac{|g_{mnv}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{\hbar^2} \quad (24.127)$$

$$2V_{\sigma\sigma}^{\text{NET}}(q + k, -k) = V_{\downarrow\uparrow}^{\text{NET}}(q + k, -k) = \frac{-2}{\hbar\beta N_p} \sum_{v \in \{1, \dots, rd_x\}} \frac{\omega_v(-\mathbf{k} + \mathbf{q})}{\omega_{-k+q}^2 + \omega_v^2(-\mathbf{k} + \mathbf{q})} \frac{|g_{mnv}(\mathbf{k}, -\mathbf{k} + \mathbf{q})|^2}{\hbar^2} \quad (24.128)$$

Proof 24.55 See Appendix, Proofs, page 543.

24.2.2. Electron-Electron-Interactions for Spin-Singlet Eliashberg-Theory

In order to prove the validity of our approach, we derive the [E]liashberg-[T]heory-equations from a limiting case of our theory.

Approximation 24.56 We approximate the interactions by focusing on the ET- $\downarrow\uparrow$ -contributions and neglecting the other terms as

$$S_{\text{el-el-int}}[\psi, \bar{\psi}] = S_{\downarrow\uparrow}^{\text{ET}}[\psi, \bar{\psi}] + S_{\Delta V}[\psi, \bar{\psi}] \approx S_{\downarrow\uparrow}^{\text{ET}}[\psi, \bar{\psi}]$$

with the Eliashberg contribution

$$S_{\downarrow\uparrow}^{\text{ET}}[\psi, \bar{\psi}] = \sum_{\substack{k, k' \in i\Omega_{-1} \times Q \\ m, n \in \{1, 2, 3, \dots\}}} \bar{\psi}_{m\uparrow}(k) \bar{\psi}_{m\downarrow}(-k) V_{\downarrow\uparrow}^{\text{ET}}(k, k') \psi_{n\downarrow}(-k') \psi_{n\uparrow}(k')$$

and the vertex

$$V_{\downarrow\uparrow}^{\text{ET}}(k, k') = \frac{-2}{\hbar \beta N_p} \sum_{v \in \{1, \dots, rd\}} \frac{\omega_v(\mathbf{k} - \mathbf{k}')}{\omega_{k-k'}^2 + \omega_v^2(\mathbf{k} - \mathbf{k}')} \frac{|g_{mv}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{\hbar^2} \quad (24.129)$$

Notation 24.57 We condense the notation even further, by introducing the multi-index

$$A \equiv (k, m, \downarrow, \uparrow) \in \Pi, \quad B \equiv (k', n, \downarrow, \uparrow) \in \Pi \quad \text{with} \quad \Pi \equiv i\Omega_{-1} \times Q \times \{1, 2, 3, \dots\} \times \{\downarrow\} \times \{\uparrow\} \quad (24.130)$$

and density-like terms

$$\bar{\rho}(A) \equiv \bar{\psi}_{m\uparrow}(k) \bar{\psi}_{m\downarrow}(-k), \quad \rho(B) \equiv \psi_{n\downarrow}(-k') \psi_{n\uparrow}(k') \quad (24.131)$$

with $\rho(A) = \psi_{m\downarrow}(-k) \psi_{m\uparrow}(k)$.

Remark 24.57.1 Notice that with this, we can rewrite the Eliashberg-interaction term as a bilinear

$$S_{\downarrow\uparrow}^{\text{ET}}[\psi, \bar{\psi}] = \sum_{A, B \in \Pi} \bar{\rho}(A) V_{\downarrow\uparrow}^{\text{ET}}(A, B) \rho(B) \quad (24.132)$$

24.2.3. Hubbard-Stratanovich-Transformation

We now exchange the quartic interaction for a quadratic-part through the Cooper-channel. We therefore have to find the inverse of $V_{\downarrow\uparrow}^{\text{ET}}$.

Hypothesis 24.58 We assume, the existence of another generalised matrix $\left(V_{\downarrow\uparrow}^{\text{ET}}\right)^{-1} \equiv \Lambda_{\downarrow\uparrow}^{\text{ET}}$ such that the inverse relation holds

$$\left(V_{\downarrow\uparrow}^{\text{ET}} \cdot \Lambda_{\downarrow\uparrow}^{\text{ET}}\right)(A, A') = \sum_B V_{\downarrow\uparrow}^{\text{ET}}(A, B) \Lambda_{\downarrow\uparrow}^{\text{ET}}(B, A') = \delta(A, A'). \quad (24.133)$$

Remark 24.58.1 For the moment, we will not bother with the concrete shape of this inverse matrix. We simply assume its existence and give it a name to perform further mathematical operations.

Lemma 24.59 Performing a Hubbard-Stratanovich-transformation of the Eliashberg-interaction term through the Cooper-channel yields

$$\begin{aligned} \exp -S_{\downarrow\uparrow}^{\text{ET}}[\psi, \bar{\psi}] &= \exp - \sum_{A, B \in \Pi^2} \bar{\rho}(A) V_{\downarrow\uparrow}^{\text{ET}}(A, B) \rho(B) \\ &= \int \mathcal{D}(\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}) \exp + \left[\sum_{A, B \in \Pi^2} \bar{\delta}(A) \Lambda_{\downarrow\uparrow}^{\text{ET}}(A, B) \delta(B) - \sum_{A \in \Pi} \left(\bar{\delta}(A) \rho(A) + \bar{\rho}(A) \delta(A) \right) \right] \\ &= \int \mathcal{D}(\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}) \exp - \left[S_{\text{aux.}}[\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}] + S_{\text{int}}[\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}, \psi, \bar{\psi}] \right]. \end{aligned}$$

Proof 24.59 Shifting variables according to

$$\delta(A) \mapsto \delta(A) + \sum_{B \in \Pi} V_{\downarrow\uparrow}^{\text{ET}}(A, B) \rho(B), \quad \bar{\delta}(A) \mapsto \bar{\delta}(A) + \sum_{B \in \Pi} \bar{\rho}(B) V_{\downarrow\uparrow}^{\text{ET}}(B, A) \quad (24.134)$$

proves the Lemma.

Fact 24.60 The full action, in absense of external sources, now reads

$$S_{\text{tot}}[\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}, \psi, \bar{\psi}] = S_{\text{el}}[\psi, \bar{\psi}] + S_{\text{aux}}[\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}] + S_{\text{int}}[\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}, \psi, \bar{\psi}] \quad (24.135)$$

with

- $S_{\text{el}}[\psi, \bar{\psi}] \equiv \sum_{\substack{\sigma \in \{\downarrow, \uparrow\} \\ k \in i\Omega_{-1} \times Q \\ m \in \{1, 2, 3, \dots\}}} \bar{\psi}_{m\sigma}(k) \left(-\frac{1}{\hbar} \mathcal{G}_0^{-\text{F}}(k) \right)_{m, \sigma} \psi_{m\sigma}(k)$
- $S_{\text{aux}}[\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}] = - \sum_{A, B \in \Pi^2} \bar{\delta}(A) \Lambda_{\downarrow\uparrow}^{\text{ET}}(A, B) \delta(B)$

$$\bullet S_{\text{int}}[\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}, \psi, \bar{\psi}] = \sum_{\substack{k \in i\Omega_{-1} \times Q \\ m \in \{1,2,3,\dots\}}} \bar{\delta}_{m\downarrow\uparrow}(k) \psi_{m\downarrow}(-k) \psi_{m\uparrow}(k) + \bar{\psi}_{m\uparrow}(k) \bar{\psi}_{m\downarrow}(-k) \delta_{m\downarrow\uparrow}(k)$$

Definition 24.61 We now introduce Nambu-spinors as

$$\eta_m^\dagger(k) = \begin{pmatrix} \bar{\psi}_{m,\uparrow}(k) & \psi_{m,\downarrow}(-k) \end{pmatrix}, \quad \eta_{m'}(k') = \begin{pmatrix} \psi_{m'\uparrow}(k') \\ \bar{\psi}_{m'\downarrow}(-k') \end{pmatrix} \quad (24.136)$$

Remark 24.61.1 Notice that this allows us to rewrite the electronic contributions as

$$\begin{aligned} S_{\text{el}}[\psi, \bar{\psi}] + S_{\text{int}}[\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}, \psi, \bar{\psi}] &= \sum_{\substack{k \in i\Omega_{-1} \times Q \\ m \in \{1,2,\dots\}}} \begin{pmatrix} \bar{\psi}_{m,\uparrow}(k) \\ \psi_{m,\downarrow}(-k) \end{pmatrix}^T \begin{pmatrix} -\frac{1}{\hbar} \mathcal{G}_0^{-F}(k) & \delta_{m\downarrow\uparrow}(k) \\ \bar{\delta}_{m\downarrow\uparrow}(k) & +\frac{1}{\hbar} \mathcal{G}_0^{-F}(-k) \end{pmatrix} \begin{pmatrix} \psi_{m\uparrow}(k) \\ \bar{\psi}_{m\downarrow}(-k) \end{pmatrix} \\ &= \sum_{\substack{k \in i\Omega_{-1} \times Q \\ m \in \{1,2,\dots\}}} \bar{\eta}_m^T(k) \begin{bmatrix} -\frac{1}{\hbar} \mathcal{G}_m^{-1} & \\ & \text{Gorkov} \end{bmatrix}_{[\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}]}(k) \eta_m(k) \end{aligned}$$

Lemma 24.62 One finds for the imaginary time components of the Nambu-spinors the identity in imaginary time space

$$\bar{\eta}_m^T(\tau, \mathbf{k}) = \begin{pmatrix} \bar{\psi}_{m\uparrow}(\tau, \mathbf{k}) & \psi_{m\downarrow}(\tau, -\mathbf{k}) \end{pmatrix} \quad \eta_m(\tau, \mathbf{k}) = \begin{pmatrix} \psi_{m\uparrow}(\tau, \mathbf{k}) \\ \bar{\psi}_{m\downarrow}(\tau, -\mathbf{k}) \end{pmatrix} \quad (24.137)$$

Proof 24.62 See Appendix, Proofs, page 544.

Corollary 24.63 We note the correlators

$$\langle \eta_{m1}(\tau, \mathbf{k}) \bar{\eta}_{m,1}(\tau', \mathbf{k}) \rangle = \langle \psi_{m\uparrow}(\tau, \mathbf{k}) \bar{\psi}_{m,\uparrow}(\tau', \mathbf{k}) \rangle = -\hbar \mathcal{G}_{\text{Gorkov},11}^{m, [\delta_{\uparrow\uparrow}, \bar{\delta}_{\uparrow\uparrow}]}(\tau - \tau', \mathbf{k}) = -\hbar \mathcal{G}_{\text{Gorkov}}^{m\uparrow}(\tau - \tau', \mathbf{k}) \quad (24.138)$$

$$\langle \eta_{m1}(\tau, \mathbf{k}) \bar{\eta}_{m,2}(\tau', \mathbf{k}) \rangle = \langle \psi_{m\uparrow}(\tau, \mathbf{k}) \psi_{m,\uparrow}(\tau', -\mathbf{k}) \rangle = -\hbar \mathcal{G}_{\text{Gorkov},12}^{m, [\delta_{\uparrow\uparrow}, \bar{\delta}_{\uparrow\uparrow}]}(\tau - \tau', \mathbf{k}) = -\hbar \mathcal{F}_m(\tau - \tau', \mathbf{k}) \quad (24.139)$$

$$\langle \eta_{m2}(\tau, \mathbf{k}) \bar{\eta}_{m,1}(\tau', \mathbf{k}) \rangle = \langle \bar{\psi}_{m\downarrow}(\tau, -\mathbf{k}) \bar{\psi}_{m,\uparrow}(\tau', \mathbf{k}) \rangle = -\hbar \mathcal{G}_{\text{Gorkov},21}^{m, [\delta_{\uparrow\uparrow}, \bar{\delta}_{\uparrow\uparrow}]}(\tau - \tau', \mathbf{k}) = -\hbar \mathcal{F}_m^\dagger(\tau - \tau', \mathbf{k}) \quad (24.140)$$

$$\langle \eta_{m2}(\tau, \mathbf{k}) \bar{\eta}_{m,2}(\tau', \mathbf{k}) \rangle = \langle \bar{\psi}_{m\downarrow}(\tau, -\mathbf{k}) \psi_{m,\downarrow}(\tau', -\mathbf{k}) \rangle = -\hbar \mathcal{G}_{\text{Gorkov},22}^{m, [\delta_{\uparrow\uparrow}, \bar{\delta}_{\uparrow\uparrow}]}(\tau - \tau', \mathbf{k}) = -\hbar \left(-\mathcal{G}_{\text{Gorkov}}^{m\downarrow}(\tau' - \tau, -\mathbf{k}) \right) \quad (24.141)$$

Proof 24.63 See Appendix, Proofs, page 544.

Lemma 24.64 Integrating out the fermions gives us an effective bosonic theory

$$S_{\text{tot}}[\delta_{\uparrow\uparrow}, \bar{\delta}_{\uparrow\uparrow}] = S_{\text{aux.}}[\delta_{\uparrow\uparrow}, \bar{\delta}_{\uparrow\uparrow}] - \text{Tr Log} - \frac{1}{\hbar} \mathcal{G}_{\text{Gorkov}}^{-1}[\delta_{\uparrow\uparrow}, \bar{\delta}_{\uparrow\uparrow}] \quad (24.142)$$

Proof 24.64 By using standard-identities about Gauss-Berezin-integrals and using the standard-identity $\text{Tr Log} \bullet = \text{Log Det} \bullet$ gives us the desired identity.

24.2.4. Spontaneous Symmetry Breaking and Gap-Equations

We now break the U(1) symmetry by applying a saddle-point approximation and keeping only one contribution, analogous to section 28.2.5.

Lemma 24.65 We note the identity

$$\frac{\partial}{\partial \delta(\bullet)} \text{Tr Log} - \frac{1}{\hbar} \mathcal{G}_{\text{Gorkov}}^{-1}[\delta_{\uparrow\uparrow}, \bar{\delta}_{\uparrow\uparrow}] = + \text{Tr} \mathcal{G}_{\text{Gorkov}}[\delta_{\uparrow\uparrow}, \bar{\delta}_{\uparrow\uparrow}] \frac{\partial}{\partial \delta(\bullet)} \mathcal{G}_{\text{Gorkov}}^{-1}[\delta_{\uparrow\uparrow}, \bar{\delta}_{\uparrow\uparrow}] \quad (24.143)$$

Proof 24.65 With Altlands prescription: We define $f(x) = \text{Log} \left(-\frac{1}{\hbar} x \right)$, such that we obtain $f'(x) = \frac{1}{x}$. This yields for a x -dependent matrix $A \equiv A(x)$ the identity

$$\partial_x \text{Tr} f(A) = \text{Tr} (A^{-1} \partial_x A). \quad (24.144)$$

Corollary 24.66 The stationary points of the action are determined by the set of coupled equations

$$\bar{\delta}_{n\downarrow\uparrow}(k') = + \sum_{\substack{k \in i\Omega_{-1} \times Q \\ v \in \{1, \dots, rd\} \\ m \in \{1, 2, 3, \dots\}}} \text{Tr}_{2 \times 2} \left(\mathcal{G}_{\substack{m \\ [\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}} \frac{\partial}{\partial \delta_{m\downarrow\uparrow}(k)} \mathcal{G}_{\substack{m \\ [\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}}^{-1} \right) \frac{2\omega_v(\mathbf{k} - \mathbf{k}')}{\omega_{k-k'}^2 + \omega_v^2(\mathbf{k} - \mathbf{k}')} \frac{|g_{mnv}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{\hbar^3 \beta N_p}$$

Proof 24.66 See Appendix, Proofs, page 545.

Lemma 24.67 One has the identities

$$\frac{\partial}{\partial \delta_{m\downarrow\uparrow}(k)} \mathcal{G}_{\substack{m \\ [\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}}^{-1} = \begin{pmatrix} 0 & -\hbar \\ 0 & 0 \end{pmatrix} \quad (24.145)$$

and

$$\mathcal{G}_{\substack{m \\ [\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}}(k) = -\frac{1}{\hbar} \frac{\begin{pmatrix} +i\omega_k + (\epsilon_m(-\mathbf{k}) - \mu)/\hbar & +\delta_{m\downarrow\uparrow}(k) \\ +\delta_{m\downarrow\uparrow}(k) & +i\omega_k - (\epsilon_m(\mathbf{k}) - \mu)/\hbar \end{pmatrix}}{+\omega_k^2 + \frac{(\epsilon_m(\mathbf{k}) - \mu)^2}{\hbar^2} + |\delta_{m\downarrow\uparrow}(k)|^2} \quad (24.146)$$

Proof 24.67 See Appendix, Proofs, page 545.

Notice that we are now in the position to prove

Lemma 24.68 Remark ?? is true, i.e. it holds that

$$\mathcal{G}_{\substack{m, 12 \\ [\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}}(A, A')^\dagger = \mathcal{G}_{\substack{m, 21 \\ [\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}}(A, A') \quad (24.147)$$

Proof 24.68 See Appendix, Proofs, page 546.

Theorem 24.69 The gap equations become

$$\bar{\delta}_{n\downarrow\uparrow}(k') = + \sum_{\substack{k \in i\Omega_{-1} \times Q \\ v \in \{1, \dots, rd\} \\ m \in \{1, 2, 3, \dots\}}} \frac{\bar{\delta}_{m\downarrow\uparrow}(k)}{\omega_k^2 + \frac{(\epsilon_m(\mathbf{k}) - \mu)^2}{\hbar^2} + |\delta_{m\downarrow\uparrow}(k)|^2} \frac{2\omega_v(\mathbf{k} - \mathbf{k}')}{\omega_{k-k'}^2 + \omega_v^2(\mathbf{k} - \mathbf{k}')} \frac{|g_{mnv}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{\hbar^3 \beta N_p}.$$

Proof 24.69 See Appendix, Proofs, page 547.

Remark 24.69.1 This result coincides perfectly with the limiting equations from the EPW-code in Equation (24.121).

25. Material Analysis with Quantum Espresso & Wannier90 & Wannier-Tools & EPW & GRIT

We consider only stoichiometric crystals, i.e. crystals with a fixed number of atoms per unit cell and a fixed number of valence electrons per atom.

Introduction

Fact 25.1 For fixed (averaged) particle-number and volume we describe a crystal as a system of interacting phonons (massless particles) and electrons (massive particles). We work in the grand canonical ensemble with natural variables depending on the nature of the subsystem of interest, meaning that ...

- *for the phononic part of the system*, we are working in the grand canonical ensemble with natural variables (T, V) ,
- *for the electronic part of the system*, we are working in the grand canonical ensemble with natural variables (μ, T, V) .

Fact 25.2 Notice first that ...

- *for spin-unpolarised systems* (i.e. no spin-dependent exchange-interaction or spin-orbit-coupling) one finds the inequality

$$\text{"number of at least partially occupied bands"} \times 2 \geq \text{"number of electrons per unit cell"}.$$

whereas spin-degenerate bands are counted once.

- In the more general notation (i.e. counting two degenerate bands as band nr. 1 and band nr. 2) or in the case of *spin-polarised systems*, one finds the inequality

$$\text{"number of at least partially occupied bands"} \geq \text{"number of electrons per unit cell"}.$$

Fact 25.3 One has to count the number of expected electrons $\langle N \rangle$. For a crystal containing r atoms in the basis and $Z_1, Z_2, \dots, Z_r$ valence electrons per atom, the number of expected electrons is given by

$$\begin{aligned} \langle N \rangle &= \text{"number of cells"} \quad \times \quad \text{"the total number of valence electrons per unit-cell"} \\ &= N_x N_y N_z \quad \cdot \quad (Z_1 + Z_2 + \dots + Z_r) \\ &\equiv N_x N_y N_z \quad \cdot \quad n_{\text{UC}} \end{aligned} \tag{25.1}$$

25.1. Materials for a Specific Heat Analysis

25.1.1. Rb Crystal - Spacegroup Im3m (Rubidium)

Quantum Espresso Inputs

Crystal Structure from CIF-File

We have the following crystal-structure and atomic basis vectors from the cif-file:

```

1 &SYSTEM
2   A          = 5.6568708402
3   (...)
4   /
5
6 CELL_PARAMETERS {alat}
7   -0.5000000000000000  0.5000000000000000  0.5000000000000000
8   0.5000000000000000 -0.5000000000000000  0.5000000000000000
9   0.5000000000000000  0.5000000000000000 -0.5000000000000000
10
11 ATOMIC_POSITIONS {crystal}
12 Rb  0.0000000000000000  0.0000000000000000  0.0000000000000000

```

Computational Input-Parameters

Excerpt from the python-code in which all the parameters are defined:

```

1 TBD

```

Step 0: vc-relax Calculation

We obtain the following final coordinates.

```

1 Begin final coordinates
2   new unit-cell volume = 611.01184 a.u.^3 ( 90.54261 Ang^3 )
3   density = 1.56747 g/cm^3
4
5 CELL_PARAMETERS (alat= 10.55412041)
6   -0.506494229  0.506494229  0.506494229
7   0.506494229 -0.506494229  0.506494229
8   0.506494229  0.506494229 -0.506494229
9
10 ATOMIC_POSITIONS (crystal)
11 Rb  0.0000000000  0.0000000000  0.0000000000
12 End final coordinates

```

such that the new input-file for the subsequent calculations reads

```

1 &SYSTEM
2   A          = 5.6575405376
3   (...)

```

```
4 /
5
6 CELL_PARAMETERS {alat}
7 -0.5000000000000000 0.5000000000000000 0.5000000000000000
8 0.5000000000000000 -0.5000000000000000 0.5000000000000000
9 0.5000000000000000 0.5000000000000000 -0.5000000000000000
10
11 ATOMIC_POSITIONS {crystal}
12 Rb 0.0000000000000000 0.0000000000000000 0.0000000000000000
```

Step 1: SCF, Band-Structure and (Partial) Density of States

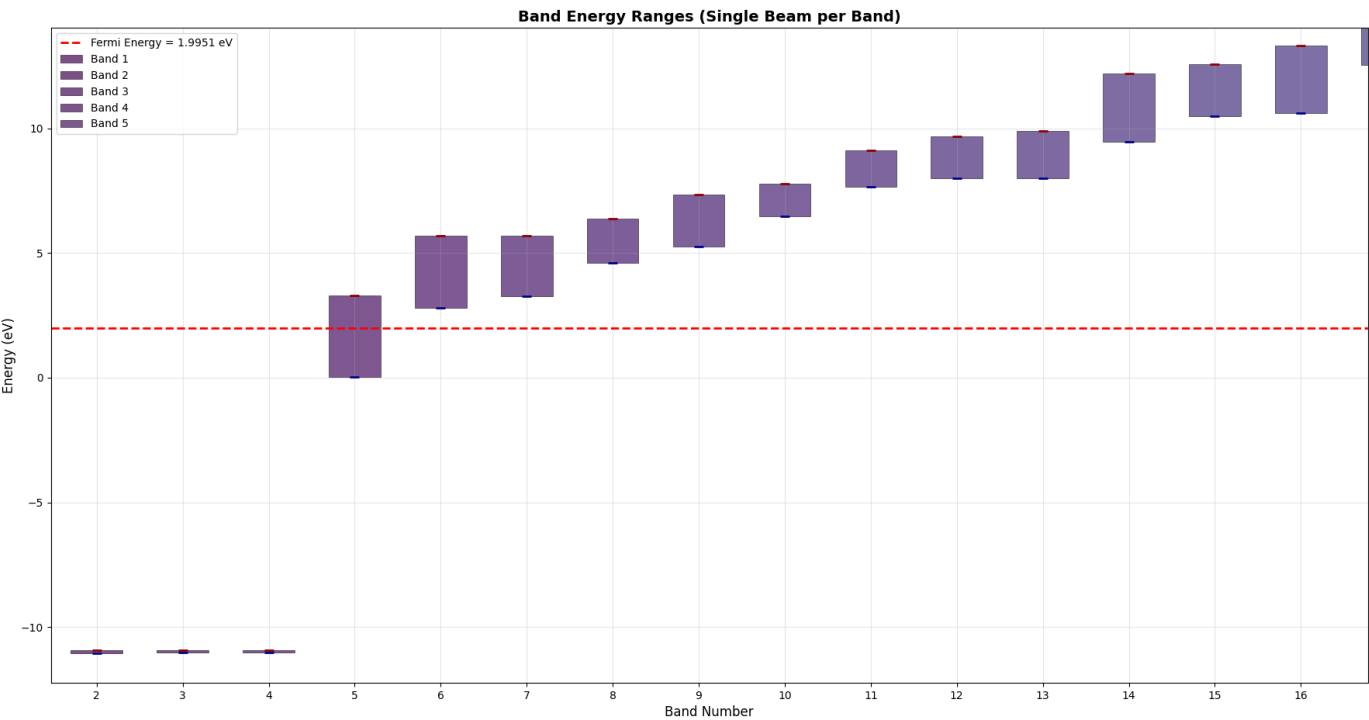


Figure 25.1.: Band-ranges and Band-indices.

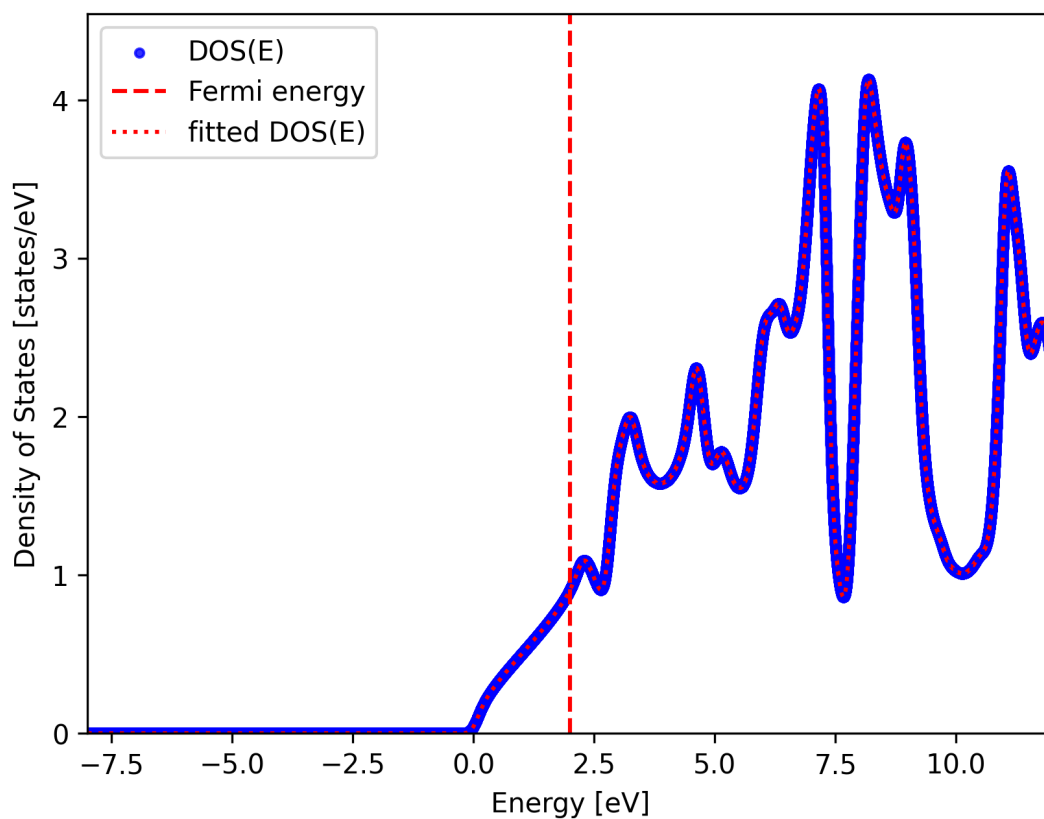


Figure 25.2.: Density of States in the LDA-Approximation.

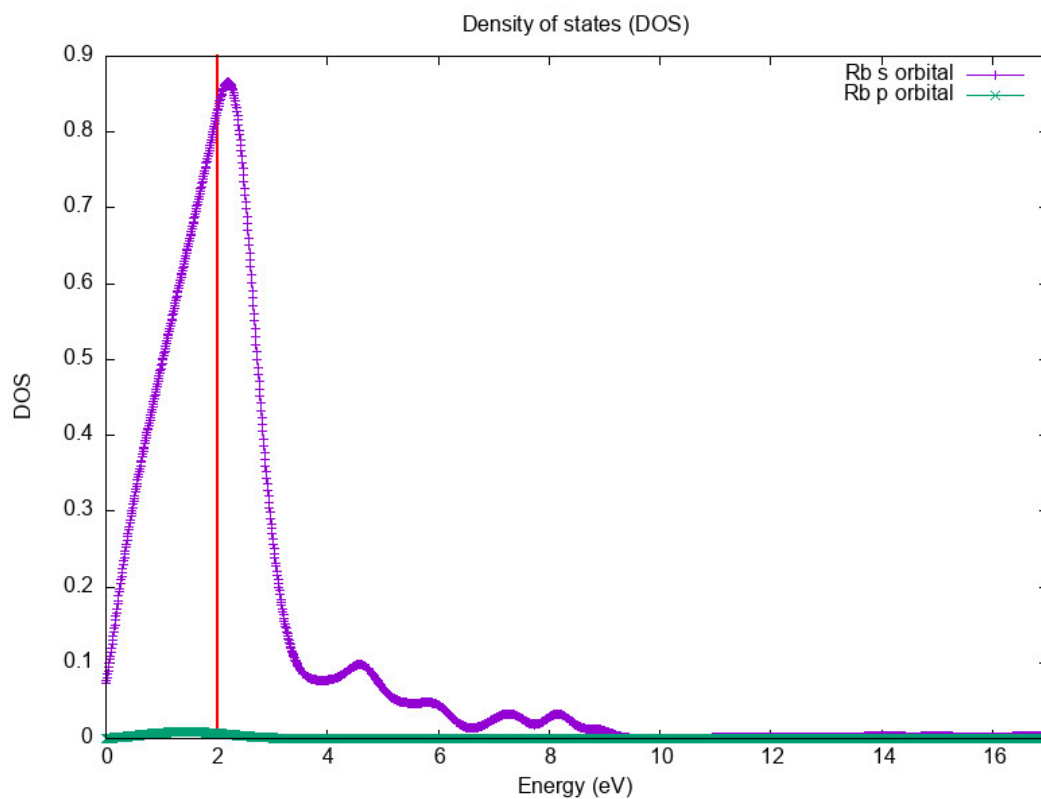


Figure 25.3.: Partial Density of States in the LDA-Approximation.

Notice that the projections give zero overlap for some highest energy, and that we don't see any d-states which is a consequence from the fact that we are using a pseudopotential.

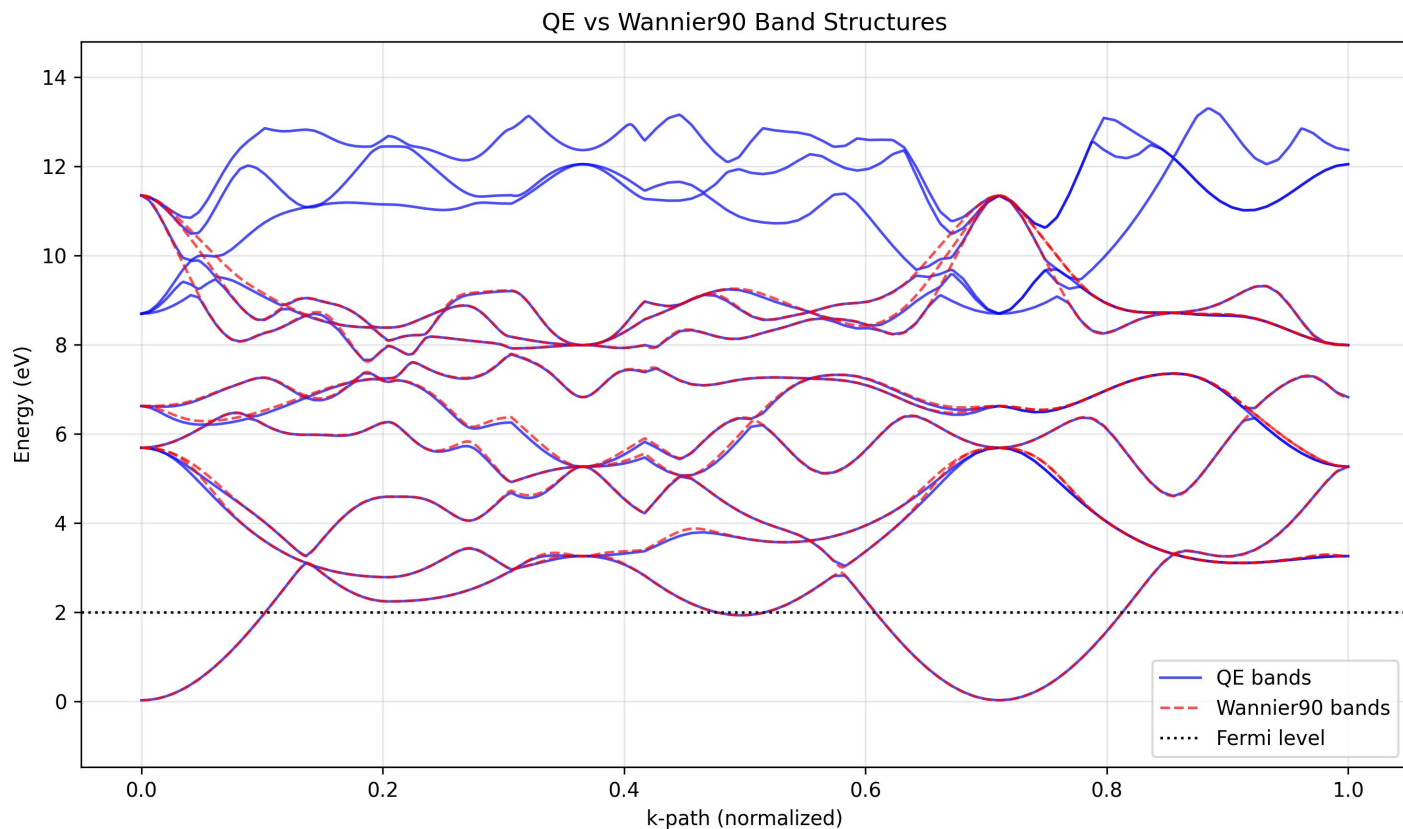


Figure 25.4.: Quantum Espresso vs. Wannier90 Bands.

TBContinued.

25.1.2.	Au	Crystal - Spacegroup Fm3m	(Gold)
25.1.3.	Ag	Crystal - Spacegroup Fm3m	(Silver)
25.1.4.	K	Crystal - Spacegroup Im3m	(Potassium)
25.1.5.	Cs	Crystal - Spacegroup Im3m	(Caesium)
25.1.6.	Cu	Crystal - Spacegroup Fm3m	(Copper)
25.1.7.	Si	Crystal - Spacegroup Fd3m	(Silicon)
25.1.8.	C	Crystal - Spacegroup Fd3m	(Diamond)
25.1.9.	KCl	Crystal - Spacegroup Fm3m	(Potassium Chloride)

25.2. Materials for a Weyl-Point Analysis

25.2.1. TaAs Crystal - Spacegroup I41md (Tantalum Arsenide)

Quantum Espresso Inputs

Crystal Structure from CIF-File

We have the following crystal-structure and atomic basis vectors from the cif-file:

```

1 &SYSTEM
2   A           = 3.43700
3   (...)
4   /
5
6 CELL_PARAMETERS {alat}
7   1.0000000000000000  0.0000000000000000  0.0000000000000000
8   0.0000000000000000  1.0000000000000000  0.0000000000000000
9   0.5000000000000000  0.5000000000000000  1.695664823974397
10
11 ATOMIC_SPECIES
12   As  74.92100  As.upf
13   Ta  180.94700  Ta.upf
14
15 ATOMIC_POSITIONS {crystal}
16 Ta  0.0000000000000000  0.0000000000000000  0.0000000000000000
17 Ta  0.7500000000000000  0.2500000000000000  0.5000000000000000
18 As  0.5830000000000000  0.5830000000000000  0.8340000000000000
19 As  0.3330000000000000  0.8330000000000000  0.3340000000000000

```

Computational Input-Parameters

Excerpt from the python-code in which all the parameters are defined:

```

1 TBD

```

Step 0: vc-relax Calculation

We didn't do a relaxation for this material, as the automation is not yet implemented.

Step 1: SCF, Band-Structure and (Partial) Density of States

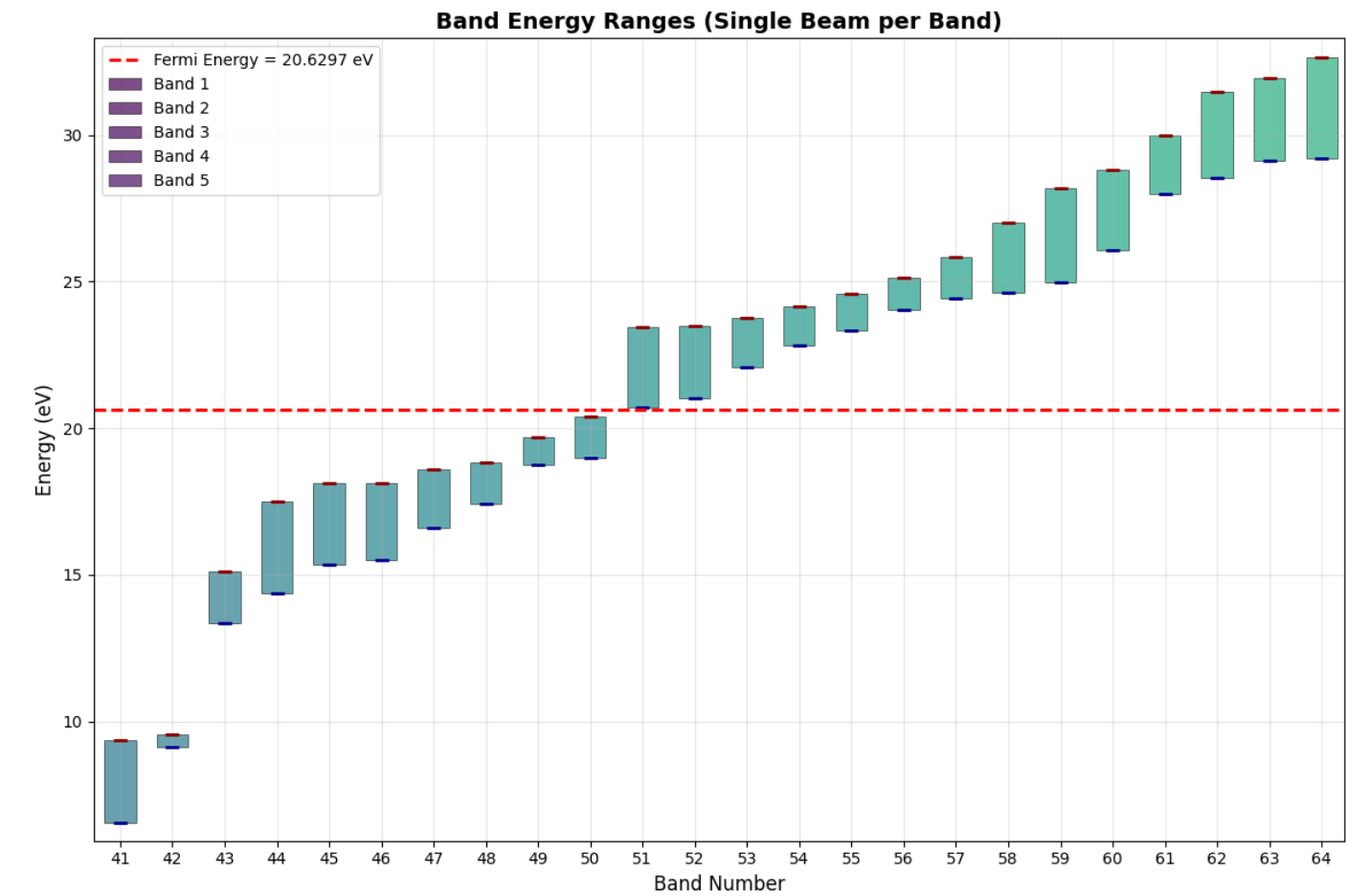


Figure 25.5.: Band-ranges and Band-indices.

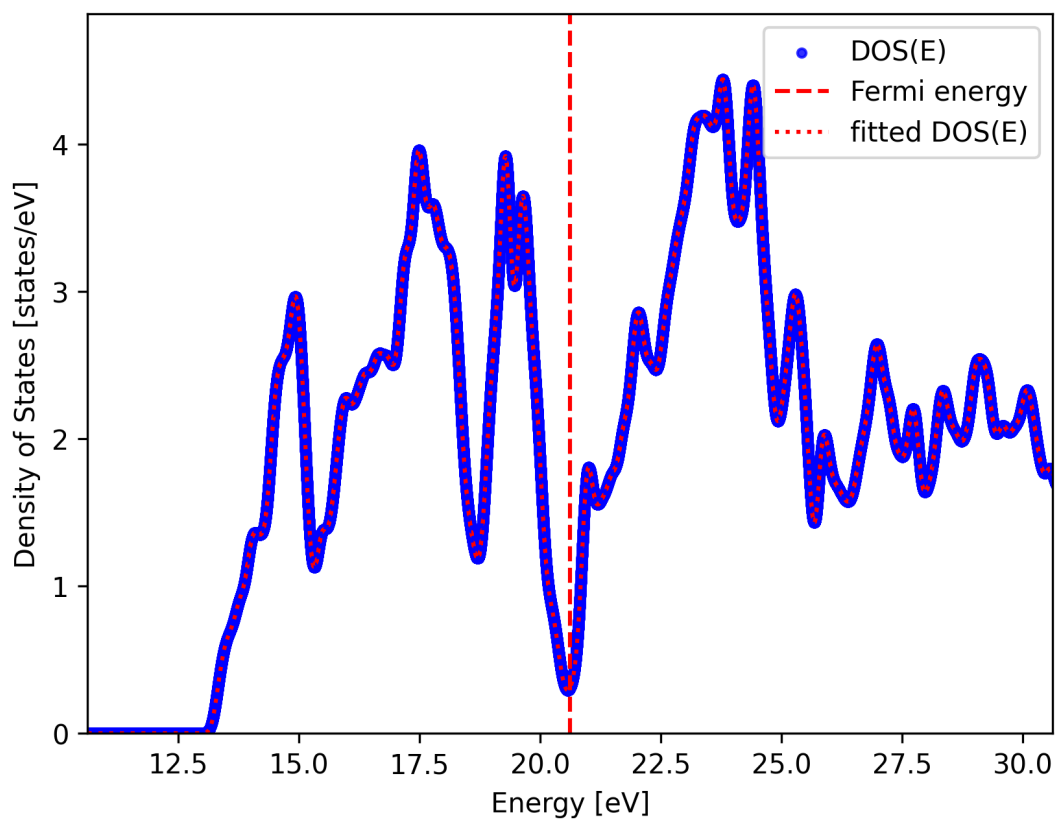


Figure 25.6.: Density of States in the LDA-Approximation.

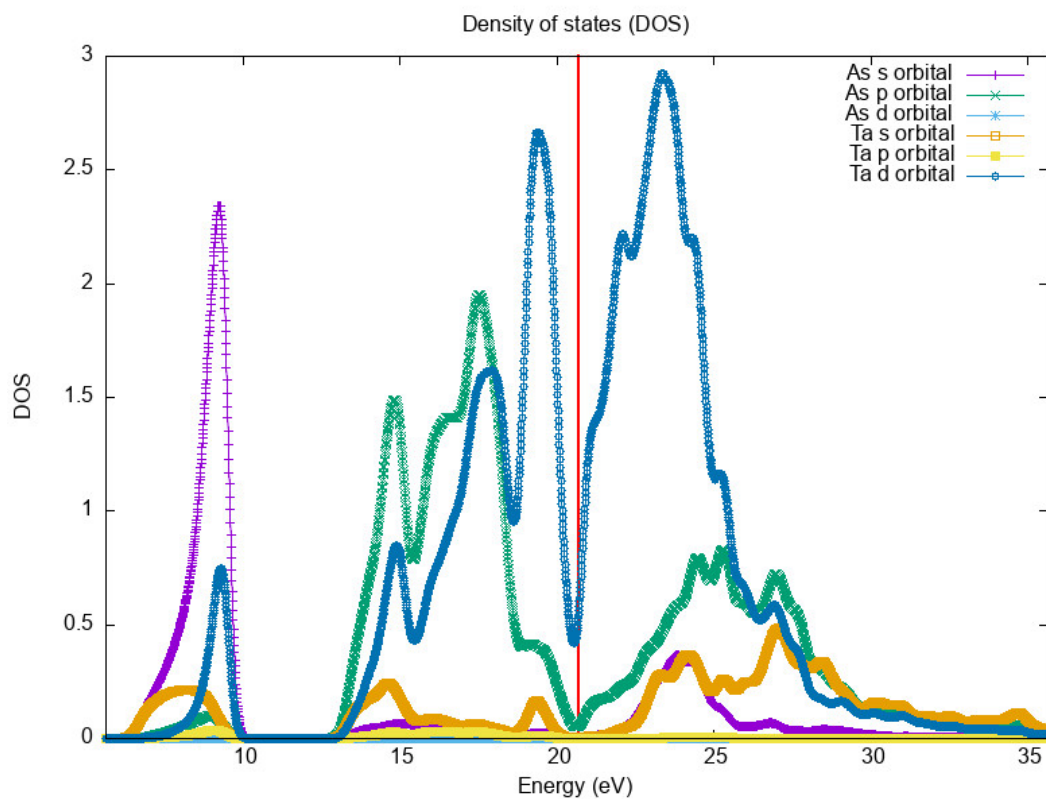


Figure 25.7.: Partial Density of States in the LDA-Approximation.

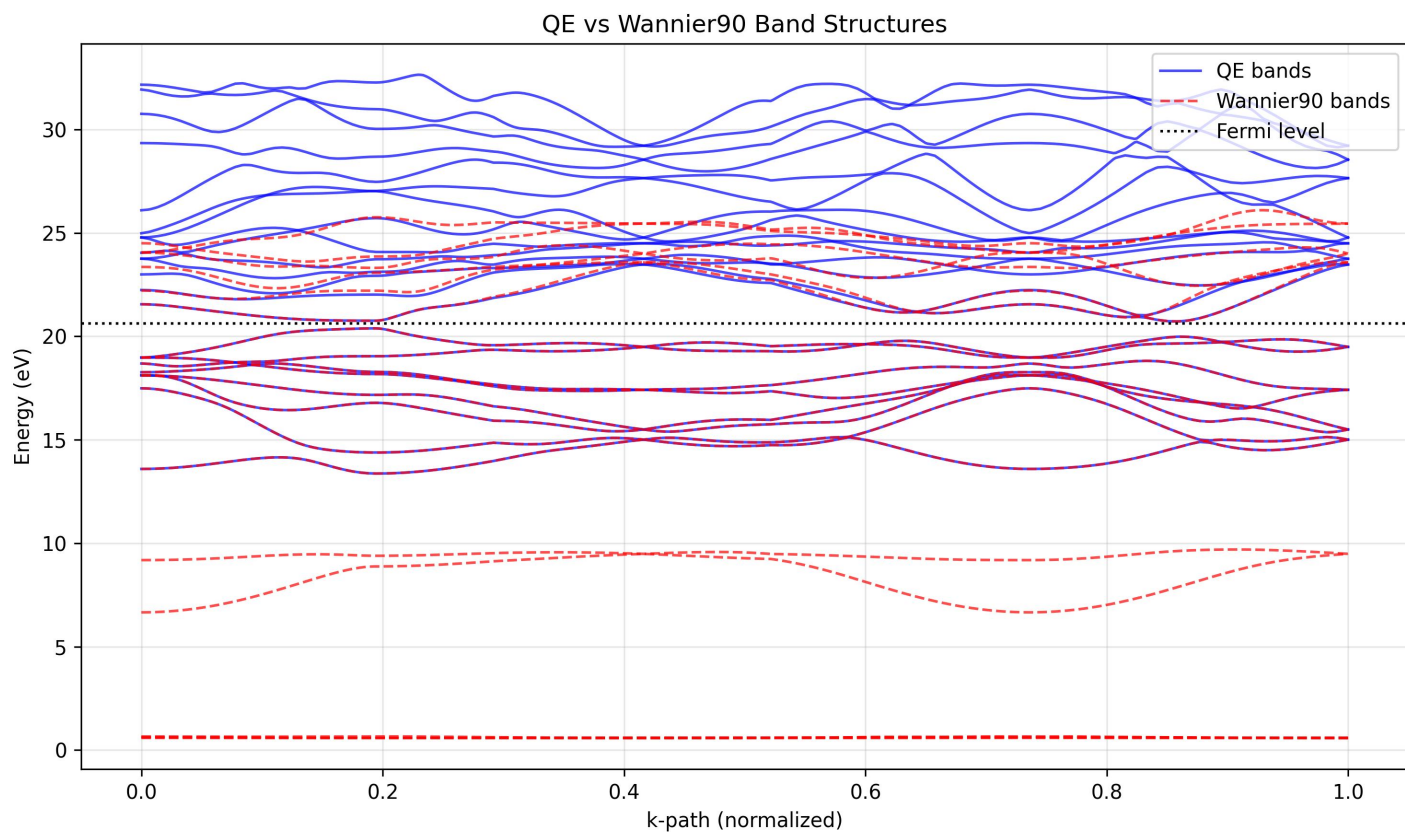


Figure 25.8.: Quantum Espresso vs. Wannier90 Bands.

25.2.2. MnTe Crystal - Spacegroup P63mmc (Manganese Telluride)

Quantum Espresso Inputs

Crystal Structure from CIF-File

We have the following crystal-structure and atomic basis vectors from the cif-file:

```

1 &SYSTEM
2   A           = 4.15020
3   (...)
4   /
5
6 CELL_PARAMETERS {alat}
7   0.866025403784439 -0.5000000000000000 0.0000000000000000
8   0.0000000000000000 1.0000000000000000 0.0000000000000000
9   0.0000000000000000 0.0000000000000000 1.619030408173100
10
11 ATOMIC_SPECIES
12   Te 127.60000 Te.upf
13   Mn 54.93800 Mn.upf
14
15   ATOMIC_POSITIONS {crystal}
16 Te 0.3333333333333333 0.6666666666666667 0.2500000000000000
17 Te 0.6666666666666667 0.3333333333333333 0.7500000000000000
18 Mn 0.0000000000000000 0.0000000000000000 0.0000000000000000
19 Mn 0.0000000000000000 0.0000000000000000 0.5000000000000000

```

Computational Input-Parameters

Excerpt from the python-code in which all the parameters are defined:

```

1 TBD

```

Step 0: vc-relax Calculation

We didn't do a relaxation for this material, as the automation is not yet implemented.

Step 1: SCF, Band-Structure and (Partial) Density of States

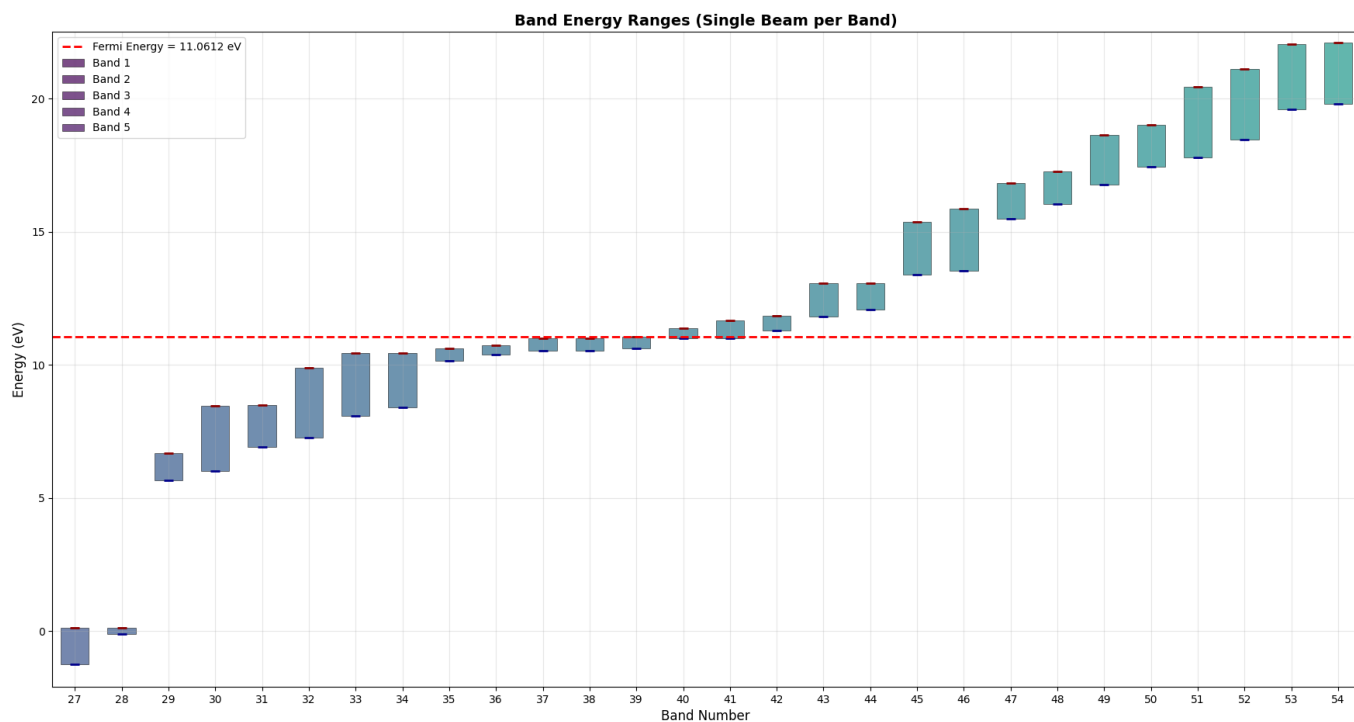


Figure 25.9.: Band-ranges and Band-indices.

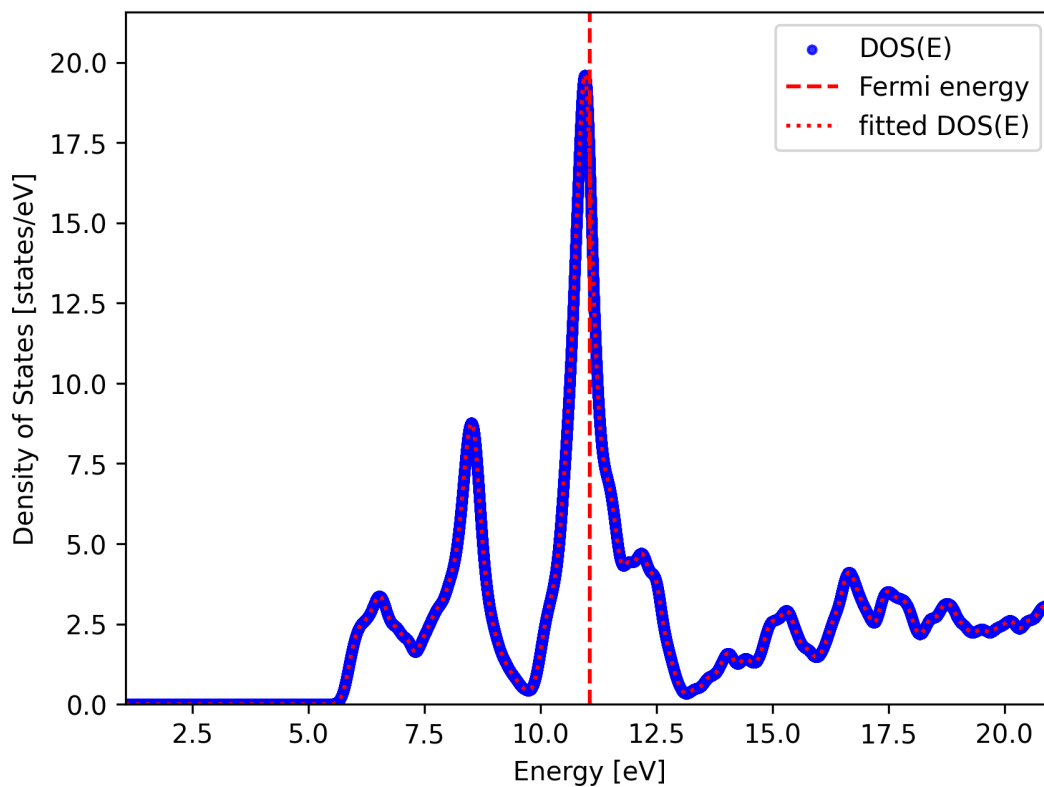


Figure 25.10.: Density of States in the LDA-Approximation.

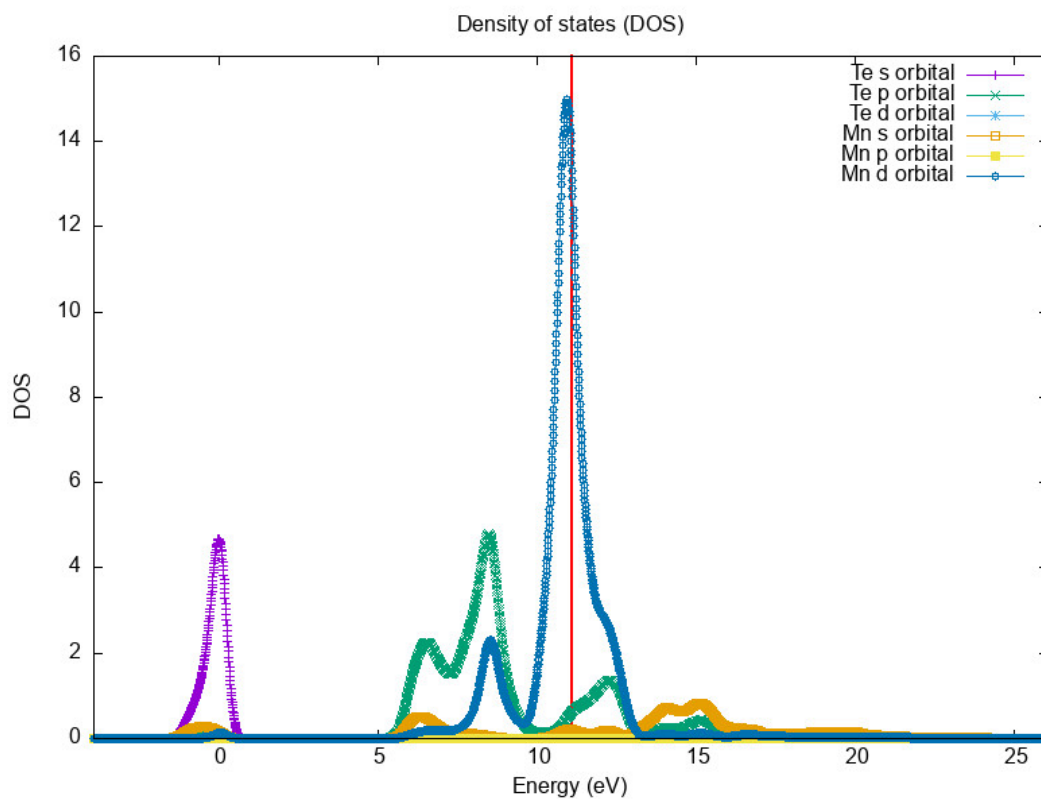


Figure 25.11.: Partial Density of States in the LDA-Approximation.

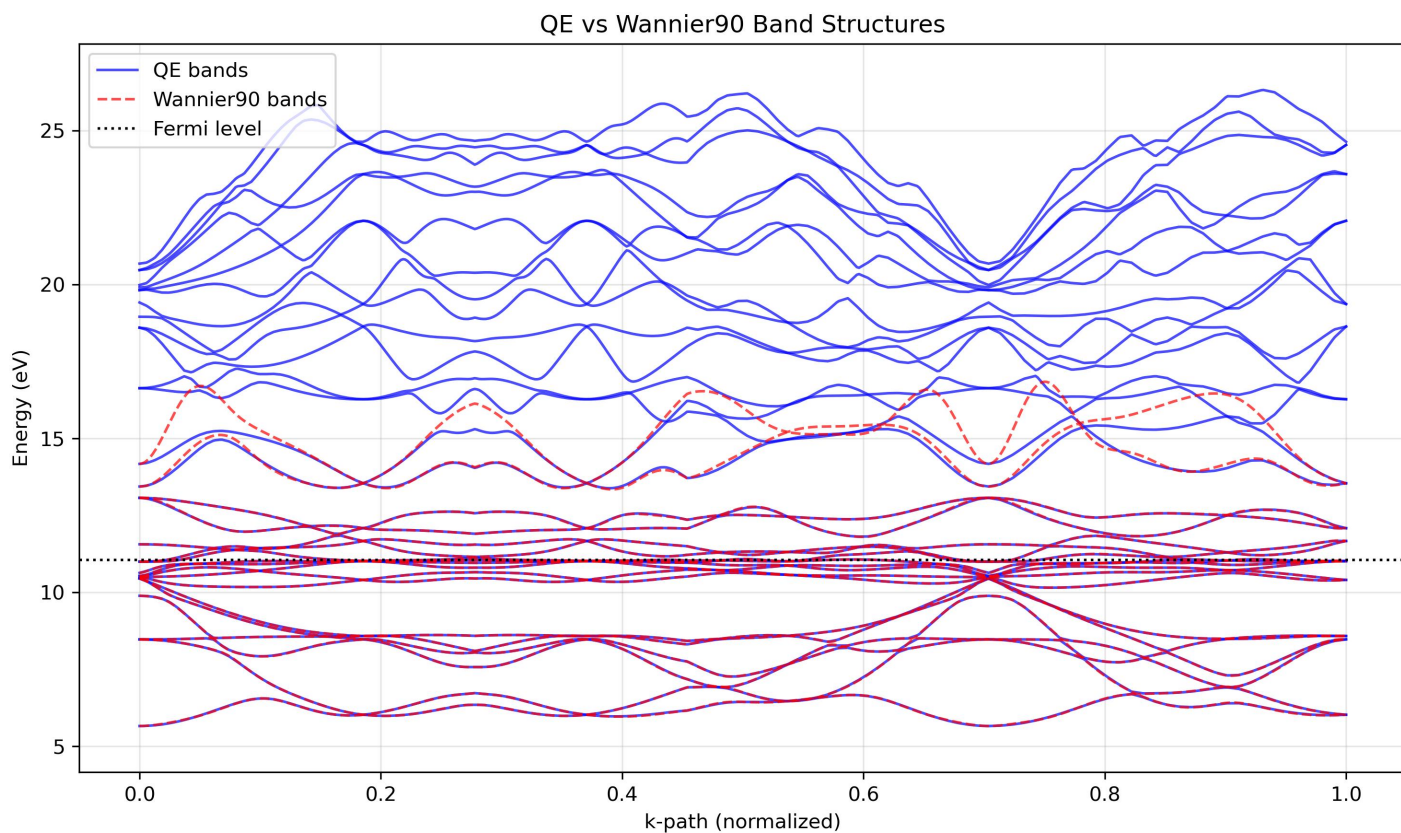


Figure 25.12.: Quantum Espresso vs. Wannier90 Bands.

25.2.3. WTe2 Crystal - Spacegroup ??? (Tungsten Ditelluride)

Quantum Espresso Inputs

Crystal Structure

We have the following crystal-structure and atomic basis vectors from the quantum-espresso input-file:

```

1 &SYSTEM
2   A           = 3.47700
3   (...)
4   /
5
6 CELL_PARAMETERS {alat}
7   1.0000000000000000  0.0000000000000000  0.0000000000000000
8   0.0000000000000000  1.797238999137187  0.0000000000000000
9   0.0000000000000000  0.0000000000000000  4.031636468219730
10
11 ATOMIC_SPECIES
12   W   183.84000   W.upf
13   Te  127.60000   Te.upf
14
15 ATOMIC_POSITIONS {crystal}
16   W   0.0000000000000000  0.6006200000000000  0.7645250000000000
17   W   0.0000000000000000  0.0398000000000000  0.2797450000000000
18   W   0.5000000000000000  0.3993800000000000  0.2645250000000000
19   W   0.5000000000000000  0.9602000000000000  0.7797450000000000
20   Te  0.0000000000000000  0.8576100000000000  0.9197750000000000
21   Te  0.5000000000000000  0.1423900000000000  0.4197750000000000
22   Te  0.0000000000000000  0.6463100000000000  0.3756450000000000
23   Te  0.5000000000000000  0.3536900000000000  0.8756450000000000
24   Te  0.0000000000000000  0.2984500000000000  0.1243550000000000
25   Te  0.5000000000000000  0.7015500000000000  0.6243550000000000
26   Te  0.0000000000000000  0.2072200000000000  0.6683950000000000
27   Te  0.5000000000000000  0.7927800000000000  0.1683950000000000

```

Computational Input-Parameters

Excerpt from the python-code in which all the parameters are defined:

```

1 TBD

```

Step 0: vc-relax Calculation

We didn't do a relaxation for this material, as the automation is not yet implemented.

Step 1: SCF, Band-Structure and (Partial) Density of States

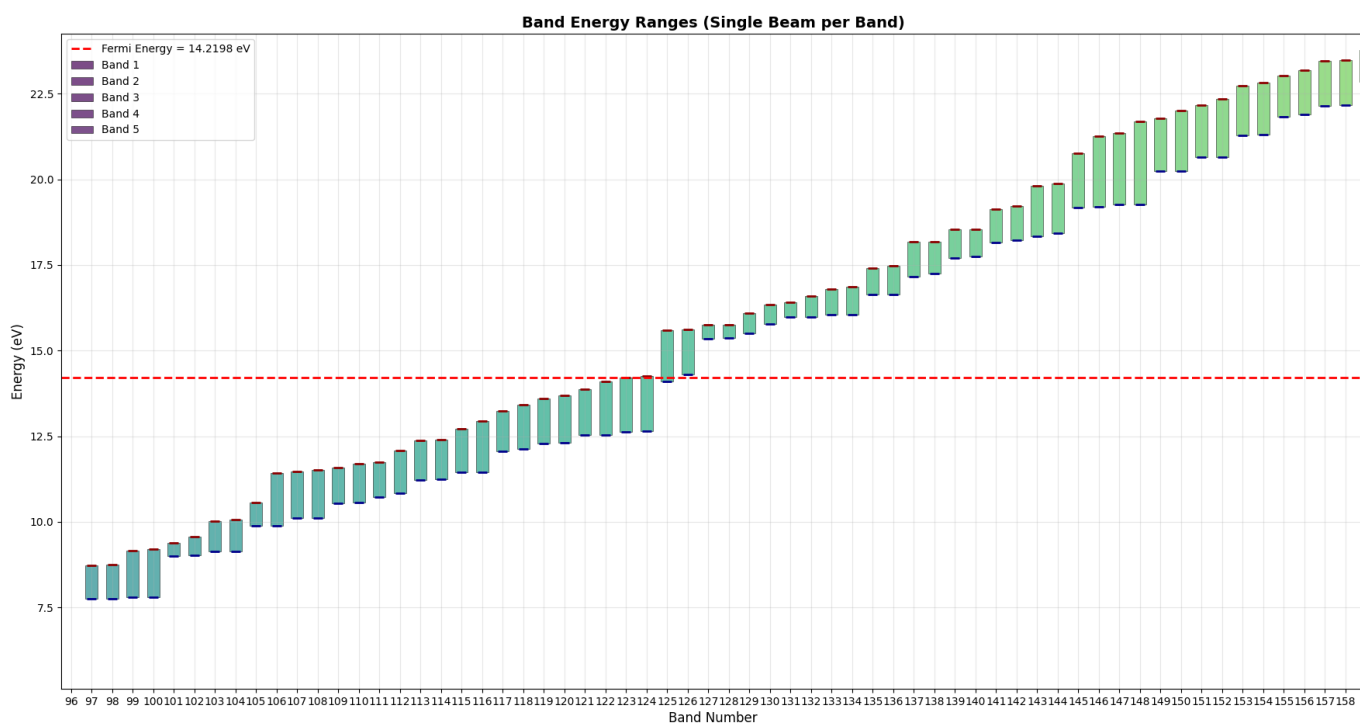


Figure 25.13.: Band-ranges and Band-indices.

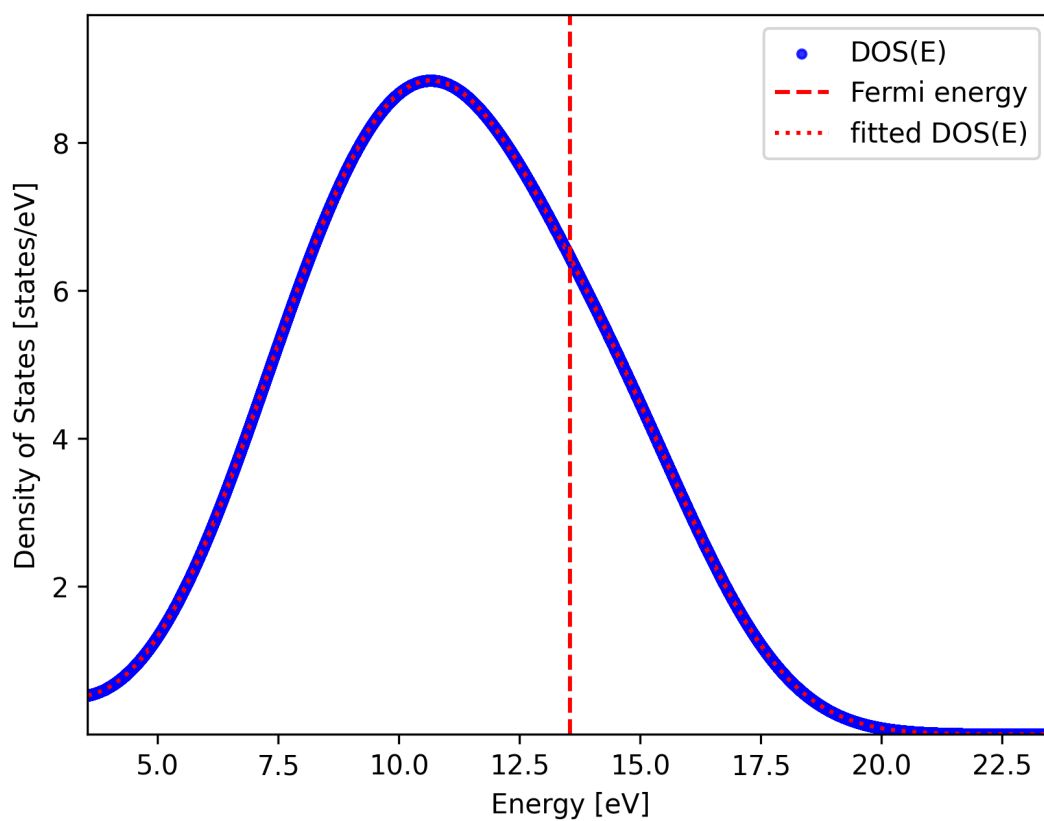


Figure 25.14.: Density of States in the LDA-Approximation.

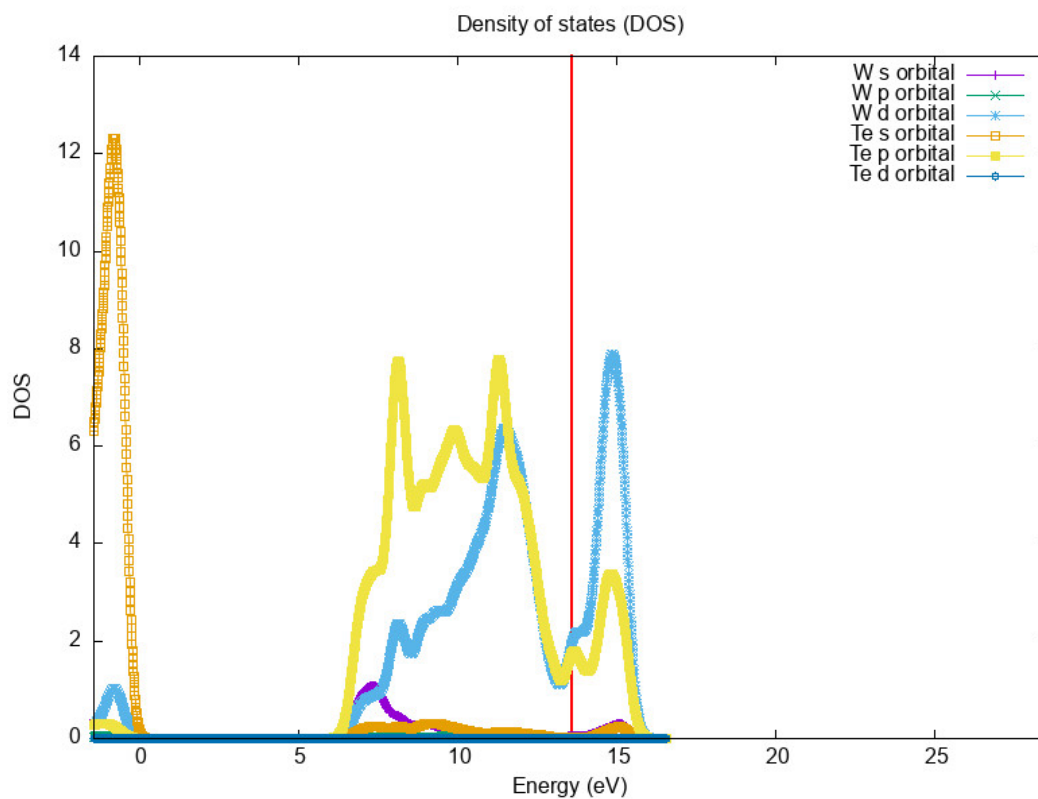


Figure 25.15.: Partial Density of States in the LDA-Approximation.

Notice that the projections give zero overlap for some highest energy which is a consequence from the fact that we are using a pseudopotential.

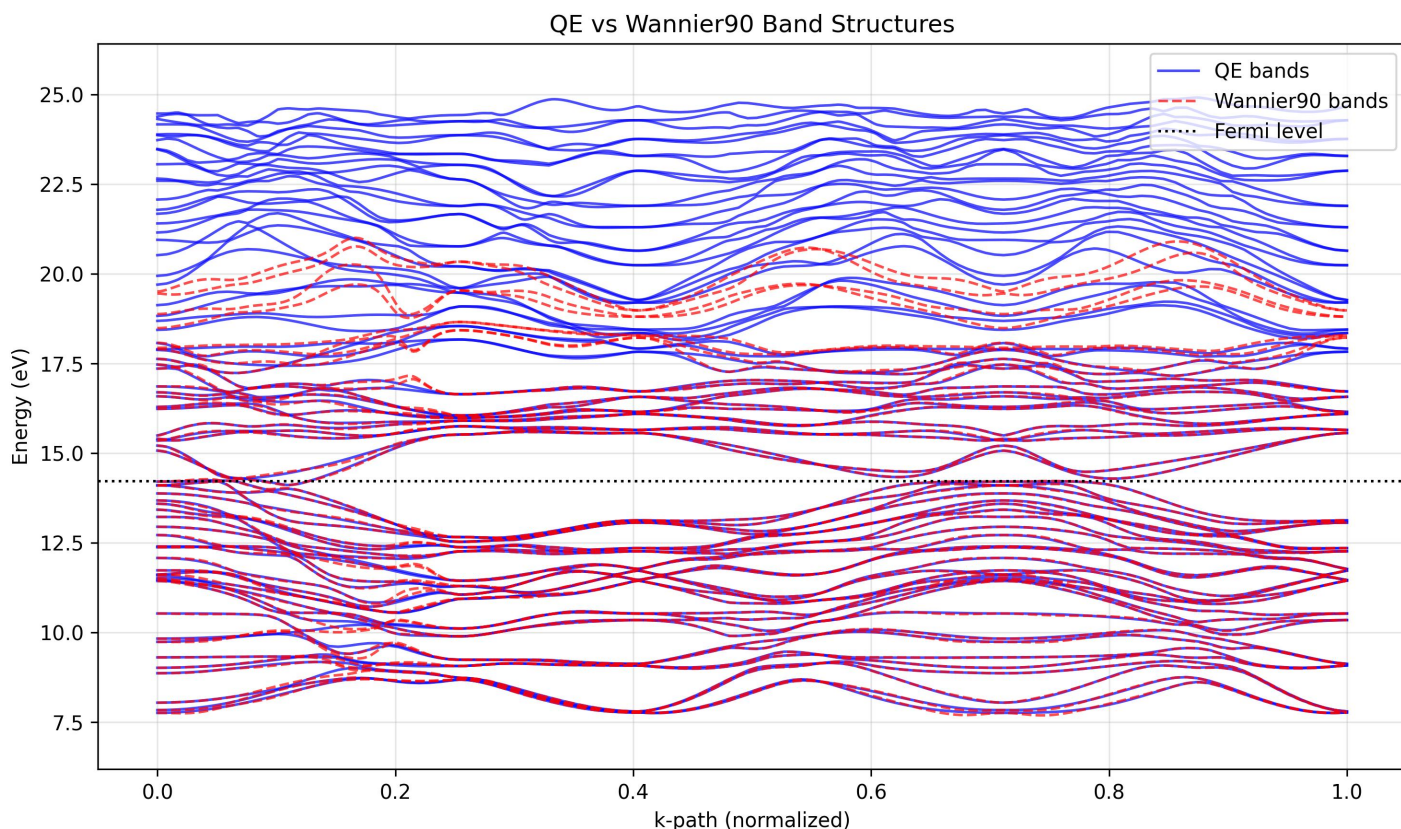


Figure 25.16.: Quantum Espresso vs. Wannier90 Bands.

25.3. Materials for a Superconductivity Analysis

25.3.1.	TaC	Crystal - Spacegroup Fm3m	(Tantalum Carbide)
25.3.2.	CaC6	Crystal - Spacegroup R-3m	(Calcium Graphite Intercalation)
25.3.3.	Nb	Crystal - Spacegroup Im3m	(Niobium)
25.3.4.	V3	Crystal - Spacegroup ???	(Vanadium Phase)
25.3.5.	Nb3Al	Crystal - Spacegroup Pm3n	(Niobium Aluminide)
25.3.6.	Nb3Sn	Crystal - Spacegroup Pm3n	(Niobium-Tin)
25.3.7.	Nb3Ga	Crystal - Spacegroup Pm3n	(Niobium Gallide)
25.3.8.	Nb3Ge	Crystal - Spacegroup Pm3n	(Niobium Germanide)
25.3.9.	MgB2	Crystal - Spacegroup P6/mmm	(Magnesium Diboride)
25.3.10.	H2S	Crystal - Spacegroup ???	(Hydrogen Sulfide)
25.3.11.	YH6	Crystal - Spacegroup ???	(Yttrium Hexahydride)
25.3.12.	LaH10	Crystal - Spacegroup ???	(Lanthanum Decahydride)

Part V.

Appendices

26. Appendix - Derivation of Self-Energies and Polarizations

26.1. Self Energy of a Diagonalized Gas with Single-Particle-Interactions

Proof 26.1 of Theorem 19.1

Computing the electronic Self-Energy

Preparations

We consider the system

$$\begin{aligned}
 (S_0 + S_{\text{int}}) [\psi, \bar{\psi}] &\equiv \sum_{\substack{\omega_n \in \Omega_\zeta \\ \alpha, \alpha' \in \mathcal{A}}} \bar{\psi}_\alpha(i\omega_n) \left[\left(-i\omega_n + \frac{\epsilon_\alpha - \mu}{\hbar} \right) \delta_{\alpha, \alpha'} + \frac{V_{\alpha\alpha'}}{\hbar} \right] \psi_{\alpha'}(i\omega_n) \\
 &\equiv \sum_{\substack{\omega_n \in \Omega_\zeta \\ \alpha, \alpha' \in \mathcal{A}}} \bar{\psi}_\alpha(i\omega_n) \left(-\frac{1}{\hbar} \mathcal{G}_{\alpha, \alpha'}^{-1}(i\omega_n) \right) \psi_{\alpha'}(i\omega_n)
 \end{aligned}$$

and compute the self-energy using the standard-trick. We will use the following trick: Inverting the Dyson-equation with the von Neumann series on the one hand, and re-expressing the imaginary-time-Green-function with help of the field integral on the other hand, gives the two equations

$$\mathcal{G}_{\alpha, \alpha'}(i\omega_n) = \sum_{n=0}^{\infty} \left[\left(\mathcal{G}_0 \cdot \Sigma \right)^n(i\omega_n) \cdot \mathcal{G}_0(i\omega_n) \right]_{\alpha, \alpha'} \quad (26.1)$$

$$\mathcal{G}_{\alpha, \alpha'}(i\omega_n) = \frac{-1}{\hbar} \langle \psi_\alpha(i\omega_n) \bar{\psi}_{\alpha'}(i\omega_n) \rangle \quad (26.2)$$

The self-energy of the electronic Green-function is known to be given as the sum of all the connected irreducible diagrams of the interaction vertex

$$\Sigma(i\omega_n) \equiv \sum_{j=1}^{\infty} \Sigma^{(j)}(i\omega_n) = \Sigma^{(1)}(i\omega_n) + \Sigma^{(2)}(i\omega_n) + \dots \quad (26.3)$$

Inserting this in the upper-equation above and keeping the first non-vanishing terms yields the sets of equations

$$\mathcal{G}_{\alpha,\alpha'}(i\omega_n) = \mathcal{G}_{\alpha,\alpha'}^{(0)}(i\omega_n) + (\mathcal{G}_0 \cdot \Sigma^{(1)} \cdot \mathcal{G}_0)_{\alpha,\alpha'}(i\omega_n) + (\mathcal{G}_0 \cdot \Sigma^{(2)} \cdot \mathcal{G}_0)_{\alpha,\alpha'}(i\omega_n) + \dots \quad (26.4)$$

$$\mathcal{G}_{\alpha,\alpha'}(i\omega_n) = \frac{-1}{\hbar} \langle \psi_\alpha(i\omega_n) \bar{\psi}_{\alpha'}(i\omega_n) \rangle \quad (26.5)$$

The lower equation can be calculated explicitly with help of the Wick-theorem, linked-cluster-theorem, and the residue-theorem, and then, by comparing the first non-vanishing coefficients with the upper-equation, we can conclude what $\Sigma^{(2)}(i\omega_n)$ must be.

Fact 1: The action, corresponding to this system, is given by the equation above.

Fact 2: The electronic Green-function is written in second order perturbation-theory as

$$\mathcal{G}_{\alpha\alpha'}(i\omega_n) = \mathcal{G}_{\alpha\alpha'}^{(0)}(i\omega_n) + \mathcal{G}_{\alpha\alpha'}^{(1)}(i\omega_n) + \mathcal{G}_{\alpha\alpha'}^{(2)}(i\omega_n) + \dots$$

The First Non-Vanishing Order

We therefore calculate

$$\mathcal{G}_{\alpha\alpha'}^{(1)}(i\omega_n) = \frac{-1}{\hbar} \sum_{\substack{\beta,\beta' \in \mathcal{A} \\ \omega_m \in \Omega_\zeta}} \frac{-V_{\beta\beta'}}{\hbar} \langle \psi_\alpha(i\omega_n) \bar{\psi}_{\alpha'}(i\omega_n) \bar{\psi}_\beta(i\omega_m) \psi_{\beta'}(i\omega_m) \rangle_0^{\text{CD}} \quad (26.6)$$

where the last expectation-value contains only [C]onnected [D]iagrams (linked-cluster-theorem). We can therefore simply calculate the free correlator and discard the contributions from unconnected diagrams.

We therefore consider the correlator

$$\begin{aligned} \langle \psi_\alpha(i\omega_n) \bar{\psi}_{\alpha'}(i\omega_n) \bar{\psi}_\beta(i\omega_m) \psi_{\beta'}(i\omega_m) \rangle_0 &= \langle \psi_\alpha(i\omega_n) \psi_{\beta'}(i\omega_m) \bar{\psi}_{\alpha'}(i\omega_n) \bar{\psi}_\beta(i\omega_m) \rangle_0 \\ &= \langle \psi(A_1) \psi(A_2) \bar{\psi}(B_1) \bar{\psi}(B_2) \rangle_0 \\ &= - \langle \psi(A_1) \psi(A_2) \bar{\psi}(B_2) \bar{\psi}(B_1) \rangle_0 \\ &= - \sum_{\pi \in S_2} \text{sign}(\pi)^{1-\theta(\zeta)} \prod_{l=1}^2 \langle \psi(A_l) \bar{\psi}(B_{\pi(l)}) \rangle_0 \\ &= -\hbar^2 \sum_{\pi \in S_2} \text{sign}(\pi)^{1-\theta(\zeta)} \prod_{l=1}^2 \mathcal{G}_0(A_l, B_{\pi(l)}) \end{aligned}$$

The electronic correlator can now be evaluated with the Wick-theorem. In order to so, we will enumerate all the permutations of the group S_3 .

$$S_2 = \{\pi_1, \pi_2\} = \left\{ \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \right\} \quad \text{with signatures} \quad \text{sgn}(\pi_1, \pi_2) = (+1, -1). \quad (26.7)$$

The first contribution is a vacuum diagram and does not contribute to the series expansion (linked cluster theorem). We therefore focus on the second term. We note that

$$A_1 = (\alpha, i\omega_n), \quad A_2 = (\beta, i\omega_m) \quad (26.8)$$

$$B_1 = (\alpha', i\omega_n), \quad B_2 = (\beta, i\omega_m) \quad (26.9)$$

and therefore we obtain for the non-vacuum contribution to the Green-function

$$\mathcal{G}_{\alpha\alpha'}^{(1)}(i\omega_n) = + \sum_{\beta, \beta' \in \mathcal{A}} \mathcal{G}_{\alpha, \beta}^0(i\omega_n) V_{\beta, \beta'} \mathcal{G}_{\beta' \alpha'}^0(i\omega_n)$$

comparison of coefficients gives the first-order correction to the self-energy

$$\Sigma_{\beta\beta'}^{(1)}(i\omega_n) = +V_{\beta, \beta'} = \text{„frequency-independent“} \quad (26.10)$$

26.2. Self Energy of a Diagonalized Gas with Quartic Interactions

Proof 26.2 of Theorem 19.2

Computing the electronic Self-Energy

Preparations

We consider the system

$$\begin{aligned} (S_0 + S_{\text{int}}) [\psi, \bar{\psi}] \equiv & \sum_{\substack{\omega_n \in \Omega_\zeta \\ \alpha, \alpha' \in \mathcal{A}}} \bar{\psi}_\alpha(i\omega_n) \left[\left(-i\omega_n + \frac{\epsilon_\alpha - \mu}{\hbar} \right) \delta_{\alpha, \alpha'} \right] \psi_{\alpha'}(i\omega_n) \\ & + \frac{1}{2} \sum_{\substack{\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathcal{A} \\ \omega_{n_1}, \omega_{n_2}, \omega_{n_3}, \omega_{n_4} \in \Omega_\zeta}} \frac{-V_{\alpha_1 \alpha_2 \alpha_3 \alpha_4} \delta_{\omega_1 + \omega_2, \omega_3 + \omega_4}}{\hbar^2 \beta} \bar{\psi}_{\alpha_1}(i\omega_1) \bar{\psi}_{\alpha_2}(i\omega_2) \psi_{\alpha_3}(i\omega_3) \psi_{\alpha_4}(i\omega_4) \end{aligned}$$

and compute the self-energy using the standard-trick. We will use the following trick: Inverting the Dyson-equation with the von Neumann series on the one hand, and re-expressing the imaginary-time-Green-function with help of the field integral on the other hand, gives the two equations

$$\mathcal{G}_{\alpha, \alpha'}(i\omega_n) = \sum_{n=0}^{\infty} \left[(\mathcal{G}_0 \cdot \Sigma)^n(i\omega_n) \cdot \mathcal{G}_0(i\omega_n) \right]_{\alpha, \alpha'} \quad (26.11)$$

$$\mathcal{G}_{\alpha, \alpha'}(i\omega_n) = \frac{-1}{\hbar} \langle \psi_\alpha(i\omega_n) \bar{\psi}_{\alpha'}(i\omega_n) \rangle \quad (26.12)$$

The self-energy of the electronic Green-function is known to be given as the sum of all the connected irreducible diagrams of the interaction vertex

$$\Sigma(i\omega_n) \equiv \sum_{j=1}^{\infty} \Sigma^{(j)}(i\omega_n) = \Sigma^{(1)}(i\omega_n) + \Sigma^{(2)}(i\omega_n) + \dots \quad (26.13)$$

Inserting this in the upper-equation above and keeping the first non-vanishing terms yields the sets of equations

$$\mathcal{G}_{\alpha, \alpha'}(i\omega_n) = \mathcal{G}_{\alpha, \alpha'}^{(0)}(i\omega_n) + (\mathcal{G}_0 \cdot \Sigma^{(1)} \cdot \mathcal{G}_0)_{\alpha, \alpha'}(i\omega_n) + (\mathcal{G}_0 \cdot \Sigma^{(2)} \cdot \mathcal{G}_0)_{\alpha, \alpha'}(i\omega_n) + \dots \quad (26.14)$$

$$\mathcal{G}_{\alpha, \alpha'}(i\omega_n) = \frac{-1}{\hbar} \langle \psi_\alpha(i\omega_n) \bar{\psi}_{\alpha'}(i\omega_n) \rangle \quad (26.15)$$

The lower equation can be calculated explicitly with help of the Wick-theorem, linked-cluster-theorem, and the residue-theorem, and then, by comparing the first non-vanishing coefficients with the upper-equation, we can conclude what $\Sigma^{(2)}(i\omega_n)$ must be.

Fact 1: The action, corresponding to this system, is given by the equation above.

Fact 2: The electronic Green-function is written in second order perturbation-theory as

$$\mathcal{G}_{\alpha\alpha'}(i\omega_n) = \mathcal{G}_{\alpha\alpha'}^{(0)}(i\omega_n) + \mathcal{G}_{\alpha\alpha'}^{(1)}(i\omega_n) + \mathcal{G}_{\alpha\alpha'}^{(2)}(i\omega_n) + \dots$$

The First Non-Vanishing Order

We therefore calculate

$$\mathcal{G}_{\alpha\alpha'}^{(1)}(i\omega_n) = \frac{-1}{\hbar} \frac{1}{2} \sum_{\substack{\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathcal{A} \\ \omega_{n_1}, \omega_{n_2}, \omega_{n_3}, \omega_{n_4} \in \Omega_\zeta}} \frac{+V_{\alpha_1\alpha_2\alpha_3\alpha_4} \delta_{\omega_{n_1}+\omega_{n_2}, \omega_{n_3}+\omega_{n_4}}}{\hbar^2 \beta} \left\langle \psi_\alpha(i\omega_n) \bar{\psi}_{\alpha'}(i\omega_n) \bar{\psi}_{\alpha_1}(i\omega_{n_1}) \bar{\psi}_{\alpha_2}(i\omega_{n_2}) \psi_{\alpha_3}(i\omega_{n_3}) \psi_{\alpha_4}(i\omega_{n_4}) \right\rangle_0 \quad (26.16)$$

We now focus on the correlator

$$\begin{aligned} \left\langle \psi_\alpha(i\omega_n) \bar{\psi}_{\alpha'}(i\omega_n) \bar{\psi}_{\alpha_1}(i\omega_1) \bar{\psi}_{\alpha_2}(i\omega_2) \psi_{\alpha_3}(i\omega_3) \psi_{\alpha_4}(i\omega_4) \right\rangle_0 &= \left\langle \psi_\alpha(i\omega_n) \psi_{\alpha_3}(i\omega_3) \psi_{\alpha_4}(i\omega_4) \bar{\psi}_{\alpha'}(i\omega_n) \bar{\psi}_{\alpha_1}(i\omega_1) \bar{\psi}_{\alpha_2}(i\omega_2) \right\rangle_0 \\ &= \left\langle \psi(A_1) \psi(A_2) \psi(A_3) \bar{\psi}(B_1) \bar{\psi}(B_2) \bar{\psi}(B_3) \right\rangle_0 \\ &= - \left\langle \psi(A_1) \psi(A_2) \psi(A_3) \bar{\psi}(B_3) \bar{\psi}(B_2) \bar{\psi}(B_1) \right\rangle_0 \\ &= - \sum_{\pi \in S_3} \text{sign}(\pi)^{1-\theta(\zeta)} \prod_{l=1}^3 \left\langle \psi(A_l) \bar{\psi}(B_{\pi(l)}) \right\rangle_0 \\ &= +\hbar^3 \sum_{\pi \in S_3} \text{sign}(\pi)^{1-\theta(\zeta)} \prod_{l=1}^3 \mathcal{G}_0(A_l, B_{\pi(l)}) \end{aligned}$$

The electronic correlator can now be evaluated with the Wick-theorem. In order to so, we will enumerate all the permutations of the group S_3 .

$$\begin{aligned} S_3 &= \{\pi_1, \pi_2, \pi_3, \pi_4, \pi_5, \pi_6\} \\ &= \left\{ \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \right\} \end{aligned}$$

with signatures

$$\text{sgn}(\pi_1, \pi_2, \pi_3, \pi_4, \pi_5, \pi_6) = (+1, -1, -1, +1, +1, -1)$$

Facts:

- The contributions from π_1 and π_2 are vacuum diagrams (which cancel out due to the linked-cluster-theorem)
- The contributions from π_3 and π_6 are "Balloon"-diagrams / "Hartree"-diagrams.
- The contributions from π_4 and π_5 are "Sunset"-diagrams / "Fock"-diagrams.

We note that

$$\begin{aligned} A_1 &= (\alpha, i\omega_n), & A_2 &= (\alpha_3, i\omega_{n_3}), & A_3 &= (\alpha_4, i\omega_{n_4}) \\ B_1 &= (\alpha', i\omega_n), & B_2 &= (\alpha_1, i\omega_{n_1}), & B_3 &= (\alpha_2, i\omega_{n_2}) \end{aligned}$$

and therefore we obtain

$$\begin{aligned}
 B_{\pi_3(1)} &\equiv B_2 \equiv (\alpha_1, i\omega_{n_1}), & B_{\pi_3(2)} &\equiv B_1 \equiv (\alpha', i\omega_n), & B_{\pi_3(3)} &\equiv B_3 \equiv (\alpha_2, i\omega_{n_2}), \\
 B_{\pi_5(1)} &\equiv B_3 \equiv (\alpha_2, i\omega_{n_2}), & B_{\pi_5(2)} &\equiv B_1 \equiv (\alpha', i\omega_n), & B_{\pi_5(3)} &\equiv B_2 \equiv (\alpha_1, i\omega_{n_1}), \\
 B_{\pi_6(1)} &\equiv B_3 \equiv (\alpha_2, i\omega_{n_2}), & B_{\pi_6(2)} &\equiv B_2 \equiv (\alpha_1, i\omega_{n_1}), & B_{\pi_6(3)} &\equiv B_1 \equiv (\alpha', i\omega_n), \\
 B_{\pi_4(1)} &\equiv B_2 \equiv (\alpha_1, i\omega_{n_1}), & B_{\pi_4(2)} &\equiv B_3 \equiv (\alpha_2, i\omega_{n_2}), & B_{\pi_4(3)} &\equiv B_1 \equiv (\alpha', i\omega_n).
 \end{aligned}$$

Balloon-Diagrams

π_3 :

This gives then for the first order correction of the Green-function

$$\begin{aligned}
 \mathcal{G}_{\alpha\alpha'}^{\pi_3}(i\omega_n) &= +\frac{1}{2} \sum_{\substack{\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathcal{A} \\ \omega_{n_1}, \omega_{n_2}, \omega_{n_3}, \omega_{n_4} \in \Omega_\zeta}} \frac{1}{\beta} V_{\alpha_1 \alpha_2 \alpha_3 \alpha_4} \delta_{\omega_{n_1} + \omega_{n_2}, \omega_{n_3} + \omega_{n_4}} \mathcal{G}_0(i\omega_n) \delta_{\omega_n, \omega_{n_1}} \mathcal{G}_0(i\omega_n) \delta_{\omega_{n_3}, \omega_n} \mathcal{G}_0(i\omega_{n_2}) \delta_{\omega_{n_2}, \omega_{n_4}} \\
 &= +\frac{1}{2} \sum_{\substack{\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathcal{A} \\ \omega_{n_1}, \omega_{n_2}, \omega_{n_3}, \omega_{n_4} \in \Omega_\zeta}} \mathcal{G}_0(i\omega_n) \delta_{\omega_n, \omega_{n_1}} \frac{1}{\beta} V_{\alpha_1 \alpha_2 \alpha_3 \alpha_4} \delta_{\omega_{n_1} + \omega_{n_2}, \omega_{n_3} + \omega_{n_4}} \mathcal{G}_0(i\omega_{n_2}) \delta_{\omega_{n_2}, \omega_{n_4}} \mathcal{G}_0(i\omega_n) \delta_{\omega_{n_3}, \omega_n} \\
 &= +\frac{1}{2} \sum_{\substack{\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathcal{A} \\ \omega_{n_1}, \omega_{n_2} \in \Omega_\zeta}} \mathcal{G}_0(i\omega_n) \delta_{\omega_n, \omega_{n_1}} \frac{1}{\beta} V_{\alpha_1 \alpha_2 \alpha_3 \alpha_4} \delta_{\omega_{n_1} + \omega_{n_2}, \omega_n + \omega_{n_2}} \mathcal{G}_0(i\omega_{n_2}) \mathcal{G}_0(i\omega_n) \\
 &= +\frac{1}{2} \sum_{\substack{\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathcal{A} \\ \omega_{n_2} \in \Omega_\zeta}} \mathcal{G}_0(i\omega_n) \frac{1}{\beta} V_{\alpha_1 \alpha_2 \alpha_3 \alpha_4} \mathcal{G}_0(i\omega_{n_2}) \mathcal{G}_0(i\omega_n)
 \end{aligned}$$

From this we can read off the self-energy

$$\Sigma_{\alpha_1 \alpha_3}^{\pi_3}(i\omega_n) = +\frac{1}{2} \sum_{\substack{\alpha_2 \in \mathcal{A} \\ \omega_{n_2} \in \Omega_\zeta}} \frac{1}{\beta} V_{\alpha_1 \alpha_2 \alpha_3 \alpha_2} \mathcal{G}_0(i\omega_{n_2}) = +\frac{1}{2} \sum_{\alpha_2 \in \mathcal{A}} V_{\alpha_1 \alpha_2 \alpha_3 \alpha_2} F_{\hbar\beta}(\epsilon_{\alpha_2} - \mu) = -\zeta \frac{1}{2} \sum_{\alpha_2 \in \mathcal{A}} V_{\alpha_1 \alpha_2 \alpha_3 \alpha_2} n_{\hbar\beta}^\zeta(\epsilon_{\alpha_2} - \mu) \quad (26.17)$$

π_6 :

This gives then for the first order correction of the Green-function

$$\begin{aligned}
 \mathcal{G}_{\alpha\alpha'}^{\pi_6}(i\omega_n) &= +\frac{1}{2} \sum_{\substack{\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathcal{A} \\ \omega_{n_1}, \omega_{n_2}, \omega_{n_3}, \omega_{n_4} \in \Omega_\zeta}} \frac{1}{\beta} V_{\alpha_1 \alpha_2 \alpha_3 \alpha_4} \delta_{\omega_{n_1} + \omega_{n_2}, \omega_{n_3} + \omega_{n_4}} \mathcal{G}_0(i\omega_n) \delta_{\omega_n, \omega_{n_2}} \mathcal{G}_0(i\omega_{n_3}) \delta_{\omega_{n_3}, \omega_{n_1}} \mathcal{G}_0(i\omega_n) \delta_{\omega_{n_4}, \omega_n} \\
 &= +\frac{1}{2} \sum_{\substack{\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathcal{A} \\ \omega_{n_3} \in \Omega_\zeta}} \frac{1}{\beta} V_{\alpha_1 \alpha_2 \alpha_3 \alpha_4} \delta_{\omega_{n_3} + \omega_n, \omega_{n_3} + \omega_n} \mathcal{G}_0(i\omega_n) \mathcal{G}_0(i\omega_{n_3}) \mathcal{G}_0(i\omega_n)
 \end{aligned}$$

From this we can read off the self-energy

$$\Sigma_{\alpha_2\alpha_4}^{\pi_6}(i\omega_n) = +\frac{1}{2} \sum_{\substack{\alpha_3 \in \mathcal{A} \\ \omega_{n_3} \in \Omega_\zeta}} \frac{1}{\beta} V_{\alpha_3\alpha_2\alpha_3\alpha_4} \mathcal{G}_{\alpha_3\alpha_3}^0(i\omega_{n_3}) = +\frac{1}{2} \sum_{\alpha_3 \in \mathcal{A}} V_{\alpha_3\alpha_2\alpha_3\alpha_4} F_{\hbar\beta}(\epsilon_{\alpha_3} - \mu) = -\zeta \frac{1}{2} \sum_{\alpha_3 \in \mathcal{A}} V_{\alpha_3\alpha_2\alpha_3\alpha_4} n_{\hbar\beta}^\zeta(\epsilon_{\alpha_3} - \mu) \quad (26.18)$$

π_4 :

This gives then for the first order correction of the Green-function

$$\begin{aligned} \mathcal{G}_{\alpha\alpha'}^{\pi_4}(i\omega_n) &= -\frac{1}{2} \sum_{\substack{\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathcal{A} \\ \omega_{n_1}, \omega_{n_2}, \omega_{n_3}, \omega_{n_4} \in \Omega_\zeta}} \frac{1}{\beta} V_{\alpha_1\alpha_2\alpha_3\alpha_4} \delta_{\omega_{n_1}+\omega_{n_2}, \omega_{n_3}+\omega_{n_4}} \mathcal{G}_{\alpha, \alpha_1}^0(i\omega_n) \delta_{\omega_n, \omega_{n_1}} \mathcal{G}_{\alpha_3\alpha_2}^0(i\omega_{n_3}) \delta_{\omega_{n_3}, \omega_{n_2}} \mathcal{G}_{\alpha_4\alpha'}^0(i\omega_n) \delta_{\omega_{n_4}, \omega_n} \\ &= -\frac{1}{2} \sum_{\substack{\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathcal{A} \\ \omega_{n_3} \in \Omega_\zeta}} \frac{1}{\beta} V_{\alpha_1\alpha_2\alpha_3\alpha_4} \delta_{\omega_n+\omega_{n_3}, \omega_{n_3}+\omega_n} \mathcal{G}_{\alpha, \alpha_1}^0(i\omega_n) \mathcal{G}_{\alpha_3\alpha_2}^0(i\omega_{n_3}) \mathcal{G}_{\alpha_4\alpha'}^0(i\omega_n) \end{aligned}$$

From this we can read off the self-energy

$$\Sigma_{\alpha_1\alpha_4}^{\pi_4}(i\omega_n) = -\frac{1}{2} \sum_{\substack{\alpha_3 \in \mathcal{A} \\ \omega_{n_3} \in \Omega_\zeta}} \frac{1}{\beta} V_{\alpha_1\alpha_3\alpha_3\alpha_4} \mathcal{G}_{\alpha_3\alpha_3}^0(i\omega_{n_3}) = -\frac{1}{2} \sum_{\alpha_3 \in \mathcal{A}} V_{\alpha_1\alpha_3\alpha_3\alpha_4} F_{\hbar\beta}(\epsilon_{\alpha_3} - \mu) = +\zeta \frac{1}{2} \sum_{\alpha_3 \in \mathcal{A}} V_{\alpha_1\alpha_3\alpha_3\alpha_4} n_{\hbar\beta}^\zeta(\epsilon_{\alpha_3} - \mu) \quad (26.19)$$

π_5 :

This gives then for the first order correction of the Green-function

$$\begin{aligned} \mathcal{G}_{\alpha\alpha'}^{\pi_5}(i\omega_n) &= -\frac{1}{2} \sum_{\substack{\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathcal{A} \\ \omega_{n_1}, \omega_{n_2}, \omega_{n_3}, \omega_{n_4} \in \Omega_\zeta}} \frac{1}{\beta} V_{\alpha_1\alpha_2\alpha_3\alpha_4} \delta_{\omega_{n_1}+\omega_{n_2}, \omega_{n_3}+\omega_{n_4}} \mathcal{G}_{\alpha, \alpha_2}^0(i\omega_n) \delta_{\omega_n, \omega_{n_2}} \mathcal{G}_{\alpha_3\alpha'}^0(i\omega_n) \delta_{\omega_{n_3}, \omega_n} \mathcal{G}_{\alpha_4\alpha_1}^0(i\omega_{n_4}) \delta_{\omega_{n_4}, \omega_{n_1}} \\ &= -\frac{1}{2} \sum_{\substack{\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathcal{A} \\ \omega_{n_4} \in \Omega_\zeta}} \frac{1}{\beta} V_{\alpha_1\alpha_2\alpha_3\alpha_4} \delta_{\omega_{n_4}+\omega_n, \omega_n+\omega_{n_4}} \mathcal{G}_{\alpha, \alpha_2}^0(i\omega_n) \mathcal{G}_{\alpha_4\alpha_1}^0(i\omega_{n_4}) \mathcal{G}_{\alpha_3\alpha'}^0(i\omega_n) \end{aligned}$$

From this we can read off the self-energy

$$\Sigma_{\alpha_2\alpha_3}^{\pi_5}(i\omega_n) = -\frac{1}{2} \sum_{\substack{\alpha_4 \in \mathcal{A} \\ \omega_{n_4} \in \Omega_\zeta}} \frac{1}{\beta} V_{\alpha_4\alpha_3\alpha_3\alpha_4} \mathcal{G}_{\alpha_4\alpha_4}^0(i\omega_{n_4}) = -\frac{1}{2} \sum_{\alpha_4 \in \mathcal{A}} V_{\alpha_4\alpha_3\alpha_3\alpha_4} F_{\hbar\beta}(\epsilon_{\alpha_4} - \mu) = +\zeta \frac{1}{2} \sum_{\alpha_4 \in \mathcal{A}} V_{\alpha_4\alpha_3\alpha_3\alpha_4} n_{\hbar\beta}^\zeta(\epsilon_{\alpha_4} - \mu) \quad (26.20)$$

26.3. Self Energy of a Spin-Independent-Interaction between an Electron-Gas and a Phonon-Gas

Proof 26.3 of Theorem 19.3

Notice first, that going over to the field-integral-description of the problem gives the action

$$\begin{aligned} S[\chi, \bar{\chi}] &= S_{\text{el}}[\psi, \bar{\psi}] + S_{\text{ph}}[\phi, \bar{\phi}] + S_{\text{el-ph-int}}[\chi, \bar{\chi}] \\ &= \int \bar{\psi}_{m\mathbf{k}\sigma}(\tau) \left(\partial_\tau + \frac{\epsilon_{m\mathbf{k}\sigma} - \mu}{\hbar} \right) \psi_{m\mathbf{k}\sigma}(\tau) + \int \bar{\phi}_{\mathbf{q}\nu}(\tau) (\partial_\tau + \omega_{\mathbf{q},\nu}) \phi_{\mathbf{q}\nu}(\tau) + S_{\text{el-ph-int}}[\chi, \bar{\chi}] \end{aligned}$$

with interaction-term

$$S_{\text{el-ph-int}}[\chi, \bar{\chi}] = \int \frac{g_{mn\nu}(\mathbf{k}, \mathbf{q})}{\hbar \sqrt{N_p}} \left(\phi_{\mathbf{q}\nu}(\tau) + \bar{\phi}_{-\mathbf{q}\nu}(\tau) \right) \bar{\psi}_{\mathbf{k}+\mathbf{q}m\sigma}(\tau) \psi_{\mathbf{k}n\sigma}(\tau)$$

We will now determine the self-energy of the electronic system.

We will use the following trick: Inverting the Dyson-equation with the von Neumann series on the one hand, and re-expressing the imaginary-time-Green-function with help of the field integral on the other hand, gives the two equations

$$\mathcal{G}_{\alpha,\alpha'}(i\omega_n) = \sum_{n=0}^{\infty} \left[(\mathcal{G}_0 \cdot \Sigma)^n(i\omega_n) \cdot \mathcal{G}_0(i\omega_n) \right]_{\alpha,\alpha'} \quad (26.21)$$

$$\mathcal{G}_{\alpha,\alpha'}(i\omega_n) = \int_0^{\hbar\beta} d\tau e^{i\omega_n\tau} \mathcal{G}_{\alpha,\alpha'}(\tau) = \frac{-1}{\hbar} \int_0^{\hbar\beta} d\tau e^{i\omega_n\tau} \langle \psi_\alpha(\tau) \bar{\psi}_{\alpha'}(0) \rangle \quad (26.22)$$

The self-energy of the electronic Green-function is known to be given as the sum of all the connected irreducible diagrams of the interaction

$$\Sigma(i\omega_n) = \sum_{j=0}^{\infty} \Sigma^{(j)}(i\omega_n) \approx 0 + \Sigma^{(2)}(i\omega_n) + 0 + \mathcal{O}(V^4), \quad (26.23)$$

whereas the first contribution will turn out to vanish since the interaction can be expanded as a sum of monomials of creation- and annihilation-operators with uneven degree. Inserting this in the upper-equation above and keeping the first non-vanishing terms yields the sets of equations

$$\mathcal{G}_{\alpha,\alpha'}(i\omega_n) = \mathcal{G}_0(i\omega_n) + (\mathcal{G}_0 \cdot \Sigma^{(2)} \cdot \mathcal{G}_0)_{\alpha,\alpha'}(i\omega_n) \quad (26.24)$$

$$\mathcal{G}_{\alpha,\alpha'}(i\omega_n) = \frac{-1}{\hbar} \int_0^{\hbar\beta} d\tau e^{i\omega_n\tau} \langle \psi_\alpha(\tau) \bar{\psi}_{\alpha'}(0) \rangle \quad (26.25)$$

The lower equation can be calculated explicitly with help of the Wick-theorem, linked-cluster-theorem, and the residue-theorem, and then, by comparing the first non-vanishing coefficients with the upper-equation, we can conclude what $\Sigma^{(2)}(i\omega_n)$ must be.

Fact 1: The action, corresponding to this system, is given by the equation above.

Fact 2: The electronic Green-function is written in second order perturbation-theory as

$$\begin{aligned} \mathcal{G}_{b,b'}^{(2)}(\tau) &= \mathcal{G}_{b,b'}^{(0)}(\tau) + \mathcal{G}_{b,b'}^{(1)}(\tau) + \mathcal{G}_{b,b'}^{(2)}(\tau) + \dots \\ &= \mathcal{G}_{b,b'}^{(0)}(\tau) + 0 + \mathcal{G}_{b,b'}^{(2)}(\tau) + \dots \end{aligned}$$

The second contribution reads

$$\begin{aligned} &\mathcal{G}_{b,b'}^{(2)}(\tau) \\ &= \frac{1}{2!} \oint_{\substack{\tau_1, \tau_2 \in [0, \hbar\beta] \\ \mu, \mu' \in \{1, \uparrow\} \\ \mathbf{k}, \mathbf{k}', \mathbf{q}, \mathbf{q}' \in \mathbf{Q} \\ m, m', n, n' \in \{1, 2, \dots\} \\ \nu, \nu' \in \{1, 2, \dots, r, d\}}} \frac{-1}{\hbar} \frac{g_{mn\nu}(\mathbf{k}, \mathbf{q}) g_{m'n'\nu'}(\mathbf{k}', \mathbf{q}')}{\hbar^2 N_p} \langle \psi_{\mathbf{p}b\sigma}(\tau) \bar{\psi}_{\mathbf{p}'b'\sigma'}(0) \left(\phi_{\mathbf{q}\nu}(\tau_1) + \bar{\phi}_{-\mathbf{q}\nu}(\tau_1) \right) \bar{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1) \psi_{\mathbf{k}n\mu}(\tau_1) \left(\phi_{\mathbf{q}'\nu'}(\tau_2) + \bar{\phi}_{-\mathbf{q}'\nu'}(\tau_2) \right) \bar{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2) \psi_{\mathbf{k}'n'\mu'}(\tau_2) \rangle_0^{\text{CD}} \end{aligned} \quad (26.26)$$

where only [C]onnected [D]iagrams are summed over (linked cluster theorem). We can therefore simply sum over all diagrams and discard the disconnected diagrams.

Fact 3: The correlator in it's decomposition into it's fermionic and bosonic component reads

$$\begin{aligned} &\frac{-1}{\hbar} \langle \psi_{\mathbf{p}b\sigma}(\tau) \bar{\psi}_{\mathbf{p}'b'\sigma'}(0) \left(\phi_{\mathbf{q}\nu}(\tau_1) + \bar{\phi}_{-\mathbf{q}\nu}(\tau_1) \right) \bar{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1) \psi_{\mathbf{k}n\mu}(\tau_1) \left(\phi_{\mathbf{q}'\nu'}(\tau_2) + \bar{\phi}_{-\mathbf{q}'\nu'}(\tau_2) \right) \bar{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2) \psi_{\mathbf{k}'n'\mu'}(\tau_2) \rangle_0^{\text{CD}} \\ &= \frac{-1}{\hbar} \langle \psi_{\mathbf{p}b\sigma}(\tau) \bar{\psi}_{\mathbf{p}'b'\sigma'}(0) \bar{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1) \psi_{\mathbf{k}n\mu}(\tau_1) \bar{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2) \psi_{\mathbf{k}'n'\mu'}(\tau_2) \rangle_0^{\text{CD}} \langle \left(\phi_{\mathbf{q}\nu}(\tau_1) + \bar{\phi}_{-\mathbf{q}\nu}(\tau_1) \right) \left(\phi_{\mathbf{q}'\nu'}(\tau_2) + \bar{\phi}_{-\mathbf{q}'\nu'}(\tau_2) \right) \rangle_0^{\text{CD}} \\ &= -\langle \psi_{\mathbf{p}b\sigma}(\tau) \psi_{\mathbf{k}n\mu}(\tau_1) \psi_{\mathbf{k}'n'\mu'}(\tau_2) \bar{\psi}_{\mathbf{p}'b'\sigma'}(0) \bar{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1) \bar{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2) \rangle_0^{\text{CD}} \cdot \mathcal{D}_{\substack{\mathbf{q}, -\mathbf{q}' \\ \nu, \nu'}}^0(\tau_1 - \tau_2) \end{aligned} \quad (26.27)$$

The electronic correlator can now be evaluated with the Wick-theorem. In order to so, we will enumerate all the permutations of the group S_3 .

$$\begin{aligned} S_3 &= \{ \pi_1, \pi_2, \pi_3, \pi_4, \pi_5, \pi_6 \} \\ &= \left\{ \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \right\} \end{aligned}$$

with signatures

$$\text{sgn}(\pi_1, \pi_2, \pi_3, \pi_4, \pi_5, \pi_6) = (+1, -1, -1, +1, +1, -1)$$

Now we write

$$\begin{aligned}
 & - \langle \psi_{\mathbf{p}b\sigma}(\tau) \psi_{\mathbf{k}n\mu}(\tau_1) \psi_{\mathbf{k}'n'\mu'}(\tau_2) \bar{\psi}_{\mathbf{p}'b'\sigma'}(0) \bar{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1) \bar{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2) \rangle_0^{\text{CD}} \\
 & \equiv - \langle \psi(A_1) \psi(A_2) \psi(A_3) \bar{\psi}(B_1) \bar{\psi}(B_2) \bar{\psi}(B_3) \rangle_0 \\
 & \equiv + \langle \psi(A_1) \psi(A_2) \psi(A_3) \bar{\psi}(B_3) \bar{\psi}(B_2) \bar{\psi}(B_1) \rangle_0 \\
 & = + \sum_{\pi \in S_3} \text{sgn}(\pi) \prod_{l=1}^3 \langle \psi(A_l) \bar{\psi}(B_{\pi(l)}) \rangle_0 \\
 & = -\hbar^3 \sum_{\pi \in S_3} \text{sgn}(\pi) \prod_{l=1}^3 \mathcal{G}_0(A_l, B_{\pi(l)})
 \end{aligned}$$

Facts:

- The contributions from π_1 and π_2 are vacuum diagrams (which cancel out due to the linked-cluster-theorem)
- The contributions from π_3 and π_6 are "Baloon"-diagrams / "Hartree"-diagrams.
- The contributions from π_4 and π_5 are "Sunset"-diagrams / "Fock"-diagrams.

Note that the two Hartree- and Fock-terms don't cancel out, but contribute equally to the series expansion (respectively) which cancels out the prefactor of $1/2!$ from the Taylor-expansion.

We note that

$$\begin{aligned}
 A_1 &= (\mathbf{p}, b, \sigma, \tau), & A_2 &= (\mathbf{k}, n, \mu, \tau_1), & A_3 &= (\mathbf{k}', n', \mu', \tau_2) \\
 B_1 &= (\mathbf{p}', b', \sigma', 0), & B_2 &= (\mathbf{k} + \mathbf{q}, m, \mu, \tau_1), & B_3 &= (\mathbf{k}' + \mathbf{q}', m', \mu', \tau_2)
 \end{aligned}$$

and therefore we obtain

$$\begin{aligned}
 B_{\pi_3(1)} &\equiv B_2 \equiv (\mathbf{q} + \mathbf{k}, m, \mu, \tau_1), & B_{\pi_3(2)} &\equiv B_1 \equiv (\mathbf{p}', b', \sigma', 0), & B_{\pi_3(3)} &\equiv B_3 \equiv (\mathbf{k}' + \mathbf{q}', m', \mu', \tau_2), \\
 B_{\pi_4(1)} &\equiv B_2 \equiv (\mathbf{k} + \mathbf{q}, m, \mu, \tau_1), & B_{\pi_4(2)} &\equiv B_3 \equiv (\mathbf{k}' + \mathbf{q}', m', \mu', \tau_2), & B_{\pi_4(3)} &\equiv B_1 \equiv (\mathbf{p}', b', \sigma', 0).
 \end{aligned}$$

This gives now for the second-order-contribution

$$\begin{aligned}
 \mathcal{G}_{b,b'}^{(2)}(\tau) &= -\hbar \oint_{\substack{\tau_1, \tau_2 \in [0, \hbar\beta] \\ \mu, \mu' \in \{\downarrow, \uparrow\} \\ \mathbf{k}, \mathbf{k}', \mathbf{q}, \mathbf{q}' \in \mathcal{Q} \\ m, m', n, n' \in \{1, 2, \dots\} \\ v, v' \in \{1, 2, \dots, rd\}}} \frac{g_{mnv}(\mathbf{k}, \mathbf{q}) g_{m'n'v'}(\mathbf{k}', \mathbf{q}')}{N_p} \mathcal{D}_0(\tau_1 - \tau_2) \sum_{\pi \in \{\pi_3, \pi_4\}} \text{sgn}(\pi) \prod_{l=1}^3 \mathcal{G}_0(A_l, B_{\pi(l)}) \\
 &= \hbar \oint_{\substack{\tau_1, \tau_2 \in [0, \hbar\beta] \\ \mu, \mu' \in \{\downarrow, \uparrow\} \\ \mathbf{k}, \mathbf{k}', \mathbf{q}, \mathbf{q}' \in \mathcal{Q} \\ m, m', n, n' \in \{1, 2, \dots\} \\ v, v' \in \{1, 2, \dots, rd\}}} + \frac{g_{mnv}(\mathbf{k}, \mathbf{q}) g_{m'n'v'}(\mathbf{k}', \mathbf{q}')}{N_p} \mathcal{G}_0(\tau - \tau_1) \mathcal{G}_0(\tau_1 - 0) \mathcal{G}_0(0) \mathcal{D}_0(\tau_1 - \tau_2) \\
 &\quad + \hbar \oint_{\substack{\tau_1, \tau_2 \in [0, \hbar\beta] \\ \mu, \mu' \in \{\downarrow, \uparrow\} \\ \mathbf{k}, \mathbf{k}', \mathbf{q}, \mathbf{q}' \in \mathcal{Q} \\ m, m', n, n' \in \{1, 2, \dots\} \\ v, v' \in \{1, 2, \dots, rd\}}} - \frac{g_{mnv}(\mathbf{k}, \mathbf{q}) g_{m'n'v'}(\mathbf{k}', \mathbf{q}')}{N_p} \mathcal{G}_0(\tau - \tau_1) \mathcal{G}_0(\tau_1 - \tau_2) \mathcal{G}_0(\tau_2 - 0) \mathcal{D}_0(\tau_1 - \tau_2)
 \end{aligned}$$

$$\equiv \mathcal{G}_{b,b'}^{\text{Hartree}}(\tau) + \mathcal{G}_{b,b'}^{\text{Fock}}(\tau)$$

$$\begin{array}{c} \mathbf{p}, \mathbf{p}' \\ \sigma, \sigma' \end{array}$$

In order to compute the self-energy, we compute the Fourier-Matsubara-transform of these functions. We therefore determine $\mathcal{G}(i\omega_n)$.

Fourier-Matsubara-Component of the Hartree-Contribution

Lemma 26.1 It holds

$$\mathcal{G}_{\sigma,\sigma'}^{\text{Hartree}}(i\omega_n) = \sum_{\substack{\mu,\mu' \in \{\downarrow,\uparrow\} \\ \mathbf{k},\mathbf{k}',\mathbf{q},\mathbf{q}' \in Q \\ m,m',n,n' \in \{1,2,\dots\} \\ v,v' \in \{1,2,\dots,rd\}}} \mathcal{G}_{\substack{\mathbf{p},\mathbf{k}+\mathbf{q} \\ b,m \\ \sigma,\mu}}^0(i\omega_n) \left(\frac{1}{\beta} \sum_{\omega_m \in \Omega_{-1}} + \frac{g_{mnv}(\mathbf{k}, \mathbf{q}) g_{m'n'v'}(\mathbf{k}', \mathbf{q}')}{N_p} \mathcal{G}_{\substack{\mathbf{k}',\mathbf{k}'+\mathbf{q}' \\ n',m' \\ \mu',\mu'}}^0(i\omega_m) \mathcal{D}_{\substack{\mathbf{q},-\mathbf{q}' \\ v,v'}}^0(0) \right) \mathcal{G}_{\substack{\mathbf{k},\mathbf{p}' \\ n,b' \\ \mu,\sigma'}}^0(i\omega_n) \quad (26.28)$$

Proof 26.4

We will now focus on the temporal components of the Hartree-contribution to the Green-function. For this we consider the function

$$\tilde{\mathcal{G}}^{\text{Hartree}}(\tau) = \iint_{[0,\hbar\beta]^2} d(\tau_1, \tau_2) \mathcal{G}_I(\tau - \tau_1) \mathcal{G}_{II}(\tau_1 - 0) \mathcal{G}_{III}(0) \mathcal{D}_{IV}(\tau_1 - \tau_2) \quad (26.29)$$

and this gives for the Matsubara-components

$$\begin{aligned} \tilde{\mathcal{G}}^{\text{Hartree}}(i\omega_n) &= \int_0^{\hbar\beta} d\tau e^{i\omega_n \tau} \tilde{\mathcal{G}}^{\text{Hartree}}(\tau) \\ &= \iiint_{[0,\hbar\beta]^3} d(\tau, \tau_1, \tau_2) e^{i\omega_n \tau} \mathcal{G}_I(\tau - \tau_1) \mathcal{G}_{II}(\tau_1 - 0) \mathcal{G}_{III}(0) \mathcal{D}_{IV}(\tau_1 - \tau_2) \\ &= \dots \end{aligned}$$

now we expand the statistical functions according to Lemma TBReferenced and we obtain

$$\begin{aligned} \dots &= \frac{1}{(\hbar\beta)^4} \sum_{\substack{\omega_{n_4} \in \Omega_{+1} \\ \omega_{n_1}, \omega_{n_2}, \omega_{n_3} \in \Omega_{-1}}} \iiint_{[0,\hbar\beta]^3} d(\tau, \tau_1, \tau_2) \mathcal{G}_{\alpha,\xi}^0(i\omega_{n_1}) \mathcal{G}_{\xi',\alpha'}^0(i\omega_{n_2}) \mathcal{G}_{\xi',\xi}^0(i\omega_{n_3}) \mathcal{D}_0(i\omega_{n_4}) e^{i(\omega_n \tau - \omega_{n_1}(\tau - \tau_1) - \omega_{n_2} \tau_1 - \omega_{n_4}(\tau_1 - \tau_2))} \\ &= \sum_{\substack{\omega_{n_4} \in \Omega_{+1} \\ \omega_{n_1}, \omega_{n_2}, \omega_{n_3} \in \Omega_{-1}}} \mathcal{G}_I^0(i\omega_{n_1}) \mathcal{G}_{II}^0(i\omega_{n_2}) \mathcal{G}_{III}^0(i\omega_{n_3}) \mathcal{D}_{IV}^0(i\omega_{n_4}) \frac{1}{(\hbar\beta)^4} \iiint_{[0,\hbar\beta]^3} d(\tau, \tau_1, \tau_2) e^{i(\omega_n \tau - \omega_{n_1}(\tau - \tau_1) - \omega_{n_2} \tau_1 - \omega_{n_4}(\tau_1 - \tau_2))} \\ &= \dots \end{aligned}$$

and we find for the integral-factor

$$\begin{aligned} \iiint_{[0, \hbar\beta]^3} d(\tau, \tau_1, \tau_2) e^{i(\omega_n \tau - \omega_{n_1}(\tau - \tau_1) - \omega_{n_2} \tau_1 - \omega_{n_4}(\tau_1 - \tau_2))} &= \int_0^{\hbar\beta} d\tau e^{i\tau(\omega_n - \omega_{n_1})} \int_0^{\hbar\beta} d\tau_1 e^{i\tau_1(\omega_{n_1} - \omega_{n_2} - \omega_{n_4})} \int_0^{\hbar\beta} d\tau_2 e^{i\tau_2 \omega_{n_4}} \\ &= (\hbar\beta)^3 \delta_{\omega_n, \omega_{n_1}} \delta_{\omega_{n_1}, \omega_{n_2}} \delta_{\omega_{n_4}, 0} \end{aligned}$$

We use the first, second and third delta-function in order to eliminate the summations over ω_{n_1} , ω_{n_2} and ω_{n_4} respectively. This yields then

$$\begin{aligned} \tilde{\mathcal{G}}^{\text{Hartree}}(i\omega_n) &= \dots = \sum_{\omega_{n_3} \in \Omega_{-1}} \frac{1}{\hbar\beta} \mathcal{G}_I(i\omega_n) \mathcal{G}_{II}(i\omega_n) \mathcal{G}_{III}(i\omega_{n_3}) \mathcal{D}_{IV}(0) \\ &\equiv \mathcal{G}_I(i\omega_n) \left(\frac{1}{\hbar\beta} \sum_{\omega_{n_3} \in \Omega_{-1}} \mathcal{G}_{III}(i\omega_{n_3}) \mathcal{D}_{IV}(0) \right) \mathcal{G}_{II}(i\omega_n) \end{aligned}$$

or in short:

$$\begin{aligned} \tilde{\mathcal{G}}^{\text{Hartree}}(\tau) &= \iint_{[0, \hbar\beta]^2} d(\tau_1, \tau_2) \mathcal{G}_I(\tau - \tau_1) \mathcal{G}_{II}(\tau_1 - 0) \mathcal{G}_{III}(0) \mathcal{D}_{IV}(\tau_1 - \tau_2) \\ \Rightarrow \tilde{\mathcal{G}}^{\text{Hartree}}(i\omega_n) &= \mathcal{G}_I(i\omega_n) \left(\frac{1}{\hbar\beta} \sum_{\omega_{n_3} \in \Omega_{-1}} \mathcal{G}_{III}(i\omega_{n_3}) \mathcal{D}_{IV}(0) \right) \mathcal{G}_{II}(i\omega_n) \end{aligned} \quad (26.30)$$

Using this identity for the Matsubara-component of a function proves the Lemma: We can compute the Fourier-Matsubara-component as

$$\begin{aligned} \mathcal{G}_{\sigma, \sigma'}^{\text{Hartree}}(i\omega_n) &= \int_0^{\hbar\beta} d\tau \mathcal{G}_{\sigma, \sigma'}^{\text{Hartree}}(\tau) e^{i\omega_n \tau} \\ &= \hbar \oint_{\substack{\tau, \tau_1, \tau_2 \in [0, \hbar\beta] \\ \mu, \mu' \in \{\downarrow, \uparrow\} \\ \mathbf{k}, \mathbf{k}', \mathbf{q}, \mathbf{q}' \in Q \\ m, m', n, n' \in \{1, 2, \dots\} \\ v, v' \in \{1, 2, \dots, rd\}}} + \frac{g_{mnv}(\mathbf{k}, \mathbf{q}) g_{m'n'v'}(\mathbf{k}', \mathbf{q}')}{N_p} \mathcal{G}_{\substack{\mathbf{p}, \mathbf{k}+\mathbf{q} \\ b, m \\ \sigma, \mu}}^0(\tau - \tau_1) \mathcal{G}_{\substack{\mathbf{k}, \mathbf{p}' \\ n, b' \\ \mu, \sigma'}}^0(\tau_1 - 0) \mathcal{G}_{\substack{\mathbf{k}', \mathbf{k}'+\mathbf{q}' \\ n', m' \\ \mu', \mu'}}^0(0) \mathcal{D}_{\substack{\mathbf{q}, -\mathbf{q}' \\ v, v'}}^0(\tau_1 - \tau_2) e^{i\omega_n \tau} \\ &= \sum_{\substack{\mu, \mu' \in \{\downarrow, \uparrow\} \\ \mathbf{k}, \mathbf{k}', \mathbf{q}, \mathbf{q}' \in Q \\ m, m', n, n' \in \{1, 2, \dots\} \\ v, v' \in \{1, 2, \dots, rd\}}} \mathcal{G}_{\substack{\mathbf{p}, \mathbf{k}+\mathbf{q} \\ b, m \\ \sigma, \mu}}^0(i\omega_n) \left(\frac{1}{\beta} \sum_{\omega_m \in \Omega_{-1}} + \frac{g_{mnv}(\mathbf{k}, \mathbf{q}) g_{m'n'v'}(\mathbf{k}', \mathbf{q}')}{N_p} \mathcal{G}_{\substack{\mathbf{k}', \mathbf{k}'+\mathbf{q}' \\ n', m' \\ \mu', \mu'}}^0(i\omega_m) \mathcal{D}_{\substack{\mathbf{q}, -\mathbf{q}' \\ v, v'}}^0(0) \right) \mathcal{G}_{\substack{\mathbf{k}, \mathbf{p}' \\ n, b' \\ \mu, \sigma'}}^0(i\omega_n) \end{aligned}$$

Fourier-Matsubara-Component of the Fock-Contribution

Lemma 26.2 It holds

$$\mathcal{G}_{\sigma,\sigma'}^{\text{Fock}}(i\omega_n) = \sum_{\substack{\mu,\mu' \in \{\downarrow,\uparrow\} \\ \mathbf{k},\mathbf{k}',\mathbf{q},\mathbf{q}' \in \mathbf{Q} \\ m,m',n,n' \in \{1,2,\dots\} \\ v,v' \in \{1,2,\dots,rd\}}} \mathcal{G}_0(i\omega_n) \left(\frac{1}{\beta} \sum_{\omega_{n_2} \in \Omega_{-1}} -\frac{g_{mnv}(\mathbf{k},\mathbf{q})g_{m'n'v'}(\mathbf{k}',\mathbf{q}')}{N_p} \mathcal{G}_0(i\omega_{n_2}) \mathcal{D}_0(i(\omega_n - \omega_{n_2})) \right) \mathcal{G}_0(i\omega_n) \quad (26.31)$$

Proof 26.5

We will now focus on the temporal components of the Fock-contribution to the Green-function. For this we consider the function

$$\tilde{\mathcal{G}}^{\text{Fock}}(\tau) = \iint_{[0,\hbar\beta]^2} d(\tau_1, \tau_2) \mathcal{G}_0(\tau - \tau_1) \mathcal{G}_0(\tau_1 - \tau_2) \mathcal{G}_0(\tau_2) \mathcal{D}_0(\tau_1 - \tau_2) \quad (26.32)$$

and this gives for the Matsubara-components

$$\begin{aligned} \tilde{\mathcal{G}}^{\text{Fock}}(i\omega_n) &= \int_0^{\hbar\beta} d\tau e^{i\omega_n \tau} \tilde{\mathcal{G}}^{\text{Fock}}(\tau) \\ &= \iiint_{[0,\hbar\beta]^3} d(\tau, \tau_1, \tau_2) e^{i\omega_n \tau} \mathcal{G}_0(\tau - \tau_1) \mathcal{G}_0(\tau_1 - \tau_2) \mathcal{G}_0(\tau_2) \mathcal{D}_0(\tau_1 - \tau_2) \\ &= \dots \end{aligned}$$

now we expand the statistical functions according to Lemma TBReferenced and we obtain

$$\begin{aligned} \dots &= \sum_{\substack{\omega_{n_1}, \omega_{n_2}, \omega_{n_3} \in \Omega_{-1} \\ \omega_{n_4} \in \Omega_{+1}}} \mathcal{G}_0(\omega_{n_1}) \mathcal{G}_0(\omega_{n_2}) \mathcal{G}_0(\omega_{n_3}) \mathcal{D}_0(\omega_{n_4}) \frac{1}{(\hbar\beta)^4} \iiint_{[0,\hbar\beta]^3} d(\tau, \tau_1, \tau_2) e^{i(\omega_n \tau - \omega_{n_1}(\tau - \tau_1) - \omega_{n_2}(\tau_1 - \tau_2) - \omega_{n_3} \tau_2 - \omega_{n_4}(\tau_1 - \tau_2))} \\ &= \dots \end{aligned}$$

and we find for the integral-factor

$$\begin{aligned} &\iiint_{[0,\hbar\beta]^3} d(\tau, \tau_1, \tau_2) e^{i(\omega_n \tau - \omega_{n_1}(\tau - \tau_1) - \omega_{n_2}(\tau_1 - \tau_2) - \omega_{n_3} \tau_2 - \omega_{n_4}(\tau_1 - \tau_2))} \\ &= \int_0^{\hbar\beta} d\tau e^{i\tau(\omega_n - \omega_{n_1})} \int_0^{\hbar\beta} d\tau_1 e^{i\tau_1(\omega_{n_1} - \omega_{n_2} - \omega_{n_4})} \int_0^{\hbar\beta} d\tau_2 e^{i\tau_2(\omega_{n_2} - \omega_{n_3} + \omega_{n_4})} \\ &= (\hbar\beta)^3 \delta_{\omega_n, \omega_{n_1}} \delta_{\omega_{n_1}, \omega_{n_2} + \omega_{n_4}} \delta_{\omega_{n_2} + \omega_{n_4}, \omega_{n_3}} \end{aligned}$$

We use the first delta-function to eliminate the summation over ω_{n_1} . The second delta-function can be simplified according to

$$\begin{aligned}
 \delta_{\omega_{n_1}, \omega_{n_2} + \omega_{n_4}} &= \delta_{\omega_{n_2}, \omega_{n_1} - \omega_{n_4}} \\
 &= \delta_{\omega_{n_2}, \omega_{n_1} - \omega_{n_3} + \omega_{n_2}} \\
 &= \delta_{\omega_{n_3}, \omega_{n_1}}
 \end{aligned}$$

and will eliminate the summation over ω_{n_3} . For the third delta-relation we simplify it to

$$\delta_{\omega_{n_2} + \omega_{n_4}, \omega_{n_3}} = \delta_{\omega_{n_4}, \omega_{n_3} - \omega_{n_2}}$$

and we will use this to eliminate the summation over ω_{n_4} .

This yields then

$$\begin{aligned}
 \tilde{\mathcal{G}}^{\text{Fock}}(i\omega_n) &= \dots = \sum_{\omega_{n_2} \in \Omega_{-1}} \frac{1}{\hbar\beta} \mathcal{G}_0(\omega_n) \mathcal{G}_0(\omega_{n_2}) \mathcal{G}_0(\omega_n) \mathcal{D}_0(i(\omega_n - \omega_{n_2})) \\
 &\equiv \mathcal{G}_0(\omega_n) \left(\frac{1}{\hbar\beta} \sum_{\omega_{n_2} \in \Omega_{-1}} \mathcal{G}_0(\omega_{n_2}) \mathcal{D}_0(i(\omega_n - \omega_{n_2})) \right) \mathcal{G}_0(\omega_n)
 \end{aligned}$$

Using this identity for the Matsubara-component of a function proves the Lemma: We can compute the Fourier-Matsubara-component as

$$\begin{aligned}
 \mathcal{G}_{\sigma, \sigma'}^{\text{Fock}}(i\omega_n) &= \int_0^{\hbar\beta} d\tau \mathcal{G}_{\sigma, \sigma'}^{\text{Fock}}(\tau) e^{i\omega_n \tau} \\
 &= \hbar \oint \tau, \tau_1, \tau_2 \in [0, \hbar\beta] \quad - \frac{g_{mn\nu}(\mathbf{k}, \mathbf{q}) g_{m'n'\nu'}(\mathbf{k}', \mathbf{q}')}{N_p} \mathcal{G}_{\substack{\mathbf{p}, \mathbf{k} + \mathbf{q} \\ b, m \\ \sigma, \mu}}^0(\tau - \tau_1) \mathcal{G}_{\substack{\mathbf{k}, \mathbf{k}' + \mathbf{q}' \\ n, m' \\ \mu, \mu'}}^0(\tau_1 - \tau_2) \mathcal{G}_{\substack{\mathbf{k}', \mathbf{p}' \\ n', b' \\ \mu', \sigma'}}^0(\tau_2 - 0) \mathcal{D}_{\substack{\mathbf{q}, -\mathbf{q}' \\ \nu, \nu'}}^0(\tau_1 - \tau_2) e^{i\omega_n \tau} \\
 &= \sum_{\substack{\mu, \mu' \in \{\downarrow, \uparrow\} \\ \mathbf{k}, \mathbf{k}', \mathbf{q}, \mathbf{q}' \in \mathbf{Q} \\ m, m', n, n' \in \{1, 2, \dots\} \\ \nu, \nu' \in \{1, 2, \dots, rd\}}} \mathcal{G}_{\substack{\mathbf{p}, \mathbf{k} + \mathbf{q} \\ b, m \\ \sigma, \mu}}^0(i\omega_n) \left(\frac{1}{\beta} \sum_{\omega_{n_2} \in \Omega_{-1}} - \frac{g_{mn\nu}(\mathbf{k}, \mathbf{q}) g_{m'n'\nu'}(\mathbf{k}', \mathbf{q}')}{N_p} \mathcal{G}_{\substack{\mathbf{k}, \mathbf{k}' + \mathbf{q}' \\ n, m' \\ \mu, \mu'}}^0(i\omega_{n_2}) \mathcal{D}_{\substack{\mathbf{q}, -\mathbf{q}' \\ \nu, \nu'}}^0(i(\omega_n - \omega_{n_2})) \right) \mathcal{G}_{\substack{\mathbf{k}', \mathbf{p}' \\ n', b' \\ \mu', \sigma'}}^0(i\omega_n).
 \end{aligned} \tag{26.33}$$

which proves the lemma.

Continuation of the Initial Proof

Continuing with the initial proof, we obtained in total

$$\begin{aligned}
 \mathcal{G}_{b,b'}^{(2)}(i\omega_n) &= \mathcal{G}_{\sigma,\sigma'}^{\text{Hartree}}(i\omega_n) + \mathcal{G}_{\sigma,\sigma'}^{\text{Fock}}(i\omega_n) \\
 &\equiv \sum_{\substack{\mu,\mu' \in \{\downarrow,\uparrow\} \\ \mathbf{k},\mathbf{k}',\mathbf{q},\mathbf{q}' \in Q \\ m,m',n,n' \in \{1,2,\dots\} \\ v,v' \in \{1,2,\dots,rd\}}} \mathcal{G}_{\substack{\mathbf{p},\mathbf{k}+\mathbf{q} \\ b,m \\ \sigma,\mu}}^0(i\omega_n) \left(\frac{1}{\beta} \sum_{\omega_m \in \Omega_{-1}} + \frac{g_{mnv}(\mathbf{k},\mathbf{q})g_{m'n'v'}(\mathbf{k}',\mathbf{q}')}{N_p} \mathcal{G}_{\substack{\mathbf{k}',\mathbf{k}'+\mathbf{q}' \\ n',m' \\ \mu',\mu'}}^0(i\omega_m) \mathcal{D}_{\substack{\mathbf{q},-\mathbf{q}' \\ v,v'}}^0(0) \right) \mathcal{G}_{\substack{\mathbf{k},\mathbf{p}' \\ n,b' \\ \mu,\sigma'}}^0(i\omega_n) \\
 &+ \sum_{\substack{\mu,\mu' \in \{\downarrow,\uparrow\} \\ \mathbf{k},\mathbf{k}',\mathbf{q},\mathbf{q}' \in Q \\ m,m',n,n' \in \{1,2,\dots\} \\ v,v' \in \{1,2,\dots,rd\}}} \mathcal{G}_{\substack{\mathbf{p},\mathbf{k}+\mathbf{q} \\ b,m \\ \sigma,\mu}}^0(i\omega_n) \left(\frac{1}{\beta} \sum_{\omega_{n_2} \in \Omega_{-1}} - \frac{g_{mnv}(\mathbf{k},\mathbf{q})g_{m'n'v'}(\mathbf{k}',\mathbf{q}')}{N_p} \mathcal{G}_{\substack{\mathbf{k},\mathbf{k}'+\mathbf{q}' \\ n,m' \\ \mu,\mu'}}^0(i\omega_{n_2}) \mathcal{D}_{\substack{\mathbf{q},-\mathbf{q}' \\ v,v'}}^0(i(\omega_n - \omega_{n_2})) \right) \mathcal{G}_{\substack{\mathbf{k}',\mathbf{p}' \\ n',b' \\ \mu',\sigma'}}^0(i\omega_n) \\
 &\stackrel{\mathbf{q} \leftrightarrow \mathbf{q}-\mathbf{k}}{=} \sum_{\substack{\mathbf{k},\mathbf{q} \in Q \\ \mu \in \{\downarrow,\uparrow\} \\ m,n \in \{1,2,\dots\}}} \mathcal{G}_{\substack{\mathbf{p},\mathbf{q} \\ b,m \\ \sigma,\mu}}^0(i\omega_n) \left(\frac{1}{\beta} \sum_{\substack{\mathbf{k}',\mathbf{q}' \in Q \\ \mu' \in \{\downarrow,\uparrow\} \\ m',n' \in \{1,2,\dots\} \\ v,v' \in \{1,2,\dots,rd\}}} \frac{g_{mnv}(\mathbf{k},\mathbf{q}-\mathbf{k})g_{m'n'v'}(\mathbf{k}',\mathbf{q}')}{N_p} \mathcal{D}_{\substack{\mathbf{q}-\mathbf{k},-\mathbf{q}' \\ v,v'}}^0(0) \sum_{\omega_m \in \Omega_{-1}} \mathcal{G}_{\substack{\mathbf{k}',\mathbf{k}'+\mathbf{q}' \\ n',m' \\ \mu',\mu'}}^0(i\omega_m) \right) \mathcal{G}_{\substack{\mathbf{k},\mathbf{p}' \\ n,b' \\ \mu,\sigma'}}^0(i\omega_n) \\
 &+ \sum_{\substack{\mathbf{k}',\mathbf{q} \in Q \\ \mu,\mu' \in \{\downarrow,\uparrow\} \\ m,n' \in \{1,2,\dots\}}} \mathcal{G}_{\substack{\mathbf{p},\mathbf{q} \\ b,m \\ \sigma,\mu}}^0(i\omega_n) \left(\sum_{\substack{\mathbf{k},\mathbf{q}' \in Q \\ m',n \in \{1,2,\dots\} \\ v,v' \in \{1,2,\dots,rd\}}} \frac{1}{\beta} \sum_{\omega_{n_2} \in \Omega_{-1}} - \frac{g_{mnv}(\mathbf{k},\mathbf{q}-\mathbf{k})g_{m'n'v'}(\mathbf{k}',\mathbf{q}')}{N_p} \mathcal{G}_{\substack{\mathbf{k},\mathbf{k}'+\mathbf{q}' \\ n,m' \\ \mu,\mu'}}^0(i\omega_{n_2}) \mathcal{D}_{\substack{\mathbf{q}-\mathbf{k},-\mathbf{q}' \\ v,v'}}^0(i(\omega_n - \omega_{n_2})) \right) \mathcal{G}_{\substack{\mathbf{k}',\mathbf{p}' \\ n',b' \\ \mu',\sigma'}}^0(i\omega_n) \\
 &= \sum_{\substack{\mathbf{k},\mathbf{q} \in Q \\ \mu \in \{\downarrow,\uparrow\} \\ m,n \in \{1,2,\dots\}}} \mathcal{G}_{\substack{\mathbf{p},\mathbf{q} \\ b,m \\ \sigma,\mu}}^0(i\omega_n) \Sigma_{\substack{\mathbf{q},\mathbf{k} \\ m,n \\ \mu,\mu}}^{\text{Hartree}}(i\omega_n) \mathcal{G}_{\substack{\mathbf{k},\mathbf{p}' \\ n,b' \\ \mu,\sigma'}}^0(i\omega_n) + \sum_{\substack{\mathbf{k}',\mathbf{q} \in Q \\ \mu \in \{\downarrow,\uparrow\} \\ m,n \in \{1,2,\dots\}}} \mathcal{G}_{\substack{\mathbf{p},\mathbf{q} \\ b,m \\ \sigma,\mu}}^0(i\omega_n) \Sigma_{\substack{\mathbf{q},\mathbf{k}' \\ m,n' \\ \mu,\mu'}}^{\text{Fock}}(i\omega_n) \mathcal{G}_{\substack{\mathbf{k}',\mathbf{p}' \\ n',b' \\ \mu',\sigma'}}^0(i\omega_n)
 \end{aligned} \tag{26.34}$$

In the next step we will perform the Matsubara-summations and simplify the self-energy expressions.

The Hartree-Contribution

Now since $\mathcal{G}_0$ is diagonal (in bands and wave-vectors) we see that only the $\mathbf{q}' = 0$ and $m' = n'$ summand contributes. Since $\mathcal{D}_0$ is diagonal in branches and wave-vectors only the $v = v'$ and $\mathbf{k} = \mathbf{q}$ summands contribute. This gives the simplification

$$\Sigma_{\substack{\mathbf{q},\mathbf{k} \\ m,n \\ \mu,\mu'}}^{\text{Hartree}}(i\omega_n) = \delta_{\mu,\mu'} \frac{1}{\beta} \sum_{\substack{\mathbf{k}' \in Q \\ \mu' \in \{\downarrow,\uparrow\} \\ m' \in \{1,2,\dots\} \\ v \in \{1,2,\dots,rd\}}} + \frac{g_{mnv}(\mathbf{k},0)g_{m'm'v}(\mathbf{k}',0)}{N_p} \mathcal{D}_{\substack{\mathbf{0},\mathbf{0} \\ v,v}}^0(0) \sum_{\omega_m \in \Omega_{-1}} \mathcal{G}_{\substack{\mathbf{k}',\mathbf{k}' \\ m',m' \\ \mu',\mu'}}^0(i\omega_m) \delta_{\mathbf{q},\mathbf{k}} \tag{26.35}$$

Therefore this contribution to the self-energy is diagonal in wave-vectors but not in the bands.

We now evaluate the summation over Matsubara-frequencies with the Residue-theorem. Omitting convergence-generating factors for notational simplicity, this gives with Corollary 15.22

$$\frac{1}{\beta} \sum_{\omega_m \in \Omega_{-1}} \mathcal{G}_{\substack{\mathbf{k}',\mathbf{k}' \\ m',m' \\ \mu',\mu'}}^0(i\omega_m) = \frac{1}{\hbar\beta} \sum_{\omega_m \in \Omega_{-1}} \frac{1}{i\omega_m - (\epsilon_{m'\mu'\mathbf{k}'} - \mu)} = F_{\beta}(\epsilon_{m'\mu'\mathbf{k}'} - \mu) \tag{26.36}$$

yielding as final expression

$$\Sigma_0^{\text{Hartree}}(i\omega_n) = \delta_{\mu,\mu'} \sum_{\substack{\mathbf{k}, \mathbf{q} \\ m, n \\ \mu, \mu'}} \frac{g_{mn\nu}(\mathbf{k}, 0) g_{m'n'\nu}(\mathbf{k}', 0)}{N_p} \mathcal{D}_0^{\mathbf{0}, \mathbf{0}}(0) F_\beta(\epsilon_{m'\mu'\mathbf{k}'} - \mu) \delta_{\mathbf{q}, \mathbf{k}} \quad (26.37)$$

The Fock-Contribution

Using now again the diagonality of the unperturbed Green-functions we obtain for the Fock-contribution to the self-energy

$$\begin{aligned} \Sigma_{\mathbf{q}, \mathbf{k}'}^{\text{Fock}}(i\omega_n) &= \sum_{\substack{\mathbf{k}, \mathbf{q}' \in Q \\ m', n \in \{1, 2, \dots\} \\ \mu, \mu' \\ v, v' \in \{1, 2, \dots, rd\}}} \frac{1}{\beta} \sum_{\omega_{n_2} \in \Omega_{-1}} - \frac{g_{mn\nu}(\mathbf{k}, \mathbf{q} - \mathbf{k}) g_{m'n'\nu'}(\mathbf{k}', \mathbf{q}')}{N_p} \mathcal{G}_{\mathbf{k}, \mathbf{k}'+\mathbf{q}'}^0(i\omega_{n_2}) \mathcal{D}_{\mathbf{q}-\mathbf{k}, -\mathbf{q}'}^0(i(\omega_n - \omega_{n_2})) \\ &= \sum_{\substack{\mathbf{q}' \in Q \\ m', n \in \{1, 2, \dots\} \\ v \in \{1, 2, \dots, rd\}}} \frac{1}{\beta} \sum_{\omega_{n_2} \in \Omega_{-1}} - \frac{g_{mn\nu}(\mathbf{k}' + \mathbf{q}', \mathbf{q} - (\mathbf{k}' + \mathbf{q}')) g_{m'n'\nu'}(\mathbf{k}', \mathbf{q}')}{N_p} \mathcal{G}_{\mathbf{k}'+\mathbf{q}', \mathbf{k}'+\mathbf{q}'}^0(i\omega_{n_2}) \mathcal{D}_{\mathbf{q}-(\mathbf{k}'+\mathbf{q}'), -\mathbf{q}'}^0(i(\omega_n - \omega_{n_2})) \\ &= \sum_{\substack{\mathbf{q}' \in Q \\ m', n \in \{1, 2, \dots\} \\ v \in \{1, 2, \dots, rd\}}} \frac{1}{\beta} \sum_{\omega_{n_2} \in \Omega_{-1}} - \frac{g_{mn\nu}(\mathbf{k}' + \mathbf{q}', \mathbf{q} - (\mathbf{k}' + \mathbf{q}')) g_{m'n'\nu'}(\mathbf{k}', \mathbf{q}')}{N_p} \mathcal{G}_{\mathbf{k}'+\mathbf{q}', \mathbf{k}'+\mathbf{q}'}^0(i\omega_{n_2}) \mathcal{D}_{-\mathbf{q}', -\mathbf{q}'}^0(i(\omega_n - \omega_{n_2})) \delta_{\mathbf{q}-(\mathbf{k}'+\mathbf{q}'), -\mathbf{q}'} \\ &= \sum_{\substack{\mathbf{q}' \in Q \\ m', n \in \{1, 2, \dots\} \\ v \in \{1, 2, \dots, rd\}}} \frac{1}{\beta} \sum_{\omega_{n_2} \in \Omega_{-1}} - \frac{g_{mn\nu}(\mathbf{k}' + \mathbf{q}', -\mathbf{q}') g_{m'n'\nu'}(\mathbf{k}', \mathbf{q}')}{N_p} \mathcal{G}_{\mathbf{k}'+\mathbf{q}', \mathbf{k}'+\mathbf{q}'}^0(i\omega_{n_2}) \mathcal{D}_{-\mathbf{q}', -\mathbf{q}'}^0(i(\omega_n - \omega_{n_2})) \delta_{\mathbf{q}, \mathbf{k}'} \\ &= \sum_{\substack{\mathbf{q}' \in Q \\ m', n \in \{1, 2, \dots\} \\ v \in \{1, 2, \dots, rd\}}} \frac{1}{\beta} \sum_{\omega_{n_2} \in \Omega_{-1}} - \frac{g_{nm\nu}^*(\mathbf{k}', \mathbf{q}') g_{m'n'\nu'}(\mathbf{k}', \mathbf{q}')}{N_p} \mathcal{G}_{\mathbf{k}'+\mathbf{q}', \mathbf{k}'+\mathbf{q}'}^0(i\omega_{n_2}) \mathcal{D}_{-\mathbf{q}', -\mathbf{q}'}^0(i(\omega_n - \omega_{n_2})) \delta_{\mathbf{q}, \mathbf{k}'} \\ &= \sum_{\substack{\mathbf{q}' \in Q \\ n \in \{1, 2, \dots\} \\ v \in \{1, 2, \dots, rd\}}} \frac{1}{\beta} \sum_{\omega_{n_2} \in \Omega_{-1}} - \frac{g_{nn'\nu'}(\mathbf{k}', \mathbf{q}') g_{nm\nu}^*(\mathbf{k}', \mathbf{q}')}{N_p} \mathcal{G}_{\mathbf{k}'+\mathbf{q}', \mathbf{k}'+\mathbf{q}'}^0(i\omega_{n_2}) \mathcal{D}_{\mathbf{q}', \mathbf{q}'}^0(i(\omega_n - \omega_{n_2})) \delta_{\mathbf{q}, \mathbf{k}'} \\ &= \sum_{\substack{\mathbf{q}' \in Q \\ n \in \{1, 2, \dots\} \\ v \in \{1, 2, \dots, rd\}}} - \frac{g_{nn'\nu'}(\mathbf{k}', \mathbf{q}') g_{nm\nu}^*(\mathbf{k}', \mathbf{q}')}{N_p} \delta_{\mathbf{q}, \mathbf{k}'} \delta_{\mu, \mu'} \frac{1}{\beta} \sum_{\omega_{n_2} \in \Omega_{-1}} \mathcal{G}_{\mathbf{k}'+\mathbf{q}', \mathbf{k}'+\mathbf{q}'}^0(i\omega_{n_2}) \mathcal{D}_{\mathbf{q}', \mathbf{q}'}^0(i(\omega_n - \omega_{n_2})) \end{aligned} \quad (26.38)$$

Note that we used $\omega_v(-\mathbf{q}) = \omega_v(\mathbf{q})$ in the last step. We now evaluate the summation over Matsubara-frequencies with the Residue-theorem. Omitting convergence-generating factors for notational simplicity, this confronts us with the task of evaluating the sum

$$\begin{aligned} \frac{1}{\beta} \sum_{\omega_{n_2} \in \Omega_{-1}} \mathcal{G}_{\mathbf{q}'+\mathbf{k}', \mathbf{q}'+\mathbf{k}'}^0(i\omega_{n_2}) \mathcal{D}_{\mathbf{q}', \mathbf{q}'}^0(i(\omega_n - \omega_{n_2})) &= \frac{1}{\hbar^2 \beta} \sum_{\omega_{n_2} \in \Omega_{-1}} \frac{1}{i\omega_{n_2} - (\epsilon_{n\mu\mathbf{q}'+\mathbf{k}'} - \mu)/\hbar} \frac{-2\omega_v(\mathbf{q}')}{(\omega_n - \omega_{n_2})^2 + \omega_v^2(\mathbf{q}')} \\ &\stackrel{\text{Corollary 15.23}}{=} \frac{-1}{\hbar} \left[\frac{B_{\hbar\beta}(\omega_v(\mathbf{q}')) + F_\beta(\xi_{\mathbf{q}'+\mathbf{k}'n\mu})}{i\omega_n - \xi_{\mathbf{q}'+\mathbf{k}'n\mu}/\hbar + \omega_v(\mathbf{q}')} + \frac{B_{\hbar\beta}(\omega_v(\mathbf{q}')) + 1 - F_\beta(\xi_{\mathbf{q}'+\mathbf{k}'n\mu})}{i\omega_n - \xi_{\mathbf{q}'+\mathbf{k}'n\mu}/\hbar - \omega_v(\mathbf{q}')} \right] \end{aligned} \quad (26.39)$$

whereas we introduced the variable $\xi_{\mathbf{q}'+\mathbf{k}'n\mu} = \epsilon_{\mathbf{q}'+\mathbf{k}'n\mu} - \mu$.

In total we obtain the final expression

$$\Sigma_{\mathbf{q}, \mathbf{k}'}^{\text{Fock}}(i\omega_n) = \sum_{\substack{\mathbf{q}' \in \mathbf{Q} \\ n \in \{1, 2, \dots\} \\ \nu \in \{1, 2, \dots, rd\}}} \frac{g_{nn'\nu}(\mathbf{k}', \mathbf{q}') g_{nm\nu}^*(\mathbf{k}', \mathbf{q}')}{N_p} \delta_{\mathbf{q}, \mathbf{k}'} \delta_{\mu, \mu'} \left[\frac{B_\beta(\hbar\omega_\nu(\mathbf{q}')) + F_\beta(\xi_{\mathbf{q}'+\mathbf{k}'n\mu})}{i\hbar\omega_n - \xi_{\mathbf{q}'+\mathbf{k}'n\mu} + \hbar\omega_\nu(\mathbf{q}')} + \frac{B_\beta(\hbar\omega_\nu(\mathbf{q}')) + 1 - F_\beta(\xi_{\mathbf{q}'+\mathbf{k}'n\mu})}{i\hbar\omega_n - \xi_{\mathbf{q}'+\mathbf{k}'n\mu} - \hbar\omega_\nu(\mathbf{q}')} \right] \quad (26.40)$$

26.4. Polarization of a Spin-Independent-Interaction between an Electron-Gas and a Phonon-Gas

Proof 26.6 of Theorem 19.4 From PHYSICAL REVIEW B 76, 165108 2007 („03“ in the phonon-folder) and / or Computer Physics Communications 209 (2016) 116–133 („04“ in the EPW-folder.)

We will use the following trick: Inverting the Dyson-equation with the von Neumann series on the one hand, and re-expressing the imaginary-time-phononic-Green-function with help of the field integral on the other hand, gives the two equations

$$\mathcal{D}_{\mathbf{p}, \mathbf{p}'}(i\omega_n) = \sum_{n=0}^{\infty} \left[(\mathcal{D}_0 \cdot \Pi)^n(i\omega_n) \cdot \mathcal{D}_0(i\omega_n) \right]_{\mathbf{p}, \mathbf{p}'} \quad (26.41)$$

$$\mathcal{D}_{\mathbf{p}, \mathbf{p}'}(i\omega_n) = \int_0^{\hbar\beta} d\tau e^{i\omega_n \tau} \mathcal{D}_{\mathbf{p}, \mathbf{p}'}(\tau) = \frac{-1}{\hbar} \int_0^{\hbar\beta} d\tau e^{i\omega_n \tau} \langle A_{\mathbf{p}\pi}(\tau) A_{-\mathbf{p}'\pi'}(0) \rangle \quad (26.42)$$

The polarization of the phononic Green-function will be a series in the interaction-parameters

$$\Pi(i\omega_n) = \sum_{j=0}^{\infty} \Pi^{(j)}(i\omega_n) \approx 0 + \Pi^{(2)}(i\omega_n) + 0 + \mathcal{O}(V^4), \quad (26.43)$$

whereas the first contribution will turn out to vanish since the interaction can be expanded as a sum of monomials of creation- and annihilation-operators with uneven degree. Inserting this in the upper-equation above and keeping the first non-vanishing terms yields the sets of equations

$$\mathcal{D}_{\mathbf{p}, \mathbf{p}'}(i\omega_n) = \mathcal{D}_{\mathbf{p}, \mathbf{p}'}^0(i\omega_n) + (\mathcal{D}_0 \cdot \Pi^{(2)} \cdot \mathcal{D}_0)_{\mathbf{p}, \mathbf{p}'}(i\omega_n) \quad (26.44)$$

$$\mathcal{D}_{\mathbf{p}, \mathbf{p}'}(i\omega_n) = \frac{-1}{\hbar} \int_0^{\hbar\beta} d\tau e^{i\omega_n \tau} \langle A_{\mathbf{p}\pi}(\tau) A_{-\mathbf{p}'\pi'}(0) \rangle \quad (26.45)$$

The lower equation can be calculated explicitly with help of the Wick-theorem, linked-cluster-theorem, and the residue-theorem, and then, by comparing the first non-vanishing coefficients with the upper-equation, we can conclude what $\Pi^{(2)}(i\omega_n)$ must be.

Fact 1: The action, corresponding to this system, is given by the equation above.

Fact 2: The phononic Green-function is written in second order perturbation-theory as

$$\begin{aligned}\mathcal{D}_{\mathbf{p},\mathbf{p}'}(\tau) &= \mathcal{D}_{\mathbf{p},\mathbf{p}'}^{(0)}(\tau) + \mathcal{D}_{\mathbf{p},\mathbf{p}'}^{(1)}(\tau) + \mathcal{D}_{\mathbf{p},\mathbf{p}'}^{(2)}(\tau) + \dots \\ &= \mathcal{D}_{\mathbf{p},\mathbf{p}'}^0(\tau) + 0 + \mathcal{D}_{\mathbf{p},\mathbf{p}'}^{(2)}(\tau) + \dots\end{aligned}$$

The second contribution reads

$$\frac{1}{2!} \sum_{\substack{\tau_1, \tau_2 \in [0, \hbar\beta] \\ \mu, \mu' \in \{\downarrow, \uparrow\} \\ \mathbf{k}, \mathbf{k}', \mathbf{q}, \mathbf{q}' \in \mathbf{Q} \\ m, m', n, n' \in \{1, 2, \dots\} \\ v, v' \in \{1, 2, \dots, rd\}}} \frac{-1}{\hbar} \frac{g_{mnv}(\mathbf{k}, \mathbf{q}) g_{m'n'v'}(\mathbf{k}', \mathbf{q}')}{\hbar^2 N_p} \langle A_{\mathbf{p}\pi}(\tau) A_{-\mathbf{p}'\pi'}(0) A_{\mathbf{q}v}(\tau_1) \bar{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1) \psi_{\mathbf{k}n\mu}(\tau_1) A_{\mathbf{q}'v'}(\tau_2) \bar{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2) \psi_{\mathbf{k}'n'\mu'}(\tau_2) \rangle_0$$

where only [C]onnected [D]iagrams are summed over (linked cluster theorem). We can therefore simply sum over all diagrams and discard the disconnected diagrams.

$$\begin{aligned}& \langle A_{\mathbf{p}\pi}(\tau) A_{-\mathbf{p}'\pi'}(0) A_{\mathbf{q}v}(\tau_1) \bar{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1) \psi_{\mathbf{k}n\mu}(\tau_1) A_{\mathbf{q}'v'}(\tau_2) \bar{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2) \psi_{\mathbf{k}'n'\mu'}(\tau_2) \rangle_0^{\text{CD}} \\ &= \langle A_{\mathbf{p}\pi}(\tau) A_{-\mathbf{p}'\pi'}(0) A_{\mathbf{q}v}(\tau_1) A_{\mathbf{q}'v'}(\tau_2) \rangle_0^{\text{CD}} \langle \bar{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1) \psi_{\mathbf{k}n\mu}(\tau_1) \bar{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2) \psi_{\mathbf{k}'n'\mu'}(\tau_2) \rangle_0^{\text{CD}}\end{aligned}$$

Computing the correlator of the phononic Green-functions yields two non-trivial contributions. Computing the correlator of the electronic Green-functions yields two more non-trivial contributions.

We therefore consider the correlator

$$\begin{aligned}\langle \bar{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1) \psi_{\mathbf{k}n\mu}(\tau_1) \bar{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2) \psi_{\mathbf{k}'n'\mu'}(\tau_2) \rangle_0 &= -\langle \psi_{\mathbf{k}n\mu}(\tau_1) \psi_{\mathbf{k}'n'\mu'}(\tau_2) \bar{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1) \bar{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2) \rangle_0 \\ &= -\langle \psi(A_1) \psi(A_2) \bar{\psi}(B_2) \bar{\psi}(B_1) \rangle_0 \\ &= -\sum_{\pi \in S_2} \text{sign}(\pi) \prod_{l=1}^2 \langle \psi(A_l) \bar{\psi}(B_{\pi(l)}) \rangle_0 \\ &= -\hbar^2 \sum_{\pi \in S_2} \text{sign}(\pi) \prod_{l=1}^2 \mathcal{G}_0(A_l, B_{\pi(l)}) \\ &= \dots\end{aligned}$$

The electronic correlator can now be evaluated with the Wick-theorem. In order to so, we will enumerate all the permutations of the group S_3 .

$$S_2 = \{\pi_1, \pi_2\} = \left\{ \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \right\} \quad \text{with signatures} \quad \text{sgn}(\pi_1, \pi_2) = (+1, -1). \quad (26.46)$$

this gives

$$\dots = -\hbar^2 \left(\mathcal{G}_{\mathbf{k}, \mathbf{k}'+\mathbf{q}}^0(\tau_1 - \tau_2) \mathcal{G}_{\mathbf{k}', \mathbf{k}+\mathbf{q}}^0(\tau_2 - \tau_1) - \mathcal{G}_{\mathbf{k}, \mathbf{k}+\mathbf{q}}^0(\tau_1 - \tau_1) \mathcal{G}_{\mathbf{k}', \mathbf{k}'+\mathbf{q}}^0(\tau_2 - \tau_2) \right)$$

Now we turn to the phononic correlators

$$\begin{aligned} \langle A_{\mathbf{p}\pi}(\tau) A_{-\mathbf{p}'\pi'}(0) A_{\mathbf{q}\nu}(\tau_1) A_{\mathbf{q}'\nu'}(\tau_2) \rangle_0 &\equiv \langle A(x_1) A(x_2) A(x_3) A(x_4) \rangle_0 \\ &= \sum_{\substack{\text{Pairings of} \\ \{x_1, \dots, x_4\}}} \langle A(x_{k_1}) A(x_{k_2}) \rangle_0 \langle A(x_{k_3}) A(x_{k_4}) \rangle_0 \\ &= \dots \end{aligned}$$

which gives [cf. Altland]

$$\begin{aligned} \dots &= \langle A(x_1) A(x_2) \rangle_0 \langle A(x_3) A(x_4) \rangle_0 + \langle A(x_1) A(x_3) \rangle_0 \langle A(x_2) A(x_4) \rangle_0 + \langle A(x_1) A(x_4) \rangle_0 \langle A(x_2) A(x_3) \rangle_0 \\ &= \langle A_{\mathbf{p}\pi}(\tau) A_{-\mathbf{p}'\pi'}(0) \rangle_0 \langle A_{\mathbf{q}\nu}(\tau_1) A_{\mathbf{q}'\nu'}(\tau_2) \rangle_0 + \langle A_{\mathbf{p}\pi}(\tau) A_{\mathbf{q}\nu}(\tau_1) \rangle_0 \langle A_{-\mathbf{p}'\pi'}(0) A_{\mathbf{q}'\nu'}(\tau_2) \rangle_0 + \langle A_{\mathbf{p}\pi}(\tau) A_{\mathbf{q}'\nu'}(\tau_2) \rangle_0 \langle A_{-\mathbf{p}'\pi'}(0) A_{\mathbf{q}\nu}(\tau_1) \rangle_0 \\ &= \hbar^2 \left(\mathcal{D}_{\substack{\pi, \pi' \\ \mathbf{p}, \mathbf{p}'}}^0(\tau) \mathcal{D}_{\substack{\nu, \nu' \\ \mathbf{q}, -\mathbf{q}'}}^0(\tau_1 - \tau_2) + \mathcal{D}_{\substack{\pi, \nu \\ \mathbf{p}, -\mathbf{q}}}^0(\tau - \tau_1) \mathcal{D}_{\substack{\pi', \nu' \\ -\mathbf{p}, -\mathbf{q}'}}^0(-\tau_2) + \mathcal{D}_{\substack{\pi, \nu' \\ \mathbf{p}, -\mathbf{q}'}}^0(\tau - \tau_2) \mathcal{D}_{\substack{\pi', \nu \\ -\mathbf{p}', -\mathbf{q}}}^0(-\tau_1) \right) \end{aligned}$$

whereas the first term will not contribute since it does not resemble a one particle irreducible diagram (or something like that... external vertices are not connected to inner ones...).

This gives us the intermediate result

$$\begin{aligned} \mathcal{D}_{\substack{\mathbf{p}, \mathbf{p}' \\ \pi, \pi'}}^{(2)}(\tau) &= \sum_{\substack{\tau_1, \tau_2 \in [0, \hbar\beta] \\ \mu, \mu' \in \{\downarrow, \uparrow\} \\ \mathbf{k}, \mathbf{k}', \mathbf{q}, \mathbf{q}' \in \mathbf{Q} \\ m, m', n, n' \in \{1, 2, \dots\} \\ \nu, \nu' \in \{1, 2, \dots, rd\}}} \frac{g_{mn\nu}(\mathbf{k}, \mathbf{q}) g_{m'n'\nu'}(\mathbf{k}', \mathbf{q}')}{2\hbar^{-1} N_p} \left(\mathcal{G}_{\substack{\mathbf{k}, \mathbf{k}'+\mathbf{q}' \\ n, m' \\ \mu, \mu'}}^0(\tau_1 - \tau_2) \mathcal{G}_{\substack{\mathbf{k}', \mathbf{k}+\mathbf{q} \\ n', m \\ \mu', \mu}}^0(\tau_2 - \tau_1) - \mathcal{G}_{\substack{\mathbf{k}, \mathbf{k}+\mathbf{q} \\ n, m \\ \mu, \mu}}^0(\tau_1 - \tau_1) \mathcal{G}_{\substack{\mathbf{k}', \mathbf{k}'+\mathbf{q}' \\ n', m' \\ \mu', \mu'}}^0(\tau_2 - \tau_2) \right) \\ &\quad \times \left(\mathcal{D}_{\substack{\pi, \nu \\ \mathbf{p}, -\mathbf{q}}}^0(\tau - \tau_1) \mathcal{D}_{\substack{\pi', \nu' \\ -\mathbf{p}, -\mathbf{q}'}}^0(-\tau_2) + \mathcal{D}_{\substack{\pi, \nu' \\ \mathbf{p}, -\mathbf{q}'}}^0(\tau - \tau_2) \mathcal{D}_{\substack{\pi', \nu \\ -\mathbf{p}', -\mathbf{q}}}^0(-\tau_1) \right) \end{aligned}$$

This yields four contributions to the phonon-propagator, which we enumerate with

- (1) – (i)
- (1) – (ii)
- (2) – (i)
- (2) – (ii)

We now consider a generic (periodic) function of the type

$$\mathcal{F}(\tau) = \iint_{[0, \hbar\beta]^2} d(\tau_1, \tau_2) \mathcal{D}(\tau - \tau_{k_1}) \mathcal{D}(\tau_{k_2}) \mathcal{G}(\tau_{k_3} - \tau_{k_4}) \mathcal{G}(\tau_{k_5} - \tau_{k_6}) \quad (26.47)$$

The Matsubara-component of this function reads

$$\begin{aligned}
 \mathcal{F}(i\omega_n) &= \iiint_{[0, \hbar\beta]^3} d(\tau, \tau_1, \tau_2) \mathcal{D}(\tau - \tau_{k_1}) \mathcal{D}(\tau_{k_2}) \mathcal{G}(\tau_{k_3} - \tau_{k_4}) \mathcal{G}(\tau_{k_5} - \tau_{k_6}) e^{i\omega_n \tau} \\
 &= \frac{1}{(\hbar\beta)^4} \sum_{\substack{\omega_{n_1}, \omega_{n_2} \in \Omega_{+1} \\ \omega_{n_3}, \omega_{n_4} \in \Omega_{-1}}} \mathcal{D}(i\omega_{n_1}) \mathcal{D}(i\omega_{n_2}) \mathcal{G}(i\omega_{n_3}) \mathcal{G}(i\omega_{n_4}) \iiint_{[0, \hbar\beta]^3} d(\tau, \tau_1, \tau_2) e^{i\tau(\omega_n - \omega_{n_1}) + i\tau_{k_1}\omega_{n_1} - i\tau_{k_2}\omega_{n_2} - i(\tau_{k_3} - \tau_{k_4})\omega_{n_3} - i\omega_{n_4}(\tau_{k_5} - \tau_{k_6})}
 \end{aligned}$$

We now compute the Matsubara-component of each contribution individually and sum up the results afterwards.

ToBeContinued.

27. Appendix - B-Sides

27.1. The Many-Body-Formalism for Indistinguishable Particles: Second Quantization - Operator Formalism

27.1.1. The Fock-Space

The construction of many-body Hilbert-spaces for indistinguishable particles is way more tedious. The initial situation (which one encounters quite naturally in typical materials) is as follows.

Notation 27.1 Let $\mathcal{H} = \text{span}_{\mathbb{C}}\{|1\rangle, |2\rangle, \dots, |d_{\mathcal{H}}\rangle\}$ be a single-particle Hilbert-space with (not necessarily, often-times orthonormal) basis and corresponding wavefunctions / state-vectors with coordinates x

$$\langle x | i \rangle \equiv \psi_i(x). \quad (27.1)$$

whereas $\langle , \rangle$ refers to the scalarproduct on $\mathcal{H}$.

Notation 27.2 In this section we denote $N \in \mathbb{N}_0$ to be a natural number.

Definition 27.3 We define the *unsymmetrized (and unphysical) N-particle-Hilbert-space* $\mathcal{H}_N$ as tensor product

$$\mathcal{H}_N = \mathcal{H} \otimes \mathcal{H} \otimes \dots \otimes \mathcal{H}. \quad (27.2)$$

Remark 27.3.1 This is not the Hilbert-space that one is allowed to work with. Only a subspace of it (depending on the "statistics" of the particles at hand) is relevant for computations.

Important Spanning Set / Basis 1 (of $\mathcal{H}_N$)

Corollary 27.4 A basis of $\mathcal{H}_N$ is given by the set

$$\mathcal{B}_{\mathcal{H}_N} \equiv \{|i_1, i_2, \dots, i_N\rangle \equiv |i_1\rangle \otimes |i_2\rangle \otimes \dots \otimes |i_N\rangle \mid i_j \in \{1, \dots, d_{\mathcal{H}}\}\} \quad (27.3)$$

Proof 27.4 See introductory lectures about tensor-products.

Remark 27.4.1 Recall that the *scalar-product* between two basis-vectors is defined as

$$\begin{aligned} (i_1, i_2, \dots, i_N | j_1, j_2, \dots, j_N) &\equiv (\langle i_1 | \otimes \langle i_2 | \otimes \dots \otimes \langle i_N |) (|j_1\rangle \otimes |j_2\rangle \otimes \dots \otimes |j_N\rangle) \\ &\equiv \langle i_1 | j_1 \rangle \langle i_2 | j_2 \rangle \dots \langle i_N | j_N \rangle \end{aligned}$$

and by sesqui-linearity thus on the entire Hilbert-space.

Remark 27.4.2 Recall that the basis yields a *completeness relation* on $\mathcal{H}_N$ given through projectors as

$$1_{\mathcal{H}_N} \equiv \sum_{i_1, i_2, \dots, i_N=1}^{d_H} |i_1, i_2, \dots, i_N\rangle \langle i_1, i_2, \dots, i_N|. \quad (27.4)$$

Notation 27.5 We write for the value of a wavefunction at some coordinate-tuple

$$\begin{aligned} \psi_{i_1, i_2, \dots, i_N}(x_1, x_2, \dots, x_N) &\equiv (x_1, x_2, \dots, x_N | i_1, i_2, \dots, i_N), \\ &\equiv (\langle x_1 | \otimes \langle x_2 | \otimes \dots \otimes \langle x_N |) (|i_1\rangle \otimes |i_2\rangle \otimes \dots \otimes |i_N\rangle), \\ &\equiv \psi_{i_1}(x_1) \psi_{i_2}(x_2) \dots \psi_{i_N}(x_N). \end{aligned}$$

Definition 27.6 For $\zeta \in \{-1, +1\}$, and $N \in \mathbb{N}_0$ we define the ζ -Fock-subspaces $\mathcal{F}_\zeta^N$ as

$$\begin{aligned} \mathcal{F}_\zeta^0 &= \mathbb{C}, \\ \mathcal{F}_\zeta^N &= \{ \psi \in \mathcal{H}_N \mid \forall \pi \in S_N : \psi(x_{\pi(1)}, \dots, x_{\pi(N)}) = \text{sign}(\pi)^{1-\theta(\zeta)} \psi(x_1, \dots, x_N) \} \quad \text{for } N \geq 1. \end{aligned} \quad (27.5)$$

Remark 27.6.1 It might seem redundant to consider a subspace with zero-particles. This has computational reasons. It will later turn out, that it is easier to compute observables, when one is working in the grand-canonical ensemble, which makes this construction (mathematically) neat.

Remark 27.6.2 We remark that the dimensions of the individual subspaces can be written explicitly in terms of N , ζ and d_H . We will compute them in Theorem 7.7. At this point we simply state the result that in the case of fermions (i.e. $\zeta = -1$) there is an $N_{\max} \equiv N_{\max}(d_H) \in \mathbb{N}$ such that $\mathcal{F}_\zeta^N = \{0\}$ for all $N > N_{\max}$.

Lemma 27.7 We define the *statistical-symmetrization-operator* as linear mapping

$$\begin{aligned} P_\zeta : \mathcal{H}_N &\rightarrow \mathcal{F}_\zeta^N \\ \psi &\mapsto P_\zeta \psi \end{aligned}$$

as

$$(P_\zeta \psi)(x_1, x_2, \dots, x_N) := \frac{1}{N!} \sum_{\pi \in S_N} \text{sign}(\pi)^{1-\theta(\zeta)} \psi(x_{\pi(1)}, x_{\pi(2)}, \dots, x_{\pi(N)}). \quad (27.6)$$

Proof 27.7 We have to prove that

$$(P_\zeta \psi)(x_{\sigma(1)}, \dots, x_{\sigma(N)}) = \text{sign}(\sigma)^{1-\theta(\zeta)} (P_\zeta \psi)(x_1, \dots, x_N) \quad (27.7)$$

which is easy to see after relabeling the summations and since $\text{sign}(\sigma) = \text{sign}(\sigma^{-1})$.

Corollary 27.8 The statistical-symmetrization-operator is a projector, i.e. it holds $P_\zeta \circ P_\zeta = P_\zeta$.

Proof 27.8 See Appendix, p. 550.

Lemma 27.9 The statistical symmetrization-operator P_ζ acts as the identity on the subspace $\mathcal{F}_\zeta^N$, i.e.

$$\forall \psi \in \mathcal{F}_\zeta^N : P_\zeta \psi \equiv 1_{\mathcal{F}_\zeta^N} \psi = \psi \quad (27.8)$$

Proof 27.9 Using the symmetry-property in Equation (27.5) we obtain a sum of $N!$ identical contributions which cancels out the prefactor in Equation (27.6).

Theorem 27.10 It follows

$$P_{+1} P_{-1} = P_{-1} P_{+1} = 0 \quad (27.9)$$

Proof 27.10 We compute for example

$$\begin{aligned} (P_{+1} P_{-1} \psi)(x_1, \dots, x_N) &= \frac{1}{N!} \sum_{\pi \in S_N} (P_{-1} \psi)(x_{\pi(1)}, \dots, x_{\pi(N)}) \\ &= \frac{1}{N!} \left(\sum_{\pi \in S_N} \text{sign}(\pi) \right) (P_{-1} \psi)(x_1, \dots, x_N) \\ &= 0 \end{aligned}$$

Lemma 27.11 For pure tensor-product states $|\psi\rangle$, i.e. states that can be written as

$$|\psi\rangle = \bigotimes_{j=1}^N |\psi_j\rangle \quad (27.10)$$

for some $|\psi_j\rangle \in \mathcal{H}$, it follows

$$P_\zeta |\psi\rangle = \frac{1}{N!} \sum_{\pi \in S_N} \text{sign}(\pi)^{1-\theta(\zeta)} \bigotimes_{j=1}^N |\psi_{\pi(j)}\rangle \quad (27.11)$$

and one obtains *Slater-determinants* and *permanents* for the coordinate representations as

$$(x_1, x_2, \dots, x_N | P_\zeta |\psi\rangle = \frac{1}{N!} \begin{vmatrix} \langle x_1 | \psi_1 \rangle_{\mathcal{H}} & \cdots & \langle x_1 | \psi_N \rangle_{\mathcal{H}} \\ \vdots & \ddots & \vdots \\ \langle x_N | \psi_1 \rangle_{\mathcal{H}} & \cdots & \langle x_N | \psi_N \rangle_{\mathcal{H}} \end{vmatrix}_\zeta. \quad (27.12)$$

Proof 27.11 See Appendix, p. 551.

Definition 27.12 For a permutation $\pi \in S_N$ we define its *action on a product-state*

$$|\psi\rangle = \bigotimes_{j=1}^N |\psi_j\rangle \quad (27.13)$$

as

$$\pi |\psi\rangle := \bigotimes_{j=1}^N |\psi_{\pi(j)}\rangle \quad (27.14)$$

Lemma 27.13 Now the statistical symmetrization-operator can be written for pure states as

$$P_\zeta := \frac{1}{N!} \sum_{\pi \in S_N} \text{sign}(\pi)^{1-\theta(\zeta)} \pi \quad (27.15)$$

Proof 27.13 Follows with Equation (27.11).

Remark 27.13.1 Notice that by linear extension this is equation can be used to act on any state in $\mathcal{H}_N$ (since every element can be expandend w.r.t. the basis in Equation (27.3)).

Lemma 27.14 The statistical symmetrization-operator P_ζ is hermitian w.r.t. the scalar-product on $\mathcal{H}_N$.

Proof 27.14 See Appendix, p. 552.

Important Spanning Set / Basis 2 (of $\mathcal{F}_\zeta^N$)

Definition 27.15 For $i_1, \dots, i_N \in \{1, 2, \dots, d_H\}$, we define *auxiliary-states* as

$$\begin{aligned} |i_1, i_2, \dots, i_N\rangle &= \frac{P_\zeta}{N!^{-1/2}} |i_1, i_2, \dots, i_N\rangle \\ &= \frac{1}{\sqrt{N!}} \sum_{\pi \in S_N} \text{sign}(\pi)^{1-\theta(\zeta)} |i_{\pi(1)}\rangle \otimes |i_{\pi(2)}\rangle \otimes \dots \otimes |i_{\pi(N)}\rangle. \end{aligned} \quad (27.16)$$

Remark 27.15.1 Notice that in the special case of $N = 1$ one finds $|i\rangle = |i\rangle$.

Remark 27.15.2 Introducing these states is a bit problematic from a pedagogical perspective, since they **only** serve in order to introduce the so-called „occupation number states“ later on (which will turn out to be the basis of states that we are **actually** interested in working with).

Lemma 27.16 The auxiliary-states have the property

$$\forall \sigma \in S_N : |i_{\sigma(1)}, i_{\sigma(2)}, \dots, i_{\sigma(N)}\rangle = \text{sign}(\sigma)^{1-\theta(\zeta)} |i_1, i_2, \dots, i_N\rangle \quad (27.17)$$

Proof 27.16 We prove this theorem by referring to the coordinate representation of the auxiliary-states. We therefore consider $\sigma \in S_N$ and we compute

$$\begin{aligned} (x_1, x_2, \dots, x_N | i_{\sigma(1)}, i_{\sigma(2)}, \dots, i_{\sigma(N)}) &= \frac{1}{\sqrt{N!}} \sum_{\pi \in S_N} \text{sign}(\pi)^{1-\theta(\zeta)} \prod_{l=1}^N \langle x_l | i_{\pi(\sigma(l))} \rangle \\ &= \text{sign}(\sigma)^{1-\theta(\zeta)} \frac{1}{\sqrt{N!}} \sum_{\pi \in S_N} \text{sign}(\pi \circ \sigma)^{1-\theta(\zeta)} \prod_{l=1}^N \langle x_l | i_{\pi(\sigma(l))} \rangle \\ &= \text{sign}(\sigma)^{1-\theta(\zeta)} (x_1, x_2, \dots, x_N | i_1, i_2, \dots, i_N) \end{aligned}$$

which proves the Lemma.

Remark 27.16.1 Notice that this implies in the fermionic case (i.e. $\zeta = -1$) that if $i_j = i_{j'}$, for some $j, j' \in \{1, \dots, N\}$, then $|i_1, i_2, \dots, i_N\rangle = 0$.

Corollary 27.17 One obtains the completeness relation on $\mathcal{F}_\zeta^N$ as

$$1_{\mathcal{F}_\zeta^N} \equiv \frac{1}{N!} \sum_{i_1, \dots, i_N=1}^{d_H} |i_1, \dots, i_N\rangle \langle i_1, \dots, i_N|. \quad (27.18)$$

Proof 27.17 We compute

$$\begin{aligned} \frac{1}{N!} \sum_{i_1, \dots, i_N=1}^{d_H} |i_1, \dots, i_N\rangle \langle i_1, \dots, i_N| &= \frac{1}{N!} \sum_{i_1, \dots, i_N=1}^{d_H} \sqrt{N!} P_\zeta |i_1, \dots, i_N\rangle \langle i_1, \dots, i_N| \sqrt{N!} P_\zeta \\ &= P_\zeta \left[\sum_{i_1, \dots, i_N=1}^{d_H} |i_1, \dots, i_N\rangle \langle i_1, \dots, i_N| \right] P_\zeta \\ &= P_\zeta 1_{\mathcal{H}_N} P_\zeta \\ &= P_\zeta \\ &= 1_{\mathcal{F}_\zeta^N}. \end{aligned}$$

whereas we used Equation (27.4), the fact that P_ζ is a projector (cf. Corollary 27.8) and that it acts as the identity on $\mathcal{F}_\zeta^N$ (cf. Lemma 27.9).

Lemma 27.18 It follows

$$\{i_1, i_2, \dots, i_N \mid j_1, j_2, \dots, j_N\} = \sum_{\pi \in S_N} \text{sign}(\pi)^{1-\theta(\zeta)} \prod_{k=1}^N \langle i_k \mid j_{\pi(k)} \rangle \quad (27.19)$$

Proof 27.18 We compute straightforwardly

$$\begin{aligned} \{i_1, i_2, \dots, i_N \mid j_1, j_2, \dots, j_N\} &= N! (i_1, i_2, \dots, i_N \mid P_\zeta^2 |j_1, j_2, \dots, j_N\rangle) \\ &= N! (i_1, i_2, \dots, i_N \mid P_\zeta |j_1, j_2, \dots, j_N\rangle) \\ &= \sum_{\pi \in S_N} \text{sign}(\pi)^{1-\theta(\zeta)} (i_1, \dots, i_N \mid (|j_{\pi(1)}\rangle \otimes |j_{\pi(2)}\rangle \otimes \dots \otimes |j_{\pi(N)}\rangle)) \\ &= \sum_{\pi \in S_N} \text{sign}(\pi)^{1-\theta(\zeta)} \prod_{k=1}^N \langle i_k \mid j_{\pi(k)} \rangle \\ &= \sum_{\pi \in S_N} \text{sign}(\pi)^{1-\theta(\zeta)} \langle i_1 \mid j_{\pi(1)} \rangle \langle i_2 \mid j_{\pi(2)} \rangle \dots \langle i_N \mid j_{\pi(N)} \rangle \end{aligned} \quad (27.20)$$

Remark 27.18.1 Notice that the auxiliary states in Definition 27.15 are not a basis, because of Lemma 27.16, but they form an overcomplete set, because of Corollary 27.17. Because of Lemma 27.18 it is evident that these states form an *overcomplete set*.

Definition 27.19 For each $k \in \{1, 2, \dots, d_H\}$, we introduce the *occupation-numbers of a state w.r.t. a basis* as functions

$$\begin{aligned} n_k : \{1, \dots, d_H\}^N &\rightarrow \mathbb{N} \\ (i_1, \dots, i_N) \equiv i &\mapsto n_k(i_1, \dots, i_N) := \sum_{l=1}^N \delta_{i_l, k} \equiv n_k, \end{aligned}$$

whereas we define $n_k(\bullet) \equiv 0$ for $N = 0$.

Remark 27.19.1 Notice that this implies with Remark 27.16.1 that in the fermionic case (i.e. $\zeta = -1$) it holds $n_k \in \{0, 1\}$ for all $k \in \{1, 2, \dots, d_H\}$.

Corollary 27.20 The overlap between different vectors is given by

$$\{i_1, i_2, \dots, i_N \mid j_1, j_2, \dots, j_N\} = \begin{cases} \text{sign}(\sigma)^{1-\theta(\zeta)} \prod_{k=1}^{d_H} n_k! & \text{if } \exists \sigma \in S_N : (i_1, i_2, \dots, i_N) = (j_{\sigma(1)}, \dots, j_{\sigma(N)}) \\ 0 & \text{otherwise.} \end{cases}$$

Proof 27.20 If the numbers $i_1, \dots, i_N$ are *not* a permutation of the numbers $j_1, \dots, j_N$, then there is in the right hand side of Equation (27.19) for each summand at least one factor which is zero, such that the entire sum vanishes. We divide the rest of the proof into two cases.

Case 1: $\zeta = -1$:

Then it follows with Remarks 27.19.1 and 27.16.1 that if the numbers are a permutation of each other, i.e. if there is some $\sigma \in S_N$ such that $(i_1, i_2, \dots, i_N) = (j_{\sigma(1)}, \dots, j_{\sigma(N)})$, then only one summand contributes in Equation (27.19), yielding the result from Equation (27.18) as

$$\begin{aligned} \{i_1, i_2, \dots, i_N \mid j_1, j_2, \dots, j_N\} &= \sum_{\pi \in S_N} \text{sign}(\pi) \prod_{k=1}^N \langle i_k \mid j_{\pi(k)} \rangle \\ &= \sum_{\pi \in S_N} \text{sign}(\pi) \prod_{k=1}^N \langle j_{\sigma(k)} \mid j_{\pi(k)} \rangle \\ &= \text{sign}(\sigma) \sum_{\pi \in S_N} \text{sign}(\sigma^{-1} \circ \pi) \prod_{k=1}^N \langle i_k \mid j_{(\sigma^{-1} \circ \pi)(k)} \rangle \\ &= \text{sign}(\sigma) \end{aligned} \tag{27.21}$$

since there is only one permutation that contributes to the sum.

Case 2: $\zeta = +1$:

For bosons, we use Lemma 27.16 in order to write $\{i_1, i_2, \dots, i_N \mid = \{j_1, j_2, \dots, j_N \mid$ and we obtain with Equation (27.18) then the overlap as

$$\begin{aligned}
 \{i_1, i_2, \dots, i_N \mid j_1, j_2, \dots, j_N\} &= \{j_1, j_2, \dots, j_N \mid j_1, j_2, \dots, j_N\} \\
 &= \sum_{\pi \in S_N} \prod_{k=1}^N \langle j_k \mid j_{\pi(k)} \rangle \\
 &= \dots
 \end{aligned} \tag{27.22}$$

the only permutations that contribute are now those, where identical j_k are being mapped onto each other. For each $k \in \{1, 2, \dots, d_H\}$ there are $n_k!$ such permutations, which yields the result.

Important Spanning Set / Basis 3 (of $\mathcal{F}_\zeta^N$)

Definition 27.21 For $i_1, \dots, i_N \in \{1, 2, \dots, d_H\}$, we define the *quasi-occupation-number-states* as

$$\begin{aligned}
 |i_1, i_2, \dots, i_N\rangle &\equiv \frac{1}{\sqrt{\prod_{k=1}^{d_H} n_k!}} |i_1, i_2, \dots, i_N\rangle \\
 &\equiv \frac{1}{\sqrt{N! \prod_{k=1}^{d_H} n_k!}} \sum_{\pi \in S_N} \text{sign}(\pi)^{1-\theta(\zeta)} |i_{\pi(1)}\rangle \otimes |i_{\pi(2)}\rangle \otimes \dots \otimes |i_{\pi(N)}\rangle
 \end{aligned} \tag{27.23}$$

with occupation-numbers as $\forall k \in \{1, 2, \dots, d_H\} : n_k \equiv n_k(i_1, \dots, i_N)$.

Remark 27.21.1 Notice that these states give *Slater-determinants* and *permanents* in the coordinate representation, i.e.

Corollary 27.22 One obtains the completeness relation

$$1_{\mathcal{F}_\zeta^N} \equiv \sum_{i_1, i_2, \dots, i_N=1}^{d_H} \frac{\prod_{k=1}^{d_H} n_k!}{N!} |i_1, i_2, \dots, i_N\rangle \langle i_1, i_2, \dots, i_N| \tag{27.24}$$

with occupation-numbers as $\forall k \in \{1, 2, \dots, d_H\} : n_k \equiv n_k(i_1, \dots, i_N)$.

Proof 27.22 Follows from the completeness relation in Corollary 27.17.

TBD: Overlap between different quasi-occupation-number-states and relation to Slater-determinants and permanents.

27.1.2. Creation- and Annihilation-Operators

Definition 27.23 In the zero-particle subspace $\mathcal{F}_\zeta^0 = \mathbb{C}$, we introduce *the vacuum-state* as

$$|0\rangle \equiv |\text{vac.}\rangle \equiv 1 \in \mathbb{C}.$$

Remark 27.23.1 Notice that with this supplement, we can extend the completeness-relation in Equation (27.24) to the entire Fock-space $\mathcal{F}_\zeta$ as

$$1_{\mathcal{F}_\zeta} = \sum_{N=0}^{\infty} \sum_{i_1, i_2, \dots, i_N=1}^{d_H} \frac{\prod_{k=1}^{d_H} n_k!}{N!} |i_1, i_2, \dots, i_N\rangle \langle i_1, i_2, \dots, i_N|. \quad (27.25)$$

with occupation-numbers as $\forall k \in \{1, 2, \dots, d_H\} : n_k \equiv n_k(i_1, \dots, i_N)$.

Definition 27.24 For each $k \in \{1, \dots, d_H\}$ we introduce a mapping as the (*particle*-)creation-operators

$$\begin{aligned} c_k^\dagger : \mathcal{F}_\zeta^N &\rightarrow \mathcal{F}_\zeta^{N+1} \\ |\psi\rangle &\mapsto c_k^\dagger |\psi\rangle \end{aligned}$$

by specifying its action on the overcomplete basis in Equation (27.15) and extending its definition by linearity on the entire Fock-space. We define for the states

$$\begin{aligned} c_k^\dagger |0\rangle &\equiv |k\rangle, \\ c_k^\dagger |i_1, i_2, \dots, i_N\rangle &\equiv |k, i_1, i_2, \dots, i_N\rangle \end{aligned} \quad (27.26)$$

Remark 27.24.1 Since these operators are mappings from the N -particle Hilbert-space to the $N + 1$ -particle Hilbert-space, the name "creation-operators" becomes evident.

Remark 27.24.2 Another thing to notice is that this definition / prefactor-convention is not unique within the literature. We are following the prefactor-convention from [Negele, Orland] in contrast to [TBD].

Corollary 27.25 It follows

$$c_k^\dagger |i_1, i_2, \dots, i_N\rangle = \sqrt{n_k + 1} |k, i_1, i_2, \dots, i_N\rangle \quad (27.27)$$

with occupation-numbers as $\forall k \in \{1, 2, \dots, d_H\} : n_k \equiv n_k(i_1, \dots, i_N)$.

Proof 27.25 Follows immediately by considering the respective definitions: We have

$$\begin{aligned} c_k^\dagger |i_1, i_2, \dots, i_N\rangle &\stackrel{\text{Eq. (27.23)}}{=} \frac{1}{\sqrt{\prod_{k=1}^{d_H} n_k!}} c_k^\dagger |i_1, i_2, \dots, i_N\rangle \\ &\stackrel{\text{Eq. (27.26)}}{=} \frac{1}{\sqrt{\prod_{k=1}^{d_H} n_k!}} |k, i_1, i_2, \dots, i_N\rangle \\ &\stackrel{\text{Eq. (27.23)}}{=} \sqrt{n_k + 1} |k, i_1, i_2, \dots, i_N\rangle \end{aligned} \quad (27.28)$$

with occupation-numbers $n_k \equiv n_k(i_1, \dots, i_N)$.

Remark 27.25.1 By repeated application of Corollary 27.25 one finds that any basis-vector can be rewritten as a product of particle creations on the vacuum-state as

$$|i_1, i_2, \dots, i_N\rangle = c_{i_1}^\dagger \cdots c_{i_N}^\dagger |0\rangle \iff |i_1, i_2, \dots, i_N\rangle = \frac{1}{\sqrt{\prod_{k=1}^{d_H} n_k!}} c_{i_1}^\dagger \cdots c_{i_N}^\dagger |0\rangle \quad (27.29)$$

with occupation-numbers $n_k \equiv n_k(i_1, \dots, i_N)$.

Corollary 27.26 One obtains the statistical commutation relations for the creation-operators

$$c_\lambda^\dagger c_\mu^\dagger - \zeta c_\mu^\dagger c_\lambda^\dagger \equiv [c_\lambda^\dagger, c_\mu^\dagger]_\zeta = 0 \quad (27.30)$$

Proof 27.26 We compute

$$\begin{aligned} c_\lambda^\dagger c_\mu^\dagger |i_1, i_2, \dots, i_N\rangle &= c_\lambda^\dagger |\mu, i_1, i_2, \dots, i_N\rangle = |\lambda, \mu, i_1, i_2, \dots, i_N\rangle \\ &= \zeta |\mu, \lambda, i_1, i_2, \dots, i_N\rangle = \zeta c_\mu^\dagger |\lambda, i_1, i_2, \dots, i_N\rangle = \zeta c_\mu^\dagger c_\lambda^\dagger |i_1, i_2, \dots, i_N\rangle \end{aligned} \quad (27.31)$$

Now rearranging terms yields the desired result.

Definition 27.27 We introduce the *particle-annihilation-operators* $\{c_1, \dots, c_{d_H}\}$ as the hermitian-adjoints of the creation-operators $\{c_1^\dagger, \dots, c_{d_H}^\dagger\}$ w.r.t. the scalar-product on the Fock-space.

Lemma 27.28 It follows

$$c_\lambda c_\mu - \zeta c_\mu c_\lambda \equiv [c_\lambda, c_\mu]_\zeta = 0 \quad (27.32)$$

Proof 27.28 Follows immediately from taking the hermitian adjoint of Equation (27.30).

Lemma 27.29 For the action of the hermitian adjoint it follows

$$c_j |i_1, i_2, \dots, i_N\rangle = \sum_{k=1}^N \zeta^{k-1} \delta_{j, i_k} |i_1, \dots, i_{k-1}, i_{k+1}, \dots, i_N\rangle \equiv \sum_{k=1}^N \zeta^{k-1} \delta_{j, i_k} |\hat{i}_k, \dots, i_N\rangle. \quad (27.33)$$

Proof 27.29 We prove this identity, by taking an arbitrary state $\{j_2, \dots, j_n\}$ and compute the overlap of the two sides of the equation. This gives

$$\begin{aligned}
 \{j_2, \dots, j_n | c_j | i_1, i_2, \dots, i_N\} &= \{j, j_2, \dots, j_n | i_1, i_2, \dots, i_N\} \\
 &\stackrel{j \equiv j_1, \text{Lemma (27.18)}}{=} \sum_{\pi \in S_N} \text{sign}(\pi)^{1-\theta(\zeta)} \prod_{k=1}^N \langle j_k | i_{\pi(k)} \rangle \\
 &= \dots
 \end{aligned} \tag{27.34}$$

In order to proceed, we have to make some considerations about the permutations $\pi \in S_N$:

Intermezzo

Every permutation in $\pi \in S_N$ is uniquely determined after one specifies

1. the image of the first element, i.e. $\pi(1) = k$ for some $k \in \{1, 2, \dots, N\}$ and
2. the permutation of the remaining $N - 1$ elements.

For some $\tau \in S_{N-1}$ that is acting bijectively on the set $\{2, \dots, N\}$, we define ...

- a *natural embedding* $\iota: S_{N-1} \rightarrow S_N$ with $\iota(\tau) = \tilde{\tau}$ as

$$\begin{aligned}
 \tilde{\tau}(1) &= 1, \\
 \tilde{\tau}(i) &= \tau(i) \quad \text{if } i > 1,
 \end{aligned} \tag{27.35}$$

- for each $k \in \{1, 2, \dots, N\}$ a *canonical insertion map* $\iota_k: S_{N-1} \rightarrow S_N$ with $\iota_k(\tau) = \pi$ as

$$\begin{aligned}
 \pi(1) &= k, \\
 \pi(i) &= \tau(i) - 1 \quad \text{if } i \geq 2 \wedge \tau(i) \leq k, \\
 \pi(i) &= \tau(i) \quad \text{if } i \geq 2 \wedge \tau(i) > k.
 \end{aligned} \tag{27.36}$$

Then it follows that

$$\pi = \iota_k(\tau) = (k \ k-1) \cdots (2 \ 1) \circ \iota(\tau) \quad \text{and} \quad \text{sign}(\pi) = \text{sign}(\iota_k(\tau)) = \text{sign}(\tau) \cdot (-1)^{k-1}. \tag{27.37}$$

This gives a disjoint union of the symmetric group S_N as

$$S_N = \bigsqcup_{k=1}^N \iota_k(S_{N-1}). \tag{27.38}$$

which allows us to reorder the sum in Equation (27.34) by summing over permutations $\tau \in S_{N-1}$

$$\begin{aligned}
 \dots &= \sum_{k=1}^N \sum_{\tau \in S_{N-1}} \text{sign}(i_k(\tau))^{1-\theta(\zeta)} \prod_{l=1}^N \langle j_l | i_{i_k(\tau)(l)} \rangle \\
 &= \sum_{k=1}^N \underbrace{(-1)^{(k-1)(1-\theta(\zeta))}}_{=\zeta^{k-1}} \sum_{\tau \in S_{N-1}} \text{sign}(\tau)^{1-\theta(\zeta)} \underbrace{\langle j_1 | i_{i_k(\tau)(1)} \rangle}_{=\langle j_1 | i_k \rangle = \delta_{j,i_k}} \prod_{l=2}^N \langle j_l | i_{i_k(\tau)(l)} \rangle \\
 &= \sum_{k=1}^N \zeta^{k-1} \delta_{j,i_k} \sum_{\tau \in S_{N-1}} \text{sign}(\tau)^{1-\theta(\zeta)} \prod_{l=2}^N \langle j_l | i_{\tau(l)} \rangle \\
 &= \sum_{k=1}^N \zeta^{k-1} \delta_{j,i_k} \{j_2, \dots, j_N | i_1, \dots, i_{k-1}, i_{k+1}, \dots, i_N\} \\
 &= \sum_{k=1}^N \zeta^{k-1} \delta_{j,i_k} \{j_2, \dots, j_N | i_1, \dots, \hat{i}_k, \dots, i_N\}
 \end{aligned} \tag{27.39}$$

Since this equation holds for any vector $\{j_2, \dots, j_N\}$, we can conclude that the identity claimed in the Lemma is correct.

Corollary 27.30 Analogously one finds

$$c_j |i_1, i_2, \dots, i_N\rangle = \frac{1}{\sqrt{n_j}} \sum_{k=1}^N \zeta^{k-1} \delta_{j,i_k} |i_1, \dots, i_{k-1}, i_{k+1}, \dots, i_N\rangle = \frac{1}{\sqrt{n_j}} \sum_{k=1}^N \zeta^{k-1} \delta_{j,i_k} |i_1, \dots, \hat{i}_k, \dots, i_N\rangle \tag{27.40}$$

Proof 27.30 Follows immediately with Equation (27.23) and the previous Lemma.

Theorem 27.31 In total, we arrive at creation- and annihilation-operators $\{c_1, c_1^\dagger, c_2, c_2^\dagger, \dots, c_{d_H}, c_{d_H}^\dagger\}$ obey the statistics-dependent commutation-relations

$$[c_i, c_j]_\zeta = [c_i^\dagger, c_j^\dagger]_\zeta = 0, \quad [c_i, c_j^\dagger]_\zeta = \delta_{i,j}$$

Proof 27.31

- TBD, cf. Negele, Orland page 15.

TBD: Basis Change etc according to page 15...

Important Spanning Set / Basis 4 (of $\mathcal{F}_\zeta^N$)

Definition 27.32 We consider a state $|i_1, i_2, \dots, i_N\rangle$. We denote its occupation-numbers $n_1, n_2, \dots, n_{d_H}$ with occupation-number-operators as usual $\forall k \in \{1, 2, \dots, d_H\} : n_k \equiv n_k(i_1, \dots, i_N)$. Then we introduce the *occupation-number-representation* of the state a.k.a. *occupation-number-states* as

$$|n_1, n_2, \dots, n_{d_H}\rangle_{\text{occ}} := |i_1, i_2, \dots, i_N\rangle \in \mathcal{F}_\zeta^N. \quad (27.41)$$

Remark 27.32.1 Usually one drops the index "occ" in the notation as it is clear from the context which states one is referring to, i.e.

$$|n_1, n_2, \dots, n_{d_H}\rangle_{\text{occ}} \equiv |n_1, n_2, \dots, n_{d_H}\rangle. \quad (27.42)$$

Remark 27.32.2 Notice that this is a *many-to-one-correspondence*.

Corollary 27.33 One obtains the simple formulae,

$$\begin{aligned} c_j |n_1, n_2, \dots, n_{d_H}\rangle &= \sqrt{n_j} |n_1, n_2, \dots, n_{j-1}, n_j - 1, n_{j+1}, \dots, n_{d_H}\rangle \\ c_j^\dagger |n_1, n_2, \dots, n_{d_H}\rangle &= \sqrt{n_j + 1} \zeta^{\sum_{l=1}^{j-1} n_l} |n_1, n_2, \dots, n_{j-1}, n_j + 1, n_{j+1}, \dots, n_{d_H}\rangle \end{aligned} \quad (27.43)$$

whereas the state on the r.h.s. is defined as 0 for $n_j = 0$.

Proof 27.33

- TBD, cf. Negele, Orland page 14 and Altland page 44.

Theorem 27.34 The occupation number states are an orthonormal basis of the Fock-space, i.e. one has the orthonormality condition

$$\langle n_1, n_2, \dots, n_{d_H} | n'_1, n'_2, \dots, n'_{d_H} \rangle = \prod_{l=1}^{d_H} \delta_{n_l, n'_l} \quad (27.44)$$

and the resolution of identity

$$\sum_{n_1, \dots, n_{d_H} \in \mathbb{N}_\zeta} |n_1, n_2, \dots, n_{d_H}\rangle \langle n_1, n_2, \dots, n_{d_H}| = 1_{\mathcal{F}_\zeta} \quad (27.45)$$

Proof 27.34

- Cf. Negele, Orland page 22, Nolting 2 page 277, Schwabl 2 page 1 onwards.

The second identity (resolution of identity) follows from Equation (27.25) and the fact that each occupation-number-state appears $\frac{N!}{\prod_k n_k!}$ times in the sum.

27.1.3. Operators on the Fock-Space

27.1.3.1. 1-Body-Operators

Corollary 27.35 We define the number-operator $n_j \equiv a_j^\dagger a_j$, which counts the "number of particles in a given state"

$$n_j |n_1, n_2, \dots, n_{d_H}\rangle = n_j |n_1, n_2, \dots, n_{d_H}\rangle \quad (27.46)$$

whereas on the l.h.s. of the equation n_j represents an operator, and on the r.h.s. of the equation it represents a number.

Proof 27.35 TBD.

Remark 27.35.1 Analog one also introduces the operator for the total number of particles

$$N = \sum_{j=1}^{d_H} n_j. \quad (27.47)$$

Theorem 27.36 One can write for a one-body-operator U that acts on occupation-number-states

$$U = \frac{1}{1!} \sum_{i,j=1}^{d_H} \langle i | U | j \rangle c_i^\dagger c_j \quad (27.48)$$

Proof 27.36

- TBD, cf. Negele, Orland page 17.

27.1.3.2. 2-Body-Operators

Theorem 27.37 One can write for a two-body-operator V that acts on occupation-number-states

$$V = \frac{1}{2!} \sum_{i,j,k,l} \langle i, j | U | k, l \rangle c_i^\dagger c_j^\dagger c_l c_k \quad (27.49)$$

Proof 27.37

- TBD, cf. Negele, Orland page 18.

27.1.3.3. n-Body-Operators

Theorem 27.38 For an n -body-operator L we write

$$L = \sum_{\substack{i_1 < \dots < i_n \\ j_1 < j_2 < \dots < j_{n-1} < j_n}} L_{j_1, \dots, j_n}^{i_1, \dots, i_n} a_{i_1}^\dagger \dots a_{i_n}^\dagger a_{j_1} \dots a_{j_n},$$

with $L_{j_1, \dots, j_n}^{i_1, \dots, i_n} \in \mathbb{C}$ being defined through the scalar-product (and omitting units for notational simplicity).

Proof 27.38

- TBD, cf. Negele, Orland page 19.

Definition 27.39 An operator R on the many-particle-Hilbert-space is called an m -body-operator if it can be written in the following way

$$R|i_1, i_2, \dots, i_N) =: \frac{1}{m!} \sum_{1 \leq n_1 \neq n_2 \neq \dots \neq n_m \leq d_H} R_{n_1, n_2, \dots, n_m} |i_1, i_2, \dots, i_N) \quad (27.50)$$

Lemma 27.40 For any permutation $\pi \in S_N$ one finds

$$(i_1, i_2, \dots, i_N | R | i'_1, i'_2, \dots, i'_N) = (i_{\pi(1)}, i_{\pi(2)}, \dots, i_{\pi(N)} | R | i'_{\pi(1)}, i'_{\pi(2)}, \dots, i'_{\pi(N)}) \quad (27.51)$$

Proof 27.40

- TBD, cf. Negele, Orland p. 10.

Corollary 27.41 Notice the identity

$$\frac{(i_1, i_2, \dots, i_N | R | i'_1, i'_2, \dots, i'_N)}{(i_1, i_2, \dots, i_N | i'_1, i'_2, \dots, i'_N)} = \frac{1}{m!} \sum_{1 \leq n_1 \neq n_2 \neq \dots \neq n_m \leq d_H} \frac{(i_1, i_2, \dots, i_N | R | i'_1, i'_2, \dots, i'_N)}{\langle i_1 | i'_1 \rangle \langle i_2 | i'_2 \rangle \dots \langle i_N | i'_N \rangle} \quad (27.52)$$

Proof 27.41

- TBD, cf. Negele, Orland p. 10.

27.2. The Reason why so few People in the DFT-Community use Second Quantization

Notation 27.42 Let $L \in \text{End}(\mathcal{H})$ be a linear operator on the single-particle-Hilbert-space $\mathcal{H}$ with second-quantized version L^{2Q} acting on the Fock-space $\mathcal{F}_\zeta$ as

$$L^{2Q} \equiv \sum_{ij} \langle i | L | j \rangle c_i^\dagger c_j \in \mathcal{A}_\zeta(\mathcal{H} \oplus \mathcal{H}^*, B_\zeta). \quad (27.53)$$

Notation 27.43 We write for the spectral decomposition of the Hamiltonian

$$H - \mu 1_{\mathcal{H}} = \sum_{n=0}^{\infty} (E_n - \mu) |n\rangle \langle n| \quad (27.54)$$

Definition 27.44 Finally, we define

$$\rho_{\mathcal{H}} \equiv F_\beta(H - \mu 1_{\mathcal{H}}) := \sum_{n=0}^{\infty} F_\beta(E_n - \mu) |n\rangle \langle n|$$

Theorem 27.45 One finds the identity

$$\text{Tr}_{\mathcal{F}}(\rho_{\mathcal{F}} A^{2Q}) = \text{Tr}_{\mathcal{H}}(\rho_{\mathcal{H}} A). \quad (27.55)$$

Proof 27.45 We first note that the trace is invariant with respect to the basis used, so we may take the basis as the set of eigenvectors of H for the trace of the single-particle Hilbert space, and we consider the occupation number basis derived from it for the trace of the Fock-space. Then

$$Z = \prod_{n=0}^{\infty} (1 + e^{-\beta(E_n - \mu)}) \quad (27.56)$$

and we can write

$$\text{Tr}_{\mathcal{F}}(\rho_{\mathcal{F}} A^{2Q}) = \sum_{n_1, n_2, \dots \in \{0,1\}} \langle \Psi_{n_1, n_2, \dots}, \rho_{\mathcal{F}} A^{2Q} \Psi_{n_1, n_2, \dots} \rangle$$

Since

$$\rho_{\mathcal{F}} \Psi_{n_1, n_2, \dots} = \frac{1}{Z} e^{-\beta \sum_{j=0}^{\infty} n_j (E_j - \mu)} \Psi_{n_1, n_2, \dots} = \frac{1}{Z} \prod_{j=0}^{\infty} e^{-\beta n_j (E_j - \mu)} \Psi_{n_1, n_2, \dots},$$

and

$$\langle \Psi_{n_1, n_2, \dots}, A^{2Q} \Psi_{n_1, n_2, \dots} \rangle = \sum_{k,m} \langle e_k, A e_m \rangle \delta_{n_k,1} \delta_{n_m,1} \delta_{k,m},$$

we have

$$\text{Tr}_{\mathcal{F}}(\rho_{\mathcal{F}} A^{2Q}) = \sum_{k=0}^{\infty} \langle e_k, A e_k \rangle \frac{1}{Z} \sum_{n_1, n_2, \dots \in \{0,1\}} \prod_{j=0}^{\infty} e^{-\beta n_j (E_j - \mu)} \delta_{n_k,1}.$$

Note that

$$\sum_{n_1, n_2, \dots \in \{0,1\}} \prod_{j=0}^{\infty} e^{-\beta n_j (E_j - \mu)} \delta_{n_k,1} = (1 + e^{-\beta(E_1 - \mu)})(1 + e^{-\beta(E_2 - \mu)}) \dots e^{-\beta(E_k - \mu)} \dots (1 + e^{-\beta(E_r - \mu)}) \dots,$$

hence

$$\frac{1}{Z} \sum_{n_1, n_2, \dots \in \{0,1\}} \prod_{j=0}^{\infty} e^{-\beta n_j (E_j - \mu)} \delta_{n_k,1} = F_{\beta}(E_k - \mu),$$

and therefore

$$\text{Tr}_{\mathcal{F}}(\rho_{\mathcal{F}} A^{2Q}) = \sum_{k=0}^{\infty} \langle e_k, A e_k \rangle F_{\beta}(E_k - \mu) = \text{Tr}_{\mathcal{H}}(\rho_{\mathcal{H}} A),$$

proving the theorem.

27.3. The Zero-Temperature Limit

In this section, we show how one obtains the usual zero-temperature-expressions for Green-functions from the finite-temperature-expressions.

Notation 27.46 We write for the temperature-dependent expectation-value in the natural ensemble

$$\langle \dots \rangle := \frac{1}{Z} \text{Tr}_{\mathcal{H}} (\dots e^{-\beta \Xi}) \quad (27.57)$$

Case 1: The Non-Degenerate Case

Hypothesis 27.47 We assume, that we can enumerate (neglecting degeneracies) the ground-state, first excited state for the operator $\Xi = H - \eta\mu N$ as

$$|0\rangle, |1\rangle, |2\rangle, \dots \quad (27.58)$$

with energies

$$E_{|0\rangle} < E_{|1\rangle} < E_{|2\rangle} < \dots \iff E_0 < E_1 < E_2 < \dots \quad (27.59)$$

We further assume, that these states form an orthonormal basis of the Hilbert-space¹, such that the basis-independence of the trace yields

$$\text{Tr}_{\mathcal{H}} \bullet = \sum_{k=0}^{\infty} \langle k | \bullet | k \rangle \quad (27.60)$$

Corollary 27.48 It holds

$$\lim_{T \rightarrow 0} \langle \dots \rangle = \langle 0 | \dots | 0 \rangle \quad (27.61)$$

and therefore in particular

$$\lim_{T \rightarrow 0} G_{\alpha, \alpha'}(t, t') = -\frac{i}{\hbar} \langle 0 | T_t a_{\alpha}(t) a_{\alpha'}^{\dagger}(t') | 0 \rangle \quad (27.62)$$

etc for the other Green-functions / expectation-values.

Proof 27.48 We compute

¹Notice that this is for example the case for usual Slater-determinants / permanents (in the case of an underlying Fock-space).

$$\begin{aligned}
 \langle \dots \rangle &\equiv -\frac{i}{\hbar} \frac{\text{Tr}_{\mathcal{H}} \left(\dots e^{-\beta(H-\eta\mu N)} \right)}{\text{Tr}_{\mathcal{H}} e^{-\beta(H-\eta\mu N)}} \\
 &= -\frac{i}{\hbar} \frac{e^{-\beta E_0} \langle 0 | \dots | 0 \rangle + e^{-\beta E_1} \langle 1 | \dots | 1 \rangle + e^{-\beta E_2} \langle 2 | \dots | 2 \rangle + \dots}{e^{-\beta E_0} + e^{-\beta E_1} + e^{-\beta E_2} + \dots} \\
 &= -\frac{i}{\hbar} \frac{\langle 0 | \dots | 0 \rangle + e^{-\beta(E_1-E_0)} \langle 1 | \dots | 1 \rangle + e^{-\beta(E_2-E_0)} \langle 2 | \dots | 2 \rangle + \dots}{1 + e^{-\beta(E_1-E_0)} + e^{-\beta(E_2-E_0)} + \dots}
 \end{aligned}$$

Therefore one obtains

$$\lim_{T \rightarrow 0} \langle \dots \rangle = -\frac{i}{\hbar} \langle 0 | \dots | 0 \rangle$$

This yields then for example for the time-ordered Green-function

$$\begin{aligned}
 G_{\alpha,\alpha'}(t,t') &\equiv -\frac{i}{\hbar} \frac{\text{Tr}_{\mathcal{H}} e^{-\beta(H-\eta\mu N)} T_t a_{\alpha}(t) a_{\alpha'}^{\dagger}(t')}{\text{Tr}_{\mathcal{H}} e^{-\beta(H-\eta\mu N)}} \\
 &= -\frac{i}{\hbar} \frac{e^{-\beta E_0} \langle 0 | T_t a_{\alpha}(t) a_{\alpha'}^{\dagger}(t') | 0 \rangle + e^{-\beta E_1} \langle 1 | T_t a_{\alpha}(t) a_{\alpha'}^{\dagger}(t') | 1 \rangle + e^{-\beta E_2} \langle 2 | T_t a_{\alpha}(t) a_{\alpha'}^{\dagger}(t') | 2 \rangle + \dots}{e^{-\beta E_0} + e^{-\beta E_1} + e^{-\beta E_2} + \dots} \\
 &= -\frac{i}{\hbar} \frac{\langle 0 | T_t a_{\alpha}(t) a_{\alpha'}^{\dagger}(t') | 0 \rangle + e^{-\beta(E_1-E_0)} \langle 1 | T_t a_{\alpha}(t) a_{\alpha'}^{\dagger}(t') | 1 \rangle + e^{-\beta(E_2-E_0)} \langle 2 | T_t a_{\alpha}(t) a_{\alpha'}^{\dagger}(t') | 2 \rangle + \dots}{1 + e^{-\beta(E_1-E_0)} + e^{-\beta(E_2-E_0)} + \dots}
 \end{aligned}$$

Therefore one obtains

$$\lim_{T \rightarrow 0} G_{\alpha,\alpha'}(t,t') = -\frac{i}{\hbar} \langle 0 | T_t a_{\alpha}(t) a_{\alpha'}^{\dagger}(t') | 0 \rangle$$

Case 2: The Degenerate Case

Corollary 27.49 In the case of degenerate states

$$|0_1\rangle, |0_2\rangle, \dots |0_{n_0}\rangle, |1_1\rangle, |1_2\rangle, \dots |1_{n_1}\rangle, |2_1\rangle, |2_2\rangle, \dots |2_{n_2}\rangle, \dots \quad (27.63)$$

with energies

$$E_{|0_1\rangle} = E_{|0_2\rangle} = \dots = E_{|0_{n_0}\rangle} < E_{|1_1\rangle} = E_{|1_2\rangle} = \dots = E_{|1_{n_1}\rangle} < E_{|2_1\rangle} = E_{|2_2\rangle} = \dots = E_{|2_{n_2}\rangle} < \dots$$

It holds

$$\lim_{T \rightarrow 0} \langle \dots \rangle = \frac{1}{n_0} \sum_{j=1}^{n_0} \langle 0_j | \dots | 0_j \rangle \quad (27.64)$$

and therefore in particular

$$\lim_{T \rightarrow 0} G_{\alpha, \alpha'}(t, t') = -\frac{i}{\hbar} \sum_{j=1}^{n_0} \langle 0_j | T_t a_{\alpha}(t) a_{\alpha'}^{\dagger}(t') | 0_j \rangle \quad (27.65)$$

etc for the other Green-functions / expectation-values.

Proof 27.49 By straightforward generalization of the computation before.

27.4. An Interlude on Thermodynamic Variables in Materials Theory

In materials theory, there is a fundamental dilemma when it comes to the modelling:

On the one hand ... the mathematical apparatus for crystals (chapters 10 - 21 or, more or less equivalently in the literature [5, 8, 35, 47]) is relatively² simple in the natural ensemble (a.k.a. the *grand canonical ensemble* for purely electronic systems), which implies that we need as *input-parameters for predictions*

1. the chemical potential μ	yielding particle number as conjugated variable	$\langle N \rangle(\mu, T, V)$
2. the volume V	yielding pressure as conjugated variable	$\langle P \rangle(\mu, T, V)$
3. the temperature T	yielding entropy as conjugated variable	$\langle S \rangle(\mu, T, V)$

On the other hand the software apparatus for crystals [17, 49] is relatively simple in the *canonical ensemble*, which implies that we need as *input-parameters for predictions*³

1. the particle number N	yielding the chemical potential as conjugated variable	$\langle \mu \rangle(N, T, V)$
2. the volume V	yielding pressure as conjugated variable	$\langle P \rangle(N, T, V)$
3. the temperature T	yielding entropy as conjugated variable	$\langle S \rangle(N, T, V)$

This becomes further complicated by the fact, that many experiments (especially those for heat capacities) are almost always being performed at constant pressure (instead of a constant volume). One often argues, that these corrections are negligible at lower temperatures ([42] p. 89), but at higher temperatures they become very relevant [19].

27.4.1. The Chemical Potential for General Systems

Fact 27.50 Following the literature⁴ [30, 35, 9, 20] one *determines* the the chemical potential $\mu \equiv \mu(V, T)$ at a given temperature (and volume) by solving the equation

$$N_{\text{desired}} = \langle N \rangle(\mu(T, V), T, V) \quad (27.66)$$

for some desired particle-number $N_{\text{desired}} \geq 0$.

Remark 27.50.1 Notice that the r.h.s. of this equation can be computed with Green-functions by utilizing Equation (14.14).

²Relative to the other available ensembles! See also [50] page 115 stating

"The thermodynamic functions are easier to calculate explicitly if the constraint of a fixed number N of particles is dropped.

It is physically realistic for a system coupled to a reservoir of particles."

or similarly in [15] page 121 stating

"In equilibrium quantum statistical mechanics, the grand canonical ensemble is most useful."

³More explicitly: The particle number enters the calculations by setting up a k -grid and therefore determining the size of the crystal.

The volume enters the calculations by setting up the lattice-constant and lattice-vectors and basis-vectors.

⁴See, for example, [30] p. 14 or [35] p. 47 and 164, or [20] p. 921, or [9] p. 149.

Remark 27.50.2 For crystalline systems, one usually chooses N_{desired} in such a way, that the total crystal is charge-neutral N_{CN} . This implies that one simply counts the number of electrons that should be in the crystal. This is usually further decomposed into number of electrons per unit-cell times the number of unit-cells as in Equation (25.1).

Fact 27.51 Oftentimes it is so, that one does not have the desire to model a charge neutral crystal:

In cases of ...

- light doping
- external electric fields

one oftentimes *chooses* a value for the chemical potential that is somewhere "close" to the Fermi-energy.

Remark 27.51.1 One also encounters often phenomenological models (tight-binding Hamiltonians and generalizations of that) which simply *set* the chemical potential to "some" value that corresponds to a desired physical situation. In these cases, it is however important to closely monitor the implied the underlying particle number (especially as a function of temperature). One sees, for example in [23] that especially for small models, the temperature-dependence of the chemical potential can be strong. This is also emphasized in [35]:

"(...) where the chemical potential for the non-interacting system, (...) need not agree with the chemical potential of the interacting system (...)."

See [35], p. 164.

27.4.2. The Chemical Potential and the Fermi-Energy of Interacting Electrons in Arbitrary Materials

We simply combine the Fluctuation-Dissipation theorem in Equation (14.14) with the Dyson-Equation 15.13 for electrons (i.e. $\eta = 1 = -\zeta$) to obtain

$$\begin{aligned} \langle N \rangle &= \frac{-1}{\pi} \lim_{\epsilon \rightarrow 0} \oint_{\substack{\alpha \in \mathcal{A} \\ E \in (-\infty, +\infty)}} F_{\beta}(E) \operatorname{Im} G_{\alpha, \alpha}^{\text{ret}}(E + i\epsilon) \\ &= \frac{-1}{\pi} \lim_{\epsilon \rightarrow 0} \oint_{\substack{\alpha \in \mathcal{A} \\ E \in (-\infty, +\infty)}} F_{\beta}(E) \operatorname{Im} \left(\frac{1}{E + i\epsilon - (H + \Sigma - \mu)} \right)_{\alpha, \alpha} \end{aligned} \quad (27.67)$$

At this point, the only thing that remains left to do, is choosing the model (i.e. Hamiltonian and implied self-energy) that one wants to work with (cf. for example Section 8.4) and inserting Hamiltonian and self-energies. Notice that this can become increasingly difficult, depending on the level of approximation that one seeks to acquire. A common approximation is, for example, the diagonal bands-approximation (cf. Remark 19.3.1).

27.4.3. The Chemical Potential and the Fermi-Energy of Interacting Electrons in Crystals

Theorem 27.52 We compute the chemical potential, by constraining the number of particles

$$\langle N \rangle = \frac{-1}{\pi} \lim_{\delta \rightarrow 0} \oint_{\substack{\mathbf{k} \in Q \\ m \in \{1, \dots, \infty\} \\ E \in (-\infty, +\infty)}} F_{\beta}(E) \operatorname{Im} G_{m,m}^{\text{ret}}(E + i\delta, \mathbf{k})$$

Proof 27.52

$$\begin{aligned} \langle N \rangle &= \sum_{\substack{\mathbf{k} \in Q \\ m \in \{1, 2, 3, \dots\}}} \langle c_{m,\mathbf{k}}^{\dagger} c_{m,\mathbf{k}} \rangle \\ &= \frac{\hbar}{i} \sum_{\substack{\mathbf{k} \in Q \\ m \in \{1, 2, 3, \dots\}}} G_{m,m}^{<}(0)_{\mathbf{k},\mathbf{k}} \\ &= \frac{\hbar}{2\pi i} \oint_{\substack{\mathbf{k} \in Q \\ m \in \{1, 2, 3, \dots\} \\ \omega \in (-\infty, \infty)}} G_{m,m}^{<}(\omega)_{\mathbf{k},\mathbf{k}} \\ &= \frac{-\hbar}{\pi} \oint_{\substack{\mathbf{k} \in Q \\ m \in \{1, 2, 3, \dots\} \\ \omega \in (-\infty, \infty)}} F_{\beta}(\omega) G_{m,m}^{\text{ret}}(\omega + i\delta)_{\mathbf{k},\mathbf{k}} \end{aligned}$$

Remark 27.52.1 Recall that the retarded Green-function is given by analytical continuation of the Matsubara-Green-function which is determined by solving the Dyson-equation

$$\mathcal{G}(\mathbf{k}, i\omega_n) = \frac{1}{\hbar i\omega_n - (H_0(\mathbf{k}) + \Sigma(\mathbf{k}, i\omega_n) - \mu)} / \hbar \quad (27.68)$$

with non-interacting band-dispersion-matrix $H_0(\mathbf{k}) \equiv \text{diag}(\epsilon_1(\mathbf{k}), \dots, \epsilon_J(\mathbf{k}))$, whereas J represents the maximal number of bands that are being considered.

27.4.4. The Chemical Potential and the Fermi-Energy of Non-Interacting Electrons in Crystals

Theorem 27.53 For a non-interacting system

$$H = \sum_{\substack{\mathbf{k} \in Q \\ m \in \{1, 2, 3, \dots\}}} \epsilon_{m,\mathbf{k}} c_{m,\mathbf{k}}^{\dagger} c_{m,\mathbf{k}} \quad (27.69)$$

without degeneracies one finds

$$|0\rangle = \prod_{j=1}^N c_{m_j, \mathbf{k}_j}^\dagger |\text{vac.}\rangle =: \prod_{\substack{\mathbf{k} \in Q \\ m \in \{1,2,3,\dots\} \\ \epsilon_{m,\mathbf{k}} < \epsilon_{\text{Fermi}}}} c_{m,\mathbf{k}}^\dagger |\text{vac.}\rangle \quad (27.70)$$

with an (a priori unknown) Fermi-energy ϵ_{Fermi} that yet has to be determined.

Proof 27.53 Non-interacting systems can be diagonalized with standard-techniques from linear algebra, such that one arrives at the diagonal form of the Hamiltonian in second quantization. The orthonormal Slater-determinants span the entire Fock-space. Since there are no degeneracies the ground state is unique. The desired particle number N therefore dictates the Fermi-energy.

Lemma 27.54 It holds in the case of non-interacting electrons in a crystal

$$\lim_{T \rightarrow 0} \mu(T) = \epsilon_{\text{Fermi}} \quad (27.71)$$

whereas ϵ_{Fermi} is the highest occupied state in the Slater-determinant $|0\rangle$.

Proof 27.54 We compute the average particle twice.

On the one hand we have

$$\langle N \rangle = \sum_{\substack{\mathbf{k} \in Q \\ m \in \{1,2,3,\dots\}}} \langle c_{m,\mathbf{k}}^\dagger c_{m,\mathbf{k}} \rangle = \frac{\hbar}{i} \sum_{\substack{\mathbf{k} \in Q \\ m \in \{1,2,3,\dots\}}} G_{\mathbf{k},\mathbf{k}}^{<,m}(0) = \sum_{\substack{\mathbf{k} \in Q \\ m \in \{1,2,3,\dots\}}} F_\beta(\epsilon_{m,\mathbf{k}} - \mu) \quad (27.72)$$

on the other hand we have

$$\langle 0|N|0\rangle = \sum_{\substack{\mathbf{k} \in Q \\ m \in \{1,2,3,\dots\}}} \langle 0|c_{m,\mathbf{k}}^\dagger c_{m,\mathbf{k}}|0\rangle = \sum_{\substack{\mathbf{k} \in Q \\ m \in \{1,2,3,\dots\}}} \theta(\epsilon_{\text{Fermi}} - \epsilon_{m,\mathbf{k}}) \quad (27.73)$$

This gives in the zero-temperature-limit the equation

$$\sum_{\substack{\mathbf{k} \in Q \\ m \in \{1,2,3,\dots\}}} \theta(\epsilon_{\text{Fermi}} - \epsilon_{m,\mathbf{k}}) = \langle 0|N|0\rangle = \lim_{T \rightarrow 0} \langle N \rangle(T) = \sum_{\substack{\mathbf{k} \in Q \\ m \in \{1,2,3,\dots\}}} \theta(\mu(T=0) - \epsilon_{m,\mathbf{k}}) \quad (27.74)$$

direct comparison of the two expression yields $\mu(T=0) = \epsilon_{\text{Fermi}}$.

27.5. Specific Heat in the Sommerfeld-Approximation and the Dulong-Petit-Law

27.5.1. Intermezzo: The [C]anonical [E]nsemble

27.5.1.1. Formalism

Notation 27.55 We consider the case in which the above mentioned system has a volume V , and is being treated in the canonical ensemble with temperature T and particle number N and Hilbert-space $\mathcal{H} \equiv \mathcal{H}_N$. This gives us the three natural variables (N, T, V) .

Fact 27.56 In the absence of „spontaneous symmetry breaking“⁵ expectation-values in the canonical-ensemble are evaluated w.r.t. the state

$$\rho_{\text{CE}} = \frac{e^{-\beta H}}{\text{Tr}_{\mathcal{H}_N} e^{-\beta H}} \equiv \frac{1}{Z_{\text{CE}}} e^{-\beta H}, \quad (27.75)$$

whereas we introduced the canonical partition-function $Z_{\text{CE}} \equiv Z_{\text{CE}}(N, T, V)$.

Remark 27.56.1 Note that this is a solution to the stationary von Neumann-equation which represents a stationary-state.

Fact 27.57 Recall, that the thermodynamic potential⁶ in the canonical ensemble is given by the so-called grand free energy

$$F \equiv F(N, T, V) = -k_B T \log_e Z_{\text{CE}}(N, T, V) \quad (27.76)$$

from which one can conclude the usual relations for the conjugated variables chemical potential, entropy and pressure

$$\frac{\partial F}{\partial N} = \mu, \quad \frac{\partial F}{\partial T} = -S, \quad \frac{\partial F}{\partial V} = -P. \quad (27.77)$$

Fact 27.58 The ionic system of a crystal is usually modelled in the canonical ensemble.

Remark 27.58.1 Looking at the appearing equations, we will realize, that one is tempted (and it is quite usual to do so) to re-interpret the ionic system as a system in the so-called „grand canonical ensemble“ with the additional condition that one sets the chemical potential to zero. However, this is just a mnemonic⁷. A non-zero chemical potential can be defined also in the case of an ionic system.

⁵This subject goes far beyond the scope of this lecture.

⁶Recall that this is a scalar-quantity which is used to represent the thermodynamic state of a system.

⁷See attached excerpts at the end of this chapter.

27.5.1.2. Application to the Ionic System

Preparation:

Before we start calculating and predicting observable quantities for the ionic system, we collect all the things we need.

- Hilbert-space: $\mathcal{H} = \bigotimes_{j=1}^{rd_x N_p} L^2(\mathbb{R}^1) \simeq L^2(\mathbb{R}^{rd_x N_p})$
 - Basis: $\mathcal{B} = \left\{ |N\rangle \equiv |n_{\lambda_1}, n_{\lambda_2}, \dots, n_{\lambda_{rd_x N_p}}\rangle \equiv \prod_{j=1}^{rd_x N_p} |n_{\lambda_j}\rangle \mid \forall j \in \{1, \dots, rd_x N_p\} : n_{\lambda_j} \in \{0, 1, 2, \dots, \infty\} \right\}$
 - Expectation-value: $\langle A \rangle = \frac{\text{Tr}_{\mathcal{H}} A e^{-\beta H_{\text{harm.}}}}{\text{Tr}_{\mathcal{H}} e^{-\beta H_{\text{harm.}}}} \equiv \frac{\sum_{N \in \mathcal{B}} \langle N | A e^{-\beta H_{\text{harm.}}} | N \rangle}{\sum_{N \in \mathcal{B}} \langle N | e^{-\beta H_{\text{harm.}}} | N \rangle}$
- (27.78)

whereas A represents an observable (i.e. an hermitian operator, usually written in second quantized form).

Theorem 27.59 One finds for the average occupation-number

$$\langle n_{\nu \mathbf{q}} \rangle = \langle b_{\nu \mathbf{q}}^\dagger b_{\nu \mathbf{q}} \rangle = B_\beta(\hbar \omega_{\nu \mathbf{q}}) \quad (27.79)$$

Proof 27.59 See Appendix, p. 553.

Corollary 27.60 It follows for the internal energy of the ionic system

$$\langle H_{\text{harm.}} \rangle(N, V, T) = \sum_{\mathbf{q} \in Q} \sum_{\nu=1}^{rd_x} \hbar \omega_{\nu \mathbf{q}} \left(\langle b_{\nu \mathbf{q}}^\dagger b_{\nu \mathbf{q}} \rangle + \frac{1}{2} \right) = \sum_{\mathbf{q} \in Q} \sum_{\nu=1}^{rd_x} \hbar \omega_{\nu \mathbf{q}} \left(B_\beta(\hbar \omega_{\nu \mathbf{q}}) + \frac{1}{2} \right)$$

Proof 27.60 Follows immediately from the linearity of the expectation-value and Theorem 27.59.

27.5.1.3. The Low-Temperature Limit

Approximation 27.61 We assume that

$$k_B T \ll \hbar \omega_{\text{opt,min}} \wedge k_B T \ll \hbar \omega_{\text{ac,max}} \quad 1 \ll \beta \hbar \omega_{\text{opt,min}} \wedge 1 \ll \beta \hbar \omega_{\text{ac,max}} \quad (27.80)$$

Corollary 27.62 The contribution from the optical branches is negligible.

Proof 27.62 For an optical phonon branch $\nu \in \{d_x + 1, \dots, rd_x\}$ we obtain

$$0 \leq \sum_{\mathbf{q} \in Q} \hbar \omega_{\nu \mathbf{q}} B_{\beta}(\hbar \omega_{\nu \mathbf{q}}) = \sum_{\mathbf{q} \in Q} \hbar \omega_{\nu \mathbf{q}} \frac{1}{e^{\beta \hbar \omega_{\nu \mathbf{q}}} - 1} \leq \sum_{\mathbf{q} \in Q} \hbar \omega_{\nu \mathbf{q}} \frac{1}{e^{\beta \hbar \omega_{\text{opt}, \min}} - 1} \approx \sum_{\mathbf{q} \in Q} \hbar \omega_{\nu \mathbf{q}} e^{-\beta \hbar \omega_{\text{opt}, \min}} \xrightarrow{T \rightarrow 0} 0 \quad (27.81)$$

Approximation 27.63 Recalling that

$$\hbar \omega_{\nu}(\mathbf{q}) \approx c_{\nu} q \quad \text{for } \nu \in \{1, \dots, d_x\} \equiv \{\text{"indices for acoustic phonons"}\} \text{ and } q \approx 0 \quad (27.82)$$

we can also evaluate the contribution from the acoustic branches.

Corollary 27.64 In the low-temperature limit, the internal energy is dominated by the acoustic phonons, i.e.

$$E \stackrel{T \approx 0}{\equiv} \langle H_{\text{harm.}} \rangle(N, V, T) \approx E_0 + \sum_{\nu=1}^{d_x} \sum_{\mathbf{q} \in Q} c_{\nu} q \frac{1}{e^{\beta c_{\nu} q} - 1} \quad (27.83)$$

Proof 27.64 ToBeFormulated Cleanly

Theorem 27.65 It follows

$$E - E_0 \approx \frac{\pi^2}{30} \frac{1}{\hbar^3 c^3} N_p V_{\text{uc}} k_B^4 T^4 \quad (27.84)$$

Proof 27.65 TBDigitalized; Cf. Bulla Script.

27.5.1.4. The High-Temperature Limit

TBD.

27.5.1.5. Summary

Corollary 27.66 One finds for the internal energy of the system

$$\begin{aligned} \langle H_{\text{harm.}} \rangle(N, V, T) &\equiv E(N, V, T) \\ &= \begin{cases} E_0 + rd_x N_p k_B T & \text{if } \forall \underline{q} \in \Xi : k_B T \gg \hbar \omega_{\underline{q}} \\ E_0 + \text{const.} \times V \times T^4 & \text{if both equations are fulfilled } \forall \underline{q} \in \Xi : k_B T \ll \hbar \omega_{\underline{q}} \approx \hbar c q. \end{cases} \end{aligned} \quad (27.85)$$

Proof 27.66 See below.

Remark 27.66.1 Notice that one gets for the specific heat capacity per volume

$$\begin{aligned} C_{N,V}(T) &\equiv \partial_T \langle H_{\text{harm.}} \rangle(N, V, T) \\ &\propto \begin{cases} \text{const.} & \text{if } \forall \underline{q} \in \Xi : k_B T \gg \hbar \omega_{\underline{q}} \\ \text{const.} T^3 & \text{if both equations are fulfilled } \forall \underline{q} \in \Xi : k_B T \ll \hbar \omega_{\underline{q}} \approx \hbar c q. \end{cases} \end{aligned} \quad (27.86)$$

which reproduces the Dulong-Petit-law for high-temperatures and the experimental behaviour advertised in the beginning of the lecture for low-temperatures.

27.5.2. Intermezzo: The [G]rand [C]anonical [E]nsemble

27.5.2.1. Formalism

Notation 27.67 We further restrict our attention to the case in which the above mentioned system has a volume V , and is being treated in the grand canonical ensemble with temperature T and chemical potential μ that is being set such that the average particle number is fixed to be a constant (see below). We further assume the system to be described the Hilbert-space⁸ $\mathcal{H}$. This gives us the three natural variables (μ, T, V) .

Remark 27.67.1 Notice that, if we are for example treating a crystal, and we count the number of electrons that we would expect to be inside the crystal, then (in general) we have to *numerically* determine the chemical potential.

Fact 27.68 In the absence of „spontaneous symmetry breaking“ expectation-values in the grand-canonical-ensemble are evaluated w.r.t. the state

$$\rho_{\text{GCE}} = \frac{e^{-\beta(H-\mu N)}}{\text{Tr}_{\mathcal{H}} e^{-\beta(H-\mu N)}} \equiv \frac{1}{Z_{\text{GCE}}} e^{-\beta(H-\mu N)}, \quad (27.87)$$

whereas we introduced the grand canonical partition-function $Z_{\text{GCE}} \equiv Z_{\text{GCE}}(\mu, T, V)$.

Remark 27.68.1 Recall that the state above represents a solution to the stationary von Neumann equation.

Remark 27.68.2 Although in the case of „spontaneous symmetry breaking“, the evaluation of observables w.r.t. the state above yields incorrect results, one can oftentimes however use the state above, and break the underlying symmetry „by hand“, in order to recover the „correct“ physical state (or, in the case that we will encounter later, the correct physical „field theory“) again.

Fact 27.69 Recall, that the thermodynamic potential in the grand canonical ensemble is given by the so-called grand canonical potential

$$\Omega \equiv \Omega(\mu, T, V) = -\frac{\text{Log}_e Z_{\text{GCE}}(\mu, T, V)}{\beta} \quad (27.88)$$

from which one can conclude the usual relations for the conjugated variables particle-number, entropy and pressure

$$\frac{\partial \Omega}{\partial \mu} = -N, \quad \frac{\partial \Omega}{\partial T} = -S, \quad \frac{\partial \Omega}{\partial V} = -P. \quad (27.89)$$

⁸For indistinguishable particles, this Hilbert-space will be equal to a Fock-space $\mathcal{H} \equiv \mathcal{F}_{\zeta}$.

27.5.2.2. Application to the Electronic System

We treat the electronic system within the LDA-approximation of Quantum-Espresso⁹.

Notation 27.70 We have the electronic density of states in a free electron gas given as obtained through quantum espresso as

$$\rho_{\text{aux}}(E) = \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\downarrow, \uparrow\} \\ n \in \{1, 2, \dots\}}} \delta(E - \epsilon_{n\mathbf{k}}) \quad (27.90)$$

Lemma 27.71 One obtains the formulae

$$\begin{aligned} \langle N \rangle &= \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\downarrow, \uparrow\} \\ n \in \{1, 2, \dots\}}} F_{\beta}(\epsilon_{n\mathbf{k}\sigma} - \mu) = \int dE F_{\beta}(E - \mu) \rho_{\text{aux}}(E) \\ \langle H_{\text{el}} \rangle &= \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\downarrow, \uparrow\} \\ n \in \{1, 2, \dots\}}} F_{\beta}(\epsilon_{n\mathbf{k}\sigma} - \mu) \epsilon_{n\mathbf{k}\sigma} = \int dE F_{\beta}(E - \mu) \rho_{\text{aux}}(E) E \end{aligned} \quad (27.91)$$

Proof 27.71 TBDigitalized.

Remark 27.71.1 Although the discrete formulae look very tempting and beautiful, they are oftentimes quite inconvenient to use in practice as they are difficult to use for reliable calculations. Therefore, it is oftentimes more convenient to instead use the integral formulae after one introduced a smearing in the density of states in Equation (27.90).

Remark 27.71.2 The smearing that one usually introduced can be explained on physical grounds. This becomes clear after we introduced the concept of self-energies for electron-phonon-systems.

Theorem 27.72 One finds for the heat-capacity $C_{V,N}(T)$ the equation

$$C_{V,N}(T) = \frac{\pi^2}{3} \rho_{\text{aux}}(E_F) k_B^2 T \equiv \gamma T \quad (27.92)$$

with heat-capacity coefficient $\gamma = \frac{\pi^2}{3} \rho_{\text{aux}}(E_F) k_B^2$.

Proof 27.72 TBDigitalized.

⁹The templates ...

- Cf. Schwabl page 186
- Bulla-Skript
- Czycholl-Book.

Corollary 27.73 Combining the results, we obtain an expression for the molar heat capacity

$$\begin{aligned}
 C_{V,N,\text{molar}}(T) &= \frac{1}{n} \cdot C_{V,N}(T) \\
 &= \frac{1}{N_x N_y N_z 1.66 \cdot 10^{-24} \text{ mol}} \cdot \frac{\pi^2}{3} \rho_{\text{aux}}(0) k_B^2 T \\
 &= \frac{\rho_{\text{aux}}(0)}{N_x N_y N_z} \cdot \frac{\pi^2}{3} \frac{k_B^2}{1.66 \cdot 10^{-24} \text{ mol}} \cdot T \\
 &= \rho_{\text{QE}}(E_F) \cdot \frac{\pi^2}{3} \frac{k_B^2}{1.66 \cdot 10^{-24} \text{ mol}} \cdot T \\
 &\equiv \gamma_m \cdot T
 \end{aligned}$$

with the final expression for γ_m as

$$\gamma_m = \rho_{\text{QE}}(E_F) \cdot \frac{\pi^2}{3} \frac{k_B^2}{1.66 \cdot 10^{-24} \text{ mol}} \quad (27.93)$$

with ρ_{QE} as in Equation (22.3).

Proof 27.73 Follows from combining Equation (27.92) with Equation (9.55).

27.5.3. Thermodynamics of a Crystal in the Non-Interacting Gases Approximation

$$\begin{aligned}
 H_{\text{tot}} &\equiv H_{\text{el}} + H_{\text{ph}} + H_{\text{el-ph-int}} \\
 &\equiv \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\downarrow, \uparrow\} \\ n \in \{1, 2, \dots\}}} \epsilon_{n\mathbf{k}} c_{n\sigma\mathbf{k}}^\dagger c_{n\sigma\mathbf{k}} + \sum_{\substack{\mathbf{k} \in Q \\ v \in \{1, 2, \dots, r d_x\}}} \hbar \omega_{v\mathbf{k}} b_{v\mathbf{k}}^\dagger b_{v\mathbf{k}} + 0
 \end{aligned}$$

with allowed wave-vectors $Q \subset$ "First Brillouin Zone". Notice that the total Hilbert-space $\mathcal{H}$ of the system is given as tensor-product

$$\mathcal{H} = \mathcal{H}_F \otimes \mathcal{H}_B$$

with basis and scalar-product inherited from the respective many-identical-particle-Hilbert-spaces as usual. Modelling the system as being connected to a heat-reservoir (the environment dictating the temperature), we work in a hybrid ensemble with thermodynamic variables (T, V, μ, N_p) , expectation-values are evaluated w.r.t. the stationary solution of the von Neumann equation. Concretely one has for an observable quantity (expressed as a polynomial expression in second quantized operators) A the expectation-value

$$\langle A \rangle \equiv \frac{\text{Tr}_{\mathcal{H}} A e^{-\beta(H_{\text{el}} - \mu N_{\text{el}} + H_{\text{ph}})}}{\text{Tr}_{\mathcal{H}} e^{-\beta(H_{\text{el}} - \mu N_{\text{el}} + H_{\text{ph}})}} \equiv \frac{\text{Tr}_{\mathcal{H}} A e^{-\beta(H_{\text{el}} - \mu N_{\text{el}} + H_{\text{ph}})}}{Z}. \quad (27.94)$$

Other thermodynamic properties (such as pressure) are determined by derivatives of the hybrid partition-function. Combining the results, we obtain for the heat capacity at low temperatures

$$C_{V,N,\text{molar}}(T) = \frac{1}{n} \cdot C_{V,N}(T) \approx \gamma_m \cdot T + A \cdot T^3$$

in excellent agreement with several experimental results.

The folklore statement that the chemical potential of phonons is zero is correct, if one treats the phononic system in the grand canonical ensemble. Notice however that in our derivation it became evident that we are *actually* working in the canonical ensemble. There is a possibility to define a chemical potential for the phononic system, which does not necessarily need to be zero. Treating the phononic system in the grand canonical ensemble and agreeing to set $\mu = 0$ is just a mnemonic, which is quite common in the literature, but it is not a fundamental statement.

Addendum: The Chemical Potential of Bosons.

Here is a (hopefully convincing) selection of excerpts supporting this claim.

Chemisches Potential: Das chemische Potential des Photongases kann aus der Gibbs-Duhem-Relation $E = TS - PV + \mu N$ berechnet werden, da es sich um ein homogenes System handelt:

als

$$H_{\text{harm}} = \sum_{\mathbf{q}} \sum_{j=1}^{d_r} \hbar \omega_j(\mathbf{q}) \left(b_{\mathbf{q}j}^\dagger b_{\mathbf{q}j} + \frac{1}{2} \right). \quad (4.41)$$

Hierbei erfüllen die Oszillator-Auf- und Absteige-Operatoren $b_{\mathbf{q}j}^{(\dagger)}$ gemäß (4.15) die Vertauschungsrelation

$$[b_{\mathbf{q}j}, b_{\mathbf{q}'j'}^\dagger] = \delta_{\mathbf{q}\mathbf{q}'} \delta_{jj'}, \quad [b_{\mathbf{q}j}, b_{\mathbf{q}'j'}] = [b_{\mathbf{q}j}^\dagger, b_{\mathbf{q}'j'}^\dagger] = 0. \quad (4.42)$$

Dies sind die Vertauschungsrelationen für Bose²-Erzeugungs- und Vernichtungs-Operatoren. Die Elementaranregungen des Hamilton-Operators in harmonischer Näherung, also die quantisierten Normalschwingungen, verhalten sich also wie die eines Systems **wechselwirkungsfreier Bosonen**. Diese nennt man auch **Phononen**. Phononen sind keine wirklichen Teilchen sondern **Quasiteilchen**. Insbesondere gibt es keine Teilchenzahlerhaltung (und deshalb auch kein **chemisches Potential**). Auch in dieser Hinsicht verhalten sich Phononen analog wie Photonen (Quanten des elektromagnetischen Feldes). In der Tat weist auch die Thermodynamik der Phononen (quantisierten Gitterwellen) eine enge Analogie zur Thermodynamik des Photongases (Hohlraum-Strahlung, quantisierte elektromagnetische Wellen) auf.

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$$\mu = \frac{1}{N} (E - TS + PV) = \frac{1}{N} \left(4 - \frac{16}{3} + \frac{4}{3} \right) \frac{\sigma V T^3}{3c} \equiv 0. \quad (4.5.22)$$

Das chemische Potential des Photongases ist für alle Temperaturen identisch 0, da die Zahl der Photonen nicht fest ist, sondern sich an die Temperatur und das Volumen anpaßt. Photonen werden von der umgebenden Materie, den Wänden des Hohlraums, absorbiert und emittiert. Generell verschwindet das chemische Potential von Teilchen und von Quasiteilchen, wie z.B. Phononen, deren Teilchenzahl durch keinen Erhaltungssatz eingeschränkt ist. Betrachten wir nämlich die freie Energie für eine fiktive feste Zahl von Photonen (Phononen etc.) $F(T, V, N_{\text{ph}})$. Da die Zahl der Photonen (Phononen) nicht eingeschränkt ist, adjustiert sie sich so, daß die freie Energie minimal wird $\left(\frac{\partial F}{\partial N_{\text{ph}}} \right)_{T,V} = 0$. Das ist aber gerade der Ausdruck für das chemische Potential, welches somit verschwindet: $\mu = 0$. Ebenso hätten wir von der Maximalität der Entropie ausgehen können $\left(\frac{\partial S}{\partial N_{\text{ph}}} \right)_{E,V} = -\frac{\mu}{T} = 0$.

Figure 27.1.: An excerpt from the book from Czycholl.

Figure 27.2.: An excerpt from the book from Schwabl.

particle reservoirs, or a subsystem is considered. The system can exchange particles with the reservoirs which are described by their chemical potentials (assuming in general several particle species present). This feature is simply included by introducing Lagrange multipliers, i.e. tacitly understanding that single-particle energies are measured relative to their chemical potential, $H \rightarrow H - \sum_s \mu_s N_s$. In this case, the partition function instead of being a function of the number of particles, is a function of the chemical potentials for the species in question

$$Z_{\text{gr}}(T, \mu_s) = \text{Tr} e^{-(H - \mu_s N_s)/kT} = e^{-\Omega(T, \mu_s)/kT} \quad (2.90)$$

specified by the average number of particles according to

$$\langle N_s \rangle = - \left(\frac{\partial \Omega}{\partial \mu_s} \right)_{T,V}. \quad (2.91)$$

For systems of particles where the total number of *particles* is not conserved, i.e. where the Hamiltonian and the total number operator do not commute, such as for phonons and photons, the chemical potential vanishes, and the grand canonical ensemble is employed. For degenerate fermions, such as electrons in a metal, the

In thermal equilibrium, the average number of particles in a state λ is given by the usual boson occupation factor:

$$\langle n_\lambda \rangle = \frac{1}{e^{\beta(\epsilon_\lambda - \mu)} - 1} \equiv N_\lambda = n_B(\epsilon_\lambda - \mu) \quad (1.118)$$

Now there is a chemical potential, μ , which can vary with temperature and concentration. It is absent in the phonon, and photon, cases because these excitations do not conserve particle number. One may make as many phonons or photons as one wishes. Another operator of interest is the density operator:

Figure 27.4.: An excerpt from the book from Mahan.

Figure 27.3.: An excerpt from the book from Rammer.

27.6. The Wave-like / Massless Nature of the Phononic Quantum Field Theory

Theorem 27.74 Going over to the field-integral-description of the problem gives the action

$$S_{\text{ph}}[\phi, \bar{\phi}] = \oint_{\substack{\mathbf{q} \in \mathbf{Q} \\ \tau \in (0, \hbar\beta) \\ v \in \{1, \dots, rd_{\mathbf{x}}\}}} \bar{\phi}_v(\tau, \mathbf{q}) (\partial_\tau + \omega_v(\mathbf{q})) \phi_v(\tau, \mathbf{q}) \quad (27.95)$$

and switching to the Matsubara-representation yields

$$S_{\text{ph}}[\phi, \bar{\phi}] = \oint_{\substack{\mathbf{q} \in \mathbf{Q} \\ \omega_n \in \Omega_{+1} \\ v \in \{1, \dots, rd_{\mathbf{x}}\}}} \bar{\phi}_v(i\omega_n, \mathbf{q}) (-i\omega_n + \omega_v(\mathbf{q})) \phi_v(i\omega_n, \mathbf{q}) \quad (27.96)$$

$$\equiv \oint_{\substack{q \in i\Omega_{+1} \times \mathbf{Q} \\ v \in \{1, \dots, rd_{\mathbf{x}}\}}} \bar{\phi}_v(q) (-i\omega_n + \omega_v(\mathbf{q})) \phi_v(q) \quad (27.97)$$

whereas we introduced the four-momentum $q \equiv (\mathbf{q}, i\omega_n)$

Proof 27.74 Follows from Corollary 18.15.

Lemma 27.75 We introduce now the auxiliary fields

$$A_v(q) = \phi_v(q) + \bar{\phi}_v(-q), \quad \Pi_v(q) = \phi_v(q) - \bar{\phi}_v(-q)$$

with inverse transformation

$$\phi(q) = \frac{A_v(q) + \Pi_v(q)}{2}, \quad \bar{\phi}(q) = \frac{A_v(-q) - \Pi_v(-q)}{2}$$

Proof 27.75 TBD.

Remark 27.75.1 Notice that the fields defined like this are not independent from each other; One finds symmetries

$$A_v(q) = \bar{A}_v(-q), \quad \Pi_v(q) = -\bar{\Pi}_v(-q). \quad (27.98)$$

In order to be able to change coordinates without double-counting, we restrict attention to the variables

$$\left\{ A_v(q), \Pi_v(q) \right\}_{\substack{q \in i\Omega_{+1} \times Q / \sim \\ v \in \{1, \dots, rd_x\}}} \quad \text{with} \quad q \sim q' : \Longleftrightarrow q = -q'. \quad (27.99)$$

Part 1: No External Sources

Theorem 27.76 Changing integration-variables according to Equations (TBD) and integrating out the Π -fields yields the action

$$S_{\text{ph}}[A, \bar{A}] = \sum_{\substack{v \in \{1, \dots, rd_x\} \\ q \in i\Omega_{+1} \times Q / \sim}} \bar{A}_v(i\omega_n, \mathbf{q}) \frac{\omega_n^2 + \omega_v(\mathbf{q})^2}{2\omega_v(\mathbf{q})} A_v(i\omega_n, \mathbf{q}) \quad (27.100)$$

Proof 27.76 See Appendix, Proofs, page 549.

Remark 27.76.1 It will turn out, that all the interacting theories that one encounters in materials theory are independent of the Π -fields, such that integrating them out is a legitimate step.

Corollary 27.77 It follows

$$\mathcal{D}_{vv'}(i\omega_n, \mathbf{q}) = -\frac{1}{\hbar} \frac{2\omega_\alpha \delta_{\alpha, \alpha'}}{\omega_n^2 + \omega_\alpha^2} \quad (27.101)$$

from matrix-inversion.

Proof 27.77 Follows from the identity in Theorem TBD.

Remark 27.77.1 Notice that we derived this identity earlier by straight-forward computation of expectation-values, in Theorem TBReferenced.

TBElaborated.

Part 2: Incorporating External Sources

Theorem 27.78 The action in presence of external sources reads

$$S_{\text{ph}}[\phi, \bar{\phi}] = \sum_{\substack{\mathbf{q} \in Q \\ \tau \in (0, \hbar\beta) \\ v \in \{1, \dots, rd_{\mathbf{x}}\}}} \bar{\phi}_v(\tau, \mathbf{q}) (\partial_\tau + \omega_v(\mathbf{q})) \phi_v(\tau, \mathbf{q}) + \sum_{\substack{\mathbf{q} \in Q \\ \tau \in (0, \hbar\beta) \\ v \in \{1, \dots, rd_{\mathbf{x}}\}}} \left[\bar{\phi}_v(\tau, \mathbf{q}) j(\tau, \mathbf{q}) + \bar{j}(\tau, \mathbf{q}) \phi_v(\tau, \mathbf{q}) \right] \quad (27.102)$$

and switching to the Matsubara-representation yields

$$S_{\text{ph}}[\phi, \bar{\phi}] = \sum_{\substack{q \in i\Omega_{+1} \times Q \\ v \in \{1, \dots, rd_{\mathbf{x}}\}}} \bar{\phi}_v(q) (-i\omega_n + \omega_v(\mathbf{q})) \phi_v(q) + \sum_{\substack{q \in i\Omega_{+1} \times Q \\ v \in \{1, \dots, rd_{\mathbf{x}}\}}} \left[\bar{\phi}_v(q) j(q) + \bar{j}(q) \phi_v(q) \right] \quad (27.103)$$

whereas we introduced the four-momentum $q \equiv (\mathbf{q}, i\omega_n)$

27.7. The Equation-of-Motion-Technique as double-edged sword

Corollary 27.79 For non-interacting-systems (i.e. quadratic Hamiltonians as in eq. (8.31)) one finds the equation of motion

$$\begin{aligned}
 (\omega + i\delta) \quad G_{\alpha\alpha'}^{\text{ret}}(\omega + i\delta) &= \delta_{\alpha\alpha'} + \sum_{\beta} \frac{h_{\alpha\beta} - \mu\delta_{\alpha\beta}}{\hbar} G_{\beta\alpha'}^{\text{ret}}(\omega + i\delta) \\
 \Leftrightarrow (\omega + i\delta) \quad \mathbf{G}^{\text{ret}}(\omega + i\delta) &= \mathbb{1} + \frac{\mathbf{H} - \mu\mathbb{1}}{\hbar} \mathbf{G}^{\text{ret}}(\omega + i\delta) \\
 \Leftrightarrow \quad \mathbf{G}^{\text{ret}}(\omega + i\delta) &= \left[\omega + i\delta - \frac{\mathbf{H} - \mu\mathbb{1}}{\hbar} \right]^{\text{inv}}
 \end{aligned} \tag{27.104}$$

Proof 27.79 TBD.

Remark 27.79.1 By straight forward calculation, one also sees, that the various Green-functions (retarded, advanced, time-ordered) obey the very same equation of motion as above.

Remark 27.79.2 The fact, that three Green-functions obey the same equation of motion makes me (personally) a little bit uncomfortable; In [Francesco 04] one comments this in the following way (with adapted notation for better understanding)

(...) *On the other hand, the detailed time-dependence of $G^{\text{ret}}(t)$, $G^{\text{adv}}(t)$ and $G(t)$ is obviously very different. From the definition of these quantities follows for example that $G^r(t < 0) \equiv 0$, while $G^a(t > 0) \equiv 0$. Thus, if one wants to determine the correct solution of the equation of motion, one must incorporate those boundary conditions appropriately. This implies that the Fourier-transformation in eq. (...) has to be performed with some care.*

And nonetheless, despite these warnings, one usually determines the retarded- and advanced-Green-functions by the equation-of-motion-technique. Update: It can be proven (from the functional integral formalism) that the equation-of-motion-technique yields the correct result for the retarded- and advanced-Green-functions. But **not** for the time-ordered Green-function.

27.8. The Curious Formalism of Position Eigenstates

In materials theory, it is unfortunately common to invoke the concept of so-called position eigenstates. The author of these notes dislikes them, as they are (in his opinion) only nice to have for aesthetical reasons, without any concrete use. We will elucidate the concept here, and then make comments about their usefulness.

Notation 27.80 We consider the situation in Notation 10.1. Let

$$\mathcal{H} = \text{span}_{\mathbb{C}}\{|\alpha\rangle \mid \alpha \in \mathcal{A}\} \tag{27.105}$$

be a single-particle Hilbert-space with orthonormal basis. Let $\{c_{\alpha}, c_{\alpha}^{\dagger}\}_{\alpha \in \mathcal{A}}$ denote the corresponding creation- and annihilation-operators obeying the statistical commutation-relations.

Definition 27.81 One introduces the so-called *position eigenstates*

$$\begin{aligned}\psi(x) &= \sum_{\alpha \in \mathcal{A}} \langle x | \alpha \rangle c_\alpha = \sum_{\alpha \in \mathcal{A}} \phi_\alpha(x) c_\alpha, \\ \psi^\dagger(x) &= \sum_{\alpha \in \mathcal{A}} \langle \alpha | x \rangle c_\alpha^\dagger = \sum_{\alpha \in \mathcal{A}} \phi_\alpha^*(x) c_\alpha^\dagger.\end{aligned}$$

Remark 27.81.1 The usual interpretation is that these operators create or annihilate a particle in a position-/spin-state x .

Lemma 27.82 One finds the commutation-relations

$$[\psi(x), \psi(x')]_\zeta = 0 = [\psi^\dagger(x), \psi^\dagger(x')]_\zeta, \quad [\psi(x), \psi^\dagger(x')]_\zeta = \delta(x - x') \quad (27.106)$$

Proof 27.82 We compute straightforwardly

$$\begin{aligned}[\psi(x), \psi^\dagger(x')]_\zeta &= \sum_{\alpha, \alpha' \in \mathcal{A}} \phi_\alpha(x) \phi_{\alpha'}^*(x) [c_\alpha, c_{\alpha'}^\dagger]_\zeta \\ &= \sum_{\alpha, \alpha' \in \mathcal{A}} \phi_\alpha(x) \phi_{\alpha'}^*(x) \delta_{\alpha, \alpha'} \\ &= \sum_{\alpha \in \mathcal{A}} \phi_\alpha(x) \phi_\alpha^*(x) \\ &= \langle x | \sum_{\alpha \in \mathcal{A}} |\alpha\rangle \langle \alpha| | x' \rangle \\ &= \delta(x - x')\end{aligned}$$

whereas we used the completeness-relation of the basis.

Lemma 27.83 The inverse transformation reads

$$c_\alpha^\dagger = \int_y \phi_\alpha(y) \psi^\dagger(y), \quad c_\alpha = \int_y \phi_\alpha^*(y) \psi(y) \quad (27.107)$$

Proof 27.83 We compute straightforwardly

$$\psi^\dagger(x) = \sum_{\alpha \in \mathcal{A}} \phi_\alpha^*(x) c_\alpha^\dagger = \sum_{\alpha \in \mathcal{A}} \int_y \phi_\alpha^*(x) \phi_\alpha(y) \psi^\dagger(y) = \int_y \sum_{\alpha \in \mathcal{A}} \langle y | \alpha \rangle \langle \alpha | x \rangle \psi^\dagger(y) = \int_y \delta(y - x) \psi^\dagger(y) = \psi^\dagger(x)$$

Lemma 27.84 If $x \equiv (\mathbf{x}, s)$ represents a combination of a spatial- and a spin-coordinate, then the kinetic energy operator $T \equiv -\frac{\hbar^2}{2m}\Delta$ and a local one-body potential operator $\mathbf{x} \mapsto U(\mathbf{x})$ and a two-body operator V can be written in second quantization as

- $T = \frac{\hbar^2}{2m} \oint_x (\nabla \psi^\dagger(x)) (\nabla \psi(x))$
- $U = \oint_x \psi^\dagger(x) U(x) \psi(x)$
- $V = \frac{1}{2} \oint_{x,x'} \psi^\dagger(x) \psi^\dagger(x') V(x, x') \psi(x') \psi(x)$

which gives an expression for the total Hamiltonian rewritten w.r.t. field-operators as

$$\begin{aligned} H &= T + U + V \\ &= \frac{\hbar^2}{2m} \oint_x \left[(\nabla \psi^\dagger(x)) (\nabla \psi(x)) + \psi^\dagger(x) U(x) \psi(x) \right] + \frac{1}{2} \oint_{x,x'} \psi^\dagger(x) \psi^\dagger(x') V(x, x') \psi(x') \psi(x) \end{aligned}$$

Proof 27.84 This is proven by straight-forward computation. For the kinetic energy operator we obtain

$$T = \sum_{i,j} c_i^\dagger T_{ij} c_j = \sum_{i,j} c_i^\dagger \langle i | -\hbar^2 \frac{\Delta}{2m} | j \rangle c_j = \sum_{i,j} \oint_x c_i^\dagger \phi_i^*(x) \left(-\hbar^2 \frac{\Delta}{2m} \right) \phi_j(x) c_j = \frac{\hbar^2}{2m} \oint_x (\nabla \psi^\dagger(x)) (\nabla \psi(x))$$

therefore we obtain for a one-body operator

$$U = \sum_{i,j} c_i^\dagger U_{ij} c_j = \sum_{i,j} c_i^\dagger \langle i | U | j \rangle c_j = \sum_{i,j} \oint_x c_i^\dagger \phi_i^*(x) U(x) \phi_j(x) c_j = \oint_x \psi^\dagger(x) U(x) \psi(x)$$

and for the two-body interaction

$$\begin{aligned} V &= \frac{1}{2} \sum_{i,j,k,m} c_i^\dagger c_j^\dagger V_{ijklm} c_m c_k = \frac{1}{2} \sum_{i,j,k,m} c_i^\dagger c_j^\dagger \langle i, j | V | k, m \rangle c_m c_k \\ &= \frac{1}{2} \sum_{i,j,k,m} \oint_{x,x'} c_i^\dagger c_j^\dagger \phi_i^*(x) \phi_j^*(x') V(x, x') \phi_k(x) \phi_m(x') c_m c_k \\ &= \frac{1}{2} \oint_{x,x'} \psi^\dagger(x) \psi^\dagger(x') V(x, x') \psi(x') \psi(x) \end{aligned}$$

Lemma 27.85 In the momentum-basis the kinetic energy operator acquires the form [Negele Eq. (1.100)].

Proof 27.85 TBD, or maybe not since it is so useless?

Definition 27.86 A two-body operator V that is diagonal in its position representation as

$$\langle \mathbf{x}_1 \mathbf{x}_2 | V | \mathbf{x}_3 \mathbf{x}_4 \rangle = \delta(\mathbf{x}_1 - \mathbf{x}_3) \delta(\mathbf{x}_2 - \mathbf{x}_4) V(\mathbf{x}_1 - \mathbf{x}_2) \quad (27.108)$$

is said to be *local*¹⁰, or *velocity independent*.

Corollary 27.87 Let V denote a two-body operator which is diagonal in the position-representation. Then, it becomes in second quantized form

$$V = \frac{1}{2} \int d^3x d^3y V(\mathbf{x} - \mathbf{y}) \psi^\dagger(\mathbf{x}) \psi^\dagger(\mathbf{y}) \psi(\mathbf{y}) \psi(\mathbf{x}) \quad (27.109)$$

Proof 27.87 TBD

¹⁰Cf. Marcos Marinos p. 46 ff

Identity for expectation-value of the Hamilton-operator as in Problem 1.6 of Negele .

Definition 27.88 Now we introduce the field operator in the Heisenberg-representation as

$$\psi(\mathbf{x}, t) = e^{iHt/\hbar} \psi(\mathbf{x}, 0) e^{-iHt/\hbar} \quad (27.110)$$

Lemma 27.89 It follows the equation of motion

$$i\partial_t \psi(\mathbf{x}, t) = \left(-\frac{\hbar^2}{2m} \Delta + U(\mathbf{x}) \right) \psi(\mathbf{x}, t) + \int d^3x' \psi^\dagger(\mathbf{x}', t) V(\mathbf{x}, \mathbf{x}') \psi(\mathbf{x}', t) \psi(\mathbf{x}, t) \quad (27.111)$$

Proof 27.89 TBDigitalized from Schwabl p. 24

Green-Functions

Definition 27.90 We introduce the Green-function of the field-operators as

$$G_{\sigma, \sigma'}(\mathbf{x} - \mathbf{x}', t - t') = \langle \psi_\sigma^\dagger(\mathbf{x}, t) \psi_{\sigma'}(\mathbf{x}', t') \rangle \quad (27.112)$$

Remark 27.90.1 From this definition, it is actually quite non-trivial that the defined Green-function only depends on space- and time-differences. The time-difference is proven analog to Lemma TBReferenced. The space-difference can be seen as follows TBDone .

- See Schwabl 2 Eq. (2.1.6), GW01 Eq. (6) for the concrete values at equal times at zero temperature.
- Flensburg (somewhere): real-space version of the green-functions in the non-interacting electron-case.
- Imaginary time Green-function in coordinate space: Henk p. 145.

Corollary 27.91 The Green-function obeys the equation of motion

$$TBD. \quad (27.113)$$

Proof 27.91 See GW01 Eq. B(9)

Lemma 27.92 The Green-function of the field-operators has the Lehmann-representation

See GW01 Eq. (7)

Self-Energy

TBDigitalized from GW01 Eq. (81), (82).

Theorem 27.93 The Green-function can be obtained from Hedins-equations

$$TBD. \quad (27.114)$$

Proof 27.93 See GW01 Appendix.

Notice that the differential equation for the finite temperature Green-function now has the form of a heat-equation

$$TBD. \quad (27.115)$$

Example: Non-Interacting Electrons

Marcos Marinos QFT-notes p. 46f: Two-Body-Interactions in real and momentum-space

TBDigitalized from Schwabl p. 35.

Example: Free Electrons with Two-Body-Interactions

Now one can also introduce the Fourier-transform of these operators as in [Negele, page 18]. This is the so-called momentum-basis. Cf. Schwabl p. 25 onwards. See also Schwabl p. 46 for the Coulomb-interaction in second quantization.

27.9. Generalized Matrices

As we will shortly see, we can catch two birds with one stone by introducing the following abbreviations.

Notation 27.94 We introduce a multi-index with a time-argument

$$A \equiv (\alpha, \tau) \in \mathcal{A} \times (0, +\hbar\beta) \equiv I_\tau$$

and consider generalized matrices with indices

$$\begin{aligned} \mathcal{G} : I_\tau \times I_\tau &\rightarrow \mathbb{C} \\ (A, A') &\mapsto \mathcal{G}(A, A') \equiv \mathcal{G}_{\alpha, \alpha'}(\tau, \tau'). \end{aligned}$$

Definition 27.95 Completely analogous to usual matrices, one defines a generalized trace and generalized matrix-multiplication¹¹ as

$$(\mathcal{G} \cdot \mathcal{G}')(A, A') \equiv \sum_{B \in I_\tau} \mathcal{G}(A, B) \mathcal{G}'(B, A')$$

Remark 27.95.1 Note that the inversion-relation now reads

$$(\mathcal{G} \cdot \mathcal{G}^{-1})(A, A') = \delta_{\alpha, \alpha'} \delta(\tau - \tau') \equiv \text{Id}$$

Definition 27.96 We call a generalized matrix a *propagator in thermal equilibrium with statistics ζ* , if there exists a periodic standard-continuation (denoted by the same symbol) on the temporal-axis

$$\mathbb{R} \ni \tau \mapsto \mathcal{G}_{\alpha, \alpha'}(\tau) \in \mathbb{C}$$

s.t. the following equations holds

$$\forall (\tau, \tau') \in \mathbb{R}^2 : \mathcal{G}_{\alpha, \alpha'}(\tau, \tau') = \mathcal{G}_{\alpha, \alpha'}(\tau - \tau') = \zeta \mathcal{G}_{\alpha, \alpha'}(\tau - \tau' + \hbar\beta)$$

Remark 27.96.1 We denote the set of all [P]ropagators [I]n [T]hermal [E]quilibrium with statistics ζ as PITE_ζ .

Remark 27.96.2 Note that the analytic standard-continuation of a PITE_ζ can be expanded with Matsubara-frequencies according to Lemma 1.11.

¹¹This is a combination of ordinary matrix-multiplication for the α in the multi-index, and a convolution for the time-argument τ .

Theorem 27.97 Let $\mathcal{G}, \mathcal{G}' \in \text{PITE}_\zeta$. Then it follows $\mathcal{G} \cdot \mathcal{G}' \in \text{PITE}_\zeta$.

Proof 27.97 One finds by immediate calculation

$$\begin{aligned}
 (\mathcal{G} \cdot \mathcal{G}')(\tau_1, \tau_2) &= + \int_0^{+\hbar\beta} d\tau \mathcal{G}(\tau_1 - \tau) \mathcal{G}'(\tau - \tau_2) \\
 &= + \int_0^{+\hbar\beta} d\tau \mathcal{G}(\tau_1 - \tau) \mathcal{G}'(\tau_1 - \tau_2 - (\tau_1 - \tau)) \\
 &= - \int_{\tau_1}^{\tau_1 - \hbar\beta} d\tau' \mathcal{G}(\tau') \mathcal{G}'(\tau_1 - \tau_2 - \tau') \\
 &= + \int_0^{+\hbar\beta} d\tau' \mathcal{G}(\tau') \mathcal{G}'(\tau_1 - \tau_2 - \tau')
 \end{aligned} \tag{27.116}$$

where we used the substitution $\tau' = \tau_1 - \tau$ in the second last step and we used Waste-Folder 1, Digital Waste-Folder 1 with Useful Links, topic 4 in the last step. Now it is clear, that the product depends only on the time-difference and that it obeys the same periodicity as the respective Green-function. Therefore there will be a periodic standard-continuation with the desired properties.

Corollary 27.98 It follows for the Matsubara-component of the product

$$(\mathcal{G} \cdot \mathcal{G}')(\mathrm{i}\omega_n) = \mathcal{G}(\mathrm{i}\omega_n) \cdot \mathcal{G}'(\mathrm{i}\omega_n). \tag{27.117}$$

Proof 27.98 By definition (1.12) of the Matsubara-component, together with the result of the preceding Theorem that $\mathcal{G} \cdot \mathcal{G}' \in \text{PITE}_\zeta$, we may write

$$(\mathcal{G} \cdot \mathcal{G}')(\mathrm{i}\omega_n) = \int_0^{\hbar\beta} d\tau_1 e^{\mathrm{i}\omega_n \tau_1} (\mathcal{G} \cdot \mathcal{G}')(\tau_1) = \int_0^{\hbar\beta} d\tau_1 e^{\mathrm{i}\omega_n \tau_1} \int_0^{\hbar\beta} d\tau \mathcal{G}(\tau) \mathcal{G}'(\tau_1 - \tau), \tag{27.118}$$

where we used Definition 27.95 for the convolution appearing under the last integral. Inserting the Matsubara-expansion of Lemma 1.11 for $\mathcal{G}'$,

$$\mathcal{G}'(\tau_1 - \tau) = \frac{1}{\hbar\beta} \sum_{\omega_m \in \Omega_\zeta} \mathcal{G}'(\mathrm{i}\omega_m) e^{-\mathrm{i}\omega_m(\tau_1 - \tau)}, \tag{27.119}$$

we obtain

$$\begin{aligned}
 (\mathcal{G} \cdot \mathcal{G}')(\mathrm{i}\omega_n) &= \frac{1}{\hbar\beta} \sum_{\omega_m \in \Omega_\zeta} \mathcal{G}'(\mathrm{i}\omega_m) \int_0^{\hbar\beta} d\tau \mathcal{G}(\tau) e^{\mathrm{i}\omega_m \tau} \int_0^{\hbar\beta} d\tau_1 e^{\mathrm{i}(\omega_n - \omega_m)\tau_1} \\
 &= \frac{1}{\hbar\beta} \sum_{\omega_m \in \Omega_\zeta} \mathcal{G}'(\mathrm{i}\omega_m) \mathcal{G}(\mathrm{i}\omega_m) \delta_{\omega_m, \omega_n} \hbar\beta,
 \end{aligned} \tag{27.120}$$

where the τ -integral was identified as $\mathcal{G}(\mathrm{i}\omega_m)$ via (1.12), and the τ_1 -integral was evaluated using Corollary 1.12 (noting $\omega_n - \omega_m \in \Omega_{+1}$, since the difference of two frequencies of the same statistics ζ is always bosonic). Carrying out the sum over ω_m against the Kronecker-delta yields

$$(\mathcal{G} \cdot \mathcal{G}')(\mathrm{i}\omega_n) = \mathcal{G}(\mathrm{i}\omega_n) \mathcal{G}'(\mathrm{i}\omega_n), \quad (27.121)$$

which proves the claim.

Remark 27.98.1 This theorem will turn out as a central result for our later discussions with the Dyson-equation of interacting systems; It implies that even if we take only a finite number of contributions from the infinite series in the Dyson-equation, we will still end up with a physically reasonable propagator.

Theorem 27.99 Under the assumption, that the Dyson-equation holds

$$\mathcal{G}(\mathrm{i}\omega_n) = \mathcal{G}_0(\mathrm{i}\omega_n) + (\mathcal{G}_0 \cdot \Sigma \cdot \mathcal{G})(\mathrm{i}\omega_n),$$

we obtain a corresponding equation with generalized matrix-product for the imaginary-time-version of the propagators, namely

$$\mathcal{G}(\tau, \tau') = \mathcal{G}_0(\tau, \tau') + (\mathcal{G}_0 \cdot \Sigma \cdot \mathcal{G})(\tau, \tau'). \quad (27.122)$$

Proof 27.99 We perform the straight-forward calculation

$$\begin{aligned} \mathcal{G}(\tau, \tau') &= \frac{1}{\hbar\beta} \sum_{\omega_n \in \Omega_\zeta} (\mathcal{G}_0(\mathrm{i}\omega_n) + \mathcal{G}_0(\mathrm{i}\omega_n) \Sigma(\mathrm{i}\omega_n) \mathcal{G}(\mathrm{i}\omega_n)) e^{-\mathrm{i}\omega_n(\tau - \tau')} \\ &= \mathcal{G}_0(\tau, \tau') + \frac{1}{\hbar\beta} \sum_{\omega_n \in \Omega_\zeta} \int_0^{+\hbar\beta} d\tau_1 \mathcal{G}_0(\tau_1) e^{\mathrm{i}\omega_n \tau_1} \Sigma(\mathrm{i}\omega_n) \int_0^{+\hbar\beta} d\tau_3 \mathcal{G}(\tau_3) e^{\mathrm{i}\omega_n \tau_3} e^{-\mathrm{i}\omega_n(\tau - \tau')} \\ &= \mathcal{G}_0(\tau, \tau') + \frac{1}{\hbar\beta} \sum_{\omega_n \in \Omega_\zeta} \int_\tau^{\tau + \hbar\beta} d\tilde{\tau}_1 \mathcal{G}_0(\tau - \tilde{\tau}_1) e^{\mathrm{i}\omega_n(\tau - \tilde{\tau}_1)} \Sigma(\mathrm{i}\omega_n) \int_0^{+\hbar\beta} d\tau_3 \mathcal{G}(\tau_3) e^{\mathrm{i}\omega_n(\tau_3 - \tau + \tau')} = \dots \end{aligned}$$

whereas we used the substitution $\tau_1 = \tau - \tilde{\tau}_1$ in the last step. We now use the substitution $\tau_3 = \tilde{\tau}_3 - \tau'$ for the second integral and subsequently Waste-Folder 1, UL, topic 4 which yields

$$\begin{aligned} \dots &= \mathcal{G}_0(\tau, \tau') + \frac{1}{\hbar\beta} \sum_{\omega_n \in \Omega_\zeta} \int_\tau^{\tau + \hbar\beta} d\tilde{\tau}_1 \mathcal{G}_0(\tau - \tilde{\tau}_1) e^{\mathrm{i}\omega_n(\tau - \tilde{\tau}_1)} \Sigma(\mathrm{i}\omega_n) \int_{\tau'}^{\tau + \hbar\beta + \tau'} d\tilde{\tau}_3 \mathcal{G}(\tilde{\tau}_3 - \tau') e^{\mathrm{i}\omega_n(\tilde{\tau}_3 - \tau' - \tau + \tau')} \\ &= \mathcal{G}_0(\tau, \tau') + \int_0^{+\hbar\beta} d\tilde{\tau}_1 \mathcal{G}_0(\tau - \tilde{\tau}_1) \frac{1}{\hbar\beta} \sum_{\omega_n \in \Omega_\zeta} \Sigma(\mathrm{i}\omega_n) e^{-\mathrm{i}\omega_n(\tilde{\tau}_1 - \tilde{\tau}_3)} \int_0^{+\hbar\beta} d\tilde{\tau}_3 \mathcal{G}(\tilde{\tau}_3 - \tau') \\ &\equiv \mathcal{G}_0(\tau, \tau') + \int_0^{+\hbar\beta} \int_0^{+\hbar\beta} d\tau_1 d\tau_2 \mathcal{G}_0(\tau, \tau_1) \Sigma(\tau_1, \tau_2) \mathcal{G}(\tau_2, \tau') \end{aligned}$$

27.10. The ("Holy") Trinity of Electronic Basis Sets in Materials Theory

Fact 27.100 There are three different levels of description for electrons in materials, which are all related to each other by a mere basis-change:

- *The Continuum-basis* (cf. TBReferenced¹²)
- *"The" Bloch-basis* (cf. TBReferenced¹³)
- *"The" Wannier-basis* (cf. TBReferenced¹⁴)

27.10.1. Continuum-Basis

TBD

27.10.2. Bloch-Basis

TBD

27.10.3. Wannier-Basis

TBD

¹²Chapter about position eigenstates

¹³LDA-approximation, EPW-system, chapter about thermal QFT

¹⁴Cf. comment in the Green-function identities and the specifications for real materials

27.10.4. The Pedestrian Way to Compute Spin- $\frac{1}{2}$ -Green-Functions

Corollary 27.101 It follows for the time-evolution

$$\begin{aligned} c_{\uparrow}(t) &= \cos \frac{\theta}{2} e^{-i\xi_{\uparrow}t/\hbar} c_{\uparrow_n} + \sin \frac{\theta}{2} e^{-i\xi_{\downarrow}t/\hbar} c_{\downarrow_n} \\ c_{\downarrow}(t) &= \sin \frac{\theta}{2} e^{i\phi} e^{-i\xi_{\uparrow}t/\hbar} c_{\uparrow_n} - \cos \frac{\theta}{2} e^{i\phi} e^{-i\xi_{\downarrow}t/\hbar} c_{\downarrow_n} \\ c_{\uparrow}^{\dagger}(t) &= \cos \frac{\theta}{2} e^{i\xi_{\uparrow}t/\hbar} c_{\uparrow_n}^{\dagger} + \sin \frac{\theta}{2} e^{i\xi_{\downarrow}t/\hbar} c_{\downarrow_n}^{\dagger} \\ c_{\downarrow}^{\dagger}(t) &= \sin \frac{\theta}{2} e^{-i\phi} e^{i\xi_{\uparrow}t/\hbar} c_{\uparrow_n}^{\dagger} - \cos \frac{\theta}{2} e^{-i\phi} e^{i\xi_{\downarrow}t/\hbar} c_{\downarrow_n}^{\dagger} \end{aligned}$$

Proof 27.101 See Appendix, Proofs, page 371.

Proof 27.101 of Corollary 27.101 Follows by straightforward application to the time-evolution operator and Remark 12.16.4.

Lemma 27.102 It holds

$$\begin{aligned} \langle c_{\uparrow_n}^{\dagger} c_{\uparrow_n} \rangle &= \frac{e^{-\beta\xi_{\uparrow}} + e^{2\mu\beta}}{1 + e^{-\beta\xi_{\uparrow}} + e^{-\beta\xi_{\downarrow}} + e^{2\mu\beta}}, & \langle c_{\uparrow_n} c_{\uparrow_n}^{\dagger} \rangle &= 1 - \langle c_{\uparrow_n}^{\dagger} c_{\uparrow_n} \rangle, \\ \langle c_{\downarrow_n}^{\dagger} c_{\downarrow_n} \rangle &= \frac{e^{-\beta\xi_{\downarrow}} + e^{2\mu\beta}}{1 + e^{-\beta\xi_{\uparrow}} + e^{-\beta\xi_{\downarrow}} + e^{2\mu\beta}}, & \langle c_{\downarrow_n} c_{\downarrow_n}^{\dagger} \rangle &= 1 - \langle c_{\downarrow_n}^{\dagger} c_{\downarrow_n} \rangle, \\ \langle c_{\uparrow_n}^{\dagger} c_{\downarrow_n} \rangle &= 0, & \langle c_{\downarrow_n} c_{\uparrow_n}^{\dagger} \rangle &= 0, \\ \langle c_{\downarrow_n}^{\dagger} c_{\uparrow_n} \rangle &= 0, & \langle c_{\uparrow_n} c_{\downarrow_n}^{\dagger} \rangle &= 0. \end{aligned}$$

Proof 27.102 See Appendix, Proofs, page 371.

Proof 27.102 of Lemma 27.102. We compute

$$\langle c_{\uparrow_n}^{\dagger} c_{\uparrow_n} \rangle = \frac{1}{Z} \sum_{\psi \in \mathbb{B}} \langle \psi | e^{-\beta(H-\mu N)} c_{\uparrow_n}^{\dagger} c_{\uparrow_n} | \psi \rangle = \frac{e^{-\beta(E_{\uparrow}-\mu)} + e^{-\beta(-2\mu)}}{1 + e^{-\beta(E_{\uparrow}-\mu)} + e^{-\beta(E_{\downarrow}-\mu)} + e^{2\beta\mu}}$$

$$\langle c_{\downarrow_n}^{\dagger} c_{\downarrow_n} \rangle = \frac{1}{Z} \sum_{\psi \in \mathbb{B}} \langle \psi | e^{-\beta(H-\mu N)} c_{\downarrow_n}^{\dagger} c_{\downarrow_n} | \psi \rangle = \frac{e^{-\beta(E_{\downarrow}-\mu)} + e^{-\beta(-2\mu)}}{1 + e^{-\beta(E_{\uparrow}-\mu)} + e^{-\beta(E_{\downarrow}-\mu)} + e^{2\beta\mu}}$$

$$\langle c_{\uparrow_n}^{\dagger} c_{\downarrow_n} \rangle = \langle c_{\downarrow_n}^{\dagger} c_{\uparrow_n} \rangle = 0$$

and the other identities follow from the CCR.

Remark 27.102.1 Notice that the expectation values in Lemma 12.17 can be rewritten as Fermi-functions in the usual way. Assuming that $E_{\uparrow} + E_{\downarrow} = 0$, we have

$$\xi_{\uparrow} + \xi_{\downarrow} = (E_{\uparrow} - \mu) + (E_{\downarrow} - \mu) = -2\mu,$$

and therefore

$$e^{2\beta\mu} = e^{-\beta(\xi_{\uparrow} + \xi_{\downarrow})} = e^{-\beta\xi_{\uparrow}} e^{-\beta\xi_{\downarrow}}.$$

Consequently,

$$\langle c_{\uparrow_n}^{\dagger} c_{\uparrow_n} \rangle = \frac{e^{-\beta\xi_{\uparrow}} + e^{2\beta\mu}}{1 + e^{-\beta\xi_{\uparrow}} + e^{-\beta\xi_{\downarrow}} + e^{2\beta\mu}} = \frac{e^{-\beta\xi_{\uparrow}} (1 + e^{-\beta\xi_{\downarrow}})}{(1 + e^{-\beta\xi_{\uparrow}}) (1 + e^{-\beta\xi_{\downarrow}})} = \frac{e^{-\beta\xi_{\uparrow}}}{1 + e^{-\beta\xi_{\uparrow}}} = \frac{1}{e^{\beta\xi_{\uparrow}} + 1} = F_{\beta}(\xi_{\uparrow}).$$

Similarly,

$$\langle c_{\downarrow_n}^{\dagger} c_{\downarrow_n} \rangle = \frac{1}{e^{\beta\xi_{\downarrow}} + 1}.$$

Thus, the occupation-number expectation values are precisely the Fermi-functions

$$\langle d_{\sigma}^{\dagger} d_{\sigma} \rangle = f_{\beta}(\xi_{\sigma}) = \frac{1}{e^{\beta\xi_{\sigma}} + 1}, \quad \sigma \in \{\uparrow, \downarrow\}.$$

Lemma 27.103 It follows

$$\begin{aligned} G_{\uparrow\uparrow}^{\leq}(t) &= \frac{i}{\hbar} e^{-i\xi_{\uparrow}t/\hbar} \cos^2 \frac{\theta}{2} \langle c_{\uparrow_n}^{\dagger} c_{\uparrow_n} \rangle &+ \frac{i}{\hbar} e^{-i\xi_{\downarrow}t/\hbar} \sin^2 \frac{\theta}{2} \langle c_{\downarrow_n}^{\dagger} c_{\downarrow_n} \rangle \\ G_{\uparrow\downarrow}^{\leq}(t) &= \frac{i}{\hbar} e^{-i\xi_{\uparrow}t/\hbar} \frac{\sin \theta}{2} e^{-i\phi} \langle c_{\uparrow_n}^{\dagger} c_{\uparrow_n} \rangle &- \frac{i}{\hbar} e^{-i\xi_{\downarrow}t/\hbar} \frac{\sin \theta}{2} e^{-i\phi} \langle c_{\downarrow_n}^{\dagger} c_{\downarrow_n} \rangle \\ G_{\downarrow\uparrow}^{\leq}(t) &= \frac{i}{\hbar} e^{-i\xi_{\uparrow}t/\hbar} \frac{\sin \theta}{2} e^{i\phi} \langle c_{\uparrow_n}^{\dagger} c_{\uparrow_n} \rangle &- \frac{i}{\hbar} e^{-i\xi_{\downarrow}t/\hbar} \frac{\sin \theta}{2} e^{i\phi} \langle c_{\downarrow_n}^{\dagger} c_{\downarrow_n} \rangle \\ G_{\downarrow\downarrow}^{\leq}(t) &= \frac{i}{\hbar} e^{-i\xi_{\uparrow}t/\hbar} \sin^2 \frac{\theta}{2} \langle c_{\uparrow_n}^{\dagger} c_{\uparrow_n} \rangle &+ \frac{i}{\hbar} e^{-i\xi_{\downarrow}t/\hbar} \cos^2 \frac{\theta}{2} \langle c_{\downarrow_n}^{\dagger} c_{\downarrow_n} \rangle \end{aligned} \tag{27.123}$$

Proof 27.103 See Appendix, Proofs, page 372.

Proof 27.103 of Lemma 27.103 We compute straightforwardly

$$\begin{aligned}
 G_{\uparrow\uparrow}^<(t) &= \frac{i}{\hbar} \langle c_{\uparrow}^{\dagger} c_{\uparrow}(t) \rangle \\
 &= \frac{i}{\hbar} \left\langle \left(\cos \frac{\theta}{2} c_{\uparrow_n}^{\dagger} + \sin \frac{\theta}{2} c_{\downarrow_n}^{\dagger} \right) \left(\cos \frac{\theta}{2} c_{\uparrow_n}(t) + \sin \frac{\theta}{2} c_{\downarrow_n}(t) \right) \right\rangle \\
 &= \cos^2 \frac{\theta}{2} \frac{i}{\hbar} \langle c_{\uparrow_n}^{\dagger} c_{\uparrow_n}(t) \rangle + \sin^2 \frac{\theta}{2} \frac{i}{\hbar} \langle c_{\downarrow_n}^{\dagger} c_{\downarrow_n}(t) \rangle \\
 &= \frac{i}{\hbar} e^{-i\xi_{\uparrow}t/\hbar} \cos^2 \frac{\theta}{2} \langle c_{\uparrow_n}^{\dagger} c_{\uparrow_n} \rangle + \frac{i}{\hbar} e^{-i\xi_{\downarrow}t/\hbar} \sin^2 \frac{\theta}{2} \langle c_{\downarrow_n}^{\dagger} c_{\downarrow_n} \rangle
 \end{aligned}$$

proving the first identity.

$$\begin{aligned}
 G_{\downarrow\downarrow}^<(t) &= \frac{i}{\hbar} \langle c_{\downarrow}^{\dagger} c_{\downarrow}(t) \rangle \\
 &= \frac{i}{\hbar} \left\langle \left(\sin \frac{\theta}{2} e^{-i\phi} c_{\uparrow_n}^{\dagger} - \cos \frac{\theta}{2} e^{-i\phi} c_{\downarrow_n}^{\dagger} \right) \left(\sin \frac{\theta}{2} e^{i\phi} c_{\uparrow_n}(t) - \cos \frac{\theta}{2} e^{i\phi} c_{\downarrow_n}(t) \right) \right\rangle \\
 &= \frac{i}{\hbar} e^{-i\xi_{\uparrow}t/\hbar} \sin^2 \frac{\theta}{2} \langle c_{\uparrow_n}^{\dagger} c_{\uparrow_n} \rangle + \frac{i}{\hbar} e^{-i\xi_{\downarrow}t/\hbar} \cos^2 \frac{\theta}{2} \langle c_{\downarrow_n}^{\dagger} c_{\downarrow_n} \rangle
 \end{aligned}$$

Proving the fourth identity. For the other identities we obtain

$$\begin{aligned}
 G_{\uparrow\downarrow}^<(t) &= \frac{i}{\hbar} \langle c_{\downarrow}^{\dagger} c_{\uparrow}(t) \rangle \\
 &= \frac{i}{\hbar} \left\langle \left(\sin \frac{\theta}{2} e^{-i\phi} c_{\uparrow_n}^{\dagger} - \cos \frac{\theta}{2} e^{-i\phi} c_{\downarrow_n}^{\dagger} \right) \left(\cos \frac{\theta}{2} c_{\uparrow_n}(t) + \sin \frac{\theta}{2} c_{\downarrow_n}(t) \right) \right\rangle \\
 &= \frac{i}{\hbar} \sin \frac{\theta}{2} \cos \frac{\theta}{2} e^{-i\phi} \langle c_{\uparrow_n}^{\dagger} c_{\uparrow_n}(t) \rangle - \frac{i}{\hbar} \sin \frac{\theta}{2} \cos \frac{\theta}{2} e^{-i\phi} \langle c_{\downarrow_n}^{\dagger} c_{\downarrow_n}(t) \rangle
 \end{aligned}$$

proving the second identity. For the last identity we obtain

$$\begin{aligned}
 G_{\downarrow\uparrow}^<(t) &= \frac{i}{\hbar} \langle c_{\uparrow}^{\dagger} c_{\downarrow}(t) \rangle \\
 &= \frac{i}{\hbar} \left\langle \left(\cos \frac{\theta}{2} c_{\uparrow_n}^{\dagger} + \sin \frac{\theta}{2} c_{\downarrow_n}^{\dagger} \right) \left(\sin \frac{\theta}{2} e^{i\phi} c_{\uparrow_n}(t) - \cos \frac{\theta}{2} e^{i\phi} c_{\downarrow_n}(t) \right) \right\rangle \\
 &= \sin \frac{\theta}{2} \cos \frac{\theta}{2} e^{i\phi} \frac{i}{\hbar} \langle c_{\uparrow_n}^{\dagger} c_{\uparrow_n}(t) \rangle - \sin \frac{\theta}{2} \cos \frac{\theta}{2} e^{i\phi} \frac{i}{\hbar} \langle c_{\downarrow_n}^{\dagger} c_{\downarrow_n}(t) \rangle
 \end{aligned}$$

proving the last identity.

Lemma 27.104 It follows

$$\begin{aligned}
 G_{\uparrow\uparrow}^>(t) &= \frac{i}{\hbar} e^{-\frac{i}{\hbar}\xi_{\uparrow}t/\hbar} \cos^2 \frac{\theta}{2} \left(\langle c_{\uparrow_n}^{\dagger} c_{\uparrow_n} \rangle - 1 \right) + \frac{i}{\hbar} e^{-i\xi_{\downarrow}t/\hbar} \sin^2 \frac{\theta}{2} \left(\langle c_{\downarrow_n}^{\dagger} c_{\downarrow_n} \rangle - 1 \right) \\
 G_{\uparrow\downarrow}^>(t) &= \frac{i}{\hbar} e^{-i\xi_{\uparrow}t/\hbar} \frac{\sin \theta}{2} e^{-i\phi} \left(\langle c_{\uparrow_n}^{\dagger} c_{\uparrow_n} \rangle - 1 \right) - \frac{i}{\hbar} e^{-i\xi_{\downarrow}t/\hbar} \frac{\sin \theta}{2} e^{-i\phi} \left(\langle c_{\downarrow_n}^{\dagger} c_{\downarrow_n} \rangle - 1 \right) \\
 G_{\downarrow\uparrow}^>(t) &= \frac{i}{\hbar} e^{-i\xi_{\uparrow}t/\hbar} \frac{\sin \theta}{2} e^{i\phi} \left(\langle c_{\uparrow_n}^{\dagger} c_{\uparrow_n} \rangle - 1 \right) - \frac{i}{\hbar} e^{-i\xi_{\downarrow}t/\hbar} \frac{\sin \theta}{2} e^{i\phi} \left(\langle c_{\downarrow_n}^{\dagger} c_{\downarrow_n} \rangle - 1 \right) \\
 G_{\downarrow\downarrow}^>(t) &= \frac{i}{\hbar} e^{-i\xi_{\uparrow}t/\hbar} \sin^2 \frac{\theta}{2} \left(\langle c_{\uparrow_n}^{\dagger} c_{\uparrow_n} \rangle - 1 \right) + \frac{i}{\hbar} e^{-i\xi_{\downarrow}t/\hbar} \cos^2 \frac{\theta}{2} \left(\langle c_{\downarrow_n}^{\dagger} c_{\downarrow_n} \rangle - 1 \right)
 \end{aligned}$$

Proof 27.104 See Appendix, Proofs, page 374.

Proof 27.104 of Lemma 27.104 We compute

$$\begin{aligned}
 G_{\uparrow\uparrow}^>(t) &= -\frac{i}{\hbar} \langle c_{\uparrow}(t) c_{\uparrow}^{\dagger} \rangle \\
 &= -\frac{i}{\hbar} \langle (\cos \frac{\theta}{2} c_{\uparrow_n}(t) + \sin \frac{\theta}{2} c_{\downarrow_n}(t)) (\cos \frac{\theta}{2} c_{\uparrow_n}^{\dagger} + \sin \frac{\theta}{2} c_{\downarrow_n}^{\dagger}) \rangle \\
 &= -\frac{i}{\hbar} \cos^2 \frac{\theta}{2} e^{-i\xi_{\uparrow}t/\hbar} \langle c_{\uparrow_n} c_{\uparrow_n}^{\dagger} \rangle - \frac{i}{\hbar} \sin^2 \frac{\theta}{2} e^{-i\xi_{\downarrow}t/\hbar} \langle c_{\downarrow_n} c_{\downarrow_n}^{\dagger} \rangle
 \end{aligned} \tag{27.124}$$

proving the first identity. For the second identity we find

$$\begin{aligned}
 G_{\downarrow\downarrow}^>(t) &= -\frac{i}{\hbar} \langle c_{\downarrow}(t) c_{\downarrow}^{\dagger} \rangle \\
 &= -\frac{i}{\hbar} \left\langle \left(\sin \frac{\theta}{2} e^{i\phi} c_{\uparrow_n}(t) - \cos \frac{\theta}{2} e^{i\phi} c_{\downarrow_n}(t) \right) \left(\sin \frac{\theta}{2} e^{-i\phi} c_{\uparrow_n}^{\dagger} - \cos \frac{\theta}{2} e^{-i\phi} c_{\downarrow_n}^{\dagger} \right) \right\rangle \\
 &= -\frac{i}{\hbar} \sin^2 \frac{\theta}{2} e^{-i\xi_{\uparrow}t/\hbar} \langle c_{\uparrow_n} c_{\uparrow_n}^{\dagger} \rangle - \frac{i}{\hbar} \cos^2 \frac{\theta}{2} e^{-i\xi_{\downarrow}t/\hbar} \langle c_{\downarrow_n} c_{\downarrow_n}^{\dagger} \rangle
 \end{aligned} \tag{27.125}$$

proving the fourth identity.

$$\begin{aligned}
 G_{\uparrow\downarrow}^>(t) &= -\frac{i}{\hbar} \langle c_{\uparrow}(t) c_{\downarrow}^{\dagger} \rangle \\
 &= -\frac{i}{\hbar} \left\langle \left(\cos \frac{\theta}{2} c_{\uparrow_n}(t) + \sin \frac{\theta}{2} c_{\downarrow_n}(t) \right) \left(\sin \frac{\theta}{2} e^{-i\phi} c_{\uparrow_n}^{\dagger} - \cos \frac{\theta}{2} e^{-i\phi} c_{\downarrow_n}^{\dagger} \right) \right\rangle \\
 &= -\frac{i}{\hbar} \cos \frac{\theta}{2} \sin \frac{\theta}{2} e^{-i\phi} e^{-\xi_{\uparrow}t/\hbar} \langle c_{\uparrow_n} c_{\uparrow_n}^{\dagger} \rangle + \frac{i}{\hbar} \cos \frac{\theta}{2} \sin \frac{\theta}{2} e^{-i\phi} e^{-\xi_{\downarrow}t/\hbar} \langle c_{\downarrow_n} c_{\downarrow_n}^{\dagger} \rangle
 \end{aligned}$$

For the last identity we find

$$\begin{aligned}
 G_{\downarrow\uparrow}^<(t) &= -\frac{i}{\hbar} \langle c_{\downarrow}(t) c_{\uparrow}^{\dagger} \rangle \\
 &= -\frac{i}{\hbar} \left\langle \left(\sin \frac{\theta}{2} e^{i\phi} c_{\uparrow_n}(t) - \cos \frac{\theta}{2} e^{i\phi} c_{\downarrow_n}(t) \right) \left(\cos \frac{\theta}{2} c_{\uparrow_n}^{\dagger} + \sin \frac{\theta}{2} c_{\downarrow_n}^{\dagger} \right) \right\rangle \\
 &= -\frac{i}{\hbar} \sin \frac{\theta}{2} \cos \frac{\theta}{2} e^{i\phi} \left(e^{i\xi_{\uparrow}t/\hbar} \langle c_{\uparrow_n} c_{\uparrow_n}^{\dagger} \rangle - e^{i\xi_{\downarrow}t/\hbar} \langle c_{\downarrow_n} c_{\downarrow_n}^{\dagger} \rangle \right)
 \end{aligned}$$

proving the last identity after using $\sin\left(\frac{\theta}{2}\right) \cos\left(\frac{\theta}{2}\right) = \frac{1}{2} \sin(\theta)$.

Corollary 27.105 Recalling Lemma 12.4

$$G(t) = \theta(t) G^>(t) + \theta(-t) G^<(t) \tag{27.126}$$

we have determined the time-ordered-Green-function, although we do not want to write it down at this moment.

Proof 27.105 See Appendix, Proofs, page 375.

Proof 27.105 of Corollary ?? We compute straightforwardly ...

Corollary 27.106 Recalling another identity from Lemma 12.4, $G^{\text{ret}}(t) = \theta(t) \cdot (G^>(t) - G^<(t))$, we obtain the retarded Green-function as a function of time:

$$\begin{aligned} G_{\uparrow\uparrow}^r(t) &= -\frac{i}{\hbar} \theta(t) e^{-i\xi_{\uparrow}t/\hbar} \cos^2 \frac{\theta}{2} - \frac{i}{\hbar} \theta(t) e^{-i\xi_{\downarrow}t/\hbar} \sin^2 \frac{\theta}{2} \\ G_{\uparrow\downarrow}^r(t) &= -\frac{i}{\hbar} \theta(t) e^{-i\xi_{\uparrow}t/\hbar} \frac{\sin \theta}{2} e^{-i\phi} + \frac{i}{\hbar} \theta(t) e^{-i\xi_{\downarrow}t/\hbar} \frac{\sin \theta}{2} e^{-i\phi} \\ G_{\downarrow\uparrow}^r(t) &= -\frac{i}{\hbar} \theta(t) e^{-i\xi_{\uparrow}t/\hbar} \frac{\sin \theta}{2} e^{i\phi} + \frac{i}{\hbar} \theta(t) e^{-i\xi_{\downarrow}t/\hbar} \frac{\sin \theta}{2} e^{i\phi} \\ G_{\downarrow\downarrow}^r(t) &= -\frac{i}{\hbar} \theta(t) e^{-i\xi_{\uparrow}t/\hbar} \sin^2 \frac{\theta}{2} - \frac{i}{\hbar} \theta(t) e^{-i\xi_{\downarrow}t/\hbar} \cos^2 \frac{\theta}{2} \end{aligned}$$

Proof 27.106 Follows immediately by insertion.

Corollary 27.107 With the Fourier-transformation we obtain the retarded Green-function as a function of frequency:

$$\begin{aligned} G_{\uparrow\uparrow}^{\text{ret}}(\omega + i\eta) &= \frac{1}{\hbar} \frac{1}{\omega + i\eta - \xi_{\uparrow}/\hbar} \cos^2 \frac{\theta}{2} + \frac{1}{\hbar} \frac{1}{\omega + i\eta - \xi_{\downarrow}/\hbar} \sin^2 \frac{\theta}{2} \\ G_{\uparrow\downarrow}^{\text{ret}}(\omega + i\eta) &= \frac{1}{\hbar} \frac{1}{\omega + i\eta - \xi_{\uparrow}/\hbar} \frac{\sin \theta}{2} e^{-i\phi} - \frac{1}{\hbar} \frac{1}{\omega + i\eta - \xi_{\downarrow}/\hbar} \frac{\sin \theta}{2} e^{-i\phi} \\ G_{\downarrow\uparrow}^{\text{ret}}(\omega + i\eta) &= \frac{1}{\hbar} \frac{1}{\omega + i\eta - \xi_{\uparrow}/\hbar} \frac{\sin \theta}{2} e^{i\phi} - \frac{1}{\hbar} \frac{1}{\omega + i\eta - \xi_{\downarrow}/\hbar} \frac{\sin \theta}{2} e^{i\phi} \\ G_{\downarrow\downarrow}^{\text{ret}}(\omega + i\eta) &= \frac{1}{\hbar} \frac{1}{\omega + i\eta - \xi_{\uparrow}/\hbar} \sin^2 \frac{\theta}{2} + \frac{1}{\hbar} \frac{1}{\omega + i\eta - \xi_{\downarrow}/\hbar} \cos^2 \frac{\theta}{2} \end{aligned} \tag{27.127}$$

Proof 27.107 See Appendix, Proofs, page 429.

Proof 27.107 of Theorem 27.108 Follows immediately by insertion.

Theorem 27.108 It follows

$$G^{\text{ret}}(\omega + i\eta) = \frac{1}{\hbar} \frac{\left(\omega + i\eta - \frac{\epsilon_0 - \mu}{\hbar} \right) \sigma_0 + \left(\frac{\vec{\epsilon}_1}{\hbar} \right) \cdot \vec{\sigma}}{\left(\omega + i\eta - \frac{\epsilon_0 - \mu}{\hbar} \right)^2 - \left(\frac{\vec{\epsilon}_1}{\hbar} \right)^2} \tag{27.128}$$

Proof 27.108 See Appendix, Proofs, page 375.

Remark 27.108.1 Note that the same result can be obtained by matrix-inversion and analytic continuation, as will be elaborated below.

28. Appendix - Applications to Toy Models

28.1. The Free Electron Gas

Possible / reasonable topics:

- fermi-energy and DOS as a function of density
- Sommerfeld-approximation for the free electron-gas and comparison with experiment.
- Band-structure and reduced zone-scheme.

28.2. Superconductivity from BCS-Theory

28.2.1. BCS-Hamiltonian in First Quantization

Notation 28.1 We consider a single electron obeying the time-independent free Schroedinger-equation

$$H_0|\psi\rangle = -\hbar^2 \frac{\Delta}{2m} |\psi\rangle = E|\psi\rangle$$

subject to periodic boundary conditions in a volume $[0, L]^{d_x} / \sim$.

Lemma 28.2 With the inclusion of spin, this gives us a basis for the single-particle Hilbert-space

$$\text{span}_{\mathbb{C}} \left\{ \psi_{\mathbf{k}\sigma} \right\}_{\substack{\mathbf{k} \in 2\pi\mathbb{Z}^{d_x}/L \\ \sigma \in \{\downarrow, \uparrow\}}} = \text{span}_{\mathbb{C}} \left\{ \psi_{\mathbf{k}\mathbf{n}\sigma} \right\}_{\substack{\mathbf{k} \in \mathbb{Q} \\ \mathbf{n} \in \mathbb{Z}^{d_x} \\ \sigma \in \{\downarrow, \uparrow\}}} = \mathcal{H} = L^2([0, L]^{d_x} / \sim) \otimes \mathbb{C}^2. \quad (28.1)$$

Proof 28.2 See [Kaul, Fischer] and [Bulla lecture notes].

Remark 28.2.1 Notice that one has to conclude from the context "which" $\mathbf{k}$ one is referring to.

Definition 28.3 We now consider an *attractive, local BCS-toy-model-interaction potential* V , which acts as a linear operator on the two-body Hilbert-space as

$$V : \mathcal{H} \otimes \mathcal{H} \rightarrow \mathcal{H} \otimes \mathcal{H}$$

$$\psi \otimes \chi \mapsto V(\psi \otimes \chi)$$

with

$$V(\psi \otimes \chi)(x, x') := -|g|\delta_{\sigma \neq \sigma'}\delta(\mathbf{r} - \mathbf{r}')\psi(x)\chi(x').$$

for all $x, x' \in [0, L]^3 \times \{\downarrow, \uparrow\}$ with $x \equiv (\mathbf{r}, \sigma)$ etc.

Remark 28.3.1 Usually one refers to V simply as *BCS-interaction*.

Remark 28.3.2 Notice that the Bra-Ket-Notation gives us the integral kernel

$$\begin{aligned} -|g|\delta_{\sigma \neq \sigma'}\delta(\mathbf{r} - \mathbf{r}')\psi(x_1)\chi(x_2) &= \langle x_1, x_2 | V | \psi, \chi \rangle \\ &= \int_{x_3, x_4 \in [0, L]^3 \times \{\downarrow, \uparrow\}} \langle x_1, x_2 | V | x_3, x_4 \rangle \langle x_3, x_4 | \psi, \chi \rangle \\ &= \int_{x_3, x_4 \in [0, L]^3 \times \{\downarrow, \uparrow\}} \langle x_1, x_2 | V | x_3, x_4 \rangle \psi(x_3)\chi(x_4) \end{aligned}$$

thus one can conclude that the integral kernel reads

$$\langle x_1, x_2 | V | x_3, x_4 \rangle = -|g|\delta_{\sigma_1 \neq \sigma_2}\delta(\mathbf{r}_1 - \mathbf{r}_2) \cdot \delta_{\sigma_1, \sigma_3}\delta(\mathbf{r}_1 - \mathbf{r}_3)\delta_{\sigma_2, \sigma_4}\delta(\mathbf{r}_2 - \mathbf{r}_4) \quad (28.2)$$

Remark 28.3.3 Notice that this is a special case of a "local two-body interaction" a.k.a. "velocity independent"¹.

We will now continue the many body problem defined by the single particle Hamiltonian H_0 above, combined with the many-particle interactions V .

28.2.2. BCS-Hamiltonian in Second Quantization

Lemma 28.4 One computes the matrix-elements of the BCS-interaction w.r.t. the basis in Equation (28.1) to be

$$\langle \mathbf{k}_1 \sigma_1, \mathbf{k}_2 \sigma_2 | V | \mathbf{k}_3 \sigma_3, \mathbf{k}_4 \sigma_4 \rangle = -\frac{|g|}{L^{d_x}} \delta_{\sigma_3 \neq \sigma_4} \delta_{\sigma_1, \sigma_3} \delta_{\sigma_2, \sigma_4} \delta_{\mathbf{k}_1 + \mathbf{k}_2, \mathbf{k}_3 + \mathbf{k}_4} \quad (28.3)$$

Proof 28.4 See Appendix, Proofs, page 554.

¹Cf. MarcosMarino p. 46, Negele Orland page 10.

Theorem 28.5 Using the plane basis in Equation (28.1), one can write the Hamiltonian in second quantization as

$$H_{\text{BCS}} = \sum_{\substack{\mathbf{k} \in 2\pi\mathbb{Z}^{d_x}/L \\ \sigma \in \{\downarrow, \uparrow\}}} \epsilon(\mathbf{k}) c_{\sigma}^{\dagger}(\mathbf{k}) c_{\sigma}(\mathbf{k}) - \frac{g}{L^{d_x}} \sum_{\mathbf{k}, \mathbf{k}', \mathbf{q} \in 2\pi\mathbb{Z}^{d_x}/L} c_{\uparrow}^{\dagger}(\mathbf{k} + \mathbf{q}) c_{\downarrow}^{\dagger}(-\mathbf{k}) c_{\downarrow}(-\mathbf{k}' + \mathbf{q}) c_{\uparrow}(\mathbf{k}') \quad (28.4)$$

with fermionic creation- and annihilation-operators as usual.

Proof 28.5 See Appendix, Proofs, page 555.

Remark 28.5.1 Notice that this expression is in harmony with the literature².

Notation 28.6 Using the basis introduced above, and notating $V \equiv L^{d_x}$, we introduce the field-operators

$$c_{\sigma}(\mathbf{r}) = \frac{1}{\sqrt{L^{d_x}}} \sum_{\mathbf{k} \in 2\pi\mathbb{Z}^{d_x}/L} e^{i\mathbf{k}\mathbf{r}} c_{\sigma}(\mathbf{k}), \quad c_{\sigma}^{\dagger}(\mathbf{r}) = \frac{1}{\sqrt{L^{d_x}}} \sum_{\mathbf{k} \in 2\pi\mathbb{Z}^{d_x}/L} e^{-i\mathbf{k}\mathbf{r}} c_{\sigma}^{\dagger}(\mathbf{k}) \quad (28.5)$$

Lemma 28.7 Note that these field-operators obey the commutation-relations

$$\{c_{\sigma}(\mathbf{r}), c_{\sigma'}^{\dagger}(\mathbf{r}')\} = \delta_{\sigma, \sigma'} \delta(\mathbf{r} - \mathbf{r}'), \quad \{c'_{\sigma}(\mathbf{r}), c_{\sigma'}^{\dagger}(\mathbf{r}')\} = \{c_{\sigma}(\mathbf{r}), c_{\sigma'}(\mathbf{r}')\} = 0 \quad (28.6)$$

whereas one has to conclude from the context, whether the symbol ψ denotes the single particle wavefunction or the many particle field-operator.

Proof 28.7 See Appendix, Proofs, page 556.

Corollary 28.8 The BCS-Hamiltonian can be rewritten with the field-operators as

$$\begin{aligned} H_{\text{BCS}} &= \oint c_{\sigma}^{\dagger}(\mathbf{r}) \left(-\hbar^2 \frac{\Delta}{2m} - \mu \right) c_{\sigma}(\mathbf{r}) - g \int d\mathbf{r} c_{\uparrow}^{\dagger}(\mathbf{r}) c_{\downarrow}^{\dagger}(\mathbf{r}) c_{\downarrow}(\mathbf{r}) c_{\uparrow}(\mathbf{r}) \\ &= \oint c_{\sigma}^{\dagger}(\mathbf{r}) \left(-\hbar^2 \frac{\Delta}{2m} - \mu \right) c_{\sigma}(\mathbf{r}) - \frac{g}{2} \int d\mathbf{r} : \left[\begin{pmatrix} c_{\uparrow}^{\dagger} & c_{\downarrow}^{\dagger} \end{pmatrix} \cdot \begin{pmatrix} c_{\uparrow} \\ c_{\downarrow} \end{pmatrix} \right]^2 (\mathbf{r}) : \end{aligned}$$

Proof 28.8 See Appendix, Proofs, page 556.

²Cf. Altland pages 267, 346, Flensberg page 321, Rickayzen page 209.

28.2.3. BCS-Action in the Field Integral Formalism

Fact 28.9 The action of the system reads

$$S_{\text{BCS}}[\psi, \bar{\psi}] = \oint_{\substack{\sigma \in \{\downarrow, \uparrow\} \\ \tau \in (0, \hbar\beta) \\ \mathbf{k} \in 2\pi\mathbb{Z}^{d_{\mathbf{x}}}/L}} \left[\bar{\psi}_{\sigma}(\tau, \mathbf{k}) \left(\partial_{\tau} + \epsilon_{\sigma}(\mathbf{k}) - \mu \right) \psi_{\sigma}(\tau, \mathbf{k}) - \frac{g}{L^{d_{\mathbf{x}}}} \oint_{\substack{\tau \in (0, \hbar\beta) \\ \mathbf{k}, \mathbf{k}', \mathbf{q} \in 2\pi\mathbb{Z}^{d_{\mathbf{x}}}/L}} \left(\bar{\psi}_{\mathbf{k}+\mathbf{q}\uparrow} \bar{\psi}_{-\mathbf{k}\downarrow} \psi_{-\mathbf{k}'+\mathbf{q}\downarrow} \psi_{\mathbf{k}'\uparrow} \right) (\tau) \right]$$

Proof 28.9 See [Draft Summary].

Notation 28.10 Recalling the shorthand-notation³ $x \equiv (\tau, \mathbf{r})$, we introduce the Fourier-transformed integration-variables as

$$\psi_{\sigma}(\tau, \mathbf{r}) = \frac{1}{\sqrt{\hbar\beta V}} \sum_{\substack{\omega_n \in \Omega_{-1} \\ \mathbf{k} \in 2\pi\mathbb{Z}^{d_{\mathbf{x}}}/L}} e^{i(\mathbf{k}\mathbf{r} - \omega_n \tau)} \psi_{\sigma}(i\omega_n, \mathbf{k}), \quad \bar{\psi}_{\sigma}(\tau, \mathbf{r}) = \frac{1}{\sqrt{\hbar\beta V}} \sum_{\substack{\omega_n \in \Omega_{-1} \\ \mathbf{k} \in 2\pi\mathbb{Z}^{d_{\mathbf{x}}}/L}} e^{-i(\mathbf{k}\mathbf{r} - \omega_n \tau)} \bar{\psi}_{\sigma}(i\omega_n, \mathbf{k}). \quad (28.7)$$

and analogously (but with bosonic Matsubara-frequencies) later on for the auxiliary fields Δ and $\bar{\Delta}$.

Fact 28.11 The corresponding action of the real-space-representation in the functional integral formalism reads⁴

$$\begin{aligned} S_{\text{BCS}}[\psi, \bar{\psi}] &= \int dx \left[\sum_{\sigma \in \{\downarrow, \uparrow\}} \bar{\psi}_{\sigma}(x) \left(\partial_{\tau} - \hbar^2 \frac{\Delta}{2m} - \mu \right) \psi_{\sigma}(x) - g \left(\bar{\psi}_{\uparrow} \bar{\psi}_{\downarrow} \psi_{\downarrow} \psi_{\uparrow} \right) (x) \right] \\ &= \oint_{x, \sigma} \bar{\psi}_{\sigma}(x) \left(\partial_{\tau} - \hbar^2 \frac{\Delta}{2m} - \mu \right) \psi_{\sigma}(x) - g \left(\bar{\psi}_{\uparrow} \bar{\psi}_{\downarrow} \psi_{\downarrow} \psi_{\uparrow} \right) (x) \end{aligned}$$

Proof 28.11 By combining the expansions in TBReferenced with Corollary TBReferenced.

Corollary 28.12 For future reference we note the representation of the action using the momentum-eigenstates

$$\begin{aligned} S_{\text{BCS}}[\psi, \bar{\psi}] &= \sum_{\substack{\sigma \in \{\downarrow, \uparrow\} \\ k \in i\Omega_{-1} \times 2\pi\mathbb{Z}^{d_{\mathbf{x}}}/L}} \bar{\psi}_{\sigma}(k) \left(-i\omega_k + \epsilon_{\sigma}(\mathbf{k}) - \mu \right) \psi_{\sigma}(k) \\ &\quad - \frac{g}{\hbar\beta L^{d_{\mathbf{x}}}} \sum_{k, k', q \in i\Omega_{-1} \times 2\pi\mathbb{Z}^{d_{\mathbf{x}}}/L} \bar{\psi}_{\uparrow}(k+q) \bar{\psi}_{\downarrow}(-k) \psi_{\downarrow}(-k'+q) \psi_{\uparrow}(k') \end{aligned}$$

³ Shit... This is in conflict to earlier where this denoted space and spin-coordinate.

⁴ Cf. this to Henk page 275.

Proof 28.12 See Appendix, Proofs, page 557.

Notation 28.13 We implement Graßmann-valued external sources into the partition-function as

$$Z \mapsto Z[j, \bar{j}] = \int \mathcal{D}_{-1}(\bar{\psi}, \psi) \exp \left(-S_{\text{BCS}} + \oint \bar{j} \psi + \bar{\psi} j \right) \quad (28.8)$$

28.2.4. Hubbard-Stratonovich-Transformation

Lemma 28.14 A Hubbard-Stratonovich-transformation reads

$$\begin{aligned} \exp -S_{\text{int}}[\bar{\psi}, \psi] &= \exp g \int dx \left(\bar{\psi}_{\uparrow} \bar{\psi}_{\downarrow} \psi_{\downarrow} \psi_{\uparrow} \right) (x) \\ &= \int \mathcal{D}_{+1}(\Delta, \bar{\Delta}) \exp - \int dx \left[\frac{|\Delta|^2(x)}{g} - \left(\bar{\Delta} \psi_{\downarrow} \psi_{\uparrow} + \Delta \bar{\psi}_{\uparrow} \bar{\psi}_{\downarrow} \right) (x) \right] \end{aligned}$$

with complex integration-variables $\Delta, \bar{\Delta}$.

Proof 28.14 See Appendix, Proofs, page 557.

Fact 28.15 After decoupling the quartic contributions we obtain the following partition-function in presence of external sources

$$Z[j, \bar{j}] = \iint \mathcal{D}_{-1}(\bar{\psi}, \psi) \mathcal{D}_{+1}(\Delta, \bar{\Delta}) \exp \left(-\tilde{S}_{\text{BCS}} + \oint \bar{j} \psi + \bar{\psi} j \right).$$

and with the new BCS-action

$$\tilde{S}_{\text{BCS}} = \oint_{x, \sigma} \bar{\psi}_{\sigma}(x) \left(\partial_{\tau} - \hbar^2 \frac{\Delta}{2m} - \mu \right) \psi_{\sigma}(x) + \int dx \left[\frac{|\Delta|^2(x)}{g} - \left(\bar{\Delta} \psi_{\downarrow} \psi_{\uparrow} + \Delta \bar{\psi}_{\uparrow} \bar{\psi}_{\downarrow} \right) (x) \right] \quad (28.9)$$

Proof 28.15 Follows by combining the previous results.

Lemma 28.16 Switching to the Fourier-representation of the action, we obtain for the individual terms

$$\bullet \oint_{x, \sigma} \bar{\psi}_{\sigma}(x) \left(\partial_{\tau} - \hbar^2 \frac{\Delta}{2m} - \mu \right) \psi_{\sigma}(x) = \sum_{\substack{\sigma \in \{\downarrow, \uparrow\} \\ k \in i\Omega_{-1} \times 2\pi \mathbb{Z}^{d_{\mathbf{x}}} / L}} \bar{\psi}_{\sigma}(k) \left(-i\omega_k + \epsilon_{\sigma}(\mathbf{k}) - \mu \right) \psi_{\sigma}(k)$$

$$\begin{aligned}
 & \bullet \int d\mathbf{x} \frac{|\Delta|^2(\mathbf{x})}{g} = \sum_{k \in i\Omega_{-1} \times 2\pi\mathbb{Z}^{d_{\mathbf{x}}}/L} \frac{|\Delta|^2(k)}{g} \\
 & \bullet \int d\mathbf{x} - \left(\bar{\Delta}\psi_{\downarrow}\psi_{\uparrow} + \Delta\bar{\psi}_{\uparrow}\bar{\psi}_{\downarrow} \right)(\mathbf{x}) = \frac{1}{\sqrt{\hbar\beta L^{d_{\mathbf{x}}}}} \sum_{k,k' \in i\Omega_{-1} \times 2\pi\mathbb{Z}^{d_{\mathbf{x}}}/L} - \left(\bar{\Delta}(k+k')\psi_{\downarrow}(k)\psi_{\uparrow}(k') + \Delta(k+k')\bar{\psi}_{\uparrow}(k)\bar{\psi}_{\downarrow}(k') \right)
 \end{aligned}$$

Proof 28.16 See the path-integral-description of BCS-theory-document, pdf-page 10.

Notation 28.17 We further introduce the *Nambu-spinors*

$$\bar{\Psi}_{\text{Nambu}}(i\omega_n, \mathbf{k}) = \left(\bar{\psi}_{\uparrow}(i\omega_n, \mathbf{k}), \psi_{\downarrow}(-i\omega_n, -\mathbf{k}) \right), \quad \Psi_{\text{Nambu}}(i\omega_n, \mathbf{k}) = \begin{pmatrix} \psi_{\uparrow}(i\omega_n, \mathbf{k}) \\ \bar{\psi}_{\downarrow}(-i\omega_n, -\mathbf{k}) \end{pmatrix} \quad (28.10)$$

Lemma 28.18 Changing the integration-variables⁵ gives the momentum-representation of the partition-function

$$Z[j, \bar{j}] = \iint \mathcal{D}_{-1}(\bar{\psi}, \psi) \mathcal{D}_{+1}(\bar{\Delta}, \Delta) \exp -\tilde{S}_{\text{BCS}} \quad (28.11)$$

with

$$\begin{aligned}
 \tilde{S}_{\text{BCS}}[\Delta, \bar{\Delta}, \psi, \bar{\psi}, j, \bar{j}] = & - \sum_{\substack{\mathbf{q} \in 2\pi\mathbb{Z}^{d_{\mathbf{x}}}/L \\ \omega_m \in \Omega_{+1}}} \frac{|\Delta(i\omega_m, \mathbf{q})|^2}{g} \\
 & + \sum_{\substack{\mathbf{k}', \mathbf{k}'' \in 2\pi\mathbb{Z}^{d_{\mathbf{x}}}/L \\ \omega_{n'}, \omega_{n''} \in \Omega_{-1}}} \bar{\Psi}_{\text{Nambu}}(i\omega_{n'}, \mathbf{k}') \left\langle i\omega_{n'}, \mathbf{k}' \left| -\frac{1}{\hbar} \mathcal{G}_{\text{Gorkov}}^{-1} \right| i\omega_{n''}, \mathbf{k}'' \right\rangle \Psi_{\text{Nambu}}(i\omega_{n''}, \mathbf{k}'') \\
 & - \left(\bar{j}\psi + \bar{j}\psi \right)
 \end{aligned}$$

and Gorkov-Green-function

$$\begin{aligned}
 \langle i\omega_n, \mathbf{k} | -\frac{1}{\hbar} \mathcal{G}_{\text{Gorkov}}^{-1} | i\omega_{n'}, \mathbf{k}' \rangle = & \langle i\omega_n, \mathbf{k} | i\omega_{n'}, \mathbf{k}' \rangle \begin{pmatrix} -i\omega_n + \left(\frac{\hbar^2 k^2}{2m} - \mu \right) & 0 \\ 0 & +i\omega_n - \left(\frac{\hbar^2 k^2}{2m} - \mu \right) \end{pmatrix} \\
 & + \frac{1}{\sqrt{\hbar\beta V}} \begin{pmatrix} 0 & -\Delta_{n+n', \mathbf{k}+\mathbf{k}'} \\ \bar{\Delta}_{n+n', \mathbf{k}+\mathbf{k}'} & 0 \end{pmatrix}
 \end{aligned}$$

Proof 28.18 See Appendix, Proofs, page 557.

⁵The unphysical, diverging normalization-factor is irrelevant, when we determine observables later on, as it will cancel out.

Corollary 28.19 Integrating out the fermionic fields gives the partition-function

$$Z[j, \bar{j}] = \int \mathcal{D}_{+1}(\bar{\Delta}, \Delta) \exp - S_{\text{BCS}}[\Delta, \bar{\Delta}, j, \bar{j}] \quad (28.12)$$

with

$$\begin{aligned} S_{\text{BCS}}[\Delta, \bar{\Delta}, j, \bar{j}] = & - \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ \omega_m \in \Omega_{+1}}} \frac{|\Delta(i\omega_m, \mathbf{q})|^2}{g} \\ & - \text{Tr} \text{Log} - \frac{1}{\hbar} \mathcal{G}_{\text{Gorkov}}^{-1}[\Delta, \bar{\Delta}] \\ & \quad \mp (\bar{j}_{\uparrow}, -j_{\downarrow}) \cdot \left(-\hbar \mathcal{G}_{\text{Gorkov}}[\Delta, \bar{\Delta}] \right) \cdot \begin{pmatrix} j_{\uparrow} \\ -j_{\downarrow} \end{pmatrix} \end{aligned}$$

Proof 28.19 We first rewrite the source-terms as linear terms in the Nambu-spinor-fields as

$$\begin{aligned} \oint \bar{j}\psi + \bar{\psi}j &= \oint_x \left(\bar{j}_{\uparrow}\psi_{\uparrow} + \bar{j}_{\downarrow}\psi_{\downarrow} + \bar{\psi}_{\uparrow}j_{\uparrow} + \bar{\psi}_{\downarrow}j_{\downarrow} \right) \\ &= \sum_k \left((\bar{\psi}_{\uparrow}, \psi_{\downarrow}) \cdot \begin{pmatrix} j_{\uparrow} \\ -j_{\downarrow} \end{pmatrix} + (\bar{j}_{\uparrow}, -j_{\downarrow}) \cdot \begin{pmatrix} \psi_{\uparrow} \\ \bar{\psi}_{\downarrow} \end{pmatrix} \right) \\ &= \sum_k \left(\bar{\Psi}_{\text{Nambu}} \cdot \begin{pmatrix} j_{\uparrow} \\ -j_{\downarrow} \end{pmatrix} + (\bar{j}_{\uparrow}, -j_{\downarrow}) \cdot \Psi_{\text{Nambu}} \right) \end{aligned}$$

Now performing the integral in the fermionic variables with the usual identities and using $\text{Log Det}(\bullet) = \text{Tr Log}(\bullet)$ gives the desired result.

28.2.5. Spontaneous Symmetry Breaking

We now perform a mean-field-approximation and subsequently a saddle-point-approximation, and break the $U(1)$ -symmetry by hand⁶, in order to identify and determine the (superconducting-)order-parameter.

Corollary 28.20 The BCS-action is invariant under the unitary transformation

$$\Delta \mapsto \Delta e^{i\theta} \quad \text{with } \theta \in [0, 2\pi). \quad (28.13)$$

Hypothesis 28.21 We make the ansatz for a mean-field-configuration as

⁶Cf. also Metsch 2020 Exercise Sheet 5 for the classical analogon.

$$\begin{aligned}\Delta(i\omega_m, \mathbf{q}) &\approx \sqrt{\hbar\beta V} \delta_{\mathbf{q},0} \delta_{\omega_m,0} \Delta \\ \bar{\Delta}(i\omega_m, \mathbf{q}) &\approx \sqrt{\hbar\beta V} \delta_{\mathbf{q},0} \delta_{\omega_m,0} \bar{\Delta}\end{aligned}$$

Remark 28.21.1 Note that this ansatz assumes, that the mean-field-configuration of the order-parameter is dominated by a single contribution which is constant in the momenta.

Corollary 28.22 Using this ansatz and minimizing the action w.r.t. this configuration in the absence of external sources yields the gap-equation

$$\frac{1}{g} = \frac{1}{\hbar\beta L^{d_x}} \sum_{k \in i\Omega_{-1} \times 2\pi\mathbb{Z}^{d_x}/L} \frac{1}{\omega_k^2 + \xi_k^2 + |\Delta|^2} \quad (28.14)$$

with $\xi_k \equiv \frac{\hbar^2 k^2}{2m} - \mu$.

Proof 28.22 Varying the action w.r.t. Δ gives

$$\left(\frac{\delta}{\delta \Delta} S \right) [\Delta, \bar{\Delta}] = \frac{\bar{\Delta}}{g} \mp \text{Tr} \left(\mathcal{G} \frac{\delta}{\delta \Delta} \mathcal{G}^{-1} \right) \quad (28.15)$$

The rest is algebra [TBDone and TBDigitalized].

Notation 28.23 One can expand the action w.r.t. a reference-field Δ as

$$\begin{aligned}S_{\text{BCS}}[\Delta + \delta\Delta, \bar{\Delta} + \delta\bar{\Delta}] &\equiv S_{\text{BCS}}[\Delta, \bar{\Delta}] \\ &+ \oint \left(\frac{\delta}{\delta \Delta} S_{\text{BCS}}[\Delta, \bar{\Delta}], \frac{\delta}{\delta \bar{\Delta}} S_{\text{BCS}}[\Delta, \bar{\Delta}] \right) \cdot \left(\frac{\delta\Delta}{\delta\bar{\Delta}} \right) \\ &+ \frac{1}{2!} \oint \oint \left(\frac{\delta\Delta}{\delta\bar{\Delta}} \right)^T \cdot \begin{pmatrix} \frac{\delta^2}{\delta\Delta\delta\bar{\Delta}} S_{\text{BCS}}[\Delta, \bar{\Delta}] & \frac{\delta^2}{\delta\Delta\delta\Delta} S_{\text{BCS}}[\Delta, \bar{\Delta}] \\ \frac{\delta^2}{\delta\bar{\Delta}\delta\bar{\Delta}} S_{\text{BCS}}[\Delta, \bar{\Delta}] & \frac{\delta^2}{\delta\bar{\Delta}\delta\Delta} S_{\text{BCS}}[\Delta, \bar{\Delta}] \end{pmatrix} \cdot \left(\frac{\delta\Delta}{\delta\bar{\Delta}} \right) \\ &+ \mathcal{O} \left(\delta\Delta^3, \delta\Delta^2\delta\bar{\Delta}^{-1}, \delta\Delta^1\delta\bar{\Delta}^{-2}, \delta\bar{\Delta}^{-3} \right).\end{aligned}$$

Expanding the action w.r.t. to one of the (infinitely many! cf. Corollary 28.20) fields that fulfills the gap-equation, the contribution that is linear in $\delta\Delta$ vanishes.

Hypothesis 28.24 We perform a saddle-point-approximation and compute the partition-function as sum over mean-field-configurations with different phases of the order parameter.

$$Z[j, \bar{j}] \approx \sum_{\theta \in [0, 2\pi)} Z_{\theta}[j, \bar{j}] \quad (28.16)$$

with

$$Z_\theta[j, \bar{j}] = \int \mathcal{D}(\delta\Delta, \delta\bar{\Delta}) \exp -S_{\text{BCS}}[\Delta + \delta\Delta, \bar{\Delta} + \delta\bar{\Delta}, j, \bar{j}]. \quad (28.17)$$

Our hypothesis is now, that in the superconducting state, not all of these (infinitely many) summands contribute "matter", but only a single one. We therefore replace the partition-function, which contains all the information about the system, as

$$Z[j, \bar{j}] \mapsto Z_\theta[j, \bar{j}] \quad (28.18)$$

We are therefore left with the computation of a single functional integral again.

Remark 28.24.1 Note that this also implies that the expectation-values to determine the Green-functions have to be taken w.r.t. this new partition-function.

Lemma 28.25 In the mean-field-approximation we evaluate the partition-function as

$$Z_\theta[j, \bar{j}] \approx \exp -S_{\text{BCS}}[\Delta, \bar{\Delta}, j, \bar{j}]. \quad (28.19)$$

whereas Δ solves the gap-equation.

Proof 28.25 TBD.

Lemma 28.26 One obtains the following decomposition of the action in the one-loop-approximation. Suppressing the dependence on the external sources, this yields

$$S_{\text{BCS}}[\Delta + \delta\Delta, \bar{\Delta} + \delta\bar{\Delta}] \approx S_{\text{BCS}}[\Delta, \bar{\Delta}] + \frac{1}{2!} \oint \oint \left(\frac{\delta\Delta}{\delta\bar{\Delta}} \right)^T \cdot \begin{pmatrix} \frac{\delta^2}{\delta\Delta\delta\bar{\Delta}} S_{\text{BCS}}[\Delta, \bar{\Delta}] & \frac{\delta^2}{\delta\bar{\Delta}\delta\bar{\Delta}} S_{\text{BCS}}[\Delta, \bar{\Delta}] \\ \frac{\delta^2}{\delta\bar{\Delta}\delta\Delta} S_{\text{BCS}}[\Delta, \bar{\Delta}] & \frac{\delta^2}{\delta\Delta\delta\Delta} S_{\text{BCS}}[\Delta, \bar{\Delta}] \end{pmatrix} \cdot \left(\frac{\delta\Delta}{\delta\bar{\Delta}} \right).$$

Proof 28.26 The one-loop-approximation relies on the expansion Now since Δ solves the gap-equation, the first contribution vanishes. The higher orders are being neglected within the saddle-point-approximation.

Theorem 28.27 One can also compute observables beyond the one-loop-approximation. Then one expands the action w.r.t. its interaction.

28.2.6. Landaus Free Energy Model

We finally want to illustrate where the usual Landau-model for the free-energy comes from.

Notation 28.28 We write the Gorkov-Green-function as

$$\mathcal{G}_{\text{Gorkov}}^{-1} \equiv \mathcal{G}_{\text{Gorkov}}^{-1} + V \equiv \mathcal{G}_{\text{Gorkov}}^{-1} \left(1 + \mathcal{G}_{\text{Gorkov}} V \right) \quad (28.20)$$

with

$$\mathcal{G}_{\text{Gorkov}}^{-1} \equiv \mathcal{G}_{\text{Gorkov}}^{-1}[\Delta = \bar{\Delta} = 0]. \quad (28.21)$$

Lemma 28.29 It holds

$$\text{Tr Log } \mathcal{G}_{\text{Gorkov}}^{-1} \equiv \text{const.} + \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \text{Tr} \left(\mathcal{G}_{\text{Gorkov}} V \right)^k \quad (28.22)$$

Proof 28.29 See Appendix, Proofs, page 558.

Theorem 28.30 Expanding the expansion up to fourth order in the field Δ gives the action

$$\begin{aligned} S_{\text{BCS}}[\Delta, \bar{\Delta}, j, \bar{j}] \approx & - \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ \omega_m \in \Omega_{+1}}} \frac{|\Delta(\text{i}\omega_m, \mathbf{q})|^2}{g} \\ & + T B D_1 |\Delta|^2 \\ & + T B D_2 |\Delta|^4 \\ & \mp (\bar{j}_{\uparrow}, -j_{\downarrow}) \cdot \left(-\frac{1}{\hbar} \mathcal{G}_{\text{Gorkov}}[\Delta, \bar{\Delta}] \right) \cdot \begin{pmatrix} j_{\uparrow} \\ -j_{\downarrow} \end{pmatrix} \end{aligned}$$

Proof 28.30 See Appendix, Proofs, page 558.

29. Appendix - Crystal Structure Determination with the Debye-Scherrer Method

Motivation and Overview

After this part of the notes you will understand ...

- how one determines the crystal structure of a material by evaluating so-called x-ray diffraction patterns like in Figure 29.1.

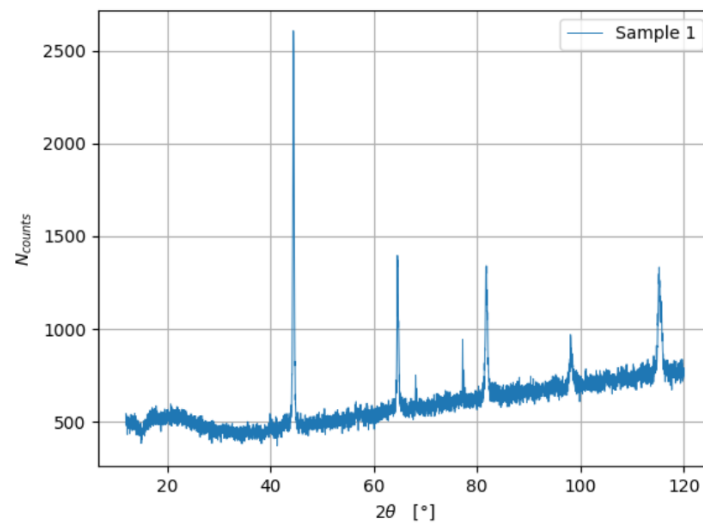


Figure 29.1.: X-Ray Diffraction Pattern of a sample that will be examined / identified in the laboratory-part of the course.

29.1. Preparation: The Miller-Indices

Notation 29.1 For the discussion of the „Miller-indices“ (to be introduced below) it is advantageous to have conventional lattice-vectors. We therefore consider a crystal with conventional lattice-vectors $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$ in this subsection.

Definition 29.2 A *lattice plane* is defined by at least three grid points that do not lie on a straight line.

Lemma 29.3 Each lattice-plane has intersection-points $\vec{S}_i \equiv m_i \mathbf{a}_i$ (whereas $m_i \in \mathbb{Q} \cup \{\infty\}$) with the axes that are associated to the primitive basis-vectors, whereas $\vec{S}_i \equiv \infty \cdot \vec{a}_i$ is defined for planes that never intersect an axis (i.e. are parallel to it).

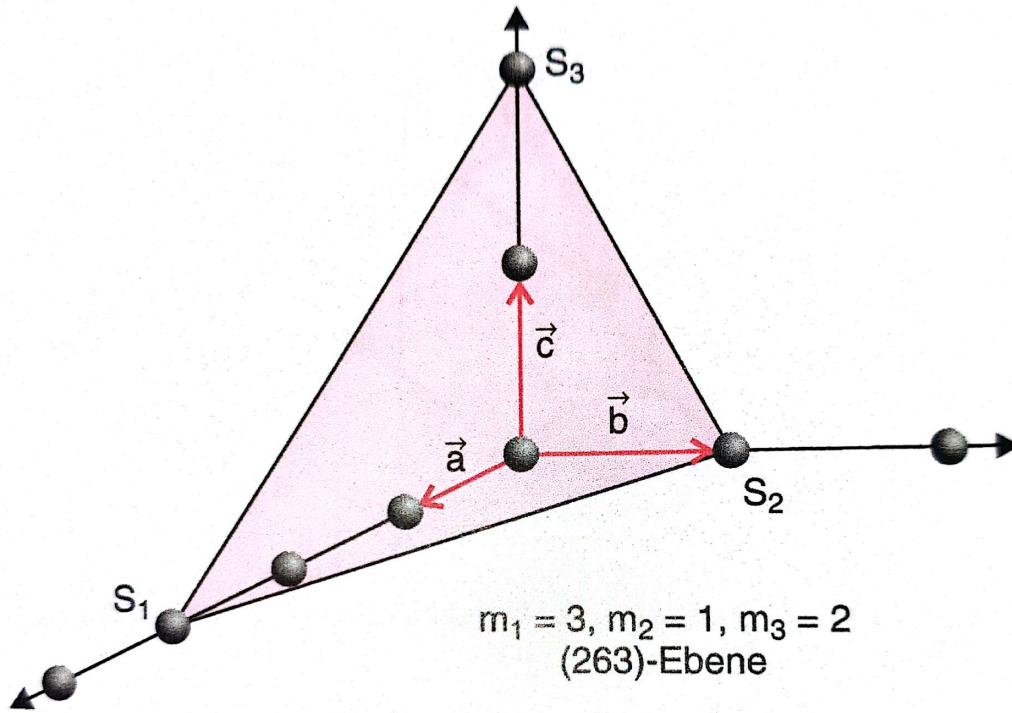


Figure 29.2.: Excerpt from Wolfgang Demtröders Experimentalphysik 3.

Proof 29.3 By taking two non-identical vectors that span the plane and simultaneously represents differences between lattice-points, we can form linear combinations such that $m_i \in \mathbb{Q}$.

Definition 29.4 Let $p \in \mathbb{N}_{>0}$ denote the smallest positive natural number s.t.

$$h = \frac{p}{m_1} \quad k = \frac{p}{m_2} \quad l = \frac{p}{m_3} \quad (29.1)$$

are three (non-partial) integers. This triple (hkl) is called the Miller's indices associated to the lattice-plane above.

Remark 29.4.1 If one considers the lattice-plane, that has the intersection-points $\vec{S}_i \equiv nm_i \mathbf{a}_i$ with $m_i \in \mathbb{Q}$ and $n \in \mathbb{Z}$, then the number $|np|$ is the smallest integer s.t.

$$\pm h = \frac{|np|}{nm_1} \quad \pm k = \frac{|np|}{nm_2} \quad \pm l = \frac{|np|}{nm_3} \quad (29.2)$$

are three (non-partial) integers. We therefore see, that the tripel $\pm(hkl)$ is not in one-to-one correspondence to a single lattice-plane, but in „infinity-to-one“ correspondence to infinitely many layers of lattice-planes.

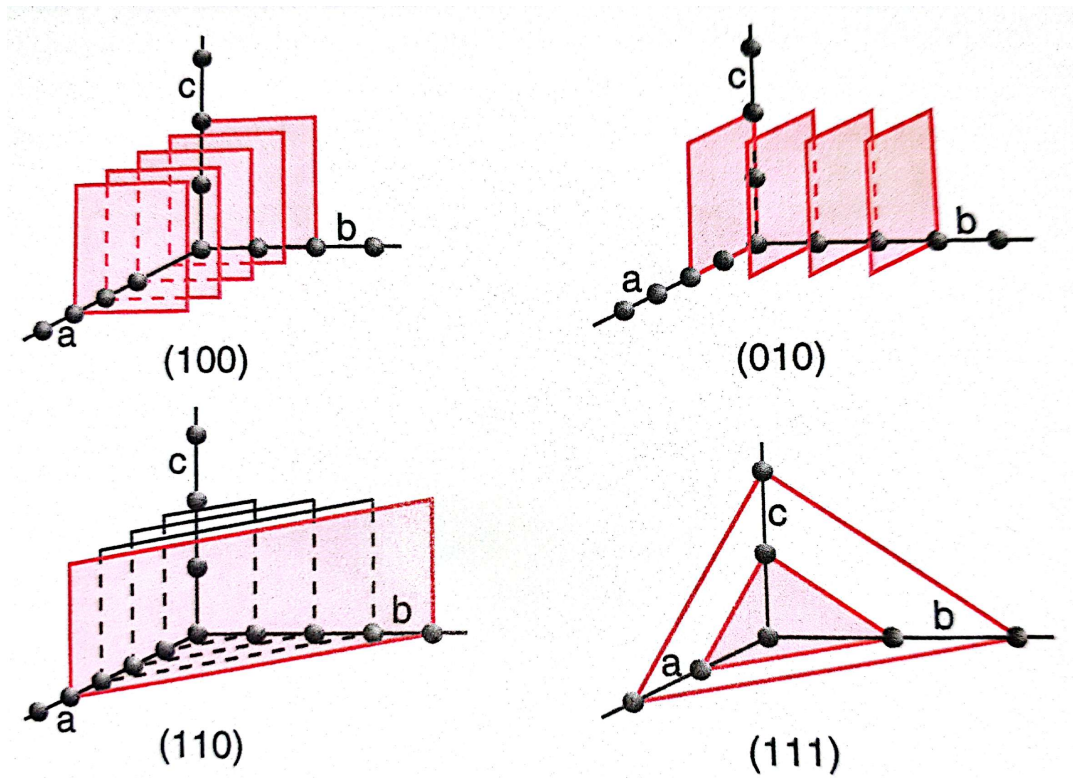


Figure 29.3.: Excerpt from Wolfgang Demtröders Experimentalphysik 3.

Lemma 29.5 The reciprocal lattice-vector $\mathbf{G} = h\mathbf{b}_1 + k\mathbf{b}_2 + l\mathbf{b}_3$ is orthogonal to the planes (hkl) .

Proof 29.5 See Appendix, p. 553.

Lemma 29.6 The absolute-value of $\mathbf{G}$ is inversely proportional to the distance d_{hkl} of two neighboring lattice-planes of the layers (hkl) as

$$|\mathbf{G}| = \frac{2\pi}{d_{hkl}}. \quad (29.3)$$

Proof 29.6 The vector $\mathbf{G}/|\mathbf{G}|$ has length one and is orthogonal to the plane. We consider now any lattice-point on the plane as „origin“. A vector, that points to a point in a neighboring plane is, for example, $\mathbf{v} \equiv \frac{\mathbf{a}_1}{h}$ whereas we assume without loss of generality that $h \neq 0$. The distance $d \equiv d_{hkl}$ between those two planes is then given as

$$d_{hkl} = \frac{\mathbf{G}}{G} \cdot \mathbf{v} = \frac{2\pi}{G}. \quad (29.4)$$

Corollary 29.7 For a three dimensional simple cubic lattice we obtain for the distance d_{hkl} between lattice-layers described by the tripel (hkl)

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}. \quad (29.5)$$

Proof 29.7 Follows from the orthogonality of the reciprocal lattice-vectors and the formula in the Lemma.

29.2. The Scattering-Amplitude

29.2.1. A Single Scattering-Center

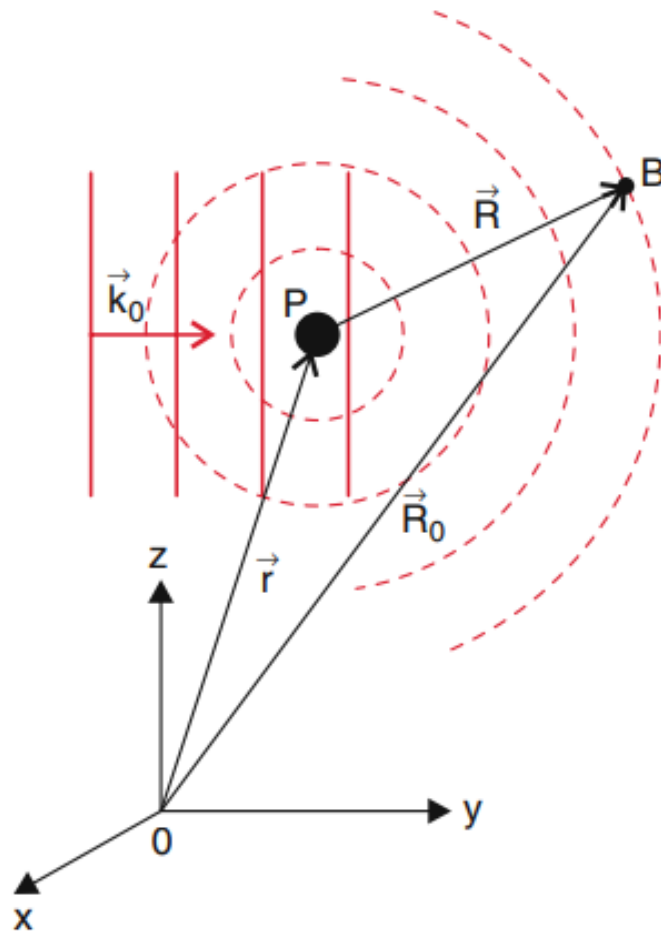


Figure 29.4.: Excerpt from Wolfgang Demtröders Experimentalphysik 3.

We consider as in the figure a scattering-center (i.e. a single atom inside a crystal) at the position described by the vector $\mathbf{r}$. If a plane wave

$$\mathbf{E}(\mathbf{x}) = \mathbf{A}_e e^{i\mathbf{k}_0 \cdot \mathbf{x} - \omega t} \quad (29.6)$$

falls into this scattering-center, a fraction f of this plane-wave is being scattered as a spherical wave¹ into all directions. In an observation-point $B(\mathbf{R}_0)$ with $\mathbf{R}_0 \gg \mathbf{r}$ we have for the amplitude A_B of the scattered wave

¹This is far from obvious. ToBeElaborated.

$$\begin{aligned}
 A_B &= f \frac{\mathbf{E}(\mathbf{x})}{R} e^{i\mathbf{k} \cdot \mathbf{R}} \\
 &= f \frac{A_e}{R} e^{i\mathbf{k}_0 \cdot \mathbf{x} - i\omega t} e^{i\mathbf{k} \cdot \mathbf{R}} \\
 &= f \frac{A_e}{R} e^{i\mathbf{k} \cdot \mathbf{R}_0} e^{i(\mathbf{k}_0 - \mathbf{k}) \cdot \mathbf{x}} e^{-i\omega t} \\
 &= C f e^{i\Delta\mathbf{k} \cdot \mathbf{x}}
 \end{aligned}$$

whereas we introduced the abbreviations $\Delta\mathbf{k} = \mathbf{k}_0 - \mathbf{k}$ and $C = (A_e/R) e^{i(\mathbf{k} \cdot \mathbf{R}_0 - \omega t)}$.

29.2.2. A Crystal with a mono-atomic basis as Scattering-Centers at Lattice-Positions

In a crystal with a mono-atomic basis, we have many scattering-centers with positions $\mathbf{R}_n \equiv \sum_j n_j \mathbf{a}_j$. This yields in a calculation analogous to the one above the amplitude A_B of the scattered waves

$$\begin{aligned}
 A_B &= C f \sum_{\mathbf{R}_n \in L_f} e^{i\Delta\mathbf{k} \cdot \mathbf{R}_n} \\
 &= C f \mathcal{F}_{\text{Lattice}}(\Delta\mathbf{k}),
 \end{aligned}$$

whereas we introduced the so-called „lattice-factor“ $\mathcal{F}_{L_f}(\Delta\mathbf{k})$. The usual way to proceed at this point is to argue that the lattice-factor is non-zero, whenever the Laue-condition

$$\Delta\mathbf{k} \cdot \mathbf{R}_n \in 2\pi\mathbb{Z} \quad \Longleftrightarrow \quad \Delta\mathbf{k} \equiv \mathbf{G}_{hkl} \in \text{RL}_{\infty} \quad (29.7)$$

is fulfilled, and averages to zero (due to a so-called „random phase cancellation“) whenever this is not the case, which reads as a formula

$$\mathcal{F}_{\text{Lattice}}(\Delta\mathbf{k}) \approx \begin{cases} N_p & \text{if } \Delta\mathbf{k} \equiv \mathbf{G}_{hkl} \in \text{RL}_{\infty} \\ 0 & \text{otherwise (due to rpc).} \end{cases} \quad (29.8)$$

It is important to note, that this equation alone gives us nothing else but the Bragg-condition, i.e. implying that the lattice-structure does imply the existence of constructive interference. However, in the next step we will determine proper tools in order to make further conclusions about the type of the crystal-structure.

29.2.3. A Crystal with a many-atomic basis as Scattering-Centers at Crystal-Positions

In a crystal with a potentially many-atomic basis, we have many scattering-centers with positions

$$\mathbf{R}_{n\kappa} \equiv \sum_j n_j \mathbf{a}_j + \sum_j c_{j\kappa} \mathbf{a}_j \equiv \mathbf{R}_n + \sum_j c_{j\kappa} \mathbf{a}_j \quad (29.9)$$

with $c_{j\kappa} < 1$. This yields in a calculation analogous to the one above the amplitude A_B of the scattered waves

$$A_B = C \sum_{\mathbf{R}_n \in L_f} e^{i\Delta\mathbf{k} \cdot \mathbf{R}_n} \sum_{\kappa \in \{1, \dots, r\}} f_{\kappa} e^{i\Delta\mathbf{k} \cdot (\sum_j c_{j\kappa} \mathbf{a}_j)} \\ = C \mathcal{F}_{\text{Lattice}}(\Delta\mathbf{k}) \mathcal{F}_{\text{Atomic}}(\Delta\mathbf{k}),$$

whereas we introduced the so-called „atomic scattering-factor“ $\mathcal{F}_{\text{Atomic}}(\Delta\mathbf{k})$.

29.3. The Debye-Scherrer Setup

Copy-Pasted from [Demtröder, Experimentalphysik 3] into DeepL and subsequently Copy-Pasted (and slightly adapted) into this document:

In the case of polycrystalline solids (e.g. pulverized crystalline material such as common salt, diamond powder, etc.), the microcrystals and their lattice planes are randomly oriented (...), and the Bragg Bragg method can therefore not be used. Here is a method developed by Peter J. W.

Debye (1884-1966) and Paul Scherrer (1890-1969) is useful here. The general idea is shown in Fig. 11.29. A thin tube containing the powdered sample is placed in the axis of a cylindrical vessel and irradiated with a monochromatic parallel X-ray beam. Only those microcrystals whose orientation is suitable to fulfill the Bragg condition contribute noticeably to the scattered radiation. This means that only at certain angles $2\theta_i$ against the direction of incidence, i.e. on cones with full aperture angles $4\theta_i$, reflected intensity is to be expected. It is measured on a film strip which is attached to the inner wall of the cylindrical container with radius RZ . On the unrolled developed film you can therefore see blackened approximated circular arcs (Fig. 11.29b), from whose radii $R_i \equiv R_C \sin(2\theta_i)$ the angles θ_i and thus, for a known wavelength λ the grid plane spacing d_{hkl} can be determined.

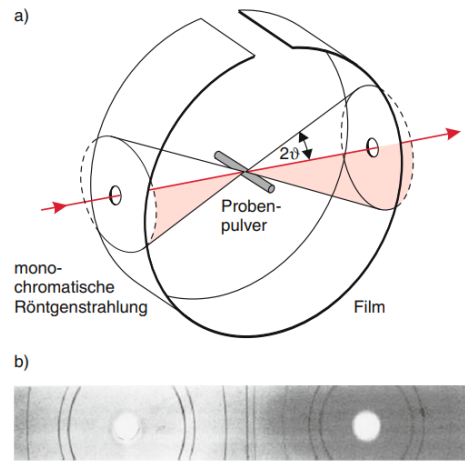


Abb. 11.29a,b. Debye-Scherrer-Verfahren: (a) Experimentelle Anordnung; (b) Aufnahme der ringförmigen Interferenzmaxima von Kupferpulver

Figure 29.5.: Excerpt from Wolfgang Demtröders Experimentalphysik 3.

29.4. Cooking-Recipe to determine the Crystal-Structure from a Debye-Scherrer-Experiment

Given a diffraction pattern N_{counts} vs 2θ , we give a a cooking-recipe on how to determine the crystal-structure, whereas we restrict our attention to cubic lattices with only a single type of atom inside the crystal.

See the Lab-Sheet for the rest.

30. Appendix - Proofs for Part I

Proof 30.1 of Lemma 1.3

Inserting the one equation into the other gives

$$\begin{aligned}
 \xi_j &= \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} \hat{\xi}_n e^{-i2\pi \frac{n}{N} j} \\
 &= \frac{1}{N} \sum_{n=0}^{N-1} \sum_{m=0}^{N-1} \xi_m e^{-i2\pi \frac{n}{N} j} e^{+i2\pi \frac{m}{N} n} \\
 &= \frac{1}{N} \sum_{n=0}^{N-1} \sum_{m=0}^{N-1} \xi_m \exp \left[2\pi i \frac{1}{N} (mn - jn) \right] \\
 &= \frac{1}{N} \sum_{m=0}^{N-1} \xi_m \sum_{n=0}^{N-1} \left(\exp \left[2\pi i \frac{1}{N} (m - j) \right] \right)^n = \xi_j
 \end{aligned} \tag{30.1}$$

whereas we used in the last step the finite geometric series

$$\sum_{n=0}^{N-1} \left(\exp \left[2\pi i \frac{1}{N} (m - j) \right] \right)^n = \begin{cases} N, & m = j \\ \frac{1 - e^{2\pi i (m-j)}}{1 - e^{2\pi i \frac{1}{N} (m-j)}} = 0, & m \neq j \end{cases} \tag{30.2}$$

Proof 30.2 of Lemma 1.11¹

Using the Fourier-expansion in Equation (1.5) we obtain²

$$\mathcal{G}_\zeta(\tau) = \frac{1}{\hbar\beta} \sum_{n=-\infty}^{+\infty} \mathcal{G}_\zeta(i\omega_n) e^{-in\pi\tau/\hbar\beta} \tag{30.3}$$

with coefficients

$$\mathcal{G}_\zeta(i\omega_n) = \frac{1}{2} \int_{-\hbar\beta}^{+\hbar\beta} d\tau \mathcal{G}_\zeta(\tau) e^{in\pi\tau/\hbar\beta} \tag{30.4}$$

Bosons: $\zeta = +1$

We use that

¹Reference: [30] p. 110 onwards.

²Notice that we have a different-prefactor-convention w.r.t. the equation before. This is *the* prefactor-convention for Matsubara-Green-functions.

$$-\hbar\beta < \tau < 0 \quad \Rightarrow \quad \mathcal{G}_\zeta(\tau) = \mathcal{G}_\zeta(\tau + \hbar\beta) \quad (30.5)$$

and therefore we obtain

$$\mathcal{G}_\zeta(i\omega_n) = \frac{1}{2} \left[\int_0^{\hbar\beta} d\tau \mathcal{G}_\zeta(\tau) e^{in\pi\tau/\hbar\beta} + \int_{-\hbar\beta}^0 d\tau \mathcal{G}_\zeta(\tau) e^{in\pi\tau/\hbar\beta} \right] \quad (30.6)$$

For the second term we change variables from $\tau \mapsto \tau + \hbar\beta$ which gives

$$\mathcal{G}_\zeta(i\omega_n) = \frac{1}{2} (1 + e^{in\pi}) \int_0^{\hbar\beta} d\tau \mathcal{G}_\zeta(\tau) e^{in\pi\tau/\hbar\beta} \quad (30.7)$$

This expression has the feature that $\mathcal{G}_\zeta(i\omega_n) = 0$ whenever n is an odd integer. For bosons we hence obtain

$$\begin{aligned} \mathcal{G}_\zeta(\tau) &= \sum_{\omega_n \in \Omega_{+1}} e^{-i\omega_n\tau} \mathcal{G}_\zeta(i\omega_n), \\ \mathcal{G}_\zeta(i\omega_n) &= \int_0^{\hbar\beta} d\tau e^{i\omega_n\tau} \mathcal{G}_\zeta(\tau), \\ \Omega_{+1} &= \{ \omega_n = 2n\pi/\hbar\beta \mid n \in \mathbb{Z} \}. \end{aligned}$$

Fermions: $\zeta = -1$

We use that

$$-\hbar\beta < \tau < 0 \quad \Rightarrow \quad \mathcal{G}_\zeta(\tau) = -\mathcal{G}_\zeta(\tau + \hbar\beta) \quad (30.8)$$

and therefore we obtain

$$\mathcal{G}_\zeta(i\omega_n) = \frac{1}{2} \left[\int_0^{\hbar\beta} d\tau \mathcal{G}_\zeta(\tau) e^{in\pi\tau/\hbar\beta} + \int_{-\hbar\beta}^0 d\tau \mathcal{G}_\zeta(\tau) e^{in\pi\tau/\hbar\beta} \right] \quad (30.9)$$

For the second term we obtain in the same manner

$$\mathcal{G}_\zeta(i\omega_n) = \frac{1}{2} (1 - e^{in\pi}) \int_0^{\hbar\beta} d\tau \mathcal{G}_\zeta(\tau) e^{in\pi\tau/\hbar\beta} \quad (30.10)$$

This expression has the feature that $\mathcal{G}_\zeta(i\omega_n) = 0$ whenever n is an even integer. For fermions we hence obtain

$$\begin{aligned} \mathcal{G}_\zeta(\tau) &= \sum_{\omega_n \in \Omega_{-1}} e^{-i\omega_n\tau} \mathcal{G}_\zeta(i\omega_n), \\ \mathcal{G}_\zeta(i\omega_n) &= \int_0^{\hbar\beta} d\tau e^{i\omega_n\tau} \mathcal{G}_\zeta(\tau), \\ \Omega_{-1} &= \{ \omega_n = (2n+1)\pi/\hbar\beta \mid n \in \mathbb{Z} \}. \end{aligned}$$

Proof 30.3 of Corollary 1.12

By straight-forward integration we obtain

$$\int_0^{\hbar\beta} d\tau e^{i\tau\omega_m} = \begin{cases} \frac{1}{i\omega_m} (e^{i\tau\omega_m} - 1) & \text{for } \omega_m \neq 0 \\ \hbar\beta & \text{for } \omega_m = 0 \end{cases} \quad (30.11)$$

hence we have proven the identity.

Proof 30.4 of Lemma 1.16

We obtain in usual Dirac-notation

$$\begin{aligned} \langle \tau | \mathcal{G}_\zeta \rangle &= \mathcal{G}_\zeta(\tau) \\ &= \sum_{\omega_n \in \Omega_\zeta} \langle \tau | i\omega_n \rangle \langle i\omega_n | \mathcal{G}_\zeta \rangle \\ &= \frac{1}{\hbar\beta} \sum_{\omega_n \in \Omega_\zeta} \mathcal{G}_\zeta(i\omega_n) e^{-i\omega_n\tau} \end{aligned}$$

and also

$$\begin{aligned} \langle \tau | \tau' \rangle &= \delta(\tau - \tau') \\ &= \frac{1}{\hbar\beta} \sum_{\omega_n \in \Omega_\zeta} e^{-i\omega_n\tau + i\omega_n\tau'} \\ &= \frac{1}{\hbar\beta} \sum_{\omega_n \in \Omega_\zeta} e^{i\omega_n(\tau - \tau')} \end{aligned}$$

Proof 30.5 of Lemma 2.11 To prove the canonical commutation relations:

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij},$$

we start with the definitions of the position and momentum operators:

$$\hat{x}_i = x_i, \quad \hat{p}_j = -i\hbar \frac{\partial}{\partial x_j}.$$

The commutator is defined as:

$$[\hat{x}_i, \hat{p}_j] = \hat{x}_i \hat{p}_j - \hat{p}_j \hat{x}_i.$$

Applying this to a test function $f(x)$, we have:

$$[\hat{x}_i, \hat{p}_j]f(x) = x_i \left(-i\hbar \frac{\partial f}{\partial x_j} \right) - \left(-i\hbar \frac{\partial}{\partial x_j} (x_i f(x)) \right).$$

Expanding the derivative $\frac{\partial}{\partial x_j} (x_i f(x))$ using the product rule:

$$\frac{\partial}{\partial x_j} (x_i f(x)) = \delta_{ij} f(x) + x_i \frac{\partial f}{\partial x_j}.$$

Substituting back, we get:

$$[\hat{x}_i, \hat{p}_j]f(x) = -i\hbar x_i \frac{\partial f}{\partial x_j} + i\hbar \left(\delta_{ij} f(x) + x_i \frac{\partial f}{\partial x_j} \right).$$

Simplifying:

$$[\hat{x}_i, \hat{p}_j]f(x) = i\hbar \delta_{ij} f(x).$$

Thus, we conclude:

$$[\hat{x}_i, \hat{p}_j] = i\hbar \delta_{ij}.$$

Proof 30.6 of Theorem 2.15 For concreteness we consider a single particle with Hamilton-function given as $H = \frac{p^2}{2m} + V(x)$. The proof can be straightforwardly generalized to many-particle systems with more complicated Hamilton-functions. Substituting the product ansatz $\psi(\mathbf{x}, t) = \psi(\mathbf{x})\phi(t)$ into the time-dependent Schrödinger equation:

$$i\hbar \frac{\partial \psi(\mathbf{x}, t)}{\partial t} = \left(-\frac{\hbar^2}{2m} \nabla_{\mathbf{x}}^2 + V(\mathbf{x}) \right) \psi(\mathbf{x}, t), \quad (30.12)$$

we obtain

$$i\hbar \frac{\partial}{\partial t} (\psi(\mathbf{x})\phi(t)) = \psi(\mathbf{x}) \left(-\frac{\hbar^2}{2m} \nabla_{\mathbf{x}}^2 \psi(\mathbf{x}) + V(\mathbf{x})\psi(\mathbf{x}) \right). \quad (30.13)$$

Using the product rule for differentiation on the left-hand side gives:

$$i\hbar \psi(\mathbf{x}) \frac{d\phi(t)}{dt} = \phi(t) \left(-\frac{\hbar^2}{2m} \nabla_{\mathbf{x}}^2 \psi(\mathbf{x}) + V(\mathbf{x})\psi(\mathbf{x}) \right). \quad (30.14)$$

Dividing through by $\psi(\mathbf{x})\phi(t)$, we get:

$$i\hbar \frac{1}{\phi(t)} \frac{d\phi(t)}{dt} = \frac{-\frac{\hbar^2}{2m} \nabla_{\mathbf{x}}^2 \psi(\mathbf{x}) + V(\mathbf{x})\psi(\mathbf{x})}{\psi(\mathbf{x})}. \quad (30.15)$$

The left-hand side is a function of t only, while the right-hand side is a function of $\mathbf{x}$ only. Hence, both must equal a constant, say E . This gives two separate equations:

$$i\hbar \frac{1}{\phi(t)} \frac{d\phi(t)}{dt} = E, \quad (30.16)$$

$$-\frac{\hbar^2}{2m} \nabla_{\mathbf{x}}^2 \psi(\mathbf{x}) + V(\mathbf{x})\psi(\mathbf{x}) = E\psi(\mathbf{x}). \quad (30.17)$$

The first equation simplifies to

$$\frac{d\phi(t)}{dt} = -\frac{iE}{\hbar} \phi(t), \quad (30.18)$$

which has the solution

$$\phi(t) = e^{-iEt/\hbar}. \quad (30.19)$$

The second equation is the time-independent Schrödinger equation:

$$E\psi(\mathbf{x}) = H\psi(\mathbf{x}). \quad (30.20)$$

Thus, the theorem is proven.

Proof 30.7 of Corollary 2.16 A stationary state is defined as a quantum state for which the expectation value of any observable $\hat{O}$ is constant in time. The expectation value is given by

$$\langle \hat{O} \rangle(t) = \int \psi^*(\mathbf{x}, t) \hat{O} \psi(\mathbf{x}, t) d^d \mathbf{x}. \quad (30.21)$$

Substituting $\psi(\mathbf{x}, t) = \psi(\mathbf{x})\phi(t)$ and $\phi(t) = e^{-iEt/\hbar}$ gives

$$\psi^*(\mathbf{x}, t) = \psi^*(\mathbf{x})e^{iEt/\hbar}, \quad \psi(\mathbf{x}, t) = \psi(\mathbf{x})e^{-iEt/\hbar}. \quad (30.22)$$

The time-dependent factors cancel out:

$$\langle \hat{O} \rangle(t) = \int \psi^*(\mathbf{x}) \hat{O} \psi(\mathbf{x}) d^d \mathbf{x}. \quad (30.23)$$

Since $\psi(\mathbf{x})$ is time-independent, the expectation value $\langle \hat{O} \rangle(t)$ is constant in time. Therefore, the state is stationary, and the corollary is proven.

Proof 30.8 of Theorem 2.25 We first note the auxiliary identities

$$\begin{aligned}\frac{d}{dt}U(t, 0) &= -\frac{i}{\hbar}HU(t, 0) \\ \frac{d}{dt}U(t, 0)^\dagger &= \frac{d}{dt}U(0, t) = \frac{i}{\hbar}U(0, t)H = \frac{i}{\hbar}U(t, 0)^\dagger H\end{aligned}$$

Then it follows

$$\begin{aligned}\frac{d}{dt}X_H(t) &= \frac{d}{dt}U(t, 0)^\dagger X(t)U(t, 0) \\ &= \left(\frac{d}{dt}U(t, 0)^\dagger\right) X(t)U(t, 0) + U(t, 0)^\dagger \left(\frac{d}{dt}X(t)\right) U(t, 0) + U(t, 0)^\dagger X(t) \left(\frac{d}{dt}U(t, 0)\right) \\ &= \frac{i}{\hbar}U(t, 0)^\dagger HX(t)U(t, 0) + U(t, 0)^\dagger \left(\frac{d}{dt}X(t)\right) U(t, 0) + U(t, 0)^\dagger X(t) \left(-\frac{i}{\hbar}HU(t, 0)\right) \\ &= \frac{i}{\hbar}U(t, 0)^\dagger (HX(t) - X(t)H)U(t, 0) + U(t, 0)^\dagger \left(\frac{d}{dt}X(t)\right) U(t, 0) \\ &= \frac{i}{\hbar}U(t, 0)^\dagger [H, X(t)]U(t, 0) + U(t, 0)^\dagger \left(\frac{d}{dt}X(t)\right) U(t, 0) \\ &= -\frac{i}{\hbar}[X, H]_H(t) + U(t, 0)^\dagger \left(\frac{d}{dt}X(t)\right) U(t, 0)\end{aligned}$$

Proof 30.9 of Corollary 3.13

To be done: Needs to be updated such that the details are fine.

- Cf. the weblink for the details of the proof:

https://en.wikipedia.org/wiki/Fourier_series#Fourier_series_of_a_Bravais-lattice-periodic_function

- TBD: Fixing the details of the proof with the notes. Cf. Waste-Folder-1

We provide the proof for the three-dimensional case³. The one- and two-dimensional cases can be proven analogously.⁴

We define a new function g as,

$$g(x_1, x_2, x_3) := f\left(x_1 \frac{\mathbf{a}_1}{a_1} + x_2 \frac{\mathbf{a}_2}{a_2} + x_3 \frac{\mathbf{a}_3}{a_3}\right).$$

This new function, $g(x_1, x_2, x_3)$, is now a function of three variables, each of which has periodicity a_1 , a_2 , and a_3 respectively:

$$g(x_1, x_2, x_3) = g(x_1 + a_1, x_2, x_3) = g(x_1, x_2 + a_2, x_3) = g(x_1, x_2, x_3 + a_3).$$

This enables us to build up a set of Fourier coefficients, each being indexed by three independent integers m_1, m_2, m_3 . If we write a series for g on the interval $[0, a_1]$ for x_1 , we can define the following:

$$h^{\text{one}}(m_1, x_2, x_3) \equiv \frac{1}{a_1} \int_0^{a_1} g(x_1, x_2, x_3) \cdot e^{-i2\pi \frac{m_1}{a_1} x_1} dx_1.$$

And then we can write:

$$g(x_1, x_2, x_3) = \sum_{m_1=-\infty}^{\infty} h^{\text{one}}(m_1, x_2, x_3) \cdot e^{i2\pi \frac{m_1}{a_1} x_1}.$$

Further defining:

$$\begin{aligned} h^{\text{two}}(m_1, m_2, x_3) &\equiv \frac{1}{a_2} \int_0^{a_2} h^{\text{one}}(m_1, x_2, x_3) \cdot e^{-i2\pi \frac{m_2}{a_2} x_2} dx_2 \\ &= \frac{1}{a_2} \int_0^{a_2} dx_2 \frac{1}{a_1} \int_0^{a_1} dx_1 g(x_1, x_2, x_3) \cdot e^{-i2\pi \frac{m_1}{a_1} x_1} \cdot e^{-i2\pi \frac{m_2}{a_2} x_2} \end{aligned}$$

We can write g once again as:

$$g(x_1, x_2, x_3) = \sum_{m_1=-\infty}^{\infty} \sum_{m_2=-\infty}^{\infty} h^{\text{two}}(m_1, m_2, x_3) \cdot e^{i2\pi \frac{m_1}{a_1} x_1} \cdot e^{i2\pi \frac{m_2}{a_2} x_2}$$

³ Cf. [85] in Waste-Folder 3.

⁴ TBD: Add the proof for the one- and two-dimensional cases.

Finally, applying the same for the third coordinate, we define:

$$\begin{aligned} h^{\text{three}}(m_1, m_2, m_3) &\equiv \frac{1}{a_3} \int_0^{a_3} h^{\text{two}}(m_1, m_2, x_3) \cdot e^{-i2\pi \frac{m_3}{a_3} x_3} dx_3 \\ &= \frac{1}{a_3} \int_0^{a_3} dx_3 \frac{1}{a_2} \int_0^{a_2} dx_2 \frac{1}{a_1} \int_0^{a_1} dx_1 g(x_1, x_2, x_3) \cdot e^{-i2\pi \frac{m_1}{a_1} x_1} \cdot e^{-i2\pi \frac{m_2}{a_2} x_2} \cdot e^{-i2\pi \frac{m_3}{a_3} x_3} \end{aligned}$$

We write g as:

$$\begin{aligned} g(x_1, x_2, x_3) &= \sum_{m_1=-\infty}^{\infty} \sum_{m_2=-\infty}^{\infty} \sum_{m_3=-\infty}^{\infty} h^{\text{three}}(m_1, m_2, m_3) \cdot e^{i2\pi \frac{m_1}{a_1} x_1} \cdot e^{i2\pi \frac{m_2}{a_2} x_2} \cdot e^{i2\pi \frac{m_3}{a_3} x_3} \\ &= \sum_{m_1, m_2, m_3 \in \mathbb{Z}} h^{\text{three}}(m_1, m_2, m_3) \cdot e^{i2\pi \left(\frac{m_1}{a_1} x_1 + \frac{m_2}{a_2} x_2 + \frac{m_3}{a_3} x_3 \right)}. \end{aligned}$$

Now, every reciprocal lattice vector can be written as

$\mathbf{G} = m_1 \mathbf{b}_1 + m_2 \mathbf{b}_2 + m_3 \mathbf{b}_3$, where m_i are integers and $\mathbf{b}_i$ are reciprocal lattice vectors to satisfy $\mathbf{b}_i \cdot \mathbf{a}_j = 2\pi \delta_{ij}$ ($\delta_{ij} = 1$ for $i = j$, and $\delta_{ij} = 0$ for $i \neq j$). Then for any arbitrary reciprocal lattice vector $\mathbf{G}$ and arbitrary position vector $\mathbf{r} = \left(x_1 \frac{\mathbf{a}_1}{a_1} + x_2 \frac{\mathbf{a}_2}{a_2} + x_3 \frac{\mathbf{a}_3}{a_3} \right)$ in the original Bravais lattice space, their scalar product is:

$$\mathbf{G} \cdot \mathbf{r} = (m_1 \mathbf{g}_1 + m_2 \mathbf{g}_2 + m_3 \mathbf{g}_3) \cdot \left(x_1 \frac{\mathbf{a}_1}{a_1} + x_2 \frac{\mathbf{a}_2}{a_2} + x_3 \frac{\mathbf{a}_3}{a_3} \right) = 2\pi \left(x_1 \frac{m_1}{a_1} + x_2 \frac{m_2}{a_2} + x_3 \frac{m_3}{a_3} \right).$$

So it is clear that in our expansion of $g(x_1, x_2, x_3) = f(\mathbf{r})$, the sum is actually over reciprocal lattice vectors:

$$f(\mathbf{r}) = \sum_{\mathbf{G}} h(\mathbf{G}) \cdot e^{i\mathbf{G} \cdot \mathbf{r}},$$

where

$$h(\mathbf{G}) = \frac{1}{a_3} \int_0^{a_3} dx_3 \frac{1}{a_2} \int_0^{a_2} dx_2 \frac{1}{a_1} \int_0^{a_1} dx_1 f \left(x_1 \frac{\mathbf{a}_1}{a_1} + x_2 \frac{\mathbf{a}_2}{a_2} + x_3 \frac{\mathbf{a}_3}{a_3} \right) \cdot e^{-i\mathbf{G} \cdot \mathbf{r}}.$$

Assuming

$$\mathbf{r} = (x, y, z) = a_1 x_1 + a_2 x_2 + a_3 x_3,$$

we can solve this system of three linear equations for x, y, z in terms of x_1, x_2, x_3 in order to calculate the volume element in the original Cartesian coordinate system. Once we have x, y, z in terms of x_1, x_2, x_3 , we can calculate the Jacobian determinant:

$$\begin{vmatrix} \frac{\partial x_1}{\partial x} & \frac{\partial x_1}{\partial y} & \frac{\partial x_1}{\partial z} \\ \frac{\partial x_2}{\partial x} & \frac{\partial x_2}{\partial y} & \frac{\partial x_2}{\partial z} \\ \frac{\partial x_3}{\partial x} & \frac{\partial x_3}{\partial y} & \frac{\partial x_3}{\partial z} \end{vmatrix}$$

which after some calculation and applying some non-trivial cross-product identities (TBDone) can be shown to be equal to:

$$\frac{a_1 a_2 a_3}{a_1 \cdot (a_2 \times a_3)}.$$

(It may be advantageous for the sake of simplifying calculations to work in such a Cartesian coordinate system in which it just so happens that a_1 is parallel to the x -axis, a_2 lies in the xy -plane, and a_3 has components of all three axes). The denominator is exactly the volume of the primitive unit cell which is enclosed by the three primitive vectors a_1, a_2, a_3 . In particular, we now know that:

$$dx_1 dx_2 dx_3 = \frac{a_1 a_2 a_3}{a_1 \cdot (a_2 \times a_3)} dx dy dz.$$

We can write now $\hat{h}(\mathbf{G})$ as an integral with the traditional coordinate system over the volume of the primitive cell, instead of with the x_1, x_2, x_3 variables:

$$\hat{h}(\mathbf{G}) = \frac{1}{a_1 \cdot (a_2 \times a_3)} \int_C e^{-i\mathbf{G} \cdot \mathbf{r}} f(\mathbf{r}) d^3r,$$

writing d^3r for the volume element $dx dy dz$; and where C is the primitive unit cell, thus, $a_1 \cdot (a_2 \times a_3)$ is the volume of the primitive unit cell.

Proof 30.10 of Theorem 4.8 We compute

$$\begin{aligned}
 0 &= (H - E) \psi(\mathbf{r}, \boldsymbol{\tau}) \\
 &= \sum_{\alpha \in \mathcal{A}} (H_0 + T_a - E) \chi_\alpha(\boldsymbol{\tau}) \phi_{\alpha, \tau}(\mathbf{r}) \\
 &= \sum_{\alpha \in \mathcal{A}} (\epsilon_{\alpha, \tau} + T_a - E) \chi_\alpha(\boldsymbol{\tau}) \phi_{\alpha, \tau}(\mathbf{r}) \\
 &= \dots
 \end{aligned}$$

We now use the general Leibniz-rule and compute

$$\Delta_{\tau_i} [\chi_\alpha(\boldsymbol{\tau}) \phi_{\alpha, \tau}(\mathbf{r})] = [\Delta_{\tau_i} \chi_\alpha(\boldsymbol{\tau})] \phi_{\alpha, \tau}(\mathbf{r}) + [\nabla_{\tau_i} \chi_\alpha(\boldsymbol{\tau})] \cdot [\nabla_{\tau_i} \phi_{\alpha, \tau}(\mathbf{r})] + \chi_\alpha(\boldsymbol{\tau}) [\Delta_{\tau_i} \phi_{\alpha, \tau}(\mathbf{r})] \quad (30.24)$$

and therefore obtain

$$\begin{aligned}
 T_a \chi_\alpha(\boldsymbol{\tau}) \phi_{\alpha, \tau}(\mathbf{r}) &= \sum_{i=1}^{N_a} -\frac{\hbar^2}{2M_i} [\Delta_{\tau_i} \chi_\alpha(\boldsymbol{\tau})] \phi_{\alpha, \tau}(\mathbf{r}) + \text{remaining terms which will be neglected} \\
 &\approx \sum_{i=1}^{N_a} -\frac{\hbar^2}{2M_i} [\Delta_{\tau_i} \chi_\alpha(\boldsymbol{\tau})] \phi_{\alpha, \tau}(\mathbf{r})
 \end{aligned} \quad (30.25)$$

Inserting this in the expression above yields

$$\dots = \sum_{\alpha \in \mathcal{A}} \left[(\epsilon_{\alpha, \tau} + T_a - E) \chi_\alpha(\boldsymbol{\tau}) \right] \phi_{\alpha, \tau}(\mathbf{r}) \quad (30.26)$$

multiplying with $\phi_{\beta, \tau}^*(\mathbf{r})$ and integrating over the electronic coordinates yields

$$\sum_{\alpha \in \mathcal{A}} \left[(\epsilon_{\alpha, \tau} + T_a - E) \chi_\alpha(\boldsymbol{\tau}) \right] \int d\mathbf{r} \phi_{\beta, \tau}^*(\mathbf{r}) \phi_{\alpha, \tau}(\mathbf{r}) = (\epsilon_{\beta, \tau} + T_a - E) \chi_\beta(\boldsymbol{\tau}) = 0 \iff (T_a + \epsilon_{\beta, \tau}) \chi_\beta(\boldsymbol{\tau}) = E \chi_\beta(\boldsymbol{\tau}) \quad (30.27)$$

Proof 30.11 of Corollary 5.18 The kinetic energy term is straightforward from Lemma 5.17. The potential energy term is a bit more involved, since we have to use the different expansion formulae Equations (5.25) and (5.27) for the coordinates. However, by direct calculation one has

$$\begin{aligned}
 V_{\text{eff}}(\tau) &\approx \frac{1}{2} \sum_{\substack{\mathbf{R}_n, \mathbf{R}_{n'} \in \mathbb{L} \\ \kappa, \kappa' \in \{1, \dots, r\} \\ \alpha, \alpha' \in \{1, 2, 3\}}} \Delta \tau_{\mathbf{n}\kappa\alpha} \left(\frac{\partial^2 V_{\text{eff}}}{\partial \tau_{\mathbf{n}\kappa\alpha} \partial \tau_{\mathbf{n}'\kappa'\alpha'}} \right) (\mathbf{R}) \Delta \tau_{\mathbf{n}'\kappa'\alpha'} \\
 &= \frac{1}{2} \sum_{\substack{\kappa, \kappa' \in \{1, \dots, r\} \\ \alpha, \alpha' \in \{1, 2, 3\}}} \sqrt{M_\kappa M_{\kappa'}} \sum_{\mathbf{R}_n, \mathbf{R}_{n'} \in \mathbb{L}} \Delta \tau_{\mathbf{n}\kappa\alpha} D_{\kappa\alpha, \kappa'\alpha'}(\mathbf{R}_n - \mathbf{R}_{n'}) \Delta \tau_{\mathbf{n}'\kappa'\alpha'} \\
 &= \frac{1}{2} \frac{1}{N^2} \sum_{\substack{\kappa, \kappa' \in \{1, \dots, r\} \\ \alpha, \alpha' \in \{1, 2, 3\}}} \sqrt{M_\kappa M_{\kappa'}} \sum_{\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3 \in \mathbb{Q}} \Delta \tau_{-\mathbf{q}_1\kappa\alpha} D_{\kappa\alpha, \kappa'\alpha'}(\mathbf{q}_2) \Delta \tau_{\mathbf{q}_3\kappa'\alpha'} \sum_{\mathbf{R}_n, \mathbf{R}_{n'} \in \mathbb{L}} e^{i\mathbf{R}_n(\mathbf{q}_1 - \mathbf{q}_2)} e^{-i\mathbf{R}_{n'}(+\mathbf{q}_3 - \mathbf{q}_2)} \\
 &= \frac{1}{2} \sum_{\substack{\kappa, \kappa' \in \{1, \dots, r\} \\ \alpha, \alpha' \in \{1, 2, 3\}}} \sqrt{M_\kappa M_{\kappa'}} \sum_{\mathbf{q} \in \mathbb{Q}} \Delta \tau_{-\mathbf{q}\kappa\alpha} D_{\kappa\alpha, \kappa'\alpha'}(\mathbf{q}) \Delta \tau_{\mathbf{q}\kappa'\alpha'} \\
 &= \frac{1}{2} \sum_{\mathbf{q} \in \mathbb{Q}} \sum_{\substack{\kappa, \kappa' \in \{1, \dots, r\} \\ \alpha, \alpha' \in \{1, 2, 3\}}} \Delta \tau_{-\mathbf{q}\kappa\alpha} \frac{D_{\kappa\alpha, \kappa'\alpha'}(\mathbf{q})}{\sqrt{M_\kappa M_{\kappa'}}^{-1}} \Delta \tau_{\mathbf{q}\kappa'\alpha'}
 \end{aligned}$$

proving the claim.

Proof 30.12 of Lemma 5.19 By direct calculation one obtains with the symmetry of the real-space-dynamical-matrix from Remark 5.6.1, together with the translation invariance from Theorem 5.15, through a long chain of equations

$$\begin{aligned}
 (D^T(\mathbf{q}))_{\kappa\alpha, \kappa'\alpha'} &= D_{\kappa'\alpha', \kappa\alpha}(\mathbf{q}) = \sum_{\mathbf{R}_n \in \mathbb{L}_f} D_{\kappa'\alpha', \kappa\alpha}(\mathbf{R}_n) e^{+i\mathbf{q} \cdot \mathbf{R}_n} = \sum_{\mathbf{R}_n \in \mathbb{L}_f} D_{\mathbf{n}\kappa'\alpha', 0\kappa\alpha} e^{+i\mathbf{q} \cdot \mathbf{R}_n} = \sum_{\mathbf{R}_n \in \mathbb{L}_f} D_{0\kappa'\alpha', -\mathbf{n}\kappa\alpha} e^{+i\mathbf{q} \cdot \mathbf{R}_n} = \dots \\
 &\dots \stackrel{\mathbf{R}_n \mapsto -\mathbf{R}_n}{=} \sum_{\mathbf{R}_n \in \mathbb{L}_f} D_{0\kappa'\alpha', \mathbf{n}\kappa\alpha} e^{-i\mathbf{q} \cdot \mathbf{R}_n} \stackrel{D \text{ symmetric}}{=} \sum_{\mathbf{R}_n \in \mathbb{L}_f} D_{\mathbf{n}\kappa\alpha, 0\kappa'\alpha'} e^{-i\mathbf{q} \cdot \mathbf{R}_n} \\
 &= \sum_{\mathbf{R}_n \in \mathbb{L}_f} D_{\kappa\alpha, \kappa'\alpha'}(\mathbf{R}_n) e^{-i\mathbf{q} \cdot \mathbf{R}_n} = (D_{\kappa\alpha, \kappa'\alpha'}(\mathbf{q}))^*
 \end{aligned}$$

Proof 30.13 of Lemma 5.23

$D_{\kappa\alpha, \kappa'\alpha'}(\mathbf{q})$ is hermitian, according to Lemma 5.19. We obtain further with Equation (5.25) and the fact that the dynamical matrix is real-valued in real-space as follows from Equation (5.2)

$$D_{\kappa\alpha, \kappa'\alpha'}(\mathbf{q}) \equiv \sum_{\mathbf{R}_n \in \mathbb{L}_f} D_{\kappa\alpha, \kappa'\alpha'}(\mathbf{R}_n) e^{i\mathbf{q} \cdot \mathbf{R}_n} \Rightarrow D_{\kappa\alpha, \kappa'\alpha'}(-\mathbf{q}) = D_{\kappa\alpha, \kappa'\alpha'}^*(\mathbf{q}) \quad (30.28)$$

Thus we obtain for $v \in \{1, \dots, rd_x\}$ the equations

$$D(\mathbf{q})e_{;v}(\mathbf{q}) = \omega_v^2(\mathbf{q})e_{;v}(\mathbf{q}) \iff D(-\mathbf{q})e_{;v}^*(\mathbf{q}) = \omega_v^2(\mathbf{q})e_{;v}^*(\mathbf{q}) \quad (30.29)$$

which lets us conclude that the eigenvectors $\{\mathbf{e}_{:,1}(-\mathbf{q}), \mathbf{e}_{:,2}(-\mathbf{q}), \dots, \mathbf{e}_{:,r_{d_x}}(-\mathbf{q})\}$ of $D(-\mathbf{q})$ are given by the complex conjugates of the eigenvectors of $D(\mathbf{q})$, and that $\omega_v^2(-\mathbf{q}) = \omega_v^2(\mathbf{q})$, proving the theorem.

Proof 30.14 of Lemma 5.25

- For the inverse-relation, we find with Lemma 5.22 that

$$\Delta\tau_{\mathbf{q}\kappa\alpha} = \sum_{v'} e_{\kappa\alpha,v'}(\mathbf{q}) u_{\mathbf{q}v'} \frac{1}{\sqrt{M_\kappa}} \quad (30.30)$$

$$= \sum_{\kappa'\alpha'} \underbrace{\sum_{v'} e_{\kappa'\alpha',v'}^*(\mathbf{q}) e_{\kappa\alpha,v'}(\mathbf{q})}_{= \delta_{\kappa'\kappa} \delta_{\alpha'\alpha}} \sqrt{\frac{M_{\kappa'}}{M_\kappa}} \Delta\tau_{\mathbf{q}\kappa'\alpha'} \quad (30.31)$$

$$= \sum_{\kappa'\alpha'} \delta_{\kappa'\kappa} \delta_{\alpha'\alpha} \sqrt{\frac{M_{\kappa'}}{M_\kappa}} \Delta\tau_{\mathbf{q}\kappa'\alpha'} \quad (30.32)$$

$$= \Delta\tau_{\mathbf{q}\kappa\alpha} \quad (30.33)$$

- The second auxiliary identity is evident from Remark 5.16.2.
- For the derivative-relations, we find that

$$u_{\mathbf{q}v} = \sum_{\kappa'\alpha'} e_{\kappa'\alpha',v}^*(\mathbf{q}) \sqrt{M_{\kappa'}} \Delta\tau_{\mathbf{q}\kappa'\alpha'} \quad (30.34)$$

which implies

$$\frac{\partial}{\partial \Delta\tau_{\mathbf{q}\kappa\alpha}} = \sum_{v'} \frac{\partial u_{\mathbf{q}v'}}{\partial \Delta\tau_{\mathbf{q}\kappa\alpha}} \frac{\partial}{\partial u_{\mathbf{q}v'}} = \sum_{v'} e_{\kappa\alpha,v'}^*(\mathbf{q}) \sqrt{M_\kappa} \frac{\partial}{\partial u_{\mathbf{q}v'}} \quad (30.35)$$

which proves the left side of the identity. Now we come to the right side, and compute straight forwardly as well

$$\Delta\tau_{\mathbf{q}\kappa\alpha} = \sum_{v'} e_{\kappa\alpha,v'}(\mathbf{q}) u_{\mathbf{q}v'} \frac{1}{\sqrt{M_\kappa}} \quad (30.36)$$

which implies

$$\frac{\partial}{\partial u_{\mathbf{q}v}} = \sum_{\kappa'\alpha'} \frac{\partial \Delta\tau_{\mathbf{q}\kappa'\alpha'}}{\partial u_{\mathbf{q}v}} \frac{\partial}{\partial \Delta\tau_{\mathbf{q}\kappa'\alpha'}} = \sum_{\kappa'\alpha'} e_{\kappa'\alpha',v}(\mathbf{q}) \frac{1}{\sqrt{M_{\kappa'}}} \frac{\partial}{\partial \Delta\tau_{\mathbf{q}\kappa'\alpha'}} \quad (30.37)$$

- For the preservation of the CCR, we first note that

$$\left[\tau_{\mathbf{n}\kappa\alpha}, -i\hbar \frac{\partial}{\partial \tau_{\mathbf{n}'\kappa'\alpha'}} \right] = +i\hbar \delta_{\mathbf{n}\mathbf{n}'} \delta_{\kappa\kappa'} \delta_{\alpha\alpha'}, \quad (30.38)$$

it follows

$$\left[u_{\mathbf{q}\nu}, -i\hbar \frac{\partial}{\partial u_{\mathbf{q}'\nu'}} \right] = \frac{1}{N_p} \sum_{\substack{n\mathbf{n}' \\ \kappa\kappa' \\ \alpha\alpha'}} e_{\kappa\alpha,\nu}^*(\mathbf{q}) \sqrt{M_\kappa} e^{-i\mathbf{q}\mathbf{R}_n} \left[(\tau_{n\kappa\alpha} - R_{n\kappa\alpha}), \quad e_{\kappa'\alpha',\nu'}(\mathbf{q}') e^{i\mathbf{q}'\mathbf{R}_{n'}} \frac{1}{\sqrt{M_{\kappa'}}} \frac{\partial}{\partial \tau_{n'\kappa'\alpha'}} (-i\hbar) \right] \quad (30.39)$$

$$= \frac{1}{N_p} \sum_{\substack{n,\mathbf{n}' \\ \kappa,\kappa' \\ \alpha,\alpha'}} e_{\kappa\alpha,\nu}^*(\mathbf{q}) \sqrt{\frac{M_\kappa}{M_{\kappa'}}} e^{i(\mathbf{q}'\mathbf{R}_{n'} - \mathbf{q}\mathbf{R}_n)} \underbrace{\left[\tau_{n\kappa\alpha}, \quad -i\hbar \frac{\partial}{\partial \tau_{n'\kappa'\alpha'}} \right]}_{= \delta_{n,\mathbf{n}'} \delta_{\kappa,\kappa'} \delta_{\alpha,\alpha'} i\hbar} e_{\kappa'\alpha',\nu'}(\mathbf{q}') \quad (30.40)$$

$$= \frac{1}{N_p} \sum_{\substack{n \\ \kappa,\alpha}} e_{\kappa\alpha,\nu}^*(\mathbf{q}) \sqrt{\frac{M_\kappa}{M_{\kappa'}}} e^{i\mathbf{R}_n(\mathbf{q}' - \mathbf{q})} e_{\kappa\alpha,\nu'}(\mathbf{q}') i\hbar \quad (30.41)$$

$$= \frac{1}{N_p} \sum_{\kappa\alpha} \underbrace{e_{\kappa\alpha,\nu}^*(\mathbf{q}) e_{\kappa\alpha,\nu'}(\mathbf{q}')}_{= \delta_{\nu\nu'}} \delta_{\mathbf{q}\mathbf{q}'} i\hbar \quad (30.42)$$

$$= \delta_{\nu\nu'} \delta_{\mathbf{q}\mathbf{q}'} i\hbar \quad (30.43)$$

Implying

$$-i\hbar \left[u_{\mathbf{q}\nu}, \frac{\partial}{\partial u_{\mathbf{q}'\nu'}} \right] = i\hbar \delta_{\nu\nu'} \delta_{\mathbf{q}\mathbf{q}'} \iff \left[u_{\mathbf{q}\nu}, \frac{\partial}{\partial u_{\mathbf{q}'\nu'}} \right] = -\delta_{\nu\nu'} \delta_{\mathbf{q}\mathbf{q}'} \quad (30.44)$$

Proof 30.15 of Lemma 5.26

- For the identity $\tau_{n\kappa\alpha} = \dots$:

We simply have to summarize all the coordinate-changes. Those were ...

$$\begin{array}{lll} \tau_{n\kappa\alpha} & \begin{array}{l} \xrightarrow{\text{Translation}} \\ \xrightarrow{\text{Fourier}} \\ \xrightarrow{\text{Eigenmodes and rescaling}} \end{array} & \begin{array}{l} \Delta\tau_{n\kappa\alpha} := \tau_{n\kappa\alpha} - R_{n\kappa\alpha} \\ \Delta\tau_{\mathbf{q}\kappa\alpha} := \frac{1}{\sqrt{N_p}} \sum_{\mathbf{n}} \Delta\tau_{n\kappa\alpha} e^{-i\mathbf{q}\mathbf{R}_n} \\ u_{\mathbf{q}\nu} := \sum_{\kappa\alpha} e_{\kappa\alpha,\nu}^*(\mathbf{q}) \sqrt{M_\kappa} \Delta\tau_{\mathbf{q}\kappa\alpha} \end{array} \end{array}$$

Now combining the equations gives

$$\tau_{\mathbf{n}\kappa\alpha} \equiv R_{\mathbf{n}\kappa\alpha} + \Delta\tau_{\mathbf{n}\kappa\alpha} \quad (30.45)$$

$$= R_{\mathbf{n}\kappa\alpha} + \frac{1}{\sqrt{N_p}} \sum_{\mathbf{q}} \underbrace{\Delta\tau_{\mathbf{q}\kappa\alpha}}_{\Delta\tau_{\mathbf{q}\kappa\alpha}} e^{-i\mathbf{q}\mathbf{R}_n} \quad (30.46)$$

$$= \sum_{\nu'} e_{\kappa\alpha,\nu'}(\mathbf{q}) u_{\mathbf{q}\nu'} \frac{1}{\sqrt{M_\kappa}} \\ = R_{\mathbf{n}\kappa\alpha} + \frac{1}{\sqrt{N_p}} \sum_{\mathbf{q}} \sum_{\nu'} e_{\kappa\alpha,\nu'}(\mathbf{q}) e^{-i\mathbf{q}\mathbf{R}_n} \frac{1}{\sqrt{M_\kappa}} u_{\mathbf{q}\nu'} \quad (30.47)$$

which is the left hand side of Equation (5.38). Combining equations from the other side, gives the right hand side of the equation.

- For the identity $u_{\mathbf{q}\nu} = \dots$: Taking the individual inverses one gets the relation

$$u_{\mathbf{q}\nu} = \sum_{\kappa'\alpha'} e_{\kappa'\alpha',\nu}^*(\mathbf{q}) \sqrt{M_{\kappa'}} \Delta\tau_{\mathbf{q}\kappa'\alpha'} \quad (30.48)$$

$$= \sum_{\kappa'\alpha'} e_{\kappa'\alpha',\nu}^*(\mathbf{q}) \sqrt{M_{\kappa'}} e^{-i\mathbf{q}\mathbf{R}_n} \frac{1}{\sqrt{N_p}} \underbrace{\Delta\tau_{\mathbf{n}\kappa'\alpha'}}_{=(\tau_{\mathbf{n}\kappa'\alpha'} - \mathbf{R}_{\mathbf{n}\kappa'\alpha'})} \quad (30.49)$$

$$= \frac{1}{\sqrt{N_p}} \sum_{\kappa'\alpha'} e_{\kappa'\alpha',\nu}^*(\mathbf{q}) \sqrt{M_{\kappa'}} e^{-i\mathbf{q}\mathbf{R}_n} (\tau_{\mathbf{n}\kappa'\alpha'} - \mathbf{R}_{\mathbf{n}\kappa'\alpha'}) \quad (30.50)$$

We can also simply insert the equation and see that it describely corrects the inverse relation:

$$u_{\mathbf{q}\nu} = \sum_{\kappa'\alpha'} e_{\kappa'\alpha',\nu}^*(\mathbf{q}) \sqrt{M_{\kappa'}} \Delta\tau_{\mathbf{q}\kappa'\alpha'} \quad (30.51)$$

$$= \sum_{\kappa'\alpha'} e_{\kappa'\alpha',\nu}^*(\mathbf{q}) \sqrt{M_{\kappa'}} \sum_{\nu''} e_{\kappa'\alpha',\nu''}(\mathbf{q}) u_{\mathbf{q}\nu''} \frac{1}{\sqrt{M_{\kappa'}}} \quad (30.52)$$

$$= \sum_{\nu''} u_{\mathbf{q}\nu''} \sqrt{\frac{M_{\kappa'}}{M_{\kappa'}}} \underbrace{\sum_{\kappa'\alpha'} e_{\kappa'\alpha',\nu}^*(\mathbf{q}) e_{\kappa'\alpha',\nu''}(\mathbf{q})}_{\delta_{\nu\nu''}} \quad (30.53)$$

$$= u_{\mathbf{q}\nu} \quad (30.54)$$

- For the identity $\frac{1}{\partial\tau_{\mathbf{n}\kappa\alpha}} = \dots$:

Formula: If $x_i = x_i(x'_1, \dots, x'_n)$ then

$$\frac{\partial}{\partial x'_i} = \sum_j \frac{\partial x_j}{\partial x'_i} \frac{\partial}{\partial x_j}$$

Therefore

$$\frac{\partial}{\partial \tau_{n\kappa\alpha}} = \frac{\partial}{\partial (\Delta \tau_{n\kappa\alpha})} \underbrace{\frac{\partial (\Delta \tau_{n\kappa\alpha})}{\partial \tau_{n\kappa\alpha}}}_{=1} = \frac{\partial}{\partial (\Delta \tau_{n\kappa\alpha})} \quad (30.55)$$

$$\frac{\partial}{\partial (\Delta \tau_{n\kappa\alpha})} = \sum_{\mathbf{q}} \frac{\partial}{\partial (\Delta \tau_{\mathbf{q}\kappa\alpha})} \underbrace{\frac{\partial (\Delta \tau_{\mathbf{q}\kappa\alpha})}{\partial (\Delta \tau_{n\kappa\alpha})}}_{=e^{-i\mathbf{q}\mathbf{R}_n} \frac{1}{\sqrt{N_p}}} = \frac{1}{\sqrt{N_p}} \sum_{\mathbf{q}} e^{-i\mathbf{q}\mathbf{R}_n} \frac{\partial}{\partial (\Delta \tau_{\mathbf{q}\kappa\alpha})} \quad (30.56)$$

$$\frac{\partial}{\partial (\Delta \tau_{\mathbf{q}\kappa\alpha})} = \sum_{\nu'} \frac{\partial}{\partial u_{\mathbf{q}\nu'}} \underbrace{\frac{\partial u_{\mathbf{q}\nu'}}{\partial (\Delta \tau_{\mathbf{q}\kappa\alpha})}}_{=\sqrt{M_\kappa} e_{\kappa\alpha,\nu'}^*(\mathbf{q})} = \sum_{\nu'} \sqrt{M_\kappa} e_{\kappa\alpha,\nu'}^*(\mathbf{q}) \frac{\partial}{\partial u_{\mathbf{q}\nu'}} \quad (30.57)$$

Combining these equations yields the desired result as

$$\frac{\partial}{\partial \tau_{n\kappa\alpha}} = \frac{\partial}{\partial (\Delta \tau_{n\kappa\alpha})} \quad (30.58)$$

$$= \frac{1}{\sqrt{N_p}} \sum_{\mathbf{q}} e^{-i\mathbf{q}\mathbf{R}_n} \underbrace{\frac{\partial}{\partial (\Delta \tau_{\mathbf{q}\kappa\alpha})}}_{= \sum_{\mu} \sqrt{M_\kappa} e_{\kappa\alpha,\mu}^*(\mathbf{q}) \frac{\partial}{\partial u_{\mathbf{q}\mu}}} \quad (30.59)$$

$$= \sum_{\mu} \sqrt{M_\kappa} e_{\kappa\alpha,\mu}^*(\mathbf{q}) \frac{\partial}{\partial u_{\mathbf{q}\mu}} = \frac{\sqrt{M_\kappa}}{\sqrt{N_p}} \sum_{\mu} e^{-i\mathbf{q}\mathbf{R}_n} e_{\kappa\alpha,\mu}^*(\mathbf{q}) \frac{\partial}{\partial u_{\mathbf{q}\mu}} \quad (30.60)$$

- For the identity $\frac{\partial}{\partial u_{\mathbf{q}\nu}} = \dots$:

$$\frac{\partial}{\partial u_{\mathbf{q}\nu}} = \sum_{\substack{\mathbf{R}_n \in L_f \\ \kappa', \alpha'}} \frac{\partial}{\partial \tau_{n\kappa'\alpha'}} \underbrace{\frac{\partial \tau_{n\kappa'\alpha'}}{\partial u_{\mathbf{q}\nu}}}_{= \frac{1}{\sqrt{N_p}} e_{\kappa'\alpha',\nu}^*(\mathbf{q}) \frac{1}{\sqrt{M_{\kappa'}}} e^{-i\mathbf{q}\mathbf{R}_n}} \quad (30.61)$$

$$= \frac{1}{\sqrt{N_p}} \sum_{\substack{\mathbf{R}_n \in L \\ \kappa', \alpha'}} e_{\kappa'\alpha',\nu}^*(\mathbf{q}) e^{-i\mathbf{q}\mathbf{R}_n} \frac{1}{\sqrt{M_{\kappa'}}} \frac{\partial}{\partial \tau_{n\kappa'\alpha'}} \quad (30.62)$$

Proof 30.16 of Corollary 5.27 We can the terms appearing in the kinetic and potential energy expression through their new coordinates as

$$\Delta \tau_{\mathbf{q}\kappa\alpha}^* = \sum_{\nu} e_{\kappa\alpha,\nu}^*(\mathbf{q}) u_{\mathbf{q}\nu}^* \frac{1}{\sqrt{M_\kappa}}, \quad \Delta \tau_{\mathbf{q}\kappa'\alpha'} = \sum_{\nu} e_{\kappa'\alpha',\nu}(\mathbf{q}) u_{\mathbf{q}\nu} \frac{1}{\sqrt{M_{\kappa'}}} \quad (30.63)$$

with $u_{\mathbf{q}\nu}$ as new coordinates.

Potential Energy Term:

Inserting this into the expression for the potential energy gives

$$\frac{1}{2} \sum_{\mathbf{q}} \sum_{\substack{\kappa\kappa' \\ \alpha\alpha'}} \Delta\tau_{\mathbf{q}\kappa\alpha}^* \frac{D_{\kappa\alpha, \kappa'\alpha'}(\mathbf{q})}{\sqrt{M_{\kappa} M_{\kappa'}}} \Delta\tau_{\mathbf{q}\kappa'\alpha'} = \frac{1}{2} \sum_{\mathbf{q}} \sum_{\substack{\kappa\kappa' \\ \alpha\alpha' \\ \nu\nu'}} e_{\kappa\alpha, \nu}^*(\mathbf{q}) u_{\mathbf{q}\nu}^* D_{\kappa\alpha, \kappa'\alpha'}(\mathbf{q}) e_{\kappa'\alpha', \nu'}(\mathbf{q}) u_{\mathbf{q}\nu'} \quad (30.64)$$

$$= \dots \quad (30.65)$$

Performing the sum over $\kappa'\alpha'$ (i.e. using Equation (5.31)) gives

$$\dots = \frac{1}{2} \sum_{\mathbf{q}} \sum_{\substack{\kappa\alpha \\ \nu\nu'}} e_{\kappa\alpha, \nu}^*(\mathbf{q}) u_{\mathbf{q}\nu}^* \omega_{\nu}^2(\mathbf{q}) e_{\kappa\alpha, \nu'}(\mathbf{q}) u_{\mathbf{q}\nu'} = \dots \quad (30.66)$$

Now we use Equation (5.32)

$$\sum_{\kappa\alpha} e_{\kappa\alpha, \nu}^*(\mathbf{q}) e_{\kappa\alpha, \nu'}(\mathbf{q}) = \delta_{\nu, \nu'} \quad (**)$$

such that we obtain

$$\dots = \frac{1}{2} \sum_{\mathbf{q}\nu} u_{\mathbf{q}\nu}^* \omega_{\nu}^2(\mathbf{q}) u_{\mathbf{q}\nu} = \frac{1}{2} \sum_{\mathbf{q}\nu} \omega_{\nu}^2(\mathbf{q}) u_{-\mathbf{q}\nu} u_{\mathbf{q}\nu} \quad (30.68)$$

Kinetic Energy Term:

Before we proceed let's pay attention to the Momentum again:

$$T_e \equiv \sum_{\substack{\mathbf{R}_n \in L \\ \kappa\alpha}} -\frac{\hbar^2}{2M_{\kappa}} \frac{\partial^2}{\partial \tau_{\mathbf{n}\kappa\alpha}^2} \quad (30.69)$$

$$\equiv \sum_{\mathbf{n}\kappa\alpha} \frac{\partial}{\partial \tau_{\mathbf{n}\kappa\alpha}} \left(-\frac{\hbar^2}{2M_{\kappa}} \right) \frac{\partial}{\partial \tau_{\mathbf{n}\kappa\alpha}} \quad (30.70)$$

Then it follows

$$T_e = \sum_{\mathbf{n}\kappa\alpha} \frac{\partial}{\partial \tau_{\mathbf{n}\kappa\alpha}} \left(-\frac{\hbar^2}{2M_{\kappa}} \right) \frac{\partial}{\partial \tau_{\mathbf{n}\kappa\alpha}} \quad (30.71)$$

With $\sum_{\kappa\alpha} \equiv \sum_{\nu}$

$$T_e = \sum_{\substack{\mathbf{q}\mathbf{q}' \\ \mu\mu' \\ nv}} \left(-\frac{\hbar^2}{2M_\kappa} \right) \left(\frac{M_\kappa}{N_p} \right) e^{-i\mathbf{q}\mathbf{R}_n} e_{\nu\mu}^* (\mathbf{q}) \frac{\partial}{\partial u_{\mathbf{q}\mu}} e^{-i\mathbf{q}'\mathbf{R}_n} e_{\nu\mu'}^* (\mathbf{q}') \frac{\partial}{\partial u_{\mathbf{q}'\mu}} \quad (30.72)$$

$$= \sum_{\substack{\mathbf{q}\mathbf{q}' \\ \mu\mu' \\ v}} \underbrace{\sum_n e^{-i\mathbf{R}_n(\mathbf{q}+\mathbf{q}')} }_{=N_p\delta_{\mathbf{q}+\mathbf{q}',0}} \left(-\frac{\hbar^2}{2N_p} \right) e_{\nu\mu}^* (\mathbf{q}) \frac{\partial}{\partial u_{\mathbf{q}\mu}} e_{\nu\mu'}^* (\mathbf{q}') \frac{\partial}{\partial u_{\mathbf{q}'\mu}} \quad (30.73)$$

$$= \sum_{\substack{\mathbf{q} \\ \mu\mu' \\ v}} -\frac{\hbar^2}{2} e_{\nu\mu}^* (\mathbf{q}) \underbrace{e_{\nu\mu'}^* (-\mathbf{q})}_{=e_{\nu\mu'}(\mathbf{q})} \frac{\partial}{\partial u_{\mathbf{q}\mu}} \frac{\partial}{\partial u_{-\mathbf{q}\mu}} \quad (30.74)$$

$$= \sum_{\substack{\mathbf{q} \\ \mu\mu'}} -\frac{\hbar^2}{2} \underbrace{\sum_v e_{\nu\mu}^* (\mathbf{q}) e_{\nu\mu'} (\mathbf{q})}_{=\delta_{\mu\mu'}} \frac{\partial}{\partial u_{\mathbf{q}\mu}} \frac{\partial}{\partial u_{-\mathbf{q}\mu}} \quad (30.75)$$

$$= \sum_{\mathbf{q}\mu} -\frac{\hbar^2}{2} \frac{\partial}{\partial u_{\mathbf{q}\mu}} \frac{\partial}{\partial u_{-\mathbf{q}\mu}} \quad (30.76)$$

Proof 30.17 of Lemma 5.29

Proof of CCR:

$$\left[a_{\mathbf{q}\mu}, a_{\mathbf{q}\mu}^\dagger \right] = \sqrt{\frac{\omega_\mu(\mathbf{q})\omega_{\mu'}(\mathbf{q}')}{(2\hbar)^2}} \left(\frac{\hbar}{\omega_\mu(\mathbf{q})} \underbrace{\left[\frac{\partial}{\partial u_{-\mathbf{q}\mu}}, u_{-\mathbf{q}'\mu'} \right]}_{=\delta_{\mu\mu'}\delta_{\mathbf{q}\mathbf{q}'}} - \frac{\hbar}{\omega_{\mu'}(\mathbf{q}')} \left[u_{\mathbf{q}\mu}, \frac{\partial}{\partial u_{\mathbf{q}\mu}} \right] \right) \quad (30.77)$$

$$= \sqrt{\frac{\omega_\mu(\mathbf{q})\omega_{\mu'}(\mathbf{q}')}{(2\hbar)^2}} \hbar \left(\frac{\delta_{\mathbf{q}\mathbf{q}'}\delta_{\mu\mu'}}{\omega_\mu(\mathbf{q})} + \frac{\delta_{\mathbf{q}\mathbf{q}'}\delta_{\mu\mu'}}{\omega_{\mu'}(\mathbf{q}')} \right) \quad (30.78)$$

$$= \delta_{\mathbf{q}\mathbf{q}'}\delta_{\mu\mu'} \quad (30.79)$$

Proof of the inversion Formulae:

$$u_{\mathbf{q}\mu} = \left(a_{\mathbf{q}\mu} + a_{-\mathbf{q}\mu}^\dagger \right) \sqrt{\frac{\hbar}{2\omega_\mu(\mathbf{q})}} \quad (30.80)$$

Proof:

$$(a_{\mathbf{q}\mu} + a_{-\mathbf{q}\mu}^\dagger) \sqrt{\frac{\hbar}{2\omega_\mu(\mathbf{q})}} \quad (30.81)$$

$$= \left(\sqrt{\frac{\omega_\mu(\mathbf{q})}{2\hbar}} \left(u_{\mathbf{q}\mu} + \frac{\hbar}{\omega_\mu(\mathbf{q})} \frac{\partial}{\partial u_{-\mathbf{q}\mu}} \right) + \sqrt{\frac{\omega_\mu(-\mathbf{q})}{2\hbar}} \left(u_{\mathbf{q}\mu} - \frac{\hbar}{\omega_\mu(-\mathbf{q})} \frac{\partial}{\partial u_{-\mathbf{q}\mu}} \right) \right) \sqrt{\frac{\hbar}{2\omega_\mu(\mathbf{q})}} \quad (30.82)$$

$$\stackrel{\text{symmetry of } \omega_\mu}{=} \sqrt{\frac{\omega_\mu(\mathbf{q})}{2\hbar}} \left(\left(u_{\mathbf{q}\mu} + \frac{\hbar}{\omega_\mu(\mathbf{q})} \frac{\partial}{\partial u_{-\mathbf{q}\mu}} \right) + \left(u_{\mathbf{q}\mu} - \frac{\hbar}{\omega_\mu(\mathbf{q})} \frac{\partial}{\partial u_{-\mathbf{q}\mu}} \right) \right) \sqrt{\frac{\hbar}{2\omega_\mu(\mathbf{q})}} \quad (30.83)$$

$$= \sqrt{\frac{\omega_\mu(\mathbf{q})}{2\hbar}} (u_{\mathbf{q}\mu} + u_{\mathbf{q}\mu}) \sqrt{\frac{\hbar}{2\omega_\mu(\mathbf{q})}} \quad (30.84)$$

$$= u_{\mathbf{q}\mu} \quad (30.85)$$

Now the other identity:

$$\frac{\partial}{\partial u_{\mathbf{q}\mu}} = (a_{-\mathbf{q}\mu} - a_{\mathbf{q}\mu}^\dagger) \sqrt{\frac{\omega_\mu(\mathbf{q})}{2\hbar}} \quad (30.86)$$

By straight forward computation we find

$$(a_{-\mathbf{q}\mu} - a_{\mathbf{q}\mu}^\dagger) \sqrt{\frac{\omega_\mu(\mathbf{q})}{2\hbar}} \quad (30.87)$$

$$= \left(\sqrt{\frac{\omega_\mu(-\mathbf{q})}{2\hbar}} \left(u_{-\mathbf{q}\mu} + \frac{\hbar}{\omega_\mu(-\mathbf{q})} \frac{\partial}{\partial u_{\mathbf{q}\mu}} \right) - \sqrt{\frac{\omega_\mu(\mathbf{q})}{2\hbar}} \left(u_{-\mathbf{q}\mu} - \frac{\hbar}{\omega_\mu(\mathbf{q})} \frac{\partial}{\partial u_{\mathbf{q}\mu}} \right) \right) \sqrt{\frac{\omega_\mu(\mathbf{q})}{2\hbar}} \quad (30.88)$$

$$\stackrel{\text{symmetry of } \omega_\mu}{=} \sqrt{\frac{\omega_\mu(\mathbf{q})}{2\hbar}} \left(\left(u_{-\mathbf{q}\mu} + \frac{\hbar}{\omega_\mu(\mathbf{q})} \frac{\partial}{\partial u_{\mathbf{q}\mu}} \right) - \left(u_{-\mathbf{q}\mu} - \frac{\hbar}{\omega_\mu(\mathbf{q})} \frac{\partial}{\partial u_{\mathbf{q}\mu}} \right) \right) \sqrt{\frac{\omega_\mu(\mathbf{q})}{2\hbar}} \quad (30.89)$$

$$= \sqrt{\frac{\omega_\mu(\mathbf{q})}{2\hbar}} \left(\frac{2\hbar}{\omega_\mu(\mathbf{q})} \frac{\partial}{\partial u_{\mathbf{q}\mu}} \right) \sqrt{\frac{\omega_\mu(\mathbf{q})}{2\hbar}} \quad (30.90)$$

$$= \frac{\partial}{\partial u_{\mathbf{q}\mu}} \quad (30.91)$$

Proof 30.18 of Corollary 5.30

$$H \approx \sum_{\mathbf{q}\mu} \left(-\frac{\hbar^2}{2} \frac{\partial}{\partial u_{\mathbf{q}\mu}} \frac{\partial}{\partial u_{-\mathbf{q}\mu}} + \frac{1}{2} \omega_{\mu}^2(\mathbf{q}) u_{\mathbf{q}\mu} u_{-\mathbf{q}\mu} \right) \quad (30.92)$$

$$= \sum_{\mathbf{q}\mu} -\frac{\hbar^2}{2} (a_{-\mathbf{q}\mu} - a_{\mathbf{q}\mu}^{\dagger}) \sqrt{\frac{\omega_{\mu}(\mathbf{q})}{2\hbar}} (a_{\mathbf{q}\mu} - a_{-\mathbf{q}\mu}^{\dagger}) \sqrt{\frac{\omega_{\mu}(-\mathbf{q})}{2\hbar}} \quad (30.93)$$

$$+ \frac{1}{2} \omega_{\mu}^2(\mathbf{q}) (a_{\mathbf{q}\mu} + a_{-\mathbf{q}\mu}^{\dagger}) \sqrt{\frac{\hbar}{2\omega_{\mu}(\mathbf{q})}} (a_{-\mathbf{q}\mu} + a_{\mathbf{q}\mu}^{\dagger}) \sqrt{\frac{\hbar}{2\omega_{\mu}(-\mathbf{q})}} \quad (30.94)$$

$$= \sum_{\mathbf{q}\mu} -\frac{\hbar\omega_{\mu}(\mathbf{q})}{4} (a_{-\mathbf{q}\mu} - a_{\mathbf{q}\mu}^{\dagger}) (a_{\mathbf{q}\mu} - a_{-\mathbf{q}\mu}^{\dagger}) \quad (30.95)$$

$$+ \frac{\hbar\omega_{\mu}(\mathbf{q})}{4} (a_{\mathbf{q}\mu} + a_{-\mathbf{q}\mu}^{\dagger}) (a_{-\mathbf{q}\mu} + a_{\mathbf{q}\mu}^{\dagger}) \quad (30.96)$$

$$= \sum_{\mathbf{q}\mu} \frac{\hbar\omega_{\mu}(\mathbf{q})}{4} \left(-a_{-\mathbf{q}\mu} a_{\mathbf{q}\mu} + a_{-\mathbf{q}\mu} a_{-\mathbf{q}\mu}^{\dagger} + a_{\mathbf{q}\mu}^{\dagger} a_{\mathbf{q}\mu} - a_{\mathbf{q}\mu}^{\dagger} a_{-\mathbf{q}\mu}^{\dagger} + a_{\mathbf{q}\mu} a_{-\mathbf{q}\mu} + a_{\mathbf{q}\mu} a_{\mathbf{q}\mu}^{\dagger} + a_{-\mathbf{q}\mu}^{\dagger} a_{-\mathbf{q}\mu} + a_{-\mathbf{q}\mu}^{\dagger} a_{\mathbf{q}\mu}^{\dagger} \right) \quad (30.97)$$

$$= \sum_{\mathbf{q}\mu} \frac{\hbar\omega_{\mu}(\mathbf{q})}{4} \left(\underbrace{a_{-\mathbf{q}\mu} a_{-\mathbf{q}\mu}^{\dagger}}_{=1+a_{-\mathbf{q}\mu}^{\dagger} a_{-\mathbf{q}\mu}} + \underbrace{a_{\mathbf{q}\mu}^{\dagger} a_{\mathbf{q}\mu}}_{=1+a_{\mathbf{q}\mu}^{\dagger} a_{\mathbf{q}\mu}} + a_{\mathbf{q}\mu} a_{\mathbf{q}\mu}^{\dagger} + a_{-\mathbf{q}\mu}^{\dagger} a_{-\mathbf{q}\mu} \right) \quad (30.98)$$

$$= \sum_{\mathbf{q}\mu} \frac{\hbar\omega_{\mu}(\mathbf{q})}{4} \left(2 + 4a_{\mathbf{q}\mu}^{\dagger} a_{\mathbf{q}\mu} \right) \quad (30.99)$$

$$= \sum_{\mathbf{q}\mu} \hbar\omega_{\mu}(\mathbf{q}) \left(a_{\mathbf{q}\mu}^{\dagger} a_{\mathbf{q}\mu} + \frac{1}{2} \right) \quad (30.100)$$

Proof 30.19 of Lemma 5.31

Combining the derived equations we obtain

$$\Delta\tau_{\mathbf{n}\nu} = \frac{1}{\sqrt{N_p}} \sum_{\mathbf{q}} e_{\nu\nu'}(\mathbf{q}) e^{-i\mathbf{q}\mathbf{R}_n} \frac{1}{\sqrt{M_{\nu}}} \underbrace{u_{\mathbf{q}\nu'}}_{= (a_{\mathbf{q}\nu'} + a_{-\mathbf{q}\nu'}^{\dagger})} \sqrt{\frac{\hbar}{2\omega_{\nu'}(\mathbf{q})}} \quad (30.101)$$

$$= \frac{1}{\sqrt{N_p}} \sum_{\mathbf{q}} \sqrt{\frac{\hbar}{2M_{\nu}\omega_{\nu'}(\mathbf{q})}} e_{\nu\nu'}(\mathbf{q}) e^{-i\mathbf{q}\mathbf{R}_n} (a_{\mathbf{q}\nu'} + a_{-\mathbf{q}\nu'}^{\dagger}) \quad (30.102)$$

31. Appendix - Proofs for Part II

Proof 31.1 of Theorem 7.7¹

The factorization of the Fock-space into polynomial expressions of different orders is a direct consequence of the construction.

The fermionic ($\zeta = -1$) case.

One way to find out the number of linearly independent elements in $\Lambda^n(\mathcal{H})$ is by direct counting. Linearity allows any element in $\omega = \Lambda^n(\mathcal{H})$ in the following way

$$\omega = \sum_{1 \leq i_1, \dots, i_n \leq d_{\mathcal{H}}} \omega_{i_1, \dots, i_n} |i_1 \otimes \dots \otimes i_n\rangle \quad (31.1)$$

Due to the defining relation $vv' + v'v = 0$ for all $v, v' \in \mathcal{H}$, any two elements $|i_1 \otimes \dots \otimes i_n\rangle, |i'_1 \otimes \dots \otimes i'_n\rangle$ are equal to zero, if one basis element appears more than one time $|i_j \otimes \dots \otimes i_j\rangle = 0$. Thus, the number of possible non-zero combinations for $|i_1 \otimes \dots \otimes i_n\rangle$ is

$$\begin{aligned} \# \left\{ |i_1 \otimes \dots \otimes i_n\rangle \mid i_1, \dots, i_n \in \{1, \dots, d_{\mathcal{H}}\} \wedge \nexists j, j' \in \{1, \dots, n\} : i_j = i_{j'} \right\} &= d_{\mathcal{H}} \cdot (d_{\mathcal{H}} - 1) \cdot \dots \cdot (d_{\mathcal{H}} - n + 1) \\ &= \frac{d_{\mathcal{H}}!}{(d_{\mathcal{H}} - n)!}. \end{aligned}$$

We now notice the following things:

- Every element $|i_1 \otimes \dots \otimes i_n\rangle$ can be re-ordered as $|\tilde{i}_1 \otimes \dots \otimes \tilde{i}_n\rangle$ such that $\tilde{i}_1 < \dots < \tilde{i}_n$.
- The number of ways to reorder elements as $|i_1 \otimes \dots \otimes i_n\rangle$ is $n!$ (due to the dimension of the symmetric group).

Hence $\omega = \Lambda^n(\mathcal{H})$ can be rewritten as

$$\omega = \sum_{1 \leq i_1 < \dots < i_n \leq d_{\mathcal{H}}} \tilde{\omega}_{i_1, \dots, i_n} |i_1 \otimes \dots \otimes i_n\rangle \quad (31.2)$$

Hence we obtain

$$\# \left\{ |i_1 \otimes \dots \otimes i_n\rangle \mid 1 \leq i_1 < \dots < i_n \leq d_{\mathcal{H}} \right\} = \binom{d_{\mathcal{H}}}{n} = \dim_{\mathbb{C}} \Lambda^n(\mathcal{H}).$$

The bosonic ($\zeta = +1$) case.

¹References: For the Bosonic part: See Zirnbauer 2018 / 2019 Sheet 09 Exercise 2 a). For the fermionic part: See Zirnbauer SS 2020, Sheet 6 Exercise 3 a).

Notice first, that the identity to prove can be written as

$$\dim_{\mathbb{C}} S^n(\mathcal{H}) = \binom{d_H + n - 1}{n} = \# \left\{ (x_1, \dots, x_{d_H}) \mid \sum_{i=1}^{d_H} x_i = n \quad \wedge \quad x_1, \dots, x_{d_H} \geq 0 \right\} \quad (31.3)$$

by induction. Since we have two integers d_H, n to vary, we need to perform the induction-start on the lines

$$\{(n, d_H) \in \mathbb{N}_+^2 \mid n \in \{1, \dots, \infty\} \wedge d_H = 1\} \quad (31.4)$$

and

$$\{(n, d_H) \in \mathbb{N}_+^2 \mid n = 1 \wedge d_H \in \{1, \dots, \infty\}\} \quad (31.5)$$

This is fulfilled

Now we prove that

$$\# \left\{ (x_1, \dots, x_{d_H}, x_{d_H+1}) \mid \sum_{i=1}^{d_H+1} x_i = n + 1 \quad \wedge \quad x_1, \dots, x_{d_H}, x_{d_H+1} \geq 0 \right\} = \binom{d_H + n + 1}{n + 1} \quad (31.6)$$

under the assumption that Equation (31.3) is true for $(n + 1, d_H)$ and $(n, d_H + 1)$. **TBD: Adding sketch.** For this, we decompose the set

$$\# \left\{ (x_1, \dots, x_{d_H}, x_{d_H+1}) \mid \sum_{i=1}^{d_H+1} x_i = n + 1 \quad \wedge \quad x_1, \dots, x_{d_H}, x_{d_H+1} \geq 0 \right\} \quad (31.7)$$

into the two cases $x_1 = 0$ and $x_1 \geq 1$.

- Case 1: We consider the set

$$\# \left\{ (x_2, \dots, x_{d_H}, x_{d_H+1}) \mid \sum_{i=2}^{d_H+1} x_i = n + 1 \quad \wedge \quad x_2, \dots, x_{d_H}, x_{d_H+1} \geq 0 \right\} = \binom{d_H + n}{n + 1} \quad (31.8)$$

- Case 2: We have now that $y_1 \equiv x_1 - 1 \geq 0$ We consider the set

$$\# \left\{ (y_1, x_2, \dots, x_{d_H}, x_{d_H+1}) \mid y_1 + \sum_{i=2}^{d_H+1} x_i = n \quad \wedge \quad y_1, x_2, \dots, x_{d_H}, x_{d_H+1} \geq 0 \right\} = \binom{d_H + n}{n} \quad (31.9)$$

Thus we obtain in total

$$\begin{aligned} \# \left\{ (x_1, \dots, x_{d_H}, x_{d_H+1}) \mid \sum_{i=1}^{d_H+1} x_i = n + 1 \quad \wedge \quad x_1, \dots, x_{d_H}, x_{d_H+1} \geq 0 \right\} &= \binom{d_H + n}{n + 1} + \binom{d_H + n}{n} \\ &= \binom{d_H + n + 1}{n + 1} \end{aligned}$$

Proving the theorem for the bosonic case.

Proof 31.2 of Corollary 7.12

We have to show, that taking two elements $v \equiv \psi$ and $v' \equiv \phi$ (which are then interpreted as many-particle wavefunctions), the condition in Equation (7.7) is still satisfied. This is done by straightforward computation

$$\begin{aligned} (\psi \circledast \phi)(x_1, x_2) - \zeta (\phi \circledast \psi)(x_1, x_2) &= \sum_{\pi \in S_2} \text{sign}(\pi)^{1-\theta(\zeta)} \psi(x_{\pi(1)}) \phi(x_{\pi(2)}) - \zeta \sum_{\pi \in S_2} \text{sign}(\pi)^{1-\theta(\zeta)} \phi(x_{\pi(1)}) \psi(x_{\pi(2)}) \\ &= 0 \end{aligned} \tag{31.10}$$

It remains to prove the associativity. This is proven more or less straight-forwardly²

Proof 31.3 of Lemma 7.14

Introducing $u \equiv v + \phi$ and $u' \equiv v' + \phi'$ we obtain by direct computation

$$\begin{aligned} &B_\zeta(u, u') + \zeta B_\zeta(u', u) \\ &= B_\zeta(v + \phi, v' + \phi') + \zeta B_\zeta(v' + \phi', v + \phi) \\ &= \phi(v') - \zeta \phi'(v) + \zeta (\phi'(v) - \zeta \phi(v')) \\ &= 0 \end{aligned}$$

Proof 31.4 of Lemma 7.17

Expanding the r.h.s. of the identity yields

$$\begin{aligned} [A, B]_\zeta C + \zeta B[A, C]_\zeta &= (AB - \zeta BA)C + \zeta B(AC - \zeta CA) \\ &= ABC - \zeta^2 BCA \\ &= ABC - BCA \\ &= [A, BC] \end{aligned}$$

Proof 31.5 of Notation 7.21 One fixes the elements $u \equiv |i\rangle + 0$ and $u' \equiv 0 + \langle j|$ and immediately obtains the relations by considering their commutators.

Proof 31.6 of Theorem 7.23 Recall that a mapping

$$\begin{aligned} \Phi : \mathcal{A}_\zeta(\mathcal{H} \oplus \mathcal{H}^*, B_\zeta) \times \mathcal{F}_\zeta &\rightarrow \mathcal{F}_\zeta \\ (g, v) &\mapsto \Phi(g, v) \equiv g \cdot v \end{aligned}$$

qualifies as a representation, iff

²See, for example, [48].

1. for all $g \in \mathcal{A}_\zeta(\mathcal{H} \oplus \mathcal{H}^*, B_\zeta)$ the map

$$\begin{aligned}\Phi(g) : \mathcal{F}_\zeta &\mapsto \mathcal{F}_\zeta \\ v &\mapsto \Phi(g, v)\end{aligned}$$

is linear over $\mathbb{C}$,

2. for all $g_1, g_2 \in \mathcal{A}_\zeta(\mathcal{H} \oplus \mathcal{H}^*, B_\zeta)$ and $v \in \mathcal{F}_\zeta$ one has

$$\begin{aligned}e \cdot v &= v \\ g_1 \cdot (g_2 \cdot v) &= (g_1 g_2) \cdot v\end{aligned}$$

whereas $e \equiv 1 \equiv 1_\mathbb{C}$ is the identity in the algebra.

The first property is obviously fulfilled, as we defined our representation on the basis-elements and extended the definition by linearity. The identity condition is also fulfilled by applying the definition for multiplication with scalars. For the third condition there are two non-trivial cases:

1. $g_1 \equiv w + 0, g_2 \equiv 0 + \phi$
2. $g_1 \equiv 0 + \phi, g_2 \equiv w + 0$

and we have to check in each case whether $v \in \mathbb{C}, v \in \mathcal{H}$ or $v \in \mathcal{F}_\zeta^n$ for $n \geq 2$.

Case 1:

In this case we obtain

$$\begin{aligned}g_1 g_2 w &= (w + 0)(0 + \phi) \cdot v \\ &\equiv \iota(w)\epsilon(\phi) \cdot v = \dots\end{aligned}$$

which gives for each type of v :

- $v \in \mathbb{C}$: $\dots = 0$.
- $v \in \mathcal{H}$: $\dots = \iota(w) \cdot \phi(v) = w \cdot_\zeta \phi(v)$
- $v \equiv v_1 \cdot_\zeta v_2 \in \mathcal{F}_\zeta^n$ for $n \geq 2$: $\dots =$

TBContinued.

Proof 31.7 of Theorem 7.24 TBD.

Proof 31.8 of Corollary 7.26 TBD. With computation-rules of determinants / permanents.

Proof 31.9 of Corollary 7.29 TBD. Recall that the hermitian adjoint of an operator w.r.t. a scalar-product is defined as TBWritten.

Proof 31.10 of Theorem 7.34 TBD. The states form obviously a basis, since every element in the Fock-space can be expanded w.r.t. the states. Now to the orthonormality: TBDone.

Proof 31.11 of Corollary 7.37 TBD.

Proof 31.12 of Lemma 7.38 TBD.

Proof 31.13 of Lemma 7.39 TBD.

Proof 31.14 of Corollary 7.37 TBD.

Proof 31.15 of Corollary 7.45 TBD.

Proof 31.16 of Corollary 7.46 TBD.

Proof 31.17 of Theorem 7.48 TBD.

Proof 31.18 of Theorem 8.9

TBUpdated w.r.t. the ensemble Hamiltonian.

We first note the auxiliary identities

$$\begin{aligned}\frac{d}{dt}U(t, 0) &= -\frac{i}{\hbar}HU(t, 0) \\ \frac{d}{dt}U(t, 0)^\dagger &= \frac{d}{dt}U(0, t) = \frac{i}{\hbar}U(0, t)H = \frac{i}{\hbar}U(t, 0)^\dagger H\end{aligned}$$

Then it follows

$$\begin{aligned}\frac{d}{dt}X_H(t) &= \frac{d}{dt}U(t, 0)^\dagger X(t)U(t, 0) \\ &= \left(\frac{d}{dt}U(t, 0)^\dagger\right) X(t)U(t, 0) + U(t, 0)^\dagger \left(\frac{d}{dt}X(t)\right) U(t, 0) + U(t, 0)^\dagger X(t) \left(\frac{d}{dt}U(t, 0)\right) \\ &= \frac{i}{\hbar}U(t, 0)^\dagger HX(t)U(t, 0) + U(t, 0)^\dagger \left(\frac{d}{dt}X(t)\right) U(t, 0) + U(t, 0)^\dagger X(t) \left(-\frac{i}{\hbar}HU(t, 0)\right) \\ &= \frac{i}{\hbar}U(t, 0)^\dagger (HX(t) - X(t)H)U(t, 0) + U(t, 0)^\dagger \left(\frac{d}{dt}X(t)\right) U(t, 0) \\ &= \frac{i}{\hbar}U(t, 0)^\dagger [H, X(t)]U(t, 0) + U(t, 0)^\dagger \left(\frac{d}{dt}X(t)\right) U(t, 0) \\ &= -\frac{i}{\hbar}[X, H]_H(t) + U(t, 0)^\dagger \left(\frac{d}{dt}X(t)\right) U(t, 0)\end{aligned}$$

Proof 31.19 of Theorem 8.18

TBUpdated w.r.t. the ensemble Hamiltonian.

Analogously to the real-time theorem we first note the auxiliary identities

$$\begin{aligned}\frac{d}{d\tau}U(\tau, 0) &= -\frac{1}{\hbar}HU(\tau, 0) \\ \frac{d}{d\tau}U(0, \tau) &= \frac{1}{\hbar}U(0, \tau)H\end{aligned}$$

Then it follows

$$\begin{aligned}\frac{d}{d\tau}X_H(\tau) &= \frac{d}{d\tau}U(0, \tau)X(\tau)U(\tau, 0) \\ &= \left(\frac{d}{d\tau}U(0, \tau)\right)X(\tau)U(\tau, 0) + U(0, \tau)\left(\frac{d}{d\tau}X(\tau)\right)U(\tau, 0) + U(0, \tau)X(\tau)\left(\frac{d}{d\tau}U(\tau, 0)\right) \\ &= \frac{1}{\hbar}U(0, \tau)HX(\tau)U(\tau, 0) + U(0, \tau)\left(\frac{d}{d\tau}X(\tau)\right)U(\tau, 0) + U(0, \tau)X(\tau)\left(-\frac{1}{\hbar}HU(\tau, 0)\right) \\ &= \frac{1}{\hbar}U(0, \tau)(HX(\tau) - X(\tau)H)U(\tau, 0) + U(0, \tau)\left(\frac{d}{d\tau}X(\tau)\right)U(\tau, 0) \\ &= \frac{1}{\hbar}U(0, \tau)[H, X(\tau)]U(\tau, 0) + U(0, \tau)\left(\frac{d}{d\tau}X(\tau)\right)U(\tau, 0) \\ &= -\frac{1}{\hbar}[X, H]_H(\tau) + U(0, \tau)\left(\frac{d}{d\tau}X(\tau)\right)U(\tau, 0)\end{aligned}$$

Proof 31.20 of Lemma 8.22

There is at least two possibilities to prove the lemma.

1. With the Equation of motion-method.
2. With the Baker-Campbell-Hausdorff-series³.

TBElaborated with Mahan eq. (3.44) and BCH-series

TBElaborated with Mahan eq. (3.67) and BCH-series

The Equation of motion-method

We compute with the Heisenberg-equation-of-motion in Equation TBReferenced, and with the identity for statistical commutators

$$\begin{aligned}\frac{d}{dt}c_{\bullet}(t) &= -\frac{i}{\hbar} \sum_{\alpha \in \mathcal{A}} (\epsilon_{\alpha} - \mu) [c_{\bullet}, c_{\alpha}^{\dagger} c_{\alpha}]_H(t) \\ &= -\frac{i}{\hbar} \sum_{\alpha \in \mathcal{A}} (\epsilon_{\alpha} - \mu) ([c_{\bullet}, c_{\alpha}^{\dagger}]_{\zeta} c_{\alpha} + \zeta c_{\alpha}^{\dagger} [c_{\bullet}, c_{\alpha}]_{\zeta})_H(t) \\ &= \frac{i}{\hbar} (\epsilon_{\bullet} - \mu) c_{\bullet}(t)\end{aligned}$$

³ TBElaborated with Mahan eq. (3.44) and Zirnbauers exercise-sheet

with the initial conditions $c_\alpha(t = 0) = c_\alpha$ we see, that

$$c_\bullet(t) = e^{-i(\epsilon_\bullet - \mu)t/\hbar} c_\bullet \quad (31.11)$$

is a solution of the differential equation. Taking the hermitian conjugate proves the other identity.

The Baker-Campbell-Hausdorff-method

TBD.

Other references related to this⁴

Proof 31.21 of Lemma 8.30

We compute the hermitian conjugate of the interaction-term as

$$\begin{aligned} (H_{\text{el-ph}})^\dagger &= \sum_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ v \in \{1, \dots, r d_x\} \\ n, m \in \{1, \dots, \infty\}}} \frac{g_{mnv}^*(\mathbf{k}, \mathbf{q})}{\sqrt{N_p}} c_{n\sigma}^\dagger(\mathbf{k}) c_{m\sigma}(\mathbf{k} + \mathbf{q}) (b_v(-\mathbf{q}) + b_v^\dagger(+\mathbf{q})) \\ &= \sum_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ v \in \{1, \dots, r d_x\} \\ n, m \in \{1, \dots, \infty\}}} \frac{g_{mnv}^*(\mathbf{k}, -\mathbf{q})}{\sqrt{N_p}} c_{n\sigma}^\dagger(\mathbf{k}) c_{m\sigma}(\mathbf{k} - \mathbf{q}) (b_v(+\mathbf{q}) + b_v^\dagger(-\mathbf{q})) \\ &= \sum_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ v \in \{1, \dots, r d_x\} \\ n, m \in \{1, \dots, \infty\}}} \frac{g_{nmv}^*(\mathbf{k} + \mathbf{q}, -\mathbf{q})}{\sqrt{N_p}} c_{m\sigma}^\dagger(\mathbf{k} + \mathbf{q}) c_{n\sigma}(\mathbf{k}) (b_v(\mathbf{q}) + b_v^\dagger(-\mathbf{q})) \end{aligned}$$

whereas we used $\mathbf{q} \mapsto -\mathbf{q}$, in the second equation sign and $\mathbf{k} \mapsto \mathbf{k} + \mathbf{q}$ and $m \leftrightarrow n$ in the third equation sign.

Therefore we obtain

$$g_{mnv}(\mathbf{k}, \mathbf{q}) = g_{nmv}^*(\mathbf{k} + \mathbf{q}, -\mathbf{q})$$

Proof 31.22 of Lemma 9.47

For this proof, see also the literature⁵. We first rewrite

$$n^\zeta(x) = \frac{1}{e^x - \zeta} = (e^x - \zeta)^{-1} \quad (31.12)$$

and then compute for the first identity

⁴ TBCopied: cf. Zirnbauer AQM, 2018, 2019, Sheet 10 Exercise 1b).

⁵ Cf. Mahan Eqs. (3.236), (3.237).

$$\frac{d}{dx} n^\zeta(x) = -1 \cdot \frac{1}{(e^x - \zeta)^2} \cdot e^x = \frac{-1}{4} \cdot \frac{2e^{x/2}}{e^x - \zeta} \cdot \frac{2e^{x/2}}{e^x - \zeta} = \frac{-1}{4} \zeta \operatorname{ch}^2\left(\frac{x}{2}\right) \quad (31.13)$$

For the second identity we simplify the appearing expressions in another way as

$$\frac{d}{dx} n^\zeta(x) = -1 \cdot \frac{1}{(e^x - \zeta)^2} \cdot e^x = - \cdot \frac{1}{e^x - \zeta} \cdot \frac{e^x - \zeta + \zeta}{e^x - \zeta} = -n^\zeta(x) \cdot (1 + \zeta n^\zeta(x)) \quad (31.14)$$

Proof 31.23 of Lemma 9.47

We compute

$$F_{\hbar\beta}(\bullet + i\omega_n) = \frac{1}{e^{\hbar\beta(\bullet + i\omega_n)} + 1} = \frac{1}{-e^{\hbar\beta\bullet} + 1} = -B_{\hbar\beta}(\bullet)$$

and

$$F_{\hbar\beta}(i\omega_n - \bullet) - B_{\hbar\beta}(\bullet) = \frac{1}{e^{\hbar\beta(i\omega_n - \bullet)} + 1} - \frac{1}{e^{\hbar\beta\bullet} - 1} = \frac{1}{1 - e^{-\hbar\beta\bullet}} - \frac{1}{e^{\hbar\beta\bullet} - 1} = \frac{e^{-\hbar\beta\bullet} - 1}{e^{-\hbar\beta\bullet} - 1} = 1$$

For the third identity **we still have to digitalize / create the proof.**

Proof 31.24 of Lemma 9.48

We define $x = \hbar\beta\omega = \frac{\hbar\omega}{k_B T}$ such that we obtain

$$\frac{dx}{dT} = \hbar\omega \frac{d\beta}{dT} = \hbar\omega \frac{d}{dT} \left(\frac{1}{k_B T} \right) = -\frac{\hbar\omega}{k_B T^2}. \quad (31.15)$$

and

$$n_{\hbar\beta}^\zeta(\omega) = n^\zeta(x) \quad (31.16)$$

and therefore

$$\frac{d}{dT} n_{\hbar\beta}^\zeta(\omega) = \frac{d}{dT} n^\zeta(x) = \frac{d}{dx} n^\zeta(x) \cdot \frac{dx}{dT} \quad (31.17)$$

Combining this with Lemma 9.45 yields the desired result.

Proof 31.25 of Corollary 9.49 We recall the definition of the Bose-function

$$B_{\hbar\beta}(z) = \frac{1}{e^{\hbar\beta z} - 1} = (e^{\hbar\beta z} - 1)^{-1} \quad (31.18)$$

and compute straightforwardly

$$\begin{aligned}
 \frac{d}{dz} B_{\hbar\beta}(z) &= - (e^{\hbar\beta z} - 1)^{-2} \cdot \underbrace{\frac{d}{dz} e^{\hbar\beta z}}_{\hbar\beta e^{\hbar\beta z}} \\
 &= -\hbar\beta \frac{e^{\hbar\beta z} - 1 + 1}{(e^{\hbar\beta z} - 1)^2} \\
 &= -\hbar\beta \left[B_{\hbar\beta}(z) + B_{\hbar\beta}^2(z) \right] \\
 &= -\hbar\beta B_{\hbar\beta}(z) \left[1 + B_{\hbar\beta}(z) \right]
 \end{aligned} \tag{31.19}$$

For the Fermi-function the computation is analogous.

Proof 31.26 of Lemma 12.3 With the cyclicity of the trace, and the Baker-Campbell-Hausdorff-formula, the proof is straightforward: [TBElaborated...]

Proof 31.27 of Lemma 12.4

See Waste-Folder 1, topic 10.

The proofs are more or less straightforward by simply writing out the underlying equations. We obtain for the individual Green-functions

$$\begin{aligned}
 G(A, A') &= -\frac{i}{\hbar} \langle \theta(t - t') c_\alpha(t) c_{\alpha'}^\dagger(t') + \zeta \theta(t' - t) c_{\alpha'}^\dagger(t') c_\alpha(t) \rangle \\
 &= \theta(t - t') G^>(A, A') + \theta(t' - t) G^<(A, A')
 \end{aligned}$$

$$\begin{aligned}
 G^\top(A, A') &= -\frac{i}{\hbar} \langle \theta(t' - t) c_\alpha(t) c_{\alpha'}^\dagger(t') + \zeta \theta(t - t') c_{\alpha'}^\dagger(t') c_\alpha(t) \rangle \\
 &= \theta(t' - t) G^>(A, A') + \theta(t - t') G^<(A, A')
 \end{aligned}$$

$$G^{\text{ret}}(A, A') = -\frac{i}{\hbar} \theta(t - t') \langle c_\alpha(t) c_{\alpha'}^\dagger(t') - \zeta c_{\alpha'}^\dagger(t') c_\alpha(t) \rangle$$

$$G^{\text{adv}}(A, A') = +\frac{i}{\hbar} \theta(t' - t) \langle c_\alpha(t) c_{\alpha'}^\dagger(t') - \zeta c_{\alpha'}^\dagger(t') c_\alpha(t) \rangle$$

(i) We obtain for the retarded Green-function

$$G_{\alpha, \alpha'}^{\text{ret}}(t, t') = -\frac{i}{\hbar} \theta(t - t') \langle c_\alpha(t) c_{\alpha'}^\dagger(t') - \zeta c_{\alpha'}^\dagger(t') c_\alpha(t) \rangle \tag{31.20}$$

and for the advanced Green-function

$$G_{\alpha,\alpha'}^{\text{adv}}(t, t') = +\frac{i}{\hbar}\theta(t' - t)\langle c_\alpha(t)c_{\alpha'}^\dagger(t') - \zeta c_{\alpha'}^\dagger(t')c_\alpha(t) \rangle \quad (31.21)$$

and hence

$$\begin{aligned} \left(G_{\alpha',\alpha}^{\text{adv}}(t', t)\right)^\dagger &= \left(+\frac{i}{\hbar}\theta(t - t')\langle c_{\alpha'}(t')c_\alpha^\dagger(t) + c_\alpha^\dagger(t)c_{\alpha'}(t') \rangle\right)^\dagger \\ &= -\frac{i}{\hbar}\theta(t - t')\langle c_\alpha(t)c_{\alpha'}^\dagger(t') + c_{\alpha'}^\dagger(t')c_\alpha(t) \rangle = G_{\alpha,\alpha'}^{\text{ret}}(t, t') \end{aligned}$$

Therefore we obtain

$$G_{\alpha,\alpha'}^{\text{ret}}(t, t') = \left(G_{\alpha',\alpha}^{\text{adv}}(t', t)\right)^\dagger \iff G^{\text{ret}}(A, A') = \left(G^{\text{adv}}(A', A)\right)^\dagger \quad (31.22)$$

(ii) We obtain further

$$\begin{aligned} (G^{++} - G^{-+})(A, A') &= -\frac{i}{\hbar}\langle (\theta(t - t') - 1) c_\alpha(t)c_{\alpha'}(t') + \zeta\theta(t' - t)c_{\alpha'}(t')c_\alpha(t) \rangle \\ &= -\frac{i}{\hbar}\langle -\theta(t' - t)c_\alpha(t)c_{\alpha'}(t') + \zeta\theta(t' - t)c_{\alpha'}(t')c_\alpha(t) \rangle \\ &= \frac{i}{\hbar}\theta(t' - t)\langle c_\alpha(t)c_{\alpha'}(t') - \zeta c_{\alpha'}(t')c_\alpha(t) \rangle \end{aligned}$$

(iii) We obtain further

$$\begin{aligned} (G^{++} - G^{--})(A, A') &= -\frac{i}{\hbar}\langle -\theta(t' - t)c_\alpha(t)c_{\alpha'}(t') + (-\zeta\theta(t - t') + \zeta) c_{\alpha'}(t')c_\alpha(t) \rangle \\ &= -\frac{i}{\hbar}\langle -\theta(t' - t)c_\alpha(t)c_{\alpha'}(t') + \zeta\theta(t' - t)c_{\alpha'}(t')c_\alpha(t) \rangle \\ &= +\frac{i}{\hbar}\theta(t' - t)\langle c_\alpha(t)c_{\alpha'}(t') - \zeta c_{\alpha'}(t')c_\alpha(t) \rangle \end{aligned}$$

(iv) We obtain further

$$\begin{aligned} (G^{++} - G^{-+})(A, A') &= -\frac{i}{\hbar}\langle \zeta c_{\alpha'}(t')c_\alpha(t) - c_\alpha(t)c_{\alpha'}(t') \rangle \\ &= \frac{i}{\hbar}\langle c_\alpha(t)c_{\alpha'}(t') - \zeta\theta(t' - t)c_{\alpha'}(t')c_\alpha(t) \rangle \end{aligned}$$

(v) We obtain further

$$\begin{aligned} (G^{++} - G^{+-})(A, A') &= -\frac{i}{\hbar} \langle \theta(t - t') c_\alpha(t) c_{\alpha'}(t') + \zeta (\theta(t' - t) - 1) c_{\alpha'}(t') c_\alpha(t) \rangle \\ &= -\frac{i}{\hbar} \theta(t - t') \langle c_\alpha(t) c_{\alpha'}(t') - \zeta c_{\alpha'}(t') c_\alpha(t) \rangle \end{aligned}$$

(vi) We obtain further

$$\begin{aligned} (G^{-+} - G^{--})(A, A') &= -\frac{i}{\hbar} \langle (1 - \theta(t' - t)) c_\alpha(t) c_{\alpha'}(t') - \zeta \theta(t - t') c_{\alpha'}(t') c_\alpha(t) \rangle \\ &= -\frac{i}{\hbar} \theta(t - t') \langle c_\alpha(t) c_{\alpha'}(t') - \zeta c_{\alpha'}(t') c_\alpha(t) \rangle \end{aligned}$$

and therefore we obtain

$$\theta(t - t') (G^{-+} - G^{+-})(A, A') = -\frac{i}{\hbar} \theta(t - t') \langle c_\alpha(t) c_{\alpha'}(t') - \zeta c_{\alpha'}(t') c_\alpha(t) \rangle$$

ToBeSorted. The last two identities follow from combining the results we just obtained **TBDone.**

Proof 31.28 of Lemma 12.6

The time-ordered Green-function

With the equation of motion technique, we consider the definition

$$\mathbb{R} \ni t \mapsto G_{\alpha, \alpha'}^\top(t) = -\frac{i}{\hbar} \frac{\text{Tr}_{\mathcal{F}_\zeta} e^{-\beta(H - \mu N)} \left[\theta(t) c_\alpha(t) c_{\alpha'}^\dagger(0) + \zeta \theta(-t) c_{\alpha'}^\dagger(0) c_\alpha(t) \right]}{\text{Tr}_{\mathcal{F}_\zeta} e^{-\beta(H - \mu N)}} \in \mathbb{C}$$

and want to calculate its derivative. We therefore calculate first

$$\begin{aligned} \partial_t T_t c_\alpha(t) c_{\alpha'}^\dagger(0) &\propto \partial_t \left[\theta(t) c_\alpha(t) c_{\alpha'}^\dagger(0) + \zeta \theta(-t) c_{\alpha'}^\dagger(0) c_\alpha(t) \right] \\ &= \delta(t) c_\alpha(t) c_{\alpha'}^\dagger(0) - \zeta \delta(-t) c_{\alpha'}^\dagger(0) c_\alpha(t) - \frac{i}{\hbar} \theta(t) [c_\alpha, H - \mu]_{\text{H}}(t) c_{\alpha'}^\dagger(0) - \frac{i}{\hbar} \theta(-t) \zeta c_{\alpha'}^\dagger(0) [c_\alpha, H - \mu]_{\text{H}}(t) \\ &= \delta(t) [c_\alpha(t), c_{\alpha'}^\dagger(0)]_\zeta - \frac{i}{\hbar} T_t ([c_\alpha, H - \mu]_{\text{H}}(t) c_{\alpha'}^\dagger(0)) \end{aligned}$$

whereas we used the statistical commutator from Definition TBD and the Heisenberg-equation-of-motions from Lemma TBD. This gives the equation of motion

$$\partial_t G_{\alpha,\alpha'}^{\mathbb{T}}(t) = -\frac{i}{\hbar} \delta(t) \delta_{\alpha,\alpha'} - \frac{1}{\hbar} \left\langle \mathbb{T}_t \left[c_\alpha, \frac{H - \mu N}{\hbar} \right]_{\mathbb{H}}(t) c_{\alpha'}^\dagger(0) \right\rangle \quad (31.23)$$

The anti-time-ordered Green-function

With the equation of motion technique, we consider the definition

$$\mathbb{R} \ni t \mapsto G_{\alpha,\alpha'}^{\mathbb{T}}(t) = -\frac{i}{\hbar} \frac{\text{Tr}_{\mathcal{F}_\zeta} e^{-\beta(H-\mu N)} \left[\theta(-t) c_\alpha(t) c_{\alpha'}^\dagger(0) + \zeta \theta(+t) c_{\alpha'}^\dagger(0) c_\alpha(t) \right]}{\text{Tr}_{\mathcal{F}_\zeta} e^{-\beta(H-\mu N)}} \in \mathbb{C}$$

and want to calculate its derivative. We therefore calculate first

$$\begin{aligned} \partial_t \mathbb{T}_t c_\alpha(t) c_{\alpha'}^\dagger(0) &\propto \partial_t \left[\theta(-t) c_\alpha(t) c_{\alpha'}^\dagger(0) + \zeta \theta(+t) c_{\alpha'}^\dagger(0) c_\alpha(t) \right] \\ &= -\delta(-t) c_\alpha(t) c_{\alpha'}^\dagger(0) + \zeta \delta(t) c_{\alpha'}^\dagger(0) c_\alpha(t) - \frac{i}{\hbar} \theta(-t) [c_\alpha, H - \mu]_{\mathbb{H}}(t) c_{\alpha'}^\dagger(0) - \frac{i}{\hbar} \theta(t) \zeta c_{\alpha'}^\dagger(0) [c_\alpha, H - \mu]_{\mathbb{H}}(t) \\ &= -\delta(-t) [c_\alpha(t), c_{\alpha'}^\dagger(0)]_\zeta - \frac{i}{\hbar} \mathbb{T}_t ([c_\alpha, H - \mu]_{\mathbb{H}}(t) c_{\alpha'}^\dagger(0)) \end{aligned}$$

whereas we used the statistical commutator from Definition TBD and the Heisenberg-equation-of-motions from Lemma TBD. This gives the equation of motion

$$\partial_t G_{\alpha,\alpha'}^{\mathbb{T}}(t) = \frac{i}{\hbar} \delta(t) \delta_{\alpha,\alpha'} - \frac{1}{\hbar} \left\langle \mathbb{T}_t \left[c_\alpha, \frac{H - \mu N}{\hbar} \right]_{\mathbb{H}}(t) c_{\alpha'}^\dagger(0) \right\rangle \quad (31.24)$$

The retarded Green-function

With the equation of motion technique, we consider the definition

$$\mathbb{R} \ni t \mapsto G_{\alpha,\alpha'}^{\text{ret}}(t) = -\frac{i}{\hbar} \theta(t) \frac{\text{Tr}_{\mathcal{F}_\zeta} e^{-\beta(H-\mu N)} \left[c_\alpha(t) c_{\alpha'}^\dagger(0) - \zeta c_{\alpha'}^\dagger(0) c_\alpha(t) \right]}{\text{Tr}_{\mathcal{F}_\zeta} e^{-\beta(H-\mu N)}} \in \mathbb{C}$$

and want to calculate its derivative. We therefore calculate first

$$\begin{aligned} \partial_t \theta(t) [c_\alpha(t), c_{\alpha'}^\dagger(0)]_\zeta &= \delta(t) [c_\alpha(t), c_{\alpha'}^\dagger(0)]_\zeta + \theta(t) [\partial_t c_\alpha(t), c_{\alpha'}^\dagger(0)]_\zeta \\ &= \delta(t) \delta_{\alpha,\alpha'} - \frac{i}{\hbar} \theta(t) [[c_\alpha, H - \mu]_{\mathbb{H}}(t), c_{\alpha'}^\dagger(0)]_\zeta \end{aligned}$$

whereas we used the statistical commutator from Definition TBD and the Heisenberg-equation-of-motions from Lemma TBD. This gives the equation of motion

$$\partial_t G_{\alpha,\alpha'}^{\text{ret}}(t) = -\frac{i}{\hbar} \delta(t) \delta_{\alpha,\alpha'} - \frac{1}{\hbar} \theta(t) \left\langle \left[[c_\alpha, H - \mu]_{\text{H}}(t), c_{\alpha'}^\dagger(0) \right]_\zeta \right\rangle \quad (31.25)$$

The advanced Green-function

With the equation of motion technique, we consider the definition

$$\mathbb{R} \ni t \mapsto G_{\alpha,\alpha'}^{\text{adv}}(t) = \frac{i}{\hbar} \theta(-t) \frac{\text{Tr}_{\mathcal{F}_\zeta} e^{-\beta(H-\mu N)} \left[c_\alpha(t) c_{\alpha'}^\dagger(0) - \zeta c_{\alpha'}^\dagger(0) c_\alpha(t) \right]}{\text{Tr}_{\mathcal{F}_\zeta} e^{-\beta(H-\mu N)}} \in \mathbb{C}$$

and want to calculate its derivative. We therefore calculate first

$$\begin{aligned} \partial_t \theta(-t) [c_\alpha(t), c_{\alpha'}^\dagger(0)]_\zeta &= -\delta(-t) [c_\alpha(t), c_{\alpha'}^\dagger(0)]_\zeta + \theta(-t) [\partial_t c_\alpha(t), c_{\alpha'}^\dagger(0)]_\zeta \\ &= -\delta(-t) \delta_{\alpha,\alpha'} - \frac{i}{\hbar} \theta(-t) \left[[c_\alpha, H - \mu]_{\text{H}}(t), c_{\alpha'}^\dagger(0) \right]_\zeta \end{aligned}$$

whereas we used the statistical commutator from Definition TBD and the Heisenberg-equation-of-motions from Lemma TBD. This gives the equation of motion

$$\partial_t G_{\alpha,\alpha'}^{\text{adv}}(t) = -\frac{i}{\hbar} \delta(-t) \delta_{\alpha,\alpha'} + \frac{1}{\hbar} \theta(-t) \left\langle \left[[c_\alpha, H - \mu]_{\text{H}}(t), c_{\alpha'}^\dagger(0) \right]_\zeta \right\rangle \quad (31.26)$$

Proof 31.29 of Corollary 12.8 Recall that we have, *for example* (we could use any other one as well) for the time-ordered-Green-function

$$G_{i,j}(t, t') = -\frac{i}{\hbar} \left\langle \text{T}_t c_i(t) c_j^\dagger(t') \right\rangle \quad (31.27)$$

and therefore we obtain

$$\begin{aligned} \tilde{G}_{\alpha,\beta}(t, t') &= -\frac{i}{\hbar} \left\langle \text{T}_t c_\alpha(t) c_\beta^\dagger(t') \right\rangle \\ &= -\frac{i}{\hbar} \sum_{i,j} U_{i,\alpha}^* U_{j,\beta} \left\langle \text{T}_t c_i(t) c_j^\dagger(t') \right\rangle \\ &= \sum_{i,j} U_{i,\alpha}^* G_{i,j}(t, t') U_{j,\beta} \end{aligned} \quad (31.28)$$

Proof 31.30 of Theorem 12.9

See Waste-Folder 1, topic 5.

We start with the lesser Green-function and we focus on the case $\alpha = \alpha'$ since it is easy to see that the expectation-value is diagonal in these indices. This gives (lightening the notation) which gives

$$G^< = -\frac{i}{\hbar} \zeta e^{-i\xi t/\hbar} \frac{1}{Z} \sum_{j \in \mathbb{N}} \langle j | e^{-\beta(\epsilon-\mu)c^\dagger c} c^\dagger c | j \rangle = \dots \quad (31.29)$$

Now we simplify

$$\begin{aligned} \frac{1}{Z} \sum_{j \in \mathbb{N}} \langle j | e^{-\beta(\epsilon-\mu)c^\dagger c} c^\dagger c | j \rangle &= \frac{\sum_{j \in \mathbb{N}} j e^{-\beta(\epsilon-\mu)j}}{\sum_{j \in \mathbb{N}} e^{-\beta(\epsilon-\mu)j}} \\ &= -\partial_x \log \sum_{j \in \mathbb{N}} e^{-\beta(\epsilon-\mu)x} \Big|_{x=\beta(\epsilon-\mu)} \\ &= -\partial_x \log \frac{1}{(1 - \zeta e^{-x})^\zeta} \Big|_{x=\beta(\epsilon-\mu)} \\ &= -\partial_x \log (1 - \zeta e^{-x})^{-\zeta} \Big|_{x=\beta(\epsilon-\mu)} \\ &= \zeta \partial_x \log (1 - \zeta e^{-x}) \Big|_{x=\beta(\epsilon-\mu)} \\ &= \zeta^2 \frac{e^{-x}}{1 - \zeta} \Big|_{x=\beta(\epsilon-\mu)} \\ &= n_{\hbar\beta}^\zeta(\xi) \end{aligned} \quad (31.30)$$

whereas in the last step we used Notation 9.44. This yields therefore

$$\dots = -\frac{i}{\hbar} \zeta e^{-i\xi t/\hbar} n_{\hbar\beta}^\zeta(\xi) \quad (31.31)$$

We compute further

$$\begin{aligned} \frac{i}{\hbar} e^{-i\xi t/\hbar} &= \frac{i}{\hbar} e^{-i\xi t/\hbar} \langle [c, c^\dagger]_\zeta \rangle \\ &= \frac{i}{\hbar} \langle c(t) c^\dagger \rangle - \zeta \frac{i}{\hbar} \langle c^\dagger c(t) \rangle \\ &= -G^>(t) + G^<(t) \end{aligned} \quad (31.32)$$

which can be rewritten as

$$\begin{aligned} G^>(t) &= G^<(t) - \frac{i}{\hbar} e^{-i\xi t/\hbar} \\ &= -\frac{i}{\hbar} e^{-i\xi t/\hbar} \left(\zeta n_{\hbar\beta}^\zeta(\xi) + 1 \right) \end{aligned} \quad (31.33)$$

This then gives eventually

$$\begin{aligned} G^\mathbb{T}(t) &= \theta(t) G^>(t) + \theta(-t) G^< \\ &= -\frac{i}{\hbar} e^{-i\xi t/\hbar} \left(\left(1 + \zeta n_{\hbar\beta}^\zeta(\xi) \right) \theta(t) + \theta(-t) \zeta n_{\hbar\beta}^\zeta(\xi) \right) \end{aligned} \quad (31.34)$$

And further we obtain

$$\begin{aligned} G^{\text{ret}}(t) &= \theta(t) (G^>(t) - G^<(t)) \\ &= \theta(t) \left(-\frac{i}{\hbar} \right) e^{-i\xi t/\hbar} \end{aligned} \quad (31.35)$$

and we also obtain

$$\begin{aligned} G^{\text{adv}}(t) &= \theta(-t) (G^<(t) - G^>(t)) \\ &= \theta(-t) \left(\frac{i}{\hbar} \right) e^{-i\xi t/\hbar} \end{aligned} \quad (31.36)$$

Now we transition to the frequency-versions of these equations and obtain

$$\begin{aligned} G^<(\omega) &= \int dt e^{i\omega t} G^<(t) \\ &= -\frac{i}{\hbar} \zeta n_{\hbar\beta}^{\zeta}(\xi) \underbrace{\int dt e^{i\omega t - i\xi t/\hbar}}_{=2\pi\delta(\omega - \xi/\hbar)} \end{aligned} \quad (31.37)$$

and likewise we obtain

$$\begin{aligned} G^>(\omega) &= \int dt e^{i\omega t} G^>(t) \\ &= -\frac{i}{\hbar} \left(1 + \zeta n_{\hbar\beta}^{\zeta}(\xi) \right) \underbrace{\int dt e^{i\omega t - i\xi t/\hbar}}_{=2\pi\delta(\omega - \xi/\hbar)} \end{aligned} \quad (31.38)$$

The identities for G^{ret} and G^{adv} follow by computing the integral. For the time-ordered Green-function we obtain

$$\begin{aligned} G_{\eta}^{\mathbb{T}}(\omega) &= \int dt e^{i\omega t} G_{\eta}^{\mathbb{T}}(t) \\ &= \int dt e^{i\omega t} G^{\mathbb{T}}(t) e^{-\eta|t|} \\ &= -\frac{i}{\hbar} \left(1 + \zeta n_{\hbar\beta}^{\zeta}(\omega) \right) \int dt e^{i\omega t} e^{-\eta|t|} e^{-i\xi t/\hbar} \theta(t) \\ &\quad + \zeta \left(-\frac{i}{\hbar} \right) n_{\hbar\beta}^{\zeta}(\omega) \int dt e^{i\omega t} e^{-\eta|t|} e^{-i\xi t/\hbar} \theta(-t) \\ &= \frac{1 - \zeta n_{\hbar\beta}^{\zeta}(\omega)}{\hbar} \frac{1}{\omega + i\eta - \xi/\hbar} - \zeta \frac{n_{\hbar\beta}^{\zeta}(\omega)}{\hbar} \frac{1}{\omega - i\eta - \xi/\hbar} \end{aligned} \quad (31.39)$$

Lastly we obtain

$$\begin{aligned} G^{\mathbb{T}}(t) &= \theta(-t) G^>(t) + \theta(t) G^<(t) \\ &= -\frac{i}{\hbar} e^{-i\xi t/\hbar} \left(\theta(-t) \left(1 + \zeta n_{\hbar\beta}^{\zeta}(\omega) \right) + \theta(t) \zeta n_{\hbar\beta}^{\zeta}(\omega) \right) \end{aligned} \quad (31.40)$$

and

$$\begin{aligned}
 G_{\eta}^{\overline{\overline{}}}(\omega) &= \int dt G_{\eta}^{\overline{\overline{}}}(t) \\
 &= \int dt G_{\eta}^{\overline{\overline{}}}(t) e^{-\eta|t|} \\
 &= -\frac{i}{\hbar} \left(1 + \zeta n_{\hbar\beta}^{\zeta}(\omega) \right) \int dt e^{i\omega t} e^{-i\xi t/\hbar} e^{-\eta|t|} \theta(-t) \\
 &= -\frac{i}{\hbar} \zeta n_{\hbar\beta}^{\zeta}(\omega) \int dt e^{i\omega t} e^{-i\xi t/\hbar} e^{-\eta|t|} \theta(t) \\
 &= -\frac{1 + \zeta n_{\hbar\beta}^{\zeta}(\omega)}{\hbar} \frac{1}{\omega - i\eta - \xi/\hbar} + \frac{\zeta n_{\hbar\beta}^{\zeta}(\omega)}{\hbar} \frac{1}{\omega + i\eta - \xi/\hbar}
 \end{aligned} \tag{31.41}$$

Proof 31.0 of Lemma 12.13 We consider the unit-vector

$$\frac{\epsilon}{\epsilon} \equiv \mathbf{n} \equiv n_x \mathbf{e}_x + n_y \mathbf{e}_y + n_z \mathbf{e}_z \in S^2 \subset \mathbb{R}^3$$

and find that

$$H \equiv H(\mathbf{n}) := \epsilon_0 \sigma_0 + \epsilon \mathbf{n} \cdot \boldsymbol{\sigma} = \epsilon_0 \sigma_0 + \epsilon (n_x \sigma_x + n_y \sigma_y + n_z \sigma_z).$$

Therefore, if $|\lambda\rangle$ is an eigenstate of $\mathbf{n} \cdot \boldsymbol{\sigma}$ with eigenvalue λ then it follows

$$\epsilon_0 \sigma_0 + \epsilon (n_x \sigma_x + n_y \sigma_y + n_z \sigma_z) |\lambda\rangle = (\epsilon_0 \sigma_0 + \epsilon \lambda) |\lambda\rangle$$

therefore $|\lambda\rangle$ is also ein eigenvalue of H with eigenvalue $(\epsilon_0 \sigma_0 + \epsilon \lambda)$.

Proof 31.0 of Lemma 12.15 We compute the eigenvalues of the 2×2 -matrix:

$$\mathbf{n} \cdot \boldsymbol{\sigma} = \begin{pmatrix} n_z & -in_y + n_x \\ +in_y + n_x & -n_z \end{pmatrix}$$

by solving the characteristic equation

$$\begin{aligned}
 \text{Det} (\mathbf{n} \cdot \boldsymbol{\sigma} - \lambda \cdot \mathbf{1}_{2 \times 2}) &= \text{Det} \begin{pmatrix} n_z - \lambda & -in_y + n_x \\ +in_y + n_x & -n_z - \lambda \end{pmatrix} \stackrel{\text{Binomial Formulae}}{=} \lambda^2 - n_z^2 - (n_x^2 + n_y^2) = \lambda^2 - \mathbf{n}^2 = 0 \\
 \iff \lambda^2 = \mathbf{n}^2 = 1 &\iff \lambda_{\pm} = \pm 1
 \end{aligned}$$

Now we compute the eigenstates of $\mathbf{n} \cdot \boldsymbol{\sigma}$:

$$\begin{aligned}
 \text{Kern}(\mathbf{n} \cdot \boldsymbol{\sigma} - (\pm 1) \cdot \mathbf{1}_{2 \times 2}) &= \text{Kern} \begin{pmatrix} n_z \mp 1 & -in_y + n_x \\ 0 & 0 \end{pmatrix} \\
 &= \left\{ \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \in \mathbb{C}^2 \mid (n_z \mp 1)v_1 + (-in_y + n_x)v_2 = 0 \right\} \\
 &= \text{Span}_{\alpha \in \mathbb{C}} \left\{ \begin{pmatrix} \alpha \\ \frac{-(n_z \mp 1)\alpha}{n_x - in_y} \end{pmatrix} \right\} \quad (\star)
 \end{aligned}$$

The eigenstate to the eigenvalue +1:

We multiply a "clever one"

$$\frac{1}{n_x - in_y} = \frac{n_x + in_y}{n_x^2 + n_y^2} = \frac{(\cos(\phi) + i \sin(\phi)) \sin(\theta)}{(\cos^2(\phi) + \sin^2(\phi)) \sin^2(\theta)} = \frac{e^{i\phi}}{\sin(\theta)}$$

In order to reproduce the control result, we consider⁶ the eigenvector in $(\star)$ with $\alpha = \cos \frac{\theta}{2}$. Then it follows for the eigenvector

$$\begin{aligned}
 |\uparrow_{\mathbf{n}}\rangle &= \cos \frac{\theta}{2} |\uparrow\rangle + \frac{(1 - n_z) \cos \frac{\theta}{2}}{n_x - in_y} |\downarrow\rangle \\
 &= \cos \frac{\theta}{2} |\uparrow\rangle + \frac{(1 - \cos(\theta)) \cos(\frac{\theta}{2})}{\sin(\theta)} e^{i\phi} |\downarrow\rangle \\
 &= \cos \frac{\theta}{2} |\uparrow\rangle + \sin \frac{\theta}{2} e^{i\phi} |\downarrow\rangle
 \end{aligned}$$

whereas in the last equation-sign we used a standard-result⁷ in the last step. This eigenstate is also normalized as one finds

$$\langle \uparrow_{\mathbf{n}} | \uparrow_{\mathbf{n}} \rangle = \left(\cos \frac{\theta}{2} \langle \uparrow | + \sin \frac{\theta}{2} e^{-i\phi} \langle \downarrow | \right) \left(\cos \frac{\theta}{2} |\uparrow\rangle + \sin \frac{\theta}{2} e^{i\phi} |\downarrow\rangle \right) = \cos^2 \frac{\theta}{2} + \sin^2 \frac{\theta}{2} = 1.$$

The eigenstate to the eigenvalue -1:

This eigenstate can be concluded by demanding orthogonality to $|\uparrow_{\mathbf{n}}\rangle$. One finds that

$$|\downarrow_{\mathbf{n}}\rangle = \sin \frac{\theta}{2} |\uparrow\rangle - \cos \frac{\theta}{2} e^{i\phi} |\downarrow\rangle$$

is orthogonal since

$$\langle \downarrow_{\mathbf{n}} | \uparrow_{\mathbf{n}} \rangle = \left(\sin \frac{\theta}{2} \langle \uparrow | - \cos \frac{\theta}{2} e^{-i\phi} \langle \downarrow | \right) \left(\cos \frac{\theta}{2} |\uparrow\rangle + \sin \frac{\theta}{2} e^{i\phi} |\downarrow\rangle \right) = \sin \frac{\theta}{2} \cos \frac{\theta}{2} - \cos \frac{\theta}{2} \sin \frac{\theta}{2} e^0 = 0$$

⁶Recall that a complex eigenvector is only unique up to a complex phase.

⁷See [53].

Analogously one finds that $|\downarrow_n\rangle$ is normalized.

Remark: In the limit $\mathbf{n} \rightarrow \mathbf{e}_z$ one obtains states, that are in the same equivalence-class of states as the usual ones of the third Pauli-matrix.

Proof 31.0 of Corollary 12.16 The equations

$$\begin{aligned} |\uparrow_n\rangle &= \cos \frac{\theta}{2} |\uparrow\rangle + \sin \frac{\theta}{2} e^{i\phi} |\downarrow\rangle \\ |\downarrow_n\rangle &= \sin \frac{\theta}{2} |\uparrow\rangle - \cos \frac{\theta}{2} e^{i\phi} |\downarrow\rangle \end{aligned}$$

can be rewritten in matrix-form as

$$\begin{aligned} \begin{pmatrix} |\uparrow_n\rangle \\ |\downarrow_n\rangle \end{pmatrix} &= \begin{pmatrix} \cos \frac{\theta}{2} & \sin \frac{\theta}{2} e^{i\phi} \\ \sin \frac{\theta}{2} & -\cos \frac{\theta}{2} e^{i\phi} \end{pmatrix} \begin{pmatrix} |\uparrow\rangle \\ |\downarrow\rangle \end{pmatrix} \\ \Leftrightarrow \begin{pmatrix} |\uparrow\rangle \\ |\downarrow\rangle \end{pmatrix} &= \begin{pmatrix} \cos \frac{\theta}{2} & \sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} e^{-i\phi} & -\cos \frac{\theta}{2} e^{-i\phi} \end{pmatrix} \begin{pmatrix} |\uparrow_n\rangle \\ |\downarrow_n\rangle \end{pmatrix} \end{aligned}$$

since the determinant of the first matrix is equal to $-e^{i\phi}$, and therefore we have

$$\begin{aligned} |\uparrow\rangle &= \cos \frac{\theta}{2} |\uparrow_n\rangle + \sin \frac{\theta}{2} |\downarrow_n\rangle \\ |\downarrow\rangle &= \sin \frac{\theta}{2} e^{-i\phi} |\uparrow_n\rangle - \cos \frac{\theta}{2} e^{-i\phi} |\downarrow_n\rangle \end{aligned}$$

Proof 31.0 of Corollary 27.107 One finds

$$\begin{aligned} (\omega - \xi_\downarrow) \cos^2 \frac{\theta}{2} + (\omega - \xi_\uparrow) \sin^2 \frac{\theta}{2} &= \omega - (\epsilon_0 - \epsilon_z) \cos^2 \frac{\theta}{2} - (\epsilon_0 + \epsilon_z) \sin^2 \frac{\theta}{2} \\ &= \omega - \epsilon_0 + \epsilon_z (\cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}) \\ &= \omega - \epsilon_0 + \epsilon_z \end{aligned}$$

whereas we used $\cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} = \cos \theta$, etc. for the other entries.

Proof 31.31 of Theorem 12.21
The Lesser Temporal Green-function

Starting from Definition 12.19

$$D_{\nu, \nu'}^<(t) = -\frac{i}{\hbar} \left\langle A_{\nu'}^\dagger(0, \mathbf{q}') A_\nu(t, \mathbf{q}) \right\rangle. \quad (31.42)$$

we simply insert Definition 12.18 for the displacement-operators

$$A_\nu(\mathbf{q}) = c_{\mathbf{q}\nu} + c_{-\mathbf{q}\nu}^\dagger, \quad A_{\nu'}^\dagger(\mathbf{q}') = c_{\mathbf{q}'\nu'}^\dagger + c_{-\mathbf{q}'\nu'}. \quad (31.43)$$

and we use that for the non-interacting phonon Hamiltonian from Remark 8.22.4

$$H_0 = \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ \nu \in \{1, 2, \dots, r d_x\}}} \Omega_{\mathbf{q}\nu} c_{\mathbf{q}\nu}^\dagger c_{\mathbf{q}\nu}, \quad \text{with } \Omega_{\mathbf{q}\nu} = \hbar \omega_{\mathbf{q}\nu}, \quad (31.44)$$

the Heisenberg operators evolve according to Lemma 8.26 as

$$c_{\mathbf{q}\nu}(t) = e^{-i\omega_{\mathbf{q}\nu}t} c_{\mathbf{q}\nu}, \quad c_{\mathbf{q}\nu}^\dagger(t) = e^{+i\omega_{\mathbf{q}\nu}t} c_{\mathbf{q}\nu}^\dagger. \quad (31.45)$$

Consequently we find for the time-evolution of the displacement-operators the equation,

$$A_\nu(t, \mathbf{q}) = e^{-i\omega_{\mathbf{q}\nu}t} c_{\mathbf{q}\nu} + e^{+i\omega_{-\mathbf{q}\nu}t} c_{-\mathbf{q}\nu}^\dagger. \quad (31.46)$$

Assuming that time-reversal symmetry is not broken, we obtain with Remark 8.22.6 that $\omega_{-\mathbf{q}\nu} = \omega_{\mathbf{q}\nu}$, such that we obtain

$$D_{\nu, \nu'}^<(t) = -\frac{i}{\hbar} \left\langle \left(c_{\mathbf{q}'\nu'}^\dagger + c_{-\mathbf{q}'\nu'} \right) \left(e^{-i\omega_{\mathbf{q}\nu}t} c_{\mathbf{q}\nu} + e^{+i\omega_{\mathbf{q}\nu}t} c_{-\mathbf{q}\nu}^\dagger \right) \right\rangle \quad (31.47)$$

$$= -\frac{i}{\hbar} \left[e^{-i\omega_{\mathbf{q}\nu}t} \left\langle c_{\mathbf{q}'\nu'}^\dagger c_{\mathbf{q}\nu} \right\rangle + e^{+i\omega_{\mathbf{q}\nu}t} \left\langle c_{-\mathbf{q}'\nu'} c_{-\mathbf{q}\nu}^\dagger \right\rangle \right]. \quad (31.48)$$

whereas we used in the last equation that we don't have any interaction-terms, the anomalous expectation values vanish, i.e.

$$\left\langle c_{\mathbf{q}'\nu'}^\dagger c_{-\mathbf{q}\nu}^\dagger \right\rangle = \left\langle c_{-\mathbf{q}'\nu'} c_{\mathbf{q}\nu} \right\rangle = 0. \quad (31.49)$$

For the other expectation-value we obtain the regular case of non-interacting bosons with Remark 12.9.1,

$$\left\langle c_{\mathbf{q}'\nu'}^\dagger c_{\mathbf{q}\nu} \right\rangle = \delta_{\mathbf{q},\mathbf{q}'} \delta_{\nu,\nu'} B_\beta(\Omega_{\mathbf{q}\nu}), \quad \text{and} \quad \left\langle c_{-\mathbf{q}'\nu'} c_{-\mathbf{q}\nu}^\dagger \right\rangle = \delta_{\mathbf{q},\mathbf{q}'} \delta_{\nu,\nu'} [B_\beta(\Omega_{\mathbf{q}\nu}) + 1], \quad (31.50)$$

whereas we used the CCR in Equation (10.4) and with the Bose-function from Notation 9.46. Therefore we obtain

$$D_{\mathbf{q},\mathbf{q}'}^<_{\nu,\nu'}(t) = -\frac{i}{\hbar} \delta_{\nu,\nu'} \delta_{\mathbf{q},\mathbf{q}'} \left[B_\beta(\Omega_{\mathbf{q}\nu}) e^{-i\omega_{\mathbf{q}\nu}t} + (B_\beta(\Omega_{\mathbf{q}\nu}) + 1) e^{+i\omega_{\mathbf{q}\nu}t} \right]. \quad (31.51)$$

Thus the Green function is diagonal in momentum and branch indices,

$$D_{\mathbf{q},\mathbf{q}'}^<_{\nu,\nu'}(t) = \delta_{\nu,\nu'} \delta_{\mathbf{q},\mathbf{q}'} D_{\mathbf{q}\nu}^<(t), \quad \text{with} \quad D_{\mathbf{q}\nu}^<(t) = -\frac{i}{\hbar} [(B_\beta(\Omega_{\mathbf{q}\nu}) + 1) e^{+i\omega_{\mathbf{q}\nu}t} + B_\beta(\Omega_{\mathbf{q}\nu}) e^{-i\omega_{\mathbf{q}\nu}t}]. \quad (31.52)$$

The Greater Temporal Green-function

Starting from Definition 12.19

$$D_{\mathbf{q},\mathbf{q}'}^>_{\nu,\nu'}(t) = -\frac{i}{\hbar} \left\langle A_\nu(t, \mathbf{q}) A_{\nu'}^\dagger(0, \mathbf{q}') \right\rangle. \quad (31.53)$$

Using now Definition 12.18 for the displacement-operators and the Heisenberg-evolution according to Lemma 8.26 as

$$A_\nu(t, \mathbf{q}) = e^{-i\omega_{\mathbf{q}\nu}t} c_{\mathbf{q}\nu} + e^{+i\omega_{\mathbf{q}\nu}t} c_{-\mathbf{q}\nu}^\dagger, \quad A_{\nu'}^\dagger(\mathbf{q}', 0) = c_{\mathbf{q}'\nu'}^\dagger + c_{-\mathbf{q}'\nu'}, \quad (31.54)$$

we find

$$D_{\mathbf{q},\mathbf{q}'}^>_{\nu,\nu'}(t) = -\frac{i}{\hbar} \left\langle \left(e^{-i\omega_{\mathbf{q}\nu}t} c_{\mathbf{q}\nu} + e^{+i\omega_{\mathbf{q}\nu}t} c_{-\mathbf{q}\nu}^\dagger \right) \left(c_{\mathbf{q}'\nu'}^\dagger + c_{-\mathbf{q}'\nu'} \right) \right\rangle. \quad (31.55)$$

The anomalous expectation values vanish again, i.e.

$$\left\langle c_{\mathbf{q}\nu} c_{-\mathbf{q}'\nu'} \right\rangle = \left\langle c_{-\mathbf{q}\nu}^\dagger c_{\mathbf{q}'\nu'}^\dagger \right\rangle = 0. \quad (31.56)$$

Therefore we obtain

$$D_{\mathbf{q},\mathbf{q}'}^>_{\nu,\nu'}(t) = -\frac{i}{\hbar} \left[e^{-i\omega_{\mathbf{q}\nu}t} \left\langle c_{\mathbf{q}\nu} c_{\mathbf{q}'\nu'}^\dagger \right\rangle + e^{+i\omega_{\mathbf{q}\nu}t} \left\langle c_{-\mathbf{q}\nu}^\dagger c_{-\mathbf{q}'\nu'} \right\rangle \right]. \quad (31.57)$$

For thermal bosons we obtain again with Remark 12.9.1

$$\left\langle c_{\mathbf{q}\nu} c_{\mathbf{q}'\nu'}^\dagger \right\rangle = \delta_{\mathbf{q},\mathbf{q}'} \delta_{\nu,\nu'} \left[B_\beta(\Omega_{\mathbf{q}\nu}) + 1 \right], \quad \left\langle c_{-\mathbf{q}\nu}^\dagger c_{-\mathbf{q}'\nu'} \right\rangle = \delta_{\nu,\nu'} \delta_{\mathbf{q},\mathbf{q}'} B_\beta(\Omega_{\mathbf{q}\nu}), \quad (31.58)$$

with the Bose-function from Notation 9.46. Hence,

$$D_{\mathbf{q},\mathbf{q}'}^>_{\nu,\nu'}(t) = -\frac{i}{\hbar} \delta_{\nu,\nu'} \delta_{\mathbf{q},\mathbf{q}'} \left[\left(B_\beta(\omega_{\mathbf{q}\nu}) + 1 \right) e^{-i\omega_{\mathbf{q}\nu}t} + B_\beta(\omega_{\mathbf{q}\nu}) e^{+i\omega_{\mathbf{q}\nu}t} \right]. \quad (31.59)$$

Thus,

$$D_{\mathbf{q},\mathbf{q}'}^>_{\nu,\nu'}(t) = \delta_{\nu,\nu'} \delta_{\mathbf{q},\mathbf{q}'} D_{\mathbf{q}\nu}^>(t), \quad \text{with } D_{\mathbf{q}\nu}^>(t) = -\frac{i}{\hbar} \left[\left(B_\beta(\Omega_{\mathbf{q}\nu}) + 1 \right) e^{-i\omega_{\mathbf{q}\nu}t} + B_\beta(\Omega_{\mathbf{q}\nu}) e^{+i\omega_{\mathbf{q}\nu}t} \right]. \quad (31.60)$$

Time- and Anti-Time-Ordered Temporal Green-Functions

For these, we simply use Lemma 12.4 and obtain the desired identities.

Retarded Temporal Green-function

Starting from Definition 12.19

$$D_{\mathbf{q},\mathbf{q}'}^{\text{ret}}_{\nu,\nu'}(t, t') = -\frac{i}{\hbar} \theta(t - t') \left\langle \left[A_\nu(t, \mathbf{q}), A_{\nu'}^\dagger(t', \mathbf{q}') \right] \right\rangle. \quad (31.61)$$

Using now Definition 12.18 for the displacement-operators and the Heisenberg-evolution according to Lemma 8.26 as

$$A_\nu(t, \mathbf{q}) = e^{-i\omega_{\mathbf{q}\nu}t} c_{\mathbf{q}\nu} + e^{+i\omega_{\mathbf{q}\nu}t} c_{-\mathbf{q}\nu}^\dagger, \quad A_{\nu'}^\dagger(t', \mathbf{q}') = e^{+i\omega_{\mathbf{q}'\nu'}t'} c_{\mathbf{q}'\nu'}^\dagger + e^{-i\omega_{\mathbf{q}'\nu'}t'} c_{-\mathbf{q}'\nu'}, \quad (31.62)$$

the commutator becomes

$$\left[A_\nu(t, \mathbf{q}), A_{\nu'}^\dagger(t', \mathbf{q}') \right] = e^{-i\omega_{\mathbf{q}\nu}t} e^{+i\omega_{\mathbf{q}'\nu'}t'} \left[c_{\mathbf{q}\nu}, c_{\mathbf{q}'\nu'}^\dagger \right] + e^{+i\omega_{\mathbf{q}\nu}t} e^{-i\omega_{\mathbf{q}'\nu'}t'} \left[c_{-\mathbf{q}\nu}^\dagger, c_{-\mathbf{q}'\nu'} \right]. \quad (31.63)$$

The mixed commutators vanish due to the CCR in Equation (10.4):

$$\left[c_{\mathbf{q}\nu}, c_{-\mathbf{q}'\nu'} \right] = \left[c_{-\mathbf{q}\nu}^\dagger, c_{\mathbf{q}'\nu'}^\dagger \right] = 0. \quad (31.64)$$

and for the other commutators we obtain

$$\left[c_{\mathbf{q}\nu}, c_{\mathbf{q}'\nu'}^\dagger \right] = \delta_{\mathbf{q},\mathbf{q}'} \delta_{\nu,\nu'}, \quad \left[c_{-\mathbf{q}\nu}^\dagger, c_{-\mathbf{q}'\nu'} \right] = -\delta_{\mathbf{q},\mathbf{q}'} \delta_{\nu,\nu'}. \quad (31.65)$$

Therefore,

$$[A_v(t, \mathbf{q}), A_{v'}^\dagger(t', \mathbf{q}')] = \delta_{v,v'} \delta_{\mathbf{q},\mathbf{q}'} \left[e^{-i\omega_{\mathbf{q}v}(t-t')} - e^{+i\omega_{\mathbf{q}v}(t-t')} \right]. \quad (31.66)$$

Using the identity

$$e^{-ix} - e^{+ix} = -2i \sin(x), \quad (31.67)$$

we obtain

$$[A_v(t, \mathbf{q}), A_{v'}^\dagger(t', \mathbf{q}')] = -2i \delta_{v,v'} \delta_{\mathbf{q},\mathbf{q}'} \sin(\omega_{\mathbf{q}v}(t-t')). \quad (31.68)$$

Substitution into the retarded Green-function in Equation (31.61) gives

$$D_{v,v',\mathbf{q},\mathbf{q}'}^{\text{ret}}(t, t') = -\frac{i}{\hbar} \theta(t-t') [-2i \delta_{\mathbf{q},\mathbf{q}'} \delta_{v,v'} \sin(\omega_{\mathbf{q}v}(t-t'))] = -\frac{2}{\hbar} \delta_{\mathbf{q},\mathbf{q}'} \delta_{v,v'} \theta(t-t') \sin[\omega_{\mathbf{q}v}(t-t')]. \quad (31.69)$$

Hence the retarded Green function is diagonal in momentum and branch indices:

$$D_{v,v',\mathbf{q},\mathbf{q}'}^{\text{ret}}(t, t') = \delta_{v,v'} \delta_{\mathbf{q},\mathbf{q}'} D_{\mathbf{q}v}^{\text{ret}}(t-t'), \quad \text{with } D_{\mathbf{q}v}^{\text{ret}}(t-t') = -\frac{2}{\hbar} \theta(t-t') \sin[\omega_{\mathbf{q}v}(t-t')]. \quad (31.70)$$

Temporal advanced Green-function Derivation

Starting from Definition 12.19

$$D_{v,v',\mathbf{q},\mathbf{q}'}^{\text{adv}}(t, t') = \frac{i}{\hbar} \theta(t'-t) \langle [A_v(t, \mathbf{q}), A_{v'}^\dagger(t', \mathbf{q}')] \rangle. \quad (31.71)$$

Notice that the only difference relative to the retarded Green-function is the prefactor. This gives thus for the advanced Green-function

$$D_{v,v',\mathbf{q},\mathbf{q}'}^{\text{adv}}(t, t') = \delta_{v,v'} \delta_{\mathbf{q},\mathbf{q}'} D_{\mathbf{q}v}^{\text{adv}}(t-t'), \quad \text{with } D_{\mathbf{q}v}^{\text{adv}}(t-t') = \frac{2}{\hbar} \theta(t'-t) \sin[\omega_{\mathbf{q}v}(t-t')]. \quad (31.72)$$

The Frequency-Version of the Lesser Green-Function

We start from the already derived temporal expression in Equation (31.52)

$$D_{\mathbf{q}v}^<(t) = -\frac{i}{\hbar} [(B_\beta(\Omega_{\mathbf{q}v}) + 1) e^{+i\omega_{\mathbf{q}v}t} + B_\beta(\Omega_{\mathbf{q}v}) e^{-i\omega_{\mathbf{q}v}t}]. \quad (31.73)$$

Applying the Fourier transform from Notation 1.17,

$$\begin{aligned}
 D_{\mathbf{q}_v}^<(\omega) &= \int_{-\infty}^{+\infty} dt e^{+i\omega t} D_{\mathbf{q}_v}^<(t) \\
 &= -\frac{i}{\hbar} \int_{-\infty}^{+\infty} dt e^{+i\omega t} \left[(B_{\beta}(\Omega_{\mathbf{q}_v}) + 1) e^{+i\omega_{\mathbf{q}_v} t} + B_{\beta}(\Omega_{\mathbf{q}_v}) e^{-i\omega_{\mathbf{q}_v} t} \right] \\
 &= -\frac{i}{\hbar} \left[(B_{\beta}(\Omega_{\mathbf{q}_v}) + 1) \int_{-\infty}^{+\infty} dt e^{+i(\omega + \omega_{\mathbf{q}_v}) t} + B_{\beta}(\Omega_{\mathbf{q}_v}) \int_{-\infty}^{+\infty} dt e^{+i(\omega - \omega_{\mathbf{q}_v}) t} \right].
 \end{aligned} \tag{31.74}$$

Using now Theorem 1.22, i.e.

$$\int_{-\infty}^{+\infty} dt e^{ixt} = 2\pi\delta(x), \tag{31.75}$$

we obtain

$$D_{\mathbf{q}_v}^<(\omega) = \dots = -\frac{2\pi i}{\hbar} \left[(B_{\beta}(\Omega_{\mathbf{q}_v}) + 1) \delta(\omega + \omega_{\mathbf{q}_v}) + B_{\beta}(\Omega_{\mathbf{q}_v}) \delta(\omega - \omega_{\mathbf{q}_v}) \right]. \tag{31.76}$$

Therefore we obtain

$$D_{\mathbf{q}_v}^<(\omega) = -\frac{2\pi i}{\hbar} \left[(B_{\beta}(\Omega_{\mathbf{q}_v}) + 1) \delta(\omega + \omega_{\mathbf{q}_v}) + B_{\beta}(\Omega_{\mathbf{q}_v}) \delta(\omega - \omega_{\mathbf{q}_v}) \right]. \tag{31.77}$$

The Frequency-Version of the Greater Green-Function

We start from the already derived temporal expression in Equation (31.60)

$$D_{\mathbf{q}_v}^>(t) = -\frac{i}{\hbar} \left[(B_{\beta}(\Omega_{\mathbf{q}_v}) + 1) e^{-i\omega_{\mathbf{q}_v} t} + B_{\beta}(\Omega_{\mathbf{q}_v}) e^{+i\omega_{\mathbf{q}_v} t} \right]. \tag{31.78}$$

Applying the Fourier transform from Notation 1.17,

$$D_{\mathbf{q}_v}^>(\omega) = \int_{-\infty}^{+\infty} dt e^{+i\omega t} D_{\mathbf{q}_v}^>(t) \tag{31.79}$$

$$= -\frac{i}{\hbar} \int_{-\infty}^{+\infty} dt e^{+i\omega t} \left[(B_{\beta}(\Omega_{\mathbf{q}_v}) + 1) e^{-i\omega_{\mathbf{q}_v} t} + B_{\beta}(\Omega_{\mathbf{q}_v}) e^{+i\omega_{\mathbf{q}_v} t} \right] \tag{31.80}$$

$$= -\frac{i}{\hbar} \left[(B_{\beta}(\Omega_{\mathbf{q}_v}) + 1) \int_{-\infty}^{+\infty} dt e^{+i(\omega - \omega_{\mathbf{q}_v}) t} + B_{\beta}(\Omega_{\mathbf{q}_v}) \int_{-\infty}^{+\infty} dt e^{+i(\omega + \omega_{\mathbf{q}_v}) t} \right] \tag{31.81}$$

$$= \dots \tag{31.82}$$

Using now Theorem 1.22

$$\int_{-\infty}^{+\infty} dt e^{ixt} = 2\pi\delta(x), \tag{31.83}$$

we obtain

$$D_{\mathbf{q}_v}^>(\omega) = \dots = -\frac{2\pi i}{\hbar} \left[(B_{\hbar\beta}(\Omega_{\mathbf{q}_v}) + 1) \delta(\omega - \omega_{\mathbf{q}_v}) + B_{\hbar\beta}(\Omega_{\mathbf{q}_v}) \delta(\omega + \omega_{\mathbf{q}_v}) \right]. \quad (31.84)$$

Therefore we obtain

$$D_{\mathbf{q}_v}^>(\omega) = -\frac{2\pi i}{\hbar} \left[(B_{\hbar\beta}(\Omega_{\mathbf{q}_v}) + 1) \delta(\omega - \omega_{\mathbf{q}_v}) + B_{\hbar\beta}(\Omega_{\mathbf{q}_v}) \delta(\omega + \omega_{\mathbf{q}_v}) \right]. \quad (31.85)$$

Frequency representation of the time-ordered Green-function

We take the Fourier-transform convention

$$D_{\mathbf{q}_v}^{\mathbb{T}}(\omega) = \lim_{\eta \rightarrow 0^+} \int_{-\infty}^{+\infty} dt e^{+i\omega t - \eta|t|} D_{\mathbf{q}_v}^{\mathbb{T}}(t). \quad (31.86)$$

The temporal Green-function is according to Equation (12.27)

$$D_{\mathbf{q}_v}^{\mathbb{T}}(t) = -\frac{i}{\hbar} \left[\theta(t) (B_{\hbar\beta}(\Omega_{\mathbf{q}_v}) + 1) e^{-i\omega_{\mathbf{q}_v} t} + B_{\hbar\beta}(\Omega_{\mathbf{q}_v}) e^{+i\omega_{\mathbf{q}_v} t} + \theta(-t) (B_{\hbar\beta}(\Omega_{\mathbf{q}_v}) + 1) e^{+i\omega_{\mathbf{q}_v} t} + B_{\hbar\beta}(\Omega_{\mathbf{q}_v}) e^{-i\omega_{\mathbf{q}_v} t} \right]. \quad (31.87)$$

Consequently,

$$\begin{aligned} D_{\mathbf{q}_v}^{\mathbb{T}}(\omega) = -\frac{i}{\hbar} \lim_{\eta \rightarrow 0^+} \left\{ (B_{\hbar\beta}(\Omega_{\mathbf{q}_v}) + 1) \int_0^{+\infty} dt e^{i(\omega - \omega_{\mathbf{q}_v})t - \eta t} \right. \\ + B_{\hbar\beta}(\Omega_{\mathbf{q}_v}) \int_0^{+\infty} dt e^{i(\omega + \omega_{\mathbf{q}_v})t - \eta t} \\ + (B_{\hbar\beta}(\Omega_{\mathbf{q}_v}) + 1) \int_{-\infty}^0 dt e^{i(\omega + \omega_{\mathbf{q}_v})t + \eta t} \\ \left. + B_{\hbar\beta}(\Omega_{\mathbf{q}_v}) \int_{-\infty}^0 dt e^{i(\omega - \omega_{\mathbf{q}_v})t + \eta t} \right\}. \end{aligned} \quad (31.88)$$

The regulated integrals are evaluated analogous to Lemma 1.19 and 1.20

$$\int_0^{+\infty} dt e^{ixt - \eta t} = \frac{1}{ix - \eta} [e^{ixt - \eta t}]_0^{\infty} = \frac{-1}{ix - \eta} \frac{-i}{-i} = \frac{i}{x + i\eta}, \quad (31.89)$$

$$(31.90)$$

$$\int_{-\infty}^0 dt e^{ixt + \eta t} = \frac{1}{ix + \eta} [e^{ixt + \eta t}]_0^{\infty} = \frac{-1}{ix + \eta} \frac{-i}{-i} = -\frac{i}{x - i\eta}. \quad (31.91)$$

Therefore,

$$D_{\mathbf{q}\nu}^{\mathbb{T}}(\omega) = \lim_{\eta \rightarrow 0^+} \frac{1}{\hbar} \left[\frac{B_{\hbar\beta}(\Omega_{\mathbf{q}\nu}) + 1}{\omega - \omega_{\mathbf{q}\nu} + i\eta} + \frac{B_{\hbar\beta}(\Omega_{\mathbf{q}\nu})}{\omega + \omega_{\mathbf{q}\nu} + i\eta} - \frac{B_{\hbar\beta}(\Omega_{\mathbf{q}\nu}) + 1}{\omega + \omega_{\mathbf{q}\nu} - i\eta} - \frac{B_{\hbar\beta}(\Omega_{\mathbf{q}\nu})}{\omega - \omega_{\mathbf{q}\nu} - i\eta} \right]. \quad (31.92)$$

Hence,

$$\begin{aligned} D_{\mathbf{q}\nu}^{\mathbb{T}}(\omega) &= \frac{1}{\hbar} \lim_{\eta \rightarrow 0^+} \left[\frac{B_{\hbar\beta}(\Omega_{\mathbf{q}\nu}) + 1}{\omega - \omega_{\mathbf{q}\nu} + i\eta} + \frac{B_{\hbar\beta}(\Omega_{\mathbf{q}\nu})}{\omega + \omega_{\mathbf{q}\nu} + i\eta} - \frac{B_{\hbar\beta}(\Omega_{\mathbf{q}\nu}) + 1}{\omega + \omega_{\mathbf{q}\nu} - i\eta} - \frac{B_{\hbar\beta}(\Omega_{\mathbf{q}\nu})}{\omega - \omega_{\mathbf{q}\nu} - i\eta} \right] \\ &= \frac{1}{\hbar} \lim_{\eta \rightarrow 0^+} \sum_{\sigma \in \{+, -\}} \left[\sigma \frac{B_{\hbar\beta}(\Omega_{\mathbf{q}\nu}) + 1}{\omega - \sigma\omega_{\mathbf{q}\nu} + \sigma i\eta} + \sigma \frac{B_{\hbar\beta}(\Omega_{\mathbf{q}\nu})}{\omega + \sigma\omega_{\mathbf{q}\nu} + \sigma i\eta} \right]. \end{aligned} \quad (31.93)$$

Frequency representation of the anti-time-ordered Green-function

We use the same Fourier-transform convention,

$$D_{\mathbf{q}\nu}^{\mathbb{T}}(\omega) = \lim_{\eta \rightarrow 0^+} \int_{-\infty}^{+\infty} dt e^{+i\omega t - \eta|t|} D_{\mathbf{q}\nu}^{\mathbb{T}}(t). \quad (31.94)$$

The temporal anti-time-ordered Green-function is according to Equation (12.19)

$$\begin{aligned} D_{\mathbf{q}\nu}^{\mathbb{T}}(t) &= -\frac{i}{\hbar} \left[\theta(-t) \left((B_{\hbar\beta}(\Omega_{\mathbf{q}\nu}) + 1) e^{-i\omega_{\mathbf{q}\nu} t} + B_{\hbar\beta}(\Omega_{\mathbf{q}\nu}) e^{+i\omega_{\mathbf{q}\nu} t} \right) \right. \\ &\quad \left. + \theta(t) \left((B_{\hbar\beta}(\Omega_{\mathbf{q}\nu}) + 1) e^{+i\omega_{\mathbf{q}\nu} t} + B_{\hbar\beta}(\Omega_{\mathbf{q}\nu}) e^{-i\omega_{\mathbf{q}\nu} t} \right) \right]. \end{aligned} \quad (31.95)$$

Thus,

$$\begin{aligned} D_{\mathbf{q}\nu}^{\mathbb{T}}(\omega) &= -\frac{i}{\hbar} \lim_{\eta \rightarrow 0^+} \left\{ (B_{\hbar\beta}(\Omega_{\mathbf{q}\nu}) + 1) \int_{-\infty}^0 dt e^{i(\omega - \omega_{\mathbf{q}\nu})t + \eta t} \right. \\ &\quad + B_{\hbar\beta}(\Omega_{\mathbf{q}\nu}) \int_{-\infty}^0 dt e^{i(\omega + \omega_{\mathbf{q}\nu})t + \eta t} \\ &\quad + (B_{\hbar\beta}(\Omega_{\mathbf{q}\nu}) + 1) \int_0^{+\infty} dt e^{i(\omega + \omega_{\mathbf{q}\nu})t - \eta t} \\ &\quad \left. + B_{\hbar\beta}(\Omega_{\mathbf{q}\nu}) \int_0^{+\infty} dt e^{i(\omega - \omega_{\mathbf{q}\nu})t - \eta t} \right\}. \end{aligned} \quad (31.96)$$

Using the regulated integrals above in Equations (31.89) and (31.91), we find

$$\begin{aligned} D_{\mathbf{q}\nu}^{\mathbb{T}}(\omega) &= \lim_{\eta \rightarrow 0^+} \frac{1}{\hbar} \left[\frac{B_{\hbar\beta}(\Omega_{\mathbf{q}\nu})}{\omega - \omega_{\mathbf{q}\nu} + i\eta} + \frac{B_{\hbar\beta}(\Omega_{\mathbf{q}\nu}) + 1}{\omega + \omega_{\mathbf{q}\nu} + i\eta} \right. \\ &\quad \left. - \frac{B_{\hbar\beta}(\Omega_{\mathbf{q}\nu}) + 1}{\omega - \omega_{\mathbf{q}\nu} - i\eta} - \frac{B_{\hbar\beta}(\Omega_{\mathbf{q}\nu})}{\omega + \omega_{\mathbf{q}\nu} - i\eta} \right]. \end{aligned} \quad (31.97)$$

Therefore,

$$\begin{aligned} D_{\mathbf{q}_v}^{\bar{\tau}}(\omega) &= \frac{1}{\hbar} \lim_{\eta \rightarrow 0^+} \left[\frac{B_{\hbar\beta}(\Omega_{\mathbf{q}_v})}{\omega - \omega_{\mathbf{q}_v} + i\eta} + \frac{B_{\hbar\beta}(\Omega_{\mathbf{q}_v}) + 1}{\omega + \omega_{\mathbf{q}_v} + i\eta} - \frac{B_{\hbar\beta}(\Omega_{\mathbf{q}_v}) + 1}{\omega - \omega_{\mathbf{q}_v} - i\eta} - \frac{B_{\hbar\beta}(\Omega_{\mathbf{q}_v})}{\omega + \omega_{\mathbf{q}_v} - i\eta} \right] \\ &= \frac{1}{\hbar} \lim_{\eta \rightarrow 0^+} \sum_{\sigma \in \{+, -\}} \left[\sigma \frac{B_{\hbar\beta}(\Omega_{\mathbf{q}_v})}{\omega - \sigma\omega_{\mathbf{q}_v} + \sigma i\eta} + \sigma \frac{B_{\hbar\beta}(\Omega_{\mathbf{q}_v}) + 1}{\omega + \sigma\omega_{\mathbf{q}_v} + \sigma i\eta} \right]. \end{aligned} \quad (31.98)$$

Frequency representation of the retarded Green-function

We use the Fourier-transform convention from Notation 1.17

$$D_{\mathbf{q}_v}^{\text{ret}}(\omega) = \lim_{\eta \rightarrow 0^+} \int_{-\infty}^{+\infty} dt e^{+i\omega t - \eta|t|} D_{\mathbf{q}_v}^{\text{ret}}(t). \quad (31.99)$$

Starting from the temporal expression in Equation (31.70)

$$D_{\mathbf{q}_v}^{\text{ret}}(t) = -\frac{2}{\hbar} \theta(t) \sin(\omega_{\mathbf{q}_v} t), \quad (31.100)$$

we obtain

$$D_{\mathbf{q}_v}^{\text{ret}}(\omega) = -\frac{2}{\hbar} \lim_{\eta \rightarrow 0^+} \int_0^{+\infty} dt e^{+i\omega t - \eta t} \sin(\omega_{\mathbf{q}_v} t). \quad (31.101)$$

Notice that, omitting the limit, this is formally the same expression as

$$D_{\mathbf{q}_v}^{\text{ret}}(\omega + i\eta) = -\frac{2}{\hbar} \int_0^{+\infty} dt e^{+i\omega t - \eta t} \sin(\omega_{\mathbf{q}_v} t). \quad (31.102)$$

for $\eta > 0$. Using now the decomposition again

$$\sin(\omega_{\mathbf{q}_v} t) = \frac{1}{2i} [e^{+i\omega_{\mathbf{q}_v} t} - e^{-i\omega_{\mathbf{q}_v} t}], \quad (31.103)$$

we find

$$D_{\mathbf{q}_v}^{\text{ret}}(\omega) = -\frac{1}{i\hbar} \lim_{\eta \rightarrow 0^+} \left[\int_0^{+\infty} dt e^{+i(\omega + \omega_{\mathbf{q}_v})t - \eta t} - \int_0^{+\infty} dt e^{+i(\omega - \omega_{\mathbf{q}_v})t - \eta t} \right]. \quad (31.104)$$

The regulated integral is as in Equation (31.89)

$$\int_0^{+\infty} dt e^{+ixt - \eta t} = \frac{i}{x + i\eta}. \quad (31.105)$$

Therefore,

$$D_{\mathbf{q}\nu}^{\text{ret}}(\omega) = -\frac{1}{i\hbar} \lim_{\eta \rightarrow 0^+} \left[\frac{i}{\omega + \omega_{\mathbf{q}\nu} + i\eta} - \frac{i}{\omega - \omega_{\mathbf{q}\nu} + i\eta} \right] \quad (31.106)$$

$$= \frac{1}{\hbar} \lim_{\eta \rightarrow 0^+} \left[\frac{1}{\omega - \omega_{\mathbf{q}\nu} + i\eta} - \frac{1}{\omega + \omega_{\mathbf{q}\nu} + i\eta} \right]. \quad (31.107)$$

Hence,

$$D_{\mathbf{q}\nu}^{\text{ret}}(\omega) = \frac{1}{\hbar} \lim_{\eta \rightarrow 0^+} \left[\frac{1}{\omega - \omega_{\mathbf{q}\nu} + i\eta} - \frac{1}{\omega + \omega_{\mathbf{q}\nu} + i\eta} \right]. \quad (31.108)$$

Notice that this can be further simplified as

$$\begin{aligned} D_{\mathbf{q}\nu}^{\text{ret}}(\omega + i\epsilon) &= \frac{1}{\hbar} \frac{1}{\omega + i\epsilon - \omega_{\nu}(\mathbf{q})} - \frac{1}{\hbar} \frac{1}{\omega + i\epsilon + \omega_{\nu}(\mathbf{q})} \\ &= \frac{1}{\hbar} \left[\frac{\omega + i\epsilon + \omega_{\nu}(\mathbf{q}) - (\omega + i\epsilon - \omega_{\nu}(\mathbf{q}))}{(\omega + i\epsilon)^2 - \omega_{\nu}^2(\mathbf{q})} \right] \\ &= \frac{1}{\hbar} \frac{2\omega_{\nu}(\mathbf{q})}{(\omega + i\epsilon)^2 - \omega_{\nu}^2(\mathbf{q})} \end{aligned} \quad (31.109)$$

Frequency representation of the advanced Green-function

The Fourier transform of the advanced Green-function is according to Notation 1.17

$$D_{\mathbf{q}\nu}^{\text{adv}}(\omega) = \lim_{\eta \rightarrow 0^+} \int_{-\infty}^{+\infty} dt e^{+i\omega t - \eta|t|} D_{\mathbf{q}\nu}^{\text{adv}}(t). \quad (31.110)$$

Starting from Equation (31.72)

$$D_{\mathbf{q}\nu}^{\text{adv}}(t) = +\frac{2}{\hbar} \theta(-t) \sin(\omega_{\mathbf{q}\nu} t), \quad (31.111)$$

we obtain

$$D_{\mathbf{q}\nu}^{\text{adv}}(\omega) = \frac{2}{\hbar} \lim_{\eta \rightarrow 0^+} \int_{-\infty}^0 dt e^{+i\omega t + \eta t} \sin(\omega_{\mathbf{q}\nu} t). \quad (31.112)$$

Using the exponential representation of the sine function,

$$D_{\mathbf{q}\nu}^{\text{adv}}(\omega) = \frac{1}{i\hbar} \lim_{\eta \rightarrow 0^+} \left[\int_{-\infty}^0 dt e^{+i(\omega + \omega_{\mathbf{q}\nu})t + \eta t} - \int_{-\infty}^0 dt e^{+i(\omega - \omega_{\mathbf{q}\nu})t + \eta t} \right]. \quad (31.113)$$

The regulated integral over negative times is according to Equation (31.91)

$$\int_{-\infty}^0 dt e^{+ixt+\eta t} = -\frac{i}{x-i\eta}. \quad (31.114)$$

Consequently,

$$D_{\mathbf{q}_v}^{\text{adv}}(\omega) = \frac{1}{i\hbar} \lim_{\eta \rightarrow 0^+} \left[-\frac{i}{\omega + \omega_{\mathbf{q}_v} - i\eta} + \frac{i}{\omega - \omega_{\mathbf{q}_v} - i\eta} \right] \quad (31.115)$$

$$= \frac{1}{\hbar} \lim_{\eta \rightarrow 0^+} \left[\frac{1}{\omega - \omega_{\mathbf{q}_v} - i\eta} - \frac{1}{\omega + \omega_{\mathbf{q}_v} - i\eta} \right]. \quad (31.116)$$

Hence,

$$\begin{aligned} D_{\mathbf{q}_v}^{\text{adv}}(\omega) &= \frac{1}{\hbar} \lim_{\eta \rightarrow 0^+} \left[\frac{1}{\omega - \omega_{\mathbf{q}_v} - i\eta} - \frac{1}{\omega + \omega_{\mathbf{q}_v} - i\eta} \right] \\ &= \frac{1}{\hbar} \lim_{\eta \rightarrow 0^+} \frac{\omega + \omega_{\mathbf{q}_v} - i\eta - (\omega - \omega_{\mathbf{q}_v} - i\eta)}{(\omega - \omega_{\mathbf{q}_v})^2 - i\eta} \\ &= \frac{1}{\hbar} \lim_{\eta \rightarrow 0^+} \frac{2\omega_{\mathbf{q}_v}}{(\omega - \omega_{\mathbf{q}_v})^2 - i\eta} \end{aligned} \quad (31.117)$$

Proof 31.32 of Theorem 13.4 We compute

$$\begin{aligned}\mathcal{G}_{X,X'}(\tau, \tau') &= -\frac{1}{\hbar} \frac{\text{Tr}_{\mathcal{H}} e^{-\beta(H-\eta\mu N)} \left[\theta(\tau - \tau') X(\tau) X'(\tau') + S_{X,X'} \theta(\tau' - \tau) X'(\tau') X(\tau) \right]}{\text{Tr}_{\mathcal{H}} e^{-\beta(H-\eta\mu N)}} \\ &= -\frac{1}{\hbar} \frac{\text{Tr}_{\mathcal{H}} e^{-\beta(H-\eta\mu N)} \left[\theta(\tau - \tau') X(\tau - \tau') X'(0) + S_{X,X'} \theta(\tau' - \tau) X'(0) X(\tau - \tau') \right]}{\text{Tr}_{\mathcal{H}} e^{-\beta(H-\eta\mu N)}}\end{aligned}$$

Therefore the Green-function depends only on time-differences and we can write

$$\mathcal{G}_{X,X'}(\tau, \tau') \equiv \mathcal{G}_{X,X'}(\tau - \tau'). \quad (31.118)$$

With the periodic standard-continuation of the θ -function, we obtain, using the cyclicity property twice, for $-\hbar\beta < \tau < 0$ on the one hand

$$\begin{aligned}\mathcal{G}_{X,X'}(\tau) &= -\frac{S_{X,X'}}{\hbar} \frac{\text{Tr}_{\mathcal{H}} e^{-\beta(H-\eta\mu N)} X'(0) X(\tau)}{\text{Tr}_{\mathcal{H}} e^{-\beta(H-\eta\mu N)}} \\ &= -\frac{S_{X,X'}}{\hbar} \frac{\text{Tr}_{\mathcal{H}} e^{-\beta(H-\eta\mu N)} X' e^{+\tau(H-\eta\mu N)/\hbar} X e^{-\tau(H-\eta\mu N)/\hbar}}{\text{Tr}_{\mathcal{H}} e^{-\beta(H-\eta\mu N)}} \\ &= -\frac{S_{X,X'}}{\hbar} \frac{\text{Tr}_{\mathcal{H}} e^{+\tau(H-\eta\mu N)/\hbar} X e^{-(\beta+\tau/\hbar)(H-\eta\mu N)} X'}{\text{Tr}_{\mathcal{H}} e^{-\beta(H-\eta\mu N)}} \\ &= \dots\end{aligned}$$

now we multiply with one: $1 = e^{\pm\beta(H-\eta\mu N)}$ in order to get

$$\dots = -\frac{S_{X,X'}}{\hbar} \frac{\text{Tr}_{\mathcal{H}} e^{-\beta(H-\eta\mu N)} e^{+(\beta+\tau/\hbar)(H-\eta\mu N)} X e^{-(\beta+\tau/\hbar)(H-\eta\mu N)} X'}{\text{Tr}_{\mathcal{H}} e^{-\beta(H-\eta\mu N)}}$$

and we ultimately compute on the other hand (again for $-\hbar\beta < \tau < 0$)

$$\begin{aligned}\mathcal{G}_{X,X'}(\tau + \hbar\beta) &= -\frac{1}{\hbar} \frac{\text{Tr}_{\mathcal{H}} e^{-\beta(H-\eta\mu N)} X(\tau + \hbar\beta) X'(0)}{\text{Tr}_{\mathcal{H}} e^{-\beta(H-\eta\mu N)}} \\ &= -\frac{1}{\hbar} \frac{\text{Tr}_{\mathcal{H}} e^{-\beta(H-\eta\mu N)} e^{+(\tau/\hbar+\beta)(H-\eta\mu N)} X e^{+(\tau/\hbar+\beta)(H-\eta\mu N)} X'}{\text{Tr}_{\mathcal{H}} e^{-\beta(H-\eta\mu N)}}\end{aligned}$$

and therefore

$$\mathcal{G}_{X,X'}(\tau + \hbar\beta) = S_{X,X'} \mathcal{G}_{X,X'}(\tau). \quad (31.119)$$

and hence there is a periodic standard-continuation, which is unique, and therefore the originally defined function.

Proof 31.33 of Lemma 13.5

We consider the definition

$$(-\hbar\beta, +\hbar\beta) \ni \tau \mapsto \mathcal{G}_{\alpha, \alpha'}(\tau) = -\frac{1}{\hbar} \frac{\text{Tr}_{\mathcal{F}_\zeta} e^{-\beta(H-\mu N)} \left[\theta(\tau) c_\alpha(\tau) c_{\alpha'}^\dagger(0) + \zeta \theta(-\tau) c_{\alpha'}^\dagger(0) c_\alpha(\tau) \right]}{\text{Tr}_{\mathcal{F}_\zeta} e^{-\beta(H-\mu N)}} \in \mathbb{C}$$

and want to calculate its derivative. We therefore calculate first

$$\begin{aligned} \partial_\tau \text{T}_\tau c_\alpha(\tau) c_{\alpha'}^\dagger(0) &\propto \partial_\tau \left[\theta(\tau) c_\alpha(\tau) c_{\alpha'}^\dagger(0) + \zeta \theta(-\tau) c_{\alpha'}^\dagger(0) c_\alpha(\tau) \right] \\ &= \delta(\tau) c_\alpha(\tau) c_{\alpha'}^\dagger(0) - \zeta \delta(-\tau) c_{\alpha'}^\dagger(0) c_\alpha(\tau) - \frac{1}{\hbar} \theta(\tau) [c_\alpha, H - \mu]_{\text{H}}(\tau) c_{\alpha'}^\dagger(0) - \frac{\zeta}{\hbar} \theta(-\tau) c_{\alpha'}^\dagger(0) [c_\alpha, H - \mu]_{\text{H}}(\tau) \\ &= \delta(\tau) [c_\alpha(\tau), c_{\alpha'}^\dagger(0)]_\zeta - \frac{1}{\hbar} \text{T}_\tau ([c_\alpha, H - \mu]_{\text{H}}(\tau) c_{\alpha'}^\dagger(0)) \end{aligned}$$

whereas we used the statistical commutator from Definition TBD and the imaginary-time Heisenberg-equation-of-motions from Lemma TBD and the fact. This gives the equation of motion

$$\partial_\tau \mathcal{G}_{\alpha\alpha'}(\tau) = -\frac{1}{\hbar} \delta(\tau) \delta_{\alpha, \alpha'} + \frac{1}{\hbar} \left\langle \text{T}_\tau \left[c_\alpha, \frac{H - \mu N}{\hbar} \right]_{\text{H}}(\tau) c_{\alpha'}^\dagger(0) \right\rangle \quad (31.120)$$

Proof 31.34 of Theorem 13.6

For Fermions: See Waste-Folder 1, Topic 4, page 8 and Mahan p. 115

For Bosons: See Waste-Folder 1, topic 9.

For both: See Flensberg, page 164f. , Negele and Orland, page 73.

Fermions

For the imaginary-time Green-function one obtains

$$\mathcal{G}(\tau) = -\frac{1}{\hbar} \theta(\tau) e^{-\xi\tau/\hbar} (1 - F_\beta(\xi)) + \frac{1}{\hbar} \theta(-\tau) e^{-\xi\tau/\hbar} F_\beta(\xi) \quad (31.121)$$

and for the coefficients of the Fourier-series one has

$$\mathcal{G}(i\omega_n) = \int_0^{\hbar\beta} d\tau e^{i\omega_n\tau} \mathcal{G}(\tau) = -\frac{1}{\hbar} (1 - F_\beta(\xi)) \int_0^{\hbar\beta} d\tau e^{\tau(i\omega_n - \xi/\hbar)} = \dots$$

Now we compute

$$\int_0^{\hbar\beta} d\tau e^{\tau(i\omega_n - \xi/\hbar)} = (-e^{\beta\xi} - 1) \frac{1}{i\omega_n - \xi/\hbar} = -(1 + e^{\beta\xi}) \frac{1}{i\omega_n - \xi/\hbar}$$

which yields

$$\dots = \frac{1}{\hbar} \frac{(1 - F_\beta(\xi))(1 + e^{\beta\xi})}{i\omega_n - \xi/\hbar} = \frac{1}{\hbar} \frac{1}{i\omega_n - \xi/\hbar}$$

whereas the last equation-sign is proven in the literature⁸.

Bosons

We compute (at the example of a non-interacting phonon-gas)

$$\mathcal{G}_{\nu,\nu'}(\tau, 0) = -\theta(\tau) \langle c_{\mathbf{q}\nu}(\tau) c_{\mathbf{q}'\nu'}^\dagger(0) \rangle - \theta(-\tau) \langle c_{\mathbf{q}'\nu'}^\dagger(0) c_{\mathbf{q}\nu}(\tau) \rangle = \dots$$

Now we use

$$c_{\mathbf{q}\nu}(\tau) = e^{\tau H/\hbar} c_{\mathbf{q}\nu} e^{-\tau H/\hbar} = e^{-\tau\omega_{\mathbf{q}\nu}} c_{\mathbf{q}\nu} \quad (31.122)$$

and

$$c_{\mathbf{q}\nu}^\dagger(\tau) = e^{\tau H/\hbar} c_{\mathbf{q}\nu}^\dagger e^{-\tau H/\hbar} = e^{+\tau\omega_{\mathbf{q}\nu}} c_{\mathbf{q}\nu}^\dagger \quad (31.123)$$

and therefore we obtain

$$-\langle c_{\mathbf{q}\nu}(\tau) c_{\mathbf{q}'\nu'}^\dagger(0) \rangle = -e^{-\tau\omega_{\mathbf{q}\nu}} \langle c_{\mathbf{q}\nu} c_{\mathbf{q}'\nu'}^\dagger \rangle = -\delta_{\mathbf{q},\mathbf{q}'} \delta_{\nu,\nu'} e^{-\tau\omega_{\mathbf{q}\nu}} (B_\beta(\hbar\omega_{\mathbf{q}\nu}) + 1) \quad (31.124)$$

and

$$-\langle c_{\mathbf{q}'\nu'}^\dagger(0) c_{\mathbf{q}\nu}(\tau) \rangle = -e^{-\tau\omega_{\mathbf{q}\nu}} \langle c_{\mathbf{q}'\nu'}^\dagger c_{\mathbf{q}\nu} \rangle = -\delta_{\mathbf{q},\mathbf{q}'} \delta_{\nu,\nu'} e^{-\tau\omega_{\mathbf{q}\nu}} B_\beta(\hbar\omega_{\mathbf{q}\nu}) \quad (31.125)$$

and therefore we obtain

$$\begin{aligned} \mathcal{G}_{\nu,\nu'}(\tau, 0) &= \dots \\ &= -\theta(\tau) \delta_{\mathbf{q},\mathbf{q}'} \delta_{\nu,\nu'} e^{-\tau\omega_{\mathbf{q}\nu}} (B_\beta(\hbar\omega_{\mathbf{q}\nu}) + 1) - \theta(-\tau) \delta_{\mathbf{q},\mathbf{q}'} \delta_{\nu,\nu'} e^{-\tau\omega_{\mathbf{q}\nu}} B_\beta(\hbar\omega_{\mathbf{q}\nu}) \end{aligned}$$

Proof 31.35 of Theorem 13.7 We will prove the theorem in two steps. First we will consider the equation of motion of the function to conclude its form. Then we will verify that the function has the correct periodicity.

⁸Cf. [30] page 115. One can also simply write the Equation out and see it immediately.

Fundamental theorems on the uniqueness of differential equations yields then that we have obtained the correct solution. We now evaluate

$$\begin{aligned}
 [c_\alpha, H - \mu] &= \sum_{\beta, \beta'} H_{\beta, \beta'} [c_\alpha, c_\beta^\dagger c_{\beta'}] + \mu \sum_{\beta} [c_\alpha, c_\beta^\dagger c_\beta] \\
 &= \sum_{\beta, \beta'} H_{\beta, \beta'} [c_\alpha, c_\beta^\dagger]_\zeta c_{\beta'} + \mu \sum_{\beta} [c_\alpha, c_\beta^\dagger]_\zeta c_\beta \\
 &= \sum_{\beta'} H_{\alpha, \beta'} c_{\beta'} + \mu c_\alpha
 \end{aligned}$$

Therefore we obtain

$$\dots = \delta(\tau) \delta_{\alpha, \alpha'} - \sum_{\beta} \frac{H_{\alpha, \beta} - \delta_{\alpha, \beta} \mu}{\hbar} T_\tau (c_\beta(\tau) c_{\alpha'}^\dagger(0))$$

which gives combined in total

$$\begin{aligned}
 \partial_\tau \mathcal{G}_{\alpha, \alpha'}(\tau) &= -\delta(\tau)/\hbar \delta_{\alpha, \alpha'} - \sum_{\beta} \frac{H_{\alpha, \beta} - \delta_{\alpha, \beta} \mu}{\hbar} \mathcal{G}_{\beta, \alpha'}(\tau) \iff \partial_\tau \mathcal{G}(\tau) = -\delta(\tau)/\hbar - \frac{H - \mu}{\hbar} \mathcal{G}(\tau) \\
 &\iff \left(\partial_\tau + \frac{H - \mu}{\hbar} \right) \mathcal{G}(\tau) = -\delta(\tau)/\hbar \\
 &\iff \dots
 \end{aligned}$$

whereas the r.h.sides are to be understood as matrix-identities. Solving this differential equation w.r.t. the boundary-conditions $\mathcal{G}(\hbar\beta) = \zeta \mathcal{G}(0)$ yields the desired result. We will now use the Dirac-identity for (anti-)periodic functions as⁹

$$\delta(\tau) = \frac{1}{\hbar\beta} \sum_{\omega_n \in \Omega_\zeta} e^{-i\omega_n \tau} \quad (31.126)$$

and use the expansion from Lemma TBD to obtain

$$\left(\partial_\tau + \frac{H - \mu}{\hbar} \right) \mathcal{G}(\tau) = -\delta(\tau)/\hbar \iff \frac{1}{\hbar} \sum_{\omega_n \in \Omega_\zeta} e^{-i\omega_n \tau} \left(-i\omega_n + \frac{H - \mu}{\hbar} \right) \mathcal{G}(i\omega_n) = \frac{1}{\hbar} \sum_{\omega_n \in \Omega_\zeta} e^{-i\omega_n \tau} (-1/\hbar)$$

using now that the coefficients of a Fourier-series are unique, one obtains by comparing coefficients

$$\left(-i\omega_n + \frac{H - \mu}{\hbar} \right) \mathcal{G}(i\omega_n) = -\frac{1}{\hbar} \iff \mathcal{G}(i\omega_n) = \frac{1}{\hbar} \left(\frac{1}{i\omega_n - \frac{H - \mu}{\hbar}} \right)$$

⁹Cf. Waste-Folder 1, topic 34.

Notice that we can't expand the r.h. sides w.r.t. Matsubara-frequencies according to Lemma TBD since we can't use the usual Kronecker-delta-relation since we are only integrating from 0 to $\hbar\beta$.

Equation of Motion Technique for Hybridized Systems

Equations of motion for generalized time-ordered, retarded and advanced Green-functions: Elk-Gasser p. 25

With the equation-of-motion-technique; See Appendix (ToBeReferenced.)

Proof 31.36 of Theorem 13.8 The first (Matsubara-)identity follows from Theorem 13.7 and the rules for inverting 2x2-matrices. For the temporal component, we need to evaluate the Matsubara-sums

$$\mathcal{G}(\tau) = \frac{1}{\hbar\beta} \sum_{\omega_n \in \Omega_{-1}} \mathcal{G}(i\omega_n) e^{-i\omega_n \tau} = \star \quad (31.127)$$

Following Notation 15.17, we consider the function

$$z \mapsto \mathcal{G}(-iz) = \frac{1}{\hbar} \frac{(z - (\epsilon_0 - \mu)\sigma_0/\hbar) + \epsilon \cdot \sigma/\hbar}{(z - \omega_+)(z - \omega_-)} e^{-z\tau} \quad (31.128)$$

with

$$\omega_{\pm} = (\pm\epsilon + \epsilon_0 - \mu) / \hbar \quad (31.129)$$

Notice that we are now in the situation of Theorem 15.20, with

$$f_{-1}(z) = g_{-1}(z)\mathcal{G}(-iz) \quad \text{and} \quad g_{-1}(z) = \hbar\beta n_{\hbar\beta}^{-1}(z) = \hbar\beta F_{\hbar\beta}(z) \quad (31.130)$$

which gives us two cases:

Case 1: $\omega_- \neq \omega_+$

$$\begin{aligned} \star &= \sum_{k=\pm} \text{Res}(f_{-1}(z), z = z_k) \\ &= \frac{1}{\hbar} \sum_{k=\pm} \frac{(z - (\epsilon_0 - \mu)\sigma_0/\hbar) \sigma_0 + \epsilon \cdot \sigma/\hbar}{(z - \omega_+)(z - \omega_-)} e^{-z\tau} (z - z_k) F_{\hbar\beta}(z) \Big|_{z=z_k} \\ &= \frac{1}{\hbar} \frac{(\omega_+ - (\epsilon_0 - \mu)\sigma_0/\hbar) \sigma_0 + \epsilon \cdot \sigma/\hbar}{(\omega_+ - \omega_-)} e^{-\omega_+ \tau} F_{\hbar\beta}(\omega_+) \\ &\quad + \frac{1}{\hbar} \frac{(\omega_- - (\epsilon_0 - \mu)\sigma_0/\hbar) \sigma_0 + \epsilon \cdot \sigma/\hbar}{(\omega_- - \omega_+)} e^{-\omega_- \tau} F_{\hbar\beta}(\omega_-) \end{aligned} \quad (31.131)$$

Case 2: $\omega_- = \omega_+ = \omega_{\bullet}$

Then we have the function

$$z \mapsto \mathcal{G}(-iz) = \frac{1}{\hbar} \frac{\sigma_0}{z - \omega_{\bullet}} e^{-z\tau} \quad (31.132)$$

and therefore we obtain with the Residue-theorem

$$\star = +1 \operatorname{Res} (f_{-1}(z), z = \omega_*) = \frac{\sigma_0}{\hbar} e^{-z\tau} F_{\hbar\beta}(z) \Big|_{z=\omega_*} \quad (31.133)$$

It is not that obvious that both expressions coincide in the limit $\epsilon \rightarrow 0$. Their agreement follows however from the differentiability of the (complex) Fermi-function.

Proofs to Section 7.2

Proof 31.37 of Theorem 13.10 By direct computation¹⁰ we obtain

$$\begin{aligned} \mathcal{D}_{0, \alpha, \alpha'}(\tau) &= \frac{-\theta(\tau)}{\hbar} \langle A_\alpha(\tau) A_{\alpha'}^\dagger(0) \rangle + \theta(-\tau) \langle A_{\alpha'}^\dagger(0) A_\alpha(\tau) \rangle \\ &= \frac{-\theta(\tau)}{\hbar} \left[\langle c_{\mathbf{q}\nu}(\tau) c_{\mathbf{q}'\nu'}^\dagger(0) \rangle + \langle c_{-\mathbf{q}\nu}^\dagger(\tau) c_{-\mathbf{q}'\nu'}(0) \rangle \right] + \frac{-\theta(-\tau)}{\hbar} \left[\langle c_{-\mathbf{q}'\nu'}(0) c_{-\mathbf{q}\nu}^\dagger(\tau) \rangle + \langle c_{\mathbf{q}'\nu'}^\dagger(0) c_{\mathbf{q}\nu}(\tau) \rangle \right] \\ &= \dots \end{aligned}$$

Now we use the identities in Equation (8.26)

$$\begin{aligned} c_{\mathbf{q}\nu}(\tau) &= c_{\mathbf{q}\nu}(0) e^{-\tau\omega_{\mathbf{q}\nu}} = c_{\mathbf{q}\nu} e^{-\tau\omega_{\mathbf{q}\nu}} & \langle c_\alpha^\dagger c_\beta \rangle &= \delta_{\alpha,\beta} B_{\hbar\beta}(\omega_\alpha) \\ c_{\mathbf{q}\nu}^\dagger(\tau) &= c_{\mathbf{q}\nu}^\dagger(0) e^{+\tau\omega_{\mathbf{q}\nu}} = c_{\mathbf{q}\nu}^\dagger e^{+\tau\omega_{\mathbf{q}\nu}} & \langle c_\beta c_\alpha^\dagger \rangle &= \langle c_\alpha^\dagger c_\beta \rangle + \delta_{\alpha,\beta} \equiv \delta_{\alpha,\beta} B_{\hbar\beta}(\omega_\alpha) + \delta_{\alpha,\beta} \end{aligned}$$

This gives (together with Remark 12.9.1)

$$\begin{aligned} \dots &= \frac{-\theta(\tau)}{\hbar} \left[e^{-\tau\omega_{\mathbf{q}\nu}} \langle c_{\mathbf{q}\nu} c_{\mathbf{q}'\nu'}^\dagger \rangle + e^{\tau\omega_{\mathbf{q}\nu}} \langle c_{-\mathbf{q}\nu}^\dagger c_{-\mathbf{q}'\nu'} \rangle \right] + \frac{-\theta(-\tau)}{\hbar} \left[e^{\tau\omega_{\mathbf{q}\nu}} \langle c_{-\mathbf{q}'\nu'} c_{-\mathbf{q}\nu}^\dagger \rangle + e^{-\tau\omega_{\mathbf{q}\nu}} \langle c_{\mathbf{q}'\nu'}^\dagger c_{\mathbf{q}\nu} \rangle \right] \\ &= \delta_{\mathbf{q},\mathbf{q}'} \delta_{\nu,\nu'} \frac{-\theta(\tau)}{\hbar} \left[e^{-\tau\omega_{\mathbf{q}\nu}} (B_{\hbar\beta}(\omega_\alpha) + 1) + e^{+\tau\omega_{\mathbf{q}\nu}} B_{\hbar\beta}(\omega_\alpha) \right] + \delta_{\mathbf{q},\mathbf{q}'} \delta_{\nu,\nu'} \frac{-\theta(-\tau)}{\hbar} \left[e^{+\tau\omega_{\mathbf{q}\nu}} (B_{\hbar\beta}(\omega_\alpha) + 1) + e^{-\tau\omega_{\mathbf{q}\nu}} B_{\hbar\beta}(\omega_\alpha) \right] \\ &= \dots \end{aligned}$$

which can be summarized as

$$\dots = -\delta_{\alpha,\alpha'} \frac{1}{\hbar} \sum_{\nu \in \{+, -\}} \theta(\nu\tau) \left[(1 + B_{\hbar\beta}(\omega_\alpha)) e^{-\nu\omega_\alpha\tau} + B_{\hbar\beta}(\omega_\alpha) e^{\nu\omega_\alpha\tau} \right] \quad (31.134)$$

The Green's function of frequency is

¹⁰Cf. [30] p. 116-117.

$$\mathcal{D}_{0, \nu, \nu'}(i\omega_n, \mathbf{q}) = \frac{1}{\hbar} \int_0^{\hbar\beta} d\tau e^{i\omega_n \tau} \mathcal{D}_{0, \nu, \nu'}(\tau, \mathbf{q}) \quad (31.135)$$

$$= -\frac{1}{\hbar} \delta_{\nu, \nu'} \left[(B_{\hbar\beta}(\omega_{\mathbf{q}_\nu}) + 1) \frac{e^{\beta(i\omega_n - \omega_{\mathbf{q}_\nu})} - 1}{i\omega_n - \omega_{\mathbf{q}_\nu}} + B_{\hbar\beta}(\omega_{\mathbf{q}_\nu}) \frac{e^{\beta(i\omega_n + \omega_{\mathbf{q}_\nu})} - 1}{i\omega_n + \omega_{\mathbf{q}_\nu}} \right]. \quad (31.136)$$

The terms in the numerators may be simplified by noting that for bosons $\exp(i\omega_n \beta) = 1$ so that the Green's function is

$$\mathcal{D}_{0, \nu, \nu'}(i\omega_n, \mathbf{q}) = -\frac{1}{\hbar} \delta_{\alpha, \alpha'} \left[(B_{\hbar\beta}(\omega_{\mathbf{q}_\nu}) + 1) \frac{e^{-\beta\omega_{\mathbf{q}_\nu}} - 1}{i\omega_n - \omega_{\mathbf{q}_\nu}} + B_{\hbar\beta}(\omega_{\mathbf{q}_\nu}) \frac{e^{\beta\omega_{\mathbf{q}_\nu}} - 1}{i\omega_n + \omega_{\mathbf{q}_\nu}} \right]. \quad (31.137)$$

Now we compute ...

- for the first numerator

$$\begin{aligned} B_{\hbar\beta}(\omega_{\mathbf{q}_\nu}) + 1 &= \frac{1}{e^{\hbar\beta\omega_{\mathbf{q}_\nu}} - 1} + 1 \\ &= \frac{1}{e^{\hbar\beta\omega_{\mathbf{q}_\nu}} - 1} + \frac{e^{\hbar\beta\omega_{\mathbf{q}_\nu}} - 1}{e^{\hbar\beta\omega_{\mathbf{q}_\nu}} - 1} \\ &= \frac{e^{\hbar\beta\omega_{\mathbf{q}_\nu}}}{e^{\hbar\beta\omega_{\mathbf{q}_\nu}} - 1} \\ &= \frac{1}{1 - e^{-\hbar\beta\omega_{\mathbf{q}_\nu}}} \end{aligned} \quad (31.138)$$

such that we obtain

$$(B_{\hbar\beta}(\omega_{\mathbf{q}_\nu}) + 1) \frac{e^{-\beta\omega_{\mathbf{q}_\nu}} - 1}{1} = -1 \quad (31.139)$$

- for the other term we obtain

$$B_{\hbar\beta}(\omega_{\mathbf{q}_\nu}) (e^{\hbar\beta\omega_{\mathbf{q}_\nu}} - 1) = \frac{1}{e^{\hbar\beta\omega_{\mathbf{q}_\nu}} - 1} (e^{\hbar\beta\omega_{\mathbf{q}_\nu}} - 1) = +1 \quad (31.140)$$

Combining these results yields therefore

$$\begin{aligned} \mathcal{D}_{0, \nu, \nu'}(i\omega_n, \mathbf{q}) &= -\frac{1}{\hbar} \left(-\frac{1}{i\omega_n - \omega_{\mathbf{q}_\nu}} + \frac{1}{i\omega_n + \omega_{\mathbf{q}_\nu}} \right) \\ &= \frac{1}{\hbar} \frac{2\omega_{\mathbf{q}_\nu}}{(i\omega_n)^2 - \omega_{\mathbf{q}_\nu}^2} \\ &= -\frac{1}{\hbar} \frac{2\omega_{\mathbf{q}_\nu}}{\omega_n^2 + \omega_{\mathbf{q}_\nu}^2}. \end{aligned} \quad (31.141)$$

Proof 31.38 of Theorem 13.11

Expanding the time-ordering operator yields the equation

$$\begin{aligned}\mathcal{D}_{\alpha\alpha'}^0(\tau) &= -\frac{1}{\hbar}\langle T_\tau A_\alpha(\tau)A_{\alpha'}^\dagger(0)\rangle_0 \\ &= -\frac{1}{\hbar}\langle \theta(\tau)A_\alpha(\tau)A_{\alpha'}^\dagger(0) + \theta(-\tau)A_{\alpha'}^\dagger(0)A_\alpha(\tau)\rangle_0\end{aligned}$$

Hence we obtain for the time-derivative

$$\begin{aligned}\partial_\tau \mathcal{D}_{\alpha\alpha'}^0(\tau) &= -\frac{1}{\hbar}\langle T_\tau \partial_\tau A_\alpha(\tau)A_{\alpha'}^\dagger(0)\rangle_0 - \frac{1}{\hbar}\delta(\tau)\langle A_\alpha(0)A_{\alpha'}^\dagger(0)\rangle_0 + \frac{1}{\hbar}\delta(-\tau)\langle A_{\alpha'}^\dagger(0)A_\alpha(0)\rangle_0 \\ &= \dots\end{aligned}$$

for the first term we have with Heisenbergs equation of motion

$$\partial_\tau A_\alpha(\tau) = -\frac{1}{\hbar}[A_\alpha, H_0]_H(\tau) \quad (31.142)$$

Now we compute the commutators

$$\begin{aligned}[c_\alpha, c_\beta^\dagger c_\beta] &= \delta_{\alpha,\beta} c_\beta \\ [c_\alpha^\dagger, c_\beta^\dagger c_\beta] &= -\delta_{\alpha,\beta} c_\beta^\dagger\end{aligned}$$

and obtain

$$\begin{aligned}[A_\nu(\mathbf{k}), H_0] &= \sum_{\mathbf{k}', \nu'} \hbar\omega_\nu(\mathbf{k}') ([c_\nu(\mathbf{k}), c_{\nu'}^\dagger(\mathbf{k}')c_{\nu'}(\mathbf{k}')] + [c_\nu^\dagger(-\mathbf{k}), c_{\nu'}^\dagger(\mathbf{k}')c_{\nu'}(\mathbf{k}')]) \\ &= \hbar\omega_\nu(\mathbf{k}) (c_\nu(\mathbf{k}) - c_\nu^\dagger(-\mathbf{k})) \\ &\equiv \hbar\omega_\nu(\mathbf{k}) P_\nu(\mathbf{k})\end{aligned}$$

hence we obtain, summarizing the terms with delta-functions with the distributivity-law

$$\dots = -\frac{1}{\hbar} \left(\frac{-1}{\hbar} \omega_\nu(\mathbf{k}) \langle T_\tau P_\nu(\mathbf{k}) A_{\nu'}^\dagger(\mathbf{k}')(0) \rangle_0 + \delta(\tau) \left[\langle A_\nu(\mathbf{k}) A_{\nu'}^\dagger(\mathbf{k}') \rangle_0 - \langle A_{\nu'}^\dagger(\mathbf{k}') A_\nu(\mathbf{k}) \rangle_0 \right] \right)$$

Now we compute the δ -functions

$$\begin{aligned}&\langle A_\nu(\mathbf{k}) A_{\nu'}^\dagger(\mathbf{k}') \rangle_0 - \langle A_{\nu'}^\dagger(\mathbf{k}') A_\nu(\mathbf{k}) \rangle_0 \\ &= \delta_{\nu,\nu'} \delta_{\mathbf{k},\mathbf{k}'} (1 - B_{\hbar\beta}(\omega_\nu(\mathbf{k})) + 0 + 0 + B_{\hbar\beta}(\omega_\nu(-\mathbf{k})) - [B_{\hbar\beta}(\omega_\nu(\mathbf{k})) + 0 + 0 + 1 - B_{\hbar\beta}(\omega_\nu(-\mathbf{k}))]) \\ &= 0\end{aligned}$$

in total we obtain thus

$$\partial_{\tau} \mathcal{D}_{\alpha\alpha'}^0(\tau) = \frac{+\omega_v(\mathbf{k})}{\hbar} \langle T_{\tau} P_v(\mathbf{k}, \tau) A_{v'}^{\dagger}(\mathbf{k}', 0) \rangle_0 \quad (31.143)$$

Hence we obtain for the second derivative analogously

$$\begin{aligned} \partial_{\tau}^2 \mathcal{D}_{\alpha\alpha'}^0(\tau) &= \frac{+\omega_v(\mathbf{k})}{\hbar} \left(\langle T_{\tau} \partial_{\tau} P_v(\mathbf{k}, \tau) A_{v'}^{\dagger}(\mathbf{k}', 0) \rangle_0 + \delta(\tau) \langle P_v(\mathbf{k}) A_{v'}^{\dagger}(\mathbf{k}') \rangle_0 - \delta(-\tau) \langle A_{v'}^{\dagger}(\mathbf{k}') P_v(\mathbf{k}) \rangle_0 \right) \\ &= \dots \end{aligned}$$

The first term becomes a commutator with Heisenbergs equation of motion

$$\partial_{\tau} P_v(\mathbf{k}, \tau) = -\frac{1}{\hbar} [P_v(\mathbf{k}), H_0]_{\text{H}}(\tau). \quad (31.144)$$

Analogously to the previous computation we obtain

$$[P_v(\mathbf{k}), H_0] = \hbar \omega_v(\mathbf{k}) A_v(\mathbf{k})$$

and for the other term we compute

$$\begin{aligned} &\langle P_v(\mathbf{k}) A_{v'}^{\dagger}(\mathbf{k}') \rangle_0 - \langle A_{v'}^{\dagger}(\mathbf{k}') P_v(\mathbf{k}) \rangle_0 \\ &= \delta_{v,v'} \delta_{\mathbf{k},\mathbf{k}'} (1 - B_{\hbar\beta}(\omega_v(\mathbf{k})) + 0 + 0 - B_{\hbar\beta}(\omega_v(-\mathbf{k})) - [B_{\hbar\beta}(\omega_v(\mathbf{k})) + 0 + 0 - (1 - B_{\hbar\beta}(\omega_v(-\mathbf{k}))]) \\ &= 2\delta_{v,v'} \delta_{\mathbf{k},\mathbf{k}'} \end{aligned}$$

Hence we obtain in total

$$\begin{aligned} \partial_{\tau}^2 \mathcal{D}_{\alpha\alpha'}^0(\tau) &= \frac{+\omega_v(\mathbf{k})}{\hbar} (-\omega_v(\mathbf{k}) \langle T_{\tau} A_v(\mathbf{k}, \tau) A_{v'}^{\dagger}(\mathbf{k}', 0) \rangle_0 + 2\delta(\tau) \delta_{v,v'} \delta_{\mathbf{k},\mathbf{k}'}) \\ \iff (\partial_{\tau}^2 - \omega_v(\mathbf{k})^2) \mathcal{D}_{\alpha\alpha'}^0(\tau) &= 2\omega_v(\mathbf{k}) \delta(\tau) \delta_{v,v'} \delta_{\mathbf{k},\mathbf{k}'} \end{aligned} \quad (31.145)$$

Proof 31.39 of Lemma 13.12

We work within the free phononic gas of Remark 8.22.4, i.e. $a_{v,\mathbf{q}} \equiv c_v(\mathbf{q})$ with

$$H_0 = \sum_{v,\mathbf{q}} \hbar \omega_v(\mathbf{q}) c_v^{\dagger}(\mathbf{q}) c_v(\mathbf{q}),$$

which is a special case of the diagonalized gas of Definition 8.21, and we recall from Definition 12.18 that $A_v(\mathbf{q}) = c_v(\mathbf{q}) + c_v^{\dagger}(-\mathbf{q})$. We abbreviate $\alpha \equiv (v, \mathbf{q})$ and split the proof into the two equality-signs.

First equality sign.

Since H_0 is diagonal in the occupation-number-eigenbasis of Definition 8.21, the thermal state $\propto e^{-\beta H_0}$ is diagonal in this basis as well. The operators $c_v(\mathbf{q})c_v(-\mathbf{q})$ and $c_v^{\dagger}(\mathbf{q})c_v^{\dagger}(-\mathbf{q})$ each lower, respectively raise, some occupation-number by a total of two quanta, hence they map every occupation-number-eigenstate onto a state orthogonal to itself. Consequently their diagonal matrix-elements vanish and so do their thermal expectation-values,

$$\langle c_v(\mathbf{q})c_v(-\mathbf{q}) \rangle = 0 = \langle c_v^\dagger(\mathbf{q})c_v^\dagger(-\mathbf{q}) \rangle.$$

Using this together with the bosonic commutation-relation $[c_v(-\mathbf{q}), c_v^\dagger(-\mathbf{q})] = 1$ we compute

$$\begin{aligned} \langle A_{v,\mathbf{q}}^\dagger A_{v,\mathbf{q}} \rangle &= \langle (c_v^\dagger(\mathbf{q}) + c_v(-\mathbf{q})) (c_v(\mathbf{q}) + c_v^\dagger(-\mathbf{q})) \rangle \\ &= \langle c_v^\dagger(\mathbf{q})c_v(\mathbf{q}) \rangle + \underbrace{\langle c_v^\dagger(\mathbf{q})c_v^\dagger(-\mathbf{q}) \rangle}_{=0} + \underbrace{\langle c_v(-\mathbf{q})c_v(\mathbf{q}) \rangle}_{=0} + \langle c_v(-\mathbf{q})c_v^\dagger(-\mathbf{q}) \rangle \\ &= \langle c_v^\dagger(\mathbf{q})c_v(\mathbf{q}) \rangle + 1 + \langle c_v^\dagger(-\mathbf{q})c_v(-\mathbf{q}) \rangle. \end{aligned}$$

By Remark 12.9.1, applied to the diagonal gas underlying Remark 8.22.4 (recall $\eta = 0$, $\zeta = +1$ for phonons), both occupation-numbers are given by Bose-functions of the respective mode-energies,

$$\langle c_v^\dagger(\mathbf{q})c_v(\mathbf{q}) \rangle = B_{\hbar\beta}(\omega_v(\mathbf{q})), \quad \langle c_v^\dagger(-\mathbf{q})c_v(-\mathbf{q}) \rangle = B_{\hbar\beta}(\omega_v(-\mathbf{q})).$$

Since the crystal is time-reversal-symmetric, Remark 8.22.6 gives $\omega_v(\mathbf{q}) = \omega_v(-\mathbf{q})$, so the two occupation-numbers above coincide and both equal $\langle a_{v,\mathbf{q}}^\dagger a_{v,\mathbf{q}} \rangle$. This yields

$$\langle A_{v,\mathbf{q}}^\dagger A_{v,\mathbf{q}} \rangle = 2 \langle a_{v,\mathbf{q}}^\dagger a_{v,\mathbf{q}} \rangle + 1,$$

which is the first equality-sign.

Second equality sign.

From Definition 12.19 (with $\alpha = \alpha'$) we have, using stationarity of the ensemble to place both time-arguments at $t = t' = 0$,

$$D_{\alpha,\alpha}^<(t=0) = -\frac{i}{\hbar} \langle A_\alpha^\dagger A_\alpha \rangle \iff \langle A_\alpha^\dagger A_\alpha \rangle = i\hbar D_{\alpha,\alpha}^<(t=0).$$

Writing this equal-time value as the Fourier-back-transform of $D_{\alpha,\alpha}^<(\omega)$,

$$D_{\alpha,\alpha}^<(t=0) = \int_{-\infty}^{+\infty} \frac{d\omega}{2\pi} D_{\alpha,\alpha}^<(\omega),$$

and inserting the fluctuation-dissipation-theorem for phonons, Theorem 14.21,

$$D_{\alpha,\alpha}^<(\omega) = 2i B_{\hbar\beta}(\omega) \operatorname{Im} \lim_{\epsilon \rightarrow 0} D_{\alpha,\alpha}^{\text{ret}}(\omega + i\epsilon),$$

we obtain

$$\begin{aligned} \langle A_\alpha^\dagger A_\alpha \rangle &= i\hbar \int_{-\infty}^{+\infty} \frac{d\omega}{2\pi} 2i B_{\hbar\beta}(\omega) \operatorname{Im} \lim_{\epsilon \rightarrow 0} D_{\alpha,\alpha}^{\text{ret}}(\omega + i\epsilon) \\ &= -2\hbar \int_{-\infty}^{+\infty} \frac{d\omega}{2\pi} B_{\hbar\beta}(\omega) \operatorname{Im} \lim_{\epsilon \rightarrow 0} D_{\alpha,\alpha}^{\text{ret}}(\omega + i\epsilon) \\ &= \lim_{\epsilon \rightarrow 0} \int_{-\infty}^{+\infty} d\omega B_{\hbar\beta}(\omega) \left(-\frac{\hbar}{\pi} \operatorname{Im} D_{\alpha,\alpha}^{\text{ret}}(\omega + i\epsilon) \right), \end{aligned}$$

whereas in the last line we used $i \cdot 2i = -2$ and cancelled the resulting factor 2 against the 2π in the denominator. Writing out $\alpha = (\nu, \mathbf{q})$ this is precisely the second equality-sign (with $\nu' \equiv \nu, \mathbf{q}' \equiv \mathbf{q}$ in the notation of the Lemma).

Combining both equality-signs concludes the proof.

Remark on the prefactor. One can independently confirm the prefactor obtained above (and thereby rule out an additional factor of 2π) using the explicit free-phonon propagator of Theorem 13.10. Analytically continuing $i\omega_n \rightarrow \omega + i\epsilon$ and using the Sokhotski-Plemelj-theorem, Equation (14.3), one finds for $\alpha = (\nu, \mathbf{q})$

$$-\frac{\hbar}{\pi} \operatorname{Im} \lim_{\epsilon \rightarrow 0} D_{\alpha, \alpha}^{\text{ret}}(\omega + i\epsilon) = \delta(\omega - \omega_\nu(\mathbf{q})) - \delta(\omega + \omega_\nu(\mathbf{q})),$$

so that

$$\int_{-\infty}^{+\infty} d\omega B_{\hbar\beta}(\omega) \left(-\frac{\hbar}{\pi} \operatorname{Im} D_{\alpha, \alpha}^{\text{ret}}(\omega + i\epsilon) \right) = B_{\hbar\beta}(\omega_\nu(\mathbf{q})) - B_{\hbar\beta}(-\omega_\nu(\mathbf{q})) = 2 B_{\hbar\beta}(\omega_\nu(\mathbf{q})) + 1,$$

which reproduces $2\langle a_{\nu, \mathbf{q}}^\dagger a_{\nu, \mathbf{q}} \rangle + 1$ from the first equality-sign, as it must. Had one kept the additional factor of $1/(2\pi)$, this cross-check would fail by exactly that factor.

Proof 31.40 of Lemma 14.4

We complete the proof, by sorting the Green-functions into four groups.

- **Group 1:** Time- and anti-time-ordered Green-function $G^{\mathbb{T}}, G^{\overline{\mathbb{T}}}$.
- **Group 2:** Retarded and advanced Green-function $G^{\text{ret}}, G^{\text{adv}}$.
- **Group 3:** Lesser and greater Green-function $G^<, G^>$ and the spectral function A .
- **Group 4:** Imaginary time Green-function $\mathcal{G}$.

For each group, we will prove some Lemmata in order to complete the main proof. Ultimately we want to compute the Fourier-expansion coefficients for each Green-function. In each case we set, without loss of generality, the second time argument to zero $t' = \tau' = 0$ in order to simplify the notation.

Lemmata for the first two groups

In the first two groups we are confronted with the obstacle, that the usual Fourier-transform will turn out to be undefined in the functional sense¹¹. The Fourier-transform only exists in the distributional sense. We therefore introduce the auxiliary functions

$$G_{X, X'}^{\blacksquare, \epsilon}(t) = G_{X, X'}^{\blacksquare}(t) e^{-\epsilon|t|} \quad \text{for } \blacksquare \in \{\mathbb{T}, \overline{\mathbb{T}}, \text{ret}, \text{adv}\} \quad (31.146)$$

then one has

$$G_{X, X'}^{\blacksquare, \epsilon}(\omega) = \int dt e^{i\omega t} G_{X, X'}^{\blacksquare, \epsilon}(t) = \int dt e^{i\omega t} G_{X, X'}^{\blacksquare}(t) e^{-\epsilon|t|} \quad (31.147)$$

In order to compute the regularized Fourier-transformations, we remind the reader of the auxiliary-identities in Lemmata 1.19, 1.19 and Fact 1.21.

¹¹Cf. [14].

Group 1

The time-ordered Green-function

For the time-ordered Green-function we obtain

$$\begin{aligned}
 G_{X,X'}^{\mathbb{T}}(t) &= G_{X,X'}^{\mathbb{T}}(t, t' = 0) \\
 &= -\frac{i}{\hbar} \langle \theta(t) X(t) X'(0) + S_{X,X'} \theta(-t) X'(0) X(t) \rangle \\
 &= -\frac{i}{\hbar} \frac{1}{Z} \sum_{n,m} \theta(t) \langle n | e^{-\beta \Xi} X(t) | m \rangle \langle m | X' | n \rangle + S_{X,X'} \theta(-t) \langle n | e^{-\beta \Xi} X' | m \rangle \langle m | X(t) | n \rangle \\
 &= -\frac{i}{\hbar} \frac{1}{Z} \sum_{n,m} \theta(t) e^{-\beta \xi_n} e^{it(\xi_n - \xi_m)/\hbar} \langle n | X | m \rangle \langle m | X' | n \rangle + S_{X,X'} \theta(-t) e^{-\beta \xi_n} e^{it(\xi_m - \xi_n)/\hbar} \langle n | X' | m \rangle \langle m | X | n \rangle \\
 &= -\frac{i}{\hbar} \frac{1}{Z} \sum_{n,m} e^{it(\xi_n - \xi_m)/\hbar} \langle n | X | m \rangle \langle m | X' | n \rangle \left(\theta(t) e^{-\beta \xi_n} + S_{X,X'} \theta(-t) e^{-\beta \xi_m} \right)
 \end{aligned}$$

We therefore see, that the Fourier-transform of the time-ordered Green-function is strictly speaking ill defined. We therefore introduce for each "small" $\epsilon > 0$, the regularized time-ordered Green-function

$$\begin{aligned}
 G_{X,X'}^{\mathbb{T}, \epsilon}(t) &= G_{X,X'}^{\mathbb{T}}(t) e^{-\epsilon|t|} \\
 &= -\frac{i}{\hbar} \frac{1}{Z} \sum_{n,m} e^{it(\xi_n - \xi_m)/\hbar - \epsilon|t|} \langle n | X | m \rangle \langle m | X' | n \rangle \left(\theta(t) e^{-\beta \xi_n} + S_{X,X'} \theta(-t) e^{-\beta \xi_m} \right)
 \end{aligned}$$

which gives for the Fourier-transform then

$$\begin{aligned}
 G_{X,X'}^{\mathbb{T}, \epsilon}(\omega) &= \int dt e^{i\omega t} G_{X,X'}^{\mathbb{T}, \epsilon}(t) \\
 &= -\frac{1}{Z} \sum_{n,m} \langle n | X | m \rangle \langle m | X' | n \rangle e^{-\beta \xi_n} \frac{i}{\hbar} \int dt e^{i\omega t} e^{it(\xi_n - \xi_m)/\hbar - \epsilon|t|} \theta(t) \\
 &\quad - S_{X,X'} \frac{1}{Z} \sum_{n,m} \langle n | X | m \rangle \langle m | X' | n \rangle e^{-\beta \xi_m} \frac{i}{\hbar} \int dt e^{i\omega t} e^{it(\xi_n - \xi_m)/\hbar - \epsilon|t|} \theta(-t) \\
 &= -\frac{1}{Z} \sum_{n,m} \langle n | X | m \rangle \langle m | X' | n \rangle e^{-\beta \xi_n} \frac{1}{\hbar} \frac{1}{\omega + i\epsilon + (\xi_n - \xi_m)/\hbar} \\
 &\quad - S_{X,X'} \frac{1}{Z} \sum_{n,m} \langle n | X | m \rangle \langle m | X' | n \rangle e^{-\beta \xi_m} \frac{1}{\hbar} \frac{1}{\omega - i\epsilon + (\xi_n - \xi_m)/\hbar} \\
 &= \frac{1}{Z} \sum_{n,m} \langle n | X | m \rangle \langle m | X' | n \rangle \frac{1}{\hbar} \left(\frac{e^{-\beta \xi_n}}{\omega + i\epsilon + (\xi_n - \xi_m)/\hbar} - S_{X,X'} \frac{e^{-\beta \xi_m}}{\omega - i\epsilon + (\xi_n - \xi_m)/\hbar} \right)
 \end{aligned}$$

The anti-time-ordered Green-function

For the anti-time-ordered Green-function we obtain

$$\begin{aligned}
 G_{X,X'}^{\bar{\bar{}}}(t) &= G_{X,X'}^{\bar{\bar{}}}(t, t' = 0) \\
 &= -\frac{i}{\hbar} \langle \theta(-t) X(t) X'(0) + S_{X,X'} \theta(+t) X'(0) X(t) \rangle \\
 &= -\frac{i}{\hbar} \frac{1}{Z} \sum_{n,m} \theta(-t) \langle n | e^{-\beta \Xi} X(t) | m \rangle \langle m | X' | n \rangle + S_{X,X'} \theta(+t) \langle n | e^{-\beta \Xi} X' | m \rangle \langle m | X(t) | n \rangle \\
 &= -\frac{i}{\hbar} \frac{1}{Z} \sum_{n,m} \theta(-t) e^{-\beta \xi_n} e^{it(\xi_n - \xi_m)/\hbar} \langle n | X | m \rangle \langle m | X' | n \rangle + S_{X,X'} \theta(+t) e^{-\beta \xi_n} e^{it(\xi_m - \xi_n)/\hbar} \langle n | X' | m \rangle \langle m | X | n \rangle \\
 &= -\frac{i}{\hbar} \frac{1}{Z} \sum_{n,m} e^{it(\xi_n - \xi_m)/\hbar} \langle n | X | m \rangle \langle m | X' | n \rangle (\theta(-t) e^{-\beta \xi_n} + S_{X,X'} \theta(+t) e^{-\beta \xi_m})
 \end{aligned}$$

We therefore see, that the Fourier-transform of the anti-time-ordered Green-function is strictly speaking ill defined. We therefore introduce for each „small“ $\epsilon > 0$, the regularized anti time-ordered Green-function

$$\begin{aligned}
 G_{X,X'}^{\bar{\bar{\epsilon}}}(t) &= G_{X,X'}^{\bar{\bar{}}}(t) e^{-\epsilon|t|} \\
 &= -\frac{i}{\hbar} \frac{1}{Z} \sum_{n,m} e^{it(\xi_n - \xi_m)/\hbar - \epsilon|t|} \langle n | X | m \rangle \langle m | X' | n \rangle (\theta(-t) e^{-\beta \xi_n} + S_{X,X'} \theta(+t) e^{-\beta \xi_m})
 \end{aligned}$$

which gives for the Fourier-transform then

$$\begin{aligned}
 G_{X,X'}^{\bar{\bar{\epsilon}}}(\omega) &= \int dt e^{i\omega t} G_{X,X'}^{\bar{\bar{\epsilon}}}(t) \\
 &= -\frac{1}{Z} \sum_{n,m} \langle n | X | m \rangle \langle m | X' | n \rangle e^{-\beta \xi_n} \frac{i}{\hbar} \int dt e^{i\omega t} e^{it(\xi_n - \xi_m)/\hbar - \epsilon|t|} \theta(-t) \\
 &\quad - S_{X,X'} \frac{1}{Z} \sum_{n,m} \langle n | X | m \rangle \langle m | X' | n \rangle e^{-\beta \xi_m} \frac{i}{\hbar} \int dt e^{i\omega t} e^{it(\xi_n - \xi_m)/\hbar - \epsilon|t|} \theta(+t) \\
 &= -\frac{1}{Z} \sum_{n,m} \langle n | X | m \rangle \langle m | X' | n \rangle e^{-\beta \xi_n} \frac{1}{\hbar} \frac{1}{\omega - i\epsilon + (\xi_n - \xi_m)/\hbar} \\
 &\quad - S_{X,X'} \frac{1}{Z} \sum_{n,m} \langle n | X | m \rangle \langle m | X' | n \rangle e^{-\beta \xi_m} \frac{1}{\hbar} \frac{1}{\omega + i\epsilon + (\xi_n - \xi_m)/\hbar} \\
 &= \frac{1}{Z} \sum_{n,m} \langle n | X | m \rangle \langle m | X' | n \rangle \frac{1}{\hbar} \left(\frac{-e^{-\beta \xi_n}}{\omega - i\epsilon + (\xi_n - \xi_m)/\hbar} + S_{X,X'} \frac{e^{-\beta \xi_m}}{\omega + i\epsilon + (\xi_n - \xi_m)/\hbar} \right)
 \end{aligned}$$

Group 2

The retarded Green-function

For the retarded Green-function we obtain

$$\begin{aligned}
 G_{X,X'}^{\text{ret}}(t) &= G_{X,X'}^{\text{ret}}(t, t' = 0) \\
 &= -\frac{i}{\hbar} \theta(t) \langle X(t) X'(0) - S_{X,X'} X'(0) X(t) \rangle \\
 &= -\frac{i}{\hbar} \theta(t) \frac{1}{Z} \sum_{n,m} \langle n | e^{-\beta \Xi} X(t) | m \rangle \langle m | X' | n \rangle - S_{X,X'} \langle n | e^{-\beta \Xi} X' | m \rangle \langle m | X(t) | n \rangle \\
 &= -\frac{i}{\hbar} \theta(t) \frac{1}{Z} \sum_{n,m} e^{-\beta \xi_n} e^{it(\xi_n - \xi_m)/\hbar} \langle n | X | m \rangle \langle m | X' | n \rangle - S_{X,X'} e^{-\beta \xi_n} e^{it(\xi_m - \xi_n)/\hbar} \langle n | X' | m \rangle \langle m | X | n \rangle \\
 &= -\frac{i}{\hbar} \frac{1}{Z} \theta(t) \sum_{n,m} e^{it(\xi_n - \xi_m)/\hbar} \langle n | X | m \rangle \langle m | X' | n \rangle (e^{-\beta \xi_n} - S_{X,X'} e^{-\beta \xi_m})
 \end{aligned}$$

We therefore see, that the Fourier-transform of the retarded Green-function is strictly speaking ill defined. We therefore introduce for each „small“ $\epsilon > 0$, the regularized retarded Green-function

$$\begin{aligned}
 G_{X,X'}^{\text{ret}, \epsilon}(t) &= G_{X,X'}^{\text{ret}}(t) e^{-\epsilon|t|} \\
 &= -\frac{i}{\hbar} \frac{1}{Z} \theta(t) \sum_{n,m} e^{it(\xi_n - \xi_m)/\hbar - \epsilon|t|} \langle n | X | m \rangle \langle m | X' | n \rangle (e^{-\beta \xi_n} - S_{X,X'} e^{-\beta \xi_m})
 \end{aligned}$$

which gives for the Fourier-transform then

$$\begin{aligned}
 G_{X,X'}^{\text{ret}, \epsilon}(\omega) &= \int dt e^{i\omega t} G_{X,X'}^{\text{ret}, \epsilon}(t) \\
 &= -\frac{1}{Z} \sum_{n,m} \langle n | X | m \rangle \langle m | X' | n \rangle (e^{-\beta \xi_n} - S_{X,X'} e^{-\beta \xi_m}) \frac{i}{\hbar} \int dt e^{i\omega t} e^{it(\xi_n - \xi_m)/\hbar - \epsilon|t|} \theta(t) \\
 &= -\frac{1}{Z} \sum_{n,m} \langle n | X | m \rangle \langle m | X' | n \rangle (e^{-\beta \xi_n} - S_{X,X'} e^{-\beta \xi_m}) \frac{-1}{\hbar} \frac{1}{\omega + i\epsilon + (\xi_n - \xi_m)/\hbar} \\
 &= \frac{1}{Z} \sum_{n,m} \langle n | X | m \rangle \langle m | X' | n \rangle (e^{-\beta \xi_n} - S_{X,X'} e^{-\beta \xi_m}) \frac{1}{\hbar} \frac{1}{\omega + i\epsilon + (\xi_n - \xi_m)/\hbar}
 \end{aligned}$$

Lastly we switch to the usual notation found in physics-textbooks and note the formal identity

$$\begin{aligned}
 G_{X,X'}^{\text{ret}, \epsilon}(\omega) &= \int dt e^{i\omega t} G_{X,X'}^{\text{ret}, \epsilon}(t) \\
 &= -\frac{1}{Z} \sum_{n,m} \langle n | X | m \rangle \langle m | X' | n \rangle (e^{-\beta \xi_n} - S_{X,X'} e^{-\beta \xi_m}) \frac{i}{\hbar} \int dt e^{i\omega t} e^{it(\xi_n - \xi_m)/\hbar - \epsilon|t|} \theta(t) \\
 &= -\frac{1}{Z} \sum_{n,m} \langle n | X | m \rangle \langle m | X' | n \rangle (e^{-\beta \xi_n} - S_{X,X'} e^{-\beta \xi_m}) \frac{i}{\hbar} \int_0^\infty dt e^{i\omega t} e^{it(\xi_n - \xi_m)/\hbar - \epsilon t}
 \end{aligned}$$

$$\begin{aligned}
 &= -\frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle (e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}) \frac{i}{\hbar} \int_0^\infty dt e^{i(\omega+i\epsilon)t} e^{it(\xi_n-\xi_m)/\hbar} \\
 &= -\frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle (e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}) \frac{i}{\hbar} \int dt e^{i(\omega+i\epsilon)t} e^{it(\xi_n-\xi_m)/\hbar} \theta(t) \\
 &= \int dt e^{i(\omega+i\epsilon)t} G_{X,X'}^{\text{ret}}(t) \\
 &= G_{X,X'}^{\text{ret}}(\omega + i\epsilon)
 \end{aligned}$$

Hence, for future reference, we have

$$G_{X,X'}^{\text{ret}}(\omega + i\epsilon) = \frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle (e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}) \frac{1}{\hbar} \frac{1}{\omega + i\epsilon + (\xi_n - \xi_m)/\hbar}$$

The advanced Green-function

For the advanced Green-function we obtain

$$\begin{aligned}
 G_{X,X'}^{\text{adv}}(t) &= G_{X,X'}^{\text{adv}}(t, t' = 0) \\
 &= +\frac{i}{\hbar} \theta(-t) \langle X(t)X'(0) - S_{X,X'} X'(0)X(t) \rangle \\
 &= +\frac{i}{\hbar} \theta(-t) \frac{1}{Z} \sum_{n,m} \langle n|e^{-\beta\Xi} X(t)|m\rangle \langle m|X'|n\rangle - S_{X,X'} \langle n|e^{-\beta\Xi} X'|m\rangle \langle m|X(t)|n\rangle \\
 &= +\frac{i}{\hbar} \theta(-t) \frac{1}{Z} \sum_{n,m} e^{-\beta\xi_n} e^{it(\xi_n-\xi_m)/\hbar} \langle n|X|m\rangle \langle m|X'|n\rangle - S_{X,X'} e^{-\beta\xi_m} e^{it(\xi_m-\xi_n)/\hbar} \langle n|X'|m\rangle \langle m|X|n\rangle \\
 &= +\frac{i}{\hbar} \frac{1}{Z} \theta(-t) \sum_{n,m} e^{it(\xi_n-\xi_m)/\hbar} \langle n|X|m\rangle \langle m|X'|n\rangle (e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m})
 \end{aligned}$$

We therefore see, that the Fourier-transform of the advanced Green-function is strictly speaking ill defined. We therefore introduce for each „small“ $\epsilon > 0$, the regularized advanced Green-function

$$\begin{aligned}
 G_{X,X'}^{\text{adv},\epsilon}(t) &= G_{X,X'}^{\text{adv}}(t) e^{-\epsilon|t|} \\
 &= +\frac{i}{\hbar} \frac{1}{Z} \theta(-t) \sum_{n,m} e^{it(\xi_n-\xi_m)/\hbar - \epsilon|t|} \langle n|X|m\rangle \langle m|X'|n\rangle (e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m})
 \end{aligned}$$

which gives for the Fourier-transform then

$$\begin{aligned}
 G_{X,X'}^{\text{adv}}(\omega) &= \int dt e^{i\omega t} G_{X,X'}^{\text{adv}}(t) \\
 &= +\frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle (e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}) \frac{i}{\hbar} \int dt e^{i\omega t} e^{it(\xi_n - \xi_m)/\hbar - \epsilon|t|} \theta(-t) \\
 &= +\frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle (e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}) \frac{1}{\hbar} \frac{1}{\omega - i\epsilon + (\xi_n - \xi_m)/\hbar} \\
 &= \frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle (e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}) \frac{1}{\hbar} \frac{1}{\omega - i\epsilon + (\xi_n - \xi_m)/\hbar}
 \end{aligned}$$

Lastly we switch to the usual notation found in physics-textbooks and note the formal identity

$$\begin{aligned}
 G_{X,X'}^{\text{adv}}(\omega) &= \int dt e^{i\omega t} G_{X,X'}^{\text{adv}}(t) \\
 &= +\frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle (e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}) \frac{i}{\hbar} \int dt e^{i\omega t} e^{it(\xi_n - \xi_m)/\hbar - \epsilon|t|} \theta(-t) \\
 &= +\frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle (e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}) \frac{i}{\hbar} \int_{-\infty}^0 dt e^{i\omega t} e^{it(\xi_n - \xi_m)/\hbar + \epsilon t} \\
 &= +\frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle (e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}) \frac{i}{\hbar} \int_{-\infty}^0 dt e^{i(\omega - i\epsilon)t} e^{it(\xi_n - \xi_m)/\hbar} \\
 &= +\frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle (e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}) \frac{i}{\hbar} \int dt e^{i(\omega - i\epsilon)t} e^{it(\xi_n - \xi_m)/\hbar} \theta(-t) \\
 &= \int dt e^{i(\omega - i\epsilon)t} G_{X,X'}^{\text{adv}}(t) \\
 &= G_{X,X'}^{\text{adv}}(\omega - i\epsilon)
 \end{aligned}$$

Hence, for future reference, we have

$$G_{X,X'}^{\text{adv}}(\omega - i\epsilon) = \frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle (e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}) \frac{1}{\hbar} \frac{1}{\omega - i\epsilon + (\xi_n - \xi_m)/\hbar}$$

Group 3

Lemmata for the third group

In the third group we are in the comfortable situation that we can refer to standard-theorems about distribution-theory, such as Theorem 1.22. Notice also that we can reformulate Fact 14.2 equivalently (in wise foresight) as

$$\frac{1}{\omega + i\epsilon + (\xi_n - \xi_m)/\hbar} = P \frac{1}{\omega + (\xi_n - \xi_m)/\hbar} - i\pi\delta(\omega + (\xi_n - \xi_m)/\hbar) \quad (31.148)$$

The lesser Green-function

For the lesser Green-function we obtain

$$\begin{aligned}
 G_{X,X'}^<(t) &= G_{X,X'}^<(t, t' = 0) \\
 &= -\frac{i}{\hbar} S_{X,X'} \langle X'(0) X(t) \rangle \\
 &= -\frac{i}{\hbar} S_{X,X'} \frac{1}{Z} \sum_{n,m} \langle n | e^{-\beta \Xi} X' | m \rangle \langle m | X(t) | n \rangle \\
 &= -\frac{i}{\hbar} S_{X,X'} \frac{1}{Z} \sum_{n,m} e^{-\beta \xi_n} e^{it(\xi_m - \xi_n)/\hbar} \langle n | X' | m \rangle \langle m | X | n \rangle \\
 &= \frac{1}{Z} \sum_{n,m} \langle n | X | m \rangle \langle m | X' | n \rangle e^{-\beta \xi_m} \left(-\frac{i}{\hbar} S_{X,X'} \right) e^{it(\xi_n - \xi_m)/\hbar}
 \end{aligned}$$

We therefore see, that the Fourier-transform of the lesser Green-function is simply the usual delta-distribution. This gives for the Fourier-transform then

$$\begin{aligned}
 G_{X,X'}^<(\omega) &= \int dt e^{i\omega t} G_{X,X'}^<(t) \\
 &= \frac{1}{Z} \sum_{n,m} \langle n | X | m \rangle \langle m | X' | n \rangle \left(-\frac{i}{\hbar} S_{X,X'} \right) e^{-\beta \xi_m} \int dt e^{i\omega t} e^{it(\xi_n - \xi_m)/\hbar} \\
 &= -\frac{1}{Z} \sum_{n,m} \langle n | X | m \rangle \langle m | X' | n \rangle \left(-2\pi i \frac{1}{\hbar} S_{X,X'} \right) e^{-\beta \xi_m} \delta(\omega + (\xi_n - \xi_m)/\hbar)
 \end{aligned}$$

The greater Green-function

For the greater Green-function we obtain

$$\begin{aligned}
 G_{X,X'}^>(t) &= G_{X,X'}^>(t, t' = 0) \\
 &= -\frac{i}{\hbar} \langle X(t) X'(0) \rangle \\
 &= -\frac{i}{\hbar} \frac{1}{Z} \sum_{n,m} \langle n | e^{-\beta \Xi} X(t) | m \rangle \langle m | X' | n \rangle \\
 &= -\frac{i}{\hbar} \frac{1}{Z} \sum_{n,m} e^{-\beta \xi_n} e^{it(\xi_n - \xi_m)/\hbar} \langle n | X | m \rangle \langle m | X' | n \rangle \\
 &= \frac{1}{Z} \sum_{n,m} \langle n | X | m \rangle \langle m | X' | n \rangle e^{-\beta \xi_n} \left(-\frac{i}{\hbar} \right) e^{it(\xi_n - \xi_m)/\hbar}
 \end{aligned}$$

We therefore see, that the Fourier-transform of the greater Green-function is simply the usual delta-distribution. This gives for the Fourier-transform then

$$\begin{aligned}
 G_{X,X'}^>(\omega) &= \int dt e^{i\omega t} G_{X,X'}^>(t) \\
 &= \frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle \left(-\frac{i}{\hbar}\right) e^{-\beta\xi_n} \int dt e^{i\omega t} e^{it(\xi_n - \xi_m)/\hbar} \\
 &= \frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle \left(-2\pi\frac{i}{\hbar}\right) e^{-\beta\xi_n} \delta(\omega + (\xi_n - \xi_m)/\hbar)
 \end{aligned}$$

The spectral function

We have for the spectral function

$$\begin{aligned}
 A_{X,X'}(\omega) &= -\frac{\hbar}{\pi} \text{Im} \lim_{\epsilon \rightarrow 0} G_{X,X'}^{\text{ret}}(\omega + i\epsilon) \\
 &= -\frac{\hbar}{\pi} \text{Im} \lim_{\epsilon \rightarrow 0} \frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle (e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}) \frac{1}{\hbar} \frac{1}{\omega + i\epsilon + (\xi_n - \xi_m)/\hbar} \\
 &= -\frac{\hbar}{\pi} \frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle (e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}) \frac{1}{\hbar} \text{Im} \left[\text{P} \frac{1}{\omega + (\xi_n - \xi_m)/\hbar} - i\pi \delta(\omega + (\xi_n - \xi_m)/\hbar) \right] \\
 &= \frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle (e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}) \delta(\omega + (\xi_n - \xi_m)/\hbar) \\
 &= \dots
 \end{aligned}$$

Notice, for later reference, that this can be simplified further as¹²

$$\begin{aligned}
 A_{X,X'}(\omega) &= \dots \\
 &= \frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle e^{-\beta\xi_n} (1 - S_{X,X'} e^{-\beta(\xi_m - \xi_n)}) \delta(\omega + (\xi_n - \xi_m)/\hbar) \\
 &= \frac{1}{Z} (1 - S_{X,X'} e^{-\hbar\beta\omega}) \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle e^{-\beta\xi_n} \delta(\omega + (\xi_n - \xi_m)/\hbar)
 \end{aligned} \tag{31.149}$$

Group 4 (The imaginary-time Green-function)

Lemma for the four group

In the third group we are in the comfortable situation that we are dealing with a periodic functions, which are particularly simple to handle due to the vast literature about them. We only need Lemma 1.13.

The imaginary-time-ordered Green-function

For the imaginary-time-ordered Green-function we obtain

¹²It is just a simple use of the delta-distribution, but yet smart to do it like this. Cf. [30] page 122.

$$\begin{aligned}
 \mathcal{G}_{X,X'}(\tau) &= \mathcal{G}_{X,X'}(\tau, \tau' = 0) \\
 &= -\frac{1}{\hbar} \langle \theta(\tau) X(\tau) X'(0) + S_{X,X'} \theta(-\tau) X'(0) X(\tau) \rangle \\
 &= -\frac{1}{\hbar} \frac{1}{Z} \sum_{n,m} \theta(\tau) \langle n | e^{-\beta \Xi} X(\tau) | m \rangle \langle m | X' | n \rangle + S_{X,X'} \theta(-\tau) \langle n | e^{-\beta \Xi} X' | m \rangle \langle m | X(\tau) | n \rangle \\
 &= -\frac{1}{\hbar} \frac{1}{Z} \sum_{n,m} \theta(\tau) e^{-\beta \xi_n} e^{\tau(\xi_n - \xi_m)/\hbar} \langle n | X | m \rangle \langle m | X' | n \rangle + S_{X,X'} \theta(-\tau) e^{-\beta \xi_n} e^{\tau(\xi_m - \xi_n)/\hbar} \langle n | X' | m \rangle \langle m | X | n \rangle \\
 &= -\frac{1}{\hbar} \frac{1}{Z} \sum_{n,m} e^{\tau(\xi_n - \xi_m)/\hbar} \langle n | X | m \rangle \langle m | X' | n \rangle \left(\theta(\tau) e^{-\beta \xi_n} + S_{X,X'} \theta(-\tau) e^{-\beta \xi_m} \right)
 \end{aligned}$$

We now compute the Matsubara-coefficients of these functions.

$$\begin{aligned}
 \mathcal{G}_{X,X'}(i\omega_n) &= \int_0^{\hbar\beta} d\tau e^{i\omega_n \tau} \mathcal{G}_{X,X'}(\tau) \\
 &= -\frac{1}{Z} \sum_{n,m} \langle n | X | m \rangle \langle m | X' | n \rangle e^{-\beta \xi_n} \frac{1}{\hbar} \int_0^{\hbar\beta} d\tau e^{i\omega_n \tau} e^{\tau(\xi_n - \xi_m)/\hbar} \\
 &= -\frac{1}{Z} \sum_{n,m} \langle n | X | m \rangle \langle m | X' | n \rangle e^{-\beta \xi_n} \frac{1}{\hbar} \frac{S_{X,X'} e^{\beta(\xi_n - \xi_m)} - 1}{i\omega_n + (\xi_n - \xi_m)/\hbar} \\
 &= \frac{1}{Z} \sum_{n,m} \langle n | X | m \rangle \langle m | X' | n \rangle \frac{1}{\hbar} \frac{e^{-\beta \xi_n} - S_{X,X'} e^{-\beta \xi_m}}{i\omega_n + (\xi_n - \xi_m)/\hbar}
 \end{aligned}$$

Summarizing the results, one finds for the different Green-functions (and the spectral function)

$$G_{X,X'}^{\mathbb{T}_\epsilon}(\omega) = \frac{1}{Z} \sum_{n,m} \langle n | X | m \rangle \langle m | X' | n \rangle \frac{1}{\hbar} \left(\frac{e^{-\beta \xi_n}}{\omega + i\epsilon + (\xi_n - \xi_m)/\hbar} - S_{X,X'} \frac{e^{-\beta \xi_m}}{\omega - i\epsilon + (\xi_n - \xi_m)/\hbar} \right) \quad (31.150)$$

$$G_{X,X'}^<(\omega) = \frac{1}{Z} \sum_{(n,m) \in \mathbb{N}^2} \langle n | X | m \rangle \langle m | X' | n \rangle \frac{-2\pi i S_{X,X'} e^{-\beta \xi_m}}{\hbar} \delta(\omega + (\xi_n - \xi_m)/\hbar) \quad (31.151)$$

$$G_{X,X'}^>(\omega) = \frac{1}{Z} \sum_{(n,m) \in \mathbb{N}^2} \langle n | X | m \rangle \langle m | X' | n \rangle \frac{2\pi i e^{-\beta \xi_n}}{\hbar} \delta(\omega + (\xi_n - \xi_m)/\hbar) \quad (31.152)$$

$$G_{X,X'}^{\bar{\mathbb{T}}_\epsilon}(\omega) = \frac{1}{Z} \sum_{n,m} \langle n | X | m \rangle \langle m | X' | n \rangle \frac{1}{\hbar} \left(\frac{-e^{-\beta \xi_n}}{\omega - i\epsilon + (\xi_n - \xi_m)/\hbar} + S_{X,X'} \frac{e^{-\beta \xi_m}}{\omega + i\epsilon + (\xi_n - \xi_m)/\hbar} \right) \quad (31.153)$$

$$G_{X,X'}^{\text{ret}}(\omega + i\epsilon) = \frac{1}{Z} \sum_{(n,m) \in \mathbb{N}^2} \langle n | X | m \rangle \langle m | X' | n \rangle \frac{e^{-\beta \xi_n} - S_{X,X'} e^{-\beta \xi_m}}{\hbar} \frac{1}{\omega + i\epsilon + (\xi_n - \xi_m)/\hbar} \quad (31.154)$$

$$G_{X,X'}^{\text{adv}}(\omega - i\epsilon) = \frac{1}{Z} \sum_{(n,m) \in \mathbb{N}^2} \langle n|X|m\rangle \langle m|X'|n\rangle \frac{e^{-\beta\xi_n} - S_{X,X'}e^{-\beta\xi_m}}{\hbar} \frac{1}{\omega - i\epsilon + (\xi_n - \xi_m)/\hbar} \quad (31.155)$$

$$\mathcal{G}_{X,X'}(i\omega_l) = \frac{1}{Z} \sum_{(n,m) \in \mathbb{N}^2} \langle n|X|m\rangle \langle m|X'|n\rangle \frac{e^{-\beta\xi_n} - S_{X,X'}e^{-\beta\xi_m}}{\hbar} \frac{1}{i\omega_l + (\xi_n - \xi_m)/\hbar} \quad (31.156)$$

$$A_{X,X'}(\omega) = \frac{1}{Z} \sum_{(n,m) \in \mathbb{N}^2} \langle n|X|m\rangle \langle m|X'|n\rangle (e^{-\beta\xi_n} - S_{X,X'}e^{-\beta\xi_m}) \delta(\omega + (\xi_n - \xi_m)/\hbar) \quad (31.157)$$

Proof 31.41 of Theorem 14.6 We divide the proof into the two equation signs.

The spectral function for the lesser Green-function

We recall from Equation (31.157) that the spectral function

$$\begin{aligned} A_{X,X'}(\omega) &= \frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle (e^{-\beta\xi_n} - S_{X,X'}e^{-\beta\xi_m}) \delta(\omega + (\xi_n - \xi_m)/\hbar) \\ &= \frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle e^{-\beta\xi_m} (e^{\beta(\xi_m - \xi_n)} - S_{X,X'}) \delta(\omega + (\xi_n - \xi_m)/\hbar) \\ &= \frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle e^{-\beta\xi_m} (e^{\beta\hbar\omega} - S_{X,X'}) \delta(\omega + (\xi_n - \xi_m)/\hbar) \\ &= \frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle e^{-\beta\xi_m} \left(n_{\hbar\beta}^{S_{X,X'}}(\omega) \right)^{-1} \delta(\omega + (\xi_n - \xi_m)/\hbar) \end{aligned}$$

With the the ζ -distribution from Equation (TBReferenced). Now comparing the Lehmann-representation of the spectral-function (with a proper prefactor) with the one for the lesser-Green-function gives

$$\begin{aligned} -2\pi \frac{i}{\hbar} S_{X,X'} n_{\hbar\beta}^{S_{X,X'}}(\omega) A_{X,X'}(\omega) &= 2i S_{X,X'} n_{\hbar\beta}^{S_{X,X'}}(\omega) \text{Im} \lim_{\epsilon \rightarrow 0} G_{X,X'}^{\text{ret}}(\omega + i\epsilon) \\ &= -2\pi \frac{i}{\hbar} S_{X,X'} \frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle e^{-\beta\xi_m} \delta(\omega + (\xi_n - \xi_m)/\hbar) \\ &= G_{X,X'}^<(\omega) \end{aligned}$$

whereas in the last equation we used Equation (31.151).

The spectral function for the greater Green-function

We recall from Equation (31.157) that the spectral function

$$\begin{aligned}
 A_{X,X'}(\omega) &= \frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle \left(e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m} \right) \delta(\omega + (\xi_n - \xi_m)/\hbar) \\
 &= \frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle e^{-\beta\xi_n} \left(1 - S_{X,X'} e^{-\beta(\xi_m - \xi_n)} \right) \delta(\omega + (\xi_n - \xi_m)/\hbar) \\
 &= \frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle e^{-\beta\xi_n} \left(1 - S_{X,X'} e^{-\beta\hbar\omega} \right) \delta(\omega + (\xi_n - \xi_m)/\hbar) \\
 &= \dots
 \end{aligned}$$

Now we compute with the ζ -distribution

$$\begin{aligned}
 1 + S_{X,X'} n_{\hbar\beta}^{S_{X,X'}}(\omega) &= 1 + \frac{S_{X,X'}}{e^{\hbar\beta\omega} - S_{X,X'}} = \frac{e^{\hbar\beta\omega} - S_{X,X'} + S_{X,X'}}{e^{\hbar\beta\omega} - S_{X,X'}} = \frac{1}{1 - S_{X,X'} e^{-\hbar\beta\omega}} = \left(1 - S_{X,X'} e^{-\hbar\beta\omega} \right)^{-1} \\
 \Leftrightarrow \left(1 + S_{X,X'} n_{\hbar\beta}^{S_{X,X'}}(\omega) \right)^{-1} &= 1 - S_{X,X'} e^{-\hbar\beta\omega}
 \end{aligned}$$

and inserting this above allows us to rewrite the expression above to the form

$$A_{X,X'}(\omega) = \dots = \frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle e^{-\beta\xi_n} \left(1 + S_{X,X'} n_{\hbar\beta}^{S_{X,X'}}(\omega) \right)^{-1} \delta(\omega + (\xi_n - \xi_m)/\hbar)$$

This yields then

$$\begin{aligned}
 -\frac{i}{\hbar} 2\pi \left(1 + S_{X,X'} n_{\hbar\beta}^{S_{X,X'}}(\omega) \right) A_{X,X'}(\omega) &= 2i \left(1 + S_{X,X'} n_{\hbar\beta}^{S_{X,X'}}(\omega) \right) \text{Im} \lim_{\epsilon \rightarrow 0} G_{X,X'}^{\text{ret}}(\omega + i\epsilon) \\
 &= \frac{-i}{\hbar} 2\pi \frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle e^{-\beta\xi_n} \delta(\omega + (\xi_n - \xi_m)/\hbar) \\
 &= G_{X,X'}^>(\omega)
 \end{aligned}$$

whereas we used Equation (31.152) in the last step.

Proof 31.42 of Theorem 14.7

We first prove the identities relating the time-ordered and the retarded Green-function. For this we recall Equations (31.150), (31.154) and the Sokhotski-Plemelj-theorem in Equation (14.3)

$$\frac{1}{\omega + i\epsilon + \text{const.}} = \text{P} \frac{1}{\omega + \text{const.}} - i\pi \delta(\omega + \text{const.}), \quad (31.158)$$

$$\frac{1}{\omega - i\epsilon + \text{const.}} = \text{P} \frac{1}{\omega + \text{const.}} + i\pi \delta(\omega + \text{const.}), \quad (31.159)$$

such that we obtain the identities

$$G_{X,X'}^{\mathbb{T}}(\omega) = \frac{1}{Z} \sum_{n,m} \langle n|X|m\rangle \langle m|X'|n\rangle \frac{1}{\hbar} \left(\frac{e^{-\beta\xi_n}}{\omega + i\epsilon + (\xi_n - \xi_m)/\hbar} - S_{X,X'} \frac{e^{-\beta\xi_m}}{\omega - i\epsilon + (\xi_n - \xi_m)/\hbar} \right) \quad (31.160)$$

$$G_{X,X'}^{\text{ret}}(\omega + i\epsilon) = \frac{1}{Z} \sum_{(n,m) \in \mathbb{N}^2} \langle n|X|m\rangle \langle m|X'|n\rangle \frac{e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}}{\hbar} \frac{1}{\omega + i\epsilon + (\xi_n - \xi_m)/\hbar} \quad (31.161)$$

We will prove the identity for each term in the individual summations, which will hence prove the theorem.

The time-ordered Green-function $G^{\mathbb{T}}$

For the time-ordered Green-function we obtain on the one hand

$$\begin{aligned} \frac{e^{-\beta\xi_n}}{\omega + i\epsilon + (\xi_n - \xi_m)/\hbar} - S_{X,X'} \frac{e^{-\beta\xi_m}}{\omega - i\epsilon + (\xi_n - \xi_m)/\hbar} &= (e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}) \mathcal{P} \frac{1}{\omega + (\xi_n - \xi_m)/\hbar} \\ &\quad + (e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}) i\pi\delta(\omega + (\xi_n - \xi_m)/\hbar) \end{aligned}$$

and hence we obtain

$$\text{Re} \left(\frac{e^{-\beta\xi_n}}{\omega + i\epsilon + (\xi_n - \xi_m)/\hbar} - S_{X,X'} \frac{e^{-\beta\xi_m}}{\omega - i\epsilon + (\xi_n - \xi_m)/\hbar} \right) = (e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}) \mathcal{P} \frac{1}{\omega + (\xi_n - \xi_m)/\hbar}$$

$$\text{Im} \left(\frac{e^{-\beta\xi_n}}{\omega + i\epsilon + (\xi_n - \xi_m)/\hbar} - S_{X,X'} \frac{e^{-\beta\xi_m}}{\omega - i\epsilon + (\xi_n - \xi_m)/\hbar} \right) = (e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}) \pi\delta(\omega + (\xi_n - \xi_m)/\hbar)$$

The retarded Green-function G^{ret}

For the retarded Green-function we obtain on the other hand

$$(e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}) \frac{1}{\omega + i\epsilon + (\xi_n - \xi_m)/\hbar} = (e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}) \left(\mathcal{P} \frac{1}{\omega + (\xi_n - \xi_m)/\hbar} - i\pi\delta(\omega + (\xi_n - \xi_m)/\hbar) \right)$$

and hence

$$\text{Re} \left((e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}) \frac{1}{\omega + i\epsilon + (\xi_n - \xi_m)/\hbar} \right) = (e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}) \mathcal{P} \frac{1}{\omega + (\xi_n - \xi_m)/\hbar}$$

$$\text{Im} \left((e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}) \frac{1}{\omega + i\epsilon + (\xi_n - \xi_m)/\hbar} \right) = -(e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}) \pi\delta(\omega + (\xi_n - \xi_m)/\hbar)$$

Comparing the Results

The real parts obviously coincide. For the imaginary part we obtain, using the delta-distribution

$$\begin{aligned}
 & - \left(e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m} \right) \cdot \left(\coth \frac{\beta\hbar\omega}{2} \right)^{S_{X,X'}} \stackrel{!}{=} - \left(e^{-\beta\xi_n} + S_{X,X'} e^{-\beta\xi_m} \right) \\
 \Leftrightarrow & \left(\coth \frac{\beta\hbar\omega}{2} \right)^{S_{X,X'}} \stackrel{!}{=} \frac{e^{-\beta\xi_n} + S_{X,X'} e^{-\beta\xi_m}}{e^{-\beta\xi_n} - S_{X,X'} e^{-\beta\xi_m}} \equiv \frac{e^{-\beta\xi_n} + S_{X,X'} e^{-\beta(\xi_n + \hbar\omega)}}{e^{-\beta\xi_n} - S_{X,X'} e^{-\beta(\xi_n + \hbar\omega)}} = \frac{1 + S_{X,X'} e^{-\beta\hbar\omega}}{1 - S_{X,X'} e^{-\beta\hbar\omega}} = \frac{e^{\beta\hbar\omega/2} + S_{X,X'} e^{-\beta\hbar\omega/2}}{e^{\beta\hbar\omega/2} - S_{X,X'} e^{-\beta\hbar\omega/2}}
 \end{aligned}$$

Recalling now the definition of the cotangens-hyperbolicus we have proven the theorem. For the advanced Green-function the theorem is proven analogously.

Proof 31.43 of Lemma 14.14

We re-express the average particle number as lesser Green-functions and then use the fluctuation-dissipation-theorem

$$\begin{aligned}
 \langle N \rangle &= \sum_{\alpha \in \mathcal{A}} \langle c_{\alpha}^{\dagger} c_{\alpha} \rangle \\
 &= -\zeta \frac{\hbar}{i} \sum_{\alpha \in \mathcal{A}} G_{\alpha,\alpha}^{<}(t=0) \\
 &= -\zeta \frac{1}{2\pi i} \oint_{\substack{\alpha \in \mathcal{A} \\ E \in (-\infty, +\infty)}} G_{\alpha,\alpha}^{<}(E) \\
 &= \frac{-1}{\pi} \lim_{\eta \rightarrow 0} \oint_{\substack{\alpha \in \mathcal{A} \\ E \in (-\infty, +\infty)}} n_{\beta}^{\zeta}(E) \operatorname{Im} G_{\alpha,\alpha}^{\operatorname{ret}}(E + i\eta)
 \end{aligned} \tag{31.162}$$

Proof 31.44 of Lemma 14.15

For the first identity we compute

$$\begin{aligned}
 \delta_{\alpha,\alpha'} &= \langle \delta_{\alpha,\alpha'} \rangle \\
 &= \langle [c_{\alpha}, c_{\alpha'}^{\dagger}]_{\zeta} \rangle \\
 &= \langle c_{\alpha} c_{\alpha'}^{\dagger} \rangle - \zeta \langle c_{\alpha'}^{\dagger} c_{\alpha} \rangle \\
 &= -\frac{\hbar}{i} \left(G_{\alpha,\alpha'}^{>}(t=0) - G_{\alpha,\alpha'}^{<}(t=0) \right) \\
 &= -\frac{\hbar}{i} \int \frac{d\omega}{2\pi} \left(G_{\alpha,\alpha'}^{>}(\omega) - G_{\alpha,\alpha'}^{<}(\omega) \right) \\
 &= -\frac{\hbar}{i} \lim_{\eta \rightarrow 0} \int \frac{d\omega}{2\pi} 2i \operatorname{Im} G_{\alpha,\alpha'}^{\operatorname{ret}}(\omega + i\eta) \\
 &= \lim_{\eta \rightarrow 0} -\frac{\hbar}{\pi} \int d\omega \operatorname{Im} G_{\alpha,\alpha'}^{\operatorname{ret}}(\omega + i\eta) \\
 &\equiv \int d\omega A_{\alpha,\alpha'}(\omega)
 \end{aligned}$$

The second identity follows from the first identity with

$$d = \sum_{\alpha \in \mathcal{A}} \delta_{\alpha, \alpha} = \dots = \int d\omega \sum_{\alpha \in \mathcal{A}} \delta_{\alpha, \alpha} A_{\alpha, \alpha}(\omega) \quad (31.163)$$

The third identity follows immediately from Equation (31.157).

Proof 31.45 of Lemma 15.2

We compute straight forwardly

$$\begin{aligned} \partial_\tau U_{\text{int}}(\tau) &= e^{\tau \Xi_0 / \hbar} \frac{\Xi_0 - \Xi}{\hbar} e^{-\tau \Xi / \hbar} \\ &= -\frac{1}{\hbar} e^{\tau \Xi_0 / \hbar} V e^{-\tau \Xi / \hbar} \\ &= -\frac{1}{\hbar} e^{\tau \Xi_0 / \hbar} V e^{-\tau \Xi_0 / \hbar} e^{+\tau \Xi_0 / \hbar} e^{-\tau \Xi / \hbar} \\ &= -\frac{1}{\hbar} V_{H_{\Xi_0}}(\tau) U_{\text{int}}(\tau). \end{aligned} \quad (31.164)$$

Proof 31.46 of Corollary 15.3

By using repeated integrations on both sides, demanding $U(0) = 1$, we obtain

$$\begin{aligned} U_{\text{int}} &= 1 - \frac{1}{\hbar} \int_0^\tau d\tau_1 V_{H_{\Xi_0}}(\tau_1) U_{\text{int}}(\tau_1) \\ &= 1 + \left(-\frac{1}{\hbar}\right) \int_0^\tau d\tau_1 V_{H_{\Xi_0}}(\tau_1) U_{\text{int}}(\tau_1) + \left(-\frac{1}{\hbar}\right)^2 \int_0^\tau d\tau_1 \int_0^{\tau_1} d\tau_2 V_{H_{\Xi_0}}(\tau_1) V_{H_{\Xi_0}}(\tau_2) U_{\text{int}}(\tau_1) \\ &= \dots \\ &= \sum_{n=0}^{\infty} \left(-\frac{1}{\hbar}\right)^n \int_0^\tau d\tau_1 \dots \int_0^{\tau_{n-1}} d\tau_n V_{H_{\Xi_0}}(\tau_1) \dots V_{H_{\Xi_0}}(\tau_n) \\ &= \sum_{n=0}^{\infty} \left(-\frac{1}{\hbar}\right)^n \frac{1}{n!} \int_0^\tau d\tau_1 \dots \int_0^\tau d\tau_n T_\tau [V_{H_{\Xi_0}}(\tau_1) \dots V_{H_{\Xi_0}}(\tau_n)] \\ &\equiv T_\tau \exp \left[-\frac{1}{\hbar} \int_0^\tau d\tilde{\tau} V_{H_{\Xi_0}}(\tilde{\tau}) \right] \end{aligned} \quad (31.165)$$

whereas in the last steps we regrouped the summation as an ordered product.

Proof 31.47 of Lemma 15.4

We compute straight forwardly

$$\begin{aligned}
 \mathcal{G}_{X,X'}(\tau) &\equiv -\frac{1}{\hbar} \langle X(\tau) X'(0) \rangle \\
 &\equiv -\frac{1}{\hbar} \frac{\text{Tr}_{\mathcal{H}} e^{-\beta \Xi} X(\tau) X'(0)}{\text{Tr}_{\mathcal{H}} e^{-\beta \Xi}} \\
 &\equiv -\frac{1}{\hbar} \frac{\text{Tr}_{\mathcal{H}} e^{-\beta \Xi} e^{+\tau \Xi / \hbar} X(\tau) e^{-\tau \Xi / \hbar} X'(0)}{\text{Tr}_{\mathcal{H}} e^{-\beta \Xi}} \\
 &\equiv -\frac{1}{\hbar} \frac{\text{Tr}_{\mathcal{H}} e^{-\beta \Xi_0} e^{+\beta \Xi_0} e^{-\beta \Xi} e^{+\tau \Xi / \hbar} e^{-\beta \Xi_0} e^{+\beta \Xi_0} X e^{-\beta \Xi_0} e^{+\beta \Xi_0} e^{-\tau \Xi / \hbar} X'(0)}{\text{Tr}_{\mathcal{H}} e^{-\beta \Xi_0} e^{+\beta \Xi_0} e^{-\beta \Xi}} \\
 &\equiv -\frac{1}{\hbar} \frac{\text{Tr}_{\mathcal{H}} e^{-\beta \Xi_0} U_{\text{int}}(\beta) U_{\text{int}}^{-1}(\tau) X_{H_{\Xi_0}}(\tau) U_{\text{int}}(\tau) X'(0)}{\text{Tr}_{\mathcal{H}} e^{-\beta \Xi_0} U(\beta)}
 \end{aligned} \tag{31.166}$$

proving the Lemma.

Proof 31.48 of Theorem 15.7

We combine the prior results as

$$\begin{aligned}
 \mathcal{G}_{X,X'}(\tau) &\equiv -\frac{1}{\hbar} \langle T_{\tau} X(\tau) X'(0) \rangle \\
 &= -\frac{1}{\hbar} \langle X(\tau) X'(0) \rangle \\
 &= -\frac{1}{\hbar} \frac{\text{Tr}_{\mathcal{H}} e^{-\beta \Xi_0} U_{\text{int}}(\beta) U_{\text{int}}^{-1}(\tau) X_{H_{\Xi_0}}(\tau) U_{\text{int}}(\tau) X'(0)}{\text{Tr}_{\mathcal{H}} e^{-\beta \Xi_0} U(\beta)} \\
 &= -\frac{1}{\hbar} \frac{\text{Tr}_{\mathcal{H}} e^{-\beta \Xi_0} S(\beta, \tau) X_{H_{\Xi_0}}(\tau) S(\tau) X'_{H_{\Xi_0}}(0)}{\text{Tr}_{\mathcal{H}} e^{-\beta \Xi_0} S(\beta)} \\
 &= -\frac{1}{\hbar} \frac{\text{Tr}_{\mathcal{H}} e^{-\beta \Xi_0} T_{\tau} S(\beta, \tau) X_{H_{\Xi_0}}(\tau) S(\tau) X'_{H_{\Xi_0}}(0)}{\text{Tr}_{\mathcal{H}} e^{-\beta \Xi_0} S(\beta)} \\
 &= -\frac{1}{\hbar} \frac{\text{Tr}_{\mathcal{H}} e^{-\beta \Xi_0} S(\beta) X_{H_{\Xi_0}}(\tau) X'_{H_{\Xi_0}}(0)}{\text{Tr}_{\mathcal{H}} e^{-\beta \Xi_0} S(\beta)} \\
 &= -\frac{1}{\hbar} \frac{\langle S(\beta) X_{H_{\Xi_0}}(\tau) X'_{H_{\Xi_0}}(0) \rangle_0}{\langle S(\beta) \rangle_0}
 \end{aligned} \tag{31.167}$$

whereas we used that we had a time-ordered product, such that we could re-introduce the time-ordering operator, and we subsequently used that inside the time-ordering operator we could re-arrange terms at the cost of potential minus signs, and that the time-ordering operator is even in the creation- and annihilation-operators.

Proof 31.49 of Lemma 15.14 Notice that for typical phononic systems that one encounters in crystals, one obtains with Theorem 13.10 the identity

$$\mathcal{D}_{\nu, \nu'}^{-1}(\mathbf{q}, i\omega_n) = -\hbar \frac{\omega_n^2 + \omega_{\nu}^2(\mathbf{q})}{2\omega_{\nu}(\mathbf{q})} \delta_{\nu, \nu'}, \tag{31.168}$$

The correct procedure is to keep (31.168) in matrix form and invert the *full* matrix. Introduce the diagonal matrices (both of size $rd_{\mathbf{x}} \times rd_{\mathbf{x}}$, at fixed $\mathbf{q}$)

$$2\omega(\mathbf{q}) := 2 \operatorname{diag}_{rd_x \times rd_x} (\omega_1(\mathbf{q}), \omega_2(\mathbf{q}), \dots, \omega_{rd_x}(\mathbf{q})), \quad \omega^2(\mathbf{q}) := \operatorname{diag}_{rd_x \times rd_x} (\omega_1^2(\mathbf{q}), \omega_2^2(\mathbf{q}), \dots, \omega_{rd_x}^2(\mathbf{q})), \quad (31.169)$$

so that (31.168) reads, as a matrix equation,

$$\mathcal{D}_0^{-1}(\mathbf{q}, i\omega_n) = -\hbar (2\omega(\mathbf{q}))^{-1} \left[(\omega_n^2 \mathbb{1}_{rd_x \times rd_x} + \omega^2(\mathbf{q})) \right]. \quad (31.170)$$

Using the fact that $\omega^{-1}(\mathbf{q}) \cdot \omega(\mathbf{q}) = \mathbb{1}$ we can factor

$$\begin{aligned} \mathcal{D}^{-1}(\mathbf{q}, i\omega_n) &= \mathcal{D}_0^{-1}(\mathbf{q}, i\omega_n) - \Pi(\mathbf{q}, i\omega_n) \\ &= -\hbar (2\omega(\mathbf{q}))^{-1} \left[(\omega_n^2 \mathbb{1} + \Omega^2(\mathbf{q})) + \frac{2}{\hbar} \omega(\mathbf{q}) \Pi(\mathbf{q}, i\omega_n) \right], \end{aligned} \quad (31.171)$$

Inverting both sides gives then

$$\begin{aligned} \mathcal{D}(\mathbf{q}, i\omega_n) &= -\frac{1}{\hbar} \left[(\omega_n^2 \mathbb{1}_{rd_x \times rd_x} + \omega^2(\mathbf{q})) + \frac{2}{\hbar} \omega(\mathbf{q}) \Pi(\mathbf{q}, i\omega_n) \right]^{-1} 2\omega(\mathbf{q}), \\ &= -\frac{1}{\hbar} \frac{2\omega(\mathbf{q})}{(\omega_n^2 \mathbb{1}_{rd_x \times rd_x} + \omega^2(\mathbf{q})) + \frac{2}{\hbar} \omega(\mathbf{q}) \Pi(\mathbf{q}, i\omega_n)} \end{aligned} \quad (31.172)$$

Proof 31.50 of Lemma 15.18 We write $z = i\omega_n + \Delta z$ and consider

$$\begin{aligned} \lim_{\Delta z \rightarrow 0} \Delta z g_\zeta(i\omega_n + \Delta z) &= \lim_{\Delta z \rightarrow 0} \Delta z \frac{\hbar\beta}{e^{i\omega_n \hbar\beta} e^{\hbar\beta \Delta z} - \zeta} \\ &= \lim_{\Delta z \rightarrow 0} \Delta z \frac{\hbar\beta}{e^{\hbar\beta \Delta z} - 1} \zeta \\ &= \lim_{\Delta z \rightarrow 0} \frac{\hbar\beta \Delta z}{\hbar\beta \Delta z + \frac{1}{2}(\hbar\beta \Delta z)^2 + \dots} \zeta \\ &= \zeta. \end{aligned}$$

Proof 31.51 of Theorem 15.20

We compute

$$\operatorname{Res}(f_\zeta(z), z = i\omega_n) = \lim_{z \rightarrow i\omega_n} (z - i\omega_n) g_\zeta(z) h(-iz) = \zeta h(\omega_n)$$

and obtain therefore

$$\begin{aligned}
 0 &= \frac{1}{2\pi i} \oint_{\gamma} dz f_{\zeta}(z) \iff 0 = \sum_{z_{\star} \in P} \text{Res}(f(z), z = z_{\star}) \\
 &\iff 0 = \sum_{z_{\star} \in Z} \text{Res}(f(z), z = z_{\star}) + \sum_{i\omega_n \in \Omega_{\zeta}} \text{Res}(f(z), z = i\omega_n) \\
 &\iff S = -\zeta \sum_{z_{\star} \in Z} \text{Res}(f(z), z = z_{\star})
 \end{aligned}$$

Proof 31.52 of Corollary 15.22

One obtains the function

$$\mathbb{C} - \{C\} \ni z \mapsto h(-iz) \equiv \frac{1}{\hbar\beta} \frac{1}{(z - C)} \in \mathbb{C} \quad (31.173)$$

Therefore, with the Lemma above, we obtain

$$\sum_{\omega_n \in \Omega_{\zeta}} \frac{1}{\hbar} \frac{1}{i\omega_n - C} = -\frac{\zeta}{\hbar\beta} g_{\zeta}(C) = -\zeta n_{\hbar\beta}^{\zeta}(C) = -\zeta n_{\hbar\beta}^{\zeta}(C) \quad (31.174)$$

Proof 31.53 of Corollary 15.23

One obtains the function

$$\mathbb{C} - \{z_1, z_2, z_3\} \ni z \mapsto h(-iz) \equiv \frac{1}{\hbar^2\beta} \frac{1}{(z - C_1)} \frac{-2C_2}{(\omega_m + iz)^2 + C_2^2} \in \mathbb{C} \quad (31.175)$$

The first pole of this function is $z_1 = C_1$. The other two poles are determined by the equation

$$\begin{aligned}
 0 &= (\omega_m + iz)^2 + C_2^2 \iff \pm iC_2 = \omega_m + iz_{\pm} \\
 &\iff iz_{\pm} = \pm iC_2 - \omega_m \\
 &\iff z_{\pm} \equiv z_{2,3} = \mp C_2 + i\omega_m
 \end{aligned}$$

Hence

$$z_1 = C_1, \quad z_2 = -C_2 + i\omega_m, \quad z_3 = +C_2 + i\omega_m \quad (31.176)$$

and since

$$\begin{aligned}
 (z - z_2)(z - z_3) &= (z + C_2 - i\omega_m)(z - C_2 - i\omega_m) \\
 &= (z - i\omega_m)^2 - C_2^2 \\
 &= -i^2 ((z - i\omega_m)^2 - C_2^2) \\
 &= -((\omega_m + iz)^2 + C_2^2)
 \end{aligned}$$

we conclude that

$$\text{const.} \equiv \frac{2C_2}{\hbar^2 \beta} \quad (31.177)$$

Therefore we can now write down the products appearing in Equation (15.29) to evaluate the Matsubara-summation

$$k = 1 : \quad n_{\hbar\beta}^\zeta(z_1) = F_{\hbar\beta}(C_1), \quad z_1 - z_2 = C_1 + C_2 - i\omega_m, \quad z_1 - z_3 = C_1 - C_2 - i\omega_m$$

$$k = 2 : \quad n_{\hbar\beta}^\zeta(z_1) = F_{\hbar\beta}(-C_2 + i\omega_m), \quad z_2 - z_1 = -C_2 + i\omega_m - C_1, \quad z_2 - z_3 = -C_2 + i\omega_m - C_2 - i\omega_m = -2C_2$$

$$k = 3 : \quad n_{\hbar\beta}^\zeta(z_1) = F_{\hbar\beta}(+C_2 + i\omega_m), \quad z_3 - z_1 = C_2 + i\omega_m - C_1, \quad z_3 - z_2 = C_2 + i\omega_m + C_2 - i\omega_m = 2C_2$$

Using the identities in Lemma 9.47 we obtain therefore

$$k = 1 : \quad n_{\hbar\beta}^\zeta(z_1) = F_{\hbar\beta}(C_1), \quad z_1 - z_2 = C_1 + C_2 - i\omega_m, \quad z_1 - z_3 = C_1 - C_2 - i\omega_m$$

$$k = 2 : \quad n_{\hbar\beta}^\zeta(z_1) = B_{\hbar\beta}(C_2) + 1, \quad z_2 - z_1 = -C_2 + i\omega_m - C_1, \quad z_2 - z_3 = -2C_2$$

$$k = 3 : \quad n_{\hbar\beta}^\zeta(z_1) = -B_{\hbar\beta}(C_2), \quad z_3 - z_1 = C_2 + i\omega_m - C_1, \quad z_3 - z_2 = 2C_2$$

This gives with Equation (15.29) the intermediate result

$$\begin{aligned} S_{-1} &= \sum_{\omega_n \in \Omega_{-1}} h(\omega_n) \\ &= +1\hbar\beta \text{ const} \sum_{k=1}^N n_{\hbar\beta}^\zeta(z_k) \prod_{\substack{l=1 \\ l \neq k}}^N \frac{1}{z_k - z_l} \\ &= \frac{2C_2}{\hbar} \left[\frac{F_{\hbar\beta}(C_1)}{(C_1 + C_2 - i\omega_m)(C_1 - C_2 - i\omega_m)} + \frac{B_{\hbar\beta}(C_2) + 1}{(-C_2 + i\omega_m - C_1)(-2C_2)} + \frac{-B_{\hbar\beta}(C_2)}{(C_2 + i\omega_m - C_1)(2C_2)} \right] \\ &= \frac{1}{\hbar} \left[\frac{2C_2 F_{\hbar\beta}(C_1)}{(i\omega_m - C_1 - C_2)(i\omega_m - C_1 + C_2)} - \frac{B_{\hbar\beta}(C_2) + 1}{-C_2 + i\omega_m - C_1} - \frac{B_{\hbar\beta}(C_2)}{C_2 + i\omega_m - C_1} \right] \\ &= \star \star \star \end{aligned}$$

We now use the partial fraction expansion (cf. WolframAlpha)

$$\frac{2C_2}{(x + C_2)(x - C_2)} = -\frac{1}{x + C_2} + \frac{1}{x - C_2} \quad (31.178)$$

and obtain

$$\begin{aligned}
\star \star \star &= \frac{1}{\hbar} \left[\frac{-F_{\hbar\beta}(C_1)}{i\omega_m - C_1 + C_2} + \frac{F_{\hbar\beta}(C_1)}{i\omega_m - C_1 - C_2} - \frac{B_{\hbar\beta}(C_2) + 1}{-C_2 + i\omega_m - C_1} - \frac{B_{\hbar\beta}(C_2)}{C_2 + i\omega_m - C_1} \right] \\
&= \frac{-1}{\hbar} \left[\frac{B_{\hbar\beta}(C_2) + F_{\hbar\beta}(C_1)}{i\omega_m - C_1 + C_2} + \frac{B_{\hbar\beta}(C_2) + 1 - F_{\hbar\beta}(C_1)}{i\omega_m - C_1 - C_2} \right]
\end{aligned}$$

32. Appendix - Proofs for Part III

Proof 32.1 of Lemma 16.6 For $\zeta = +1$ the invertibility follows from basic algebraic rules. $\zeta = -1$, we prove that Equation (16.7) is indeed the inverse element of f . For this, we compute

$$\begin{aligned}
 f^{-1}(\psi, \bar{\psi})f(\psi, \bar{\psi}) &= \frac{1}{f_0} \sum_{n=0}^{N_f-1} \left(-\frac{f_{\text{nil}}(\psi, \bar{\psi})}{f_0} \right)^n (f_0 + f_{\text{nil}}) \\
 &= \sum_{n=0}^{N_f-1} \left(-\frac{f_{\text{nil}}(\psi, \bar{\psi})}{f_0} \right)^n + \frac{f_{\text{nil}}}{f_0} \sum_{n=0}^{N_f-1} \left(-\frac{f_{\text{nil}}(\psi, \bar{\psi})}{f_0} \right)^n \\
 &= \sum_{n=0}^{N_f-1} \left(-\frac{f_{\text{nil}}(\psi, \bar{\psi})}{f_0} \right)^n - \sum_{n=1}^{N_f} \left(-\frac{f_{\text{nil}}(\psi, \bar{\psi})}{f_0} \right)^n \\
 &= 1 + 0 + 0
 \end{aligned} \tag{32.1}$$

Proof 32.2 of Lemma 16.14 The bosonic case is trivial. In the fermionic case we obtain by direct computation for the first identity

$$\exp \sum_k \bar{j}_k \psi_k = \prod_k \exp \bar{j}_k \psi_k = \exp \bar{j}_i \psi_i \prod_{\substack{k \\ k \neq i}} \exp \bar{j}_k \psi_k \tag{32.2}$$

which can be combined with

$$\frac{\partial}{\partial \bar{j}_i} \exp \bar{j}_i \psi_i = \frac{\partial}{\partial \bar{j}_i} (1 + \bar{j}_i \psi_i) = \psi_i = \psi_i (1 + \bar{j}_i \psi_i) \tag{32.3}$$

and

$$\frac{\partial}{\partial j_{i'}} \exp \bar{\psi}_{i'} j_{i'} = \frac{\partial}{\partial j_{i'}} (1 + \bar{\psi}_{i'} j_{i'}) = -\bar{\psi}_{i'} = -\bar{\psi}_{i'} (1 + \bar{\psi}_{i'} j_{i'}) \tag{32.4}$$

Proof 32.3 of Lemma 16.32

We will provide two proofs of this identity. One direct proof (presented here), and another proof based on symmetry-principles, presented in Chapter 17. We simply compute

$$\begin{aligned}
 W[j, \bar{j}] &= \log \left\langle e^{-(j^\dagger \psi + \psi^\dagger j)} \right\rangle \\
 &= \log \left\langle \exp - \sum_{\alpha} (j^\dagger \psi + \psi^\dagger j) \right\rangle \\
 &= \log \left\langle \left(\prod_{\alpha} \exp - j^\dagger \psi \right) \left(\prod_{\alpha} \exp - \psi^\dagger j \right) \right\rangle
 \end{aligned} \tag{32.5}$$

Now taking the expectation-value we see that W is *even* in the fields j and $\bar{j}$ and therefore ϕ and $\bar{\phi}$ are odd in the fields.

Proof 32.4 of Lemma 16.33

Iteratively we can construct the desired expansion. We lighten the notation and write

$$W = \bar{j} G j + \frac{1}{4} W^{(4)} \bar{j} j j j + \dots \tag{32.6}$$

This gives then (in the fermionic (the bosonic case works analogously) case) for the functional derivatives

$$\phi = \frac{\partial W}{\partial \bar{j}} = G j + \frac{1}{2} W^{(4)} \bar{j} j j j + \dots \tag{32.7}$$

and

$$\bar{\phi} = -\frac{\partial W}{\partial j} = \bar{j} G + \frac{1}{2} W^{(4)} \bar{j} j j j + \dots \tag{32.8}$$

We therefore find the coupled nonlinear system

$$\begin{aligned}
 \phi &= G j + \mathcal{O}(\bar{j} j j j) \\
 \bar{\phi} &= \bar{j} G + \mathcal{O}(\bar{j} j j j)
 \end{aligned} \tag{32.9}$$

so unlike the scalar field theorz case that one knows from textbooks, both equations must be inverted *simultaneously*. We seek to construct therefore

$$j \equiv j[\phi, \bar{\phi}], \quad \bar{j} \equiv \bar{j}[\phi, \bar{\phi}] \tag{32.10}$$

The clue (that is missing from every book that the author opened so far) is now to understand this as an inversion of a *power series*.

The linear Part.

We consider first the linear part

$$\phi = G j, \quad \bar{\phi} = \bar{j} G \tag{32.11}$$

which can be inverted as

$$j = G^{-1}\phi, \quad \bar{j} = \bar{\phi}G^{-1} \quad (32.12)$$

and therefore, the requirement for the inversion of the linear equation is the invertibility of the connected Green-function. Once the linear part is invertible, the nonlinear corrections are constructed *order by order*.

The next Order.

We write

$$\begin{aligned} j &= G^{-1}\phi + A(\bar{\phi}, \phi, \phi) + \dots \\ \bar{j} &= \bar{\phi}G^{-1} + B(\bar{\phi}, \bar{\phi}, \phi) + \dots \end{aligned} \quad (32.13)$$

whereas structurally one has something of the form

$$A \sim \bar{\phi}\phi\phi, \quad B \sim \bar{\phi}\bar{\phi}\phi \quad (32.14)$$

Next we plug this into the equation for ϕ

$$\phi = Gj + \frac{1}{2}W^{(4)}\bar{j}jj + \dots \quad (32.15)$$

then using

$$j = G^{-1}\phi + A + \dots \quad (32.16)$$

yields

$$Gj = G(G^{-1}\phi + A) + \dots \quad (32.17)$$

and therefore

$$Gj = \phi + GA + \dots \quad (32.18)$$

This term GA needs to cancel out now with the other second order term. At the order at which A first appears, we only need the leading linear expression for j and $\bar{j}$:

$$j = G^{-1}\phi, \quad \bar{j} = \bar{\phi}G^{-1} \quad (32.19)$$

and therefore

$$\bar{j}jj = (\bar{\phi}G^{-1})(G^{-1}\phi)(G^{-1}\phi) \quad (32.20)$$

and therefore we obtain the equation

$$\frac{1}{2}W^{(4)}\bar{j}jj = \frac{1}{2}W^{(4)}(\bar{\phi}G^{-1})(G^{-1}\phi)^2 \quad (32.21)$$

Now we aim to match the terms, and therefore we are looking into the defining equations, meaning

$$\phi = \phi + GA + \frac{1}{2}W^{(4)}\left(\bar{\phi}G^{-1}\right)\left(G^{-1}\phi\right)^2 \quad (32.22)$$

Now we cancel ϕ on both sides, such that

$$0 = GA + \frac{1}{2}W^{(4)}\left(\bar{\phi}G^{-1}\right)\left(G^{-1}\phi\right)^2. \quad (32.23)$$

Now we solve this equation for A, meaning that we multiply by G^{-1} giving us

$$A = -G^{-1}\frac{1}{2}W^{(4)}\left(\bar{\phi}G^{-1}\right)\left(G^{-1}\phi\right)^2 \quad (32.24)$$

Thus, to this order, we obtain

$$j = G^{-1}\phi - G^{-1}\frac{1}{2}W^{(4)}\left(\bar{\phi}G^{-1}\right)\left(G^{-1}\phi\right)^2 + \dots \quad (32.25)$$

which is the first nonlinear correction to the inverse relation $j[\phi, \bar{\phi}]$.

We will do it similarly for $\bar{j}$

Performing analogous calculations using the equation for $\bar{\phi}$ one obtains

$$B = -\frac{1}{2}\left(\bar{\phi}G^{-1}\right)^2\left(G^{-1}\phi\right)W^{(4)} \cdot G^{-1} \quad (32.26)$$

hence we obtain

$$\bar{j} = \bar{\phi}G^{-1} - \frac{1}{2}\left(\bar{\phi}G^{-1}\right)^2\left(G^{-1}\phi\right)W^{(4)}G^{-1} + \dots \quad (32.27)$$

Proof 32.5 of Lemma 16.37

We use Equation (16.46) and simplify the notation as

$$F[\pi, \bar{\pi}] \equiv \sum_{\substack{i_1, i_2 \\ i'_1, i'_2}} \bar{\pi}(i_1)\bar{\pi}(i_2)\tilde{f}(i_1, i_2, i'_1, i'_2)\pi(i'_2)\pi(i'_1) \equiv \sum_{\substack{1, 2 \\ 1', 2'}} \bar{\pi}(1)\bar{\pi}(2)\tilde{f}(1, 2, 1', 2')\pi(2')\pi(1') = \dots \quad (32.28)$$

now we decompose the interaction f as

$$f \equiv f^{\text{sym, sym}} + f^{\text{sym, as}} + f^{\text{as, sym}} + f^{\text{as, as}} \quad (32.29)$$

with

$$\begin{aligned}
 \bullet \quad f^{\text{sym, sym}}(1, 2, 3, 4) &\equiv \frac{1}{4} \left(\tilde{f}(1, 2, 3, 4) + \tilde{f}(2, 1, 3, 4) + \tilde{f}(1, 2, 4, 3) + \tilde{f}(2, 1, 4, 3) \right) \\
 \bullet \quad f^{\text{as, sym}}(1, 2, 3, 4) &\equiv \frac{1}{4} \left(\tilde{f}(1, 2, 3, 4) - \tilde{f}(2, 1, 3, 4) + \tilde{f}(1, 2, 4, 3) - \tilde{f}(2, 1, 4, 3) \right) \\
 \bullet \quad f^{\text{sym, as}}(1, 2, 3, 4) &\equiv \frac{1}{4} \left(\tilde{f}(1, 2, 3, 4) + \tilde{f}(2, 1, 3, 4) - \tilde{f}(1, 2, 4, 3) - \tilde{f}(2, 1, 4, 3) \right) \\
 \bullet \quad f^{\text{as, as}}(1, 2, 3, 4) &\equiv \frac{1}{4} \left(\tilde{f}(1, 2, 3, 4) - \tilde{f}(2, 1, 3, 4) - \tilde{f}(1, 2, 4, 3) + \tilde{f}(2, 1, 4, 3) \right)
 \end{aligned} \tag{32.30}$$

such that we have

$$f^{\bullet, \bullet'}(\sigma(1), \sigma(2), \sigma'(3), \sigma'(4)) = (\text{sgn}(\sigma))^{n(\bullet)} (\text{sgn}(\sigma'))^{n(\bullet')} f(1, 2, 3, 4) \tag{32.31}$$

with

$$n(\bullet) = \begin{cases} 1, & \bullet = \text{as} \\ 0, & \bullet = \text{sym} \end{cases} \tag{32.32}$$

whereas we obtain

$$f = \sum_{\bullet, \bullet' \in \text{as, sym}} f^{\bullet, \bullet'} \tag{32.33}$$

This gives, for example for $\zeta = -1$

$$\sum_{1,2,3,4} \bar{\pi}_1 \bar{\pi}_2 f^{\text{sym, as}}(1, 2, 3, 4) \psi_4 \psi_3 \stackrel{1 \rightarrow 2, 2 \rightarrow 1}{=} \sum_{1,2,3,4} \underbrace{\bar{\pi}_2 \bar{\pi}_1}_{-\bar{\pi}_1 \bar{\pi}_2} \underbrace{f^{\text{sym, as}}(2, 1, 3, 4)}_{f^{\text{sym, as}}(1, 2, 3, 4)} \psi_4 \psi_3 = - \sum_{1,2,3,4} \bar{\pi}_1 \bar{\pi}_2 f^{\text{sym, as}}(1, 2, 3, 4) \psi_4 \psi_3 = 0 \tag{32.34}$$

and similar for $\zeta = +1$ (except that one needs to use the (anti)-symmetry of the respective other part of the coupling-function).

Proof 32.6 of Lemma 16.39.

We compute the derivatives explicitly and find the identity

$$\begin{aligned}
 &\frac{\partial^{n+m}}{\partial \bar{\pi}(i_1) \cdots \partial \bar{\pi}(i_n) \partial \pi(i'_m) \cdots \partial \pi(i'_1)} F[\bar{\pi}, \pi] \Big|_{\bar{\pi}=\pi=0} \\
 &= \frac{\partial^{n+m}}{\partial \bar{\pi}(i_1) \cdots \partial \bar{\pi}(i_n) \partial \pi(i'_m) \cdots \partial \pi(i'_1)} \sum_{n,m=0}^{\infty} \sum_{\substack{i_1 < \dots < i_n \\ i'_1 < i'_2 < \dots < i'_m}} \bar{\pi}(i_1) \cdots \bar{\pi}(i_n) f(i_1, \dots, i_n, i'_1, \dots, i'_m) \pi(i'_m) \cdots \pi(i'_1) \\
 &= \zeta^{nm} \sum_{n,m=0}^{\infty} \sum_{\substack{i_1 < \dots < i_n \\ i'_1 < i'_2 < \dots < i'_m}} \frac{\partial^n}{\partial \bar{\pi}(i_1) \cdots \partial \bar{\pi}(i_n)} \bar{\pi}(i_1) \cdots \bar{\pi}(i_n) f(i_1, \dots, i_n, i'_1, \dots, i'_m) \frac{\partial^n}{\partial \pi(i'_m) \cdots \partial \pi(i'_1)} \pi(i'_m) \cdots \pi(i'_1) \\
 &= \zeta^{nm+n(n-1)/2+m(m-1)/2} f(i_1, \dots, i_n, i'_1, \dots, i'_m)
 \end{aligned} \tag{32.35}$$

Proof 32.7 of Lemma 16.35. One has¹

$$\begin{aligned}
 \frac{\partial}{\partial \phi(\bullet)} \Gamma[\phi, \bar{\phi}] &= - \sum_x \left(\frac{\partial j(x)}{\partial \phi(\bullet)}, \frac{\partial \bar{j}(x)}{\partial \phi(\bullet)} \right) \cdot \left(\frac{\partial}{\partial \bar{j}(x)} W[j, \bar{j}] \right) + \zeta \bar{j}(\bullet) + \sum_x \left(\frac{\partial \bar{j}(x)}{\partial \phi(\bullet)} \phi(x) + \zeta \bar{\phi}(x) \frac{\partial j(x)}{\partial \phi(\bullet)} \right) \\
 &= - \sum_x \left(\frac{\partial j(x)}{\partial \phi(\bullet)}, \frac{\partial \bar{j}(x)}{\partial \phi(\bullet)} \right) \cdot \left(\frac{\zeta \bar{\phi}(x)}{\phi(x)} \right) + \zeta \bar{j}(\bullet) + \sum_x \left(\frac{\partial \bar{j}(x)}{\partial \phi(\bullet)} \phi(x) + \zeta \bar{\phi}(x) \frac{\partial j(x)}{\partial \phi(\bullet)} \right) \\
 &= - \sum_x \left(\frac{\partial j(x)}{\partial \phi(\bullet)} \zeta \bar{\phi}(x) + \frac{\partial \bar{j}(x)}{\partial \phi(\bullet)} \phi(x) \right) + \zeta \bar{j}(\bullet) + \sum_x \left(\frac{\partial \bar{j}(x)}{\partial \phi(\bullet)} \phi(x) + \zeta \bar{\phi}(x) \frac{\partial j(x)}{\partial \phi(\bullet)} \right) \\
 &= - \sum_x \left(\frac{\partial j(x)}{\partial \phi(\bullet)} \zeta \bar{\phi}(x) \right) + \zeta \bar{j}(\bullet) + \sum_x \left(\zeta \bar{\phi}(x) \frac{\partial j(x)}{\partial \phi(\bullet)} \right) \\
 &= \zeta \bar{j}(\bullet),
 \end{aligned}$$

and

$$\begin{aligned}
 \frac{\partial}{\partial \bar{\phi}(\bullet)} \Gamma[\phi, \bar{\phi}] &= - \sum_x \left(\frac{\partial j(x)}{\partial \bar{\phi}(\bullet)}, \frac{\partial \bar{j}(x)}{\partial \bar{\phi}(\bullet)} \right) \cdot \left(\frac{\partial}{\partial \bar{j}(x)} W[j, \bar{j}] \right) + j(\bullet) + \sum_x \left(\frac{\partial \bar{j}(x)}{\partial \bar{\phi}(\bullet)} \phi(x) + \zeta \bar{\phi}(x) \frac{\partial j(x)}{\partial \bar{\phi}(\bullet)} \right) \\
 &= - \sum_x \left(\frac{\partial j(x)}{\partial \bar{\phi}(\bullet)}, \frac{\partial \bar{j}(x)}{\partial \bar{\phi}(\bullet)} \right) \cdot \left(\frac{\zeta \bar{\phi}(x)}{\phi(x)} \right) + j(\bullet) + \sum_x \left(\frac{\partial \bar{j}(x)}{\partial \bar{\phi}(\bullet)} \phi(x) + \zeta \bar{\phi}(x) \frac{\partial j(x)}{\partial \bar{\phi}(\bullet)} \right) \\
 &= +j(\bullet).
 \end{aligned}$$

Proof 32.8 of Lemma 16.40

We use Corollary 16.15 and find

$$\begin{aligned}
 \frac{\partial^n}{\partial \bar{j}(i_1) \cdots \partial \bar{j}(i_n)} \frac{\partial^n}{\partial j(i'_1) \cdots \partial j(i'_n)} \frac{Z[\bar{j}, j]}{Z} &= \frac{\partial^n}{\partial \bar{j}(i_1) \cdots \partial \bar{j}(i_n)} \zeta^n \left\langle \bar{\psi}(i'_n) \cdots \bar{\psi}(i'_1) \exp \left[j^\dagger \psi + \psi^\dagger j \right] \right\rangle \\
 &= \pm \zeta^n \left\langle \bar{\psi}(i'_n) \cdots \bar{\psi}(i'_1) \frac{\partial^n}{\partial \bar{j}(i_1) \cdots \partial \bar{j}(i_n)} \exp \left[j^\dagger \psi + \psi^\dagger j \right] \right\rangle \\
 &= \pm \zeta^n \left\langle \bar{\psi}(i'_n) \cdots \bar{\psi}(i'_1) \psi(i_1) \cdots \psi(i_n) \exp \left[j^\dagger \psi + \psi^\dagger j \right] \right\rangle \\
 &= + \zeta^n \left\langle \psi(i_1) \cdots \psi(i_n) \bar{\psi}(i'_n) \cdots \bar{\psi}(i'_1) \exp \left[j^\dagger \psi + \psi^\dagger j \right] \right\rangle
 \end{aligned}$$

Therefore we obtain

$$\begin{aligned}
 (-\zeta/c)^n \frac{\partial^n}{\partial \bar{j}(i_1) \cdots \partial \bar{j}(i_n)} \frac{\partial^n}{\partial j(i'_1) \cdots \partial j(i'_n)} \frac{Z[\bar{j}, j]}{Z} &= + (-1/c)^n \left\langle \psi(i_1) \cdots \psi(i_n) \bar{\psi}(i'_n) \cdots \bar{\psi}(i'_1) \exp \left[j^\dagger \psi + \psi^\dagger j \right] \right\rangle \\
 &= \dots
 \end{aligned}$$

¹Cf. the derivative-rule for Grassmann-vector-valued functions. Cf. the differentiation-rule from Zinn-Justin on page 8!

comparing this now with Equation (16.33) proves the Lemma

$$\dots = \mathcal{G}(i_1, \dots, i_n \mid i'_1, \dots, i'_n). \quad (32.36)$$

Proof 32.9 of Corollary 16.43

We compute

$$\begin{aligned}
 \mathcal{G}_c(i_1|i'_1) &= -\left(\frac{\zeta}{c}\right) \left(\frac{\partial^2}{\partial \bar{j}(i_1) \partial j(i'_1)} W \right) [j, \bar{j}]_{|j=\bar{j}=0} \\
 &= -\left(\frac{\zeta}{c}\right) \partial_{\bar{j}(i_1)} \partial_{j(i'_1)} \log \left\langle \exp \left[j^\dagger \psi + \psi^\dagger j \right] \right\rangle_{|j=\bar{j}=0} \\
 &= -\left(\frac{\zeta}{c}\right) \partial_{\bar{j}(i_1)} \left\langle \exp \left[j^\dagger \psi + \psi^\dagger j \right] \right\rangle^{-1} \left\langle \zeta \bar{\psi}(i'_1) \exp \left[j^\dagger \psi + \psi^\dagger j \right] \right\rangle_{|j=\bar{j}=0} \\
 &= -\left(\frac{1}{c}\right) \partial_{\bar{j}(i_1)} \left\langle \exp \left[j^\dagger \psi + \psi^\dagger j \right] \right\rangle^{-1} \left\langle \bar{\psi}(i'_1) \exp \left[j^\dagger \psi + \psi^\dagger j \right] \right\rangle_{|j=\bar{j}=0} \\
 &= -\left(\frac{1}{c}\right) \left[\left\langle \exp \left[j^\dagger \psi + \psi^\dagger j \right] \right\rangle^{-2} \left\langle \psi(i_1) \exp \left[j^\dagger \psi + \psi^\dagger j \right] \right\rangle \left\langle \bar{\psi}(i'_1) \exp \left[j^\dagger \psi + \psi^\dagger j \right] \right\rangle \right. \\
 &\quad \left. + \left\langle \exp \left[j^\dagger \psi + \psi^\dagger j \right] \right\rangle^{-2} \left\langle \zeta \bar{\psi}(i'_1) \psi(i_1) \exp \left[j^\dagger \psi + \psi^\dagger j \right] \right\rangle \right]_{|j=\bar{j}=0} \\
 &= -\frac{\zeta}{c} \langle \bar{\psi}(i'_1) \psi(i_1) \rangle \\
 &= -\frac{1}{c} \langle \psi(i_1) \bar{\psi}(i'_1) \rangle \\
 &= \mathcal{G}(i_1|i'_1)
 \end{aligned} \tag{32.37}$$

For the other formula: Cf. Negele and Orland somewhere around page 107, 108.

Proof 32.10 of Lemma 16.44 We compute straight-forwardly

$$\begin{aligned}
 \frac{\partial}{\partial \phi(\bullet)} F[\bar{j}, j] &= \sum_x \left(\frac{\partial F}{\partial \bar{j}(x)}, \frac{\partial F}{\partial j(x)} \right) \cdot \left(\frac{\frac{\partial \bar{j}(x)}{\partial \phi(\bullet)}}{\frac{\partial j(x)}{\partial \phi(\bullet)}} \right) \\
 &= \sum_x \frac{\partial F}{\partial \bar{j}(x)} \frac{\partial \bar{j}(x)}{\partial \phi(\bullet)} + \frac{\partial F}{\partial j(x)} \frac{\partial j(x)}{\partial \phi(\bullet)} \\
 &= \sum_x \zeta \frac{\partial F}{\partial \bar{j}(x)} \frac{\partial^2 \Gamma}{\partial \phi(\bullet) \partial \phi(x)} + \frac{\partial F}{\partial j(x)} \frac{\partial^2 \Gamma}{\partial \phi(\bullet) \partial \phi(x)}
 \end{aligned} \tag{32.38}$$

and

$$\begin{aligned}
 \frac{\partial}{\partial \bar{\phi}(\bullet)} F[\bar{j}, j] &= \sum_x \left(\frac{\partial F}{\partial \bar{j}(x)}, \frac{\partial F}{\partial j(x)} \right) \cdot \left(\frac{\frac{\partial \bar{j}(x)}{\partial \bar{\phi}(\bullet)}}{\frac{\partial j(x)}{\partial \bar{\phi}(\bullet)}} \right) \\
 &= \sum_x \frac{\partial F}{\partial \bar{j}(x)} \frac{\partial \bar{j}(x)}{\partial \bar{\phi}(\bullet)} + \frac{\partial F}{\partial j(x)} \frac{\partial j(x)}{\partial \bar{\phi}(\bullet)} \\
 &= \sum_x \zeta \frac{\partial F}{\partial \bar{j}(x)} \frac{\partial^2 \Gamma}{\partial \bar{\phi}(\bullet) \partial \phi(x)} + \frac{\partial F}{\partial j(x)} \frac{\partial^2 \Gamma}{\partial \bar{\phi}(\bullet) \partial \phi(x)}
 \end{aligned} \tag{32.39}$$

Proof 32.11 of Corollary 16.45 We use the Lemma and compute ...

The first equation:

$$\begin{aligned}\delta_{1,3} &= \frac{\partial\phi(3)}{\partial\phi(1)} = \frac{\partial}{\partial\phi(1)} \frac{\partial W}{\partial\bar{j}(3)} \\ &= \sum_2 \zeta \left(\frac{\partial^2 W}{\partial\bar{j}(2)\partial\bar{j}(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial\phi(1)\partial\phi(2)} \right) + \left(\frac{\partial^2 W}{\partial j(2)\partial\bar{j}(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial\phi(1)\partial\bar{\phi}(2)} \right)\end{aligned}\quad (32.40)$$

The second equation:

$$\begin{aligned}\delta_{1,3} &= \frac{\partial\bar{\phi}(3)}{\partial\bar{\phi}(1)} = \zeta \frac{\partial}{\partial\phi(1)} \frac{\partial W}{\partial j(3)} \\ &= \zeta \sum_2 \zeta \left(\frac{\partial^2 W}{\partial\bar{j}(2)\partial j(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial\bar{\phi}(1)\partial\phi(2)} \right) + \left(\frac{\partial^2 W}{\partial j(2)\partial j(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial\bar{\phi}(1)\partial\bar{\phi}(2)} \right) \\ &= \sum_2 \left(\frac{\partial^2 W}{\partial\bar{j}(2)\partial j(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial\bar{\phi}(1)\partial\phi(2)} \right) + \zeta \left(\frac{\partial^2 W}{\partial j(2)\partial j(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial\bar{\phi}(1)\partial\bar{\phi}(2)} \right)\end{aligned}\quad (32.41)$$

The third equation:

$$\begin{aligned}0 &= \frac{\partial\phi(3)}{\partial\bar{\phi}(1)} = \frac{\partial}{\partial\bar{\phi}(1)} \frac{\partial W}{\partial\bar{j}(3)} \\ &= \sum_2 \zeta \left(\frac{\partial^2 W}{\partial\bar{j}(2)\partial\bar{j}(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial\bar{\phi}(1)\partial\phi(2)} \right) + \left(\frac{\partial^2 W}{\partial j(2)\partial\bar{j}(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial\bar{\phi}(1)\partial\bar{\phi}(2)} \right)\end{aligned}\quad (32.42)$$

The fourth equation:

$$\begin{aligned}0 &= \frac{\partial\bar{\phi}(3)}{\partial\phi(1)} = \zeta \frac{\partial}{\partial\phi(1)} \frac{\partial W}{\partial j(3)} \\ &= \sum_2 \left(\frac{\partial^2 W}{\partial\bar{j}(2)\partial j(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial\phi(1)\partial\phi(2)} \right) + \zeta \left(\frac{\partial^2 W}{\partial j(2)\partial j(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial\phi(1)\partial\bar{\phi}(2)} \right)\end{aligned}\quad (32.43)$$

Now summarizing² the equations we find

$$\sum_{i_2} \begin{pmatrix} \frac{\partial^2 W}{\partial\bar{j}(i_3)\partial j(i_2)} & \zeta \frac{\partial^2 W}{\partial\bar{j}(i_3)\partial\bar{j}(i_2)} \\ \zeta \frac{\partial^2 W}{\partial j(i_3)\partial j(i_2)} & \frac{\partial^2 W}{\partial j(i_3)\partial\bar{j}(i_2)} \end{pmatrix} \begin{pmatrix} \frac{\partial^2 \Gamma}{\partial\bar{\phi}(i_2)\partial\phi(i_1)} & \frac{\partial^2 \Gamma}{\partial\bar{\phi}(i_2)\partial\bar{\phi}(i_1)} \\ \frac{\partial^2 \Gamma}{\partial\phi(i_2)\partial\phi(i_1)} & \frac{\partial^2 \Gamma}{\partial\phi(i_2)\partial\bar{\phi}(i_1)} \end{pmatrix} = \delta_{i_1, i_3} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\quad (32.44)$$

Proof 32.12 of Theorem 16.47

We start with Equation (32.41)

²The first equation gives the 1,1 component, the second to the 2,2 component, the third for the 1,2 and the fourth to the 2,1 component.

$$\delta_{1,3} = \sum_2 \left(\frac{\partial^2 W}{\partial \bar{j}(2) \partial j(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial \bar{\phi}(1) \partial \phi(2)} \right) + \zeta \left(\frac{\partial^2 W}{\partial j(2) \partial j(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial \bar{\phi}(1) \partial \bar{\phi}(2)} \right) \quad (32.45)$$

and now we take the derivative w.r.t. $\phi(4)$ on both sides

$$\begin{aligned} 0 = \partial_{\phi(4)} \delta_{1,3} &= \sum_2 \left(\partial_{\phi(4)} \frac{\partial^2 W}{\partial \bar{j}(2) \partial j(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial \bar{\phi}(1) \partial \phi(2)} \right) + \left(\frac{\partial^2 W}{\partial \bar{j}(2) \partial j(3)} \right) \left(\partial_{\phi(4)} \frac{\partial^2 \Gamma}{\partial \bar{\phi}(1) \partial \phi(2)} \right) \\ &+ \zeta \left(\partial_{\phi(4)} \frac{\partial^2 W}{\partial j(2) \partial j(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial \bar{\phi}(1) \partial \bar{\phi}(2)} \right) + \zeta \left(\frac{\partial^2 W}{\partial j(2) \partial j(3)} \right) \left(\partial_{\phi(4)} \frac{\partial^2 \Gamma}{\partial \bar{\phi}(1) \partial \bar{\phi}(2)} \right) \\ &= \dots \end{aligned} \quad (32.46)$$

now we consider the following two terms individually:

$$\partial_{\phi(4)} \frac{\partial^2 W}{\partial \bar{j}(2) \partial j(3)} = \sum_5 \zeta \left(\frac{\partial^3 W}{\partial \bar{j}(5) \partial \bar{j}(2) \partial j(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial \phi(4) \partial \phi(5)} \right) + \left(\frac{\partial^3 W}{\partial j(5) \partial \bar{j}(2) \partial j(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial \phi(4) \partial \bar{\phi}(5)} \right) \quad (32.47)$$

and

$$\partial_{\phi(4)} \frac{\partial^2 W}{\partial j(2) \partial j(3)} = \sum_5 \zeta \left(\frac{\partial^3 W}{\partial \bar{j}(5) \partial j(2) \partial j(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial \phi(4) \partial \phi(5)} \right) + \left(\frac{\partial^3 W}{\partial j(5) \partial j(2) \partial j(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial \phi(4) \partial \bar{\phi}(5)} \right) \quad (32.48)$$

inserting this above yields

$$\begin{aligned} 0 = \dots &= \sum_2 \left[\left(\frac{\partial^2 W}{\partial \bar{j}(2) \partial j(3)} \right) \left(\frac{\partial^3 \Gamma}{\partial \phi(4) \partial \bar{\phi}(1) \partial \bar{\phi}(2)} \right) + \zeta \left(\frac{\partial^2 W}{\partial j(2) \partial j(3)} \right) \left(\frac{\partial^3 \Gamma}{\partial \phi(4) \partial \bar{\phi}(1) \partial \bar{\phi}(2)} \right) \right. \\ &+ \sum_5 \left[\zeta \frac{\partial^3 W}{\partial \bar{j}(5) \partial \bar{j}(2) \partial j(3)} \cdot \frac{\partial^2 \Gamma}{\partial \phi(4) \partial \bar{\phi}(5)} + \frac{\partial^3 W}{\partial j(5) \partial \bar{j}(2) \partial j(3)} \cdot \frac{\partial^2 \Gamma}{\partial \phi(4) \partial \bar{\phi}(5)} \right] \cdot \frac{\partial^2 \Gamma}{\partial \bar{\phi}(1) \partial \phi(2)} \\ &+ \sum_2 \left[\frac{\partial^3 W}{\partial \bar{j}(5) \partial j(2) \partial j(3)} \frac{\partial^2 \Gamma}{\partial \phi(4) \partial \phi(5)} + \zeta \frac{\partial^3 W}{\partial j(5) \partial j(2) \partial j(3)} \frac{\partial^2 \Gamma}{\partial \phi(4) \partial \bar{\phi}(5)} \right] \cdot \frac{\partial^2 \Gamma}{\partial \bar{\phi}(1) \partial \bar{\phi}(2)} \Big] \end{aligned} \quad (32.49)$$

Now we take the derivative w.r.t. $\bar{\phi}(6)$. We consider each of the six terms (seperated by plus-symbols respectively) separately:

1. We compute

$$\begin{aligned} &\partial_{\bar{\phi}(6)} \sum_2 \left(\frac{\partial^2 W}{\partial \bar{j}(2) \partial j(3)} \right) \left(\frac{\partial^3 \Gamma}{\partial \phi(4) \partial \bar{\phi}(1) \partial \phi(2)} \right) \\ &= \sum_2 \left(\partial_{\bar{\phi}(6)} \frac{\partial^2 W}{\partial \bar{j}(2) \partial j(3)} \right) \left(\frac{\partial^3 \Gamma}{\partial \phi(4) \partial \bar{\phi}(1) \partial \phi(2)} \right) + \left(\frac{\partial^2 W}{\partial \bar{j}(2) \partial j(3)} \right) \left(\frac{\partial^4 \Gamma}{\partial \bar{\phi}(6) \partial \phi(4) \partial \bar{\phi}(1) \partial \phi(2)} \right) \\ &\equiv \sum_2 0 + \left(\frac{\partial^2 W}{\partial \bar{j}(2) \partial j(3)} \right) \left(\frac{\partial^4 \Gamma}{\partial \bar{\phi}(6) \partial \phi(4) \partial \bar{\phi}(1) \partial \phi(2)} \right) \end{aligned} \quad (32.50)$$

2. This term doesn't contribute because of $\partial_{j(2)}\partial_{j(3)}$
3. This term doesn't contribute because of $\partial_{\phi(4)}\partial_{\phi(5)}$
4. We compute straight-forwardly

$$\begin{aligned}
 & \partial_{\phi(6)} \sum_{2,5} \left(\frac{\partial^3 W}{\partial j(5)\partial \bar{j}(2)\partial j(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial \phi(4)\partial \bar{\phi}(5)} \right) \left(\frac{\partial^2 \Gamma}{\partial \bar{\phi}(1)\partial \phi(2)} \right) \\
 &= \sum_{2,5} \left(\partial_{\phi(6)} \frac{\partial^3 W}{\partial j(5)\partial \bar{j}(2)\partial j(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial \phi(4)\partial \bar{\phi}(5)} \right) \left(\frac{\partial^2 \Gamma}{\partial \bar{\phi}(1)\partial \phi(2)} \right) \\
 &+ \left(\frac{\partial^3 W}{\partial j(5)\partial \bar{j}(2)\partial j(3)} \right) \left[\frac{\partial^3 \Gamma}{\partial \bar{\phi}(6)\partial \phi(4)\partial \bar{\phi}(5)} \cdot \frac{\partial^2 \Gamma}{\partial \bar{\phi}(1)\partial \phi(2)} + \frac{\partial^2 \Gamma}{\partial \phi(4)\partial \bar{\phi}(5)} \cdot \frac{\partial^3 \Gamma}{\partial \bar{\phi}(6)\partial \bar{\phi}(1)\partial \phi(2)} \right] \\
 &\equiv \dots
 \end{aligned} \tag{32.51}$$

now we will simplify this, using that the terms in the second line dont contribute in the absence of external sources and spontaneous symmetry breaking, implying that one obtains

$$\dots = \sum_{2,5,7} \left[\zeta \frac{\partial^4 W}{\partial \bar{j}(7)\partial j(5)\partial \bar{j}(2)\partial j(3)} \cdot \frac{\partial^2 \Gamma}{\partial \bar{\phi}(6)\partial \phi(7)} + \dots \right] \cdot \frac{\partial^2 \Gamma}{\partial \phi(4)\partial \bar{\phi}(5)} \cdot \frac{\partial^2 \Gamma}{\partial \bar{\phi}(1)\partial \phi(2)} \tag{32.52}$$

5. This term doesn't contribute because of $\partial_{\phi(1)}\partial_{\phi(2)}$
6. This term doesn't contribute because of $\partial_{\bar{\phi}(1)}\partial_{\bar{\phi}(2)}$

Now we collect the individual (non-vanishing) contributions and obtain

$$\begin{aligned}
 0 &= \sum_2 \left(\frac{\partial^2 W}{\partial \bar{j}(2)\partial j(3)} \right) \left(\frac{\partial^4 \Gamma}{\partial \bar{\phi}(6)\partial \phi(4)\partial \bar{\phi}(1)\partial \phi(2)} \right) \\
 &+ \zeta \sum_{2,5,7} \left(\frac{\partial^4 W}{\partial \bar{j}(7)\partial j(5)\partial \bar{j}(2)\partial j(3)} \right) \left(\frac{\partial^2 \Gamma}{\partial \phi(4)\partial \bar{\phi}(5)} \right) \left(\frac{\partial^2 \Gamma}{\partial \bar{\phi}(1)\partial \phi(2)} \right) \left(\frac{\partial^2 \Gamma}{\partial \bar{\phi}(6)\partial \phi(7)} \right) \\
 &= \sum_2 \left(\frac{\partial^2 W}{\partial \bar{j}(2)\partial j(3)} \right) \left(\frac{\zeta \partial^4 \Gamma}{\partial \bar{\phi}(6)\partial \bar{\phi}(1)\partial \phi(4)\partial \phi(2)} \right) \\
 &+ \zeta \sum_{2,5,7} \left(\frac{\zeta \partial^4 W}{\partial \bar{j}(7)\partial \bar{j}(2)\partial j(5)\partial j(3)} \right) \left(\frac{\zeta \partial^2 \Gamma}{\partial \bar{\phi}(5)\partial \phi(4)} \right) \left(\frac{\partial^2 \Gamma}{\partial \bar{\phi}(1)\partial \phi(2)} \right) \left(\frac{\partial^2 \Gamma}{\partial \bar{\phi}(6)\partial \phi(7)} \right) \\
 &= \sum_2 \left(-\frac{\hbar}{\zeta} \mathcal{G}_c(2, 3) \right) \zeta \left(-\frac{\zeta}{\hbar} \right)^2 V(6, 1|2, 4) \\
 &+ \zeta \sum_{2,5,7} \zeta \left(-\frac{\hbar}{\zeta} \right)^2 \mathcal{G}_c(7, 2|3, 5) \zeta \left(-\frac{\zeta}{\hbar} V(5, 4) \right) \left(-\frac{\zeta}{\hbar} V(1, 2) \right) \left(-\frac{\zeta}{\hbar} V(6, 7) \right) \\
 &= -\frac{1}{\hbar} \sum_2 \mathcal{G}_c(2, 3) V(6, 1|2, 4) \\
 &- \frac{1}{\hbar} \sum_{2,5,7} \mathcal{G}_c(7, 2|3, 5) V(5, 4) V(1, 2) V(6, 7)
 \end{aligned} \tag{32.53}$$

rearranging terms and multiplying both sides with $\hbar$ gives therefore

$$\sum_2 \mathcal{G}_c(2, 3)V(6, 1|2, 4) = - \sum_{2,5,7} \mathcal{G}_c(7, 2|3, 5)V(5, 4)V(1, 2)V(6, 7) \quad (32.54)$$

Notice that this equation holds for all 1, 3, 4, 6. Now we make the replacements

- $1 \mapsto \tilde{1}$
- $3 \mapsto \tilde{2}$
- $4 \mapsto \tilde{3}$
- $6 \mapsto \tilde{4}$

and obtain therefore

$$\sum_2 \mathcal{G}_c(2, \tilde{2})V(\tilde{4}, \tilde{1}|2, \tilde{3}) = - \sum_{2,5,7} \mathcal{G}_c(7, 2|\tilde{2}, 5)V(5, \tilde{3})V(\tilde{1}, 2)V(\tilde{4}, 7) \quad (32.55)$$

or, equivalently

$$\sum_2 \mathcal{G}_c(2, i_2)V(i_4, i_1|2, i_3) = - \sum_{2,5,7} \mathcal{G}_c(7, 2|i_2, 5)V(5, i_3)V(i_1, 2)V(i_4, 7) \quad (32.56)$$

Now we multiply both sides with $\mathcal{G}_c(i_3|x_1)$ and sum over i_3 , whereas x_1 still has to be determined, and we obtain

$$\begin{aligned} \sum_{2,i_3} \mathcal{G}_c(2, i_2)V(i_4, i_1|2, i_3)\mathcal{G}_c(i_3|x_1) &= - \sum_{2,5,7} \mathcal{G}_c(7, 2|i_2, 5)V(i_1, 2)V(i_4, 7) \sum_{i_3} V(5, i_3)\mathcal{G}_c(i_3|x_1) \\ &= - \sum_{2,5,7} \mathcal{G}_c(7, 2|i_2, 5)V(i_1, 2)V(i_4, 7)\delta_{5,x_1} \\ &= - \sum_{2,7} \mathcal{G}_c(7, 2|i_2, x_1)V(i_1, 2)V(i_4, 7) \end{aligned} \quad (32.57)$$

Now we multiply both sides with $\mathcal{G}_c(x_2|i_1)\mathcal{G}_c(x_3|i_4)$ and sum over i_1 , whereas i_4 still has to be determined, and we obtain

$$\begin{aligned} \sum_{2,i_1,i_3,i_4} \mathcal{G}_c(2, i_2)V(i_4, i_1|2, i_3)\mathcal{G}_c(i_3|x_1)\mathcal{G}_c(x_2|i_1)\mathcal{G}_c(x_3|i_4) &= - \sum_{2,7} \mathcal{G}_c(7, 2|i_2, x_1) \sum_{i_1} \mathcal{G}_c(x_2|i_1)V(i_1, 2) \sum_{i_4} \mathcal{G}_c(x_3|i_4)V(i_4, 7) \\ &= - \sum_{2,7} \mathcal{G}_c(7, 2|i_2, x_1)\delta_{x_2,2}\delta_{x_3,7} \\ &= -\mathcal{G}_c(x_3, x_2|i_2, x_1) \end{aligned} \quad (32.58)$$

Now we replace

1. $x_3 \mapsto y_1$
2. $x_2 \mapsto y_2$
3. $i_2 \mapsto y_3$
4. $x_1 \mapsto y_4$

and we obtain

$$\begin{aligned}
 \mathcal{G}_c(y_1, y_2 | y_3, y_4) &= - \sum_{2, i_1, i_3, i_4} \mathcal{G}_c(2, y_3) V(i_4, i_1 | 2, i_3) \mathcal{G}_c(i_3 | y_4) \mathcal{G}_c(y_2 | i_1) \mathcal{G}_c(y_1 | i_4) \\
 &= - \sum_{2, i_1, i_3, i_4} \mathcal{G}_c(y_1 | i_4) \mathcal{G}_c(y_2 | i_1) V(i_4, i_1 | 2, i_3) \mathcal{G}_c(2, y_3) \mathcal{G}_c(i_3 | y_4) \\
 &= \dots
 \end{aligned} \tag{32.59}$$

and now, we replace finally

1. $i_4 \mapsto l_1$
2. $i_1 \mapsto l_2$
3. $2 \mapsto l'_1$
4. $i_3 \mapsto l'_2$

and obtain

$$\mathcal{G}_c(y_1, y_2 | y_3, y_4) = - \sum_{l_1, l_2, l'_1, l'_2} \mathcal{G}_c(y_1 | l_1) \mathcal{G}_c(y_2 | l_2) V(l_1, l_2 | l'_1, l'_2) \mathcal{G}_c(l'_1, y_3) \mathcal{G}_c(l'_2 | y_4) \tag{32.60}$$

proving the theorem.

Proof 32.13 of Approximation 16.55.

The problem is here, that we need a very clean definition of the left right derivative in the series-expansion

Starting with eq. (16.44) and lightening the notation, we obtain

$$\begin{aligned}\Gamma[\phi, \bar{\phi}] &= -W[j, \bar{j}] + \sum \bar{j}\phi + \bar{\phi}j \\ &= -\log Z[j, \bar{j}] + \sum \bar{j}\phi + \bar{\phi}j,\end{aligned}$$

We do the following things:

1. We multiply this equation by -1 and apply the exponential-function to both sides,
2. we insert the definition of the source-dependent partition function $Z[j, \bar{j}]$ from eq. (16.37),
3. we use the invariance of the functional integral by translations (Grassmann-valued *and* complex) $\psi - \phi \rightarrow \Delta$, etc.

This gives in equations

$$\begin{aligned}\exp\left(-\Gamma[\phi, \bar{\phi}]\right) &= Z[j, \bar{j}] \exp\left(-\sum \bar{j}\phi + \bar{\phi}j\right) \\ &= \int \mathcal{D}(\psi, \bar{\psi}) \exp\left(-S[\psi, \bar{\psi}] + \sum \bar{j}(\psi - \phi) + (\bar{\psi} - \bar{\phi})j\right) \\ &= \int \mathcal{D}(\Delta, \bar{\Delta}) \exp\left(-S[\phi + \Delta, \bar{\phi} + \bar{\Delta}] + \sum \bar{j}\Delta + \bar{\Delta}j\right).\end{aligned}$$

The action S can now be expanded in the variation analogously to Notation 16.36, whereas here we use schematic notation and we emphasize that the left-right-derivative carries non-trivial signs, giving us the equation

$$\begin{aligned}S[\phi + \Delta, \bar{\phi} + \bar{\Delta}] &= S[\phi, \bar{\phi}] \\ &+ \sum \left(\frac{\partial}{\partial \phi} S[\phi, \bar{\phi}], \frac{\partial}{\partial \bar{\phi}} S[\phi, \bar{\phi}] \right) \cdot \left(\frac{\Delta}{\Delta} \right) \\ &+ \frac{1}{2!} \sum \sum \left(\frac{\Delta}{\Delta} \right)^T \cdot \left(\begin{array}{cc} \frac{\partial^2}{\partial \phi \partial \phi} S[\phi, \bar{\phi}] & \frac{\partial^2}{\partial \phi \partial \bar{\phi}} S[\phi, \bar{\phi}] \\ \frac{\partial^2}{\partial \bar{\phi} \partial \phi} S[\phi, \bar{\phi}] & \frac{\partial^2}{\partial \bar{\phi} \partial \bar{\phi}} S[\phi, \bar{\phi}] \end{array} \right) \cdot \left(\frac{\Delta}{\Delta} \right) \\ &+ \mathcal{O}\left(\Delta^3, \Delta^2 \bar{\Delta}^1, \Delta^1 \bar{\Delta}^2, \bar{\Delta}^3\right).\end{aligned}$$

With this we get

$$\exp\left(-\Gamma[\phi, \bar{\phi}]\right) = \exp\left(-S[\phi, \bar{\phi}]\right) \int \mathcal{D}(\Delta, \bar{\Delta}) \exp P(\Delta, \bar{\Delta}, \phi, \bar{\phi}) \quad (32.61)$$

with a polynomial P in the integration-variables and in the fields. In the so-called *Tree-Level-approximation* one now approximates³

³The meaning of the symbol “ $\approx$ ” is always referring to “closeness” (in some sense that needs to be specified) to experimental data.

$$\int \mathcal{D}(\Delta, \bar{\Delta}) \exp P(\Delta, \bar{\Delta}, \phi, \bar{\phi}) \approx \mathcal{O}(1).$$

So, one assumes that the functional integral (which, when evaluated, is again a polynomial expression in the classical fields) gives essentially ϕ -independent contributions. After applying the logarithm and using the inverse property (also for Graßmann-valued polynomial expressions?!) we thus obtain

$$\Gamma[\phi, \bar{\phi}] \approx S[\phi, \bar{\phi}] + \text{const.}$$

Remembering now that the physically relevant part of the effective action is in the numerical values of the non-trivial correlators (and thus in the coefficients of the non-constant monomials in Γ (cf. the series-expansion in Equation (16.45))), then, one usually simply notes for convenience:

$$\Gamma[\phi, \bar{\phi}] \approx S[\phi, \bar{\phi}].$$

Proof 32.14 of Approximation 16.57.

The *One-Loop-approximation* builds on the Tree-Level-approximation. If we use Corollary 16.56, then (in the context of the Tree-Level-approximation) in eq. (32.61) the terms linear in Δ and $\bar{\Delta}$ cancel out. Then we can use this to connect to eq. (32.61), by neglecting the contributions to the polynomial P of order $\Delta^n \bar{\Delta}^m$ with $n + m \geq 3$:

$$\begin{aligned} \exp(-\Gamma[\phi, \bar{\phi}]) &\approx \exp(-S[\phi, \bar{\phi}]) \int \mathcal{D}(\Delta, \bar{\Delta}) \exp -\frac{1}{2} \sum \sum \left(\frac{\Delta}{\bar{\Delta}}\right)^T \cdot \begin{pmatrix} \frac{\partial^2}{\partial \phi \partial \phi} S[\phi, \bar{\phi}] & \frac{\partial^2}{\partial \phi \partial \bar{\phi}} S[\phi, \bar{\phi}] \\ \frac{\partial^2}{\partial \bar{\phi} \partial \phi} S[\phi, \bar{\phi}] & \frac{\partial^2}{\partial \bar{\phi} \partial \bar{\phi}} S[\phi, \bar{\phi}] \end{pmatrix} \cdot \begin{pmatrix} \Delta \\ \bar{\Delta} \end{pmatrix} \\ &= \exp(-S[\phi, \bar{\phi}]) \text{Pf} \begin{pmatrix} \frac{\partial^2}{\partial \phi \partial \phi} S[\phi, \bar{\phi}] & \frac{\partial^2}{\partial \phi \partial \bar{\phi}} S[\phi, \bar{\phi}] \\ \frac{\partial^2}{\partial \bar{\phi} \partial \phi} S[\phi, \bar{\phi}] & \frac{\partial^2}{\partial \bar{\phi} \partial \bar{\phi}} S[\phi, \bar{\phi}] \end{pmatrix}. \end{aligned}$$

If we now take the logarithm on both sides and use the inversion-property between exponential-function and logarithm (!) and fundamental identities for the matrix logarithm (also for Grassmann-valued matrices?!), we get

$$\begin{aligned} -\Gamma[\phi, \bar{\phi}] &\approx -S[\phi, \bar{\phi}] + \log \text{Pf} \begin{pmatrix} \frac{\partial^2}{\partial \phi \partial \phi} S[\phi, \bar{\phi}] & \frac{\partial^2}{\partial \phi \partial \bar{\phi}} S[\phi, \bar{\phi}] \\ \frac{\partial^2}{\partial \bar{\phi} \partial \phi} S[\phi, \bar{\phi}] & \frac{\partial^2}{\partial \bar{\phi} \partial \bar{\phi}} S[\phi, \bar{\phi}] \end{pmatrix} \\ &\stackrel{?!}{=} -S[\phi, \bar{\phi}] + \frac{1}{2} \log \text{Det} \begin{pmatrix} \frac{\partial^2}{\partial \phi \partial \phi} S[\phi, \bar{\phi}] & \frac{\partial^2}{\partial \phi \partial \bar{\phi}} S[\phi, \bar{\phi}] \\ \frac{\partial^2}{\partial \bar{\phi} \partial \phi} S[\phi, \bar{\phi}] & \frac{\partial^2}{\partial \bar{\phi} \partial \bar{\phi}} S[\phi, \bar{\phi}] \end{pmatrix} \\ &\stackrel{?!}{=} -S[\phi, \bar{\phi}] + \frac{1}{2} \text{Tr} \log \begin{pmatrix} \frac{\partial^2}{\partial \phi \partial \phi} S[\phi, \bar{\phi}] & \frac{\partial^2}{\partial \phi \partial \bar{\phi}} S[\phi, \bar{\phi}] \\ \frac{\partial^2}{\partial \bar{\phi} \partial \phi} S[\phi, \bar{\phi}] & \frac{\partial^2}{\partial \bar{\phi} \partial \bar{\phi}} S[\phi, \bar{\phi}] \end{pmatrix} \\ \Leftrightarrow +\Gamma[\phi, \bar{\phi}] &\approx +S[\phi, \bar{\phi}] - \frac{1}{2} \text{Tr} \log \begin{pmatrix} \frac{\partial^2}{\partial \phi \partial \phi} S[\phi, \bar{\phi}] & \frac{\partial^2}{\partial \phi \partial \bar{\phi}} S[\phi, \bar{\phi}] \\ \frac{\partial^2}{\partial \bar{\phi} \partial \phi} S[\phi, \bar{\phi}] & \frac{\partial^2}{\partial \bar{\phi} \partial \bar{\phi}} S[\phi, \bar{\phi}] \end{pmatrix}. \end{aligned}$$

Proof 32.15 of Approximation 16.58.

One separates the second variation according to

$$\begin{pmatrix} \frac{\partial^2}{\partial \phi \partial \phi} S[\phi, \bar{\phi}] & \frac{\partial^2}{\partial \phi \partial \bar{\phi}} S[\phi, \bar{\phi}] \\ \frac{\partial^2}{\partial \bar{\phi} \partial \phi} S[\phi, \bar{\phi}] & \frac{\partial^2}{\partial \bar{\phi} \partial \bar{\phi}} S[\phi, \bar{\phi}] \end{pmatrix} = \begin{pmatrix} 0 & -\mathbf{x} \mathcal{G}_0^{-T} \\ \mathbf{x} \mathcal{G}_0^{-1} & 0 \end{pmatrix} + \begin{pmatrix} \frac{\partial^2}{\partial \phi \partial \phi} S_{\text{int}}[\phi, \bar{\phi}] & \frac{\partial^2}{\partial \phi \partial \bar{\phi}} S_{\text{int}}[\phi, \bar{\phi}] \\ \frac{\partial^2}{\partial \bar{\phi} \partial \phi} S_{\text{int}}[\phi, \bar{\phi}] & \frac{\partial^2}{\partial \bar{\phi} \partial \bar{\phi}} S_{\text{int}}[\phi, \bar{\phi}] \end{pmatrix}$$

$$\equiv \mathbf{x} \widehat{\mathcal{G}}_0^{-1} + \widehat{V}[\phi, \bar{\phi}].$$

Using now that

$$\mathbf{x} \widehat{\mathcal{G}}_0 = \begin{pmatrix} 0 & -\mathbf{x} \mathcal{G}_0^{-T} \\ \mathbf{x} \mathcal{G}_0^{-1} & 0 \end{pmatrix}^{-1} = \begin{pmatrix} 0 & \mathbf{x} \mathcal{G}_0 \\ -\mathbf{x} \mathcal{G}_0^T & 0 \end{pmatrix},$$

it follows, after simplifying the notation $\widehat{V}[\phi, \bar{\phi}] \equiv \widehat{V}$, that

$$\begin{aligned} \text{Tr} \log [\mathbf{x} \widehat{\mathcal{G}}_0^{-1} + \widehat{V}] &= \text{Tr} \log [\mathbf{x} \widehat{\mathcal{G}}_0^{-1} \cdot (1 + \mathbf{x} \widehat{\mathcal{G}}_0 \widehat{V})] \\ &\stackrel{?!}{=} \text{Tr} \log [\mathbf{x} \widehat{\mathcal{G}}_0^{-1}] + \text{Tr} \log [1 + \mathbf{x} \widehat{\mathcal{G}}_0 \cdot \widehat{V}] \\ &= \mathcal{O}(1) + \text{Tr} \log [1 + \mathbf{x} \widehat{\mathcal{G}}_0 \cdot \widehat{V}]. \end{aligned}$$

The first term again provides a (in ϕ and $\bar{\phi}$) constant (and thus physically irrelevant) contribution. The second term provides (physically relevant) corrections. Therefore, we focus on the second term and expand the logarithm in its series representation for “small” corrections to the identity:

$$\begin{aligned} \text{Tr} \log [1 + \mathbf{x} \widehat{\mathcal{G}}_0 \cdot \widehat{V}] &\approx \text{Tr} \left[\mathbf{x} \widehat{\mathcal{G}}_0 \cdot \widehat{V} - \frac{(\mathbf{x} \widehat{\mathcal{G}}_0 \cdot \widehat{V})^2}{2} + \dots \right] \\ &= \text{Tr} [\mathbf{x} \widehat{\mathcal{G}}_0 \cdot \widehat{V}] - \frac{1}{2} \text{Tr} [\mathbf{x}^2 \widehat{\mathcal{G}}_0 \cdot \widehat{V} \cdot \widehat{\mathcal{G}}_0 \cdot \widehat{V}] + \dots \end{aligned}$$

First summand. One computes

$$\begin{aligned} \mathbf{x} \widehat{\mathcal{G}}_0 \widehat{V} &= \begin{pmatrix} 0 & \mathbf{x} \mathcal{G}_0 \\ -\mathbf{x} \mathcal{G}_0^T & 0 \end{pmatrix} \cdot \begin{pmatrix} \frac{\partial^2}{\partial \phi \partial \phi} S_{\text{int}}[\phi, \bar{\phi}] & \frac{\partial^2}{\partial \phi \partial \bar{\phi}} S_{\text{int}}[\phi, \bar{\phi}] \\ \frac{\partial^2}{\partial \bar{\phi} \partial \phi} S_{\text{int}}[\phi, \bar{\phi}] & \frac{\partial^2}{\partial \bar{\phi} \partial \bar{\phi}} S_{\text{int}}[\phi, \bar{\phi}] \end{pmatrix} \\ &= \begin{pmatrix} \mathbf{x} \mathcal{G}_0 \cdot \frac{\partial^2}{\partial \phi \partial \phi} S_{\text{int}}[\phi, \bar{\phi}] & \mathbf{x} \mathcal{G}_0 \cdot \frac{\partial^2}{\partial \phi \partial \bar{\phi}} S_{\text{int}}[\phi, \bar{\phi}] \\ -\mathbf{x} \mathcal{G}_0^T \cdot \frac{\partial^2}{\partial \bar{\phi} \partial \phi} S_{\text{int}}[\phi, \bar{\phi}] & -\mathbf{x} \mathcal{G}_0^T \cdot \frac{\partial^2}{\partial \bar{\phi} \partial \bar{\phi}} S_{\text{int}}[\phi, \bar{\phi}] \end{pmatrix}, \end{aligned}$$

where the formal products are to be evaluated by insertion of corresponding “unities”.

Because of the symmetry of the statistical derivatives, one easily concludes that

$$\frac{\partial^2}{\partial\phi\partial\bar\phi}S_{\text{int}}[\phi,\bar\phi] = \mathbf{x} \left(\frac{\partial^2}{\partial\bar\phi\partial\phi}S_{\text{int}}[\phi,\bar\phi] \right)^T, \quad \text{and} \quad B^T A^T = (AB)^T. \quad (32.62)$$

Due to the cyclicity of the trace we arrive at the first term in eq. (16.79).

Second summand. We write

$$\begin{aligned} \widehat{\mathcal{G}}_0 \widehat{V} \widehat{\mathcal{G}}_0 \widehat{V} &= \begin{pmatrix} \mathcal{G}_0 \cdot \frac{\partial^2}{\partial\bar\phi\partial\phi}S_{\text{int}}[\phi,\bar\phi] & \mathcal{G}_0 \cdot \frac{\partial^2}{\partial\bar\phi\partial\bar\phi}S_{\text{int}}[\phi,\bar\phi] \\ -\mathcal{G}_0^T \cdot \frac{\partial^2}{\partial\phi\partial\phi}S_{\text{int}}[\phi,\bar\phi] & -\mathcal{G}_0^T \cdot \frac{\partial^2}{\partial\phi\partial\bar\phi}S_{\text{int}}[\phi,\bar\phi] \end{pmatrix} \cdot \begin{pmatrix} \mathcal{G}_0 \cdot \frac{\partial^2}{\partial\bar\phi\partial\phi}S_{\text{int}}[\phi,\bar\phi] & \mathcal{G}_0 \cdot \frac{\partial^2}{\partial\bar\phi\partial\bar\phi}S_{\text{int}}[\phi,\bar\phi] \\ -\mathcal{G}_0^T \cdot \frac{\partial^2}{\partial\phi\partial\phi}S_{\text{int}}[\phi,\bar\phi] & -\mathcal{G}_0^T \cdot \frac{\partial^2}{\partial\phi\partial\bar\phi}S_{\text{int}}[\phi,\bar\phi] \end{pmatrix} \\ &\equiv \begin{pmatrix} A+B & \cdots \\ \cdots & C+D \end{pmatrix}, \end{aligned}$$

(the secondary diagonals are irrelevant if one takes the trace and only provide corrections on higher orders of perturbation theory) with

$$\begin{aligned} A &= \mathcal{G}_0 \cdot \frac{\partial^2}{\partial\bar\phi\partial\phi}S_{\text{int}}[\phi,\bar\phi] \cdot \mathcal{G}_0 \cdot \frac{\partial^2}{\partial\bar\phi\partial\phi}S_{\text{int}}[\phi,\bar\phi], \\ B &= -\mathcal{G}_0 \cdot \frac{\partial^2}{\partial\bar\phi\partial\bar\phi}S_{\text{int}}[\phi,\bar\phi] \cdot \mathcal{G}_0^T \cdot \frac{\partial^2}{\partial\phi\partial\phi}S_{\text{int}}[\phi,\bar\phi], \\ C &= -\mathcal{G}_0^T \cdot \frac{\partial^2}{\partial\phi\partial\phi}S_{\text{int}}[\phi,\bar\phi] \cdot \mathcal{G}_0 \cdot \frac{\partial^2}{\partial\bar\phi\partial\bar\phi}S_{\text{int}}[\phi,\bar\phi], \\ D &= \mathcal{G}_0^T \cdot \frac{\partial^2}{\partial\phi\partial\bar\phi}S_{\text{int}}[\phi,\bar\phi] \cdot \mathcal{G}_0 \cdot \frac{\partial^2}{\partial\phi\partial\bar\phi}S_{\text{int}}[\phi,\bar\phi]. \end{aligned}$$

Now, one has $\text{Tr } B = \text{Tr } C$. Due to the cyclicity of the trace and $\text{Tr } A = \text{Tr } D$ because of eq. (32.62). Eq. (16.79) is now proven.

This entire section of proofs needs to be revised, cause we adjusted the prefactor convention for the functional derivati

Proof 32.16 of Lemma 17.2.

For the first equation we compute

$$\begin{aligned} W[j, \bar{j}] &= -\log \int \mathcal{D}(\psi, \bar{\psi}) \exp \left(-S[\psi, \bar{\psi}] + \sum (\bar{j}\psi + \bar{\psi}j) \right) \\ &= -\log \int \mathcal{D}(M \cdot \psi, \bar{M} \cdot \bar{\psi}) \exp \left(-S[M \cdot \psi, \bar{M} \cdot \bar{\psi}] + \sum_x \bar{j}(x) (M \cdot \psi)(x) + (\bar{M} \cdot \bar{\psi})(x) j(x) \right) \\ &= \dots \end{aligned}$$

using now

$$\oint_x \bar{j}(x) (M \cdot \psi)(x) = \sum_x \sum_y \bar{j}(x) M(x, y) \psi(y) = \sum_y \sum_x M^T(y, x) \bar{j}(x) \psi(y) = \sum_y (M^T \cdot \bar{j})(y) \psi(y)$$

and analogously one has

$$\sum_x (\bar{M} \cdot \bar{\psi})(x) j(x) = \sum_y \bar{\psi}(y) (\bar{M}^T \cdot j)(y).$$

Then, it follows (together with Definition 17.1)

$$\begin{aligned} \dots &= -\log \int \mathcal{D}(\psi, \bar{\psi}) \exp \left(-S[\psi, \bar{\psi}] + \sum_y (M^T \cdot \bar{j})(y) \psi(y) + \bar{\psi}(y) (\bar{M}^T \cdot j)(y) \right) \\ &= W[\bar{M}^T \cdot j, M^T \cdot \bar{j}]. \end{aligned}$$

For the second equation one has

$$\begin{aligned} \phi[\bullet, \bar{M}^T \cdot j, M^T \cdot \bar{j}] &= -\frac{\vec{\partial}}{\partial(M^T \cdot \bar{j})(\bullet)} W[\bar{M}^T \cdot j, M^T \cdot \bar{j}] \\ &= -\frac{\vec{\partial}}{\partial(M^T \cdot \bar{j})(\bullet)} W[j, \bar{j}] \\ &= \dots \end{aligned}$$

Now one calculates

$$\bar{j}(x) = (M^{-T} \cdot M^T \cdot \bar{j})(x) = \sum_y M^{-T}(x, y) (M^T \cdot \bar{j})(y)$$

and therefore

$$\frac{\vec{\partial}}{\partial(M^T \cdot \bar{j})(\bullet)} = \sum_x \left(\frac{\vec{\partial}}{\partial(M^T \cdot \bar{j})(\bullet)} \bar{j}(x) \right) \frac{\vec{\partial}}{\partial \bar{j}(x)} = \sum_x M^{-T}(x, \bullet) \frac{\vec{\partial}}{\partial \bar{j}(x)} = \sum_x M^{-1}(\bullet, x) \frac{\vec{\partial}}{\partial \bar{j}(x)}.$$

It then follows with Notation 16.31

$$\dots = - \sum_x M^{-1}(\bullet, x) \left(\frac{\vec{\partial}}{\partial \bar{j}(x)} W[j, \bar{j}] \right) = \sum_x M^{-1}(\bullet, x) \phi[x, j, \bar{j}] = (M^{-1} \cdot \phi) [\bullet, j, \bar{j}],$$

Analogously, one obtains

$$\bar{\phi}[\bullet, \bar{M}^T \cdot j, M^T \cdot \bar{j}] = \frac{\vec{\partial}}{\partial (\bar{M}^T \cdot j)(\bullet)} W[j, \bar{j}] = \sum_x \bar{M}^{-1}(\bullet, x) \left(\frac{\vec{\partial}}{\delta j(x)} W[j, \bar{j}] \right) = (\bar{M}^{-1} \cdot \bar{\phi}) [\bullet, j, \bar{j}].$$

Proof 32.17 of Theorem 17.3.

With Lemma 17.2 one can prove the theorem very easily:

$$\begin{aligned} \Gamma[M^{-1} \cdot \phi, \bar{M}^{-1} \cdot \bar{\phi}] &= W[\bar{M}^T \cdot j, M^T \cdot \bar{j}] + \sum_x (M^T \cdot \bar{j})(x) (M^{-1} \cdot \phi)(x) + (\bar{M}^{-1} \cdot \bar{\phi})(x) (\bar{M}^T \cdot j)(x) \\ &= W[j, \bar{j}] + \sum_x \bar{j}(x) \phi(x) + \bar{\phi}(x) j(x) \\ &= \Gamma[\phi, \bar{\phi}]. \end{aligned}$$

Proof 32.18 of Theorem 18.4

TBDigitalized. (Negele, Orland p. 29, Altland p.163 for fermions, and Altland p. 159 for bosons.)

Proof 32.19 of Theorem 18.5

Proof 32.20 of Lemma 18.6

TBDigitalized, cf. Negele Orland page 30 or Altland page 181 for fermions, Negele Orland page 21 or Altland p. 159 for bosons.

Proof 32.21 of Theorem 18.7

TBD.

Proof 32.22 of Theorem 18.8

- NO page 22, Altland page 160 (bosons)
- NO page 39, Altland page 181 f (fermions)

Proof 32.23 of Lemma 18.9

Follows by construction via the recipe of second quantization.

Proof 32.24 of Lemma 18.10

We obtain by direct computation ...

- Altland page 182.

Proof 32.25 of Lemma 18.11

TBDigitalized, cf. Negele Orland page 24 f.

Proof 32.26 of Lemma 18.12

By a direct computation we obtain

- TBDigitalized.

Proof 32.27 of Theorem 18.13

Herleitung des „functional field integral“

$$\begin{aligned}
 \mathcal{Z} &= \text{tr} e^{-\beta(\hat{H} - \mu\hat{N})} = \sum_n \langle n | e^{-\beta(\hat{H} - \mu\hat{N})} | n \rangle \\
 &= \int d(\bar{\psi}, \psi) e^{-\bar{\psi} \cdot \psi} \sum_n \langle n | \psi \rangle \langle \psi | e^{-\beta(\hat{H} - \mu\hat{N})} | n \rangle \\
 &\stackrel{\langle n | \psi \rangle \langle \psi | n \rangle = \langle \psi | n \rangle \langle n | \psi \rangle}{=} \int d(\bar{\psi}^0, \psi^0) e^{-\bar{\psi}^0 \cdot \psi^0} \langle \psi^0 | e^{-\beta(\hat{H} - \mu\hat{N})} | \psi^0 \rangle \\
 &= \int \prod_{j=0}^{N-1} d\bar{\psi}^j d\psi^j e^{-\bar{\psi}^j \cdot \psi^j} \langle \bar{\psi}^{j+1} | e^{-\beta(\hat{H} - \mu\hat{N})/N + O(\beta/N)^2} | \psi^j \rangle \\
 &\quad \bar{\psi}^N = \bar{\psi}^0, \quad \psi^N = \psi^0 \\
 &\stackrel{\delta = \beta/N}{=} \exp \left[-\delta \left(\frac{\bar{\psi}^j - \bar{\psi}^{j+1}}{\delta} \cdot \psi^j + H(\bar{\psi}^{j+1}, \psi^j) - \mu N(\bar{\psi}^{j+1}, \psi^j) \right) \right] \\
 &\quad + O(\delta^2) \\
 &\xrightarrow{N \rightarrow \infty} \int d(\bar{\psi}, \psi) \exp - \int_0^\beta d\tau \left(\bar{\psi} \partial_\tau \psi + H(\bar{\psi}, \psi) - \mu N(\bar{\psi}, \psi) \right)
 \end{aligned}$$

Proof 32.28 of Lemma 18.14

Shit... We need a finite set of Matsubara-frequencies...

We assume, without⁴ loss of generality, that N (which is going to infinity) is even for bosons and it is odd for fermions. This allows us to introduce the *finite set of Matsubara-frequencies* $\Omega_{\zeta, N}$ as

$$\Omega_{\zeta, N} = \begin{cases} \left\{ 2n\pi/\hbar\beta \mid n \in \{-N, -N+1, \dots, -1, 0, +1, \dots, N\} \right\} & \text{for bosons} \\ \left\{ (2n+1)\pi/\hbar\beta \mid n \in \{-N, -N+1, \dots, -1, 0, +1, \dots, N\} \right\} & \text{for fermions} \end{cases} \quad (32.63)$$

Something in this direction...

We introduce (nostalgically) a discretization for the temporal component and denote the abbreviations

$$\psi(\tau) \equiv \psi(\tau^j) \equiv \psi^j = \frac{1}{\sqrt{\hbar\beta}} \sum_{i\omega_n \in i\Omega_\zeta} \psi(i\omega_n) e^{-i\omega_n \hbar\beta j/N}, \quad \bar{\psi}^j = \frac{1}{\sqrt{\hbar\beta}} \sum_{i\omega_m \in i\Omega_\zeta} \bar{\psi}(i\omega_m) e^{i\omega_m \hbar\beta j/N} \quad (32.64)$$

⁴Note the entire proof remains perfectly valid for the respective other case; The notation becomes simply more tedious.

which we write equivalently, remembering that for each timestep there is an entire basis-index-set attached to the fields, as

$$\psi_\alpha^j = \frac{1}{\sqrt{\hbar\beta}} \sum_{i\omega_n \in i\Omega_\zeta} \psi_\alpha(i\omega_n) e^{-i\omega_n \hbar\beta j/N}, \quad \bar{\psi}_\alpha^j = \frac{1}{\sqrt{\hbar\beta}} \sum_{i\omega_m \in i\Omega_\zeta} \bar{\psi}_\alpha(i\omega_m) e^{i\omega_m \hbar\beta j/N} \quad (32.65)$$

with $\alpha \in \mathcal{A}$. Before the substitution we have an integral measure as

$$\mathcal{D}(\bar{\psi}, \psi) = \lim_{N \rightarrow \infty} \prod_{n=1}^N d(\bar{\psi}^n, \psi^n) = \lim_{N \rightarrow \infty} \prod_{n=1}^N \prod_{\alpha \in \mathcal{A}} \frac{d\bar{\psi}_\alpha^n d\psi_\alpha^n}{\pi^{(1+\zeta)/2}} \quad (32.66)$$

It obviously suffices to examine the transformation for a single $\alpha \in \mathcal{A}$.

We consider the bosonic case!

Recall the transformation-formula

$$\int_{g(\Omega)} f(x) dx = \int_{\Omega} f(g(y)) |\det J_g(y)| dy \quad (32.67)$$

and now we introduce

$$x = \begin{pmatrix} \psi^1 \\ \vdots \\ \psi^N \\ \bar{\psi}^1 \\ \vdots \\ \bar{\psi}^N \end{pmatrix} \quad (32.68)$$

and

$$g(y) = \begin{pmatrix} M & 0 \\ 0 & M \end{pmatrix} \begin{pmatrix} \psi^1 \\ \vdots \\ \psi^N \\ \bar{\psi}^1 \\ \vdots \\ \bar{\psi}^N \end{pmatrix} = \begin{pmatrix} \psi^1 \\ \vdots \\ \psi^N \\ \bar{\psi}^1 \\ \vdots \\ \bar{\psi}^N \end{pmatrix} \quad (32.69)$$

The equation $g(y) = x$ can now be rewritten as

$$\begin{pmatrix} \psi^1 \\ \vdots \\ \psi^N \\ \bar{\psi}^1 \\ \vdots \\ \bar{\psi}^N \end{pmatrix} = \begin{pmatrix} f(1,1) & f(1,2) & \cdots & f(1,N) & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots & \vdots & 0 & & \vdots \\ f(N,1) & f(N,2) & \cdots & f(N,N) & 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & f(1,1) & f(1,2) & \cdots & f(1,N) \\ \vdots & 0 & & \vdots & \vdots & & & \vdots \\ 0 & 0 & \cdots & 0 & f(N,1) & f(N,2) & \cdots & f(N,N) \end{pmatrix} \begin{pmatrix} \psi^1 \\ \vdots \\ \psi^N \\ \bar{\psi}^1 \\ \vdots \\ \bar{\psi}^N \end{pmatrix} \quad (32.70)$$

with

$$f(j, n) = \frac{1}{\sqrt{\hbar\beta}} e^{-i\omega_n \hbar\beta j/N}. \quad (32.71)$$

Then we obtain for the transformation matrix M the identity

$$\begin{aligned}
 M_{k,:} \cdot (M^\dagger)_{:,l} &= \sum_{n=1}^N M_{k,n} (M^\dagger)_{n,l} \\
 &= \sum_{n=1}^N M_{k,n} M_{l,n}^* \\
 &= \sum_{n=1}^N f(k,n) f(l,n)^* \\
 &= \frac{1}{\hbar\beta} \sum_{n=1}^N e^{-i\omega_n \hbar\beta k/N} e^{+i\omega_n \hbar\beta l/N} \\
 &= \frac{1}{\hbar\beta} \sum_{n=1}^N e^{-i\omega_n \hbar\beta(l-k)/N} \\
 &= \frac{1}{\hbar\beta} \sum_{n=1}^N e^{-i(2n+1-\theta(\zeta))(\pi/N)(l-k)} \\
 &= \frac{1}{\hbar\beta} e^{i(1-\theta(\zeta))(\pi/N)(l-k)} \underbrace{\sum_{n=1}^N e^{i2n(\pi/N)(l-k)}}_{N\delta_{l,k}} \\
 &= \frac{N}{\hbar\beta} \delta_{l,k}
 \end{aligned} \tag{32.72}$$

Therefore, the matrix $\sqrt{\hbar\beta/N} \cdot M$ is unitary! We therefore find

$$\left| \det \begin{pmatrix} M & 0 \\ 0 & M \end{pmatrix} \right| = \left(\sqrt{\frac{\hbar\beta}{N}} \right)^{-1} \cdot |\det \text{"unitary matrix"}| = \left(\left(\frac{\hbar\beta}{N} \right)^N \right)^{-1} = \left(\frac{N}{\hbar\beta} \right)^N \tag{32.73}$$

Further one finds for bosons

$$\int d(v^\dagger, v) e^{-v^\dagger A v} = \pi^N (\det A)^{-1} = \pi^N (\det A)^{-\zeta} \tag{32.74}$$

therefore, the equation

$$\int d(\bar{\psi}, \psi) e^{-\bar{\psi}_n \epsilon \psi_n} = (\beta\epsilon)^{-\zeta} \tag{32.75}$$

is being reproduced except for a diverging factor. Whereas ψ_n represents a single Grassmann-variable here, i.e.

$$\overset{\text{here}}{\psi_n} \neq \psi_{j,n} \tag{32.76}$$

We consider the fermionic case.

Recall the transformation-rule

$$\bar{v} = \left(\overline{M}\right)^{\text{inv}} \bar{\phi}, \quad v = M^{\text{inv}} \phi \quad \Rightarrow \quad \int d(\bar{\phi}, \phi) f(\bar{\phi}, \phi) = \det \left(M^{\text{inv}} \overline{M}^{\text{inv}} \right) \int d(\bar{v}, v) f(\bar{\phi}(\bar{v}), \phi(v)) \quad (32.77)$$

Here we have

$$\bar{\psi}_{\cdot} = \left(\overline{M}\right)^{\text{inv}} \bar{\psi}^{\cdot}, \quad \psi_{\cdot} = M^{\text{inv}} \psi^{\cdot} \quad \Rightarrow \quad \int d(\bar{\phi}, \phi) f(\bar{\phi}, \phi) = \det \left(\frac{\hbar\beta}{N} \right)^N \int d(\bar{v}, v) f(\bar{\phi}(\bar{v}), \phi(v)) \quad (32.78)$$

whereas we used

$$M^{\text{inv}} = M^{\dagger} \cdot \frac{\beta}{N} = \sqrt{\frac{\beta}{N}} \left(\sqrt{\frac{\beta}{N}} \right)^{\dagger} \quad \Rightarrow \quad \det M^{\text{inv}} = \left(\frac{\beta}{N} \right)^N \quad (32.79)$$

Proof 32.29 of Corollary 18.15

The discrete version of the partition-function reads

$$Z = \lim_{N \rightarrow \infty} \int_{\substack{\psi^N = \zeta \psi^0 \\ \bar{\psi}^N = \zeta \bar{\psi}^0}} \prod_{j=0}^N d_{\zeta}(\bar{\psi}^j, \psi^j) \exp -\delta \sum_{j=0}^{N-1} -\frac{\bar{\psi}^{j+1} - \bar{\psi}^j}{\delta} \cdot \psi^j + H(\bar{\psi}^{j+1}, \psi^j) - \mu N(\bar{\psi}^{j+1}, \psi^j)$$

Now we examine each of the terms in the exponents individually, when we transform coordinates according to

$$\psi^j = \frac{1}{\hbar\beta} \sum_{\omega_n} \psi_{\omega_n} e^{-i\omega_n \tau_j}, \quad \bar{\psi}^j = \frac{1}{\hbar\beta} \sum_{\omega_n} \bar{\psi}_{\omega_n} e^{+i\omega_n \tau_j} \quad (32.80)$$

for

$$n \in \left\{ -\frac{N-1}{2}, \dots, -1, 0, +1, \dots, \frac{N-1}{2} \right\} \quad (32.81)$$

We also notice that

$$\sum_{j=0}^{N-1} e^{i\tau_j(\omega_n - \omega_m)} = N\delta_{n,m} \quad (32.82)$$

The Derivative

Noticing that $1 \equiv e^{i\omega_n \tau_0}$ and remembering that $\delta = \hbar\beta/N$ we obtain

$$\begin{aligned}
 & \delta \sum_{j=0}^{N-1} -\frac{1}{\delta} \left(\frac{1}{\sqrt{\hbar\beta}} \sum_{\omega_n} \bar{\psi}_{\omega_n} e^{i\omega_n \tau_{j+1}} - \bar{\psi}_{\omega_n} e^{i\omega_n \tau_j} \right) \frac{1}{\sqrt{\hbar\beta}} \sum_{\omega_m} \psi_{\omega_m} e^{-i\omega_m \tau_j} \\
 &= \delta \sum_{j=0}^{N-1} -\frac{1}{\delta \hbar \beta} \sum_{\omega_n} \bar{\psi}_{\omega_n} (e^{i\omega_n \tau_1} - 1) e^{i\omega_n \tau_j} \sum_{\omega_m} \psi_{\omega_m} e^{-i\omega_m \tau_j} \\
 &\approx \delta \sum_{j=0}^{N-1} -\frac{1}{\delta \hbar \beta} \sum_{\omega_n} \bar{\psi}_{\omega_n} i\omega_n \delta e^{i\omega_n \tau_j} \sum_{\omega_m} \psi_{\omega_m} e^{-i\omega_m \tau_j} \\
 &= \frac{\delta}{\hbar \beta} \sum_{\omega_n, \omega_m} -i\omega_n \bar{\psi}_{\omega_n} \psi_{\omega_m} \sum_{j=0}^{N-1} e^{i\tau_j(\omega_n - \omega_m)} \\
 &= \sum_{\omega_n} -i\omega_n \bar{\psi}_{\omega_n} \psi_{\omega_n}
 \end{aligned}$$

The other Quadratic Terms

$$\begin{aligned}
 \delta \sum_{j=0}^{N-1} \bar{\psi}(\tau_{j+1}) \psi(\tau_j) &= \frac{\delta}{\hbar \beta} \sum_{j=0}^{N-1} \sum_{\omega_n, \omega_m} \bar{\psi}_{\omega_n} e^{i\omega_n \tau_{j+1}} \psi_{\omega_m} e^{-i\omega_m \tau_j} \\
 &= \sum_{\omega_n, \omega_m} \bar{\psi}_{\omega_n} \psi_{\omega_m} \frac{\delta}{\hbar \beta} \left(\sum_{j=0}^{N-1} e^{i(\omega_n - \omega_m) \tau_j} \right) e^{i\omega_n \tau_1} \\
 &= \sum_{\omega_n} \bar{\psi}_{\omega_n} \psi_{\omega_n} e^{i\omega_n \delta}
 \end{aligned}$$

Proof 32.30 of Lemma 18.16 For the non-interacting case see, for example, [5] Chapter 4 or [35] Equation (2.137b), or [47] chapter 7.2 p. 139, 145, 149 or [13] p. 37, 38. For the interacting case, we refer to [35] again.

Proof 32.31 of Theorem 18.17

- Cf. Negele p. 70 following.
- Cf. Diehl QFT-1-Script 20, 21, page 98
- Cf. Coleman p. 428.

Proof 32.32 of Theorem 18.18

We compute straightforwardly

$$\begin{aligned}
 \mathcal{G}_{\alpha, \alpha'}^0(\tau) &= -\frac{1}{\hbar} \langle \psi_{\alpha}(\tau) \bar{\psi}_{\alpha'}(0) \rangle \\
 &= -\frac{1}{\hbar} \frac{1}{\hbar \beta} \sum_{\omega_n, \omega_m \in \Omega_{\zeta}} \langle \psi_{\alpha}(\omega_n) \bar{\psi}_{\alpha'}(\omega_m) \rangle e^{-i\omega_n \tau} \\
 &= \dots
 \end{aligned}$$

now we use that the action is diagonal w.r.t. Matsubara-frequencies s.t. we obtain

$$\langle \psi_\alpha(\omega) \bar{\psi}_{\alpha'}(\omega_m) \rangle = \delta_{n,m} \langle \psi_\alpha(\omega_n) \bar{\psi}_{\alpha'}(\omega_n) \rangle \quad (32.83)$$

and therefore we obtain

$$\dots = \frac{1}{\hbar\beta} \sum_{\omega_n \in \Omega_\zeta} \langle \psi_\alpha(\omega_n) \bar{\psi}_{\alpha'}(\omega_m) \rangle e^{-i\omega_n \tau} \equiv \frac{1}{\hbar\beta} \sum_{\omega_n \in \Omega_\zeta} \mathcal{G}_{\alpha,\alpha'}^0(i\omega_n) e^{-i\omega_n \tau} \quad (32.84)$$

Therefore we obtain for the non-interacting Green-function

$$\mathcal{G}_{\alpha,\alpha'}^0(i\omega_n) \equiv -\frac{1}{\hbar} \langle \psi_\alpha(\omega_n) \bar{\psi}_{\alpha'}(\omega_n) \rangle_0 = -\frac{1}{\hbar} \left(\left[-i\omega_n + \frac{H - \mu}{\hbar} \right]^{\text{inv}} \right)_{\alpha,\alpha'} = \frac{1}{\hbar} \left(\left[i\omega_n - \frac{H - \mu}{\hbar} \right]^{\text{inv}} \right)_{\alpha,\alpha'} \quad (32.85)$$

which completes the proof.

33. Appendix - Proofs for Part VI

Proof 33.1 of Lemma 21.15

The only non-trivial identity is the interaction-term. Using Lemma 18.14 and Corollary 1.12 we obtain

$$\begin{aligned}
 S_{\text{el-ph-int}}[\chi, \bar{\chi}] &= \oint \sum_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ \tau \in (0, \hbar\beta) \\ v \in \{1, 2, \dots, rd_x\} \\ m, n \in \{1, 2, 3, \dots\}}} \frac{g_{mnv}(\mathbf{k}, \mathbf{q})}{\hbar \sqrt{N_p}} \left(\phi_v(\tau, \mathbf{q}) + \bar{\phi}_v(\tau, -\mathbf{q}) \right) \bar{\psi}_{m\sigma}(\tau, \mathbf{k} + \mathbf{q}) \psi_{n\sigma}(\tau, \mathbf{k}) \\
 &= \oint \sum_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ \tau \in (0, \hbar\beta) \\ v \in \{1, 2, \dots, rd_x\} \\ m, n \in \{1, 2, 3, \dots\}}} \frac{g_{mnv}(\mathbf{k}, \mathbf{q})}{\hbar \sqrt{N_p}} \phi_v(\tau, \mathbf{q}) \bar{\psi}_{m\sigma}(\tau, \mathbf{k} + \mathbf{q}) \psi_{n\sigma}(\tau, \mathbf{k}) \\
 &\quad + \oint \sum_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ \tau \in (0, \hbar\beta) \\ v \in \{1, 2, \dots, rd_x\} \\ m, n \in \{1, 2, 3, \dots\}}} \frac{g_{mnv}(\mathbf{k}, -\mathbf{q})}{\hbar \sqrt{N_p}} \bar{\phi}_v(\tau, \mathbf{q}) \bar{\psi}_{m\sigma}(\tau, \mathbf{k} - \mathbf{q}) \psi_{n\sigma}(\tau, \mathbf{k}) \\
 &= \frac{1}{(\hbar\beta)^{3/2}} \sum_{\substack{\omega_q \in \Omega_{+1} \\ \omega_n, \omega_k \in i\Omega_{-1}}} \sum_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ v \in \{1, 2, \dots, rd_x\} \\ m, n \in \{1, 2, 3, \dots\}}} \frac{g_{mnv}(\mathbf{k}, \mathbf{q})}{\hbar \sqrt{N_p}} \phi_v(i\omega_q, \mathbf{q}) \bar{\psi}_{m\sigma}(i\omega_n, \mathbf{k} + \mathbf{q}) \psi_{n\sigma}(i\omega_k, \mathbf{k}) \int_0^{\hbar\beta} d\tau e^{i\tau(-\omega_q + \omega_n - \omega_k)} \\
 &\quad + \frac{1}{(\hbar\beta)^{3/2}} \sum_{\substack{\omega_q \in \Omega_{+1} \\ \omega_n, \omega_k \in i\Omega_{-1}}} \sum_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ v \in \{1, 2, \dots, rd_x\} \\ m, n \in \{1, 2, 3, \dots\}}} \frac{g_{mnv}(\mathbf{k}, -\mathbf{q})}{\hbar \sqrt{N_p}} \bar{\phi}_v(i\omega_q, \mathbf{q}) \bar{\psi}_{m\sigma}(i\omega_n, \mathbf{k} - \mathbf{q}) \psi_{n\sigma}(i\omega_k, \mathbf{k}) \int_0^{\hbar\beta} d\tau e^{i\tau(+\omega_q + \omega_n - \omega_k)} \\
 &= \frac{1}{\sqrt{\hbar\beta}} \sum_{\substack{\omega_q \in \Omega_{+1} \\ \omega_n, \omega_k \in i\Omega_{-1}}} \sum_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ v \in \{1, 2, \dots, rd_x\} \\ m, n \in \{1, 2, 3, \dots\}}} \frac{g_{mnv}(\mathbf{k}, \mathbf{q})}{\hbar \sqrt{N_p}} \phi_v(i\omega_q, \mathbf{q}) \bar{\psi}_{m\sigma}(i\omega_n, \mathbf{k} + \mathbf{q}) \psi_{n\sigma}(i\omega_k, \mathbf{k}) \delta_{\omega_n, \omega_k + \omega_q} \\
 &\quad + \frac{1}{\sqrt{\hbar\beta}} \sum_{\substack{\omega_q \in \Omega_{+1} \\ \omega_n, \omega_k \in i\Omega_{-1}}} \sum_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ v \in \{1, 2, \dots, rd_x\} \\ m, n \in \{1, 2, 3, \dots\}}} \frac{g_{mnv}(\mathbf{k}, -\mathbf{q})}{\hbar \sqrt{N_p}} \bar{\phi}_v(i\omega_q, \mathbf{q}) \bar{\psi}_{m\sigma}(i\omega_n, \mathbf{k} - \mathbf{q}) \psi_{n\sigma}(i\omega_k, \mathbf{k}) \delta_{\omega_n, \omega_k - \omega_q} \\
 &= \frac{1}{\sqrt{\hbar\beta}} \sum_{\substack{k \in i\Omega_{-1} \times Q \\ q \in i\Omega_{+1} \times Q \\ v \in \{1, 2, \dots, rd_x\} \\ m, n \in \{1, 2, 3, \dots\}}} \frac{g_{mnv}(\mathbf{k}, \mathbf{q})}{\hbar \sqrt{N_p}} \left(\phi_v(q) + \bar{\phi}_v(-q) \right) \bar{\psi}_{m\sigma}(q + k) \psi_{n\sigma}(k)
 \end{aligned}$$

Proof 33.2 of Lemma 21.16

By straightforward computation we obtain

$$\begin{aligned}
 S_{\text{el-ph-int}}[\chi, \bar{\chi}] &= \frac{1}{\sqrt{\hbar\beta}} \sum_{\substack{\omega_q \in \Omega_{+1} \\ \omega_k \in \text{i}\Omega_{-1}}} \sum \frac{g_{mn\nu}(\mathbf{k}, \mathbf{q})}{\hbar\sqrt{N_p}} \phi_\nu(\text{i}\omega_q, \mathbf{q}) \bar{\psi}_{m\sigma}(\text{i}(\omega_k + \omega_q), \mathbf{k} + \mathbf{q}) \psi_{n\sigma}(\text{i}\omega_k, \mathbf{k}) \\
 &+ \frac{1}{\sqrt{\hbar\beta}} \sum_{\substack{\omega_q \in \Omega_{+1} \\ \omega_k \in \text{i}\Omega_{-1}}} \sum \frac{g_{mn\nu}(\mathbf{k}, -\mathbf{q})}{\hbar\sqrt{N_p}} \bar{\phi}_\nu(\mathbf{q}, \text{i}\omega_q) \bar{\psi}_{m\sigma}(\mathbf{k} - \mathbf{q}, \text{i}(\omega_k - \omega_q)) \psi_{n\sigma}(\mathbf{k}, \text{i}\omega_k) \\
 &= \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ \omega_q \in \Omega_{+1} \\ \nu \in \{1, \dots, rd_{\mathbf{x}}\}}} \phi_\nu(\text{i}\omega_q, \mathbf{q}) \rho_\nu(\text{i}\omega_q, \mathbf{q}) + \bar{\phi}_\nu(\text{i}\omega_q, \mathbf{q}) \rho_\nu(-\text{i}\omega_q, -\mathbf{q})
 \end{aligned}$$

with density-like-interaction

$$\rho_\nu(\text{i}\omega_q, \mathbf{q}) = \frac{1}{\sqrt{\hbar\beta N_p}} \sum_{\substack{\mathbf{k} \in \mathbf{Q} \\ \sigma \in \{\downarrow, \uparrow\} \\ \omega_k \in \text{i}\Omega_{-1} \\ n, m \in \{1, \dots, \infty\}}} \frac{g_{mn\nu}(\mathbf{k}, \mathbf{q})}{\hbar} \bar{\psi}_{m\sigma}(\text{i}(\omega_k + \omega_q), \mathbf{k} + \mathbf{q}) \psi_{n\sigma}(\text{i}\omega_k, \mathbf{k}) \quad (33.1)$$

Proof 33.3 of Lemma 21.18

We keep in mind that

$$S_{\text{ph}}[\phi, \bar{\phi}] = \sum_{\substack{q \in \text{i}\Omega_{+1} \times \mathbf{Q} \\ \nu \in \{1, 2, \dots, rd_{\mathbf{x}}\}}} \bar{\phi}_\nu(q) \left(-\frac{1}{\hbar} \mathcal{G}_{0,\nu}^{-\text{B}}(q) \right) \phi_\nu(q) \quad (33.2)$$

and

$$S_{\text{el-ph-int}}[\chi, \bar{\chi}] = \sum_{\substack{q \in \text{i}\Omega_{+1} \times \mathbf{Q} \\ \nu \in \{1, \dots, rd_{\mathbf{x}}\}}} \phi_\nu(q) \left(\rho_\nu(q) - \bar{j}_\nu(q) \right) + \bar{\phi}_\nu(q) \left(\rho_\nu(-q) - j_\nu(q) \right)$$

Now we change the variables according to

$$\begin{aligned}
 \phi_\nu(q) &\mapsto \phi_\nu(q) + \hbar \mathcal{G}_{0,\nu}(q) \left(\rho_\nu(-q) - j_\nu(q) \right) ng \\
 \bar{\phi}_\nu(q) &\mapsto \bar{\phi}_\nu(q) + \hbar \mathcal{G}_{0,\nu}(q) \left(\rho_\nu(q) - \bar{j}_\nu(q) \right)
 \end{aligned}$$

This yields

$$\begin{aligned}
 (S_{\text{ph}} + S_{\text{el-ph}}) [\bar{\chi}, \chi] = & \sum_{\substack{q \in i\Omega_{+1} \times Q \\ \nu \in \{1, \dots, rd_x\}}} \left[\bar{\phi}_\nu(q) + \hbar \mathcal{G}_0^{\text{B}}(q) \left(\rho_\nu(q) - \bar{j}_\nu(q) \right) \right] \left(-\frac{1}{\hbar} \mathcal{G}_0^{-\text{B}}(q) \right) \left[\phi_\nu(q) + \hbar \mathcal{G}_0^{\text{B}}(q) \left(\rho_\nu(-q) - j_\nu(q) \right) \right] \\
 & + \sum_{\substack{q \in i\Omega_{+1} \times Q \\ \nu \in \{1, \dots, rd_x\}}} \left[\phi_\nu(q) + \hbar \mathcal{G}_0^{\text{B}}(q) \left(\rho_\nu(-q) - j_\nu(q) \right) \right] \left(\rho_\nu(q) - \bar{j}_\nu(q) \right) \\
 & + \sum_{\substack{q \in i\Omega_{+1} \times Q \\ \nu \in \{1, \dots, rd_x\}}} \left[\bar{\phi}_\nu(q) + \hbar \mathcal{G}_0^{\text{B}}(q) \left(\rho_\nu(q) - \bar{j}_\nu(q) \right) \right] \left(\rho_\nu(-q) - j_\nu(q) \right)
 \end{aligned} \tag{33.3}$$

Now we obtain 8 terms in the total action, the first four from the first line, the next two (number 5 and 6) from the second line, and the last two (number 7 and 8) from the third line. We list them below:

$$S_1 \equiv \sum_{\substack{q \in i\Omega_{+1} \times Q \\ \nu \in \{1, \dots, rd_x\}}} \bar{\phi}_\nu(q) \left(-\frac{1}{\hbar} \mathcal{G}_0^{-\text{B}}(q) \right) \phi_\nu(q) \tag{33.4}$$

$$S_2 \equiv -\hbar \sum_{\substack{q \in i\Omega_{+1} \times Q \\ \nu \in \{1, \dots, rd_x\}}} \left(\rho_\nu(q) - \bar{j}_\nu(q) \right) \mathcal{G}_0^{\text{B}}(q) \left(\rho_\nu(-q) - j_\nu(q) \right) \tag{33.5}$$

$$S_3 \equiv - \sum_{\substack{q \in i\Omega_{+1} \times Q \\ \nu \in \{1, \dots, rd_x\}}} \left(\rho_\nu(q) - \bar{j}_\nu(q) \right) \phi_\nu(q) \tag{33.6}$$

$$S_4 \equiv - \sum_{\substack{q \in i\Omega_{+1} \times Q \\ \nu \in \{1, \dots, rd_x\}}} \bar{\phi}_\nu(q) \left(\rho_\nu(-q) - j_\nu(q) \right)$$

$$S_5 \equiv \sum_{\substack{q \in i\Omega_{+1} \times Q \\ \nu \in \{1, \dots, rd_x\}}} \phi_\nu(q) \left(\rho_\nu(q) - \bar{j}_\nu(q) \right) \tag{33.7}$$

$$S_6 \equiv \hbar \sum_{\substack{q \in i\Omega_{+1} \times Q \\ \nu \in \{1, \dots, rd_x\}}} \left(\rho_\nu(-q) - j_\nu(q) \right) \mathcal{G}_0^{\text{B}}(q) \left(\rho_\nu(q) - \bar{j}_\nu(q) \right) \tag{33.8}$$

$$S_7 \equiv \sum_{\substack{q \in i\Omega_{+1} \times Q \\ \nu \in \{1, \dots, rd_x\}}} \bar{\phi}_\nu(q) \left(\rho_\nu(-q) - j_\nu(q) \right) \tag{33.9}$$

$$S_8 \equiv \hbar \sum_{\substack{q \in i\Omega_{+1} \times Q \\ \nu \in \{1, \dots, rd_x\}}} \left(\rho_\nu(q) - \bar{j}_\nu(q) \right) \mathcal{G}_0^{\text{B}}(q) \left(\rho_\nu(-q) - j_\nu(q) \right) \tag{33.10}$$

Now we obtain

$$"2 + 6 = 0 = 3 + 5 = 4 + 7" \tag{33.11}$$

such that only 1 and 8 contribute. Then, when one integrates out the phonons (with Lemma 16.18) and we discard the divergent factor because it cancels out when we compute Green-functions, one obtains the effective electron-electron interaction, giving us

$$\begin{aligned} S_{\text{el-el-int}}[\psi, \bar{\psi}] &= +\hbar \sum_{\substack{q \in i\Omega_{+1} \times Q \\ v \in \{1, \dots, rd_x\}}} \left(\rho_v(q) - \bar{j}_B(q) \right) \mathcal{G}_0^B(q) \left(\rho_v(-q) - j_B(q) \right) \\ &= +\hbar \sum_{\substack{q \in i\Omega_{+1} \times Q \\ v \in \{1, \dots, rd_x\}}} \overline{\left(\rho_v(-q) - j_B(q) \right)} \mathcal{G}_0^B(q) \left(\rho_v(-q) - j_B(q) \right) \end{aligned}$$

For the latter identity: We compute straightforwardly for the source-independent term we obtain

$$\begin{aligned} S_{\text{el-el-int}}[\psi, \bar{\psi}] &= +\hbar \sum_{\substack{q \in Q \\ \omega_q \in \Omega_{+1} \\ v \in \{1, \dots, rd_x\}}} \rho_v(i\omega_q, \mathbf{q}) \mathcal{G}_0^B(\mathbf{q}, i\omega_q) \rho_v(-i\omega_q, -\mathbf{q}) \\ &= + \sum_{\substack{q \in Q \\ \omega_q \in \Omega_{+1} \\ v \in \{1, \dots, rd_x\}}} \frac{\rho_v(i\omega_q, +\mathbf{q}) \rho_v(-i\omega_q, -\mathbf{q})}{i\omega_q - \omega_v(\mathbf{q})} \\ &= - \sum_{\substack{q \in Q \\ \omega_q \in \Omega_{+1} \\ v \in \{1, \dots, rd_x\}}} \frac{\rho_v(+i\omega_q, +\mathbf{q}) \rho_v(-i\omega_q, -\mathbf{q})}{\omega_q^2 + \omega_v^2(\mathbf{q})} (i\omega_q + \omega_v(\mathbf{q})) \\ &= - \sum_{\substack{q \in Q \\ \omega_q \in \Omega_{+1} \\ v \in \{1, \dots, rd_x\}}} \frac{\omega_v(\mathbf{q})}{\omega_q^2 + \omega_v^2(\mathbf{q})} \rho_v(+i\omega_q, +\mathbf{q}) \rho_v(-i\omega_q, -\mathbf{q}) \\ &\equiv - \sum_{\substack{q \in i\Omega_{+1} \times Q \\ v \in \{1, \dots, rd_x\}}} \frac{\omega_v(\mathbf{q})}{\omega_q^2 + \omega_v^2(\mathbf{q})} \rho_v(+q) \rho_v(-q) \end{aligned}$$

whereas in the second last line we used $\omega_v(\mathbf{q}) = \omega_v(-\mathbf{q})$.

Proof 33.4 of Lemma 21.19

Now we simplify the interaction-term

$$\begin{aligned} &\rho_v(+i\omega_q, +\mathbf{q}) \rho_v(-i\omega_q, -\mathbf{q}) \\ &= \frac{1}{\hbar \beta N_p} \sum_{\substack{\mathbf{k}, \mathbf{k}' \in Q \\ \omega_k, \omega_{k'} \in i\Omega_{+1} \\ \sigma, \sigma' \in \{\downarrow, \uparrow\} \\ n, n', m, m' \in \{1, \dots, \infty\}}} \frac{g_{mnv}(\mathbf{k}, \mathbf{q}) g_{m'n'v}(\mathbf{k}', -\mathbf{q})}{\hbar^2} \bar{\psi}_{m\sigma}(i(\omega_k + \omega_q), \mathbf{k} + \mathbf{q}) \psi_{n\sigma}(i\omega_k, \mathbf{k}) \bar{\psi}_{m'\sigma'}(i(\omega_{k'} - \omega_q), \mathbf{k}' - \mathbf{q}) \psi_{n'\sigma'}(i\omega_{k'}, \mathbf{k}') \end{aligned} \quad (33.12)$$

introducing the four-momentum we can write this shorter as

$$\begin{aligned}\rho_v(q)\rho_v(-q) &= \frac{1}{\hbar\beta N_p} \sum_{\substack{k,k' \in Q \times \Omega_{-1} \\ \sigma,\sigma' \in \{\downarrow,\uparrow\} \\ n,n',m,m' \in \{1,\dots,\infty\}}} \frac{g_{mnv}(\mathbf{k}, \mathbf{q})g_{m'n'v}(\mathbf{k}', -\mathbf{q})}{\hbar^2} \bar{\psi}_{m\sigma}(k+q)\psi_{n\sigma}(k)\bar{\psi}_{m'\sigma'}(k'-q)\psi_{n'\sigma'}(k') \\ &= \frac{-1}{\hbar\beta N_p} \sum_{\substack{k,k' \in Q \times \Omega_{-1} \\ \sigma,\sigma' \in \{\downarrow,\uparrow\} \\ n,n',m,m' \in \{1,\dots,\infty\}}} \frac{g_{mnv}(\mathbf{k}, \mathbf{q})g_{m'n'v}(\mathbf{k}', -\mathbf{q})}{\hbar^2} \bar{\psi}_{m\sigma}(k+q)\bar{\psi}_{m'\sigma'}(k'-q)\psi_{n\sigma}(k)\psi_{n'\sigma'}(k')\end{aligned}$$

Therefore we obtain for the source-independent interaction-term

$$\begin{aligned}S_{\text{el-el-int}}[\psi, \bar{\psi}] &= \frac{+1}{\hbar\beta N_p} \sum_{\substack{\mathbf{q} \in Q \\ i\omega_q \in i\Omega_{+1} \\ v \in \{1,\dots,rd_x\} \\ k,k' \in Q \times \Omega_{-1} \\ \sigma,\sigma' \in \{\downarrow,\uparrow\} \\ n,n',m,m' \in \{1,\dots,\infty\}}} \frac{\omega_v(\mathbf{q})}{\omega_q^2 + \omega_v^2(\mathbf{q})} \frac{g_{mnv}(\mathbf{k}, \mathbf{q})g_{m'n'v}(\mathbf{k}', -\mathbf{q})}{\hbar^2} \bar{\psi}_{m\sigma}(k+q)\bar{\psi}_{m'\sigma'}(k'-q)\psi_{n\sigma}(k)\psi_{n'\sigma'}(k') \\ &= \sum_{\substack{\mathbf{q} \in Q \\ \omega_q \in \Omega_{+1} \\ v \in \{1,\dots,rd_x\} \\ k,k' \in Q \times \Omega_{-1} \\ \sigma,\sigma' \in \{\downarrow,\uparrow\} \\ n,n',m,m' \in \{1,\dots,\infty\}}} \bar{\psi}_{m\sigma}(k+q)\bar{\psi}_{m'\sigma'}(k'-q) \frac{+1}{\hbar\beta N_p} \frac{\omega_v(\mathbf{q})}{\omega_q^2 + \omega_v^2(\mathbf{q})} \frac{g_{mnv}(\mathbf{k}, \mathbf{q})g_{m'n'v}(\mathbf{k}', -\mathbf{q})}{\hbar^2} \psi_{n\sigma}(k)\psi_{n'\sigma'}(k') \\ &\stackrel{\mathbf{k}' \mapsto \mathbf{k}' + \mathbf{q}}{=} \sum_{\substack{\mathbf{q} \in Q \\ \omega_q \in \Omega_{+1} \\ v \in \{1,\dots,rd_x\} \\ k,k' \in Q \times \Omega_{-1} \\ \sigma,\sigma' \in \{\downarrow,\uparrow\} \\ n,n',m,m' \in \{1,\dots,\infty\}}} \bar{\psi}_{m\sigma}(k+q)\bar{\psi}_{m'\sigma'}(k') \frac{-1}{\hbar\beta N_p} \frac{\omega_v(\mathbf{q})}{\omega_q^2 + \omega_v^2(\mathbf{q})} \frac{g_{mnv}(\mathbf{k}, \mathbf{q})g_{m'n'v}(\mathbf{k}' + \mathbf{q}, -\mathbf{q})}{\hbar^2} \psi_{n'\sigma'}(k' + q)\psi_{n\sigma}(k) \\ &\stackrel{\text{Lemma 8.30}}{=} \sum_{\substack{\mathbf{q} \in Q \\ \omega_q \in \Omega_{+1} \\ v \in \{1,\dots,rd_x\} \\ k,k' \in Q \times \Omega_{-1} \\ \sigma,\sigma' \in \{\downarrow,\uparrow\} \\ n,n',m,m' \in \{1,\dots,\infty\}}} \bar{\psi}_{m\sigma}(k+q)\bar{\psi}_{m'\sigma'}(k') \frac{-1}{\hbar\beta N_p} \frac{\omega_v(\mathbf{q})}{\omega_q^2 + \omega_v^2(\mathbf{q})} \frac{g_{mnv}(\mathbf{k}, \mathbf{q})g_{n'm'v}^*(\mathbf{k}', \mathbf{q})}{\hbar^2} \psi_{n'\sigma'}(k' + q)\psi_{n\sigma}(k)\end{aligned}\tag{33.13}$$

Now we introduce¹ the new fermionic Matsubara summation-variable $i\Omega_{-1} \times Q \ni p \equiv k + q$ in order to replace the summation over $q \equiv p - k$. This yields

¹More concretely: For each k we introduce a p such that the equation $p \equiv k + q$ is fulfilled.

$$\begin{aligned}
 & S_{\text{el-el-int}}[\psi, \bar{\psi}] \\
 &= \sum_{\substack{v \in \{1, \dots, rd_x\} \\ p, k, k' \in i\Omega_{-1} \times Q \\ \sigma, \sigma' \in \{\downarrow, \uparrow\} \\ n, n', m, m' \in \{1, \dots, \infty\}}} \bar{\psi}_{m\sigma}(p) \bar{\psi}_{m'\sigma'}(k') \frac{-1}{\hbar \beta N_p} \frac{\omega_v(\mathbf{p} - \mathbf{k})}{\omega_{p-k}^2 + \omega_v^2(\mathbf{p} - \mathbf{k})} \frac{g_{mnv}(\mathbf{k}, \mathbf{p} - \mathbf{k}) g_{n'm'v}^*(\mathbf{k}', \mathbf{p} - \mathbf{k})}{\hbar^2} \psi_{n'\sigma'}(k' + p - k) \psi_{n\sigma}(k) \\
 &= \sum_{\substack{p, k, k' \in i\Omega_{-1} \times Q \\ \sigma, \sigma' \in \{\downarrow, \uparrow\} \\ n, n', m, m' \in \{1, \dots, \infty\}}} \bar{\psi}_{m\sigma}(p) \bar{\psi}_{m'\sigma'}(k') \frac{-1}{\hbar \beta N_p} \sum_{v \in \{1, \dots, rd_x\}} \frac{\omega_v(\mathbf{p} - \mathbf{k})}{\omega_{p-k}^2 + \omega_v^2(\mathbf{p} - \mathbf{k})} \frac{g_{mnv}(\mathbf{k}, \mathbf{p} - \mathbf{k}) g_{n'm'v}^*(\mathbf{k}', \mathbf{p} - \mathbf{k})}{\hbar^2} \psi_{n'\sigma'}(k' + p - k) \psi_{n\sigma}(k) \\
 &= \sum_{\substack{p, p', k, k' \in i\Omega_{-1} \times Q \\ \sigma, \sigma' \in \{\downarrow, \uparrow\} \\ n, n', m, m' \in \{1, \dots, \infty\}}} \bar{\psi}_{m\sigma}(p) \bar{\psi}_{m'\sigma'}(k') V_{m, m'; n', n}(p, k'; p', k) \psi_{n'\sigma'}(p') \psi_{n\sigma}(k)
 \end{aligned} \tag{33.14}$$

with a quartic vertex over which we sum without any restrictions

$$V_{m, m'; n', n}(p, k'; p', k) \equiv \frac{-1}{\hbar \beta N_p} \sum_{v \in \{1, \dots, rd_x\}} \frac{\omega_v(\mathbf{p} - \mathbf{k})}{\omega_{p-k}^2 + \omega_v^2(\mathbf{p} - \mathbf{k})} \frac{g_{mnv}(\mathbf{k}, \mathbf{p} - \mathbf{k}) g_{n'm'v}^*(\mathbf{k}', \mathbf{p} - \mathbf{k})}{\hbar^2} \delta_{p', k' + p - k} \tag{33.15}$$

Now we will simplify this expression, by relabelling the indices. This yields, after replacing $k' \longleftrightarrow p'$, the following expression

$$S_{\text{el-el-int}}[\psi, \bar{\psi}] = \sum_{\substack{p, k, k' \in Q \times \Omega_{-1} \\ \sigma, \sigma' \in \{\downarrow, \uparrow\} \\ n, n', m, m' \in \{1, \dots, \infty\}}} \bar{\psi}_{m\sigma}(p) \bar{\psi}_{m'\sigma'}(p') V_{m, m'; n', n}(p, p'; k', k) \psi_{n'\sigma'}(k') \psi_{n\sigma}(k)$$

with

$$V_{m, m'; n', n}(p, p'; k', k) = \frac{-1}{\hbar \beta N_p} \sum_{v \in \{1, \dots, rd_x\}} \frac{\omega_v(\mathbf{p} - \mathbf{k})}{\omega_{p_0-k_0}^2 + \omega_v^2(\mathbf{p} - \mathbf{k})} \frac{g_{mnv}(\mathbf{k}, \mathbf{p} - \mathbf{k}) g_{n'm'v}^*(\mathbf{p}', \mathbf{p} - \mathbf{k})}{\hbar^2} \delta_{k', p' + p - k} \tag{33.16}$$

Now we rewrite this even further to find a bilinearform

$$\begin{aligned}
 & S_{\text{el-el-int}}[\psi, \bar{\psi}] \\
 &= \sum_{\substack{p, k, k' \in Q \times \Omega_{-1} \\ \sigma, \sigma', \pi, \pi' \in \{\downarrow, \uparrow\} \\ n, n', m, m' \in \{1, \dots, \infty\}}} \bar{\psi}_{m\sigma}(p) \bar{\psi}_{m'\sigma'}(p') V_{m, m'; n', n}^{\sigma, \sigma', \pi', \pi}(p, p'; k', k) \psi_{n'\pi'}(k') \psi_{n\pi}(k)
 \end{aligned}$$

with

$$V_{m,m';n',n}^{\sigma,\sigma',\pi',\pi}(p,p';k',k) = \frac{-1}{\hbar\beta N_p} \sum_{\nu \in \{1,\dots,rd_x\}} \frac{\omega_\nu(\mathbf{p}-\mathbf{k})}{\omega_{p_0-k_0}^2 + \omega_\nu^2(\mathbf{p}-\mathbf{k})} \frac{g_{mn\nu}(\mathbf{k},\mathbf{p}-\mathbf{k})g_{n'm'\nu}^*(\mathbf{p}',\mathbf{p}-\mathbf{k})}{\hbar^2} \delta_{k'+k,p'+p} \delta_{\sigma,\pi} \delta_{\sigma',\pi'} \quad (33.17)$$

Now we condense the notation even further, by introducing the multi-index

$$\alpha \equiv (m_\alpha, \sigma_\alpha, \omega_\alpha, \mathbf{k}_\alpha) \in \{1, 2, \dots\} \times \{\downarrow, \uparrow\} \times \Omega_{-1} \times \mathbb{Q} \equiv \Xi, \quad (33.18)$$

which then yields (notice the exchanged order of the Grassmann-fields)

$$S_{\text{el-el-int}}[\psi, \bar{\psi}] = \sum_{\alpha_1, \dots, \alpha_4 \in \Xi} \bar{\psi}_{\alpha_1} \bar{\psi}_{\alpha_2} V_{\alpha_1, \alpha_2; \alpha_3, \alpha_4} \psi_{\alpha_4} \psi_{\alpha_3}$$

with Lemma 8.30

$$\begin{aligned} & V_{\alpha_1, \alpha_2; \alpha_3, \alpha_4} \\ &= \frac{1}{\hbar\beta N_p} \sum_{\nu \in \{1, \dots, rd_x\}} \frac{\omega_\nu(\mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4})}{\omega_{\alpha_1-\alpha_4}^2 + \omega_\nu^2(\mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4})} \frac{g_{m_{\alpha_1}m_{\alpha_4}\nu}(\mathbf{k}_{\alpha_4}, \mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4})g_{m_{\alpha_3}m_{\alpha_2}\nu}^*(\mathbf{k}_{\alpha_2}, \mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4})}{\hbar^2} \delta_{k_{\alpha_3}+k_{\alpha_4}, k_{\alpha_2}+k_{\alpha_1}} \delta_{\sigma_{\alpha_1}, \sigma_{\alpha_4}} \delta_{\sigma_{\alpha_2}, \sigma_{\alpha_3}} \\ &= \frac{1}{\hbar\beta N_p} \sum_{\nu \in \{1, \dots, rd_x\}} \frac{\omega_\nu(\mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4})}{\omega_{\alpha_1-\alpha_4}^2 + \omega_\nu^2(\mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4})} \frac{g_{m_{\alpha_1}m_{\alpha_4}\nu}(\mathbf{k}_{\alpha_4}, \mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4})g_{m_{\alpha_2}m_{\alpha_3}\nu}(\mathbf{k}_{\alpha_2} + \mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4}, -\mathbf{k}_{\alpha_1} + \mathbf{k}_{\alpha_4})}{\hbar^2} \delta_{k_{\alpha_3}+k_{\alpha_4}, k_{\alpha_2}+k_{\alpha_1}} \delta_{\sigma_{\alpha_1}, \sigma_{\alpha_4}} \delta_{\sigma_{\alpha_2}, \sigma_{\alpha_3}} \end{aligned} \quad (33.19)$$

Proof 33.5 of Corollary 21.20 We compute

$$\left(\sum_{\alpha_i} \psi_{\alpha_1}^\dagger \psi_{\alpha_2}^\dagger V_{\alpha_1, \alpha_2; \alpha_3, \alpha_4} \psi_{\alpha_4} \psi_{\alpha_3} \right)^\dagger = \sum_{\alpha_i} \psi_{\alpha_3}^\dagger \psi_{\alpha_4}^\dagger V_{\alpha_1, \alpha_2; \alpha_3, \alpha_4}^* \psi_{\alpha_2} \psi_{\alpha_1} = \sum_{\alpha_i} \psi_{\alpha_1}^\dagger \psi_{\alpha_2}^\dagger V_{\alpha_4, \alpha_3; \alpha_2, \alpha_1} \psi_{\alpha_4} \psi_{\alpha_3}$$

Therefore this is hermitian, if the following equation is fulfilled²

$$\begin{aligned} & \sum_{\nu \in \{1, \dots, rd_x\}} \frac{\omega_\nu(\mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4})}{\omega_{\alpha_1-\alpha_4}^2 + \omega_\nu^2(\mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4})} \frac{g_{m_{\alpha_1}m_{\alpha_4}\nu}(\mathbf{k}_{\alpha_4}, \mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4})g_{m_{\alpha_3}m_{\alpha_2}\nu}^*(\mathbf{k}_{\alpha_2}, \mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4})}{\hbar^3 \beta N_p} \\ & \stackrel{!}{=} \sum_{\nu \in \{1, \dots, rd_x\}} \frac{\omega_\nu(\mathbf{k}_{\alpha_4} - \mathbf{k}_{\alpha_1})}{\omega_{\alpha_4-\alpha_1}^2 + \omega_\nu^2(\mathbf{k}_{\alpha_4} - \mathbf{k}_{\alpha_1})} \frac{g_{m_{\alpha_4}m_{\alpha_1}\nu}^*(\mathbf{k}_{\alpha_1}, \mathbf{k}_{\alpha_4} - \mathbf{k}_{\alpha_1})g_{m_{\alpha_2}m_{\alpha_3}\nu}(\mathbf{k}_{\alpha_3}, \mathbf{k}_{\alpha_4} - \mathbf{k}_{\alpha_1})}{\hbar^3 \beta N_p} \end{aligned}$$

Now since

² Maybe we need to go through this proof again, since we symmetrized the expression.

$$g_{m_{\alpha_2} m_{\alpha_3} \nu}(\mathbf{k}_{\alpha_3}, \mathbf{k}_{\alpha_4} - \mathbf{k}_{\alpha_1}) = g_{m_{\alpha_3} m_{\alpha_2} \nu}^*(\mathbf{k}_{\alpha_3} + \mathbf{k}_{\alpha_4} - \mathbf{k}_{\alpha_1}, \mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4}) = g_{m_{\alpha_3} m_{\alpha_2} \nu}^*(\mathbf{k}_{\alpha_2}, \mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4})$$

etc, which proves the corollary.

Proof 33.6 of Corollary 21.22

Electrons:

One obtains through direct computation using Equation (18.37) and using further the identity relating Green-functions to functional derivatives

$$\begin{aligned} \langle n_{n\sigma}(\mathbf{k}) \rangle &= \langle c_{n\sigma}^\dagger(\mathbf{k}) c_{n\sigma}(\mathbf{k}) \rangle \\ &= \frac{\delta^2}{\delta \bar{j}_{n,\sigma}(\tau=0-, \mathbf{k}) \delta j_{n,\sigma}(\tau'=0, \mathbf{k})} \frac{Z[j_F, \bar{j}_F, j_B, \bar{j}_B]}{Z} \Big|_{\bar{j}=j=0} \\ &= \frac{1}{\hbar\beta} \lim_{\delta \downarrow 0} \sum_{\omega_m \in \Omega_{-1}} e^{+i\omega_m \delta} \frac{\partial^2}{\partial \bar{j}_{n,\sigma}(i\omega_m, \mathbf{k}) \partial j_{n,\sigma}(i\omega_m, \mathbf{k})} \frac{Z[j_F, \bar{j}_F, j_B, \bar{j}_B]}{Z} \Big|_{\bar{j}=j=0} \end{aligned}$$

whereas we used that the partition-function is diagonal w.r.t. the Matsubara-frequencies. After recalling the definition of the total electronic number operator as

$$\langle N \rangle = \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\downarrow, \uparrow\} \\ n \in \{1, 2, \dots, \infty\}}} \langle n_{n\sigma}(\mathbf{k}) \rangle.$$

Follows from using the Lemma ?? as

$$\begin{aligned} \langle H_{el} \rangle &= \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\downarrow, \uparrow\} \\ n \in \{1, \dots, \infty\}}} \epsilon_{n\sigma}(\mathbf{k}) \langle c_{n\sigma}^\dagger(\mathbf{k}) c_{n\sigma}(\mathbf{k}) \rangle \\ &= \frac{1}{\hbar\beta} \lim_{\delta \downarrow 0} \sum_{\substack{\sigma \in \{\downarrow, \uparrow\} \\ k \in i\Omega_{-1} \times Q \\ n \in \{1, 2, \dots, \infty\}}} \epsilon_{n\sigma}(\mathbf{k}) e^{+i\omega_k \delta} \frac{\partial^2}{\partial \bar{j}_{n,\sigma}(k) \partial j_{n,\sigma}(k)} \frac{Z[j_F, \bar{j}_F, j_B, \bar{j}_B]}{Z} \Big|_{\bar{j}=j=0} \end{aligned}$$

Phonons:

The equations for the phonons are derived analogously.

Interactions:

We first rewrite the interaction term as

$$\begin{aligned}
 \left\langle c_{\mathbf{k}+\mathbf{q}m\sigma}^\dagger c_{\mathbf{k}n\sigma} \left(b_{\mathbf{q}\nu} + b_{-\mathbf{q}\nu}^\dagger \right) \right\rangle &= \left\langle \bar{\psi}_{\mathbf{k}+\mathbf{q}m\sigma}(0) \psi_{\mathbf{k}n\sigma}(0) \left(\phi_{\mathbf{q}\nu}(0) + \bar{\phi}_{-\mathbf{q}\nu}(0) \right) \right\rangle \\
 &= \frac{\delta^2}{\delta \bar{j}(\tau=0-, \mathbf{k}n\sigma) \delta j(\tau=0, \mathbf{k} + \mathbf{q}m\sigma)} \left(\frac{\delta}{\delta \bar{j}(\tau=0-, \mathbf{q}\nu) + \delta j(\tau=0, -\mathbf{q}\nu)} \right) \frac{Z[j, \bar{j}]}{Z} \Big|_{j=\bar{j}=0}
 \end{aligned} \tag{33.20}$$

Now we compute functional derivatives again. First we note that the functional derivatives can be expanded as

$$\frac{\delta}{\delta \bar{j}(\tau=0, \mathbf{q}\nu)} = \frac{1}{\sqrt{\hbar\beta}} \sum_{\omega_m \in \Omega_{+1}} \frac{\partial}{\partial \bar{j}(i\omega_m, \mathbf{q}\nu)}, \tag{33.21}$$

and

$$\frac{\delta}{\delta j(\tau=0, -\mathbf{q}\nu)} = \frac{1}{\sqrt{\hbar\beta}} \sum_{\omega_m \in \Omega_{+1}} \frac{\partial}{\partial j(-i\omega_m, -\mathbf{q}\nu)}. \tag{33.22}$$

We also expand the other functional derivatives as

$$\frac{\delta^2}{\delta \bar{j}(\tau=0-, \mathbf{k}n\sigma) \delta j(\tau=0, \mathbf{k} + \mathbf{q}m\sigma)} = \frac{1}{\hbar\beta} \lim_{\delta \downarrow 0} \sum_{\omega_n, \omega_{n'} \in \Omega_{-1}} e^{i\omega_n \delta} \frac{\partial^2}{\partial \bar{j}_{n,\sigma}(i\omega_n, \mathbf{k}) \partial j_{m,\sigma}(i\omega_{n'}, \mathbf{k} + \mathbf{q})} \tag{33.23}$$

which gives the desired result.

Proof 33.7 of Lemma 22.3 We use Lemma 18.16 and obtain the first equality. For the second equation, we notice that the expression can be rewritten with respect to the density of states as

$$\begin{aligned}
 \Omega_{0,\text{el}}(\mu, T, V) &= \frac{-1}{\beta} \sum_{\substack{\mathbf{k} \in \mathbf{Q} \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \text{Log} \left(1 + e^{-\beta(\epsilon_{m\mathbf{k}\sigma} - \mu)} \right) \\
 &= \frac{-1}{\beta} \int dE \sum_{\substack{\mathbf{k} \in \mathbf{Q} \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \text{Log} \left(1 + e^{-\beta(E - \mu)} \right) \delta(E - \epsilon_{m\mathbf{k}\sigma}) \\
 &= \frac{-1}{\beta} \int dE \text{Log} \left(1 + e^{-\beta(E - \mu)} \right) \rho_0(E - \mu) \\
 &= \frac{-1}{\beta} \int dE \text{Log} \left(1 + e^{-\beta E} \right) \rho_0(E)
 \end{aligned}$$

Proof 33.8 of Theorem 22.15

The grand canonical potential $\Omega(\mu, T, V)$ obeys the equation³

³ See also Mahan 3.6.1 for an expansion of the thermodynamic potential.

$$\Omega - \Omega_0 = -\frac{1}{\beta} \text{Log} \frac{Z}{Z_0} = -\frac{1}{\beta} \times \text{„all connected diagrams“} \quad (33.24)$$

Thus we compute the connected diagrams of the system.

Computing the connected Diagrams through a Series-Expansion

We can therefore simply compute the partition-function in a perturbative expansion, take from the appearing Feynman-diagrams the connected ones and add the expression for the non-interacting grand canonical potential on top of it.

$$\begin{aligned} & \frac{Z}{Z_0} \\ &= \langle \exp -S_{\text{int}}[\bar{\chi}, \chi] \rangle_0 \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \langle S_{\text{int}}^n[\bar{\chi}, \chi] \rangle_0 \\ &= 1 + 0 \\ &+ \frac{1}{2} \sum_{\substack{\tau_1, \tau_2 \in [0, \hbar\beta] \\ \mu, \mu' \in \{\downarrow, \uparrow\} \\ \mathbf{k}, \mathbf{k}', \mathbf{q}, \mathbf{q}' \in \mathbb{Q} \\ m, m', n, n' \in \{1, 2, \dots\} \\ v, v' \in \{1, 2, \dots, r d_x\}}} \frac{g_{mnv}(\mathbf{k}, \mathbf{q}) g_{m'n'v'}(\mathbf{k}', \mathbf{q}')}{\hbar^2 N_p} \left\langle \left(\phi_{\mathbf{q}v}(\tau_1) + \bar{\phi}_{-\mathbf{q}v}(\tau_1) \right) \bar{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1) \psi_{\mathbf{k}n\mu}(\tau_1) \left(\phi_{\mathbf{q}'v'}(\tau_2) + \bar{\phi}_{-\mathbf{q}'v'}(\tau_2) \right) \bar{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2) \psi_{\mathbf{k}'n'\mu'}(\tau_2) \right\rangle_0 \\ &= \dots \end{aligned}$$

We now factorize and compute the correlators

$$\begin{aligned} & \left\langle \left(\phi_{\mathbf{q}v}(\tau_1) + \bar{\phi}_{-\mathbf{q}v}(\tau_1) \right) \bar{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1) \psi_{\mathbf{k}n\mu}(\tau_1) \left(\phi_{\mathbf{q}'v'}(\tau_2) + \bar{\phi}_{-\mathbf{q}'v'}(\tau_2) \right) \bar{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2) \psi_{\mathbf{k}'n'\mu'}(\tau_2) \right\rangle_0 \\ &= \langle \bar{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1) \psi_{\mathbf{k}n\mu}(\tau_1) \bar{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2) \psi_{\mathbf{k}'n'\mu'}(\tau_2) \rangle_0 \left\langle \left(\phi_{\mathbf{q}v}(\tau_1) + \bar{\phi}_{-\mathbf{q}v}(\tau_1) \right) \left(\phi_{\mathbf{q}'v'}(\tau_2) + \bar{\phi}_{-\mathbf{q}'v'}(\tau_2) \right) \right\rangle_0 \\ &= -\hbar \langle \bar{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1) \psi_{\mathbf{k}n\mu}(\tau_1) \bar{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2) \psi_{\mathbf{k}'n'\mu'}(\tau_2) \rangle_0 \mathcal{D}_{\substack{\mathbf{q}, -\mathbf{q}' \\ v, v'}}^0(\tau_1 - \tau_2) \\ &= +\hbar \langle \psi_{\mathbf{k}n\mu}(\tau_1) \psi_{\mathbf{k}'n'\mu'}(\tau_2) \bar{\psi}_{\mathbf{k}+\mathbf{q}m\mu}(\tau_1) \bar{\psi}_{\mathbf{k}'+\mathbf{q}'m'\mu'}(\tau_2) \rangle_0 \mathcal{D}_{\substack{\mathbf{q}, -\mathbf{q}' \\ v, v'}}^0(\tau_1 - \tau_2) \\ &= +\hbar^3 \left(\mathcal{G}_{\substack{\mathbf{k}, \mathbf{k}'+\mathbf{q}' \\ n, m' \\ \mu, \mu'}}^0(\tau_1 - \tau_2) \mathcal{G}_{\substack{\mathbf{k}', \mathbf{k}+\mathbf{q} \\ n', m \\ \mu', \mu}}^0(\tau_2 - \tau_1) - \mathcal{G}_{\substack{\mathbf{k}, \mathbf{k}+\mathbf{q} \\ n, m \\ \mu, \mu}}^0(\tau_1 - \tau_1) \mathcal{G}_{\substack{\mathbf{k}', \mathbf{k}'+\mathbf{q}' \\ n', m' \\ \mu', \mu'}}^0(\tau_2 - \tau_2) \right) \mathcal{D}_{\substack{\mathbf{q}, -\mathbf{q}' \\ v, v'}}^0(\tau_1 - \tau_2) \end{aligned}$$

Reinserting this in the equation above yields

$$\begin{aligned}
 \dots &= 1 \\
 &+ \frac{\hbar}{2N_p} \sum_{\substack{\tau_1, \tau_2 \in [0, \hbar\beta] \\ \mu \in \{\downarrow, \uparrow\} \\ \mathbf{k}, \mathbf{k}', \mathbf{q} \in \mathbf{Q} \\ m, n \in \{1, 2, \dots\} \\ v \in \{1, 2, \dots, rd_x\}}} g_{mnv}(\mathbf{k}, \mathbf{q}) g_{nmv}(\mathbf{k}', -\mathbf{q}) \mathcal{G}_{\mathbf{k}, \mathbf{k}' - \mathbf{q}}^0(\tau_1 - \tau_2) \mathcal{G}_{\mathbf{k}', \mathbf{k} + \mathbf{q}}^0(\tau_2 - \tau_1) \mathcal{D}_{\mathbf{q}v}^0(\tau_1 - \tau_2) \\
 &- \frac{\hbar}{2N_p} \sum_{\substack{\tau_1, \tau_2 \in [0, \hbar\beta] \\ \mu, \mu' \in \{\downarrow, \uparrow\} \\ \mathbf{k}, \mathbf{k}', \mathbf{q} \in \mathbf{Q} \\ m, m' \in \{1, 2, \dots\} \\ v, v' \in \{1, 2, \dots, rd_x\}}} g_{mmv}(\mathbf{k}, \mathbf{q}) g_{m'm'v'}(\mathbf{k}', -\mathbf{q}) \mathcal{G}_{\mathbf{k}, \mathbf{k} + \mathbf{q}}^0(\tau_1 - \tau_1) \mathcal{G}_{\mathbf{k}', \mathbf{k}' - \mathbf{q}}^0(\tau_2 - \tau_2) \mathcal{D}_{\mathbf{q}, v}^0(\tau_1 - \tau_2)
 \end{aligned}$$

This can be simplified by using the Kronecker-relation for the delta-functions

$$\begin{aligned}
 \dots &= 1 + \frac{\hbar}{2N_p} \sum_{\substack{\tau_1, \tau_2 \in [0, \hbar\beta] \\ \mu \in \{\downarrow, \uparrow\} \\ \mathbf{k}', \mathbf{q} \in \mathbf{Q} \\ m, n \in \{1, 2, \dots\} \\ v \in \{1, 2, \dots, rd_x\}}} g_{mnv}(\mathbf{k}' - \mathbf{q}, \mathbf{q}) g_{mnv}^*(\mathbf{k}' - \mathbf{q}, +\mathbf{q}) \mathcal{G}_{\mathbf{k}' - \mathbf{q}, \mathbf{k}' - \mathbf{q}}^0(\tau_1 - \tau_2) \mathcal{G}_{\mathbf{k}', \mathbf{k}'}^0(\tau_2 - \tau_1) \mathcal{D}_{\mathbf{q}v}^0(\tau_1 - \tau_2) \\
 &- \frac{\hbar}{2N_p} \sum_{\substack{\tau_1, \tau_2 \in [0, \hbar\beta] \\ \mu, \mu' \in \{\downarrow, \uparrow\} \\ \mathbf{k}, \mathbf{k}' \in \mathbf{Q} \\ m, m' \in \{1, 2, \dots\} \\ v, v' \in \{1, 2, \dots, rd_x\}}} g_{mmv}(\mathbf{k}, 0) g_{m'm'v'}(\mathbf{k}', 0) \mathcal{G}_{\mathbf{k}, \mathbf{k}}^0(\tau_1 - \tau_1) \mathcal{G}_{\mathbf{k}', \mathbf{k}'}^0(\tau_2 - \tau_2) \mathcal{D}_{\mathbf{q}=0, v}^0(\tau_1 - \tau_2) = \dots
 \end{aligned}$$

Now using the hermiticity of g and changing summation $\mathbf{q} \mapsto -\mathbf{q}$ and simplifying the notation we obtain

$$\begin{aligned}
 \dots &= 1 + \frac{\hbar}{2N_p} \sum_{\substack{\tau_1, \tau_2 \in [0, \hbar\beta] \\ \mu \in \{\downarrow, \uparrow\} \\ \mathbf{k}', \mathbf{q} \in \mathbf{Q} \\ m, n \in \{1, 2, \dots\} \\ v \in \{1, 2, \dots, rd_x\}}} |g_{mnv}(\mathbf{k}' + \mathbf{q}, -\mathbf{q})|^2 \mathcal{G}_{n, \mu}^0(\mathbf{k}' + \mathbf{q}, \tau_1 - \tau_2) \mathcal{G}_{m, \mu}^0(\mathbf{k}', \tau_2 - \tau_1) \mathcal{D}_v^0(\mathbf{q}, \tau_1 - \tau_2) \\
 &- \frac{\hbar}{2N_p} \sum_{\substack{\tau_1, \tau_2 \in [0, \hbar\beta] \\ \mu, \mu' \in \{\downarrow, \uparrow\} \\ \mathbf{k}, \mathbf{k}' \in \mathbf{Q} \\ m, m' \in \{1, 2, \dots\} \\ v, v' \in \{1, 2, \dots, rd_x\}}} g_{mmv}(\mathbf{k}, 0) g_{m'm'v'}(\mathbf{k}', 0) \mathcal{G}_{m, \mu}^0(\mathbf{k}, \tau_1 - \tau_1) \mathcal{G}_{m', \mu'}^0(\mathbf{k}', \tau_2 - \tau_2) \mathcal{D}_v^0(\mathbf{q} = 0, \tau_1 - \tau_2) = \dots
 \end{aligned}$$

ToBeUpdated, because one of the terms doesn't necessarily vanish.⁴

[See Waste-Folder 1, topic 18 for the final expressions.]

⁴Cf. the GPAW-documentation:

For the three translational modes at $\mathbf{q} = 0$ the matrix elements $g_{mnv}(\mathbf{k}, \mathbf{q}) = 0$, as a consequence of the acoustic sum rule.

See <https://gpaw.readthedocs.io/documentation/elph/elph.html>

See further Felicianos article:

In the case of the three translational modes at $|\mathbf{q}| = 0$ we set the matrix elements $g_{mnv}(\mathbf{k}, \mathbf{q})$ to zero, as a consequence of the acoustic sum rule (...).

Remark: Notice that one could also use now

$$|g_{mn\nu}(\mathbf{k}' + \mathbf{q}, -\mathbf{q})| = |g_{nm\nu}(\mathbf{k}', \mathbf{q})| \quad (33.25)$$

Proof 33.9 of Lemma 23.3

We recall

$$\langle H_{0,\text{el}} \rangle_0(\mu, T, V) = \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, 2, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} F_\beta(\epsilon_{m,\sigma,\mathbf{k}} - \mu) \quad (33.26)$$

and then compute the temperature-derivative as

$$\partial_T \langle H_{0,\text{el}} \rangle_0(\mu, T, V) = \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, 2, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} \partial_T F_1(\beta(\epsilon_{m,\sigma,\mathbf{k}} - \mu)) = \dots \quad (33.27)$$

Now we use chain and product rule to obtain

$$\begin{aligned} \partial_T F_1(\beta(\epsilon_{m,\sigma,\mathbf{k}} - \mu)) &= (\partial_T F_1)(\beta(\epsilon_{m,\sigma,\mathbf{k}} - \mu)) \cdot \partial_T \beta(\epsilon_{m,\sigma,\mathbf{k}} - \mu) \\ &\stackrel{\text{Lemma 9.45}}{=} -\frac{1}{4} \text{sech}^2\left(\frac{\epsilon_{m,\sigma,\mathbf{k}} - \mu}{2k_B T}\right) \cdot (\partial_T \beta)(\epsilon_{m,\sigma,\mathbf{k}} - \mu) \\ &= -\frac{1}{4} \text{sech}^2\left(\frac{\epsilon_{m,\sigma,\mathbf{k}} - \mu}{2k_B T}\right) \cdot \left(\frac{-1}{k_B T^2}\right) (\epsilon_{m,\sigma,\mathbf{k}} - \mu) \end{aligned} \quad (33.28)$$

which allows us to write the semi-final result as

$$\partial_T \langle H_{0,\text{el}} \rangle_0(\mu, T, V) = \dots = \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \frac{\epsilon_{m\mathbf{k}\sigma}}{4} \text{sech}^2\left(\frac{\epsilon_{m\mathbf{k}\sigma} - \mu}{2k_B T}\right) \left[\frac{\epsilon_{m\mathbf{k}\sigma} - \mu}{k_B T^2}\right] = \star \star \star \quad (33.29)$$

Now we introduce

$$\rho(E) = \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, 2, \dots, \infty\}}} \delta(E - \xi_{m\mathbf{k}\sigma}) = \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, 2, \dots, \infty\}}} \delta(E + \mu - \epsilon_{m\mathbf{k}\sigma}) \quad (33.30)$$

and

$$\rho_{\text{aux}}(E) = \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, 2, \dots, \infty\}}} \delta(E - \epsilon_{m\mathbf{k}\sigma}) \quad (33.31)$$

which implies $\rho_{\text{aux}}(E + \mu) = \rho(E)$. This allows us to rewrite the expression as

$$\begin{aligned}
 \partial_T \langle H_{0,\text{el}} \rangle_0 &= \star = \int dE \rho(E) \frac{E + \mu}{4} \operatorname{sech}^2 \left(\frac{E}{2k_B T} \right) \left[\frac{E}{k_B T^2} \right] \\
 &= \int dE \rho_{\text{aux}}(E) \frac{E}{4} \operatorname{sech}^2 \left(\frac{E - \mu}{2k_B T} \right) \left[\frac{E - \mu}{k_B T^2} \right] \\
 &= \star\star
 \end{aligned} \tag{33.32}$$

We also recall the definition of the density of states from quantum espresso as

$$\rho_{\text{QE}}(E) = \frac{1}{N_p} \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, 2, \dots, \infty\}}} \delta(E - \epsilon_{m\mathbf{k}\sigma}) = \frac{1}{N_p} \rho_{\text{aux}}(E) \tag{33.33}$$

such that we obtain the final result

$$\frac{1}{N_x N_y N_z} \partial_T \langle H_{0,\text{el}} \rangle_0(\mu, T, V) = \int dE \rho_{\text{QE}}(E) \frac{E}{4} \operatorname{sech}^2 \left(\frac{E - \mu}{2k_B T} \right) \left[\frac{E - \mu(T)}{k_B T^2} \right] \tag{33.34}$$

which can also be (equivalently) rewritten as

$$\begin{aligned}
 \partial_T \frac{\langle H_{\text{el}} \rangle_0}{N_x N_y N_z} &= \int dE \frac{E}{4} \operatorname{sech}^2 \left(\frac{E - \mu(T)}{2k_B T} \right) \left[\frac{E - \mu(T)}{k_B T^2} \right] \rho_{\text{QE}}(E) \\
 &= \frac{1}{N_x N_y N_z} \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \frac{\epsilon_{m\mathbf{k}\sigma}}{4} \operatorname{sech}^2 \left(\frac{\epsilon_{m\mathbf{k}\sigma} - \mu(T)}{2k_B T} \right) \left[\frac{\epsilon_{m\mathbf{k}\sigma} - \mu(T)}{k_B T^2} \right]
 \end{aligned} \tag{33.35}$$

Proof 33.10 of Lemma 23.4 We compute the derivative w.r.t. the chemical potential and obtain the desired result with Lemma 9.45 and the chain-rule. This gives basically immediately the desired result

$$\partial_\mu \frac{\langle H_{\text{el}} \rangle_0}{N_x N_y N_z} = \frac{\beta}{4} \int dE E \operatorname{sech}^2 \left(\frac{E - \mu}{2k_B T} \right) \rho_{\text{QE}}(E) \tag{33.36}$$

Proof 33.11 of Theorem 23.10

We use Equation (21.20) and compute

$$\frac{\partial}{\partial j_{n\sigma}(k)} Z[j, \bar{j}] = - \int \mathcal{D}(\bar{\psi}, \psi) \bar{\psi}_{n\sigma}(k) \exp \left(\bar{j}_{\text{el}} \psi + \bar{\psi} j_{\text{el}} \right) \exp - \left(S_{\text{el}} + S_{\text{el-el-int}}^{[j_{\text{ph}} \bar{j}_{\text{ph}}]} \right) [\psi, \bar{\psi}] \tag{33.37}$$

and this yields for the double-derivative

$$\frac{\partial^2}{\partial \bar{j}_{n\sigma}(k) \partial j_{n\sigma}(k)} Z[j, \bar{j}] = - \int \mathcal{D}(\bar{\psi}, \psi) \psi_{n\sigma}(k) \bar{\psi}_{n\sigma}(k) \exp \left(\bar{j}_{\text{el}} \psi + \bar{\psi} j_{\text{el}} \right) \exp - \left(S_{\text{el}} + S_{\text{el-el-int}}^{[j_{\text{ph}}, \bar{j}_{\text{ph}}]} \right) [\psi, \bar{\psi}] \quad (33.38)$$

hence we obtain

$$\frac{\partial^2}{\partial \bar{j}_{n\sigma}(k) \partial j_{n\sigma}(k)} \frac{Z[j, \bar{j}]}{Z} \Big|_{j=\bar{j}=0} = - \langle \psi_{n\sigma}(k) \bar{\psi}_{n\sigma}(k) \rangle = + \mathcal{G}_{\sigma, \sigma}^{n, n}(k, k) \quad (33.39)$$

whereas in the last step we used Equation [TBReferenced](#).

I think the other identity is proven in Waste-Folder 1

Proof 33.12 of Theorem 23.9

We perform a calculation analogously to Equation (31.162)

$$\begin{aligned} \langle H_{\text{el}} \rangle &= \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} \langle n_{m\mathbf{k}\sigma} \rangle \\ &= \frac{\hbar}{i} \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\}}} \epsilon_{m\mathbf{k}\sigma} G_{\mathbf{k}, \mathbf{k}}^{<, m}(t=0) \\ &= \frac{1}{2\pi i} \oint_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\} \\ E \in (-\infty, +\infty)}} \epsilon_{m\mathbf{k}\sigma} G_{\mathbf{k}, \mathbf{k}}^{<, m}(E) \\ &= \frac{-1}{\pi} \lim_{\eta \rightarrow 0} \oint_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\} \\ E \in (-\infty, +\infty)}} \epsilon_{m\mathbf{k}\sigma} F_{\beta}(E) \text{Im} G_{\mathbf{k}, \mathbf{k}}^{\text{ret}, m}(E + i\eta). \end{aligned} \quad (33.40)$$

Proof 33.13 of Theorem 23.11

We divide the proof in consecutive steps.

1. We first compute the first derivative as

$$\frac{\partial}{\partial j_v(q)} Z[j, \bar{j}] = \int \mathcal{D}(\bar{\psi}, \psi) \left(\frac{-\partial}{\partial j_v(q)} S_{\text{el-el-int}}^{[j, \bar{j}]}[\psi, \bar{\psi}] \right) \exp - S_{[j, \bar{j}]}[\psi, \bar{\psi}] \quad (33.41)$$

2. We then compute the second derivative with Equation (21.20) as

$$\frac{\partial^2}{\partial \bar{j}_v(q) \partial j_v(q)} Z[j, \bar{j}] = \int \mathcal{D}(\bar{\psi}, \psi) \left[\frac{-\partial^2}{\partial \bar{j}_v(q) \partial j_v(q)} S_{\text{el-el-int}}^{[j, \bar{j}]}[\psi, \bar{\psi}] + \left(\frac{-\partial}{\partial j_v(q)} S_{\text{el-el-int}}^{[j, \bar{j}]}[\psi, \bar{\psi}] \right) \left(\frac{-\partial}{\partial \bar{j}_v(q)} S_{\text{el-el-int}}^{[j, \bar{j}]}[\psi, \bar{\psi}] \right) \right] \exp - S_{[j, \bar{j}]}[\psi, \bar{\psi}] \quad (33.42)$$

3. We then compute the individual terms with Equation (21.16)

$$\frac{\partial}{\partial j_v(q)} S_{\text{el-el-int}}^{[j, \bar{j}]}[\psi, \bar{\psi}] = \hbar \bar{j}_v(q) \mathcal{G}_{0, v}(q) - \hbar \rho_v(q) \mathcal{G}_{0, v}(q) \quad (33.43)$$

4. and for the other derivative we have with Equation (21.16)

$$\frac{\partial}{\partial \bar{j}_v(q)} S_{\text{el-el-int}}[\psi, \bar{\psi}]_{[j, \bar{j}]} = \hbar \mathcal{G}_{0,v}(q) j_v(q) - \hbar \mathcal{G}_{0,v}(q) \rho_v(-q) \quad (33.44)$$

5. and for the double-derivative we have

$$\frac{\partial^2}{\partial \bar{j}_v(q) \partial j_v(q)} S_{\text{el-el-int}}[\psi, \bar{\psi}]_{[j, \bar{j}]} = \hbar \mathcal{G}_{0,v}(q) \quad (33.45)$$

Combining these results, we obtain

$$\begin{aligned} \frac{\partial^2}{\partial \bar{j}_v(q) \partial j_v(q)} \frac{Z[j, \bar{j}]}{Z} \Big|_{j=\bar{j}=0} &= -\hbar \mathcal{G}_{0,v}(q) + \mathcal{G}_{0,v}^2(q) \frac{1}{Z} \int \mathcal{D}(\bar{\psi}, \psi) (\hbar^2 \rho_v(q) \rho_v(-q)) \exp -S_{[j, \bar{j}]}[\psi, \bar{\psi}] \\ &= -\hbar \mathcal{G}_{0,v}(q) + \mathcal{G}_{0,v}^2(q) \hbar^2 \langle \rho_v(q) \rho_v(-q) \rangle \\ &= \star \star \star \end{aligned} \quad (33.46)$$

Now we focus on the second term, which can be expanded using

$$\rho_v(q) = \frac{1}{\sqrt{\hbar \beta N_p}} \sum_{\substack{\sigma \in \{\downarrow, \uparrow\} \\ k \in i\Omega_{-1} \times Q \\ n, m \in \{1, \dots, \infty\}}} \frac{g_{mnv}(\mathbf{k}, \mathbf{q})}{\hbar} \bar{\psi}_{m\sigma}(k+q) \psi_{n\sigma}(k) \quad (33.47)$$

which implies for the product

$$\begin{aligned} \mathcal{G}_{0,v}^2(q) \hbar^2 \langle \rho_v(q) \rho_v(-q) \rangle &= \frac{\mathcal{G}_{0,v}^2(q)}{\hbar \beta N_p} \sum_{\substack{\sigma, \sigma' \in \{\downarrow, \uparrow\} \\ k, k' \in i\Omega_{-1} \times Q \\ n, n', m, m' \in \{1, \dots, \infty\}}} g_{mnv}(\mathbf{k}, \mathbf{q}) g_{m'n'v}(\mathbf{k}', -\mathbf{q}) \langle \bar{\psi}_{m\sigma}(k+q) \psi_{n\sigma}(k) \bar{\psi}_{m'\sigma'}(k'-q) \psi_{n'\sigma'}(k') \rangle \\ &= \frac{\mathcal{G}_{0,v}^2(q)}{\hbar \beta N_p} \sum_{\substack{\sigma, \sigma' \in \{\downarrow, \uparrow\} \\ k, k' \in i\Omega_{-1} \times Q \\ n, n', m, m' \in \{1, \dots, \infty\}}} g_{mnv}(\mathbf{k}, \mathbf{q}) g_{m'n'v}(\mathbf{k}', -\mathbf{q}) \langle \psi_{n\sigma}(k) \psi_{n'\sigma'}(k') \bar{\psi}_{m'\sigma'}(k'-q) \bar{\psi}_{m\sigma}(k+q) \rangle \end{aligned} \quad (33.48)$$

Now we use the Bethe-Salpeter-equation **TBReferenced** and Equation **Exp. Value as GF** to approximate⁵ the expectation-value as

$$\begin{aligned} &\langle \psi_{n\sigma}(k) \psi_{n'\sigma'}(k') \bar{\psi}_{m'\sigma'}(k'-q) \bar{\psi}_{m\sigma}(k+q) \rangle \\ &\approx \hbar^2 \left(\mathcal{G}_{n,m}(k, k+q) \mathcal{G}_{n',m'}(k', k'-q) - \mathcal{G}_{n,m'}(k, k'-q) \mathcal{G}_{n',m}(k', k+q) \right) + \mathcal{O}(g^2) \\ &= \hbar^2 (\delta_{q,0} \delta_{n,m} \delta_{n',m'} \mathcal{G}_{n,\sigma}(k) \mathcal{G}_{n',\sigma'}(k') - \delta_{k+q,k'} \delta_{n,m'} \delta_{n',m} \delta_{\sigma,\sigma'} \mathcal{G}_{n,\sigma}(k) \mathcal{G}_{n',\sigma'}(k')) + \mathcal{O}(g^2) \end{aligned} \quad (33.49)$$

Plugging this into the equation above yields the final results

⁵ In order to go beyond this approximation: See TBDigitalized, TBD6.

$$\begin{aligned}
 \star \star \star &= -\hbar \mathcal{G}_{0,v}(q) + \frac{\hbar}{\beta N_p} \mathcal{G}_{0,v}^2(q) \delta_{q,0} \sum_{\substack{k,k' \in i\Omega_{-1} \times Q \\ n,n' \in \{1,\dots,\infty\} \\ \sigma,\sigma' \in \{\downarrow,\uparrow\}}} g_{nnv}(\mathbf{k}, 0) g_{n'n'v}(\mathbf{k}', 0) \mathcal{G}_{n,\sigma}(k) \mathcal{G}_{n',\sigma'}(k') \\
 &\quad - \frac{\hbar}{\beta N_p} \mathcal{G}_{0,v}^2(q) \sum_{\substack{k \in i\Omega_{-1} \times Q \\ n,n' \in \{1,\dots,\infty\} \\ \sigma \in \{\downarrow,\uparrow\}}} g_{n'nv}(\mathbf{k}, \mathbf{q}) g_{nn'v}(\mathbf{k} + \mathbf{q}, -\mathbf{q}) \mathcal{G}_{n,\sigma}(k) \mathcal{G}_{n',\sigma}(k + q) + \mathcal{O}(g^4) \\
 &= -\hbar \mathcal{G}_{0,v}(q) + \frac{\hbar}{\beta N_p} \mathcal{G}_{0,v}^2(q) \delta_{q,0} \sum_{\substack{k,k' \in i\Omega_{-1} \times Q \\ n,n' \in \{1,\dots,\infty\} \\ \sigma,\sigma' \in \{\downarrow,\uparrow\}}} g_{nnv}(\mathbf{k}, 0) g_{n'n'v}(\mathbf{k}', 0) \mathcal{G}_{n,\sigma}(k) \mathcal{G}_{n',\sigma'}(k') \\
 &\quad - \frac{\hbar}{\beta N_p} \mathcal{G}_{0,v}^2(q) \sum_{\substack{k \in i\Omega_{-1} \times Q \\ n,n' \in \{1,\dots,\infty\} \\ \sigma \in \{\downarrow,\uparrow\}}} |g_{n'nv}(\mathbf{k}, \mathbf{q})|^2 \mathcal{G}_{n,\sigma}(k) \mathcal{G}_{n',\sigma}(k + q) + \mathcal{O}(g^4)
 \end{aligned} \tag{33.50}$$

since we can now replace $\mathcal{G}$ with $\mathcal{G}_0$ since the corrections will only contribute to higher orders in g , which proves the theorem.

Proof 33.14 of Lemma 23.12

We combine Theorem 23.11 with Equation (21.24) to obtain

$$\begin{aligned}
 \langle H_{\text{ph}} \rangle &\approx -\frac{1}{\hbar\beta} \sum_{\substack{q \in i\Omega_{+1} \times Q \\ v \in \{1,2,\dots,rd_x\}}} e^{+i\omega_q 0^+} \Omega_v(\mathbf{q}) \hbar \mathcal{G}_{0,v}(q) \\
 &\quad + \frac{1}{\hbar\beta} \sum_{\substack{q \in i\Omega_{+1} \times Q \\ v \in \{1,2,\dots,rd_x\}}} e^{+i\omega_q 0^+} \Omega_v(\mathbf{q}) \frac{\hbar}{\beta N_p} \mathcal{G}_{0,v}^2(q) \delta_{q,0} \sum_{\substack{k,k' \in i\Omega_{-1} \times Q \\ n,n' \in \{1,\dots,\infty\} \\ \sigma,\sigma' \in \{\downarrow,\uparrow\}}} g_{nnv}(\mathbf{k}, 0) g_{n'n'v}(\mathbf{k}', 0) \mathcal{G}_{n,\sigma}(k) \mathcal{G}_{n',\sigma'}(k') \\
 &\quad - \frac{1}{\hbar\beta} \sum_{\substack{q \in i\Omega_{+1} \times Q \\ v \in \{1,2,\dots,rd_x\}}} e^{+i\omega_q 0^+} \Omega_v(\mathbf{q}) \frac{\hbar}{\beta N_p} \mathcal{G}_{0,v}^2(q) \sum_{\substack{k \in i\Omega_{-1} \times Q \\ n,n' \in \{1,\dots,\infty\} \\ \sigma \in \{\downarrow,\uparrow\}}} |g_{n'nv}(\mathbf{k}, \mathbf{q})|^2 \mathcal{G}_{n,\sigma}(k) \mathcal{G}_{n',\sigma}(k + q)
 \end{aligned} \tag{33.51}$$

Now simplifying the appearing terms and neglecting terms of the order $\mathcal{O}(g^4)$ and rearranging the terms yields the desired result.

Proof 33.15 of Corollary 23.13

First Term We compute the phononic, non-interacting contribution and introduce $q \equiv (i\omega_n, \mathbf{q})$

$$\begin{aligned}
 \frac{1}{\hbar\beta} \sum_{i\omega_n \in i\Omega_{+1}} (-\hbar \mathcal{G}_{0,v}(q)) &= \frac{-1}{\hbar\beta} \sum_{i\omega_n \in i\Omega_{+1}} \frac{1}{i\omega_n - \omega_v(\mathbf{q})} \\
 &\stackrel{\text{Corollary 15.22}}{=} B_{\hbar\beta}(\omega_v(\mathbf{q}))
 \end{aligned} \tag{33.52}$$

Second Term For the mix-terms we compute

$$\begin{aligned}
 \frac{1}{\beta^2} \sum_{\substack{\omega_k \in \Omega_{-1} \\ \omega_q \in \Omega_{+1}}} \mathcal{G}_{0,\nu}^2(q) \mathcal{G}_{0,n,\sigma}(k) \mathcal{G}_{0,n',\sigma}(k+q) &= \frac{1}{\beta^2 \hbar^4} \sum_{\substack{\omega_k \in \Omega_{-1} \\ \omega_q \in \Omega_{+1}}} \frac{1}{(\mathrm{i}\omega_q - \omega_\nu(\mathbf{q}))^2} \cdot \frac{1}{\mathrm{i}\omega_k - \xi_{n,\sigma,\mathbf{k}}/\hbar} \cdot \frac{1}{\mathrm{i}\omega_{k+q} - \xi_{n',\sigma,\mathbf{k}+\mathbf{q}}/\hbar} \\
 &= \dots
 \end{aligned} \tag{33.53}$$

We now decompose $\Omega_{-1} \ni \omega_{q+k} \equiv \omega_q + \omega_k$, then we can evaluate the sum over ω_k first as

$$\begin{aligned}
 \dots &= \frac{1}{\hbar^3 \beta} \sum_{\omega_q \in \Omega_{+1}} \frac{1}{(\mathrm{i}\omega_q - \omega_\nu(\mathbf{q}))^2} \cdot \underbrace{\frac{1}{\hbar \beta} \sum_{\omega_k \in \Omega_{-1}} \frac{1}{\mathrm{i}\omega_k - \xi_{n,\sigma,\mathbf{k}}/\hbar} \cdot \frac{1}{\mathrm{i}\omega_k + \mathrm{i}\omega_q - \xi_{n',\sigma,\mathbf{k}+\mathbf{q}}/\hbar}}_{=S_{-1} \equiv \sum_{\omega_k \in \Omega_{-1}} h(\omega_k)} \\
 &= \dots
 \end{aligned} \tag{33.54}$$

We now consider the function

$$\begin{aligned}
 \mathbb{C} - Z \ni z \mapsto h(-\mathrm{i}z) &= \frac{1}{\hbar \beta} \cdot \frac{1}{z - \xi_{n,\sigma,\mathbf{k}}/\hbar} \cdot \frac{1}{z + \mathrm{i}\omega_q - \xi_{n',\sigma,\mathbf{k}+\mathbf{q}}/\hbar} \\
 &= \frac{1}{\hbar \beta} \cdot \frac{1}{z - z_1} \cdot \frac{1}{z - z_2}
 \end{aligned} \tag{33.55}$$

with first order poles

$$z_1 = \xi_{n,\sigma,\mathbf{k}}/\hbar, \quad z_2 = \xi_{n',\sigma,\mathbf{k}+\mathbf{q}}/\hbar - \mathrm{i}\omega_q \tag{33.56}$$

This gives with Theorem 15.20 (a version of the Residue-theorem)

$$S_{-1} = \sum_{\omega_k \in \Omega_{-1}} h(\omega_k) = -\zeta \sum_{l=1}^2 \text{Res}(f_\zeta(z), z = z_l)|_{\zeta=-1} \tag{33.57}$$

with auxiliary function

$$f_\zeta(z)|_{\zeta=-1} = g_\zeta(z)h(-\mathrm{i}z)|_{\zeta=-1} = \hbar \beta n_{\hbar\beta}^\zeta(z)h(-\mathrm{i}z)|_{\zeta=-1} \tag{33.58}$$

This gives for the first residue

$$\begin{aligned}
 \text{Res}(f_\zeta(z), z = z_1)|_{\zeta=-1} &= \lim_{z \rightarrow z_1} (z - z_1) f_\zeta(z)|_{\zeta=-1} \\
 &= \lim_{z \rightarrow z_1} (z - z_2)^{-1} F_{\hbar\beta}(z) \\
 &= \frac{F_\beta(\xi_{n,\sigma,\mathbf{k}})}{\xi_{n,\sigma,\mathbf{k}}/\hbar - \xi_{n',\sigma,\mathbf{k}+\mathbf{q}}/\hbar + \mathrm{i}\omega_q}
 \end{aligned} \tag{33.59}$$

For the second residue we obtain likewise

$$\begin{aligned}
 \text{Res}(f_\zeta(z), z = z_2)|_{\zeta=-1} &= \lim_{z \rightarrow z_2} (z - z_2) f_\zeta(z)|_{\zeta=-1} \\
 &= \lim_{z \rightarrow z_2} (z - z_1)^{-1} F_{\hbar\beta}(z) \\
 &= \frac{F_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}} - i\hbar\omega_q)}{\xi_{n',\sigma,\mathbf{k}+\mathbf{q}}/\hbar - \xi_{n,\sigma,\mathbf{k}}/\hbar - i\omega_q} \\
 &\stackrel{\star}{=} \frac{F_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}})}{\xi_{n',\sigma,\mathbf{k}+\mathbf{q}}/\hbar - \xi_{n,\sigma,\mathbf{k}}/\hbar - i\omega_q}
 \end{aligned} \tag{33.60}$$

whereas in $\star$ we used Remark 9.47.1. Summing up the residues yields then

$$\begin{aligned}
 \dots &= \frac{F_\beta(\xi_{n,\sigma,\mathbf{k}}) - F_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}})}{\hbar^2} \underbrace{\frac{1}{\hbar\beta} \sum_{\omega_q \in \Omega_{+1}} \frac{1}{(i\omega_q - \omega_v(\mathbf{q}))^2} \cdot \frac{1}{i\omega_q + \xi_{n,\sigma,\mathbf{k}}/\hbar - \xi_{n',\sigma,\mathbf{k}+\mathbf{q}}/\hbar}}_{=S_{+1} \equiv \sum_{\omega_q \in \Omega_{+1}} h(\omega_q)} \\
 &= \dots
 \end{aligned} \tag{33.61}$$

We now, again, consider the function

$$\begin{aligned}
 \mathbb{C} - Z \ni z \mapsto h(-iz) &= \frac{1}{\hbar\beta} \cdot \frac{1}{(z - \omega_v(\mathbf{q}))^2} \cdot \frac{1}{z + \xi_{n,\sigma,\mathbf{k}}/\hbar - \xi_{n',\sigma,\mathbf{k}+\mathbf{q}}/\hbar} \\
 &= \frac{1}{\hbar\beta} \cdot \frac{1}{(z - z_2)^2} \cdot \frac{1}{z - z_1}
 \end{aligned} \tag{33.62}$$

with poles of different order

- $z_1 = \xi_{n',\sigma,\mathbf{k}+\mathbf{q}}/\hbar - \xi_{n,\sigma,\mathbf{k}}/\hbar$ (first order pole)
- $z_2 = \omega_v(\mathbf{q})$ (second order pole)

This gives with Theorem 15.20

$$S_{+1} = \sum_{\omega_q \in \Omega_{+1}} h(\omega_q) = - \sum_{l=1}^2 \text{Res}(f_{+1}(z), z = z_l) = \star \star \star \tag{33.63}$$

with auxiliary function

$$f_{+1}(z) = \hbar\beta B_{\hbar\beta}(z) h(-iz) = \hbar\beta B_{\hbar\beta}(z) \cdot \frac{1}{\hbar\beta} \cdot \frac{1}{(z - z_2)^2} \cdot \frac{1}{z - z_1} = B_{\hbar\beta}(z) \cdot \frac{1}{(z - z_2)^2} \cdot \frac{1}{z - z_1} \tag{33.64}$$

This gives for the first residue

$$-\text{Res}(f_{+1}(z), z = z_1) = - \lim_{z \rightarrow z_1} (z - z_2)^{-2} B_{\hbar\beta}(z) = \frac{-B_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}} - \xi_{n,\sigma,\mathbf{k}})}{(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}}/\hbar - \xi_{n,\sigma,\mathbf{k}}/\hbar - \omega_v(\mathbf{q}))^2} \tag{33.65}$$

Now we compute the second residue

$$-\text{Res}(f_{+1}(z), z = z_2) = - \lim_{z \rightarrow z_2} \frac{d}{dz} [(z - z_1)^{-1} B_{\hbar\beta}(z)] = \square\square\square \tag{33.66}$$

which gives with Lemma 9.61 the result

$$\begin{aligned} \square\square\square &= \lim_{z \rightarrow z_2} - \left[- (z - z_1)^{-2} B_{\hbar\beta}(z) + (z - z_1)^{-1} \cdot (-\hbar\beta) B_{\hbar\beta}(z) (1 + B_{\hbar\beta}(z)) \right] \\ &= \frac{B_\beta(\Omega_v(\mathbf{q}))}{(\omega_v(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}}/\hbar - \xi_{n',\sigma,\mathbf{k}+\mathbf{q}}/\hbar)^2} + \frac{\hbar\beta B_\beta(\Omega_v(\mathbf{q})) (1 + B_\beta(\Omega_v(\mathbf{q})))}{\omega_v(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}}/\hbar - \xi_{n',\sigma,\mathbf{k}+\mathbf{q}}/\hbar} \end{aligned} \quad (33.67)$$

This gives the result

$$\star \star \star = \text{TBD}. \quad (33.68)$$

Using now Equation (TBDReferenced) we obtain the final expression as

$$\begin{aligned} &\frac{1}{\beta^2} \sum_{\substack{\omega_k \in \Omega_{-1} \\ \omega_q \in \Omega_{+1}}} \mathcal{G}_{0,v}^2(q) \mathcal{G}_{0,n,\sigma}(k) \mathcal{G}_{0,n',\sigma}(k+q) \\ &= \dots \\ &= \frac{F_\beta(\xi_{n,\sigma,\mathbf{k}}) - F_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}})}{\hbar^2} \left[\frac{B_\beta(\Omega_v(\mathbf{q})) - B_\beta(\xi_{n',\sigma,\mathbf{k}+\mathbf{q}} - \xi_{n,\sigma,\mathbf{k}})}{(\omega_v(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}}/\hbar - \xi_{n',\sigma,\mathbf{k}+\mathbf{q}}/\hbar)^2} + \frac{\hbar\beta B_\beta(\Omega_v(\mathbf{q})) (1 + B_\beta(\Omega_v(\mathbf{q})))}{\omega_v(\mathbf{q}) + \xi_{n,\sigma,\mathbf{k}}/\hbar - \xi_{n',\sigma,\mathbf{k}+\mathbf{q}}/\hbar} \right] \end{aligned} \quad (33.69)$$

now we use Lemma 9.48 to obtain

$$B_\beta(\Omega_v(\mathbf{q})) (1 + B_\beta(\Omega_v(\mathbf{q}))) = 4^{-1} \text{csch}^2 \left(\frac{\beta\Omega_v(\mathbf{q})}{2} \right) \quad (33.70)$$

which proves the corollary. Notice that Equation (33.69) has the right unit, namely Energy⁻².

Proof 33.16 of Lemma 23.14

We first simplify Equation (21.25): We use Equation (21.20) to integrate out the phonons, such that we can obtain

$$\frac{\partial}{\partial \bar{j}(i\omega_m, \mathbf{q}\nu)} Z[j, \bar{j}] = \int \mathcal{D}(\bar{\psi}, \psi) \left(\frac{-\partial}{\partial \bar{j}(i\omega_m, \mathbf{q}\nu)} S_{\text{el-el-int}}^{[j_{\text{ph}}, \bar{j}_{\text{ph}}]}[\psi, \bar{\psi}] \right) \exp - S_{j, \bar{j}}[\psi, \bar{\psi}] \quad (33.71)$$

and

$$\frac{\partial}{\partial j(-i\omega_m, -\mathbf{q}\nu)} Z[j, \bar{j}] = \int \mathcal{D}(\bar{\psi}, \psi) \left(\frac{-\partial}{\partial j(-i\omega_m, -\mathbf{q}\nu)} S_{\text{el-el-int}}^{[j_{\text{ph}}, \bar{j}_{\text{ph}}]}[\psi, \bar{\psi}] \right) \exp - S_{j, \bar{j}}[\psi, \bar{\psi}] \quad (33.72)$$

Now we use Equation (21.16) to obtain the explicit form of these derivatives as

$$\frac{-\partial}{\partial j(i\omega_m, \mathbf{q}\nu)} S_{\text{el-el-int}}^{[j_{\text{ph}}, \bar{j}_{\text{ph}}]}[\psi, \bar{\psi}] = -\hbar \bar{j}_v(q) \mathcal{G}_{0,v}(q) + \hbar \rho_v(q) \mathcal{G}_{0,v}(q) \quad (33.73)$$

and

$$\frac{-\partial}{\partial \bar{j}(\mathrm{i}\omega_m, \mathbf{q}\nu)} S_{\text{el-el-int}}[\psi, \bar{\psi}] = -\hbar \mathcal{G}_{0,\nu}(-q) j_\nu(-q) + \hbar \mathcal{G}_{0,\nu}(-q) \rho_\nu(q) \quad (33.74)$$

Now we can use Equation (21.20) again to obtain

$$\frac{\partial}{\partial j_{m,\sigma}(\mathrm{i}\omega_{n'}, \mathbf{k} + \mathbf{q})} Z[j, \bar{j}] = \int \mathcal{D}(\bar{\psi}, \psi) \underbrace{\left(\frac{\partial}{\partial j_{m,\sigma}(\mathrm{i}\omega_{n'}, \mathbf{k} + \mathbf{q})} \sum \bar{\psi} j_{\text{el}} \right)}_{=-\psi_{m\sigma}(\mathrm{i}\omega_{n'}, \mathbf{k} + \mathbf{q})} \exp - S_{[j, \bar{j}]}[\psi, \bar{\psi}] \quad (33.75)$$

This implies that

$$\frac{\partial^2}{\partial \bar{j}_{n,\sigma}(\mathrm{i}\omega_n, \mathbf{k}) \partial j_{m,\sigma}(\mathrm{i}\omega_{n'}, \mathbf{k} + \mathbf{q})} \frac{Z[j, \bar{j}]}{Z} \Big|_{j=\bar{j}=0} = \langle \bar{\psi}_{m\sigma}(\mathrm{i}\omega_{n'}, \mathbf{k} + \mathbf{q}) \psi_{n\sigma}(\mathrm{i}\omega_n, \mathbf{k}) \rangle \quad (33.76)$$

Now we combine this with Equation (20.24) to obtain the desired result:

$$\begin{aligned} \langle H_{\text{int}} \rangle &= \sum_{\substack{\mathbf{k}, \mathbf{q} \in Q \\ \sigma \in \{\downarrow, \uparrow\} \\ \nu \in \{1, 2, \dots, rd_x\} \\ n, m \in \{1, \dots, \infty\} \\ \omega_n \in \Omega_{+1} \\ \omega_{n'}, \omega_{n''} \in \Omega_{-1}}} \frac{g_{mn\nu}(\mathbf{k}, \mathbf{q}) \hbar}{\sqrt{N_p} \sqrt{\hbar \beta}^3} \left\langle \bar{\psi}_{m\sigma}(\mathrm{i}\omega_{n'}, \mathbf{k} + \mathbf{q}) \psi_{n\sigma}(\mathrm{i}\omega_n, \mathbf{k}) \left[\rho_\nu(\mathrm{i}\omega_m, \mathbf{q}) \mathcal{G}_{0,\nu}(\mathrm{i}\omega_m, \mathbf{q}) + \mathcal{G}_{0,\nu}(-\mathrm{i}\omega_m, -\mathbf{q}) \rho_\nu(\mathrm{i}\omega_m, \mathbf{q}) \right] \right\rangle \\ &= \sum_{\substack{k \in \mathrm{i}\Omega_{-1} \times Q \\ q \in \mathrm{i}\Omega_{+1} \times Q \\ \sigma \in \{\downarrow, \uparrow\} \\ \nu \in \{1, 2, \dots, rd_x\} \\ n, m \in \{1, \dots, \infty\} \\ \omega_{n'} \in \Omega_{-1}}} \frac{g_{mn\nu}(\mathbf{k}, \mathbf{q}) \hbar}{\sqrt{N_p} \sqrt{\hbar \beta}^3} \left\langle \bar{\psi}_{m\sigma}(\mathrm{i}\omega_{n'}, \mathbf{k} + \mathbf{q}) \psi_{n\sigma}(\mathrm{i}\omega_n, \mathbf{k}) \left[\rho_\nu(q) \mathcal{G}_{0,\nu}(q) + \mathcal{G}_{0,\nu}(-q) \rho_\nu(q) \right] \right\rangle \end{aligned} \quad (33.77)$$

Proof 33.17 of Corollary 23.15

We decompose Equation (23.16) as

$$\begin{aligned} \langle H_{\text{int}} \rangle &= \sum_{\substack{k \in \mathrm{i}\Omega_{-1} \times Q \\ q \in \mathrm{i}\Omega_{+1} \times Q \\ \sigma \in \{\downarrow, \uparrow\} \\ \nu \in \{1, 2, \dots, rd_x\} \\ n, m \in \{1, \dots, \infty\} \\ \omega_{n'} \in \Omega_{-1}}} \frac{g_{mn\nu}(\mathbf{k}, \mathbf{q}) \hbar}{\sqrt{N_p} \sqrt{\hbar \beta}^3} \left\langle \bar{\psi}_{m\sigma}(\mathrm{i}\omega_{n'}, \mathbf{k} + \mathbf{q}) \psi_{n\sigma}(\mathrm{i}\omega_n, \mathbf{k}) \left[\rho_\nu(q) \mathcal{G}_{0,\nu}(q) + \mathcal{G}_{0,\nu}(-q) \rho_\nu(q) \right] \right\rangle \\ &\equiv \text{I} + \text{II} \end{aligned} \quad (33.78)$$

We focus on the first term I and compute

$$\begin{aligned} \text{I} &= \sum_{\substack{k, \tilde{k} \in \mathrm{i}\Omega_{-1} \times Q \\ q \in \mathrm{i}\Omega_{+1} \times Q \\ \sigma, \tilde{\sigma} \in \{\downarrow, \uparrow\} \\ \nu \in \{1, 2, \dots, rd_x\} \\ n, \tilde{n}, m, \tilde{m} \in \{1, \dots, \infty\} \\ \omega_{n'} \in \Omega_{-1}}} \frac{g_{mn\nu}(\mathbf{k}, \mathbf{q}) g_{\tilde{m}\tilde{n}\nu}(\tilde{\mathbf{k}}, \mathbf{q})}{N_p (\hbar \beta)^2} \left\langle \bar{\psi}_{m\sigma}(\mathrm{i}\omega_{n'}, \mathbf{k} + \mathbf{q}) \psi_{n\sigma}(k) \bar{\psi}_{\tilde{m}\tilde{\sigma}}(\tilde{k} + q) \psi_{\tilde{n}\tilde{\sigma}}(\tilde{k}) \right\rangle \mathcal{G}_{0,\nu}(q) \\ &= \dots \end{aligned} \quad (33.79)$$

Now we simplify the correlator as

$$\begin{aligned}
 & \left\langle \bar{\psi}_{m\sigma}(i\omega_{n'}, \mathbf{k} + \mathbf{q}) \psi_{n\sigma}(k) \bar{\psi}_{\tilde{m}\tilde{\sigma}}(\tilde{k} + q) \psi_{\tilde{n}\tilde{\sigma}}(\tilde{k}) \right\rangle \\
 &= - \left\langle \psi_{n\sigma}(k) \psi_{\tilde{n}\tilde{\sigma}}(\tilde{k}) \bar{\psi}_{m\sigma}(i\omega_{n'}, \mathbf{k} + \mathbf{q}) \bar{\psi}_{\tilde{m}\tilde{\sigma}}(\tilde{k} + q) \right\rangle \\
 &= - \left[\left\langle \psi_{n\sigma}(k) \bar{\psi}_{\tilde{m}\tilde{\sigma}}(\tilde{k} + q) \right\rangle_0 \left\langle \psi_{\tilde{n}\tilde{\sigma}}(\tilde{k}) \bar{\psi}_{m\sigma}(i\omega_{n'}, \mathbf{k} + \mathbf{q}) \right\rangle_0 - \left\langle \psi_{n\sigma}(k) \bar{\psi}_{m\sigma}(i\omega_{n'}, \mathbf{k} + \mathbf{q}) \right\rangle_0 \left\langle \psi_{\tilde{n}\tilde{\sigma}}(\tilde{k}) \bar{\psi}_{\tilde{m}\tilde{\sigma}}(\tilde{k} + q) \right\rangle_0 + \mathcal{O}(g^2) \right] \\
 &= -\hbar^2 \left[\mathcal{G}_{\substack{n,\tilde{m} \\ \sigma,\tilde{\sigma}}}^0(k, \tilde{k} + q) \mathcal{G}_{\substack{\tilde{n},m \\ \tilde{\sigma},\sigma}}^0 \left(\begin{smallmatrix} i\omega_{\tilde{k}}, i\omega_{n'} \\ \tilde{\mathbf{k}}, \mathbf{k} + \mathbf{q} \end{smallmatrix} \right) - \mathcal{G}_{\substack{n,m \\ \sigma,\sigma}}^0 \left(\begin{smallmatrix} i\omega_k, i\omega_{n'} \\ \mathbf{k}, \mathbf{k} + \mathbf{q} \end{smallmatrix} \right) \mathcal{G}_{\substack{\tilde{n},\tilde{m} \\ \tilde{\sigma},\tilde{\sigma}}}^0(\tilde{k}, \tilde{k} + q) \right]
 \end{aligned} \tag{33.80}$$

This gives a natural decomposition for

$$I = \dots = I1 + I2 \tag{33.81}$$

Now we simplify

$$\begin{aligned}
 I1 &= \sum_{\substack{k,\tilde{k} \in i\Omega_{-1} \times Q \\ (i\omega_q, \mathbf{q}) \equiv q \in i\Omega_{+1} \times Q \\ \sigma, \tilde{\sigma} \in \{\downarrow, \uparrow\} \\ v \in \{1, 2, \dots, rd_x\} \\ n, \tilde{n}, m, \tilde{m} \in \{1, \dots, \infty\} \\ \omega_{n'} \in \Omega_{-1}}} \frac{g_{mnv}(\mathbf{k}, \mathbf{q}) g_{\tilde{m}\tilde{n}v}(\tilde{\mathbf{k}}, \mathbf{q})}{N_p \beta^2} \cdot (-1) \cdot \left[\delta_{n,\tilde{m}} \delta_{\sigma,\tilde{\sigma}} \delta_{k-q,\tilde{k}} \mathcal{G}_{0,n,\sigma}(k) \delta_{\tilde{n},m} \delta_{\omega_{\tilde{k}}, \omega_{n'}} \delta_{\tilde{\mathbf{k}}, \mathbf{k}+\mathbf{q}} \mathcal{G}_{0,m,\sigma}(i\omega_k, \mathbf{k}) \right] \\
 &= \star \star \star
 \end{aligned} \tag{33.82}$$

Now we use the individual delta-functions to ...

1. kill $\tilde{m}$,
2. kill $\tilde{\sigma}$,
3. kill $\tilde{k}$,
4. kill $\tilde{n}$,
5. kill $\omega_{n'}$ as $\tilde{k}$ is already killed,
6. kill $\mathbf{q}$ as $\tilde{k}$ is already killed.

This allows us to simplify

$$\begin{aligned}
 \star \star \star &= \sum_{\substack{k \in i\Omega_{-1} \times Q \\ i\omega_q \in \Omega_{+1} \\ \sigma \in \{\downarrow, \uparrow\} \\ v \in \{1, 2, \dots, r d_x\} \\ n, m \in \{1, \dots, \infty\}}} \frac{g_{mnv}(\mathbf{k}, 0) g_{nmv}(\mathbf{k}, 0)}{N_p \beta^2} \cdot (-1) \cdot \mathcal{G}_{0,n,\sigma}(k) \mathcal{G}_{0,m,\sigma}(i\omega_{k-q}, \mathbf{k}) \\
 &= \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\downarrow, \uparrow\} \\ v \in \{1, 2, \dots, r d_x\} \\ n, m \in \{1, \dots, \infty\}}} \frac{-|g_{mnv}(\mathbf{k}, 0)|^2}{N_p \beta^2 \hbar^2} \sum_{\substack{i\omega_k \in \Omega_{-1} \\ i\omega_q \in \Omega_{+1}}} \frac{1}{i\omega_k - \xi_{n,\sigma}/\hbar} \cdot \frac{1}{i\omega_k - i\omega_q - \xi_{m,\sigma}/\hbar} \\
 &= \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\downarrow, \uparrow\} \\ v \in \{1, 2, \dots, r d_x\} \\ n, m \in \{1, \dots, \infty\}}} \frac{-|g_{mnv}(\mathbf{k}, 0)|^2}{N_p} \frac{1}{\hbar \beta} \sum_{\omega_k \in \Omega_{-1}} \frac{1}{i\omega_k - \xi_{n,\sigma}/\hbar} \cdot \underbrace{\frac{1}{\hbar \beta} \sum_{\omega_q \in \Omega_{+1}} \frac{1}{i\omega_q - i\omega_k + \xi_{m,\sigma}/\hbar}}_{=-B_\beta(\xi_{m,\sigma} - i\hbar\omega_k)}
 \end{aligned} \tag{33.83}$$

whereas we used corollary 16.15 (or something).

Intermezzo: We now simplify $B_\beta(\bullet - i\hbar\omega_k) \dots$ **TBContinued.**

Proof 33.18 of Theorem 23.16

For **the electronic contribution** of the expectation-value we simply refer to Theorem 23.9.

For **the phononic contribution**, we simplify refer to Corollary 23.13.

For **the interaction contribution**, we refer to Theorem **TBCreated after PP3 has been digitalized.**

Proof 33.19 of Lemma 23.19 We compute the derivative of Equation (22.26) as

$$\begin{aligned}
 (\partial_T F)(\mu, T) &= \langle N \rangle(\mu, T) \\
 &= \partial_T \frac{-1}{\pi} \lim_{\delta \rightarrow 0} \oint_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\} \\ E \in (-\infty, +\infty)}} F_\beta(E) \operatorname{Im} G_{m,m}^{\text{ret}}(E + i\delta, \mathbf{k}) \\
 &= \frac{-1}{\pi} \lim_{\delta \rightarrow 0} \oint_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\} \\ E \in (-\infty, +\infty)}} \left[(\partial_T F_\beta(E)) \operatorname{Im} G_{m,m}^{\text{ret}}(E + i\delta, \mathbf{k}) + F_\beta(E) \operatorname{Im} \partial_T G_{m,m}^{\text{ret}}(E + i\delta, \mathbf{k}) \right] \\
 &= \dots
 \end{aligned} \tag{33.84}$$

Now we use Lemma 9.48, i.e. we use the Equation

$$\partial_T F_\beta(E) = \frac{E}{k_B T^2} F_\beta(E) (1 - F_\beta(E)) \tag{33.85}$$

and for the other derivative we have

$$\partial_T G_{m,\sigma,\mathbf{k}}^{\text{ret}}(E + i\delta) = \partial_T \frac{1}{E + i\delta - \left(\epsilon_{m\mathbf{k}\sigma} + \Sigma_{m,\sigma,\mathbf{k}}^{\text{ret}}(E + i\delta) - \mu \right)} = \frac{\partial_T \Sigma_{m,\sigma,\mathbf{k}}^{\text{ret}}(E + i\delta)}{E + i\delta - \left(\epsilon_{m\mathbf{k}\sigma} + \Sigma_{m,\sigma,\mathbf{k}}^{\text{ret}}(E + i\delta) - \mu \right)^2} \tag{33.86}$$

Now we compute the derivative and use the diagonal bands approximation for the self-energy and use Lemma 9.48

$$\begin{aligned}
 \partial_T \Sigma_{m\sigma\mathbf{k}}^{\text{ret}}(E + i\delta) &= \sum_{\substack{\mathbf{q} \in Q \\ n \in \{1, \dots, \infty\} \\ v \in \{1, \dots, d_x\}}} \frac{|g_{nmv}(\mathbf{k}, \mathbf{q})|^2}{N_p} \left[\partial_T \frac{B_\beta(\Omega_v(\mathbf{q})) + F_\beta(\xi_{n\sigma\mathbf{k}+\mathbf{q}})}{E + i\delta - \xi_{n\sigma\mathbf{k}+\mathbf{q}} + \Omega_v(\mathbf{q})} + \partial_T \frac{B_\beta(\Omega_v(\mathbf{q})) + 1 - F_\beta(\xi_{n\sigma\mathbf{k}+\mathbf{q}})}{E + i\delta - \xi_{n\sigma\mathbf{k}+\mathbf{q}} - \Omega_v(\mathbf{q})} \right] \\
 &= \sum_{\substack{\mathbf{q} \in Q \\ n \in \{1, \dots, \infty\} \\ v \in \{1, \dots, d_x\}}} \frac{|g_{nmv}(\mathbf{k}, \mathbf{q})|^2}{4k_B T^2 N_p} \left[\frac{\Omega_v(\mathbf{q}) \operatorname{csch}^2(\beta\Omega_v(\mathbf{q})/2) + \xi_{n\sigma\mathbf{k}+\mathbf{q}} \operatorname{sech}^2(\beta\xi_{n\sigma\mathbf{k}+\mathbf{q}}/2)}{E + i\delta - \xi_{n\sigma\mathbf{k}+\mathbf{q}} + \Omega_v(\mathbf{q})} + \frac{\Omega_v(\mathbf{q}) \operatorname{csch}^2(\beta\Omega_v(\mathbf{q})/2) - \xi_{n\sigma\mathbf{k}+\mathbf{q}} \operatorname{sech}^2(\beta\xi_{n\sigma\mathbf{k}+\mathbf{q}}/2)}{E + i\delta - \xi_{n\sigma\mathbf{k}+\mathbf{q}} - \Omega_v(\mathbf{q})} \right]
 \end{aligned} \tag{33.87}$$

now we multiply both sides of this equation with $k_B T^2$. For the Hartree-contribution we obtain with Equation (19.10) the equation

$$\begin{aligned}
 \partial_T \Sigma_{m\sigma\mathbf{k}}^{\text{Hartree}}(\bullet) &= - \sum_{\substack{\mathbf{q} \in Q \\ \sigma' \in \{\downarrow, \uparrow\} \\ n \in \{1, 2, \dots\} \\ v \in \{1, 2, \dots, d_x\}}} \frac{g_{mmv}(\mathbf{k}, 0) g_{nnv}(\mathbf{q}, 0)}{N_p} \frac{1}{\Omega_v(\mathbf{q}' \equiv 0)} \partial_T (F_\beta(\xi_{n\sigma'\mathbf{q}})) \\
 &= \dots
 \end{aligned} \tag{33.88}$$

now we use Lemma 9.48 and obtain finally

$$\dots = - \sum_{\substack{\mathbf{q} \in Q \\ \sigma' \in \{\downarrow, \uparrow\} \\ n \in \{1, 2, \dots\} \\ v \in \{1, 2, \dots, r d_x\}}} \frac{g_{mmv}(\mathbf{k}, 0) g_{nnv}(\mathbf{q}, 0)}{N_p} \frac{1}{\Omega_v(\mathbf{q}' \equiv 0)} \frac{\xi_{n\sigma' \mathbf{q}}}{4k_B T^2} \partial_T \left(\text{sech}^2 \frac{\xi_{n\sigma' \mathbf{q}}}{2k_B T} \right)^2 \quad (33.89)$$

Proof 33.20 of Lemma 23.20 We rewrite the particle-number equation as

$$\langle N \rangle(\mu, T) = \frac{-1}{\pi} \lim_{\delta \rightarrow 0} \oint \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\} \\ E \in (-\infty, +\infty)}} F_\beta(E) \text{Im} \frac{1}{E + \mu + i\delta - \left(\epsilon_{m\sigma \mathbf{k}} + \tilde{\Sigma}_{m\sigma \mathbf{k}}^{\text{ret}}(E + \mu + i\delta) \right)} = \dots \quad (33.90)$$

whereas we introduced

$$\tilde{\Sigma}_{m\sigma \mathbf{k}}^{\text{ret}}(E + \mu + i\delta) = \tilde{\Sigma}_{m\sigma \mathbf{k}}^{\text{Hartree}}(E + \mu + i\delta) + \tilde{\Sigma}_{m\sigma \mathbf{k}}^{\text{Fock}}(E + \mu + i\delta) \quad (33.91)$$

with Hartree-contribution

$$\tilde{\Sigma}_{m\sigma \mathbf{k}}^{\text{Hartree}}(\bullet) = - \sum_{\substack{\mathbf{q} \in Q \\ \sigma' \in \{\downarrow, \uparrow\} \\ n \in \{1, 2, \dots\} \\ v \in \{1, 2, \dots, r d_x\}}} \frac{g_{mmv}(\mathbf{k}, 0) g_{nnv}(\mathbf{q}, 0)}{N_p} \frac{F_\beta(\epsilon_{n\sigma' \mathbf{q}} - \mu)}{\Omega_v(\mathbf{q}' = 0)} \quad (33.92)$$

and Fock-contribution

$$\tilde{\Sigma}_{m\sigma \mathbf{k}}^{\text{Fock}}(E + \mu + i\delta) = \sum_{\substack{\mathbf{q} \in Q \\ n \in \{1, 2, \dots\} \\ v \in \{1, 2, \dots, r d_x\}}} \frac{|g_{nmv}(\mathbf{k}, \mathbf{q})|^2}{N_p} \left[\frac{B_\beta(\Omega_v(\mathbf{q})) + F_\beta(\xi_{\mathbf{q}+\mathbf{k}n\sigma})}{E + \mu + i\delta - \epsilon_{\mathbf{q}+\mathbf{k}n\sigma} + \Omega_v(\mathbf{q})} + \frac{B_\beta(\Omega_v(\mathbf{q})) + 1 - F_\beta(\xi_{\mathbf{q}+\mathbf{k}n\sigma})}{E + \mu + i\delta - \epsilon_{\mathbf{q}+\mathbf{k}n\sigma} - \Omega_v(\mathbf{q})} \right] \quad (33.93)$$

now, using the substitution-rule allows us to rewrite the above Equation as

$$\langle N \rangle(\mu, T) = \dots = \frac{-1}{\pi} \lim_{\delta \rightarrow 0} \oint \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\} \\ E \in (-\infty, +\infty)}} F_\beta(E - \mu) \text{Im} \frac{1}{E + i\delta - \left(\epsilon_{m\sigma \mathbf{k}} + \tilde{\Sigma}_{m\sigma \mathbf{k}}^{\text{ret}}(E + i\delta) \right)} \quad (33.94)$$

Now we use Equation (TBReferenced) together with Lemma TBReferenced, and introduce the function

$$g(\mu) = \beta(E - \mu) \quad \text{which implies} \quad \left(\frac{dg}{d\mu} \right)(\mu) = -\beta \quad (33.95)$$

such that we obtain

$$F_\beta(E - \mu) = F(g(\mu)) \quad (33.96)$$

which implies with the chain-rule

$$\partial_\mu (F \circ g)(\mu) = (\partial_\mu F)(g(\mu)) \cdot \left(\frac{dg}{d\mu} \right)(\mu) = \frac{1}{4} \operatorname{sech}^2 \left(\beta \frac{E - \mu}{2} \right) \cdot \beta \quad (33.97)$$

Analogously we obtain

$$\partial_\mu \widetilde{\Sigma}_{m\sigma\mathbf{k}}^{\text{ret}}(E + i\delta) = \partial_\mu \widetilde{\Sigma}_{m\sigma\mathbf{k}}^{\text{Hartree}}(E + i\delta) + \partial_\mu \widetilde{\Sigma}_{m\sigma\mathbf{k}}^{\text{Fock}}(E + i\delta) \quad (33.98)$$

with Hartree-contribution

$$\partial_\mu \widetilde{\Sigma}_{m\sigma\mathbf{k}}^{\text{Hartree}}(\bullet) = - \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ \sigma' \in \{\downarrow, \uparrow\} \\ n \in \{1, 2, \dots\} \\ \nu \in \{1, 2, \dots, rd_x\}}} \frac{g_{mm\nu}(\mathbf{k}, 0) g_{nn\nu}(\mathbf{q}, 0)}{4N_p} \frac{\beta}{\Omega_\nu(\mathbf{q}' = 0)} \operatorname{sech}^2(\beta \xi_{n\sigma'\mathbf{q}}/2) \quad (33.99)$$

and Fock-contribution

$$\partial_\mu \widetilde{\Sigma}_{m\sigma\mathbf{k}}^{\text{Fock}}(E + i\delta) = \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ n \in \{1, 2, \dots\} \\ \nu \in \{1, 2, \dots, rd_x\}}} \frac{|g_{nm\nu}(\mathbf{k}, \mathbf{q})|^2}{4N_p k_B T} \operatorname{sech}^2(\beta \xi_{\mathbf{q}+\mathbf{k}n\sigma}/2) \left[\frac{1}{E + i\delta - \epsilon_{\mathbf{q}+\mathbf{k}n\sigma} + \Omega_\nu(\mathbf{q})} - \frac{1}{E + i\delta - \epsilon_{\mathbf{q}+\mathbf{k}n\sigma} - \Omega_\nu(\mathbf{q})} \right] \quad (33.100)$$

Now we can compute the derivative as

$$\langle N \rangle(\mu, T) = \frac{-1}{\pi} \lim_{\delta \rightarrow 0} \oint \sum_{\substack{\mathbf{k} \in \mathbf{Q} \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\} \\ E \in (-\infty, +\infty)}} F_\beta(E - \mu) \operatorname{Im} \left[E + i\delta - \left(\epsilon_{m\sigma\mathbf{k}} + \widetilde{\Sigma}_{m\sigma\mathbf{k}}^{\text{ret}}(E + i\delta) \right) \right]^{-1} \quad (33.101)$$

which implies

$$\begin{aligned} \partial_\mu \langle N \rangle(\mu, T) &= \frac{-1}{\pi} \lim_{\delta \rightarrow 0} \oint \sum_{\substack{\mathbf{k} \in \mathbf{Q} \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\} \\ E \in (-\infty, +\infty)}} \frac{\beta}{4} \operatorname{sech}^2 \left(\beta \frac{E - \mu}{2} \right) \operatorname{Im} \left[E + i\delta - \left(\epsilon_{m\sigma\mathbf{k}} + \widetilde{\Sigma}_{m\sigma\mathbf{k}}^{\text{ret}}(E + i\delta) \right) \right]^{-1} \\ &\quad + \frac{-1}{\pi} \lim_{\delta \rightarrow 0} \oint \sum_{\substack{\mathbf{k} \in \mathbf{Q} \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\} \\ E \in (-\infty, +\infty)}} F_\beta(E - \mu) \cdot (-1)^2 \cdot \operatorname{Im} \frac{\partial_\mu \widetilde{\Sigma}_{m\sigma\mathbf{k}}^{\text{ret}}(E + i\delta)}{\left(E + i\delta - \left(\epsilon_{m\sigma\mathbf{k}} + \widetilde{\Sigma}_{m\sigma\mathbf{k}}^{\text{ret}}(E + i\delta) \right) \right)^2} \\ &= \frac{-1}{\pi} \lim_{\delta \rightarrow 0} \oint \sum_{\substack{\mathbf{k} \in \mathbf{Q} \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\} \\ E \in (-\infty, +\infty)}} \frac{\beta}{4} \left[\operatorname{sech}^2(\beta E/2) \operatorname{Im} \frac{1}{E + i\delta - (\xi_{m\sigma\mathbf{k}} + \Sigma_{m\sigma\mathbf{k}}^{\text{ret}}(E + i\delta))} \right. \\ &\quad \left. + F_\beta(E) \operatorname{Im} \frac{\sigma_{m\sigma\mathbf{k}}(E + i\delta)}{\left[E + i\delta - (\xi_{m\sigma\mathbf{k}} + \Sigma_{m\sigma\mathbf{k}}^{\text{ret}}(E + i\delta)) \right]^2} \right] \end{aligned} \quad (33.102)$$

whereas we introduced the notation

$$\sigma_{m\sigma\mathbf{k}}(E + i\delta) = \sigma_{m\sigma\mathbf{k}}^{\text{Hartree}}(E + i\delta) + \sigma_{m\sigma\mathbf{k}}^{\text{Fock}}(E + i\delta), \quad (33.103)$$

with the Hartree-contribution

$$\sigma_{m\sigma\mathbf{k}}^{\text{Hartree}}(E + i\delta) = - \sum_{\substack{\mathbf{q} \in Q \\ \sigma' \in \{\downarrow, \uparrow\} \\ n \in \{1, 2, \dots\} \\ v \in \{1, 2, \dots, rd_x\}}} \frac{g_{mmv}(\mathbf{k}, 0)g_{mv}(\mathbf{q}, 0)}{N_p} \frac{1}{\Omega_v(\mathbf{q}' = 0)} \text{sech}^2(\beta\xi_{n\sigma'\mathbf{q}}/2) \quad (33.104)$$

and the Fock-contribution

$$\sigma_{m\sigma\mathbf{k}}^{\text{Fock}}(E + i\delta) = \sum_{\substack{\mathbf{q} \in Q \\ n \in \{1, 2, \dots\} \\ v \in \{1, 2, \dots, rd_x\}}} \frac{|g_{nmv}(\mathbf{k}, \mathbf{q})|^2}{N_p} \text{sech}^2(\beta\xi_{\mathbf{q}+\mathbf{k}n\sigma}/2) \left[\frac{1}{E + i\delta - \xi_{\mathbf{q}+\mathbf{k}n\sigma} + \Omega_v(\mathbf{q})} - \frac{1}{E + i\delta - \xi_{\mathbf{q}+\mathbf{k}n\sigma} - \Omega_v(\mathbf{q})} \right] \quad (33.105)$$

Proof 33.21 of Lemma 23.21

We compute the derivative of Equation (23.18) as

$$\begin{aligned} \partial_T \langle H_{\text{el}} \rangle &= \partial_T \frac{-1}{\pi} \lim_{\delta \rightarrow 0} \oint \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\} \\ E \in (-\infty, +\infty)}} \epsilon_{m\mathbf{k}\sigma} F_\beta(E) \text{Im} G_{m,m}^{\text{ret}}(E + i\delta, \mathbf{k}) \\ &= \frac{-1}{\pi} \lim_{\delta \rightarrow 0} \oint \sum_{\substack{\mathbf{k} \in Q \\ \sigma \in \{\uparrow, \downarrow\} \\ m \in \{1, \dots, \infty\} \\ E \in (-\infty, +\infty)}} \epsilon_{m\mathbf{k}\sigma} \left[(\partial_T F_\beta(E)) \text{Im} G_{m,m}^{\text{ret}}(E + i\delta, \mathbf{k}) + F_\beta(E) \text{Im} \partial_T G_{m,m}^{\text{ret}}(E + i\delta, \mathbf{k}) \right] \\ &= \dots \end{aligned} \quad (33.106)$$

Now we use Lemma 9.48, i.e. we use the Equation

$$d_T F_\beta(E) = \frac{E}{k_B T^2} F_\beta(E) (1 - F_\beta(E)) \quad (33.107)$$

and for the other derivative we have

$$\partial_T G_{m\sigma\mathbf{k}}^{\text{ret}}(E + i\delta) = \partial_T \frac{1}{E + i\delta - (\epsilon_{m\mathbf{k}\sigma} + \Sigma_{m\sigma\mathbf{k}}^{\text{ret}}(E + i\delta) - \mu)} = \frac{\partial_T \Sigma_{m\sigma\mathbf{k}}^{\text{ret}}(E + i\delta)}{E + i\delta - (\epsilon_{m\mathbf{k}\sigma} + \Sigma_{m\sigma\mathbf{k}}^{\text{ret}}(E + i\delta) - \mu)^2} \quad (33.108)$$

Now we compute the derivative and use the diagonal bands approximation for the self-energy and use Lemma 9.48

$$\begin{aligned} \partial_T \Sigma_{m\sigma\mathbf{k}}^{\text{ret}}(E + i\delta) &= \sum_{\substack{\mathbf{q} \in Q \\ n \in \{1, \dots, \infty\} \\ v \in \{1, \dots, rd_x\}}} \frac{|g_{nmv}(\mathbf{k}, \mathbf{q})|^2}{N_p} \left[\partial_T \frac{B_\beta(\Omega_v(\mathbf{q})) + F_\beta(\xi_{n\sigma\mathbf{k}+\mathbf{q}})}{E + i\delta - \xi_{n\sigma\mathbf{k}+\mathbf{q}} + \Omega_v(\mathbf{q})} + \partial_T \frac{B_\beta(\Omega_v(\mathbf{q})) + 1 - F_\beta(\xi_{n\sigma\mathbf{k}+\mathbf{q}})}{E + i\delta - \xi_{n\sigma\mathbf{k}+\mathbf{q}} - \Omega_v(\mathbf{q})} \right] \\ &= \sum_{\substack{\mathbf{q} \in Q \\ n \in \{1, \dots, \infty\} \\ v \in \{1, \dots, rd_x\}}} \frac{|g_{nmv}(\mathbf{k}, \mathbf{q})|^2}{4k_B T^2 N_p} \left[\frac{\Omega_v(\mathbf{q}) \text{csch}^2(\beta\Omega_v(\mathbf{q})/2) + \xi_{n\sigma\mathbf{k}+\mathbf{q}} \text{sech}^2(\beta\xi_{n\sigma\mathbf{k}+\mathbf{q}}/2)}{E + i\delta - \xi_{n\sigma\mathbf{k}+\mathbf{q}} + \Omega_v(\mathbf{q})} + \frac{\Omega_v(\mathbf{q}) \text{csch}^2(\beta\Omega_v(\mathbf{q})/2) - \xi_{n\sigma\mathbf{k}+\mathbf{q}} \text{sech}^2(\beta\xi_{n\sigma\mathbf{k}+\mathbf{q}}/2)}{E + i\delta - \xi_{n\sigma\mathbf{k}+\mathbf{q}} - \Omega_v(\mathbf{q})} \right] \end{aligned} \quad (33.109)$$

Proof 33.22 of Lemma 23.23

Taking the temperature-derivative of

$$\langle H_{\text{ph}} \rangle_0 = \sum_{\substack{\mathbf{q} \in \mathbf{Q} \\ v \in \{1, 2, \dots, r d_x\}}} \Omega_v(\mathbf{q}) B_\beta(\Omega_v(\mathbf{q})) \quad (33.110)$$

and using Lemma 9.48 we obtain the desired result.

Now we take care of the corrections. We compute the derivative

$$\begin{aligned} d_T \langle \delta H_{\text{ph}} \rangle &\approx \sum_{v \in \{1, 2, \dots, r d_x\}} A_v d_T (B_v)^2 \\ &- d_T \sum_{\substack{\mathbf{k}, \mathbf{q} \in \mathbf{Q} \\ v \in \{1, 2, \dots, r d_x\} \\ n, n' \in \{1, \dots, \infty\} \\ \sigma \in \{\downarrow, \uparrow\}}} \frac{\Omega_v(\mathbf{q})}{N_p} |g_{n'n v}(\mathbf{k}, \mathbf{q})|^2 (F_\beta(\xi_{n, \sigma, \mathbf{k}}) - F_\beta(\xi_{n', \sigma, \mathbf{k} + \mathbf{q}})) \left[\frac{B_\beta(\Omega_v(\mathbf{q})) - B_\beta(\xi_{n', \sigma, \mathbf{k} + \mathbf{q}} - \xi_{n, \sigma, \mathbf{k}})}{(\Omega_v(\mathbf{q}) + \xi_{n, \sigma, \mathbf{k}} - \xi_{n', \sigma, \mathbf{k} + \mathbf{q}})^2} + \frac{1}{4} \frac{\beta \operatorname{csch}(\beta \Omega_v(\mathbf{q})/2)^2}{\Omega_v(\mathbf{q}) + \xi_{n, \sigma, \mathbf{k}} - \xi_{n', \sigma, \mathbf{k} + \mathbf{q}}} \right] \end{aligned}$$

we neglect the first term for the beginning, since it only contributes for crystals with more than one atom in the UC and focus on the second term, which can be computed as

$$\begin{aligned} &- d_T \sum_{\substack{\mathbf{k}, \mathbf{q} \in \mathbf{Q} \\ v \in \{1, 2, \dots, r d_x\} \\ n, n' \in \{1, \dots, \infty\} \\ \sigma \in \{\downarrow, \uparrow\}}} \frac{\Omega_v(\mathbf{q})}{N_p} |g_{n'n v}(\mathbf{k}, \mathbf{q})|^2 (F_\beta(\xi_{n, \sigma, \mathbf{k}}) - F_\beta(\xi_{n', \sigma, \mathbf{k} + \mathbf{q}})) \left[\frac{B_\beta(\Omega_v(\mathbf{q})) - B_\beta(\xi_{n', \sigma, \mathbf{k} + \mathbf{q}} - \xi_{n, \sigma, \mathbf{k}})}{(\Omega_v(\mathbf{q}) + \xi_{n, \sigma, \mathbf{k}} - \xi_{n', \sigma, \mathbf{k} + \mathbf{q}})^2} + \frac{1}{4} \frac{\beta \operatorname{csch}(\beta \Omega_v(\mathbf{q})/2)^2}{\Omega_v(\mathbf{q}) + \xi_{n, \sigma, \mathbf{k}} - \xi_{n', \sigma, \mathbf{k} + \mathbf{q}}} \right] \\ &= - \sum_{\substack{\mathbf{k}, \mathbf{q} \in \mathbf{Q} \\ v \in \{1, 2, \dots, r d_x\} \\ n, n' \in \{1, \dots, \infty\} \\ \sigma \in \{\downarrow, \uparrow\}}} \frac{\Omega_v(\mathbf{q})}{N_p} |g_{n'n v}(\mathbf{k}, \mathbf{q})|^2 \left[d_T (F_\beta(\xi_{n, \sigma, \mathbf{k}}) - F_\beta(\xi_{n', \sigma, \mathbf{k} + \mathbf{q}})) \right] \left[\frac{B_\beta(\Omega_v(\mathbf{q})) - B_\beta(\xi_{n', \sigma, \mathbf{k} + \mathbf{q}} - \xi_{n, \sigma, \mathbf{k}})}{(\Omega_v(\mathbf{q}) + \xi_{n, \sigma, \mathbf{k}} - \xi_{n', \sigma, \mathbf{k} + \mathbf{q}})^2} + \frac{1}{4} \frac{\beta \operatorname{csch}(\beta \Omega_v(\mathbf{q})/2)^2}{\Omega_v(\mathbf{q}) + \xi_{n, \sigma, \mathbf{k}} - \xi_{n', \sigma, \mathbf{k} + \mathbf{q}}} \right] \\ &- \sum_{\substack{\mathbf{k}, \mathbf{q} \in \mathbf{Q} \\ v \in \{1, 2, \dots, r d_x\} \\ n, n' \in \{1, \dots, \infty\} \\ \sigma \in \{\downarrow, \uparrow\}}} \frac{\Omega_v(\mathbf{q})}{N_p} |g_{n'n v}(\mathbf{k}, \mathbf{q})|^2 (F_\beta(\xi_{n, \sigma, \mathbf{k}}) - F_\beta(\xi_{n', \sigma, \mathbf{k} + \mathbf{q}})) \left[d_T \frac{B_\beta(\Omega_v(\mathbf{q})) - B_\beta(\xi_{n', \sigma, \mathbf{k} + \mathbf{q}} - \xi_{n, \sigma, \mathbf{k}})}{(\Omega_v(\mathbf{q}) + \xi_{n, \sigma, \mathbf{k}} - \xi_{n', \sigma, \mathbf{k} + \mathbf{q}})^2} + \frac{1}{4} d_T \frac{\beta \operatorname{csch}(\beta \Omega_v(\mathbf{q})/2)^2}{\Omega_v(\mathbf{q}) + \xi_{n, \sigma, \mathbf{k}} - \xi_{n', \sigma, \mathbf{k} + \mathbf{q}}} \right] \end{aligned}$$

Using now again Lemma 9.48 we obtain the stated result for the first two terms. For the last term, we note the identity

$$\frac{d}{dx} \operatorname{csch}(x) = -\coth(x) \operatorname{csch}(x) \quad (33.111)$$

which gives

$$\begin{aligned} &d_T \left(\frac{1}{k_B T} \cdot \operatorname{csch}^2 \left(\frac{\Omega_v(\mathbf{q})}{2k_B T} \right) \right) \\ &= \frac{-1}{k_B T^2} \operatorname{csch}^2 \left(\frac{\Omega_v(\mathbf{q})}{2k_B T} \right) + \frac{1}{k_B T} \cdot 2 \cdot \operatorname{csch} \left(\frac{\Omega_v(\mathbf{q})}{2k_B T} \right) \cdot \left(-\coth \left(\frac{\Omega_v(\mathbf{q})}{2k_B T} \right) \cdot \operatorname{csch} \left(\frac{\Omega_v(\mathbf{q})}{2k_B T} \right) \cdot \left(\frac{-\Omega_v(\mathbf{q})}{2k_B T^2} \right) \right) \\ &= \frac{-1}{k_B T^2} \operatorname{csch}^2 \left(\frac{\Omega_v(\mathbf{q})}{2k_B T} \right) \cdot \left[1 - \frac{\Omega_v(\mathbf{q})}{k_B T} \coth \left(\frac{\Omega_v(\mathbf{q})}{2k_B T} \right) \right] \end{aligned} \quad (33.112)$$

Proof 33.23 of Lemma 24.10

In order to prove this, we just use the very definitions

Case 1: $\tau > 0$

When we can omit the time-ordering operator and we obtain

$$\begin{aligned}
 \widehat{\mathcal{G}}_{\text{Gorkov}}^m(\tau, \mathbf{k}) &= -\frac{1}{\hbar} \langle T_\tau c_{m,\downarrow}^\dagger(\tau, -\mathbf{k}) c_{m,\uparrow}^\dagger(0, \mathbf{k}) \rangle_{\text{SC}} \\
 &= -\frac{1}{\hbar} \langle c_{m,\downarrow}^\dagger(\tau, -\mathbf{k}) c_{m,\uparrow}^\dagger(0, \mathbf{k}) \rangle_{\text{SC}} \\
 &\stackrel{\text{Eq. (24.13)}}{=} -\frac{1}{\hbar} \langle c_{m,\downarrow}^\dagger(0, -\mathbf{k}) c_{m,\uparrow}^\dagger(-\tau, \mathbf{k}) \rangle_{\text{SC}} \\
 &= \left[-\frac{1}{\hbar} \langle c_{m,\uparrow}(+\tau, \mathbf{k}) c_{m,\downarrow}(0, -\mathbf{k}) \rangle_{\text{SC}} \right]^\dagger \\
 &= \left[\widehat{\mathcal{G}}_{\text{Gorkov}}^m(\tau, \mathbf{k}) \right]^\dagger
 \end{aligned}$$

Case 2: $\tau < 0$

Analogous. **TBDone.**

Proof 33.24 of Lemma 24.12

Starting from the definition

$$\widehat{\mathcal{G}}_{\text{Gorkov}}^m(\tau, \mathbf{k}) = \begin{pmatrix} \mathcal{G}_m(\tau, \mathbf{k}) & \mathcal{F}_m(\tau, \mathbf{k}) \\ \mathcal{F}_m^\dagger(\tau, \mathbf{k}) & -\mathcal{G}_m(-\tau, -\mathbf{k}) \end{pmatrix}$$

we compute the Matsubara-component of this function. For reasons that are difficult to motivate in advance, it is advantageous to use the definition from Theorem 1.7 in this calculation

$$\widehat{\mathcal{G}}_{\text{Gorkov}}^m(i\omega_n, \mathbf{k}) = \frac{1}{2} \int_{-\hbar\beta}^{\hbar\beta} d\tau e^{i\omega_n\tau} \begin{pmatrix} \mathcal{G}_m(\tau, \mathbf{k}) & \mathcal{F}_m(\tau, \mathbf{k}) \\ \mathcal{F}_m^\dagger(\tau, \mathbf{k}) & -\mathcal{G}_m(-\tau, -\mathbf{k}) \end{pmatrix}$$

Now the only component that requires further calculations is the lower right component of this matrix. By changing variables as $\tau \mapsto -\tau$ we compute

$$\begin{aligned}
 -\left(\widehat{\mathcal{G}}_{\text{Gorkov}}^m(i\omega_n, \mathbf{k}) \right)_{2,2} &= \frac{1}{2} \int_{-\hbar\beta}^{\hbar\beta} d\tau e^{i\omega_n\tau} \mathcal{G}_m(-\tau, -\mathbf{k}) \\
 &= \frac{1}{2} \int_{+\hbar\beta}^{-\hbar\beta} d\tau e^{-i\omega_n\tau} \mathcal{G}_m(+\tau, -\mathbf{k}) \cdot (-1) \\
 &= \frac{1}{2} \int_{-\hbar\beta}^{+\hbar\beta} d\tau e^{-i\omega_n\tau} \mathcal{G}_m(+\tau, -\mathbf{k}) \\
 &= \mathcal{G}_m(-i\omega_n, -\mathbf{k})
 \end{aligned}$$

Combining this with the very definition, we obtain the desired equation.

Proof 33.25 of Lemma 24.13 Recall from Theorem 13.6 that

$$\mathcal{G}_m(i\omega_n, \mathbf{k}) = \frac{1}{\hbar} \frac{1}{i\omega_n - (\epsilon_m(\mathbf{k}) - \mu) / \hbar}$$

and

$$\begin{aligned} -\mathcal{G}_m(-i\omega_n, -\mathbf{k}) &= \frac{-1}{\hbar} \frac{1}{-i\omega_n - (\epsilon_m(-\mathbf{k}) - \mu) / \hbar} \\ &= \frac{1}{\hbar} \frac{1}{i\omega_n + (\epsilon_m(-\mathbf{k}) - \mu) / \hbar} \end{aligned}$$

Thus we obtain

$$\mathcal{G}_m^{-1}(i\omega_n, \mathbf{k}) = \hbar (i\omega_n - (\epsilon_m(\mathbf{k}) - \mu) / \hbar)$$

and

$$-\hat{\mathcal{G}}_m^{-1}(-i\omega_n, -\mathbf{k}) = \hbar (i\omega_n + (\epsilon_m(-\mathbf{k}) - \mu) / \hbar)$$

and using the Pauli-matrices and the theorem about inversion-symmetric and spin-degenerate band-structures we obtain

$$\hat{\mathcal{G}}_{0,m}^{-1}(i\omega_n, \mathbf{k}) = i\hbar\omega_n\sigma_0 - (\epsilon_m(\mathbf{k}) - \mu)\sigma_3 \quad (33.113)$$

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proving the lemma.

Proof 33.26 of Corollary 24.15 We use Equation (24.21) and introduce the self-energy

$$\hat{\Sigma}_{\text{Gorkov}}^m(k) = \begin{pmatrix} \Sigma_{11}^m(k) & \Sigma_{12}^m(k) \\ \Sigma_{21}^m(k) & \Sigma_{22}^m(k) \end{pmatrix} \quad (33.114)$$

This yields for the product

$$\begin{aligned}
 (\widehat{\mathcal{G}}_0 \widehat{\Sigma} \widehat{\mathcal{G}})(k) &= \widehat{\mathcal{G}}_0(k) \begin{pmatrix} \Sigma_m(k) & \Sigma_m(k) \\ \Sigma_m(k) & \Sigma_m(k) \end{pmatrix} \begin{pmatrix} \mathcal{G}_{\text{Gorkov}}^m(k) & \mathcal{F}_m(k) \\ \mathcal{F}_m^*(k) & -\mathcal{G}_{\text{Gorkov}}^m(-k) \end{pmatrix} \\
 &= \widehat{\mathcal{G}}_0(k) \begin{pmatrix} \Sigma_m(k) \mathcal{G}_{\text{Gorkov}}^m(k) + \Sigma_m(k) \mathcal{F}_m^*(k) & \Sigma_m(k) \mathcal{F}_m(k) - \Sigma_m(k) \mathcal{G}_{\text{Gorkov}}^m(-k) \\ \Sigma_m(k) \mathcal{G}_{\text{Gorkov}}^m(k) + \Sigma_m(k) \mathcal{F}_m^*(k) & \Sigma_m(k) \mathcal{F}_m(k) - \Sigma_m(k) \mathcal{G}_{\text{Gorkov}}^m(-k) \end{pmatrix} \\
 &= \begin{pmatrix} \mathcal{G}_0(k) & 0 \\ 0 & -\mathcal{G}_0(-k) \end{pmatrix} \begin{pmatrix} \Sigma_m(k) \mathcal{G}_{\text{Gorkov}}^m(k) + \Sigma_m(k) \mathcal{F}_m^*(k) & \Sigma_m(k) \mathcal{F}_m(k) - \Sigma_m(k) \mathcal{G}_{\text{Gorkov}}^m(-k) \\ \Sigma_m(k) \mathcal{G}_{\text{Gorkov}}^m(k) + \Sigma_m(k) \mathcal{F}_m^*(k) & \Sigma_m(k) \mathcal{F}_m(k) - \Sigma_m(k) \mathcal{G}_{\text{Gorkov}}^m(-k) \end{pmatrix}
 \end{aligned}$$

whereas we used Equation (24.19). Using now Equation (24.21) yields the desired result.

Proof 33.27 of Lemma 24.18

Recalling that $U(\xi\tau, 0) = \exp -\frac{1}{\hbar} \xi\tau \widetilde{H} = \exp -\xi(\tau - 0) \widetilde{H} / \hbar$, with $\widetilde{H} = H - \mu N$ we obtain analogously the auxiliary identities

$$\begin{aligned}
 \frac{d}{d\tau} U(\xi\tau, 0) &= -\frac{\xi}{\hbar} \widetilde{H} U(\xi\tau, 0) \\
 \frac{d}{d\tau} U(0, \xi\tau) &= \frac{\xi}{\hbar} U(0, \xi\tau) \widetilde{H}
 \end{aligned}$$

Remark: There are more identities like this, but we only need these here. Then it follows

$$\begin{aligned}
 \frac{d}{d\tau} X_H(\xi\tau) &= \frac{d}{dt} U(0, \xi\tau) X(\xi\tau) U(\xi\tau, 0) \\
 &= \left(\frac{d}{d\tau} U(0, \xi\tau) \right) X(\xi\tau) U(\xi\tau, 0) + U(0, \xi\tau) \left(\frac{d}{d\tau} X(\xi\tau) \right) U(\xi\tau, 0) + U(0, \xi\tau) X(\xi\tau) \left(\frac{d}{d\tau} U(\xi\tau, 0) \right) \\
 &= \frac{\xi}{\hbar} U(0, \xi\tau) \widetilde{H} X(\xi\tau) U(\xi\tau, 0) + U(0, \xi\tau) \left(\frac{d}{d\tau} X(\xi\tau) \right) U(\xi\tau, 0) + U(0, \xi\tau) X(\xi\tau) \left(-\frac{\xi}{\hbar} \widetilde{H} U(\xi\tau, 0) \right) \\
 &= \frac{\xi}{\hbar} U(0, \xi\tau) \left(\widetilde{H} X(\xi\tau) - X(\xi\tau) \widetilde{H} \right) U(\xi\tau, 0) + U(0, \xi\tau) \left(\frac{d}{dt} X(\xi\tau) \right) U(\xi\tau, 0) \\
 &= \frac{\xi}{\hbar} U(0, \xi\tau) \left[\widetilde{H}, X(\xi\tau) \right] U(\xi\tau, 0) + U(0, \xi\tau) \left(\frac{d}{d\tau} X(\xi\tau) \right) U(\xi\tau, 0) \\
 &= -\frac{\xi}{\hbar} \left[X, \widetilde{H} \right]_H(\xi\tau) + U(0, \xi\tau) \left(\frac{d}{d\tau} X(\xi\tau) \right) U(\xi\tau, 0)
 \end{aligned}$$

Proof 33.28 of Lemma 24.19 We compute the equation of motion for the annihilation-operators using the Heisenberg-equation of motion (24.31) for arbitrary $\bullet \in Q$ with H from Equation (24.1)

$$\begin{aligned}
 \partial_\tau b_\nu(\xi\tau, \bullet) &= -\frac{\xi}{\hbar} [b_\nu(\bullet), H - \mu N]_{\text{H}}(\xi\tau) \\
 &= -\xi \sum_{\substack{\mathbf{q}' \in Q \\ \nu' \in \{1, \dots, rd_x\}}} \omega_{\nu'}(\mathbf{q}') [b_\nu(\bullet), b_{\nu'}^\dagger(\mathbf{q}') b_{\nu'}(\mathbf{q}')]_{\text{H}}(\xi\tau) + 0 \\
 &\quad - \xi \sum_{\substack{\mathbf{k}', \mathbf{q}' \in Q \\ \sigma' \in \{\uparrow, \downarrow\} \\ \nu' \in \{1, \dots, rd_x\} \\ m', n' \in \{1, \dots, \infty\}}} \frac{g_{m'n'\nu'}(\mathbf{k}', \mathbf{q}')}{\hbar \sqrt{N_p}} c_{m'\sigma'}^\dagger(\xi\tau, \mathbf{k}' + \mathbf{q}') c_{n'\sigma'}(\xi\tau, \mathbf{k}') ([b_\nu(\bullet), b_{\nu'}^\dagger(-\mathbf{q}')]_{\text{H}}(\xi\tau) + 0) \\
 &= \dots
 \end{aligned}$$

Now we use Equation (10.4) and Lemma 7.17, i.e.

$$[A, BC] = [A, B]_\zeta C + \zeta B[A, C]_\zeta \quad (33.115)$$

to simplify the second and third row as

$$\begin{aligned}
 [b_\nu(\bullet), b_{\nu'}^\dagger(\mathbf{q}') b_{\nu'}(\mathbf{q}')] &= \delta_{\nu, \nu'} \delta_{\bullet, \mathbf{q}'} b_{\nu'}(\mathbf{q}') \\
 [b_\nu(\bullet), b_{\nu'}^\dagger(-\mathbf{q}')] &= \delta_{\nu, \nu'} \delta_{\bullet, -\mathbf{q}'}
 \end{aligned}$$

such that we obtain

$$\dots = -\xi \omega_\nu(\bullet) b_\nu(\xi\tau, \bullet) - \xi \sum_{\substack{\mathbf{k}' \in Q \\ \sigma' \in \{\uparrow, \downarrow\} \\ m', n' \in \{1, \dots, \infty\}}} \frac{g_{m'n'\nu}(\mathbf{k}', -\bullet)}{\hbar \sqrt{N_p}} c_{m'\sigma'}^\dagger(\xi\tau, \mathbf{k}' - \bullet) c_{n'\sigma'}(\xi\tau, \mathbf{k}')$$

replacing now the summation indices (i.e. dropping the primes), we obtain the desired result.

In the same fashion we obtain

$$\begin{aligned}
 \partial_\tau b_\nu^\dagger(\xi\tau, -\bullet) &= -\frac{\xi}{\hbar} [b_\nu^\dagger(-\bullet), H - \mu N]_{\text{H}}(\xi\tau) \\
 &= -\xi \sum_{\substack{\mathbf{q}' \in Q \\ \nu' \in \{1, \dots, rd_x\}}} \omega_{\nu'}(\mathbf{q}') [b_\nu^\dagger(-\bullet), b_{\nu'}^\dagger(\mathbf{q}') b_{\nu'}(\mathbf{q}')]_{\text{H}}(\xi\tau) + 0 \\
 &\quad - \xi \sum_{\substack{\mathbf{k}', \mathbf{q}' \in Q \\ \sigma' \in \{\uparrow, \downarrow\} \\ \nu' \in \{1, \dots, rd_x\} \\ m', n' \in \{1, \dots, \infty\}}} \frac{g_{m'n'\nu'}(\mathbf{k}', \mathbf{q}')}{\hbar \sqrt{N_p}} c_{m'\sigma'}^\dagger(\xi\tau, \mathbf{k}' + \mathbf{q}') c_{n'\sigma'}(\xi\tau, \mathbf{k}') ([b_\nu^\dagger(-\bullet), b_{\nu'}(+\mathbf{q}')]_{\text{H}}(\xi\tau) + 0) \\
 &= \dots
 \end{aligned}$$

Now we use again Equation (10.4) and Lemma 7.17

$$\begin{aligned} [b_v^\dagger(-\bullet), b_{v'}^\dagger(\mathbf{q}')b_{v'}(\mathbf{q}')] &= -\delta_{v,v'}\delta_{-\bullet,\mathbf{q}'}b_{v'}^\dagger(\mathbf{q}') \\ [b_v^\dagger(-\bullet), b_{v'}(\mathbf{q}')] &= -\delta_{v,v'}\delta_{-\bullet,\mathbf{q}'} \end{aligned}$$

(notice that the extra minus sign appears due to the commutation-relations), such that we obtain

$$\dots = +\xi\omega_v(-\bullet)b_v^\dagger(\xi\tau, -\bullet) + \xi \sum_{\substack{\mathbf{k}' \in Q \\ \sigma' \in \{\uparrow, \downarrow\} \\ m', n' \in \{1, \dots, \infty\}}} \frac{g_{m'n'v}(\mathbf{k}', -\bullet)}{\hbar\sqrt{N_p}} c_{m'\sigma'}^\dagger(\xi\tau, \mathbf{k}' - \bullet) c_{n'\sigma'}(\xi\tau, \mathbf{k}')$$

using now the inversion-symmetry of ω and dropping the primes we obtain the desired result.

Proof 33.29 of Lemma 24.20 Recall from Lemma 13.5 that the equation of motion for the Green-function in imaginary time reads

$$\partial_\tau \mathcal{G}_{\text{Gorkov}}^n(\tau, \mathbf{k}) = -\frac{1}{\hbar}\delta(\tau) - \frac{1}{\hbar} \left\langle T_\tau \left[\frac{H - \mu N}{\hbar}, c_{n,\uparrow}(\mathbf{k}) \right]_H(\tau) c_{n,\uparrow}^\dagger(\mathbf{k}, 0) \right\rangle_{\text{SC}} \quad (33.116)$$

whereas N refers to the total electronic occupation-number operator. We now compute the commutator for the EPW-system as

$$\begin{aligned} \left[\frac{H - \mu N}{\hbar}, c_{n,\uparrow}(\mathbf{k}) \right] &= \sum_{\substack{\mathbf{k}' \in Q \\ \sigma' \in \{\uparrow, \downarrow\} \\ n' \in \{1, \dots, \infty\}}} \frac{1}{\hbar} (\epsilon_{n'}(\mathbf{k}') - \mu) [c_{n'\sigma'}^\dagger(\mathbf{k}') c_{n'\sigma'}(\mathbf{k}'), c_{n,\uparrow}(\mathbf{k})] \\ &+ \sum_{\substack{\mathbf{q}', \mathbf{k}' \in Q \\ \sigma' \in \{\uparrow, \downarrow\} \\ v' \in \{1, \dots, d_x\} \\ n', m' \in \{1, \dots, \infty\}}} \frac{g_{m'n'v'}(\mathbf{k}', \mathbf{q}')}{\hbar\sqrt{N_p}} [c_{m'\sigma'}^\dagger(\mathbf{k}' + \mathbf{q}') c_{n'\sigma'}(\mathbf{k}') (b_{v'}(\mathbf{q}') + b_{v'}^\dagger(-\mathbf{q}')), c_{n,\uparrow}(\mathbf{k})] \\ &= \dots \end{aligned}$$

Now we evaluate the individual commutators. For the first line we have

$$[c_{n'\sigma'}^\dagger(\mathbf{k}') c_{n'\sigma'}(\mathbf{k}'), c_{n,\uparrow}(\mathbf{k})] = -[c_{n,\uparrow}(\mathbf{k}), c_{n'\sigma'}^\dagger(\mathbf{k}') c_{n'\sigma'}(\mathbf{k}')] = -\delta_{n',n} \delta_{\sigma',\uparrow} \delta_{\mathbf{k}',\mathbf{k}} c_{n'\sigma'}(\mathbf{k}') \quad (33.117)$$

and

$$\begin{aligned} [c_{m'\sigma'}^\dagger(\mathbf{k}' + \mathbf{q}') c_{n'\sigma'}(\mathbf{k}') (b_{v'}(\mathbf{q}') + b_{v'}^\dagger(-\mathbf{q}')), c_{n,\uparrow}(\mathbf{k})] &= -[c_{n,\uparrow}(\mathbf{k}), c_{m'\sigma'}^\dagger(\mathbf{k}' + \mathbf{q}') c_{n'\sigma'}(\mathbf{k}')] (b_{v'}(\mathbf{q}') + b_{v'}^\dagger(-\mathbf{q}')) \\ &= -\delta_{m',n} \delta_{\sigma',\uparrow} \delta_{\mathbf{k}'+\mathbf{q}',\mathbf{k}} c_{n'\sigma'}(\mathbf{k}') (b_{v'}(\mathbf{q}') + b_{v'}^\dagger(-\mathbf{q}')) \end{aligned}$$

whereas we used Lemma 7.17 in both cases, and thus we obtain, after terminating the sums over $\mathbf{q}'$, m' and σ' , the equation

$$\dots = - \left(\frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar} \right) c_{n\uparrow}(\mathbf{k}) - \sum_{\substack{\mathbf{k}' \in Q \\ \nu' \in \{1, \dots, rd_x\} \\ n' \in \{1, \dots, \infty\}}} \frac{g_{nn'\nu'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')}{\hbar \sqrt{N_p}} (b_{\nu'}(\mathbf{k} - \mathbf{k}') + b_{\nu'}^\dagger(\mathbf{k}' - \mathbf{k})) c_{n'\uparrow}(\mathbf{k}')$$

Plugging this into the equation of motion, Equation (33.116), yields

$$\begin{aligned} \partial_\tau \mathcal{G}_{\text{Gorkov}}^n(\tau, \mathbf{k}) &= -\delta(\tau)/\hbar + \frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar^2} \left\langle T_\tau c_{n\uparrow}(\tau, \mathbf{k}) c_{n\uparrow}^\dagger(0, \mathbf{k}) \right\rangle_{\text{SC}} \\ &+ \sum_{\substack{\mathbf{k}' \in Q \\ \nu' \in \{1, \dots, rd_x\} \\ n' \in \{1, \dots, \infty\}}} \frac{g_{nn'\nu'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')}{\hbar^2 \sqrt{N_p}} \left\langle T_\tau (b_{\nu'}(\tau, \mathbf{k} - \mathbf{k}') + b_{\nu'}^\dagger(\tau, \mathbf{k}' - \mathbf{k})) c_{n'\uparrow}(\tau, \mathbf{k}') c_{n\uparrow}^\dagger(\mathbf{k}, 0) \right\rangle_{\text{SC}} \end{aligned}$$

which can be simplified to

$$\begin{aligned} &\left(\partial_\tau + \frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar} \right) \mathcal{G}_{\text{Gorkov}}^n(\tau, \mathbf{k}) \\ &= -\delta(\tau)/\hbar + \sum_{\substack{\mathbf{k}' \in Q \\ \nu' \in \{1, \dots, rd_x\} \\ n' \in \{1, \dots, \infty\}}} \frac{g_{nn'\nu'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')}{\hbar^2 \sqrt{N_p}} \left\langle T_\tau (b_{\nu'}(\tau, \mathbf{k} - \mathbf{k}') + b_{\nu'}^\dagger(\tau, \mathbf{k}' - \mathbf{k})) c_{n'\uparrow}(\tau, \mathbf{k}') c_{n\uparrow}^\dagger(\mathbf{k}, 0) \right\rangle_{\text{SC}} \end{aligned} \quad (33.118)$$

Proof 33.30 of Lemma 24.22 Notice first that Equation (24.33) can be expanded as

$$\begin{aligned} \mathcal{G}_{\nu', n, n'}^{\text{aux}}(\tau, \tau', \mathbf{k}, \mathbf{k}') &= -\frac{1}{\hbar} \left[+\theta(\tau') \left\langle (b_{\nu'}(\tau, \mathbf{k} - \mathbf{k}') + b_{\nu'}^\dagger(\tau, \mathbf{k}' - \mathbf{k})) c_{n', \uparrow}(\tau', \mathbf{k}') c_{n, \uparrow}^\dagger(0, \mathbf{k}) \right\rangle_{\text{SC}} \right. \\ &\quad \left. - \theta(-\tau') \left\langle (b_{\nu'}(\tau, \mathbf{k} - \mathbf{k}') + b_{\nu'}^\dagger(\tau, \mathbf{k}' - \mathbf{k})) c_{n, \uparrow}^\dagger(0, \mathbf{k}) c_{n', \uparrow}(\tau', \mathbf{k}') \right\rangle_{\text{SC}} \right]. \end{aligned} \quad (33.119)$$

Since the time-ordering-operator decomposes into the bosonic and fermionic part and it is sufficient⁶ to focus on the bosonic part. Thus taking the derivative w.r.t. τ yields doesn't involve any δ -functions. One finds by straightforward computation with Equation (33.119) that⁷

$$\begin{aligned} \partial_\tau \mathcal{G}_{\nu', n, n'}^{\text{aux}}(\tau, \tau', \mathbf{k}, \mathbf{k}') &= -\frac{1}{\hbar} \left\langle T_\tau (\partial_\tau b_{\nu'}(\tau, \mathbf{k} - \mathbf{k}')) c_{n', \uparrow}(\tau', \mathbf{k}') c_{n, \uparrow}^\dagger(0, \mathbf{k}) \right\rangle_{\text{SC}} \\ &\quad - \frac{1}{\hbar} \left\langle T_\tau (\partial_\tau b_{\nu'}^\dagger(\tau, \mathbf{k}' - \mathbf{k})) c_{n', \uparrow}(\tau' \pm \epsilon, \mathbf{k}') c_{n, \uparrow}^\dagger(0, \mathbf{k}) \right\rangle_{\text{SC}} = \dots \end{aligned}$$

and then by using Lemma 24.19 we obtain

⁶ Maybe this should be added as a corollary. Cf. definition 8.14.

⁷ Notice that the order doesn't matter cause within T_τ it has even statistics.

$$\begin{aligned}
 & \dots \\
 &= -\frac{1}{\hbar} \left\langle T_\tau c_{n',\uparrow}(\tau', \mathbf{k}') c_{n,\uparrow}^\dagger(0, \mathbf{k}) \left[-\omega_{v'}(\mathbf{k} - \mathbf{k}') b_{v'}(\tau, \mathbf{k} - \mathbf{k}') - \sum_{\substack{\tilde{\mathbf{k}} \in Q \\ \tilde{\sigma} \in \{\uparrow, \downarrow\} \\ \tilde{m}, \tilde{n} \in \{1, \dots, \infty\}}} \frac{g_{\tilde{m}\tilde{n}v'}(\tilde{\mathbf{k}}, \mathbf{k}' - \mathbf{k})}{\hbar \sqrt{N_p}} c_{\tilde{m}\tilde{\sigma}}^\dagger(\tau, \tilde{\mathbf{k}} + \mathbf{k}' - \mathbf{k}) c_{\tilde{n}\tilde{\sigma}}(\tau, \tilde{\mathbf{k}}) \right] \right\rangle \\
 & - \frac{1}{\hbar} \left\langle T_\tau c_{n',\uparrow}(\tau', \mathbf{k}') c_{n,\uparrow}^\dagger(0, \mathbf{k}) \left[\omega_{v'}(\mathbf{k}' - \mathbf{k}) b_{v'}^\dagger(\tau, \mathbf{k}' - \mathbf{k}) + \sum_{\substack{\tilde{\mathbf{k}} \in Q \\ \tilde{\sigma} \in \{\uparrow, \downarrow\} \\ \tilde{m}, \tilde{n} \in \{1, \dots, \infty\}}} \frac{g_{\tilde{m}\tilde{n}v'}(\tilde{\mathbf{k}}, \mathbf{k}' - \mathbf{k})}{\hbar \sqrt{N_p}} c_{\tilde{m}\tilde{\sigma}}^\dagger(\tau, \tilde{\mathbf{k}} + \mathbf{k}' - \mathbf{k}) c_{\tilde{n}\tilde{\sigma}}(\tau, \tilde{\mathbf{k}}) \right] \right\rangle \\
 &= \dots
 \end{aligned} \tag{SC}$$

and see that the contributions containing the electron-phonon-coupling g cancel out and we obtain

$$\begin{aligned}
 \partial_\tau \mathcal{G}_{v',n,n'}^{\text{aux}}(\tau, \tau', \mathbf{k}, \mathbf{k}') &= +\frac{\omega_{v'}(\mathbf{k} - \mathbf{k}')}{\hbar} \left\langle T_\tau b_{v'}(\tau, \mathbf{k} - \mathbf{k}') c_{n',\uparrow}(\tau', \mathbf{k}') c_{n,\uparrow}^\dagger(0, \mathbf{k}) \right\rangle_{\text{SC}} \\
 & - \frac{\omega_{v'}(\mathbf{k}' - \mathbf{k})}{\hbar} \left\langle T_\tau b_{v'}^\dagger(\tau, \mathbf{k}' - \mathbf{k}) c_{n',\uparrow}(\tau', \mathbf{k}') c_{n,\uparrow}^\dagger(0, \mathbf{k}) \right\rangle_{\text{SC}} \\
 &= \frac{\omega_{v'}(\mathbf{k} - \mathbf{k}')}{\hbar} \left\langle T_\tau (b_{v'}(\tau, \mathbf{k} - \mathbf{k}') - b_{v'}^\dagger(\tau, \mathbf{k}' - \mathbf{k})) c_{n',\uparrow}(\tau', \mathbf{k}') c_{n,\uparrow}^\dagger(0, \mathbf{k}) \right\rangle_{\text{SC}}
 \end{aligned}$$

whereas we used the inversion-symmetry $\omega_v(+\bullet) = \omega_v(-\bullet)$ in the last equation-sign and hence we obtain the desired result. For the second derivative we obtain

$$\begin{aligned}
 \partial_\tau^2 \mathcal{G}_{v',n,n'}^{\text{aux}}(\tau, \tau', \mathbf{k}, \mathbf{k}') &= \frac{\omega_{v'}(\mathbf{k} - \mathbf{k}')}{\hbar} \left\langle T_\tau \partial_\tau (b_{v'}(\tau, \mathbf{k} - \mathbf{k}') - b_{v'}^\dagger(\tau, \mathbf{k}' - \mathbf{k})) c_{n',\uparrow}(\tau', \mathbf{k}') c_{n,\uparrow}^\dagger(0, \mathbf{k}) \right\rangle_{\text{SC}} \\
 &= \text{I} + \text{II} \\
 &= \dots
 \end{aligned} \tag{33.120}$$

using now Lemma 24.19 again we obtain for the individual contributions

$$\begin{aligned}
 \text{I} &= \frac{\omega_{v'}(\mathbf{k} - \mathbf{k}')}{\hbar} \left\langle T_\tau \left[-\omega_{v'}(\mathbf{k} - \mathbf{k}') b_{v'}(\tau, \mathbf{k} - \mathbf{k}') \right. \right. \\
 & \quad \left. \left. - \sum_{\substack{\tilde{\mathbf{k}} \in Q \\ \tilde{\sigma} \in \{\uparrow, \downarrow\} \\ \tilde{m}, \tilde{n} \in \{1, \dots, \infty\}}} \frac{g_{\tilde{m}\tilde{n}v'}(\tilde{\mathbf{k}}, \mathbf{k}' - \mathbf{k})}{\hbar \sqrt{N_p}} c_{\tilde{m}\tilde{\sigma}}^\dagger(\tau, \tilde{\mathbf{k}} + \mathbf{k}' - \mathbf{k}) c_{\tilde{n}\tilde{\sigma}}(\tau, \tilde{\mathbf{k}}) \right] c_{n',\uparrow}(\tau', \mathbf{k}') c_{n,\uparrow}^\dagger(0, \mathbf{k}) \right\rangle_{\text{SC}}
 \end{aligned}$$

and

$$\begin{aligned} \Pi = & -\frac{\omega_{\nu'}(\mathbf{k}-\mathbf{k}')}{\hbar} \left\langle T_{\tau} \left[+\omega_{\nu'}(\mathbf{k}'-\mathbf{k})b_{\nu'}^{\dagger}(\tau, \mathbf{k}'-\mathbf{k}) \right. \right. \\ & \left. \left. + \sum_{\substack{\tilde{\mathbf{k}} \in Q \\ \tilde{\sigma} \in \{\uparrow, \downarrow\} \\ \tilde{m}, \tilde{n} \in \{1, \dots, \infty\}}} \frac{g_{\tilde{m}\tilde{n}\nu'}(\tilde{\mathbf{k}}, \mathbf{k}'-\mathbf{k})}{\hbar\sqrt{N_p}} c_{\tilde{m}\tilde{\sigma}}^{\dagger}(\tau, \tilde{\mathbf{k}}+\mathbf{k}'-\mathbf{k})c_{\tilde{n}\tilde{\sigma}}(\tau, \tilde{\mathbf{k}}) \right] c_{n', \uparrow}(\tau', \mathbf{k}')c_{n, \uparrow}^{\dagger}(0, \mathbf{k}) \right\rangle_{\text{SC}} \end{aligned}$$

and collecting the individual contributions we obtain

$$\begin{aligned} \partial_{\tau}^2 \mathcal{G}_{\nu', n, n'}^{\text{aux}}(\tau, \tau', \mathbf{k}, \mathbf{k}') &= \omega_{\nu'}^2(\mathbf{k}-\mathbf{k}') \mathcal{G}_{\nu', n, n'}^{\text{aux}}(\tau, \tau', \mathbf{k}, \mathbf{k}') \\ &- \frac{2\omega_{\nu'}(\mathbf{k}-\mathbf{k}')}{\hbar} \sum_{\substack{\tilde{\mathbf{k}} \in Q \\ \tilde{\sigma} \in \{\uparrow, \downarrow\} \\ \tilde{m}, \tilde{n} \in \{1, \dots, \infty\}}} \frac{g_{\tilde{m}\tilde{n}\nu'}(\tilde{\mathbf{k}}, \mathbf{k}'-\mathbf{k})}{\hbar\sqrt{N_p}} \left\langle T_{\tau} c_{\tilde{m}\tilde{\sigma}}^{\dagger}(\tau, \tilde{\mathbf{k}}+\mathbf{k}'-\mathbf{k})c_{\tilde{n}\tilde{\sigma}}(\tau, \tilde{\mathbf{k}})c_{n', \uparrow}(\tau', \mathbf{k}')c_{n, \uparrow}^{\dagger}(0, \mathbf{k}) \right\rangle_{\text{SC}} \end{aligned} \quad (33.121)$$

Now rearranging terms and relabelling the summation-indices we obtain

$$\begin{aligned} & (\partial_{\tau}^2 - \omega_{\nu'}^2(\mathbf{k}-\mathbf{k}')) \mathcal{G}_{\nu', n, n'}^{\text{aux}}(\tau, \tau', \mathbf{k}, \mathbf{k}') \\ &= \frac{-2\omega_{\nu'}(\mathbf{k}-\mathbf{k}')}{\hbar} \sum_{\substack{\mathbf{q} \in Q \\ mm' \in \{1, \dots, \infty\} \\ \sigma \in \{\uparrow, \downarrow\}}} \frac{g_{mm'\nu'}(\mathbf{q}, \mathbf{k}'-\mathbf{k})}{\hbar N_p} \left\langle T_{\tau} c_{m, \sigma}^{\dagger}(\tau, \mathbf{q}+\mathbf{k}'-\mathbf{k})c_{m', \sigma}(\tau, \mathbf{q})c_{n', \uparrow}(\tau', \mathbf{k}')c_{n, \uparrow}^{\dagger}(0, \mathbf{k}) \right\rangle_{\text{SC}} \end{aligned}$$

Proof 33.31 of Corollary 24.23

First, we exchange⁸ the roles of τ and τ' in the equations (24.35) and (24.36). This yields for the two equations

$$\begin{aligned} \partial_{\tau'} \mathcal{G}_{\nu', n, n'}^{\text{aux}}(\tau', \tau, \mathbf{k}, \mathbf{k}') &= +\frac{\omega_{\nu'}(\mathbf{k}-\mathbf{k}')}{\hbar} \left\langle T_{\tau} (b_{\nu'}(\tau', \mathbf{k}-\mathbf{k}') - b_{\nu'}^{\dagger}(\tau', \mathbf{k}'-\mathbf{k})) c_{n', \uparrow}(\tau, \mathbf{k}')c_{n, \uparrow}^{\dagger}(0, \mathbf{k}) \right\rangle_{\text{SC}} \\ &= \frac{-2\omega_{\nu'}(\mathbf{k}-\mathbf{k}')}{\hbar} \sum_{\substack{\mathbf{q} \in Q \\ m, m' \in \{1, \dots, \infty\} \\ \sigma \in \{\uparrow, \downarrow\}}} \frac{g_{mm'\nu'}(\mathbf{q}, \mathbf{k}'-\mathbf{k})}{\hbar\sqrt{N_p}} \left\langle T_{\tau} c_{m, \sigma}^{\dagger}(\tau', \mathbf{q}+\mathbf{k}'-\mathbf{k})c_{m', \sigma}(\tau', \mathbf{q})c_{n', \uparrow}(\tau, \mathbf{k}')c_{n, \uparrow}^{\dagger}(0, \mathbf{k}) \right\rangle_{\text{SC}} \end{aligned} \quad (33.122)$$

Therefore, we start with equation (33.122) which is

⁸Notice that this has no influence on the time-ordering-operator which has the inconvenient notation of a τ in its index.

$$\begin{aligned}
 & (\partial_{\tau'}^2 - \omega_{\nu'}^2(\mathbf{k} - \mathbf{k}')) \mathcal{G}_{\nu', n, n'}^{\text{aux}}(\tau', \tau, \mathbf{k}, \mathbf{k}') \\
 &= \frac{-2\omega_{\nu'}(\mathbf{k} - \mathbf{k}')}{\hbar} \sum_{\substack{\mathbf{q} \in \mathcal{Q} \\ m, m' \in \{1, \dots, \infty\} \\ \sigma \in \{\uparrow, \downarrow\}}} \frac{g_{mm'\nu'}(\mathbf{q}, \mathbf{k}' - \mathbf{k})}{\hbar \sqrt{N_p}} \left\langle T_{\tau} c_{m, \sigma}^{\dagger}(\tau', \mathbf{q} + \mathbf{k}' - \mathbf{k}) c_{m', \sigma}(\tau', \mathbf{q}) c_{n', \uparrow}(\tau, \mathbf{k}') c_{n, \uparrow}^{\dagger}(0, \mathbf{k}) \right\rangle_{\text{SC}}
 \end{aligned}$$

we multiply both sides with $\mathcal{D}_{0, \nu'}(\tau - \tau')$, integrate τ' from 0 to $\hbar\beta$, perform two integration by parts and using the Equation (13.23) from Theorem 13.11.

$$(\partial_{\tau}^2 - \omega_{\nu'}^2(\mathbf{k} - \mathbf{k}')) \mathcal{D}_{0, \nu'}(\tau - \tau') = 2\omega_{\nu'}(\mathbf{k} - \mathbf{k}') \delta(\tau - \tau') / \hbar$$

Notice that the left hand side of this has the unit $\frac{1}{\text{time}^2} \frac{1}{\text{Energy} \times \text{time}}$ in complete harmony with the r.h.s. of this equation. This yields

$$\begin{aligned}
 & \int_0^{\hbar\beta} d\tau' \mathcal{G}_{\nu', n, n'}^{\text{aux}}(\tau', \tau, \mathbf{k}, \mathbf{k}') (\partial_{\tau'}^2 - \omega_{\nu'}^2(\mathbf{k} - \mathbf{k}')) \mathcal{D}_{0, \nu'}(\tau - \tau') \\
 &= \frac{+2\omega_{\nu'}(\mathbf{k} - \mathbf{k}')}{\hbar} \mathcal{G}_{\nu', n, n'}^{\text{aux}}(\tau, \tau, \mathbf{k}, \mathbf{k}) \\
 &= \int_0^{\hbar\beta} d\tau' \mathcal{D}_{0, \nu'}(\tau - \tau') \frac{-2\omega_{\nu'}(\mathbf{k} - \mathbf{k}')}{\hbar} \sum_{\substack{\mathbf{q} \in \mathcal{Q} \\ m, m' \in \{1, \dots, \infty\} \\ \sigma \in \{\uparrow, \downarrow\}}} \frac{g_{mm'\nu'}(\mathbf{q}, \mathbf{k}' - \mathbf{k})}{\hbar \sqrt{N_p}} \left\langle T_{\tau} c_{m, \sigma}^{\dagger}(\tau', \mathbf{q} + \mathbf{k}' - \mathbf{k}) c_{m', \sigma}(\tau', \mathbf{q}) c_{n', \uparrow}(\tau, \mathbf{k}') c_{n, \uparrow}^{\dagger}(0, \mathbf{k}) \right\rangle_{\text{SC}}
 \end{aligned} \tag{33.123}$$

simplifying both sides gives then the final result

$$\begin{aligned}
 & \mathcal{G}_{\nu', n, n'}^{\text{aux}}(\tau, \tau, \mathbf{k}, \mathbf{k}') \\
 &= \oint_{\substack{\mathbf{q} \in \mathcal{Q} \\ \tau' \in [0, \hbar\beta] \\ m, m' \in \{1, \dots, \infty\} \\ \sigma \in \{\uparrow, \downarrow\}}} (-1) \mathcal{D}_{0, \nu'}(\tau - \tau') \frac{g_{mm'\nu'}(\mathbf{q}, \mathbf{k}' - \mathbf{k})}{\hbar \sqrt{N_p}} \left\langle T_{\tau} c_{m, \sigma}^{\dagger}(\tau', \mathbf{q} + \mathbf{k}' - \mathbf{k}) c_{m', \sigma}(\tau', \mathbf{q}) c_{n', \uparrow}(\tau, \mathbf{k}') c_{n, \uparrow}^{\dagger}(0, \mathbf{k}) \right\rangle
 \end{aligned} \tag{33.124}$$

Notice that this equation has again the right units (1 / Energy x time x time on both sides).

Proof 33.32 of Theorem 24.24 We combine the approximation of the Bethe-Salpether-equation with Remark 24.23.1: The Bethe-Salpeter-equation (15.19), in which one neglects the vertex-correction, reads

$$\mathcal{G}(i_1, i_2 | i'_1, i'_2) \approx \mathcal{G}(i_1 | i'_1) \mathcal{G}(i_2 | i'_2) + \zeta \mathcal{G}(i_1 | i'_2) \mathcal{G}(i_2 | i'_1) \tag{33.125}$$

The left hand side of this equation is

$$\mathcal{G}(i_1, i_2 | i'_1, i'_2) = \left(-\frac{1}{\hbar} \right)^2 \left\langle T_{\tau} c_{i_1} c_{i_2} c_{i'_2}^{\dagger} c_{i'_1}^{\dagger} \right\rangle \tag{33.126}$$

Now, following Definition 24.61, we introduce the following notation

$$c_{m\sigma}^\dagger(\tau', \mathbf{q} + \mathbf{k}' - \mathbf{k}) = \begin{cases} \eta_{m,2}(\tau', -\mathbf{q} - \mathbf{k}' + \mathbf{k}) & \text{if } \sigma = \downarrow \\ \bar{\eta}_{m,1}(\tau', \mathbf{q} + \mathbf{k}' - \mathbf{k}) & \text{if } \sigma = \uparrow \end{cases} \quad (33.127)$$

and

$$c_{m'\sigma}(\tau, \mathbf{q}) = \begin{cases} \bar{\eta}_{m',2}(\tau, -\mathbf{q}) & \text{if } \sigma = \downarrow \\ \eta_{m',1}(\tau, \mathbf{q}) & \text{if } \sigma = \uparrow \end{cases} \quad (33.128)$$

and hence also $c_{n'\uparrow}(\tau, \mathbf{k}') \equiv \eta_{n',1}(\tau, \mathbf{k}')$, and $c_{n\uparrow}^\dagger(0, \mathbf{k}) \equiv \bar{\eta}_{n,1}(0, \mathbf{k})$.

This gives thus for the sum from Equation (24.37)

$$\begin{aligned} & \sum_{\sigma \in \{\downarrow, \uparrow\}} \left\langle T_\tau c_{m,\sigma}^\dagger(\tau', \mathbf{q} + \mathbf{k}' - \mathbf{k}) c_{m',\sigma}(\tau', \mathbf{q}) c_{n',\uparrow}(\tau, \mathbf{k}') c_{n,\uparrow}^\dagger(0, \mathbf{k}) \right\rangle \\ &= \left\langle T_\tau \eta_{m,2}(\tau', -\mathbf{q} - \mathbf{k}' + \mathbf{k}) \bar{\eta}_{m',2}(\tau', -\mathbf{q}) \eta_{n',1}(\tau, \mathbf{k}') \bar{\eta}_{n,1}(0, \mathbf{k}) \right\rangle \\ &+ \left\langle T_\tau \bar{\eta}_{m,1}(\tau', +\mathbf{q} + \mathbf{k}' - \mathbf{k}) \eta_{m',1}(\tau', +\mathbf{q}) \eta_{n',1}(\tau, \mathbf{k}') \bar{\eta}_{n,1}(0, \mathbf{k}) \right\rangle \\ &= \left\langle T_\tau \eta_{n',1}(\tau, \mathbf{k}') \eta_{m,2}(\tau', -\mathbf{q} - \mathbf{k}' + \mathbf{k}) \bar{\eta}_{m',2}(\tau', -\mathbf{q}) \bar{\eta}_{n,1}(0, \mathbf{k}) \right\rangle \\ &+ \left\langle T_\tau \eta_{m',1}(\tau', +\mathbf{q}) \eta_{n',1}(\tau, \mathbf{k}') \bar{\eta}_{m,1}(\tau', +\mathbf{q} + \mathbf{k}' - \mathbf{k}) \bar{\eta}_{n,1}(0, \mathbf{k}) \right\rangle \\ &\equiv \text{I} + \text{II} \end{aligned} \quad (33.129)$$

Now we use the decoupling for the $\sigma = \downarrow$ contribution (for the first of the two consecutive lines)

$$\begin{aligned} \text{I} &= \hbar^2 \mathcal{G}(i_1 = [n', 1, \tau, \mathbf{k}'], i_2 = [m, 2, \tau', -\mathbf{q} - \mathbf{k}' + \mathbf{k}] \mid i'_1 = [n, 1, 0, \mathbf{k}], i'_2 = [m', 2, \tau', -\mathbf{q}]) \\ &\approx \hbar^2 \mathcal{G}_{1,1}^{n',n}(\tau - 0; \mathbf{k}', \mathbf{k}) \mathcal{G}_{2,2}^{m,m'}(0; -\mathbf{q} - \mathbf{k}' + \mathbf{k}, -\mathbf{q}) - \hbar^2 \mathcal{G}_{1,2}^{n',m'}(\tau - \tau'; \mathbf{k}', -\mathbf{q}) \mathcal{G}_{2,1}^{m,n}(\tau'; -\mathbf{q} - \mathbf{k}' + \mathbf{k}, \mathbf{k}) \\ &= \hbar^2 \delta_{n,n'} \delta_{m,m'} \delta_{\mathbf{k},\mathbf{k}'} \mathcal{G}_{1,1}^n(\tau, \mathbf{k}) \mathcal{G}_{2,2}^m(0, -\mathbf{q}) - \hbar^2 \delta_{n',m'} \delta_{m,n} \delta_{\mathbf{k}',-\mathbf{q}} \mathcal{G}_{1,2}^{n'}(\tau - \tau', \mathbf{k}') \mathcal{G}_{2,1}^m(\tau', \mathbf{k}) \\ &= \hbar^2 - \delta_{n,n'} \delta_{m,m'} \delta_{\mathbf{k},\mathbf{k}'} \mathcal{G}_{\text{Gorkov}}^n(\tau, \mathbf{k}) \mathcal{G}_{\text{Gorkov}}^m(0, +\mathbf{q}) - \hbar^2 \delta_{n',m'} \delta_{m,n} \delta_{\mathbf{k}',-\mathbf{q}} \mathcal{F}_{n'}(\tau - \tau', \mathbf{k}') \mathcal{F}_n^\dagger(\tau', \mathbf{k}) \end{aligned}$$

whereas we used Definition 24.11 in the last equation. Now inserting this into the r.h.s. of Equation (24.37) yields (in addition to the delta-functions) two terms

$$\begin{aligned} (\text{I}, 1) &= -1 \cdot \oint_{\substack{\tau' \in (0, \hbar\beta) \\ \mathbf{q}, \mathbf{k}' \in \mathbf{Q} \\ m, m', n' \in \{1, \dots, \infty\} \\ v' \in \{1, \dots, d_x\}}} \frac{g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}') g_{mm'v'}(\mathbf{q}, \mathbf{k}' - \mathbf{k})}{\hbar^2 N_p} \mathcal{D}_{0,v'}(\tau - \tau') \delta_{n,n'} \delta_{m,m'} \delta_{\mathbf{k},\mathbf{k}'} \mathcal{G}_{\text{Gorkov}}^n(\tau, \mathbf{k}) \mathcal{G}_{\text{Gorkov}}^m(0, +\mathbf{q}) \hbar^2 \\ &= -1 \cdot \oint_{\substack{\tau' \in (0, \hbar\beta) \\ \mathbf{q} \in \mathbf{Q} \\ m \in \{1, \dots, \infty\} \\ v' \in \{1, \dots, d_x\}}} \frac{g_{nnv'}(\mathbf{k}', 0) g_{mmv'}(\mathbf{q}, 0)}{N_p \hbar^2} \mathcal{D}_{0,v'}(\tau - \tau') \mathcal{G}_{\text{Gorkov}}^n(\tau, \mathbf{k}) \mathcal{G}_{\text{Gorkov}}^m(0, +\mathbf{q}) \hbar^2 \end{aligned} \quad (33.130)$$

and the second term can be treated similarly, yielding

$$\begin{aligned}
 (\text{I}, 2) &= -1 \cdot \sum_{\substack{\tau' \in (0, \hbar\beta) \\ \mathbf{q}, \mathbf{k}' \in \mathbf{Q} \\ m, m', n' \in \{1, \dots, \infty\} \\ v' \in \{1, \dots, rd_x\}}} \frac{g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}') g_{mm'v'}(\mathbf{q}, \mathbf{k}' - \mathbf{k})}{\hbar^2 N_p} \mathcal{D}_{0,v'}(\tau - \tau') \delta_{n',m'} \delta_{m,n} \delta_{\mathbf{k}', -\mathbf{q}} \mathcal{F}_{n'}(\tau - \tau', \mathbf{k}') \mathcal{F}_n^\dagger(\tau', \mathbf{k}) \hbar^2 \\
 &= -1 \cdot \sum_{\substack{\tau' \in (0, \hbar\beta) \\ \mathbf{q} \in \mathbf{Q} \\ n' \in \{1, \dots, \infty\} \\ v' \in \{1, \dots, rd_x\}}} \frac{g_{nn'v'}(-\mathbf{q}, \mathbf{k} + \mathbf{q}) g_{nn'v'}(\mathbf{q}, -\mathbf{q} - \mathbf{k})}{N_p \hbar^2} \mathcal{D}_{0,v'}(\tau - \tau') \mathcal{F}_{n'}(\tau - \tau', -\mathbf{q}) \mathcal{F}_n^\dagger(\tau', \mathbf{k}) \hbar^2 \\
 &= -1 \cdot \sum_{\substack{\tau' \in (0, \hbar\beta) \\ \mathbf{q} \in \mathbf{Q} \\ n' \in \{1, \dots, \infty\} \\ v' \in \{1, \dots, rd_x\}}} \frac{|g_{nn'v'}(\mathbf{q}, -\mathbf{q} - \mathbf{k})|^2}{N_p} \mathcal{D}_{0,v'}(\tau - \tau') \mathcal{F}_{n'}(\tau - \tau', -\mathbf{q}) \mathcal{F}_n^\dagger(\tau', \mathbf{k})
 \end{aligned} \tag{33.131}$$

whereas we terminated the summations over m and m' in the second last equation-sign and we used the inversion-symmetry for the couplings in the last equation-sign. Now we do the same procedure for the other term in Equation (33.129), i.e. the second line, giving

$$\begin{aligned}
 \Pi &= \hbar^2 \mathcal{G}(i_1 = [m', 1, \tau', \mathbf{q}], i_2 = [n', 1, \tau, \mathbf{k}'] \mid i'_1 = [n, 1, 0, \mathbf{k}], i'_2 = [m, 1, \tau', +\mathbf{q} + \mathbf{k}' - \mathbf{k}]) \\
 &\approx \hbar^2 \mathcal{G}_{m',n}^{1,1}(\tau' - 0) \mathcal{G}_{n',m}^{1,1}(\tau - \tau') - \hbar^2 \mathcal{G}_{m',m}^{1,1}(0) \mathcal{G}_{n',n}^{1,1}(\tau - 0) \\
 &= \hbar^2 \delta_{m',n} \delta_{n',m} \delta_{\mathbf{q},\mathbf{k}} \delta_{\mathbf{q},\mathbf{k}'} \mathcal{G}_{n'}^{1,1}(\tau') \mathcal{G}_m^{1,1}(\tau - \tau') - \hbar^2 \delta_{m',m} \delta_{n',n} \delta_{\mathbf{q},+\mathbf{q}} \delta_{\mathbf{k}',\mathbf{k}} \mathcal{G}_m^{1,1}(0) \mathcal{G}_n^{1,1}(\tau - 0) \\
 &= \hbar^2 \delta_{m',n} \delta_{n',m} \delta_{\mathbf{q},\mathbf{k}} \delta_{\mathbf{q},\mathbf{k}'} \mathcal{G}_{\text{Gorkov}}^n(\tau', \mathbf{k}) \mathcal{G}_{\text{Gorkov}}^m(\tau - \tau', +\mathbf{k}') - \hbar^2 \delta_{m',m} \delta_{n',n} \delta_{\mathbf{q},+\mathbf{q}} \delta_{\mathbf{k}',\mathbf{k}} \mathcal{G}_{\text{Gorkov}}^m(0, +\mathbf{q}) \mathcal{G}_{\text{Gorkov}}^n(\tau, \mathbf{k})
 \end{aligned}$$

whereas we used Definition 24.11 in the last equation.

Now we insert this into Equation (24.37), giving two additional terms

$$\begin{aligned}
 (\text{II}, 3) &= \sum_{\substack{\tau' \in (0, \hbar\beta) \\ \mathbf{q}, \mathbf{k}' \in \mathbf{Q} \\ m, m', n' \in \{1, \dots, \infty\} \\ v' \in \{1, \dots, rd_x\}}} \frac{g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}') g_{mm'v'}(\mathbf{q}, \mathbf{k}' - \mathbf{k})}{\hbar^2 N_p} \mathcal{D}_{0,v'}(\tau - \tau') \delta_{m',n} \delta_{n',m} \delta_{\mathbf{q},\mathbf{k}} \delta_{\mathbf{q},\mathbf{k}'} \mathcal{G}_{\text{Gorkov}}^n(\tau', \mathbf{k}) \mathcal{G}_{\text{Gorkov}}^m(\tau - \tau', +\mathbf{k}') \hbar^2 \\
 &= \sum_{\substack{\tau' \in (0, \hbar\beta) \\ \mathbf{k}' \in \mathbf{Q} \\ n' \in \{1, \dots, \infty\} \\ v' \in \{1, \dots, rd_x\}}} \frac{g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}') g_{nn'v'}(\mathbf{k}, \mathbf{k}' - \mathbf{k})}{N_p} \mathcal{D}_{0,v'}(\tau - \tau') \mathcal{G}_{\text{Gorkov}}^n(\tau', \mathbf{k}) \mathcal{G}_{\text{Gorkov}}^{n'}(\tau - \tau', +\mathbf{k}') \\
 &= \sum_{\substack{\tau' \in (0, \hbar\beta) \\ \mathbf{k}' \in \mathbf{Q} \\ n' \in \{1, \dots, \infty\} \\ v' \in \{1, \dots, rd_x\}}} \frac{|g_{nn'v'}(\mathbf{k}, \mathbf{k}' - \mathbf{k})|^2}{N_p} \mathcal{D}_{0,v'}(\tau - \tau') \mathcal{G}_{\text{Gorkov}}^n(\tau', \mathbf{k}) \mathcal{G}_{\text{Gorkov}}^{n'}(\tau - \tau', +\mathbf{k}')
 \end{aligned} \tag{33.132}$$

whereas in the second last equation, we terminated the summations over m and m' and $\mathbf{q}$, and in the last equation we used the hermiticity of the couplings and something went wrong because we lost a Kronecker-delta. . And for the other term we obtain

$$\begin{aligned}
 (\text{II}, 4) &= -1 \cdot \sum_{\substack{\tau' \in (0, \hbar\beta) \\ \mathbf{q}, \mathbf{k}' \in \mathcal{Q} \\ m, m', n' \in \{1, \dots, \infty\} \\ v' \in \{1, \dots, rd_x\}}} \frac{g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}') g_{mm'v'}(\mathbf{q}, \mathbf{k}' - \mathbf{k})}{\hbar^2 N_p} \mathcal{D}_{0,v'}(\tau - \tau') \delta_{m',m} \delta_{n',n} \delta_{\mathbf{q},+\mathbf{q}} \delta_{\mathbf{k}',\mathbf{k}} \mathcal{G}_{\text{Gorkov}}^m(0, +\mathbf{q}) \mathcal{G}_{\text{Gorkov}}^{n'}(\tau, \mathbf{k}) \\
 &= -1 \cdot \sum_{\substack{\tau' \in (0, \hbar\beta) \\ \mathbf{q} \in \mathcal{Q} \\ m \in \{1, \dots, \infty\} \\ v' \in \{1, \dots, rd_x\}}} \frac{g_{nnv'}(\mathbf{k}, 0) g_{mmv'}(\mathbf{q}, 0)}{N_p} \mathcal{D}_{0,v'}(\tau - \tau') \mathcal{G}_{\text{Gorkov}}^m(0, +\mathbf{q}) \mathcal{G}_{\text{Gorkov}}^n(\tau, \mathbf{k})
 \end{aligned} \tag{33.133}$$

whereas we terminated the summations over m' , n' and $\mathbf{k}'$ in the last equation.

Standard Eliashberg-theory now assumes $g(\mathbf{k}, 0) = 0$, i.e. restricting attention to systems with only one atom per unit cell.

Now we collect the non-vanishing contributions and obtain

$$\begin{aligned}
 &\left(\partial_\tau + \frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar} \right) \mathcal{G}_{\text{Gorkov}}^n(\tau, \mathbf{k}) \\
 &= -\frac{1}{\hbar} \delta(\tau) - \sum_{\substack{\tau' \in (0, \hbar\beta) \\ \mathbf{q} \in \mathcal{Q} \\ n' \in \{1, \dots, \infty\} \\ v' \in \{1, \dots, rd_x\}}} \left(|g_{nn'v'}(-\mathbf{q}, \mathbf{k} + \mathbf{q})|^2 / (N_p \hbar) \right) \mathcal{D}_{0,v'}(\tau - \tau') \mathcal{F}_{n'}(\tau - \tau', -\mathbf{q}) \mathcal{F}_n^\dagger(\tau', \mathbf{k}) \hbar^2 \\
 &+ \sum_{\substack{\tau' \in (0, \hbar\beta) \\ \mathbf{k}' \in \mathcal{Q} \\ n' \in \{1, \dots, \infty\} \\ v' \in \{1, \dots, rd_x\}}} \left(|g_{n'n v'}(\mathbf{k}, \mathbf{k}' - \mathbf{k})|^2 / (N_p \hbar) \right) \mathcal{D}_{0,v'}(\tau - \tau') \mathcal{G}_{\text{Gorkov}}^n(\tau', \mathbf{k}) \mathcal{G}_{\text{Gorkov}}^{n'}(\tau - \tau', +\mathbf{k}') \hbar^2 \\
 &\stackrel{\mathbf{q} \mapsto -\mathbf{k}'}{=} -\frac{1}{\hbar} \delta(\tau) - \sum_{\substack{\tau' \in (0, \hbar\beta) \\ \mathbf{k}' \in \mathcal{Q} \\ n' \in \{1, \dots, \infty\} \\ v' \in \{1, \dots, rd_x\}}} \left(|g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2 / (N_p \hbar) \right) \mathcal{D}_{0,v'}(\tau - \tau') \mathcal{F}_{n'}(\tau - \tau', \mathbf{k}') \mathcal{F}_n^\dagger(\tau', \mathbf{k}) \hbar^2 \\
 &+ \sum_{\substack{\tau' \in (0, \hbar\beta) \\ \mathbf{k}' \in \mathcal{Q} \\ n' \in \{1, \dots, \infty\} \\ v' \in \{1, \dots, rd_x\}}} \left(|g_{n'n v'}(\mathbf{k}, \mathbf{k}' - \mathbf{k})|^2 / (N_p \hbar) \right) \mathcal{D}_{0,v'}(\tau - \tau') \mathcal{G}_{\text{Gorkov}}^n(\tau', \mathbf{k}) \mathcal{G}_{\text{Gorkov}}^{n'}(\tau - \tau', +\mathbf{k}') \hbar^2 \\
 &= \dots
 \end{aligned} \tag{33.134}$$

and this gives then

$$\begin{aligned}
 \dots &= -\frac{1}{\hbar} \delta(\tau) \\
 &+ \sum_{\substack{\tau' \in (0, \hbar\beta) \\ \mathbf{k}' \in \mathcal{Q} \\ n' \in \{1, \dots, \infty\} \\ v' \in \{1, \dots, rd_x\}}} \hbar^2 \left(|g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2 / (N_p \hbar) \right) \mathcal{D}_{0,v'}(\tau - \tau') \left[\mathcal{G}_{\text{Gorkov}}^n(\tau', \mathbf{k}) \mathcal{G}_{\text{Gorkov}}^{n'}(\tau - \tau', +\mathbf{k}') - \mathcal{F}_{n'}(\tau - \tau', \mathbf{k}') \mathcal{F}_n^\dagger(\tau', \mathbf{k}) \right]
 \end{aligned} \tag{33.135}$$

Proof 33.33 of Theorem 24.25 We rewrite the equation of motion, i.e. Equation (33.140) in Matsubara-space. We start with the l.h.s. and obtain with Lemma 1.11 and Theorem 13.6

$$\begin{aligned} \left(\partial_\tau + \frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar} \right) \mathcal{G}_{\text{Gorkov}}^n(\tau, \mathbf{k}) &= \frac{1}{\hbar\beta} \sum_{\omega_{n_1} \in \Omega_{-1}} \left(-i\omega_{n_1} + \frac{\epsilon_n(\mathbf{k}) - \mu}{\hbar} \right) \mathcal{G}_{\text{Gorkov}}^n(\mathbf{k}, \omega_{n_1}) e^{-i\omega_{n_1} \tau} \\ &= \frac{1}{\hbar\beta} \sum_{\omega_{n_1} \in \Omega_{-1}} \left(-\hbar \mathcal{G}_{\text{Gorkov}}^0(i\omega_{n_1}, \mathbf{k}) \right)^{-1} \mathcal{G}_{\text{Gorkov}}^n(i\omega_{n_1}, \mathbf{k}) e^{-i\omega_{n_1} \tau} \end{aligned}$$

On the r.h.s. we expand each term individually and obtain with Theorem 1.16

$$-\frac{1}{\hbar} \delta(\tau) = \frac{1}{\hbar\beta} \sum_{\omega_{n_1} \in \Omega_{-1}} -\frac{1}{\hbar} e^{-i\omega_{n_1} \tau} \quad (33.136)$$

for the delta-function and for the other term number (II,3):

$$\begin{aligned} \text{II,3} &= + \frac{1}{(\hbar\beta)^3} \sum_{\substack{\mathbf{k}' \in \mathcal{Q} \\ v' \in \{1, \dots, rd_x\} \\ n' \in \{1, \dots, \infty\} \\ \tau' \in (0, \hbar\beta)}} \frac{|g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{\hbar^2 N_p} \hbar^2 \sum_{\substack{\omega_{n_1}, \omega_{n_2} \in \Omega_{-1} \\ \omega_{n_3} \in \Omega_1}} e^{-i\omega_{n_3}(\tau - \tau') + i\omega_{n_1}(\tau' - \tau) - i\omega_{n_2}\tau'} \mathcal{D}_{0,v'}(\mathbf{k} - \mathbf{k}', i\omega_{n_3}) \mathcal{G}_{\text{Gorkov}}^{n'}(i\omega_{n_1}, \mathbf{k}') \mathcal{G}_{\text{Gorkov}}^n(i\omega_{n_2}, \mathbf{k}) \\ &= \dots \end{aligned}$$

Now we collect the terms in τ' and perform the integration with Corollary 1.12

$$\int_0^{\hbar\beta} d\tau' e^{i\tau'(\omega_{n_3} + \omega_{n_1} - \omega_{n_2})} = \hbar\beta \delta_{\omega_{n_3} + \omega_{n_1}, \omega_{n_2}} \quad (33.137)$$

Noting further that

$$e^{i\tau(-\omega_{n_3} - \omega_{n_1})} \Big|_{\omega_{n_3} + \omega_{n_1} = \omega_{n_2} \Leftrightarrow \omega_{n_3} = \omega_{n_2} - \omega_{n_1}} = e^{-i\tau\omega_{n_2}} \quad (33.138)$$

This allows us to terminate the summation over ω_{n_3} and continue as

$$\text{I} = \dots = \frac{\hbar^2}{\hbar\beta} \sum_{\substack{\omega_{n_1}, \omega_{n_2} \in \Omega_{-1} \\ \mathbf{k}' \in \mathcal{Q} \\ n' \in \{1, \dots, \infty\} \\ v' \in \{1, \dots, rd_x\}}} \left(\frac{|g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{N_p} \right) \left(+ \frac{1}{\hbar\beta} \right) e^{-i\omega_{n_2}\tau} \mathcal{D}_{0,v'}(\mathbf{k} - \mathbf{k}', i\omega_{n_2} + i\omega_{n_1}) \mathcal{G}_{\text{Gorkov}}^{n'}(i\omega_{n_1}, \mathbf{k}') \mathcal{G}_{\text{Gorkov}}^n(i\omega_{n_2}, \mathbf{k})$$

Now we shift our attention to the term II (still?) and obtain for the other term number 2 (still?):

$$\text{I,2} = \frac{-1}{(\hbar\beta)^3} \sum_{\substack{\mathbf{k}' \in \mathcal{Q} \\ v' \in \{1, \dots, rd_x\} \\ n' \in \{1, \dots, \infty\} \\ \tau' \in (0, \hbar\beta)}} \frac{|g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{N_p \hbar^2} \hbar^2 \sum_{\substack{\omega_{n_1}, \omega_{n_2} \in \Omega_{-1} \\ \omega_{n_3} \in \Omega_1}} e^{-i\omega_{n_3}(\tau - \tau') - i\omega_{n_1}(\tau - \tau') - i\omega_{n_2}\tau'} \mathcal{D}_{0,v'}(\omega_{n_3}) \mathcal{F}_n^\dagger(\mathbf{k}, \omega_{n_2}) \mathcal{F}_{n'}(\mathbf{k}', \omega_{n_1})$$

Now we collect the terms in τ' and perform the integration

$$\int_0^{\hbar\beta} d\tau' e^{i\tau'(\omega_{n_3}+\omega_{n_1}-\omega_{n_2})} = \hbar\beta\delta_{\omega_{n_3},\omega_{n_2}-\omega_{n_1}} \quad (33.139)$$

Noting further that

$$e^{i\tau(-\omega_{n_3}-\omega_{n_1})} = e^{-i\tau(-\omega_{n_1}+\omega_{n_2})-\omega_{n_1}\tau i} = e^{-i\tau\omega_{n_2}}$$

which gives the result

$$I_{1,2} = \dots = \frac{-1}{\hbar\beta} \sum_{\substack{\omega_{n_1}, \omega_{n_2} \in \Omega_{-1} \\ n' \in \{1, \dots, \infty\} \\ v' \in \{1, \dots, rd_x\} \\ \mathbf{k}' \in Q}} \hbar^2 \left(\frac{|g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{\hbar^2 N_p} \right) \left(\frac{1}{\hbar\beta} \right) e^{-i\omega_{n_2}\tau} \mathcal{D}_{0,v'}(\omega_{n_2} - \omega_{n_1}) \mathcal{F}_n^\dagger(i\omega_{n_2}, \mathbf{k}) \mathcal{F}_{n'}(i\omega_{n_1}, \mathbf{k}')$$

Combining the results gives the final result

$$\begin{aligned} & + \frac{1}{\hbar\beta} \sum_{\omega_{n_1} \in \Omega_{-1}} \left(-\hbar \mathcal{G}_{0,n}^{\text{Gorkov}}(i\omega_{n_1}, \mathbf{k}) \right)^{-1} \mathcal{G}_n^{\text{Gorkov}}(i\omega_{n_1}, \mathbf{k}) e^{-i\omega_{n_1}\tau} \\ & = \frac{1}{\hbar\beta} \sum_{\omega_{n_1} \in \Omega_{-1}} -\frac{1}{\hbar} e^{-i\omega_{n_1}\tau} \\ & + \frac{1}{\hbar\beta} \sum_{\substack{\omega_{n_1}, \omega_{n_2} \in \Omega_{-1} \\ n' \in \{1, \dots, \infty\} \\ v' \in \{1, \dots, rd_x\} \\ \mathbf{k}' \in Q}} \hbar^2 \left(\frac{|g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{\hbar^2 N_p} \right) \left(+\frac{1}{\hbar\beta} \right) e^{-i\omega_{n_2}\tau} \mathcal{D}_{0,v'}(\omega_{n_2} + i\omega_{n_1}) \mathcal{G}_{n'}^{\text{Gorkov}}(i\omega_{n_1}, \mathbf{k}') \mathcal{G}_n^{\text{Gorkov}}(i\omega_{n_2}, \mathbf{k}) \\ & - \frac{1}{\hbar\beta} \sum_{\substack{\omega_{n_1}, \omega_{n_2} \in \Omega_{-1} \\ n' \in \{1, \dots, \infty\} \\ v' \in \{1, \dots, rd_x\} \\ \mathbf{k}' \in Q}} \hbar^2 \left(\frac{|g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{\hbar^2 N_p} \right) \left(\frac{1}{\hbar\beta} \right) e^{-i\omega_{n_2}\tau} \mathcal{D}_{0,v'}(\omega_{n_2} - \omega_{n_1}) \mathcal{F}_n^\dagger(i\omega_{n_2}, \mathbf{k}) \mathcal{F}_{n'}(i\omega_{n_1}, \mathbf{k}') \end{aligned}$$

now relabeling $\omega_{n_1} \leftrightarrow \omega_{n_2}$ and comparing coefficients gives the desired result.

Going to Matsubara-space gives the equation of motion for the Green-function as

$$\begin{aligned} & \left(-\hbar \mathcal{G}_{0,n}^{\text{Gorkov}}(i\omega_{n_1}, \mathbf{k}) \right)^{-1} \mathcal{G}_n^{\text{Gorkov}}(i\omega_{n_1}, \mathbf{k}) \\ & = -\frac{1}{\hbar} + \sum_{\substack{\omega_{n_2} \in \Omega_{-1} \\ \mathbf{k}' \in Q \\ n' \in \{1, \dots, \infty\} \\ v' \in \{1, \dots, rd_x\}}} \hbar^2 \frac{|g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{\hbar^2 N_p} \left(+\frac{1}{\hbar\beta} \right) \mathcal{D}_{0,v'}(i(\omega_{n_1} - \omega_{n_2})) \left[\mathcal{G}_{n'}^{\text{Gorkov}}(i\omega_{n_2}, \mathbf{k}') \mathcal{G}_n^{\text{Gorkov}}(i\omega_{n_1}, \mathbf{k}) - \mathcal{F}_n^\dagger(i\omega_{n_1}, \mathbf{k}) \mathcal{F}_{n'}(i\omega_{n_2}, \mathbf{k}') \right] \end{aligned} \quad (33.140)$$

which can be rewritten as

$$\begin{aligned}
 & \mathcal{G}_{\text{Gorkov}}^n(i\omega_{n_1}, \mathbf{k}) \\
 &= \mathcal{G}_{\text{Gorkov}}^{0,n}(i\omega_{n_1}, \mathbf{k}) \left[1 + \sum_{\substack{\omega_{n_2} \in \Omega_{-1} \\ \mathbf{k}' \in \mathbf{Q} \\ n' \in \{1, \dots, \infty\} \\ v' \in \{1, \dots, rd_x\}}} \frac{-|g_{nn'v'}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{N_p} \left(+\frac{1}{\beta} \right) \mathcal{D}_{0,v'}(i(\omega_{n_1} - \omega_{n_2})) \left[\mathcal{G}_{\text{Gorkov}}^{n'}(i\omega_{n_2}, \mathbf{k}') \mathcal{G}_{\text{Gorkov}}^n(i\omega_{n_1}, \mathbf{k}) - \mathcal{F}_n^\dagger(i\omega_{n_1}, \mathbf{k}) \mathcal{F}_{n'}(i\omega_{n_2}, \mathbf{k}') \right] \right]
 \end{aligned}
 \tag{33.141}$$

[Cf. continuation on TBDigitalized for SC. page 1]

Proof 33.34 of Corollary 24.44

We use the decompositions in Equation (24.107) for self-energy and Equation (24.19) for the non-interacting Gorkov-Green-Function and insert them into the Dyson-equation (24.20) to obtain

$$\hat{\mathcal{G}}_{\text{Gorkov}}^{-1}(\mathbf{i}\omega_n, \mathbf{k}) = \hbar\omega_n Z_m(\mathbf{i}\omega_n, \mathbf{k})\sigma_0 - \phi_m(\mathbf{i}\omega_n, \mathbf{k})\sigma_1 - (\chi_m(\mathbf{i}\omega_n, \mathbf{k}) + (\epsilon_m(\mathbf{k}) - \mu))\sigma_3 \quad (33.142)$$

which can be inverted with the inversion-formula for 2x2-matrices⁹ to

$$\hat{\mathcal{G}}_{\text{Gorkov}}(\mathbf{i}\omega_n, \mathbf{k}) = \frac{\hbar\omega_n Z_m(\mathbf{i}\omega_n, \mathbf{k})\sigma_0 + \phi_m(\mathbf{i}\omega_n, \mathbf{k})\sigma_1 + (\chi_m(\mathbf{i}\omega_n, \mathbf{k}) + (\epsilon_m(\mathbf{k}) - \mu))\sigma_3}{\text{Det } \hat{\mathcal{G}}_{\text{Gorkov}}^{-1}(\mathbf{i}\omega_n, \mathbf{k})} \quad (33.143)$$

Noticing

$$\begin{aligned} \text{Det } \hat{\mathcal{G}}_{\text{Gorkov}}^{-1}(\mathbf{i}\omega_n, \mathbf{k}) &= -\hbar^2\omega_n^2 Z_m(\mathbf{k}, \mathbf{i}\omega_n)^2 - (\chi_m(\mathbf{k}, \mathbf{i}\omega_n) + (\epsilon_m(\mathbf{k}) - \mu))^2 - (\phi_m(\mathbf{k}, \mathbf{i}\omega_n))^2 \\ &= -\left(\hbar^2\omega_n^2 Z_m(\mathbf{k}, \mathbf{i}\omega_n)^2 + (\chi_m(\mathbf{k}, \mathbf{i}\omega_n) + (\epsilon_m(\mathbf{k}) - \mu))^2 + \phi_m(\mathbf{k}, \mathbf{i}\omega_n)^2\right) \\ &\equiv -\theta_m(\mathbf{i}\omega_n, \mathbf{k}) \end{aligned}$$

Inserting this above yields the desired result

$$\hat{\mathcal{G}}_{\text{Gorkov}}(\mathbf{i}\omega_n, \mathbf{k}) = -\frac{\hbar\omega_n Z_m(\mathbf{i}\omega_n, \mathbf{k})\sigma_0 + \phi_m(\mathbf{i}\omega_n, \mathbf{k})\sigma_1 + (\chi_m(\mathbf{i}\omega_n, \mathbf{k}) + (\epsilon_m(\mathbf{k}) - \mu))\sigma_3}{\theta_m(\mathbf{i}\omega_n, \mathbf{k})} \quad (33.144)$$

Proof 33.35 of Lemma 24.46 We need to compute

$$\sigma_3 \hat{\mathcal{G}}_{\text{Gorkov}}^m(\mathbf{i}\omega_{n'}, \mathbf{q} + \mathbf{k})\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \mathcal{G}_{\text{Gorkov}}^m(\bullet) & \mathcal{F}_m(\bullet) \\ \mathcal{F}_m^*(\bullet) & -\mathcal{G}_{\text{Gorkov}}^m(-\bullet) \end{pmatrix}_{\bullet=(\mathbf{i}\omega_{n'}, \mathbf{q}+\mathbf{k})} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (33.145)$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \mathcal{G}_{\text{Gorkov}}^m(\bullet) & -\mathcal{F}_m(\bullet) \\ \mathcal{F}_m^*(\bullet) & \mathcal{G}_{\text{Gorkov}}^m(-\bullet) \end{pmatrix}_{\bullet=(\mathbf{i}\omega_{n'}, \mathbf{q}+\mathbf{k})} \quad (33.146)$$

$$= \begin{pmatrix} \mathcal{G}_{\text{Gorkov}}^m(\bullet) & -\mathcal{F}_m(\bullet) \\ -\mathcal{F}_m^*(\bullet) & -\mathcal{G}_{\text{Gorkov}}^m(-\bullet) \end{pmatrix}_{\bullet=(\mathbf{i}\omega_{n'}, \mathbf{q}+\mathbf{k})} \quad (33.147)$$

Proof 33.36 of Theorem 24.50

Inserting Equation (24.108) for the Green-function into Equation (24.111) for the self-energy, and comparing coefficients with Equation (24.107) for the self-energy, we obtain the Eliashberg-equations

⁹Cf. Waste-Folder 1 Topic 22: For $A = a_0\sigma_0 + \mathbf{a} \cdot \boldsymbol{\sigma}$

$$\begin{aligned}
 \hat{\Sigma}_{n,m}^{\text{ep}}(\mathbf{i}\omega_n, \mathbf{k}) &= T \sum_{\substack{\mathbf{k}' \in Q \\ \omega_{n'} \in \Omega_{-1}}} \hbar \mathbf{i}\omega_{n'} \frac{Z_m(\mathbf{i}\omega_{n'}, \mathbf{k}')}{\theta_m(\mathbf{i}\omega_{n'}, \mathbf{k}')} \sigma_3 \sigma_0 \sigma_3 \sum_{\nu} |g_{mn\nu}(\mathbf{k}, \mathbf{k}')|^2 \mathcal{D}_{\nu}(\mathbf{i}\omega_n - \mathbf{i}\omega_{n'}, \mathbf{k} - \mathbf{k}') \\
 &\quad - T \sum_{\substack{\mathbf{k}' \in Q \\ \omega_{n'} \in \Omega_{-1}}} \frac{\phi_m(\mathbf{i}\omega_{n'}, \mathbf{k}')}{\theta_m(\mathbf{i}\omega_{n'}, \mathbf{k}')} \sigma_1 \sum_{\nu} |g_{mn\nu}(\mathbf{k}, \mathbf{k}')|^2 \mathcal{D}_{\nu}(\mathbf{i}\omega_n - \mathbf{i}\omega_{n'}, \mathbf{k} - \mathbf{k}') \\
 &\quad + T \sum_{\substack{\mathbf{k}' \in Q \\ \omega_{n'} \in \Omega_{-1}}} \frac{\chi_m(\mathbf{i}\omega_{n'}, \mathbf{k}') + (\epsilon_m(\mathbf{k}) - \mu)}{\theta_m(\mathbf{i}\omega_{n'}, \mathbf{k}')} \sigma_3 \sum_{\nu} |g_{mn\nu}(\mathbf{k}, \mathbf{k}')|^2 \mathcal{D}_{\nu}(\mathbf{i}\omega_n - \mathbf{i}\omega_{n'}, \mathbf{k} - \mathbf{k}') \\
 &\stackrel{!!!}{=} \mathbf{i}\omega_n \left(1 - Z_m(\mathbf{i}\omega_n, \mathbf{k})\right) \sigma_0 + \phi_m(\mathbf{i}\omega_n, \mathbf{k}) \sigma_1 + \chi_m(\mathbf{i}\omega_n, \mathbf{k}) \sigma_3
 \end{aligned}$$

In this equation, all of the momentum-indices are messed up!

Notice further, that we used the following identities for the Pauli-matrices **ToBeWrittenDown**.

Proof 33.37 of Lemma 24.51 Cf. 06 Superhydrides Success of Eliashberg. **ToBeDone**. -

Proof 33.38 of Corollary 24.52

The Eliashberg-Equations from 03-EPW reads for ϕ :

$$\bar{\phi}_n(\mathbf{i}\omega_j, \mathbf{k}) = \frac{T}{N_F} \sum_{\substack{\mathbf{q} \in Q \\ \omega_{j'} \in \Omega_{-1} \\ m \in \{1, \dots, \infty\}}} \frac{\bar{\phi}_m(\mathbf{i}\omega_{j'}, \mathbf{k} + \mathbf{q})}{\theta_m(\mathbf{i}\omega_{j'}, \mathbf{k} + \mathbf{q})} \lambda_{n,m}(\mathbf{i}\omega_j - \mathbf{i}\omega_{j'}, \mathbf{k}, \mathbf{k} + \mathbf{q}) \quad (33.148)$$

with

$$\theta_m(\mathbf{i}\omega_n, \mathbf{k}) = \hbar^2 \omega_n^2 Z_m(\mathbf{k}, \mathbf{i}\omega_n)^2 + \left(\chi_m(\mathbf{k}, \mathbf{i}\omega_n) + (\epsilon_m(\mathbf{k}) - \mu) \right)^2 + \phi_m(\mathbf{i}\omega_n, \mathbf{k})^2 \quad (33.149)$$

and

$$\lambda_{n,m}(\omega_n) = N_F \sum_{\substack{\omega \in (0, \infty) \\ \nu \in \{1, \dots, d_x\}}} \frac{2\omega}{\omega_n^2 + \omega^2} |g_{mn\nu}(\mathbf{k}, \mathbf{q})|^2 \delta(\omega - \omega_{\nu}(\mathbf{q})) \quad (33.150)$$

Considering now the limiting case $Z \equiv 1$, $\chi \equiv 0$ we obtain

$$\begin{aligned}
 \bar{\phi}_n(i\omega_j, \mathbf{k}) &= \sum_{\substack{\mathbf{q} \in Q \\ \omega_{j'} \in \Omega_{-1} \\ m \in \{1, \dots, \infty\}}} \frac{\bar{\phi}_m(i\omega_{j'}, \mathbf{k} + \mathbf{q})}{\hbar^2 \omega_{j'}^2 + (\epsilon_m(\mathbf{k} + \mathbf{q}) - \mu)^2 + |\phi_m(i\omega_{j'}, \mathbf{k} + \mathbf{q})|^2} |g_{mnv}(\mathbf{k}, \mathbf{q})|^2 \frac{2\omega_v(\mathbf{q})}{(\omega_j - \omega_{j'})^2 + \omega_v(\mathbf{q})^2} \frac{1}{\hbar\beta} \\
 &\stackrel{\mathbf{q} \mapsto \mathbf{q} - \mathbf{k}}{=} \sum_{\substack{\mathbf{q} \in Q \\ \omega_{j'} \in \Omega_{-1} \\ m \in \{1, \dots, \infty\}}} \frac{\bar{\phi}_m(i\omega_{j'}, \mathbf{q})}{\hbar^2 \omega_{j'}^2 + (\epsilon_m(\mathbf{q}) - \mu)^2 + |\phi_m(i\omega_{j'}, \mathbf{q})|^2} |g_{mnv}(\mathbf{k}, \mathbf{q} - \mathbf{k})|^2 \frac{2\omega_v(\mathbf{q} - \mathbf{k})}{(\omega_j - \omega_{j'})^2 + \omega_v(\mathbf{q} - \mathbf{k})^2} \frac{1}{\hbar\beta} \\
 &\stackrel{q \equiv (\mathbf{q}, i\omega_{j'}), k \equiv (\mathbf{k}, i\omega_j)}{=} \sum_{\substack{q \in Q \times \Omega_{-1} \\ m \in \{1, \dots, \infty\}}} \frac{\bar{\phi}_m(q)}{\hbar^2 \omega_q^2 + (\epsilon_m(q) - \mu)^2 + |\phi_m(q)|^2} |g_{mnv}(\mathbf{k}, \mathbf{q} - \mathbf{k})|^2 \frac{2\omega_v(\mathbf{q} - \mathbf{k})}{\omega_{q-k}^2 + \omega_v(\mathbf{q} - \mathbf{k})^2} \frac{1}{\hbar\beta}
 \end{aligned}$$

which can be summarized as

$$\bar{\phi}_n(k) = \sum_{\substack{q \in Q \times \Omega_{-1} \\ m \in \{1, \dots, \infty\}}} \frac{\bar{\phi}_m(q)}{\hbar^2 \omega_q^2 + (\epsilon_m(q) - \mu)^2 + |\phi_m(q)|^2} |g_{mnv}(\mathbf{k}, \mathbf{q} - \mathbf{k})|^2 \frac{2\omega_v(\mathbf{q} - \mathbf{k})}{\omega_{q-k}^2 + \omega_v(\mathbf{q} - \mathbf{k})^2} \frac{1}{\hbar\beta}$$

Note: The template for this was 03-EPW.

Proof 33.39 of Theorem 24.53

There are many couplings contributing. We will „divide and conquer“ the problem. The Kronecker-deltas for the spin yield the enumeration

- $\alpha_1 \equiv (m_{\alpha_1}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_1}, \sigma, k_{\alpha_1})$
- $\alpha_2 \equiv (m_{\alpha_2}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_2}, \pi, k_{\alpha_2})$
- $\alpha_3 \equiv (m_{\alpha_3}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_3}, \pi, k_{\alpha_3})$
- $\alpha_4 \equiv (m_{\alpha_4}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_4}, \sigma, k_{\alpha_4})$

Spin-Singlet Contributions

Case 1: $\sigma = \uparrow$ and $\pi = \downarrow$. Then we obtain

- $\alpha_1 \equiv (m_{\alpha_1}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_1}, \uparrow, k_{\alpha_1})$
- $\alpha_2 \equiv (m_{\alpha_2}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_2}, \downarrow, k_{\alpha_2})$
- $\alpha_3 \equiv (m_{\alpha_3}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_3}, \downarrow, k_{\alpha_3})$
- $\alpha_4 \equiv (m_{\alpha_4}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_4}, \uparrow, k_{\alpha_4})$

Then we can proceed by enumerating the momenta in the following way

- $\alpha_1 \equiv (m_{\alpha_1}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_1}, \uparrow, k_{\alpha_1})$
- $\alpha_2 \equiv (m_{\alpha_2}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_2}, \downarrow, -k)$
- $\alpha_3 \equiv (m_{\alpha_3}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_3}, \downarrow, k_{\alpha_3})$

- $\alpha_4 \equiv (m_{\alpha_4}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_4}, \uparrow, k')$

Momentum conservation yields then

- $\alpha_1 \equiv (m_{\alpha_1}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_1}, \uparrow, k_{\alpha_1})$
- $\alpha_2 \equiv (m_{\alpha_2}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_2}, \downarrow, -k)$
- $\alpha_3 \equiv (m_{\alpha_3}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_3}, \downarrow, k_{\alpha_1} - k - k')$
- $\alpha_4 \equiv (m_{\alpha_4}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_4}, \uparrow, k')$

Now we replace the summation over $k_{\alpha_1} \mapsto k_{\alpha_1} + k$.

- $\alpha_1 \equiv (m_{\alpha_1}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_1}, \uparrow, k_{\alpha_1} + k)$
- $\alpha_2 \equiv (m_{\alpha_2}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_2}, \downarrow, -k)$
- $\alpha_3 \equiv (m_{\alpha_3}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_3}, \downarrow, k_{\alpha_1} - k')$
- $\alpha_4 \equiv (m_{\alpha_4}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_4}, \uparrow, k')$

And now we write $k_{\alpha_1} \equiv q$ and obtain finally

- $\alpha_1 \equiv (m_{\alpha_1}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_1}, \uparrow, q + k)$
- $\alpha_2 \equiv (m_{\alpha_2}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_2}, \downarrow, -k)$
- $\alpha_3 \equiv (m_{\alpha_3}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_3}, \downarrow, q - k')$
- $\alpha_4 \equiv (m_{\alpha_4}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_4}, \uparrow, k')$

This gives for the matrix-element

$$\begin{aligned}
 & V_{\alpha_1, \alpha_2; \alpha_3, \alpha_4} \\
 &= \frac{-1}{\hbar \beta N_p} \sum_{v \in \{1, \dots, r d_x\}} \frac{\omega_v(\mathbf{k} + \mathbf{q} - \mathbf{k}')}{\omega_{k+q-k'}^2 + \omega_v^2(\mathbf{k} + \mathbf{q} - \mathbf{k}')} \frac{g_{m_{\alpha_1} m_{\alpha_4} v}(\mathbf{k}', \mathbf{k} + \mathbf{q} - \mathbf{k}') g_{m_{\alpha_3} m_{\alpha_2} v}^*(-\mathbf{k}, \mathbf{k} + \mathbf{q} - \mathbf{k}')}{\hbar^2} \\
 &= \frac{-1}{\hbar \beta N_p} \sum_{v \in \{1, \dots, r d_x\}} \frac{\omega_v(\mathbf{k} + \mathbf{q} - \mathbf{k}')}{\omega_{k+q-k'}^2 + \omega_v^2(\mathbf{k} + \mathbf{q} - \mathbf{k}')} \frac{g_{m_{\alpha_1} m_{\alpha_4} v}(\mathbf{k}', \mathbf{k} + \mathbf{q} - \mathbf{k}') g_{m_{\alpha_2} m_{\alpha_3} v}(\mathbf{q} - \mathbf{k}', -(\mathbf{k} + \mathbf{q} - \mathbf{k}'))}{\hbar^2}
 \end{aligned}$$

Case 2: $\sigma = \downarrow$ and $\pi = \uparrow$. Then we obtain

- $\alpha_1 \equiv (m_{\alpha_1}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_1}, \downarrow, k_{\alpha_1})$
- $\alpha_2 \equiv (m_{\alpha_2}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_2}, \uparrow, k_{\alpha_2})$
- $\alpha_3 \equiv (m_{\alpha_3}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_3}, \uparrow, k_{\alpha_3})$
- $\alpha_4 \equiv (m_{\alpha_4}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_4}, \downarrow, k_{\alpha_4})$

Then we can proceed by enumerating the momenta in the following way

- $\alpha_1 \equiv (m_{\alpha_1}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_1}, \downarrow, -k)$
- $\alpha_2 \equiv (m_{\alpha_2}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_2}, \uparrow, k_{\alpha_2})$
- $\alpha_3 \equiv (m_{\alpha_3}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_3}, \uparrow, k')$

$$\bullet \alpha_4 \equiv (m_{\alpha_4}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_4}, \downarrow, k_{\alpha_4})$$

Momentum conservation yields then

$$\begin{aligned} \bullet \alpha_1 &\equiv (m_{\alpha_1}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_1}, \downarrow, -k) \\ \bullet \alpha_2 &\equiv (m_{\alpha_2}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_2}, \uparrow, k_{\alpha_2}) \\ \bullet \alpha_3 &\equiv (m_{\alpha_3}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_3}, \uparrow, k') \\ \bullet \alpha_4 &\equiv (m_{\alpha_4}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_4}, \downarrow, k_{\alpha_2} - k - k') \end{aligned}$$

Now we replace the summation over $k_{\alpha_2} \mapsto k_{\alpha_2} + k$.

$$\begin{aligned} \bullet \alpha_1 &\equiv (m_{\alpha_1}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_1}, \downarrow, -k) \\ \bullet \alpha_2 &\equiv (m_{\alpha_2}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_2}, \uparrow, k_{\alpha_2} + k) \\ \bullet \alpha_3 &\equiv (m_{\alpha_3}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_3}, \uparrow, k') \\ \bullet \alpha_4 &\equiv (m_{\alpha_4}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_4}, \downarrow, k_{\alpha_2} - k') \end{aligned}$$

And now we write $k_{\alpha_2} \equiv q$ and obtain

$$\begin{aligned} \bullet \alpha_1 &\equiv (m_{\alpha_1}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_1}, \downarrow, -k) \\ \bullet \alpha_2 &\equiv (m_{\alpha_2}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_2}, \uparrow, q + k) \\ \bullet \alpha_3 &\equiv (m_{\alpha_3}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_3}, \uparrow, k') \\ \bullet \alpha_4 &\equiv (m_{\alpha_4}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_4}, \downarrow, q - k') \end{aligned}$$

finally we exchange $m_{\alpha_1} \leftrightarrow m_{\alpha_2}$ and (simultaneously) $m_{\alpha_3} \leftrightarrow m_{\alpha_4}$ which leads to

$$\begin{aligned} \bullet \alpha_1 &\equiv (m_{\alpha_2}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_2}, \downarrow, -k) \\ \bullet \alpha_2 &\equiv (m_{\alpha_1}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_1}, \uparrow, q + k) \\ \bullet \alpha_3 &\equiv (m_{\alpha_4}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_4}, \uparrow, k') \\ \bullet \alpha_4 &\equiv (m_{\alpha_3}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_3}, \downarrow, q - k') \end{aligned}$$

This gives for the matrix-element

$$\begin{aligned} &V_{\alpha_1, \alpha_2; \alpha_3, \alpha_4} \\ &= \frac{-1}{\hbar \beta N_p} \sum_{p \in \{1, \dots, r d_x\}} \frac{\omega_v(-\mathbf{k} - \mathbf{q} + \mathbf{k}')}{\omega_{-k-q+k'}^2 + \omega_v^2(-\mathbf{k} - \mathbf{q} + \mathbf{k}')} \frac{g_{m_{\alpha_2} m_{\alpha_3} v}(\mathbf{q} - \mathbf{k}', -\mathbf{k} - \mathbf{q} + \mathbf{k}') g_{m_{\alpha_4} m_{\alpha_1} v}^*(\mathbf{q} + \mathbf{k}, -\mathbf{k} - \mathbf{q} + \mathbf{k}')}{\hbar^2} \\ &= \frac{-1}{\hbar \beta N_p} \sum_{p \in \{1, \dots, r d_x\}} \frac{\omega_v(-\mathbf{k} - \mathbf{q} + \mathbf{k}')}{\omega_{-k-q+k'}^2 + \omega_v^2(-\mathbf{k} - \mathbf{q} + \mathbf{k}')} \frac{g_{m_{\alpha_2} m_{\alpha_3} v}(\mathbf{q} - \mathbf{k}', -\mathbf{k} - \mathbf{q} + \mathbf{k}') g_{m_{\alpha_1} m_{\alpha_4} v}(\mathbf{k}', +\mathbf{k} + \mathbf{q} - \mathbf{k}')}{\hbar^2} \end{aligned}$$

Spin-Triplet-Contributions

Case 1: $\sigma = \uparrow$ and $\pi = \uparrow$. Then we obtain by the exact same calculations as above

- $\alpha_1 \equiv (m_{\alpha_1}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_1}, \uparrow, q + k)$
- $\alpha_2 \equiv (m_{\alpha_2}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_2}, \uparrow, -k)$
- $\alpha_3 \equiv (m_{\alpha_3}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_3}, \uparrow, q - k')$
- $\alpha_4 \equiv (m_{\alpha_4}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_4}, \uparrow, k')$

This gives for the matrix-element

$$\begin{aligned}
 & V_{\alpha_1, \alpha_2; \alpha_3, \alpha_4}^{\uparrow \uparrow} \\
 &= \frac{-1}{\hbar \beta N_p} \sum_{v \in \{1, \dots, r d_x\}} \frac{\omega_v(\mathbf{k} + \mathbf{q} - \mathbf{k}')}{\omega_{k+q-k'}^2 + \omega_v^2(\mathbf{k} + \mathbf{q} - \mathbf{k}')} \frac{g_{m_{\alpha_1} m_{\alpha_4} v}(\mathbf{k}', \mathbf{k} + \mathbf{q} - \mathbf{k}') g_{m_{\alpha_3} m_{\alpha_2} v}^*(-\mathbf{k}, \mathbf{k} + \mathbf{q} - \mathbf{k}')}{\hbar^2} \\
 &= \frac{-1}{\hbar \beta N_p} \sum_{v \in \{1, \dots, r d_x\}} \frac{\omega_v(\mathbf{k} + \mathbf{q} - \mathbf{k}')}{\omega_{k+q-k'}^2 + \omega_v^2(\mathbf{k} + \mathbf{q} - \mathbf{k}')} \frac{g_{m_{\alpha_1} m_{\alpha_4} v}(\mathbf{k}', \mathbf{k} + \mathbf{q} - \mathbf{k}') g_{m_{\alpha_2} m_{\alpha_3} v}(\mathbf{q} - \mathbf{k}', -(\mathbf{k} + \mathbf{q} - \mathbf{k}'))}{\hbar^2}
 \end{aligned}$$

Case 2: $\sigma = \downarrow$ and $\pi = \downarrow$. Then we obtain by the exact same calculations as above

- $\alpha_1 \equiv (m_{\alpha_1}, \sigma_{\alpha_1}, k_{\alpha_1}) = (m_{\alpha_1}, \downarrow, q + k)$
- $\alpha_2 \equiv (m_{\alpha_2}, \sigma_{\alpha_2}, k_{\alpha_2}) = (m_{\alpha_2}, \downarrow, -k)$
- $\alpha_3 \equiv (m_{\alpha_3}, \sigma_{\alpha_3}, k_{\alpha_3}) = (m_{\alpha_3}, \downarrow, q - k')$
- $\alpha_4 \equiv (m_{\alpha_4}, \sigma_{\alpha_4}, k_{\alpha_4}) = (m_{\alpha_4}, \downarrow, k')$

This gives for the matrix-element

$$\begin{aligned}
 & V_{\alpha_1, \alpha_2; \alpha_3, \alpha_4}^{\downarrow \downarrow} \\
 &= \frac{-1}{\hbar \beta N_p} \sum_{v \in \{1, \dots, r d_x\}} \frac{\omega_v(\mathbf{k} + \mathbf{q} - \mathbf{k}')}{\omega_{k+q-k'}^2 + \omega_v^2(\mathbf{k} + \mathbf{q} - \mathbf{k}')} \frac{g_{m_{\alpha_1} m_{\alpha_4} v}(\mathbf{k}', \mathbf{k} + \mathbf{q} - \mathbf{k}') g_{m_{\alpha_3} m_{\alpha_2} v}^*(-\mathbf{k}, \mathbf{k} + \mathbf{q} - \mathbf{k}')}{\hbar^2} \\
 &= \frac{-1}{\hbar \beta N_p} \sum_{v \in \{1, \dots, r d_x\}} \frac{\omega_v(\mathbf{k} + \mathbf{q} - \mathbf{k}')}{\omega_{k+q-k'}^2 + \omega_v^2(\mathbf{k} + \mathbf{q} - \mathbf{k}')} \frac{g_{m_{\alpha_1} m_{\alpha_4} v}(\mathbf{k}', \mathbf{k} + \mathbf{q} - \mathbf{k}') g_{m_{\alpha_2} m_{\alpha_3} v}(\mathbf{q} - \mathbf{k}', -(\mathbf{k} + \mathbf{q} - \mathbf{k}'))}{\hbar^2}
 \end{aligned}$$

Proof 33.40 of Theorem 24.55 We decompose the spin-singlet contributions as

$$S[\psi, \bar{\psi}] \equiv S^{\text{ET}}[\psi, \bar{\psi}] + S^{\text{NET}}[\psi, \bar{\psi}] + S^{\text{RPC}}[\psi, \bar{\psi}] \quad (33.151)$$

whereas for the ([N]on-)[E]liashberg-[T]heory contribution

$$\begin{aligned}
 S^{\text{ET}}[\psi, \bar{\psi}] &= \sum_{\substack{k, k' \in i\Omega_{-1} \times Q \\ m, n \in \{1, 2, 3, \dots\}}} \bar{\psi}_{m\uparrow}(k) \bar{\psi}_{m\downarrow}(-k) V_{\downarrow\uparrow}^{\text{ET}}(k, k') \psi_{n\downarrow}(-k') \psi_{n\uparrow}(k') \\
 &+ \sum_{\substack{\sigma \in \{\downarrow, \uparrow\} \\ k, k' \in i\Omega_{-1} \times Q \\ m, n \in \{1, 2, 3, \dots\}}} \bar{\psi}_{m\sigma}(k) \bar{\psi}_{m\sigma}(-k) V_{\sigma\sigma}^{\text{ET}}(k, k') \psi_{n\sigma}(-k') \psi_{n\sigma}(k')
 \end{aligned}$$

$$\begin{aligned}
 S_{\downarrow\uparrow}^{\text{NET}}[\psi, \bar{\psi}] &= \sum_{\substack{q, k \in i\Omega_{-1} \times Q \\ m, n \in \{1, 2, 3, \dots\}}} \bar{\psi}_{m\uparrow}(q+k) \bar{\psi}_{n\downarrow}(-k) V_{\downarrow\uparrow}^{\text{NET}}(q, k) \psi_{m\downarrow}(q+k) \psi_{n\uparrow}(-k) \\
 &+ \sum_{\substack{\sigma \in \{\downarrow, \uparrow\} \\ q, k \in i\Omega_{-1} \times Q \\ m, n \in \{1, 2, 3, \dots\}}} \bar{\psi}_{m\sigma}(q+k) \bar{\psi}_{n\sigma}(-k) V_{\sigma\sigma}^{\text{NET}}(q, k) \psi_{m\sigma}(q+k) \psi_{n\sigma}(-k)
 \end{aligned}$$

we have, when we take the double contributions into account, the vertices

$$2V_{\sigma\sigma}^{\text{ET}}(k, k') = V_{\downarrow\uparrow}^{\text{ET}}(k, k') = \frac{-2}{\hbar\beta N_p} \sum_{v \in \{1, \dots, rd_x\}} \frac{\omega_v(\mathbf{k} - \mathbf{k}')}{\omega_{k-k'}^2 + \omega_v^2(\mathbf{k} - \mathbf{k}')} \frac{|g_{mnv}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{\hbar^2} \quad (33.152)$$

$$2V_{\sigma\sigma}^{\text{NET}}(q, k) = V_{\downarrow\uparrow}^{\text{NET}}(q, k) = \frac{-2}{\hbar\beta N_p} \sum_{v \in \{1, \dots, rd_x\}} \frac{\omega_v(2\mathbf{k} + \mathbf{q})}{\omega_{2k+q}^2 + \omega_v^2(2\mathbf{k} + \mathbf{q})} \frac{|g_{mnv}(-\mathbf{k}, 2\mathbf{k} + \mathbf{q})|^2}{\hbar^2} \quad (33.153)$$

and the [R]andom [P]hase [C]ancellations are all the remaining contributions (which have complex vertices).

Proof 33.41 of Lemma 24.62

We compute straightforwardly

$$\eta_m(\tau, \mathbf{k}) = \frac{1}{\sqrt{\hbar\beta}} \sum_{\omega_n \in i\Omega_{-1}} \eta_m(\mathbf{k}, i\omega_n) e^{-i\omega_n \tau} = \frac{1}{\sqrt{\hbar\beta}} \sum_{i\omega_n \in i\Omega_{-1}} \left(\frac{\psi_{m\uparrow}(i\omega_n, \mathbf{k})}{\bar{\psi}_{m\downarrow}(-i\omega_n, -\mathbf{k})} \right) e^{-i\omega_n \tau} = \left(\frac{\psi_{m\uparrow}(\tau, \mathbf{k})}{\bar{\psi}_{m\downarrow}(\tau, -\mathbf{k})} \right)$$

whereas in the last step we replaced $\omega_n \mapsto -\omega_n$ in the second component. Likewise, one obtains also

$$\bar{\eta}_m^T(\tau, \mathbf{k}) = \left(\frac{\bar{\psi}_{m\uparrow}(\tau, \mathbf{k})}{\psi_{m\downarrow}(\tau, -\mathbf{k})} \right) \quad (33.154)$$

Proof 33.42 of Corollary 24.63 We note that the action can be rewritten in imaginary-time space as

$$\begin{aligned}
 S_{\text{int}}[\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}, \psi, \bar{\psi}] &= \sum_{\substack{k \in i\Omega_{-1} \times Q \\ m \in \{1,2,3,\dots\}}} \bar{\eta}_m^T(k) \left(-\frac{1}{\hbar} \mathcal{G}_{\substack{m \\ [\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}}^{-1}(k) \right) \eta_m(k) \\
 &= \sum_{\substack{\mathbf{k} \in Q \\ m \in \{1,2,3,\dots\} \\ \tau, \tau' \in (0, \beta\hbar)}} \bar{\eta}_m^T(\mathbf{k}, \tau) \left(-\frac{1}{\hbar} \frac{1}{\hbar\beta} \sum_{\substack{i\omega_n \in i\Omega_{-1} \\ [\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}} \mathcal{G}_m^{-1}(i\omega_n, \mathbf{k}) e^{-i\omega_n(\tau' - \tau)} \right) \eta_m(\tau', \mathbf{k}) \\
 &\equiv \sum_{\substack{\mathbf{k} \in Q \\ m \in \{1,2,3,\dots\} \\ \tau, \tau' \in (0, \beta\hbar)}} \bar{\eta}_m^T(\tau, \mathbf{k}) \left(-\frac{1}{\hbar} \mathcal{G}_{\substack{m \\ [\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}}^{-1}(\tau - \tau', \mathbf{k}) \right) \eta_m(\tau', \mathbf{k})
 \end{aligned}$$

Now the identities follow by simply using the usual relations from [Draft Summary].

Proof 33.43 of Corollary 24.66

Searching for stationary points yields the equation

$$\begin{aligned}
 0 &= \left(\frac{\partial}{\partial \delta(\Gamma)} S_{\text{tot}} \right) [\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}] \\
 &= - \sum_{A \in \Pi} \bar{\delta}(A) \Lambda_{\downarrow\uparrow}^{\text{ET}}(A, \Gamma) - \text{Tr}_{\substack{[\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}} \mathcal{G}_{\substack{[\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}} \frac{\partial}{\partial \delta(\Gamma)} \mathcal{G}_{\substack{[\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}}^{-1}
 \end{aligned} \tag{33.155}$$

Multiplying now from both sides with the inverse of $\Lambda_{\downarrow\uparrow}^{\text{ET}}$ from the right yields the gap-equation

$$\bar{\delta}(B) = - \sum_{B \in \Pi} \text{Tr} \left(\mathcal{G}_{\substack{[\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}} \frac{\partial}{\partial \delta(A)} \mathcal{G}_{\substack{[\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}}^{-1} \right) V_{\downarrow\uparrow}^{\text{ET}}(A, B) \tag{33.156}$$

Referring now to the multi-indices in Equation (24.130) and the matrix-elements in Equation (24.129), and partially performing the trace over the non-vanishing contributions, we obtain for each band and each four-momentum the desired equations.

Proof 33.44 of Lemma 24.67

Recalling that the Gorkov-Green-function block-diagonal w.r.t. bands and wave-vectors with individual blocks

$$-\frac{1}{\hbar} \mathcal{G}_{\substack{m \\ [\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}}^{-1}(k) = \begin{pmatrix} -\frac{1}{\hbar} \mathcal{G}_0^{-F}(k) & \delta_{m\downarrow\uparrow}(k) \\ \bar{\delta}_{m\uparrow} & +\frac{1}{\hbar} \mathcal{G}_0^{-F}(-k) \\ \delta_{m\downarrow\uparrow}(k) & m\downarrow \end{pmatrix} \tag{33.157}$$

which allows us on the one hand to compute the derivative

$$\frac{\partial}{\partial \delta_{m\downarrow\uparrow}(k)} \mathcal{G}_{\substack{m \\ [\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}}^{-1} = \begin{pmatrix} 0 & -\hbar \\ 0 & 0 \end{pmatrix} \tag{33.158}$$

and by inversion of the equation

$$-\hbar \mathcal{G}_{\substack{m \\ [\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}}(k) = \begin{pmatrix} -\frac{1}{\hbar} \mathcal{G}_0^{-F}(k) & \delta_{m\downarrow\uparrow}(k) \\ \bar{\delta}_{m\downarrow\uparrow}(k) & +\frac{1}{\hbar} \mathcal{G}_0^{-F}(-k) \end{pmatrix}^{-1} \iff \mathcal{G}_{\substack{m \\ [\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}}(k) = -\frac{1}{\hbar} \begin{pmatrix} -\frac{1}{\hbar} \mathcal{G}_0^{-F}(k) & \delta_{m\downarrow\uparrow}(k) \\ \bar{\delta}_{m\downarrow\uparrow}(k) & +\frac{1}{\hbar} \mathcal{G}_0^{-F}(-k) \end{pmatrix}^{-1} \quad (33.159)$$

Recalling that one has for the non-interacting Green-functions the relations (we are considering a spin-degenerate system)

$$\begin{aligned} -\frac{1}{\hbar} \mathcal{G}_0^{-F}(k) &= -i\omega_n + (\epsilon_m(\mathbf{k}) - \mu)/\hbar \\ +\frac{1}{\hbar} \mathcal{G}_0^{-F}(-k) &= -i\omega_n - (\epsilon_m(-\mathbf{k}) - \mu)/\hbar \end{aligned}$$

we obtain for the inverse with the rule for 2×2 matrices

$$\mathcal{G}_{\substack{m \\ [\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}}(k) = -\frac{1}{\hbar} \frac{\begin{pmatrix} -i\omega_k - (\epsilon_m(-\mathbf{k}) - \mu)/\hbar & -\delta_{m\downarrow\uparrow}(k) \\ -\bar{\delta}_{m\downarrow\uparrow}(k) & -i\omega_k + (\epsilon_m(\mathbf{k}) - \mu)/\hbar \end{pmatrix}}{\text{Det} \begin{pmatrix} -\frac{1}{\hbar} \mathcal{G}_0^{-F}(k) & \delta_{m\downarrow\uparrow}(k) \\ \bar{\delta}_{m\downarrow\uparrow}(k) & +\frac{1}{\hbar} \mathcal{G}_0^{-F}(-k) \end{pmatrix}} \quad (33.160)$$

we therefore obtain ONLY IN THE CASE OF INVERSION SYMMETRIC CRYSTALS¹⁰

$$\text{Det} \begin{pmatrix} -\frac{1}{\hbar} \mathcal{G}_0^{-F}(k) & \delta_{m\downarrow\uparrow}(k) \\ \bar{\delta}_{m\downarrow\uparrow}(k) & +\frac{1}{\hbar} \mathcal{G}_0^{-F}(-k) \end{pmatrix} = -\omega_k^2 - \frac{(\epsilon_m(\mathbf{k}) - \mu)^2}{\hbar^2} - |\delta_{m\downarrow\uparrow}(k)|^2 \quad (33.161)$$

which simplifies the expression above to

$$\mathcal{G}_{\substack{m \\ [\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}}(k) = -\frac{1}{\hbar} \frac{\begin{pmatrix} +i\omega_k + (\epsilon_m(-\mathbf{k}) - \mu)/\hbar & +\delta_{m\downarrow\uparrow}(k) \\ +\bar{\delta}_{m\downarrow\uparrow}(k) & +i\omega_k - (\epsilon_m(\mathbf{k}) - \mu)/\hbar \end{pmatrix}}{+\omega_k^2 + \frac{(\epsilon_m(\mathbf{k}) - \mu)^2}{\hbar^2} + |\delta_{m\downarrow\uparrow}(k)|^2} \quad (33.162)$$

Proof 33.45 of Lemma 24.68 We first notice that

$$\mathcal{G}_{\substack{m,12 \\ [\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}}(k) = \mathcal{G}_{\substack{m,21 \\ [\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}] \\ \text{Gorkov}}}(k)^* \quad (33.163)$$

¹⁰In this case one finds $\epsilon_m(-\mathbf{k}) = \epsilon_m(+\mathbf{k})$

and obtain therefore

$$\begin{aligned} \mathcal{G}_{m,12}^{[\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}]}(\tau - \tau') &= \frac{1}{\hbar\beta} \sum_{i\omega_n \in i\Omega_{-1}} \mathcal{G}_{m,12}^{[\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}]}(i\omega_n, \mathbf{k}) e^{-i\omega_n(\tau - \tau')} \\ &= \frac{1}{\hbar\beta} \sum_{i\omega_n \in i\Omega_{-1}} \frac{\delta_{m\downarrow\uparrow}(k)}{\omega_n^2 + (\epsilon_m(\mathbf{k}) - \mu)^2 / \hbar^2 + |\delta_{m\downarrow\uparrow}(k)|^2} e^{-i\omega_n(\tau - \tau')} \end{aligned}$$

And on the other hand we obtain

$$\begin{aligned} \mathcal{G}_{m,21}^{[\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}]}(\tau' - \tau) &= \frac{1}{\hbar\beta} \sum_{i\omega_n \in i\Omega_{-1}} \mathcal{G}_{m,21}^{[\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}]}(i\omega_n, \mathbf{k}) e^{-i\omega_n(\tau' - \tau)} \\ &= \frac{1}{\hbar\beta} \sum_{i\omega_n \in i\Omega_{-1}} \frac{\bar{\delta}_{m\downarrow\uparrow}(k)}{\omega_n^2 + (\epsilon_m(\mathbf{k}) - \mu)^2 / \hbar^2 + |\delta_{m\downarrow\uparrow}(k)|^2} e^{+i\omega_n(\tau - \tau')} \\ &= \mathcal{G}_{m,12}^{[\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}]}(\tau - \tau')^* \end{aligned}$$

Thus, we obtain in the sense of generalized matrices the desired identity.

Proof 33.46 of Theorem 24.69

Combining the results [TBReferenced](#) and [TBReferenced](#) gives us

$$\begin{aligned} \left(\mathcal{G}_{m,12}^{[\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}]} \frac{\partial}{\partial \delta_{m\downarrow\uparrow}(k)} \mathcal{G}_{m,21}^{[\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}]} \right) &= -\frac{1}{\hbar} \frac{\begin{pmatrix} +i\omega_k + (\epsilon_m(-\mathbf{k}) - \mu)/\hbar & +\delta_{m\downarrow\uparrow}(k) \\ +\bar{\delta}_{m\downarrow\uparrow}(k) & +i\omega_k - (\epsilon_m(\mathbf{k}) - \mu)/\hbar \end{pmatrix}}{+\omega_k^2 + \frac{(\epsilon_m(\mathbf{k}) - \mu)^2}{\hbar^2} + |\delta_{m\downarrow\uparrow}(k)|^2} \cdot \begin{pmatrix} 0 & -\hbar \\ 0 & 0 \end{pmatrix} \\ &= + \frac{\begin{pmatrix} 0 & +i\omega_k + (\epsilon_m(-\mathbf{k}) - \mu)/\hbar \\ 0 & \bar{\delta}_{m\downarrow\uparrow}(k) \end{pmatrix}}{+\omega_k^2 + \frac{(\epsilon_m(\mathbf{k}) - \mu)^2}{\hbar^2} + |\delta_{m\downarrow\uparrow}(k)|^2} \end{aligned} \quad (33.164)$$

which gives for the trace

$$\text{Tr}_{2 \times 2} \left(\mathcal{G}_{m,12}^{[\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}]} \frac{\partial}{\partial \delta_{m\downarrow\uparrow}(k)} \mathcal{G}_{m,21}^{[\delta_{\downarrow\uparrow}, \bar{\delta}_{\downarrow\uparrow}]} \right) = + \frac{\bar{\delta}_{m\downarrow\uparrow}(k)}{\omega_k^2 + \frac{(\epsilon_m(\mathbf{k}) - \mu)^2}{\hbar^2} + |\delta_{m\downarrow\uparrow}(k)|^2} \quad (33.165)$$

Now we arrive at the final form of the gap-equations

$$\bar{\delta}_{n\downarrow\uparrow}(k') = + \sum_{\substack{k \in i\Omega_{-1} \times Q \\ v \in \{1, \dots, rd\} \\ m \in \{1, 2, 3, \dots\}}} \frac{\bar{\delta}_{m\downarrow\uparrow}(k)}{\omega_k^2 + \frac{(\epsilon_m(\mathbf{k}) - \mu)^2}{\hbar^2} + |\delta_{m\downarrow\uparrow}(k)|^2} \frac{2\omega_v(\mathbf{k} - \mathbf{k}')}{\omega_{k-k'}^2 + \omega_v^2(\mathbf{k} - \mathbf{k}')} \frac{|g_{mnv}(\mathbf{k}', \mathbf{k} - \mathbf{k}')|^2}{\hbar^3 \beta N_p}$$

34. Appendix - Proofs for Part V

Proof 34.1 of Theorem 27.76

We rewrite the action w.r.t. to the new fields

$$\begin{aligned}
 S_{\text{ph}}[\phi, \bar{\phi}] &= \sum \bar{\phi}_v(q) (-i\omega_n + \omega_{\mathbf{q},v}) \phi_v(q) \\
 &= \frac{1}{4} \sum_q (A_v - \Pi_v)(-q) [-i\omega_n + \omega_v(\mathbf{q})] (A_v + \Pi_v)(q) \\
 &= \frac{1}{4} \sum_q A_v(-q) [-i\omega_n + \omega_v(\mathbf{q})] A_v(q) \\
 &\quad - \frac{1}{4} \sum_q \Pi_v(-q) [-i\omega_n + \omega_v(\mathbf{q})] \Pi_v(+q) \\
 &\quad + \frac{1}{4} \sum_q A_v(-q) [-i\omega_n + \omega_v(\mathbf{q})] \Pi_v(q) \\
 &\quad - \frac{1}{4} \sum_q \Pi_v(-q) [-i\omega_n + \omega_v(\mathbf{q})] A_v(q) \\
 &\equiv \text{I} + \text{II} + \text{III} + \text{IV} \\
 &= \dots
 \end{aligned}$$

Now we will examine each term individually.

$$\text{IV} = -\frac{1}{4} \sum_q A_v(-q) [+i\omega_n + \omega_v(\mathbf{q})] \Pi_v(+q) \quad (34.1)$$

Therefore the terms proportional to $\omega_v(\mathbf{q})$ in III and IV will cancel out. Due to the symmetries $q \leftrightarrow -q$ there are more cancellations happening. This gives

$$\begin{aligned}
 \dots &= \frac{1}{4} \sum_q A_v(-q) [\omega_v(\mathbf{q})] A_v(q) \\
 &\quad - \frac{1}{4} \sum_q \Pi_v(-q) [\omega_v(\mathbf{q})] \Pi_v(+q) \\
 &\quad + \frac{1}{4} \sum_q A_v(-q) [-i\omega_n] \Pi_v(q) \\
 &\quad - \frac{1}{4} \sum_q A_v(-q) [i\omega_n] \Pi_v(+q) \\
 &\equiv \widetilde{\text{I}} + \widetilde{\text{II}} + \widetilde{\text{III}} + \widetilde{\text{IV}} \\
 &= \dots
 \end{aligned}$$

Each term gets evaluated

$$\widetilde{\text{I}} = \frac{1}{2} \sum_{q \in \Omega_{+1} \times Q / \sim} A_v(-q) [\omega_v(\mathbf{q})] A_v(q) \quad (34.2)$$

$$\widetilde{\text{II}} = -\frac{1}{2} \sum_{q \in \Omega_{+1} \times Q / \sim} \Pi_v(-q) [\omega_v(\mathbf{q})] \Pi_v(+q) \quad (34.3)$$

$$\begin{aligned} \widetilde{\text{III}} + \widetilde{\text{IV}} &= -\frac{1}{2} \sum_q A_v(-q) [i\omega_n] \Pi_v(q) \\ &= -\frac{1}{2} \sum_{q \in \Omega_{+1} \times Q / \sim} A_v(-q) [i\omega_n] \Pi_v(q) + A_v(+q) [-i\omega_n] \Pi_v(-q) \end{aligned}$$

Now we focus on the terms that are proportional to Π and find by performing the integral

$$\begin{aligned} \widetilde{\text{II}} + \widetilde{\text{III}} + \widetilde{\text{IV}} &= \sum_{q \in \Omega_{+1} \times Q / \sim} \left\{ \Pi_v(-q) \frac{-\omega_v(\mathbf{q})}{2} \Pi_v(+q) + A_v(-q) \frac{-i\omega_n}{2} \Pi_v(q) + A_v(+q) \frac{+i\omega_n}{2} \Pi_v(-q) \right\} \\ &\equiv \sum_{q \in \Omega_{+1} \times Q / \sim} -A_v(-q) \frac{-i\omega_n}{2} \frac{2}{-\omega_v(\mathbf{q})} \frac{+i\omega_n}{2} A_v(+q) \\ &\equiv \sum_{q \in \Omega_{+1} \times Q / \sim} A_v(-q) \frac{\omega_n^2}{2\omega_v(\mathbf{q})} A_v(+q) \end{aligned}$$

and therefore

$$\begin{aligned} \dots &= \sum_{q \in \Omega_{+1} \times Q / \sim} A_v(-q) \left[\omega_v(\mathbf{q}) + \frac{\omega_n^2}{2\omega_v(\mathbf{q})} \right] A_v(+q) \\ &= \sum_{q \in \Omega_{+1} \times Q / \sim} A_v(-q) \left[\frac{\omega_n^2 + \omega_v^2(\mathbf{q})}{2\omega_v(\mathbf{q})} \right] A_v(+q) \\ &= \sum_{q \in \Omega_{+1} \times Q / \sim} A_v(-q) \left[-\frac{1}{\hbar} \mathcal{D}_{0,v}^{-1}(q) \right] A_v(+q) \end{aligned}$$

Proof 34.2 of Corollary 27.8 We compute

$$\begin{aligned}
 (P_\zeta^2 \psi)(x_1, \dots, x_n) &= P_\zeta \frac{1}{N!} \sum_{\pi \in S_N} \text{sign}(\pi)^{1-\theta(\zeta)} \psi(x_{\pi(1)}, \dots, x_{\pi(N)}) \\
 &= \frac{1}{N!^2} \sum_{\pi, \sigma \in S_N} \text{sign}(\pi)^{1-\theta(\zeta)} \text{sign}(\sigma)^{1-\theta(\zeta)} \psi(x_{\sigma(\pi(1))}, \dots, x_{\sigma(\pi(N))}) \\
 &= \frac{1}{N!^2} \sum_{\pi, \sigma \in S_N} \text{sign}(\sigma \circ \pi)^{1-\theta(\zeta)} \psi(x_{\sigma(\pi(1))}, \dots, x_{\sigma(\pi(N))}) \\
 &= \dots
 \end{aligned}$$

Now we use, that for each $\pi \in S_N$ one has $S_N = S_N \circ \pi$ (as a set). Hence we can write

$$\begin{aligned}
 \dots &= \frac{1}{N!^2} \sum_{\pi \in S_N} \sum_{\tilde{\sigma} \in S_N} \text{sign}(\tilde{\sigma})^{1-\theta(\zeta)} \psi(x_{\tilde{\sigma}(1)}, \dots, x_{\tilde{\sigma}(N)}) \\
 &= \frac{1}{N!} \sum_{\sigma \in S_N} \text{sign}(\sigma)^{1-\theta(\zeta)} \psi(x_{\sigma(1)}, \dots, x_{\sigma(N)}) \\
 &= (P_\zeta \psi)(x_1, \dots, x_n)
 \end{aligned}$$

Proof 34.3 of Lemma 27.11

We have to show that the two expressions coincide. The wavefunction associated with Equation (27.10) is given as

$$(x_1, \dots, x_N | \psi) = \prod_{j=1}^N \langle x_j | \psi_j \rangle \equiv \psi(x_1, \dots, x_N) \quad (34.4)$$

Therefore we obtain on the one hand

$$\begin{aligned}
 (P_\zeta \psi)(x_1, \dots, x_N) &= \frac{1}{N!} \sum_{\pi \in S_N} \text{sign}(\pi)^{1-\theta(\zeta)} \psi(x_{\pi(1)}, \dots, x_{\pi(N)}) \\
 &= \frac{1}{N!} \sum_{\pi \in S_N} \text{sign}(\pi)^{1-\theta(\zeta)} \prod_{j=1}^N \langle x_{\pi(j)} | \psi_j \rangle \\
 &= \frac{1}{N!} \sum_{\pi \in S_N} \text{sign}(\pi)^{1-\theta(\zeta)} \prod_{j=1}^N \psi_j(x_{\pi(j)})
 \end{aligned}$$

and we obtain on the other hand with Equation (27.11)

$$(x_1, \dots, x_N | P_\zeta | \psi) = \frac{1}{N!} \sum_{\pi \in S_N} \text{sign}(\pi)^{1-\theta(\zeta)} \prod_{j=1}^N \psi_{\pi(j)}(x_j)$$

the two expressions coincide which can be seen by reordering the products and using $\text{sign}(\sigma) = \text{sign}(\sigma^{-1})$.

Proof 34.4 of Lemma 27.14 We have to show, that

$$\forall \psi, \chi \in \mathcal{H}_N : \langle \psi | P_\zeta \chi \rangle = \langle P_\zeta \psi | \chi \rangle \quad (34.5)$$

It suffices to prove this identity for basis-elements. Therefore we compute on the one hand

$$\begin{aligned} (i_1, \dots, i_N | P_\zeta | j_1, \dots, j_N) &= \frac{1}{N!} \sum_{\pi \in S_N} \text{sign}(\pi)^{1-\theta(\zeta)} (i_1, \dots, i_N | j_{\pi(1)}, \dots, j_{\pi(N)}) \\ &= \frac{1}{N!} \sum_{\pi \in S_N} \text{sign}(\pi)^{1-\theta(\zeta)} \prod_{l=1}^N \langle i_l | j_{\pi(l)} \rangle \\ &= \frac{1}{N!} \sum_{\pi \in S_N} \text{sign}(\pi)^{1-\theta(\zeta)} \prod_{l=1}^N \delta_{i_l, j_{\pi(l)}} \end{aligned}$$

and we obtain on the other hand

$$(i_1, \dots, i_N | P_\zeta^\dagger | j_1, \dots, j_N) = \frac{1}{N!} \sum_{\pi \in S_N} \text{sign}(\pi)^{1-\theta(\zeta)} \prod_{l=1}^N \delta_{i_{\pi(l)}, j_l}$$

Just as in the prove of the Lemma before, the two expressions coincide, which can also be seen by expanding the products reordering the appearing factors and using $\text{sign}(\sigma) = \text{sign}(\sigma^{-1})$.

Proof 34.5 of Theorem 27.59 We adopt the more general notation from the earlier section. This yields for the Hamiltonian in the harmonic approximation

$$H \equiv \sum_{\lambda \in \mathcal{A}} \hbar \omega_{\lambda} b_{\lambda}^{\dagger} b_{\lambda} \quad (34.6)$$

in the grand-canonical Ensemble, we obtain, referring to an enumeration $\lambda_1, \lambda_2, \dots, \lambda_{rd_{\mathbf{x}}N_p}$ of $\mathcal{A}$, the identities

$$\langle b_{\lambda}^{\dagger} b_{\lambda} \rangle = \frac{\text{Tr}_{\mathcal{H}} (e^{-\beta H} b_{\lambda}^{\dagger} b_{\lambda})}{\text{Tr}_{\mathcal{H}} (e^{-\beta H})} = \frac{\sum_{n_{\lambda_1}=0}^{\infty} \sum_{n_{\lambda_2}=0}^{\infty} \cdots \sum_{n_{\lambda_{rd_{\mathbf{x}}N_p}}=0}^{\infty} \langle n_{\lambda_1}, \dots, n_{\lambda_{rd_{\mathbf{x}}N_p}} | e^{-\beta H} b_{\lambda}^{\dagger} b_{\lambda} | n_{\lambda_1}, \dots, n_{\lambda_{rd_{\mathbf{x}}N_p}} \rangle}{\sum_{n_{\lambda_1}=0}^{\infty} \sum_{n_{\lambda_2}=0}^{\infty} \cdots \sum_{n_{\lambda_{rd_{\mathbf{x}}N_p}}=0}^{\infty} \langle n_{\lambda_1}, \dots, n_{\lambda_{rd_{\mathbf{x}}N_p}} | e^{-\beta H} | n_{\lambda_1}, \dots, n_{\lambda_{rd_{\mathbf{x}}N_p}} \rangle} = \star$$

using now the hermiticity of the Hamiltonian, we see that we can replace $b_{\lambda_i}^{\dagger} b_{\lambda_i}$ by the numbers n_{λ_i} and factorize the expression, such that everything except for one pair of sums cancels out and we obtain

$$\star = \frac{\sum_{n_{\lambda}=0}^{\infty} e^{-\beta \hbar \omega_{\lambda} n_{\lambda}}}{\sum_{n_{\lambda}=0}^{\infty} e^{-\beta n_{\lambda} \hbar \omega_{\lambda}}} \equiv - \left(\frac{f'(x)}{f(x)} \right)_{|x=\beta \hbar \omega_{\lambda}}$$

We therefore examine the function

$$f(x) = \sum_{n_{\lambda}=0}^{\infty} e^{-n_{\lambda} x} = \frac{1}{1 - e^{-x}}$$

and obtain the derivative

$$f'(x) = -\frac{e^{-x}}{(1 - e^{-x})^2}$$

such that we obtain

$$\frac{-f'(x)}{f(x)} = \frac{e^{-x}}{(1 - e^{-x})^2} \cdot \frac{1 - e^{-x}}{1} = \frac{e^{-x}}{1 - e^{-x}} = \frac{1}{e^{+x} - 1} = B_1(x)$$

Proof 34.6 of Lemma 29.5 As long as $m_{1,2,3} \neq \infty$ (or, equivalently $h, k, l \neq 0$) we can consider the three special vectors

$$\mathbf{T}_1 = m_1 \mathbf{a}_1 - m_2 \mathbf{a}_2, \quad \mathbf{T}_2 = m_2 \mathbf{a}_2 - m_3 \mathbf{a}_3, \quad \mathbf{T}_3 = m_3 \mathbf{a}_3 - m_1 \mathbf{a}_1. \quad (34.7)$$

According to the mnemonic

$$\text{Plane} = \text{Support-Vector} + c_1 \text{Directional-Vector-1} + c_2 \text{Directional-Vector-2} \quad c_i \in \mathbb{R} \quad (34.8)$$

we have to show that the scalar-product $\mathbf{T}_i \cdot \mathbf{G}$ vanishes.

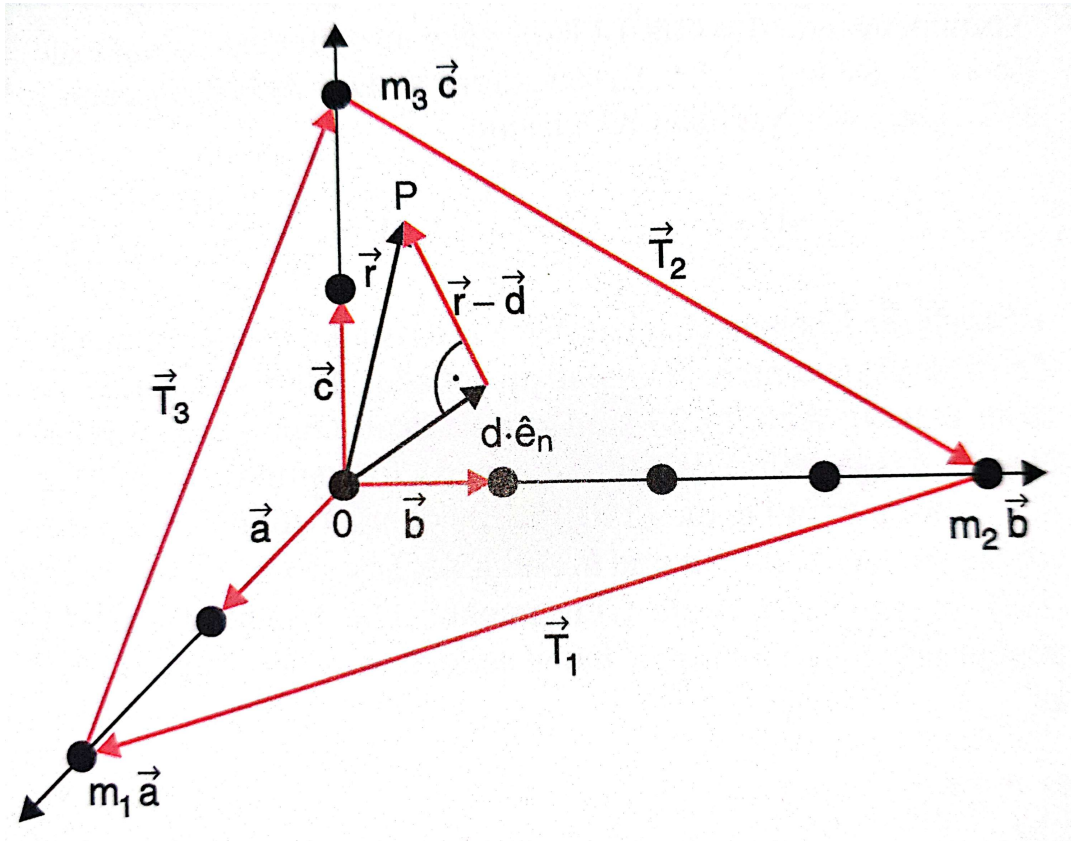


Figure 34.1.: Excerpt from Wolfgang Demtröders Experimentalphysik 3.

And indeed, a straight-forward calculation yields

$$\begin{aligned} \mathbf{T}_1 \cdot \mathbf{G} &= (m_1 \mathbf{a}_1 - m_2 \mathbf{a}_2) \cdot (h \mathbf{b}_1 + k \mathbf{b}_2 + l \mathbf{b}_3) \\ &= 2\pi (m_1 h - m_2 k) \\ &= 0 \end{aligned}$$

and analog for $\mathbf{T}_{1,2}$.

Now we consider the case that at least one of the m_i is equal to infinity. Without loss of generality, we consider $m_1 = \infty$ (equivalently one has $h = 0$). Then we consider $\tilde{\mathbf{T}}_1 = \mathbf{a}_1$ and $\mathbf{T}_2 = m_2 \mathbf{a}_2 - m_3 \mathbf{a}_3$ as directional-vectors of the planes and the scalar-product vanishes again.

Now we consider the case that two of the m_i are equal to infinity. Without loss of generality, we consider $m_1 = m_2 = \infty$ (equivalently one has $h = k = 0$). Then we consider $\tilde{\mathbf{T}}_1 = \mathbf{a}_1$ and $\tilde{\mathbf{T}}_2 = \mathbf{a}_2$ as directional-vectors of the planes and the scalar-product vanishes again.

Adding Pictures.

Proof 34.7 of Lemma 28.4

We compute the matrix-elements to be

$$\begin{aligned}
 \langle \mathbf{k}_1 \sigma_1, \mathbf{k}_2 \sigma_2 | V | \mathbf{k}_3 \sigma_3, \mathbf{k}_4 \sigma_4 \rangle &= -|g| \delta_{\sigma_3 \neq \sigma_4} \delta_{\sigma_1, \sigma_3} \delta_{\sigma_2, \sigma_4} \int d^3 \mathbf{r} d^3 \mathbf{r}' \delta(\mathbf{r} - \mathbf{r}') \psi_{\mathbf{k}_1}^*(\mathbf{r}) \psi_{\mathbf{k}_2}^*(\mathbf{r}') \psi_{\mathbf{k}_3}(\mathbf{r}) \psi_{\mathbf{k}_4}(\mathbf{r}') \\
 &= -|g| \delta_{\sigma_3 \neq \sigma_4} \delta_{\sigma_1, \sigma_3} \delta_{\sigma_2, \sigma_4} \int d^3 \mathbf{r} \psi_{\mathbf{k}_1}^*(\mathbf{r}) \psi_{\mathbf{k}_2}^*(\mathbf{r}) \psi_{\mathbf{k}_3}(\mathbf{r}) \psi_{\mathbf{k}_4}(\mathbf{r}) \\
 &= -\frac{|g|}{L^{d_x}} \delta_{\sigma_3 \neq \sigma_4} \delta_{\sigma_1, \sigma_3} \delta_{\sigma_2, \sigma_4} \delta_{\mathbf{k}_1 + \mathbf{k}_2, \mathbf{k}_3 + \mathbf{k}_4}
 \end{aligned}$$

Proof 34.8 of Theorem 28.5 We have computed the matrix-elements and will now insert them into the definition of the second quantized version of V . This yields

$$\begin{aligned}
 V^{2Q} &= \frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \sum_{\sigma_1, \sigma_2, \sigma_3, \sigma_4} \frac{-|g|}{L^{d_x}} \delta_{\sigma_3 \neq \sigma_4} \delta_{\sigma_1, \sigma_3} \delta_{\sigma_2, \sigma_4} \delta_{\mathbf{k}_1 + \mathbf{k}_2, \mathbf{k}_3 + \mathbf{k}_4} c_{\mathbf{k}_1 \sigma_1}^\dagger c_{\mathbf{k}_2 \sigma_2}^\dagger c_{\mathbf{k}_4 \sigma_4} c_{\mathbf{k}_3 \sigma_3} \\
 &= \frac{-|g|}{2L^{d_x}} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \sum_{\sigma_1, \sigma_2} \delta_{\sigma_1 \neq \sigma_2} \delta_{\mathbf{k}_1 + \mathbf{k}_2, \mathbf{k}_3 + \mathbf{k}_4} c_{\mathbf{k}_1 \sigma_1}^\dagger c_{\mathbf{k}_2 \sigma_2}^\dagger c_{\mathbf{k}_4 \sigma_2} c_{\mathbf{k}_3 \sigma_1} \\
 &= \frac{-|g|}{2L^{d_x}} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \delta_{\mathbf{k}_1 + \mathbf{k}_2, \mathbf{k}_3 + \mathbf{k}_4} \left(c_{\mathbf{k}_1 \uparrow}^\dagger c_{\mathbf{k}_2 \downarrow}^\dagger c_{\mathbf{k}_4 \downarrow} c_{\mathbf{k}_3 \uparrow} + c_{\mathbf{k}_1 \downarrow}^\dagger c_{\mathbf{k}_2 \uparrow}^\dagger c_{\mathbf{k}_4 \uparrow} c_{\mathbf{k}_3 \downarrow} \right) \\
 &= \frac{-|g|}{L^{d_x}} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \delta_{\mathbf{k}_1 + \mathbf{k}_2, \mathbf{k}_3 + \mathbf{k}_4} c_{\mathbf{k}_1 \uparrow}^\dagger c_{\mathbf{k}_2 \downarrow}^\dagger c_{\mathbf{k}_4 \downarrow} c_{\mathbf{k}_3 \uparrow}
 \end{aligned}$$

Lastly we need to relabel the summation-indices to obtain the desired form of the Hamiltonian. For this we do the following manipulations

$$\begin{aligned}
 V^{2Q} &= \frac{-|g|}{L^{d_x}} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \delta_{\mathbf{k}_1 + \mathbf{k}_2, \mathbf{k}_3 + \mathbf{k}_4} c_{\mathbf{k}_1 \uparrow}^\dagger c_{\mathbf{k}_2 \downarrow}^\dagger c_{\mathbf{k}_4 \downarrow} c_{\mathbf{k}_3 \uparrow} \\
 &\stackrel{k_3 \leftrightarrow k_4}{=} \frac{-|g|}{L^{d_x}} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \delta_{\mathbf{k}_1 + \mathbf{k}_2, \mathbf{k}_3 + \mathbf{k}_4} c_{\mathbf{k}_1 \uparrow}^\dagger c_{\mathbf{k}_2 \downarrow}^\dagger c_{\mathbf{k}_3 \downarrow} c_{\mathbf{k}_4 \uparrow} \\
 &= \frac{-|g|}{L^{d_x}} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_4} c_{\mathbf{k}_1 \uparrow}^\dagger c_{\mathbf{k}_2 \downarrow}^\dagger c_{\mathbf{k}_1 + \mathbf{k}_2 - \mathbf{k}_4 \downarrow} c_{\mathbf{k}_4 \uparrow} \\
 &\stackrel{k_2 \rightarrow -k_2}{=} \frac{-|g|}{L^{d_x}} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_4} c_{\mathbf{k}_1 \uparrow}^\dagger c_{-\mathbf{k}_2 \downarrow}^\dagger c_{\mathbf{k}_1 - \mathbf{k}_2 - \mathbf{k}_4 \downarrow} c_{\mathbf{k}_4 \uparrow} \\
 &\stackrel{k_2 \equiv k, k_4 \equiv k'}{=} \frac{-|g|}{L^{d_x}} \sum_{\mathbf{k}_1, \mathbf{k}, \mathbf{k}'} c_{\mathbf{k}_1 \uparrow}^\dagger c_{-\mathbf{k} \downarrow}^\dagger c_{\mathbf{k}_1 - \mathbf{k} - \mathbf{k}' \downarrow} c_{\mathbf{k}' \uparrow} \\
 &\stackrel{k_1 \equiv q}{=} \frac{-|g|}{L^{d_x}} \sum_{\mathbf{q}, \mathbf{k}, \mathbf{k}'} c_{\mathbf{q} \uparrow}^\dagger c_{-\mathbf{k} \downarrow}^\dagger c_{\mathbf{q} - \mathbf{k} - \mathbf{k}' \downarrow} c_{\mathbf{k}' \uparrow} \\
 &\stackrel{q \rightarrow q+k}{=} \frac{-|g|}{L^{d_x}} \sum_{\mathbf{q}, \mathbf{k}, \mathbf{k}'} c_{\mathbf{q} + \mathbf{k} \uparrow}^\dagger c_{-\mathbf{k} \downarrow}^\dagger c_{\mathbf{q} - \mathbf{k}' \downarrow} c_{\mathbf{k}' \uparrow}
 \end{aligned}$$

proving the theorem.

Proof 34.9 of Lemma 28.7

See Draft-Summary.

Proof 34.10 of Corollary 28.8

The First Identity

We have the notation

$$\langle \mathbf{x} | \mathbf{q} \rangle \equiv \frac{1}{\sqrt{L^{d_x}}} e^{i\mathbf{q} \cdot \mathbf{x}} \equiv \psi_{\mathbf{q}}(\mathbf{x}) \quad (34.9)$$

for the spatial part of wavefunctions. The only term that was not proven in the [Draft Summary] is the interaction-term.

Hence we compute

$$\begin{aligned} \int d\mathbf{r} c_{\uparrow}^{\dagger}(\mathbf{r}) c_{\downarrow}^{\dagger}(\mathbf{r}) c_{\downarrow}(\mathbf{r}) c_{\uparrow}(\mathbf{r}) &= \sum_{\mathbf{k}_1, \dots, \mathbf{k}_4} \int d\mathbf{r} \bar{\psi}_{\mathbf{k}_1 \uparrow}(\mathbf{r}) \bar{\psi}_{\mathbf{k}_2 \downarrow}(\mathbf{r}) \psi_{\mathbf{k}_3 \downarrow}(\mathbf{r}) \psi_{\mathbf{k}_4 \uparrow}(\mathbf{r}) c_{\uparrow}^{\dagger}(\mathbf{k}_1) c_{\downarrow}^{\dagger}(\mathbf{k}_2) c_{\downarrow}(\mathbf{k}_3) c_{\uparrow}(\mathbf{k}_4) \\ &= \dots \end{aligned}$$

the integral factor can¹ be simplified to

$$\int d\mathbf{r} \bar{\psi}_{\mathbf{k}_1 \uparrow}(\mathbf{r}) \bar{\psi}_{\mathbf{k}_2 \downarrow}(\mathbf{r}) \psi_{\mathbf{k}_3 \downarrow}(\mathbf{r}) \psi_{\mathbf{k}_4 \uparrow}(\mathbf{r}) = \frac{1}{L^{2d_x}} \int d\mathbf{r} e^{i(\mathbf{k}_3 + \mathbf{k}_4 - (\mathbf{k}_1 + \mathbf{k}_2)) \cdot \mathbf{r}} = \frac{1}{L^{d_x}} \delta_{\mathbf{k}_1 + \mathbf{k}_2, \mathbf{k}_3 + \mathbf{k}_4} \quad (34.10)$$

hence we obtain

$$\begin{aligned} \dots &= \frac{1}{L^{d_x}} \sum_{\mathbf{k}_1, \dots, \mathbf{k}_3} c_{\uparrow}^{\dagger}(\mathbf{k}_1) c_{\downarrow}^{\dagger}(\mathbf{k}_2) c_{\downarrow}(\mathbf{k}_3) c_{\uparrow}(\mathbf{k}_1 + \mathbf{k}_2 - \mathbf{k}_3) \\ &\stackrel{\mathbf{k}_2 \rightarrow -\mathbf{k}_2}{=} \frac{1}{L^{d_x}} \sum_{\mathbf{k}_1, \dots, \mathbf{k}_3} c_{\uparrow}^{\dagger}(\mathbf{k}_1) c_{\downarrow}^{\dagger}(-\mathbf{k}_2) c_{\downarrow}(\mathbf{k}_3) c_{\uparrow}(\mathbf{k}_1 - (\mathbf{k}_2 + \mathbf{k}_3)) \\ &\stackrel{\mathbf{k}_2 \rightarrow \mathbf{k}}{=} \frac{1}{L^{d_x}} \sum_{\mathbf{k}_1, \mathbf{k}, \mathbf{k}_3} c_{\uparrow}^{\dagger}(\mathbf{k}_1) c_{\downarrow}^{\dagger}(-\mathbf{k}) c_{\downarrow}(\mathbf{k}_3) c_{\uparrow}(\mathbf{k}_1 - (\mathbf{k} + \mathbf{k}_3)) \\ &\stackrel{\mathbf{k}_1 \rightarrow \mathbf{k}_1 + \mathbf{k}}{=} \frac{1}{L^{d_x}} \sum_{\mathbf{k}_1, \mathbf{k}, \mathbf{k}_3} c_{\uparrow}^{\dagger}(\mathbf{k}_1 + \mathbf{k}) c_{\downarrow}^{\dagger}(-\mathbf{k}) c_{\downarrow}(\mathbf{k}_3) c_{\uparrow}(\mathbf{k}_1 - \mathbf{k}_3) \\ &\stackrel{\mathbf{k}_1 \equiv \mathbf{q}}{=} \frac{1}{L^{d_x}} \sum_{\mathbf{q}, \mathbf{k}, \mathbf{k}_3} c_{\uparrow}^{\dagger}(\mathbf{k} + \mathbf{q}) c_{\downarrow}^{\dagger}(-\mathbf{k}) c_{\downarrow}(\mathbf{k}_3) c_{\uparrow}(\mathbf{q} - \mathbf{k}_3) \\ &\stackrel{\mathbf{k}_3 \rightarrow -\mathbf{k}_3 + \mathbf{q}}{=} \frac{1}{L^{d_x}} \sum_{\mathbf{q}, \mathbf{k}, \mathbf{k}_3} c_{\uparrow}^{\dagger}(\mathbf{k} + \mathbf{q}) c_{\downarrow}^{\dagger}(-\mathbf{k}) c_{\downarrow}(\mathbf{q} - \mathbf{k}_3) c_{\uparrow}(\mathbf{k}_3) \\ &\stackrel{\mathbf{k}_3 \equiv \mathbf{k}'}{=} \frac{1}{L^{d_x}} \sum_{\mathbf{q}, \mathbf{k}, \mathbf{k}'} c_{\uparrow}^{\dagger}(\mathbf{k} + \mathbf{q}) c_{\downarrow}^{\dagger}(-\mathbf{k}) c_{\downarrow}(\mathbf{q} - \mathbf{k}') c_{\uparrow}(\mathbf{k}') \end{aligned}$$

¹Cf. page 64 in Supermathematics.

The Second Identity

TBD.

Proof 34.11 of Corollary 28.12

See Waste-Folder 6, Topic 3 for a proof.

We compute with results from [Draft Summary]

$$\begin{aligned} S_{\text{int}}[\bar{\psi}, \psi] &\equiv \sum_{\omega_{n_1}, \omega_{n_2}, \omega_{n_3}, \omega_{n_4}} \delta_{\omega_{n_1} + \omega_{n_2}, \omega_{n_3} + \omega_{n_4}} \bar{\psi}(\omega_{n_1}) \bar{\psi}(\omega_{n_2}) \psi(\omega_{n_3}) \psi(\omega_{n_4}) \\ &= \dots \end{aligned}$$

now we change the enumeration of the second sum as $\omega_{n_2} \mapsto -\omega_{n_2}$ and obtain hence $\omega_{n_1} = \omega_{n_3} + \omega_{n_4} + \omega_{n_2}$ which gives

$$\begin{aligned} \dots &= \sum_{\omega_{n_1}, \omega_{n_2}, \omega_{n_3}, \omega_{n_4}} \delta_{\omega_{n_1} + \omega_{n_2}, \omega_{n_3} + \omega_{n_4}} \bar{\psi}(\omega_{n_2} + \omega_{n_3} + \omega_{n_4}) \bar{\psi}(\omega_{n_2}) \psi(\omega_{n_3}) \psi(\omega_{n_4}) \\ &= \dots \end{aligned}$$

Now we shift the third summation according to $\omega_{n_3} \mapsto \omega_{n_3} - \omega_{n_4}$ and obtain finally

$$\dots = \sum_{\omega_{n_1}, \omega_{n_2}, \omega_{n_3}, \omega_{n_4}} \delta_{\omega_{n_1} + \omega_{n_2}, \omega_{n_3} + \omega_{n_4}} \bar{\psi}(\omega_{n_2} + \omega_{n_3}) \bar{\psi}(\omega_{n_2}) \psi(\omega_{n_3} - \omega_{n_4}) \psi(\omega_{n_4})$$

Proof 34.12 of Lemma 28.14 See Supermathematics, equation (3.31) and See Waste-Folder 6, topic 5, page 6 for the boundary-conditions.

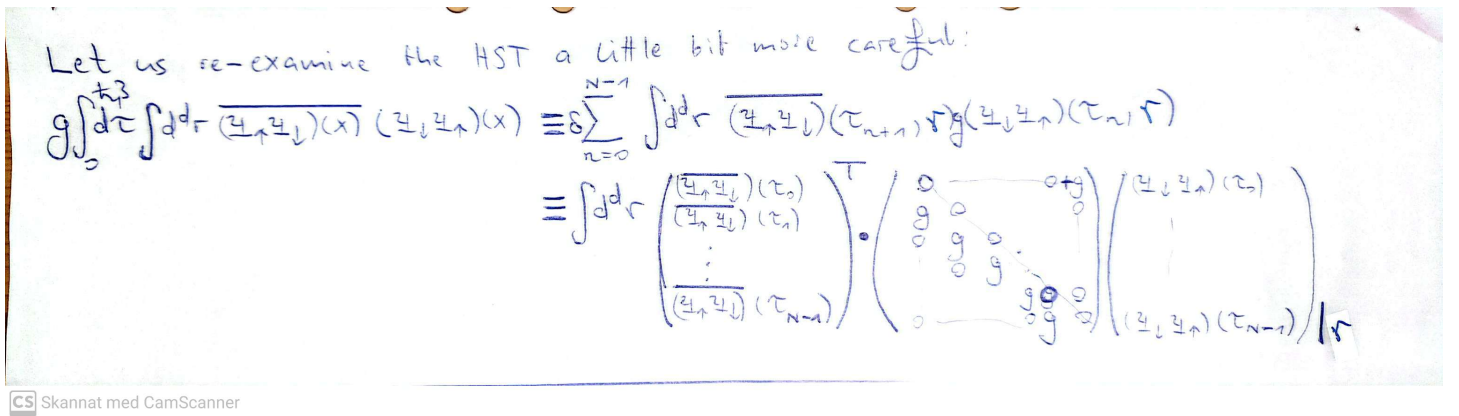


Figure 34.2.: Notes to clarify why periodic boundary-conditions emerge.

Proof 34.13 of Lemma 28.18

TBD

Proof 34.14 of Lemma 28.29

By direct computation, using the von Neumann series

$$\begin{aligned} \mathrm{Tr} \, \mathrm{Log} \, \mathcal{G}_{\mathrm{Gorkov}}^{-1} &\equiv \mathrm{Tr} \, \mathrm{Log} \, \mathcal{G}_{\mathrm{Gorkov}}^{-1} + \mathrm{Tr} \, \mathrm{Log} \left(1 + \mathcal{G}_{\mathrm{Gorkov}} \, {}_0 V \right)^{-1} \\ &\equiv \mathrm{const.} + \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \mathrm{Tr} \left(\mathcal{G}_{\mathrm{Gorkov}} \, {}_0 V \right)^k. \end{aligned}$$

Proof 34.15 of Theorem 28.30

TBD

Proof 34.16 of Corollary 28.22

TBD

Proof 34.17 of Theorem ?? (Linked-Cluster-Theorem)

TBUpdated: Ordering of the indices.

We rewrite the partition-function in its symmetrized form

With the expansion

$$\begin{aligned} \exp -S_{\text{int}}[\xi] &= \sum_{n=0}^{\infty} \frac{(-1/\hbar)^n}{n!} \prod_{\bullet=1}^n \int_0^{\hbar\beta} d\tau_{\bullet} V(\xi(\tau_{\bullet})) \\ &= \sum_{n=0}^{\infty} \frac{(-1/\hbar)^n}{n!} \prod_{\bullet=1}^n \oint V_{\alpha_1^{\bullet}, \dots, \alpha_m^{\bullet}} \xi_{\alpha_1^{\bullet}}(\tau_{\bullet}) \cdots \xi_{\alpha_m^{\bullet}}(\tau_{\bullet}) \end{aligned}$$

one finds

$$\begin{aligned} \mathcal{G}_{\alpha, \alpha'}(\tau) &= \frac{-1}{\hbar} \langle \xi_{\alpha}(\tau) \xi_{\alpha'}(0) \rangle \\ &= \frac{-1}{\hbar} \frac{\sum_{n=0}^{\infty} \frac{(-1/\hbar)^n}{n!} \langle \xi_{\alpha}(\tau) \xi_{\alpha'}(0) \prod_{\bullet=1}^n \oint V_{\alpha_1^{\bullet}, \dots, \alpha_m^{\bullet}} \xi_{\alpha_1^{\bullet}}(\tau_{\bullet}) \cdots \xi_{\alpha_m^{\bullet}}(\tau_{\bullet}) \rangle_0}{\sum_{n=0}^{\infty} \frac{(-1/\hbar)^n}{n!} \langle \prod_{\bullet=1}^n \oint V_{\alpha_1^{\bullet}, \dots, \alpha_m^{\bullet}} \xi_{\alpha_1^{\bullet}}(\tau_{\bullet}) \cdots \xi_{\alpha_m^{\bullet}}(\tau_{\bullet}) \rangle_0} \end{aligned}$$

We will now examine the numerator of this expression and see that it factorizes into two factors, with one being equal to the denominator; This is what Alexander Altland called a „miraculous cancellation mechanism“. It follows by combinatorial counting

$$\begin{aligned} &\text{numerator} \\ &= \sum_{n=0}^{\infty} \frac{(-1/\hbar)^n}{n!} \langle \xi_{\alpha}(\tau) \xi_{\alpha'}(0) \prod_{\bullet=1}^n \oint V_{\alpha_1^{\bullet}, \dots, \alpha_m^{\bullet}} \xi_{\alpha_1^{\bullet}}(\tau_{\bullet}) \cdots \xi_{\alpha_m^{\bullet}}(\tau_{\bullet}) \rangle_0 \\ &= \sum_{n=0}^{\infty} \sum_{p=0}^n \frac{(-1/\hbar)^n}{(n-p)!p!} \langle \xi_{\alpha}(\tau) \xi_{\alpha'}(0) \prod_{\bullet=1}^{n-p} \oint V_{\alpha_1^{\bullet}, \dots, \alpha_m^{\bullet}} \xi_{\alpha_1^{\bullet}}(\tau_{\bullet}) \cdots \xi_{\alpha_m^{\bullet}}(\tau_{\bullet}) \rangle_0^{\text{n.v.}} \langle \prod_{\bullet=1}^p \oint V_{\alpha_1^{\bullet}, \dots, \alpha_m^{\bullet}} \xi_{\alpha_1^{\bullet}}(\tau_{\bullet}) \cdots \xi_{\alpha_m^{\bullet}}(\tau_{\bullet}) \rangle_0 \\ &= \sum_{n=0}^{\infty} \frac{(-1/\hbar)^n}{n!} \langle \xi_{\alpha}(\tau) \xi_{\alpha'}(0) \prod_{\bullet=1}^n \oint V_{\alpha_1^{\bullet}, \dots, \alpha_m^{\bullet}} \xi_{\alpha_1^{\bullet}}(\tau_{\bullet}) \cdots \xi_{\alpha_m^{\bullet}}(\tau_{\bullet}) \rangle_0^{\text{n.v.}} \sum_{p=0}^{\infty} \frac{(-1/\hbar)^p}{p!} \langle \prod_{\bullet=1}^p \oint V_{\alpha_1^{\bullet}, \dots, \alpha_m^{\bullet}} \xi_{\alpha_1^{\bullet}}(\tau_{\bullet}) \cdots \xi_{\alpha_m^{\bullet}}(\tau_{\bullet}) \rangle_0 \end{aligned}$$

and it follows that

$$\mathcal{G}_{\alpha, \alpha'}(\tau) = -\frac{1}{\hbar} \sum_{n=0}^{\infty} \frac{(-1/\hbar)^n}{n!} \langle \xi_{\alpha}(\tau) \xi_{\alpha'}(0) \prod_{\bullet=1}^n \oint V_{\alpha_1^{\bullet}, \dots, \alpha_m^{\bullet}} \xi_{\alpha_1^{\bullet}}(\tau_{\bullet}) \cdots \xi_{\alpha_m^{\bullet}}(\tau_{\bullet}) \rangle_0^{\text{n.v.}} \quad (34.11)$$

ToBeRedone with Graph Theory, Cf. the book from Salmhofer

Part VI.

"Experimental" (Work in Progresss)

35. Experimental: The Internal Energy using the full Bethe-Salpeter-Equation in the Tree-Level-Approximation

35.1. Correcting the Internal Energy for the Phononic Contribution

We will now elaborate how we can compute the correction-terms from the Bethe-Salpeter-equation in the specific heat calculation

$$\begin{aligned}
 \mathcal{G}_{0,v}^2(q) \hbar^2 \langle \rho_v(q) \rho_v(-q) \rangle &= \frac{\mathcal{G}_{0,v}^2(q)}{\hbar \beta N_p} \sum_{\substack{\sigma, \sigma' \in \{\downarrow, \uparrow\} \\ k, k' \in i\Omega_{-1} \times Q \\ n, n', m, m' \in \{1, \dots, \infty\}}} g_{mnv}(\mathbf{k}, \mathbf{q}) g_{m'n'v}(\mathbf{k}', -\mathbf{q}) \langle \bar{\psi}_{m\sigma}(k+q) \psi_{n\sigma}(k) \bar{\psi}_{m'\sigma'}(k'-q) \psi_{n'\sigma'}(k') \rangle \\
 &= \frac{\mathcal{G}_{0,v}^2(q)}{\hbar \beta N_p} \sum_{\substack{\sigma, \sigma' \in \{\downarrow, \uparrow\} \\ k, k' \in i\Omega_{-1} \times Q \\ n, n', m, m' \in \{1, \dots, \infty\}}} g_{mnv}(\mathbf{k}, \mathbf{q}) g_{m'n'v}(\mathbf{k}', -\mathbf{q}) \langle \psi_{n\sigma}(k) \psi_{n'\sigma'}(k') \bar{\psi}_{m'\sigma'}(k'-q) \bar{\psi}_{m\sigma}(k+q) \rangle \\
 &= \dots
 \end{aligned} \tag{35.1}$$

We focus on the contribution that we neglected earlier (from the Bethe-Salpeter-equation **TBReferenced** and Equation **Exp. Value as GF**) and compute

$$\begin{aligned}
 &\langle \psi_{n\sigma}(k) \psi_{n'\sigma'}(k') \bar{\psi}_{m'\sigma'}(k'-q) \bar{\psi}_{m\sigma}(k+q) \rangle \\
 &= \hbar^2 \mathcal{G} (n\sigma k, n'\sigma' k' | m\sigma k+q, m'\sigma' k'-q) \\
 &= \text{"Wick-Theorem-contributions"} \\
 &= \hbar^2 \sum_{\substack{\sigma_1, \sigma_2, \sigma'_1, \sigma'_2 \in \{\downarrow, \uparrow\} \\ n_1, n_2, n'_1, n'_2 \in \{1, \dots, \infty\} \\ k_1, k_2, k'_1, k'_2 \in i\Omega_{-1} \times Q}} \mathcal{G}_{n, n_1}^0 \mathcal{G}_{\sigma, \sigma_1}^0 \mathcal{G}_{\sigma', \sigma'_1}^0 V_{n_1, n_2, n'_1, n'_2} \mathcal{G}_{\sigma_1, \sigma_2, \sigma'_1, \sigma'_2}^0 \mathcal{G}_{k_1, k_2, k'_1, k'_2}^0 \mathcal{G}_{\sigma'_1, \sigma}^0 \mathcal{G}_{\sigma'_2, \sigma'}^0
 \end{aligned} \tag{35.2}$$

inserting this into the equation above yields

$$\begin{aligned}
 \dots &= -\hbar^2 \frac{\mathcal{G}_{0,v}^2(q)}{\hbar \beta N_p} \sum_{\substack{\sigma, \sigma' \in \{\downarrow, \uparrow\} \\ k, k' \in i\Omega_{-1} \times Q \\ n, n', m, m' \in \{1, \dots, \infty\}}} g_{mnv}(\mathbf{k}, \mathbf{q}) g_{m'n'v}(\mathbf{k}', -\mathbf{q}) \sum_{\substack{\sigma_1, \sigma_2, \sigma'_1, \sigma'_2 \in \{\downarrow, \uparrow\} \\ n_1, n_2, n'_1, n'_2 \in \{1, \dots, \infty\} \\ k_1, k_2, k'_1, k'_2 \in i\Omega_{-1} \times Q}} \mathcal{G}_{n, n_1}^0 \mathcal{G}_{\sigma, \sigma_1}^0 \mathcal{G}_{\sigma', \sigma'_1}^0 V_{n_1, n_2, n'_1, n'_2}^{\text{as}} \mathcal{G}_{\sigma_1, \sigma_2, \sigma'_1, \sigma'_2}^0 \mathcal{G}_{k_1, k_2, k'_1, k'_2}^0 \mathcal{G}_{\sigma'_1, \sigma}^0 \mathcal{G}_{\sigma'_2, \sigma'}^0 \\
 &= \dots
 \end{aligned} \tag{35.3}$$

now we terminate the summations over $n_1, n_2, n'_1, n'_2, \sigma_1, \sigma_2, \sigma'_1, \sigma'_2$ and k_1, k_2, k'_1, k'_2 and obtain

$$\dots = \frac{-\hbar \mathcal{G}_{0,v}^2(q)}{\beta N_p} \sum_{\substack{\sigma, \sigma' \in \{\downarrow, \uparrow\} \\ k, k' \in i\Omega_{-1} \times Q \\ n, n', m, m' \in \{1, \dots, \infty\}}} g_{mnv}(\mathbf{k}, \mathbf{q}) g_{m'n'v}(\mathbf{k}', -\mathbf{q}) \mathcal{G}_{0,v} \mathcal{G}_{0,v} V_{n,n',m,m'}^{\text{as}} \mathcal{G}_{0,v} \mathcal{G}_{0,v} \quad (35.4)$$

In order to get the total contribution, we sum over v and q and multiply with $1/(\hbar\beta)$ obtain

$$\begin{aligned} & \text{"BS-Correction-Terms"} \\ &= \frac{1}{\hbar\beta} \sum_{\substack{v \in \{1, \dots, rd_x\} \\ q \in i\Omega_{+1} \times Q}} \frac{-\hbar \mathcal{G}_{0,v}^2(q)}{\beta N_p} \sum_{\substack{\sigma, \sigma' \in \{\downarrow, \uparrow\} \\ k, k' \in i\Omega_{-1} \times Q \\ n, n', m, m' \in \{1, \dots, \infty\}}} g_{mnv}(\mathbf{k}, \mathbf{q}) g_{m'n'v}(\mathbf{k}', -\mathbf{q}) \mathcal{G}_{0,v} \mathcal{G}_{0,v} V_{n,n',m,m'}^{\text{as}} \mathcal{G}_{0,v} \mathcal{G}_{0,v} \\ &= \frac{-1}{\beta^2 N_p} \sum_{\substack{\sigma, \sigma' \in \{\downarrow, \uparrow\} \\ v \in \{1, \dots, rd_x\} \\ q \in i\Omega_{+1} \times Q \\ k, k' \in i\Omega_{-1} \times Q \\ n, n', m, m' \in \{1, \dots, \infty\}}} g_{mnv}(\mathbf{k}, \mathbf{q}) g_{m'n'v}(\mathbf{k}', -\mathbf{q}) \mathcal{G}_{0,v}^2(q) \mathcal{G}_{0,v} \mathcal{G}_{0,v} V_{n,n',m,m'}^{\text{as}} \mathcal{G}_{0,v} \mathcal{G}_{0,v} \end{aligned} \quad (35.5)$$

Now we use Equation (21.18) and compute the antisymmetrized part of the vertex as ...

$$\begin{aligned} & V_{\alpha_1, \alpha_2; \alpha_3, \alpha_4}^{\text{as}} \\ &= \text{AS} \frac{1}{\hbar\beta N_p} \sum_{v \in \{1, \dots, rd_x\}} \frac{\omega_v(\mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4})}{\omega_{\alpha_1 - \alpha_4}^2 + \omega_v^2(\mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4})} \frac{g_{m_{\alpha_1} m_{\alpha_4} v}(\mathbf{k}_{\alpha_4}, \mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4}) g_{m_{\alpha_3} m_{\alpha_2} v}^*(\mathbf{k}_{\alpha_2}, \mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4})}{\hbar^2} \delta_{k_{\alpha_3} + k_{\alpha_4}, k_{\alpha_2} + k_{\alpha_1}} \delta_{\sigma_{\alpha_1}, \sigma_{\alpha_4}} \delta_{\sigma_{\alpha_2}, \sigma_{\alpha_3}} \\ &= \text{AS} \frac{1}{\hbar\beta N_p} \sum_{v \in \{1, \dots, rd_x\}} \frac{\omega_v(\mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4})}{\omega_{\alpha_1 - \alpha_4}^2 + \omega_v^2(\mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4})} \frac{g_{m_{\alpha_1} m_{\alpha_4} v}(\mathbf{k}_{\alpha_4}, \mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4}) g_{m_{\alpha_2} m_{\alpha_3} v}(\mathbf{k}_{\alpha_2} + \mathbf{k}_{\alpha_1} - \mathbf{k}_{\alpha_4}, -\mathbf{k}_{\alpha_1} + \mathbf{k}_{\alpha_4})}{\hbar^2} \delta_{k_{\alpha_3} + k_{\alpha_4}, k_{\alpha_2} + k_{\alpha_1}} \delta_{\sigma_{\alpha_1}, \sigma_{\alpha_4}} \delta_{\sigma_{\alpha_2}, \sigma_{\alpha_3}} \end{aligned} \quad (35.6)$$

35.2. Correcting the Internal Energy for the Interacting Contribution

TBD.

36. Experimental: More Algebraic Concepts in Quantum Field Theory

36.1. Hubbard–Stratonovich Transformations and Fierz Identities

The purpose of this section is to formulate the Hubbard–Stratonovich transformation and the Fierz identities in a form in which every sign, normalization, and index permutation is visible. The two constructions are closely related, but they should not be confused. A Hubbard–Stratonovich transformation is an exact Gaussian integral identity. A Fierz identity is an exact change of basis in the finite-dimensional vector space of invariant tensors. The first introduces an auxiliary field; the second rewrites the same quartic polynomial in a different contraction channel.

The distinction is particularly important for fermions. The elementary variables anticommute, whereas a fermionic bilinear is even and consequently commutes with every other even bilinear. Most apparent disagreements in the literature can be traced to one of three choices: the order in which the Grassmann variables are written, whether a factor $1/2$ is absorbed into a kernel, and the contour used for the auxiliary-field integral.

36.1.1. Algebraic setting and conventions

Notation 36.1 Throughout this section, $\mathcal{I}$ denotes a finite set of one-particle labels and $D \equiv |\mathcal{I}|$. A label may contain position, imaginary time, momentum, orbital, flavour, or spin. Repeated labels are summed unless a displayed sum is written explicitly. The fermionic fields are

$$\psi = (\psi_a)_{a \in \mathcal{I}}, \quad \bar{\psi} = (\bar{\psi}_a)_{a \in \mathcal{I}}, \quad \psi_a \psi_b = -\psi_b \psi_a, \quad \bar{\psi}_a \psi_b = -\psi_b \bar{\psi}_a. \quad (36.1)$$

Every matrix acting on $\mathcal{I}$ is an ordinary complex matrix and therefore commutes with the Grassmann algebra. We write tr for the finite-dimensional trace and use $i^2 = -1$.

Definition 36.2 [Grassmann parity] A polynomial in the fermionic variables is called *even* if every monomial contains an even number of Grassmann generators, and *odd* if every monomial contains an odd number. The parity is denoted by $|F| \in \mathbb{Z}_2$. Homogeneous polynomials satisfy the graded commutation rule

$$FG = (-1)^{|F||G|}GF. \quad (36.2)$$

In particular, all density and pairing composites used below are even and may be treated as commuting sources in an ordinary bosonic Gaussian integral.

Lemma 36.3 [The elementary fermionic reordering identity] For arbitrary labels $a, b, c, d \in \mathcal{I}$,

$$(\bar{\psi}_a \psi_b)(\bar{\psi}_c \psi_d) = -\bar{\psi}_a \bar{\psi}_c \psi_b \psi_d = -(\bar{\psi}_a \psi_d)(\bar{\psi}_c \psi_b). \quad (36.3)$$

Proof 36.3 Move ψ_b through $\bar{\psi}_c$ once. This produces the first minus sign. To obtain the last expression, move ψ_d through ψ_b and then move it through the two barred variables in the reverse comparison; equivalently, expand the last product directly:

$$(\bar{\psi}_a \psi_d)(\bar{\psi}_c \psi_b) = -\bar{\psi}_a \bar{\psi}_c \psi_d \psi_b = +\bar{\psi}_a \bar{\psi}_c \psi_b \psi_d.$$

Hence the original product is the negative of the exchanged product.

Definition 36.4 [Quartic interaction tensor] A general fermionic two-body interaction can be written as

$$S_4[\bar{\psi}, \psi] \equiv \frac{1}{4} \bar{\psi}_a \bar{\psi}_b U_{ab;cd} \psi_d \psi_c. \quad (36.4)$$

The factor $1/4 = (2!)^{-2}$ is convenient when the tensor is summed over unrestricted labels. The physically relevant part of the interaction tensor is antisymmetric in the incoming and outgoing pairs:

$$U_{ba;cd} = -U_{ab;cd}, \quad U_{ab;dc} = -U_{ab;cd}. \quad (36.5)$$

No generality is lost by imposing these relations, because only the antisymmetric projection of an arbitrary tensor contributes to (36.4).

Theorem 36.5 [Antisymmetrization of a quartic kernel] Let $\tilde{U}_{ab;cd}$ be an arbitrary four-index tensor and define

$$U_{ab;cd} \equiv \frac{1}{4} \left(\tilde{U}_{ab;cd} - \tilde{U}_{ba;cd} - \tilde{U}_{ab;dc} + \tilde{U}_{ba;dc} \right). \quad (36.6)$$

Then U satisfies (36.5) and

$$\bar{\psi}_a \bar{\psi}_b \tilde{U}_{ab;cd} \psi_d \psi_c = \bar{\psi}_a \bar{\psi}_b U_{ab;cd} \psi_d \psi_c. \quad (36.7)$$

Proof 36.5 Interchanging a and b changes the sign of $\bar{\psi}_a \bar{\psi}_b$, and interchanging c and d changes the sign of $\psi_d \psi_c$. Averaging over both permutations therefore leaves the quartic polynomial invariant and produces precisely the four terms in the definition of U . Applying either permutation to that expression proves the two antisymmetry relations.

36.1.2. The finite-dimensional Hubbard–Stratonovich identity

Definition 36.6 [Normalized real Gaussian measure] Let $V \in \text{Mat}_{N \times N}(\mathbb{R})$ be real, symmetric, and positive definite. For a commuting vector $\varphi \in \mathbb{R}^N$, define

$$\int d\mu_V(\varphi) F(\varphi) \equiv \frac{1}{\sqrt{\det(2\pi V)}} \int_{\mathbb{R}^N} d^N \varphi \exp\left(-\frac{1}{2} \varphi^\top V^{-1} \varphi\right) F(\varphi). \quad (36.8)$$

Thus $\int d\mu_V = 1$. The transpose in this section acts only on the finite label space; Grassmann conjugation is not being performed by $\top$.

Lemma 36.7 [The ordinary Gaussian generating function] For every commuting source $J \in \mathbb{C}^N$,

$$\int d\mu_V(\varphi) \exp(J^\top \varphi) = \exp\left(\frac{1}{2} J^\top V J\right). \quad (36.9)$$

Proof 36.7 Complete the square:

$$-\frac{1}{2}\varphi^\top V^{-1}\varphi + J^\top \varphi = -\frac{1}{2}(\varphi - VJ)^\top V^{-1}(\varphi - VJ) + \frac{1}{2}J^\top VJ. \quad (36.10)$$

Translation invariance of the Lebesgue integral removes the shift VJ , and the remaining normalized Gaussian integral equals one.

Theorem 36.8 [Real Hubbard–Stratonovich transformation] Let V be as in (36.8), let $\lambda > 0$, and let $\rho \in \mathbb{C}^N$ be a vector of mutually commuting quantities. Then

$$\exp\left(-\frac{\lambda}{2}\rho^\top V\rho\right) = \int d\mu_{\lambda V}(\varphi) \exp(-i\varphi^\top \rho) = \frac{1}{\sqrt{\det(2\pi\lambda V)}} \int_{\mathbb{R}^N} d^N\varphi \exp\left(-\frac{1}{2\lambda}\varphi^\top V^{-1}\varphi - i\varphi^\top \rho\right). \quad (36.11)$$

In particular, if ρ is a vector of fermionic density bilinears, the quartic expression on the left is replaced by a quadratic fermion–auxiliary-field coupling on the right.

Proof 36.8 Apply the Gaussian identity (36.9) to the covariance λV and the source $J = -i\rho$. Since $(-i)^2 = -1$, one obtains

$$\int d\mu_{\lambda V}(\varphi) \exp(-i\rho^\top \varphi) = \exp\left(-\frac{\lambda}{2}\rho^\top V\rho\right).$$

Rewriting the normalized measure $d\mu_{\lambda V}$ in terms of $d^N\varphi$ gives exactly (36.11).

Corollary 36.9 [The shift form of the transformation] Under the assumptions of the theorem, the same identity follows from

$$-\frac{1}{2\lambda}\varphi^\top V^{-1}\varphi - i\varphi^\top \rho = -\frac{1}{2\lambda}(\varphi + i\lambda V\rho)^\top V^{-1}(\varphi + i\lambda V\rho) - \frac{\lambda}{2}\rho^\top V\rho. \quad (36.12)$$

The formal stationary auxiliary field is therefore

$$\varphi_{\text{stat}} = -i\lambda V\rho. \quad (36.13)$$

Proof 36.9 Expand the right-hand side of (36.12). The cross term is $-i\varphi^\top \rho$ and the constant term is $-\lambda\rho^\top V\rho/2$. Differentiating the exponent before the shift with respect to φ gives $-V^{-1}\varphi/\lambda - i\rho = 0$, which is (36.13).

Remark 36.9.1 [Why the factor i appears] For a positive interaction kernel V , the real Gaussian has a decaying weight only when the quartic term on the left of (36.11) has a minus sign. The factor i is what turns the positive Gaussian covariance into a negative quadratic form for the fermionic composite. If the desired quartic term has the opposite sign, the coupling to the auxiliary field can be chosen real. Thus an imaginary Yukawa coupling is not an error; it records the choice of convergent contour and the sign of the interaction.

Theorem 36.10 [Hubbard–Stratonovich transformation for a regulated functional integral] Let x denote a finite regulator of all space, time, and internal labels, and let $V(x, y)$ be an invertible positive kernel. For an even composite field $\rho(x)$,

$$\exp\left[-\frac{\lambda}{2}\int dx dy \rho(x)V(x, y)\rho(y)\right] = \mathcal{N}_{\lambda V} \int \mathcal{D}\varphi \exp\left[-\frac{1}{2\lambda}\int dx dy \varphi(x)V^{-1}(x, y)\varphi(y) - i\int dx \varphi(x)\rho(x)\right]. \quad (36.14)$$

Here $\mathcal{N}_{\lambda V}$ is fixed by requiring the identity to equal one for $\rho = 0$. The continuum formula is defined by taking the regulator away after the finite-dimensional identity has been established.

Proof 36.10 Replace the integrals by sums on the regulated set of labels. Equation (36.14) is then exactly the finite-dimensional identity (36.11). The continuum notation is a compact notation for this sequence of finite-dimensional identities; no additional algebraic step is involved.

Definition 36.11 [Exact auxiliary-field representation] Suppose an action has the form

$$S[\bar{\psi}, \psi] = S_0[\bar{\psi}, \psi] + \frac{\lambda}{2} \rho^\top V \rho. \quad (36.15)$$

An *exact Hubbard–Stratonovich representation* of its partition function is the equality

$$Z = \mathcal{N}_{\lambda V} \int \mathcal{D}\varphi \mathcal{D}(\bar{\psi}, \psi) \exp \left[-S_0[\bar{\psi}, \psi] - \frac{1}{2\lambda} \varphi^\top V^{-1} \varphi - i\varphi^\top \rho \right]. \quad (36.16)$$

The auxiliary field is bosonic, even when ρ is constructed from fermions.

Theorem 36.12 [Exactness and reversibility] The transformation (36.16) is reversible: integrating out φ gives back (36.15), while keeping φ and integrating out the fermions gives an exact bosonic effective action.

Proof 36.12 The first statement is the Gaussian identity (36.11). For the second, define

$$\exp[-S_{\text{eff}}[\varphi]] \equiv \mathcal{N}_{\lambda V} \exp \left(-\frac{1}{2\lambda} \varphi^\top V^{-1} \varphi \right) \int \mathcal{D}(\bar{\psi}, \psi) \exp [-S_0 - i\varphi^\top \rho]. \quad (36.17)$$

Substitution into the partition function produces $Z = \int \mathcal{D}\varphi \exp[-S_{\text{eff}}[\varphi]]$. Every equality is an integral identity, so no approximation has yet been made.

36.1.3. Density, exchange, and Cooper channels

Definition 36.13 [Particle–hole and particle–particle composites] For a fermionic field with internal labels $a, b \in \mathcal{I}$, define the particle–hole composite

$$\rho_{ab} \equiv \bar{\psi}_a \psi_b. \quad (36.18)$$

The same composite can be called a *direct* density when the kernel contracts (a, b) with the corresponding incoming pair, or an *exchange* density when its kernel contracts crossed labels. The particle–particle composites are

$$P_{ab} \equiv \psi_a \psi_b = -P_{ba}, \quad \bar{P}_{ab} \equiv \bar{\psi}_b \bar{\psi}_a = -\bar{P}_{ba}. \quad (36.19)$$

A density-channel interaction and a Cooper-channel interaction are consequently written as

$$S_{\text{ph}} = \frac{1}{2} \rho_{ab} V_{ab;cd} \rho_{cd}, \quad (36.20)$$

$$S_{\text{pp}} = \frac{1}{2} \bar{P}_{ab} W_{ab;cd} P_{cd}, \quad (36.21)$$

where W is understood as a kernel on the antisymmetric pair space.

Lemma 36.14 [Pair-space factors] If $W_{ab;cd}$ is antisymmetric in both pairs, then

$$\frac{1}{2} \bar{P}_{ab} W_{ab;cd} P_{cd} = -\frac{1}{8} \bar{\psi}_a \bar{\psi}_b (W_{ab;cd} - W_{ba;cd} - W_{ab;dc} + W_{ba;dc}) \psi_c \psi_d. \quad (36.22)$$

The factor $1/2$ removes the double counting of each antisymmetric pair. If unrestricted four-index sums are used instead, the factor $1/4$ in (36.4) is the equivalent convention.

Proof 36.14 Insert $P_{cd} = \psi_c \psi_d$ and $\bar{P}_{ab} = \bar{\psi}_b \bar{\psi}_a$, then average over $a \leftrightarrow b$ and $c \leftrightarrow d$. The two antisymmetry relations produce the four displayed terms. Since the pair tensor has two equivalent orderings, precisely one factor $1/2$ is removed.

Theorem 36.15 [Channel-wise Hubbard–Stratonovich decoupling] Let ρ_I be any chosen finite list of even composites, and let V_{IJ} be an invertible symmetric kernel. If

$$S_{\text{int}} = \frac{\lambda}{2} \rho_I V_{IJ} \rho_J, \quad (36.23)$$

then the direct, exchange, or any other chosen channel has the exact decoupling

$$\exp(-S_{\text{int}}) = \mathcal{N}_{\lambda V} \int \mathcal{D}\varphi_I \exp \left[-\frac{1}{2\lambda} \varphi_I (V^{-1})_{IJ} \varphi_J - i \varphi_I \rho_I \right]. \quad (36.24)$$

For a Cooper channel one uses the complex version below, with $\rho_I = P_I$ and $\bar{\rho}_I = \bar{P}_I$.

Proof 36.15 Regard the composite index I as one ordinary finite-dimensional index and apply the identity (36.11). The proof does not depend on whether I originated from a density, an exchange, or an antisymmetric pair of labels.

Definition 36.16 [Normalized complex Gaussian measure] Let V be Hermitian and positive definite on $\mathbb{C}^N$. For a complex auxiliary field $\Delta \in \mathbb{C}^N$, write

$$\int d\mu_V^{\mathbb{C}}(\Delta) F(\Delta, \bar{\Delta}) \equiv \frac{\det(V^{-1})}{\pi^N} \int d^{2N} \Delta \exp \left(-\bar{\Delta}^{\text{T}} V^{-1} \Delta \right) F(\Delta, \bar{\Delta}). \quad (36.25)$$

Theorem 36.17 [Complex Hubbard–Stratonovich transformation] For even complex composites $P, \bar{P}$ and $\lambda > 0$,

$$\exp \left(-\lambda \bar{P}^{\text{T}} V P \right) = \int d\mu_V^{\mathbb{C}}(\Delta) \exp \left[i \sqrt{\lambda} \left(\bar{\Delta}^{\text{T}} P + \bar{P}^{\text{T}} \Delta \right) \right]. \quad (36.26)$$

Without the factors i , the right-hand side gives $\exp(+\lambda \bar{P}^{\text{T}} V P)$ instead.

Proof 36.17 The complex Gaussian generating function is

$$\int d\mu_V^{\mathbb{C}}(\Delta) \exp \left(\bar{\eta}^{\text{T}} \Delta + \bar{\Delta}^{\text{T}} \eta \right) = \exp \left(\bar{\eta}^{\text{T}} V \eta \right). \quad (36.27)$$

Set $\eta = i\sqrt{\lambda}P$ and $\bar{\eta} = i\sqrt{\lambda}\bar{P}$. Their product is $-\lambda\bar{P}^\top VP$, which proves (36.26). The same result follows by shifting $\Delta \mapsto \Delta - i\sqrt{\lambda}VP$ and $\bar{\Delta} \mapsto \bar{\Delta} - i\sqrt{\lambda}\bar{P}V$.

Remark 36.17.1 [The three standard channels] In the direct particle–hole channel one chooses $\rho_{ab} = \bar{\psi}_a\psi_b$ and a kernel $V_{ab;cd}$. In the exchange particle–hole channel the same bilinear is used but the kernel is written with crossed outgoing labels, for example $V_{ab;dc}$. In the Cooper channel one chooses $P_{ab} = \psi_a\psi_b$ and its barred partner. These are not three different interactions. They are three coordinate systems for the same four-fermion tensor, related by permutations and, after a basis is chosen, by Fierz transformations.

36.1.4. Fierz identities as completeness identities

Definition 36.18 [Dual matrix bases] Let $\{\Gamma^A\}_{A=1}^{D^2}$ be a basis of $\text{Mat}_{D \times D}(\mathbb{C})$. Its dual basis $\{\Gamma_A\}$ is defined by

$$\text{tr}(\Gamma_A \Gamma^B) = \delta_A^B. \quad (36.28)$$

Every matrix M then has the expansion

$$M = \Gamma^A \text{tr}(\Gamma_A M). \quad (36.29)$$

We also define the bilinear associated with a matrix M by

$$B_M \equiv \bar{\psi} M \psi = \bar{\psi}_a M_{ab} \psi_b. \quad (36.30)$$

Lemma 36.19 [Completeness relation] A dual pair of matrix bases satisfies

$$\Gamma_{ij}^A (\Gamma_A)_{kl} = \delta_{il} \delta_{jk}. \quad (36.31)$$

Equivalently, the crossed products $\Gamma_{ad}^A \Gamma_{cb}^B$ form a basis of four-index tensors.

Proof 36.19 Choose the matrix-unit basis E_{ij} as a temporary basis. Its dual basis is E_{ji} and

$$\sum_{i,j} (E_{ij})_{mn} (E_{ji})_{pq} = \sum_{i,j} \delta_{mi} \delta_{nj} \delta_{pj} \delta_{qi} = \delta_{mq} \delta_{np}.$$

The relation is basis-independent because changing the matrix basis and changing the dual basis by the inverse transformation leaves the contraction unchanged.

Theorem 36.20 [General fermionic Fierz identity] For arbitrary matrices $M, N \in \text{Mat}_{D \times D}(\mathbb{C})$,

$$B_M B_N = - \sum_{A,B} \text{tr}(\Gamma_A M \Gamma_B N) B_{\Gamma^A} B_{\Gamma^B}. \quad (36.32)$$

Proof 36.20 Expand the four-index coefficient in the crossed basis using completeness:

$$M_{ab} N_{cd} = \sum_{A,B} C_{AB} (\Gamma^A)_{ad} (\Gamma^B)_{cb}. \quad (36.33)$$

Contract both sides with $(\Gamma_A)_{da}(\Gamma_B)_{bc}$ and sum all one-particle labels. The coefficient is

$$C_{AB} = (\Gamma_A)_{da} M_{ab} (\Gamma_B)_{bc} N_{cd} = \text{tr}(\Gamma_A M \Gamma_B N).$$

Multiplying (36.33) by $\bar{\psi}_a \psi_b \bar{\psi}_c \psi_d$ and using the elementary reordering identity (36.3) produces one minus sign and yields (36.32).

Corollary 36.21 [Fierz transformation of a four-index tensor] Let $T_{ab;cd}$ be a particle–hole tensor. Its exchange-channel representative is obtained by the index permutation

$$(\mathcal{F}T)_{ad;cb} \equiv T_{ab;cd}. \quad (36.34)$$

In a matrix basis, the coefficients of this permutation are the traces

$$\mathcal{F}_{AB;CD} = \text{tr}(\Gamma_A \Gamma^C \Gamma_B \Gamma^D). \quad (36.35)$$

The Fierz map is an involution:

$$\mathcal{F}^2 = \text{id}. \quad (36.36)$$

Proof 36.21 Equation (36.35) is the coefficient formula in the proof of the Fierz identity (36.32). The permutation interchanges the two crossed labels twice when it is applied twice, so it returns $T_{ab;cd}$. This proves (36.36) without choosing a basis.

Remark 36.21.1 [What a Fierz identity does not do] A Fierz identity does not change the interaction and does not select a phase. It only changes the coordinates used to describe a fixed tensor. Selecting a density, spin, or pairing decoupling becomes a physical approximation only after an operation such as a saddle-point approximation, a restricted ansatz, or a truncation of the effective action has been made.

36.1.5. Explicit Fierz examples

Theorem 36.22 [Pauli completeness] For a two-component spinor, let σ^i be the Pauli matrices and let σ^0 be the 2×2 identity. Then

$$\delta_{ab} \delta_{cd} = \frac{1}{2} (\delta_{ad} \delta_{cb} + \sigma_{ad}^i \sigma_{cb}^i), \quad \sigma_{ab}^i \sigma_{cd}^i = 2\delta_{ad} \delta_{cb} - \delta_{ab} \delta_{cd}. \quad (36.37)$$

Proof 36.22 Use the orthonormal basis $\Gamma^0 = \sigma^0/\sqrt{2}$ and $\Gamma^i = \sigma^i/\sqrt{2}$, for which $\text{tr}(\Gamma^A \Gamma^B) = \delta^{AB}$. The general completeness relation (36.31) gives the first identity. Multiplying that identity by two and solving for the Pauli term gives the second one.

Corollary 36.23 [Spin-density Fierz identity] Define the charge and spin bilinears

$$n \equiv \bar{\psi} \psi, \quad s^i \equiv \bar{\psi} \sigma^i \psi. \quad (36.38)$$

In the Grassmann action convention used here,

$$n^2 = -\frac{1}{2} (n^2 + s^i s^i), \quad s^i s^i = -3n^2. \quad (36.39)$$

Proof 36.23 Apply the Fierz identity (36.32) with $M = N = \sigma^0$ and the orthonormal Pauli basis. The trace coefficient is δ_{AB} , while $B_{\Gamma^0} = n/\sqrt{2}$ and $B_{\Gamma^i} = s^i/\sqrt{2}$. Thus

$$n^2 = -\left[\left(\frac{n}{\sqrt{2}} \right)^2 + \sum_i \left(\frac{s^i}{\sqrt{2}} \right)^2 \right],$$

which is (36.39).

Definition 36.24 [Two-spin Hubbard monomial] For $\psi = (\psi_\uparrow, \psi_\downarrow)$, define

$$Q \equiv \bar{\psi}_\uparrow \bar{\psi}_\downarrow \psi_\downarrow \psi_\uparrow, \quad P \equiv \psi_\downarrow \psi_\uparrow, \quad \bar{P} \equiv \bar{\psi}_\uparrow \bar{\psi}_\downarrow. \quad (36.40)$$

Theorem 36.25 [The Hubbard interaction in several exact forms] With $n = \bar{\psi}\psi$ and $s^i = \bar{\psi}\sigma^i\psi$,

$$Q = \frac{1}{2} n^2 = -\frac{1}{6} s^i s^i = \bar{P} P. \quad (36.41)$$

More generally, for every $\alpha \in \mathbb{C}$,

$$Q = \frac{\alpha}{2} n^2 - \frac{1-\alpha}{6} s^i s^i. \quad (36.42)$$

Proof 36.25 Put $q_\uparrow = \bar{\psi}_\uparrow \psi_\uparrow$ and $q_\downarrow = \bar{\psi}_\downarrow \psi_\downarrow$. Since $q_\uparrow^2 = q_\downarrow^2 = 0$ and the q 's are even,

$$n^2 = (q_\uparrow + q_\downarrow)^2 = 2q_\uparrow q_\downarrow = 2Q.$$

The spin-density identity gives $s^i s^i = -3n^2 = -6Q$. Finally,

$$\bar{P} P = \bar{\psi}_\uparrow \bar{\psi}_\downarrow \psi_\downarrow \psi_\uparrow = Q.$$

Equation (36.42) is an arbitrary linear combination of the first two equalities.

Remark 36.25.1 [The Fierz parameter and the Fierz ambiguity] Equation (36.42) is an exact one-parameter family. In the exact functional integral every value of α gives the same partition function after the corresponding auxiliary fields are integrated out. A mean-field ansatz that keeps only a charge field, only a spin field, or only a pairing field generally gives an α -dependent result. This dependence is called the Fierz ambiguity of the approximation, not an ambiguity of the original Hubbard interaction.

Theorem 36.26 [Spinless fermions have no local quartic interaction] If $D = 1$ and the interaction is local in the one-particle label, then every local quartic monomial vanishes:

$$\bar{\psi} \bar{\psi} \psi \psi = 0. \quad (36.43)$$

Proof 36.26 The Grassmann relations imply $\psi^2 = 0$ and $\bar{\psi}^2 = 0$. Hence the displayed monomial contains two identical generators of each type and vanishes. A nonzero local quartic interaction requires at least two distinguishable internal states, or a nonlocal label such as two different orbitals or sites.

Theorem 36.27 [The $SU(N)$ charge-generator identity] Let T^a , $a = 1, \dots, N^2 - 1$, be the generators of $SU(N)$ in the fundamental representation, normalized by

$$\text{tr}(T^a T^b) = \frac{1}{2} \delta^{ab}. \quad (36.44)$$

For $n = \bar{\psi}_i \psi_i$ and $J^a = \bar{\psi}_i T_{ij}^a \psi_j$,

$$\delta_{ij} \delta_{kl} = \frac{1}{N} \delta_{il} \delta_{kj} + 2 T_{il}^a T_{kj}^a, \quad (36.45)$$

and the fermionic Fierz relation is

$$n^2 = -\frac{1}{N} n^2 - 2 J^a J^a, \quad J^a J^a = -\frac{N+1}{2N} n^2. \quad (36.46)$$

Proof 36.27 Use the matrix basis $\Gamma^0 = \mathbf{1}/\sqrt{N}$ and $\Gamma^a = \sqrt{2} T^a$. It is orthonormal under the trace pairing, so its completeness relation is (36.45). The general Fierz identity with $M = N = \mathbf{1}$ gives

$$n^2 = -\left[\left(\frac{n}{\sqrt{N}} \right)^2 + 2 J^a J^a \right],$$

which is (36.46) after solving for $J^a J^a$.

Remark 36.27.1 [The $N = 2$ check] For $T^a = \sigma^a/2$, one has $J^a = s^a/2$ and (36.46) becomes $s^a s^a = -3n^2$, exactly as in (36.39). The $SU(N)$ form is often the cleanest starting point for large- N expansions because the relative factor between the singlet and adjoint channels is explicit.

Theorem 36.28 [A pairing identity from antisymmetric tensors] Let $C_{ab} = -C_{ba}$ be an antisymmetric matrix and define $P_C \equiv C_{ab} \psi_a \psi_b$. Then

$$P_C = \frac{1}{2} C_{ab} (\psi_a \psi_b - \psi_b \psi_a) \quad (36.47)$$

and only the antisymmetric part of a pairing kernel contributes to $\bar{P}_C P_C$. For a two-component spinor, the antisymmetric space is one-dimensional and is generated by

$$\varepsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad P_\varepsilon = 2\psi_\uparrow \psi_\downarrow = -2\psi_\downarrow \psi_\uparrow. \quad (36.48)$$

Proof 36.28 Decompose an arbitrary matrix C into symmetric and antisymmetric parts. The symmetric part multiplies $\psi_a \psi_b = -\psi_b \psi_a$ and therefore vanishes. The second statement follows because every 2×2 antisymmetric matrix is proportional to ε .

36.1.6. Worked Hubbard–Stratonovich decouplings

Theorem 36.29 [Charge-channel decoupling of the repulsive Hubbard monomial] For $U > 0$ and the monomial Q in (36.40),

$$\exp(-UQ) = \frac{1}{\sqrt{2\pi U}} \int_{\mathbb{R}} d\phi \exp \left[-\frac{\phi^2}{2U} - i\phi n \right]. \quad (36.49)$$

The quartic Hubbard term has become a quadratic coupling $-i\phi\bar{\psi}\psi$.

Proof 36.29 Use $Q = n^2/2$ from (36.41) and apply the HST identity (36.11) with $N = 1$, $V = 1$, $\lambda = U$, and $\rho = n$.

Theorem 36.30 [Spin-channel decoupling of the repulsive Hubbard monomial] For $U > 0$,

$$\exp(-UQ) = \mathcal{N}_U \int_{\mathbb{R}^3} d^3\phi \exp \left[-\frac{3}{2U}\phi^2 + \phi \cdot s \right], \quad (36.50)$$

where $s = (s^1, s^2, s^3)$ and $\mathcal{N}_U = (3/(2\pi U))^{3/2}$.

Proof 36.30 Equation (36.41) gives $-UQ = +Us^i s^i/6$. The ordinary three-dimensional Gaussian identity

$$\left(\frac{a}{\pi} \right)^{3/2} \int d^3\phi \exp(-a\phi^2 + \phi \cdot s) = \exp \left(\frac{s^2}{4a} \right)$$

with $a = 3/(2U)$ produces (36.50).

Theorem 36.31 [Cooper-channel decoupling] With $P = \psi_{\downarrow}\psi_{\uparrow}$ and $\bar{P} = \bar{\psi}_{\uparrow}\bar{\psi}_{\downarrow}$,

$$\exp(-U\bar{P}P) = \frac{1}{\pi U} \int d^2\Delta \exp \left[-\frac{|\Delta|^2}{U} + i(\bar{\Delta}P + \bar{P}\Delta) \right]. \quad (36.51)$$

If the interaction in the action is attractive, $-g\bar{P}P$ with $g > 0$, the same formula is used with the factors i removed:

$$\exp(+g\bar{P}P) = \frac{1}{\pi g} \int d^2\Delta \exp \left[-\frac{|\Delta|^2}{g} + \bar{\Delta}P + \bar{P}\Delta \right]. \quad (36.52)$$

Proof 36.31 Apply the complex HST identity (36.26) with one complex composite. For the repulsive sign use $\lambda = U$ and the imaginary coupling. For the attractive sign use the real-coupling version stated after the identity. The normalization follows from $\int d^2\Delta e^{-|\Delta|^2/U} = \pi U$.

Remark 36.31.1 [Mean-field interpretation] In the attractive Cooper decoupling, the auxiliary field Δ couples linearly to the pair amplitude. At a translationally invariant saddle point, Δ is a constant complex number and its phase is the phase of the superconducting order parameter. The Hubbard–Stratonovich identity itself does not assume that Δ is constant; that restriction is a subsequent saddle-point ansatz.

Theorem 36.32 [Exact quadratic fermionic action after decoupling] Let $\rho_A = \bar{\psi}\Gamma^A\psi$ and consider a quadratic fermionic action $S_0 = \bar{\psi}K\psi$. The charge-channel decoupling gives

$$S_{\text{HS}}[\bar{\psi}, \psi, \varphi] = \bar{\psi}K\psi + \frac{1}{2\lambda}\varphi_A(V^{-1})_{AB}\varphi_B + i\varphi_A\bar{\psi}\Gamma^A\psi. \quad (36.53)$$

For fixed φ this is quadratic in the fermions, and therefore

$$Z = \mathcal{N}_{\lambda V} \int \mathcal{D}\varphi \exp \left[-\frac{1}{2\lambda}\varphi V^{-1}\varphi \right] \det (K + i\varphi_A\Gamma^A). \quad (36.54)$$

Proof 36.32 The first equation is the direct substitution of the real HST identity. The finite-dimensional Berezin Gaussian identity is

$$\int \mathcal{D}(\bar{\psi}, \psi) \exp(-\bar{\psi}M\psi) = \det M.$$

Apply it with $M = K + i\varphi_A\Gamma^A$. The determinant is exact and contains all fermionic fluctuations around every auxiliary-field configuration.

Corollary 36.33 [Exact saddle-point equation] Define

$$S_{\text{eff}}[\varphi] \equiv \frac{1}{2\lambda}\varphi V^{-1}\varphi - \text{Tr} \log (K + i\varphi_A\Gamma^A), \quad (36.55)$$

where Tr includes all one-particle labels. A stationary field obeys

$$\frac{1}{\lambda}(V^{-1})_{AB}\varphi_B - i \text{Tr} \left[(K + i\varphi_C\Gamma^C)^{-1} \Gamma^A \right] = 0. \quad (36.56)$$

Proof 36.33 Use $\delta \text{Tr} \log M = \text{Tr}(M^{-1}\delta M)$ and $\delta M = i\delta\varphi_A\Gamma^A$. Differentiation of S_{eff} gives (36.56). The equation is an approximation only when one declares the integral over φ to be dominated by this saddle.

Theorem 36.34 [Exact auxiliary-field expectation identity] In the exact HST representation,

$$\langle \varphi_A \rangle = -i\lambda V_{AB} \langle \rho_B \rangle. \quad (36.57)$$

The relation holds before any saddle-point approximation, whenever the integrations and boundary terms are well defined.

Proof 36.34 Integrate the total derivative

$$0 = \int \mathcal{D}\varphi \frac{\partial}{\partial \varphi_A} \exp \left[-\frac{1}{2\lambda}\varphi V^{-1}\varphi - i\varphi\rho \right].$$

The derivative gives $-(V^{-1})_{AB}\varphi_B/\lambda - i\rho_A$. Taking the normalized expectation value and multiplying by $-\lambda V$ proves (36.57).

36.1.7. Fierz covariance and practical bookkeeping

Theorem 36.35 [Covariance under a change of composite basis] Let R be invertible and define a new composite basis by $\rho' = R\rho$. If

$$V' = R^{-\top} V R^{-1}, \quad \varphi' = R^{-\top} \varphi, \quad (36.58)$$

then

$$\rho'^{\top} V' \rho' = \rho^{\top} V \rho, \quad \varphi'^{\top} V'^{-1} \varphi' = \varphi^{\top} V^{-1} \varphi, \quad \varphi'^{\top} \rho' = \varphi^{\top} \rho. \quad (36.59)$$

Hence the exact HST representation is covariant under Fierz changes of basis.

Proof 36.35 Substitute (36.58). Since $V'^{-1} = R V R^{\top}$, the three equalities follow by cancellation of $R^{-1} R$ and $R^{\top} R^{-\top}$. The Jacobian of a finite-dimensional linear field transformation is a constant and can be absorbed into the Gaussian normalization.

Lemma 36.36 [The factor-of-two diagnostic] The following two expressions are equal only if the indicated symmetry and normalization conventions are used consistently:

$$\frac{1}{2} \rho_I V_{IJ} \rho_J \quad \text{and} \quad \frac{1}{4} \bar{\psi}_a \bar{\psi}_b U_{ab;cd} \psi_d \psi_c. \quad (36.60)$$

The first expression treats I as a composite index and includes one quadratic-form factor $1/2$. The second expression treats (a, b) and (c, d) as unrestricted antisymmetric pairs and includes one factor $1/2$ for each pair.

Proof 36.36 A quadratic form counts the pair (I, J) and (J, I) twice, hence the factor $1/2$. An unrestricted antisymmetric pair sum counts (a, b) and (b, a) twice on each side, hence $(2!)^{-2} = 1/4$. Passing from one notation to the other changes the number of explicit permutations but not the polynomial. Omitting one of these factors doubles the interaction and changes the HST covariance by the same factor.

Remark 36.36.1 [Reality and convergence] For a real auxiliary field the quadratic kernel in the Gaussian weight must be positive on the chosen contour. An indefinite interaction can be treated by diagonalizing the kernel and rotating the integration contour in the negative eigendirections. For a complex kernel the formula is first established in a convergence domain and then continued analytically. In numerical determinant methods, the corresponding choice of imaginary versus real Yukawa coupling can control whether the auxiliary-field measure is manifestly positive or suffers from a sign problem.

Remark 36.36.2 [Operator order versus Grassmann order] The symbols $\bar{\psi}_a \psi_b$ in a coherent-state path integral are Grassmann variables, not operator products. Consequently, an identity such as $n^2 = 2Q$ is an identity of Grassmann polynomials with the ordering displayed in (36.40). Translating the same statement to normal-ordered operators requires applying the coherent-state correspondence and keeping track of operator ordering separately. Mixing the two algebras is a frequent source of an apparent sign disagreement.

Theorem 36.37 [Summary of the exact decoupling logic] Let a fermionic action contain a quartic polynomial S_4 . The following sequence is exact:

1. antisymmetrize its four-index tensor;

2. choose a composite basis, such as charge, spin, exchange, or Cooper pairs;
3. use completeness and Fierz identities to express S_4 in that basis;
4. introduce one bosonic Hubbard–Stratonovich field for each retained composite;
5. integrate out the fermions, if desired, to obtain a determinant or Pfaffian effective action.

An approximation enters only after this sequence, for example by restricting the auxiliary field to a saddle point, discarding fluctuations, or truncating the generated effective action.

Proof 36.37 Steps one and three are finite-dimensional tensor identities, proved by antisymmetrization and completeness. Step four is the identity (36.11) or (36.26). Step five is the Berezin Gaussian identity. Since each step is an equality, their composition is an equality. Replacing an integral by a saddle-point value or truncating its determinant expansion is not an identity and must therefore be stated as an additional approximation.

37. Experimental: Feynman-Diagramology

In quantum field theory, there are *four* famous theorems about "Feynman-diagrams", of which three are never proven in the generality in which one claims their applicability. Those are ...

- The partition-function is the sum of all "diagrams" without "external vertices".
- The free energy is the sum of all "connected diagrams" without "external vertices".
- The self-energy is the sum of all "one-particle-irreducible two-point¹ diagrams".
- The n -body Green-functions are computed as sums of all "connected diagrams" with n "external vertices" (a.k.a. "Linked Cluster Theorem").

It should be emphasized at this point, that the entire Feynman-diagrammatic-machinery lacks mathematical precision (at least in the context of condensed matter theory). We will fix this here.

37.1. Graph Theory

37.1.1. Fundamentals

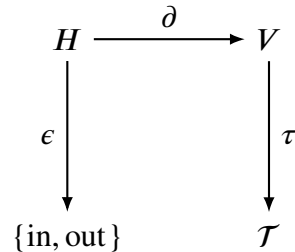
Definition 37.1 A *statistical pre-graph* is a quintuple

$$g = (V, H, \partial, \epsilon, \tau), \tag{37.1}$$

where

- $V = \{v_1, v_2, \dots, v_{|V|}\}$ is a finite set of elements which we call *vertices*,
- $H = \{h_1, h_2, \dots, h_{|H|}\}$ is a finite set of elements which we call *oriented half-edges*,
- $(\partial, \epsilon) : H \rightarrow V \times \{\text{in}, \text{out}\}$ assigns to each half-edge its *associated vertex* and *direction*,
- $\tau : V \rightarrow \mathcal{T}$ assigns each vertex a *type* from a finite set of *vertex-types*.

Remark 37.1.1 The diagram elucidates the definitions.



¹I.e. diagrams with two external legs.

Remark 37.1.2 Instead of laboriously defining the set of half-edges and their image under the map ∂ , one often simply draws the half-graph as below.

Definition 37.2 For a vertex $v \in V$, its *valency* is defined by

$$\text{val}(v) = \left| \partial^{-1}(v) \right|. \quad (37.2)$$

Fact 37.3 Vertices of valency 0 will never appear in the physical theory. Vertices of valency 1, and 2 and higher are usually denoted as ...

- A vertex v with $\text{val}(v) = 1$ is attached to exactly one half-edge. These are "*source vertices*" or "*external vertices*", whereas we will clarify this distinction later on, when we introduce the underlying symbols. All remaining edges are called *internal edges*.
- A vertex v with $\text{val}(v) = 2$ is (depending on the context) called a *mass* or a *potential* or an *impurity vertex* or a *one-body interaction-vertex*.
- A vertex v with $\text{val}(v) = 4$ is called a *two-body interaction-vertex*.
- A vertex v with $\text{val}(v) = 6$ is called a *three-body interaction-vertex*.
- ...
- A vertex v with $\text{val}(v) \geq 2$ can (more generally) be called a *many-body interaction-vertex* or (inclusively) *internal vertex*

Notation 37.4 The table summarizes the symbols that we will use to denote the different types of vertices and their properties.

Symbol(v)	$\tau(v)$	$(\epsilon \circ \partial^{-1})(v)$	$ \partial^{-1}(v) $
•	external vertex	{in}, {out}	1
×	source vertex	{out}	1
◦	drain / sink vertex	{in}	1
$\boxed{m}$	mass vertex	{in, out}	2
$\boxed{p}$	potential vertex	{in, out}	2
$\boxed{\text{imp}}$	impurity vertex	{in, out}	2
	interaction vertex		3, 4, 5 ...
	interaction vertex		3, 4, 5 ...

Examples:

Some **common examples** that one might be anticipating, if one has already seen some so-called "Feynman-diagrams", are the following:

- a) Two vertices, one with outgoing one with incoming half-edge (valency 1 respectively). Cf. TBDrawn 1

- b) Three vertices, one with outgoing one with incoming half-edge (valency 1 respectively) and one with two half-edges, one incoming and one outgoing (valency 2). **Cf. TBDrawn 1**
- c) Three vertices, one with outgoing one with incoming half-edge (valency 1 respectively) and one with two half-edges, two incoming and two outgoing (valency 4). **Cf. TBDrawn 1**
- d) Four vertices, one with outgoing one with incoming half-edge (valency 1 respectively) and one with two half-edges, two incoming and two outgoing (valency 4), and one with two half-edges, one incoming and one outgoing (valency 2). **Cf. TBDrawn 1**

Other, more **exotic examples**, that will however also become relevant, are the following:

- a) Four vertices, two with outgoing two with incoming half-edges (valency 1 respectively). **Cf. TBDrawn 2**
- b) Three vertices, one with outgoing one with incoming half-edges (valency 1 respectively) and one with three half-edges ...
 - three incoming, **Cf. TBDrawn 2**
 - two incoming and one outgoing, **Cf. TBDrawn 2**
 - one incoming and two outgoing, **Cf. TBDrawn 2**
 - three outgoing. **Cf. TBDrawn 2**
- c) Three vertices, one with outgoing one with incoming half-edge (valency 1 respectively) and one with two half-edges, two incoming and two outgoing (valency 4). **Cf. TBDrawn 2**

We now extend the definition above.

Definition 37.5 A *statistical graph* is a statistical pre-graph $g = (V, H, \partial, \epsilon, \tau)$ equipped with an involution

$$\iota : H \rightarrow H,$$

that we call *pairing*, satisfying

$$\iota^2 = \text{id}_H, \quad \epsilon \neq \epsilon \circ \iota \quad (37.3)$$

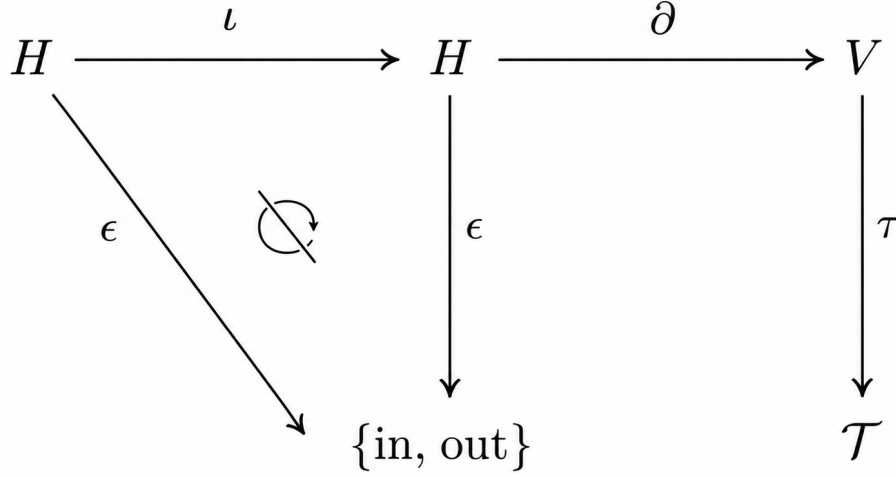
Remark 37.5.1 We will notate the statistical graph as a sextuple $G = (V, H, \partial, \epsilon, \tau, \iota)$.

Remark 37.5.2 The involution tells us which half-edges are *glued* together to form a so-called "edge" (to be defined cleanly below). The condition $\iota^2 = \text{id}_H$ means that if $\iota(h) = h'$, then h is said to be *glued* to h' .

Remark 37.5.3 Notice that the second conditions prohibits a half-edge from being glued to itself.

Remark 37.5.4 Notice further that it might not be always possible to find a pairing for a given statistical pre-graph. In that case, we say that the statistical pre-graph is *unpairable*.

Remark 37.5.5 Notice that the definition implies that one finds the following diagram



Lemma 37.6 Let

$$g = (V, H, \partial, \epsilon, \tau) \quad (37.4)$$

be a statistical pregraph, and define

$$H_{\text{in}}(g) := \{h \in H \mid \epsilon(h) = \text{in}\}, \quad H_{\text{out}}(g) := \{h \in H \mid \epsilon(h) = \text{out}\}. \quad (37.5)$$

A pairing of g exists if and only if

$$|H_{\text{in}}(g)| = |H_{\text{out}}(g)|. \quad (37.6)$$

Proof 37.6 Suppose first that a pairing ι exists. Since every incoming half-edge is paired with an outgoing half-edge, the restriction

$$\iota|_{H_{\text{in}}(g)} : H_{\text{in}}(g) \rightarrow H_{\text{out}}(g) \quad (37.7)$$

is a bijection. Therefore the two sets have equal cardinality. Conversely, suppose that the two sets have equal cardinality. Since they are finite, there exists a bijection

$$b : H_{\text{out}}(g) \rightarrow H_{\text{in}}(g). \quad (37.8)$$

Define $\iota : H \rightarrow H$ by

$$\iota(h) := \begin{cases} b(h) & \text{if } h \in H_{\text{out}}(g), \\ b^{-1}(h) & \text{if } h \in H_{\text{in}}(g). \end{cases} \quad (37.9)$$

Then $\iota^2 = \text{id}_H$, and ι exchanges incoming and outgoing half-edges. Hence ι is a pairing of g .

Corollary 37.7 The number of pairings of a statistical pregraph g is

$$n_{\text{pairs}}(g) = \begin{cases} |H_{\text{in}}(g)|! & \text{if } |H_{\text{in}}(g)| = |H_{\text{out}}(g)|, \\ 0 & \text{otherwise.} \end{cases} \quad (37.10)$$

Proof 37.7 If the cardinalities differ, no pairing exists by Lemma 37.6.

Suppose that both sets have cardinality m . Every pairing is uniquely determined by its restriction to a bijection

$$H_{\text{out}}(g) \rightarrow H_{\text{in}}(g), \quad (37.11)$$

and every such bijection defines a pairing. Since there are $m!$ bijections between two sets of cardinality m , one obtains

$$n_{\text{pairs}}(g) = m!. \quad (37.12)$$

Notation 37.8 For a given statistical pre-graph $g = (V, H, \partial, \epsilon, \tau)$, we denote the set of all statistical graphs that can be constructed from g by equipping it with a pairing ι as $C(g)$, i.e.

$$C(g) = \{(V, H, \partial, \epsilon, \tau, \iota) \mid \iota \text{ is a pairing on } H\} \equiv \bigcup_{j=1}^{n_{\text{pairs}}} \{G_j(g)\}. \quad (37.13)$$

Remark 37.8.1 If no such pairing exists, then we say that the statistical pre-graph is *unpairable* and we set $C(g) = \emptyset$. Oftentimes we also call $\emptyset$ the *null-graph*

Definition 37.9 Let $G = (V, H, \partial, \epsilon, \tau, \iota)$ be a statistical graph. We define a relation $\sim_\iota$ on H by

$$h \sim_\iota h' \iff h' \in \{h, \iota(h)\}. \quad (37.14)$$

Remark 37.9.1 Notice that this implies that half-edges that were glued together are equivalent under this relation.

Definition 37.10 For a statistical graph, we define the *set of edges* as the quotient

$$E(G) := H / \sim_\iota = \{\{h, \iota(h)\} \mid h \in H\}. \quad (37.15)$$

An element

$$e = [h]_{\sim_l} = \{h, \iota(h)\} \in E(G) \quad (37.16)$$

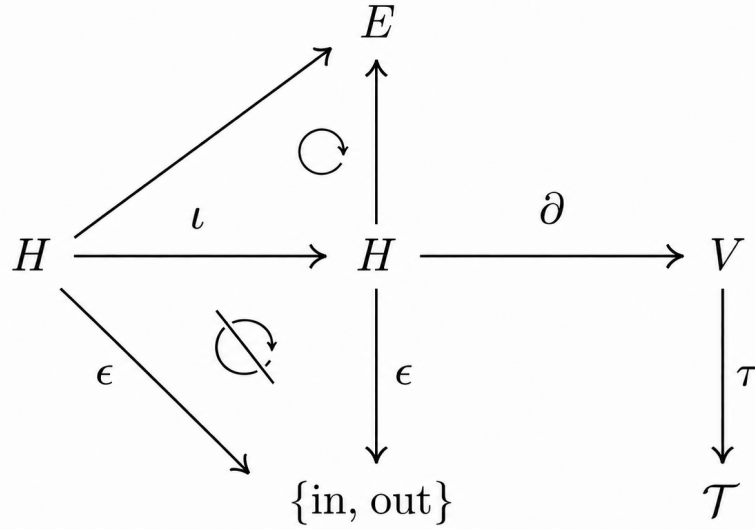
is called an *edge* of G .

Lemma 37.11 Notice that one finds that

$$\forall e \in E(G) : \iota(e) = e. \quad (37.17)$$

Proof 37.11 Follows from set-parenthesis allowing the rearrangement of the order, without changing the set.

Remark 37.11.1 Notice that the definition implies that one finds the following diagram



TBD: Re-creating this picture as a tikzpicture.

Definition 37.12 Let

$$G = (V, H, \partial, \epsilon, \tau, \iota) \quad (37.18)$$

be a statistical graph and let $E(G) = H / \sim_l$ be its set of edges. For every edge $e \in E(G)$, there are unique half-edges $h_{\text{out}}(e), h_{\text{in}}(e) \in H$ satisfying

$$\epsilon(h_{\text{out}}(e)) = \text{out}, \quad \epsilon(h_{\text{in}}(e)) = \text{in}. \quad (37.19)$$

Definition 37.13 Let $G = (V, H, \partial, \epsilon, \tau, \iota)$ be a statistical graph. Two vertices $v, w \in V$ are *adjacent* if and only if there exists two half-edges $h_{\text{in}}, h_{\text{out}} \in H$ such that

$$\partial(h_{\text{in}}) = v, \text{ and } (\partial \circ \iota)(h_{\text{out}}) = w \quad \text{or} \quad \partial(h_{\text{out}}) = w, \text{ and } (\partial \circ \iota)(h_{\text{in}}) = v. \quad (37.20)$$

Definition 37.14 We define the maps

$$s_G, t_G : E(G) \rightarrow V \quad (37.21)$$

by

$$s_G(e) := \partial(h_{\text{out}}(e)), \quad t_G(e) := \partial(h_{\text{in}}(e)). \quad (37.22)$$

Definition 37.15 We then define the *Feynman-map* by

$$\text{Feynman} : G = (V, H, \partial, \epsilon, \tau, \iota) \mapsto \Gamma(G) := (V, E(G), s_G, t_G, \tau). \quad (37.23)$$

We call $\Gamma(G)$ the *statistical Feynman-diagram associated with G* .

Remark 37.15.1 If we combine every vertex with its type and define

$$\mathcal{V} := \{(v, \tau(v)) \mid v \in V\}, \quad (37.24)$$

then we may equivalently write the statistical Feynman-diagram as

$$\Gamma = (\mathcal{V}, E, s, t). \quad (37.25)$$

Definition 37.16 Let

$$\Gamma = (V, E, s, t, \tau) \quad (37.26)$$

be a statistical Feynman-diagram. We call two vertices $v, w \in V$ *adjacent* if there exists an edge $e \in E$ such that

$$(s(e), t(e)) = \{v, w\} \quad (37.27)$$

Remark 37.16.1 The Feynman-map acts elementwise on the set of statistical graphs generated by a pregraph g . Thus,

$$C(g) = \{G_i(g)\}_{i=1}^{n_{\text{pairs}}(g)} \xrightarrow{\text{Feynman}} \{\Gamma_i(g) := \text{Feynman}(G_i(g))\}_{i=1}^{n_{\text{pairs}}(g)}. \quad (37.28)$$

Hence every pairing of g first gives a statistical graph and subsequently a labelled statistical Feynman-diagram. Only after passing to isomorphism classes will diagrams that differ merely by a structure-preserving relabelling be identified.

So far we have only been considering Feynman-diagrams that originate from a single pre-graph.

Definition 37.17 Let

$$\Gamma_1 = (V_1, E_1, s_1, t_1, \tau_1), \quad \Gamma_2 = (V_2, E_2, s_2, t_2, \tau_2) \quad (37.29)$$

be two statistical Feynman-diagrams with

$$V_1 \cap V_2 = \emptyset, \quad E_1 \cap E_2 = \emptyset. \quad (37.30)$$

We define their *disjoint union* by

$$\Gamma_1 \sqcup \Gamma_2 := (V_1 \sqcup V_2, E_1 \sqcup E_2, s_{1 \sqcup 2}, t_{1 \sqcup 2}, \tau_{1 \sqcup 2}), \quad (37.31)$$

where

$$s_{1 \sqcup 2}(e) := \begin{cases} s_1(e) & \text{if } e \in E_1, \\ s_2(e) & \text{if } e \in E_2, \end{cases} \quad t_{1 \sqcup 2}(e) := \begin{cases} t_1(e) & \text{if } e \in E_1, \\ t_2(e) & \text{if } e \in E_2, \end{cases} \quad (37.32)$$

and

$$\tau_{1 \sqcup 2}(v) := \begin{cases} \tau_1(v) & \text{if } v \in V_1, \\ \tau_2(v) & \text{if } v \in V_2. \end{cases} \quad (37.33)$$

Definition 37.18 A non-empty statistical Feynman-diagram

$$\Gamma = (V, E, s, t, \tau) \quad (37.34)$$

is called *connected* if there do not exist two non-empty statistical Feynman-diagrams Γ_1 and Γ_2 such that

$$\Gamma = \Gamma_1 \sqcup \Gamma_2. \quad (37.35)$$

If such a decomposition exists, then Γ is called *disconnected*.

Definition 37.19 Let

$$\Gamma = (V, E, s, t, \tau) \quad (37.36)$$

be a statistical Feynman-diagram. A *walk of length* $n \in \mathbb{N}_0$ from $u \in V$ to $v \in V$ is a finite sequence

$$(v_0, e_1, v_1, e_2, \dots, e_n, v_n) \quad (37.37)$$

satisfying

$$v_0 = u, \quad v_n = v, \quad (37.38)$$

and, for every $l \in \{1, \dots, n\}$,

$$\{s(e_l), t(e_l)\} = \{v_{l-1}, v_l\}. \quad (37.39)$$

For $n = 0$, the one-element sequence (u) is called the *trivial walk* from u to itself. A walk is called a *path* if its vertices $v_0, \dots, v_n$ are pairwise distinct, except that the trivial path of length zero is allowed.

Definition 37.20 Let

$$\Gamma = (V, E, s, t, \tau) \quad (37.40)$$

be a statistical Feynman-diagram. For $u, v \in V$, we define

$$u \sim_\Gamma v \iff \text{there exists a finite walk in } \Gamma \text{ from } u \text{ to } v. \quad (37.41)$$

Notation 37.21 For every $v \in V$, we denote its equivalence class under $\sim_\Gamma$ by

$$C_v(\Gamma) := [v]_{\sim_\Gamma} = \{u \in V \mid u \sim_\Gamma v\}. \quad (37.42)$$

Theorem 37.22 The relation $\sim_\Gamma$ defined in Equation (37.41) is an equivalence relation on the vertex set V of a statistical Feynman-diagram

$$\Gamma = (V, E, s, t, \tau). \quad (37.43)$$

Proof 37.22 We verify the three defining properties.

Reflexivity: For every $u \in V$, the trivial walk (u) is a walk from u to itself. Hence

$$u \sim_\Gamma u. \quad (37.44)$$

Symmetry: Suppose that $u \sim_\Gamma v$. Then there exists a walk

$$(v_0, e_1, v_1, \dots, e_n, v_n) \quad (37.45)$$

with $v_0 = u$ and $v_n = v$. Since walks may traverse edges in either direction, the reversed sequence

$$(v_n, e_n, v_{n-1}, \dots, e_1, v_0) \quad (37.46)$$

is a walk from v to u . Therefore

$$v \sim_{\Gamma} u. \quad (37.47)$$

Transitivity: Suppose that $u \sim_{\Gamma} v$ and $v \sim_{\Gamma} w$. Concatenating a walk from u to v with a walk from v to w gives a finite walk from u to w . Hence

$$u \sim_{\Gamma} w. \quad (37.48)$$

Thus $\sim_{\Gamma}$ is reflexive, symmetric, and transitive.

Corollary 37.23 The relation $\sim_{\Gamma}$ induces a partition of the vertex set V into equivalence classes:

$$V / \sim_{\Gamma} = \{C_i\}_{i \in I}, \quad (37.49)$$

where $I \subseteq V$ may be chosen to contain exactly one representative from each equivalence class.

Proof 37.23 Since $\sim_{\Gamma}$ is an equivalence relation, its equivalence classes are non-empty, pairwise disjoint, and their union is V .

Notation 37.24 For every equivalence class $C_i \in V / \sim_{\Gamma}$, define

$$E_i := \{e \in E \mid s(e) \in C_i \text{ and } t(e) \in C_i\}. \quad (37.50)$$

We denote the corresponding statistical *Feynman-subdiagram* by

$$\Gamma_i := (C_i, E_i, s_i, t_i, \tau_i), \quad (37.51)$$

where

$$s_i := s|_{E_i}, \quad t_i := t|_{E_i}, \quad \tau_i := \tau|_{C_i}. \quad (37.52)$$

Theorem 37.25 Every statistical Feynman-diagram

$$\Gamma = (V, E, s, t, \tau) \quad (37.53)$$

admits a unique decomposition into connected components:

$$\Gamma = \bigsqcup_{i \in I} \Gamma_i, \quad (37.54)$$

where the vertex set of Γ_i is the equivalence class C_i under $\sim_{\Gamma}$.

Proof 37.25 The equivalence classes $\{C_i\}_{i \in I}$ form a partition of V . Moreover, the endpoints of every edge $e \in E$ are adjacent and therefore belong to the same equivalence class. Consequently, the sets $\{E_i\}_{i \in I}$ form a partition of E . The restrictions s_i , t_i , and τ_i therefore reconstruct s , t , and τ under the disjoint union. Hence

$$\Gamma = \bigsqcup_{i \in I} \Gamma_i. \quad (37.55)$$

Each Γ_i is connected because any two vertices in C_i are joined by a finite walk. Finally, in any decomposition of Γ into connected components, two vertices belong to the same component if and only if they are connected by a finite walk. Thus the vertex sets of the components must be precisely the equivalence classes C_i . The decomposition is therefore unique up to the ordering of its components.

[Maybe we will add things here later on.]²

Definition 37.26 Let

$$\Gamma_i = (V_i, E_i, s_i, t_i, \tau_i), \quad i \in \{1, 2\}, \quad (37.56)$$

be two statistical Feynman-diagrams. An *isomorphism*

$$\varphi : \Gamma_1 \rightarrow \Gamma_2 \quad (37.57)$$

is a pair of bijections

$$\varphi = (\varphi_V, \varphi_E), \quad \varphi_V : V_1 \rightarrow V_2, \quad \varphi_E : E_1 \rightarrow E_2, \quad (37.58)$$

satisfying

$$\tau_1 = \tau_2 \circ \varphi_V, \quad (37.59)$$

together with

$$\varphi_V \circ s_1 = s_2 \circ \varphi_E, \quad \varphi_V \circ t_1 = t_2 \circ \varphi_E. \quad (37.60)$$

If such an isomorphism exists, we call Γ_1 and Γ_2 *isomorphic* and write

$$\Gamma_1 \cong \Gamma_2. \quad (37.61)$$

²Cf. Book Graph Theory from Robin J. Wilson or arXiv:1301.6918v1.

Might be later relevant:

- matrix-representations of graphs
- adjacency matrices
- vector-space of a Graph
- Cf. page 42 in the book of Salmhofer.
- page 110, 111 in Negele, Orland: Definition of a Tree-Diagram.
- page 110: reference to SSB and their implications for expectation-values.
- Definition of a pairing from Etingof.
- Cf. also Mahan page 85 for a decomposition of graphs.
- Cf. also Peskin page 305 for some nice pictures of Feynman-Diagrams.

An isomorphism from a statistical Feynman-diagram Γ to itself is called an *automorphism*³ of Γ .

Remark 37.26.1 Every isomorphism preserves adjacency. Indeed, if an edge $e \in E_1$ connects two vertices $v, w \in V_1$, then $\varphi_E(e) \in E_2$ connects $\varphi_V(v)$ and $\varphi_V(w)$ with the same orientation. The converse is generally false: preservation of adjacency alone does not preserve the number of parallel edges or distinguish different permutations of edges. This is why an isomorphism consists of both a vertex bijection φ_V and an edge bijection φ_E .

Notation 37.27 For a statistical Feynman-diagram

$$\Gamma = (V, E, s, t, \tau), \quad (37.62)$$

we denote its set of automorphisms by

$$\text{Aut}(\Gamma) := \{ \varphi = (\varphi_V, \varphi_E) \mid \varphi : \Gamma \rightarrow \Gamma \text{ is an isomorphism} \}. \quad (37.63)$$

Lemma 37.28 The set of automorphisms $\text{Aut}(\Gamma)$ forms a group under composition of maps.

Proof 37.28 Let

$$\varphi = (\varphi_V, \varphi_E), \quad \Psi = (\psi_V, \psi_E) \quad (37.64)$$

be two elements of $\text{Aut}(\Gamma)$. Define their composition componentwise by

$$\varphi \circ \Psi := (\varphi_V \circ \psi_V, \varphi_E \circ \psi_E). \quad (37.65)$$

Since compositions of bijections are bijections, both components of $\varphi \circ \Psi$ are bijective. Moreover,

$$\tau = \tau \circ (\varphi_V \circ \psi_V), \quad (37.66)$$

and

$$(\varphi_V \circ \psi_V) \circ s = s \circ (\varphi_E \circ \psi_E), \quad (\varphi_V \circ \psi_V) \circ t = t \circ (\varphi_E \circ \psi_E). \quad (37.67)$$

Hence $\varphi \circ \Psi \in \text{Aut}(\Gamma)$. Composition is associative because composition of maps is associative. The identity element is

$$\text{id}_\Gamma := (\text{id}_V, \text{id}_E). \quad (37.68)$$

Finally, if $\varphi = (\varphi_V, \varphi_E) \in \text{Aut}(\Gamma)$, then

³Cf. also 01-Feynman-Diagrams (Michael Polyak Notes), under Equation (18).

$$\varphi^{-1} := (\varphi_V^{-1}, \varphi_E^{-1}) \quad (37.69)$$

also preserves τ , s , and t , and therefore belongs to $\text{Aut}(\Gamma)$. Thus every element has an inverse, and $\text{Aut}(\Gamma)$ is a group.

Lemma 37.29 Isomorphism defines an equivalence relation on every collection of statistical Feynman-diagrams. Explicitly, define

$$\Gamma_1 \sim_{\text{iso}} \Gamma_2 \iff \Gamma_1 \cong \Gamma_2. \quad (37.70)$$

Then $\sim_{\text{iso}}$ is an equivalence relation.

Proof 37.29 The relation is reflexive because

$$(\text{id}_V, \text{id}_E) \quad (37.71)$$

is an isomorphism from every statistical Feynman-diagram to itself. It is symmetric because, if

$$\varphi = (\varphi_V, \varphi_E) : \Gamma_1 \rightarrow \Gamma_2 \quad (37.72)$$

is an isomorphism, then

$$\varphi^{-1} = (\varphi_V^{-1}, \varphi_E^{-1}) : \Gamma_2 \rightarrow \Gamma_1 \quad (37.73)$$

is also an isomorphism. Finally, it is transitive because the componentwise composition of an isomorphism from Γ_1 to Γ_2 with an isomorphism from Γ_2 to Γ_3 is an isomorphism from Γ_1 to Γ_3 .

Remark 37.29.1 Let

$$\{\Gamma_i(g)\}_{i=1}^{n_{\text{pairs}}(g)} \quad (37.74)$$

be the family of labelled statistical Feynman-diagrams generated by a pregraph g . Its quotient under diagram isomorphism is

$$\{\Gamma_i(g)\}_{i=1}^{n_{\text{pairs}}(g)} / \sim_{\text{iso}} = \{[\Gamma_j(g)]_{\text{iso}}\}_{j \in \tilde{I}}, \quad (37.75)$$

where

$$[\Gamma_j(g)]_{\text{iso}} := \{\Gamma_i(g) \mid \Gamma_i(g) \cong \Gamma_j(g)\}, \quad (37.76)$$

and $\tilde{I}$ contains one representative index from each isomorphism class.

37.1.2. These Wicked Diagrams...

We now define a mapping that associates a given statistical pre-graph with a set of statistical graphs, and consequently a set of statistical Feynman-diagrams.

Theorem 37.30 Let

$$g = (V, H, \partial, \epsilon, \tau) \quad (37.77)$$

be a statistical pregraph. Considering all pairings of g gives the sequence

$$g \mapsto C(g) = \left\{ G_i(g) \right\}_{i=1}^{n_{\text{pairs}}(g)} \xrightarrow{\text{Feynman}} D_{\text{lab}}(g) := \left\{ \Gamma_i(g) \right\}_{i=1}^{n_{\text{pairs}}(g)}, \quad (37.78)$$

where

$$\Gamma_i(g) := \text{Feynman} \left(G_i(g) \right). \quad (37.79)$$

Thus, the number of labelled statistical Feynman-diagrams generated from g , counted before passing to isomorphism classes, is

$$|D_{\text{lab}}(g)| = |C(g)| = n_{\text{pairs}}(g). \quad (37.80)$$

Explicitly,

$$n_{\text{pairs}}(g) = \begin{cases} |H_{\text{in}}(g)|! & \text{if } |H_{\text{in}}(g)| = |H_{\text{out}}(g)|, \\ 0 & \text{otherwise.} \end{cases} \quad (37.81)$$

Graphical Examples

TBD.

In the next sections, we will use the above definitions to formulate the diagrammatic expansion of the partition function, the free energy, and the Green-functions. We will also discuss how to compute the self-energy and the vertex functions using Feynman-diagrams. For simplicity, we will focus on systems with particle-number preserving interactions for the beginning.

37.2. Pure Systems

Notation 37.31 We use the notation of Notation 16.24. Thus

$$S[\psi, \bar{\psi}] = S_0[\psi, \bar{\psi}] + S_{\text{int}}[\psi, \bar{\psi}] \quad \text{with} \quad S_0[\psi, \bar{\psi}] = \psi^\dagger A \psi \quad (37.82)$$

and

$$C(i|i') := \langle \psi_i \bar{\psi}_{i'} \rangle_0 = A_{ii'}^{-1} \quad (37.83)$$

denotes the *free covariance*.

Remark 37.31.1 Notice that the free covariance is merely (in the physics notation) the Green-function. One has by Equation (16.29),

$$C(i|i') \equiv -c \mathcal{G}_0(i|i'). \quad (37.84)$$

Remark 37.31.2 The letter C is used here only to keep the diagrammatic combinatorics separate from the conventional Green-function prefactors. The edges of a diagram naturally carry the covariance $C = A^{-1}$; after the conventional normalization one rewrites them as free Green-functions.

Notation 37.32 We consider an index-set $\mathcal{V} = \{1, \dots, n_{\text{int}}\}$ for our interactions and we write the interaction as a finite sum of ordered monomials

$$S_{\text{int}}[\psi, \bar{\psi}] = \sum_{v \in \mathcal{V}} M_v[\psi, \bar{\psi}] \quad (37.85)$$

where every M_v is an ordered sum of monomials with degree $d_v + d'_v$ in the generators of $\mathcal{P}_\zeta$. If

$$\begin{aligned} M_v[\psi, \bar{\psi}] &\equiv \sum_{\substack{1 \leq j_1 < \dots < j_{d_v} \leq D \\ 1 \leq j'_1 < j'_2 < \dots < j'_{d'_v} \leq D}} \bar{\psi}(j_1) \cdots \bar{\psi}(j_{d_v}) m_v(j_1, \dots, j_{d_v}, j'_1, \dots, j'_{d'_v}) \psi(j'_{d'_v}) \cdots \psi(j'_1) \\ &= \frac{1}{d_v! d'_v!} \sum_{\substack{1 \leq j_1 < \dots < j_{d_v} \leq D \\ 1 \leq j'_1 < j'_2 < \dots < j'_{d'_v} \leq D}} \bar{\psi}(j_1) \cdots \bar{\psi}(j_{d_v}) m_v(j_1, \dots, j_{d_v}, j'_1, \dots, j'_{d'_v}) \psi(j'_{d'_v}) \cdots \psi(j'_1) \end{aligned} \quad (37.86)$$

whereas the coefficients $m_v(j_1, \dots, j_{d_v}, j'_1, \dots, j'_{d'_v})$ are allowed to be ...

- complex numbers (simple case),
- elements in a polynomial algebra (general, and more complicated case).

Definition 37.33 For fixed *external indices*

$$(I|I') = (i_1, \dots, i_n | i'_n, \dots, i'_1) \quad (37.87)$$

we consider the ordered monomial

$$X_{(I|I')}[\psi, \bar{\psi}] := \psi_{i_1} \cdots \psi_{i_n} \bar{\psi}_{i'_1} \cdots \bar{\psi}_{i'_n}. \quad (37.88)$$

Fact 37.34 When it comes to the computation of ...

- partition-functions,
- free energies,
- Green-functions,

... we encounter the problem of computing the expectation-value of a polynomial expression of the form

$$P[\psi, \bar{\psi}] \equiv (X_{(I|I')} e^{-S_{\text{int}}}) [\psi, \bar{\psi}] = \left(\sum_{k=0}^{\infty} \frac{(-1)^k}{k!} X_{(I|I')} S_{\text{int}}^k \right) [\psi, \bar{\psi}] \quad (37.89)$$

and since with Equation (37.85) one has

$$S_{\text{int}}^k [\psi, \bar{\psi}] = \left(\sum_{v \in \mathcal{V}} M_v [\psi, \bar{\psi}] \right)^k = \sum_{v_1, \dots, v_k \in \mathcal{V}} M_{v_1} [\psi, \bar{\psi}] \cdots M_{v_k} [\psi, \bar{\psi}] \quad (37.90)$$

The goal is now to map polynomial expressions of the form

$$(X_{(I|I')} M_{v_1} \cdots M_{v_k}) [\psi, \bar{\psi}] \quad (37.91)$$

to pre-graphs (and therefore, subsequently, to Feynman-diagrams). We seek to do so, as we will encounter such polynomial expressions in the perturbative expansion of the partition function, the free energy, and the Green-functions. The following translation-key will allow us to do so. To this end, we now use Equation (37.86) and obtain the frighteningly long expression

$$\begin{aligned} & (X_{(I|I')} M_{v_1} \cdots M_{v_k}) [\psi, \bar{\psi}] \\ &= \psi_{i_1} \cdots \psi_{i_n} \bar{\psi}_{i'_1} \cdots \bar{\psi}_{i'_n} \prod_{l=1}^k \frac{1}{d_{v_l}! d'_{v_l}!} \sum_{\substack{1 \leq j_{1,l} < \dots < j_{d_{v_l},l} \leq D \\ 1 \leq j'_{1,l} < \dots < j'_{d'_{v_l},l} \leq D}} \bar{\psi}(j_{1,l}) \cdots \bar{\psi}(j_{d_{v_l},l}) m_{v_l}(j_{1,l}, \dots, j_{d_{v_l},l}, j'_{1,l}, \dots, j'_{d'_{v_l},l}) \psi(j'_{d'_{v_l},l}) \cdots \psi(j'_{1,l}) \\ &= \sum_{\substack{1 \leq j_{1,1} < \dots < j_{d_{v_1},1} \leq D \\ 1 \leq j'_{1,1} < \dots < j'_{d'_{v_1},1} \leq D \\ \vdots \\ 1 \leq j_{1,k} < \dots < j_{d_{v_k},k} \leq D \\ 1 \leq j'_{1,k} < \dots < j'_{d'_{v_k},k} \leq D}} \psi_{i_1} \cdots \psi_{i_n} \bar{\psi}_{i'_1} \cdots \bar{\psi}_{i'_n} \prod_{l=1}^k \bar{\psi}(j_{1,l}) \cdots \bar{\psi}(j_{d_{v_l},l}) \frac{m_{v_l}(j_{1,l}, \dots, j_{d_{v_l},l}, j'_{1,l}, \dots, j'_{d'_{v_l},l})}{d_{v_l}! d'_{v_l}!} \psi(j'_{d'_{v_l},l}) \cdots \psi(j'_{1,l}) \\ &= \sum_{\substack{1 \leq j_{1,1} < \dots < j_{d_{v_1},1} \leq D \\ 1 \leq j'_{1,1} < \dots < j'_{d'_{v_1},1} \leq D \\ \vdots \\ 1 \leq j_{1,k} < \dots < j_{d_{v_k},k} \leq D \\ 1 \leq j'_{1,k} < \dots < j'_{d'_{v_k},k} \leq D}} \left(\prod_{l=1}^k \frac{m_{v_l}(j_{1,l}, \dots, j_{d_{v_l},l}, j'_{1,l}, \dots, j'_{d'_{v_l},l})}{d_{v_l}! d'_{v_l}!} \right) \psi_{i_1} \cdots \psi_{i_n} \bar{\psi}_{i'_1} \cdots \bar{\psi}_{i'_n} \prod_{l=1}^k \bar{\psi}(j_{1,l}) \cdots \bar{\psi}(j_{d_{v_l},l}) \psi(j'_{d'_{v_l},l}) \cdots \psi(j'_{1,l}) \\ &= \dots \end{aligned} \quad (37.92)$$

TBImplemented: A potential sign-factor that originates from commuting the m_v .

Remark 37.34.1 Notice that we have now for each monomial appearing in the sum of Equation (37.92) in total ...

- $N_{\text{tot}} \equiv 2n + \sum_{l=1}^k (d_{v_l} + d'_{v_l})$ variables
- $N_{\text{non-conj}} \equiv n + \sum_{l=1}^k d'_{v_l}$ are non-conjugated variables
- $N_{\text{conj}} \equiv n + \sum_{l=1}^k d_{v_l}$ are conjugated variables

Definition 37.35 We define the *pregraph* associated to the polynomial expression in Equation (37.92) as the statistical pre-graph with ...

- n external vertices with an outgoing half-edge
- n external vertices with an incoming half-edge
- k internal vertices of type $v_1, \dots, v_k$, where the vertex of type v_l carries d_{v_l} incoming half-edges and d'_{v_l} outgoing half-edges, for $l = 1, \dots, k$.

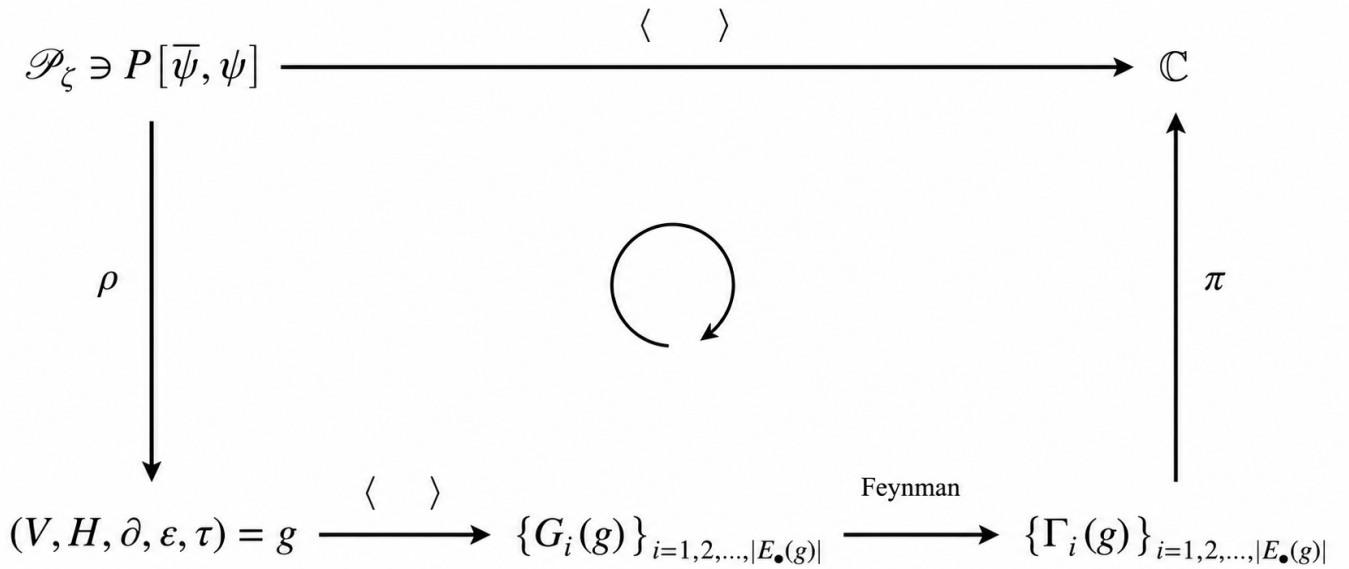


Figure 37.1.: TBD: Implementing this as a tikzpicture.

Fact 37.36 Notice that we can rewrite the polynomial expression in Equation (37.92) as

$$\begin{aligned}
 \dots &= \sum_{\substack{1 \leq j_{1,1} < \dots < j_{d_{v_1},l} \leq D \\ 1 \leq j'_{1,1} < j'_{2,1} < \dots < j'_{d_{v_1},l} \leq D \\ \vdots \\ 1 \leq j_{1,k} < \dots < j_{d_{v_k},k} \leq D \\ 1 \leq j'_{1,k} < j'_{2,k} < \dots < j'_{d_{v_k},k} \leq D}} \left(\prod_{l=1}^k \frac{m_{v_l}(j_{1,l}, \dots, j_{d_{v_l},l}, j'_{1,l}, \dots, j'_{d_{v_l},l})}{d_{v_l}! d_{v_l}'!} \right) \psi_{i_1} \dots \psi_{i_n} \bar{\psi}_{i'_1} \dots \bar{\psi}_{i'_n} \prod_{l=1}^k \bar{\psi}(j_{1,l}) \dots \bar{\psi}(j_{d_{v_l},l}) \psi(j'_{d_{v_l},l}) \dots \psi(j'_{1,l}) \\
 &= \sum_{\substack{1 \leq j_{1,1} < \dots < j_{d_{v_1},l} \leq D \\ 1 \leq j'_{1,1} < j'_{2,1} < \dots < j'_{d_{v_1},l} \leq D \\ \vdots \\ 1 \leq j_{1,k} < \dots < j_{d_{v_k},k} \leq D \\ 1 \leq j'_{1,k} < j'_{2,k} < \dots < j'_{d_{v_k},k} \leq D}} \left(\prod_{l=1}^k \frac{m_{v_l}(j_{1,l}, \dots, j_{d_{v_l},l}, j'_{1,l}, \dots, j'_{d_{v_l},l})}{d_{v_l}! d_{v_l}'!} \right) \psi_{i_1} \dots \psi_{i_n} \bar{\psi}_{i'_1} \dots \bar{\psi}_{i'_n} \left(\prod_{l=1}^k \bar{\psi}(j_{1,l}) \dots \bar{\psi}(j_{d_{v_l},l}) \right) \left(\prod_{l=1}^k \psi(j'_{d_{v_l},l}) \dots \psi(j'_{1,l}) \right) \\
 &= \dots
 \end{aligned} \tag{37.93}$$

in order to get all the conjugated variables to the right and all the not-conjugated variables to the left, we need to apply the commutation-relations. This gives for $\psi(j'_{d_{v_1},1})$ the prefactor $\zeta^{n+\sum_{l=1}^k d_{v_l}}$ this gives in total (applying the same scheme to all of the other variables) the prefactor

$$\zeta^{d'_{v_1}(n+\sum_{l=1}^k d_{v_l})+d'_{v_2}(n+\sum_{l=1}^k d_{v_l})+\dots+d'_{v_k}(n+\sum_{l=1}^k d_{v_l})} = \zeta^{\sum_{i=1}^k d'_{v_i}(n+\sum_{l=1}^k d_{v_l})} \tag{37.94}$$

such that

$$\dots = \zeta^{\sum_{i=1}^k d'_{v_i}(n+\sum_{l=1}^k d_{v_l})} \sum_{\substack{1 \leq j_{1,1} < \dots < j_{d_{v_1},l} \leq D \\ 1 \leq j'_{1,1} < j'_{2,1} < \dots < j'_{d_{v_1},l} \leq D \\ \vdots \\ 1 \leq j_{1,k} < \dots < j_{d_{v_k},k} \leq D \\ 1 \leq j'_{1,k} < j'_{2,k} < \dots < j'_{d_{v_k},k} \leq D}} \left(\prod_{l=1}^k \frac{m_{v_l}(j_{1,l}, \dots, j_{d_{v_l},l}, j'_{1,l}, \dots, j'_{d_{v_l},l})}{d_{v_l}! d_{v_l}'!} \right) \left[\psi_{i_1} \dots \psi_{i_n} \left(\prod_{l=1}^k \psi(j'_{d_{v_l},l}) \dots \psi(j'_{1,l}) \right) \bar{\psi}_{i'_1} \dots \bar{\psi}_{i'_n} \left(\prod_{l=1}^k \bar{\psi}(j_{1,l}) \dots \bar{\psi}(j_{d_{v_l},l}) \right) \right] \tag{37.95}$$

37.2.1. "Experimental Section": The Diagrammatic Expansions

TBD.

38. Experimental: Wannierizations

38.1. Wannierizations

Cf. <https://chatgpt.com/share/6a7cf205-a79c-83eb-8b26-ad5740690599?ogimg=plain>

Definition 38.1 We introduce the *Wannier*-functions associated with a unitary transformation-matrices $U_{mn}(\mathbf{k})$ as

$$w_{n\mathbf{R}}(\mathbf{r}) = \frac{1}{N_k} \sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{R}} \sum_m U_{mn}^{(\mathbf{k})} \psi_{m\mathbf{k}}(\mathbf{r}).$$

Remark 38.1.1 Notice that one finds due to the translational symmetry

$$w_{n\mathbf{R}}(\mathbf{r}) = w_{n0}(\mathbf{r} - \mathbf{R}) \quad (38.1)$$

So every Wannier-function in another cell is just a translated copy of the *home-cell Wannier-function*.

Notation 38.2 Each of these Wannier-functions has an associated *Wannier-center* defined through

$$\boldsymbol{\tau}_n = \langle w_{n0} | \mathbf{r} | w_{n0} \rangle \quad (38.2)$$

with

$$\{\boldsymbol{\tau}_1, \boldsymbol{\tau}_2, \boldsymbol{\tau}_3, \dots\} \subset \text{UC} \quad (38.3)$$

Corollary 38.3 We enumerate the Wannier-functions with their band- and lattice-index as $(n, \mathbf{R})$ and obtain for the Coulomb-matrix-element in second quantization

$$U_{n_1\mathbf{R}_1, n_2\mathbf{R}_2, n_3\mathbf{R}_3, n_4\mathbf{R}_4} = \iint d\mathbf{r} d\mathbf{r}' w_{n_1\mathbf{R}_1}^*(\mathbf{r}) w_{n_2\mathbf{R}_2}^*(\mathbf{r}') \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} w_{n_3\mathbf{R}_3}(\mathbf{r}') w_{n_4\mathbf{R}_4}(\mathbf{r}).$$

such that the whole Coulomb-interaction can be written as

$$\sum_{\substack{n_1, n_2, n_3, n_4 \in \{1, 2, 3, \dots\} \\ \mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3, \mathbf{R}_4 \in L_f}} U_{n_1\mathbf{R}_1, n_2\mathbf{R}_2, n_3\mathbf{R}_3, n_4\mathbf{R}_4} c_{n_1\mathbf{R}_1}^\dagger c_{n_2\mathbf{R}_2}^\dagger c_{n_4\mathbf{R}_4} c_{n_3\mathbf{R}_3} \quad (38.4)$$

Proof 38.3 Follows from the second quantization-procedure.

Lemma 38.4 One finds for the Coulomb matrix element

$$U_{n_1 \mathbf{R}_1, n_2 \mathbf{R}_2, n_3 \mathbf{R}_3, n_4 \mathbf{R}_4} = U_{n_1 \mathbf{0}, n_2 (\mathbf{R}_2 - \mathbf{R}_1), n_3 (\mathbf{R}_3 - \mathbf{R}_1), n_4 (\mathbf{R}_4 - \mathbf{R}_1)}. \quad (38.5)$$

Hence, the Coulomb matrix elements depend only on the relative positions of the Wannier functions.

Proof 38.4 The Wannier functions obey the translational property

$$w_{n\mathbf{R}}(\mathbf{r}) = w_{n\mathbf{0}}(\mathbf{r} - \mathbf{R}). \quad (38.6)$$

Starting from the Coulomb matrix element

$$U_{n_1 \mathbf{R}_1, n_2 \mathbf{R}_2, n_3 \mathbf{R}_3, n_4 \mathbf{R}_4} = \iint d\mathbf{r} d\mathbf{r}' w_{n_1 \mathbf{R}_1}^*(\mathbf{r}) w_{n_2 \mathbf{R}_2}^*(\mathbf{r}') \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} w_{n_3 \mathbf{R}_3}(\mathbf{r}') w_{n_4 \mathbf{R}_4}(\mathbf{r}), \quad (38.7)$$

we perform the change of variables

$$\mathbf{r} = \tilde{\mathbf{r}} + \mathbf{R}_1, \quad \mathbf{r}' = \tilde{\mathbf{r}}' + \mathbf{R}_1. \quad (38.8)$$

Using translational invariance, the Wannier functions transform as

$$w_{n_i \mathbf{R}_i}(\tilde{\mathbf{r}} + \mathbf{R}_1) = w_{n_i, \mathbf{R}_i - \mathbf{R}_1}(\tilde{\mathbf{r}}), \quad (38.9)$$

and analogously for the functions depending on

$\tilde{\mathbf{r}}'$. Furthermore,

$$|(\tilde{\mathbf{r}} + \mathbf{R}_1) - (\tilde{\mathbf{r}}' + \mathbf{R}_1)| = |\tilde{\mathbf{r}} - \tilde{\mathbf{r}}'|. \quad (38.10)$$

The integration measure is unchanged, $d\mathbf{r} d\mathbf{r}' = d\tilde{\mathbf{r}} d\tilde{\mathbf{r}}'$.

Consequently,

$$U_{n_1 \mathbf{R}_1, n_2 \mathbf{R}_2, n_3 \mathbf{R}_3, n_4 \mathbf{R}_4} = U_{n_1 \mathbf{0}, n_2 (\mathbf{R}_2 - \mathbf{R}_1), n_3 (\mathbf{R}_3 - \mathbf{R}_1), n_4 (\mathbf{R}_4 - \mathbf{R}_1)}, \quad (38.11)$$

which proves the assertion.

Definition 38.5

We introduce the *pair densities*

$$\rho_{1,4}(\mathbf{r}) = w_{n_1\mathbf{R}_1}^*(\mathbf{r})w_{n_4\mathbf{R}_4}(\mathbf{r}) \quad (38.12)$$

and

$$\rho_{2,3}(\mathbf{r}') = w_{n_2\mathbf{R}_2}^*(\mathbf{r}')w_{n_3\mathbf{R}_3}(\mathbf{r}'). \quad (38.13)$$

In terms of these pair densities, the Coulomb matrix element becomes

$$U_{1,2,3,4} = \iint d^3\mathbf{r} d^3\mathbf{r}' \rho_{1,4}(\mathbf{r}) \frac{e^2}{|\mathbf{r} - \mathbf{r}'|} \rho_{2,3}(\mathbf{r}'). \quad (38.14)$$

Theorem 38.6 The Coulomb matrix element can be rewritten in reciprocal space as

$$U_{1,2,3,4} = e^2 \int \frac{d^3\mathbf{q}}{(2\pi)^3} \frac{4\pi}{q^2} \rho_{1,4}(-\mathbf{q}) \rho_{2,3}(\mathbf{q}), \quad (38.15)$$

where the Fourier transform of a pair density is defined by

$$\rho_{i,j}(\mathbf{q}) = \int d^3\mathbf{r} e^{-i\mathbf{q}\cdot\mathbf{r}} \rho_{i,j}(\mathbf{r}). \quad (38.16)$$

Proof 38.6 We begin with the Fourier representation of the Coulomb kernel,

$$\frac{1}{|\mathbf{r} - \mathbf{r}'|} = \int \frac{d^3\mathbf{q}}{(2\pi)^3} \frac{4\pi}{q^2} e^{i\mathbf{q}\cdot(\mathbf{r}-\mathbf{r}')}. \quad (38.17)$$

Inserting this expression into the Coulomb matrix element gives

$$U_{1,2,3,4} = e^2 \iint d^3\mathbf{r} d^3\mathbf{r}' \rho_{1,4}(\mathbf{r}) \left[\int \frac{d^3\mathbf{q}}{(2\pi)^3} \frac{4\pi}{q^2} e^{i\mathbf{q}\cdot(\mathbf{r}-\mathbf{r}')} \right] \rho_{2,3}(\mathbf{r}'). \quad (38.18)$$

Using

$$e^{i\mathbf{q}\cdot(\mathbf{r}-\mathbf{r}')} = e^{i\mathbf{q}\cdot\mathbf{r}} e^{-i\mathbf{q}\cdot\mathbf{r}'}, \quad (38.19)$$

the integrations over $\mathbf{r}$ and $\mathbf{r}'$ separate, such that

$$U_{1,2,3,4} = e^2 \int \frac{d^3\mathbf{q}}{(2\pi)^3} \frac{4\pi}{q^2} \left[\int d^3\mathbf{r} \rho_{1,4}(\mathbf{r}) e^{i\mathbf{q}\cdot\mathbf{r}} \right] \left[\int d^3\mathbf{r}' \rho_{2,3}(\mathbf{r}') e^{-i\mathbf{q}\cdot\mathbf{r}'} \right]. \quad (38.20)$$

By the definition of the Fourier-transformed pair densities,

$$\int d^3\mathbf{r} \rho_{1,4}(\mathbf{r})e^{i\mathbf{q}\cdot\mathbf{r}} = \rho_{1,4}(-\mathbf{q}), \quad (38.21)$$

and

$$\int d^3\mathbf{r}' \rho_{2,3}(\mathbf{r}')e^{-i\mathbf{q}\cdot\mathbf{r}'} = \rho_{2,3}(\mathbf{q}). \quad (38.22)$$

Hence,

$$U_{1,2,3,4} = e^2 \int \frac{d^3\mathbf{q}}{(2\pi)^3} \frac{4\pi}{q^2} \rho_{1,4}(-\mathbf{q})\rho_{2,3}(\mathbf{q}). \quad (38.23)$$

39. Experimental: Superconductivity

39.1. Functionally Generalized Eliashberg-Theory

[TBCopied from my other files.]

How about THIS:

- first modelling the doping with a virtual crystal approximation in DFT + LDA
- then doing DFPT to get the phonons and related couplings constants (e.g. phonon-phonon and electron-phonon constants) within this approximation
- then doing a wannierization to get a basis in real-space.
- Then computing the matrix elements for hartree, exchange correlation, Coulomb and electron-phonon and phonon-phonon in the wannier-basis
- Then doing an FRG in order to get effective coupling-constants in the Wannier-basis
- then transforming back to the reciprocal basis in order to get effective four-point-vertices
- then doing a mean-field and Eliashberg-type theory in order to get superconducting temperatures and gaps

40. Experimental: Logarithm of a Grassmann-Polynomial

We start from

$$\log (1+\eta)=\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \eta^n \quad (40.1)$$

This gives therefore

$$\partial_{\eta} \log (1+\eta)=\sum_{n=1}^{\infty}(-1)^{n+1} \eta^{n-1}=\ldots \quad (40.2)$$

now shifting the index as $m=n-1$ gives

$$\ldots=\sum_{m=0}^{\infty}(-1)^m \eta^m \quad (40.3)$$

Notice that this is exactly the expression for the geometric series as

$$(1+\eta)^{-1}=\sum_{m=0}^{\infty}(-1)^m \eta^m \quad (40.4)$$

Therefore we see that

$$\partial_{\eta} \log (1+\eta)=(1+\eta)^{-1} . \quad (40.5)$$

41. Experimental: Conductivities from Linear Response

41.1. Electrical Conductivity

41.2. Thermal Conductivity

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